Neutron Documentation

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Neutron development team

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CONTENTS

Neutron is an OpenStack project to provide network connectivity as a service between interface devices (e.g., vNICs) managed by other OpenStack services (e.g., nova). It implements the OpenStack Networking API.

This documentation is generated by the Sphinx toolkit and lives in the source tree. Additional documentation on Neutron and other components of OpenStack can be found on the OpenStack wiki and the *Neutron section of the wiki*. The Neutron Development wiki is also a good resource for new contributors.

Enjoy!

CHAPTER

ONE

OVERVIEW

The OpenStack project is an open source cloud computing platform that supports all types of cloud environments. The project aims for simple implementation, massive scalability, and a rich set of features. Cloud computing experts from around the world contribute to the project.

OpenStack provides an Infrastructure-as-a-Service (IaaS) solution through a variety of complementary services. Each service offers an Application Programming Interface (API) that facilitates this integration.

This guide covers step-by-step deployment of the major OpenStack services using a functional example architecture suitable for new users of OpenStack with sufficient Linux experience. This guide is not intended to be used for production system installations, but to create a minimum proof-of-concept for the purpose of learning about OpenStack.

After becoming familiar with basic installation, configuration, operation, and troubleshooting of these OpenStack services, you should consider the following steps toward deployment using a production architecture:

- Determine and implement the necessary core and optional services to meet performance and redundancy requirements.
- Increase security using methods such as firewalls, encryption, and service policies.
- Implement a deployment tool such as Ansible, Chef, Puppet, or Salt to automate deployment and management of the production environment.

1.1 Example architecture

The example architecture requires at least two nodes (hosts) to launch a basic virtual machine (VM) or instance. Optional services such as Block Storage and Object Storage require additional nodes.

Important

The example architecture used in this guide is a minimum configuration, and is not intended for production system installations. It is designed to provide a minimum proof-of-concept for the purpose of learning about OpenStack. For information on creating architectures for specific use cases, or how to determine which architecture is required, see the Architecture Design Guide.

This example architecture differs from a minimal production architecture as follows:

- Networking agents reside on the controller node instead of one or more dedicated network nodes.
- Overlay (tunnel) traffic for self-service networks traverses the management network instead of a dedicated network.

For more information on production architectures, see the Architecture Design Guide, OpenStack Operations Guide, and *OpenStack Networking Guide*.

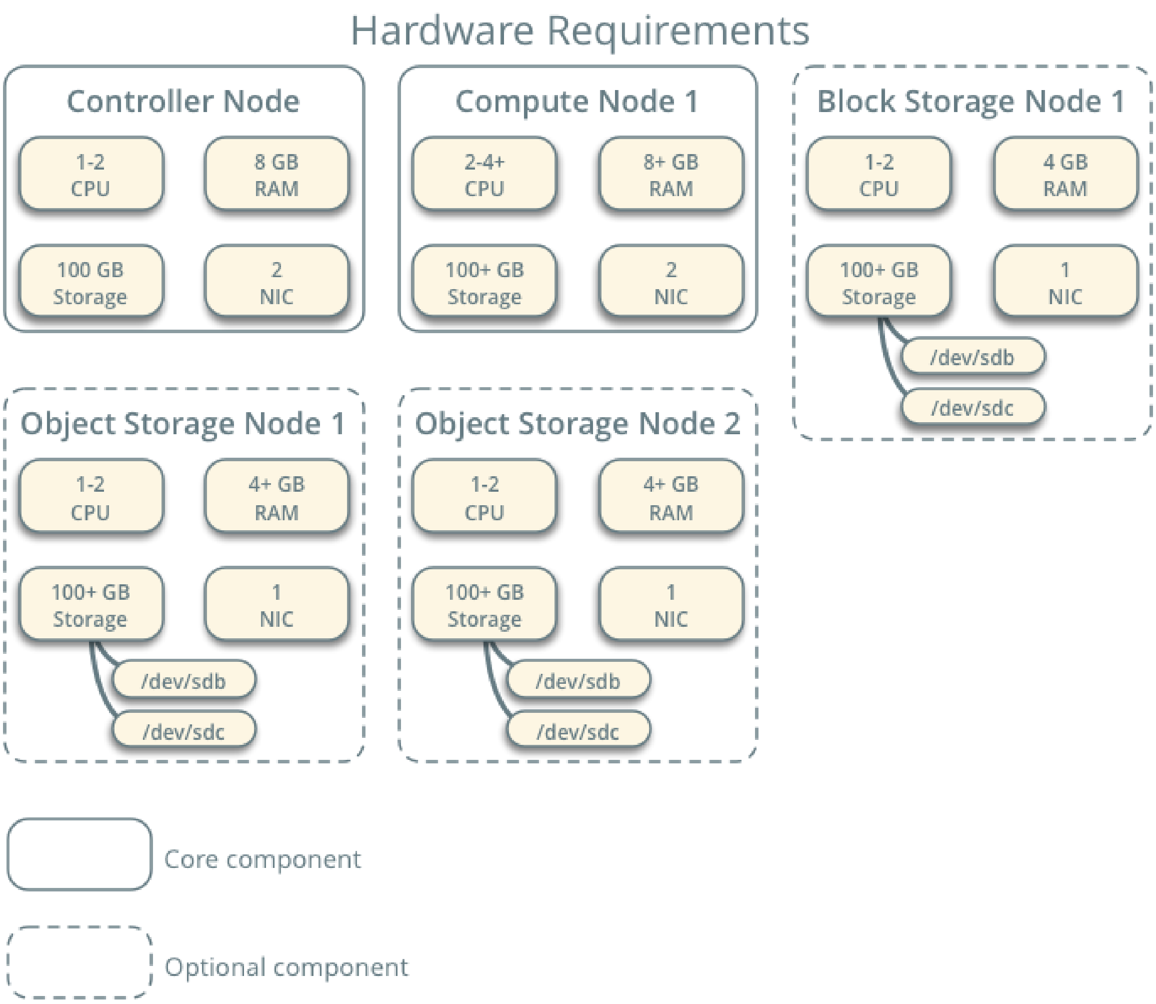


Fig. 1: **Hardware requirements**

1.1.1 Controller

The controller node runs the Identity service, Image service, management portions of Compute, management portion of Networking, various Networking agents, and the Dashboard. It also includes supporting services such as an SQL database, message queue, and Network Time Protocol (NTP).

Optionally, the controller node runs portions of the Block Storage, Object Storage, Orchestration, and Telemetry services.

The controller node requires a minimum of two network interfaces.

1.1.2 Compute

The compute node runs the hypervisor portion of Compute that operates instances. By default, Compute uses the kernel-based VM (KVM) hypervisor. The compute node also runs a Networking service agent that connects instances to virtual networks and provides firewalling services to instances via security groups.

You can deploy more than one compute node. Each node requires a minimum of two network interfaces.

1.1.3 Block Storage

The optional Block Storage node contains the disks that the Block Storage and Shared File System services provision for instances.

For simplicity, service traffic between compute nodes and this node uses the management network. Production environments should implement a separate storage network to increase performance and security.

You can deploy more than one block storage node. Each node requires a minimum of one network interface.

1.1.4 Object Storage

The optional Object Storage node contain the disks that the Object Storage service uses for storing accounts, containers, and objects.

For simplicity, service traffic between compute nodes and this node uses the management network. Production environments should implement a separate storage network to increase performance and security.

This service requires two nodes. Each node requires a minimum of one network interface. You can deploy more than two object storage nodes.

1.2 Networking

Choose one of the following virtual networking options.

1.2.1 Networking Option 1: Provider networks

The provider networks option deploys the OpenStack Networking service in the simplest way possible with primarily layer-2 (bridging/switching) services and VLAN segmentation of networks. Essentially, it bridges virtual networks to physical networks and relies on physical network infrastructure for layer-3 (routing) services. Additionally, a DHCP<Dynamic Host Configuration Protocol (DHCP) service provides IP address information to instances.

The OpenStack user requires more information about the underlying network infrastructure to create a virtual network to exactly match the infrastructure.

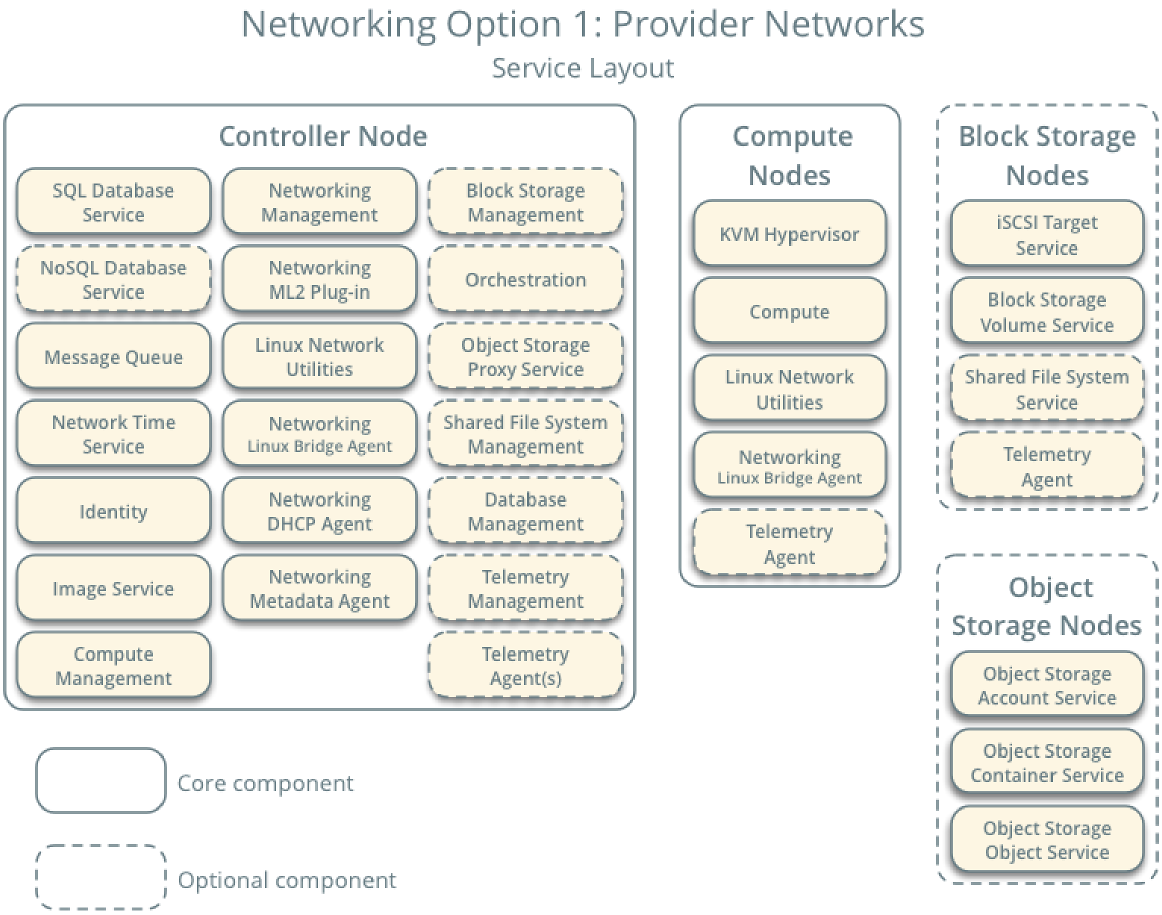
Warning

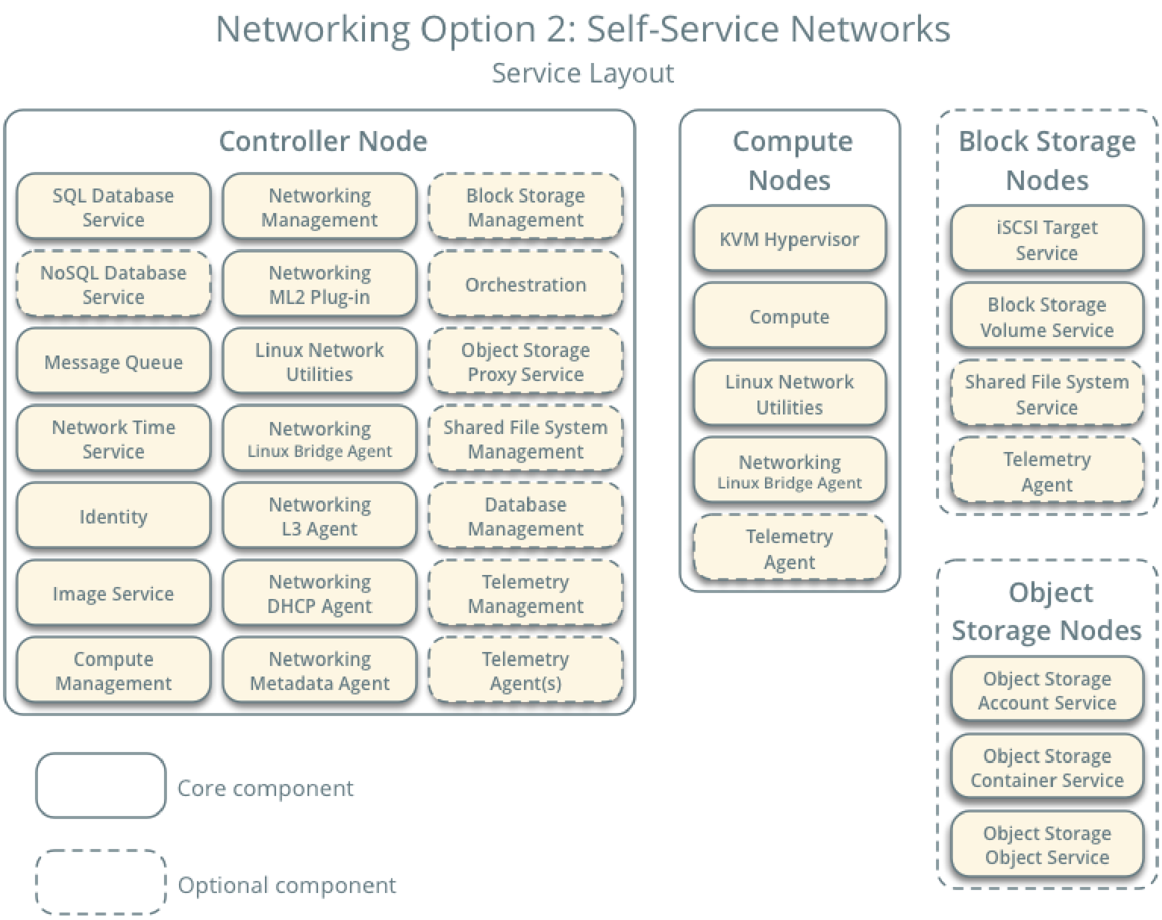
This option lacks support for self-service (private) networks, layer-3 (routing) services, and advanced services such as FireWall-as-a-Service (FWaaS). Consider the self-service networks option below if you desire these features.

1.2.2 Networking Option 2: Self-service networks

The self-service networks option augments the provider networks option with layer-3 (routing) services that enable self-service networks using overlay segmentation methods such as Virtual Extensible LAN (VXLAN). Essentially, it routes virtual networks to physical networks using Network Address Translation (NAT). Additionally, this option provides the foundation for advanced services such as FWaaS.

The OpenStack user can create virtual networks without the knowledge of underlying infrastructure on the data network. This can also include VLAN networks if the layer-2 plug-in is configured accordingly.


Networking Option 1: Provider Networks
Service Layout
Controller Node
SQL Database Service
Networking Management
Block Storage Management
NoSQL Database Service
Networking ML2 Plug-in
Orchestration
Message Queue
Linux Network Utilities
Object Storage Proxy Service
Network Time Service
Networking Linux Bridge Agent
Shared File System Management
Identity
Networking DHCP Agent
Database Management
Image Service
Networking Metadata Agent
Telemetry Management
Compute Management
Telemetry Agent(s)
Compute Nodes
KVM Hypervisor
Compute
Linux Network Utilities
Networking Linux Bridge Agent
Telemetry Agent
Block Storage Nodes
iSCSI Target Service
Block Storage Volume Service
Shared File System Service
Telemetry Agent
Object Storage Nodes
Object Storage Account Service
Object Storage Container Service
Object Storage Object Service
Core component
Optional component


Networking Option 2: Self-Service Networks
Service Layout
Controller Node
SQL Database Service
Networking Management
Block Storage Management
NoSQL Database Service
Networking ML2 Plug-in
Orchestration
Message Queue
Linux Network Utilities
Object Storage Proxy Service
Network Time Service
Networking Linux Bridge Agent
Shared File System Management
Identity
Networking L3 Agent
Database Management
Image Service
Networking DHCP Agent
Telemetry Management
Compute Management
Networking Metadata Agent
Telemetry Agent(s)
Compute Nodes
KVM Hypervisor
Compute
Linux Network Utilities
Networking Linux Bridge Agent
Telemetry Agent
Block Storage Nodes
iSCSI Target Service
Block Storage Volume Service
Shared File System Service
Telemetry Agent
Object Storage Nodes
Object Storage Account Service
Object Storage Container Service
Object Storage Object Service
Core component
Optional component

CHAPTER

TWO

NETWORKING SERVICE OVERVIEW

OpenStack Networking (neutron) allows you to create and attach interface devices managed by other OpenStack services to networks. Plug-ins can be implemented to accommodate different networking equipment and software, providing flexibility to OpenStack architecture and deployment.

It includes the following components:

neutron-server
: Accepts and routes API requests to the appropriate OpenStack Networking plug-in for action.

OpenStack Networking plug-ins and agents
: Plug and unplug ports, create networks or subnets, and provide IP addressing. These plug-ins and agents differ depending on the vendor and technologies used in the particular cloud. OpenStack Networking ships with plug-ins and agents for Open vSwitch and Open Virtual Network (OVN), as well as for SR-IOV and Macvtap.

  The common agents are L3 (layer 3), DHCP (dynamic host IP addressing), and a plug-in agent.

Messaging queue
: Used by most OpenStack Networking installations to route information between the neutron-server and various agents. Also acts as a database to store networking state for particular plug-ins.

OpenStack Networking mainly interacts with OpenStack Compute to provide networks and connectivity for its instances.

CHAPTER

THREE

NETWORKING (NEUTRON) CONCEPTS

OpenStack Networking (neutron) manages all networking facets for the Virtual Networking Infrastructure (VNI) and the access layer aspects of the Physical Networking Infrastructure (PNI) in your OpenStack environment. OpenStack Networking enables projects to create advanced virtual network topologies which may include services such as a firewall, and a virtual private network (VPN).

Networking provides networks, subnets, and routers as object abstractions. Each abstraction has functionality that mimics its physical counterpart: networks contain subnets, and routers route traffic between different subnets and networks.

Any given Networking set up has at least one external network. Unlike the other networks, the external network is not merely a virtually defined network. Instead, it represents a view into a slice of the physical, external network accessible outside the OpenStack installation. IP addresses on the external network are accessible by anybody physically on the outside network.

In addition to external networks, any Networking set up has one or more internal networks. These software-defined networks connect directly to the VMs. Only the VMs on any given internal network, or those on subnets connected through interfaces to a similar router, can access VMs connected to that network directly.

For the outside network to access VMs, and vice versa, routers between the networks are needed. Each router has one gateway that is connected to an external network and one or more interfaces connected to internal networks. Like a physical router, subnets can access machines on other subnets that are connected to the same router, and machines can access the outside network through the gateway for the router.

Additionally, you can allocate IP addresses on external networks to ports on the internal network. Whenever something is connected to a subnet, that connection is called a port. You can associate external network IP addresses with ports to VMs. This way, entities on the outside network can access VMs.

Networking also supports *security groups*. Security groups enable administrators to define firewall rules in groups. A VM can belong to one or more security groups, and Networking applies the rules in those security groups to block or unblock ports, port ranges, or traffic types for that VM.

Each plug-in that Networking uses has its own concepts. While not vital to operating the VNI and OpenStack environment, understanding these concepts can help you set up Networking. All Networking installations use a core plug-in and a security group plug-in (or just the No-Op security group plug-in). Additionally, Firewall-as-a-Service (FWaaS) is available.

CHAPTER

FOUR

INSTALL AND CONFIGURE FOR RED HAT ENTERPRISE LINUX AND CENTOS

4.1 Host networking

After installing the operating system on each node for the architecture that you choose to deploy, you must configure the network interfaces. We recommend that you disable any automated network management tools and manually edit the appropriate configuration files for your distribution. For more information on how to configure networking on your distribution, see the documentation .

All nodes require Internet access for administrative purposes such as package installation, security updates, Domain Name System (DNS), and Network Time Protocol (NTP). In most cases, nodes should obtain Internet access through the management network interface. To highlight the importance of network separation, the example architectures use private address space for the management network and assume that the physical network infrastructure provides Internet access via Network Address Translation (NAT) or other methods. The example architectures use routable IP address space for the provider (external) network and assume that the physical network infrastructure provides direct Internet access.

In the provider networks architecture, all instances attach directly to the provider network. In the self-service (private) networks architecture, instances can attach to a self-service or provider network. Self-service networks can reside entirely within OpenStack or provide some level of external network access using Network Address Translation (NAT) through the provider network.

The example architectures assume use of the following networks:

- Management on 10.0.0.0/24 with gateway 10.0.0.1

  This network requires a gateway to provide Internet access to all nodes for administrative purposes such as package installation, security updates, Domain Name System (DNS), and Network Time Protocol (NTP).

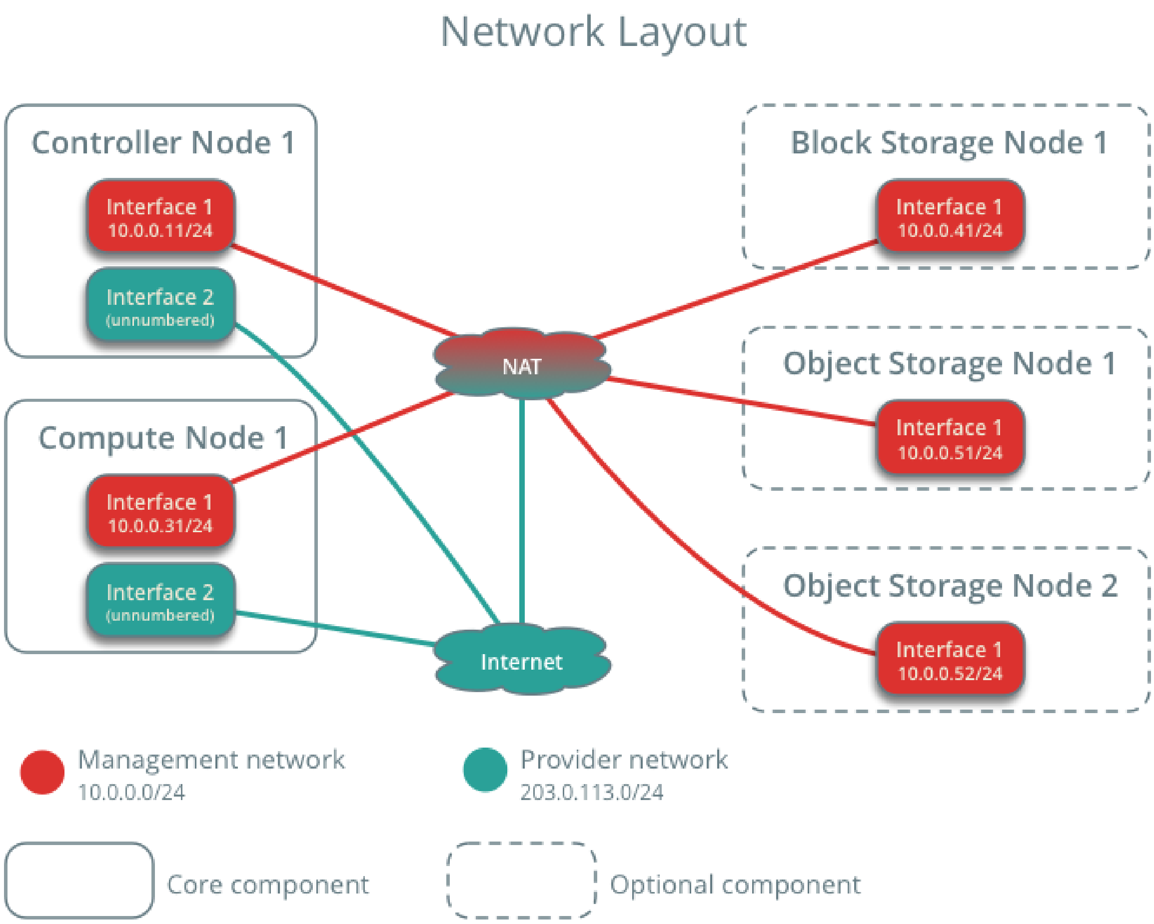
- Provider on 203.0.113.0/24 with gateway 203.0.113.1

  This network requires a gateway to provide Internet access to instances in your OpenStack environment.

You can modify these ranges and gateways to work with your particular network infrastructure.

Network interface names vary by distribution. Traditionally, interfaces use `eth` followed by a sequential number. To cover all variations, this guide refers to the first interface as the interface with the lowest number and the second interface as the interface with the highest number.

Unless you intend to use the exact configuration provided in this example architecture, you must modify the networks in this procedure to match your environment. Each node must resolve the other nodes by name in addition to IP address. For example, the `controller` name must resolve to `10.0.0.11`, the IP address of the management interface on the controller node.


Network Layout
Controller Node 1
Interface 1
10.0.0.11/24
Interface 2
(unnumbered)
Compute Node 1
Interface 1
10.0.0.31/24
Interface 2
(unnumbered)
NAT
Internet
Block Storage Node 1
Interface 1
10.0.0.41/24
Object Storage Node 1
Interface 1
10.0.0.51/24
Object Storage Node 2
Interface 1
10.0.0.52/24
Management network
10.0.0.0/24
Provider network
203.0.113.0/24
Core component
Optional component

Warning

Reconfiguring network interfaces will interrupt network connectivity. We recommend using a local terminal session for these procedures.

Note

Your distribution enables a restrictive firewall by default. During the installation process, certain steps will fail unless you alter or disable the firewall. For more information about securing your environment, refer to the OpenStack Security Guide.

4.1.1 Controller node

Configure network interfaces

1. Configure the first interface as the management interface:

   IP address: 10.0.0.11

   Network mask: 255.255.255.0 (or /24)

   Default gateway: 10.0.0.1

2. The provider interface uses a special configuration without an IP address assigned to it. Configure the second interface as the provider interface:

   Replace `INTERFACE_NAME` with the actual interface name. For example, *eth1* or *ens224*.

- Edit the `/etc/sysconfig/network-scripts/ifcfg-INTERFACE_NAME` file to contain the following:

  Do not change the `HWADDR` and `UUID` keys.

```
DEVICE=INTERFACE_NAME
TYPE=Ethernet
ONBOOT="yes"
BOOTPROTO="none"
```

1. Reboot the system to activate the changes.

Configure name resolution

1. Set the hostname of the node to `controller`.
2. Edit the `/etc/hosts` file to contain the following:

```
# controller
10.0.0.11       controller

# compute1
10.0.0.31       compute1

# block1
10.0.0.41       block1
```

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```
# object1
10.0.0.51       object1

# object2
10.0.0.52       object2
```

Warning

Some distributions add an extraneous entry in the `/etc/hosts` file that resolves the actual hostname to another loopback IP address such as `127.0.1.1`. You must comment out or remove this entry to prevent name resolution problems. **Do not remove the 127.0.0.1 entry.**

Note

This guide includes host entries for optional services in order to reduce complexity should you choose to deploy them.

4.1.2 Compute node

Configure network interfaces

1. Configure the first interface as the management interface:

   IP address: 10.0.0.31

   Network mask: 255.255.255.0 (or /24)

   Default gateway: 10.0.0.1

   Note

   Additional compute nodes should use 10.0.0.32, 10.0.0.33, and so on.

2. The provider interface uses a special configuration without an IP address assigned to it. Configure the second interface as the provider interface:

   Replace `INTERFACE_NAME` with the actual interface name. For example, *eth1* or *ens224*.

- Edit the `/etc/sysconfig/network-scripts/ifcfg-INTERFACE_NAME` file to contain the following:

  Do not change the `HWADDR` and `UUID` keys.

```
DEVICE=INTERFACE_NAME
TYPE=Ethernet
ONBOOT="yes"
BOOTPROTO="none"
```

1. Reboot the system to activate the changes.

Configure name resolution

1. Set the hostname of the node to `compute1`.
2. Edit the `/etc/hosts` file to contain the following:

```
# controller
10.0.0.11       controller

# compute1
10.0.0.31       compute1

# block1
10.0.0.41       block1

# object1
10.0.0.51       object1

# object2
10.0.0.52       object2
```

> **Warning**
>
> Some distributions add an extraneous entry in the `/etc/hosts` file that resolves the actual hostname to another loopback IP address such as `127.0.1.1`. You must comment out or remove this entry to prevent name resolution problems. **Do not remove the 127.0.0.1 entry.**

> **Note**
>
> This guide includes host entries for optional services in order to reduce complexity should you choose to deploy them.

4.1.3 Block storage node (Optional)

If you want to deploy the Block Storage service, configure one additional storage node.

Configure network interfaces

- Configure the management interface:
  - IP address: `10.0.0.41`
  - Network mask: `255.255.255.0` (or `/24`)
  - Default gateway: `10.0.0.1`

Configure name resolution

1. Set the hostname of the node to `block1`.
2. Edit the `/etc/hosts` file to contain the following:

```
# controller
10.0.0.11       controller

# compute1
10.0.0.31       compute1

# block1
10.0.0.41       block1

# object1
10.0.0.51       object1

# object2
10.0.0.52       object2
```

Warning

Some distributions add an extraneous entry in the `/etc/hosts` file that resolves the actual hostname to another loopback IP address such as `127.0.1.1`. You must comment out or remove this entry to prevent name resolution problems. **Do not remove the 127.0.0.1 entry.**

Note

This guide includes host entries for optional services in order to reduce complexity should you choose to deploy them.

3. Reboot the system to activate the changes.

4.1.4 Verify connectivity

We recommend that you verify network connectivity to the Internet and among the nodes before proceeding further.

1. From the *controller* node, test access to the Internet:

```
# ping -c 4 openstack.org

PING openstack.org (174.143.194.225) 56(84) bytes of data.
64 bytes from 174.143.194.225: icmp_seq=1 ttl=54 time=18.3 ms
64 bytes from 174.143.194.225: icmp_seq=2 ttl=54 time=17.5 ms
64 bytes from 174.143.194.225: icmp_seq=3 ttl=54 time=17.5 ms
64 bytes from 174.143.194.225: icmp_seq=4 ttl=54 time=17.4 ms

--- openstack.org ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3022ms
rtt min/avg/max/mdev = 17.489/17.715/18.346/0.364 ms
```

2. From the *controller* node, test access to the management interface on the *compute* node:

```
# ping -c 4 compute1

PING compute1 (10.0.0.31) 56(84) bytes of data.
64 bytes from compute1 (10.0.0.31): icmp_seq=1 ttl=64 time=0.263 ms
64 bytes from compute1 (10.0.0.31): icmp_seq=2 ttl=64 time=0.202 ms
64 bytes from compute1 (10.0.0.31): icmp_seq=3 ttl=64 time=0.203 ms
64 bytes from compute1 (10.0.0.31): icmp_seq=4 ttl=64 time=0.202 ms

--- compute1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3000ms
rtt min/avg/max/mdev = 0.202/0.217/0.263/0.030 ms
```

3. From the *compute* node, test access to the Internet:

```
# ping -c 4 openstack.org

PING openstack.org (174.143.194.225) 56(84) bytes of data.
64 bytes from 174.143.194.225: icmp_seq=1 ttl=54 time=18.3 ms
64 bytes from 174.143.194.225: icmp_seq=2 ttl=54 time=17.5 ms
64 bytes from 174.143.194.225: icmp_seq=3 ttl=54 time=17.5 ms
64 bytes from 174.143.194.225: icmp_seq=4 ttl=54 time=17.4 ms

--- openstack.org ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3022ms
rtt min/avg/max/mdev = 17.489/17.715/18.346/0.364 ms
```

4. From the *compute* node, test access to the management interface on the *controller* node:

```
# ping -c 4 controller

PING controller (10.0.0.11) 56(84) bytes of data.
64 bytes from controller (10.0.0.11): icmp_seq=1 ttl=64 time=0.263 ms
64 bytes from controller (10.0.0.11): icmp_seq=2 ttl=64 time=0.202 ms
64 bytes from controller (10.0.0.11): icmp_seq=3 ttl=64 time=0.203 ms
64 bytes from controller (10.0.0.11): icmp_seq=4 ttl=64 time=0.202 ms

--- controller ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3000ms
rtt min/avg/max/mdev = 0.202/0.217/0.263/0.030 ms
```

Note

Your distribution enables a restrictive firewall by default. During the installation process, certain steps will fail unless you alter or disable the firewall. For more information about securing your environment, refer to the OpenStack Security Guide.

4.2 Install and configure controller node

4.2.1 Prerequisites

Before you configure the OpenStack Networking (neutron) service, you must create a database, service credentials, and API endpoints.

1. To create the database, complete these steps:

   - Use the database access client to connect to the database server as the `root` user:

     ```
     $ mysql -u root -p
     ```

   - Create the `neutron` database:

     ```
     MariaDB [(none)]> CREATE DATABASE neutron;
     ```

   - Grant proper access to the `neutron` database, replacing `NEUTRON_DBPASS` with a suitable password:

     ```
     MariaDB [(none)]> GRANT ALL PRIVILEGES ON neutron.* TO 'neutron'@
     ↪'localhost' \
       IDENTIFIED BY 'NEUTRON_DBPASS';
     MariaDB [(none)]> GRANT ALL PRIVILEGES ON neutron.* TO 'neutron'@'%'␣
     ↪\
       IDENTIFIED BY 'NEUTRON_DBPASS';
     ```

   - Exit the database access client.

2. Source the `admin` credentials to gain access to admin-only CLI commands:

   ```
   $ . admin-openrc
   ```

3. To create the service credentials, complete these steps:

   - Create the `neutron` user:

     ```
     $ openstack user create --domain default --password-prompt neutron

     User Password:
     Repeat User Password:
     +---------------------+----------------------------------+
     | Field               | Value                            |
     +---------------------+----------------------------------+
     | domain_id           | default                          |
     | enabled             | True                             |
     | id                  | fdb0f541e28141719b6a43c8944bf1fb |
     | name                | neutron                          |
     | options             | {}                               |
     | password_expires_at | None                             |
     +---------------------+----------------------------------+
     ```

   - Add the `admin` role to the `neutron` user:

```
$ openstack role add --project service --user neutron admin
```

Note

This command provides no output.

- Create the `neutron` service entity:

```
$ openstack service create --name neutron \
  --description "OpenStack Networking" network

+-------------+----------------------------------+
| Field       | Value                            |
+-------------+----------------------------------+
| description | OpenStack Networking             |
| enabled     | True                             |
| id          | f71529314dab4a4d8eca427e701d209e |
| name        | neutron                          |
| type        | network                          |
+-------------+----------------------------------+
```

4. Create the Networking service API endpoints:

```
$ openstack endpoint create --region RegionOne \
  network public http://controller:9696

+--------------+----------------------------------+
| Field        | Value                            |
+--------------+----------------------------------+
| enabled      | True                             |
| id           | 85d80a6d02fc4b7683f611d7fc1493a3 |
| interface    | public                           |
| region       | RegionOne                        |
| region_id    | RegionOne                        |
| service_id   | f71529314dab4a4d8eca427e701d209e |
| service_name | neutron                          |
| service_type | network                          |
| url          | http://controller:9696           |
+--------------+----------------------------------+

$ openstack endpoint create --region RegionOne \
  network internal http://controller:9696

+--------------+----------------------------------+
| Field        | Value                            |
+--------------+----------------------------------+
| enabled      | True                             |
| id           | 09753b537ac74422a68d2d791cf3714f |
| interface    | internal                         |
```

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```
| region       | RegionOne                        |
| region_id    | RegionOne                        |
| service_id   | f71529314dab4a4d8eca427e701d209e |
| service_name | neutron                          |
| service_type | network                          |
| url          | http://controller:9696           |
+--------------+----------------------------------+

$ openstack endpoint create --region RegionOne \
  network admin http://controller:9696

+--------------+----------------------------------+
| Field        | Value                            |
+--------------+----------------------------------+
| enabled      | True                             |
| id           | 1ee14289c9374dffb5db92a5c112fc4e |
| interface    | admin                            |
| region       | RegionOne                        |
| region_id    | RegionOne                        |
| service_id   | f71529314dab4a4d8eca427e701d209e |
| service_name | neutron                          |
| service_type | network                          |
| url          | http://controller:9696           |
+--------------+----------------------------------+
```

4.2.2 Configure networking options

You can deploy the Networking service using one of two architectures represented by options 1 and 2.

Option 1 deploys the simplest possible architecture that only supports attaching instances to provider (external) networks. No self-service (private) networks, routers, or floating IP addresses. Only the `admin` or other privileged user can manage provider networks.

Option 2 augments option 1 with layer-3 services that support attaching instances to self-service networks. The `demo` or other unprivileged user can manage self-service networks including routers that provide connectivity between self-service and provider networks. Additionally, floating IP addresses provide connectivity to instances using self-service networks from external networks such as the Internet.

Self-service networks typically use overlay networks. Overlay network protocols such as VXLAN include additional headers that increase overhead and decrease space available for the payload or user data. Without knowledge of the virtual network infrastructure, instances attempt to send packets using the default Ethernet maximum transmission unit (MTU) of 1500 bytes. The Networking service automatically provides the correct MTU value to instances via DHCP. However, some cloud images do not use DHCP or ignore the DHCP MTU option and require configuration using metadata or a script.

Note

Option 2 also supports attaching instances to provider networks.

Choose one of the following networking options to configure services specific to it. Afterwards, return here and proceed to *Configure the metadata agent*.

Networking Option 1: Provider networks

Install and configure the Networking components on the *controller* node.

Install the components

```
# dnf install openstack-neutron openstack-neutron-ml2 \
  openstack-neutron-openvswitch
```

Configure the server component

The Networking server component configuration includes the database, authentication mechanism, message queue, topology change notifications, and plug-in.

Note

Default configuration files vary by distribution. You might need to add these sections and options rather than modifying existing sections and options. Also, an ellipsis (`...`) in the configuration snippets indicates potential default configuration options that you should retain.

- Edit the `/etc/neutron/neutron.conf` file and complete the following actions:
  - In the `[database]` section, configure database access:

    ```
    [database]
    # ...
    connection = mysql+pymysql://neutron:NEUTRON_DBPASS@controller/
    ↪neutron
    ```

    Replace `NEUTRON_DBPASS` with the password you chose for the database.

    Note

    Comment out or remove any other `connection` options in the `[database]` section.

  - In the `[DEFAULT]` section, enable the Modular Layer 2 (ML2) plug-in and disable additional plug-ins:

    ```
    [DEFAULT]
    # ...
    core_plugin = ml2
    service_plugins =
    ```

  - In the `[DEFAULT]` section, configure `RabbitMQ` message queue access:

    ```
    [DEFAULT]
    # ...
    transport_url = rabbit://openstack:RABBIT_PASS@controller
    ```

    Replace `RABBIT_PASS` with the password you chose for the `openstack` account in RabbitMQ.

- In the [DEFAULT] and [keystone_authtoken] sections, configure Identity service access:

```
[DEFAULT]
# ...
auth_strategy = keystone

[keystone_authtoken]
# ...
www_authenticate_uri = http://controller:5000
auth_url = http://controller:5000
memcached_servers = controller:11211
auth_type = password
project_domain_name = Default
user_domain_name = Default
project_name = service
username = neutron
password = NEUTRON_PASS
```

  Replace NEUTRON_PASS with the password you chose for the neutron user in the Identity service.

  Note

  Comment out or remove any other options in the [keystone_authtoken] section.

- In the [DEFAULT] and [nova] sections, configure Networking to notify Compute of network topology changes:

```
[DEFAULT]
# ...
notify_nova_on_port_status_changes = true
notify_nova_on_port_data_changes = true

[nova]
# ...
auth_url = http://controller:5000
auth_type = password
project_domain_name = Default
user_domain_name = Default
region_name = RegionOne
project_name = service
username = nova
password = NOVA_PASS
```

  Replace NOVA_PASS with the password you chose for the nova user in the Identity service.

- In the [oslo_concurrency] section, configure the lock path:

```
[oslo_concurrency]
# ...
lock_path = /var/lib/neutron/tmp
```

Configure the Modular Layer 2 (ML2) plug-in

The ML2 plug-in uses the Linux bridge mechanism to build layer-2 (bridging and switching) virtual networking infrastructure for instances.

- Edit the /etc/neutron/plugins/ml2/ml2_conf.ini file and complete the following actions:
  - In the [ml2] section, enable flat and VLAN networks:

    ```
    [ml2]
    # ...
    type_drivers = flat,vlan
    ```

  - In the [ml2] section, disable self-service networks:

    ```
    [ml2]
    # ...
    project_network_types =
    ```

  - In the [ml2] section, enable the Linux bridge mechanism:

    ```
    [ml2]
    # ...
    mechanism_drivers = openvswitch
    ```

    Warning

    After you configure the ML2 plug-in, removing values in the type_drivers option can lead to database inconsistency.

  - In the [ml2] section, enable the port security extension driver:

    ```
    [ml2]
    # ...
    extension_drivers = port_security
    ```

  - In the [ml2_type_flat] section, configure the provider virtual network as a flat network:

    ```
    [ml2_type_flat]
    # ...
    flat_networks = provider
    ```

Configure the Open vSwitch agent

The Linux bridge agent builds layer-2 (bridging and switching) virtual networking infrastructure for instances and handles security groups.

- Edit the /etc/neutron/plugins/ml2/openvswitch_agent.ini file and complete the following actions:
  - In the [ovs] section, map the provider virtual network to the provider physical bridge:

```
[ovs]
bridge_mappings = provider:PROVIDER_BRIDGE_NAME
```

Replace PROVIDER_BRIDGE_NAME with the name of the bridge connected to the underlying provider physical network. See *Host networking* and *Open vSwitch: Provider networks* for more information.

- Ensure PROVIDER_BRIDGE_NAME external bridge is created and PROVIDER_INTERFACE_NAME is added to that bridge

```
# ovs-vsctl add-br $PROVIDER_BRIDGE_NAME
# ovs-vsctl add-port $PROVIDER_BRIDGE_NAME $PROVIDER_INTERFACE_NAME
```

- In the [securitygroup] section, enable security groups and configure the Open vSwitch native or the hybrid iptables firewall driver:

```
[securitygroup]
# ...
enable_security_group = true
firewall_driver = openvswitch
#firewall_driver = iptables_hybrid
```

- In the case of using the hybrid iptables firewall driver, ensure your Linux operating system kernel supports network bridge filters by verifying all the following sysctl values are set to 1:

```
net.bridge.bridge-nf-call-iptables
net.bridge.bridge-nf-call-ip6tables
```

To enable networking bridge support, typically the br_netfilter kernel module needs to be loaded. Check your operating systems documentation for additional details on enabling this module.

Configure the DHCP agent

The DHCP agent provides DHCP services for virtual networks.

- Edit the /etc/neutron/dhcp_agent.ini file and complete the following actions:
  - In the [DEFAULT] section, configure the Dnsmasq DHCP driver, and enable isolated metadata so instances on provider networks can access metadata over the network:

```
[DEFAULT]
# ...
dhcp_driver = neutron.agent.linux.dhcp.Dnsmasq
enable_isolated_metadata = true
```

Create the provider network

Follow this provider network document from the General Installation Guide.

Return to *Networking controller node configuration*.

Networking Option 2: Self-service networks

Install and configure the Networking components on the *controller* node.

Install the components

```
# dnf install openstack-neutron openstack-neutron-ml2 \
  openstack-neutron-openvswitch ebtables
```

Configure the server component

- Edit the /etc/neutron/neutron.conf file and complete the following actions:
  - In the [database] section, configure database access:

    ```
    [database]
    # ...
    connection = mysql+pymysql://neutron:NEUTRON_DBPASS@controller/
    ↪neutron
    ```

    Replace NEUTRON_DBPASS with the password you chose for the database.

    Note

    Comment out or remove any other connection options in the [database] section.

  - In the [DEFAULT] section, enable the Modular Layer 2 (ML2) plug-in and router service:

    ```
    [DEFAULT]
    # ...
    core_plugin = ml2
    service_plugins = router
    ```

  - In the [DEFAULT] section, configure RabbitMQ message queue access:

    ```
    [DEFAULT]
    # ...
    transport_url = rabbit://openstack:RABBIT_PASS@controller
    ```

    Replace RABBIT_PASS with the password you chose for the openstack account in RabbitMQ.

  - In the [DEFAULT] and [keystone_authtoken] sections, configure Identity service access:

    ```
    [DEFAULT]
    # ...
    auth_strategy = keystone

    [keystone_authtoken]
    # ...
    www_authenticate_uri = http://controller:5000
    auth_url = http://controller:5000
    ```

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```
memcached_servers = controller:11211
auth_type = password
project_domain_name = Default
user_domain_name = Default
project_name = service
username = neutron
password = NEUTRON_PASS
```

Replace `NEUTRON_PASS` with the password you chose for the `neutron` user in the Identity service.

Note

Comment out or remove any other options in the `[keystone_authtoken]` section.

- In the `[DEFAULT]` and `[nova]` sections, configure Networking to notify Compute of network topology changes:

```
[DEFAULT]
# ...
notify_nova_on_port_status_changes = true
notify_nova_on_port_data_changes = true

[nova]
# ...
auth_url = http://controller:5000
auth_type = password
project_domain_name = Default
user_domain_name = Default
region_name = RegionOne
project_name = service
username = nova
password = NOVA_PASS
```

Replace `NOVA_PASS` with the password you chose for the `nova` user in the Identity service.

- In the `[oslo_concurrency]` section, configure the lock path:

```
[oslo_concurrency]
# ...
lock_path = /var/lib/neutron/tmp
```

Configure the Modular Layer 2 (ML2) plug-in

The ML2 plug-in uses the Linux bridge mechanism to build layer-2 (bridging and switching) virtual networking infrastructure for instances.

- Edit the `/etc/neutron/plugins/ml2/ml2_conf.ini` file and complete the following actions:

  - In the `[ml2]` section, enable flat, VLAN, and VXLAN networks:

```
[ml2]
# ...
type_drivers = flat,vlan,vxlan
```

- In the [ml2] section, enable VXLAN self-service networks:

```
[ml2]
# ...
project_network_types = vxlan
```

- In the [ml2] section, enable the Linux bridge and layer-2 population mechanisms:

```
[ml2]
# ...
mechanism_drivers = openvswitch,l2population
```

Warning

After you configure the ML2 plug-in, removing values in the `type_drivers` option can lead to database inconsistency.

Note

The Linux bridge agent only supports VXLAN overlay networks.

- In the [ml2] section, enable the port security extension driver:

```
[ml2]
# ...
extension_drivers = port_security
```

- In the [ml2_type_flat] section, configure the provider virtual network as a flat network:

```
[ml2_type_flat]
# ...
flat_networks = provider
```

- In the [ml2_type_vxlan] section, configure the VXLAN network identifier range for self-service networks:

```
[ml2_type_vxlan]
# ...
vni_ranges = 1:1000
```

Configure the Open vSwitch agent

The Linux bridge agent builds layer-2 (bridging and switching) virtual networking infrastructure for instances and handles security groups.

- Edit the `/etc/neutron/plugins/ml2/openvswitch_agent.ini` file and complete the following actions:
  - In the `[ovs]` section, map the provider virtual network to the provider physical bridge and configure the IP address of the physical network interface that handles overlay networks:

    ```
    [ovs]
    bridge_mappings = provider:PROVIDER_BRIDGE_NAME
    local_ip = OVERLAY_INTERFACE_IP_ADDRESS
    ```

    Replace `PROVIDER_BRIDGE_NAME` with the name of the bridge connected to the underlying provider physical network. See *Host networking* and *Open vSwitch: Provider networks* for more information.

    Also replace `OVERLAY_INTERFACE_IP_ADDRESS` with the IP address of the underlying physical network interface that handles overlay networks. The example architecture uses the management interface to tunnel traffic to the other nodes. Therefore, replace `OVERLAY_INTERFACE_IP_ADDRESS` with the management IP address of the controller node. See *Host networking* for more information.
  - Ensure `PROVIDER_BRIDGE_NAME` external bridge is created and `PROVIDER_INTERFACE_NAME` is added to that bridge

    ```
    # ovs-vsctl add-br $PROVIDER_BRIDGE_NAME
    # ovs-vsctl add-port $PROVIDER_BRIDGE_NAME $PROVIDER_INTERFACE_NAME
    ```
  - In the `[agent]` section, enable VXLAN overlay networks and enable layer-2 population:

    ```
    [agent]
    tunnel_types = vxlan
    l2_population = true
    ```
  - In the `[securitygroup]` section, enable security groups and configure the Open vSwitch native or the hybrid iptables firewall driver:

    ```
    [securitygroup]
    # ...
    enable_security_group = true
    firewall_driver = openvswitch
    #firewall_driver = iptables_hybrid
    ```
  - In the case of using the hybrid iptables firewall driver, ensure your Linux operating system kernel supports network bridge filters by verifying all the following `sysctl` values are set to 1:

    ```
    net.bridge.bridge-nf-call-iptables
    net.bridge.bridge-nf-call-ip6tables
    ```

    To enable networking bridge support, typically the `br_netfilter` kernel module needs to be loaded. Check your operating systems documentation for additional details on enabling this module.

Configure the layer-3 agent

The Layer-3 (L3) agent provides routing and NAT services for self-service virtual networks.

- Edit the `/etc/neutron/l3_agent.ini` file in case additional customization is needed.

Configure the DHCP agent

The DHCP agent provides DHCP services for virtual networks.

- Edit the `/etc/neutron/dhcp_agent.ini` file and complete the following actions:
  - In the `[DEFAULT]` section, configure Dnsmasq DHCP driver, and enable isolated metadata so instances on provider networks can access metadata over the network:

```
[DEFAULT]
# ...
dhcp_driver = neutron.agent.linux.dhcp.Dnsmasq
enable_isolated_metadata = true
```

Return to *Networking controller node configuration*.

4.2.3 Configure the metadata agent

The metadata agent provides configuration information such as credentials to instances.

- Edit the `/etc/neutron/metadata_agent.ini` file and complete the following actions:
  - In the `[DEFAULT]` section, configure the metadata host and shared secret:

```
[DEFAULT]
# ...
nova_metadata_host = controller
metadata_proxy_shared_secret = METADATA_SECRET
```

    Replace `METADATA_SECRET` with a suitable secret for the metadata proxy.

4.2.4 Configure the Compute service to use the Networking service

Note

The Nova compute service must be installed to complete this step. For more details see the compute install guide found under the *Installation Guides* section of the docs website.

- Edit the `/etc/nova/nova.conf` file and perform the following actions:
  - In the `[neutron]` section, configure access parameters, enable the metadata proxy, and configure the secret:

```
[neutron]
# ...
auth_url = http://controller:5000
auth_type = password
project_domain_name = Default
```

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```
user_domain_name = Default
region_name = RegionOne
project_name = service
username = neutron
password = NEUTRON_PASS
service_metadata_proxy = true
metadata_proxy_shared_secret = METADATA_SECRET
```

Replace `NEUTRON_PASS` with the password you chose for the `neutron` user in the Identity service.

Replace `METADATA_SECRET` with the secret you chose for the metadata proxy.

See the compute service configuration guide for the full set of options including overriding the service catalog endpoint URL if necessary.

4.2.5 Finalize installation

1. The Networking service initialization scripts expect a symbolic link `/etc/neutron/plugin.ini` pointing to the ML2 plug-in configuration file, `/etc/neutron/plugins/ml2/ml2_conf.ini`. If this symbolic link does not exist, create it using the following command:

```
# ln -s /etc/neutron/plugins/ml2/ml2_conf.ini /etc/neutron/plugin.ini
```

2. Populate the database:

```
# su -s /bin/sh -c "neutron-db-manage --config-file /etc/neutron/neutron.
↪conf \
  --config-file /etc/neutron/plugins/ml2/ml2_conf.ini upgrade head"␣
↪neutron
```

Note

Database population occurs later for Networking because the script requires complete server and plug-in configuration files.

3. Restart the Compute API service:

```
# systemctl restart openstack-nova-api.service
```

4. Start the Networking services and configure them to start when the system boots.

For both networking options:

```
# systemctl enable neutron-server.service \
  neutron-openvswitch-agent.service neutron-dhcp-agent.service \
  neutron-metadata-agent.service
# systemctl start neutron-server.service \
  neutron-openvswitch-agent.service neutron-dhcp-agent.service \
  neutron-metadata-agent.service
```

For networking option 2, also enable and start the layer-3 service:

```
# systemctl enable neutron-l3-agent.service
# systemctl start neutron-l3-agent.service
```

4.3 Install and configure compute node

The compute node handles connectivity and security groups for instances.

4.3.1 Install the components

```
# dnf install openstack-neutron-openvswitch
```

4.3.2 Configure the common component

The Networking common component configuration includes the authentication mechanism, message queue, and plug-in.

Note

Default configuration files vary by distribution. You might need to add these sections and options rather than modifying existing sections and options. Also, an ellipsis (`...`) in the configuration snippets indicates potential default configuration options that you should retain.

- Edit the `/etc/neutron/neutron.conf` file and complete the following actions:
  - In the `[database]` section, comment out any `connection` options because compute nodes do not directly access the database.
  - In the `[DEFAULT]` section, configure `RabbitMQ` message queue access:

    ```
    [DEFAULT]
    # ...
    transport_url = rabbit://openstack:RABBIT_PASS@controller
    ```

    Replace `RABBIT_PASS` with the password you chose for the `openstack` account in RabbitMQ.

- In the `[oslo_concurrency]` section, configure the lock path:

```
[oslo_concurrency]
# ...
lock_path = /var/lib/neutron/tmp
```

4.3.3 Configure networking options

Choose the same networking option that you chose for the controller node to configure services specific to it. Afterwards, return here and proceed to *Configure the Compute service to use the Networking service*.

Networking Option 1: Provider networks

Configure the Networking components on a *compute* node.

Configure the Open vSwitch agent

The Open vSwitch agent builds layer-2 (bridging and switching) virtual networking infrastructure for instances and handles security groups.

- Edit the `/etc/neutron/plugins/ml2/openvswitch_agent.ini` file and complete the following actions:
  - In the `[ovs]` section, map the provider virtual network to the provider physical bridge:

    ```
    [ovs]
    bridge_mappings = provider:PROVIDER_BRIDGE_NAME
    ```

    Replace `PROVIDER_BRIDGE_NAME` with the name of the bridge connected to the underlying provider physical network. See *Host networking* and *Open vSwitch: Provider networks* for more information.
  - Ensure `PROVIDER_BRIDGE_NAME` external bridge is created and `PROVIDER_INTERFACE_NAME` is added to that bridge

    ```
    # ovs-vsctl add-br $PROVIDER_BRIDGE_NAME
    # ovs-vsctl add-port $PROVIDER_BRIDGE_NAME $PROVIDER_INTERFACE_NAME
    ```

  - In the `[securitygroup]` section, enable security groups and configure the Open vSwitch native or the hybrid iptables firewall driver:

    ```
    [securitygroup]
    # ...
    enable_security_group = true
    firewall_driver = openvswitch
    #firewall_driver = iptables_hybrid
    ```

  - In the case of using the hybrid iptables firewall driver, ensure your Linux operating system kernel supports network bridge filters by verifying all the following `sysctl` values are set to 1:

    ```
    net.bridge.bridge-nf-call-iptables
    net.bridge.bridge-nf-call-ip6tables
    ```

    To enable networking bridge support, typically the `br_netfilter` kernel module needs to be loaded. Check your operating systems documentation for additional details on enabling this module.

Return to *Networking compute node configuration*

Networking Option 2: Self-service networks

Configure the Networking components on a *compute* node.

Configure the Open vSwitch agent

The Open vSwitch agent builds layer-2 (bridging and switching) virtual networking infrastructure for instances and handles security groups.

- Edit the /etc/neutron/plugins/ml2/openvswitch_agent.ini file and complete the following actions:
  - In the [ovs] section, map the provider virtual network to the provider physical bridge and configure the IP address of the physical network interface that handles overlay networks:

    ```
    [ovs]
    bridge_mappings = provider:PROVIDER_BRIDGE_NAME
    local_ip = OVERLAY_INTERFACE_IP_ADDRESS
    ```

    Replace PROVIDER_BRIDGE_NAME with the name of the bridge connected to the underlying provider physical network. See *Host networking* and *Open vSwitch: Provider networks* for more information.

    Also replace OVERLAY_INTERFACE_IP_ADDRESS with the IP address of the underlying physical network interface that handles overlay networks. The example architecture uses the management interface to tunnel traffic to the other nodes. Therefore, replace OVERLAY_INTERFACE_IP_ADDRESS with the management IP address of the compute node. See *Host networking* for more information.

  - Ensure PROVIDER_BRIDGE_NAME external bridge is created and PROVIDER_INTERFACE_NAME is added to that bridge

    ```
    # ovs-vsctl add-br $PROVIDER_BRIDGE_NAME
    # ovs-vsctl add-port $PROVIDER_BRIDGE_NAME $PROVIDER_INTERFACE_NAME
    ```

  - In the [agent] section, enable VXLAN overlay networks and enable layer-2 population:

    ```
    [agent]
    tunnel_types = vxlan
    l2_population = true
    ```

  - In the [securitygroup] section, enable security groups and configure the Open vSwitch native or the hybrid iptables firewall driver:

    ```
    [securitygroup]
    # ...
    enable_security_group = true
    firewall_driver = openvswitch
    #firewall_driver = iptables_hybrid
    ```

  - In the case of using the hybrid iptables firewall driver, ensure your Linux operating system kernel supports network bridge filters by verifying all the following sysctl values are set to 1:

    ```
    net.bridge.bridge-nf-call-iptables
    net.bridge.bridge-nf-call-ip6tables
    ```

    To enable networking bridge support, typically the br_netfilter kernel module needs to be loaded. Check your operating systems documentation for additional details on enabling this module.

Return to *Networking compute node configuration*.

4.3.4 Configure the Compute service to use the Networking service

- Edit the /etc/nova/nova.conf file and complete the following actions:
  - In the [neutron] section, configure access parameters:

```
[neutron]
# ...
auth_url = http://controller:5000
auth_type = password
project_domain_name = Default
user_domain_name = Default
region_name = RegionOne
project_name = service
username = neutron
password = NEUTRON_PASS
```

Replace NEUTRON_PASS with the password you chose for the neutron user in the Identity service.

See the compute service configuration guide for the full set of options including overriding the service catalog endpoint URL if necessary.

4.3.5 Finalize installation

1. Restart the Compute service:

```
# systemctl restart openstack-nova-compute.service
```

2. Start the Linux bridge agent and configure it to start when the system boots:

```
# systemctl enable neutron-openvswitch-agent.service
# systemctl start neutron-openvswitch-agent.service
```

4.4 Verify operation

Note

Perform these commands on the controller node.

1. Source the admin credentials to gain access to admin-only CLI commands:

```
$ . admin-openrc
```

2. List loaded extensions to verify successful launch of the neutron-server process:

```
$ openstack extension list --network
```

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```
+---------------------------+---------------------------+-----------------
↪-----------+
| Name                      | Alias                     | Description    ␣
↪           |
+---------------------------+---------------------------+-----------------
↪-----------+
| Default Subnetpools       | default-subnetpools       | Provides␣
↪ability to mark   |
|                           |                           | and use a␣
↪subnetpool as    |
|                           |                           | the default    ␣
↪           |
| Availability Zone         | availability_zone         | The␣
↪availability zone      |
|                           |                           | extension.     ␣
↪           |
| Network Availability Zone | network_availability_zone | Availability␣
↪zone support   |
|                           |                           | for network.   ␣
↪           |
| Port Binding              | binding                   | Expose port␣
↪bindings of a   |
|                           |                           | virtual port to␣
↪external    |
|                           |                           | application    ␣
↪           |
| agent                     | agent                     | The agent␣
↪management        |
|                           |                           | extension.     ␣
↪           |
| Subnet Allocation         | subnet_allocation         | Enables␣
↪allocation of      |
|                           |                           | subnets from a␣
↪subnet pool |
| DHCP Agent Scheduler      | dhcp_agent_scheduler      | Schedule␣
↪networks among    |
|                           |                           | dhcp agents    ␣
↪           |
| Neutron external network  | external-net              | Adds external␣
↪network      |
|                           |                           | attribute to␣
↪network        |
|                           |                           | resource.      ␣
↪           |
| Neutron Service Flavors   | flavors                   | Flavor␣
↪specification for   |
|                           |                           | Neutron␣
↪advanced services  |
| Network MTU               | net-mtu                   | Provides MTU␣
```

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```
↪attribute for |
|                              |                              | a network␣
↪resource.         |
| Network IP Availability     | network-ip-availability     | Provides IP␣
↪availability    |
|                              |                              | data for each␣
↪network and   |
|                              |                              | subnet.          ␣
↪              |
| Quota management support    | quotas                       | Expose␣
↪functions for        |
|                              |                              | quotas␣
↪management per       |
|                              |                              | project          ␣
↪              |
| Provider Network            | provider                     | Expose mapping␣
↪of virtual   |
|                              |                              | networks to␣
↪physical        |
|                              |                              | networks         ␣
↪              |
| Multi Provider Network      | multi-provider               | Expose mapping␣
↪of virtual   |
|                              |                              | networks to␣
↪multiple        |
|                              |                              | physical␣
↪networks           |
| Address scope               | address-scope                | Address scopes␣
↪extension.   |
| Subnet service types        | subnet-service-types         | Provides␣
↪ability to set     |
|                              |                              | the subnet␣
↪service_types    |
|                              |                              | field            ␣
↪              |
| Resource timestamps         | standard-attr-timestamp      | Adds created_at␣
↪and          |
|                              |                              | updated_at␣
↪fields to all    |
|                              |                              | Neutron␣
↪resources that      |
|                              |                              | have Neutron␣
↪standard       |
|                              |                              | attributes.      ␣
↪              |
| Neutron Service Type        | service-type                 | API for␣
↪retrieving service |
| Management                   |                              | providers for␣
↪Neutron       |
```

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```
|                            |                            | advanced␣
↪services           |
| resources: subnet,         |                            | more L2 and L3␣
↪resources.  |
| subnetpool, port, router   |                            |               ␣
↪            |
| Neutron Extra DHCP opts    | extra_dhcp_opt             | Extra options  ␣
↪            |
|                            |                            | configuration␣
↪for DHCP.    |
|                            |                            | For example PXE␣
↪boot        |
|                            |                            | options to DHCP␣
↪clients     |
|                            |                            | can be␣
↪specified (e.g.      |
|                            |                            | tftp-server,␣
↪server-ip-     |
|                            |                            | address,␣
↪bootfile-name)     |
| Resource revision numbers  | standard-attr-revisions    | This extension␣
↪will        |
|                            |                            | display the␣
↪revision       |
|                            |                            | number of␣
↪neutron            |
|                            |                            | resources.     ␣
↪            |
| Pagination support         | pagination                 | Extension that␣
↪indicates   |
|                            |                            | that pagination␣
↪is          |
|                            |                            | enabled.       ␣
↪            |
| Sorting support            | sorting                    | Extension that␣
↪indicates   |
|                            |                            | that sorting is␣
↪enabled.    |
| security-group             | security-group             | The security␣
↪groups         |
|                            |                            | extension.     ␣
↪            |
| RBAC Policies              | rbac-policies              | Allows creation␣
↪and         |
|                            |                            | modification of␣
↪policies    |
|                            |                            | that control␣
↪project access|
|                            |                            | to resources.  ␣
```

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```
↪           |
| standard-attr-description | standard-attr-description | Extension to␣
↪add           |
|                           |                           | descriptions to␣
↪standard    |
|                           |                           | attributes     ␣
↪           |
| Port Security             | port-security             | Provides port␣
↪security     |
| Allowed Address Pairs     | allowed-address-pairs     | Provides␣
↪allowed address    |
|                           |                           | pairs          ␣
↪           |
| project_id field enabled  | project-id                | Extension that␣
↪indicates    |
|                           |                           | that project_id␣
↪field is    |
|                           |                           | enabled.       ␣
↪           |
+---------------------------+---------------------------+-----------------
↪-----------+
```

Note

Actual output may differ slightly from this example.

You can perform further testing of your networking using the neutron-sanity-check command line client.

Use the verification section for the networking option that you chose to deploy.

4.4.1 Networking Option 1: Provider networks

- List agents to verify successful launch of the neutron agents:

```
$ openstack network agent list

+--------------------------------------+--------------------+------------
↪+-------------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host       ␣
↪| Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+------------
↪+-------------------+-------+-------+---------------------------+
| 0400c2f6-4d3b-44bc-89fa-99093432f3bf | Metadata agent     | controller␣
↪| None              | True  | UP    | neutron-metadata-agent    |
| 83cf853d-a2f2-450a-99d7-e9c6fc08f4c3 | DHCP agent         | controller␣
↪| nova              | True  | UP    | neutron-dhcp-agent        |
| ec302e51-6101-43cf-9f19-88a78613cbee | Open vSwitch agent | compute    ␣
↪| None              | True  | UP    | neutron-openvswitch-agent |
| fcb9bc6e-22b1-43bc-9054-272dd517d025 | Open vSwitch agent | controller␣
```

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```
↪| None              | True  | UP    | neutron-openvswitch-agent |
+--------------------------------------+--------------------+------------
↪+-------------------+-------+-------+---------------------------+
```

The output should indicate three agents on the controller node and one agent on each compute node.

4.4.2 Networking Option 2: Self-service networks

- List agents to verify successful launch of the neutron agents:

```
$ openstack network agent list

+--------------------------------------+--------------------+------------
↪+-------------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host       ␣
↪| Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+------------
↪+-------------------+-------+-------+---------------------------+
| f49a4b81-afd6-4b3d-b923-66c8f0517099 | Metadata agent     | controller␣
↪| None              | True  | UP    | neutron-metadata-agent    |
| 27eee952-a748-467b-bf71-941e89846a92 | Open vSwitch agent | controller␣
↪| None              | True  | UP    | neutron-openvswitch-agent |
| 08905043-5010-4b87-bba5-aedb1956e27a | Open vSwitch agent | compute1   ␣
↪| None              | True  | UP    | neutron-openvswitch-agent |
| 830344ff-dc36-4956-84f4-067af667a0dc | L3 agent           | controller␣
↪| nova              | True  | UP    | neutron-l3-agent          |
| dd3644c9-1a3a-435a-9282-eb306b4b0391 | DHCP agent         | controller␣
↪| nova              | True  | UP    | neutron-dhcp-agent        |
+--------------------------------------+--------------------+------------
↪+-------------------+-------+-------+---------------------------+
```

The output should indicate four agents on the controller node and one agent on each compute node.

CHAPTER

FIVE

INSTALL AND CONFIGURE FOR UBUNTU

5.1 Host networking

After installing the operating system on each node for the architecture that you choose to deploy, you must configure the network interfaces. We recommend that you disable any automated network management tools and manually edit the appropriate configuration files for your distribution. For more information on how to configure networking on your distribution, see the documentation.

All nodes require Internet access for administrative purposes such as package installation, security updates, Domain Name System (DNS), and Network Time Protocol (NTP). In most cases, nodes should obtain Internet access through the management network interface. To highlight the importance of network separation, the example architectures use private address space for the management network and assume that the physical network infrastructure provides Internet access via Network Address Translation (NAT) or other methods. The example architectures use routable IP address space for the provider (external) network and assume that the physical network infrastructure provides direct Internet access.

In the provider networks architecture, all instances attach directly to the provider network. In the self-service (private) networks architecture, instances can attach to a self-service or provider network. Self-service networks can reside entirely within OpenStack or provide some level of external network access using Network Address Translation (NAT) through the provider network.

The example architectures assume use of the following networks:

- Management on 10.0.0.0/24 with gateway 10.0.0.1

  This network requires a gateway to provide Internet access to all nodes for administrative purposes such as package installation, security updates, Domain Name System (DNS), and Network Time Protocol (NTP).

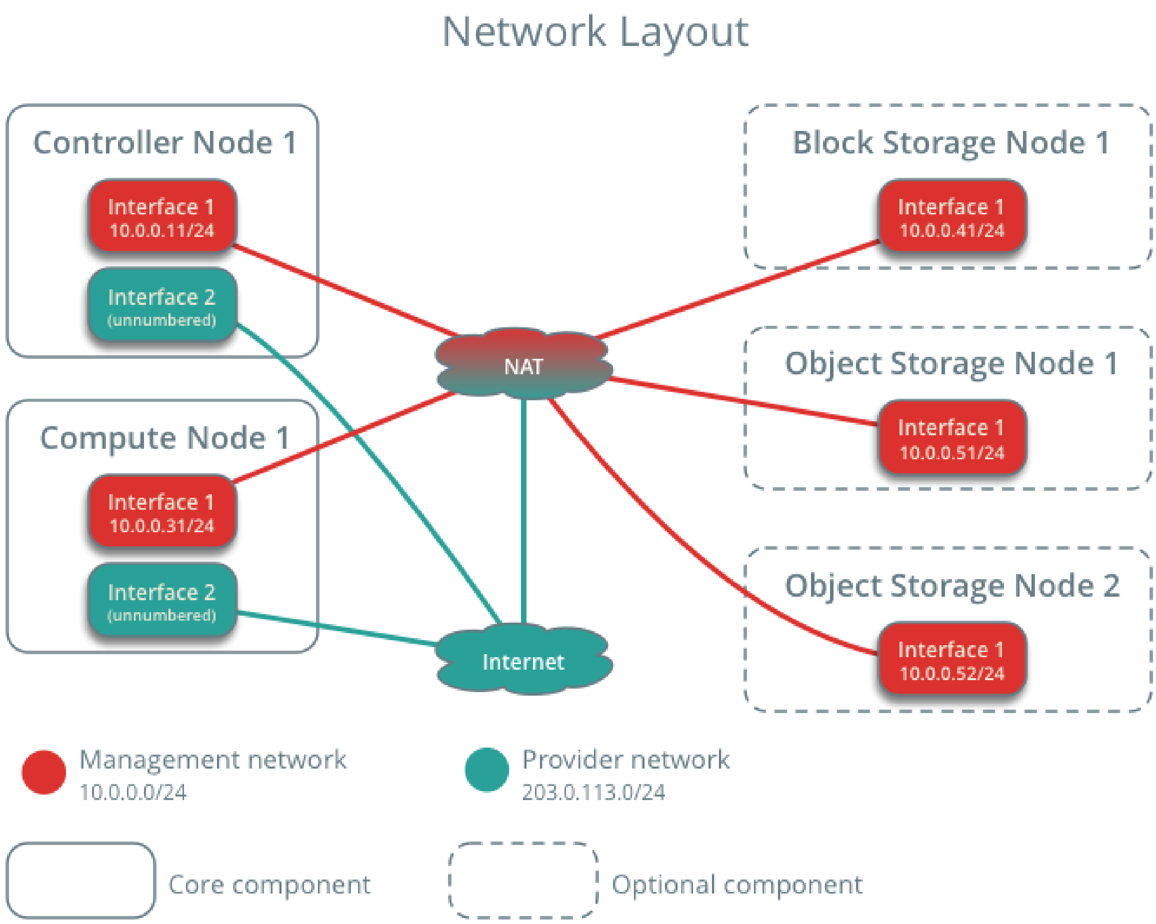
- Provider on 203.0.113.0/24 with gateway 203.0.113.1

  This network requires a gateway to provide Internet access to instances in your OpenStack environment.

You can modify these ranges and gateways to work with your particular network infrastructure.

Network interface names vary by distribution. Traditionally, interfaces use `eth` followed by a sequential number. To cover all variations, this guide refers to the first interface as the interface with the lowest number and the second interface as the interface with the highest number.

Unless you intend to use the exact configuration provided in this example architecture, you must modify the networks in this procedure to match your environment. Each node must resolve the other nodes by name in addition to IP address. For example, the `controller` name must resolve to `10.0.0.11`, the IP address of the management interface on the controller node.



Network Layout
Controller Node 1
Interface 1
10.0.0.11/24
Interface 2
(unnumbered)
Compute Node 1
Interface 1
10.0.0.31/24
Interface 2
(unnumbered)
NAT
Internet
Block Storage Node 1
Interface 1
10.0.0.41/24
Object Storage Node 1
Interface 1
10.0.0.51/24
Object Storage Node 2
Interface 1
10.0.0.52/24
Management network
10.0.0.0/24
Provider network
203.0.113.0/24
Core component
Optional component

Warning

Reconfiguring network interfaces will interrupt network connectivity. We recommend using a local terminal session for these procedures.

Note

Your distribution does not enable a restrictive firewall by default. For more information about securing your environment, refer to the OpenStack Security Guide.

5.1.1 Controller node

Configure network interfaces

1. Configure the first interface as the management interface:

   IP address: 10.0.0.11

   Network mask: 255.255.255.0 (or /24)

   Default gateway: 10.0.0.1

2. The provider interface uses a special configuration without an IP address assigned to it. Configure the second interface as the provider interface:

   Replace `INTERFACE_NAME` with the actual interface name. For example, *eth1* or *ens224*.

- Edit the `/etc/network/interfaces` file to contain the following:

```
# The provider network interface
auto INTERFACE_NAME
iface INTERFACE_NAME inet manual
up ip link set dev $IFACE up
down ip link set dev $IFACE down
```

1. Reboot the system to activate the changes.

Configure name resolution

1. Set the hostname of the node to `controller`.

2. Edit the `/etc/hosts` file to contain the following:

```
# controller
10.0.0.11       controller

# compute1
10.0.0.31       compute1

# block1
10.0.0.41       block1

# object1
```

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```
10.0.0.51       object1

# object2
10.0.0.52       object2
```

Warning

Some distributions add an extraneous entry in the `/etc/hosts` file that resolves the actual hostname to another loopback IP address such as `127.0.1.1`. You must comment out or remove this entry to prevent name resolution problems. **Do not remove the 127.0.0.1 entry.**

Note

This guide includes host entries for optional services in order to reduce complexity should you choose to deploy them.

5.1.2 Compute node

Configure network interfaces

1. Configure the first interface as the management interface:

   IP address: 10.0.0.31

   Network mask: 255.255.255.0 (or /24)

   Default gateway: 10.0.0.1

   Note

   Additional compute nodes should use 10.0.0.32, 10.0.0.33, and so on.

2. The provider interface uses a special configuration without an IP address assigned to it. Configure the second interface as the provider interface:

   Replace `INTERFACE_NAME` with the actual interface name. For example, *eth1* or *ens224*.

- Edit the `/etc/network/interfaces` file to contain the following:

```
# The provider network interface
auto INTERFACE_NAME
iface  INTERFACE_NAME inet manual
up ip link set dev $IFACE up
down ip link set dev $IFACE down
```

1. Reboot the system to activate the changes.

Configure name resolution

1. Set the hostname of the node to `compute1`.
2. Edit the `/etc/hosts` file to contain the following:

```
# controller
10.0.0.11       controller

# compute1
10.0.0.31       compute1

# block1
10.0.0.41       block1

# object1
10.0.0.51       object1

# object2
10.0.0.52       object2
```

Warning

Some distributions add an extraneous entry in the `/etc/hosts` file that resolves the actual hostname to another loopback IP address such as `127.0.1.1`. You must comment out or remove this entry to prevent name resolution problems. **Do not remove the 127.0.0.1 entry.**

Note

This guide includes host entries for optional services in order to reduce complexity should you choose to deploy them.

5.1.3 Verify connectivity

We recommend that you verify network connectivity to the Internet and among the nodes before proceeding further.

1. From the *controller* node, test access to the Internet:

```
# ping -c 4 openstack.org

PING openstack.org (174.143.194.225) 56(84) bytes of data.
64 bytes from 174.143.194.225: icmp_seq=1 ttl=54 time=18.3 ms
64 bytes from 174.143.194.225: icmp_seq=2 ttl=54 time=17.5 ms
64 bytes from 174.143.194.225: icmp_seq=3 ttl=54 time=17.5 ms
64 bytes from 174.143.194.225: icmp_seq=4 ttl=54 time=17.4 ms

--- openstack.org ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3022ms
rtt min/avg/max/mdev = 17.489/17.715/18.346/0.364 ms
```

2. From the *controller* node, test access to the management interface on the *compute* node:

```
# ping -c 4 compute1

PING compute1 (10.0.0.31) 56(84) bytes of data.
64 bytes from compute1 (10.0.0.31): icmp_seq=1 ttl=64 time=0.263 ms
64 bytes from compute1 (10.0.0.31): icmp_seq=2 ttl=64 time=0.202 ms
64 bytes from compute1 (10.0.0.31): icmp_seq=3 ttl=64 time=0.203 ms
64 bytes from compute1 (10.0.0.31): icmp_seq=4 ttl=64 time=0.202 ms

--- compute1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3000ms
rtt min/avg/max/mdev = 0.202/0.217/0.263/0.030 ms
```

3. From the *compute* node, test access to the Internet:

```
# ping -c 4 openstack.org

PING openstack.org (174.143.194.225) 56(84) bytes of data.
64 bytes from 174.143.194.225: icmp_seq=1 ttl=54 time=18.3 ms
64 bytes from 174.143.194.225: icmp_seq=2 ttl=54 time=17.5 ms
64 bytes from 174.143.194.225: icmp_seq=3 ttl=54 time=17.5 ms
64 bytes from 174.143.194.225: icmp_seq=4 ttl=54 time=17.4 ms

--- openstack.org ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3022ms
rtt min/avg/max/mdev = 17.489/17.715/18.346/0.364 ms
```

4. From the *compute* node, test access to the management interface on the *controller* node:

```
# ping -c 4 controller

PING controller (10.0.0.11) 56(84) bytes of data.
64 bytes from controller (10.0.0.11): icmp_seq=1 ttl=64 time=0.263 ms
64 bytes from controller (10.0.0.11): icmp_seq=2 ttl=64 time=0.202 ms
64 bytes from controller (10.0.0.11): icmp_seq=3 ttl=64 time=0.203 ms
64 bytes from controller (10.0.0.11): icmp_seq=4 ttl=64 time=0.202 ms

--- controller ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3000ms
rtt min/avg/max/mdev = 0.202/0.217/0.263/0.030 ms
```

Note

Your distribution does not enable a restrictive firewall by default. For more information about securing your environment, refer to the OpenStack Security Guide.

5.2 Install and configure controller node

5.2.1 Prerequisites

Before you configure the OpenStack Networking (neutron) service, you must create a database, service credentials, and API endpoints.

1. To create the database, complete these steps:

   - Use the database access client to connect to the database server as the `root` user:

     ```
     $ mysql -u root -p
     ```

   - Create the `neutron` database:

     ```
     MariaDB [(none)]> CREATE DATABASE neutron;
     ```

   - Grant proper access to the `neutron` database, replacing `NEUTRON_DBPASS` with a suitable password:

     ```
     MariaDB [(none)]> GRANT ALL PRIVILEGES ON neutron.* TO 'neutron'@
     ↪'localhost' \
       IDENTIFIED BY 'NEUTRON_DBPASS';
     MariaDB [(none)]> GRANT ALL PRIVILEGES ON neutron.* TO 'neutron'@'%'␣
     ↪\
       IDENTIFIED BY 'NEUTRON_DBPASS';
     ```

   - Exit the database access client.

2. Source the `admin` credentials to gain access to admin-only CLI commands:

   ```
   $ . admin-openrc
   ```

3. To create the service credentials, complete these steps:

   - Create the `neutron` user:

     ```
     $ openstack user create --domain default --password-prompt neutron

     User Password:
     Repeat User Password:
     +---------------------+----------------------------------+
     | Field               | Value                            |
     +---------------------+----------------------------------+
     | domain_id           | default                          |
     | enabled             | True                             |
     | id                  | fdb0f541e28141719b6a43c8944bf1fb |
     | name                | neutron                          |
     | options             | {}                               |
     | password_expires_at | None                             |
     +---------------------+----------------------------------+
     ```

   - Add the `admin` role to the `neutron` user:

```
$ openstack role add --project service --user neutron admin
```

> **Note**
>
> This command provides no output.

- Create the `neutron` service entity:

```
$ openstack service create --name neutron \
  --description "OpenStack Networking" network

+-------------+----------------------------------+
| Field       | Value                            |
+-------------+----------------------------------+
| description | OpenStack Networking             |
| enabled     | True                             |
| id          | f71529314dab4a4d8eca427e701d209e |
| name        | neutron                          |
| type        | network                          |
+-------------+----------------------------------+
```

4. Create the Networking service API endpoints:

```
$ openstack endpoint create --region RegionOne \
  network public http://controller:9696

+--------------+----------------------------------+
| Field        | Value                            |
+--------------+----------------------------------+
| enabled      | True                             |
| id           | 85d80a6d02fc4b7683f611d7fc1493a3 |
| interface    | public                           |
| region       | RegionOne                        |
| region_id    | RegionOne                        |
| service_id   | f71529314dab4a4d8eca427e701d209e |
| service_name | neutron                          |
| service_type | network                          |
| url          | http://controller:9696           |
+--------------+----------------------------------+

$ openstack endpoint create --region RegionOne \
  network internal http://controller:9696

+--------------+----------------------------------+
| Field        | Value                            |
+--------------+----------------------------------+
| enabled      | True                             |
| id           | 09753b537ac74422a68d2d791cf3714f |
| interface    | internal                         |
```

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```
| region       | RegionOne                        |
| region_id    | RegionOne                        |
| service_id   | f71529314dab4a4d8eca427e701d209e |
| service_name | neutron                          |
| service_type | network                          |
| url          | http://controller:9696           |
+--------------+----------------------------------+

$ openstack endpoint create --region RegionOne \
  network admin http://controller:9696

+--------------+----------------------------------+
| Field        | Value                            |
+--------------+----------------------------------+
| enabled      | True                             |
| id           | 1ee14289c9374dffb5db92a5c112fc4e |
| interface    | admin                            |
| region       | RegionOne                        |
| region_id    | RegionOne                        |
| service_id   | f71529314dab4a4d8eca427e701d209e |
| service_name | neutron                          |
| service_type | network                          |
| url          | http://controller:9696           |
+--------------+----------------------------------+
```

5.2.2 Configure networking options

You can deploy the Networking service using one of two architectures represented by options 1 and 2.

Option 1 deploys the simplest possible architecture that only supports attaching instances to provider (external) networks. No self-service (private) networks, routers, or floating IP addresses. Only the `admin` or other privileged user can manage provider networks.

Option 2 augments option 1 with layer-3 services that support attaching instances to self-service networks. The `demo` or other unprivileged user can manage self-service networks including routers that provide connectivity between self-service and provider networks. Additionally, floating IP addresses provide connectivity to instances using self-service networks from external networks such as the Internet.

Self-service networks typically use overlay networks. Overlay network protocols such as VXLAN include additional headers that increase overhead and decrease space available for the payload or user data. Without knowledge of the virtual network infrastructure, instances attempt to send packets using the default Ethernet maximum transmission unit (MTU) of 1500 bytes. The Networking service automatically provides the correct MTU value to instances via DHCP. However, some cloud images do not use DHCP or ignore the DHCP MTU option and require configuration using metadata or a script.

Note

Option 2 also supports attaching instances to provider networks.

Choose one of the following networking options to configure services specific to it. Afterwards, return here and proceed to *Configure the metadata agent*.

Networking Option 1: Provider networks

Install and configure the Networking components on the *controller* node.

Install the components

```
# apt install neutron-server neutron-plugin-ml2 \
  neutron-openvswitch-agent neutron-dhcp-agent \
  neutron-metadata-agent
```

Configure the server component

The Networking server component configuration includes the database, authentication mechanism, message queue, topology change notifications, and plug-in.

Note

Default configuration files vary by distribution. You might need to add these sections and options rather than modifying existing sections and options. Also, an ellipsis (`...`) in the configuration snippets indicates potential default configuration options that you should retain.

- Edit the `/etc/neutron/neutron.conf` file and complete the following actions:
  - In the `[database]` section, configure database access:

    ```
    [database]
    # ...
    connection = mysql+pymysql://neutron:NEUTRON_DBPASS@controller/
    ↪neutron
    ```

    Replace `NEUTRON_DBPASS` with the password you chose for the database.

    Note

    Comment out or remove any other `connection` options in the `[database]` section.

  - In the `[DEFAULT]` section, enable the Modular Layer 2 (ML2) plug-in and disable additional plug-ins:

    ```
    [DEFAULT]
    # ...
    core_plugin = ml2
    service_plugins =
    ```

  - In the `[DEFAULT]` section, configure `RabbitMQ` message queue access:

    ```
    [DEFAULT]
    # ...
    transport_url = rabbit://openstack:RABBIT_PASS@controller
    ```

    Replace `RABBIT_PASS` with the password you chose for the `openstack` account in RabbitMQ.

– In the [DEFAULT] and [keystone_authtoken] sections, configure Identity service access:

```
[DEFAULT]
# ...
auth_strategy = keystone

[keystone_authtoken]
# ...
www_authenticate_uri = http://controller:5000
auth_url = http://controller:5000
memcached_servers = controller:11211
auth_type = password
project_domain_name = Default
user_domain_name = Default
project_name = service
username = neutron
password = NEUTRON_PASS
```

Replace NEUTRON_PASS with the password you chose for the neutron user in the Identity service.

Note

Comment out or remove any other options in the [keystone_authtoken] section.

– In the [DEFAULT] and [nova] sections, configure Networking to notify Compute of network topology changes:

```
[DEFAULT]
# ...
notify_nova_on_port_status_changes = true
notify_nova_on_port_data_changes = true

[nova]
# ...
auth_url = http://controller:5000
auth_type = password
project_domain_name = Default
user_domain_name = Default
region_name = RegionOne
project_name = service
username = nova
password = NOVA_PASS
```

Replace NOVA_PASS with the password you chose for the nova user in the Identity service.

- In the [oslo_concurrency] section, configure the lock path:

```
[oslo_concurrency]
# ...
lock_path = /var/lib/neutron/tmp
```

Configure the Modular Layer 2 (ML2) plug-in

The ML2 plug-in uses the Linux bridge mechanism to build layer-2 (bridging and switching) virtual networking infrastructure for instances.

- Edit the /etc/neutron/plugins/ml2/ml2_conf.ini file and complete the following actions:
  - In the [ml2] section, enable flat and VLAN networks:

    ```
    [ml2]
    # ...
    type_drivers = flat,vlan
    ```

  - In the [ml2] section, disable self-service networks:

    ```
    [ml2]
    # ...
    project_network_types =
    ```

  - In the [ml2] section, enable the Linux bridge mechanism:

    ```
    [ml2]
    # ...
    mechanism_drivers = openvswitch
    ```

    Warning

    After you configure the ML2 plug-in, removing values in the type_drivers option can lead to database inconsistency.

  - In the [ml2] section, enable the port security extension driver:

    ```
    [ml2]
    # ...
    extension_drivers = port_security
    ```

  - In the [ml2_type_flat] section, configure the provider virtual network as a flat network:

    ```
    [ml2_type_flat]
    # ...
    flat_networks = provider
    ```

Configure the Open vSwitch agent

The Linux bridge agent builds layer-2 (bridging and switching) virtual networking infrastructure for instances and handles security groups.

- Edit the /etc/neutron/plugins/ml2/openvswitch_agent.ini file and complete the following actions:
  - In the [ovs] section, map the provider virtual network to the provider physical bridge:

```
[ovs]
bridge_mappings = provider:PROVIDER_BRIDGE_NAME
```

Replace PROVIDER_BRIDGE_NAME with the name of the bridge connected to the underlying provider physical network. See *Host networking* and *Open vSwitch: Provider networks* for more information.

- Ensure PROVIDER_BRIDGE_NAME external bridge is created and PROVIDER_INTERFACE_NAME is added to that bridge

```
# ovs-vsctl add-br $PROVIDER_BRIDGE_NAME
# ovs-vsctl add-port $PROVIDER_BRIDGE_NAME $PROVIDER_INTERFACE_NAME
```

- In the [securitygroup] section, enable security groups and configure the Open vSwitch native or the hybrid iptables firewall driver:

```
[securitygroup]
# ...
enable_security_group = true
firewall_driver = openvswitch
#firewall_driver = iptables_hybrid
```

- In the case of using the hybrid iptables firewall driver, ensure your Linux operating system kernel supports network bridge filters by verifying all the following sysctl values are set to 1:

```
net.bridge.bridge-nf-call-iptables
net.bridge.bridge-nf-call-ip6tables
```

To enable networking bridge support, typically the br_netfilter kernel module needs to be loaded. Check your operating systems documentation for additional details on enabling this module.

Configure the DHCP agent

The DHCP agent provides DHCP services for virtual networks.

- Edit the /etc/neutron/dhcp_agent.ini file and complete the following actions:
  - In the [DEFAULT] section, configure the Dnsmasq DHCP driver, and enable isolated metadata so instances on provider networks can access metadata over the network:

```
[DEFAULT]
# ...
dhcp_driver = neutron.agent.linux.dhcp.Dnsmasq
enable_isolated_metadata = true
```

Create the provider network

Follow this provider network document from the General Installation Guide.

Return to *Networking controller node configuration*.

Networking Option 2: Self-service networks

Install and configure the Networking components on the *controller* node.

Install the components

```
# apt install neutron-server neutron-plugin-ml2 \
  neutron-openvswitch-agent neutron-l3-agent neutron-dhcp-agent \
  neutron-metadata-agent
```

Configure the server component

- Edit the `/etc/neutron/neutron.conf` file and complete the following actions:
  - In the `[database]` section, configure database access:

    ```
    [database]
    # ...
    connection = mysql+pymysql://neutron:NEUTRON_DBPASS@controller/
    ↪neutron
    ```

    Replace `NEUTRON_DBPASS` with the password you chose for the database.

    Note

    Comment out or remove any other `connection` options in the `[database]` section.

  - In the `[DEFAULT]` section, enable the Modular Layer 2 (ML2) plug-in and router service:

    ```
    [DEFAULT]
    # ...
    core_plugin = ml2
    service_plugins = router
    ```

  - In the `[DEFAULT]` section, configure `RabbitMQ` message queue access:

    ```
    [DEFAULT]
    # ...
    transport_url = rabbit://openstack:RABBIT_PASS@controller
    ```

    Replace `RABBIT_PASS` with the password you chose for the `openstack` account in RabbitMQ.

  - In the `[DEFAULT]` and `[keystone_authtoken]` sections, configure Identity service access:

    ```
    [DEFAULT]
    # ...
    auth_strategy = keystone

    [keystone_authtoken]
    # ...
    www_authenticate_uri = http://controller:5000
    ```

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```
auth_url = http://controller:5000
memcached_servers = controller:11211
auth_type = password
project_domain_name = Default
user_domain_name = Default
project_name = service
username = neutron
password = NEUTRON_PASS
```

Replace `NEUTRON_PASS` with the password you chose for the `neutron` user in the Identity service.

Note

Comment out or remove any other options in the `[keystone_authtoken]` section.

– In the `[DEFAULT]` and `[nova]` sections, configure Networking to notify Compute of network topology changes:

```
[DEFAULT]
# ...
notify_nova_on_port_status_changes = true
notify_nova_on_port_data_changes = true

[nova]
# ...
auth_url = http://controller:5000
auth_type = password
project_domain_name = Default
user_domain_name = Default
region_name = RegionOne
project_name = service
username = nova
password = NOVA_PASS
```

Replace `NOVA_PASS` with the password you chose for the `nova` user in the Identity service.

• In the `[oslo_concurrency]` section, configure the lock path:

```
[oslo_concurrency]
# ...
lock_path = /var/lib/neutron/tmp
```

Configure the Modular Layer 2 (ML2) plug-in

The ML2 plug-in uses the Linux bridge mechanism to build layer-2 (bridging and switching) virtual networking infrastructure for instances.

• Edit the `/etc/neutron/plugins/ml2/ml2_conf.ini` file and complete the following actions:

– In the `[ml2]` section, enable flat, VLAN, and VXLAN networks:

```
[ml2]
# ...
type_drivers = flat,vlan,vxlan
```

– In the `[ml2]` section, enable VXLAN self-service networks:

```
[ml2]
# ...
project_network_types = vxlan
```

– In the `[ml2]` section, enable the Linux bridge and layer-2 population mechanisms:

```
[ml2]
# ...
mechanism_drivers = openvswitch,l2population
```

Warning

After you configure the ML2 plug-in, removing values in the `type_drivers` option can lead to database inconsistency.

Note

The Linux bridge agent only supports VXLAN overlay networks.

– In the `[ml2]` section, enable the port security extension driver:

```
[ml2]
# ...
extension_drivers = port_security
```

– In the `[ml2_type_flat]` section, configure the provider virtual network as a flat network:

```
[ml2_type_flat]
# ...
flat_networks = provider
```

– In the `[ml2_type_vxlan]` section, configure the VXLAN network identifier range for self-service networks:

```
[ml2_type_vxlan]
# ...
vni_ranges = 1:1000
```

Configure the Open vSwitch agent

The Linux bridge agent builds layer-2 (bridging and switching) virtual networking infrastructure for instances and handles security groups.

- Edit the /etc/neutron/plugins/ml2/openvswitch_agent.ini file and complete the following actions:
  - In the [ovs] section, map the provider virtual network to the provider physical bridge and configure the IP address of the physical network interface that handles overlay networks:

    ```
    [ovs]
    bridge_mappings = provider:PROVIDER_BRIDGE_NAME
    local_ip = OVERLAY_INTERFACE_IP_ADDRESS
    ```

    Replace PROVIDER_BRIDGE_NAME with the name of the bridge connected to the underlying provider physical network. See *Host networking* and *Open vSwitch: Provider networks* for more information.

    Also replace OVERLAY_INTERFACE_IP_ADDRESS with the IP address of the underlying physical network interface that handles overlay networks. The example architecture uses the management interface to tunnel traffic to the other nodes. Therefore, replace OVERLAY_INTERFACE_IP_ADDRESS with the management IP address of the controller node. See *Host networking* for more information.
  - Ensure PROVIDER_BRIDGE_NAME external bridge is created and PROVIDER_INTERFACE_NAME is added to that bridge

    ```
    # ovs-vsctl add-br $PROVIDER_BRIDGE_NAME
    # ovs-vsctl add-port $PROVIDER_BRIDGE_NAME $PROVIDER_INTERFACE_NAME
    ```
  - In the [agent] section, enable VXLAN overlay networks and enable layer-2 population:

    ```
    [agent]
    tunnel_types = vxlan
    l2_population = true
    ```
  - In the [securitygroup] section, enable security groups and configure the Open vSwitch native or the hybrid iptables firewall driver:

    ```
    [securitygroup]
    # ...
    enable_security_group = true
    firewall_driver = openvswitch
    #firewall_driver = iptables_hybrid
    ```
  - In the case of using the hybrid iptables firewall driver, ensure your Linux operating system kernel supports network bridge filters by verifying all the following sysctl values are set to 1:

    ```
    net.bridge.bridge-nf-call-iptables
    net.bridge.bridge-nf-call-ip6tables
    ```

    To enable networking bridge support, typically the br_netfilter kernel module needs to be loaded. Check your operating systems documentation for additional details on enabling this module.

Configure the layer-3 agent

The Layer-3 (L3) agent provides routing and NAT services for self-service virtual networks.

- Edit the `/etc/neutron/l3_agent.ini` file in case additional customization is needed.

Configure the DHCP agent

The DHCP agent provides DHCP services for virtual networks.

- Edit the `/etc/neutron/dhcp_agent.ini` file and complete the following actions:
  - In the `[DEFAULT]` section, configure Dnsmasq DHCP driver, and enable isolated metadata so instances on provider networks can access metadata over the network:

```
[DEFAULT]
# ...
dhcp_driver = neutron.agent.linux.dhcp.Dnsmasq
enable_isolated_metadata = true
```

Return to *Networking controller node configuration*.

5.2.3 Configure the metadata agent

The metadata agent provides configuration information such as credentials to instances.

- Edit the `/etc/neutron/metadata_agent.ini` file and complete the following actions:
  - In the `[DEFAULT]` section, configure the metadata host and shared secret:

```
[DEFAULT]
# ...
nova_metadata_host = controller
metadata_proxy_shared_secret = METADATA_SECRET
```

    Replace `METADATA_SECRET` with a suitable secret for the metadata proxy.

5.2.4 Configure the Compute service to use the Networking service

Note

The Nova compute service must be installed to complete this step. For more details see the compute install guide found under the *Installation Guides* section of the docs website.

- Edit the `/etc/nova/nova.conf` file and perform the following actions:
  - In the `[neutron]` section, configure access parameters, enable the metadata proxy, and configure the secret:

```
[neutron]
# ...
auth_url = http://controller:5000
auth_type = password
project_domain_name = Default
```

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```
user_domain_name = Default
region_name = RegionOne
project_name = service
username = neutron
password = NEUTRON_PASS
service_metadata_proxy = true
metadata_proxy_shared_secret = METADATA_SECRET
```

Replace `NEUTRON_PASS` with the password you chose for the `neutron` user in the Identity service.

Replace `METADATA_SECRET` with the secret you chose for the metadata proxy.

See the compute service configuration guide for the full set of options including overriding the service catalog endpoint URL if necessary.

5.2.5 Finalize installation

1. Populate the database:

```
# su -s /bin/sh -c "neutron-db-manage --config-file /etc/neutron/neutron.
↪conf \
  --config-file /etc/neutron/plugins/ml2/ml2_conf.ini upgrade head"␣
↪neutron
```

Note

Database population occurs later for Networking because the script requires complete server and plug-in configuration files.

2. Restart the Compute API service:

```
# service nova-api restart
```

3. Restart the Networking services.

For both networking options:

```
# service neutron-server restart
# service neutron-openvswitch-agent restart
# service neutron-dhcp-agent restart
# service neutron-metadata-agent restart
```

For networking option 2, also restart the layer-3 service:

```
# service neutron-l3-agent restart
```

5.3 Install and configure compute node

The compute node handles connectivity and security groups for instances.

5.3.1 Install the components

```
# apt install neutron-openvswitch-agent
```

5.3.2 Configure the common component

The Networking common component configuration includes the authentication mechanism, message queue, and plug-in.

Note

Default configuration files vary by distribution. You might need to add these sections and options rather than modifying existing sections and options. Also, an ellipsis (`...`) in the configuration snippets indicates potential default configuration options that you should retain.

- Edit the `/etc/neutron/neutron.conf` file and complete the following actions:
  - In the `[database]` section, comment out any `connection` options because compute nodes do not directly access the database.
  - In the `[DEFAULT]` section, configure `RabbitMQ` message queue access:

    ```
    [DEFAULT]
    # ...
    transport_url = rabbit://openstack:RABBIT_PASS@controller
    ```

    Replace `RABBIT_PASS` with the password you chose for the `openstack` account in RabbitMQ.

- In the `[oslo_concurrency]` section, configure the lock path:

  ```
  [oslo_concurrency]
  # ...
  lock_path = /var/lib/neutron/tmp
  ```

5.3.3 Configure networking options

Choose the same networking option that you chose for the controller node to configure services specific to it. Afterwards, return here and proceed to *Configure the Compute service to use the Networking service*.

Networking Option 1: Provider networks

Configure the Networking components on a *compute* node.

Configure the Open vSwitch agent

The Open vSwitch agent builds layer-2 (bridging and switching) virtual networking infrastructure for instances and handles security groups.

- Edit the /etc/neutron/plugins/ml2/openvswitch_agent.ini file and complete the following actions:
  - In the [ovs] section, map the provider virtual network to the provider physical bridge:

    ```
    [ovs]
    bridge_mappings = provider:PROVIDER_BRIDGE_NAME
    ```

    Replace PROVIDER_BRIDGE_NAME with the name of the bridge connected to the underlying provider physical network. See *Host networking* and *Open vSwitch: Provider networks* for more information.

  - Ensure PROVIDER_BRIDGE_NAME external bridge is created and PROVIDER_INTERFACE_NAME is added to that bridge

    ```
    # ovs-vsctl add-br $PROVIDER_BRIDGE_NAME
    # ovs-vsctl add-port $PROVIDER_BRIDGE_NAME $PROVIDER_INTERFACE_NAME
    ```

  - In the [securitygroup] section, enable security groups and configure the Open vSwitch native or the hybrid iptables firewall driver:

    ```
    [securitygroup]
    # ...
    enable_security_group = true
    firewall_driver = openvswitch
    #firewall_driver = iptables_hybrid
    ```

  - In the case of using the hybrid iptables firewall driver, ensure your Linux operating system kernel supports network bridge filters by verifying all the following sysctl values are set to 1:

    ```
    net.bridge.bridge-nf-call-iptables
    net.bridge.bridge-nf-call-ip6tables
    ```

    To enable networking bridge support, typically the br_netfilter kernel module needs to be loaded. Check your operating systems documentation for additional details on enabling this module.

Return to *Networking compute node configuration*

Networking Option 2: Self-service networks

Configure the Networking components on a *compute* node.

Configure the Open vSwitch agent

The Open vSwitch agent builds layer-2 (bridging and switching) virtual networking infrastructure for instances and handles security groups.

- Edit the /etc/neutron/plugins/ml2/openvswitch_agent.ini file and complete the following actions:
  - In the [ovs] section, map the provider virtual network to the provider physical bridge and configure the IP address of the physical network interface that handles overlay networks:

```
[ovs]
bridge_mappings = provider:PROVIDER_BRIDGE_NAME
local_ip = OVERLAY_INTERFACE_IP_ADDRESS
```

Replace PROVIDER_BRIDGE_NAME with the name of the bridge connected to the underlying provider physical network. See *Host networking* and *Open vSwitch: Provider networks* for more information.

Also replace OVERLAY_INTERFACE_IP_ADDRESS with the IP address of the underlying physical network interface that handles overlay networks. The example architecture uses the management interface to tunnel traffic to the other nodes. Therefore, replace OVERLAY_INTERFACE_IP_ADDRESS with the management IP address of the compute node. See *Host networking* for more information.

- Ensure PROVIDER_BRIDGE_NAME external bridge is created and PROVIDER_INTERFACE_NAME is added to that bridge

```
# ovs-vsctl add-br $PROVIDER_BRIDGE_NAME
# ovs-vsctl add-port $PROVIDER_BRIDGE_NAME $PROVIDER_INTERFACE_NAME
```

- In the [agent] section, enable VXLAN overlay networks and enable layer-2 population:

```
[agent]
tunnel_types = vxlan
l2_population = true
```

- In the [securitygroup] section, enable security groups and configure the Open vSwitch native or the hybrid iptables firewall driver:

```
[securitygroup]
# ...
enable_security_group = true
firewall_driver = openvswitch
#firewall_driver = iptables_hybrid
```

- In the case of using the hybrid iptables firewall driver, ensure your Linux operating system kernel supports network bridge filters by verifying all the following sysctl values are set to 1:

```
net.bridge.bridge-nf-call-iptables
net.bridge.bridge-nf-call-ip6tables
```

To enable networking bridge support, typically the br_netfilter kernel module needs to be loaded. Check your operating systems documentation for additional details on enabling this module.

Return to *Networking compute node configuration*.

5.3.4 Configure the Compute service to use the Networking service

- Edit the /etc/nova/nova.conf file and complete the following actions:
  - In the [neutron] section, configure access parameters:

```
[neutron]
# ...
auth_url = http://controller:5000
auth_type = password
project_domain_name = Default
user_domain_name = Default
region_name = RegionOne
project_name = service
username = neutron
password = NEUTRON_PASS
```

Replace `NEUTRON_PASS` with the password you chose for the `neutron` user in the Identity service.

See the compute service configuration guide for the full set of options including overriding the service catalog endpoint URL if necessary.

5.3.5 Finalize installation

1. Restart the Compute service:

```
# service nova-compute restart
```

2. Restart the Linux bridge agent:

```
# service neutron-openvswitch-agent restart
```

CHAPTER

SIX

OVN INSTALL DOCUMENTATION

6.1 Manual install & Configuration

Note

These instructions are intended for advanced users only, and could be incomplete. Please consult your distro-specific documentation for more details.

It is also assumed you have already installed neutron components, see the latest Install Tutorials and Guides for more information.

This document discusses what is required for manual installation or integration into a production OpenStack deployment tool of conventional architectures that include the following types of nodes:

- Controller - Runs OpenStack control plane services such as REST APIs and databases.
- Network - Runs the layer-2, layer-3 (routing), DHCP, and metadata agents for the Networking service. Some agents optional. Usually provides connectivity between provider (public) and project (private) networks via NAT and floating IP addresses.

Note

Some tools deploy these services on controller nodes.

- Compute - Runs the hypervisor and layer-2 agent for the Networking service.

6.1.1 Packaging

The Networking service integration for OVN is now one of the in-tree Neutron drivers, so should be delivered with the `neutron` package, beginning with the Ussuri release.

For deployment tools using distribution packages, the names of them are different depending on the distribution.

1. RHEL/Fedora and compatible distributions include the `ovn-central` and `ovn-host` packages, which automatically install `openvswitch` as a dependency.
2. Ubuntu/Debian distributions include the `ovn-central`, `ovn-host`, `ovn-common` and `ovn-docker` packages, which automatically install the appropriate Open vSwitch dependencies as needed.

6.1.2 Controller nodes

Each controller node runs the Open vSwitch (OVS) service (including dependent services such as `ovsdb-server`) and `ovn-northd`. Only a single instance of the `ovsdb-server` and `ovn-northd` services can operate in a deployment. However, deployment tools can implement active/passive high-availability using a management tool that monitors service health and automatically starts these services on another node after failure of the primary node. See the *Frequently Asked Questions* for more information.

1. Install the `ovn-central` and `openvswitch` packages (RHEL/Fedora).
2. Install the `ovn-central` and `openvswitch-common` packages (Ubuntu/Debian).
3. Start the OVS service. The central OVS service starts the `ovsdb-server` service that manages OVN databases.

   Using the *systemd* unit:

   ```
   # systemctl start openvswitch (RHEL/Fedora)
   # systemctl start openvswitch-switch (Ubuntu/Debian)
   ```

4. Configure the `ovsdb-server` component. By default, the `ovsdb-server` service only permits local access to databases via Unix socket. However, OVN services on compute nodes require access to these databases.

   - Permit remote database access.

     ```
     # ovn-nbctl set-connection ptcp:6641:0.0.0.0 -- \
                 set connection . inactivity_probe=60000
     # ovn-sbctl set-connection ptcp:6642:0.0.0.0 -- \
                 set connection . inactivity_probe=60000
     # if using the VTEP functionality:
     #   ovs-appctl -t ovsdb-server ovsdb-server/add-remote ptcp:6640:0.0.
     ↪0.0
     ```

     Replace `0.0.0.0` with the IP address of the management network interface on the controller node to avoid listening on all interfaces.

     Note

     Permit remote access to TCP ports: 6640 (OVS) to VTEPS (if you use vteps), 6642 (SBDB) to hosts running neutron-server, gateway nodes that run ovn-controller, and compute node services like ovn-controller and ovn-metadata-agent. 6641 (NBDB) to hosts running neutron-server.

     - Since we are using `options:redirect-type` set to `bridged` for Logical Router Ports in VLAN and FLAT networks, OVN redirects packets to the gateway chassis using the localnet port of the routers peer logical switch, instead of a tunnel. If `external-ids:ovn-chassis-mac-mappings` is not configured reply packets from VM on compute node leave physnet with source MAC address of Neutron Logical Router Port. Which cause MAC jump over computes and gateways. To avoid this setting `external-ids:ovn-chassis-mac-mappings` for each physnet with unique MAC address on each compute is required.

Below you can find example how to set `ovn-chassis-mac-mappings` for compute host with two physical networks `physnet1` and `physnet2`. For example you can take MAC address from physical interface that is plugged into physnet.

```
# ovs-vsctl set open . external-ids:ovn-chassis-mac-
↪mappings=\
    physnet1:aa:bb:cc:dd:ee:ff,physnet2:aa:bb:cc:dd:ee:fe
```

5. Start the `ovn-northd` service.

   Using the *systemd* unit:

```
# systemctl start ovn-northd
```

6. Configure the Networking server component. The Networking service implements OVN as an ML2 driver. Edit the `/etc/neutron/neutron.conf` file:

   - Enable the ML2 core plug-in.

```
[DEFAULT]
...
core_plugin = ml2
```

   - Enable the OVN layer-3 service.

```
[DEFAULT]
...
service_plugins = ovn-router
```

7. Configure the ML2 plug-in. Edit the `/etc/neutron/plugins/ml2/ml2_conf.ini` file:

   - Configure the OVN mechanism driver, network type drivers, self-service (project) network types, and enable the port security extension.

```
[ml2]
...
mechanism_drivers = ovn
type_drivers = local,flat,vlan,geneve
project_network_types = geneve
extension_drivers = port_security
overlay_ip_version = 4
```

Note

To enable VLAN self-service networks, make sure that OVN version 2.11 (or higher) is used, then add `vlan` to the `project_network_types` option. The first network type in the list becomes the default self-service network type.

To use IPv6 for all overlay (tunnel) network endpoints, set the `overlay_ip_version` option to `6`.

   - Configure the Geneve ID range and maximum header size. The IP version overhead (20 bytes for IPv4 (default) or 40 bytes for IPv6) is added to the maximum header size based on the

ML2 overlay_ip_version option.

```
[ml2_type_geneve]
...
vni_ranges = 1:65536
max_header_size = 38
```

Note

The Networking service uses the vni_ranges option to allocate network segments. However, OVN ignores the actual values. Thus, the ID range only determines the quantity of Geneve networks in the environment. For example, a range of 5001:6000 defines a maximum of 1000 Geneve networks. On the other hand, these values are still relevant in Neutron context so 1:1000 and 5001:6000 are *not* simply interchangeable.

Warning

The default for max_header_size, 30, is too low for OVN. OVN requires at least 38.

- Optionally, enable support for VXLAN type networks. Because of limited space in VXLAN VNI to pass over the needed information that requires OVN to identify a packet, the header size to contain the segmentation ID is reduced to 12 bits, that allows a maximum number of 4096 networks. The same limitation applies to the number of ports in each network, that are also identified with a 12 bits header chunk, limiting their number to 4096 ports. Please check[1] for more information.

```
[ml2]
...
type_drivers = geneve,vxlan

[ml2_type_vxlan]
vni_ranges = 1001:1100
```

- Optionally, enable support for VLAN provider and self-service networks on one or more physical networks. If you specify only the physical network, only administrative (privileged) users can manage VLAN networks. Additionally specifying a VLAN ID range for a physical network enables regular (non-privileged) users to manage VLAN networks. The Networking service allocates the VLAN ID for each self-service network using the VLAN ID range for the physical network.

```
[ml2_type_vlan]
...
network_vlan_ranges = PHYSICAL_NETWORK:MIN_VLAN_ID:MAX_VLAN_ID
```

Replace PHYSICAL_NETWORK with the physical network name and optionally define the minimum and maximum VLAN IDs. Use a comma to separate each physical network.

For example, to enable support for administrative VLAN networks on the physnet1 network and self-service VLAN networks on the physnet2 network using VLAN IDs 1001 to 2000:

[1] https://mail.openvswitch.org/pipermail/ovs-dev/2020-September/375189.html

```
network_vlan_ranges = physnet1,physnet2:1001:2000
```

- Enable security groups.

```
[securitygroup]
...
enable_security_group = true
```

> **Note**
>
> The `firewall_driver` option under `[securitygroup]` is ignored since the OVN ML2 driver itself handles security groups.

- Configure OVS database access and L3 scheduler

```
[ovn]
...
ovn_nb_connection = tcp:IP_ADDRESS:6641
ovn_sb_connection = tcp:IP_ADDRESS:6642
ovn_l3_scheduler = OVN_L3_SCHEDULER
```

> **Note**
>
> Replace `IP_ADDRESS` with the IP address of the controller node that runs the `ovsdb-server` service. Replace `OVN_L3_SCHEDULER` with `leastloaded` if you want the scheduler to select a compute node with the least number of gateway ports or `chance` if you want the scheduler to randomly select a compute node from the available list of compute nodes.

- Set ovn-cms-options with enable-chassis-as-gw in Open_vSwitch tables external_ids column. Then if this chassis has proper bridge mappings, it will be selected for scheduling gateway routers.

```
# ovs-vsctl set open . external-ids:ovn-cms-options=enable-chassis-
↪as-gw
```

8. Start, or restart, the `neutron-server` service.

   Using the *systemd* unit:

```
# systemctl start neutron-server
```

6.1.3 Network nodes

Deployments using OVN native layer-3 and DHCP services do not require conventional network nodes because connectivity to external networks (including VTEP gateways) and routing occurs on compute nodes.

6.1.4 Compute nodes

Each compute node runs the OVS and `ovn-controller` services. The `ovn-controller` service replaces the conventional OVS layer-2 agent.

1. Install the `ovn-host`, `openvswitch` and `neutron-ovn-metadata-agent` packages (RHEL/Fedora).
2. Install the `ovn-host`, `openvswitch-switch` and `neutron-ovn-metadata-agent` packages (Ubuntu/Debian).
3. Start the OVS service.

   Using the *systemd* unit:

   ```
   # systemctl start openvswitch (RHEL/Fedora)
   # systemctl start openvswitch-switch (Ubuntu/Debian)
   ```

4. Configure the OVS service.
   - Use OVS databases on the controller node.

     ```
     # ovs-vsctl set open . external-ids:ovn-remote=tcp:IP_ADDRESS:6642
     ```

     Replace `IP_ADDRESS` with the IP address of the controller node that runs the `ovsdb-server` service.
   - Enable one or more overlay network protocols. At a minimum, OVN requires enabling the `geneve` protocol. Deployments using VTEP gateways should also enable the `vxlan` protocol.

     ```
     # ovs-vsctl set open . external-ids:ovn-encap-type=geneve,vxlan
     ```

     Note

     Deployments without VTEP gateways can safely enable both protocols.
   - Configure the overlay network local endpoint IP address.

     ```
     # ovs-vsctl set open . external-ids:ovn-encap-ip=IP_ADDRESS
     ```

     Replace `IP_ADDRESS` with the IP address of the overlay network interface on the compute node.
5. Start the `ovn-controller` and `neutron-ovn-metadata-agent` services.

   Using the *systemd* unit:

   ```
   # systemctl start ovn-controller neutron-ovn-metadata-agent
   ```

6.1.5 Verify operation

1. Each compute node should contain an `ovn-controller` instance.

   ```
   # ovn-sbctl show
     <output>
   ```

6.1.6 References

CHAPTER SEVEN

OPEN VSWITCH (OVS) AND OVN REQUIREMENTS

Neutron uses Open vSwitch with the ML2/openswitch and ML2/ovn mechanism drivers. The latter also uses OVN. These are binary dependencies of a Neutron deployment with some version constraints. In the table below we hope to document the known minimum versions of OVS and OVN per Neutron release. Or at least the minimum versions we know we used in our CI system while developing.

We also have CI jobs to test with the main branch of OVS and OVN therefore we feel no need to document the known maximum versions - the newest OVS/OVN release should work.

Warning

In releases 2024.1 and earlier it happened that different groups of CI jobs (or rather Zuul job templates: base, grenade, rally) used different OVS and OVN versions. Here we document only the base version, that is the version used in single and multinode tempest jobs. Stadium projects like ovn-bgp-agent may also have different requirements.

Table 1: OpenStack Neutron OVS/OVN Minimum Requirement Matrix

OpenStack Release	Neutron Release	`OVS_BRANCH`	`OVN_BRANCH`
2025.2 (flamingo)	27.0	branch-3.3	branch-24.03
2025.1 (epoxy)	26.0	branch-3.3	branch-24.03
2024.2 (dalmatian)	25.0	branch-3.3	branch-24.03
2024.1 (caracal)	24.0	a4b04276ab5934d087669ff2d191a23931335c8	v21.06.0
2023.1 (antelope)	22.0	a4b04276ab5934d087669ff2d191a23931335c8	v21.06.0
zed	21.0	a4b04276ab5934d087669ff2d191a23931335c8	v21.06.0
yoga	20.0	v2.16.0	v21.06.0
xena	19.0	v2.16.0	v21.06.0
wallaby	18.0	0047ca3a0290f1ef954f2c76b31477cf4b9755f5	v20.06.1
victoria	17.0	51e9479da62edb04a5be47a7655de75c299b9fa	v20.06.1

This chapter explains how to install and configure the Networking service (neutron) using the *provider networks* or *self-service networks* option.

For more information about the Networking service including virtual networking components, layout, and traffic flows, see the *OpenStack Networking Guide*.

CHAPTER

EIGHT

OPENSTACK NETWORKING GUIDE

This guide targets OpenStack administrators seeking to deploy and manage OpenStack Networking (neutron).

8.1 Introduction

The OpenStack Networking service (neutron) provides an API that allows users to set up and define network connectivity and addressing in the cloud. The project code-name for Networking services is neutron. OpenStack Networking handles the creation and management of a virtual networking infrastructure, including networks, switches, subnets, and routers for devices managed by the OpenStack Compute service (nova). Advanced services such as firewalls or virtual private network (VPN) can also be used.

OpenStack Networking consists of the neutron-server, a database for persistent storage, and any number of plug-in agents, which provide other services such as interfacing with native Linux networking mechanisms, external devices, or SDN controllers.

OpenStack Networking is entirely standalone and can be deployed to a dedicated host. If your deployment uses a controller host to run centralized Compute components, you can deploy the Networking server to that specific host instead.

OpenStack Networking integrates with various OpenStack components:

- OpenStack Identity service (keystone) is used for authentication and authorization of API requests.
- OpenStack Compute service (nova) is used to plug each virtual NIC on the VM into a particular network.
- OpenStack Dashboard (horizon) is used by administrators and project users to create and manage network services through a web-based graphical interface.

Note

The network address ranges used in this guide are chosen in accordance with RFC 5737 and RFC 3849, and as such are restricted to the following:

IPv4:

- 192.0.2.0/24
- 198.51.100.0/24
- 203.0.113.0/24

IPv6:

- 2001:DB8::/32

The network address ranges in the examples of this guide should not be used for any purpose other than documentation.

Note

To reduce clutter, this guide removes command output without relevance to the particular action.

8.1.1 Basic networking

Ethernet

Ethernet is a networking protocol, specified by the IEEE 802.3 standard. Most wired network interface cards (NICs) communicate using Ethernet.

In the OSI model of networking protocols, Ethernet occupies the second layer, which is known as the data link layer. When discussing Ethernet, you will often hear terms such as *local network*, *layer 2*, *L2*, *link layer* and *data link layer*.

In an Ethernet network, the hosts connected to the network communicate by exchanging *frames*. Every host on an Ethernet network is uniquely identified by an address called the media access control (MAC) address. In particular, every virtual machine instance in an OpenStack environment has a unique MAC address, which is different from the MAC address of the compute host. A MAC address has 48 bits and is typically represented as a hexadecimal string, such as `08:00:27:b9:88:74`. The MAC address is hard-coded into the NIC by the manufacturer, although modern NICs allow you to change the MAC address programmatically. In Linux, you can retrieve the MAC address of a NIC using the **ip** command:

```
$ ip link show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP␣
↪mode DEFAULT group default qlen 1000
    link/ether 08:00:27:b9:88:74 brd ff:ff:ff:ff:ff:ff
```

Conceptually, you can think of an Ethernet network as a single bus that each of the network hosts connects to. In early implementations, an Ethernet network consisted of a single coaxial cable that hosts would tap into to connect to the network. However, network hosts in modern Ethernet networks connect directly to a network device called a *switch*. Still, this conceptual model is useful, and in network diagrams (including those generated by the OpenStack dashboard) an Ethernet network is often depicted as if it was a single bus. Youll sometimes hear an Ethernet network referred to as a *layer 2 segment*.

In an Ethernet network, every host on the network can send a frame directly to every other host. An Ethernet network also supports broadcasts so that one host can send a frame to every host on the network by sending to the special MAC address `ff:ff:ff:ff:ff:ff`. *ARP* and *DHCP* are two notable protocols that use Ethernet broadcasts. Because Ethernet networks support broadcasts, you will sometimes hear an Ethernet network referred to as a *broadcast domain*.

When a NIC receives an Ethernet frame, by default the NIC checks to see if the destination MAC address matches the address of the NIC (or the broadcast address), and the Ethernet frame is discarded if the MAC address does not match. For a compute host, this behavior is undesirable because the frame may be intended for one of the instances. NICs can be configured for *promiscuous mode*, where they pass all Ethernet frames to the operating system, even if the MAC address does not match. Compute hosts should always have the appropriate NICs configured for promiscuous mode.

As mentioned earlier, modern Ethernet networks use switches to interconnect the network hosts. A switch is a box of networking hardware with a large number of ports that forward Ethernet frames from one connected host to another. When hosts first send frames over the switch, the switch doesnt know which MAC address is associated with which port. If an Ethernet frame is destined for an unknown MAC address, the switch broadcasts the frame to all ports. The switch learns which MAC addresses are at which ports by observing the traffic. Once it knows which MAC address is associated with a port, it can send Ethernet frames to the correct port instead of broadcasting. The switch maintains the mappings of MAC addresses to switch ports in a table called a *forwarding table* or *forwarding information base* (FIB). Switches can be daisy-chained together, and the resulting connection of switches and hosts behaves like a single network.

VLANs

VLAN is a networking technology that enables a single switch to act as if it was multiple independent switches. Specifically, two hosts that are connected to the same switch but on different VLANs do not see each others traffic. OpenStack is able to take advantage of VLANs to isolate the traffic of different projects, even if the projects happen to have instances running on the same compute host. Each VLAN has an associated numerical ID, between 1 and 4094. We say VLAN 15 to refer to the VLAN with a numerical ID of 15.

To understand how VLANs work, lets consider VLAN applications in a traditional IT environment, where physical hosts are attached to a physical switch, and no virtualization is involved. Imagine a scenario where you want three isolated networks but you only have a single physical switch. The network administrator would choose three VLAN IDs, for example, 10, 11, and 12, and would configure the switch to associate switchports with VLAN IDs. For example, switchport 2 might be associated with VLAN 10, switchport 3 might be associated with VLAN 11, and so forth. When a switchport is configured for a specific VLAN, it is called an *access port*. The switch is responsible for ensuring that the network traffic is isolated across the VLANs.

Now consider the scenario that all of the switchports in the first switch become occupied, and so the organization buys a second switch and connects it to the first switch to expand the available number of switchports. The second switch is also configured to support VLAN IDs 10, 11, and 12. Now imagine host A connected to switch 1 on a port configured for VLAN ID 10 sends an Ethernet frame intended for host B connected to switch 2 on a port configured for VLAN ID 10. When switch 1 forwards the Ethernet frame to switch 2, it must communicate that the frame is associated with VLAN ID 10.

If two switches are to be connected together, and the switches are configured for VLANs, then the switchports used for cross-connecting the switches must be configured to allow Ethernet frames from any VLAN to be forwarded to the other switch. In addition, the sending switch must tag each Ethernet frame with the VLAN ID so that the receiving switch can ensure that only hosts on the matching VLAN are eligible to receive the frame.

A switchport that is configured to pass frames from all VLANs and tag them with the VLAN IDs is called a *trunk port*. IEEE 802.1Q is the network standard that describes how VLAN tags are encoded in Ethernet frames when trunking is being used.

Note that if you are using VLANs on your physical switches to implement project isolation in your OpenStack cloud, you must ensure that all of your switchports are configured as trunk ports.

It is important that you select a VLAN range not being used by your current network infrastructure. For example, if you estimate that your cloud must support a maximum of 100 projects, pick a VLAN range outside of that value, such as VLAN 200299. OpenStack, and all physical network infrastructure that handles project networks, must then support this VLAN range.

Trunking is used to connect between different switches. Each trunk uses a tag to identify which VLAN is in use. This ensures that switches on the same VLAN can communicate.

Subnets and ARP

While NICs use MAC addresses to address network hosts, TCP/IP applications use IP addresses. The Address Resolution Protocol (ARP) bridges the gap between Ethernet and IP by translating IP addresses into MAC addresses.

IP addresses are broken up into two parts: a *network number* and a *host identifier*. Two hosts are on the same *subnet* if they have the same network number. Recall that two hosts can only communicate directly over Ethernet if they are on the same local network. ARP assumes that all machines that are in the same subnet are on the same local network. Network administrators must take care when assigning IP addresses and netmasks to hosts so that any two hosts that are in the same subnet are on the same local network, otherwise ARP does not work properly.

To calculate the network number of an IP address, you must know the *netmask* associated with the address. A netmask indicates how many of the bits in the 32-bit IP address make up the network number.

There are two syntaxes for expressing a netmask:

- dotted quad
- classless inter-domain routing (CIDR)

Consider an IP address of 192.0.2.5, where the first 24 bits of the address are the network number. In dotted quad notation, the netmask would be written as `255.255.255.0`. CIDR notation includes both the IP address and netmask, and this example would be written as `192.0.2.5/24`.

Note

Creating CIDR subnets including a multicast address or a loopback address cannot be used in an OpenStack environment. For example, creating a subnet which is part of `224.0.0.0/4` or `127.0.0.0/8` address blocks is not supported.

Sometimes we want to refer to a subnet, but not any particular IP address on the subnet. A common convention is to set the host identifier to all zeros to make reference to a subnet. For example, if a hosts IP address is `192.0.2.24/24`, then we would say the subnet is `192.0.2.0/24`.

To understand how ARP translates IP addresses to MAC addresses, consider the following example. Assume host *A* has an IP address of `192.0.2.5/24` and a MAC address of `fc:99:47:49:d4:a0`, and wants to send a packet to host *B* with an IP address of `192.0.2.7`. Note that the network number is the same for both hosts, so host *A* is able to send frames directly to host *B*.

The first time host *A* attempts to communicate with host *B*, the destination MAC address is not known. Host *A* makes an ARP request to the local network. The request is a broadcast with a message like this:

To: everybody (ff:ff:ff:ff:ff:ff). I am looking for the computer who has IP address 192.0.2.7. Signed: MAC address fc:99:47:49:d4:a0.

Host *B* responds with a response like this:

To: fc:99:47:49:d4:a0. I have IP address 192.0.2.7. Signed: MAC address 54:78:1a:86:00:a5.

Host *A* then sends Ethernet frames to host *B*.

You can initiate an ARP request manually using the **arping** command. For example, to send an ARP request to IP address `192.0.2.132`:

```
$ arping -I eth0 192.0.2.132
ARPING 192.0.2.132 from 192.0.2.131 eth0
Unicast reply from 192.0.2.132 [54:78:1A:86:1C:0B]  0.670ms
Unicast reply from 192.0.2.132 [54:78:1A:86:1C:0B]  0.722ms
Unicast reply from 192.0.2.132 [54:78:1A:86:1C:0B]  0.723ms
Sent 3 probes (1 broadcast(s))
Received 3 response(s)
```

To reduce the number of ARP requests, operating systems maintain an ARP cache that contains the mappings of IP addresses to MAC address. On a Linux machine, you can view the contents of the ARP cache by using the **arp** command:

```
$ arp -n
Address                  HWtype  HWaddress           Flags Mask            ␣
↪Iface
192.0.2.3                ether   52:54:00:12:35:03   C                     ␣
↪eth0
192.0.2.2                ether   52:54:00:12:35:02   C                     ␣
↪eth0
```

DHCP

Hosts connected to a network use the Dynamic Host Configuration Protocol (DHCP) to dynamically obtain IP addresses. A DHCP server hands out the IP addresses to network hosts, which are the DHCP clients.

DHCP clients locate the DHCP server by sending a *UDP* packet from port 68 to address `255.255.255.255` on port 67. Address `255.255.255.255` is the local network broadcast address: all hosts on the local network see the UDP packets sent to this address. However, such packets are not forwarded to other networks. Consequently, the DHCP server must be on the same local network as the client, or the server will not receive the broadcast. The DHCP server responds by sending a UDP packet from port 67 to port 68 on the client. The exchange looks like this:

1. The client sends a discover (Im a client at MAC address `08:00:27:b9:88:74`, I need an IP address)
2. The server sends an offer (OK `08:00:27:b9:88:74`, Im offering IP address `192.0.2.112`)
3. The client sends a request (Server `192.0.2.131`, I would like to have IP `192.0.2.112`)
4. The server sends an acknowledgement (OK `08:00:27:b9:88:74`, IP `192.0.2.112` is yours)

OpenStack uses a third-party program called dnsmasq to implement the DHCP server. Dnsmasq writes to the syslog, where you can observe the DHCP request and replies:

```
Apr 23 15:53:46 c100-1 dhcpd: DHCPDISCOVER from 08:00:27:b9:88:74 via eth2
Apr 23 15:53:46 c100-1 dhcpd: DHCPOFFER on 192.0.2.112 to 08:00:27:b9:88:74␣
↪via eth2
Apr 23 15:53:48 c100-1 dhcpd: DHCPREQUEST for 192.0.2.112 (192.0.2.131) from␣
↪08:00:27:b9:88:74 via eth2
Apr 23 15:53:48 c100-1 dhcpd: DHCPACK on 192.0.2.112 to 08:00:27:b9:88:74 via␣
↪eth2
```

When troubleshooting an instance that is not reachable over the network, it can be helpful to examine this log to verify that all four steps of the DHCP protocol were carried out for the instance in question.

IP

The Internet Protocol (IP) specifies how to route packets between hosts that are connected to different local networks. IP relies on special network hosts called *routers* or *gateways*. A router is a host that is connected to at least two local networks and can forward IP packets from one local network to another. A router has multiple IP addresses: one for each of the networks it is connected to.

In the OSI model of networking protocols IP occupies the third layer, known as the network layer. When discussing IP, you will often hear terms such as *layer 3*, *L3*, and *network layer*.

A host sending a packet to an IP address consults its *routing table* to determine which machine on the local network(s) the packet should be sent to. The routing table maintains a list of the subnets associated with each local network that the host is directly connected to, as well as a list of routers that are on these local networks.

On a Linux machine, any of the following commands displays the routing table:

```
$ ip route show
$ route -n
$ netstat -rn
```

Here is an example of output from **ip route show**:

```
$ ip route show
default via 192.0.2.2 dev eth0
192.0.2.0/24 dev eth0  proto kernel  scope link  src 192.0.2.15
198.51.100.0/25 dev eth1  proto kernel  scope link  src 198.51.100.100
198.51.100.192/26 dev virbr0  proto kernel  scope link  src 198.51.100.193
```

Line 1 of the output specifies the location of the default route, which is the effective routing rule if none of the other rules match. The router associated with the default route (`192.0.2.2` in the example above) is sometimes referred to as the *default gateway*. A *DHCP* server typically transmits the IP address of the default gateway to the DHCP client along with the clients IP address and a netmask.

Line 2 of the output specifies that IPs in the `192.0.2.0/24` subnet are on the local network associated with the network interface eth0.

Line 3 of the output specifies that IPs in the `198.51.100.0/25` subnet are on the local network associated with the network interface eth1.

Line 4 of the output specifies that IPs in the `198.51.100.192/26` subnet are on the local network associated with the network interface virbr0.

The output of the **route -n** and **netstat -rn** commands are formatted in a slightly different way. This example shows how the same routes would be formatted using these commands:

```
$ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         192.0.2.2       0.0.0.0         UG        0 0          0 eth0
192.0.2.0       0.0.0.0         255.255.255.0   U         0 0          0 eth0
198.51.100.0    0.0.0.0         255.255.255.128 U         0 0          0 eth1
```

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```
198.51.100.192  0.0.0.0         255.255.255.192 U         0 0          0␣
↪virbr0
```

The **ip route get** command outputs the route for a destination IP address. From the below example, destination IP address `192.0.2.14` is on the local network of eth0 and would be sent directly:

```
$ ip route get 192.0.2.14
192.0.2.14 dev eth0  src 192.0.2.15
```

The destination IP address `203.0.113.34` is not on any of the connected local networks and would be forwarded to the default gateway at `192.0.2.2`:

```
$ ip route get 203.0.113.34
203.0.113.34 via 192.0.2.2 dev eth0  src 192.0.2.15
```

It is common for a packet to hop across multiple routers to reach its final destination. On a Linux machine, the `traceroute` and more recent `mtr` programs prints out the IP address of each router that an IP packet traverses along its path to its destination.

TCP/UDP/ICMP

For networked software applications to communicate over an IP network, they must use a protocol layered atop IP. These protocols occupy the fourth layer of the OSI model known as the *transport layer* or *layer 4*. See the Protocol Numbers web page maintained by the Internet Assigned Numbers Authority (IANA) for a list of protocols that layer atop IP and their associated numbers.

The *Transmission Control Protocol* (TCP) is the most commonly used layer 4 protocol in networked applications. TCP is a *connection-oriented* protocol: it uses a client-server model where a client connects to a server, where *server* refers to the application that receives connections. The typical interaction in a TCP-based application proceeds as follows:

1. Client connects to server.
2. Client and server exchange data.
3. Client or server disconnects.

Because a network host may have multiple TCP-based applications running, TCP uses an addressing scheme called *ports* to uniquely identify TCP-based applications. A TCP port is associated with a number in the range 1-65535, and only one application on a host can be associated with a TCP port at a time, a restriction that is enforced by the operating system.

A TCP server is said to *listen* on a port. For example, an SSH server typically listens on port 22. For a client to connect to a server using TCP, the client must know both the IP address of a servers host and the servers TCP port.

The operating system of the TCP client application automatically assigns a port number to the client. The client owns this port number until the TCP connection is terminated, after which the operating system reclaims the port number. These types of ports are referred to as *ephemeral ports*.

IANA maintains a registry of port numbers for many TCP-based services, as well as services that use other layer 4 protocols that employ ports. Registering a TCP port number is not required, but registering a port number is helpful to avoid collisions with other services. See firewalls and default ports in OpenStack Installation Guide for the default TCP ports used by various services involved in an OpenStack deployment.

The most common application programming interface (API) for writing TCP-based applications is called *Berkeley sockets*, also known as *BSD sockets* or, simply, *sockets*. The sockets API exposes a *stream oriented* interface for writing TCP applications. From the perspective of a programmer, sending data over a TCP connection is similar to writing a stream of bytes to a file. It is the responsibility of the operating systems TCP/IP implementation to break up the stream of data into IP packets. The operating system is also responsible for automatically retransmitting dropped packets, and for handling flow control to ensure that transmitted data does not overrun the senders data buffers, receivers data buffers, and network capacity. Finally, the operating system is responsible for re-assembling the packets in the correct order into a stream of data on the receivers side. Because TCP detects and retransmits lost packets, it is said to be a *reliable* protocol.

The *User Datagram Protocol* (UDP) is another layer 4 protocol that is the basis of several well-known networking protocols. UDP is a *connectionless* protocol: two applications that communicate over UDP do not need to establish a connection before exchanging data. UDP is also an *unreliable* protocol. The operating system does not attempt to retransmit or even detect lost UDP packets. The operating system also does not provide any guarantee that the receiving application sees the UDP packets in the same order that they were sent in.

UDP, like TCP, uses the notion of ports to distinguish between different applications running on the same system. Note, however, that operating systems treat UDP ports separately from TCP ports. For example, it is possible for one application to be associated with TCP port 16543 and a separate application to be associated with UDP port 16543.

Like TCP, the sockets API is the most common API for writing UDP-based applications. The sockets API provides a *message-oriented* interface for writing UDP applications: a programmer sends data over UDP by transmitting a fixed-sized message. If an application requires retransmissions of lost packets or a well-defined ordering of received packets, the programmer is responsible for implementing this functionality in the application code.

DHCP, the Domain Name System (DNS), the Network Time Protocol (NTP), and *Virtual extensible local area network (VXLAN)* are examples of UDP-based protocols used in OpenStack deployments.

UDP has support for one-to-many communication: sending a single packet to multiple hosts. An application can broadcast a UDP packet to all of the network hosts on a local network by setting the receiver IP address as the special IP broadcast address `255.255.255.255`. An application can also send a UDP packet to a set of receivers using *IP multicast*. The intended receiver applications join a multicast group by binding a UDP socket to a special IP address that is one of the valid multicast group addresses. The receiving hosts do not have to be on the same local network as the sender, but the intervening routers must be configured to support IP multicast routing. VXLAN is an example of a UDP-based protocol that uses IP multicast.

The *Internet Control Message Protocol* (ICMP) is a protocol used for sending control messages over an IP network. For example, a router that receives an IP packet may send an ICMP packet back to the source if there is no route in the routers routing table that corresponds to the destination address (ICMP code 1, destination host unreachable) or if the IP packet is too large for the router to handle (ICMP code 4, fragmentation required and dont fragment flag is set).

The **`ping`** and **`mtr`** Linux command-line tools are two examples of network utilities that use ICMP.

8.1.2 Network components

Switches

Switches are Multi-Input Multi-Output (MIMO) devices that enable packets to travel from one node to another. Switches connect hosts that belong to the same layer-2 network. Switches enable forwarding of the packet received on one port (input) to another port (output) so that they reach the desired destination node. Switches operate at layer-2 in the networking model. They forward the traffic based on the destination Ethernet address in the packet header.

Routers

Routers are special devices that enable packets to travel from one layer-3 network to another. Routers enable communication between two nodes on different layer-3 networks that are not directly connected to each other. Routers operate at layer-3 in the networking model. They route the traffic based on the destination IP address in the packet header.

Firewalls

Firewalls are used to regulate traffic to and from a host or a network. A firewall can be either a specialized device connecting two networks or a software-based filtering mechanism implemented on an operating system. Firewalls are used to restrict traffic to a host based on the rules defined on the host. They can filter packets based on several criteria such as source IP address, destination IP address, port numbers, connection state, and so on. It is primarily used to protect the hosts from unauthorized access and malicious attacks.

Load balancers

Load balancers can be software-based or hardware-based devices that allow traffic to evenly be distributed across several servers. By distributing the traffic across multiple servers, it avoids overload of a single server thereby preventing a single point of failure in the product. This further improves the performance, network throughput, and response time of the servers. Load balancers are typically used in a 3-tier architecture. In this model, a load balancer receives a request from the front-end web server, which then forwards the request to one of the available back-end database servers for processing. The response from the database server is passed back to the web server for further processing.

8.1.3 Overlay (tunnel) protocols

Tunneling is a mechanism that makes transfer of payloads feasible over an incompatible delivery network. It allows the network user to gain access to denied or insecure networks. Data encryption may be employed to transport the payload, ensuring that the encapsulated user network data appears as public even though it is private and can easily pass the conflicting network.

Generic routing encapsulation (GRE)

Generic routing encapsulation (GRE) is a protocol that runs over IP and is employed when delivery and payload protocols are compatible but payload addresses are incompatible. For instance, a payload might think it is running on a datalink layer but it is actually running over a transport layer using datagram protocol over IP. GRE creates a private point-to-point connection and works by encapsulating a payload. GRE is a foundation protocol for other tunnel protocols but the GRE tunnels provide only weak authentication.

Virtual extensible local area network (VXLAN)

The purpose of VXLAN is to provide scalable network isolation. VXLAN is a Layer 2 overlay scheme on a Layer 3 network. It allows an overlay layer-2 network to spread across multiple underlay layer-3 network domains. Each overlay is termed a VXLAN segment. Only VMs within the same VXLAN segment can communicate.

Generic Network Virtualization Encapsulation (GENEVE)

Geneve is designed to recognize and accommodate changing capabilities and needs of different devices in network virtualization. It provides a framework for tunneling rather than being prescriptive about the entire system. Geneve defines the content of the metadata flexibly that is added during encapsulation and tries to adapt to various virtualization scenarios. It uses UDP as its transport protocol and is dynamic in size using extensible option headers. Geneve supports unicast, multicast, and broadcast.

8.1.4 Network namespaces

A namespace is a way of scoping a particular set of identifiers. Using a namespace, you can use the same identifier multiple times in different namespaces. You can also restrict an identifier set visible to particular processes.

For example, Linux provides namespaces for networking and processes, among other things. If a process is running within a process namespace, it can only see and communicate with other processes in the same namespace. So, if a shell in a particular process namespace ran **`ps waux`**, it would only show the other processes in the same namespace.

Linux network namespaces

In a network namespace, the scoped identifiers are network devices; so a given network device, such as `eth0`, exists in a particular namespace. Linux starts up with a default network namespace, so if your operating system does not do anything special, that is where all the network devices will be located. But it is also possible to create further non-default namespaces, and create new devices in those namespaces, or to move an existing device from one namespace to another.

Each network namespace also has its own routing table, and in fact this is the main reason for namespaces to exist. A routing table is keyed by destination IP address, so network namespaces are what you need if you want the same destination IP address to mean different things at different times - which is something that OpenStack Networking requires for its feature of providing overlapping IP addresses in different virtual networks.

Each network namespace also has its own set of iptables (for both IPv4 and IPv6). So, you can apply different security to flows with the same IP addressing in different namespaces, as well as different routing.

Any given Linux process runs in a particular network namespace. By default this is inherited from its parent process, but a process with the right capabilities can switch itself into a different namespace; in practice this is mostly done using the **`ip netns exec NETNS COMMAND...`** invocation, which starts `COMMAND` running in the namespace named `NETNS`. Suppose such a process sends out a message to IP address A.B.C.D, the effect of the namespace is that A.B.C.D will be looked up in that namespaces routing table, and that will determine the network device that the message is transmitted through.

Virtual routing and forwarding (VRF)

Virtual routing and forwarding is an IP technology that allows multiple instances of a routing table to coexist on the same router at the same time. It is another name for the network namespace functionality described above.

8.1.5 Network address translation

Network Address Translation (NAT) is a process for modifying the source or destination addresses in the headers of an IP packet while the packet is in transit. In general, the sender and receiver applications are not aware that the IP packets are being manipulated.

NAT is often implemented by routers, and so we will refer to the host performing NAT as a *NAT router*. However, in OpenStack deployments it is typically Linux servers that implement the NAT functionality, not hardware routers. These servers use the iptables software package to implement the NAT functionality.

There are multiple variations of NAT, and here we describe three kinds commonly found in OpenStack deployments.

SNAT

In *Source Network Address Translation* (SNAT), the NAT router modifies the IP address of the sender in IP packets. SNAT is commonly used to enable hosts with *private addresses* to communicate with servers on the public Internet.

RFC 1918 reserves the following three subnets as private addresses:

- `10.0.0.0/8`
- `172.16.0.0/12`
- `192.168.0.0/16`

These IP addresses are not publicly routable, meaning that a host on the public Internet can not send an IP packet to any of these addresses. Private IP addresses are widely used in both residential and corporate environments.

Often, an application running on a host with a private IP address will need to connect to a server on the public Internet. An example is a user who wants to access a public website such as www.openstack.org. If the IP packets reach the web server at www.openstack.org with a private IP address as the source, then the web server cannot send packets back to the sender.

SNAT solves this problem by modifying the source IP address to an IP address that is routable on the public Internet. There are different variations of SNAT; in the form that OpenStack deployments use, a NAT router on the path between the sender and receiver replaces the packets source IP address with the routers public IP address. The router also modifies the source TCP or UDP port to another value, and the router maintains a record of the senders true IP address and port, as well as the modified IP address and port.

When the router receives a packet with the matching IP address and port, it translates these back to the private IP address and port, and forwards the packet along.

Because the NAT router modifies ports as well as IP addresses, this form of SNAT is sometimes referred to as *Port Address Translation* (PAT). It is also sometimes referred to as *NAT overload*.

OpenStack uses SNAT to enable applications running inside of instances to connect out to the public Internet.

DNAT

In *Destination Network Address Translation* (DNAT), the NAT router modifies the IP address of the destination in IP packet headers.

OpenStack uses DNAT to route packets from instances to the OpenStack metadata service. Applications running inside of instances access the OpenStack metadata service by making HTTP GET requests to a web server with IP address 169.254.169.254. In an OpenStack deployment, there is no host with this IP address. Instead, OpenStack uses DNAT to change the destination IP of these packets so they reach the network interface that a metadata service is listening on.

One-to-one NAT

In *one-to-one NAT*, the NAT router maintains a one-to-one mapping between private IP addresses and public IP addresses. OpenStack uses one-to-one NAT to implement floating IP addresses.

8.1.6 OpenStack Networking

OpenStack Networking allows you to create and manage network objects, such as networks, subnets, and ports, which other OpenStack services can use. Plug-ins can be implemented to accommodate different networking equipment and software, providing flexibility to OpenStack architecture and deployment.

The Networking service, code-named neutron, provides an API that lets you define network connectivity and addressing in the cloud. The Networking service enables operators to leverage different networking technologies to power their cloud networking. The Networking service also provides an API to configure and manage a variety of network services ranging from L3 forwarding and Network Address Translation (NAT) to perimeter firewalls, and virtual private networks.

It includes the following components:

API server
: The OpenStack Networking API includes support for Layer 2 networking and IP Address Management (IPAM), as well as an extension for a Layer 3 router construct that enables routing between Layer 2 networks and gateways to external networks. OpenStack Networking includes a growing list of plug-ins that enable interoperability with various commercial and open source network technologies, including routers, switches, virtual switches and software-defined networking (SDN) controllers.

OpenStack Networking plug-in and agents
: Plugs and unplugs ports, creates networks or subnets, and provides IP addressing. The chosen plug-in and agents differ depending on the vendor and technologies used in the particular cloud. It is important to mention that only one plug-in can be used at a time.

Messaging queue
: Accepts and routes RPC requests between agents to complete API operations. Message queue is used in the ML2 plug-in for RPC between the neutron server and neutron agents that run on each hypervisor, in the ML2 mechanism drivers for Open vSwitch and Linux bridge.

Concepts

To configure rich network topologies, you can create and configure networks and subnets and instruct other OpenStack services like Compute to attach virtual devices to ports on these networks. OpenStack Compute is a prominent consumer of OpenStack Networking to provide connectivity for its instances. In particular, OpenStack Networking supports each project having multiple private networks and enables projects to choose their own IP addressing scheme, even if those IP addresses overlap with those that other projects use. There are two types of network, project and provider networks. It is possible to share any of these types of networks among projects as part of the network creation process.

Provider networks

Provider networks offer layer-2 connectivity to instances with optional support for DHCP and metadata services. These networks connect, or map, to existing layer-2 networks in the data center, typically using VLAN (802.1q) tagging to identify and separate them.

Provider networks generally offer simplicity, performance, and reliability at the cost of flexibility. By default only administrators can create or update provider networks because they require configuration of physical network infrastructure. It is possible to change the user who is allowed to create or update provider networks with the following parameters of `policy.yaml`:

- `create_network:provider:physical_network`
- `update_network:provider:physical_network`

> **Warning**
>
> The creation and modification of provider networks enables use of physical network resources, such as VLAN-s. Enable these changes only for trusted projects.

Also, provider networks only handle layer-2 connectivity for instances, thus lacking support for features such as routers and floating IP addresses.

In many cases, operators who are already familiar with virtual networking architectures that rely on physical network infrastructure for layer-2, layer-3, or other services can seamlessly deploy the OpenStack Networking service. In particular, provider networks appeal to operators looking to migrate from the Compute networking service (nova-network) to the OpenStack Networking service. Over time, operators can build on this minimal architecture to enable more cloud networking features.

In general, the OpenStack Networking software components that handle layer-3 operations impact performance and reliability the most. To improve performance and reliability, provider networks move layer-3 operations to the physical network infrastructure.

In one particular use case, the OpenStack deployment resides in a mixed environment with conventional virtualization and bare-metal hosts that use a sizable physical network infrastructure. Applications that run inside the OpenStack deployment might require direct layer-2 access, typically using VLANs, to applications outside of the deployment.

Routed provider networks

Routed provider networks offer layer-3 connectivity to instances. These networks map to existing layer-3 networks in the data center. More specifically, the network maps to multiple layer-2 segments, each of which is essentially a provider network. Each has a router gateway attached to it which routes traffic between them and externally. The Networking service does not provide the routing.

Routed provider networks offer performance at scale that is difficult to achieve with a plain provider network at the expense of guaranteed layer-2 connectivity.

Neutron port could be associated with only one network segment, but there is an exception for OVN distributed services like OVN Metadata.

See *Routed provider networks* for more information.

Self-service networks

Self-service networks primarily enable general (non-privileged) projects to manage networks without involving administrators. These networks are entirely virtual and require virtual routers to interact with provider and external networks such as the Internet. Self-service networks also usually provide DHCP and metadata services to instances.

In most cases, self-service networks use overlay protocols such as VXLAN or GRE because they can support many more networks than layer-2 segmentation using VLAN tagging (802.1q). Furthermore, VLANs typically require additional configuration of physical network infrastructure.

IPv4 self-service networks typically use private IP address ranges (RFC1918) and interact with provider networks via source NAT on virtual routers. Floating IP addresses enable access to instances from provider networks via destination NAT on virtual routers. IPv6 self-service networks always use public IP address ranges and interact with provider networks via virtual routers with static routes.

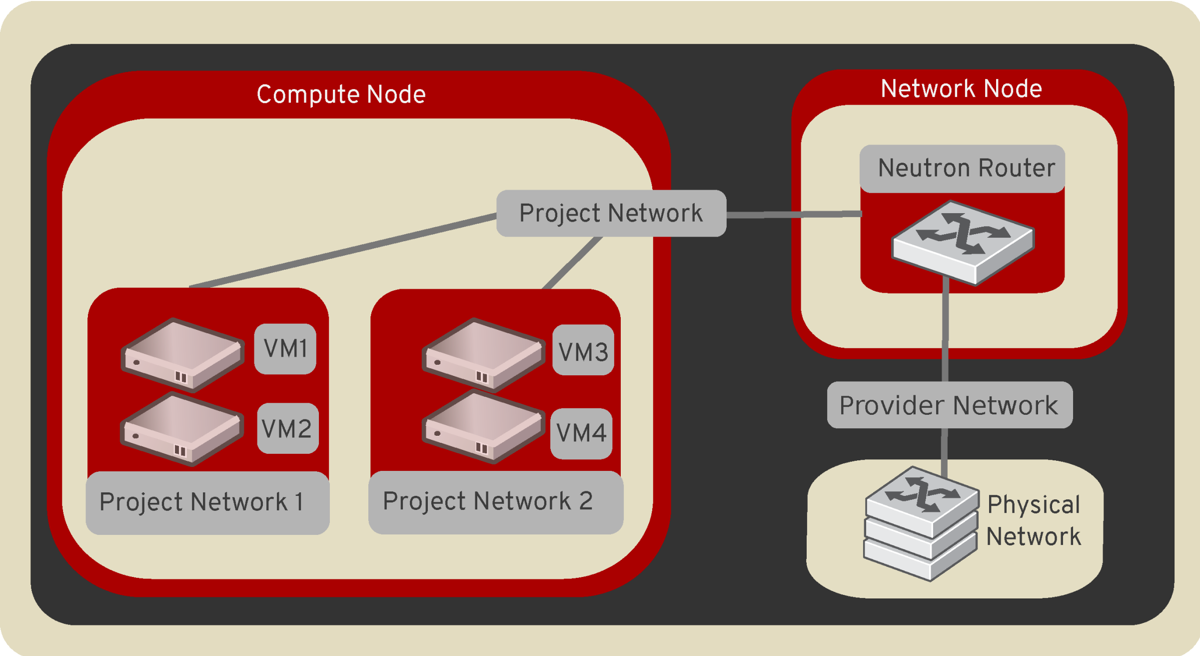
The Networking service implements routers using a layer-3 agent that typically resides at least one network node. Contrary to provider networks that connect instances to the physical network infrastructure at layer-2, self-service networks must traverse a layer-3 agent. Thus, oversubscription or failure of a layer-3 agent or network node can impact a significant quantity of self-service networks and instances using them. Consider implementing one or more high-availability features to increase redundancy and performance of self-service networks.

Users create project networks for connectivity within projects. By default, they are fully isolated and are not shared with other projects. OpenStack Networking supports the following types of network isolation and overlay technologies.

Flat
: All instances reside on the same network, which can also be shared with the hosts. No VLAN tagging or other network segregation takes place.

VLAN
: Networking allows users to create multiple provider or project networks using VLAN IDs (802.1Q tagged) that correspond to VLANs present in the physical network. This allows instances to communicate with each other across the environment. They can also communicate with dedicated servers, firewalls, and other networking infrastructure on the same layer 2 VLAN.

GRE and VXLAN
: VXLAN and GRE are encapsulation protocols that create overlay networks to activate and control communication between compute instances. A Networking router is required to allow traffic to flow outside of the GRE or VXLAN project network. A router is also required to connect directly-connected project networks with external networks, including the Internet. The router provides the ability to connect to instances directly from an external network using floating IP addresses.



Subnets

A block of IP addresses and associated configuration state. This is also known as the native IPAM (IP Address Management) provided by the networking service for both project and provider networks. Subnets are used to allocate IP addresses when new ports are created on a network.

Subnet pools

End users normally can create subnets with any valid IP addresses without other restrictions. However, in some cases, it is nice for the admin or the project to pre-define a pool of addresses from which to create subnets with automatic allocation.

Using subnet pools constrains what addresses can be used by requiring that every subnet be within the defined pool. It also prevents address reuse or overlap by two subnets from the same pool.

See *Subnet Pools* for more information.

Ports

A port is a connection point for attaching a single device, such as the NIC of a virtual server, to a virtual network. The port also describes the associated network configuration, such as the MAC and IP addresses to be used on that port.

Routers

Routers provide virtual layer-3 services such as routing and NAT between self-service and provider networks or among self-service networks belonging to a project. The Networking service uses a layer-3 agent to manage routers via namespaces.

Security groups

Security groups provide a container for virtual firewall rules that control ingress (inbound to instances) and egress (outbound from instances) network traffic at the port level. Security groups use a default deny policy and only contain rules that allow specific traffic. Each port can reference one or more security groups in an additive fashion. The firewall driver translates security group rules to a configuration for the underlying packet filtering technology such as `iptables`.

Each project contains a `default` security group that by default allows all egress traffic and denies all ingress traffic. You can change the rules in the `default` security group. Admin user can also define own set of security group rules which will be added by default to each new `default` and each new non-default (custom) security group created for every project in the cloud. There is `security-group-default-rules` API extension which allows to define such own set of the default security group rules. If you launch an instance without specifying a security group, the `default` security group automatically applies to it. Similarly, if you create a port without specifying a security group, the `default` security group automatically applies to it.

> **Note**
>
> If you use the metadata service, removing the default egress rules denies access to TCP port 80 on 169.254.169.254, thus preventing instances from retrieving metadata.

Security group rules are stateful. Thus, allowing ingress TCP port 22 for secure shell automatically creates rules that allow return egress traffic and ICMP error messages involving those TCP connections.

By default, all security groups contain a series of basic (sanity) and anti-spoofing rules that perform the following actions:

- Allow egress traffic only if it uses the source MAC and IP addresses of the port for the instance, source MAC and IP combination in `allowed-address-pairs`, or valid MAC address (port or `allowed-address-pairs`) and associated EUI64 link-local IPv6 address.
- Allow egress DHCP discovery and request messages that use the source MAC address of the port for the instance and the unspecified IPv4 address (0.0.0.0).
- Allow ingress DHCP and DHCPv6 responses from the DHCP server on the subnet so instances can acquire IP addresses.
- Deny egress DHCP and DHCPv6 responses to prevent instances from acting as DHCP(v6) servers.
- Allow ingress/egress ICMPv6 MLD, neighbor solicitation, and neighbor discovery messages so instances can discover neighbors and join multicast groups.
- Deny egress ICMPv6 router advertisements to prevent instances from acting as IPv6 routers and forwarding IPv6 traffic for other instances.
- Allow egress ICMPv6 MLD reports (v1 and v2) and neighbor solicitation messages that use the source MAC address of a particular instance and the unspecified IPv6 address (::). Duplicate address detection (DAD) relies on these messages.
- Allow egress non-IP traffic from the MAC address of the port for the instance and any additional MAC addresses in `allowed-address-pairs` on the port for the instance.

Those rules mentioned above are added automatically by neutron and cannot be changed using `default security group rules` API provided by the `security-group-default-rules` extensions.

Although non-IP traffic, security groups do not implicitly allow all ARP traffic. Separate ARP filtering rules prevent instances from using ARP to intercept traffic for another instance. You cannot disable or remove these rules.

You can disable security groups including basic and anti-spoofing rules by setting the port attribute `port_security_enabled` to `False`.

Extensions

The OpenStack Networking service is extensible. Extensions serve two purposes: they allow the introduction of new features in the API without requiring a version change and they allow the introduction of vendor specific niche functionality. Applications can programmatically list available extensions by performing a GET on the `/extensions` URI. Note that this is a versioned request; that is, an extension available in one API version might not be available in another.

DHCP

The optional DHCP service manages IP addresses for instances on provider and self-service networks. The Networking service implements the DHCP service using an agent that manages `qdhcp` namespaces and the `dnsmasq` service.

Metadata

The optional metadata service provides an API for instances to obtain metadata such as SSH keys.

Service and component hierarchy

Server

- Provides API, manages database, etc.

Plug-ins

- Manages agents

Agents

- Provides layer 2/3 connectivity to instances
- Handles physical-virtual network transition
- Handles metadata, etc.

Layer 2 (Ethernet and Switching)

- Linux Bridge
- OVS

Layer 3 (IP and Routing)

- L3
- DHCP

Miscellaneous

- Metadata

Services

Routing services

VPNaaS

The Virtual Private Network-as-a-Service (VPNaaS) is a neutron extension that introduces the VPN feature set.

LBaaS

The Load-Balancer-as-a-Service (LBaaS) API provisions and configures load balancers. The reference implementation is based on the HAProxy software load balancer. See the Octavia project for more information.

FWaaS

The Firewall-as-a-Service (FWaaS) API allows to apply firewalls to OpenStack objects such as projects, routers, and router ports.

8.1.7 Neutron API policies and supported roles

As part of the `Consistent and Secure Default RBAC` community goal[1] Neutron implemented support for various scopes and personas in all of the API policies which are defined in the Neutron code.

Roles supported by the default Neutron API policies

Roles supported by the default Neutron API policies are:

- PROJECT_READER - this role is intended to have read-only access to the project owned resources.
- PROJECT_MEMBER - this role inherits all of the privileges from the PROJECT_READER role and also has access to `create`, `update` and `delete` project-owned resources.
- PROJECT_MANAGER - this role inherits all of the privileges from the PROJECT_MEMBER role and additionally is allowed to do more operations on the project-owned resources.
- ADMIN - this role is the same as it was in the old default policies. A user with granted ADMIN role is allowed to do almost every possible modification on all resources, even those which belong to different projects.
- SERVICE - this is a special role designed to be used for service-to-service communication only, for example, between Nova and Neutron. It does not inherit any privileges from any other roles mentioned above.

Default API policies defined in Neutron

By default, all of the existing API policies can be used with `project` scoped tokens only. Tokens with `service` scope are not supported by any of the policies defined in the Neutron code.

[1] https://governance.openstack.org/tc/goals/selected/consistent-and-secure-rbac.html

Default API policies

Default API policies defined in the Neutron code can be found in the *Policy Reference* document.

References

8.1.8 Firewall-as-a-Service (FWaaS)

The Firewall-as-a-Service (FWaaS) plug-in applies firewalls to OpenStack objects such as projects, routers, and router ports.

The central concepts with OpenStack firewalls are the notions of a firewall policy and a firewall rule. A policy is an ordered collection of rules. A rule specifies a collection of attributes (such as port ranges, protocol, and IP addresses) that constitute match criteria and an action to take (allow or deny) on matched traffic. A policy can be made public, so it can be shared across projects.

Firewalls are implemented in various ways, depending on the driver used. For example, an iptables driver implements firewalls using iptable rules. An OpenVSwitch driver implements firewall rules using flow entries in flow tables.

FWaaS v2

The newer FWaaS implementation, v2, provides a much more granular service. The notion of a firewall has been replaced with firewall group to indicate that a firewall consists of two policies: an ingress policy and an egress policy. A firewall group is applied not at the router level (all ports on a router) but at the port level. Currently, router ports can be specified. For Ocata, VM ports can also be specified.

FWaaS v1

FWaaS v1 was deprecated in the Newton cycle and removed entirely in the Stein cycle.

FWaaS Feature Matrix

The following table shows FWaaS v2 features.

Feature	Supported
Supports L3 firewalling for routers	NO*
Supports L3 firewalling for router ports	YES
Supports L2 firewalling (VM ports)	YES
CLI support	YES
Horizon support	YES

* A firewall group can be applied to all ports on a given router in order to effect this.

For further information, see the FWaaS v2 configuration guide.

8.2 Configuration

8.2.1 Active-active L3 Gateway with Multihoming

Why

By default, Neutron routers are set up with a single external gateway port connected to a single layer 2 broadcast domain. To allow layer 3 connectivity to the outside world, a single static default route is added per address family, pointing at the IP address provided by the network administrator.

In such a configuration high availability is achieved by ensuring the same layer 2 broadcast domain is available to all gateway chassis, allowing the network equipment to be configured to provide the gateway IP as a virtual IP address serviced by multiple routers.

Providing a single layer 2 broadcast domain to many hosts in a large data center network can be undesired, this feature may provide a way to implement external gateway high availability at the layer 3 level.

Both approaches have their benefits and drawbacks, so make sure to familiarize yourself with the *limitations* and *scale considerations* before deciding whether this feature meets your requirements.

Prerequisites

The network equipment involved in routing to/from the cloud needs to support Bidirectional Forwarding Detection (BFD) for static routes. For the purpose of this document we will be using FRR support for BFD static route monitoring.

There are further requirements for the network equipment acting as border gateways, which may include provision of direct links, configuration of IGP, redistribution of static routes and so on, however these details are outside the scope of this document.

Supported drivers and versions

- OpenStack 2024.1 or newer.
- OVN 22.03 or newer.

Note

At the time of this writing only the ML2/OVN driver supports this feature.

Limitations

- There is currently no integration with dynamic routing protocols such as BGP for this feature, next-hop liveness detection is provided by BFD.
- The feature can not be used together with Network address translation (NAT).

Warning

The feature can not be used together with NAT, routers and gateways must be created with the `--disable-snat` argument, and instances must use unique local (ULA) or globally routable addresses.

Scale considerations

- Enabling BFD for default routes will establish one BFD session per router gateway port. Each participant in a BFD Session typically transmit one message per second. The BFD Control packets are subject to slow path processing and it is advised to ensure the control plane capacity in network equipment aligns with the expected number of router gateway ports.

How

As laid out in the Active-active L3 Gateway with Multihoming specification, the components involved in achieving high availability at the layer 3 level are:

- *Adding multiple gateway ports to a router*, providing interfaces in multiple layer 2 broadcast domains and/or layer 3 subnets.
- *Adding multiple default routes to a router*, each with different output port and next-hop addresses, effectively enabling Equal-cost multi-path routing (ECMP).
- *Enabling BFD for next-hop liveness detection*.
- *Avoiding the use of NAT*.

There are also a set of *use cases* with examples below.

Adding multiple gateway ports to a router

A router can be set up with multiple gateway ports at router creation time by passing multiple `--external-gateway` arguments. You can also specify which IP address to use by passing the `--fixed-ip` with both the `subnet` and `ip-address` keys populated. The `subnet` provided must be attached to one of the networks provided to the `--external-gateway` arguments.

An existing router can be modified to have multiple gateway ports by using the `openstack router add gateway` command with router and network as arguments and optionally specifying the IP address by passing the `--fixed-ip` argument.

By default only one default route will be created.

Adding multiple default routes to a router

Whether to create multiple default routes is controlled by the `enable-default-route-ecmp` router property. It can be set per router at router creation time by passing the `--enable-default-route-ecmp` argument or by updating an existing router using the `openstack router set` command.

The default behavior for new routers can be controlled using the enable_default_route_ecmp configuration option.

Note

Adding multiple default routes without also *enabling BFD for next-hop liveness detection* is not recommended, as it will lead to degraded performance in the event of failure.

Enabling BFD for next-hop liveness detection

Whether to enable monitoring of next-hop liveness through BFD for default routes is controlled by the `enable-default-route-bfd` router property. It can be set per router at router creation time by passing the `--enable-default-route-bfd` argument or by updating an existing router using the `openstack router set` command.

The default behavior for new routers can be controlled using the enable_default_route_bfd configuration option.

It is recommended to enable this when *adding multiple default routes to a router* as failure to do so will lead to degraded performance in the event of failure.

Avoiding the use of NAT

OVN relies on connection tracking to keep required state for ongoing connections to implement NAT, and this state is local to each gateway chassis.

When you set up high availability at the layer 3 level, traffic can take multiple paths, even individual packets in a single flow.

Packets of an individual flow taking multiple paths does not work well with the local state of gateway chassis. To give an example; if traffic from a flow exits chassis A and then return traffic enters on chassis B, chassis B will not know to whom the packet belongs when NAT is enabled.

Use cases

Independent network paths for gateways without need for shared L2

```
                 +--------+
                 | spine  |
                 +--------+

     +-------+                +-------+
     | leaf  |                | leaf  |
     +-------+                +-------+

+-----------------+      +-----------------+
|      rack       |      |      rack       |
| +--------------+ |     | +--------------+ |
| | hypervisor   | |     | | hypervisor   | |
| +--------------+ |     | +--------------+ |
| | network-gw1 | |      | | network-gw2 | |
| +--------------+ |     | +--------------+ |
|      | phy1:net1 |     |      | phy2:net2 |
| +-------------+  |     | +-------------+  |
| | Border GW1 |   |     | | Border GW2 |   |
| +-------------+  |     | +-------------+  |
+-----------------+      +-----------------+
```

Example

First create the external networks:

```
$ source openrc admin

$ openstack network create \
    --external \
    --provider-network-type flat \
    --provider-physical-network phy1 \
    net1

$ openstack network create \
    --external \
    --provider-network-type flat \
    --provider-physical-network phy2 \
```

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```
net2
```

Then create subnets for the external networks:

```
$ source openrc admin

$ openstack subnet create \
    --subnet-range 192.0.2.0/24 \
    --no-dhcp \
    --network net1 \
    --gateway 192.0.2.2 \
    subnet1

$ openstack subnet create \
    --subnet-range 198.51.100.0/24 \
    --no-dhcp \
    --network net2 \
    --gateway 198.51.100.2 \
    subnet2
```

Then create the router with gateway ports in both external networks:

```
$ source openrc admin

$ openstack router create \
    --disable-snat \
    --external-gateway net1 \
    --fixed-ip subnet=subnet1,ip-address=192.0.2.100 \
    --external-gateway net2 \
    --fixed-ip subnet=subnet2,ip-address=198.51.100.100 \
    --enable-default-route-bfd \
    --enable-default-route-ecmp \
    router1
```

The end user can then create a subnet for use by a project:

```
$ source openrc demo

$ openstack network create project-network

$ openstack subnet create \
    --subnet-range 203.0.113.0/24 \
    --network project-network \
    project-subnet
```

And finally attach the project subnet to the router:

```
$ source openrc demo

$ openstack router add subnet router1 project-subnet
```

The border router configuration might look like this:

```
hostname border-router-1
!
ip route 203.0.113.0/24 192.0.2.100 bfd
!
bfd
 profile default
  transmit-interval 1000
  receive-interval 1000
 exit
 !
 peer 192.0.2.100 local-address 192.0.2.2 interface eth2
  profile default
 exit
 !
exit
!
end
```

```
hostname border-router-2
!
ip route 203.0.113.0/24 198.51.100.100 bfd
!
bfd
 profile default
  transmit-interval 1000
  receive-interval 1000
 exit
 !
 peer 198.51.100.100 local-address 198.51.100.2 interface eth2
  profile default
 exit
 !
exit
!
end
```

In a successful configuration the BFD status might look like this:

```
$ sudo ovn-nbctl find bfd dst_ip=192.0.2.2
_uuid               : b7efc8ac-cfd0-4f43-9dd2-2d38baa43571
detect_mult         : []
dst_ip              : "192.0.2.2"
external_ids        : {}
logical_port        : lrp-ad4ab4e8-1353-4230-8525-5e22fab7277e
min_rx              : []
min_tx              : []
options             : {}
status              : up
```

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```
$ sudo ovn-nbctl find bfd dst_ip=198.51.100.2
_uuid               : 905f1f69-0901-4d19-bfcb-40729532ff85
detect_mult         : []
dst_ip              : "198.51.100.2"
external_ids        : {}
logical_port        : lrp-7fd315dc-a76f-4468-86a5-2c65f55153e4
min_rx              : []
min_tx              : []
options             : {}
status              : up
```

```
border-router-1# sh bfd peer
BFD Peers:
  peer 192.0.2.100 local-address 192.0.2.2 vrf default interface eth2
    ID: 2436324418
    Remote ID: 300179009
    Active mode
    Status: up
    Uptime: 25 minute(s), 25 second(s)
    Diagnostics: ok
    Remote diagnostics: ok
    Peer Type: configured
    RTT min/avg/max: 0/0/0 usec
    Local timers:
        Detect-multiplier: 3
        Receive interval: 1000ms
        Transmission interval: 1000ms
        Echo receive interval: 50ms
        Echo transmission interval: disabled
    Remote timers:
        Detect-multiplier: 5
        Receive interval: 1000ms
        Transmission interval: 1000ms
        Echo receive interval: disabled
```

```
border-router-2# sh bfd peer
BFD Peers:
  peer 198.51.100.100 local-address 198.51.100.2 vrf default interface eth2
    ID: 3137653350
    Remote ID: 35580729
    Active mode
    Status: up
    Uptime: 26 minute(s), 2 second(s)
    Diagnostics: ok
    Remote diagnostics: ok
    Peer Type: configured
    RTT min/avg/max: 0/0/0 usec
    Local timers:
        Detect-multiplier: 3
```

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```
        Receive interval: 1000ms
        Transmission interval: 1000ms
        Echo receive interval: 50ms
        Echo transmission interval: disabled
    Remote timers:
        Detect-multiplier: 5
        Receive interval: 1000ms
        Transmission interval: 1000ms
        Echo receive interval: disabled
```

Load sharing

Expanding on the above example, load sharing can also be accomplished by adding multiple gateway ports in each subnet.

Assuming there are enough chassis available, Neutron will make sure to schedule multiple Logical Router Ports (LRP) for a single router so that different chassis serve as the primary gateway chassis.

```
$ openstack router add gateway \
    --fixed-ip subnet=subnet1,ip-address=192.0.2.101 \
    router1 \
    net1

 $ openstack router add gateway \
     --fixed-ip subnet=subnet2,ip-address=198.51.100.101 \
     router1 \
     net2
```

```
hostname border-router-1
!
ip route 203.0.113.0/24 192.0.2.101 bfd
!
bfd
 peer 192.0.2.101 local-address 192.0.2.2 interface eth2
  profile default
 exit
 !
exit
!
end
```

```
hostname border-router-2
!
ip route 203.0.113.0/24 198.51.100.101 bfd
!
bfd
 peer 198.51.100.101 local-address 198.51.100.2 interface eth2
  profile default
 exit
```

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```
 !
exit
!
end
```

```
$ sudo ovn-nbctl find bfd dst_ip=192.0.2.2
_uuid               : b7efc8ac-cfd0-4f43-9dd2-2d38baa43571
detect_mult         : []
dst_ip              : "192.0.2.2"
external_ids        : {}
logical_port        : lrp-ad4ab4e8-1353-4230-8525-5e22fab7277e
min_rx              : []
min_tx              : []
options             : {}
status              : up

_uuid               : efbcf3c2-0c34-4fbc-89ed-b742baa25f9b
detect_mult         : []
dst_ip              : "192.0.2.2"
external_ids        : {}
logical_port        : lrp-7aa481e9-732f-4700-acfb-37de5eb1984a
min_rx              : []
min_tx              : []
options             : {}
status              : up

$ sudo ovn-nbctl find bfd dst_ip=198.51.100.2
_uuid               : 905f1f69-0901-4d19-bfcb-40729532ff85
detect_mult         : []
dst_ip              : "198.51.100.2"
external_ids        : {}
logical_port        : lrp-7fd315dc-a76f-4468-86a5-2c65f55153e4
min_rx              : []
min_tx              : []
options             : {}
status              : up

_uuid               : 2214892e-5df3-47a4-b8e0-24fe7446129c
detect_mult         : []
dst_ip              : "198.51.100.2"
external_ids        : {}
logical_port        : lrp-2f0aae53-8561-46af-a741-13963368ef2a
min_rx              : []
min_tx              : []
options             : {}
status              : up
```

```
border-router-1# sh bfd peer
BFD Peers:
```

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```
  peer 192.0.2.101 local-address 192.0.2.2 vrf default interface eth2
     ID: 2436324418
     Remote ID: 300179009
     Active mode
     Status: up
     Uptime: 37 minute(s), 54 second(s)
     Diagnostics: ok
     Remote diagnostics: ok
     Peer Type: configured
     RTT min/avg/max: 0/0/0 usec
     Local timers:
         Detect-multiplier: 3
         Receive interval: 1000ms
         Transmission interval: 1000ms
         Echo receive interval: 50ms
         Echo transmission interval: disabled
     Remote timers:
         Detect-multiplier: 5
         Receive interval: 1000ms
         Transmission interval: 1000ms
         Echo receive interval: disabled

  peer 192.0.2.101 local-address 192.0.2.2 vrf default interface eth2
     ID: 295648861
     Remote ID: 877647437
     Active mode
     Status: up
     Uptime: 37 minute(s), 55 second(s)
     Diagnostics: ok
     Remote diagnostics: ok
     Peer Type: configured
     RTT min/avg/max: 0/0/0 usec
     Local timers:
         Detect-multiplier: 3
         Receive interval: 1000ms
         Transmission interval: 1000ms
         Echo receive interval: 50ms
         Echo transmission interval: disabled
     Remote timers:
         Detect-multiplier: 5
         Receive interval: 1000ms
         Transmission interval: 1000ms
         Echo receive interval: disabled
```

```
border-router-2# sh bfd peer
BFD Peers:
  peer 198.51.100.100 local-address 198.51.100.2 vrf default interface eth2
     ID: 90369368
     Remote ID: 3549983429
```

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```
    Active mode
    Status: up
    Uptime: 37 minute(s), 57 second(s)
    Diagnostics: ok
    Remote diagnostics: ok
    Peer Type: configured
    RTT min/avg/max: 0/0/0 usec
    Local timers:
        Detect-multiplier: 3
        Receive interval: 1000ms
        Transmission interval: 1000ms
        Echo receive interval: 50ms
        Echo transmission interval: disabled
    Remote timers:
        Detect-multiplier: 5
        Receive interval: 1000ms
        Transmission interval: 1000ms
        Echo receive interval: disabled

peer 198.51.100.101 local-address 198.51.100.2 vrf default interface eth2
    ID: 3137653350
    Remote ID: 35580729
    Active mode
    Status: up
    Uptime: 37 minute(s), 57 second(s)
    Diagnostics: ok
    Remote diagnostics: ok
    Peer Type: configured
    RTT min/avg/max: 0/0/0 usec
    Local timers:
        Detect-multiplier: 3
        Receive interval: 1000ms
        Transmission interval: 1000ms
        Echo receive interval: 50ms
        Echo transmission interval: disabled
    Remote timers:
        Detect-multiplier: 5
        Receive interval: 1000ms
        Transmission interval: 1000ms
        Echo receive interval: disabled
```

8.2.2 Address Scopes

Address scopes build from subnet pools. While subnet pools provide a mechanism for controlling the allocation of addresses to subnets, address scopes show where addresses can be routed between networks, preventing the use of overlapping addresses in any two subnets. Because all addresses allocated in the address scope do not overlap, neutron routers do not NAT between your projects network and your external network. As long as the addresses within an address scope match, the Networking service performs simple routing between networks.

Accessing address scopes

Anyone with access to the Networking service can create their own address scopes. However, network administrators can create shared address scopes, allowing other projects to create networks within that address scope.

Access to addresses in a scope are managed through subnet pools. Subnet pools can either be created in an address scope, or updated to belong to an address scope.

With subnet pools, all addresses in use within the address scope are unique from the point of view of the address scope owner. Therefore, add more than one subnet pool to an address scope if the pools have different owners, allowing for delegation of parts of the address scope. Delegation prevents address overlap across the whole scope. Otherwise, you receive an error if two pools have the same address ranges.

Each router interface is associated with an address scope by looking at subnets connected to the network. When a router connects to an external network with matching address scopes, network traffic routes between without Network address translation (NAT). The router marks all traffic connections originating from each interface with its corresponding address scope. If traffic leaves an interface in the wrong scope, the router blocks the traffic.

Backwards compatibility

Networks created before the Mitaka release do not contain explicitly named address scopes, unless the network contains subnets from a subnet pool that belongs to a created or updated address scope. The Networking service preserves backwards compatibility with pre-Mitaka networks through special address scope properties so that these networks can perform advanced routing:

1. Unlimited address overlap is allowed.
2. Neutron routers, by default, will NAT traffic from internal networks to external networks.
3. Pre-Mitaka address scopes are not visible through the API. You cannot list address scopes or show details. Scopes exist implicitly as a catch-all for addresses that are not explicitly scoped.

Create shared address scopes as an administrative user

This section shows how to set up shared address scopes to allow simple routing for project networks with the same subnet pools.

Note

Irrelevant fields have been trimmed from the output of these commands for brevity.

1. Create IPv6 and IPv4 address scopes:

```
$ openstack address scope create --share --ip-version 6 address-scope-ip6

+------------+--------------------------------------+
| Field      | Value                                |
+------------+--------------------------------------+
| headers    |                                      |
| id         | 28424dfc-9abd-481b-afa3-1da97a8fead7 |
| ip_version | 6                                    |
| name       | address-scope-ip6                    |
```

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```
| project_id | 098429d072d34d3596c88b7dbf7e91b6     |
| shared     | True                                 |
+------------+--------------------------------------+
```

```
$ openstack address scope create --share --ip-version 4 address-scope-ip4

+------------+--------------------------------------+
| Field      | Value                                |
+------------+--------------------------------------+
| headers    |                                      |
| id         | 3193bd62-11b5-44dc-acf8-53180f21e9f2 |
| ip_version | 4                                    |
| name       | address-scope-ip4                    |
| project_id | 098429d072d34d3596c88b7dbf7e91b6     |
| shared     | True                                 |
+------------+--------------------------------------+
```

2. Create subnet pools specifying the name (or UUID) of the address scope that the subnet pool belongs to. If you have existing subnet pools, use the **openstack subnet pool set** command to put them in a new address scope:

```
$ openstack subnet pool create --address-scope address-scope-ip6 \
--share --pool-prefix 2001:db8:a583::/48 --default-prefix-length 64 \
subnet-pool-ip6
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| address_scope_id  | 28424dfc-9abd-481b-afa3-1da97a8fead7 |
| created_at        | 2016-12-13T22:53:30Z                 |
| default_prefixlen | 64                                   |
| default_quota     | None                                 |
| description       |                                      |
| id                | a59ff52b-0367-41ff-9781-6318b927dd0e |
| ip_version        | 6                                    |
| is_default        | False                                |
| max_prefixlen     | 128                                  |
| min_prefixlen     | 64                                   |
| name              | subnet-pool-ip6                      |
| prefixes          | 2001:db8:a583::/48                   |
| project_id        | 098429d072d34d3596c88b7dbf7e91b6     |
| revision_number   | 1                                    |
| shared            | True                                 |
| tags              | []                                   |
| updated_at        | 2016-12-13T22:53:30Z                 |
+-------------------+--------------------------------------+
```

```
$ openstack subnet pool create --address-scope address-scope-ip4 \
--share --pool-prefix 203.0.113.0/24 --default-prefix-length 26 \
subnet-pool-ip4
```

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```
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| address_scope_id  | 3193bd62-11b5-44dc-acf8-53180f21e9f2 |
| created_at        | 2016-12-13T22:55:09Z                 |
| default_prefixlen | 26                                   |
| default_quota     | None                                 |
| description       |                                      |
| id                | d02af70b-d622-426f-8e60-ed9df2a8301f |
| ip_version        | 4                                    |
| is_default        | False                                |
| max_prefixlen     | 32                                   |
| min_prefixlen     | 8                                    |
| name              | subnet-pool-ip4                      |
| prefixes          | 203.0.113.0/24                       |
| project_id        | 098429d072d34d3596c88b7dbf7e91b6     |
| revision_number   | 1                                    |
| shared            | True                                 |
| tags              | []                                   |
| updated_at        | 2016-12-13T22:55:09Z                 |
+-------------------+--------------------------------------+
```

3. Make sure that subnets on an external network are created from the subnet pools created above:

```
$ openstack subnet show ipv6-public-subnet
+-------------------+------------------------------------------+
| Field             | Value                                    |
+-------------------+------------------------------------------+
| allocation_pools  | 2001:db8:a583::2-2001:db8:a583:0:ffff:ff |
|                   | ff:ffff:ffff                             |
| cidr              | 2001:db8:a583::/64                       |
| created_at        | 2016-12-10T21:36:04Z                     |
| description       |                                          |
| dns_nameservers   |                                          |
| enable_dhcp       | False                                    |
| gateway_ip        | 2001:db8:a583::1                         |
| host_routes       |                                          |
| id                | b333bf5a-758c-4b3f-97ec-5f12d9bfceb7     |
| ip_version        | 6                                        |
| ipv6_address_mode | None                                     |
| ipv6_ra_mode      | None                                     |
| name              | ipv6-public-subnet                       |
| network_id        | 05a8d31e-330b-4d96-a3fa-884b04abfa4c     |
| project_id        | 098429d072d34d3596c88b7dbf7e91b6         |
| revision_number   | 2                                        |
| segment_id        | None                                     |
| service_types     |                                          |
| subnetpool_id     | a59ff52b-0367-41ff-9781-6318b927dd0e     |
| tags              | []                                       |
| updated_at        | 2016-12-10T21:36:04Z                     |
```

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```
+-------------------+-------------------------------------------+
```

```
$ openstack subnet show public-subnet
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 203.0.113.2-203.0.113.62             |
| cidr              | 203.0.113.0/26                       |
| created_at        | 2016-12-10T21:35:52Z                 |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | False                                |
| gateway_ip        | 203.0.113.1                          |
| host_routes       |                                      |
| id                | 7fd48240-3acc-4724-bc82-16c62857edec |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | public-subnet                        |
| network_id        | 05a8d31e-330b-4d96-a3fa-884b04abfa4c |
| project_id        | 098429d072d34d3596c88b7dbf7e91b6     |
| revision_number   | 2                                    |
| segment_id        | None                                 |
| service_types     |                                      |
| subnetpool_id     | d02af70b-d622-426f-8e60-ed9df2a8301f |
| tags              | []                                   |
| updated_at        | 2016-12-10T21:35:52Z                 |
+-------------------+--------------------------------------+
```

Routing with address scopes for non-privileged users

This section shows how non-privileged users can use address scopes to route straight to an external network without NAT.

1. Create a couple of networks to host subnets:

```
$ openstack network create network1
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2016-12-13T23:21:01Z                 |
| description               |                                      |
| headers                   |                                      |
| id                        | 1bcf3fe9-a0cb-4d88-a067-a4d7f8e635f0 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| mtu                       | 1450                                 |
```

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```
| name                      | network1                             |
| port_security_enabled     | True                                 |
| project_id                | 098429d072d34d3596c88b7dbf7e91b6     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 94                                   |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2016-12-13T23:21:01Z                 |
+---------------------------+--------------------------------------+
```

```
$ openstack network create network2
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2016-12-13T23:21:45Z                 |
| description               |                                      |
| headers                   |                                      |
| id                        | 6c583603-c097-4141-9c5c-288b0e49c59f |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| mtu                       | 1450                                 |
| name                      | network2                             |
| port_security_enabled     | True                                 |
| project_id                | 098429d072d34d3596c88b7dbf7e91b6     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 81                                   |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2016-12-13T23:21:45Z                 |
+---------------------------+--------------------------------------+
```

2. Create a subnet not associated with a subnet pool or an address scope:

```
$ openstack subnet create --network network1 --subnet-range \
198.51.100.0/26 subnet-ip4-1
+-------------------+--------------------------------------+
```

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```
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 198.51.100.2-198.51.100.62           |
| cidr              | 198.51.100.0/26                      |
| created_at        | 2016-12-13T23:24:16Z                 |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 198.51.100.1                         |
| headers           |                                      |
| host_routes       |                                      |
| id                | 66874039-d31b-4a27-85d7-14c89341bbb7 |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | subnet-ip4-1                         |
| network_id        | 1bcf3fe9-a0cb-4d88-a067-a4d7f8e635f0 |
| project_id        | 098429d072d34d3596c88b7dbf7e91b6     |
| revision_number   | 2                                    |
| service_types     |                                      |
| subnetpool_id     | None                                 |
| tags              | []                                   |
| updated_at        | 2016-12-13T23:24:16Z                 |
+-------------------+--------------------------------------+
```

```
$ openstack subnet create --network network1 --ipv6-ra-mode slaac \
--ipv6-address-mode slaac --ip-version 6 --subnet-range \
2001:db8:80d2:c4d3::/64 subnet-ip6-1
+-------------------+-----------------------------------------+
| Field             | Value                                   |
+-------------------+-----------------------------------------+
| allocation_pools  | 2001:db8:80d2:c4d3::2-2001:db8:80d2:c4d |
|                   | 3:ffff:ffff:ffff:ffff                   |
| cidr              | 2001:db8:80d2:c4d3::/64                 |
| created_at        | 2016-12-13T23:28:28Z                    |
| description       |                                         |
| dns_nameservers   |                                         |
| enable_dhcp       | True                                    |
| gateway_ip        | 2001:db8:80d2:c4d3::1                   |
| headers           |                                         |
| host_routes       |                                         |
| id                | a7551b23-2271-4a88-9c41-c84b048e0722    |
| ip_version        | 6                                       |
| ipv6_address_mode | slaac                                   |
| ipv6_ra_mode      | slaac                                   |
| name              | subnet-ip6-1                            |
| network_id        | 1bcf3fe9-a0cb-4d88-a067-a4d7f8e635f0    |
| project_id        | 098429d072d34d3596c88b7dbf7e91b6        |
| revision_number   | 2                                       |
```

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```
| service_types     |                                          |
| subnetpool_id     | None                                     |
| tags              | []                                       |
| updated_at        | 2016-12-13T23:28:28Z                     |
+-------------------+------------------------------------------+
```

3. Create a subnet using a subnet pool associated with an address scope from an external network:

```
$ openstack subnet create --subnet-pool subnet-pool-ip4 \
--network network2 subnet-ip4-2
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 203.0.113.2-203.0.113.62             |
| cidr              | 203.0.113.0/26                       |
| created_at        | 2016-12-13T23:32:12Z                 |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 203.0.113.1                          |
| headers           |                                      |
| host_routes       |                                      |
| id                | 12be8e8f-5871-4091-9e9e-4e0651b9677e |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | subnet-ip4-2                         |
| network_id        | 6c583603-c097-4141-9c5c-288b0e49c59f |
| project_id        | 098429d072d34d3596c88b7dbf7e91b6     |
| revision_number   | 2                                    |
| service_types     |                                      |
| subnetpool_id     | d02af70b-d622-426f-8e60-ed9df2a8301f |
| tags              | []                                   |
| updated_at        | 2016-12-13T23:32:12Z                 |
+-------------------+--------------------------------------+
```

```
$ openstack subnet create --ip-version 6 --ipv6-ra-mode slaac \
--ipv6-address-mode slaac --subnet-pool subnet-pool-ip6 \
--network network2 subnet-ip6-2
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 2001:db8:a583::2-2001:db8:a583:0:fff |
|                   | f:ffff:ffff:ffff                     |
| cidr              | 2001:db8:a583::/64                   |
| created_at        | 2016-12-13T23:31:17Z                 |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
```

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```
| gateway_ip        | 2001:db8:a583::1                     |
| headers           |                                      |
| host_routes       |                                      |
| id                | b599c2be-e3cd-449c-ba39-3cfcc744c4be |
| ip_version        | 6                                    |
| ipv6_address_mode | slaac                                |
| ipv6_ra_mode      | slaac                                |
| name              | subnet-ip6-2                         |
| network_id        | 6c583603-c097-4141-9c5c-288b0e49c59f |
| project_id        | 098429d072d34d3596c88b7dbf7e91b6     |
| revision_number   | 2                                    |
| service_types     |                                      |
| subnetpool_id     | a59ff52b-0367-41ff-9781-6318b927dd0e |
| tags              | []                                   |
| updated_at        | 2016-12-13T23:31:17Z                 |
+-------------------+--------------------------------------+
```

By creating subnets from scoped subnet pools, the network is associated with the address scope.

```
$ openstack network show network2
+---------------------------+------------------------------+
| Field                     | Value                        |
+---------------------------+------------------------------+
| admin_state_up            | UP                           |
| availability_zone_hints   |                              |
| availability_zones        | nova                         |
| created_at                | 2016-12-13T23:21:45Z         |
| description               |                              |
| id                        | 6c583603-c097-4141-9c5c-     |
|                           | 288b0e49c59f                 |
| ipv4_address_scope        | 3193bd62-11b5-44dc-          |
|                           | acf8-53180f21e9f2            |
| ipv6_address_scope        | 28424dfc-9abd-481b-          |
|                           | afa3-1da97a8fead7            |
| mtu                       | 1450                         |
| name                      | network2                     |
| port_security_enabled     | True                         |
| project_id                | 098429d072d34d3596c88b7dbf7e |
|                           | 91b6                         |
| provider:network_type     | vxlan                        |
| provider:physical_network | None                         |
| provider:segmentation_id  | 81                           |
| revision_number           | 10                           |
| router:external           | Internal                     |
| shared                    | False                        |
| status                    | ACTIVE                       |
| subnets                   | 12be8e8f-5871-4091-9e9e-     |
|                           | 4e0651b9677e, b599c2be-e3cd- |
|                           | 449c-ba39-3cfcc744c4be       |
| tags                      | []                           |
```

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```
| updated_at                | 2016-12-13T23:32:12Z          |
+---------------------------+-------------------------------+
```

4. Connect a router to each of the project subnets that have been created, for example, using a router called `router1`:

```
$ openstack router add subnet router1 subnet-ip4-1
$ openstack router add subnet router1 subnet-ip4-2
$ openstack router add subnet router1 subnet-ip6-1
$ openstack router add subnet router1 subnet-ip6-2
```

Checking connectivity

This example shows how to check the connectivity between networks with address scopes.

1. Launch two instances, `instance1` on `network1` and `instance2` on `network2`. Associate a floating IP address to both instances.

2. Adjust security groups to allow pings and SSH (both IPv4 and IPv6):

```
$ openstack server list
+--------------+------------+-----------------------------------------------------------
↪----------------------------+--------+----------+
| ID           | Name       | Networks                                                 ␣
↪                            | Image  | Flavor   |
+--------------+------------+-----------------------------------------------------------
↪----------------------------+--------+----------+
| 97e49c8e-... | instance1 |␣
↪network1=2001:db8:80d2:c4d3:f816:3eff:fe52:b69f, 198.51.100.3, 203.0.
↪113.3| cirros | m1.tiny  |
| ceba9638-... | instance2 | network2=203.0.113.3,␣
↪2001:db8:a583:0:f816:3eff:fe42:1eeb, 203.0.113.4    | centos | m1.small␣
↪|
+--------------+------------+-----------------------------------------------------------
↪----------------------------+--------+----------+
```

Regardless of address scopes, the floating IPs can be pinged from the external network:

```
$ ping -c 1 203.0.113.3
1 packets transmitted, 1 received, 0% packet loss, time 0ms
$ ping -c 1 203.0.113.4
1 packets transmitted, 1 received, 0% packet loss, time 0ms
```

You can now ping `instance2` directly because `instance2` shares the same address scope as the external network:

Note

BGP routing can be used to automatically set up a static route for your instances.

```
# ip route add 203.0.113.0/26 via 203.0.113.2
$ ping -c 1 203.0.113.3
1 packets transmitted, 1 received, 0% packet loss, time 0ms
```

```
# ip route add 2001:db8:a583::/64 via 2001:db8::1
$ ping6 -c 1 2001:db8:a583:0:f816:3eff:fe42:1eeb
1 packets transmitted, 1 received, 0% packet loss, time 0ms
```

You cannot ping `instance1` directly because the address scopes do not match:

```
# ip route add 198.51.100.0/26 via 203.0.113.2
$ ping -c 1 198.51.100.3
1 packets transmitted, 0 received, 100% packet loss, time 0ms
```

```
# ip route add 2001:db8:80d2:c4d3::/64 via 2001:db8::1
$ ping6 -c 1 2001:db8:80d2:c4d3:f816:3eff:fe52:b69f
1 packets transmitted, 0 received, 100% packet loss, time 0ms
```

If the address scopes match between networks then pings and other traffic route directly through. If the scopes do not match between networks, the router either drops the traffic or applies NAT to cross scope boundaries.

8.2.3 Automatic allocation of network topologies

The auto-allocation feature introduced in Mitaka simplifies the procedure of setting up an external connectivity for end-users, and is also known as **Get Me A Network**.

Previously, a user had to configure a range of networking resources to boot a server and get access to the Internet. For example, the following steps are required:

- Create a network
- Create a subnet
- Create a router
- Uplink the router on an external network
- Downlink the router on the previously created subnet

These steps need to be performed on each logical segment that a VM needs to be connected to, and may require networking knowledge the user might not have.

This feature is designed to automate the basic networking provisioning for projects. The steps to provision a basic network are run during instance boot, making the networking setup hands-free.

To make this possible, provide a default external network and default subnetpools (one for IPv4, or one for IPv6, or one of each) so that the Networking service can choose what to do in lieu of input. Once these are in place, users can boot their VMs without specifying any networking details. The Compute service will then use this feature automatically to wire user VMs.

Enabling the deployment for auto-allocation

To use this feature, the neutron service must have the following extensions enabled:

- `auto_allocate`
- `ovn-router` or `router` for non-OVN scenarios.

Before the end-user can use the auto-allocation feature, the operator must create the resources that will be used for the auto-allocated network topology creation. To perform this task, proceed with the following steps:

1. Set up a default external network

   Assuming the external network to be used for the auto-allocation feature is named `public`, make it the `default` external network with the following command:

   ```
   $ openstack network set public --default
   ```

   Note

   The flag `--default` (and `--no-default` flag) is only effective with external networks and has no effects on regular (or internal) networks.

2. Create default subnetpools

   The auto-allocation feature requires at least one default subnetpool. One for IPv4, or one for IPv6, or one of each.

   ```
   $ openstack subnet pool create --share --default \
     --pool-prefix 192.0.2.0/24 --default-prefix-length 26 \
     shared-default

   +-------------------+--------------------------------------+
   | Field             | Value                                |
   +-------------------+--------------------------------------+
   | address_scope_id  | None                                 |
   | created_at        | 2017-01-12T15:10:34Z                 |
   | default_prefixlen | 26                                   |
   | default_quota     | None                                 |
   | description       |                                      |
   | headers           |                                      |
   | id                | b41b7b9c-de57-4c19-b1c5-731985bceb7f |
   | ip_version        | 4                                    |
   | is_default        | True                                 |
   | max_prefixlen     | 32                                   |
   | min_prefixlen     | 8                                    |
   | name              | shared-default                       |
   | prefixes          | 192.0.2.0/24                         |
   | project_id        | 86acdbd1d72745fd8e8320edd7543400     |
   | revision_number   | 1                                    |
   | shared            | True                                 |
   | tags              | []                                   |
   ```

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```
  | updated_at        | 2017-01-12T15:10:34Z                 |
  +-------------------+--------------------------------------+

$ openstack subnet pool create --share --default \
  --pool-prefix 2001:db8:8000::/48 --default-prefix-length 64 \
  default-v6

+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| address_scope_id  | None                                 |
| created_at        | 2017-01-12T15:14:35Z                 |
| default_prefixlen | 64                                   |
| default_quota     | None                                 |
| description       |                                      |
| headers           |                                      |
| id                | 6f387016-17f0-4564-96ad-e34775b6ea14 |
| ip_version        | 6                                    |
| is_default        | True                                 |
| max_prefixlen     | 128                                  |
| min_prefixlen     | 64                                   |
| name              | default-v6                           |
| prefixes          | 2001:db8:8000::/48                   |
| project_id        | 86acdbd1d72745fd8e8320edd7543400     |
| revision_number   | 1                                    |
| shared            | True                                 |
| tags              | []                                   |
| updated_at        | 2017-01-12T15:14:35Z                 |
+-------------------+--------------------------------------+
```

Get Me A Network

In a deployment where the operator has set up the resources as described above, they can get or create their auto-allocated network topology as follows:

```
$ openstack network auto allocated topology create --or-show
+------------+--------------------------------------+
| Field      | Value                                |
+------------+--------------------------------------+
| id         | a380c780-d6cd-4510-a4c0-1a6ec9b85a29 |
| project_id | cfd1889ac7d64ad891d4f20aef9f8d7c     |
+------------+--------------------------------------+
```

Note

When the `--or-show` option is used the command returns the topology information if it already exists, or creates it if it does not.

Operators (and users with admin role) can get or create the auto-allocated topology for a project by specifying the project ID:

```
$ openstack network auto allocated topology create --project \
  cfd1889ac7d64ad891d4f20aef9f8d7c --or-show
+------------+--------------------------------------+
| Field      | Value                                |
+------------+--------------------------------------+
| id         | a380c780-d6cd-4510-a4c0-1a6ec9b85a29 |
| project_id | cfd1889ac7d64ad891d4f20aef9f8d7c     |
+------------+--------------------------------------+
```

The ID returned by this command is a network which can be used for booting a VM.

```
$ openstack server create --flavor m1.small --image \
  cirros-0.3.5-x86_64-uec --nic \
  net-id=8b835bfb-cae2-4acc-b53f-c16bb5f9a7d0 vm1
```

The auto-allocated topology for a user never changes. In practice, when a user boots a server omitting the `--nic` option, and there is more than one network available, the Compute service will invoke the API behind `auto allocated topology create`, fetch the network UUID, and pass it on during the boot process.

Alternately one can delete their auto-allocated network topology as follows:

```
$ openstack network auto allocated topology delete
```

Validating the requirements for auto-allocation

To validate that the required resources are correctly set up for auto-allocation, without actually provisioning anything, use the `--check-resources` option:

```
$ openstack network auto allocated topology create --check-resources
Deployment error: No default router:external network.

$ openstack network set public --default

$ openstack network auto allocated topology create --check-resources
Deployment error: No default subnetpools defined.

$ openstack subnet pool set shared-default --default

$ openstack network auto allocated topology create --check-resources
+---------+-------+
| Field   | Value |
+---------+-------+
| dry-run | pass  |
+---------+-------+
```

The validation option behaves identically for all users. However, it is considered primarily an admin or service utility since it is the operator who must set up the requirements.

Project resources created by auto-allocation

The auto-allocation feature creates one network topology in every project where it is used. The auto-allocated network topology for a project contains the following resources:

Resource	Name
network	auto_allocated_network
subnet (IPv4)	auto_allocated_subnet_v4
subnet (IPv6)	auto_allocated_subnet_v6
router	auto_allocated_router

Compatibility notes

Nova uses the `auto allocated topology` feature with API micro version 2.37 or later. This is because, unlike the neutron feature which was implemented in the Mitaka release, the integration for nova was completed during the Newton release cycle. Note that the CLI option `--nic` can be omitted regardless of the microversion used as long as there is no more than one network available to the project, in which case nova fails with a 400 error because it does not know which network to use. Furthermore, nova does not start using the feature, regardless of whether or not a user requests micro version 2.37 or later, unless all of the `nova-compute` services are running Newton-level code.

8.2.4 Availability Zones

An availability zone groups network nodes that run services like DHCP, L3, FW, and others. It is defined as an agents attribute on the network node. This allows users to associate an availability zone with their resources so that the resources get high availability.

Use case

An availability zone is used to make network resources highly available. The operators group the nodes that are attached to different power sources under separate availability zones and configure scheduling for resources with high availability so that they are scheduled on different availability zones.

Required extensions

The core plug-in must support the `availability_zone` extension. The core plug-in also must support the `network_availability_zone` extension to schedule a network according to availability zones. The `Ml2Plugin` supports it. The router service plug-in must support the `router_availability_zone` extension to schedule a router according to the availability zones. The `L3RouterPlugin` supports it.

```
$ openstack extension list --network -c Alias -c Name
+---------------------------+---------------------------+
| Name                      | Alias                     |
+---------------------------+---------------------------+
...
| Network Availability Zone | network_availability_zone |
...
| Availability Zone         | availability_zone         |
...
| Router Availability Zone  | router_availability_zone  |
...
+---------------------------+---------------------------+
```

Availability zone of agents

The `availability_zone` attribute can be defined in `dhcp-agent` and `l3-agent`. To define an availability zone for each agent, set the value into `[AGENT]` section of `/etc/neutron/dhcp_agent.ini` or `/etc/neutron/l3_agent.ini`:

```
[AGENT]
availability_zone = zone-1
```

To confirm the agents availability zone:

```
$ openstack network agent show 116cc128-4398-49af-a4ed-3e95494cd5fc
+---------------------+---------------------------------------------------+
| Field               | Value                                             |
+---------------------+---------------------------------------------------+
| admin_state_up      | UP                                                |
| agent_type          | DHCP agent                                        |
| alive               | True                                              |
| availability_zone   | zone-1                                            |
| binary              | neutron-dhcp-agent                                |
| configurations      | dhcp_driver='neutron.agent.linux.dhcp.Dnsmasq',   |
|                     | dhcp_lease_duration='86400',                      |
|                     | log_agent_heartbeats='False', networks='2',       |
|                     | notifies_port_ready='True', ports='6', subnets='4 |
| created_at          | 2016-12-14 00:25:54                               |
| description         | None                                              |
| heartbeat_timestamp | 2016-12-14 06:20:24                               |
| host                | ankur-desktop                                     |
| id                  | 116cc128-4398-49af-a4ed-3e95494cd5fc              |
| started_at          | 2016-12-14 00:25:54                               |
| topic               | dhcp_agent                                        |
+---------------------+---------------------------------------------------+

$ openstack network agent show 9632309a-2aa4-4304-8603-c4de02c4a55f
+---------------------+-------------------------------------------------+
| Field               | Value                                           |
+---------------------+-------------------------------------------------+
| admin_state_up      | UP                                              |
| agent_type          | L3 agent                                        |
| alive               | True                                            |
| availability_zone   | zone-1                                          |
| binary              | neutron-l3-agent                                |
| configurations      | agent_mode='legacy', ex_gw_ports='2',           |
|                     | floating_ips='0',                               |
|                     | handle_internal_only_routers='True',            |
|                     | interface_driver='openvswitch', interfaces='4', |
|                     | log_agent_heartbeats='False', routers='2'       |
| created_at          | 2016-12-14 00:25:58                             |
| description         | None                                            |
| heartbeat_timestamp | 2016-12-14 06:20:28                             |
| host                | ankur-desktop                                   |
```

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```
| id                   | 9632309a-2aa4-4304-8603-c4de02c4a55f                 |
| started_at           | 2016-12-14 00:25:58                                  |
| topic                | l3_agent                                             |
+----------------------+------------------------------------------------------+
```

Availability zone related attributes

The following attributes are added into network and router:

Attribute name	Access	Required	Input type	Description
availability_zone_hints	RW(POST only)	No	list of string	availability zone candidates for the resource
availability_zones	RO	N/A	list of string	availability zones for the resource

Use `availability_zone_hints` to specify the zone in which the resource is hosted:

```
$ openstack network create --availability-zone-hint zone-1 \
--availability-zone-hint zone-2 net1
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   | zone-1                               |
|                           | zone-2                               |
| availability_zones        |                                      |
| created_at                | 2016-12-14T06:23:36Z                 |
| description               |                                      |
| headers                   |                                      |
| id                        | ad88e059-e7fa-4cf7-8857-6731a2a3a554 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| mtu                       | 1450                                 |
| name                      | net1                                 |
| port_security_enabled     | True                                 |
| project_id                | cfd1889ac7d64ad891d4f20aef9f8d7c     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 77                                   |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2016-12-14T06:23:37Z                 |
+---------------------------+--------------------------------------+
```

```
$ openstack router create --ha --availability-zone-hint zone-1 \
--availability-zone-hint zone-2 router1
+-------------------------+--------------------------------------+
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints | zone-1                               |
|                         | zone-2                               |
| availability_zones      |                                      |
| created_at              | 2016-12-14T06:25:40Z                 |
| description             |                                      |
| distributed             | False                                |
| external_gateway_info   | null                                 |
| flavor_id               | None                                 |
| ha                      | False                                |
| headers                 |                                      |
| id                      | ced10262-6cfe-47c1-8847-cd64276a868c |
| name                    | router1                              |
| project_id              | cfd1889ac7d64ad891d4f20aef9f8d7c     |
| revision_number         | 3                                    |
| routes                  |                                      |
| status                  | ACTIVE                               |
| tags                    | []                                   |
| updated_at              | 2016-12-14T06:25:40Z                 |
+-------------------------+--------------------------------------+
```

Availability zone is selected from `default_availability_zones` in `/etc/neutron/neutron.conf` if a resource is created without `availability_zone_hints`:

```
default_availability_zones = zone-1,zone-2
```

To confirm the availability zone defined by the system:

```
$ openstack availability zone list
+-----------+-------------+
| Zone Name | Zone Status |
+-----------+-------------+
| zone-1    | available   |
| zone-2    | available   |
| zone-1    | available   |
| zone-2    | available   |
+-----------+-------------+
```

Look at the `availability_zones` attribute of each resource to confirm in which zone the resource is hosted:

```
$ openstack network show net1
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
```

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```
| availability_zone_hints   | zone-1                               |
|                           | zone-2                               |
| availability_zones        | zone-1                               |
|                           | zone-2                               |
| created_at                | 2016-12-14T06:23:36Z                 |
| description               |                                      |
| headers                   |                                      |
| id                        | ad88e059-e7fa-4cf7-8857-6731a2a3a554 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| mtu                       | 1450                                 |
| name                      | net1                                 |
| port_security_enabled     | True                                 |
| project_id                | cfd1889ac7d64ad891d4f20aef9f8d7c     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 77                                   |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2016-12-14T06:23:37Z                 |
+---------------------------+--------------------------------------+
```

```
$ openstack router show router1
+-------------------------+--------------------------------------+
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints | zone-1                               |
|                         | zone-2                               |
| availability_zones      | zone-1                               |
|                         | zone-2                               |
| created_at              | 2016-12-14T06:25:40Z                 |
| description             |                                      |
| distributed             | False                                |
| external_gateway_info   | null                                 |
| flavor_id               | None                                 |
| ha                      | False                                |
| headers                 |                                      |
| id                      | ced10262-6cfe-47c1-8847-cd64276a868c |
| name                    | router1                              |
| project_id              | cfd1889ac7d64ad891d4f20aef9f8d7c     |
| revision_number         | 3                                    |
| routes                  |                                      |
| status                  | ACTIVE                               |
| tags                    | []                                   |
```

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```
| updated_at             | 2016-12-14T06:25:40Z               |
+------------------------+------------------------------------+
```

Note

The `availability_zones` attribute does not have a value until the resource is scheduled. Once the Networking service schedules the resource to zones according to `availability_zone_hints`, `availability_zones` shows in which zone the resource is hosted practically. The `availability_zones` may not match `availability_zone_hints`. For example, even if you specify a zone with `availability_zone_hints`, all agents of the zone may be dead before the resource is scheduled. In general, they should match, unless there are failures or there is no capacity left in the zone requested.

Availability zone aware scheduler

Network scheduler

Set `AZAwareWeightScheduler` to `network_scheduler_driver` in `/etc/neutron/neutron.conf` so that the Networking service schedules a network according to the availability zone:

```
network_scheduler_driver = neutron.scheduler.dhcp_agent_scheduler.
↪AZAwareWeightScheduler
dhcp_load_type = networks
```

The Networking service schedules a network to one of the agents within the selected zone as with `WeightScheduler`. In this case, scheduler refers to `dhcp_load_type` as well.

Router scheduler

Set `AZLeastRoutersScheduler` to `router_scheduler_driver` in file `/etc/neutron/neutron.conf` so that the Networking service schedules a router according to the availability zone:

```
router_scheduler_driver = neutron.scheduler.l3_agent_scheduler.
↪AZLeastRoutersScheduler
```

The Networking service schedules a router to one of the agents within the selected zone as with `LeastRouterScheduler`.

Achieving high availability with availability zone

Although, the Networking service provides high availability for routers and high availability and fault tolerance for networks DHCP services, availability zones provide an extra layer of protection by segmenting a Networking service deployment in isolated failure domains. By deploying HA nodes across different availability zones, it is guaranteed that network services remain available in face of zone-wide failures that affect the deployment.

This section explains how to get high availability with the availability zone for L3 and DHCP. You should naturally set above configuration options for the availability zone.

L3 high availability

Set the following configuration options in file /etc/neutron/neutron.conf so that you get L3 high availability.

```
l3_ha = True
max_l3_agents_per_router = 3
```

HA routers are created on availability zones you selected when creating the router.

DHCP high availability

Set the following configuration options in file /etc/neutron/neutron.conf so that you get DHCP high availability.

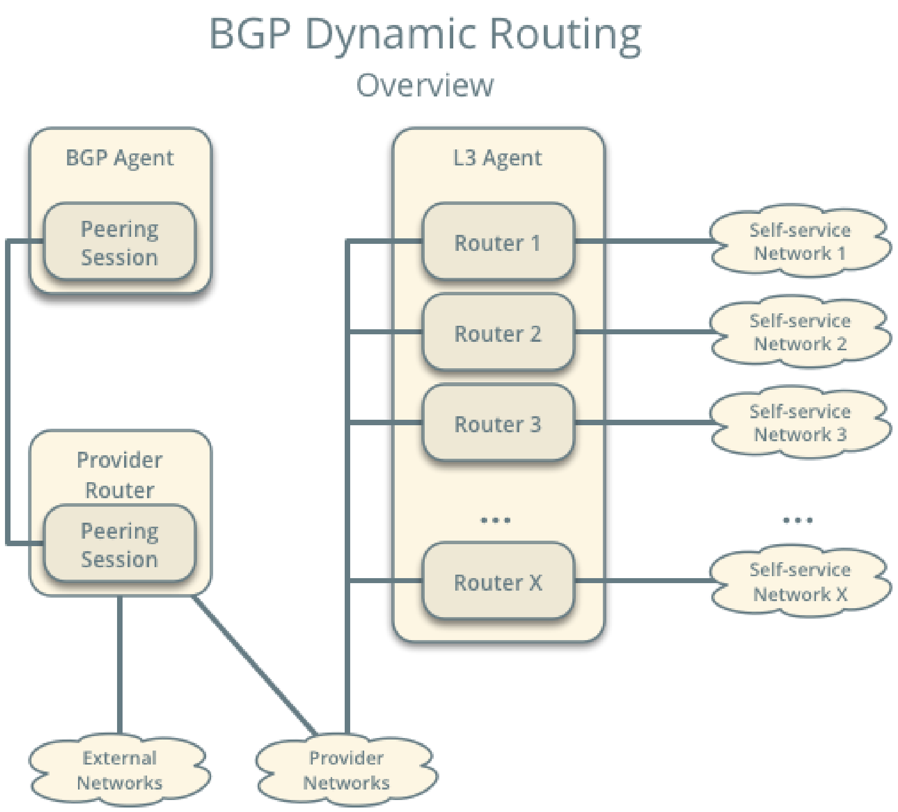
```
dhcp_agents_per_network = 2
```

DHCP services are created on availability zones you selected when creating the network.

8.2.5 BGP Dynamic Routing

BGP dynamic routing enables advertisement of self-service (private) network prefixes to physical network devices that support BGP such as routers, thus removing the conventional dependency on static routes. The feature relies on *address scopes* and requires knowledge of their operation for proper deployment.

BGP dynamic routing consists of a service plug-in and an agent. The service plug-in implements the Networking service extension and the agent manages BGP peering sessions. A cloud administrator creates and configures a BGP speaker using the CLI or API and manually schedules it to one or more hosts running the agent. Agents can reside on hosts with or without other Networking service agents. Prefix advertisement depends on the binding of external networks to a BGP speaker and the address scope of external and internal IP address ranges or subnets.



> **Note**
>
> Although self-service networks generally use private IP address ranges (RFC1918) for IPv4 subnets, BGP dynamic routing can advertise any IPv4 address ranges.

Example configuration

The example configuration involves the following components:

- One BGP agent.
- One address scope containing IP address range 203.0.113.0/24 for provider networks, and IP address ranges 192.0.2.0/25 and 192.0.2.128/25 for self-service networks.
- One provider network using IP address range 203.0.113.0/24.
- Three self-service networks.
  - Self-service networks 1 and 2 use IP address ranges inside of the address scope.
  - Self-service network 3 uses a unique IP address range 198.51.100.0/24 to demonstrate that the BGP speaker does not advertise prefixes outside of address scopes.
- Three routers. Each router connects one self-service network to the provider network.
  - Router 1 contains IP addresses 203.0.113.11 and 192.0.2.1
  - Router 2 contains IP addresses 203.0.113.12 and 192.0.2.129
  - Router 3 contains IP addresses 203.0.113.13 and 198.51.100.1

- One preexisting peering network 10.0.0.0/24 on the host running the neutron BGP dynamic routing agent to facilitate BGP communication with its peer. 10.0.0.1 is the address for the host and 10.0.0.2 the address for the peer.

Note

The example configuration assumes sufficient knowledge about the Networking service, routing, and BGP. For basic deployment of the Networking service, consult one of the *Deployment examples*. For more information on BGP, see RFC 4271.

Controller node

- In the `neutron.conf` file, enable the conventional layer-3 and BGP dynamic routing service plug-ins:

```
[DEFAULT]
service_plugins = neutron_dynamic_routing.services.bgp.bgp_plugin.
↪BgpPlugin,neutron.services.l3_router.l3_router_plugin.L3RouterPlugin
```

Agent nodes

- In the `bgp_dragent.ini` file:
  - Configure the driver.

    ```
    [BGP]
    bgp_speaker_driver = neutron_dynamic_routing.services.bgp.agent.
    ↪driver.os_ken.driver.OsKenBgpDriver
    ```

    Note

    The agent currently only supports the os-ken BGP driver.

  - Configure the router ID.

    ```
    [BGP]
    bgp_router_id = ROUTER_ID
    ```

    Replace `ROUTER_ID` with a suitable unique 32-bit number, typically an IPv4 address on the host running the agent. For example, 10.0.0.1.

Verify service operation

1. Source the administrative project credentials.
2. Verify presence and operation of each BGP dynamic routing agent.

```
$ openstack network agent list --agent-type bgp
+--------------------------------------+---------------------------+------
↪------+-------------------+-------+-------+---------------------+
```

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```
| ID                                   | Agent Type                | Host␣
↪      | Availability Zone | Alive | State | Binary                |
+--------------------------------------+---------------------------+------
↪------+-------------------+-------+-------+--------------------+
| 37729181-2224-48d8-89ef-16eca8e2f77e | BGP dynamic routing agent |␣
↪controller | None              | :-)   | UP    | neutron-bgp-dragent |
+--------------------------------------+---------------------------+------
↪------+-------------------+-------+-------+--------------------+
```

Create the address scope and subnet pools

1. Create an address scope. The provider (external) and self-service networks must belong to the same address scope for the agent to advertise those self-service network prefixes.

```
$ openstack address scope create --share --ip-version 4 bgp

+------------+--------------------------------------+
| Field      | Value                                |
+------------+--------------------------------------+
| headers    |                                      |
| id         | f71c958f-dbe8-49a2-8fb9-19c5f52a37f1 |
| ip_version | 4                                    |
| name       | bgp                                  |
| project_id | 86acdbd1d72745fd8e8320edd7543400     |
| shared     | True                                 |
+------------+--------------------------------------+
```

2. Create subnet pools. The provider and self-service networks use different pools.

   - Create the provider network pool.

```
$ openstack subnet pool create --pool-prefix 203.0.113.0/24 \
  --address-scope bgp provider

+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| address_scope_id  | f71c958f-dbe8-49a2-8fb9-19c5f52a37f1 |
| created_at        | 2017-01-12T14:58:57Z                 |
| default_prefixlen | 8                                    |
| default_quota     | None                                 |
| description       |                                      |
| headers           |                                      |
| id                | 63532225-b9a0-445a-9935-20a15f9f68d1 |
| ip_version        | 4                                    |
| is_default        | False                                |
| max_prefixlen     | 32                                   |
| min_prefixlen     | 8                                    |
| name              | provider                             |
| prefixes          | 203.0.113.0/24                       |
```

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```
| project_id        | 86acdbd1d72745fd8e8320edd7543400     |
| revision_number   | 1                                    |
| shared            | False                                |
| tags              | []                                   |
| updated_at        | 2017-01-12T14:58:57Z                 |
+-------------------+--------------------------------------+
```

- Create the self-service network pool.

```
$ openstack subnet pool create --pool-prefix 192.0.2.0/25 \
  --pool-prefix 192.0.2.128/25 --address-scope bgp \
  --share selfservice

+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| address_scope_id  | f71c958f-dbe8-49a2-8fb9-19c5f52a37f1 |
| created_at        | 2017-01-12T15:02:31Z                 |
| default_prefixlen | 8                                    |
| default_quota     | None                                 |
| description       |                                      |
| headers           |                                      |
| id                | 8d8270b1-b194-4b7e-914c-9c741dcbd49b |
| ip_version        | 4                                    |
| is_default        | False                                |
| max_prefixlen     | 32                                   |
| min_prefixlen     | 8                                    |
| name              | selfservice                          |
| prefixes          | 192.0.2.0/25, 192.0.2.128/25         |
| project_id        | 86acdbd1d72745fd8e8320edd7543400     |
| revision_number   | 1                                    |
| shared            | True                                 |
| tags              | []                                   |
| updated_at        | 2017-01-12T15:02:31Z                 |
+-------------------+--------------------------------------+
```

Create the provider and self-service networks

1. Create the provider network.

```
$ openstack network create provider --external --provider-physical-
↪network \
  provider --provider-network-type flat
Created a new network:
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
```

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```
| availability_zones        |                                      |
| created_at                | 2016-12-21T08:47:41Z                 |
| description               |                                      |
| headers                   |                                      |
| id                        | 190ca651-2ee3-4a4b-891f-dedda47974fe |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| mtu                       | 1450                                 |
| name                      | provider                             |
| port_security_enabled     | True                                 |
| project_id                | c961a8f6d3654657885226378ade8220     |
| provider:network_type     | flat                                 |
| provider:physical_network | provider                             |
| provider:segmentation_id  | 66                                   |
| revision_number           | 3                                    |
| router:external           | External                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2016-12-21T08:47:41Z                 |
+---------------------------+--------------------------------------+
```

2. Create a subnet on the provider network using an IP address range from the provider subnet pool.

```
$ openstack subnet create --subnet-pool provider \
  --prefix-length 24 --gateway 203.0.113.1 --network provider \
  --allocation-pool start=203.0.113.11,end=203.0.113.254 provider
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 203.0.113.11-203.0.113.254           |
| cidr              | 203.0.113.0/24                       |
| created_at        | 2016-03-17T23:17:16                  |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 203.0.113.1                          |
| host_routes       |                                      |
| id                | 8ed65d41-2b2a-4f3a-9f92-45adb266e01a |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | provider                             |
| network_id        | 68ec148c-181f-4656-8334-8f4eb148689d |
| project_id        | b3ac05ef10bf441fbf4aa17f16ae1e6d     |
| segment_id        | None                                 |
| service_types     |                                      |
| subnetpool_id     | 3771c0e7-7096-46d3-a3bd-699c58e70259 |
```

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```
| tags              |                                      |
| updated_at        | 2016-03-17T23:17:16                  |
+-------------------+--------------------------------------+
```

Note

The IP address allocation pool starting at `.11` improves clarity of the diagrams. You can safely omit it.

3. Create the self-service networks.

```
$ openstack network create selfservice1
Created a new network:
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2016-12-21T08:49:38Z                 |
| description               |                                      |
| headers                   |                                      |
| id                        | 9d842606-ef3d-4160-9ed9-e03fa63aed96 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| mtu                       | 1450                                 |
| name                      | selfservice1                         |
| port_security_enabled     | True                                 |
| project_id                | c961a8f6d3654657885226378ade8220     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 106                                  |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2016-12-21T08:49:38Z                 |
+---------------------------+--------------------------------------+

$ openstack network create selfservice2
Created a new network:
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
```

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```
| availability_zones        |                                      |
| created_at                | 2016-12-21T08:50:05Z                 |
| description               |                                      |
| headers                   |                                      |
| id                        | f85639e1-d23f-438e-b2b1-f40570d86b1c |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| mtu                       | 1450                                 |
| name                      | selfservice2                         |
| port_security_enabled     | True                                 |
| project_id                | c961a8f6d3654657885226378ade8220     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 21                                   |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2016-12-21T08:50:05Z                 |
+---------------------------+--------------------------------------+

$ openstack network create selfservice3
Created a new network:
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2016-12-21T08:50:35Z                 |
| description               |                                      |
| headers                   |                                      |
| id                        | eeccdb82-5cf4-4999-8ab3-e7dc99e7d43b |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| mtu                       | 1450                                 |
| name                      | selfservice3                         |
| port_security_enabled     | True                                 |
| project_id                | c961a8f6d3654657885226378ade8220     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 86                                   |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
```

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```
| tags                       | []                                      |
| updated_at                 | 2016-12-21T08:50:35Z                    |
+----------------------------+-----------------------------------------+
```

4. Create a subnet on the first two self-service networks using an IP address range from the self-service subnet pool.

```
$ openstack subnet create --network selfservice1 --subnet-pool␣
↪selfservice \
  --prefix-length 25 selfservice1
+-------------------+----------------------------------------------------+
| Field             | Value                                              |
+-------------------+----------------------------------------------------+
| allocation_pools  | 192.0.2.2-192.0.2.127                              |
| cidr              | 192.0.2.0/25                                       |
| created_at        | 2016-03-17T23:20:20                                |
| description       |                                                    |
| dns_nameservers   |                                                    |
| enable_dhcp       | True                                               |
| gateway_ip        | 198.51.100.1                                       |
| host_routes       |                                                    |
| id                | 8edd3dc2-df40-4d71-816e-a4586d61c809               |
| ip_version        | 4                                                  |
| ipv6_address_mode |                                                    |
| ipv6_ra_mode      |                                                    |
| name              | selfservice1                                       |
| network_id        | be79de1e-5f56-11e6-9dfb-233e41cec48c               |
| project_id        | b3ac05ef10bf441fbf4aa17f16ae1e6d                   |
| revision_number   | 1                                                  |
| subnetpool_id     | c7e9737a-cfd3-45b5-a861-d1cee1135a92               |
| tags              | []                                                 |
| tenant_id         | b3ac05ef10bf441fbf4aa17f16ae1e6d                   |
| updated_at        | 2016-03-17T23:20:20                                |
+-------------------+----------------------------------------------------+

$ openstack subnet create --network selfservice2 --subnet-pool␣
↪selfservice \
  --prefix-length 25 selfservice2
+-------------------+------------------------------------------------+
| Field             | Value                                          |
+-------------------+------------------------------------------------+
| allocation_pools  | 192.0.2.130-192.0.2.254                        |
| cidr              | 192.0.2.128/25                                 |
| created_at        | 2016-03-17T23:20:20                            |
| description       |                                                |
| dns_nameservers   |                                                |
| enable_dhcp       | True                                           |
| gateway_ip        | 192.0.2.129                                    |
| host_routes       |                                                |
| id                | 8edd3dc2-df40-4d71-816e-a4586d61c809           |
```

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```
| ip_version        | 4                                         |
| ipv6_address_mode |                                           |
| ipv6_ra_mode      |                                           |
| name              | selfservice2                              |
| network_id        | c1fd9846-5f56-11e6-a8ac-0f998d9cc0a2      |
| project_id        | b3ac05ef10bf441fbf4aa17f16ae1e6d          |
| revision_number   | 1                                         |
| subnetpool_id     | c7e9737a-cfd3-45b5-a861-d1cee1135a92      |
| tags              | []                                        |
| tenant_id         | b3ac05ef10bf441fbf4aa17f16ae1e6d          |
| updated_at        | 2016-03-17T23:20:20                       |
+-------------------+-------------------------------------------+
```

5. Create a subnet on the last self-service network using an IP address range outside of the address scope.

```
$ openstack subnet create --network selfservice3 --prefix 198.51.100.0/24␣
↪subnet3
+-------------------+---------------------------------------------+
| Field             | Value                                       |
+-------------------+---------------------------------------------+
| allocation_pools  | 198.51.100.2-198.51.100.254                 |
| cidr              | 198.51.100.0/24                             |
| created_at        | 2016-03-17T23:20:20                         |
| description       |                                             |
| dns_nameservers   |                                             |
| enable_dhcp       | True                                        |
| gateway_ip        | 198.51.100.1                                |
| host_routes       |                                             |
| id                | cd9f9156-5f59-11e6-aeec-172ec7ee939a        |
| ip_version        | 4                                           |
| ipv6_address_mode |                                             |
| ipv6_ra_mode      |                                             |
| name              | selfservice3                                |
| network_id        | c283dc1c-5f56-11e6-bfb6-efc30e1eb73b        |
| project_id        | b3ac05ef10bf441fbf4aa17f16ae1e6d            |
| revision_number   | 1                                           |
| subnetpool_id     |                                             |
| tags              | []                                          |
| tenant_id         | b3ac05ef10bf441fbf4aa17f16ae1e6d            |
| updated_at        | 2016-03-17T23:20:20                         |
+-------------------+---------------------------------------------+
```

Create and configure the routers

1. Create the routers.

```
$ openstack router create router1
+-------------------------+-------------------------------------+
```

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```
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints |                                      |
| availability_zones      |                                      |
| created_at              | 2017-01-10T13:15:19Z                 |
| description             |                                      |
| distributed             | False                                |
| external_gateway_info   | null                                 |
| flavor_id               | None                                 |
| ha                      | False                                |
| headers                 |                                      |
| id                      | 3f6f4ef8-63be-11e6-bbb3-2fbcef363ab8 |
| name                    | router1                              |
| project_id              | b3ac05ef10bf441fbf4aa17f16ae1e6d     |
| revision_number         | 1                                    |
| routes                  |                                      |
| status                  | ACTIVE                               |
| tags                    | []                                   |
| updated_at              | 2017-01-10T13:15:19Z                 |
+-------------------------+--------------------------------------+

$ openstack router create router2
+-------------------------+--------------------------------------+
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints |                                      |
| availability_zones      |                                      |
| created_at              | 2017-01-10T13:15:19Z                 |
| description             |                                      |
| distributed             | False                                |
| external_gateway_info   | null                                 |
| flavor_id               | None                                 |
| ha                      | False                                |
| headers                 |                                      |
| id                      | 3fd21a60-63be-11e6-9c95-5714c208c499 |
| name                    | router2                              |
| project_id              | b3ac05ef10bf441fbf4aa17f16ae1e6d     |
| revision_number         | 1                                    |
| routes                  |                                      |
| status                  | ACTIVE                               |
| tags                    | []                                   |
| updated_at              | 2017-01-10T13:15:19Z                 |
+-------------------------+--------------------------------------+

$ openstack router create router3
+-------------------------+--------------------------------------+
| Field                   | Value                                |
```

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(continued from previous page)

```
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints |                                      |
| availability_zones      |                                      |
| created_at              | 2017-01-10T13:15:19Z                 |
| description             |                                      |
| distributed             | False                                |
| external_gateway_info   | null                                 |
| flavor_id               | None                                 |
| ha                      | False                                |
| headers                 |                                      |
| id                      | 40069a4c-63be-11e6-9ecc-e37c1eaa7e84 |
| name                    | router3                              |
| project_id              | b3ac05ef10bf441fbf4aa17f16ae1e6d     |
| revision_number         | 1                                    |
| routes                  |                                      |
| status                  | ACTIVE                               |
| tags                    | []                                   |
| updated_at              | 2017-01-10T13:15:19Z                 |
+-------------------------+--------------------------------------+
```

2. For each router, add one self-service subnet as an interface on the router.

```
$ openstack router add subnet router1 selfservice1

$ openstack router add subnet router2 selfservice2

$ openstack router add subnet router3 selfservice3
```

3. Add the provider network as a gateway on each router.

```
$ openstack router set --external-gateway provider router1

$ openstack router set --external-gateway provider router2

$ openstack router set --external-gateway provider router3
```

Create and configure the BGP speaker

The BGP speaker advertises the next-hop IP address for eligible self-service networks and floating IP addresses for instances using those networks.

1. Create the BGP speaker.

```
$ openstack bgp speaker create --ip-version 4 \
  --local-as LOCAL_AS bgpspeaker
Created a new bgp_speaker:
+-----------------------------------+------------------------------------
↪-+
| Field                             | Value                              ␣
```

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```
↪ |
+-----------------------------------+-----------------------------------
↪-+
| advertise_floating_ip_host_routes | True                                 ␣
↪ |
| advertise_tenant_networks         | True                                 ␣
↪ |
| id                                | 5f227f14-4f46-4eca-9524-
↪fc5a1eabc358 |
| ip_version                        | 4                                    ␣
↪ |
| local_as                          | 64496                                ␣
↪ |
| name                              | bgpspeaker                           ␣
↪ |
| networks                          |                                      ␣
↪ |
| peers                             |                                      ␣
↪ |
| tenant_id                         | b3ac05ef10bf441fbf4aa17f16ae1e6d     ␣
↪ |
+-----------------------------------+-----------------------------------
↪-+
```

Replace LOCAL_AS with an appropriate local autonomous system number. The example configuration uses AS 64496.

2. A BGP speaker requires association with a provider network to determine eligible prefixes. The association builds a list of all virtual routers with gateways on provider and self-service networks in the same address scope so the BGP speaker can advertise self-service network prefixes with the corresponding router as the next-hop IP address. Associate the BGP speaker with the provider network.

```
$ openstack bgp speaker add network bgpspeaker provider
Added network provider to BGP speaker bgpspeaker.
```

3. Verify association of the provider network with the BGP speaker.

```
$ openstack bgp speaker show bgpspeaker
+-----------------------------------+-----------------------------------
↪-+
| Field                             | Value                                ␣
↪ |
+-----------------------------------+-----------------------------------
↪-+
| advertise_floating_ip_host_routes | True                                 ␣
↪ |
| advertise_tenant_networks         | True                                 ␣
↪ |
| id                                | 5f227f14-4f46-4eca-9524-
```

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```
↪fc5a1eabc358 |
| ip_version                       | 4                                    ␣
↪ |
| local_as                         | 64496                                ␣
↪ |
| name                             | bgpspeaker                           ␣
↪ |
| networks                         | 68ec148c-181f-4656-8334-
↪8f4eb148689d |
| peers                            |                                      ␣
↪ |
| tenant_id                        | b3ac05ef10bf441fbf4aa17f16ae1e6d     ␣
↪ |
+----------------------------------+------------------------------------
↪-+
```

4. Verify the prefixes and next-hop IP addresses that the BGP speaker advertises.

```
$ openstack bgp speaker list advertised routes bgpspeaker
+-----------------+--------------+
| Destination     | Nexthop      |
+-----------------+--------------+
| 192.0.2.0/25    | 203.0.113.11 |
| 192.0.2.128/25  | 203.0.113.12 |
+-----------------+--------------+
```

5. Create a BGP peer.

```
$ openstack bgp peer create --peer-ip 10.0.0.2 \
  --remote-as REMOTE_AS bgppeer
Created a new bgp_peer:
+-----------+--------------------------------------+
| Field     | Value                                |
+-----------+--------------------------------------+
| auth_type | none                                 |
| id        | 35c89ca0-ac5a-4298-a815-0b073c2362e9 |
| name      | bgppeer                              |
| peer_ip   | 10.0.0.2                             |
| remote_as | 64497                                |
| tenant_id | b3ac05ef10bf441fbf4aa17f16ae1e6d     |
+-----------+--------------------------------------+
```

Replace `REMOTE_AS` with an appropriate remote autonomous system number. The example configuration uses AS 64497 which triggers EBGP peering.

Note

The host containing the BGP agent must have layer-3 connectivity to the provider router.

6. Add a BGP peer to the BGP speaker.

```
$ openstack bgp speaker add peer bgpspeaker bgppeer
Added BGP peer bgppeer to BGP speaker bgpspeaker.
```

7. Verify addition of the BGP peer to the BGP speaker.

```
$ openstack bgp speaker show bgpspeaker
+-----------------------------------+--------------------------------------
↪-+
| Field                             | Value                                ␣
↪ |
+-----------------------------------+--------------------------------------
↪-+
| advertise_floating_ip_host_routes | True                                 ␣
↪ |
| advertise_tenant_networks         | True                                 ␣
↪ |
| id                                | 5f227f14-4f46-4eca-9524-
↪fc5a1eabc358 |
| ip_version                        | 4                                    ␣
↪ |
| local_as                          | 64496                                ␣
↪ |
| name                              | bgpspeaker                           ␣
↪ |
| networks                          | 68ec148c-181f-4656-8334-
↪8f4eb148689d |
| peers                             | 35c89ca0-ac5a-4298-a815-
↪0b073c2362e9 |
| tenant_id                         | b3ac05ef10bf441fbf4aa17f16ae1e6d     ␣
↪ |
+-----------------------------------+--------------------------------------
↪-+
```

> **Note**
>
> After creating a peering session, you cannot change the local or remote autonomous system numbers.

Schedule the BGP speaker to an agent

1. Unlike most agents, BGP speakers require manual scheduling to an agent. BGP speakers only form peering sessions and begin prefix advertisement after scheduling to an agent. Schedule the BGP speaker to agent `37729181-2224-48d8-89ef-16eca8e2f77e`.

```
$ openstack bgp dragent add speaker 37729181-2224-48d8-89ef-16eca8e2f77e␣
↪bgpspeaker
Associated BGP speaker bgpspeaker to the Dynamic Routing agent.
```

2. Verify scheduling of the BGP speaker to the agent.

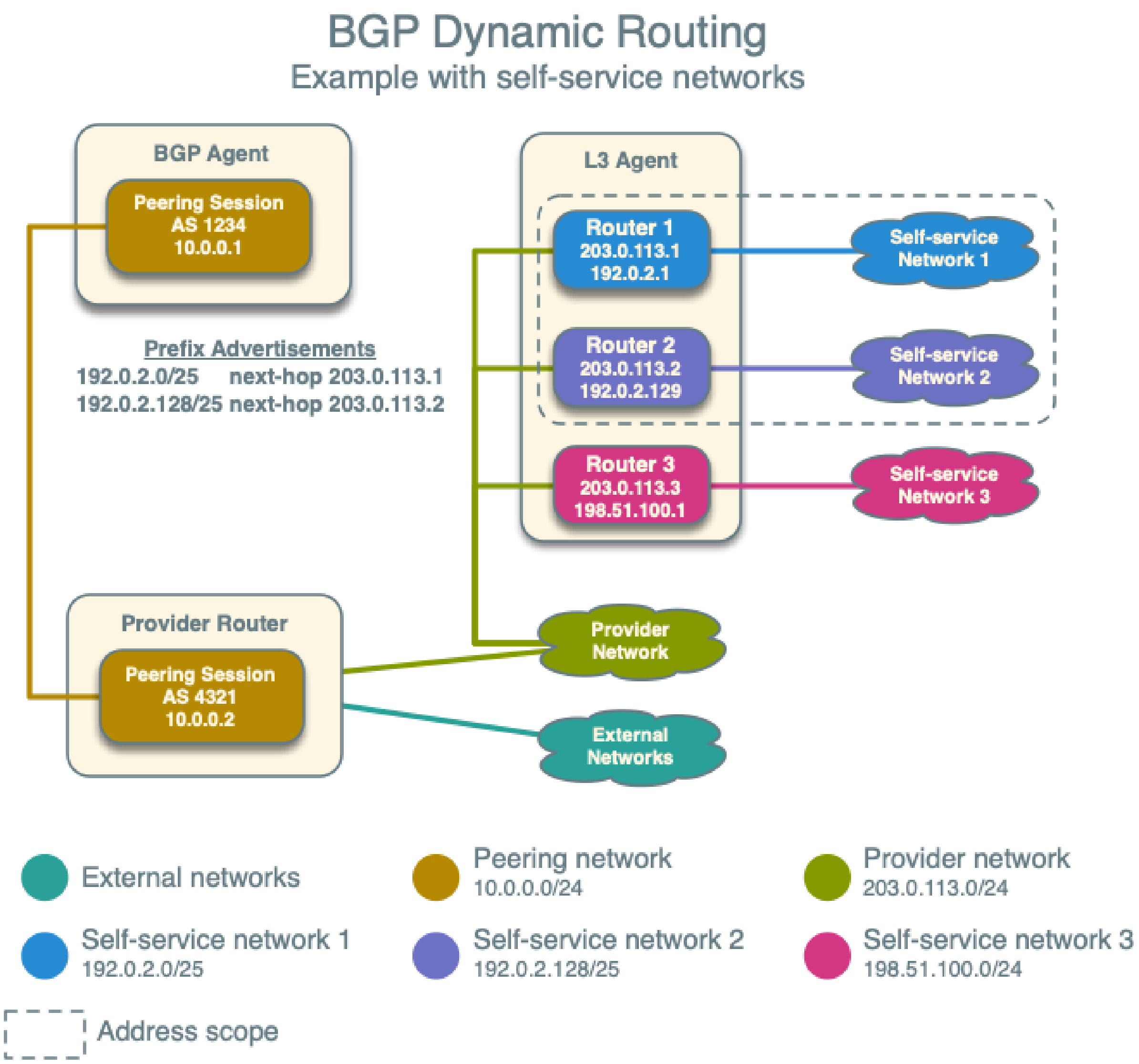
```
$ openstack bgp dragent list --bgp-speaker bgpspeaker
+--------------------------------------+------------+-------+-------+
| ID                                   | Host       | State | Alive |
+--------------------------------------+------------+-------+-------+
| 37729181-2224-48d8-89ef-16eca8e2f77e | controller | True  | :-)   |
+--------------------------------------+------------+-------+-------+
```

Prefix advertisement

BGP dynamic routing advertises prefixes for self-service networks and host routes for floating IP addresses.

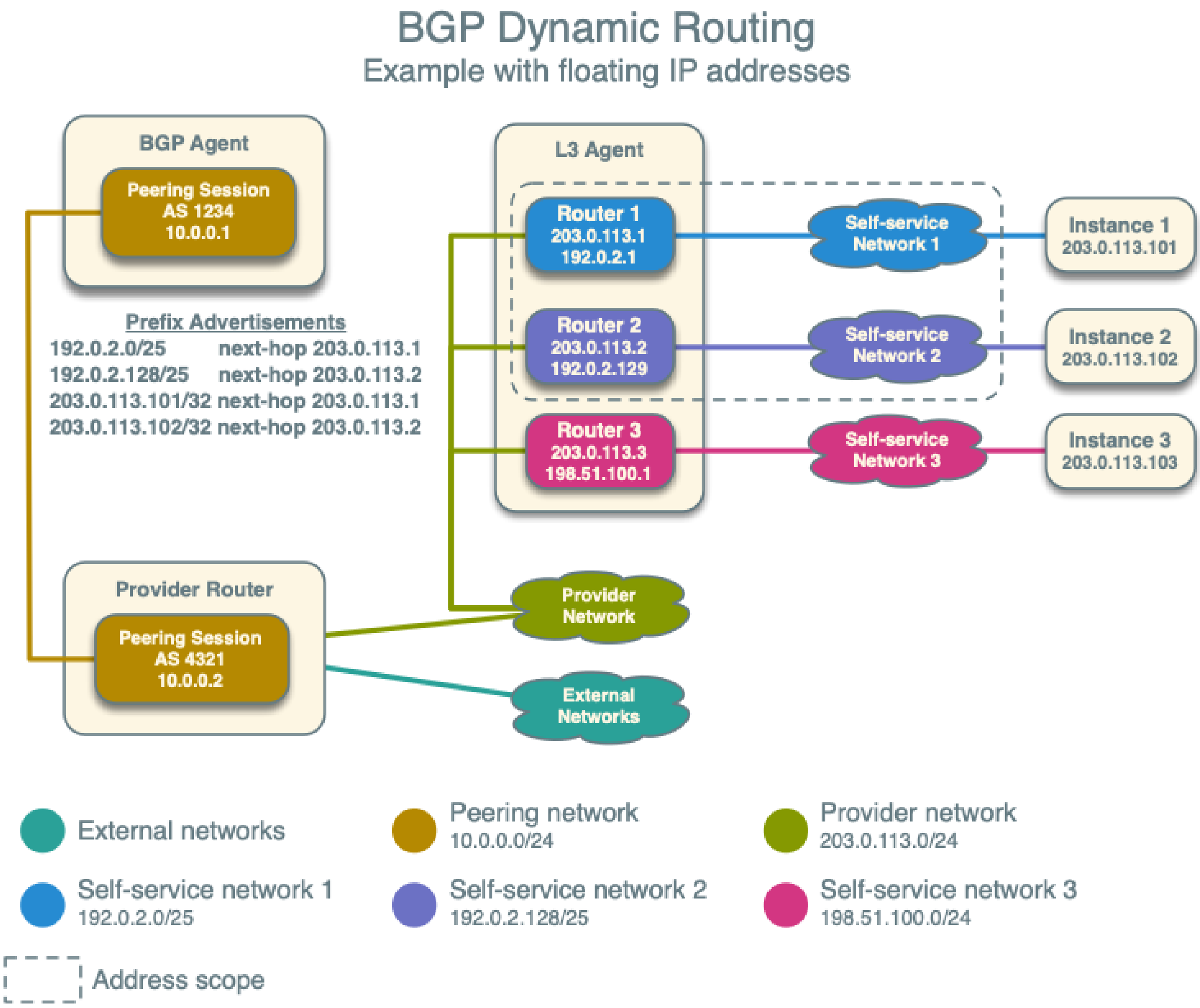
Advertisement of a self-service network requires satisfying the following conditions:

- The external and self-service network reside in the same address scope.
- The router contains an interface on the self-service subnet and a gateway on the external network.
- The BGP speaker associates with the external network that provides a gateway on the router.
- The BGP speaker has the `advertise_tenant_networks` attribute set to `True`.



Advertisement of a floating IP address requires satisfying the following conditions:

- The router with the floating IP address binding contains a gateway on an external network with the BGP speaker association.
- The BGP speaker has the `advertise_floating_ip_host_routes` attribute set to `True`.



Operation with Distributed Virtual Routers (DVR)

For both floating IP and IPv4 fixed IP addresses, the BGP speaker advertises the floating IP agent gateway on the corresponding compute node as the next-hop IP address. When using IPv6 fixed IP addresses, the BGP speaker advertises the DVR SNAT node as the next-hop IP address.

For example, consider the following components:

1. A provider network using IP address range 203.0.113.0/24, and supporting floating IP addresses 203.0.113.101, 203.0.113.102, and 203.0.113.103.
2. A self-service network using IP address range 198.51.100.0/24.
3. Instances with fixed IPs 198.51.100.11, 198.51.100.12, and 198.51.100.13
4. The SNAT gateway resides on 203.0.113.11.
5. The floating IP agent gateways (one per compute node) reside on 203.0.113.12, 203.0.113.13, and 203.0.113.14.

6. Three instances, one per compute node, each with a floating IP address.
7. `advertise_tenant_networks` is set to `False` on the BGP speaker

```
$ openstack bgp speaker list advertised routes bgpspeaker
+------------------+---------------+
| Destination      | Nexthop       |
+------------------+---------------+
| 198.51.100.0/24  | 203.0.113.11 |
| 203.0.113.101/32 | 203.0.113.12 |
| 203.0.113.102/32 | 203.0.113.13 |
| 203.0.113.103/32 | 203.0.113.14 |
+------------------+---------------+
```

When floating IPs are disassociated and `advertise_tenant_networks` is set to `True`, the following routes will be advertised:

```
$ openstack bgp speaker list advertised routes bgpspeaker
+------------------+---------------+
| Destination      | Nexthop       |
+------------------+---------------+
| 198.51.100.0/24  | 203.0.113.11 |
| 198.51.100.11/32 | 203.0.113.12 |
| 198.51.100.12/32 | 203.0.113.13 |
| 198.51.100.13/32 | 203.0.113.14 |
+------------------+---------------+
```

You can also identify floating IP agent gateways in your environment to assist with verifying operation of the BGP speaker.

```
$ openstack port list --device-owner network:floatingip_agent_gateway
+--------------------------------------+------+-------------------+-----------
↪---------------------------------------------------------------------------
↪----------------+
| ID                                   | Name | MAC Address       | Fixed IP␣
↪Addresses                                                                  ␣
↪                |
+--------------------------------------+------+-------------------+-----------
↪---------------------------------------------------------------------------
↪----------------+
| 87cf2970-4970-462e-939e-00e808295dfa |      | fa:16:3e:7c:68:e3 | ip_
↪address='203.0.113.12', subnet_id='8ed65d41-2b2a-4f3a-9f92-45adb266e01a'   ␣
↪                      |
| 8d218440-0d2e-49d0-8a7b-3266a6146dc1 |      | fa:16:3e:9d:78:cf | ip_
↪address='203.0.113.13', subnet_id='8ed65d41-2b2a-4f3a-9f92-45adb266e01a'   ␣
↪                      |
| 87cf2970-4970-462e-939e-00e802281dfa |      | fa:16:3e:6b:18:e0 | ip_
↪address='203.0.113.14', subnet_id='8ed65d41-2b2a-4f3a-9f92-45adb266e01a'   ␣
↪                      |
+--------------------------------------+------+-------------------+-----------
↪---------------------------------------------------------------------------
↪----------------+
```

IPv6

BGP dynamic routing supports peering via IPv6 and advertising IPv6 prefixes.

- To enable peering via IPv6, create a BGP peer and use an IPv6 address for `peer_ip`.
- To enable advertising IPv6 prefixes, create an address scope with `ip_version=6` and a BGP speaker with `ip_version=6`.

Note

DVR lacks support for routing directly to a fixed IPv6 address via the floating IP agent gateway port and thus prevents the BGP speaker from advertising /128 host routes.

High availability

BGP dynamic routing supports scheduling a BGP speaker to multiple agents which effectively multiplies prefix advertisements to the same peer. If an agent fails, the peer continues to receive advertisements from one or more operational agents.

1. Show available dynamic routing agents.

```
$ openstack network agent list --agent-type bgp
+--------------------------------------+---------------------------+------
↪----+-------------------+-------+-------+---------------------------+
| ID                                   | Agent Type                | Host␣
↪    | Availability Zone | Alive | State | Binary                    |
+--------------------------------------+---------------------------+------
↪----+-------------------+-------+-------+---------------------------+
| 37729181-2224-48d8-89ef-16eca8e2f77e | BGP dynamic routing agent | bgp-
↪ha1  | None              | :-)   | UP    | neutron-bgp-dragent       |
| 1a2d33bb-9321-30a2-76ab-22eff3d2f56a | BGP dynamic routing agent | bgp-
↪ha2  | None              | :-)   | UP    | neutron-bgp-dragent       |
+--------------------------------------+---------------------------+------
↪----+-------------------+-------+-------+---------------------------+
```

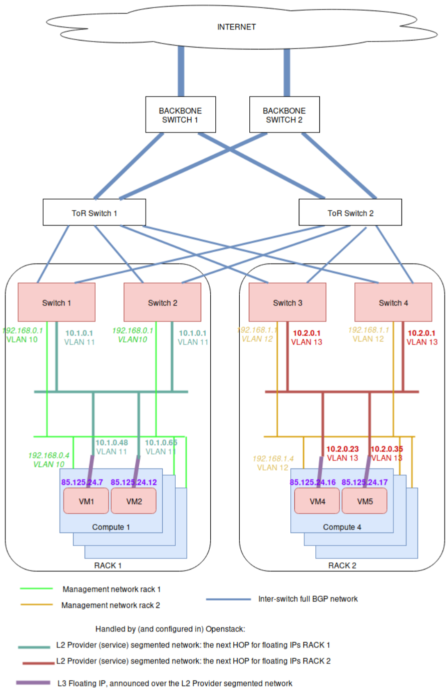
2. Schedule BGP speaker to multiple agents.

```
$ openstack bgp dragent add speaker 37729181-2224-48d8-89ef-16eca8e2f77e␣
↪bgpspeaker
Associated BGP speaker bgpspeaker to the Dynamic Routing agent.

$ openstack bgp dragent add speaker 1a2d33bb-9321-30a2-76ab-22eff3d2f56a␣
↪bgpspeaker
Associated BGP speaker bgpspeaker to the Dynamic Routing agent.

$ openstack bgp dragent list --bgp-speaker bgpspeaker
+--------------------------------------+---------+-------+-------+
| ID                                   | Host    | State | Alive |
+--------------------------------------+---------+-------+-------+
| 37729181-2224-48d8-89ef-16eca8e2f77e | bgp-ha1 | True  | :-)   |
| 1a2d33bb-9321-30a2-76ab-22eff3d2f56a | bgp-ha2 | True  | :-)   |
+--------------------------------------+---------+-------+-------+
```

8.2.6 BGP Floating IPs over L2 Segmented Networks

The general principle is that L2 connectivity will be bound to a single rack. Everything outside the switches of the rack will be routed using BGP. To perform the BGP announcement, neutron-dynamic-routing is leveraged.

To achieve this, on each rack, servers are setup with a different management network using a vlan ID per rack (light green and orange network below). Note that a unique vlan ID per rack isnt mandatory, its also possible to use the same vlan ID on all racks. The point here is only to isolate L2 segments (typically, routing between the switch of each racks will be done over BGP, without L2 connectivity).



On the OpenStack side, a provider network must be setup, which is using a different subnet range and vlan ID for each rack. This includes:

- an address scope
- some network segments for that network, which are attached to a named physical network
- a subnet pool using that address scope
- one provider network subnet per segment (each subnet+segment pair matches one rack physical network name)

A segment is attached to a specific vlan and physical network name. In the above figure, the provider network is represented by 2 subnets: the dark green and the red ones. The dark green subnet is on one network segment, and the red one on another. Both subnet are of the subnet service type network:floatingip_agent_gateway, so that they cannot be used by virtual machines directly.

On top of all of this, a floating IP subnet without a segment is added, which spans in all of the racks. This subnet must have the below service types:

- network:routed
- network:floatingip
- network:router_gateway

Since the network:routed subnet isnt bound to a segment, it can be used on all racks. As the service types network:floatingip and network:router_gateway are used for the provider network, the subnet can only be used for floating IPs and router gateways, meaning that the subnet using segments will be used as floating IP gateways (ie: the next HOP to reach these floating IP / router external gateways).

Configuring the Neutron API side

On the controller side (ie: API and RPC server), only the Neutron Dynamic Routing Python library must be installed (for example, in the Debian case, that would be the neutron-dynamic-routing-common and python3-neutron-dynamic-routing packages). On top of that, segments and bgp must be added to the list of plugins in service_plugins. For example in neutron.conf:

```
[DEFAULT]
service_plugins=router,metering,qos,trunk,segments,bgp
```

The BGP agent

The neutron-bgp-agent must be installed. Best is to install it twice per rack, on any machine (it doesnt mater much where). Then each of these BGP agents will establish a session with one switch, and advertise all of the BGP configuration.

Setting-up BGP peering with the switches

A peer that represents the network equipment must be created. Then a matching BGP speaker needs to be created. Then, the BGP speaker must be associated to a dynamic-routing-agent (in our example, the dynamic-routing agents run on compute 1 and 4). Finally, the peer is added to the BGP speaker, so the speaker initiates a BGP session to the network equipment.

```
$ # Create a BGP peer to represent the switch 1,
$ # which runs FRR on 10.1.0.253 with AS 64498
$ openstack bgp peer create \
    --peer-ip 10.1.0.253 \
    --remote-as 64498 \
    rack1-switch-1

$ # Create a BGP speaker on compute-1
$ BGP_SPEAKER_ID_COMPUTE_1=$(openstack bgp speaker create \
    --local-as 64499 --ip-version 4 mycloud-compute-1.example.com \
    --format value -c id)

$ # Get the agent ID of the dragent running on compute 1
$ BGP_AGENT_ID_COMPUTE_1=$(openstack network agent list \
    --host mycloud-compute-1.example.com --agent-type bgp \
    --format value -c ID)

$ # Add the BGP speaker to the dragent of compute 1
$ openstack bgp dragent add speaker \
    ${BGP_AGENT_ID_COMPUTE_1} ${BGP_SPEAKER_ID_COMPUTE_1}

$ # Add the BGP peer to the speaker of compute 1
$ openstack bgp speaker add peer \
    mycloud-compute-1.example.com rack1-switch-1
```

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```
$ # Tell the speaker not to advertize tenant networks
$ openstack bgp speaker set \
    --no-advertise-tenant-networks mycloud-compute-1.example.com
```

It is possible to repeat this operation for a 2nd machine on the same rack, if the deployment is using bonding (and then, LACP between both switches), as per the figure above. It also can be done on each rack. One way to deploy is to select two computers in each rack (for example, one compute node and one network node), and install the neutron-dynamic-routing-agent on each of them, so they can talk to both switches of the rack. All of this depends on what the configuration is on the switch side. It may be that you only need to talk to two ToR racks in the whole deployment. The thing you must know is that you can deploy as many dynamic-routing agent as needed, and that one agent can talk to a single device.

Setting-up physical network names

Before setting-up the provider network, the physical network name must be set in each host, according to the rack names. On the compute or network nodes, this is done in /etc/neutron/plugins/ml2/openvswitch_agent.ini using the bridge_mappings directive:

```
[ovs]
bridge_mappings = physnet-rack1:br-ex
```

All of the physical networks created this way must be added in the configuration of the neutron-server as well (ie: this is used by both neutron-api and neutron-rpc-server). For example, with 3 racks, heres how /etc/neutron/plugins/ml2/ml2_conf.ini should look like:

```
[ml2_type_flat]
flat_networks = physnet-rack1,physnet-rack2,physnet-rack3

[ml2_type_vlan]
network_vlan_ranges = physnet-rack1,physnet-rack2,physnet-rack3
```

Once this is done, the provider network can be created, using physnet-rack1 as physical network.

Setting-up the provider network

Everything that is in the provider networks scope will be advertised through BGP. Here is how to create the network scope:

```
$ # Create the address scope
$ openstack address scope create --share --ip-version 4 provider-addr-scope
```

Then, the network can be ceated using the physical network name set above:

```
$ # Create the provider network that spawns over all racks
$ openstack network create --external --share \
    --provider-physical-network physnet-rack1 \
    --provider-network-type vlan \
    --provider-segment 11 \
    provider-network
```

This automatically creates a network AND a segment. Though by default, this segment has no name, which isnt convenient. This name can be changed though:

```
$ # Get the network ID:
$ PROVIDER_NETWORK_ID=$(openstack network show provider-network \
      --format value -c id)

$ # Get the segment ID:
$ FIRST_SEGMENT_ID=$(openstack network segment list \
      --format csv -c ID -c Network | \
      q -H -d, "SELECT ID FROM - WHERE Network='${PROVIDER_NETWORK_ID}'")

$ # Set the 1st segment name, matching the rack name
$ openstack network segment set --name segment-rack1 ${FIRST_SEGMENT_ID}
```

Setting-up the 2nd segment

The 2nd segment, which will be attached to our provider network, is created this way:

```
$ # Create the 2nd segment, matching the 2nd rack name
$ openstack network segment create \
      --physical-network physnet-rack2 \
      --network-type vlan \
      --segment 13 \
      --network provider-network \
      segment-rack2
```

Setting-up the provider subnets for the BGP next HOP routing

These subnets will be in use in different racks, depending on what physical network is in use in the machines. In order to use the address scope, subnet pools must be used. Here is how to create the subnet pool with the two ranges to use later when creating the subnets:

```
$ # Create the provider subnet pool which includes all ranges for all racks
$ openstack subnet pool create \
      --pool-prefix 10.1.0.0/24 \
      --pool-prefix 10.2.0.0/24 \
      --address-scope provider-addr-scope \
      --share \
      provider-subnet-pool
```

Then, this is how to create the two subnets. In this example, we are keeping the addresses in .1 for the gateway, .2 for the DHCP server, and .253 +.254, as these addresses will be used by the switches for the BGP announcements:

```
$ # Create the subnet for the physnet-rack-1, using the segment-rack-1, and
$ # the subnet_service_type network:floatingip_agent_gateway
$ openstack subnet create \
      --service-type 'network:floatingip_agent_gateway' \
      --subnet-pool provider-subnet-pool \
      --subnet-range 10.1.0.0/24 \
```

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```
    --allocation-pool start=10.1.0.3,end=10.1.0.252 \
    --gateway 10.1.0.1 \
    --network provider-network \
    --network-segment segment-rack1 \
    provider-subnet-rack1

$ # The same, for the 2nd rack
$ openstack subnet create \
    --service-type 'network:floatingip_agent_gateway' \
    --subnet-pool provider-subnet-pool \
    --subnet-range 10.2.0.0/24 \
    --allocation-pool start=10.2.0.3,end=10.2.0.252 \
    --gateway 10.2.0.1 \
    --network provider-network \
    --network-segment segment-rack2 \
    provider-subnet-rack2
```

Note the service types. network:floatingip_agent_gateway makes sure that these subnets will be in use only as gateways (ie: the next BGP hop). The above can be repeated for each new rack.

Adding a subnet for VM floating IPs and router gateways

This is to be repeated each time a new subnet must be created for floating IPs and router gateways. First, the range is added in the subnet pool, then the subnet itself is created:

```
$ # Add a new prefix in the subnet pool for the floating IPs:
$ openstack subnet pool set \
    --pool-prefix 203.0.113.0/24 \
    provider-subnet-pool

$ # Create the floating IP subnet
$ openstack subnet create vm-fip \
    --service-type 'network:routed' \
    --service-type 'network:floatingip' \
    --service-type 'network:router_gateway' \
    --subnet-pool provider-subnet-pool \
    --subnet-range 203.0.113.0/24 \
    --network provider-network
```

The service-type network:routed ensures were using BGP through the provider network to advertize the IPs. network:floatingip and network:router_gateway limits the use of these IPs to floating IPs and router gateways.

Setting-up BGP advertizing

The provider network needs to be added to each of the BGP speakers. This means each time a new rack is setup, the provider network must be added to the 2 BGP speakers of that rack.

```
$ # Add the provider network to the BGP speakers.
$ openstack bgp speaker add network \
    mycloud-compute-1.example.com provider-network
```

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```
$ openstack bgp speaker add network \
    mycloud-compute-4.example.com provider-network
```

In this example, weve selected two compute nodes that are also running an instance of the neutron-dynamic-routing-agent daemon.

Per project operation

This can be done by each customer. A subnet pool isnt mandatory, but it is nice to have. Typically, the customer network will not be advertized through BGP (but this can be done if needed).

```
$ # Create the tenant private network
$ openstack network create tenant-network

$ # Self-service network pool:
$ openstack subnet pool create \
    --pool-prefix 192.168.130.0/23 \
    --share \
    tenant-subnet-pool

$ # Self-service subnet:
$ openstack subnet create \
    --network tenant-network \
    --subnet-pool tenant-subnet-pool \
    --prefix-length 24 \
    tenant-subnet-1

$ # Create the router
$ openstack router create tenant-router

$ # Add the tenant subnet to the tenant router
$ openstack router add subnet \
    tenant-router tenant-subnet-1

$ # Set the router's default gateway. This will use one public IP.
$ openstack router set \
    --external-gateway provider-network tenant-router

$ # Create a first VM on the tenant subnet
$ openstack server create --image debian-10.5.0-openstack-amd64.qcow2 \
    --flavor cpu2-ram6-disk20 \
    --nic net-id=tenant-network \
    --key-name yubikey-zigo \
    test-server-1

$ # Eventually, add a floating IP
$ openstack floating ip create provider-network
+---------------------+--------------------------------------+
| Field               | Value                                |
+---------------------+--------------------------------------+
```

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```
| created_at          | 2020-12-15T11:48:36Z                 |
| description         |                                      |
| dns_domain          | None                                 |
| dns_name            | None                                 |
| fixed_ip_address    | None                                 |
| floating_ip_address | 203.0.113.17                         |
| floating_network_id | 859f5302-7b22-4c50-92f8-1f71d6f3f3f4 |
| id                  | 01de252b-4b78-4198-bc28-1328393bf084 |
| name                | 203.0.113.17                         |
| port_details        | None                                 |
| port_id             | None                                 |
| project_id          | d71a5d98aef04386b57736a4ea4f3644     |
| qos_policy_id       | None                                 |
| revision_number     | 0                                    |
| router_id           | None                                 |
| status              | DOWN                                 |
| subnet_id           | None                                 |
| tags                | []                                   |
| updated_at          | 2020-12-15T11:48:36Z                 |
+---------------------+--------------------------------------+
$ openstack server add floating ip test-server-1 203.0.113.17
```

Cumulus switch configuration

Because of the way Neutron works, for each new port associated with an IP address, a GARP is issued, to inform the switch about the new MAC / IP association. Unfortunately, this confuses the switches where they may think they should use local ARP table to route the packet, rather than giving it to the next HOP to route. The definitive solution would be to patch Neutron to make it stop sending GARP for any port on a subnet with the network:routed service type. Such patch would be hard to write, though lucky, theres a fix that works (at least with Cumulus switches). Heres how.

In /etc/network/switchd.conf we change this:

```
# configure a route instead of a neighbor with the same ip/mask
#route.route_preferred_over_neigh = FALSE
route.route_preferred_over_neigh = TRUE
```

and then simply restart switchd:

```
systemctl restart switchd
```

This reboots the switch ASIC of the switch, so it may be a dangerous thing to do with no switch redundancy (so be careful when doing it). The completely safe procedure, if having 2 switches per rack, looks like this:

```
# save clagd priority
OLDPRIO=$(clagctl status | sed -r -n  's/.*Our.*Role: ([0-9]+) 0.*/\1/p')
# make sure that this switch is not the primary clag switch. otherwise the
# secondary switch will also shutdown all interfaces when loosing contact
# with the primary switch.
```

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```
clagctl priority 16535

# tell neighbors to not route through this router
vtysh
vtysh# router bgp 64499
vtysh# bgp graceful-shutdown
vtysh# exit
systemctl restart switchd
clagctl priority $OLDPRIO
```

Verification

If everything goes well, the floating IPs are advertized over BGP through the provider network. Here is an example with 4 VMs deployed on 2 racks. Neutron is here picking-up IPs on the segmented network as Nexthop.

```
$ # Check the advertized routes:
$ openstack bgp speaker list advertised routes \
    mycloud-compute-4.example.com
+-----------------+-----------+
| Destination     | Nexthop   |
+-----------------+-----------+
| 203.0.113.17/32 | 10.1.0.48 |
| 203.0.113.20/32 | 10.1.0.65 |
| 203.0.113.40/32 | 10.2.0.23 |
| 203.0.113.55/32 | 10.2.0.35 |
+-----------------+-----------+
```

8.2.7 Agents and Services

A usual neutron setup consists of multiple services and agents running on one or multiple nodes (though some setups may not need any agents). Each of these services provide some of the networking or API services. Among those of special interest are:

1. The neutron-server that provides API endpoints and serves as a single point of access to the database. It usually runs on the controller nodes.
2. Layer2 agent that can utilize Open vSwitch or other vendor-specific technology to provide network segmentation and isolation for project networks. The L2 agent should run on every node where it is deemed responsible for wiring and securing virtual interfaces (usually both compute and network nodes).
3. Layer3 agent that runs on network node and provides east-west and north-south routing plus some advanced services such as FWaaS or VPNaaS.

Configuration options

The neutron configuration options are segregated between neutron-server and agents. Both services and agents may load the main `neutron.conf` since this file should contain the oslo.messaging configuration for internal neutron RPCs and may contain host specific configuration, such as file paths. The `neutron.conf` contains the database, keystone, nova credentials, and endpoints strictly for neutron-server to use.

In addition, neutron-server may load a plugin-specific configuration file, yet the agents should not. As the plugin configuration is primarily site wide options and the plugin provides the persistence layer for neutron, agents should be instructed to act upon these values through RPC.

Each individual agent may have its own configuration file. This file should be loaded after the main `neutron.conf` file, so the agent configuration takes precedence. The agent-specific configuration may contain configurations which vary between hosts in a neutron deployment such as the `local_ip` for an L2 agent. If any agent requires access to additional external services beyond the neutron RPC, those endpoints should be defined in the agent-specific configuration file (for example, nova metadata for metadata agent).

Agents admin state specific config options

When creating a new agent the `admin_state_up` field will be set to the value of `enable_new_agents` config option, the default value of this config option is `true`:

```
[DEFAULT]
enable_new_agents = true
```

It is possible to set the `admin_state_up` value of an agent to `False` via the API, or CLI:

```
$ openstack network agent set agent-uuid --disable
```

The effect of this varies by agent type:

L2 agents

The `admin_state_up` field of the agent in the Neutron database is set to `False`, but the agent is still capable of binding ports. This is true for openvswitch-agent and sriov-agent.

> **Note**
>
> In case of OVN based deployment Neutron doesnt keep track of OVN controllers in the `agents` db table, so setting the `admin_state_up` is not allowed as Neutron has no control over OVN entities. The possiblity to delete an OVN agent via Neutron REST API, is to clean up bad chassis information.

Metadata agent

Setting `admin_state_up` to False has no effect to the Metadata agent.

DHCP agent

DHCP agent scheduler will schedule networks to agents whose `admin_state_up` is `True`.

L3 agent

L3 scheduler will schedule routers to L3 agents whose `admin_state_up` field is `True`.

External processes run by agents

Some neutron agents, like DHCP, Metadata or L3, often run external processes to provide some of their functionalities. It may be keepalived, dnsmasq, haproxy or some other process. Neutron agents are responsible for spawning and killing such processes when necessary. By default, to kill such processes, agents use a simple `kill` command, but in some cases, like for example when those additional services are running inside containers, it may be not a good solution. To address this problem, operators should use the `AGENT` config group option `kill_scripts_path` to configure a path to where `kill scripts` for such processes live. By default, it is set to `/etc/neutron/kill_scripts/`. If option `kill_scripts_path` is changed in the config to the different location, `exec_dirs` in `/etc/neutron/rootwrap.conf` should be changed accordingly. If `kill_scripts_path` is set, every time neutron has to kill a process, for example `dnsmasq`, it will look in this directory for a file with the name `<process_name>-kill`. So for `dnsmasq` process it will look for a `dnsmasq-kill` script. If such a file exists there, it will be called instead of using the `kill` command.

Kill scripts are called with two parameters:

```
<process>-kill <sig> <pid>
```

where: `<sig>` is the signal, same as with the `kill` command, for example 9 or `SIGKILL`; and `<pid>` is pid of the process to kill.

This external script should then handle killing of the given process as neutron will not call the `kill` command for it anymore.

8.2.8 DNS Integration

This page serves as a guide for how to use the DNS integration functionality of the Networking service and its interaction with the Compute service.

The integration of the Networking service with an external DNSaaS (DNS-as-a-Service) is described in *DNS Integration with an External Service*.

Users can control the behavior of the Networking service in regards to DNS using two attributes associated with ports, networks, and floating IPs. The following table shows the attributes available for each one of these resources:

Resource	dns_name	dns_domain
Ports	Yes	Yes
Networks	No	Yes
Floating IPs	Yes	Yes

Note

The `DNS Integration` extension enables all the attribute and resource combinations shown in the previous table, except for `dns_domain` for ports, which requires the `dns_domain for ports` extension.

Note

Since the `DNS Integration` extension is a subset of `dns_domain for ports`, if `dns_domain` functionality for ports is required, only the latter extension has to be configured.

Note

When the `dns_domain for ports` extension is configured, `DNS Integration` is also included when the Neutron server responds to a request to list the active API extensions. This preserves backwards API compatibility.

Note

For `Floating IPs` external DNSaaS is also required as described in *DNS Integration with an External Service*.

The Networking service internal DNS resolution

The Networking service enables users to control the name assigned to ports by the internal DNS. To enable this functionality, do the following:

1. Edit the `/etc/neutron/neutron.conf` file and assign a value different to `openstacklocal` (its default value) to the `dns_domain` parameter in the `[default]` section. As an example:

```
dns_domain = example.org.
```

2. Add `dns` (for the `DNS Integration` extension) or `dns_domain_ports` (for the `dns_domain for ports` extension) to `extension_drivers` in the `[ml2]` section of `/etc/neutron/plugins/ml2/ml2_conf.ini`. The following is an example:

```
[ml2]
extension_drivers = port_security,dns_domain_ports
```

After re-starting the `neutron-server`, users will be able to assign a `dns_name` attribute to their ports.

Valid `extension_drivers` values related to the DNS integration are:

- `dns`
- `dns_domain_ports`
- `subnet_dns_publish_fixed_ip`
- `dns_domain_keywords`

Note

The enablement of this functionality is prerequisite for the enablement of the Networking service integration with an external DNS service, which is described in detail in *DNS Integration with an External Service*.

The following illustrates the creation of a port with `my-port` in its `dns_name` attribute.

Note

The name assigned to the port by the Networking service internal DNS is now visible in the response in the `dns_assignment` attribute.

```
$ openstack port create --network my-net --dns-name my-port test
+-----------------------+-----------------------------------------------------------------------------
↪-------------------------+
| Field                 | Value
↪                         |
+-----------------------+-----------------------------------------------------------------------------
↪-------------------------+
| admin_state_up        | UP
↪                         |
| allowed_address_pairs |
↪                         |
| binding_host_id       |
↪                         |
| binding_profile       |
↪                         |
| binding_vif_details   |
↪                         |
| binding_vif_type      | unbound
↪                         |
| binding_vnic_type     | normal
↪                         |
| created_at            | 2016-02-05T21:35:04Z
↪                         |
| data_plane_status     | None
↪                         |
| description           |
↪                         |
| device_id             |
↪                         |
| device_owner          |
↪                         |
| dns_assignment        | fqdn='my-port.example.org.', hostname='my-port', ip_
↪address='192.0.2.67'     |
| dns_domain            | None
↪                         |
| dns_name              | my-port
↪                         |
| extra_dhcp_opts       |
↪                         |
| fixed_ips             | ip_address='192.0.2.67', subnet_id='6141b474-56cd-
↪430f-b731-71660bb79b79'     |
| id                    | fb3c10f4-017e-420c-9be1-8f8c557ae21f
↪                         |
| mac_address           | fa:16:3e:aa:9b:e1
↪                         |
| name                  | test
↪                         |
| network_id            | bf2802a0-99a0-4e8c-91e4-107d03f158ea
↪                         |
| port_security_enabled | True
```

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```
↪                                                           |
| project_id            | d5660cb1e6934612a01b4fb2fb630725                       ␣
↪                                                           |
| qos_policy_id         | None                                                   ␣
↪                                                           |
| revision_number       | 1                                                      ␣
↪                                                           |
| security_group_ids    | 1f0ddd73-7e3c-48bd-a64c-7ded4fe0e635                   ␣
↪                                                           |
| status                | DOWN                                                   ␣
↪                                                           |
| tags                  |                                                        ␣
↪                                                           |
| trunk_details         | None                                                   ␣
↪                                                           |
| updated_at            | 2016-02-05T21:35:04Z                                   ␣
↪                                                           |
+-----------------------+-------------------------------------------------------
↪--------------------------+
```

When this functionality is enabled, it is leveraged by the Compute service when creating instances. When allocating ports for an instance during boot, the Compute service populates the `dns_name` attributes of these ports with the `hostname` attribute of the instance, which is a DNS sanitized version of its display name. As a consequence, at the end of the boot process, the allocated ports will be known in the dnsmasq associated to their networks by their instance `hostname`.

The following is an example of an instance creation, showing how its `hostname` populates the `dns_name` attribute of the allocated port:

```
$ openstack server create --image cirros --flavor 42 \
  --nic net-id=37aaff3a-6047-45ac-bf4f-a825e56fd2b3 my_vm
+--------------------------------------+-----------------------------------
↪--------------------------+
| Field                                | Value                              ␣
↪                           |
+--------------------------------------+-----------------------------------
↪--------------------------+
| OS-DCF:diskConfig                    | MANUAL                             ␣
↪                           |
| OS-EXT-AZ:availability_zone          |                                    ␣
↪                           |
| OS-EXT-STS:power_state               | 0                                  ␣
↪                           |
| OS-EXT-STS:task_state                | scheduling                         ␣
↪                           |
| OS-EXT-STS:vm_state                  | building                           ␣
↪                           |
| OS-SRV-USG:launched_at               | -                                  ␣
↪                           |
| OS-SRV-USG:terminated_at             | -                                  ␣
```

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```
↪                                |
| accessIPv4                          |                                  ␣
↪                                |
| accessIPv6                          |                                  ␣
↪                                |
| adminPass                           | dB45Zvo8Jpfe                     ␣
↪                                |
| config_drive                        |                                  ␣
↪                                |
| created                             | 2016-02-05T21:35:04Z             ␣
↪                                |
| flavor                              | m1.nano (42)                     ␣
↪                                |
| hostId                              |                                  ␣
↪                                |
| id                                  | 66c13cb4-3002-4ab3-8400-7efc2659c363␣
↪                                |
| image                               | cirros-0.3.5-x86_64-uec(b9d981eb-
↪d21c-4ce2-9dbc-dd38f3d9015f)   |
| key_name                            | -                                ␣
↪                                |
| locked                              | False                            ␣
↪                                |
| metadata                            | {}                               ␣
↪                                |
| name                                | my_vm                            ␣
↪                                |
| os-extended-volumes:volumes_attached | []                              ␣
↪                                |
| progress                            | 0                                ␣
↪                                |
| security_groups                     | default                          ␣
↪                                |
| status                              | BUILD                            ␣
↪                                |
| tenant_id                           | d5660cb1e6934612a01b4fb2fb630725 ␣
↪                                |
| updated                             | 2016-02-05T21:35:04Z             ␣
↪                                |
| user_id                             | 8bb6e578cba24e7db9d3810633124525 ␣
↪                                |
+--------------------------------------+-----------------------------------
↪--------------------------+

$ openstack port list --device-id 66c13cb4-3002-4ab3-8400-7efc2659c363
+--------------------------------------+------+-------------------+----------
↪-----------------------------------------------------------------------------
↪+--------+
| ID                                   | Name | MAC Address       | Fixed IP␣
```

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```
↪Addresses                                                                        ␣
↪ | Status |
+--------------------------------------+------+-------------------+-----------
↪---------------------------------------------------------------------------
↪+--------+
| b3ecc464-1263-44a7-8c38-2d8a52751773 |      | fa:16:3e:a8:ce:b8 | ip_
↪address='203.0.113.8', subnet_id='277eca5d-9869-474b-960e-6da5951d09f7'    ␣
↪       | ACTIVE |
|                                      |      |                   | ip_
↪address='2001:db8:10::8', subnet_id='eab47748-3f0a-4775-a09f-b0c24bb64bc4' ␣
↪       |        |
+--------------------------------------+------+-------------------+-----------
↪---------------------------------------------------------------------------
↪+--------+

$ openstack port show b3ecc464-1263-44a7-8c38-2d8a52751773
+-----------------------+------------------------------------------------------
↪-------------------------------------------------------+
| Field                 | Value                                                   ␣
↪                                                       |
+-----------------------+------------------------------------------------------
↪-------------------------------------------------------+
| admin_state_up        | UP                                                      ␣
↪                                                       |
| allowed_address_pairs |                                                         ␣
↪                                                       |
| binding_host_id       | vultr.guest                                             ␣
↪                                                       |
| binding_profile       |                                                         ␣
↪                                                       |
| binding_vif_details   | datapath_type='system', ovs_hybrid_plug='True',␣
↪port_filter='True'                                            |
| binding_vif_type      | ovs                                                     ␣
↪                                                       |
| binding_vnic_type     | normal                                                  ␣
↪                                                       |
| created_at            | 2016-02-05T21:35:04Z                                    ␣
↪                                                       |
| data_plane_status     | None                                                    ␣
↪                                                       |
| description           |                                                         ␣
↪                                                       |
| device_id             | 66c13cb4-3002-4ab3-8400-7efc2659c363                    ␣
↪                                                       |
| device_owner          | compute:None                                            ␣
↪                                                       |
| dns_assignment        | fqdn='my-vm.example.org.', hostname='my-vm', ip_
↪address='203.0.113.8'                                          |
|                       | fqdn='my-vm.example.org.', hostname='my-vm', ip_
```

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```
↪address='2001:db8:10::8'                           |
| dns_domain            | example.org.                                                     ␣
↪                                                    |
| dns_name              | my-vm                                                            ␣
↪                                                    |
| extra_dhcp_opts       |                                                                  ␣
↪                                                    |
| fixed_ips             | ip_address='203.0.113.8', subnet_id='277eca5d-9869-
↪474b-960e-6da5951d09f7'                              |
|                       | ip_address='2001:db8:10::8', subnet_id='eab47748-
↪3f0a-4775-a09f-b0c24bb64bc4'                          |
| id                    | b3ecc464-1263-44a7-8c38-2d8a52751773                             ␣
↪                                                    |
| mac_address           | fa:16:3e:a8:ce:b8                                                ␣
↪                                                    |
| name                  |                                                                  ␣
↪                                                    |
| network_id            | 37aaff3a-6047-45ac-bf4f-a825e56fd2b3                             ␣
↪                                                    |
| port_security_enabled | True                                                             ␣
↪                                                    |
| project_id            | d5660cb1e6934612a01b4fb2fb630725                                 ␣
↪                                                    |
| qos_policy_id         | None                                                             ␣
↪                                                    |
| revision_number       | 1                                                                ␣
↪                                                    |
| security_group_ids    | 1f0ddd73-7e3c-48bd-a64c-7ded4fe0e635                             ␣
↪                                                    |
| status                | ACTIVE                                                           ␣
↪                                                    |
| tags                  |                                                                  ␣
↪                                                    |
| trunk_details         | None                                                             ␣
↪                                                    |
| updated_at            | 2016-02-05T21:35:04Z                                             ␣
↪                                                    |
+-----------------------+-----------------------------------------------------------------
↪----------------------------------------------------+
```

In the above example notice that:

- The name given to the instance by the user, `my_vm`, is sanitized by the Compute service and becomes `my-vm` as the ports `dns_name`.
- The ports `dns_assignment` attribute shows that its FQDN is `my-vm.example.org.` in the Networking service internal DNS, which is the result of concatenating the ports `dns_name` with the value configured in the `dns_domain` parameter in `neutron.conf`, as explained previously.
- The `dns_assignment` attribute also shows that the ports `hostname` in the Networking service internal DNS is `my-vm`.

- Instead of having the Compute service create the port for the instance, the user might have created it and assigned a value to its `dns_name` attribute. In this case, the value assigned to the `dns_name` attribute must be equal to the value that Compute service will assign to the instances `hostname`, in this example `my-vm`. Otherwise, the instance boot will fail.

Note

When the Networking service integration with an external DNS service is enabled, a ports FQDN in the `dns_assignment` attribute will not be calculated as described above in some well defined cases. For a description of these cases please see *The ports dns_assignment attribute with use case 3*.

8.2.9 DNS Integration with an External Service

This page serves as a guide for how to use the DNS integration functionality of the Networking service with an external DNSaaS (DNS-as-a-Service).

As a prerequisite this needs the internal DNS functionality offered by the Networking service to be enabled, see *DNS Integration*.

Configuring OpenStack Networking for integration with an external DNS service

The first step to configure the integration with an external DNS service is to enable the functionality described in *The Networking service internal DNS resolution*. Once this is done, the user has to take the following steps and restart `neutron-server`.

1. Edit the `[default]` section of `/etc/neutron/neutron.conf` and specify the external DNS service driver to be used in parameter `external_dns_driver`. The valid options are defined in namespace `neutron.services.external_dns_drivers`. The following example shows how to set up the driver for the OpenStack DNS service:

```
external_dns_driver = designate
```

2. If the OpenStack DNS service is the target external DNS, the `[designate]` section of `/etc/neutron/neutron.conf` must define the following parameters:

   - `url`: the OpenStack DNS service public endpoint URL. Note that this must always be the versioned endpoint currently.
   - `auth_type`: the authorization plugin to use. Usually this should be `password`, see https://docs.openstack.org/keystoneauth/latest/authentication-plugins.html for other options.
   - `auth_url`: the Identity service authorization endpoint url. This endpoint will be used by the Networking service to authenticate as an user to create and update reverse lookup (PTR) zones.
   - `username`: the username to be used by the Networking service to create and update reverse lookup (PTR) zones.
   - `password`: the password of the user to be used by the Networking service to create and update reverse lookup (PTR) zones.
   - `project_name`: the name of the project to be used by the Networking service to create and update reverse lookup (PTR) zones.
   - `project_domain_name`: the name of the domain for the project to be used by the Networking service to create and update reverse lookup (PTR) zones.

- `user_domain_name`: the name of the domain for the user to be used by the Networking service to create and update reverse lookup (PTR) zones.
- `region_name`: the name of the region to be used by the Networking service to create and update reverse lookup (PTR) zones.
- `allow_reverse_dns_lookup`: a boolean value specifying whether to enable or not the creation of reverse lookup (PTR) records.
- `ipv4_ptr_zone_prefix_size`: the size in bits of the prefix for the IPv4 reverse lookup (PTR) zones.
- `ipv6_ptr_zone_prefix_size`: the size in bits of the prefix for the IPv6 reverse lookup (PTR) zones.
- `ptr_zone_email`: the email address to use when creating new reverse lookup (PTR) zones. The default is `admin@<dns_domain>` where `<dns_domain>` is the domain for the first record being created in that zone.
- `insecure`: whether to disable SSL certificate validation. By default, certificates are validated.
- `cafile`: Path to a valid Certificate Authority (CA) certificate. Optional, the system CAs are used as default.

The following is an example:

```
[designate]
url = http://192.0.2.240:9001/v2
auth_type = password
auth_url = http://192.0.2.240:5000
username = neutron
password = PASSWORD
project_name = service
project_domain_name = Default
user_domain_name = Default
allow_reverse_dns_lookup = True
ipv4_ptr_zone_prefix_size = 24
ipv6_ptr_zone_prefix_size = 116
ptr_zone_email = admin@example.org
cafile = /etc/ssl/certs/my_ca_cert
```

Once the `neutron-server` has been configured and restarted, users will have functionality that covers three use cases, described in the following sections. In each of the use cases described below:

- The examples assume the OpenStack DNS service as the external DNS.
- A, AAAA and PTR records will be created in the DNS service.
- Before executing any of the use cases, the user must create in the DNS service under their project a DNS zone where the A and AAAA records will be created. For the description of the use cases below, it is assumed the zone `example.org.` was created previously.
- The PTR records will be created in zones owned by the project specified for `project_name` above.

Use case 1: Floating IPs are published with associated port DNS attributes

In this use case, the address of a floating IP is published in the external DNS service in conjunction with the dns_name of its associated port and the dns_domain of the ports network. The steps to execute in this use case are the following:

1. Assign a valid domain name to the networks dns_domain attribute. This name must end with a period (.).
2. Boot an instance or alternatively, create a port specifying a valid value to its dns_name attribute. If the port is going to be used for an instance boot, the value assigned to dns_name must be equal to the hostname that the Compute service will assign to the instance. Otherwise, the boot will fail.
3. Create a floating IP and associate it to the port.

Following is an example of these steps:

```
$ openstack network set --dns-domain example.org. 38c5e950-b450-4c30-83d4-
↪ee181c28aad3

$ openstack network show 38c5e950-b450-4c30-83d4-ee181c28aad3
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        | nova                                 |
| created_at                | 2016-05-04T19:27:34Z                 |
| description               |                                      |
| dns_domain                | example.org.                         |
| id                        | 38c5e950-b450-4c30-83d4-ee181c28aad3 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | None                                 |
| is_vlan_transparent       | None                                 |
| mtu                       | 1450                                 |
| name                      | private                              |
| port_security_enabled     | True                                 |
| project_id                | d5660cb1e6934612a01b4fb2fb630725     |
| provider:network_type     | vlan                                 |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 24                                   |
| qos_policy_id             | None                                 |
| revision_number           | 1                                    |
| router:external           | Internal                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   | 43414c53-62ae-49bc-aa6c-c9dd7705818a |
|                           | 5b9282a1-0be1-4ade-b478-7868ad2a16ff |
| tags                      |                                      |
| updated_at                | 2016-05-04T19:27:34Z                 |
```

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```
+---------------------------+--------------------------------------+

$ openstack server create --image cirros --flavor 42 \
  --nic net-id=38c5e950-b450-4c30-83d4-ee181c28aad3 my_vm
+--------------------------------------+------------------------------------
↪-------------------------+
| Field                                | Value                              ␣
↪                         |
+--------------------------------------+------------------------------------
↪-------------------------+
| OS-DCF:diskConfig                    | MANUAL                             ␣
↪                         |
| OS-EXT-AZ:availability_zone          |                                    ␣
↪                         |
| OS-EXT-STS:power_state               | 0                                  ␣
↪                         |
| OS-EXT-STS:task_state                | scheduling                         ␣
↪                         |
| OS-EXT-STS:vm_state                  | building                           ␣
↪                         |
| OS-SRV-USG:launched_at               | -                                  ␣
↪                         |
| OS-SRV-USG:terminated_at             | -                                  ␣
↪                         |
| accessIPv4                           |                                    ␣
↪                         |
| accessIPv6                           |                                    ␣
↪                         |
| adminPass                            | oTLQLR3Kezmt                       ␣
↪                         |
| config_drive                         |                                    ␣
↪                         |
| created                              | 2016-02-15T19:27:34Z               ␣
↪                         |
| flavor                               | m1.nano (42)                       ␣
↪                         |
| hostId                               |                                    ␣
↪                         |
| id                                   | 43f328bb-b2d1-4cf1-a36f-3b2593397cb1␣
↪                         |
| image                                | cirros-0.3.5-x86_64-uec (b9d981eb-
↪d21c-4ce2-9dbc-dd38f3d9015f) |
| key_name                             | -                                  ␣
↪                         |
| locked                               | False                              ␣
↪                         |
| metadata                             | {}                                 ␣
↪                         |
| name                                 | my_vm                              ␣
```

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```
↪                          |
| os-extended-volumes:volumes_attached | []                                   ␣
↪                          |
| progress                             | 0                                    ␣
↪                          |
| security_groups                      | default                              ␣
↪                          |
| status                               | BUILD                                ␣
↪                          |
| tenant_id                            | d5660cb1e6934612a01b4fb2fb630725     ␣
↪                          |
| updated                              | 2016-02-15T19:27:34Z                 ␣
↪                          |
| user_id                              | 8bb6e578cba24e7db9d3810633124525     ␣
↪                          |
+--------------------------------------+--------------------------------------
↪-------------------------+

$ openstack server list
+--------------------------------------+-------+--------+---------------------
↪--------------------------------------+--------+---------+
| ID                                   | Name  | Status | Networks            ␣
↪                                      | Image  | Flavor  |
+--------------------------------------+-------+--------+---------------------
↪--------------------------------------+--------+---------+
| 43f328bb-b2d1-4cf1-a36f-3b2593397cb1 | my_vm | ACTIVE |␣
↪private=fda4:653e:71b0:0:f816:3eff:fe16:b5f2, 192.0.2.15 | cirros | m1.nano␣
↪|
+--------------------------------------+-------+--------+---------------------
↪--------------------------------------+--------+---------+

$ openstack port list --device-id 43f328bb-b2d1-4cf1-a36f-3b2593397cb1
+--------------------------------------+------+-------------------+-----------
↪-----------------------------------------------------------------------------
↪---------------------+--------+
| ID                                   | Name | MAC Address       | Fixed IP␣
↪Addresses                                                                    ␣
↪                     | Status |
+--------------------------------------+------+-------------------+-----------
↪-----------------------------------------------------------------------------
↪---------------------+--------+
| da0b1f75-c895-460f-9fc1-4d6ec84cf85f |      | fa:16:3e:16:b5:f2 | ip_
↪address='192.0.2.15', subnet_id='5b9282a1-0be1-4ade-b478-7868ad2a16ff'      ␣
↪                           | ACTIVE |
|                                      |      |                   | ip_
↪address='fda4:653e:71b0:0:f816:3eff:fe16:b5f2', subnet_id='43414c53-62ae-
↪49bc-aa6c-c9dd7705818a'          |        |
+--------------------------------------+------+-------------------+-----------
↪-----------------------------------------------------------------------------
```

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```
↪---------------------+--------+

$ openstack port show da0b1f75-c895-460f-9fc1-4d6ec84cf85f
+-----------------------+-------------------------------------------------------
↪------------------------------------------------------+
| Field                 | Value                                                 ␣
↪                                                     |
+-----------------------+-------------------------------------------------------
↪------------------------------------------------------+
| admin_state_up        | UP                                                    ␣
↪                                                     |
| allowed_address_pairs |                                                       ␣
↪                                                     |
| binding_host_id       | vultr.guest                                           ␣
↪                                                     |
| binding_profile       |                                                       ␣
↪                                                     |
| binding_vif_details   | datapath_type='system', ovs_hybrid_plug='True',␣
↪port_filter='True'                                           |
| binding_vif_type      | ovs                                                   ␣
↪                                                     |
| binding_vnic_type     | normal                                                ␣
↪                                                     |
| created_at            | 2016-02-15T19:27:34Z                                  ␣
↪                                                     |
| data_plane_status     | None                                                  ␣
↪                                                     |
| description           |                                                       ␣
↪                                                     |
| device_id             | 43f328bb-b2d1-4cf1-a36f-3b2593397cb1                  ␣
↪                                                     |
| device_owner          | compute:None                                          ␣
↪                                                     |
| dns_assignment        | fqdn='my-vm.example.org.', hostname='my-vm', ip_
↪address='192.0.2.15'                                         |
|                       | fqdn='my-vm.example.org.', hostname='my-vm', ip_
↪address='fda4:653e:71b0:0:f816:3eff:fe16:b5f2'               |
| dns_domain            | example.org.                                          ␣
↪                                                     |
| dns_name              | my-vm                                                 ␣
↪                                                     |
| extra_dhcp_opts       |                                                       ␣
↪                                                     |
| fixed_ips             | ip_address='192.0.2.15', subnet_id='5b9282a1-0be1-
↪4ade-b478-7868ad2a16ff'                                     |
|                       | ip_address='fda4:653e:71b0:0:f816:3eff:fe16:b5f2',␣
↪subnet_id='43414c53-62ae-49bc-aa6c-c9dd7705818a'           |
| id                    | da0b1f75-c895-460f-9fc1-4d6ec84cf85f                  ␣
↪                                                     |
```

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```
| mac_address           | fa:16:3e:16:b5:f2                                                    ␣
↪                                                                   |
| name                  |                                                                      ␣
↪                                                                   |
| network_id            | 38c5e950-b450-4c30-83d4-ee181c28aad3                                 ␣
↪                                                                   |
| port_security_enabled | True                                                                 ␣
↪                                                                   |
| project_id            | d5660cb1e6934612a01b4fb2fb630725                                     ␣
↪                                                                   |
| qos_policy_id         | None                                                                 ␣
↪                                                                   |
| revision_number       | 1                                                                    ␣
↪                                                                   |
| security_group_ids    | 1f0ddd73-7e3c-48bd-a64c-7ded4fe0e635                                 ␣
↪                                                                   |
| status                | ACTIVE                                                               ␣
↪                                                                   |
| tags                  |                                                                      ␣
↪                                                                   |
| trunk_details         | None                                                                 ␣
↪                                                                   |
| updated_at            | 2016-02-15T19:27:34Z                                                 ␣
↪                                                                   |
+-----------------------+-----------------------------------------------------------------------
↪--------------------------------------------------------+

$ openstack recordset list example.org.
+--------------------------------------+---------------------+------+----------
↪------------------------------------------------------------+---------+-----
↪---+
| id                                   | name                | type | records ␣
↪                                                          | status |␣
↪action |
+--------------------------------------+---------------------+------+----------
↪------------------------------------------------------------+---------+-----
↪---+
| a5fe696d-203f-4018-b0d8-590221adb513 | example.org.        | NS   | ns1.
↪devstack.org.                                                | ACTIVE␣
↪| NONE   |
| e7c05a5d-83a0-4fe5-8bd5-ab058a3326aa | example.org.        | SOA  | ns1.
↪devstack.org. malavall.us.ibm.com. 1513767794 3532 600 86400 3600 | ACTIVE␣
↪| NONE   |
+--------------------------------------+---------------------+------+----------
↪------------------------------------------------------------+---------+-----
↪---+

$ openstack floating ip create 41fa3995-9e4a-4cd9-bb51-3e5424f2ff2a \
  --port da0b1f75-c895-460f-9fc1-4d6ec84cf85f
```

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```
+---------------------+--------------------------------------+
| Field               | Value                                |
+---------------------+--------------------------------------+
| created_at          | 2016-02-15T20:27:34Z                 |
| description         |                                      |
| dns_domain          |                                      |
| dns_name            |                                      |
| fixed_ip_address    | 192.0.2.15                           |
| floating_ip_address | 198.51.100.4                         |
| floating_network_id | 41fa3995-9e4a-4cd9-bb51-3e5424f2ff2a |
| id                  | e78f6eb1-a35f-4a90-941d-87c888d5fcc7 |
| name                | 198.51.100.4                         |
| port_id             | da0b1f75-c895-460f-9fc1-4d6ec84cf85f |
| project_id          | d5660cb1e6934612a01b4fb2fb630725     |
| qos_policy_id       | None                                 |
| revision_number     | 1                                    |
| router_id           | 970ebe83-c4a3-4642-810e-43ab7b0c2b5f |
| status              | DOWN                                 |
| subnet_id           | None                                 |
| tags                | []                                   |
| updated_at          | 2016-02-15T20:27:34Z                 |
+---------------------+--------------------------------------+

$ openstack recordset list example.org.
+--------------------------------------+--------------------+------+----------
↪------------------------------------------------------------+--------+-----
↪---+
| id                                   | name               | type | records ␣
↪                                                            | status |␣
↪action |
+--------------------------------------+--------------------+------+----------
↪------------------------------------------------------------+--------+-----
↪---+
| a5fe696d-203f-4018-b0d8-590221adb513 | example.org.       | NS   | ns1.
↪devstack.org.                                                | ACTIVE␣
↪| NONE   |
| e7c05a5d-83a0-4fe5-8bd5-ab058a3326aa | example.org.       | SOA  | ns1.
↪devstack.org. malavall.us.ibm.com. 1513768814 3532 600 86400 3600 | ACTIVE␣
↪| NONE   |
| 5ff53fd0-3746-48da-b9c9-77ed3004ec67 | my-vm.example.org. | A    | 198.51.
↪100.4                                                      | ACTIVE |␣
↪NONE   |
+--------------------------------------+--------------------+------+----------
↪------------------------------------------------------------+--------+-----
↪---+
```

In this example, notice that the data is published in the DNS service when the floating IP is associated to the port.

Following are the PTR records created for this example. Note that for IPv4, the value of `ipv4_ptr_zone_prefix_size` is 24. Also, since the zone for the PTR records is created in the service project, you need to use admin credentials in order to be able to view it.

```
$ openstack recordset list --all-projects 100.51.198.in-addr.arpa.
+--------------------------------------+----------------------------------+---
↪-----------------------+------+----------------------------------------
↪-------------------------+--------+--------+
| id                                   | project_id                       |␣
↪name                    | type | data                                    ␣
↪                         | status | action |
+--------------------------------------+----------------------------------+---
↪-------------------------+-----------------------------------------------
↪-------------------------+--------+--------+
| 2dd0b894-25fa-4563-9d32-9f13bd67f329 | 07224d17d76d42499a38f00ba4339710 |␣
↪100.51.198.in-addr.arpa.    | NS   | ns1.devstack.org.                      ␣
↪                            | ACTIVE | NONE   |
| 47b920f1-5eff-4dfa-9616-7cb5b7cb7ca6 | 07224d17d76d42499a38f00ba4339710 |␣
↪100.51.198.in-addr.arpa.    | SOA  | ns1.devstack.org. admin.example.org.␣
↪1455564862 3600 600 86400 3600 | ACTIVE | NONE   |
| fb1edf42-abba-410c-8397-831f45fd0cd7 | 07224d17d76d42499a38f00ba4339710 | 4.
↪100.51.198.in-addr.arpa. | PTR  | my-vm.example.org.                     ␣
↪                         | ACTIVE | NONE   |
+--------------------------------------+----------------------------------+---
↪-----------------------+------+----------------------------------------
↪-------------------------+--------+--------+
```

Use case 2: Floating IPs are published in the external DNS service

In this use case, the user assigns dns_name and dns_domain attributes to a floating IP when it is created. The floating IP data becomes visible in the external DNS service as soon as it is created. The floating IP can be associated with a port on creation or later on. The following example shows a user booting an instance and then creating a floating IP associated to the port allocated for the instance:

```
$ openstack network show 38c5e950-b450-4c30-83d4-ee181c28aad3
+---------------------------+----------------------------------------------
↪-------------------------+
| Field                     | Value                                        ␣
↪                          |
+---------------------------+----------------------------------------------
↪-------------------------+
| admin_state_up            | UP                                           ␣
↪                          |
| availability_zone_hints   |                                              ␣
↪                          |
| availability_zones        | nova                                         ␣
↪                          |
| created_at                | 2016-05-04T19:27:34Z                         ␣
↪                          |
| description               |                                              ␣
↪                          |
| dns_domain                | example.org.                                 ␣
↪                          |
```

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```
| id                        | 38c5e950-b450-4c30-83d4-ee181c28aad3                       ␣
↪                         |
| ipv4_address_scope        | None                                                       ␣
↪                         |
| ipv6_address_scope        | None                                                       ␣
↪                         |
| is_default                | None                                                       ␣
↪                         |
| is_vlan_transparent       | None                                                       ␣
↪                         |
| mtu                       | 1450                                                       ␣
↪                         |
| name                      | private                                                    ␣
↪                         |
| port_security_enabled     | True                                                       ␣
↪                         |
| project_id                | d5660cb1e6934612a01b4fb2fb630725                           ␣
↪                         |
| provider:network_type     | vlan                                                       ␣
↪                         |
| provider:physical_network | None                                                       ␣
↪                         |
| provider:segmentation_id  | 24                                                         ␣
↪                         |
| qos_policy_id             | None                                                       ␣
↪                         |
| revision_number           | 1                                                          ␣
↪                         |
| router:external           | Internal                                                   ␣
↪                         |
| segments                  | None                                                       ␣
↪                         |
| shared                    | False                                                      ␣
↪                         |
| status                    | ACTIVE                                                     ␣
↪                         |
| subnets                   | 43414c53-62ae-49bc-aa6c-c9dd7705818a, 5b9282a1-
↪0be1-4ade-b478-7868ad2a16ff |
| tags                      |                                                            ␣
↪                         |
| updated_at                | 2016-05-04T19:27:34Z                                       ␣
↪                         |
+---------------------------+-------------------------------------------------
↪--------------------------+

$ openstack server create --image cirros --flavor 42 \
  --nic net-id=38c5e950-b450-4c30-83d4-ee181c28aad3 my_vm
+--------------------------------------+---------------------------------------
↪--------------------------+
```

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```
| Field                                | Value                                    ␣
↪                          |
+--------------------------------------+----------------------------------------
↪--------------------------+
| OS-DCF:diskConfig                    | MANUAL                                   ␣
↪                          |
| OS-EXT-AZ:availability_zone          |                                          ␣
↪                          |
| OS-EXT-STS:power_state               | 0                                        ␣
↪                          |
| OS-EXT-STS:task_state                | scheduling                               ␣
↪                          |
| OS-EXT-STS:vm_state                  | building                                 ␣
↪                          |
| OS-SRV-USG:launched_at               | -                                        ␣
↪                          |
| OS-SRV-USG:terminated_at             | -                                        ␣
↪                          |
| accessIPv4                           |                                          ␣
↪                          |
| accessIPv6                           |                                          ␣
↪                          |
| adminPass                            | HLXGznYqXM4J                             ␣
↪                          |
| config_drive                         |                                          ␣
↪                          |
| created                              | 2016-02-15T19:42:44Z                     ␣
↪                          |
| flavor                               | m1.nano (42)                             ␣
↪                          |
| hostId                               |                                          ␣
↪                          |
| id                                   | 71fb4ac8-eed8-4644-8113-0641962bb125␣
↪                          |
| image                                | cirros-0.3.5-x86_64-uec (b9d981eb-
↪d21c-4ce2-9dbc-dd38f3d9015f) |
| key_name                             | -                                        ␣
↪                          |
| locked                               | False                                    ␣
↪                          |
| metadata                             | {}                                       ␣
↪                          |
| name                                 | my_vm                                    ␣
↪                          |
| os-extended-volumes:volumes_attached | []                                       ␣
↪                          |
| progress                             | 0                                        ␣
↪                          |
| security_groups                      | default                                  ␣
```

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```
↪                         |
| status                             | BUILD                                     ␣
↪                         |
| tenant_id                          | d5660cb1e6934612a01b4fb2fb630725          ␣
↪                         |
| updated                            | 2016-02-15T19:42:44Z                      ␣
↪                         |
| user_id                            | 8bb6e578cba24e7db9d3810633124525          ␣
↪                         |
+------------------------------------+-------------------------------------------
↪-------------------------+

$ openstack server list
+--------------------------------------+-------+--------+-------------------------
↪------------------------------------+--------+---------+
| ID                                   | Name  | Status | Networks                 ␣
↪                                    | Image  | Flavor  |
+--------------------------------------+-------+--------+-------------------------
↪------------------------------------+--------+---------+
| 71fb4ac8-eed8-4644-8113-0641962bb125 | my_vm | ACTIVE |␣
↪private=fda4:653e:71b0:0:f816:3eff:fe24:8614, 192.0.2.16 | cirros | m1.nano␣
↪|
+--------------------------------------+-------+--------+-------------------------
↪------------------------------------+--------+---------+

$ openstack port list --device-id 71fb4ac8-eed8-4644-8113-0641962bb125
+--------------------------------------+------+-------------------+-------------
↪--------------------------------------------------------------------------------
↪-------------+--------+
| ID                                   | Name | MAC Address       | Fixed IP␣
↪Addresses                                                                       ␣
↪             | Status |
+--------------------------------------+------+-------------------+-------------
↪--------------------------------------------------------------------------------
↪-------------+--------+
| 1e7033fb-8e9d-458b-89ed-8312cafcfdcb |      | fa:16:3e:24:86:14 | ip_
↪address='192.0.2.16', subnet_id='5b9282a1-0be1-4ade-b478-7868ad2a16ff'         ␣
↪                       | ACTIVE |
|                                      |      |                   | ip_
↪address='fda4:653e:71b0:0:f816:3eff:fe24:8614', subnet_id='43414c53-62ae-
↪49bc-aa6c-c9dd7705818a' |        |
+--------------------------------------+------+-------------------+-------------
↪--------------------------------------------------------------------------------
↪-------------+--------+

$ openstack port show 1e7033fb-8e9d-458b-89ed-8312cafcfdcb
+-----------------------+-------------------------------------------------------
↪-------------------------------------------------------+
| Field                 | Value                                                    ␣
```

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```
↪                                                         |
+-----------------------+-----------------------------------------------------------
↪-------------------------------------------------------+
| admin_state_up        | UP                                                       ␣
↪                                                         |
| allowed_address_pairs |                                                          ␣
↪                                                         |
| binding_host_id       | vultr.guest                                              ␣
↪                                                         |
| binding_profile       |                                                          ␣
↪                                                         |
| binding_vif_details   | datapath_type='system', ovs_hybrid_plug='True',␣
↪port_filter='True'                                           |
| binding_vif_type      | ovs                                                      ␣
↪                                                         |
| binding_vnic_type     | normal                                                   ␣
↪                                                         |
| created_at            | 2016-02-15T19:42:44Z                                     ␣
↪                                                         |
| data_plane_status     | None                                                     ␣
↪                                                         |
| description           |                                                          ␣
↪                                                         |
| device_id             | 71fb4ac8-eed8-4644-8113-0641962bb125                     ␣
↪                                                         |
| device_owner          | compute:None                                             ␣
↪                                                         |
| dns_assignment        | fqdn='my-vm.example.org.', hostname='my-vm', ip_
↪address='192.0.2.16'                                         |
|                       | fqdn='my-vm.example.org.', hostname='my-vm', ip_
↪address='fda4:653e:71b0:0:f816:3eff:fe24:8614'               |
| dns_domain            | example.org.                                             ␣
↪                                                         |
| dns_name              | my-vm                                                    ␣
↪                                                         |
| extra_dhcp_opts       |                                                          ␣
↪                                                         |
| fixed_ips             | ip_address='192.0.2.16', subnet_id='5b9282a1-0be1-
↪4ade-b478-7868ad2a16ff'                                   |
|                       | ip_address='fda4:653e:71b0:0:f816:3eff:fe24:8614',␣
↪subnet_id='43414c53-62ae-49bc-aa6c-c9dd7705818a'          |
| id                    | 1e7033fb-8e9d-458b-89ed-8312cafcfdcb                     ␣
↪                                                         |
| mac_address           | fa:16:3e:24:86:14                                        ␣
↪                                                         |
| name                  |                                                          ␣
↪                                                         |
| network_id            | 38c5e950-b450-4c30-83d4-ee181c28aad3                     ␣
↪                                                         |
```

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```
| port_security_enabled | True                                          ␣
↪                                                                       |
| project_id            | d5660cb1e6934612a01b4fb2fb630725              ␣
↪                                                                       |
| qos_policy_id         | None                                          ␣
↪                                                                       |
| revision_number       | 1                                             ␣
↪                                                                       |
| security_group_ids    | 1f0ddd73-7e3c-48bd-a64c-7ded4fe0e635          ␣
↪                                                                       |
| status                | ACTIVE                                        ␣
↪                                                                       |
| tags                  |                                               ␣
↪                                                                       |
| trunk_details         | None                                          ␣
↪                                                                       |
| updated_at            | 2016-02-15T19:42:44Z                          ␣
↪                                                                       |
+-----------------------+-----------------------------------------------
↪-------------------------------------------------------+

$ openstack recordset list example.org.
+--------------------------------------+--------------------+------+---------
↪-----------------------------------------------------------+--------+-----
↪---+
| id                                   | name               | type | records ␣
↪                                                           | status |␣
↪action |
+--------------------------------------+--------------------+------+---------
↪-----------------------------------------------------------+--------+-----
↪---+
| 56ca0b88-e343-4c98-8faa-19746e169baf | example.org.       | NS   | ns1.
↪devstack.org.                                                      | ACTIVE␣
↪| NONE   |
| 10a36008-6ecf-47c3-b321-05652a929b04 | example.org.       | SOA  | ns1.
↪devstack.org. malavall.us.ibm.com. 1455565110 3532 600 86400 3600 | ACTIVE␣
↪| NONE   |
+--------------------------------------+--------------------+------+---------
↪-----------------------------------------------------------+--------+-----
↪---+

$ openstack floating ip create --dns-domain example.org. --dns-name my-
↪floatingip 41fa3995-9e4a-4cd9-bb51-3e5424f2ff2a
+---------------------+--------------------------------------+
| Field               | Value                                |
+---------------------+--------------------------------------+
| created_at          | 2019-06-12T15:54:45Z                 |
| description         |                                      |
| dns_domain          | example.org.                         |
```

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```
| dns_name            | my-floatingip                        |
| fixed_ip_address    | None                                 |
| floating_ip_address | 198.51.100.5                         |
| floating_network_id | 41fa3995-9e4a-4cd9-bb51-3e5424f2ff2a |
| id                  | 3ae82f53-3349-4aac-810e-ed2a8f6374b8 |
| name                | 198.51.100.53                        |
| port_details        | None                                 |
| port_id             | None                                 |
| project_id          | d5660cb1e6934612a01b4fb2fb630725     |
| qos_policy_id       | None                                 |
| revision_number     | 0                                    |
| router_id           | None                                 |
| status              | DOWN                                 |
| subnet_id           | None                                 |
| tags                | []                                   |
| updated_at          | 2019-06-12T15:54:45Z                 |
+---------------------+--------------------------------------+

$ openstack recordset list example.org.
+--------------------------------------+----------------------------+------+--
↪---------------------------------------------------------------------+------
↪--+--------+
| id                                   | name                       | type |␣
↪records                                                               |␣
↪status | action |
+--------------------------------------+----------------------------+------+--
↪---------------------------------------------------------------------+------
↪--+--------+
| 56ca0b88-e343-4c98-8faa-19746e169baf | example.org.               | NS   |␣
↪ns1.devstack.org.                                                     |␣
↪ACTIVE | NONE   |
| 10a36008-6ecf-47c3-b321-05652a929b04 | example.org.               | SOA  |␣
↪ns1.devstack.org. malavall.us.ibm.com. 1455565110 3532 600 86400 3600 |␣
↪ACTIVE | NONE   |
| 8884c56f-3ef5-446e-ae4d-8053cc8bc2b4 | my-floatingip.example.org. | A    |␣
↪198.51.100.53                                                         |␣
↪ACTIVE | NONE   |
+--------------------------------------+----------------------------+------+--
↪---------------------------------------------------------------------+------
↪--+--------+
```

Note that in this use case:

- The `dns_name` and `dns_domain` attributes of a floating IP must be specified together on creation. They cannot be assigned to the floating IP separately and they cannot be changed after the floating IP has been created.
- The `dns_name` and `dns_domain` of a floating IP have precedence, for purposes of being published in the external DNS service, over the `dns_name` of its associated port and the `dns_domain` of the ports network, whether they are specified or not. Only the `dns_name` and the `dns_domain` of the floating IP are published in the external DNS service.

Following are the PTR records created for this example. Note that for IPv4, the value of `ipv4_ptr_zone_prefix_size` is 24. Also, since the zone for the PTR records is created in the project specified in the `[designate]` section in the config above, usually the `service` project, you need to use admin credentials in order to be able to view it.

```
$ openstack recordset list --all-projects 100.51.198.in-addr.arpa.
+--------------------------------------+----------------------------------+---
↪-------------------------+------+--------------------------------------------
↪-------------------------+--------+--------+
| id                                   | project_id                       |␣
↪name                      | type | data                                      ␣
↪                          | status | action |
+--------------------------------------+----------------------------------+---
↪-------------------------------+---------------------------------------------
↪-------------------------+--------+--------+
| 2dd0b894-25fa-4563-9d32-9f13bd67f329 | 07224d17d76d42499a38f00ba4339710 |␣
↪100.51.198.in-addr.arpa.   | NS   | ns1.devstack.org.                        ␣
↪                           | ACTIVE | NONE   |
| 47b920f1-5eff-4dfa-9616-7cb5b7cb7ca6 | 07224d17d76d42499a38f00ba4339710 |␣
↪100.51.198.in-addr.arpa.   | SOA  | ns1.devstack.org. admin.example.org.␣
↪1455564862 3600 600 86400 3600 | ACTIVE | NONE   |
| 589a0171-e77a-4ab6-ba6e-23114f2b9366 | 07224d17d76d42499a38f00ba4339710 | 5.
↪100.51.198.in-addr.arpa. | PTR  | my-floatingip.example.org.                ␣
↪                         | ACTIVE | NONE   |
+--------------------------------------+----------------------------------+---
↪-------------------------+------+--------------------------------------------
↪-------------------------+--------+--------+
```

Use case 3: Ports are published directly in the external DNS service

In this case, the user is creating ports or booting instances on a network that is accessible externally. There are multiple possible scenarios here depending on which of the DNS extensions is enabled in the Neutron configuration. These extensions are described in the following in descending order of priority.

Use case 3a: The `subnet_dns_publish_fixed_ip` extension

When the `subnet_dns_publish_fixed_ip` extension is enabled, it is possible to make a selection per subnet whether DNS records should be published for fixed IPs that are assigned to ports from that subnet. This happens via the `dns_publish_fixed_ips` attribute that this extension adds to the definition of the subnet resource. It is a boolean flag with a default value of `False` but it can be set to `True` when creating or updating subnets. When the flag is `True`, all fixed IPs from this subnet are published in the external DNS service, while at the same time IPs from other subnets having the flag set to `False` are not published, even if they otherwise would meet the criteria from the other use cases below.

A typical scenario for this use case is a dual stack deployment, where a tenant network would be configured with both an IPv4 and an IPv6 subnet. The IPv4 subnet will usually be using some RFC1918 address space and being NATted towards the outside on the attached router, therefore the fixed IPs from this subnet will not be globally routed and they also should not be published in the DNS service. (One can still bind floating IPs to these fixed IPs and DNS records for those floating IPs can still be published as described above in use cases 1 and 2).

But for the IPv6 subnet, no NAT will happen, instead the subnet will be configured with some globally routable prefix and thus the user will want to publish DNS records for fixed IPs from this subnet. This can be achieved by setting the `dns_publish_fixed_ips` attribute for the IPv6 subnet to `True` while leaving the flag set to `False` for the IPv4 subnet. Example:

```
$ openstack network create dualstack
... output omitted ...
$ openstack subnet create --network dualstack dualstackv4 --subnet-range 192.
↪0.2.0/24
... output omitted ...
$ openstack subnet create --network dualstack dualstackv6 --protocol ipv6 --
↪subnet-range 2001:db8:42:42::/64 --dns-publish-fixed-ip
... output omitted ...
$ openstack zone create example.org. --email mail@example.org
... output omitted ...
$ openstack recordset list example.org.
+--------------------------------------+--------------+------+---------------
↪-----------------------------------------------------+--------+--------+
| id                                   | name         | type | records      ␣
↪                                                      | status | action |
+--------------------------------------+--------------+------+---------------
↪-----------------------------------------------------+--------+--------+
| 404e9846-1482-433b-8bbc-67677e587d28 | example.org. | NS   | ns1.devstack.
↪org.                                                      | ACTIVE | NONE   |
| de73576a-f9c7-4892-934c-259b77ff02c0 | example.org. | SOA  | ns1.devstack.
↪org. mail.example.org. 1575897792 3559 600 86400 3600 | ACTIVE | NONE   |
+--------------------------------------+--------------+------+---------------
↪-----------------------------------------------------+--------+--------+
$ openstack port create port1 --dns-domain example.org. --dns-name port1 --
↪network dualstack
+-------------------------+-----------------------------------------------
↪------------------------------------------------------------------------
↪-----------------------------------------+
| Field                   | Value                                         ␣
↪                                                                        ␣
↪                                         |
+-------------------------+-----------------------------------------------
↪------------------------------------------------------------------------
↪-----------------------------------------+
| admin_state_up          | UP                                            ␣
↪                                                                        ␣
↪                                         |
| allowed_address_pairs   |                                               ␣
↪                                                                        ␣
↪                                         |
| binding_host_id         | None                                          ␣
↪                                                                        ␣
↪                                         |
| binding_profile         | None                                          ␣
↪                                                                        ␣
↪                                         |
| binding_vif_details     | None                                          ␣
↪                                                                        ␣
```

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```
↪                                                  |
| binding_vif_type          | None                                                     ␣
↪                                                                                      ␣
↪                                                  |
| binding_vnic_type         | normal                                                   ␣
↪                                                                                      ␣
↪                                                  |
| created_at                | 2019-12-09T13:23:52Z                                     ␣
↪                                                                                      ␣
↪                                                  |
| data_plane_status         | None                                                     ␣
↪                                                                                      ␣
↪                                                  |
| description               |                                                          ␣
↪                                                                                      ␣
↪                                                  |
| device_id                 |                                                          ␣
↪                                                                                      ␣
↪                                                  |
| device_owner              |                                                          ␣
↪                                                                                      ␣
↪                                                  |
| dns_assignment            | fqdn='port1.openstackgate.local.', hostname='port1
↪', ip_address='192.0.2.100'                                                           ␣
↪                                                  |
|                           | fqdn='port1.openstackgate.local.', hostname='port1
↪', ip_address='2001:db8:42:42::2a2'                                                   ␣
↪                                                  |
| dns_domain                | example.org.                                             ␣
↪                                                                                      ␣
↪                                                  |
| dns_name                  | port1                                                    ␣
↪                                                                                      ␣
↪                                                  |
| extra_dhcp_opts           |                                                          ␣
↪                                                                                      ␣
↪                                                  |
| fixed_ips                 | ip_address='192.0.2.100', subnet_id='47cc9a39-
↪c88b-4082-a52c-1237c2a1d479'                                                          ␣
↪                                                      |
|                           | ip_address='2001:db8:42:42::2a2', subnet_id=
↪'f9c04195-1000-4575-a203-3c174772617f'                                                ␣
↪                                                        |
| id                        | f8bc991b-1f84-435a-a5f8-814bd8b9ae9f                     ␣
↪                                                                                      ␣
↪                                                  |
| location                  | cloud='devstack', project.domain_id='default',␣
↪project.domain_name=, project.id='86de4dab952d48f79e625b106f7a75f7',␣
↪project.name='demo', region_name='RegionOne', zone= |
```

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```
| mac_address             | fa:16:3e:13:7a:56                                    ␣
↪                                                                              ␣
↪                                     |
| name                    | port1                                                ␣
↪                                                                              ␣
↪                                     |
| network_id              | fa8118ed-b7c2-41b8-89bc-97e46f0491ac                 ␣
↪                                                                              ␣
↪                                     |
| port_security_enabled   | True                                                 ␣
↪                                                                              ␣
↪                                     |
| project_id              | 86de4dab952d48f79e625b106f7a75f7                     ␣
↪                                                                              ␣
↪                                     |
| propagate_uplink_status | None                                                 ␣
↪                                                                              ␣
↪                                     |
| qos_policy_id           | None                                                 ␣
↪                                                                              ␣
↪                                     |
| resource_request        | None                                                 ␣
↪                                                                              ␣
↪                                     |
| revision_number         | 1                                                    ␣
↪                                                                              ␣
↪                                     |
| security_group_ids      | f0b02df0-a0b9-4ce8-b067-8b61a8679e9d                 ␣
↪                                                                              ␣
↪                                     |
| status                  | DOWN                                                 ␣
↪                                                                              ␣
↪                                     |
| tags                    |                                                      ␣
↪                                                                              ␣
↪                                     |
| trunk_details           | None                                                 ␣
↪                                                                              ␣
↪                                     |
| updated_at              | 2019-12-09T13:23:53Z                                 ␣
↪                                                                              ␣
↪                                     |
+-------------------------+----------------------------------------------------
↪----------------------------------------------------------------------------
↪------------------------------------------+
$ openstack recordset list example.org.
+-----------------------------------+------------------+------+---------
↪-----------------------------------------------------------+--------+--------
↪+
```

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```
| id                                   | name              | type | records ␣
↪                                                         | status | action␣
↪|
+--------------------------------------+-------------------+------+----------
↪---------------------------------------------------------+--------+--------
↪+
| 404e9846-1482-433b-8bbc-67677e587d28 | example.org.      | NS   | ns1.
↪devstack.org.                                            | ACTIVE |␣
↪NONE   |
| de73576a-f9c7-4892-934c-259b77ff02c0 | example.org.      | SOA  | ns1.
↪devstack.org. mail.example.org. 1575897833 3559 600 86400 3600 | ACTIVE |␣
↪NONE   |
| 85ce74a5-7dd6-42d3-932c-c9a029dea05e | port1.example.org. | AAAA |␣
↪2001:db8:42:42::2a2                                              | ACTIVE␣
↪| NONE   |
+--------------------------------------+-------------------+------+----------
↪---------------------------------------------------------+--------+--------
↪+
```

Use case 3b: The `dns_domain_ports` extension

If the `dns_domain for ports` extension has been configured, the user can create a port specifying a non-blank value in its `dns_domain` attribute. If the port is created in an externally accessible network, DNS records will be published for this port:

```
$ openstack port create --network 37aaff3a-6047-45ac-bf4f-a825e56fd2b3 --dns-
↪name my-vm --dns-domain port-domain.org. test
+-------------------------+-------------------------------------------------
↪--------------------------+
| Field                   | Value                                           ␣
↪                           |
+-------------------------+-------------------------------------------------
↪--------------------------+
| admin_state_up          | UP                                              ␣
↪                           |
| allowed_address_pairs   |                                                 ␣
↪                           |
| binding_host_id         | None                                            ␣
↪                           |
| binding_profile         | None                                            ␣
↪                           |
| binding_vif_details     | None                                            ␣
↪                           |
| binding_vif_type        | None                                            ␣
↪                           |
| binding_vnic_type       | normal                                          ␣
↪                           |
| created_at              | 2019-06-12T15:43:29Z                            ␣
↪                           |
```

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```
| data_plane_status       | None                                                    ␣
↪                           |
| description             |                                                         ␣
↪                           |
| device_id               |                                                         ␣
↪                           |
| device_owner            |                                                         ␣
↪                           |
| dns_assignment          | fqdn='my-vm.example.org.', hostname='my-vm', ip_
↪address='203.0.113.9'        |
|                         | fqdn='my-vm.example.org.', hostname='my-vm', ip_
↪address='2001:db8:10::9'     |
| dns_domain              | port-domain.org.                                        ␣
↪                           |
| dns_name                | my-vm                                                   ␣
↪                           |
| extra_dhcp_opts         |                                                         ␣
↪                           |
| fixed_ips               | ip_address='203.0.113.9', subnet_id='277eca5d-
↪9869-474b-960e-6da5951d09f7'    |
|                         | ip_address='2001:db8:10::9', subnet_id='eab47748-
↪3f0a-4775-a09f-b0c24bb64bc4' |
| id                      | 57541c27-f8a9-41f1-8dde-eb10155496e6                    ␣
↪                           |
| mac_address             | fa:16:3e:55:d6:c7                                       ␣
↪                           |
| name                    | test                                                    ␣
↪                           |
| network_id              | 37aaff3a-6047-45ac-bf4f-a825e56fd2b3                    ␣
↪                           |
| port_security_enabled   | True                                                    ␣
↪                           |
| project_id              | 07b21ad4-edb6-420b-bd76-9bb4aab0d135                    ␣
↪                           |
| propagate_uplink_status | None                                                    ␣
↪                           |
| qos_policy_id           | None                                                    ␣
↪                           |
| resource_request        | None                                                    ␣
↪                           |
| revision_number         | 1                                                       ␣
↪                           |
| security_group_ids      | 82227b10-d135-4bca-b41f-63c1f2286b3e                    ␣
↪                           |
| status                  | DOWN                                                    ␣
↪                           |
| tags                    |                                                         ␣
↪                           |
| trunk_details           | None                                                    ␣
```

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```
↪                                   |
| updated_at              | 2019-06-12T15:43:29Z                                     ␣
↪                                   |
+-------------------------+-------------------------------------------------------
↪---------------------------+
```

In this case, the ports `dns_name` (`my-vm`) will be published in the `port-domain.org.` zone, as shown here:

```
$ openstack recordset list port-domain.org.
+--------------------------------------+-------------------------+------+-----
↪--------------------------------------------------------------------+--------
↪+--------+
| id                                   | name                    | type |␣
↪records                                                             |␣
↪status | action |
+--------------------------------------+-------------------------+------+-----
↪--------------------------------------------------------------------+--------
↪+--------+
| 03e5a35b-d984-4d10-942a-2de8ccb9b941 | port-domain.org.        | SOA  | ns1.
↪devstack.org. malavall.us.ibm.com. 1503272259 3549 600 86400 3600 | ACTIVE␣
↪| NONE   |
| d2dd1dfe-531d-4fea-8c0e-f5b559942ac5 | port-domain.org.        | NS   | ns1.
↪devstack.org.                                                     | ACTIVE␣
↪| NONE   |
| 67a8e83d-7e3c-4fb1-9261-0481318bb7b5 | my-vm.port-domain.org.  | A    | 203.
↪0.113.9                                                           | ACTIVE␣
↪| NONE   |
| 5a4f671c-9969-47aa-82e1-e05754021852 | my-vm.port-domain.org.  | AAAA |␣
↪2001:db8:10::9                                                      |␣
↪ACTIVE | NONE   |
+--------------------------------------+-------------------------+------+-----
↪--------------------------------------------------------------------+--------
↪+--------+
```

Note

If both the port and its network have a valid non-blank string assigned to their `dns_domain` attributes, the ports `dns_domain` takes precedence over the networks.

Note

The name assigned to the ports `dns_domain` attribute must end with a period (`.`).

Note

In the above example, the `port-domain.org.` zone must be created before Neutron can publish any port data to it.

Note

See *Configuration of the externally accessible network for use cases 3b and 3c* for detailed instructions on how to create the externally accessible network.

Use case 3c: The dns extension

If the user wants to publish a port in the external DNS service in a zone specified by the `dns_domain` attribute of the network, these are the steps to be taken:

1. Assign a valid domain name to the networks `dns_domain` attribute. This name must end with a period (`.`).
2. Boot an instance specifying the externally accessible network. Alternatively, create a port on the externally accessible network specifying a valid value to its `dns_name` attribute. If the port is going to be used for an instance boot, the value assigned to `dns_name` must be equal to the `hostname` that the Compute service will assign to the instance. Otherwise, the boot will fail.

Once these steps are executed, the ports DNS data will be published in the external DNS service. This is an example:

```
$ openstack network list
+--------------------------------------+----------+-------------------------
↪-----------------------------------------------+
| ID                                   | Name     | Subnets                 ␣
↪                                               |
+--------------------------------------+----------+-------------------------
↪-----------------------------------------------+
| 41fa3995-9e4a-4cd9-bb51-3e5424f2ff2a | public   | a67cfdf7-9d5d-406f-8a19-
↪3f38e4fc3e74, cbd8c6dc-ca81-457e-9c5d-f8ece7ef67f8   |
| 37aaff3a-6047-45ac-bf4f-a825e56fd2b3 | external | 277eca5d-9869-474b-960e-
↪6da5951d09f7, eab47748-3f0a-4775-a09f-b0c24bb64bc4   |
| bf2802a0-99a0-4e8c-91e4-107d03f158ea | my-net   | 6141b474-56cd-430f-b731-
↪71660bb79b79                                         |
| 38c5e950-b450-4c30-83d4-ee181c28aad3 | private  | 43414c53-62ae-49bc-aa6c-
↪c9dd7705818a, 5b9282a1-0be1-4ade-b478-7868ad2a16ff   |
+--------------------------------------+----------+-------------------------
↪-----------------------------------------------+

$ openstack network set --dns-domain example.org. 37aaff3a-6047-45ac-bf4f-
↪a825e56fd2b3

$ openstack network show 37aaff3a-6047-45ac-bf4f-a825e56fd2b3
+---------------------------+-----------------------------------------------
↪---------------------------+
| Field                     | Value                                         ␣
↪                           |
+---------------------------+-----------------------------------------------
```

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```
↪---------------------------+
| admin_state_up            | UP                                       ␣
↪                           |
| availability_zone_hints   |                                          ␣
↪                           |
| availability_zones        | nova                                     ␣
↪                           |
| created_at                | 2016-02-14T19:42:44Z                     ␣
↪                           |
| description               |                                          ␣
↪                           |
| dns_domain                | example.org.                             ␣
↪                           |
| id                        | 37aaff3a-6047-45ac-bf4f-a825e56fd2b3     ␣
↪                           |
| ipv4_address_scope        | None                                     ␣
↪                           |
| ipv6_address_scope        | None                                     ␣
↪                           |
| is_default                | None                                     ␣
↪                           |
| is_vlan_transparent       | None                                     ␣
↪                           |
| mtu                       | 1450                                     ␣
↪                           |
| name                      | external                                 ␣
↪                           |
| port_security_enabled     | True                                     ␣
↪                           |
| project_id                | 04fc2f83966245dba907efb783f8eab9         ␣
↪                           |
| provider:network_type     | vlan                                     ␣
↪                           |
| provider:physical_network | None                                     ␣
↪                           |
| provider:segmentation_id  | 2016                                     ␣
↪                           |
| qos_policy_id             | None                                     ␣
↪                           |
| revision_number           | 4                                        ␣
↪                           |
| router:external           | Internal                                 ␣
↪                           |
| segments                  | None                                     ␣
↪                           |
| shared                    | True                                     ␣
↪                           |
| status                    | ACTIVE                                   ␣
↪                           |
```

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```
| subnets                 | eab47748-3f0a-4775-a09f-b0c24bb64bc4, 277eca5d-
↪9869-474b-960e-6da5951d09f7 |
| tags                    |                                                ␣
↪                          |
| updated_at              | 2016-02-15T13:42:44Z                           ␣
↪                          |
+-------------------------+------------------------------------------------
↪-------------------------+

$ openstack recordset list example.org.
+--------------------------------------+--------------+------+--------------
↪-------------------------------------------------------+--------+--------+
| id                                   | name         | type | records      ␣
↪                                                        | status | action |
+--------------------------------------+--------------+------+--------------
↪-------------------------------------------------------+--------+--------+
| a5fe696d-203f-4018-b0d8-590221adb513 | example.org. | NS   | ns1.devstack.
↪org.                                                      | ACTIVE | NONE   |
| e7c05a5d-83a0-4fe5-8bd5-ab058a3326aa | example.org. | SOA  | ns1.devstack.
↪org. malavall.us.ibm.com. 1513767619 3532 600 86400 3600 | ACTIVE | NONE   |
+--------------------------------------+--------------+------+--------------
↪-------------------------------------------------------+--------+--------+

$ openstack port create --network 37aaff3a-6047-45ac-bf4f-a825e56fd2b3 --dns-
↪name my-vm test
+-----------------------+--------------------------------------------------
↪-------------------------+
| Field                 | Value                                            ␣
↪                          |
+-----------------------+--------------------------------------------------
↪-------------------------+
| admin_state_up        | UP                                               ␣
↪                          |
| allowed_address_pairs |                                                  ␣
↪                          |
| binding_host_id       |                                                  ␣
↪                          |
| binding_profile       |                                                  ␣
↪                          |
| binding_vif_details   |                                                  ␣
↪                          |
| binding_vif_type      | unbound                                          ␣
↪                          |
| binding_vnic_type     | normal                                           ␣
↪                          |
| created_at            | 2016-02-15T16:42:44Z                             ␣
↪                          |
| data_plane_status     | None                                             ␣
↪                          |
```

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```
| description           |
                                    |
| device_id             |
                                    |
| device_owner          |
                                    |
| dns_assignment        | fqdn='my-vm.example.org.', hostname='my-vm', ip_
address='203.0.113.9'           |
|                       | fqdn='my-vm.example.org.', hostname='my-vm', ip_
address='2001:db8:10::9'        |
| dns_domain            | None
                                    |
| dns_name              | my-vm
                                    |
| extra_dhcp_opts       |
                                    |
| fixed_ips             | ip_address='203.0.113.9', subnet_id='277eca5d-9869-
474b-960e-6da5951d09f7'     |
|                       | ip_address='2001:db8:10::9', subnet_id=eab47748-
3f0a-4775-a09f-b0c24bb64bc4 |
| id                    | 04be331b-dc5e-410a-9103-9c8983aeb186
                                    |
| mac_address           | fa:16:3e:0f:4b:e4
                                    |
| name                  | test
                                    |
| network_id            | 37aaff3a-6047-45ac-bf4f-a825e56fd2b3
                                    |
| port_security_enabled | True
                                    |
| project_id            | d5660cb1e6934612a01b4fb2fb630725
                                    |
| qos_policy_id         | None
                                    |
| revision_number       | 1
                                    |
| security_group_ids    | 1f0ddd73-7e3c-48bd-a64c-7ded4fe0e635
                                    |
| status                | DOWN
                                    |
| tags                  |
                                    |
| trunk_details         | None
                                    |
| updated_at            | 2016-02-15T16:42:44Z
                                    |
+-----------------------+-----------------------------------------------
--------------------------+
```

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```
$ openstack recordset list example.org.
+--------------------------------------+--------------------+------+----------
↪------------------------------------------------------------+--------+-----
↪---+
| id                                   | name               | type | records ␣
↪                                                            | status |␣
↪action |
+--------------------------------------+--------------------+------+----------
↪------------------------------------------------------------+--------+-----
↪---+
| a5fe696d-203f-4018-b0d8-590221adb513 | example.org.       | NS   | ns1.
↪devstack.org.                                               | ACTIVE␣
↪| NONE   |
| e7c05a5d-83a0-4fe5-8bd5-ab058a3326aa | example.org.       | SOA  | ns1.
↪devstack.org. malavall.us.ibm.com. 1513767794 3532 600 86400 3600 | ACTIVE␣
↪| NONE   |
| fa753ab8-bffa-400d-9ef8-d4a3b1a7ffbf | my-vm.example.org. | A    | 203.0.
↪113.9                                                      | ACTIVE |␣
↪NONE   |
| 04abf9f8-c7a3-43f6-9a55-95cee9b144a9 | my-vm.example.org. | AAAA |␣
↪2001:db8:10::9                                                      |␣
↪ACTIVE | NONE   |
+--------------------------------------+--------------------+------+----------
↪------------------------------------------------------------+--------+-----
↪---+

$ openstack server create --image cirros --flavor 42 \
  --nic port-id=04be331b-dc5e-410a-9103-9c8983aeb186 my_vm
+--------------------------------------+--------------------------------------
↪--------------------------+
| Field                                | Value                               ␣
↪                          |
+--------------------------------------+--------------------------------------
↪--------------------------+
| OS-DCF:diskConfig                    | MANUAL                              ␣
↪                          |
| OS-EXT-AZ:availability_zone          |                                     ␣
↪                          |
| OS-EXT-STS:power_state               | 0                                   ␣
↪                          |
| OS-EXT-STS:task_state                | scheduling                          ␣
↪                          |
| OS-EXT-STS:vm_state                  | building                            ␣
↪                          |
| OS-SRV-USG:launched_at               | -                                   ␣
↪                          |
| OS-SRV-USG:terminated_at             | -                                   ␣
↪                          |
| accessIPv4                           |                                     ␣
```

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```
↪                               |
| accessIPv6                           |                                                      ␣
↪                               |
| adminPass                            | TDc9EpBT3B9W                                         ␣
↪                               |
| config_drive                         |                                                      ␣
↪                               |
| created                              | 2016-02-15T19:10:43Z                                 ␣
↪                               |
| flavor                               | m1.nano (42)                                         ␣
↪                               |
| hostId                               |                                                      ␣
↪                               |
| id                                   | 62c19691-d1c7-4d7b-a88e-9cc4d95d4f41␣
↪                               |
| image                                | cirros-0.3.5-x86_64-uec (b9d981eb-
↪d21c-4ce2-9dbc-dd38f3d9015f) |
| key_name                             | -                                                    ␣
↪                               |
| locked                               | False                                                ␣
↪                               |
| metadata                             | {}                                                   ␣
↪                               |
| name                                 | my_vm                                                ␣
↪                               |
| os-extended-volumes:volumes_attached | []                                                   ␣
↪                               |
| progress                             | 0                                                    ␣
↪                               |
| security_groups                      | default                                              ␣
↪                               |
| status                               | BUILD                                                ␣
↪                               |
| tenant_id                            | d5660cb1e6934612a01b4fb2fb630725                     ␣
↪                               |
| updated                              | 2016-02-15T19:10:43Z                                 ␣
↪                               |
| user_id                              | 8bb6e578cba24e7db9d3810633124525                     ␣
↪                               |
+--------------------------------------+------------------------------------------------------
↪---------------------------+

$ openstack server list
+--------------------------------------+-------+--------+-------------------------------------
↪-----------------+---------+---------+
| ID                                   | Name  | Status | Networks                            ␣
↪                 | Image   | Flavor  |
+--------------------------------------+-------+--------+-------------------------------------
↪-----------------+---------+---------+
```

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```
| 62c19691-d1c7-4d7b-a88e-9cc4d95d4f41 | my_vm | ACTIVE | external=203.0.113.
↪9, 2001:db8:10::9 | cirros | m1.nano |
+--------------------------------------+-------+--------+---------------------
↪-----------------+--------+---------+
```

In this example the port is created manually by the user and then used to boot an instance. Notice that:

- The ports data was visible in the DNS service as soon as it was created.
- See *Performance considerations* for an explanation of the potential performance impact associated with this use case.

Following are the PTR records created for this example. Note that for IPv4, the value of `ipv4_ptr_zone_prefix_size` is 24. In the case of IPv6, the value of `ipv6_ptr_zone_prefix_size` is 116.

```
$ openstack recordset list --all-projects 113.0.203.in-addr.arpa.
+--------------------------------------+----------------------------------+---
↪------------------------+------+---------------------------------------------
↪-------------------------+--------+--------+
| id                                   | project_id                       |␣
↪name                      | type | records                                     ␣
↪                         | status | action |
+--------------------------------------+----------------------------------+---
↪------------------------+------+---------------------------------------------
↪-------------------------+--------+--------+
| 32f1c05b-7c5d-4230-9088-961a0a462d28 | 07224d17d76d42499a38f00ba4339710 |␣
↪113.0.203.in-addr.arpa.   | SOA  | ns1.devstack.org. admin.example.org.␣
↪1455563035 3600 600 86400 3600 | ACTIVE | NONE   |
| 3d402c43-b215-4a75-a730-51cbb8999cb8 | 07224d17d76d42499a38f00ba4339710 |␣
↪113.0.203.in-addr.arpa.   | NS   | ns1.devstack.org.                           ␣
↪                          | ACTIVE | NONE   |
| 8e4e618c-24b0-43db-ab06-91b741a91c10 | 07224d17d76d42499a38f00ba4339710 | 9.
↪113.0.203.in-addr.arpa. | PTR  | my-vm.example.org.                            ␣
↪                        | ACTIVE | NONE   |
+--------------------------------------+----------------------------------+---
↪------------------------+------+---------------------------------------------
↪-------------------------+--------+--------+

$ openstack recordset list --all-projects  0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.
↪0.1.0.0.8.b.d.0.1.0.0.2.ip6.arpa.
+--------------------------------------+----------------------------------+---
↪---------------------------------------------------------------------------+---
↪---+-----------------------------------------------------------------------+--
↪-----+--------+
| id                                   | project_id                       |␣
↪name                                                                         |␣
↪type | records                                                               ␣
↪| status | action |
+--------------------------------------+----------------------------------+---
↪---------------------------------------------------------------------------+---
```

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```
↪---+-------------------------------------------------------------------+--
↪------+--------+
| d8923354-13eb-4bd9-914a-0a2ae5f95989 | 07224d17d76d42499a38f00ba4339710 | 0.
↪0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.1.0.0.8.b.d.0.1.0.0.2.ip6.arpa.      |␣
↪SOA  | ns1.devstack.org. admin.example.org. 1455563036 3600 600 86400 3600␣
↪| ACTIVE | NONE   |
| 72e60acd-098d-41ea-9771-5b6546c9c06f | 07224d17d76d42499a38f00ba4339710 | 0.
↪0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.1.0.0.8.b.d.0.1.0.0.2.ip6.arpa.      |␣
↪NS   | ns1.devstack.org.                                                  ␣
↪| ACTIVE | NONE   |
| 877e0215-2ddf-4d01-a7da-47f1092dfd56 | 07224d17d76d42499a38f00ba4339710 | 9.
↪0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.1.0.0.8.b.d.0.1.0.0.2.ip6.arpa. |␣
↪PTR  | my-vm.example.org.                                                 ␣
↪| ACTIVE | NONE   |
+--------------------------------------+----------------------------------+---
↪-----------------------------------------------------------------------+---
↪---+-------------------------------------------------------------------+--
↪------+--------+
```

See *Configuration of the externally accessible network for use cases 3b and 3c* for detailed instructions on how to create the externally accessible network.

Performance considerations

Only for *Use case 3: Ports are published directly in the external DNS service*, if the port binding extension is enabled in the Networking service, the Compute service will execute one additional port update operation when allocating the port for the instance during the boot process. This may have a noticeable adverse effect in the performance of the boot process that should be evaluated before adoption of this use case.

Configuration of the externally accessible network for use cases 3b and 3c

For use cases 3b and 3c, the externally accessible network must meet the following requirements:

- The network may not have attribute `router:external` set to `True`.
- The network type can be FLAT, VLAN, GRE, VXLAN or GENEVE.
- For network types VLAN, GRE, VXLAN or GENEVE, the segmentation ID must be outside the ranges assigned to project networks.

This usually implies that these use cases only work for networks specifically created for this purpose by an admin, they do not work for networks which tenants can create on their own.

The ports `dns_assignment` attribute with use case 3

The `dns_assignment` attribute is not calculated as described in *DNS Integration* when a port is created under use case 3. Instead of concatenating the ports `dns_name` with the value configured in the `dns_domain` parameter in `neutron.conf`, the `dns_name` is concatenated with the `dns_domain` of either the port or the network, depending on whether the use case is 3a, 3b or 3c. For example:

```
$ openstack network show external -c dns_domain -f value
dns-domain-1.org.
```

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```
$ cat /etc/neutron/neutron.conf | grep dns_domain
dns_domain = my-domain.org
$ openstack recordset list dns-domain-1.org.
+--------------------------------------+--------------------+------+-----------
↪-----------------------------------------------------------+--------+------
↪--+
| id                                   | name               | type | records  ␣
↪                                                            | status |␣
↪action |
+--------------------------------------+--------------------+------+-----------
↪-----------------------------------------------------------+--------+------
↪--+
| 2b3e9ea4-8035-4595-955d-ec8c55816111 | dns-domain-1.org.  | SOA  | ns1.
↪devstack.org. mlavalle.redhat.com. 1642583355 3517 600 86400 3600 | ACTIVE␣
↪| NONE   |
| 801dd911-43e6-430a-a80b-ea09af76a9a4 | dns-domain-1.org.  | NS   | ns1.
↪devstack.org.                                                | ACTIVE␣
↪| NONE   |
+--------------------------------------+--------------------+------+-----------
↪-----------------------------------------------------------+--------+------
↪--+
$ openstack port create --dns-name a-port --network external a_port
+-------------------------+-----------------------------------------------------
↪-----------------------------------------------------------------------------
↪-------------------------------------------------------------+
| Field                   | Value                                               ␣
↪                                                                             ␣
↪                                                             |
+-------------------------+-----------------------------------------------------
↪-----------------------------------------------------------------------------
↪-------------------------------------------------------------+
| admin_state_up          | UP                                                  ␣
↪                                                                             ␣
↪                                                             |
| allowed_address_pairs   |                                                     ␣
↪                                                                             ␣
↪                                                             |
| binding_host_id         | None                                                ␣
↪                                                                             ␣
↪                                                             |
| binding_profile         | None                                                ␣
↪                                                                             ␣
↪                                                             |
| binding_vif_details     | None                                                ␣
↪                                                                             ␣
↪                                                             |
| binding_vif_type        | None                                                ␣
↪                                                                             ␣
↪                                                             |
```

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```
| binding_vnic_type       | normal                                                 ␣
↪                                                                                  ␣
↪                                                                    |
| created_at              | 2022-01-19T09:15:34Z                                   ␣
↪                                                                                  ␣
↪                                                                    |
| data_plane_status       | None                                                   ␣
↪                                                                                  ␣
↪                                                                    |
| description             |                                                        ␣
↪                                                                                  ␣
↪                                                                    |
| device_id               |                                                        ␣
↪                                                                                  ␣
↪                                                                    |
| device_owner            |                                                        ␣
↪                                                                                  ␣
↪                                                                    |
| device_profile          | None                                                   ␣
↪                                                                                  ␣
↪                                                                    |
| dns_assignment          | fqdn='a-port.dns-domain-1.org.', hostname='a-port
↪', ip_address='172.31.251.113'                                                    ␣
↪                                                                     |
|                         | fqdn='a-port.dns-domain-1.org.', hostname='a-port
↪', ip_address='fd5e:7a6b:1a62::a3'                                                ␣
↪                                                                     |
| dns_domain              |                                                        ␣
↪                                                                                  ␣
↪                                                                    |
| dns_name                | a-port                                                 ␣
↪                                                                                  ␣
↪                                                                    |
| extra_dhcp_opts         |                                                        ␣
↪                                                                                  ␣
↪                                                                    |
| fixed_ips               | ip_address='172.31.251.113', subnet_id='6795a775-
↪4a76-49b0-bac6-ba8a8b62bd22'                                                      ␣
↪                                                                     |
|                         | ip_address='fd5e:7a6b:1a62::a3', subnet_id=
↪'97b719f8-2307-4162-bf08-523d9c0fc6a9'                                            ␣
↪                                                                           |
| id                      | 080f01e7-78b9-43ed-a807-1e0d59099bed                   ␣
↪                                                                                  ␣
↪                                                                    |
| ip_allocation           | None                                                   ␣
↪                                                                                  ␣
↪                                                                    |
| location                | Munch({'cloud': '', 'region_name': 'RegionOne',
```

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```
↪'zone': None, 'project': Munch({'id': 'afc55714081b4ef29f99ec128cb1fa30',
↪'name': 'demo', 'domain_id': 'default', 'domain_name': None})}) |
| mac_address           | fa:16:3e:4d:fa:33                                    ␣
↪                                                                             ␣
↪                                                              |
| name                  | a_port                                               ␣
↪                                                                             ␣
↪                                                              |
| network_id            | 2d696f10-97a4-454c-a411-ea8d4d685636                 ␣
↪                                                                             ␣
↪                                                              |
| numa_affinity_policy  | None                                                 ␣
↪                                                                             ␣
↪                                                              |
| port_security_enabled | True                                                 ␣
↪                                                                             ␣
↪                                                              |
| project_id            | afc55714081b4ef29f99ec128cb1fa30                     ␣
↪                                                                             ␣
↪                                                              |
| propagate_uplink_status | None                                               ␣
↪                                                                             ␣
↪                                                              |
| qos_network_policy_id | None                                                 ␣
↪                                                                             ␣
↪                                                              |
| qos_policy_id         | None                                                 ␣
↪                                                                             ␣
↪                                                              |
| resource_request      | None                                                 ␣
↪                                                                             ␣
↪                                                              |
| revision_number       | 1                                                    ␣
↪                                                                             ␣
↪                                                              |
| security_group_ids    | 1a350403-cad3-4a2d-b30a-73815bd96e0f                 ␣
↪                                                                             ␣
↪                                                              |
| status                | DOWN                                                 ␣
↪                                                                             ␣
↪                                                              |
| tags                  |                                                      ␣
↪                                                                             ␣
↪                                                              |
| trunk_details         | None                                                 ␣
↪                                                                             ␣
↪                                                              |
| updated_at            | 2022-01-19T09:15:34Z                                 ␣
↪                                                                             ␣
```

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```
↪                                                              |
+--------------------------+------------------------------------------------
↪---------------------------------------------------------------------------
↪-------------------------------------------------------------+
$ openstack recordset list dns-domain-1.org.
+--------------------------------------+---------------------------+------+----
↪------------------------------------------------------------------+--------
↪+--------+
| id                                   | name                      | type |␣
↪records                                                                   |␣
↪status | action |
+--------------------------------------+---------------------------+------+----
↪------------------------------------------------------------------+--------
↪+--------+
| 2b3e9ea4-8035-4595-955d-ec8c55816111 | dns-domain-1.org.         | SOA  |␣
↪ns1.devstack.org. mlavalle.redhat.com. 1642583736 3517 600 86400 3600 |␣
↪ACTIVE | NONE   |
| 801dd911-43e6-430a-a80b-ea09af76a9a4 | dns-domain-1.org.         | NS   |␣
↪ns1.devstack.org.                                                         |␣
↪ACTIVE | NONE   |
| 61597628-dafe-4aba-8e30-1d45b4e59874 | a-port.dns-domain-1.org.  | AAAA |␣
↪fd5e:7a6b:1a62::a3                                                        |␣
↪ACTIVE | NONE   |
| 83fe489f-2ebc-4911-a67e-bef688833f31 | a-port.dns-domain-1.org.  | A    |␣
↪172.31.251.113                                                            |␣
↪ACTIVE | NONE   |
+--------------------------------------+---------------------------+------+----
↪------------------------------------------------------------------+--------
↪+--------+
```

In this manner, the `FQDN` in the `dns_assignment` attribute is compatible with what is being published by the external DNS service.

8.2.10 DNS Resolution for Instances

The Networking service offers several methods to configure name resolution (DNS) for instances. Most deployments should implement case 1 or 2a. Case 2b requires security considerations to prevent leaking internal DNS information to instances.

Note

All of these setups require the configured DNS resolvers to be reachable from the virtual network in question. So unless the resolvers are located inside the virtual network itself, this implies the need for a router to be attached to that network having an external gateway configured.

Case 1: Each virtual network uses unique DNS resolver(s)

In this case, the DHCP agent offers one or more unique DNS resolvers to instances via DHCP on each virtual network. You can configure a DNS resolver when creating or updating a subnet. To configure more than one DNS resolver, repeat the option multiple times.

- Configure a DNS resolver when creating a subnet.

  ```
  $ openstack subnet create --dns-nameserver DNS_RESOLVER
  ```

  Replace `DNS_RESOLVER` with the IP address of a DNS resolver reachable from the virtual network. Repeat the option if you want to specify multiple IP addresses. For example:

  ```
  $ openstack subnet create --dns-nameserver 203.0.113.8 --dns-nameserver
  ↪198.51.100.53
  ```

  > **Note**
  >
  > This command requires additional options outside the scope of this content.

- Add a DNS resolver to an existing subnet.

  ```
  $ openstack subnet set --dns-nameserver DNS_RESOLVER SUBNET_ID_OR_NAME
  ```

  Replace `DNS_RESOLVER` with the IP address of a DNS resolver reachable from the virtual network and `SUBNET_ID_OR_NAME` with the UUID or name of the subnet. For example, using the `selfservice` subnet:

  ```
  $ openstack subnet set --dns-nameserver 203.0.113.9 selfservice
  ```

- Remove all DNS resolvers from a subnet.

  ```
  $ openstack subnet set --no-dns-nameservers SUBNET_ID_OR_NAME
  ```

  Replace `SUBNET_ID_OR_NAME` with the UUID or name of the subnet. For example, using the `selfservice` subnet:

  ```
  $ openstack subnet set --no-dns-nameservers selfservice
  ```

  > **Note**
  >
  > You can use this option in combination with the previous one in order to replace all existing DNS resolver addresses with new ones.

You can also set the DNS resolver address to `0.0.0.0` for IPv4 subnets, or `::` for IPv6 subnets, which are special values that indicate to the DHCP agent that it should not announce any DNS resolver at all on the subnet.

> **Note**

When DNS resolvers are explicitly specified for a subnet this way, that setting will take precedence over the options presented in case 2.

Case 2: DHCP agents forward DNS queries from instances

In this case, the DHCP agent offers the list of all DHCP agents IP addresses on a subnet as DNS resolver(s) to instances via DHCP on that subnet.

The DHCP agent then runs a masquerading forwarding DNS resolver with two possible options to determine where the DNS queries are sent to.

Note

The DHCP agent will answer queries for names and addresses of instances running within the virtual network directly instead of forwarding them.

Case 2a: Queries are forwarded to an explicitly configured set of DNS resolvers

In the `dhcp_agent.ini` file, configure one or more DNS resolvers. To configure more than one DNS resolver, use a comma between the values.

```
[DEFAULT]
dnsmasq_dns_servers = DNS_RESOLVER
```

Replace `DNS_RESOLVER` with a list of IP addresses of DNS resolvers reachable from all virtual networks. For example:

```
[DEFAULT]
dnsmasq_dns_servers = 203.0.113.8, 198.51.100.53
```

Note

You must configure this option for all eligible DHCP agents and restart them to activate the values.

Case 2b: Queries are forwarded to DNS resolver(s) configured on the host

In this case, the DHCP agent forwards queries from the instances to the DNS resolver(s) configured in the `resolv.conf` file on the host running the DHCP agent. This requires these resolvers being reachable from all virtual networks.

In the `dhcp_agent.ini` file, enable using the DNS resolver(s) configured on the host.

```
[DEFAULT]
dnsmasq_local_resolv = True
```

Note

You must configure this option for all eligible DHCP agents and restart them to activate this setting.

8.2.11 Distributed Virtual Routing with VRRP

Open vSwitch: High availability using DVR supports augmentation using Virtual Router Redundancy Protocol (VRRP). Using this configuration, virtual routers support both the `--distributed` and `--ha` options.

Similar to legacy HA routers, DVR/SNAT HA routers provide a quick fail over of the SNAT service to a backup DVR/SNAT router on an l3-agent running on a different node.

SNAT high availability is implemented in a manner similar to the *Open vSwitch: High availability using VRRP* example where `keepalived` uses VRRP to provide quick failover of SNAT services.

During normal operation, the primary router periodically transmits *heartbeat* packets over a hidden project network that connects all HA routers for a particular project.

If the DVR/SNAT backup router stops receiving these packets, it assumes failure of the primary DVR/SNAT router and promotes itself to primary router by configuring IP addresses on the interfaces in the `snat` namespace. In environments with more than one backup router, the rules of VRRP are followed to select a new primary router.

Warning

There is a known bug with `keepalived` v1.2.15 and earlier which can cause packet loss when `max_l3_agents_per_router` is set to 3 or more. Therefore, we recommend that you upgrade to `keepalived` v1.2.16 or greater when using this feature.

Configuration example

The basic deployment model consists of one controller node, two or more network nodes, and multiple computes nodes.

Controller node configuration

1. Add the following to `/etc/neutron/neutron.conf`:

```
[DEFAULT]
core_plugin = ml2
service_plugins = router
router_distributed = True
l3_ha = True
l3_ha_net_cidr = 169.254.192.0/18
max_l3_agents_per_router = 3
```

When the `router_distributed = True` flag is configured, routers created by all users are distributed. Without it, only privileged users can create distributed routers by using `--distributed True`.

Similarly, when the `l3_ha = True` flag is configured, routers created by all users default to HA.

It follows that with these two flags set to `True` in the configuration file, routers created by all users will default to distributed HA routers (DVR HA).

The same can explicitly be accomplished by a user with administrative credentials setting the flags in the **`openstack router create`** command:

```
$ openstack router create name-of-router --distributed --ha
```

Note

The *max_l3_agents_per_router* determine the number of backup DVR/SNAT routers which will be instantiated.

2. Add the following to `/etc/neutron/plugins/ml2/ml2_conf.ini`:

```
[ml2]
type_drivers = flat,vxlan
project_network_types = vxlan
mechanism_drivers = openvswitch,l2population
extension_drivers = port_security

[ml2_type_flat]
flat_networks = external

[ml2_type_vxlan]
vni_ranges = MIN_VXLAN_ID:MAX_VXLAN_ID
```

Replace `MIN_VXLAN_ID` and `MAX_VXLAN_ID` with VXLAN ID minimum and maximum values suitable for your environment.

Note

The first value in the `project_network_types` option becomes the default project network type when a regular user creates a network.

Network nodes

1. Configure the Open vSwitch agent. Add the following to `/etc/neutron/plugins/ml2/openvswitch_agent.ini`:

```
[ovs]
local_ip = TUNNEL_INTERFACE_IP_ADDRESS
bridge_mappings = external:br-ex

[agent]
enable_distributed_routing = True
tunnel_types = vxlan
l2_population = True
```

Replace `TUNNEL_INTERFACE_IP_ADDRESS` with the IP address of the interface that handles VXLAN project networks.

2. Configure the L3 agent. Add the following to `/etc/neutron/l3_agent.ini`:

```
[DEFAULT]
ha_vrrp_auth_password = password
agent_mode = dvr_snat
```

Compute nodes

1. Configure the Open vSwitch agent. Add the following to `/etc/neutron/plugins/ml2/openvswitch_agent.ini`:

   ```
   [ovs]
   local_ip = TUNNEL_INTERFACE_IP_ADDRESS
   bridge_mappings = external:br-ex

   [agent]
   enable_distributed_routing = True
   tunnel_types = vxlan
   l2_population = True

   [securitygroup]
   firewall_driver = iptables_hybrid
   ```

2. Configure the L3 agent. Add the following to `/etc/neutron/l3_agent.ini`:

   ```
   [DEFAULT]
   agent_mode = dvr
   ```

   Replace `TUNNEL_INTERFACE_IP_ADDRESS` with the IP address of the interface that handles VXLAN project networks.

Keepalived VRRP health check

The health of your `keepalived` instances can be automatically monitored via a bash script that verifies connectivity to all available and configured gateway addresses. In the event that connectivity is lost, the master router is rescheduled to another node.

If all routers lose connectivity simultaneously, the process of selecting a new master router will be repeated in a round-robin fashion until one or more routers have their connectivity restored.

To enable this feature, edit the `l3_agent.ini` file:

```
ha_vrrp_health_check_interval = 30
```

Where `ha_vrrp_health_check_interval` indicates how often in seconds the health check should run. The default value is `0`, which indicates that the check should not run at all.

Known limitations

- There are certain scenarios where l2pop and distributed HA routers do not interact in an expected manner. These situations are the same that affect HA only routers and l2pop.

8.2.12 Experimental Features Framework

Some Neutron features are not supported because the community doesnt have the resources and/or technical expertise to maintain them anymore. As they arise, the Neutron team designates these features as experimental. Deployers can continue using these features at their own risk, by explicitly enabling them in the `experimental` section of `neutron.conf`.

Note

Of course, the Neutron core team would love to return experimetal features to the supported status, if interested parties step up to maintain them. If you are interested in maintaining any of the experimental features listed below, please contact the PTL shown in the Neutron project page.

The following table shows the Neutron features currently designated as experimental:

Table 1: **Neutron Experimental features**

Feature	Option in neutron.conf to enable
IPv6 Prefix Delegation	ipv6_pd_enabled

This is an example of how to enable the use of an experimental feature:

```
[experimental]
ipv6_pd_enabled = true
```

8.2.13 Floating IP Port Forwarding

Floating IP port forwarding enables users to forward traffic from a TCP/UDP/other protocol port of a floating IP to a TCP/UDP/other protocol port associated to one of the fixed IPs of a Neutron port. This is accomplished by associating `port_forwarding` sub-resource to a floating IP.

CRUD operations for port forwarding are implemented by a Neutron API extension and a service plug-in. Please refer to the Neutron API Reference documentation for details on the CRUD operations.

Configuring floating IP port forwarding

To configure floating IP port forwarding, take the following steps:

- Add the `port_forwarding` service to the `service_plugins` setting in `/etc/neutron/neutron.conf`. For example:

  ```
  service_plugins = router,segments,port_forwarding
  ```

- Set the `extensions` option in the `[agent]` section of `/etc/neutron/l3_agent.ini` to include `port_forwarding`. This has to be done in each network and compute node where the L3 agent is running. For example:

  ```
  extensions = port_forwarding
  ```

Note

The `router` service plug-in manages floating IPs and routers. As a consequence, it has to be configured along with the `port_forwarding` service plug-in.

Note

After updating the options in the configuration files, the neutron-server and every neutron-l3-agent need to be restarted for the new values to take effect.

After configuring floating IP port forwarding, the `floating-ip-port-forwarding` extension alias will be included in the output of the following command:

```
$ openstack extension list --network
```

8.2.14 IPAM Configuration

Starting with the Liberty release, OpenStack Networking includes a pluggable interface for the IP Address Management (IPAM) function. This interface creates a driver framework for the allocation and de-allocation of subnets and IP addresses, enabling the integration of alternate IPAM implementations or third-party IP Address Management systems.

The basics

In Liberty and Mitaka, the IPAM implementation within OpenStack Networking provided a pluggable and non-pluggable flavor. As of Newton, the non-pluggable flavor is no longer available. Instead, it is completely replaced with a reference driver implementation of the pluggable framework. All data will be automatically migrated during the upgrade process, unless you have previously configured a pluggable IPAM driver. In that case, no migration is necessary.

To configure a driver other than the reference driver, specify it in the `neutron.conf` file. Do this after the migration is complete. There is no need to specify any value if you wish to use the reference driver.

```
ipam_driver = ipam-driver-name
```

There is no need to specify any value if you wish to use the reference driver, though specifying `internal` will explicitly choose the reference driver. The documentation for any alternate drivers will include the value to use when specifying that driver.

Known limitations

- The driver interface is designed to allow separate drivers for each subnet pool. However, the current implementation allows only a single IPAM driver system-wide.
- Third-party drivers must provide their own migration mechanisms to convert existing OpenStack installations to their IPAM.

8.2.15 IPv6

This section describes the OpenStack Networking reference implementation for IPv6, including the following items:

- How to enable dual-stack (IPv4 and IPv6 enabled) instances.
- How those instances receive an IPv6 address.
- How those instances communicate across a router to other subnets or the internet.
- How those instances interact with other OpenStack services.

Enabling a dual-stack network in OpenStack Networking simply requires creating a subnet with the `ip_version` field set to `6`, then the IPv6 attributes (`ipv6_ra_mode` and `ipv6_address_mode`) set. The `ipv6_ra_mode` and `ipv6_address_mode` will be described in detail in the next section. Finally, the subnets `cidr` needs to be provided.

This section does not include the following items:

- Single stack IPv6 project networking
- OpenStack control communication between servers and services over an IPv6 network.
- Connection to the OpenStack APIs via an IPv6 transport network
- IPv6 multicast
- IPv6 support in conjunction with any out of tree routers, switches, services or agents whether in physical or virtual form factors.

Neutron subnets and the IPv6 API attributes

As of Juno, the OpenStack Networking service (neutron) provides two new attributes to the subnet object, which allows users of the API to configure IPv6 subnets.

There are two IPv6 attributes:

- `ipv6_ra_mode`
- `ipv6_address_mode`

These attributes can be set to the following values:

- `slaac`
- `dhcpv6-stateful`
- `dhcpv6-stateless`

The attributes can also be left unset.

IPv6 addressing

The `ipv6_address_mode` attribute is used to control how addressing is handled by OpenStack. There are a number of different ways that guest instances can obtain an IPv6 address, and this attribute exposes these choices to users of the Networking API.

Router advertisements

The `ipv6_ra_mode` attribute is used to control router advertisements for a subnet.

The IPv6 Protocol uses Internet Control Message Protocol packets (ICMPv6) as a way to distribute information about networking. ICMPv6 packets with the type flag set to 134 are called Router Advertisement messages, which contain information about the router and the route that can be used by guest instances to send network traffic.

The `ipv6_ra_mode` is used to specify if the Networking service should generate Router Advertisement messages for a subnet.

ipv6_ra_mode and ipv6_address_mode combinations

ipv6 ra mode	ipv6 address mode	neutron-generated advertisements (radvd) A,M,O	External Router A,M,O	Description
N/S	*N/S*	Off	Not Defined	Backwards compatibility with pre-Juno IPv6 behavior.
N/S	slaac	Off	1,0,0	Guest instance obtains IPv6 address from non-OpenStack router using SLAAC.
N/S	dhcpv6-stateful	Off	0,1,1	Not currently implemented in the reference implementation.
N/S	dhcpv6-stateless	Off	1,0,1	Not currently implemented in the reference implementation.
slaac	*N/S*	1,0,0	Off	Not currently implemented in the reference implementation.
dhcpv6-stateful	*N/S*	0,1,0	Off	Not currently implemented in the reference implementation.
dhcpv6-stateless	*N/S*	1,0,1	Off	Not currently implemented in the reference implementation.
slaac	slaac	1,0,0	Off	Guest instance obtains IPv6 address from OpenStack managed radvd using SLAAC.
dhcpv6-stateful	dhcpv6-stateful	0,1,0	Off	Guest instance obtains IPv6 address from dnsmasq using DHCPv6 stateful and optional info from dnsmasq using DHCPv6.
dhcpv6-stateless	dhcpv6-stateless	1,0,1	Off	Guest instance obtains IPv6 address from OpenStack managed radvd using SLAAC and optional info from dnsmasq using DHCPv6.
slaac	dhcpv6-stateful			*Invalid combination.*
slaac	dhcpv6-stateless			*Invalid combination.*
dhcpv6-stateful	slaac			*Invalid combination.*
dhcpv6-stateful	dhcpv6-stateless			*Invalid combination.*
dhcpv6-stateless	slaac			*Invalid combination.*
dhcpv6-stateless	dhcpv6-stateful			*Invalid combination.*

A - Autonomous Address Configuration Flag, M - Managed Address Configuration Flag, O - Other Configuration Flag

Note

If the M flag is set to 1, the O flag can be either 1 or 0. This is because the O flag can be ignored when the M flag is set to 1, as mentioned in RFC 4861 below:

If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

For this reason, the neutron-generated advertisements will have the M flag set to 1 and the O flag set to 0.

Project network considerations

Dataplane

All dataplane modules, including OVN and Open vSwitch, support forwarding IPv6 packets amongst the guests and router ports. Similar to IPv4, there is no special configuration or setup required to enable the dataplane to properly forward packets from the source to the destination using IPv6. Note that these dataplanes will forward Link-local Address (LLA) packets between hosts on the same network just fine without any participation or setup by OpenStack components after the ports are all connected and MAC addresses learned.

Warning

The only exception to this is the setting of the MTU value on the network an IPv6 subnet is created on. If the MTU is less than 1280 octets (the minimum link MTU value specified in RFC 8200), then it could lead to issues configuring both IPv6 and IPv4 addresses on the network, leaving the subnets unusable. For that reason, the API validates the MTU value when subnets are created to avoid this issue.

Addresses for subnets

There are three methods currently implemented for a subnet to get its `cidr` in OpenStack:

1. Direct assignment during subnet creation via command line or Horizon
2. Referencing a subnet pool during subnet creation
3. Using a Prefix Delegation (PD) client to request a prefix for a subnet from a PD server

In the future, additional techniques could be used to allocate subnets to projects, for example, use of an external IPAM module.

Address modes for ports

Note

An external DHCPv6 server in theory could override the full address OpenStack assigns based on the EUI-64 address, but that would not be wise as it would not be consistent through the system.

IPv6 supports three different addressing schemes for address configuration and for providing optional network information.

Stateless Address Auto Configuration (SLAAC)
Address configuration using Router Advertisements.

DHCPv6-stateless
Address configuration using Router Advertisements and optional information using DHCPv6.

DHCPv6-stateful
Address configuration and optional information using DHCPv6.

OpenStack can be setup such that OpenStack Networking directly provides Router Advertisements, DHCP relay and DHCPv6 address and optional information for their networks or this can be delegated to external routers and services based on the drivers that are in use. There are two neutron subnet attributes - `ipv6_ra_mode` and `ipv6_address_mode` that determine how IPv6 addressing and network information is provided to project instances:

- `ipv6_ra_mode`: Determines who sends Router Advertisements.
- `ipv6_address_mode`: Determines how instances obtain IPv6 address, default gateway, or optional information.

For the above two attributes to be effective, `enable_dhcp` of the subnet object must be set to True.

> **Warning**
>
> When updating a network which already has bound ports with a subnet in which Autonomous Address Configuration is enabled (Stateless Address Auto Configuration, DHCPv6-stateless) the ports will be updated with the new address. This will not happen if the subnet is DHCPv6-stateful. The same is true for the case when the ports are bound with an IPv6 subnet (the network has no other IPv4 subnet), and an IPv4 subnet is added later, the ports will not be updated.
>
> For more details see the bug https://bugs.launchpad.net/neutron/+bug/1719806.
>
> A workaround is to manually update the port with fixed_ips and add the subnet in the request.

Using SLAAC for addressing

When using SLAAC, the currently supported combinations for `ipv6_ra_mode` and `ipv6_address_mode` are as follows.

ipv6_ra_mo	ipv6_addres	Result
Not specified.	SLAAC	Addresses are assigned using EUI-64, and an external router will be used for routing.
SLAAC	SLAAC	Address are assigned using EUI-64, and OpenStack Networking provides routing.

Setting SLAAC for `ipv6_ra_mode` configures the neutron router with an radvd agent to send Router Advertisements. The list below captures the values set for the address configuration flags in the Router Advertisement messages in this scenario.

- Autonomous Address Configuration Flag = 1
- Managed Address Configuration Flag = 0
- Other Configuration Flag = 0

New or existing neutron networks that contain a SLAAC enabled IPv6 subnet will result in all neutron ports attached to the network receiving IPv6 addresses. This is because when Router Advertisement messages are multicast on a neutron network, they are received by all IPv6 capable ports on the network, and each port will then configure an IPv6 address based on the information contained in the Router Advertisement messages. In some cases, an IPv6 SLAAC address will be added to a port, in addition to other IPv4 and IPv6 addresses that the port already has been assigned.

Note

If a router is not created and added to the subnet, SLAAC addressing will not succeed for instances since no Router Advertisement messages will be generated.

DHCPv6

For DHCPv6, the currently supported combinations are as follows:

ipv6_ra_mo	ipv6_addres	Result
DHCPv6-stateless	DHCPv6-stateless	Addresses are assigned through Router Advertisements (see SLAAC above) and optional information is delivered through DHCPv6.
DHCPv6-stateful	DHCPv6-stateful	Addresses and optional information are assigned using DHCPv6.

Setting DHCPv6-stateless for `ipv6_ra_mode` configures the neutron router with an radvd agent to send Router Advertisements. The list below captures the values set for the address configuration flags in the Router Advertisement messages in this scenario. Similarly, setting DHCPv6-stateless for `ipv6_address_mode` configures neutron DHCP implementation to provide the additional network information.

- Autonomous Address Configuration Flag = 1
- Managed Address Configuration Flag = 0
- Other Configuration Flag = 1

Setting DHCPv6-stateful for `ipv6_ra_mode` configures the neutron router with an radvd agent to send Router Advertisements. The list below captures the values set for the address configuration flags in the Router Advertisements messages in this scenario. Similarly, setting DHCPv6-stateful for `ipv6_address_mode` configures neutron DHCP implementation to provide addresses and additional network information through DHCPv6.

- Autonomous Address Configuration Flag = 0
- Managed Address Configuration Flag = 1
- Other Configuration Flag = 0

Note

If a router is not created and added to the subnet, DHCPv6 addressing will not succeed for instances since no Router Advertisement messages will be generated.

Note

If the M flag is set to 1, the O flag can be either 1 or 0. This is because the O flag can be ignored when the M flag is set to 1, as mentioned in RFC 4861 below:

If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

For this reason, the neutron-generated advertisements will have the M flag set to 1 and the O flag set to 0.

Router support

The behavior of the neutron router for IPv6 is different than for IPv4 in a few ways.

Internal router ports, that act as default gateway ports for a network, will share a common port for all IPv6 subnets associated with the network. This implies that there will be an IPv6 internal router interface with multiple IPv6 addresses from each of the IPv6 subnets associated with the network and a separate IPv4 internal router interface for the IPv4 subnet. On the other hand, external router ports are allowed to have a dual-stack configuration with both an IPv4 and an IPv6 address assigned to them.

Neutron project networks that are assigned Global Unicast Address (GUA) prefixes and addresses dont require NAT on the neutron router external gateway port to access the outside world. As a consequence of the lack of NAT the external router port doesnt require a GUA to send and receive to the external networks. This implies a GUA IPv6 subnet prefix is not necessarily needed for the neutron external network. By default, a IPv6 LLA associated with the external gateway port can be used for routing purposes. To handle this scenario, the implementation of router-gateway-set API in neutron has been modified so that an IPv6 subnet is not required for the external network that is associated with the neutron router. The LLA address of the upstream router can be learned in two ways.

1. In the absence of an upstream Router Advertisement message, the `ipv6_gateway` flag can be set with the external router gateway LLA in the neutron L3 agent configuration file. This also requires that no subnet is associated with that port.
2. The upstream router can send a Router Advertisement and the neutron router will automatically learn the next-hop LLA, provided again that no subnet is assigned and the `ipv6_gateway` flag is not set.

Effectively the `ipv6_gateway` flag takes precedence over a Router Advertisements that is received from the upstream router. If it is desired to use a GUA next hop that is accomplished by allocating a subnet to the external router port and assigning the upstream routers GUA address as the gateway for the subnet.

Note

It should be possible for projects to communicate with each other on an isolated network (a network without a router port) using LLA with little to no participation on the part of OpenStack. The authors of this section have not proven that to be true for all scenarios.

Note

When using the neutron L3 agent in a configuration where it is auto-configuring an IPv6 address via SLAAC, and the agent is learning its default IPv6 route from the ICMPv6 Router Advertisement, it may be necessary to set the `net.ipv6.conf.<physical_interface>.accept_ra` sysctl to the value `2` in order for routing to function correctly. For a more detailed description, please see the bug.

Neutrons Distributed Router feature and IPv6

IPv6 does work when the Distributed Virtual Router functionality is enabled, but all ingress/egress traffic is via the centralized router (hence, not distributed). More work is required to fully enable this functionality.

Advanced services

VPNaaS

VPNaaS supports IPv6, but support in Kilo and prior releases will have some bugs that may limit how it can be used. More thorough and complete testing and bug fixing is being done as part of the Liberty release. IPv6-based VPN-as-a-Service is configured similar to the IPv4 configuration. Either or both the `peer_address` and the `peer_cidr` can specified as an IPv6 address. The choice of addressing modes and router modes described above should not impact support.

FWaaS

FWaaS allows creation of IPv6 based rules.

NAT & Floating IPs

At the current time OpenStack Networking does not provide any facility to support any flavor of NAT with IPv6. Unlike IPv4 there is no current embedded support for floating IPs with IPv6. It is assumed that the IPv6 addressing amongst the projects is using GUAs with no overlap across the projects.

Security considerations

For more information about security considerations, see the `Security groups` section in *OpenStack Networking*.

Configuring interfaces of the guest

OpenStack currently doesnt support the Privacy Extensions defined by RFC 4941, or the Opaque Identifier generation methods defined in RFC 7217. The interface identifier and DUID used must be directly derived from the MAC address as described in RFC 2373. The compute instances must not be set up to utilize either of these methods when generating their interface identifier, or they might not be able to communicate properly on the network. For example, in Linux guests, these are controlled via these two `sysctl` variables:

- `net.ipv6.conf.*.use_tempaddr` (Privacy Extensions)

This allows the use of non-changing interface identifiers for IPv6 addresses according to RFC3041 semantics. It should be disabled (zero) so that stateless addresses are constructed using a stable, EUI64-based value.

- `net.ipv6.conf.*.addr_gen_mode`

This defines how link-local and auto-configured IPv6 addresses are generated. It should be set to zero (default) so that IPv6 addresses are generated using an EUI64-based value.

Note

Support for `addr_gen_mode` was added in kernel version 4.11.

Other types of guests might have similar configuration options, please consult your distribution documentation for more information.

Unlike IPv4, the MTU of a given network can be conveyed in both the Router Advertisement messages sent by the router, as well as in DHCP messages.

OpenStack control & management network considerations

As of the Kilo release, considerable effort has gone in to ensuring the project network can handle dual stack IPv6 and IPv4 transport across the variety of configurations described above. OpenStack control network can be run in a dual stack configuration and OpenStack API endpoints can be accessed via an IPv6 network. At this time, Open vSwitch (OVS) tunnel types - STT, VXLAN, GRE, support both IPv4 and IPv6 endpoints.

Prefix delegation

Warning

This feature is experimental with low test coverage. There is currently no reference implementation that would implement the feature in the tree. A third party driver may have to be used to utilize it.

From the Liberty release onwards, OpenStack Networking supports IPv6 prefix delegation. This section describes the configuration and workflow steps necessary to use IPv6 prefix delegation to provide automatic allocation of subnet CIDRs. This allows you as the OpenStack administrator to rely on an external (to the OpenStack Networking service) DHCPv6 server to manage your project network prefixes.

Configuring OpenStack Networking for prefix delegation

To enable prefix delegation, edit the `/etc/neutron/neutron.conf` file.

```
ipv6_pd_enabled = True
```

Note

Drivers may require extra configuration.

This tells OpenStack Networking to use the prefix delegation mechanism for subnet allocation when the user does not provide a CIDR or subnet pool id when creating a subnet.

Requirements

To use this feature, you need a prefix delegation capable DHCPv6 server that is reachable from your OpenStack Networking node(s). This could be software running on the OpenStack Networking node(s) or elsewhere, or a physical router. For the purposes of this guide we are using the open-source DHCPv6 server, Dibbler. Dibbler is available in many Linux package managers, or from source at tomaszmrugalski/dibbler.

This guide assumes that you are running a Dibbler server on the network node where the external network bridge exists. If you already have a prefix delegation capable DHCPv6 server in place, then you can skip the following section.

Configuring the Dibbler server

After installing Dibbler, edit the /etc/dibbler/server.conf file:

```
script "/var/lib/dibbler/pd-server.sh"

iface "br-ex" {
    pd-class {
        pd-pool 2001:db8:2222::/48
        pd-length 64
    }
}
```

The options used in the configuration file above are:

- script Points to a script to be run when a prefix is delegated or released. This is only needed if you want instances on your subnets to have external network access. More on this below.
- iface The name of the network interface on which to listen for prefix delegation messages.
- pd-pool The larger prefix from which you want your delegated prefixes to come. The example given is sufficient if you do not need external network access, otherwise a unique globally routable prefix is necessary.
- pd-length The length that delegated prefixes will be. This must be 64 to work with the current OpenStack Networking reference implementation.

To provide external network access to your instances, your Dibbler server also needs to create new routes for each delegated prefix. This is done using the script file named in the config file above. Edit the /var/lib/dibbler/pd-server.sh file:

```
if [ "$PREFIX1" != "" ]; then
    if [ "$1" == "add" ]; then
        sudo ip -6 route add ${PREFIX1}/64 via $REMOTE_ADDR dev $IFACE
    fi
    if [ "$1" == "delete" ]; then
        sudo ip -6 route del ${PREFIX1}/64 via $REMOTE_ADDR dev $IFACE
    fi
fi
```

The variables used in the script file above are:

- $PREFIX1 The prefix being added/deleted by the Dibbler server.
- $1 The operation being performed.
- $REMOTE_ADDR The IP address of the requesting Dibbler client.
- $IFACE The network interface upon which the request was received.

The above is all you need in this scenario, but more information on installing, configuring, and running Dibbler is available in the Dibbler user guide, at Dibbler a portable DHCPv6.

To start your Dibbler server, run:

```
# dibbler-server run
```

Or to run in headless mode:

```
# dibbler-server start
```

When using DevStack, it is important to start your server after the `stack.sh` script has finished to ensure that the required network interfaces have been created.

User workflow

First, create a network and IPv6 subnet:

```
$ openstack network create ipv6-pd
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2017-01-25T19:26:01Z                 |
| description               |                                      |
| headers                   |                                      |
| id                        | 4b782725-6abe-4a2d-b061-763def1bb029 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| mtu                       | 1450                                 |
| name                      | ipv6-pd                              |
| port_security_enabled     | True                                 |
| project_id                | 61b7eba037fd41f29cfba757c010faff     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 46                                   |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2017-01-25T19:26:01Z                 |
+---------------------------+--------------------------------------+

$ openstack subnet create --ip-version 6 --ipv6-ra-mode slaac \
--ipv6-address-mode slaac --use-prefix-delegation \
--network ipv6-pd ipv6-pd-1
+------------------------+--------------------------------------+
| Field                  | Value                                |
+------------------------+--------------------------------------+
| allocation_pools       | ::1-::ffff:ffff:ffff:ffff            |
| cidr                   | ::/64                                |
| created_at             | 2017-01-25T19:31:53Z                 |
| description            |                                      |
| dns_nameservers        |                                      |
| dns_publish_fixed_ip   | None                                 |
```

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```
| enable_dhcp            | True                                 |
| gateway_ip             | ::                                   |
| headers                |                                      |
| host_routes            |                                      |
| id                     | 1319510d-c92c-4532-bf5d-8bcf3da761a1 |
| ip_version             | 6                                    |
| ipv6_address_mode      | slaac                                |
| ipv6_ra_mode           | slaac                                |
| name                   | ipv6-pd-1                            |
| network_id             | 4b782725-6abe-4a2d-b061-763def1bb029 |
| project_id             | 61b7eba037fd41f29cfba757c010faff     |
| revision_number        | 2                                    |
| service_types          |                                      |
| subnetpool_id          | prefix_delegation                    |
| tags                   |                                      |
| updated_at             | 2017-01-25T19:31:53Z                 |
+------------------------+--------------------------------------+
```

The subnet is initially created with a temporary CIDR before one can be assigned by prefix delegation. Any number of subnets with this temporary CIDR can exist without raising an overlap error. The subnetpool_id is automatically set to `prefix_delegation`.

To trigger the prefix delegation process, create a router interface between this subnet and a router with an active interface on the external network:

```
$ openstack router add subnet router1 ipv6-pd-1
```

The prefix delegation mechanism then sends a request via the external network to your prefix delegation server, which replies with the delegated prefix. The subnet is then updated with the new prefix, including issuing new IP addresses to all ports:

```
$ openstack subnet show ipv6-pd-1
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 2001:db8:2222:6977::2-2001:db8:2222: |
|                   | 6977:ffff:ffff:ffff:ffff             |
| cidr              | 2001:db8:2222:6977::/64              |
| created_at        | 2017-01-25T19:31:53Z                 |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 2001:db8:2222:6977::1                |
| host_routes       |                                      |
| id                | 1319510d-c92c-4532-bf5d-8bcf3da761a1 |
| ip_version        | 6                                    |
| ipv6_address_mode | slaac                                |
| ipv6_ra_mode      | slaac                                |
| name              | ipv6-pd-1                            |
| network_id        | 4b782725-6abe-4a2d-b061-763def1bb029 |
```

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```
| project_id        | 61b7eba037fd41f29cfba757c010faff     |
| revision_number   | 4                                    |
| service_types     |                                      |
| subnetpool_id     | prefix_delegation                    |
| tags              | []                                   |
| updated_at        | 2017-01-25T19:35:26Z                 |
+-------------------+--------------------------------------+
```

If the prefix delegation server is configured to delegate globally routable prefixes and setup routes, then any instance with a port on this subnet should now have external network access.

Deleting the router interface causes the subnet to be reverted to the temporary CIDR, and all ports have their IPs updated. Prefix leases are released and renewed automatically as necessary.

References

The following presentation from the Barcelona Summit provides a great guide for setting up IPv6 with OpenStack: Deploying IPv6 in OpenStack Environments.

8.2.16 Macvtap Mechanism Driver

The Macvtap mechanism driver for the ML2 plug-in generally increases network performance of instances.

Consider the following attributes of this mechanism driver to determine practicality in your environment:

- Supports only instance ports. Ports for DHCP and layer-3 (routing) services must use another mechanism driver such as Open vSwitch (OVS).
- Supports only untagged (flat) and tagged (VLAN) networks.
- Lacks support for security groups including basic (sanity) and anti-spoofing rules.
- Lacks support for layer-3 high-availability mechanisms such as Virtual Router Redundancy Protocol (VRRP) and Distributed Virtual Routing (DVR).
- Only compute resources can be attached via macvtap. Attaching other resources like DHCP, Routers and others is not supported. Therefore run either OVS in VLAN or flat mode on the controller node.
- Instance migration requires the same values for the `physical_interface_mapping` configuration option on each compute node. For more information, see https://bugs.launchpad.net/neutron/+bug/1550400.

Prerequisites

You can add this mechanism driver to an existing environment using the OVS mechanism driver with only provider networks or provider and self-service networks. You can change the configuration of existing compute nodes or add compute nodes with the Macvtap mechanism driver. The example configuration assumes addition of compute nodes with the Macvtap mechanism driver to the *Open vSwitch: Self-service networks* deployment example.

Add one or more compute nodes with the following components:

- Three network interfaces: management, provider, and overlay.

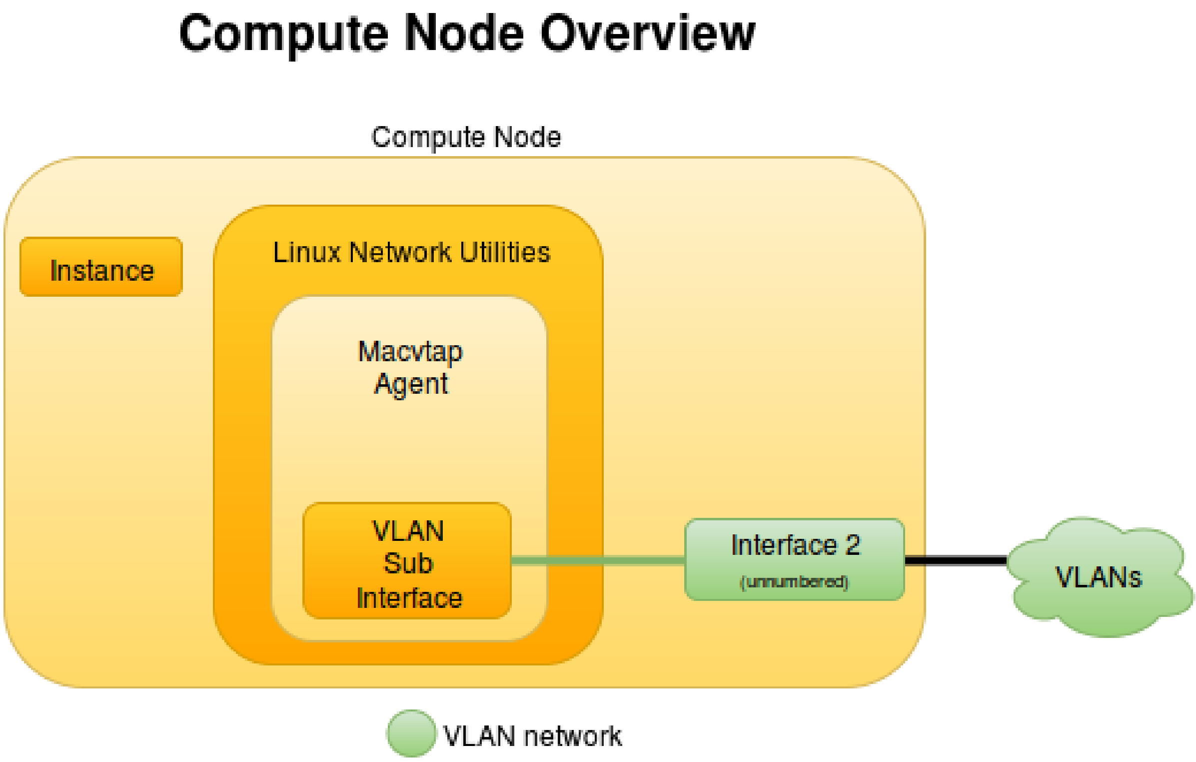
- OpenStack Networking Macvtap layer-2 agent and any dependencies.

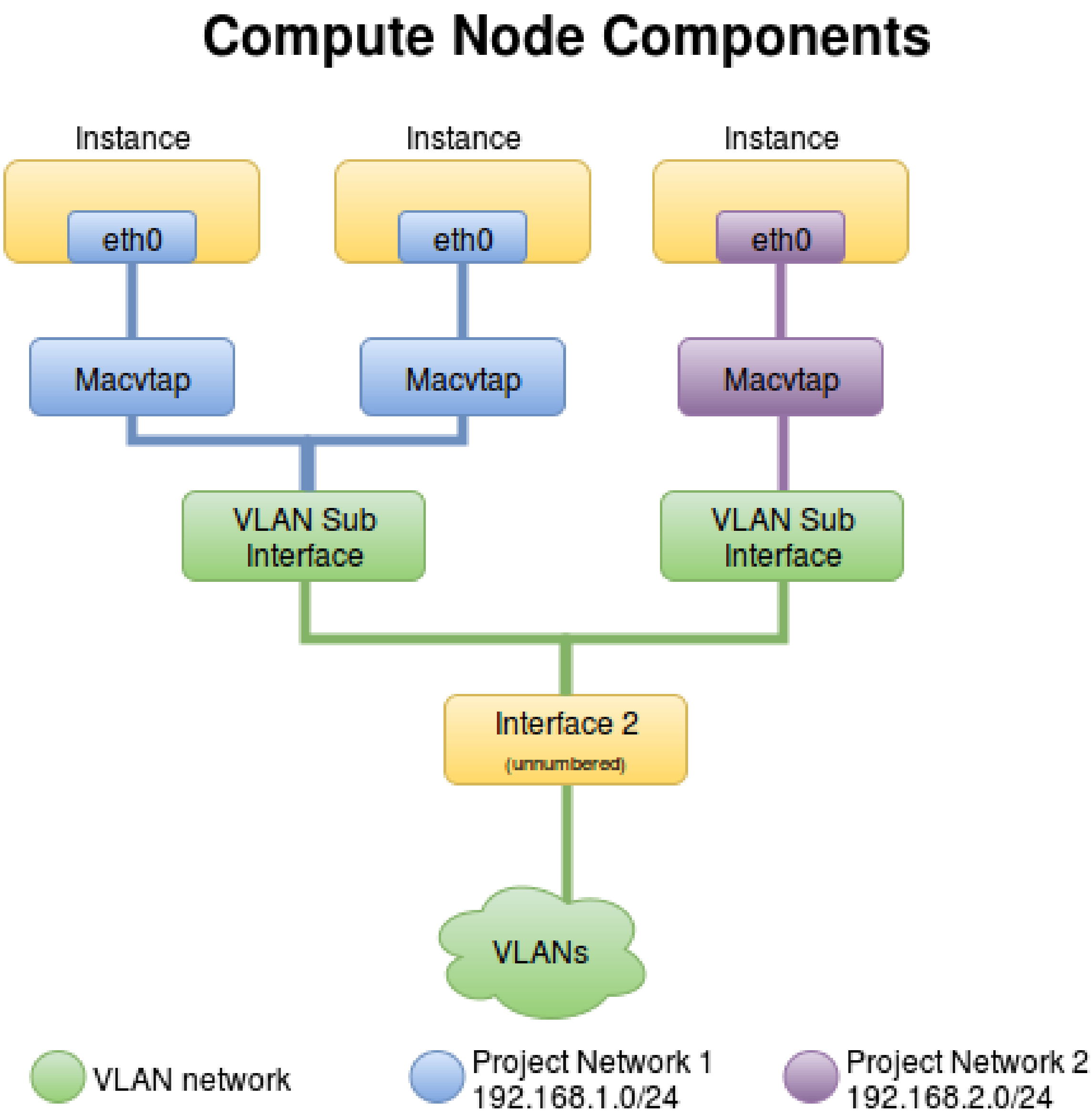
Note

To support integration with the deployment examples, this content configures the Macvtap mechanism driver to use the overlay network for untagged (flat) or tagged (VLAN) networks in addition to overlay networks such as VXLAN. Your physical network infrastructure must support VLAN (802.1q) tagging on the overlay network.

Architecture

The Macvtap mechanism driver only applies to compute nodes. Otherwise, the environment resembles the prerequisite deployment example.





Example configuration

Use the following example configuration as a template to add support for the Macvtap mechanism driver to an existing operational environment.

Controller node

1. In the `ml2_conf.ini` file:

   - Add `macvtap` to mechanism drivers.

   ```
   [ml2]
   mechanism_drivers = macvtap
   ```

   - Configure network mappings.

   ```
   [ml2_type_flat]
   flat_networks = provider,macvtap
   ```

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```
[ml2_type_vlan]
network_vlan_ranges = provider,macvtap:VLAN_ID_START:VLAN_ID_END
```

Note: Use of `macvtap` is arbitrary. Only the self-service deployment examples require VLAN ID ranges. Replace `VLAN_ID_START` and `VLAN_ID_END` with appropriate numerical values.

2. Restart the following services:
   - Server

Network nodes

No changes.

Compute nodes

1. Install the Networking service Macvtap layer-2 agent.
2. In the `neutron.conf` file, configure common options:

```
[DEFAULT]
core_plugin = ml2
auth_strategy = keystone

[database]
# ...

[keystone_authtoken]
# ...

[nova]
# ...

[agent]
# ...
```

   See the Installation Tutorials and Guides and Configuration Reference for your OpenStack release to obtain the appropriate additional configuration for the `[DEFAULT]`, `[database]`, `[keystone_authtoken]`, `[nova]`, and `[agent]` sections.

3. In the `macvtap_agent.ini` file, configure the layer-2 agent.

```
[macvtap]
physical_interface_mappings = macvtap:MACVTAP_INTERFACE

[securitygroup]
firewall_driver = noop
```

Replace MACVTAP_INTERFACE with the name of the underlying interface that handles Macvtap mechanism driver interfaces. If using a prerequisite deployment example, replace MACVTAP_INTERFACE with the name of the underlying interface that handles overlay networks. For example, eth1.

4. Start the following services:
   - Macvtap agent

Verify service operation

1. Source the administrative project credentials.
2. Verify presence and operation of the agents:

```
$ openstack network agent list
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host     |␣
↪Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
| 31e1bc1b-c872-4429-8fc3-2c8eba52634e | Metadata agent     | compute1 |␣
↪None              | True  | UP    | neutron-metadata-agent    |
| 378f5550-feee-42aa-a1cb-e548b7c2601f | Open vSwitch agent | compute1 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
| 7d2577d0-e640-42a3-b303-cb1eb077f2b6 | L3 agent           | compute1 |␣
↪nova              | True  | UP    | neutron-l3-agent          |
| d5d7522c-ad14-4c63-ab45-f6420d6a81dd | Metering agent     | compute1 |␣
↪None              | True  | UP    | neutron-metering-agent    |
| e838ef5c-75b1-4b12-84da-7bdbd62f1040 | DHCP agent         | compute1 |␣
↪nova              | True  | UP    | neutron-dhcp-agent        |
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
```

Create initial networks

This mechanism driver simply changes the virtual network interface driver for instances. Thus, you can reference the `Create initial networks` content for the prerequisite deployment example.

Verify network operation

This mechanism driver simply changes the virtual network interface driver for instances. Thus, you can reference the `Verify network operation` content for the prerequisite deployment example.

Network traffic flow

This mechanism driver simply removes handling security groups on the compute nodes. Thus, you can reference the network traffic flow scenario for the prerequisite deployment example.

8.2.17 Metadata Service Caching

The OpenStack Networking service proxies requests that VMs send to the Compute service to obtain their metadata. This functionality is provided by the `neutron-metadata-agent`, `neutron-ovn-metadata-agent` or `neutron-ovn-agent` with the metadata extension, depending on the ML2 backend used in the deployment. To obtain metadata from the Compute service, the instance ID needs to be sent to the `nova-metadata-api`. These two metadata agents provide the same functionality, but do it in slightly different ways, the difference being how the metadata agents find out the ID of the instance which is asking for metadata:

- `neutron-metadata-agent` uses RPC to ask the neutron-server process for details about a port with a specific fixed IP address connected to the given network or router (proxy service is spawned for each Neutron router or Neutron network),
- `neutron-ovn-metadata-agent` and `neutron-ovn-agent` check the instance ID in the port details of the OVN Southbound DB.

For large scale deployments which are using the `neutron-metadata-agent` this may cause significant load on the RPC bus and neutron-server, since by default for each request to the metadata service (`169.254.169.254`), the proxy will need to send an RPC query to retrieve the port details, and cloud-init is making many requests to this service during the VM boot process. To avoid this high load on the RPC bus, the `neutron-metadata-agent` allows using a caching mechanism for port details. Neutron uses oslo cache for this and it is configured through the following parameters in the `cache` section of the `metadata_agent.ini` file:

- `enabled`: enables the caching mechanism.
- `backend`: backend module to be used for caching.
- `expiration_time`: TTL, in seconds, for cached items. In case of `neutron-metadata-agent` it is recommended to use some low value here, for example, 10 seconds. Usually cloud-init will make many requests to the metadata service in a short time during boot of a VM, so caching port details for just a few seconds should be enough to avoid many RPC requests. On the other hand, using too big a value may result in having cached details for a port which has already been deleted, as a fixed IP address can be quickly re-associated to a new port in Neutron.

The oslo.cache module provides many more configuration options which can be used to tune this caching mechanism. All of them are described in the oslo.cache documentation.

8.2.18 Metadata Service Query Rate-limiting

The OpenStack Networking service proxies the requests that VMs send to the Compute service to obtain their metadata. The Networking service offers cloud administrators the ability to limit the rate at which VMs query the Computes metadata service, in order to protect the OpenStack deployment from DoS or misbehaved instances.

Metadata requests rate limiting is configured through the following parameters in the `metadata_rate_limiting` section of `neutron.conf`:

- `rate_limit_enabled`: enables rate limiting of metadata requests. It is a boolean that is set to `False` by default.
- `ip_versions`: list of comma separated strings that specify the metadata address versions (4 and/or 6) for which rate limiting must be enabled. The default is to configure rate limiting only for the IPv4 address.
- `base_window_duration`: defines in seconds the duration of the base time sliding window in which query requests will be rate limited. The default value is 10 seconds.

- `base_query_rate_limit`: maximum number of requests to be allowed during the base time window. The default value is 10 requests.
- `burst_window_duration`: this parameter can be used to define, in seconds, a shorter sliding window of time during which a requests rate higher than the base one will be allowed. The default value is 10 seconds.
- `burst_query_rate_limit`: maximum number of requests to be allowed during the burst time window. The default value is 10 requests.

Note

These parameters are used to configure HAProxy servers to perform the rate limiting. These servers run inside L3 routers and DHCP agents in the OVS backend and the metadata agent in the OVN backend.

Note

At the moment, rate limiting can only be configured either for IPv4 or IPv6 but not both at the same time, due to a limitation in the open source version of HAProxy.

Note

From the point of view of the Networking services, the base and burst windows are just two different sliding periods of time during which to enforce two different metadata requests rate limits. The Networking service doesnt enforce that the burst window should be shorter or that the burst rate should be higher. It is recommended, though, that cloud administrators use the burst window to allow, for shorter periods of time, a higher requests rate than the allowed during the base window, if there is a need to do so.

In the following `neutron.conf` snippet, the Networking service is configured to allow VMs to query the IPv4 metadata service address 6 times over a 60 seconds period, while allowing a higher rate of 2 queries during shorter periods of 10 seconds each:

```
[metadata_rate_limiting]
rate_limit_enabled = True
ip_versions = 4
base_window_duration = 60
base_query_rate_limit = 6
burst_window_duration = 10
burst_query_rate_limit = 2
```

8.2.19 ML2 Plug-in

Architecture

The Modular Layer 2 (ML2) neutron plug-in is a framework allowing OpenStack Networking to simultaneously use the variety of layer 2 networking technologies found in complex real-world data centers. The ML2 framework distinguishes between the two kinds of drivers that can be configured:

- Type drivers

  Define how an OpenStack network is technically realized. Example: VXLAN

  Each available network type is managed by an ML2 type driver. Type drivers maintain any needed type-specific network state. They validate the type specific information for provider networks and are responsible for the allocation of a free segment in project networks.

- Mechanism drivers

  Define the mechanism to access an OpenStack network of a certain type. Example: Open vSwitch mechanism driver.

  The mechanism driver is responsible for taking the information established by the type driver and ensuring that it is properly applied given the specific networking mechanisms that have been enabled.

  Mechanism drivers can utilize L2 agents (via RPC) and/or interact directly with external devices or controllers.

Multiple mechanism and type drivers can be used simultaneously to access different ports of the same virtual network.

Todo

Picture showing relationships

ML2 driver support matrix

Table 2: Mechanism drivers and L2 agents

type driver / mech driver	Flat	VLAN	VXLAN	GRE	Geneve
Open vSwitch	yes	yes	yes	yes	yes
OVN	yes	yes	yes (requires OVN 20.09+)	no	yes
SRIOV	yes	yes	no	no	no
MacVTap	yes	yes	no	no	no
L2 population	no	no	yes	yes	yes

Note

L2 population is a special mechanism driver that optimizes BUM (Broadcast, unknown destination address, multicast) traffic in the overlay networks VXLAN, GRE and Geneve. It needs to be used in conjunction with the Open vSwitch mechanism driver and cannot be used as standalone mechanism driver. For more information, see the *Mechanism drivers* section below.

Configuration

Network type drivers

To enable type drivers in the ML2 plug-in. Edit the `/etc/neutron/plugins/ml2/ml2_conf.ini` file:

```
[ml2]
type_drivers = flat,vlan,vxlan,gre
```

Note

For more detailssee the Bug 1567792.

For more details, see the Networking configuration options of Configuration Reference.

The following type drivers are available

- Flat
- VLAN
- GRE
- VXLAN

Provider network types

Provider networks provide connectivity like project networks. But only administrative (privileged) users can manage those networks because they interface with the physical network infrastructure. More information about provider networks see *OpenStack Networking*.

- Flat

  The administrator needs to configure a list of physical network names that can be used for provider networks. For more details, see the related section in the Configuration Reference.

- VLAN

  The administrator needs to configure a list of physical network names that can be used for provider networks. For more details, see the related section in the Configuration Reference.

- GRE

  No additional configuration required.

- VXLAN

  The administrator can configure the VXLAN multicast group that should be used.

  Note

  VXLAN multicast group configuration is not applicable for the Open vSwitch agent.

Project network types

Project networks provide connectivity to instances for a particular project. Regular (non-privileged) users can manage project networks within the allocation that an administrator or operator defines for them. More information about project and provider networks see *OpenStack Networking*.

Project network configurations are made in the `/etc/neutron/plugins/ml2/ml2_conf.ini` configuration file on the neutron server:

- VLAN

  The administrator needs to configure the range of VLAN IDs that can be used for project network allocation. For more details, see the related section in the Configuration Reference.

- GRE

  The administrator needs to configure the range of tunnel IDs that can be used for project network allocation. For more details, see the related section in the Configuration Reference.

- VXLAN

  The administrator needs to configure the range of VXLAN IDs that can be used for project network allocation. For more details, see the related section in the Configuration Reference.

Note

Flat networks for project allocation are not supported. They only can exist as a provider network.

Mechanism drivers

To enable mechanism drivers in the ML2 plug-in, edit the `/etc/neutron/plugins/ml2/ml2_conf.ini` file on the neutron server:

```
[ml2]
mechanism_drivers = ovs,l2pop
```

Note

For more details, see the Bug 1567792.

For more details, see the Configuration Reference.

- Open vSwitch

  No additional configurations required for the mechanism driver. Additional agent configuration is required. For details, see the related *L2 agent* section below.

- OVN

  The administrator must configure some additional configuration options for the mechanism driver. When this driver is used, architecture of the Neutron application in the cluster is different from what it is with other drivers like e.g. Open vSwitch. For details, see *OVN reference architecture*.

- SRIOV

  The SRIOV driver accepts all PCI vendor devices.

- MacVTap

  No additional configurations required for the mechanism driver. Additional agent configuration is required. Please see the related section.

- L2 population

  The administrator can configure some optional configuration options. For more details, see the related section in the Configuration Reference.

- Specialized
  - Open source

    External open source mechanism drivers exist as well as the neutron integrated reference implementations. Configuration of those drivers is not part of this document. For example:

    * OpenDaylight
    * OpenContrail
  - Proprietary (vendor)

    External mechanism drivers from various vendors exist as well as the neutron integrated reference implementations.

    Configuration of those drivers is not part of this document.

Supported VNIC types

The `vnic_type_prohibit_list` option is used to remove values from the mechanism drivers `supported_vnic_types` list.

Table 3: Mechanism drivers and supported VNIC types

mech driver / supported_vnic_types	supported VNIC types	prohibiting available
OVN	normal, direct, direct_macvtap, direct_physical	no
MacVTap	macvtap	no
Open vSwitch	normal, direct	yes (ovs_driver vnic_type_prohibit_list, see: Configuration Reference)
SRIOV	direct, macvtap, direct_physical	yes (sriov_driver vnic_type_prohibit_list, see: Configuration Reference)

Extension Drivers

The ML2 plug-in also supports extension drivers that allows other pluggable drivers to extend the core resources implemented in the ML2 plug-in (`networks`, `ports`, etc.). Examples of extension drivers include support for QoS, port security, etc. For more details see the `extension_drivers` configuration option in the Configuration Reference.

Agents

L2 agent

An L2 agent serves layer 2 (Ethernet) network connectivity to OpenStack resources. It typically runs on each Network Node and on each Compute Node.

- Open vSwitch agent

  The Open vSwitch agent configures the Open vSwitch to realize L2 networks for OpenStack resources.

Configuration for the Open vSwitch agent is typically done in the `openvswitch_agent.ini` configuration file. Make sure that on agent start you pass this configuration file as argument.

For a detailed list of configuration options, see the related section in the Configuration Reference.

- SRIOV Nic Switch agent

  The sriov nic switch agent configures PCI virtual functions to realize L2 networks for OpenStack instances. Network attachments for other resources like routers, DHCP, and so on are not supported.

  Configuration for the SRIOV nic switch agent is typically done in the `sriov_agent.ini` configuration file. Make sure that on agent start you pass this configuration file as argument.

  For a detailed list of configuration options, see the related section in the Configuration Reference.

- MacVTap agent

  The MacVTap agent uses kernel MacVTap devices for realizing L2 networks for OpenStack instances. Network attachments for other resources like routers, DHCP, and so on are not supported.

  Configuration for the MacVTap agent is typically done in the `macvtap_agent.ini` configuration file. Make sure that on agent start you pass this configuration file as argument.

  For a detailed list of configuration options, see the related section in the Configuration Reference.

L3 agent

The L3 agent offers advanced layer 3 services, like virtual Routers and Floating IPs. It requires an L2 agent running in parallel.

Configuration for the L3 agent is typically done in the `l3_agent.ini` configuration file. Make sure that on agent start you pass this configuration file as argument.

For a detailed list of configuration options, see the related section in the Configuration Reference.

DHCP agent

The DHCP agent is responsible for DHCP (Dynamic Host Configuration Protocol) and RADVD (Router Advertisement Daemon) services. It requires a running L2 agent on the same node.

Configuration for the DHCP agent is typically done in the `dhcp_agent.ini` configuration file. Make sure that on agent start you pass this configuration file as argument.

For a detailed list of configuration options, see the related section in the Configuration Reference.

Metadata agent

The Metadata agent allows instances to access cloud-init meta data and user data via the network. It requires a running L2 agent on the same node.

Configuration for the Metadata agent is typically done in the `metadata_agent.ini` configuration file. Make sure that on agent start you pass this configuration file as argument.

For a detailed list of configuration options, see the related section in the Configuration Reference.

L3 metering agent

The L3 metering agent enables layer3 traffic metering. It requires a running L3 agent on the same node.

Configuration for the L3 metering agent is typically done in the `metering_agent.ini` configuration file. Make sure that on agent start you pass this configuration file as argument.

For a detailed list of configuration options, see the related section in the Configuration Reference.

Security

L2 agents support some important security configurations.

- Security Groups

  For more details, see the related section in the Configuration Reference.

- Arp Spoofing Prevention

  Configured in the *L2 agent* configuration.

Reference implementations

Overview

In this section, the combination of a mechanism driver and an L2 agent is called reference implementation. The following table lists these implementations:

Table 4: Mechanism drivers and L2 agents

Mechanism Driver	L2 agent
Open vSwitch	Open vSwitch agent
OVN	No (there is ovn-controller running on nodes)
SRIOV	SRIOV nic switch agent
MacVTap	MacVTap agent
L2 population	Open vSwitch agent

The following tables shows which reference implementations support which non-L2 neutron agents:

Table 5: Reference implementations and other agents

Reference Implementation	L3 agent	DHCP agent	Metadata agent	L3 Metering agent
Open vSwitch & Open vSwitch agent	yes	yes	yes	yes
OVN	no (own L3 implementation)	no (DHCP provided by OVN, fully distributed)	yes (running on compute nodes, fully distributed)	no
SRIOV & SRIOV nic switch agent	no	no	no	no
MacVTap & MacVTap agent	no	no	no	no

Note

L2 population is not listed here, as it is not a standalone mechanism. If other agents are supported depends on the conjunctive mechanism driver that is used for binding a port.

More information about L2 population see the OpenStack Manuals.

Buying guide

This guide characterizes the L2 reference implementations that currently exist.

- Open vSwitch mechanism and Open vSwitch agent

  Can be used for instance network attachments as well as for attachments of other network resources like routers, DHCP, and so on.

- OVN mechanism driver

  Can be used for instance network attachments as well as for attachments of other network resources like routers, metadata ports, and so on.

- SRIOV mechanism driver and SRIOV NIC switch agent

  Can only be used for instance network attachments (device_owner = compute).

  Is deployed besides an other mechanism driver and L2 agent such as OVS. It offers instances direct access to the network adapter through a PCI Virtual Function (VF). This gives an instance direct access to hardware capabilities and high performance networking.

  The cloud consumer can decide via the neutron APIs VNIC_TYPE attribute, if an instance gets a normal OVS port or an SRIOV port.

  Due to direct connection, some features are not available when using SRIOV. For example, DVR, security groups, migration.

  For more information see the *SR-IOV*.

- MacVTap mechanism driver and MacVTap agent

  Can only be used for instance network attachments (device_owner = compute) and not for attachment of other resources like routers, DHCP, and so on.

  It is positioned as alternative to Open vSwitch support on the compute node for internal deployments.

  MacVTap offers a direct connection with very little overhead between instances and down to the adapter. You can use MacVTap agent on the compute node when you require a network connection that is performance critical. It does not require specific hardware (like with SRIOV).

  Due to the direct connection, some features are not available when using it on the compute node. For example, DVR, security groups and arp-spoofing protection.

8.2.20 MTU Considerations

The Networking service uses the MTU of the underlying physical network to calculate the MTU for virtual network components including instance network interfaces. By default, it assumes a standard 1500-byte MTU for the underlying physical network.

The Networking service only references the underlying physical network MTU. Changing the underlying physical network device MTU requires configuration of physical network devices such as switches and routers.

Jumbo frames

The Networking service supports underlying physical networks using jumbo frames and also enables instances to use jumbo frames minus any overlay protocol overhead. For example, an underlying physical network with a 9000-byte MTU yields a 8950-byte MTU for instances using a VXLAN network with IPv4 endpoints. Using IPv6 endpoints for overlay networks adds 20 bytes of overhead for any protocol.

The Networking service supports the following underlying physical network architectures. Case 1 refers to the most common architecture. In general, architectures should avoid cases 2 and 3.

Note

After you adjust MTU configuration options in `neutron.conf` and `ml2_conf.ini`, you should update `mtu` attribute for all existing networks that need a new MTU. (Network MTU update is available for all core plugins that implement the `net-mtu-writable` API extension.)

Case 1

For typical underlying physical network architectures that implement a single MTU value, you can leverage jumbo frames using two options, one in the `neutron.conf` file and the other in the `ml2_conf.ini` file. Most environments should use this configuration.

For example, referencing an underlying physical network with a 9000-byte MTU:

1. In the `neutron.conf` file:

```
[DEFAULT]
global_physnet_mtu = 9000
```

2. In the `ml2_conf.ini` file:

```
[ml2]
path_mtu = 9000
```

Case 2

Some underlying physical network architectures contain multiple layer-2 networks with different MTU values. You can configure each flat or VLAN provider network in the bridge or interface mapping options of the layer-2 agent to reference a unique MTU value.

For example, referencing a 4000-byte MTU for `provider2`, a 1500-byte MTU for `provider3`, and a 9000-byte MTU for other networks using the Open vSwitch agent:

1. In the `neutron.conf` file:

```
[DEFAULT]
global_physnet_mtu = 9000
```

2. In the `openvswitch_agent.ini` file:

```
[ovs]
bridge_mappings = provider1:eth1,provider2:eth2,provider3:eth3
```

3. In the `ml2_conf.ini` file:

```
[ml2]
physical_network_mtus = provider2:4000,provider3:1500
path_mtu = 9000
```

Case 3

Some underlying physical network architectures contain a unique layer-2 network for overlay networks using protocols such as VXLAN and GRE.

For example, referencing a 4000-byte MTU for overlay networks and a 9000-byte MTU for other networks:

1. In the `neutron.conf` file:

```
[DEFAULT]
global_physnet_mtu = 9000
```

2. In the `ml2_conf.ini` file:

```
[ml2]
path_mtu = 4000
```

> **Note**
>
> Other networks including provider networks and flat or VLAN self-service networks assume the value of the `global_physnet_mtu` option.

Instance network interfaces (VIFs)

The DHCP agent provides an appropriate MTU value to instances using IPv4, while the L3 agent provides an appropriate MTU value to instances using IPv6. IPv6 uses RA via the L3 agent because the DHCP agent only supports IPv4. Instances using IPv4 and IPv6 should obtain the same MTU value regardless of method.

> **Note**
>
> If you are using an MTU value on your network below 1280, please read the warning listed in the IPv6 configuration guide before creating any subnets.

Networks with enabled vlan transparency

In case of networks with enabled vlan transparency, if additional vlan tag is configured inside guest VM, MTU has to be lowered by 4 bytes to make space for additional vlan tag in the packets header. For example, if networks MTU is set to `1500`, value configured for the interfaces in the guest vm should be manually set to `1496` or less bytes.

8.2.21 NDP Proxy

If NDP proxy is set on a router, it is used to publish IPv6 addresses to external routers. Its purpose is similar to floating IP, but it forwards the traffic directly by using route rules and has no NAT action. Read the related specification for more details.

Configuration of NDP proxy

To configure NDP proxy, take the following steps:

- On the controller nodes:

  Add the `ndp_proxy` service to the `service_plugins` setting in the `[DEFAULT]` section of `/etc/neutron/neutron.conf`. For example:

  ```
  [DEFAULT]
  service_plugins = router,ndp_proxy
  ```

  > **Note**
  >
  > The `router` service plug-in has to be configured along with the `ndp_proxy` service plug-in.

- On the network nodes or the compute nodes (for the dvr mode router):

  Set the `extensions` option in the `[agent]` section of `/etc/neutron/l3_agent.ini` to include `ndp_proxy`. This has to be done in each network and compute node where the L3 agent is running. For example:

  ```
  extensions = ndp_proxy
  ```

> **Note**
>
> After updating the options in the configuration files, the neutron-server and every neutron-l3-agent need to be restarted for the new values to take effect.
>
> After configuring NDP proxy, the `ndp-proxy` extension alias will be included in the output of the following command:
>
> For API extension:
>
> ```
> $ openstack extension list --network
> ```
>
> For agent extension:
>
> ```
> $ openstack network agent show <l3-agent-id>
> ```

> **Note**
>
> We introduced a new command `ndsend` for the NDP proxy feature, the command can send Neighbor Advertisement about IPv6 to upstream router. With this command, we can make the upstream router rapidly perceive the change of internal IPv6 address (such as, port migrated to other node). Read the manual page for more details about this command.

Currently, you need to install this command manually in every L3 agent node. For Ubuntu, the command is provided by the `vzctl` pkg, the install command: `sudo apt install vzctl`.

- On the upstream router (the datacenters physical router):

  Generally, the admin operator should plan one or more IPv6 subnetpools to use when NDP proxy is enabled, so that all internal subnets can be allocated from a single, integrated subnetpool. In order to make NDP proxy work correctly, the admin operator needs to set direct routes for these subnetpools.

  Such as, we have a IPv6 subnetpool, its CIDR is 2001:db8::/96. The direct route like below should be set:

  ```
  2001:db8::/96 dev <ext-gw>
  ```

  The `ext-gw` is the gateway interface of the clouds external network.

User workflow

The basic steps to publish an IPv6 address to an external network (such as: public network) are the following:

Note

In order to prevent a potential security risk, the NDP proxy feature requires that an IPv6 address scope be used to ensure the uniqueness of the IPv6 address which is published externally.

1. Create an IPv6 address scope

```
$ openstack address scope create test-ipv6-as --ip-version 6
+------------+--------------------------------------+
| Field      | Value                                |
+------------+--------------------------------------+
| id         | 24761ec5-b659-4358-b9ab-495ead15fa7a |
| ip_version | 6                                    |
| name       | test-ipv6-as                         |
| project_id | bcb0c7a5338b4a46959e47971c58f0f1     |
| shared     | False                                |
+------------+--------------------------------------+
```

2. Create an IPv6 subnet pool

```
$ openstack subnet pool create test-subnetpool --address-scope test-ipv6-
↪as \
        --pool-prefix 2001:db8::/96 --default-prefix-length 112
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| address_scope_id  | 24761ec5-b659-4358-b9ab-495ead15fa7a |
| created_at        | 2022-09-05T06:16:31Z                 |
| default_prefixlen | 112                                  |
| default_quota     | None                                 |
```

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```
| description       |                                      |
| id                | 4af07f59-45b8-424d-98c5-35d20ba61526 |
| ip_version        | 6                                    |
| is_default        | False                                |
| max_prefixlen     | 128                                  |
| min_prefixlen     | 64                                   |
| name              | test-subnetpool                      |
| prefixes          | 2001:db8::/96                        |
| project_id        | bcb0c7a5338b4a46959e47971c58f0f1     |
| revision_number   | 0                                    |
| shared            | False                                |
| tags              |                                      |
| updated_at        | 2022-01-01T06:42:08Z                 |
+-------------------+--------------------------------------+
```

3. Create an external network

```
$ openstack network create --external --provider-network-type flat \
    --provider-physical-network public public
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2022-09-05T06:18:31Z                 |
| description               |                                      |
| dns_domain                | None                                 |
| id                        | 98b0f468-7be0-4530-919d-c4d9417c3abf |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| is_vlan_transparent       | None                                 |
| mtu                       | 1500                                 |
| name                      | public                               |
| port_security_enabled     | True                                 |
| project_id                | bcb0c7a5338b4a46959e47971c58f0f1     |
| provider:network_type     | flat                                 |
| provider:physical_network | public                               |
| provider:segmentation_id  | None                                 |
| qos_policy_id             | None                                 |
| revision_number           | 1                                    |
| router:external           | External                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      |                                      |
| updated_at                | 2022-01-01T06:45:08Z                 |
+---------------------------+--------------------------------------+
```

4. Create an external subnet

```
$ openstack subnet create --network public --subnet-pool test-subnetpool \
      --prefix-length 112 --ip-version 6 --no-dhcp ext-sub
+----------------------+--------------------------------------+
| Field                | Value                                |
+----------------------+--------------------------------------+
| allocation_pools     | 2001:db8::2-2001:db8::ffff           |
| cidr                 | 2001:db8::/112                       |
| created_at           | 2022-09-05T06:21:37Z                 |
| description          |                                      |
| dns_nameservers      |                                      |
| dns_publish_fixed_ip | None                                 |
| enable_dhcp          | False                                |
| gateway_ip           | 2001:db8::1                          |
| host_routes          |                                      |
| id                   | ec11de28-9b84-4cee-b6a1-0ed56135bcd8 |
| ip_version           | 6                                    |
| ipv6_address_mode    | None                                 |
| ipv6_ra_mode         | None                                 |
| name                 | ext-sub                              |
| network_id           | 98b0f468-7be0-4530-919d-c4d9417c3abf |
| project_id           | bcb0c7a5338b4a46959e47971c58f0f1     |
| revision_number      | 0                                    |
| segment_id           | None                                 |
| service_types        |                                      |
| subnetpool_id        | 4af07f59-45b8-424d-98c5-35d20ba61526 |
| tags                 |                                      |
| updated_at           | 2022-01-01T06:47:08Z                 |
+----------------------+--------------------------------------+
```

5. Create a router:

```
$ openstack router create test-router
+-------------------------+--------------------------------------+
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints |                                      |
| availability_zones      |                                      |
| created_at              | 2022-01-01T06:50:44Z                 |
| description             |                                      |
| distributed             | False                                |
| enable_ndp_proxy        | False                                |
| external_gateway_info   | null                                 |
| flavor_id               | None                                 |
| ha                      | False                                |
| id                      | 3aab8554-e5c4-4262-ab95-b92857c641de |
| name                    | test-router                          |
| project_id              | bcb0c7a5338b4a46959e47971c58f0f1     |
| revision_number         | 1                                    |
```

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```
| routes                  |                                      |
| status                  | ACTIVE                               |
| tags                    |                                      |
| updated_at              | 2022-01-01T06:50:44Z                 |
+-------------------------+--------------------------------------+
```

6. Set external gateway for the router:

```
$ openstack router set test-router --external-gateway public
```

Note

If the external network has no IPv6 subnet and the `ipv6_gateway` is configured on the `neutron-l3-agent`, you may want to set `use_lla_address` to True at `/etc/neutron/neutron.conf`, otherwise the following command will raise a 403 error.

7. Enable NDP proxy support on the router:

```
$ openstack router set test-router --enable-ndp-proxy
```

Warning

If you are using another method (such as: *BGP*, *Prefix delegation* etc.) to publish the internal IPv6 address, the command will break dataplane traffic.

8. Create an internal network and IPv6 subnet and add the subnet to the above router:

```
$ openstack network create int-net
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2022-01-01T07:11:08Z                 |
| description               |                                      |
| dns_domain                | None                                 |
| id                        | e527b38e-9e2a-439b-adf8-4ee1aa4f03b1 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| is_vlan_transparent       | None                                 |
| mtu                       | 1450                                 |
| name                      | int-net                              |
| port_security_enabled     | True                                 |
| project_id                | bcb0c7a5338b4a46959e47971c58f0f1     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
```

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```
| provider:segmentation_id  | 575                                  |
| qos_policy_id             | None                                 |
| revision_number           | 1                                    |
| router:external           | Internal                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      |                                      |
| updated_at                | 2022-01-01T07:11:08Z                 |
+---------------------------+--------------------------------------+
$ openstack subnet create --network int-net --subnet-pool test-subnetpool␣
↪\
  --prefix-length 112 --ip-version 6 \
  --ipv6-ra-mode dhcpv6-stateful \
  --ipv6-address-mode dhcpv6-stateful int-sub
+----------------------+--------------------------------------+
| Field                | Value                                |
+----------------------+--------------------------------------+
| allocation_pools     | 2001:db8::1:2-2001:db8::1:ffff       |
| cidr                 | 2001:db8::1:0/112                    |
| created_at           | 2022-09-05T06:24:13Z                 |
| description          |                                      |
| dns_nameservers      |                                      |
| dns_publish_fixed_ip | None                                 |
| enable_dhcp          | True                                 |
| gateway_ip           | 2001:db8::1:1                        |
| host_routes          |                                      |
| id                   | 9bcf194c-d44f-4e6f-90da-98510ddef283 |
| ip_version           | 6                                    |
| ipv6_address_mode    | dhcpv6-stateful                      |
| ipv6_ra_mode         | dhcpv6-stateful                      |
| name                 | int-sub                              |
| network_id           | e527b38e-9e2a-439b-adf8-4ee1aa4f03b1 |
| project_id           | bcb0c7a5338b4a46959e47971c58f0f1     |
| revision_number      | 0                                    |
| segment_id           | None                                 |
| service_types        |                                      |
| subnetpool_id        | 4af07f59-45b8-424d-98c5-35d20ba61526 |
| tags                 |                                      |
| updated_at           | 2022-01-02T08:20:26Z                 |
+----------------------+--------------------------------------+
$ openstack router add subnet test-router int-sub
```

9. Launch an instance:

```
$ openstack server create --flavor m1.tiny --image cirros-0.5.2-x86_64-
↪disk --network int-net test-server
+-------------------------------------+-------------------------------------
↪----------------------------+
```

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```
| Field                               | Value                                                        ␣
↪                              |
+-------------------------------------+-----------------------------------
↪------------------------------+
| OS-DCF:diskConfig                   | MANUAL                                                       ␣
↪                              |
| OS-EXT-AZ:availability_zone         |                                                              ␣
↪                              |
| OS-EXT-SRV-ATTR:host                | None                                                         ␣
↪                              |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None                                                         ␣
↪                              |
| OS-EXT-SRV-ATTR:instance_name       |                                                              ␣
↪                              |
| OS-EXT-STS:power_state              | NOSTATE                                                      ␣
↪                              |
| OS-EXT-STS:task_state               | scheduling                                                   ␣
↪                              |
| OS-EXT-STS:vm_state                 | building                                                     ␣
↪                              |
| OS-SRV-USG:launched_at              | None                                                         ␣
↪                              |
| OS-SRV-USG:terminated_at            | None                                                         ␣
↪                              |
| accessIPv4                          |                                                              ␣
↪                              |
| accessIPv6                          |                                                              ␣
↪                              |
| addresses                           |                                                              ␣
↪                              |
| adminPass                           | 97UvRLgdFozR                                                 ␣
↪                              |
| config_drive                        |                                                              ␣
↪                              |
| created                             | 2022-01-02T08:22:35Z                                         ␣
↪                              |
| flavor                              | m1.tiny (1)                                                  ␣
↪                              |
| hostId                              |                                                              ␣
↪                              |
| id                                  | 189a104c-36cd-479a-8702-
↪8111eb34fdb6                              |
| image                               | cirros-0.5.2-x86_64-disk␣
↪(2b2d2975-7ffc-463b-8c0e-993122f38b77) |
| key_name                            | None                                                         ␣
↪                              |
| name                                | test-server                                                  ␣
↪                              |
| progress                            | 0                                                            ␣
```

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```
↪                                   |
| project_id                          | bcb0c7a5338b4a46959e47971c58f0f1 ␣
↪                                   |
| properties                          |                                  ␣
↪                                   |
| security_groups                     | name='default'                   ␣
↪                                   |
| status                              | BUILD                            ␣
↪                                   |
| updated                             | 2022-01-02T08:22:34Z             ␣
↪                                   |
| user_id                             | 27e0947bb4fe47e4981da31d4a18ddf7 ␣
↪                                   |
| volumes_attached                    |                                  ␣
↪                                   |
+-------------------------------------+-----------------------------------
↪-----------------------------+
```

10. Create NDP proxy for the instances port:

    Query the port of the instance

```
$ openstack port list --server test-server
+--------------------------------------+------+-------------------+-------
↪-------------------------------------------------------------------------
↪-+--------+
| ID                                   | Name | MAC Address       | Fixed␣
↪IP Addresses                                                             ␣
↪ | Status |
+--------------------------------------+------+-------------------+-------
↪-------------------------------------------------------------------------
↪-+--------+
| bdd64aa0-437a-4db6-bbca-99869426c908 |      | fa:16:3e:ac:15:b8 | ip_
↪address='2001:db8::1:284', subnet_id='9bcf194c-d44f-4e6f-90da-
↪98510ddef283' | ACTIVE |
+--------------------------------------+------+-------------------+-------
↪-------------------------------------------------------------------------
↪-+--------+
```

Create NDP proxy for the port

```
$ openstack router ndp proxy create test-router --port bdd64aa0-437a-4db6-
↪bbca-99869426c908 --name test-np
+-----------------+--------------------------------------+
| Field           | Value                                |
+-----------------+--------------------------------------+
| created_at      | 2022-01-02T08:25:31Z                 |
| description     |                                      |
| id              | 73889fee-e322-443f-941e-142e4fc5f898 |
| ip_address      | 2001:db8::1:284                      |
```

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```
| name            | test-np                              |
| port_id         | bdd64aa0-437a-4db6-bbca-99869426c908 |
| project_id      | bcb0c7a5338b4a46959e47971c58f0f1     |
| revision_number | 0                                    |
| router_id       | 3aab8554-e5c4-4262-ab95-b92857c641de |
| updated_at      | 2022-01-02T08:25:31Z                 |
+-----------------+--------------------------------------+
```

11. Then ping the ports address from the upstream router:

```
$ ping 2001:db8::1:284
PING 2001:db8::1:284(2001:db8::1:284) 56 data bytes
64 bytes from 2001:db8::1:284: icmp_seq=1 ttl=64 time=0.365 ms
64 bytes from 2001:db8::1:284: icmp_seq=2 ttl=64 time=0.385 ms
```

Note

You may also need to add a security group rule that allows ICMPv6 traffic towards the instance.

Known limitations

- Using NDP proxies in combination with the OVN backend is not supported.

8.2.22 Network Segment Ranges

The network segment range service exposes the segment range management to be administered via the Neutron API. In addition, it introduces the ability for the administrator to control the segment ranges globally or on a per-tenant basis.

Why you need it

Before Stein, network segment ranges were configured as an entry in ML2 config file `ml2_conf.ini` that was statically defined for tenant network allocation and therefore had to be managed as part of the host deployment and management. When a regular tenant user creates a network, Neutron assigns the next free segmentation ID (VLAN ID, VNI etc.) from the configured segment ranges. Only an administrator can assign a specific segment ID via the provider extension.

The network segment range management service provides the following capabilities that the administrator may be interested in:

1. To check out the network segment ranges defined by the operators in the ML2 config file so that the admin can use this information to make segment range allocation.
2. To dynamically create and assign network segment ranges, which can help with the distribution of the underlying network connection mapping for privacy or dedicated business connection needs. This includes:
   - global shared network segment ranges
   - tenant-specific network segment ranges
3. To dynamically update a network segment range to offer the ability to adapt to the connection mapping changes.

4. To dynamically manage a network segment range when there are no segment ranges defined within the ML2 config file `ml2_conf.ini` and no restart of the Neutron server is required in this situation.
5. To check the availability and usage statistics of network segment ranges.

How it works

A network segment range manages a set of segments from which self-service networks can be allocated. The network segment range management service is admin-only.

As a regular project in an OpenStack cloud, you can not create a network segment range of your own and you just create networks in regular way.

If you are an admin, you can create a network segment range which can be shared (i.e. used by any regular project) or tenant-specific (i.e. assignment on a per-tenant basis). Your network segment ranges will not be visible to any other regular projects. Other CRUD operations are also supported.

When a tenant allocates a segment, it will first be allocated from an available segment range assigned to the tenant, and then a shared range if no tenant specific allocation is possible.

Default network segment ranges

A set of `default` network segment ranges are created out of the values defined in the ML2 config file: `network_vlan_ranges` for ml2_type_vlan, `vni_ranges` for ml2_type_vxlan, `tunnel_id_ranges` for ml2_type_gre and `vni_ranges` for ml2_type_geneve. They will be reloaded when Neutron server starts or restarts. The `default` network segment ranges are `read-only`, but will be treated as any other `shared` ranges on segment allocation.

The administrator can use the default network segment range information to make shared and/or per-tenant range creation and assignment.

Example configuration

Controller node

1. Enable the network segment range service plugin by appending `network_segment_range` to the list of `service_plugins` in the `neutron.conf` file on all nodes running the `neutron-server` service:

   ```
   [DEFAULT]
   # ...
   service_plugins = ...,network_segment_range,...
   ```

2. Restart the `neutron-server` service.

Verify service operation

1. Source the administrative project credentials and list the enabled extensions.
2. Use the command **`openstack extension list --network`** to verify that the `Neutron Network Segment Range` extension with Alias `network-segment-range` is enabled.

```
$ openstack extension list --network
+--------------------------------+-----------------------+---------------------
↪-------------------------------------+
```

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```
| Name                          | Alias                 | Description          ␣
↪                                      |
+-------------------------------+-----------------------+---------------------
↪-------------------------------------+
| ......                        | ......                | ......               ␣
↪                                      |
+-------------------------------+-----------------------+---------------------
↪-------------------------------------+
| Neutron Network Segment Range | network-segment-range | Provides support␣
↪for the network segment range management |
+-------------------------------+-----------------------+---------------------
↪-------------------------------------+
| ......                        | ......                | ......               ␣
↪                                      |
+-------------------------------+-----------------------+---------------------
↪-------------------------------------+
```

Workflow

At a high level, the basic workflow for a network segment range creation is the following:

1. The Cloud administrator:
   - Lists the existing network segment ranges.
   - Creates a shared or a tenant-specific network segment range based on the requirement.
2. A regular tenant creates a network in regular way. The network created will automatically allocate a segment from the segment ranges assigned to the tenant or shared if no tenant specific range available.

At a high level, the basic workflow for a network segment range update is the following:

1. The Cloud administrator:
   - Lists the existing network segment ranges and identifies the one that needs to be updated.
   - Updates the network segment range based on the requirement.
2. A regular tenant creates a network in regular way. The network created will automatically allocate a segment from the updated network segment ranges available.

List the network segment ranges or show a network segment range

As admin, list the existing network segment ranges:

```
$ openstack network segment range list
+--------------------------------------+-------------------+---------+--------
↪+----------------------------------+--------------+-------------------+------
↪------+------------+
| ID                                   | Name              | Default | Shared␣
↪| Project ID                       | Network Type | Physical Network |␣
↪Minimum ID | Maximum ID |
+--------------------------------------+-------------------+---------+--------
```

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```
↪+----------------------------------+-------------+-------------------+------
↪------+------------+
| 20ce94e1-4e51-4aa0-a5f1-26bdfb5bd90e |                   | True     | True   ␣
↪| None                             | vxlan       | None              |      ␣
↪    1 |        200 |
| 4b7af684-ec97-422d-ba38-8b9c2919ae67 | test_range_3      | False    | False  ␣
↪| 7011dc7fccac4efda89dc3b7f0d0975a | gre         | None              |      ␣
↪  100 |        120 |
| a021e582-6b0f-49f5-90cb-79a670c61973 |                   | True     | True   ␣
↪| None                             | vlan        | default           |      ␣
↪    1 |        100 |
| a3373630-969b-4ce9-bae7-dff0f8fa2f92 | test_range_2      | False    | True   ␣
↪| None                             | vxlan       | None              |      ␣
↪  501 |        505 |
| a5707a8f-76f0-4f90-9aa7-c42bf54e94b5 |                   | True     | True   ␣
↪| None                             | gre         | None              |      ␣
↪    1 |        150 |
| aad1b55b-43f1-46f9-8c35-85f270863ed6 |                   | True     | True   ␣
↪| None                             | geneve      | None              |      ␣
↪    1 |        120 |
| e3233178-2866-4f40-b794-7c6fecdc8655 | test_range_1      | False    | False  ␣
↪| 7011dc7fccac4efda89dc3b7f0d0975a | vlan        | group0-data0      |      ␣
↪   11 |         11 |
+--------------------------------------+-------------------+----------+--------
↪+----------------------------------+-------------+-------------------+------
↪------+------------+
```

The network segment ranges with `Default` as `True` are the ranges specified by the operators in the ML2 config file. Besides, there are also shared and tenant specific network segment ranges created by the admin previously.

The admin is also able to check/show the detailed information (e.g. availability and usage statistics) of a network segment range:

```
$ openstack network segment range show test_range_1
+------------------+-----------------------------------------------+
| Field            | Value                                         |
+------------------+-----------------------------------------------+
| available        | []                                            |
| default          | False                                         |
| id               | e3233178-2866-4f40-b794-7c6fecdc8655          |
| location         | None                                          |
| maximum          | 11                                            |
| minimum          | 11                                            |
| name             | test_range_1                                  |
| network_type     | vlan                                          |
| physical_network | group0-data0                                  |
| project_id       | 7011dc7fccac4efda89dc3b7f0d0975a              |
| shared           | False                                         |
| used             | {u'7011dc7fccac4efda89dc3b7f0d0975a': ['11']} |
```

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```
+------------------+----------------------------------------------------+
```

Create or update the network segment range

As admin, create a network segment range based on your requirement:

```
$ openstack network segment range create --private --project demo \
--network-type vxlan --minimum 120 --maximum 140 test_range_4
+------------------+--------------------------------------+
| Field            | Value                                |
+------------------+--------------------------------------+
| available        | ['120-140']                          |
| default          | False                                |
| id               | c016dcda-5bc3-4e98-b41f-6773e92fcd2d |
| location         | None                                 |
| maximum          | 140                                  |
| minimum          | 120                                  |
| name             | test_range_4                         |
| network_type     | vxlan                                |
| physical_network | None                                 |
| project_id       | 7011dc7fccac4efda89dc3b7f0d0975a     |
| shared           | False                                |
| used             | {}                                   |
+------------------+--------------------------------------+
```

Update a network segment range based on your requirement:

```
$ openstack network segment range set --minimum 100 --maximum 150 \
test_range_4
```

Create a tenant network

Now, as project demo (to source the client environment script demo-openrc for demo project according to https://docs.openstack.org/keystone/latest/install/keystone-openrc-rdo.html), create a network in a regular way.

```
$ source demo-openrc
$ openstack network create test_net
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2019-02-25T23:20:36Z                 |
| description               |                                      |
| dns_domain                |                                      |
| id                        | 39e5b95c-ad7a-40b5-9ec1-a4b4a8a43f14 |
| ipv4_address_scope        | None                                 |
```

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```
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| is_vlan_transparent       | None                                 |
| location                  | None                                 |
| mtu                       | 1450                                 |
| name                      | test_net                             |
| port_security_enabled     | True                                 |
| project_id                | 7011dc7fccac4efda89dc3b7f0d0975a     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | None                                  |
| qos_policy_id             | None                                 |
| revision_number           | 2                                    |
| router:external           | Internal                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      |                                      |
| updated_at                | 2019-02-25T23:20:36Z                 |
+---------------------------+--------------------------------------+
```

Then, switch back to the admin to check the segmentation ID of the tenant network created.

```
$ source admin-openrc
$ openstack network show test_net
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2019-02-25T23:20:36Z                 |
| description               |                                      |
| dns_domain                |                                      |
| id                        | 39e5b95c-ad7a-40b5-9ec1-a4b4a8a43f14 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| is_vlan_transparent       | None                                 |
| location                  | None                                 |
| mtu                       | 1450                                 |
| name                      | test_net                             |
| port_security_enabled     | True                                 |
| project_id                | 7011dc7fccac4efda89dc3b7f0d0975a     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 137                                  |
| qos_policy_id             | None                                 |
| revision_number           | 2                                    |
```

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```
| router:external          | Internal                         |
| segments                 | None                             |
| shared                   | False                            |
| status                   | ACTIVE                           |
| subnets                  |                                  |
| tags                     |                                  |
| updated_at               | 2019-02-25T23:20:36Z             |
+--------------------------+----------------------------------+
```

The tenant network created automatically allocates a segment with segmentation ID `137` from the network segment range with segmentation ID range `120-140` that is assigned to the tenant.

If no more available segment in the network segment range assigned to this tenant, then the segment allocation would refer to the `shared` segment ranges to check whether theres one segment available. If still there is no segment available, the allocation will fail as follows:

```
$ openstack network create test_net
$ Unable to create the network. No tenant network is available for
  allocation.
```

In this case, the admin is advised to check the availability and usage statistics of the related network segment ranges in order to take further actions (e.g. enlarging a segment range etc.).

Known limitations

- This service plugin is only compatible with ML2 core plugin for now. However, it is possible for other core plugins to support this feature with a follow-on effort.

8.2.23 Open vSwitch with DPDK Datapath

This page serves as a guide for how to use the OVS with DPDK datapath functionality available in the Networking service as of the Mitaka release.

The basics

Open vSwitch (OVS) provides support for a Data Plane Development Kit (DPDK) datapath since OVS 2.2, and a DPDK-backed `vhost-user` virtual interface since OVS 2.4. The DPDK datapath provides lower latency and higher performance than the standard kernel OVS datapath, while DPDK-backed `vhost-user` interfaces can connect guests to this datapath. For more information on DPDK, refer to the DPDK website.

OVS with DPDK, or OVS-DPDK, can be used to provide high-performance networking between instances on OpenStack compute nodes.

Prerequisites

Using DPDK in OVS requires the following minimum software versions:

- OVS 2.4
- DPDK 2.0
- QEMU 2.1.0
- libvirt 1.2.13

Support of `vhost-user` multiqueue that enables use of multiqueue with `virtio-net` and `igb_uio` is available if the following newer versions are used:

- OVS 2.5
- DPDK 2.2
- QEMU 2.5
- libvirt 1.2.17

In both cases, install and configure Open vSwitch with DPDK support for each node. For more information, see the OVS-DPDK installation guide (select an appropriate OVS version in the *Branch* drop-down menu).

Neutron Open vSwitch vhost-user Support for configuration of neutron OVS agent.

In case you wish to configure multiqueue, see the OVS configuration chapter on vhost-user in QEMU documentation.

The technical background of multiqueue is explained in the corresponding blueprint.

Additionally, OpenStack supports `vhost-user` reconnect feature starting from the Ocata release, as implementation of fix for bug 1604924. Starting from OpenStack Ocata release this feature is used without any configuration necessary in case the following minimum software versions are used:

- OVS 2.6
- DPDK 16.07
- QEMU 2.7

The support of this feature is not yet present in the ML2 OVN mechanism driver.

Using vhost-user interfaces

Once OVS and neutron are correctly configured with DPDK support, `vhost-user` interfaces are completely transparent to the guest (except in case of multiqueue configuration described below). However, guests must request huge pages. This can be done through flavors. For example:

```
$ openstack flavor set m1.large --property hw:mem_page_size=large
```

For more information about the syntax for `hw:mem_page_size`, refer to the Flavors guide.

Note

`vhost-user` requires file descriptor-backed shared memory. Currently, the only way to request this is by requesting large pages. This is why instances spawned on hosts with OVS-DPDK must request large pages. The aggregate flavor affinity filter can be used to associate flavors with large page support to hosts with OVS-DPDK support.

Create and add `vhost-user` network interfaces to instances in the same fashion as conventional interfaces. These interfaces can use the kernel `virtio-net` driver or a DPDK-compatible driver in the guest

```
$ openstack server create --nic net-id=$net_id ... testserver
```

Using vhost-user multiqueue

To use this feature, the following should be set in the flavor extra specs (flavor keys):

```
$ openstack flavor set $m1.large --property hw:vif_multiqueue_enabled=true
```

This setting can be overridden by the image metadata property if the feature is enabled in the extra specs:

```
$ openstack image set --property hw_vif_multiqueue_enabled=true IMAGE_NAME
```

Support of `virtio-net` multiqueue needs to be present in kernel of guest VM and is available starting from Linux kernel 3.8.

Check pre-set maximum for number of combined channels in channel configuration. Configuration of OVS and flavor done successfully should result in maximum being more than 1):

```
$ ethtool -l INTERFACE_NAME
```

To increase number of current combined channels run following command in guest VM:

```
$ ethtool -L INTERFACE_NAME combined QUEUES_NR
```

The number of queues should typically match the number of vCPUs defined for the instance. In newer kernel versions this is configured automatically.

Known limitations

- This feature is only supported when using the libvirt compute driver, and the KVM/QEMU hypervisor.
- Huge pages are required for each instance running on hosts with OVS-DPDK. If huge pages are not present in the guest, the interface will appear but will not function.
- Expect performance degradation of services using tap devices: these devices do not support DPDK. Example services include DVR and FWaaS.
- When the `ovs_use_veth` option is set to `True`, any traffic sent from a DHCP namespace will have an incorrect TCP checksum. This means that if `enable_isolated_metadata` is set to `True` and metadata service is reachable through the DHCP namespace, responses from metadata will be dropped due to an invalid checksum. In such cases, `ovs_use_veth` should be switched to `False` and Open vSwitch (OVS) internal ports should be used instead.

8.2.24 Open vSwitch Hardware Offloading

The purpose of this page is to describe how to enable Open vSwitch hardware offloading functionality available in OpenStack (using OpenStack Networking). This functionality was first introduced in the OpenStack Pike release. This page intends to serve as a guide for how to configure OpenStack Networking and OpenStack Compute to enable Open vSwitch hardware offloading.

The basics

Open vSwitch is a production quality, multilayer virtual switch licensed under the open source Apache 2.0 license. It is designed to enable massive network automation through programmatic extension, while still supporting standard management interfaces and protocols. Open vSwitch (OVS) allows Virtual Machines (VM) to communicate with each other and with the outside world. The OVS software based solution is CPU intensive, affecting system performance and preventing fully utilizing available bandwidth.

Term	Definition
PF	Physical Function. The physical Ethernet controller that supports SR-IOV.
VF	Virtual Function. The virtual PCIe device created from a physical Ethernet controller.
Representor Port	Virtual network interface similar to SR-IOV port that represents Nova instance.
First Compute Node	OpenStack Compute Node that can host Compute instances (Virtual Machines).
Second Compute Node	OpenStack Compute Node that can host Compute instances (Virtual Machines).

Supported Ethernet controllers

The following manufacturers are known to work:

- Mellanox ConnectX-4 NIC (VLAN Offload)
- Mellanox ConnectX-4 Lx/ConnectX-5 NICs (VLAN/VXLAN Offload)
- Broadcom NetXtreme-S series NICs
- Broadcom NetXtreme-E series NICs

For information on **Mellanox Ethernet Cards**, see Mellanox: Ethernet Cards - Overview.

Prerequisites

- Linux Kernel >= 4.13
- Open vSwitch >= 2.8
- iproute >= 4.12
- Mellanox or Broadcom NIC

> **Note**
>
> Mellanox NIC FW that supports Open vSwitch hardware offloading:
>
> ConnectX-5 >= 16.21.0338
>
> ConnectX-4 >= 12.18.2000
>
> ConnectX-4 Lx >= 14.21.0338

Using Open vSwitch hardware offloading

In order to enable Open vSwitch hardware offloading, the following steps are required:

1. Enable SR-IOV
2. Configure NIC to switchdev mode (relevant Nodes)
3. Enable Open vSwitch hardware offloading

Note

Throughout this guide, `enp3s0f0` is used as the PF and `eth3` is used as the representor port. These ports may vary in different environments.

Note

Throughout this guide, we use `systemctl` to restart OpenStack services. This is correct for `systemd` OS. Other methods to restart services should be used in other environments.

Create Compute virtual functions

Create the VFs for the network interface that will be used for SR-IOV. We use `enp3s0f0` as PF, which is also used as the interface for the VLAN provider network and has access to the private networks of all nodes.

Note

The following steps detail how to create VFs using Mellanox ConnectX-4 and SR-IOV Ethernet cards on an Intel system. Steps may be different for the hardware of your choice.

1. Ensure SR-IOV and VT-d are enabled on the system. Enable IOMMU in Linux by adding `intel_iommu=on` to kernel parameters, for example, using GRUB.
2. On each Compute node, create the VFs:

```
# echo '4' > /sys/class/net/enp3s0f0/device/sriov_numvfs
```

Note

A network interface can be used both for PCI passthrough, using the PF, and SR-IOV, using the VFs. If the PF is used, the VF number stored in the `sriov_numvfs` file is lost. If the PF is attached again to the operating system, the number of VFs assigned to this interface will be zero. To keep the number of VFs always assigned to this interface, update a relevant file according to your OS. See some examples below:

In Ubuntu, modifying the `/etc/network/interfaces` file:

```
auto enp3s0f0
iface enp3s0f0 inet dhcp
pre-up echo '4' > /sys/class/net/enp3s0f0/device/sriov_numvfs
```

In Red Hat, modifying the `/sbin/ifup-local` file:

```
#!/bin/sh
if [[ "$1" == "enp3s0f0" ]]
then
    echo '4' > /sys/class/net/enp3s0f0/device/sriov_numvfs
fi
```

Warning

Alternatively, you can create VFs by passing the `max_vfs` to the kernel module of your network interface. However, the `max_vfs` parameter has been deprecated, so the PCI /sys interface is the preferred method.

You can determine the maximum number of VFs a PF can support:

```
# cat /sys/class/net/enp3s0f0/device/sriov_totalvfs
8
```

3. Verify that the VFs have been created and are in up state:

Note

The PCI bus number of the PF (03:00.0) and VFs (03:00.2 .. 03:00.5) will be used later.

```
# lspci | grep Ethernet
03:00.0 Ethernet controller: Mellanox Technologies MT27800 Family␣
↪[ConnectX-5]
03:00.1 Ethernet controller: Mellanox Technologies MT27800 Family␣
↪[ConnectX-5]
03:00.2 Ethernet controller: Mellanox Technologies MT27800 Family␣
↪[ConnectX-5 Virtual Function]
03:00.3 Ethernet controller: Mellanox Technologies MT27800 Family␣
↪[ConnectX-5 Virtual Function]
03:00.4 Ethernet controller: Mellanox Technologies MT27800 Family␣
↪[ConnectX-5 Virtual Function]
03:00.5 Ethernet controller: Mellanox Technologies MT27800 Family␣
↪[ConnectX-5 Virtual Function]
```

```
# ip link show enp3s0f0
8: enp3s0f0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP␣
↪mode DEFAULT qlen 1000
   link/ether a0:36:9f:8f:3f:b8 brd ff:ff:ff:ff:ff:ff
   vf 0 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 1 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 2 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 3 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
```

If the interfaces are down, set them to up before launching a guest, otherwise the instance will fail to spawn:

```
# ip link set enp3s0f0 up
```

Configure Open vSwitch hardware offloading

1. Change the e-switch mode from legacy to switchdev on the PF device. This will also create the VF representor network devices in the host OS.

   ```
   # echo 0000:03:00.2 > /sys/bus/pci/drivers/mlx5_core/unbind
   ```

   This tells the driver to unbind VF 03:00.2

   > **Note**
   >
   > This should be done for all relevant VFs (in this example 0000:03:00.2 .. 0000:03:00.5)

2. Enable Open vSwitch hardware offloading, set PF to switchdev mode and bind VFs back.

   ```
   # sudo devlink dev eswitch set pci/0000:03:00.0 mode switchdev
   # sudo ethtool -K enp3s0f0 hw-tc-offload on
   # echo 0000:03:00.2 > /sys/bus/pci/drivers/mlx5_core/bind
   ```

   > **Note**
   >
   > This should be done for all relevant VFs (in this example 0000:03:00.2 .. 0000:03:00.5)

3. Restart Open vSwitch

   ```
   # sudo systemctl enable openvswitch.service
   # sudo ovs-vsctl set Open_vSwitch . other_config:hw-offload=true
   # sudo systemctl restart openvswitch.service
   ```

   > **Note**
   >
   > The given aging of OVS is given in milliseconds and can be controlled with:

   ```
   # ovs-vsctl set Open_vSwitch . other_config:max-idle=30000
   ```

Configure Nodes (VLAN Configuration)

1. Update `/etc/neutron/plugins/ml2/ml2_conf.ini` on Controller nodes

   ```
   [ml2]
   project_network_types = vlan
   type_drivers = vlan
   mechanism_drivers = openvswitch
   ```

2. Update `/etc/neutron/neutron.conf` on Controller nodes

   ```
   [DEFAULT]
   core_plugin = ml2
   ```

3. Update `/etc/nova/nova.conf` on Controller nodes

```
[filter_scheduler]
enabled_filters = PciPassthroughFilter
```

4. Update `/etc/nova/nova.conf` on Compute nodes

```
[pci]
#VLAN Configuration passthrough_whitelist example
passthrough_whitelist ={"'"address"'":"'"*:'"03:00"'.*"'","'"physical_
↪network"'":"'"physnet2"'"}
```

Configure Nodes (VXLAN Configuration)

1. Update `/etc/neutron/plugins/ml2/ml2_conf.ini` on Controller nodes

```
[ml2]
project_network_types = vxlan
type_drivers = vxlan
mechanism_drivers = openvswitch
```

2. Update `/etc/neutron/neutron.conf` on Controller nodes

```
[DEFAULT]
core_plugin = ml2
```

3. Update `/etc/nova/nova.conf` on Controller nodes

```
[filter_scheduler]
enabled_filters = PciPassthroughFilter
```

4. Update `/etc/nova/nova.conf` on Compute nodes

Note

VXLAN configuration requires physical_network to be null.

```
[pci]
#VLAN Configuration passthrough_whitelist example
passthrough_whitelist ={"'"address"'":"'"*:'"03:00"'.*"'","'"physical_
↪network"'":null}
```

5. Restart nova and neutron services

```
# sudo systemctl restart openstack-nova-compute.service
# sudo systemctl restart openstack-nova-scheduler.service
# sudo systemctl restart neutron-server.service
```

Validate Open vSwitch hardware offloading

Note

In this example we will bring up two instances on different Compute nodes and send ICMP echo packets between them. Then we will check TCP packets on a representor port and we will see that only the first packet will be shown there. All the rest will be offloaded.

1. Create a port `direct` on `private` network

```
# openstack port create --network private --vnic-type=direct direct_port1
```

2. Create an instance using the direct port on First Compute Node

```
# openstack server create --flavor m1.small --image cloud_image --nic␣
↪port-id=direct_port1 vm1
```

3. Repeat steps above and create a second instance on Second Compute Node

```
# openstack port create --network private --vnic-type=direct direct_port2
# openstack server create --flavor m1.small --image mellanox_fedora --nic␣
↪port-id=direct_port2 vm2
```

Note

You can use availability-zone nova:compute_node_1 option to set the desired Compute Node

4. Connect to instance1 and send ICMP Echo Request packets to instance2

```
# vncviewer localhost:5900
vm_1# ping vm2
```

5. Connect to Second Compute Node and find representor port of the instance

Note

Find a representor port first, in our case its eth3

```
compute_node2# ip link show enp3s0f0
6: enp3s0f0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq master␣
↪ovs-system state UP mode DEFAULT group default qlen 1000
   link/ether ec:0d:9a:46:9e:84 brd ff:ff:ff:ff:ff:ff
   vf 0 MAC 00:00:00:00:00:00, spoof checking off, link-state enable,␣
↪trust off, query_rss off
   vf 1 MAC 00:00:00:00:00:00, spoof checking off, link-state enable,␣
↪trust off, query_rss off
   vf 2 MAC 00:00:00:00:00:00, spoof checking off, link-state enable,␣
↪trust off, query_rss off
```

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```
    vf 3 MAC fa:16:3e:b9:b8:ce, vlan 57, spoof checking on, link-state␣
↪enable, trust off, query_rss off

compute_node2# ls -l /sys/class/net/
lrwxrwxrwx 1 root root 0 Sep 11 10:54 eth0 -> ../../devices/virtual/net/
↪eth0
lrwxrwxrwx 1 root root 0 Sep 11 10:54 eth1 -> ../../devices/virtual/net/
↪eth1
lrwxrwxrwx 1 root root 0 Sep 11 10:54 eth2 -> ../../devices/virtual/net/
↪eth2
lrwxrwxrwx 1 root root 0 Sep 11 10:54 eth3 -> ../../devices/virtual/net/
↪eth3

compute_node2# sudo ovs-dpctl show
system@ovs-system:
  lookups: hit:1684 missed:1465 lost:0
  flows: 0
  masks: hit:8420 total:1 hit/pkt:2.67
  port 0: ovs-system (internal)
  port 1: br-enp3s0f0 (internal)
  port 2: br-int (internal)
  port 3: br-ex (internal)
  port 4: enp3s0f0
  port 5: tapfdc744bb-61 (internal)
  port 6: qr-a7b1e843-4f (internal)
  port 7: qg-79a77e6d-8f (internal)
  port 8: qr-f55e4c5f-f3 (internal)
  port 9: eth3
```

6. Check traffic on the representor port. Verify that only the first ICMP packet appears.

```
compute_node2# tcpdump -nnn -i eth3

tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth3, link-type EN10MB (Ethernet), capture size 262144 bytes
17:12:41.220447 ARP, Request who-has 172.0.0.10 tell 172.0.0.13, length 46
17:12:41.220684 ARP, Reply 172.0.0.10 is-at fa:16:3e:f2:8b:23, length 42
17:12:41.260487 IP 172.0.0.13 > 172.0.0.10: ICMP echo request, id 1263,␣
↪seq 1, length 64
17:12:41.260778 IP 172.0.0.10 > 172.0.0.13: ICMP echo reply, id 1263, seq␣
↪1, length 64
17:12:46.268951 ARP, Request who-has 172.0.0.13 tell 172.0.0.10, length 42
17:12:46.271771 ARP, Reply 172.0.0.13 is-at fa:16:3e:1a:10:05, length 46
17:12:55.354737 IP6 fe80::f816:3eff:fe29:8118 > ff02::1: ICMP6, router␣
↪advertisement, length 64
17:12:56.106705 IP 0.0.0.0.68 > 255.255.255.255.67: BOOTP/DHCP, Request␣
↪from 62:21:f0:89:40:73, length 300
```

8.2.25 Open vSwitch Native Firewall Driver

Historically, Open vSwitch (OVS) could not interact directly with *iptables* to implement security groups. Thus, the OVS agent and Compute service use a Linux bridge between each instance (VM) and the OVS integration bridge `br-int` to implement security groups. The Linux bridge device contains the *iptables* rules pertaining to the instance. In general, additional components between instances and physical network infrastructure cause scalability and performance problems. To alleviate such problems, the OVS agent includes an optional firewall driver that natively implements security groups as flows in OVS rather than the Linux bridge device and *iptables*. This increases scalability and performance.

Configuring heterogeneous firewall drivers

L2 agents can be configured to use differing firewall drivers. There is no requirement that they all be the same. If an agent lacks a firewall driver configuration, it will default to what is configured on its server. This also means there is no requirement that the server has any firewall driver configured at all, as long as the agents are configured correctly.

Prerequisites

The native OVS firewall implementation requires kernel and user space support for *conntrack*, thus requiring minimum versions of the Linux kernel and Open vSwitch. All cases require Open vSwitch version 2.5 or newer.

- Kernel version 4.3 or newer includes *conntrack* support.
- Kernel version 3.3, but less than 4.3, does not include *conntrack* support and requires building the OVS modules.

It also requires the conntrack kernel module(s) to be loaded, which varies depending on the kernel version.

- Kernel version 4.19 or newer requires the *nf_conntrack* module.
- Kernel versions 4.18 or older require the *nf_conntrack_ipv4* and *nf_conntrack_ipv6* modules.

Enable the native OVS firewall driver

- On nodes running the Open vSwitch agent, edit the `openvswitch_agent.ini` file and enable the firewall driver.

  ```
  [securitygroup]
  firewall_driver = openvswitch
  ```

For more information, see the *Open vSwitch Firewall Driver* and the video.

Using GRE tunnels inside VMs with OVS firewall driver

If GRE tunnels from VM to VM are going to be used, the native OVS firewall implementation requires `nf_conntrack_proto_gre` module to be loaded in the kernel on nodes running the Open vSwitch agent. It can be loaded with the command:

```
# modprobe nf_conntrack_proto_gre
```

Some Linux distributions have files that can be used to automatically load kernel modules at boot time, for example, `/etc/modules`. Check with your distribution for further information.

This isnt necessary to use `gre` tunnel network type Neutron.

Differences between OVS and iptables firewall drivers

Both OVS and iptables firewall drivers should always behave in the same way if the same rules are configured for the security group. But in some cases that is not true and there may be slight differences between those drivers.

Case	OVS	iptables
Traffic marked as INVALID by conntrack but matching some of the SG rules (please check[1] and[2] for details)	Blocked	Allowed because it first matches SG rule, never reaches rule to drop invalid packets
Multicast traffic sent in the group 224.0.0.X (please check[3] for details)	Allowed always	Blocked, Can be enabled by SG rule.

Open Flow rules processing considerations

The native Open vSwitch firewall driver increases the number of Open Flow rules to be installed in the integration bridge, that could be up to thousands of entries, depending on the number or rules, rule type and number of ports in the compute node.

By default, these rules are written into the integration bridge in batches. The `_constants.AGENT_RES_PROCESSING_STEP` constant defines how many rules are written in a single operation. It is set to 100.

As seen in LP#1934917, during the Open Flow processing (that could be better displayed during the OVS agent initial transient period), there could be some inconsistencies in the port rules. In order to avoid them, the configuration variable `OVS.openflow_processed_per_port` allows to process all Open Flow rules related to a single port in a single transaction.

The following script provides a tool to measure, in each deployment, the processing time when using `OVS.openflow_processed_per_port` or the default `_constants.AGENT_RES_PROCESSING_STEP`:

```
# (1) Create a network with a single IPv4 subnet
openstack network create net-scale
openstack subnet create --subnet-range 10.250.0.0/16 --network net-scale snet-
↪scale

# (2) Create 400 ports bound to one host
for i in {1..400}
do
    openstack port create \
      --security-group <security_group_id> \
      --device-owner testing:scale \
      --binding-profile host_id=<compute_node_host_name> \
      --network net-scale test-large-scale-port-$i
done

# (3) Create 1000 security group rules, belonging to the same security
```

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[1] https://bugs.launchpad.net/neutron/+bug/1460741
[2] https://bugs.launchpad.net/neutron/+bug/1896587
[3] https://bugs.launchpad.net/neutron/+bug/1889631

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```
#     group <security_group_id>
for i in {3000..4000}
do
  curl -g -i -X POST http://controller:9696/v2.0/security-group-rules \
  -H "User-Agent: python-neutronclient" -H "Content-Type: application/json" \
  -H "Accept: application/json" -H "X-Auth-Token: <token>" \
  -d '{
  "security_group_rule": {
    "direction": "ingress", "protocol": "tcp",
    "ethertype": "IPv4", "port_range_max": "'$i'",
    "port_range_min": "3000",
    "security_group_id": <security_group_id>}
  }' 2>&1 > /dev/null
done

# (4) Setup the port to the host <compute_node_host_name>
# "grep" the test port list into file port_list.
$ for p in `openstack port list -f value -c id -c name -c mac_address -c␣
↪fixed_ips | grep test-large-scale-port`
  do
      mac=`echo $p | cut -f3 -d" "`
      ip_addr=`echo $p | cut -f7 -d" " | cut -f2 -d"'"`
      dev_id=`echo $p | cut -f1 -d" " | cut -b 1-11`
      dev_name="tp-$dev_id"
      echo "===" $mac "===" $ip_addr "===" $dev_id "===" $dev_name
      ovs-vsctl  --may-exist add-port br-int ${dev_name} -- set Interface \
        ${dev_name} type=internal \
        -- set Interface ${dev_name} external-ids:attached-mac="${mac}" \
        -- set Interface ${dev_name} external-ids:iface-id="${p}" \
        -- set Interface ${dev_name} external-ids:iface-status=active
      sleep 0.2

      ip link set dev ${dev_name} address ${mac}
      ip addr add ${ip_addr} dev ${dev_name}
      ip link set ${dev_name} up
  done

# (5) Restart the OVS agent and check that all flows are in place.
# (6) Check the OVS agent restart time, checking the "iteration" time and
#     number.
```

Permitted ethertypes

The OVS Firewall blocks traffic that does not have either the IPv4 or IPv6 ethertypes at present. This is a behavior change compared to the iptables_hybrid firewall, which only operates on IP packets and thus does not address other ethertypes. With the configuration option `permitted_ethertypes` it is possible to define a set of allowed ethertypes. Any traffic with these allowed ethertypes with destination to a local port or generated from a local port and MAC address, will be allowed.

References

8.2.26 Packet Logging Framework

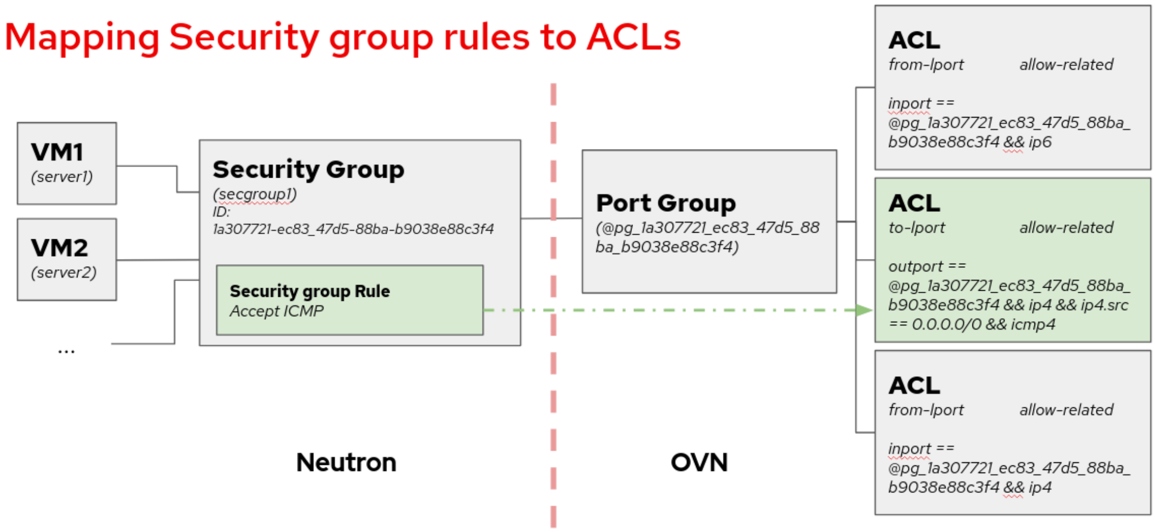
Packet logging service is designed as a Neutron plug-in that captures network packets for relevant resources (e.g. security group or firewall group) when the registered events occur.

ML2/OVN Driver

Supported loggable resource types

From the Wallaby release the ML2/OVN driver supports the `security_group` resource.

The following diagram shows a mapping from Neutron security group framework to the ACLs, which are the resources where we enable the logging when using ML2/OVN. Each security group rule maps to an ACL associated to a port group that contains all the ports belonging to the security group.



For more details on the developing peculiarities of this implementation, you can check the contributors documentation.

Service Configuration

To enable the logging service, add `log` to the `service_plugins` setting in `/etc/neutron/neutron.conf`:

```
service_plugins = router,metering,log
```

It is possible to set parameters in ml2_conf.ini to tune how we want to log the packets by modifying `rate_limit` and `burst_limit` in section `[network_log]` in `/etc/neutron/plugins/ml2/ml2_conf.ini`:

- `rate_limit` - Limit the packet rate of the logs that are sent to the OVN controller, in packets per second. The higher the number, the more logs we will get in the log file.
- `burst_limit` - Increase the packet rate limit by the specified value for a short period of time.

```
[network_log]
rate_limit = 150
burst_limit = 50
```

Note

There is a minimum value for these parameters. For `rate_limit` it is 100 and for `burst_limit` it is 25.

In order to make the changes to rate and burst effective, restart the neutron-server service. To ensure the configuration for rate and burst was updated, check the meter-band table on the OVN Northbound database. You need to create at least one log object to see the meter band entry created.

```
$ ovn-nbctl list meter-band
```

Service workflow

Create a logging resource with security group as resource type:

```
$ openstack network log create --resource-type security_group \
--resource sg1 --event ALL log1

  +-----------------+--------------------------------------+
  | Field           | Value                                |
  +-----------------+--------------------------------------+
  | Description     |                                      |
  | Enabled         | True                                 |
  | Event           | ALL                                  |
  | ID              | 67b1f618-0b89-4b9c-b3e4-9378b4472175 |
  | Name            | log1                                 |
  | Project         | 74731b187a824a8d9b85a12b6eacbcbb     |
  | Resource        | 387494cb-392a-4760-8c36-09be2fdb0b49 |
  | Target          | None                                 |
  | Type            | security_group                       |
  | created_at      | 2023-07-31T09:44:34Z                 |
  | revision_number | 0                                    |
  | tenant_id       | 74731b187a824a8d9b85a12b6eacbcbb     |
  | updated_at      | 2023-07-31T09:44:34Z                 |
  +-----------------+--------------------------------------+
```

Note

Due to the internal design of the ML2/OVN driver, there is one ACL that aggregates all dropped traffic, instead of having one drop ACL per security group. Since the smallest logging unit in OVN is the ACL, that means that if we choose to log DROP traffic, we will get traffic logged from all security groups.

If we choose to log ALL traffic, we will get the accepted traffic from the selected security group, but the dropped traffic from all security groups.

This can change in following releases if the ACL management is redesigned in OVN.

> **Warning**
>
> We cannot assign individual ports when using ML2/OVN, so the `--target` parameter is not used.

Just as with ML2/OVS, we can enable or disable logging objects at runtime. If we have two objects targeted to log the same resource, as long as one of them is enabled, the resource will be logged on the logfile.

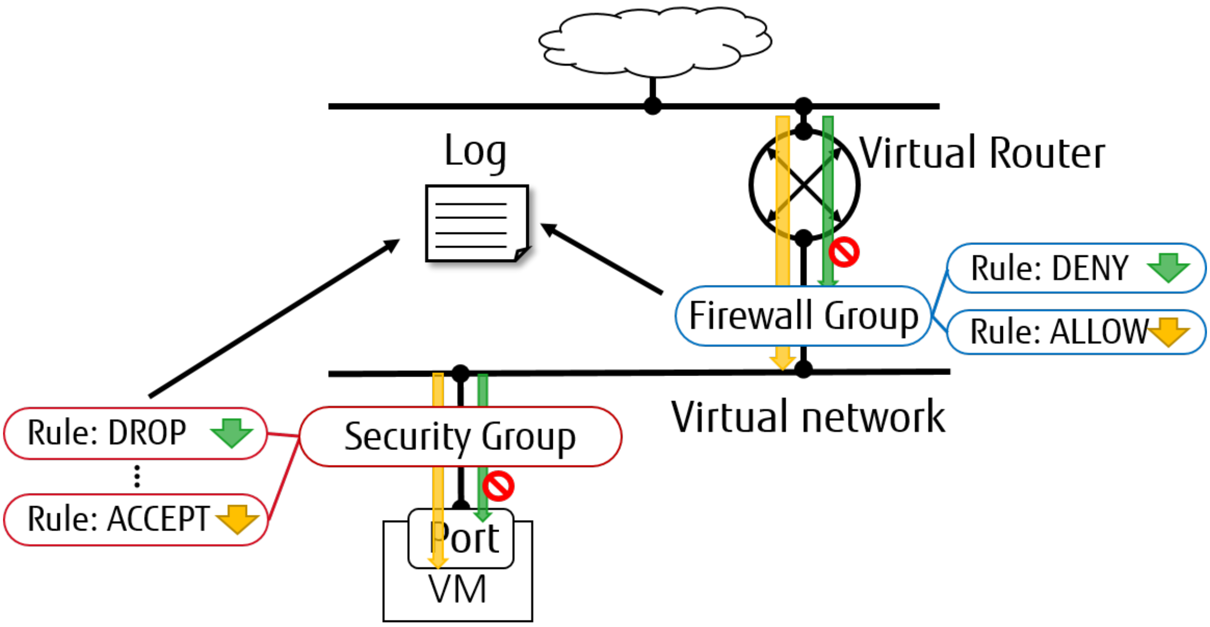
Understanding the logging

In ML2/OVN we find the packet monitoring logging recorded on each `ovn-controller.log` file within the compute nodes. This means that we will have as many logfiles as compute nodes, because each OVN controller has the capacity of logging only the traffic they manage. The location of the OVN controller log may differ depending on the distribution, please consult your installation documentation for more details. The format of the logging is:

```
2023-01-08T17:57:28.283002425+00:00 stderr F
2023-01-08T17:57:28Z|00094|acl_log(ovn_pinctrl0)|INFO|
name="neutron-e9ebf19c-3d84-49ae-a81e-7a01035a8768", verdict=allow,
severity=info, direction=to-lport: icmp, vlan_tci=0x0000,
dl_src=fa:16:3e:d3:b4:48, dl_dst=fa:16:3e:9a:d9:7d, nw_src=10.0.0.67,
nw_dst=192.168.100.11, nw_tos=0, nw_ecn=0, nw_ttl=63, nw_frag=no,
icmp_type=8, icmp_code=0
```

In this example, the name is `neutron-<security group log object ID>`. We can also see the verdict, the severity, the direction of the datagram and the content.

ML2/OVS Driver



Supported loggable resource types

From Rocky Release, the ML2/OVS driver supports both `security_group` and `firewall_group` as resource types in the Neutron packet logging framework.

Service Configuration

To enable the logging service, follow the below steps.

1. On Neutron controller node, add `log` to `service_plugins` setting in `/etc/neutron/neutron.conf` file. For example:

   ```
   service_plugins = router,metering,log
   ```

2. To enable logging service for `security_group` in Layer 2, add `log` to option `extensions` in section `[agent]` in `/etc/neutron/plugins/ml2/ml2_conf.ini` for controller node and in `/etc/neutron/plugins/ml2/openvswitch_agent.ini` for compute/network nodes. For example:

   ```
   [agent]
   extensions = log
   ```

   Note

   Fwaas v2 log is currently only supported by openvswitch.

3. To enable logging service for `firewall_group` in Layer 3, add `fwaas_v2_log` to option `extensions` in section `[AGENT]` in `/etc/neutron/l3_agent.ini` for network nodes. For example:

   ```
   [AGENT]
   extensions = fwaas_v2,fwaas_v2_log
   ```

4. On compute/network nodes, add configuration for logging service to `[network_log]` in `/etc/neutron/plugins/ml2/openvswitch_agent.ini` and in `/etc/neutron/l3_agent.ini` as shown below:

   ```
   [network_log]
   rate_limit = 100
   burst_limit = 25
   #local_output_log_base = <None>
   ```

   In which, `rate_limit` is used to configure the maximum number of packets to be logged per second (packets per second). When a high rate triggers `rate_limit`, logging queues packets to be logged. `burst_limit` is used to configure the maximum of queued packets. And logged packets can be stored anywhere by using `local_output_log_base`.

   Note

   - It requires at least `100` for `rate_limit` and at least `25` for `burst_limit`.
   - If `rate_limit` is unset, logging will log unlimited.

- If we dont specify `local_output_log_base`, logged packets will be stored in system journal like `/var/log/syslog` by default.

Trusted projects policy.yaml configuration

With the default `/etc/neutron/policy.yaml`, administrators must set up resource logging on behalf of the cloud projects.

If projects are trusted to administer their own loggable resources in their cloud, neutrons policy file `policy.yaml` can be modified to allow this.

Modify `/etc/neutron/policy.yaml` entries as follows:

```
"get_loggable_resources": "rule:regular_user"
"create_log": "rule:regular_user"
"get_log": "rule:regular_user"
"get_logs": "rule:regular_user"
"update_log": "rule:regular_user"
"delete_log": "rule:regular_user"
```

Service workflow for Operator

1. To check the loggable resources that are supported by framework:

```
$ openstack network loggable resources list
+-----------------+
| Supported types |
+-----------------+
| security_group  |
| firewall_group  |
+-----------------+
```

Note

- In VM ports, logging for `security_group` in currently works with `openvswitch` firewall driver.
- Logging for `firewall_group` works on internal router ports only. VM ports would be supported in the future.

2. Log creation:

   - Create a logging resource with an appropriate resource type

```
$ openstack network log create --resource-type security_group \
  --description "Collecting all security events" \
  --event ALL Log_Created
+-----------------+------------------------------------------------+
| Field           | Value                                          |
+-----------------+------------------------------------------------+
```

(continues on next page)

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```
| Description     | Collecting all security events                |
| Enabled         | True                                          |
| Event           | ALL                                           |
| ID              | 8085c3e6-0fa2-4954-b5ce-ff6207931b6d          |
| Name            | Log_Created                                   |
| Project         | 02568bd62b414221956f15dbe9527d16              |
| Resource        | None                                          |
| Target          | None                                          |
| Type            | security_group                                |
| created_at      | 2017-07-05T02:56:43Z                          |
| revision_number | 0                                             |
| tenant_id       | 02568bd62b414221956f15dbe9527d16              |
| updated_at      | 2017-07-05T02:56:43Z                          |
+-----------------+-----------------------------------------------+
```

Warning

In the case of `--resource` and `--target` are not specified from the request, these arguments will be assigned to `ALL` by default. Hence, there is an enormous range of log events will be created.

- Create logging resource with a given resource (sg1 or fwg1)

```
$ openstack network log create my-log --resource-type security_group --resource sg1
$ openstack network log create my-log --resource-type firewall_group --resource fwg1
```

- Create logging resource with a given target (portA)

```
$ openstack network log create my-log --resource-type security_group --target portA
```

- Create logging resource for only the given target (portB) and the given resource (sg1 or fwg1)

```
$ openstack network log create my-log --resource-type security_group --target portB --resource sg1
$ openstack network log create my-log --resource-type firewall_group --target portB --resource fwg1
```

Note

- The `Enabled` field is set to `True` by default. If enabled, logged events are written to the destination if `local_output_log_base` is configured or `/var/log/syslog` in default.
- The `Event` field will be set to `ALL` if `--event` is not specified from log creation request.

3. Enable/Disable log

We can `enable` or `disable` logging objects at runtime. It means that it will apply to all registered ports with the logging object immediately. For example:

```
$ openstack network log set --disable Log_Created
$ openstack network log show Log_Created
+-----------------+------------------------------------------------+
| Field           | Value                                          |
+-----------------+------------------------------------------------+
| Description     | Collecting all security events                 |
| Enabled         | False                                          |
| Event           | ALL                                            |
| ID              | 8085c3e6-0fa2-4954-b5ce-ff6207931b6d           |
| Name            | Log_Created                                    |
| Project         | 02568bd62b414221956f15dbe9527d16               |
| Resource        | None                                           |
| Target          | None                                           |
| Type            | security_group                                 |
| created_at      | 2017-07-05T02:56:43Z                           |
| revision_number | 1                                              |
| tenant_id       | 02568bd62b414221956f15dbe9527d16               |
| updated_at      | 2017-07-05T03:12:01Z                           |
+-----------------+------------------------------------------------+
```

Logged events description

Currently, packet logging framework supports to collect `ACCEPT` or `DROP` or both events related to registered resources. As mentioned above, Neutron packet logging framework offers two loggable resources through the `log` service plug-in: `security_group` and `firewall_group`.

The general characteristics of each event will be shown as the following:

- Log every `DROP` event: Every `DROP` security events will be generated when an incoming or outgoing session is blocked by the security groups or firewall groups
- Log an `ACCEPT` event: The `ACCEPT` security event will be generated only for each `NEW` incoming or outgoing session that is allowed by security groups or firewall groups. More details for the `ACCEPT` events are shown as bellow:
  - North/South `ACCEPT`: For a North/South session there would be a single `ACCEPT` event irrespective of direction.
  - East/West `ACCEPT`/`ACCEPT`: In an intra-project East/West session where the originating port allows the session and the destination port allows the session, i.e. the traffic is allowed, there would be two `ACCEPT` security events generated, one from the perspective of the originating port and one from the perspective of the destination port.
  - East/West `ACCEPT`/`DROP`: In an intra-project East/West session initiation where the originating port allows the session and the destination port does not allow the session there would be `ACCEPT` security events generated from the perspective of the originating port and `DROP` security events generated from the perspective of the destination port.

1. The security events that are collected by security group should include:
   - A timestamp of the flow.
   - A status of the flow `ACCEPT`/`DROP`.

- An indication of the originator of the flow, e.g which project or log resource generated the events.
- An identifier of the associated instance interface (neutron port id).
- A layer 2, 3 and 4 information (mac, address, port, protocol, etc).
- Security event record format:

  Logged data of an ACCEPT event would look like:

```
May 5 09:05:07 action=ACCEPT project_
↪id=736672c700cd43e1bd321aeaf940365c
log_resource_ids=['4522efdf-8d44-4e19-b237-64cafc49469b', '42332d89-
↪df42-4588-a2bb-3ce50829ac51']
vm_port=e0259ade-86de-482e-a717-f58258f7173f
ethernet(dst='fa:16:3e:ec:36:32',ethertype=2048,src=
↪'fa:16:3e:50:aa:b5'),
ipv4(csum=62071,dst='10.0.0.4',flags=2,header_length=5,
↪identification=36638,offset=0,
option=None,proto=6,src='172.24.4.10',tos=0,total_length=60,ttl=63,
↪version=4),
tcp(ack=0,bits=2,csum=15097,dst_port=80,offset=10,
↪option=[TCPOptionMaximumSegmentSize(kind=2,length=4,max_seg_
↪size=1460),
TCPOptionSACKPermitted(kind=4,length=2), TCPOptionTimestamps(kind=8,
↪length=10,ts_ecr=0,ts_val=196418896),
TCPOptionNoOperation(kind=1,length=1), TCPOptionWindowScale(kind=3,
↪length=3,shift_cnt=3)],
seq=3284890090,src_port=47825,urgent=0,window_size=14600)
```

  Logged data of a DROP event:

```
May 5 09:05:07 action=DROP project_
↪id=736672c700cd43e1bd321aeaf940365c
log_resource_ids=['4522efdf-8d44-4e19-b237-64cafc49469b'] vm_
↪port=e0259ade-86de-482e-a717-f58258f7173f
ethernet(dst='fa:16:3e:ec:36:32',ethertype=2048,src=
↪'fa:16:3e:50:aa:b5'),
ipv4(csum=62071,dst='10.0.0.4',flags=2,header_length=5,
↪identification=36638,offset=0,
option=None,proto=6,src='172.24.4.10',tos=0,total_length=60,ttl=63,
↪version=4),
tcp(ack=0,bits=2,csum=15097,dst_port=80,offset=10,
↪option=[TCPOptionMaximumSegmentSize(kind=2,length=4,max_seg_
↪size=1460),
TCPOptionSACKPermitted(kind=4,length=2), TCPOptionTimestamps(kind=8,
↪length=10,ts_ecr=0,ts_val=196418896),
TCPOptionNoOperation(kind=1,length=1), TCPOptionWindowScale(kind=3,
↪length=3,shift_cnt=3)],
seq=3284890090,src_port=47825,urgent=0,window_size=14600)
```

2. The events that are collected by firewall group should include:

- A timestamp of the flow.
- A status of the flow ACCEPT/DROP.
- The identifier of log objects that are collecting this event
- An identifier of the associated instance interface (neutron port id).
- A layer 2, 3 and 4 information (mac, address, port, protocol, etc).
- Security event record format:

  Logged data of an ACCEPT event would look like:

```
Jul 26 14:46:20:
action=ACCEPT, log_resource_ids=[u'2e030f3a-e93d-4a76-bc60-
↪1d11c0f6561b'], port=9882c485-b808-4a34-a3fb-b537642c66b2
pkt=ethernet(dst='fa:16:3e:8f:47:c5',ethertype=2048,src=
↪'fa:16:3e:1b:3e:67')
ipv4(csum=47423,dst='10.10.1.16',flags=2,header_length=5,
↪identification=27969,offset=0,option=None,proto=1,src='10.10.0.5',
↪tos=0,total_length=84,ttl=63,version=4)
icmp(code=0,csum=41376,data=echo(data='\xe5\xf2\xfej\x00\x00\x00\x00\
↪x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00
\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\
↪x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00
\x00\x00\x00\x00\x00\x00\x00',id=29185,seq=0),type=8)
```

  Logged data of a DROP event:

```
Jul 26 14:51:20:
action=DROP, log_resource_ids=[u'2e030f3a-e93d-4a76-bc60-1d11c0f6561b
↪'], port=9882c485-b808-4a34-a3fb-b537642c66b2
pkt=ethernet(dst='fa:16:3e:32:7d:ff',ethertype=2048,src=
↪'fa:16:3e:28:83:51')
ipv4(csum=17518,dst='10.10.0.5',flags=2,header_length=5,
↪identification=57874,offset=0,option=None,proto=1,src='10.10.1.16',
↪tos=0,total_length=84,ttl=63,version=4)
icmp(code=0,csum=23772,data=echo(data='\x8a\xa0\xac|\x00\x00\x00\x00\
↪x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00
\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\
↪x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00
\x00\x00\x00\x00\x00\x00\x00',id=25601,seq=5),type=8)
```

Note

No other extraneous events are generated within the security event logs, e.g. no debugging data, etc.

8.2.27 Quality of Service (QoS)

QoS is defined as the ability to guarantee certain network requirements like bandwidth, latency, jitter, and reliability in order to satisfy a Service Level Agreement (SLA) between an application provider and end users.

Network devices such as switches and routers can mark traffic so that it is handled with a higher priority to fulfill the QoS conditions agreed under the SLA. In other cases, certain network traffic such as Voice over IP (VoIP) and video streaming needs to be transmitted with minimal bandwidth constraints. On a system without network QoS management, all traffic will be transmitted in a best-effort manner making it impossible to guarantee service delivery to customers.

QoS is an advanced service plug-in. QoS is decoupled from the rest of the OpenStack Networking code on multiple levels and it is available through the ml2 extension driver.

Details about the DB models, API extension, and use cases are out of the scope of this guide but can be found in the Neutron QoS specification.

Supported QoS rule types

QoS supported rule types are now available as `VALID_RULE_TYPES` in QoS rule types:

- bandwidth_limit: Bandwidth limitations on networks, ports or floating IPs.
- packet_rate_limit: Packet rate limitations on certain types of traffic.
- dscp_marking: Marking network traffic with a DSCP value.
- minimum_bandwidth: Minimum bandwidth constraints on certain types of traffic.
- minimum_packet_rate: Minimum packet rate constraints on certain types of traffic.

Any QoS driver can claim support for some QoS rule types by providing a driver property called `supported_rules`, the QoS driver manager will recalculate rule types dynamically that the QoS driver supports. In the most simple case, the property can be represented by a simple Python list defined on the class.

The following table shows the Networking back ends, QoS supported rules, and traffic directions (from the VM point of view).

Table 6: **Networking back ends, supported rules, and traffic direction**

Rule \ back end	Open vSwitch	SR-IOV	OVN
Bandwidth limit	Egress \ Ingress	Egress (1)	Egress \ Ingress
Packet rate limit	Egress \ Ingress	•	•
Minimum bandwidth	Egress \ Ingress (2)	Egress \ Ingress (2)	Egress \ Ingress (2)
Minimum packet rate	•	•	•
DSCP marking	Egress	•	Egress

Note

(1) Max burst parameter is skipped because it is not supported by the IP tool.

(2) Placement based enforcement works for both egress and ingress directions, but dataplane enforcement depends on the backend.

Table 7: **Neutron backends, supported directions and enforcement types for Minimum Bandwidth rule**

Enforcement type Backend	Open vSwitch	SR-IOV	OVN
Dataplane	Egress (3)	Egress (1)	Egress (4)
Placement	Egress/Ingress (2)	Egress/Ingress (2)	Egress/Ingress (4)

Note

(1) Since Newton

(2) Since Stein

(3) Open vSwitch minimum bandwidth support is only implemented for egress direction and only for networks without tunneled traffic (only VLAN and flat network types).

(4) Since Zed

Note

The SR-IOV agent does not support dataplane enforcement for ports with `direct-physical` vnic_type. However since Yoga the Placement enforcement is supported for this vnic_type too.

Table 8: **Neutron backends, supported directions and enforcement types for Minimum Packet Rate rule**

Enforcement type Backend	Open vSwitch	SR-IOV	OVN
Dataplane	•	•	•
Placement	Any(1)/Egress/Ingress (2)	•	•

Note

(1) Minimum packet rate rule supports `any` direction that can be used with non-hardware-offloaded OVS deployments, where packets processed from both ingress and egress directions are handled by the same set of CPU cores.

(2) Since Yoga.

For an ml2 plug-in, the list of supported QoS rule types and parameters is defined as a common subset of rules supported by all active mechanism drivers. A QoS rule is always attached to a QoS policy. When a rule is created or updated:

- The QoS plug-in will check if this rule and parameters are supported by any active mechanism driver if the QoS policy is not attached to any port or network.

- The QoS plug-in will check if this rule and parameters are supported by the mechanism drivers managing those ports if the QoS policy is attached to any port or network.

Valid DSCP Marks

Valid DSCP mark values are even numbers between 0 and 56, except 2-6, 42, and 50-54. The full list of valid DSCP marks is:

0, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 44, 46, 48, 56

L3 QoS support

The Neutron L3 services have implemented their own QoS extensions. It is possible to apply QoS policies to the floating IPs and to the routers; in the last one (routers), the QoS policy will be applied on the gateway port.

The rule support depends on the ML2 backend used.

ML2/OVS

The ML2/OVS L3 QoS supports only rate limit rules:

- Floating IP bandwidth limit: rate limit is applied per floating IP address independently.
- Gateway IP bandwidth limit: the rate limit is applied in the router namespace gateway port (or in the SNAT namespace in case of DVR edge router). The rate limit applies to the gateway IP; that means all traffic using this gateway IP will be limited. This rate limit does not apply to the floating IP traffic.

ML2/OVN

The ML2/OVN L3 QoS supports both rate limit and DSCP rules. Both floating IP and gateway port QoS policies are applied using the QoS metering rules.

- Floating IP: the traffic should match the gateway port and the floating IP address. The port can be centralized or distributed.
- Gateway port: the traffic should match the gateway chassis port.

Because both floating IP and router can have QoS policies and both QoS policies will match the same traffic, the floating IP policy has a higher priority and will match this one only. In case of having port policies, that will apply to the virtual machine private port, the private port rule will be applied first. This is the QoS rules precedence and result:

- Rate limit rules: if both private port QoS and router/floating IP rules are applied, because both will be executed, the minimum rate limit value will apply.
- DSCP rules: if both private port QoS and router/floating IP rules are applied, the router/floating IP DSCP mark will be applied on the egress packet.

Note

In case of having both router and floating IP QoS policies, the floating IP QoS policy has precedence always over the router QoS one. See [OVN] Change the OVN QoS rule priority for floating IPs.

L3 services that provide QoS extensions

- L3 router: implements the rate limit using Linux TC.
- OVN L3: implements the rate limit using the OVN QoS metering rules.

The following table shows the L3 service, the QoS supported extension, and traffic directions (from the VM point of view) for **bandwidth limiting**.

Table 9: **L3 service, supported extension and traffic direction for bandwidth limiting**

L3 service	L3 router	OVN L3
Floating IP	Egress \ Ingress	Egress \ Ingress
Gateway IP	Egress \ Ingress	Egress \ Ingress

The following table shows the L3 service, the QoS supported extension, and traffic directions (from the VM point of view) for **DSCP marking**.

Table 10: **L3 service, supported extension and traffic direction for DSCP marking**

L3 service	L3 router	OVN L3
Floating IP	•	Egress
Gateway IP	•	Egress

Configuration

To enable the service on a cloud with the architecture described in Networking architecture, follow the steps below:

On the controller nodes:

1. Add the QoS service to the `service_plugins` setting in `/etc/neutron/neutron.conf`. For example:

```
service_plugins = router,metering,qos
```

2. Optionally, set the needed `notification_drivers` in the `[qos]` section in `/etc/neutron/neutron.conf` (`message_queue` is the default).

3. Optionally, in order to enable the floating IP QoS extension `qos-fip`, set the `service_plugins` option in `/etc/neutron/neutron.conf` to include both `router` and `qos`. For example:

```
service_plugins = router,qos
```

4. In `/etc/neutron/plugins/ml2/ml2_conf.ini`, add `qos` to `extension_drivers` in the `[ml2]` section. For example:

```
[ml2]
extension_drivers = port_security,qos
```

5. Edit the configuration file for the agent you are using and set the `extensions` to include `qos` in the `[agent]` section of the configuration file. The agent configuration file will reside in `/etc/neutron/plugins/ml2/<agent_name>_agent.ini` where `agent_name` is the name of the agent being used (for example `openvswitch`). For example:

```
[agent]
extensions = qos
```

On the network and compute nodes:

1. Edit the configuration file for the agent you are using and set the `extensions` to include `qos` in the `[agent]` section of the configuration file. The agent configuration file will reside in `/etc/neutron/plugins/ml2/<agent_name>_agent.ini` where `agent_name` is the name of the agent being used (for example `openvswitch`). For example:

```
[agent]
extensions = qos
```

2. Optionally, for Neutron openvswitch agent, set the `qos_meter_bandwidth` in the `[ovs]` section to enable the OVS meter bandwith limit. For example:

```
[ovs]
qos_meter_bandwidth = True
```

3. Optionally, in order to enable QoS for floating IPs, set the `extensions` option in the `[agent]` section of `/etc/neutron/l3_agent.ini` to include `fip_qos`. If `dvr` is enabled, this has to be done for all the L3 agents. For example:

```
[agent]
extensions = fip_qos
```

Note

Floating IP associated to neutron port or to port forwarding can all have bandwidth limit since Stein release. These neutron server side and agent side extension configs will enable it once for all.

1. Optionally, in order to enable QoS for router gateway IPs, set the `extensions` option in the `[agent]` section of `/etc/neutron/l3_agent.ini` to include `gateway_ip_qos`. Set this to all the `dvr_snat` or `legacy` L3 agents. For example:

```
[agent]
extensions = gateway_ip_qos
```

   And `gateway_ip_qos` should work together with the `fip_qos` in L3 agent for centralized routers, then all L3 IPs with binding QoS policy can be limited under the QoS bandwidth limit rules:

```
[agent]
extensions = fip_qos, gateway_ip_qos
```

2. As rate limit doesnt work on Open vSwitchs `internal` ports, optionally, as a workaround, to make QoS bandwidth limit work on routers gateway ports, set `ovs_use_veth` to `True` in `DEFAULT` section in `/etc/neutron/l3_agent.ini`

```
[DEFAULT]
ovs_use_veth = True
```

Note

QoS currently works with ml2 only (SR-IOV and Open vSwitch are drivers enabled for QoS).

DSCP marking on outer header for overlay networks

When using overlay networks (e.g., VxLAN), the DSCP marking rule only applies to the inner header, and during encapsulation, the DSCP mark is not automatically copied to the outer header.

1. In order to set the DSCP value of the outer header, modify the `dscp` configuration option in `/etc/neutron/plugins/ml2/<agent_name>_agent.ini` where `<agent_name>` is the name of the agent being used (e.g., `openvswitch`):

   ```
   [agent]
   dscp = 8
   ```

2. In order to copy the DSCP field of the inner header to the outer header, change the `dscp_inherit` configuration option to true in `/etc/neutron/plugins/ml2/<agent_name>_agent.ini` where `<agent_name>` is the name of the agent being used (e.g., `openvswitch`):

   ```
   [agent]
   dscp_inherit = true
   ```

   If the `dscp_inherit` option is set to true, the previous `dscp` option is overwritten.

Trusted projects policy.yaml configuration

If projects are trusted to administrate their own QoS policies in your cloud, neutrons file `policy.yaml` can be modified to allow this.

Modify `/etc/neutron/policy.yaml` policy entries as follows:

```
"get_policy": "rule:regular_user"
"create_policy": "rule:regular_user"
"update_policy": "rule:regular_user"
"delete_policy": "rule:regular_user"
"get_rule_type": "rule:regular_user"
```

To enable bandwidth limit rule:

```
"get_policy_bandwidth_limit_rule": "rule:regular_user"
"create_policy_bandwidth_limit_rule": "rule:regular_user"
"delete_policy_bandwidth_limit_rule": "rule:regular_user"
"update_policy_bandwidth_limit_rule": "rule:regular_user"
```

To enable DSCP marking rule:

```
"get_policy_dscp_marking_rule": "rule:regular_user"
"create_policy_dscp_marking_rule": "rule:regular_user"
"delete_policy_dscp_marking_rule": "rule:regular_user"
"update_policy_dscp_marking_rule": "rule:regular_user"
```

To enable minimum bandwidth rule:

```
"get_policy_minimum_bandwidth_rule": "rule:regular_user"
"create_policy_minimum_bandwidth_rule": "rule:regular_user"
"delete_policy_minimum_bandwidth_rule": "rule:regular_user"
"update_policy_minimum_bandwidth_rule": "rule:regular_user"
```

To enable minimum packet rate rule:

```
"get_policy_minimum_packet_rate_rule": "rule:regular_user"
"create_policy_minimum_packet_rate_rule": "rule:regular_user"
"delete_policy_minimum_packet_rate_rule": "rule:regular_user"
"update_policy_minimum_packet_rate_rule": "rule:regular_user"
```

User workflow

QoS policies are only created by admins with the default `policy.yaml`. Therefore, you should have the cloud operator set them up on behalf of the cloud projects.

If projects are trusted to create their own policies, check the trusted projects `policy.yaml` configuration section.

First, create a QoS policy and its bandwidth limit rule:

```
$ openstack network qos policy create bw-limiter
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| description       |                                      |
| id                | 5df855e9-a833-49a3-9c82-c0839a5f103f |
| is_default        | False                                |
| name              | bw-limiter                           |
| project_id        | 4db7c1ed114a4a7fb0f077148155c500     |
| rules             | []                                   |
| shared            | False                                |
+-------------------+--------------------------------------+


$ openstack network qos rule create --type bandwidth-limit --max-kbps 3000 \
    --max-burst-kbits 2400 --egress bw-limiter
+----------------+--------------------------------------+
| Field          | Value                                |
+----------------+--------------------------------------+
| direction      | egress                               |
| id             | 92ceb52f-170f-49d0-9528-976e2fee2d6f |
| max_burst_kbps | 2400                                 |
| max_kbps       | 3000                                 |
```

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```
| name            | None                                  |
| project_id      |                                       |
+-----------------+---------------------------------------+
```

Note

The QoS implementation requires a burst value to ensure proper behavior of bandwidth limit rules in the Open vSwitch agent. Configuring the proper burst value is very important. If the burst value is set too low, bandwidth usage will be throttled even with a proper bandwidth limit setting. This issue is discussed in various documentation sources, for example in Junipers documentation. For TCP traffic it is recommended to set burst value as 80% of desired bandwidth limit value. For example, if the bandwidth limit is set to 1000kbps then enough burst value will be 800kbit. If the configured burst value is too low, achieved bandwidth limit will be lower than expected. If the configured burst value is too high, too few packets could be limited and achieved bandwidth limit would be higher than expected. If you do not provide a value, it defaults to 80% of the bandwidth limit which works for typical TCP traffic.

Second, associate the created policy with an existing neutron port. In order to do this, user extracts the port id to be associated to the already created policy. In the next example, we will assign the `bw-limiter` policy to the VM with IP address `192.0.2.1`.

```
$ openstack port list
+--------------------------------------+-----------------------------------+
| ID                                   | Fixed IP Addresses                |
+--------------------------------------+-----------------------------------+
| 0271d1d9-1b16-4410-bd74-82cdf6dcb5b3 | { ... , "ip_address": "192.0.2.1"}|
| 88101e57-76fa-4d12-b0e0-4fc7634b874a | { ... , "ip_address": "192.0.2.3"}|
| e04aab6a-5c6c-4bd9-a600-33333551a668 | { ... , "ip_address": "192.0.2.2"}|
+--------------------------------------+-----------------------------------+

$ openstack port set --qos-policy bw-limiter \
    88101e57-76fa-4d12-b0e0-4fc7634b874a
```

In order to detach a port from the QoS policy, simply update again the port configuration.

```
$ openstack port unset --qos-policy 88101e57-76fa-4d12-b0e0-4fc7634b874a
```

Ports can be created with a policy attached to them too.

```
$ openstack port create --qos-policy bw-limiter --network private port1
+-----------------------+--------------------------------------------------+
| Field                 | Value                                            |
+-----------------------+--------------------------------------------------+
| admin_state_up        | UP                                               |
| allowed_address_pairs |                                                  |
| binding_host_id       |                                                  |
| binding_profile       |                                                  |
| binding_vif_details   |                                                  |
| binding_vif_type      | unbound                                          |
```

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```
| binding_vnic_type     | normal                                        |
| created_at            | 2017-05-15T08:43:00Z                          |
| data_plane_status     | None                                          |
| description           |                                               |
| device_id             |                                               |
| device_owner          |                                               |
| dns_assignment        | None                                          |
| dns_name              | None                                          |
| extra_dhcp_opts       |                                               |
| fixed_ips             | ip_address='10.0.10.4', subnet_id='292f8c1e-...' |
| id                    | f51562ee-da8d-42de-9578-f6f5cb248226          |
| ip_address            | None                                          |
| mac_address           | fa:16:3e:d9:f2:ba                             |
| name                  | port1                                         |
| network_id            | 55dc2f70-0f92-4002-b343-ca34277b0234          |
| option_name           | None                                          |
| option_value          | None                                          |
| port_security_enabled | False                                         |
| project_id            | 4db7c1ed114a4a7fb0f077148155c500              |
| qos_policy_id         | 5df855e9-a833-49a3-9c82-c0839a5f103f          |
| revision_number       | 6                                             |
| security_group_ids    | 0531cc1a-19d1-4cc7-ada5-49f8b08245be          |
| status                | DOWN                                          |
| subnet_id             | None                                          |
| tags                  | []                                            |
| trunk_details         | None                                          |
| updated_at            | 2017-05-15T08:43:00Z                          |
+-----------------------+-----------------------------------------------+
```

You can attach networks to a QoS policy. The meaning of this is that any compute port connected to the network will use the network policy by default unless the port has a specific policy attached to it. Internal network owned ports like DHCP and internal router ports are excluded from network policy application.

In order to attach a QoS policy to a network, update an existing network, or initially create the network attached to the policy.

```
$ openstack network set --qos-policy bw-limiter private
```

The created policy can be associated with an existing floating IP. In order to do this, user extracts the floating IP id to be associated to the already created policy. In the next example, we will assign the `bw-limiter` policy to the floating IP address `172.16.100.18`.

```
$ openstack floating ip list
+--------------------------------------+---------------------+----------------
↪--+------+-----+
| ID                                   | Floating IP Address | Fixed IP␣
↪Address | Port | ... |
+--------------------------------------+---------------------+----------------
↪--+------+-----+
| 1163d127-6df3-44bb-b69c-c0e916303eb3 | 172.16.100.9        | None           ␣
```

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```
↪   | None | ... |
| d0ed7491-3eb7-4c4f-a0f0-df04f10a067c | 172.16.100.18        | None                ␣
↪   | None | ... |
| f5a9ed48-2e9f-411c-8787-2b6ecd640090 | 172.16.100.2         | None                ␣
↪   | None | ... |
+--------------------------------------+----------------------+----------------
↪--+------+-----+
```

```
$ openstack floating ip set --qos-policy bw-limiter d0ed7491-3eb7-4c4f-a0f0-
↪df04f10a067c
```

In order to detach a floating IP from the QoS policy, simply update the floating IP configuration.

```
$ openstack floating ip set --no-qos-policy d0ed7491-3eb7-4c4f-a0f0-
↪df04f10a067c
```

Or use the `unset` action.

```
$ openstack floating ip unset --qos-policy d0ed7491-3eb7-4c4f-a0f0-
↪df04f10a067c
```

Floating IPs can be created with a policy attached to them too.

```
$ openstack floating ip create --qos-policy bw-limiter public
+---------------------+--------------------------------------+
| Field               | Value                                |
+---------------------+--------------------------------------+
| created_at          | 2017-12-06T02:12:09Z                 |
| description         |                                      |
| fixed_ip_address    | None                                 |
| floating_ip_address | 172.16.100.12                        |
| floating_network_id | 4065eb05-cccb-4048-988c-e8c5480a746f |
| id                  | 6a0efeef-462b-4312-b4ad-627cde8a20e6 |
| name                | 172.16.100.12                        |
| port_id             | None                                 |
| project_id          | 916e39e8be52433ba040da3a3a6d0847     |
| qos_policy_id       | 5df855e9-a833-49a3-9c82-c0839a5f103f |
| revision_number     | 1                                    |
| router_id           | None                                 |
| status              | DOWN                                 |
| updated_at          | 2017-12-06T02:12:09Z                 |
+---------------------+--------------------------------------+
```

The QoS bandwidth limit rules attached to a floating IP will become active when you associate the latter with a port. For example, to associate the previously created floating IP `172.16.100.12` to the instance port with uuid `a7f25e73-4288-4a16-93b9-b71e6fd00862` and fixed IP `192.168.222.5`:

```
$ openstack floating ip set --port a7f25e73-4288-4a16-93b9-b71e6fd00862 \
    0eeb1f8a-de96-4cd9-a0f6-3f535c409558
```

Note

The QoS policy attached to a floating IP is not applied to a port, it is applied to an associated floating IP only. Thus the ID of QoS policy attached to a floating IP will not be visible in a ports `qos_policy_id` field after asscoating a floating IP to the port. It is only visible in the floating IP attributes.

Note

For now, the L3 agent floating IP QoS extension only supports `bandwidth_limit` rules. Other rule types (like DSCP marking) will be silently ignored for floating IPs. A QoS policy that does not contain any `bandwidth_limit` rules will have no effect when attached to a floating IP.

If floating IP is bound to a port, and both have binding QoS bandwidth rules, the L3 agent floating IP QoS extension ignores the behavior of the port QoS, and installs the rules from the QoS policy associated to the floating IP on the appropriate device in the router namespace.

Each project can have at most one default QoS policy, although it is not mandatory. If a default QoS policy is defined, all new networks created within this project will have this policy assigned, as long as no other QoS policy is explicitly attached during the creation process. If the default QoS policy is unset, no change to existing networks will be made.

In order to set a QoS policy as default, the parameter `--default` must be used. To unset this QoS policy as default, the parameter `--no-default` must be used.

```
$ openstack network qos policy create --default bw-limiter
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| description       |                                      |
| id                | 5df855e9-a833-49a3-9c82-c0839a5f103f |
| is_default        | True                                 |
| name              | bw-limiter                           |
| project_id        | 4db7c1ed114a4a7fb0f077148155c500     |
| rules             | []                                   |
| shared            | False                                |
+-------------------+--------------------------------------+

$ openstack network qos policy set --no-default bw-limiter
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| description       |                                      |
| id                | 5df855e9-a833-49a3-9c82-c0839a5f103f |
| is_default        | False                                |
| name              | bw-limiter                           |
| project_id        | 4db7c1ed114a4a7fb0f077148155c500     |
| rules             | []                                   |
| shared            | False                                |
+-------------------+--------------------------------------+
```

Create qos policy with packet rate limit rules:

```
$ openstack network qos policy create pps-limiter
+-------------+--------------------------------------+
| Field       | Value                                |
+-------------+--------------------------------------+
| description |                                      |
| id          | 97f0ac37-7dd6-4579-8359-3bef0751a505 |
| is_default  | False                                |
| name        | pps-limiter                          |
| project_id  | 1d70739f831b421fb38a27adb368fc17     |
| rules       | []                                   |
| shared      | False                                |
| tags        | []                                   |
+-------------+--------------------------------------+

$ openstack network qos rule create --max-kpps 1000 --max-burst-kpps 100 --
↪ingress --type packet-rate-limit pps-limiter
+----------------+--------------------------------------+
| Field          | Value                                |
+----------------+--------------------------------------+
| direction      | ingress                              |
| id             | 4a1cb166-9661-48d7-bddb-00b7d75846cd |
| max_burst_kpps | 100                                  |
| max_kpps       | 1000                                 |
| name           | None                                 |
| project_id     |                                      |
+----------------+--------------------------------------+

$ openstack network qos rule create --max-kpps 1000 --max-burst-kpps 100 --
↪egress --type packet-rate-limit pps-limiter
+----------------+--------------------------------------+
| Field          | Value                                |
+----------------+--------------------------------------+
| direction      | egress                               |
| id             | 6abd67f7-0bde-4ad3-ac54-b0a6103b0449 |
| max_burst_kpps | 100                                  |
| max_kpps       | 1000                                 |
| name           | None                                 |
| project_id     |                                      |
+----------------+--------------------------------------+
```

Note

The unit for the rate and burst is kilo (1000) packets per second.

Now apply the packet rate limit QoS policy to a Port:

```
$ openstack port set --qos-policy pps-limiter 251948bd-08e4-4569-a47f-
↪ecbc1fd4af4d
```

Note

Packet rate limit is only supported by the ml2 ovs driver. And it leverages the meter actions of the ovs kernel datapath or the userspace ovs dpdk datapath. The meter action is only supported when the datapath is in user mode or ovs kernel datapath with kernerl version >= 4.15.

Administrator enforcement

Administrators are able to enforce policies on project ports or networks. As long as the policy is not shared, the project is not be able to detach any policy attached to a network or port.

If the policy is shared, the project is able to attach or detach such policy from its own ports and networks.

Rule modification

You can modify rules at runtime. Rule modifications will be propagated to any attached port.

```
$ openstack network qos rule set --max-kbps 2000 --max-burst-kbits 1600 \
    --ingress bw-limiter 92ceb52f-170f-49d0-9528-976e2fee2d6f

$ openstack network qos rule show \
    bw-limiter 92ceb52f-170f-49d0-9528-976e2fee2d6f
+----------------+--------------------------------------+
| Field          | Value                                |
+----------------+--------------------------------------+
| direction      | ingress                              |
| id             | 92ceb52f-170f-49d0-9528-976e2fee2d6f |
| max_burst_kbps | 1600                                 |
| max_kbps       | 2000                                 |
| name           | None                                 |
| project_id     |                                      |
+----------------+--------------------------------------+
```

Just like with bandwidth limiting, create a policy for DSCP marking rule:

```
$ openstack network qos policy create dscp-marking
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| description       |                                      |
| id                | d1f90c76-fbe8-4d6f-bb87-a9aea997ed1e |
| is_default        | False                                |
| name              | dscp-marking                         |
| project_id        | 4db7c1ed114a4a7fb0f077148155c500     |
| rules             | []                                   |
| shared            | False                                |
+-------------------+--------------------------------------+
```

You can create, update, list, delete, and show DSCP markings with the neutron client:

```
$ openstack network qos rule create --type dscp-marking --dscp-mark 26 \
    dscp-marking
+----------------+--------------------------------------+
| Field          | Value                                |
+----------------+--------------------------------------+
| dscp_mark      | 26                                   |
| id             | 115e4f70-8034-4176-8fe9-2c47f8878a7d |
| name           | None                                 |
| project_id     |                                      |
+----------------+--------------------------------------+
```

```
$ openstack network qos rule set --dscp-mark 22 \
    dscp-marking 115e4f70-8034-4176-8fe9-2c47f8878a7d

$ openstack network qos rule list dscp-marking
+--------------------------------------+----------------------------------+
| ID                                   | DSCP Mark                        |
+--------------------------------------+----------------------------------+
| 115e4f70-8034-4176-8fe9-2c47f8878a7d | 22                               |
+--------------------------------------+----------------------------------+

$ openstack network qos rule show \
    dscp-marking 115e4f70-8034-4176-8fe9-2c47f8878a7d
+----------------+--------------------------------------+
| Field          | Value                                |
+----------------+--------------------------------------+
| dscp_mark      | 22                                   |
| id             | 115e4f70-8034-4176-8fe9-2c47f8878a7d |
| name           | None                                 |
| project_id     |                                      |
+----------------+--------------------------------------+

$ openstack network qos rule delete \
    dscp-marking 115e4f70-8034-4176-8fe9-2c47f8878a7d
```

You can also include minimum bandwidth rules in your policy:

```
$ openstack network qos policy create bandwidth-control
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| description       |                                      |
| id                | 8491547e-add1-4c6c-a50e-42121237256c |
| is_default        | False                                |
| name              | bandwidth-control                    |
| project_id        | 7cc5a84e415d48e69d2b06aa67b317d8     |
| revision_number   | 1                                    |
| rules             | []                                   |
| shared            | False                                |
+-------------------+--------------------------------------+
```

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```
$ openstack network qos rule create \
  --type minimum-bandwidth --min-kbps 1000 --egress bandwidth-control
+------------+--------------------------------------+
| Field      | Value                                |
+------------+--------------------------------------+
| direction  | egress                               |
| id         | da858b32-44bc-43c9-b92b-cf6e2fa836ab |
| min_kbps   | 1000                                 |
| name       | None                                 |
| project_id |                                      |
+------------+--------------------------------------+
```

A policy with a minimum bandwidth ensures best efforts are made to provide no less than the specified bandwidth to each port on which the rule is applied. However, as this feature is not yet integrated with the Compute scheduler, minimum bandwidth cannot be guaranteed.

It is also possible to combine several rules in one policy, as long as the type or direction of each rule is different. For example, You can specify two `bandwidth-limit` rules, one with `egress` and one with `ingress` direction.

```
$ openstack network qos rule create --type bandwidth-limit \
    --max-kbps 50000 --max-burst-kbits 50000 --egress bandwidth-control
+----------------+--------------------------------------+
| Field          | Value                                |
+----------------+--------------------------------------+
| direction      | egress                               |
| id             | 0db48906-a762-4d32-8694-3f65214c34a6 |
| max_burst_kbps | 50000                                |
| max_kbps       | 50000                                |
| name           | None                                 |
| project_id     |                                      |
+----------------+--------------------------------------+

$ openstack network qos rule create --type bandwidth-limit \
    --max-kbps 10000 --max-burst-kbits 10000 --ingress bandwidth-control
+----------------+--------------------------------------+
| Field          | Value                                |
+----------------+--------------------------------------+
| direction      | ingress                              |
| id             | faabef24-e23a-4fdf-8e92-f8cb66998834 |
| max_burst_kbps | 10000                                |
| max_kbps       | 10000                                |
| name           | None                                 |
| project_id     |                                      |
+----------------+--------------------------------------+

$ openstack network qos rule create --type minimum-bandwidth \
    --min-kbps 1000 --egress bandwidth-control
+------------+--------------------------------------+
```

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```
| Field       | Value                                |
+-------------+--------------------------------------+
| direction   | egress                               |
| id          | da858b32-44bc-43c9-b92b-cf6e2fa836ab |
| min_kbps    | 1000                                 |
| name        | None                                 |
| project_id  |                                      |
+-------------+--------------------------------------+

$ openstack network qos policy show bandwidth-control
+-------------------+-------------------------------------------------------------
↪----------+
| Field             | Value                                                       ␣
↪          |
+-------------------+-------------------------------------------------------------
↪----------+
| description       |                                                             ␣
↪          |
| id                | 8491547e-add1-4c6c-a50e-42121237256c                        ␣
↪          |
| is_default        | False                                                       ␣
↪          |
| name              | bandwidth-control                                           ␣
↪          |
| project_id        | 7cc5a84e415d48e69d2b06aa67b317d8                            ␣
↪          |
| revision_number   | 4                                                           ␣
↪          |
| rules             | [{u'max_kbps': 50000, u'direction': u'egress',              ␣
↪          |
|                   |   u'type': u'bandwidth_limit',                              ␣
↪          |
|                   |   u'id': u'0db48906-a762-4d32-8694-3f65214c34a6',           ␣
↪          |
|                   |   u'max_burst_kbps': 50000,                                 ␣
↪          |
|                   |   u'qos_policy_id': u'8491547e-add1-4c6c-a50e-
↪42121237256c'},       |
|                   | [{u'max_kbps': 10000, u'direction': u'ingress',             ␣
↪          |
|                   |   u'type': u'bandwidth_limit',                              ␣
↪          |
|                   |   u'id': u'faabef24-e23a-4fdf-8e92-f8cb66998834',           ␣
↪          |
|                   |   u'max_burst_kbps': 10000,                                 ␣
↪          |
|                   |   u'qos_policy_id': u'8491547e-add1-4c6c-a50e-
↪42121237256c'},       |
|                   |  {u'direction':                                             ␣
```

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```
↪          |
|                 |   u'egress', u'min_kbps': 1000, u'type': u'minimum_
↪bandwidth',     |
|                 |   u'id': u'da858b32-44bc-43c9-b92b-cf6e2fa836ab',     ␣
↪          |
|                 |   u'qos_policy_id': u'8491547e-add1-4c6c-a50e-
↪42121237256c'}]      |
| shared          | False                                                  ␣
↪          |
+-----------------+-----------------------------------------------------
↪----------+
```

8.2.28 Quality of Service (QoS): Guaranteed Minimum Bandwidth

Most Networking Quality of Service (QoS) features are implemented solely by OpenStack Neutron and they are already documented in the *QoS configuration chapter of the Networking Guide*. Some more complex QoS features necessarily involve the scheduling of a cloud server, therefore their implementation is shared between OpenStack Nova, Neutron and Placement. As of the OpenStack Stein release the Guaranteed Minimum Bandwidth feature is like the latter.

This Networking Guide chapter does not aim to replace Nova or Placement documentation in any way, but it still hopes to give an overall OpenStack-level guide to understanding and configuring a deployment to use the Guaranteed Minimum Bandwidth feature.

A guarantee of minimum available bandwidth can be enforced on two levels:

- Scheduling a server on a compute host where the bandwidth is available. To be more precise: scheduling one or more ports of a server on a compute hosts physical network interfaces where the bandwidth is available.
- Queueing network packets on a physical network interface to provide the guaranteed bandwidth.

In short the enforcement has two levels:

- (server) placement and
- data plane.

Since the data plane enforcement is already documented in the *QoS chapter*, here we only document the placement-level enforcement.

Limitations

- A pre-created port with a `minimum-bandwidth` rule must be passed when booting a server (`openstack server create`). Passing a network with a minimum-bandwidth rule at boot is not supported because of technical reasons (in this case the port is created too late for Neutron to affect scheduling).
- In Stein there is no support for networks with multiple physnets. However some simpler multi-segment networks are still supported:
  - Networks with multiple segments all having the same physnet name.
  - Networks with only one physnet segment (the other segments being tunneled segments).

- If you mix ports with and without bandwidth guarantees on the same physical interface then the ports without a guarantee may starve. Therefore mixing them is not recommended. Instead it is recommended to separate them by Nova host aggregates.
- Changing the guarantee of a QoS policy (adding/deleting a `minimum_bandwidth` rule, or changing the `min_kbps` field of a `minimum_bandwidth` rule) is only possible while the policy is not in effect. That is ports of the QoS policy are not yet used by Nova. Requests to change guarantees of in-use policies are rejected.
- Changing the QoS policy of the port with new `minimum_bandwidth` rules changes placement `allocations` from Wallaby release. If the VM was booted with port without QoS policy and `minimum_bandwidth` rules the port update succeeds but placement allocations will not change. The same is true if the port has no `binding:profile`, thus no placement allocation record exists for it. But if the VM was booted with a port with QoS policy and `minimum_bandwidth` rules the update is possible and the allocations are changed in placement as well.

Note

As it is possible to update a port to remove the QoS policy, updating it back to have QoS policy with `minimum_bandwidth` rule will not result in `placement allocation` record, only the dataplane enforcement will happen.

Note

updating the `minimum_bandwidth` rule of a QoS policy that is attached to a port which is bound to a VM is still not possible.

- The first data-plane-only Guaranteed Minimum Bandwidth implementation (for SR-IOV egress traffic) was released in the Newton release of Neutron. Because of the known lack of placement-level enforcement it was marked as best effort (5th bullet point). Since placement-level enforcement was not implemented bandwidth may have become overallocated and the system level resource inventory may have become inconsistent. Therefore for users of the data-plane-only implementation a migration/healing process is mandatory (see section *On Healing of Allocations*) to bring the system level resource inventory to a consistent state. Further operations that would reintroduce inconsistency (e.g. migrating a server with `minimum_bandwidth` QoS rule, but no resource allocation in Placement) are rejected now in a backward-incompatible way.
- The Guaranteed Minimum Bandwidth feature is not complete in the Stein release. Not all Nova server lifecycle operations can be executed on a server with bandwidth guarantees. Since Stein (Nova API microversion 2.72+) you can boot and delete a server with a guarantee and detach a port with a guarantee. Since Train you can also migrate and resize a server with a guarantee. Support for further server move operations (for example evacuate, live-migrate and unshelve after shelve-offload) is to be implemented later. For the definitive documentation please refer to the Port with Resource Request chapter of the OpenStack Compute API Guide.
- If an SR-IOV physical function is configured for use by the neutron-openvswitch-agent, and the same physical functions virtual functions are configured for use by the neutron-sriov-agent then the available bandwidth must be statically split between the corresponding resource providers by administrative choice. For example a 10 Gbps SR-IOV capable physical NIC could be treated as two independent NICs - a 5 Gbps NIC (technically the physical function of the NIC) added to an Open vSwitch bridge, and another 5 Gbps NIC whose virtual functions can be handed out to servers by neutron-sriov-agent.

- Neutron allows physnet names to be case sensitive. So physnet0 and Physnet0 are treated as different physnets. Physnets are mapped to traits in Placement for scheduling purposes. However Placement traits are case insensitive and normalized to full capital. Therefore the scheduling treats physnet0 and Physnet0 as the same physnet. It is advised not to use physnet names that are only differ by case.
- There are hardware platforms (e.g.: Cavium ThunderX) where its possible to have virtual functions which are network devices that are not associated to a physical function. As bandwidth resources are tracked per physical function, for such hardware the placement enforcement of the QoS minimum bandwidth rules cannot be supported. Creating a server with ports using such QoS policy targeting such hardware backend will result in a `NoValidHost` error during scheduling.
- When QoS is used with a trunk, Placement enforcement is applied only to the trunks parent port. Subports are not going to have Placement allocation. As a workaround, parent ports QoS policy should take into account subports needs and request enough minimum bandwidth resources to accommodate every port in the trunk.

Placement pre-requisites

Placement must support microversion 1.29. This was first released in Rocky.

Nova pre-requisites

Nova must support microversion 2.72. This was first released in Stein.

Not all Nova virt drivers are supported, please refer to the Virt Driver Support section of the Nova Admin Guide.

Neutron pre-requisites

Neutron must support the following API extensions:

- `agent-resources-synced`
- `port-resource-request`
- `qos-bw-minimum-ingress`

These were all first released in Stein.

Supported drivers and agents

In release Stein the following agent-based ML2 mechanism drivers are supported:

- Open vSwitch (`openvswitch`) vnic_types: `normal`, `direct`
- SR-IOV (`sriovnicswitch`) vnic_types: `direct`, `macvtap`, `direct-physical`
- OVN (`ovn`) vnic_types: `normal`

Note

SR-IOV (`sriovnicswitch`) agent does not handle `direct-physical` ports. However the agent can report the bandwidth capacity of a network device that will be used by a `direct-physical` port.

Since 2023.1 (Antelope), Open vSwitch and OVN mechanism drivers can specify the available bandwidth for tunnelled networks (SR-IOV does not support these network types yet). The key rp_tunnelled is used to model those networks that are not backed by a physical network. This bandwidth models the limits of the VTEP/TEP interface used to send the tunnelled traffic (VXLAN, Geneve).

neutron-server config

The `placement` service plugin synchronizes the agents resource provider information from neutron-server to Placement.

Since neutron-server talks to Placement you need to configure how neutron-server should find Placement and authenticate to it.

`/etc/neutron/neutron.conf` (on controller nodes):

```
[DEFAULT]
service_plugins = placement,...
auth_strategy = keystone

[placement]
auth_type = password
auth_url = https://controller/identity
password = secret
project_domain_name = Default
project_name = service
user_domain_name = Default
username = placement
```

If a vnic_type is supported by default by multiple ML2 mechanism drivers (e.g. `vnic_type=direct` by both `openvswitch` and `sriovnicswitch`) and multiple agents resources are also meant to be tracked by Placement, then the admin must decide which driver to take ports of that vnic_type by prohibiting the vnic_type for the unwanted drivers. Use *ovs_driver.vnic_type_prohibit_list* in this case. Valid values are all the `supported_vnic_types` of the respective mechanism drivers.

`/etc/neutron/plugins/ml2/ml2_conf.ini` (on controller nodes):

```
[ovs_driver]
vnic_type_prohibit_list = direct

[sriov_driver]
#vnic_type_prohibit_list = direct
```

neutron-openvswitch-agent config

Set the agent configuration as the authentic source of the resources available. Set it on a per-bridge basis by *ovs.resource_provider_bandwidths*. The format is: `bridge:egress:ingress,...` You may set only one direction and omit the other.

Note

`egress` / `ingress` is meant from the perspective of a cloud server. That is `egress` = cloud server upload, `ingress` = download.

Egress and ingress available bandwidth values are in `kilobit/sec (kbps)`.

If desired, resource provider inventory fields can be tweaked on a per-agent basis by setting *ovs.resource_provider_inventory_defaults*. Valid values are all the optional parameters of the update resource provider inventory call.

/etc/neutron/plugins/ml2/ovs_agent.ini (on compute and network nodes):

```
[ovs]
bridge_mappings = physnet0:br-physnet0,...
resource_provider_bandwidths = br-physnet0:10000000:10000000,rp_
↪tunnelled:20000000:20000000,...
#resource_provider_inventory_defaults = step_size:1000,...
```

Note

rp_tunnelled is not a bridge nor an interface present in the host. The ML2/OVS agent will read the host local resource_provider_bandwidths and will assign, by default, the rp_tunnelled resource provider to the local host where is running. In other words, it is not needed to populate resource_provider_hypervisors with the host assigned to this specific resource provider.

neutron-sriov-agent config

The configuration of neutron-sriov-agent is analog to that of neutron-openvswitch-agent. However look out for:

- The different .ini section names as you can see below.
- That neutron-sriov-agent allows a physnet to be backed by multiple physical devices.
- Of course refer to SR-IOV physical functions instead of bridges in *sriov_nic.resource_provider_bandwidths*.

/etc/neutron/plugins/ml2/sriov_agent.ini (on compute nodes):

```
[sriov_nic]
physical_device_mappings = physnet0:ens5,physnet0:ens6,...
resource_provider_bandwidths = ens5:40000000:40000000,ens6:40000000:40000000,.
↪..
#resource_provider_inventory_defaults = step_size:1000,...
```

OVN chassis config

Bandwidth config values are stored in each SB chassis register, in external_ids:ovn-cms-options. The configuration options are the same as in SR-IOV and OVS agents. This is how the values are registered:

```
$ root@dev20:~# ovs-vsctl list Open_vSwitch
  ...
  external_ids        : {hostname=dev20.fistro.com, \
                         ovn-cms-options="resource_provider_bandwidths=br-
↪ex:1001:2000;br-ex2:3000:4000;rp_tunnelled:5000:6000, \
                                   resource_provider_inventory_
↪defaults=allocation_ratio:1.0;min_unit:10, \
                                   resource_provider_hypervisors=br-
```

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```
↪ex:dev20.fistro.com;br-ex2:dev20.fistro.com;rp_tunnelled:dev20.fistro.com",␣
↪\
                         rundir="/var/run/openvswitch", \
                         system-id="029e7d3d-d2ab-4f2c-bc92-ec58c94a8fc1"}
  ...
```

Each configuration option defined in external_ids:ovn-cms-options is divided by commas.

This information is retrieved from the OVN SB database during the Neutron server initialization and when the Chassis registers are updated.

The initial Placement configuration is retrieved when the Neutron API receives a Chassis create event, that happens when the IDL is connected to the database server. When a creation event is received, the Neutron API reads the configuration, builds a `PlacementState` instance and sends it to the Placement API.

The second method to update the Placement information is when a Chassis registers is updated. The `OVNClientPlacementExtension` extension registers an event handler that attends the OVN SB Chassis bandwidth configuration changes. This event handler builds a `PlacementState` instance and sends it to the Placement API. If a new chassis is added or an existing one changes its resource provider configuration, this event updates it in the Placement database.

Propagation of resource information

The flow of information is different for available and used resources.

The authentic source of available resources is neutron agent configuration - where the resources actually exist, as described in the agent configuration sections above. This information is propagated in the following chain: `neutron-l2-agent -> neutron-server -> Placement`.

From neutron agent to server the information is included in the `configurations` field of the agent heartbeat message sent on the message queue periodically.

```
# as admin
$ openstack network agent list --agent-type open-vswitch --host devstack0
+--------------------------------------+--------------------+-----------+-----
↪--------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host      |␣
↪Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+-----------+-----
↪--------------+-------+-------+---------------------------+
| 5e57b85f-b017-419a-8745-9c406e149f9e | Open vSwitch agent | devstack0 |␣
↪None              | :-)   | UP    | neutron-openvswitch-agent |
+--------------------------------------+--------------------+-----------+-----
↪--------------+-------+-------+---------------------------+

# output shortened and pretty printed
# note: 'configurations' on the wire, but 'configuration' in the cli
$ openstack network agent show -f value -c configuration 5e57b85f-b017-419a-
↪8745-9c406e149f9e
{'bridge_mappings': {'physnet0': 'br-physnet0'},
 'resource_provider_bandwidths': {'br-physnet0': {'egress': 10000000,
```

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```
                                        'ingress': 10000000}
                              'rp_tunnelled': {'egress': 20000000,
                                        'ingress': 20000000}},
 'resource_provider_inventory_defaults': {'allocation_ratio': 1.0,
                                          'min_unit': 1,
                                          'reserved': 0,
                                          'step_size': 1},
 ...
}
```

Re-reading the resource related subset of configuration on `SIGHUP` is not implemented. The agent must be restarted to pick up and send changed configuration.

Neutron-server propagates the information further to Placement for the resources of each agent via Placements HTTP REST API. To avoid overloading Placement this synchronization generally does not happen on every received heartbeat message. Instead the re-synchronization of the resources of one agent is triggered by:

- The creation of a network agent record (as queried by `openstack network agent list`). Please note that deleting an agent record and letting the next heartbeat to re-create it can be used to trigger synchronization without restarting an agent.
- The restart of that agent (technically `start_flag` being present in the heartbeat message).

Both of these can be used by an admin to force a re-sync if needed.

The success of a synchronization attempt from neutron-server to Placement is persisted into the relevant agents `resources_synced` attribute. For example:

```
# as admin
$ openstack network agent show -f value -c resources_synced 5e57b85f-b017-
↪419a-8745-9c406e149f9e
True
```

`resources_synced` may take the value True, False and None:

- None: No sync was attempted (normal for agents not reporting Placement-backed resources).
- True: The last sync attempt was completely successful.
- False: The last sync attempt was partially or utterly unsuccessful.

In case `resources_synced` is not True for an agent, neutron-server does try to re-sync on receiving every heartbeat message from that agent. Therefore it should be able to recover from transient errors of Neutron-Placement communication (e.g. Placement being started later than Neutron).

It is important to note that the restart of neutron-server does not trigger any kind of re-sync to Placement (to avoid an update storm).

As mentioned before, the information flow for resources requested and (if proper) allocated is different. It involves a conversation between Nova, Neutron and Placement.

1. Neutron exposes a ports resource needs in terms of resource classes and traits as the admin-only `resource_request` attribute of that port.
2. Nova reads this and incorporates it as a numbered request group into the cloud servers overall allocation candidate request to Placement.

3. Nova selects (schedules) and allocates one candidate returned by Placement.
4. Nova informs Neutron when binding the port of which physical network interface resource provider had been selected for the ports resource request in the `binding:profile.allocation` sub-attribute of that port.

For details please see slides 13-15 of a (pre-release) demo that was presented on the Berlin Summit in November 2018.

Since Yoga, the `resource_request` attribute of the port changed. With the extension `port-resource-request-groups`, Neutron informs that the blob passed to Nova can contain several bandwidth requests. Please check resource_request sanitization.

Sample usage

Physnets and QoS policies (together with their rules) are usually pre-created by a cloud admin:

```
# as admin

$ openstack network create net0 \
    --provider-network-type vlan \
    --provider-physical-network physnet0 \
    --provider-segment 100

$ openstack subnet create subnet0 \
    --network net0 \
    --subnet-range 10.0.4.0/24

$ openstack network qos policy create policy0

$ openstack network qos rule create policy0 \
    --type minimum-bandwidth \
    --min-kbps 1000000 \
    --egress

$ openstack network qos rule create policy0 \
    --type minimum-bandwidth \
    --min-kbps 1000000 \
    --ingress
```

Then a normal user can use the pre-created policy to create ports and boot servers with those ports:

```
# as an unprivileged user

# an ordinary soft-switched port: ``--vnic-type normal`` is the default
$ openstack port create port-normal-qos \
    --network net0 \
    --qos-policy policy0

# alternatively an SR-IOV port, unused in this example
$ openstack port create port-direct-qos \
    --network net0 \
    --vnic-type direct \
```

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```
    --qos-policy policy0

$ openstack server create server0 \
    --flavor cirros256 \
    --image cirros-0.5.1-x86_64-disk \
    --port port-normal-qos
```

On Healing of Allocations

Since Placement carries a global view of a cloud deployments resources (what is available, what is used) it may in some conditions get out of sync with reality.

One important case is when the data-plane-only Minimum Guaranteed Bandwidth feature was used before Stein (first released in Newton). Since before Stein guarantees were not enforced during server placement the available resources may have become overallocated without notice. In this case Placements view and the reality of resource usage should be made consistent during/after an upgrade to Stein.

Another case stems from OpenStack not having distributed transactions to allocate resources provided by multiple OpenStack components (here Nova and Neutron). There are known race conditions in which Placements view may get out of sync with reality. The design knowingly minimizes the race condition windows, but there are known problems:

- If a QoS policy is modified after Nova read a ports `resource_request` but before the port is bound its state before the modification will be applied.
- If a bound port with a resource allocation is deleted. The ports allocation is leaked. https://bugs.launchpad.net/nova/+bug/1820588

Note

Deleting a bound port has no known use case. Please consider detaching the interface first by `openstack server remove port` instead.

Incorrect allocations may be fixed by:

- Moving the server, which will delete the wrong allocation and create the correct allocation as soon as move operations are implemented (not in Stein unfortunately). Moving servers fixes local overallocations.
- The need for an upgrade-helper allocation healing tool is being tracked in bug 1819923.
- Manually, by using openstack resource provider allocation set /delete.

Debugging

- Are all components running at least the Stein release?
- Is the `placement` service plugin enabled in neutron-server?
- Is `resource_provider_bandwidths` configured for the relevant neutron agent?
- Is `resource_provider_bandwidths` aligned with `bridge_mappings` or `physical_device_mappings`?
- Was the agent restarted since changing the configuration file?

- Is `resource_provider_bandwidths` reaching neutron-server?

```
# as admin
$ openstack network agent show ... | grep configurations
```

Please find an example in section *Propagation of resource information*.

- Did neutron-server successfully sync to Placement?

```
# as admin
$ openstack network agent show ... | grep resources_synced
```

Please find an example in section *Propagation of resource information*.

- Is the resource provider tree correct? Is the root a compute host? One level below the agents? Two levels below the physical network interfaces?

```
$ openstack --os-placement-api-version 1.17 resource provider list
+--------------------------------------+------------------------------------------
↪----+------------+--------------------------------------+------------------
↪------------------+
| uuid                                 | name                                     ␣
↪    | generation | root_provider_uuid                   | parent_provider_
↪uuid                 |
+--------------------------------------+------------------------------------------
↪----+------------+--------------------------------------+------------------
↪------------------+
| 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | devstack0                                ␣
↪    |          2 | 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | None                 ␣
↪                  |
| 4a8a819d-61f9-5822-8c5c-3e9c7cb942d6 | devstack0:NIC Switch agent               ␣
↪    |          0 | 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | 3b36d91e-bf60-
↪460f-b1f8-3322dee5cdfd |
| 1c7e83f0-108d-5c35-ada7-7ebebbe43aad | devstack0:NIC Switch agent:ens5          ␣
↪    |          2 | 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | 4a8a819d-61f9-
↪5822-8c5c-3e9c7cb942d6 |
| 89ca1421-5117-5348-acab-6d0e2054239c | devstack0:Open vSwitch agent             ␣
↪    |          0 | 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | 3b36d91e-bf60-
↪460f-b1f8-3322dee5cdfd |
| f9c9ce07-679d-5d72-ac5f-31720811629a | devstack0:Open vSwitch agent:br-
↪physnet0 |          2 | 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | 89ca1421-
↪5117-5348-acab-6d0e2054239c |
| 521f53a6-c8c0-583c-98da-7a47f39ff887 | devstack0:Open vSwitch agent:rp_
↪tunnelled|          2 | 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | 89ca1421-
↪5117-5348-acab-6d0e2054239c |
+--------------------------------------+------------------------------------------
↪----+------------+--------------------------------------+------------------
↪------------------+
```

- Does Placement have the expected traits?

```
# as admin
$ openstack --os-placement-api-version 1.17 trait list | awk '/CUSTOM_/ {␣
```

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```
↪print $2 }' | sort
CUSTOM_PHYSNET_PHYSNET0
CUSTOM_TUNNELLED_NETWORKS
CUSTOM_VNIC_TYPE_DIRECT
CUSTOM_VNIC_TYPE_DIRECT_PHYSICAL
CUSTOM_VNIC_TYPE_MACVTAP
CUSTOM_VNIC_TYPE_NORMAL
```

- Do the physical network interface resource providers have the proper trait associations and inventories?

```
# as admin
$ openstack --os-placement-api-version 1.17 resource provider trait list RP-
↪UUID
$ openstack --os-placement-api-version 1.17 resource provider inventory list␣
↪RP-UUID
```

- Does the QoS policy have a `minimum-bandwidth` rule?
- Does the port have the proper policy?
- Does the port have a `resource_request`?

```
# as admin
$ openstack port show port-normal-qos | grep resource_request
```

- Was the server booted with a port (as opposed to a network)?
- Did nova allocate resources for the server in Placement?

```
# as admin
$ openstack --os-placement-api-version 1.17 resource provider allocation show␣
↪SERVER-UUID
```

- Does the allocation have a part on the expected physical network interface resource provider?

```
# as admin
$ openstack --os-placement-api-version 1.17 resource provider show --
↪allocations RP-UUID
```

- Did placement manage to produce an allocation candidate list to nova during scheduling?
- Did nova manage to schedule the server?
- Did nova tell neutron which physical network interface resource provider was allocated to satisfy the bandwidth request?

```
# as admin
$ openstack port show port-normal-qos | grep binding.profile.*allocation
```

- Did neutron manage to bind the port?

Links

- Pre-release feature demo presented on the Berlin Summit in November 2018
- Nova documentation on using a port with `resource_request`
  - API Guide
  - Admin Guide
- Neutron spec: QoS minimum bandwidth allocation in Placement API
  - on specs.openstack.org
  - on review.opendev.org
- Nova spec: Network Bandwidth resource provider
  - on specs.openstack.org
  - on review.opendev.org
- Nova spec: QoS minimum guaranteed packet rate
  - on specs.openstack.org
- Relevant OpenStack Networking API references
  - https://docs.openstack.org/api-ref/network/v2/#agent-resources-synced-extension
  - https://docs.openstack.org/api-ref/network/v2/#port-resource-request
  - https://docs.openstack.org/api-ref/network/v2/#qos-minimum-bandwidth-rules
- Microversion histories
  - Compute 2.72
  - Placement 1.29
- Implementation
  - on review.opendev.org
- Known Bugs
  - Missing tool to heal allocations
  - Bandwidth resource is leaked

8.2.29 Quality of Service (QoS): Guaranteed Minimum Packet Rate

Similarly to how bandwidth can be a limiting factor of a network interface, packet processing capacity tend to be a limiting factor of the soft switching solutions like OVS. At the same time certain applications are dependent on not just guaranteed bandwidth, but also on guaranteed packet rate to function properly. OpenStack already supports bandwidth guarantees via the minimum bandwidth QoS policy rules, which is described in detail in *Quality of Service (QoS): Guaranteed Minimum Bandwidth*. Its recommended to read Guaranteed Minimum Bandwidth guide first, but its not strictly required.

Just like *Quality of Service (QoS): Guaranteed Minimum Bandwidth* guide, this chapter does not aim to replace Nova or Placement documentation in any way, but gives a brief overview of the feature and explains how it can be configured.

In a similar way to guaranteed bandwidth, we can distinguish two levels of enforcement for guaranteeing packet processing capacity constraint:

- placement: Avoiding over-subscription when placing (scheduling) VMs and their ports.
- data plane: Enforcing the guarantee on the soft switch

> **Note**
>
> At the time of writing this guide, only placement enforcement is supported. For detailed list of supported enforcement types and backends, please refer to *QoS configuration chapter of the Networking Guide*.

The solution needs to differentiate between two different deployment scenarios:

1) The packet processing functionality is implemented on the compute host CPUs and therefore packets processed from both ingress and egress directions are handled by the same set of CPU cores. This is the case in the non-hardware-offloaded OVS deployments. In this scenario OVS represents a single packet processing resource pool, which is represented with a single resource class called `NET_PACKET_RATE_KILOPACKET_PER_SEC`.

2) The packet processing functionality is implemented in a specialized hardware where the incoming and outgoing packets are handled by independent hardware resources. This is the case for hardware-offloaded OVS. In this scenario a single OVS has two independent resource pools one for the incoming packets and one for the outgoing packets. Therefore these needs to be represented with two different resource classes `NET_PACKET_RATE_EGR_KILOPACKET_PER_SEC` and `NET_PACKET_RATE_IGR_KILOPACKET_PER_SEC`.

Limitations

Since Guaranteed Minimum Packet Rate and Guaranteed Minimum Bandwidth features have a lot in common, they also share certain limitations.

- A pre-created port with a `minimum-packet-rate` rule must be passed when booting a server (`openstack server create`). Passing a network with a minimum-packet-rate rule at boot is not supported because of technical reasons (in this case the port is created too late for Neutron to affect scheduling).
- Changing the guarantee of a QoS policy (adding/deleting a `minimum_packet_rate` rule, or changing the `min_kpps` field of a `minimum_packet_rate` rule) is only possible while the policy is not in effect. That is ports of the QoS policy are not yet bound by Nova. Requests to change guarantees of in-use policies are rejected.
- Changing the QoS policy of the port with new `minimum_packet_rate` rules changes placement `allocations` from Yoga release. If the VM was booted with port without QoS policy and `minimum_packet_rate` rules the port update succeeds but placement allocations will not change. The same is true if the port had no allocation record in Placement before QoS policy update. But if the VM was booted with a port with QoS policy and `minimum_packet_rate` rules the update is possible and the allocations are changed in placement as well.

> **Note**
>
> As it is possible to update a port to remove the QoS policy, updating it back to have QoS policy with `minimum_packet_rate` rule will not result in `placement allocation` record. In this case only dataplane enforcement will happen.

> **Note**
>
> Updating the `minimum_packet_rate` rule of a QoS policy that is attached to a port which is bound to a VM is still not possible.

- When QoS is used with a trunk, Placement enforcement is applied only to the trunks parent port. Subports are not going to have Placement allocation. As a workaround, parent port QoS policy should take into account subports needs and request enough minimum packet rate resources to accommodate every port in the trunk.

Placement pre-requisites

Placement must support microversion 1.36. This was first released in Train.

Nova pre-requisites

Nova must support top of microversion 2.72, additionally the Nova Xena release is needed to support the new `port-resource-request-groups` Neutron API extension.

Not all Nova virt drivers are supported, please refer to the Virt Driver Support section of the Nova Admin Guide.

Neutron pre-requisites

Neutron must support the following API extensions:

- `qos-pps-minimum`
- `port-resource-request-groups`

These were all first released in Yoga.

Neutron DB sanitization

The `resource_request` field of the Neutron port is used to express the resource needs of the port. The information in this field is calculated from the QoS policy rules attached to the port. Initially, only the minimum bandwidth rule was used as a source of requested resources. The format of `resource_request` looked like this:

```
{
    "required": [<CUSTOM_PHYSNET_ traits>, <CUSTOM_VNIC_TYPE traits>],
    "resources":
    {
        <NET_BW_[E|I]GR_KILOBIT_PER_SEC resource class name>:
            <requested bandwidth amount from the QoS policy>
    }
},
```

This structure allowed to describe only one group of resources and traits, which was sufficient at the time. However, with the introduction of QoS minimum packet rate rule, ports can now have multiple sources of requested resources and traits. Because of that, the format of `resource_request` field was incapable of expressing such request and it had to be changed.

To solve this issue, `port-resource-request-groups` extension was added in Neutron Yoga release. It provides support for the new format of `resource_request` field, that allows to request multiple groups of resources and traits from the same RP subtree. The new format looks like this:

```
{
    "request_groups":
    [
        {
            "id": <min-pps-group-uuid>
            "required": [<CUSTOM_VNIC_TYPE traits>],
            "resources":
            {
                NET_PACKET_RATE_[E|I]GR_KILOPACKET_PER_SEC:
                    <amount requested via the QoS policy>
            }
        },
        {
            "id": <min-bw-group-uuid>
            "required": [<CUSTOM_PHYSNET_ traits>,
                         <CUSTOM_VNIC_TYPE traits>],
            "resources":
            {
                <NET_BW_[E|I]GR_KILOBIT_PER_SEC resource class name>:
                    <requested bandwidth amount from the QoS policy>
            }
        }
    ],
    "same_subtree":
    [
        <min-pps-group-uuid>,
        <min-bw-group-uuid>
    ]
}
```

The main drawback about the new structure of `resource_request` field is lack of backwards compatibility. This can cause issues if `ml2_port_bindings` table in Neutron DB contains port bindings that were created before the introduction of `port-resource-request-groups` extension. Because `port-resource-request-groups` extension is enabled by default in Yoga release, its necessary to perform DB sanitization before upgrading Neutron to Yoga.

DB sanitization will ensure that every row of `ml2_port_bindings` table uses the new format. Upgrade check can be run before DB sanitization, to see if there are any rows in the DB that require sanitization.

```
$ neutron-status upgrade check
# If 'Port Binding profile sanity check' fails, DB sanitization is needed
$ neutron-sanitize-port-binding-profile-allocation --config-file /etc/neutron/
↪neutron.conf
```

Supported drivers and agents

In release Yoga the following agent-based ML2 mechanism drivers are supported:

- Open vSwitch (`openvswitch`) vnic_types: `normal`, `direct`

neutron-server config

QoS minimum packet rate rule requires exactly the same configuration in the `neutron-server` as QoS minimum bandwidth rule. Please refer to `neutron-server config` section of *Quality of Service (QoS): Guaranteed Minimum Bandwidth guide* for more details.

neutron-openvswitch-agent config

Set the agent configuration as the authentic source of the resources available. Depending on OVS deployment type, packet processing capacity can be configured with:

- *`ovs.resource_provider_packet_processing_without_direction`* Format for this option is <`hypervisor`>`:`<`packet_rate`>. This option should be used for non-hardware-offloaded OVS deployments.
- *`ovs.resource_provider_packet_processing_with_direction`*

  Format for this option is <`hypervisor`>`:`<`egress_packet_rate`>`:`<`ingress_packet_rate`>. You may set only one direction and omit the other. This option should be used for hardware-offloaded OVS deployments.

Regardless if direction-less or direction-oriented packet processing mode is used, configuration is always applied to the whole OVS instance.

Note

`egress` / `ingress` is meant from the VM point of view. That is `egress` = cloud server upload, `ingress` = download.

Egress and ingress available packet rate values are in `kilo packet/sec (kpps)`.

Direction-less and direction-oriented modes are mutually exclusive options. Only one can be used at a time.

The hypervisor name is optional, and needs to be set only in the rare case cases. For more information, please refer to Neutron agent documentation.

If desired, resource provider inventory fields can be tweaked on a per-agent basis by setting *`ovs.resource_provider_packet_processing_inventory_defaults`*. Valid values are all the optional parameters of the update resource provider inventory call.

`/etc/neutron/plugins/ml2/ovs_agent.ini` (on compute and network nodes):

```
[ovs]
resource_provider_packet_processing_with_direction = :10000000:10000000,...
#resource_provider_packet_processing_inventory_defaults = step_size:1000,...
```

Propagation of resource information

Propagation of resource information is explained in detail in *Quality of Service (QoS): Guaranteed Minimum Bandwidth guide*.

Sample usage

Network and QoS policies (together with their rules) are usually pre-created by a cloud admin:

```
# as admin

$ openstack network create net0

$ openstack subnet create subnet0 \
    --network net0 \
    --subnet-range 10.0.4.0/24

$ openstack network qos policy create policy0

$ openstack network qos rule create policy0 \
    --type minimum-packet-rate \
    --min-kpps 1000000 \
    --egress

$ openstack network qos rule create policy0 \
    --type minimum-packet-rate \
    --min-kpps 1000000 \
    --ingress
```

Then a normal user can use the pre-created policy to create ports and boot servers with those ports:

```
# as an unprivileged user

# an ordinary soft-switched port: ``--vnic-type normal`` is the default
$ openstack port create port-normal-qos \
    --network net0 \
    --qos-policy policy0

$ openstack server create server0 \
    --os-compute-api-version 2.72 \
    --flavor cirros256 \
    --image cirros-0.5.2-x86_64-disk \
    --port port-normal-qos
```

On Healing of Allocations

Since Placement carries a global view of a cloud deployments resources (what is available, what is used) it may in some conditions get out of sync with reality.

One important case stems from OpenStack not having distributed transactions to allocate resources provided by multiple OpenStack components (here Nova and Neutron). There are known race conditions in which Placements view may get out of sync with reality. The design knowingly minimizes the race condition windows, but there are known problems:

- If a QoS policy is modified after Nova read a ports `resource_request` but before the port is bound its state before the modification will be applied.
- If a bound port with a resource allocation is deleted. The ports allocation is leaked. https://bugs.launchpad.net/nova/+bug/1820588

Note

Deleting a bound port has no known use case. Please consider detaching the interface first by `openstack server remove port` instead.

Incorrect allocations may be fixed by:

- Moving the server, which will delete the wrong allocation and create the correct allocation. Moving servers fixes local overallocations.
- With placement heal_allocations tool.
- Manually, by using openstack resource provider allocation set /delete.

Debugging

- Is Nova running at least Xena release and Neutron at least the Yoga release?
- Are `qos-pps-minimum` and `port-resource-request-groups` extensions available?

```
$ openstack extension show qos-pps-minimum
$ openstack extension show port-resource-request-groups
```

- Is the `placement` service plugin enabled in neutron-server?
- Is `resource_provider_packet_processing_with_direction` or `resource_provider_packet_processing_without_direction` configured for the relevant neutron agent?
- Was the agent restarted since changing the configuration file?
- Is `resource_provider_packet_processing_with_direction` or `resource_provider_packet_processing_without_direction` reaching neutron-server?

```
# as admin
$ openstack network agent show ... -c configuration -f json
```

Please find an example in section *Propagation of resource information*.

- Did neutron-server successfully sync to Placement?

```
# as admin
$ openstack network agent show ... | grep resources_synced
```

Please find an example in section *Propagation of resource information*.

- Is the resource provider tree correct? Is the root a compute host? One level below the agents?

```
$ openstack --os-placement-api-version 1.17 resource provider list
+------------------------------------+-----------------------------------
```

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```
↪----+-----------+-------------------------------------+------------------
↪------------------+
| uuid                                 | name                                                
↪    | generation | root_provider_uuid                   | parent_provider_
↪uuid                 |
+--------------------------------------+-------------------------------------
↪----+-----------+-------------------------------------+------------------
↪------------------+
| 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | devstack0                                           
↪    |          2 | 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | None                               
↪                   |
| 89ca1421-5117-5348-acab-6d0e2054239c | devstack0:Open vSwitch agent                        
↪    |          0 | 3b36d91e-bf60-460f-b1f8-3322dee5cdfd | 3b36d91e-bf60-
↪460f-b1f8-3322dee5cdfd |
+--------------------------------------+-------------------------------------
↪----+-----------+-------------------------------------+------------------
↪------------------+
```

- Does Placement have the expected traits?

```
# as admin
$ openstack --os-placement-api-version 1.17 trait list | awk '/CUSTOM_/ { 
↪print $2 }' | sort
CUSTOM_VNIC_TYPE_NORMAL
CUSTOM_VNIC_TYPE_SMART_NIC
CUSTOM_VNIC_TYPE_VDPA
```

- Do the OVS agent resource provider have the proper trait associations and inventories?

```
# as admin
$ openstack --os-placement-api-version 1.17 resource provider trait list <RP-
↪UUID>
$ openstack --os-placement-api-version 1.17 resource provider inventory list
↪<RP-UUID>
```

- Does the QoS policy have a `minimum-packet-rate` rule?
- Does the port have the proper policy?
- Does the port have a `resource_request`?

```
# as admin
$ openstack port show port-normal-qos | grep resource_request
```

- Was the server booted with a port (as opposed to a network)?
- Did nova allocate resources for the server in Placement?

```
# as admin
$ openstack --os-placement-api-version 1.17 resource provider allocation show
↪<SERVER-UUID>
```

- Does the allocation have a part on the expected OVS agent resource provider?

```
# as admin
$ openstack --os-placement-api-version 1.17 resource provider show --
↪allocations <RP-UUID>
```

- Did placement manage to produce an allocation candidate list to nova during scheduling?
- Did nova manage to schedule the server?
- Did nova tell neutron which OVS agent resource provider was allocated to satisfy the packet rate request?

```
# as admin
$ openstack port show port-normal-qos | grep binding.profile.*allocation
```

- Did neutron manage to bind the port?

Links

- Nova documentation on using a port with `resource_request`
  - API Guide
  - Admin Guide
- Neutron spec: QoS minimum guaranteed packet rate
  - on specs.openstack.org
  - on review.opendev.org
- Nova spec: QoS minimum guaranteed packet rate
  - on specs.openstack.org
  - on review.opendev.org
- Relevant OpenStack Networking API references
  - https://docs.openstack.org/api-ref/network/v2/#agent-resources-synced-extension
  - https://docs.openstack.org/api-ref/network/v2/#port-resource-request
  - https://docs.openstack.org/api-ref/network/v2/#port-resource-request-groups
  - https://docs.openstack.org/api-ref/network/v2/#qos-minimum-packet-rate-rules
- Microversion histories
  - Compute 2.72
  - Placement 1.36
- Implementation
  - on review.opendev.org
- Known Bugs
  - Bandwidth resource is leaked this issue also affects packet rate resources.

8.2.30 Role-Based Access Control (RBAC)

The Role-Based Access Control (RBAC) policy framework enables both operators and users to grant access to resources for specific projects.

Supported objects for sharing with specific projects

Currently, the access that can be granted using this feature is supported by:

- Regular port creation permissions on networks (since Liberty).
- Binding QoS policies permissions to networks or ports (since Mitaka).
- Attaching router gateways to networks (since Mitaka).
- Binding security groups to ports (since Stein).
- Assigning address scopes to subnet pools (since Ussuri).
- Assigning subnet pools to subnets (since Ussuri).
- Assigning address groups to security group rules (since Wallaby).

Sharing an object with specific projects

Sharing an object with a specific project is accomplished by creating a policy entry that permits the target project the `access_as_shared` action on that object.

Sharing a network with specific projects

Create a network to share:

```
$ openstack network create secret_network
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2017-01-25T20:16:40Z                 |
| description               |                                      |
| dns_domain                | None                                 |
| id                        | f55961b9-3eb8-42eb-ac96-b97038b568de |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | None                                 |
| mtu                       | 1450                                 |
| name                      | secret_network                       |
| port_security_enabled     | True                                 |
| project_id                | 61b7eba037fd41f29cfba757c010faff     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 9                                    |
| qos_policy_id             | None                                 |
| revision_number           | 3                                    |
| router:external           | Internal                             |
```

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```
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2017-01-25T20:16:40Z                 |
+---------------------------+--------------------------------------+
```

Create the policy entry using the **openstack network rbac create** command (in this example, the ID of the project we want to share with is `b87b2fc13e0248a4a031d38e06dc191d`):

```
$ openstack network rbac create --target-project \
b87b2fc13e0248a4a031d38e06dc191d --action access_as_shared \
--type network f55961b9-3eb8-42eb-ac96-b97038b568de
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| action            | access_as_shared                     |
| id                | f93efdbf-f1e0-41d2-b093-8328959d469e |
| name              | None                                 |
| object_id         | f55961b9-3eb8-42eb-ac96-b97038b568de |
| object_type       | network                              |
| project_id        | 61b7eba037fd41f29cfba757c010faff     |
| target_project_id | b87b2fc13e0248a4a031d38e06dc191d     |
+-------------------+--------------------------------------+
```

The `target-project` parameter specifies the project that requires access to the network. The `action` parameter specifies what the project is allowed to do. The `type` parameter says that the target object is a network. The final parameter is the ID of the network we are granting access to.

Project `b87b2fc13e0248a4a031d38e06dc191d` will now be able to see the network when running **openstack network list** and **openstack network show** and will also be able to create ports on that network. No other users (other than admins and the owner) will be able to see the network.

Note

Subnets inherit the RBAC policy entries of their network.

To remove access for that project, delete the policy that allows it using the **openstack network rbac delete** command:

```
$ openstack network rbac delete f93efdbf-f1e0-41d2-b093-8328959d469e
```

If that project has ports on the network, the server will prevent the policy from being deleted until the ports have been deleted:

```
$ openstack network rbac delete f93efdbf-f1e0-41d2-b093-8328959d469e
RBAC policy on object f93efdbf-f1e0-41d2-b093-8328959d469e
cannot be removed because other objects depend on it.
```

This process can be repeated any number of times to share a network with an arbitrary number of projects.

Sharing a QoS policy with specific projects

Create a QoS policy to share:

```
$ openstack network qos policy create secret_policy
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| description       |                                      |
| id                | 1f730d69-1c45-4ade-a8f2-89070ac4f046 |
| name              | secret_policy                        |
| project_id        | 61b7eba037fd41f29cfba757c010faff     |
| revision_number   | 1                                    |
| rules             | []                                   |
| shared            | False                                |
| tags              | []                                   |
+-------------------+--------------------------------------+
```

Create the RBAC policy entry using the **openstack network rbac create** command (in this example, the ID of the project we want to share with is `be98b82f8fdf46b696e9e01cebc33fd9`):

```
$ openstack network rbac create --target-project \
be98b82f8fdf46b696e9e01cebc33fd9 --action access_as_shared \
--type qos_policy 1f730d69-1c45-4ade-a8f2-89070ac4f046
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| action            | access_as_shared                     |
| id                | 8828e38d-a0df-4c78-963b-e5f215d3d550 |
| name              | None                                 |
| object_id         | 1f730d69-1c45-4ade-a8f2-89070ac4f046 |
| object_type       | qos_policy                           |
| project_id        | 61b7eba037fd41f29cfba757c010faff     |
| target_project_id | be98b82f8fdf46b696e9e01cebc33fd9     |
+-------------------+--------------------------------------+
```

The `target-project` parameter specifies the project that requires access to the QoS policy. The `action` parameter specifies what the project is allowed to do. The `type` parameter says that the target object is a QoS policy. The final parameter is the ID of the QoS policy we are granting access to.

Project `be98b82f8fdf46b696e9e01cebc33fd9` will now be able to see the QoS policy when running **openstack network qos policy list** and **openstack network qos policy show** and will also be able to bind it to its ports or networks. No other users (other than admins and the owner) will be able to see the QoS policy.

To remove access for that project, delete the RBAC policy that allows it using the **openstack network rbac delete** command:

```
$ openstack network rbac delete 8828e38d-a0df-4c78-963b-e5f215d3d550
```

If that project has ports or networks with the QoS policy applied to them, the server will not delete the RBAC policy until the QoS policy is no longer in use:

```
$ openstack network rbac delete 8828e38d-a0df-4c78-963b-e5f215d3d550
RBAC policy on object 8828e38d-a0df-4c78-963b-e5f215d3d550
cannot be removed because other objects depend on it.
```

This process can be repeated any number of times to share a qos-policy with an arbitrary number of projects.

Sharing a security group with specific projects

Create a security group to share:

```
$ openstack security group create my_security_group
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| created_at        | 2019-02-07T06:09:59Z                 |
| description       | my_security_group                    |
| id                | 5ba835b7-22b0-4be6-bdbe-e0722d1b5f24 |
| location          | None                                 |
| name              | my_security_group                    |
| project_id        | 077e8f39d3db4c9e998d842b0503283a     |
| revision_number   | 1                                    |
| rules             | ...                                  |
| tags              | []                                   |
| updated_at        | 2019-02-07T06:09:59Z                 |
+-------------------+--------------------------------------+
```

Create the RBAC policy entry using the **openstack network rbac create** command (in this example, the ID of the project we want to share with is `32016615de5d43bb88de99e7f2e26a1e`):

```
$ openstack network rbac create --target-project \
32016615de5d43bb88de99e7f2e26a1e --action access_as_shared \
--type security_group 5ba835b7-22b0-4be6-bdbe-e0722d1b5f24
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| action            | access_as_shared                     |
| id                | 8828e38d-a0df-4c78-963b-e5f215d3d550 |
| name              | None                                 |
| object_id         | 5ba835b7-22b0-4be6-bdbe-e0722d1b5f24 |
| object_type       | security_group                       |
| project_id        | 077e8f39d3db4c9e998d842b0503283a     |
| target_project_id | 32016615de5d43bb88de99e7f2e26a1e     |
+-------------------+--------------------------------------+
```

The `target-project` parameter specifies the project that requires access to the security group. The `action` parameter specifies what the project is allowed to do. The `type` parameter says that the target object is a security group. The final parameter is the ID of the security group we are granting access to.

Project `32016615de5d43bb88de99e7f2e26a1e` will now be able to see the security group when running **openstack security group list** and **openstack security group show** and will also be able to bind it to its ports. No other users (other than admins and the owner) will be able to see the security group.

To remove access for that project, delete the RBAC policy that allows it using the **openstack network rbac delete** command:

```
$ openstack network rbac delete 8828e38d-a0df-4c78-963b-e5f215d3d550
```

If that project has ports with the security group applied to them, the server will not delete the RBAC policy until the security group is no longer in use:

```
$ openstack network rbac delete 8828e38d-a0df-4c78-963b-e5f215d3d550
RBAC policy on object 8828e38d-a0df-4c78-963b-e5f215d3d550
cannot be removed because other objects depend on it.
```

This process can be repeated any number of times to share a security-group with an arbitrary number of projects.

Creating an instance which uses a security group shared through RBAC, but only specifying the network ID when calling Nova will not work currently. In such cases Nova will check if the given security group exists in Neutron before it creates a port in the given network. The problem with that is that Nova asks only for the security groups filtered by the project_id thus it will not get the shared security group back from the Neutron API. See bug 1942615 for details. To workaround the issue, the user needs to create a port in Neutron first, and then pass that port to Nova:

```
$ openstack port create --network net1 --security-group
5ba835b7-22b0-4be6-bdbe-e0722d1b5f24 shared-sg-port

$ openstack server create --image cirros-0.5.1-x86_64-disk --flavor m1.tiny
--port shared-sg-port vm-with-shared-sg
```

Sharing an address scope with specific projects

Create an address scope to share:

```
$ openstack address scope create my_address_scope
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| id                | c19cb654-3489-4160-9c82-8a3015483643 |
| ip_version        | 4                                    |
| location          | ...                                  |
| name              | my_address_scope                     |
| project_id        | 34304bc4f233470fa4a2448d153b6324     |
| shared            | False                                |
+-------------------+--------------------------------------+
```

Create the RBAC policy entry using the **openstack network rbac create** command (in this example, the ID of the project we want to share with is `32016615de5d43bb88de99e7f2e26a1e`):

```
$ openstack network rbac create --target-project \
32016615de5d43bb88de99e7f2e26a1e --action access_as_shared \
--type address_scope c19cb654-3489-4160-9c82-8a3015483643
+-------------------+--------------------------------------+
```

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```
| Field             | Value                                |
+-------------------+--------------------------------------+
| action            | access_as_shared                     |
| id                | d54b1482-98c4-44aa-9115-ede80387ffe0 |
| location          | ...                                  |
| name              | None                                 |
| object_id         | c19cb654-3489-4160-9c82-8a3015483643 |
| object_type       | address_scope                        |
| project_id        | 34304bc4f233470fa4a2448d153b6324     |
| target_project_id | 32016615de5d43bb88de99e7f2e26a1e     |
+-------------------+--------------------------------------+
```

The `target-project` parameter specifies the project that requires access to the address scope. The `action` parameter specifies what the project is allowed to do. The `type` parameter says that the target object is an address scope. The final parameter is the ID of the address scope we are granting access to.

Project `32016615de5d43bb88de99e7f2e26a1e` will now be able to see the address scope when running **openstack address scope list** and **openstack address scope show** and will also be able to assign it to its subnet pools. No other users (other than admins and the owner) will be able to see the address scope.

To remove access for that project, delete the RBAC policy that allows it using the **openstack network rbac delete** command:

```
$ openstack network rbac delete d54b1482-98c4-44aa-9115-ede80387ffe0
```

If that project has subnet pools with the address scope applied to them, the server will not delete the RBAC policy until the address scope is no longer in use:

```
$ openstack network rbac delete d54b1482-98c4-44aa-9115-ede80387ffe0
RBAC policy on object c19cb654-3489-4160-9c82-8a3015483643
cannot be removed because other objects depend on it.
```

This process can be repeated any number of times to share an address scope with an arbitrary number of projects.

Sharing a subnet pool with specific projects

Create a subnet pool to share:

```
$ openstack subnet pool create my_subnetpool --pool-prefix 203.0.113.0/24
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| address_scope_id  | None                                 |
| created_at        | 2020-03-16T14:23:01Z                 |
| default_prefixlen | 8                                    |
| default_quota     | None                                 |
| description       |                                      |
| id                | 11f79287-bc17-46b2-bfd0-2562471eb631 |
| ip_version        | 4                                    |
```

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```
| is_default        | False                                |
| location          | ...                                  |
| max_prefixlen     | 32                                   |
| min_prefixlen     | 8                                    |
| name              | my_subnetpool                        |
| project_id        | 290ccedbcf594ecc8e76eff06f964f7e     |
| revision_number   | 0                                    |
| shared            | False                                |
| tags              |                                      |
| updated_at        | 2020-03-16T14:23:01Z                 |
+-------------------+--------------------------------------+
```

Create the RBAC policy entry using the **openstack network rbac create** command (in this example, the ID of the project we want to share with is 32016615de5d43bb88de99e7f2e26a1e):

```
$ openstack network rbac create --target-project \
32016615de5d43bb88de99e7f2e26a1e --action access_as_shared \
--type subnetpool 11f79287-bc17-46b2-bfd0-2562471eb631
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| action            | access_as_shared                     |
| id                | d54b1482-98c4-44aa-9115-ede80387ffe0 |
| location          | ...                                  |
| name              | None                                 |
| object_id         | 11f79287-bc17-46b2-bfd0-2562471eb631 |
| object_type       | subnetpool                           |
| project_id        | 290ccedbcf594ecc8e76eff06f964f7e     |
| target_project_id | 32016615de5d43bb88de99e7f2e26a1e     |
+-------------------+--------------------------------------+
```

The `target-project` parameter specifies the project that requires access to the subnet pool. The `action` parameter specifies what the project is allowed to do. The `type` parameter says that the target object is a subnet pool. The final parameter is the ID of the subnet pool we are granting access to.

Project 32016615de5d43bb88de99e7f2e26a1e will now be able to see the subnet pool when running **openstack subnet pool list** and **openstack subnet pool show** and will also be able to assign it to its subnets. No other users (other than admins and the owner) will be able to see the subnet pool.

To remove access for that project, delete the RBAC policy that allows it using the **openstack network rbac delete** command:

```
$ openstack network rbac delete d54b1482-98c4-44aa-9115-ede80387ffe0
```

If that project has subnets with the subnet pool applied to them, the server will not delete the RBAC policy until the subnet pool is no longer in use:

```
$ openstack network rbac delete d54b1482-98c4-44aa-9115-ede80387ffe0
RBAC policy on object 11f79287-bc17-46b2-bfd0-2562471eb631
cannot be removed because other objects depend on it.
```

This process can be repeated any number of times to share a subnet pool with an arbitrary number of projects.

Sharing an address group with specific projects

Create an address group to share:

```
$ openstack address group create test-ag --address 10.1.1.1
+-------------+--------------------------------------+
| Field       | Value                                |
+-------------+--------------------------------------+
| addresses   | ['10.1.1.1/32']                      |
| description |                                      |
| id          | cdb6eb3e-f9a0-4d52-8478-358eaa2c4737 |
| name        | test-ag                              |
| project_id  | 66c77cf262454777a8f455cce48c12c0     |
+-------------+--------------------------------------+
```

Create the RBAC policy entry using the **openstack network rbac create** command (in this example, the ID of the project we want to share with is `bbd82892525d4372911390b984ed3265`):

```
$ openstack network rbac create --target-project \
bbd82892525d4372911390b984ed3265 --action access_as_shared \
--type address_group cdb6eb3e-f9a0-4d52-8478-358eaa2c4737
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| action            | access_as_shared                     |
| id                | c7414ac2-9a6b-420b-84c5-4158a6cca4f9 |
| name              | None                                 |
| object_id         | cdb6eb3e-f9a0-4d52-8478-358eaa2c4737 |
| object_type       | address_group                        |
| project_id        | 66c77cf262454777a8f455cce48c12c0     |
| target_project_id | bbd82892525d4372911390b984ed3265     |
+-------------------+--------------------------------------+
```

The `target-project` parameter specifies the project that requires access to the address group. The `action` parameter specifies what the project is allowed to do. The `type` parameter says that the target object is an address group. The final parameter is the ID of the address group we are granting access to.

Project `bbd82892525d4372911390b984ed3265` will now be able to see the address group when running **openstack address group list** and **openstack address group show** and will also be able to assign it to its security group rules. No other users (other than admins and the owner) will be able to see the address group.

To remove access for that project, delete the RBAC policy that allows it using the **openstack network rbac delete** command:

```
$ openstack network rbac delete c7414ac2-9a6b-420b-84c5-4158a6cca4f9
```

If that project has security group rules with the address group applied to them, the server will not delete the RBAC policy until the address group is no longer in use:

```
$ openstack network rbac delete c7414ac2-9a6b-420b-84c5-4158a6cca4f9
RBAC policy on object cdb6eb3e-f9a0-4d52-8478-358eaa2c4737
cannot be removed because other objects depend on it
```

This process can be repeated any number of times to share an address group with an arbitrary number of projects.

How the shared flag relates to these entries

As introduced in other guide entries, neutron provides a means of making an object (`address-scope`, `network`, `qos-policy`, `security-group`, `subnetpool`) available to every project. This is accomplished using the `shared` flag on the supported object:

```
$ openstack network create global_network --share
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2017-01-25T20:32:06Z                 |
| description               |                                      |
| dns_domain                | None                                 |
| id                        | 84a7e627-573b-49da-af66-c9a65244f3ce |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | None                                 |
| mtu                       | 1450                                 |
| name                      | global_network                       |
| port_security_enabled     | True                                 |
| project_id                | 61b7eba037fd41f29cfba757c010faff     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 7                                    |
| qos_policy_id             | None                                 |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| segments                  | None                                 |
| shared                    | True                                 |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2017-01-25T20:32:07Z                 |
+---------------------------+--------------------------------------+
```

This is the equivalent of creating a policy on the network that permits every project to perform the action `access_as_shared` on that network. Neutron treats them as the same thing, so the policy entry for that network should be visible using the **`openstack network rbac list`** command:

```
$ openstack network rbac list
+-------------------------------------+-------------+-----------------------------
```

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```
↪-+
| ID                             | Object Type | Object ID                      ␣
↪ |
+--------------------------------+-------------+------------------------------
↪-+
| 58a5ee31-2ad6-467d-            | qos_policy  | 1f730d69-1c45-4ade-           ␣
↪ |
| 8bb8-8c2ae3dd1382              |             | a8f2-89070ac4f046             ␣
↪ |
| 27efbd79-f384-4d89-9dfc-       | network     | 84a7e627-573b-49da-           ␣
↪ |
| 6c4a606ceec6                   |             | af66-c9a65244f3ce             ␣
↪ |
+--------------------------------+-------------+------------------------------
↪-+
```

Use the **openstack network rbac show** command to see the details:

```
$ openstack network rbac show 27efbd79-f384-4d89-9dfc-6c4a606ceec6
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| action            | access_as_shared                     |
| id                | 27efbd79-f384-4d89-9dfc-6c4a606ceec6 |
| name              | None                                 |
| object_id         | 84a7e627-573b-49da-af66-c9a65244f3ce |
| object_type       | network                              |
| project_id        | 61b7eba037fd41f29cfba757c010faff     |
| target_project_id | *                                    |
+-------------------+--------------------------------------+
```

The output shows that the entry allows the action `access_as_shared` on object `84a7e627-573b-49da-af66-c9a65244f3ce` of type `network` to target_project `*`, which is a wildcard that represents all projects.

Currently, the `shared` flag is just a mapping to the underlying RBAC policies for a network. Setting the flag to `True` on a network creates a wildcard RBAC entry. Setting it to `False` removes the wildcard entry.

When you run **openstack network list** or **openstack network show**, the `shared` flag is calculated by the server based on the calling project and the RBAC entries for each network. For QoS objects use **openstack network qos policy list** or **openstack network qos policy show** respectively. If there is a wildcard entry, the `shared` flag is always set to `True`. If there are only entries that share with specific projects, only the projects the object is shared to will see the flag as `True` and the rest will see the flag as `False`.

Allowing a network to be used as an external network

To make a network available as an external network for specific projects rather than all projects, use the `access_as_external` action.

1. Create a network that you want to be available as an external network:

```
$ openstack network create secret_external_network
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2017-01-25T20:36:59Z                 |
| description               |                                      |
| dns_domain                | None                                 |
| id                        | 802d4e9e-4649-43e6-9ee2-8d052a880cfb |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | None                                 |
| mtu                       | 1450                                 |
| name                      | secret_external_network              |
| port_security_enabled     | True                                 |
| project_id                | 61b7eba037fd41f29cfba757c010faff     |
| proider:network_type      | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 21                                   |
| qos_policy_id             | None                                 |
| revision_number           | 3                                    |
| router:external           | Internal                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
| updated_at                | 2017-01-25T20:36:59Z                 |
+---------------------------+--------------------------------------+
```

2. Create a policy entry using the **openstack network rbac create** command (in this example, the ID of the project we want to share with is `838030a7bf3c4d04b4b054c0f0b2b17c`):

```
$ openstack network rbac create --target-project \
838030a7bf3c4d04b4b054c0f0b2b17c --action access_as_external \
--type network 802d4e9e-4649-43e6-9ee2-8d052a880cfb
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| action            | access_as_external                   |
| id                | afdd5b8d-b6f5-4a15-9817-5231434057be |
| name              | None                                 |
| object_id         | 802d4e9e-4649-43e6-9ee2-8d052a880cfb |
| object_type       | network                              |
| project_id        | 61b7eba037fd41f29cfba757c010faff     |
| target_project_id | 838030a7bf3c4d04b4b054c0f0b2b17c     |
+-------------------+--------------------------------------+
```

The `target-project` parameter specifies the project that requires access to the network. The `action` parameter specifies what the project is allowed to do. The `type` parameter indicates that the target object is a network. The final parameter is the ID of the network we are granting external access to.

Now project `838030a7bf3c4d04b4b054c0f0b2b17c` is able to see the network when running **openstack network list** and **openstack network show** and can attach router gateway ports to that network. No other users (other than admins and the owner) are able to see the network.

To remove access for that project, delete the policy that allows it using the **openstack network rbac delete** command:

```
$ openstack network rbac delete afdd5b8d-b6f5-4a15-9817-5231434057be
```

If that project has router gateway ports attached to that network, the server prevents the policy from being deleted until the ports have been deleted:

```
$ openstack network rbac delete afdd5b8d-b6f5-4a15-9817-5231434057be
RBAC policy on object afdd5b8d-b6f5-4a15-9817-5231434057be
cannot be removed because other objects depend on it.
```

This process can be repeated any number of times to make a network available as external to an arbitrary number of projects.

If a network is marked as external during creation, it now implicitly creates a wildcard RBAC policy granting everyone access to preserve previous behavior before this feature was added.

```
$ openstack network create global_external_network --external
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2017-01-25T20:41:44Z                 |
| description               |                                      |
| dns_domain                | None                                 |
| id                        | 72a257a2-a56e-4ac7-880f-94a4233abec6 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | None                                 |
| mtu                       | 1450                                 |
| name                      | global_external_network              |
| port_security_enabled     | True                                 |
| project_id                | 61b7eba037fd41f29cfba757c010faff     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 69                                   |
| qos_policy_id             | None                                 |
| revision_number           | 4                                    |
| router:external           | External                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
```

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```
| tags                      | []                                    |
| updated_at                | 2017-01-25T20:41:44Z                  |
+---------------------------+---------------------------------------+
```

In the output above the standard `router:external` attribute is `External` as expected. Now a wildcard policy is visible in the RBAC policy listings:

```
$ openstack network rbac list --long -c ID -c Action
+--------------------------------------+--------------------+
| ID                                   | Action             |
+--------------------------------------+--------------------+
| b694e541-bdca-480d-94ec-eda59ab7d71a | access_as_external |
+--------------------------------------+--------------------+
```

You can modify or delete this policy with the same constraints as any other RBAC `access_as_external` policy.

Preventing regular users from sharing objects with each other

The default `policy.yaml` file will not allow regular users to share objects with every other project using a wildcard; however, it will allow them to share objects with specific project IDs.

If an operator wants to prevent normal users from doing this, the `"create_rbac_policy":` entry in `policy.yaml` can be adjusted from `""` to `"rule:admin_only"`.

Improve database RBAC query operations

Since[1], present in Yoga version, Neutron has indexes for target_tenant (now target_project) and action columns in all RBAC related tables. That improves the SQL queries involving the RBAC tables[2]. Any system before Yoga wont have these indexes but the system administrator can manually add them to the Neutron database following the next steps:

- Find the RBAC tables:

```
$ tables=`mysql -e "use ovs_neutron; show tables;" | grep rbac`
```

- Insert the indexes for the target_tenant and action columns:

    $ for table in $tables do; mysql -e

        alter table $table add key (action);
        alter table $table add key (target_tenant);; done

In order to prevent errors during a system upgrade,[3] was implemented and backported up to Yoga. This patch checks if any index is already present in the Neutron tables and avoids executing the index creation command again.

[1] https://review.opendev.org/c/openstack/neutron/+/810072

[2] https://github.com/openstack/neutron-lib/blob/890d62a3df3f35bb18bf1a11e79a9e97e7dd2d2c/neutron_lib/db/model_query.py#L123-L131

[3] https://review.opendev.org/c/openstack/neutron/+/884617

8.2.31 Routed provider networks

Note

Use of this feature requires the OpenStack client version 3.3 or newer.

Before routed provider networks, the Networking service could not present a multi-segment layer-3 network as a single entity. Thus, each operator typically chose one of the following architectures:

- Single large layer-2 network
- Multiple smaller layer-2 networks

Single large layer-2 networks become complex at scale and involve significant failure domains.

Multiple smaller layer-2 networks scale better and shrink failure domains, but leave network selection to the user. Without additional information, users cannot easily differentiate these networks.

A routed provider network enables a single provider network to represent multiple layer-2 networks (broadcast domains) or segments and enables the operator to present one network to users. However, the particular IP addresses available to an instance depend on the segment of the network available on the particular compute node. Neutron port could be associated with only one network segment, but there is an exception for OVN distributed services like OVN Metadata.

Similar to conventional networking, layer-2 (switching) handles transit of traffic between ports on the same segment and layer-3 (routing) handles transit of traffic between segments.

Each segment requires at least one subnet that explicitly belongs to that segment. The association between a segment and a subnet distinguishes a routed provider network from other types of networks. The Networking service enforces that either zero or all subnets on a particular network associate with a segment. For example, attempting to create a subnet without a segment on a network containing subnets with segments generates an error.

The Networking service does not provide layer-3 services between segments. Instead, it relies on physical network infrastructure to route subnets. Thus, both the Networking service and physical network infrastructure must contain configuration for routed provider networks, similar to conventional provider networks. In the future, implementation of dynamic routing protocols may ease configuration of routed networks.

Prerequisites

Routed provider networks require additional prerequisites over conventional provider networks. We recommend using the following procedure:

1. Begin with segments. The Networking service defines a segment using the following components:

   - Unique physical network name
   - Segmentation type
   - Segmentation ID

   For example, `provider1`, `VLAN`, and `2016`. See the API reference for more information.

   Within a network, use a unique physical network name for each segment which enables reuse of the same segmentation details between subnets. For example, using the same VLAN ID across all segments of a particular provider network. Similar to conventional provider networks, the operator must provision the layer-2 physical network infrastructure accordingly.

2. Implement routing between segments.

   The Networking service does not provision routing among segments. The operator must implement routing among segments of a provider network. Each subnet on a segment must contain the gateway address of the router interface on that particular subnet. For example:

Segment	Version	Addresses	Gateway
segment1	4	203.0.113.0/24	203.0.113.1
segment1	6	fd00:203:0:113::/64	fd00:203:0:113::1
segment2	4	198.51.100.0/24	198.51.100.1
segment2	6	fd00:198:51:100::/64	fd00:198:51:100::1

3. Map segments to compute nodes.

   Routed provider networks imply that compute nodes reside on different segments. The operator must ensure that every compute host that is supposed to participate in a router provider network has direct connectivity to one of its segments.

Host	Rack	Physical Network
compute0001	rack 1	segment 1
compute0002	rack 1	segment 1
compute0101	rack 2	segment 2
compute0102	rack 2	segment 2
compute0102	rack 2	segment 2

4. Deploy DHCP agents.

   Unlike conventional provider networks, a DHCP agent cannot support more than one segment within a network. The operator must deploy at least one DHCP agent per segment. Consider deploying DHCP agents on compute nodes containing the segments rather than one or more network nodes to reduce node count.

Host	Rack	Physical Network
network0001	rack 1	segment 1
network0002	rack 2	segment 2

5. Configure communication of the Networking service with the Compute scheduler.

   An instance with an interface with an IPv4 address in a routed provider network must be placed by the Compute scheduler in a host that has access to a segment with available IPv4 addresses. To make this possible, the Networking service communicates to the Compute scheduler the inventory of IPv4 addresses associated with each segment of a routed provider network. The operator must configure the authentication credentials that the Networking service will use to communicate with the Compute schedulers placement API. Please see below an example configuration.

Note

Coordination between the Networking service and the Compute scheduler is not necessary for IPv6 subnets as a consequence of their large address spaces.

Note

The coordination between the Networking service and the Compute scheduler requires the following minimum API micro-versions.

- Compute service API: 2.41
- Placement API: 1.1

Example configuration

Controller node

1. Enable the segments service plug-in by appending `segments` to the list of `service_plugins` in the `neutron.conf` file on all nodes running the `neutron-server` service:

```
[DEFAULT]
# ...
service_plugins = ...,segments
```

2. Add a `placement` section to the `neutron.conf` file with authentication credentials for the Compute service placement API:

```
[placement]
www_authenticate_uri = http://192.0.2.72/identity
project_domain_name = Default
project_name = service
user_domain_name = Default
password = apassword
username = nova
auth_url = http://192.0.2.72/identity_admin
auth_type = password
region_name = RegionOne
```

3. Restart the `neutron-server` service.

4. (Optional) Configure the Nova scheduler to filter based upon routed network host aggregates. Without this option set, once ports are attached to instances and have IP addresses assigned, Nova may schedule instances to hosts which do not have access to the required segment. See the Nova configuration reference for more information.

Network or compute nodes

- Configure the layer-2 agent on each node to map one or more segments to the appropriate physical network bridge or interface and restart the agent.

Create a routed provider network

The following steps create a routed provider network with two segments. Each segment contains one IPv4 subnet and one IPv6 subnet.

1. Source the administrative project credentials.
2. Create a VLAN provider network which includes a default segment. In this example, the network uses the `provider1` physical network with VLAN ID 2016.

```
$ openstack network create --share --provider-physical-network provider1 \
  --provider-network-type vlan --provider-segment 2016 multisegment1
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| id                        | 6ab19caa-dda9-4b3d-abc4-5b8f435b98d9 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| l2_adjacency              | True                                 |
| mtu                       | 1500                                 |
| name                      | multisegment1                        |
| port_security_enabled     | True                                 |
| provider:network_type     | vlan                                 |
| provider:physical_network | provider1                            |
| provider:segmentation_id  | 2016                                 |
| revision_number           | 1                                    |
| router:external           | Internal                             |
| shared                    | True                                 |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
+---------------------------+--------------------------------------+
```

3. Rename the default segment to `segment1`.

```
$ openstack network segment list --network multisegment1
+--------------------------------------+----------+----------------------
↪--------------+--------------+---------+
| ID                                   | Name     | Network              ␣
↪              | Network Type | Segment |
+--------------------------------------+----------+----------------------
↪--------------+--------------+---------+
| 43e16869-ad31-48e4-87ce-acf756709e18 | None     | 6ab19caa-dda9-4b3d-
↪abc4-5b8f435b98d9 | vlan         |    2016 |
+--------------------------------------+----------+----------------------
↪--------------+--------------+---------+
```

```
$ openstack network segment set --name segment1 43e16869-ad31-48e4-87ce-
↪acf756709e18
```

Note

This command provides no output.

4. Create a second segment on the provider network. In this example, the segment uses the `provider2` physical network with VLAN ID 2017.

```
$ openstack network segment create --physical-network provider2 \
  --network-type vlan --segment 2017 --network multisegment1 segment2
+------------------+--------------------------------------+
| Field            | Value                                |
+------------------+--------------------------------------+
| description      | None                                 |
| headers          |                                      |
| id               | 053b7925-9a89-4489-9992-e164c8cc8763 |
| name             | segment2                             |
| network_id       | 6ab19caa-dda9-4b3d-abc4-5b8f435b98d9 |
| network_type     | vlan                                 |
| physical_network | provider2                            |
| revision_number  | 1                                    |
| segmentation_id  | 2017                                 |
| tags             | []                                   |
+------------------+--------------------------------------+
```

5. Verify that the network contains the `segment1` and `segment2` segments.

```
$ openstack network segment list --network multisegment1
+--------------------------------------+----------+---------------------
↪--------------+--------------+---------+
| ID                                   | Name     | Network             ␣
↪              | Network Type | Segment |
+--------------------------------------+----------+---------------------
↪--------------+--------------+---------+
| 053b7925-9a89-4489-9992-e164c8cc8763 | segment2 | 6ab19caa-dda9-4b3d-
↪abc4-5b8f435b98d9 | vlan         |    2017 |
| 43e16869-ad31-48e4-87ce-acf756709e18 | segment1 | 6ab19caa-dda9-4b3d-
↪abc4-5b8f435b98d9 | vlan         |    2016 |
+--------------------------------------+----------+---------------------
↪--------------+--------------+---------+
```

6. Create subnets on the `segment1` segment. In this example, the IPv4 subnet uses 203.0.113.0/24 and the IPv6 subnet uses fd00:203:0:113::/64.

```
$ openstack subnet create \
  --network multisegment1 --network-segment segment1 \
  --ip-version 4 --subnet-range 203.0.113.0/24 \
  multisegment1-segment1-v4
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
```

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```
| allocation_pools  | 203.0.113.2-203.0.113.254            |
| cidr              | 203.0.113.0/24                       |
| enable_dhcp       | True                                 |
| gateway_ip        | 203.0.113.1                          |
| id                | c428797a-6f8e-4cb1-b394-c404318a2762 |
| ip_version        | 4                                    |
| name              | multisegment1-segment1-v4            |
| network_id        | 6ab19caa-dda9-4b3d-abc4-5b8f435b98d9 |
| revision_number   | 1                                    |
| segment_id        | 43e16869-ad31-48e4-87ce-acf756709e18 |
| tags              | []                                   |
+-------------------+--------------------------------------+

$ openstack subnet create \
  --network multisegment1 --network-segment segment1 \
  --ip-version 6 --subnet-range fd00:203:0:113::/64 \
  --ipv6-address-mode slaac multisegment1-segment1-v6
+-------------------+------------------------------------------------------
↪-+
| Field             | Value                                                ␣
↪ |
+-------------------+------------------------------------------------------
↪-+
| allocation_pools  | fd00:203:0:113::2-
↪fd00:203:0:113:ffff:ffff:ffff:ffff |
| cidr              | fd00:203:0:113::/64                                  ␣
↪ |
| enable_dhcp       | True                                                 ␣
↪ |
| gateway_ip        | fd00:203:0:113::1                                    ␣
↪ |
| id                | e41cb069-9902-4c01-9e1c-268c8252256a                 ␣
↪ |
| ip_version        | 6                                                    ␣
↪ |
| ipv6_address_mode | slaac                                                ␣
↪ |
| ipv6_ra_mode      | None                                                 ␣
↪ |
| name              | multisegment1-segment1-v6                            ␣
↪ |
| network_id        | 6ab19caa-dda9-4b3d-abc4-5b8f435b98d9                 ␣
↪ |
| revision_number   | 1                                    |
| segment_id        | 43e16869-ad31-48e4-87ce-acf756709e18                 ␣
↪ |
| tags              | []                                                   ␣
↪ |
+-------------------+------------------------------------------------------
↪-+
```

Note

By default, IPv6 subnets on provider networks rely on physical network infrastructure for stateless address autoconfiguration (SLAAC) and router advertisement.

7. Create subnets on the `segment2` segment. In this example, the IPv4 subnet uses 198.51.100.0/24 and the IPv6 subnet uses fd00:198:51:100::/64.

```
$ openstack subnet create \
  --network multisegment1 --network-segment segment2 \
  --ip-version 4 --subnet-range 198.51.100.0/24 \
  multisegment1-segment2-v4
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 198.51.100.2-198.51.100.254          |
| cidr              | 198.51.100.0/24                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 198.51.100.1                         |
| id                | 242755c2-f5fd-4e7d-bd7a-342ca95e50b2 |
| ip_version        | 4                                    |
| name              | multisegment1-segment2-v4            |
| network_id        | 6ab19caa-dda9-4b3d-abc4-5b8f435b98d9 |
| revision_number   | 1                                    |
| segment_id        | 053b7925-9a89-4489-9992-e164c8cc8763 |
| tags              | []                                   |
+-------------------+--------------------------------------+

$ openstack subnet create \
  --network multisegment1 --network-segment segment2 \
  --ip-version 6 --subnet-range fd00:198:51:100::/64 \
  --ipv6-address-mode slaac multisegment1-segment2-v6
+-------------------+------------------------------------------------------
↪---+
| Field             | Value                                                ␣
↪   |
+-------------------+------------------------------------------------------
↪---+
| allocation_pools  | fd00:198:51:100::2-
↪fd00:198:51:100:ffff:ffff:ffff:ffff |
| cidr              | fd00:198:51:100::/64                                 ␣
↪   |
| enable_dhcp       | True                                                 ␣
↪   |
| gateway_ip        | fd00:198:51:100::1                                   ␣
↪   |
| id                | b884c40e-9cfe-4d1b-a085-0a15488e9441                 ␣
↪   |
| ip_version        | 6                                                    ␣
↪   |
```

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```
| ipv6_address_mode | slaac                                                ␣
↪   |
| ipv6_ra_mode      | None                                                 ␣
↪   |
| name              | multisegment1-segment2-v6                            ␣
↪   |
| network_id        | 6ab19caa-dda9-4b3d-abc4-5b8f435b98d9                 ␣
↪   |
| revision_number   | 1                                                    ␣
↪   |
| segment_id        | 053b7925-9a89-4489-9992-e164c8cc8763                 ␣
↪   |
| tags              | []                                                   ␣
↪   |
+-------------------+------------------------------------------------------
↪---+
```

8. Verify that each IPv4 subnet associates with at least one DHCP agent.

```
$ openstack network agent list --agent-type dhcp --network multisegment1
+--------------------------------------+------------+-------------+-------
↪------------+-------+-------+--------------------+
| ID                                   | Agent Type | Host        |␣
↪Availability Zone | Alive | State | Binary             |
+--------------------------------------+------------+-------------+-------
↪------------+-------+-------+--------------------+
| c904ed10-922c-4c1a-84fd-d928abaf8f55 | DHCP agent | compute0001 | nova ␣
↪            | :-)   | UP    | neutron-dhcp-agent |
| e0b22cc0-d2a6-4f1c-b17c-27558e20b454 | DHCP agent | compute0101 | nova ␣
↪            | :-)   | UP    | neutron-dhcp-agent |
+--------------------------------------+------------+-------------+-------
↪------------+-------+-------+--------------------+
```

9. Verify that inventories were created for each segment IPv4 subnet in the Compute service placement API (for the sake of brevity, only one of the segments is shown in this example).

```
$ SEGMENT_ID=053b7925-9a89-4489-9992-e164c8cc8763
$ openstack resource provider inventory list $SEGMENT_ID
+----------------+------------------+----------+----------+-----------+---
↪-------+-------+
| resource_class | allocation_ratio | max_unit | reserved | step_size |␣
↪min_unit | total |
+----------------+------------------+----------+----------+-----------+---
↪-------+-------+
| IPV4_ADDRESS   |              1.0 |        1 |        2 |         1 | ␣
↪     1 |    30 |
+----------------+------------------+----------+----------+-----------+---
↪-------+-------+
```

10. Verify that host aggregates were created for each segment in the Compute service (for the sake of brevity, only one of the segments is shown in this example).

```
$ openstack aggregate list
+----+-------------------------------------------------------+----------
↪---------+
| Id | Name                                                  |␣
↪Availability Zone |
+----+-------------------------------------------------------+----------
↪---------+
| 10 | Neutron segment id 053b7925-9a89-4489-9992-e164c8cc8763 | None     ␣
↪          |
+----+-------------------------------------------------------+----------
↪---------+
```

11. Launch one or more instances. Each instance obtains IP addresses according to the segment it uses on the particular compute node.

Note

If a fixed IP is specified by the user in the port create request, that particular IP is allocated immediately to the port. However, creating a port and passing it to an instance yields a different behavior than conventional networks. If the fixed IP is not specified on the port create request, the Networking service defers assignment of IP addresses to the port until the particular compute node becomes apparent. For example:

```
$ openstack port create --network multisegment1 port1
+-----------------------+--------------------------------------+
| Field                 | Value                                |
+-----------------------+--------------------------------------+
| admin_state_up        | UP                                   |
| binding_vnic_type     | normal                               |
| id                    | 6181fb47-7a74-4add-9b6b-f9837c1c90c4 |
| ip_allocation         | deferred                             |
| mac_address           | fa:16:3e:34:de:9b                    |
| name                  | port1                                |
| network_id            | 6ab19caa-dda9-4b3d-abc4-5b8f435b98d9 |
| port_security_enabled | True                                 |
| revision_number       | 1                                    |
| security_groups       | e4fcef0d-e2c5-40c3-a385-9c33ac9289c5 |
| status                | DOWN                                 |
| tags                  | []                                   |
+-----------------------+--------------------------------------+
```

Migrating non-routed networks to routed

Migration of existing non-routed networks is only possible if there is only one segment and one subnet on the network. To migrate a candidate network, update the subnet and set `id` of the existing network segment as `segment_id`.

Note

In the case where there are multiple subnets or segments it is not possible to safely migrate. The reason for this is that in non-routed networks addresses from the subnets allocation pools are assigned to ports without considering to which network segment the port is bound.

Example

The following steps migrate an existing non-routed network with one subnet and one segment to a routed one.

1. Source the administrative project credentials.
2. Get the id of the current network segment on the network that is being migrated.

```
$ openstack network segment list --network my_network
+--------------------------------------+------+------------------------
↪------------+--------------+----------+
| ID                                   | Name | Network                ␣
↪            | Network Type | Segment |
+--------------------------------------+------+------------------------
↪------------+--------------+----------+
| 81e5453d-4c9f-43a5-8ddf-feaf3937e8c7 | None | 45e84575-2918-471c-95c0-
↪018b961a2984 | flat         | None    |
+--------------------------------------+------+------------------------
↪------------+--------------+----------+
```

3. Get the id or name of the current subnet on the network.

```
$ openstack subnet list --network my_network
+--------------------------------------+-----------+--------------------
↪----------------+----------------+
| ID                                   | Name      | Network            ␣
↪                | Subnet         |
+--------------------------------------+-----------+--------------------
↪----------------+----------------+
| 71d931d2-0328-46ae-93bc-126caf794307 | my_subnet | 45e84575-2918-471c-
↪95c0-018b961a2984 | 172.24.4.0/24 |
+--------------------------------------+-----------+--------------------
↪----------------+----------------+
```

4. Verify the current segment_id of the subnet is None.

```
$ openstack subnet show my_subnet --c segment_id
+------------+-------+
| Field      | Value |
+------------+-------+
| segment_id | None  |
+------------+-------+
```

5. Update the segment_id of the subnet.

```
$ openstack subnet set --network-segment 81e5453d-4c9f-43a5-8ddf-
↪feaf3937e8c7 my_subnet
```

6. Verify that the subnet is now associated with the desired network segment.

```
$ openstack subnet show my_subnet --c segment_id
+------------+--------------------------------------+
| Field      | Value                                |
+------------+--------------------------------------+
| segment_id | 81e5453d-4c9f-43a5-8ddf-feaf3937e8c7 |
+------------+--------------------------------------+
```

Routed provider networks as external networks for tenant routed networks

Note

This section applies only to legacy routers, not DVR nor HA routers. A legacy router has a single instance that is hosted in one single host.

One of the consequences of this feature is the externalization of any routing operation. The communication (routing) between segments is done using the underlying network infrastructure, not managed by Neutron.

Could be the case that the user needs to split the communication between several hosts. It is possible to create tenant networks and connect them using a router. To access to the routed provider network, it should be connected as router gateway.

```
Tenant net1


                                                   Routed provided network
                                    GW port
Tenant net2
```

The routed provider network, acting as router gateway, contains all subnets associated to the segments. In a deployment without routed provided networks, the gateway port has L2 connectivity to all subnet CIDRs. In this case, the gateway port has only connectivity to the attached segment subnets and its L2 broadcast domains.

The L3 agent will create, inside the router namespace, a default route in the gateway port fixed IP CIDR. For each other subnet not belonging to the ports fixed IP address, an onlink route is created. These routes use the gateway port as routing device and allow to route any packet with destination on these CIDRs through this port.

The problem in the case of connecting the gatewat port to a routed provider network is that it will have broadcast connectivity only to those subnets that belong to the host segment:

- One of those subnets will provide the port IP address. The gateway IP address of this subnet will be the default route, through the gateway port.
- Any other subnet belonging to this segment will create a onlink route, using the gateway port as route device.

For example, lets consider the following configuration:

- Two tenant networks with CIDRs 10.1.0.0/24 and 10.2.0.0/24.

- A RPN with two segments; each segment with two subnets: segment 1 with 10.51.0.0/24 and 10.52.0.0/24, segment 2 with 10.53.0.0/24 and 10.54.0.0/24.
- The router is connected to the first segment and the gateway port has an IP address in the range of 10.51.0.0/24. This is why the default route uses an IP address in this range.

Without considering that the gateway network is a router provided network, this is the routing table set in the router namespace:

```
$ ip netns exec $r ip r
default via 10.51.0.1 dev qg-gwport proto static
10.1.0.0/24 dev qr-tenant1 proto kernel scope link src 10.1.0.1
10.2.0.0/24 dev qr-tenant2 proto kernel scope link src 10.2.0.1
10.51.0.0/24 dev qg-gwport proto kernel scope link src 10.100.0.15
10.52.0.0/24 dev qg-gwport proto static scope link
10.53.0.0/24 dev qg-gwport proto static scope link  <-- should be removed,␣
↪belongs to segment 2
10.54.0.0/24 dev qg-gwport proto static scope link  <-- should be removed,␣
↪belongs to segment 2
```

Those packets sent to 10.53.0.0/24 and 10.54.0.0/24 (the second RPN subnet CIDRs), dont have L2 connectivity and the ARP packets wont be replied. In the case of having a RPN as gateway network, all packets exiting the router through the gateway, must be sent to the gateway IP address, in this case 10.51.0.1. This is why the L3 plugin does not send the information of other segments subnets L3 agent when:

- The network is the router gateway.
- The segments plugin is enabled; this plugin is needed for routed provided networks.
- The network is connected to a segment.

Multiple routed provider segments per host

Starting with 2023.1 (Antelope), the support of routed provider networks has been enhanced to handle multiple segments per host. The main consequence will be for an operator to extend the IP pool without creating multiple networks and/or increasing broadcast domain.

Note

The present support is only available for OVS agent at this point.

1. On a given provider network, create a second segment. In this example, the second segment uses the `provider1` physical network with VLAN ID 2020.

```
$ openstack network segment create --physical-network provider1 \
  --network-type vlan --segment 2020 --network multisegment1 segment1-2
+------------------+--------------------------------------+
| Field            | Value                                |
+------------------+--------------------------------------+
| description      | None                                 |
| headers          |                                      |
| id               | 333b7925-9a89-4489-9992-e164c8cc8764 |
```

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```
| name              | segment1-2                           |
| network_id        | 6ab19caa-dda9-4b3d-abc4-5b8f435b98d9 |
| network_type      | vlan                                 |
| physical_network  | provider1                            |
| revision_number   | 1                                    |
| segmentation_id   | 2020                                 |
| tags              | []                                   |
+-------------------+--------------------------------------+
```

2. Create subnets on the `segment1-2` segment. In this example, the IPv4 subnet uses 203.0.114.0/24.

```
$ openstack subnet create \
  --network multisegment1 --network-segment segment1-2 \
  --ip-version 4 --subnet-range 203.0.114.0/24 \
  multisegment1-segment1-2
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 203.0.114.2-203.0.114.254            |
| cidr              | 203.0.114.0/24                       |
| enable_dhcp       | True                                 |
| gateway_ip        | 203.0.114.1                          |
| id                | c428797a-6f8e-4cb1-b394-c404318a2762 |
| ip_version        | 4                                    |
| name              | multisegment1-segment1-2             |
| network_id        | 6ab19caa-dda9-4b3d-abc4-5b8f435b98d9 |
| revision_number   | 1                                    |
| segment_id        | 333b7925-9a89-4489-9992-e164c8cc8764 |
| tags              | []                                   |
+-------------------+--------------------------------------+
```

Considering that, for a subnet of the given provider network `provider1` running out of available IP, Neutron will automatically switch to the subnet `multisegment1-segment1-2`.

8.2.32 Router flavors with the L3 OVN service plugin

In this chapter we give examples on how to create routers with user-defined flavors.

Note

The following example refers to a dummy user-defined service provider, which in a real situation must be replaced with user provided code.

1. Add service providers to neutron.conf. The second provider is a high availability version of the first one:

```
[service_providers]
service_provider = L3_ROUTER_NAT:user-defined:neutron.services.ovn_l3.
↪service_providers.user_defined.UserDefined
```

2. Re-start the neutron server and verify the user-defined provider has been loaded:

```
$ openstack network service provider list
+---------------+----------------+---------+
| Service Type  | Name           | Default |
+---------------+----------------+---------+
| L3_ROUTER_NAT | user-defined   | False   |
| L3_ROUTER_NAT | ovn            | True    |
+---------------+----------------+---------+
```

3. Create service profiles for the router flavors:

```
$ openstack network flavor profile create --description "User-defined␣
↪router flavor profile" --enable --driver neutron.services.ovn_l3.
↪service_providers.user_defined.UserDefined
+-------------+--------------------------------------------------------------
↪---------+
| Field       | Value                                                        ␣
↪         |
+-------------+--------------------------------------------------------------
↪---------+
| description | User-defined router flavor profile                           ␣
↪         |
| driver      | neutron.services.ovn_l3.service_providers.user_defined.
↪UserDefined |
| enabled     | True                                                         ␣
↪         |
| id          | a717c92c-63f7-47e8-9efb-6ad0d61c4875                         ␣
↪         |
| meta_info   |                                                              ␣
↪         |
| project_id  | None                                                         ␣
↪         |
+-------------+--------------------------------------------------------------
↪---------+
```

4. Create the router flavors:

```
$ openstack network flavor create --service-type L3_ROUTER_NAT --
↪description "User-defined flavor for routers in the L3 OVN plugin" user-
↪defined-router-flavor
+---------------------+------------------------------------------------------
↪---+
| Field               | Value                                                ␣
↪   |
+---------------------+------------------------------------------------------
↪---+
| description         | User-defined flavor for routers in the L3 OVN␣
↪plugin |
| enabled             | True                                                 ␣
↪   |
| id                  | 65df2587-c535-4c3a-af2f-86b2968a3191                 ␣
```

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```
↪    |
| name               | user-defined-router-flavor                     ␣
↪    |
| service_profile_ids | []                                             ␣
↪    |
| service_type       | L3_ROUTER_NAT                                   ␣
↪    |
+--------------------+----------------------------------------------
↪---+
```

5. Add service profile to the router flavors:

```
$ openstack network flavor add profile user-defined-router-flavor␣
↪a717c92c-63f7-47e8-9efb-6ad0d61c4875
```

6. Create routers specifying user-defined flavors. Please note the *ha* characteristics of the routers created:

```
$ openstack router create router-of-user-defined-flavor-noha --no-ha --
↪external-gateway public --flavor-id 65df2587-c535-4c3a-af2f-
↪86b2968a3191 --max-width 100
+-------------------------+---------------------------------------------
↪-----------------------+
| Field                   | Value                                        ␣
↪                        |
+-------------------------+---------------------------------------------
↪-----------------------+
| admin_state_up          | UP                                           ␣
↪                        |
| availability_zone_hints |                                              ␣
↪                        |
| availability_zones      |                                              ␣
↪                        |
| created_at              | 2024-03-27T00:31:56Z                         ␣
↪                        |
| description             |                                              ␣
↪                        |
| enable_default_route_bfd | False                                       ␣
↪                        |
| enable_default_route_ecmp | False                                      ␣
↪                        |
| enable_ndp_proxy        | None                                         ␣
↪                        |
| external_gateway_info   | {"network_id": "f1898eb8-54af-4704-8ce2-
↪cf58d37cd1e1",             |
|                         | "external_fixed_ips": [{"subnet_id":         ␣
↪                        |
|                         | "5f2b4aac-7ef4-4e8a-bd80-a5e1e640e16b", "ip_
↪address":               |
|                         | "172.24.8.113"}, {"subnet_id":               ␣
```

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```
↪                          |
|                           | "07227d2b-f102-4788-97f8-a8e8f1b0f6ae", "ip_
↪address":                 |
|                           | "2001:db8::234"}], "enable_snat": true}     ␣
↪                          |
| external_gateways         | [{'network_id': 'f1898eb8-54af-4704-8ce2-
↪cf58d37cd1e1',                |
|                           | 'external_fixed_ips': [{'ip_address': '172.
↪24.8.113', 'subnet_id':   |
|                           | '5f2b4aac-7ef4-4e8a-bd80-a5e1e640e16b'}, {
↪'ip_address':                |
|                           | '2001:db8::234', 'subnet_id':               ␣
↪                          |
|                           | '07227d2b-f102-4788-97f8-a8e8f1b0f6ae'}]}]  ␣
↪                          |
| flavor_id                 | 65df2587-c535-4c3a-af2f-86b2968a3191        ␣
↪                          |
| ha                        | False                                       ␣
↪                          |
| id                        | 66399600-d4c6-4d25-a05f-10789bf86b2d        ␣
↪                          |
| name                      | router-of-user-defined-flavor-noha          ␣
↪                          |
| project_id                | d458a40ca6d54aa6b2b92721badc9f48            ␣
↪                          |
| revision_number           | 3                                           ␣
↪                          |
| routes                    |                                             ␣
↪                          |
| status                    | ACTIVE                                      ␣
↪                          |
| tags                      |                                             ␣
↪                          |
| tenant_id                 | d458a40ca6d54aa6b2b92721badc9f48            ␣
↪                          |
| updated_at                | 2024-03-27T00:31:56Z                        ␣
↪                          |
+---------------------------+--------------------------------------------
↪--------------------------+

$ openstack router create router-of-user-defined-flavor-ha --ha --
↪external-gateway public --flavor-id 65df2587-c535-4c3a-af2f-
↪86b2968a3191 --max-width 100
+---------------------------+--------------------------------------------
↪--------------------------+
| Field                     | Value                                       ␣
↪                          |
+---------------------------+--------------------------------------------
↪--------------------------+
```

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```
| admin_state_up            | UP                                                     ␣
↪                           |
| availability_zone_hints   |                                                        ␣
↪                           |
| availability_zones        |                                                        ␣
↪                           |
| created_at                | 2024-03-27T00:38:47Z                                   ␣
↪                           |
| description               |                                                        ␣
↪                           |
| enable_default_route_bfd  | False                                                  ␣
↪                           |
| enable_default_route_ecmp | False                                                  ␣
↪                           |
| enable_ndp_proxy          | None                                                   ␣
↪                           |
| external_gateway_info     | {"network_id": "f1898eb8-54af-4704-8ce2-
↪cf58d37cd1e1",                |
|                           | "external_fixed_ips": [{"subnet_id":                   ␣
↪                           |
|                           | "5f2b4aac-7ef4-4e8a-bd80-a5e1e640e16b", "ip_
↪address":                  |
|                           | "172.24.8.212"}, {"subnet_id":                         ␣
↪                           |
|                           | "07227d2b-f102-4788-97f8-a8e8f1b0f6ae", "ip_
↪address":                  |
|                           | "2001:db8::20a"}], "enable_snat": true}                ␣
↪                           |
| external_gateways         | [{'network_id': 'f1898eb8-54af-4704-8ce2-
↪cf58d37cd1e1',                |
|                           | 'external_fixed_ips': [{'ip_address': '172.
↪24.8.212', 'subnet_id':    |
|                           | '5f2b4aac-7ef4-4e8a-bd80-a5e1e640e16b'}, {
↪'ip_address':                |
|                           | '2001:db8::20a', 'subnet_id':                          ␣
↪                           |
|                           | '07227d2b-f102-4788-97f8-a8e8f1b0f6ae'}]}]             ␣
↪                           |
| flavor_id                 | 65df2587-c535-4c3a-af2f-86b2968a3191                   ␣
↪                           |
| ha                        | True                                                   ␣
↪                           |
| id                        | 036e639b-f087-418d-9087-5a94c45453b9                   ␣
↪                           |
| name                      | router-of-user-defined-flavor-ha                       ␣
↪                           |
| project_id                | d458a40ca6d54aa6b2b92721badc9f48                       ␣
↪                           |
| revision_number           | 3                                                      ␣
```

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```
↪                         |
| routes                  |                                                        ␣
↪                         |
| status                  | ACTIVE                                                 ␣
↪                         |
| tags                    |                                                        ␣
↪                         |
| tenant_id               | d458a40ca6d54aa6b2b92721badc9f48                       ␣
↪                         |
| updated_at              | 2024-03-27T00:38:48Z                                   ␣
↪                         |
+-------------------------+-------------------------------------------------------
↪------------------------+

$ openstack router create router-of-user-defined-flavor-noha-implicit --
↪external-gateway public --flavor-id 65df2587-c535-4c3a-af2f-
↪86b2968a3191 --max-width 100
+-------------------------+-------------------------------------------------------
↪------------------------+
| Field                   | Value                                                  ␣
↪                         |
+-------------------------+-------------------------------------------------------
↪------------------------+
| admin_state_up          | UP                                                     ␣
↪                         |
| availability_zone_hints |                                                        ␣
↪                         |
| availability_zones      |                                                        ␣
↪                         |
| created_at              | 2024-03-27T00:40:52Z                                   ␣
↪                         |
| description             |                                                        ␣
↪                         |
| enable_default_route_bfd | False                                                 ␣
↪                         |
| enable_default_route_ecmp | False                                                ␣
↪                         |
| enable_ndp_proxy        | None                                                   ␣
↪                         |
| external_gateway_info   | {"network_id": "f1898eb8-54af-4704-8ce2-
↪cf58d37cd1e1",              |
|                         | "external_fixed_ips": [{"subnet_id":                   ␣
↪                         |
|                         | "5f2b4aac-7ef4-4e8a-bd80-a5e1e640e16b", "ip_
↪address":                |
|                         | "172.24.8.80"}, {"subnet_id":                          ␣
↪                         |
|                         | "07227d2b-f102-4788-97f8-a8e8f1b0f6ae", "ip_
↪address":                |
```

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```
|                          | "2001:db8::19c"}], "enable_snat": true}    ␣
↪                          |
| external_gateways        | [{'network_id': 'f1898eb8-54af-4704-8ce2-
↪cf58d37cd1e1',               |
|                          | 'external_fixed_ips': [{'ip_address': '172.
↪24.8.80', 'subnet_id':     |
|                          | '5f2b4aac-7ef4-4e8a-bd80-a5e1e640e16b'}, {
↪'ip_address':             |
|                          | '2001:db8::19c', 'subnet_id':              ␣
↪                          |
|                          | '07227d2b-f102-4788-97f8-a8e8f1b0f6ae'}]}] ␣
↪                          |
| flavor_id                | 65df2587-c535-4c3a-af2f-86b2968a3191       ␣
↪                          |
| ha                       | False                                      ␣
↪                          |
| id                       | ad2ab001-fc3a-4a3b-a9f0-8ad4f41f54dc       ␣
↪                          |
| name                     | router-of-user-defined-flavor-noha-implicit␣
↪                          |
| project_id               | d458a40ca6d54aa6b2b92721badc9f48           ␣
↪                          |
| revision_number          | 3                                          ␣
↪                          |
| routes                   |                                            ␣
↪                          |
| status                   | ACTIVE                                     ␣
↪                          |
| tags                     |                                            ␣
↪                          |
| tenant_id                | d458a40ca6d54aa6b2b92721badc9f48           ␣
↪                          |
| updated_at               | 2024-03-27T00:40:53Z                       ␣
↪                          |
+--------------------------+-------------------------------------------
↪-------------------------+
```

7. Create an OVN flavor router to verify it co-exists with the user-defined flavors:

```
$ openstack router create ovn-flavor-router --external-gateway public --
↪max-width 100
+-------------------------+--------------------------------------------
↪-------------------------+
| Field                   | Value                                      ␣
↪                          |
+-------------------------+--------------------------------------------
↪-------------------------+
| admin_state_up          | UP                                         ␣
↪                          |
| availability_zone_hints |                                            ␣
```

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```
↪                          |
| availability_zones      |                                                      ␣
↪                          |
| created_at              | 2023-05-25T23:34:20Z                                 ␣
↪                          |
| description             |                                                      ␣
↪                          |
| enable_ndp_proxy        | None                                                 ␣
↪                          |
| external_gateway_info   | {"network_id": "ba485dc9-2459-41c1-9d4f-
↪71914a7fba2a",                  |
|                         | "external_fixed_ips": [{"subnet_id":                 ␣
↪                          |
|                         | "2e3adb94-c544-4916-a9fb-27a9dea21820", "ip_
↪address": "172.24.8.195"}, |
|                         | {"subnet_id": "996ed143-917b-4783-8349-
↪03c6a6d9603e", "ip_address":    |
|                         | "2001:db8::263"}], "enable_snat": true}              ␣
↪                          |
| flavor_id               | None                                                 ␣
↪                          |
| ha                      | True                                                 ␣
↪                          |
| id                      | 21889ed3-b8df-4b0e-9a64-92ba9fab655d                 ␣
↪                          |
| name                    | ovn-flavor-router                                    ␣
↪                          |
| project_id              | b807321af03f44dc808ff06bbc845804                     ␣
↪                          |
| revision_number         | 3                                                    ␣
↪                          |
| routes                  |                                                      ␣
↪                          |
| status                  | ACTIVE                                               ␣
↪                          |
| tags                    |                                                      ␣
↪                          |
| tenant_id               | e6d6b109d16b4e5e857a10034f4ba558                     ␣
↪                          |
| updated_at              | 2023-07-20T23:34:21Z                                 ␣
↪                          |
+-------------------------+---------------------------------------------------
↪-------------------------+
```

Note

OVN routers are natively highly available at the OVN/OVS level, through the use of BFD monitoring. Neutron doesnt get involved in the high availability aspect beyond router scheduling. For this reason, the *ha* attribute is associated to routers of the default OVN flavor and is always set to *True*. This is done for consistency with user defined flavors routers for which the *ha* attribute will be *True* or *False*, depending on the characteristics of the router.

8. List routers to verify:

```
$ openstack router list
+--------------------------------------+--------------------------------
↪----------+--------+-------+----------------------------------+-------+
| ID                                   | Name                          ␣
↪          | Status | State | Project                          | HA    |
+--------------------------------------+--------------------------------
↪----------+--------+-------+----------------------------------+-------+
| 21889ed3-b8df-4b0e-9a64-92ba9fab655d | ovn-flavor-router             ␣
↪          | ACTIVE | UP    | b807321af03f44dc808ff06bbc845804 | True  |
| 66399600-d4c6-4d25-a05f-10789bf86b2d | router-of-user-defined-flavor-
↪noha          | ACTIVE | UP    | d458a40ca6d54aa6b2b92721badc9f48 |␣
↪False |
| 036e639b-f087-418d-9087-5a94c45453b9 | router-of-user-defined-flavor-ha␣
↪          | ACTIVE | UP    | d458a40ca6d54aa6b2b92721badc9f48 | True  |
| ad2ab001-fc3a-4a3b-a9f0-8ad4f41f54dc | router-of-user-defined-flavor-
↪noha-implicit | ACTIVE | UP    | d458a40ca6d54aa6b2b92721badc9f48 |␣
↪False |
+--------------------------------------+--------------------------------
↪----------+--------+-------+----------------------------------+-------+
```

8.2.33 SR-IOV

The purpose of this page is to describe how to enable SR-IOV functionality available in OpenStack (using OpenStack Networking). This functionality was first introduced in the OpenStack Juno release. This page intends to serve as a guide for how to configure OpenStack Networking and OpenStack Compute to create SR-IOV ports.

The basics

PCI-SIG Single Root I/O Virtualization and Sharing (SR-IOV) functionality is available in OpenStack since the Juno release. The SR-IOV specification defines a standardized mechanism to virtualize PCIe devices. This mechanism can virtualize a single PCIe Ethernet controller to appear as multiple PCIe devices. Each device can be directly assigned to an instance, bypassing the hypervisor and virtual switch layer. As a result, users are able to achieve low latency and near-line wire speed.

The following terms are used throughout this document:

Term	Definition
PF	Physical Function. The physical Ethernet controller that supports SR-IOV.
VF	Virtual Function. The virtual PCIe device created from a physical Ethernet controller.

SR-IOV agent

The SR-IOV agent allows you to set the admin state of ports, configure port security (enable and disable spoof checking), and configure QoS rate limiting and minimum bandwidth. You must include the SR-IOV agent on each compute node using SR-IOV ports.

> **Note**
>
> The SR-IOV agent was optional before Mitaka, and was not enabled by default before Liberty.

> **Note**
>
> The ability to control port security and QoS rate limit settings was added in Liberty.

Supported Ethernet controllers

The following manufacturers are known to work:

- Intel
- Mellanox
- QLogic
- Broadcom

For information on **Mellanox SR-IOV Ethernet ConnectX cards**, see the Mellanox: How To Configure SR-IOV VFs on ConnectX-4 or newer.

For information on **QLogic SR-IOV Ethernet cards**, see the Users Guide OpenStack Deployment with SR-IOV Configuration.

For information on **Broadcom NetXtreme Series Ethernet cards**, see the Broadcom NetXtreme Product Page.

Using SR-IOV interfaces

In order to enable SR-IOV, the following steps are required:

1. Create Virtual Functions (Compute)
2. Configure allow list for PCI devices in nova-compute (Compute)
3. Configure neutron-server (Controller)
4. Configure nova-scheduler (Controller)
5. Enable neutron sriov-agent (Compute)

We recommend using VLAN provider networks for segregation. This way you can combine instances without SR-IOV ports and instances with SR-IOV ports on a single network.

> **Note**
>
> Throughout this guide, `eth3` is used as the PF and `physnet2` is used as the provider network configured as a VLAN range. These ports may vary in different environments.

Create Virtual Functions (Compute)

Create the VFs for the network interface that will be used for SR-IOV. We use eth3 as PF, which is also used as the interface for the VLAN provider network and has access to the private networks of all machines.

Note

The steps detail how to create VFs using Mellanox ConnectX-4 and newer/Intel SR-IOV Ethernet cards on an Intel system. Steps may differ for different hardware configurations.

1. Ensure SR-IOV and VT-d are enabled in BIOS.
2. Enable IOMMU in Linux by adding `intel_iommu=on` to the kernel parameters, for example, using GRUB.
3. On each compute node, create the VFs via the PCI SYS interface:

```
# echo '8' > /sys/class/net/eth3/device/sriov_numvfs
```

Note

On some PCI devices, observe that when changing the amount of VFs you receive the error `Device or resource busy`. In this case, you must first set `sriov_numvfs` to `0`, then set it to your new value.

Note

A network interface could be used both for PCI passthrough, using the PF, and SR-IOV, using the VFs. If the PF is used, the VF number stored in the `sriov_numvfs` file is lost. If the PF is attached again to the operating system, the number of VFs assigned to this interface will be zero. To keep the number of VFs always assigned to this interface, modify the interfaces configuration file adding an `ifup` script command.

On Ubuntu, modify the `/etc/network/interfaces` file:

```
auto eth3
iface eth3 inet dhcp
pre-up echo '4' > /sys/class/net/eth3/device/sriov_numvfs
```

On RHEL and derivatives, modify the `/sbin/ifup-local` file:

```
#!/bin/sh
if [[ "$1" == "eth3" ]]
then
    echo '4' > /sys/class/net/eth3/device/sriov_numvfs
fi
```

Warning

Alternatively, you can create VFs by passing the `max_vfs` to the kernel module of your network interface. However, the `max_vfs` parameter has been deprecated, so the PCI SYS interface is the preferred method.

You can determine the maximum number of VFs a PF can support:

```
# cat /sys/class/net/eth3/device/sriov_totalvfs
63
```

4. Verify that the VFs have been created and are in up state. For example:

```
# lspci | grep Ethernet
82:00.0 Ethernet controller: Intel Corporation 82599ES 10-Gigabit SFI/
↪SFP+ Network Connection (rev 01)
82:00.1 Ethernet controller: Intel Corporation 82599ES 10-Gigabit SFI/
↪SFP+ Network Connection (rev 01)
82:10.0 Ethernet controller: Intel Corporation 82599 Ethernet Controller␣
↪Virtual Function (rev 01)
82:10.2 Ethernet controller: Intel Corporation 82599 Ethernet Controller␣
↪Virtual Function (rev 01)
82:10.4 Ethernet controller: Intel Corporation 82599 Ethernet Controller␣
↪Virtual Function (rev 01)
82:10.6 Ethernet controller: Intel Corporation 82599 Ethernet Controller␣
↪Virtual Function (rev 01)
82:11.0 Ethernet controller: Intel Corporation 82599 Ethernet Controller␣
↪Virtual Function (rev 01)
82:11.2 Ethernet controller: Intel Corporation 82599 Ethernet Controller␣
↪Virtual Function (rev 01)
82:11.4 Ethernet controller: Intel Corporation 82599 Ethernet Controller␣
↪Virtual Function (rev 01)
82:11.6 Ethernet controller: Intel Corporation 82599 Ethernet Controller␣
↪Virtual Function (rev 01)
```

```
# ip link show eth3
8: eth3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP␣
↪mode DEFAULT qlen 1000
   link/ether a0:36:9f:8f:3f:b8 brd ff:ff:ff:ff:ff:ff
   vf 0 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 1 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 2 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 3 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 4 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 5 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 6 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
   vf 7 MAC 00:00:00:00:00:00, spoof checking on, link-state auto
```

If the interfaces are down, set them to up before launching a guest, otherwise the instance will fail to spawn:

```
# ip link set eth3 up
```

5. Persist created VFs on reboot:

```
# echo "echo '7' > /sys/class/net/eth3/device/sriov_numvfs" >> /etc/rc.
↪local
```

Note

The suggested way of making PCI SYS settings persistent is through the `sysfsutils` tool. However, this is not available by default on many major distributions.

Configuring allow list for PCI devices nova-compute (Compute)

1. Configure which PCI devices the `nova-compute` service may use. Edit the `nova.conf` file:

```
[pci]
passthrough_whitelist = { "devname": "eth3", "physical_network": "physnet2
↪"}
```

This tells the Compute service that all VFs belonging to `eth3` are allowed to be passed through to instances and belong to the provider network `physnet2`.

Alternatively the `[pci] passthrough_whitelist` parameter also supports allowing devices by:

- PCI address: The address uses the same syntax as in `lspci` and an asterisk (*) can be used to match anything.

```
[pci]
passthrough_whitelist = { "address": "[[[[<domain>]:]<bus>]:][<slot>
↪][.[<function>]]", "physical_network": "physnet2" }
```

For example, to match any domain, bus `0a`, slot `00`, and all functions:

```
[pci]
passthrough_whitelist = { "address": "*:0a:00.*", "physical_network
↪": "physnet2" }
```

- PCI `vendor_id` and `product_id` as displayed by the Linux utility `lspci`.

```
[pci]
passthrough_whitelist = { "vendor_id": "<id>", "product_id": "<id>",
↪"physical_network": "physnet2" }
```

If the device defined by the PCI address or `devname` corresponds to an SR-IOV PF, all VFs under the PF will match the entry. Multiple `[pci] passthrough_whitelist` entries per host are supported.

In order to enable SR-IOV to request trusted mode, the `[pci] passthrough_whitelist` parameter also supports a `trusted` tag.

Note

This capability is only supported starting with version 18.0.0 (Rocky) release of the compute service configured to use the libvirt driver.

Important

There are security implications of enabling trusted ports. The trusted VFs can be set into VF promiscuous mode which will enable it to receive unmatched and multicast traffic sent to the physical function.

For example, to allow users to request SR-IOV devices with trusted capabilities on device `eth3`:

```
[pci]
passthrough_whitelist = { "devname": "eth3", "physical_network": "physnet2
↪", "trusted":"true" }
```

The ports will have to be created with a binding profile to match the `trusted` tag, see *Launching instances with SR-IOV ports*.

2. Restart the `nova-compute` service for the changes to go into effect.

Configure neutron-server (Controller)

Note

This section does not apply to remote-managed ports of SmartNIC DPU devices which also use SR-IOV at the host side but do not rely on the `sriovnicswitch` mechanism driver.

1. Add `sriovnicswitch` as mechanism driver. Edit the `ml2_conf.ini` file on each controller:

```
[ml2]
mechanism_drivers = openvswitch,sriovnicswitch
```

2. Ensure your physnet is configured for the chosen network type. Edit the `ml2_conf.ini` file on each controller:

```
[ml2_type_vlan]
network_vlan_ranges = physnet2
```

3. Add the `plugin.ini` file as a parameter to the `neutron-server` service. Edit the appropriate initialization script to configure the `neutron-server` service to load the plugin configuration file:

```
--config-file /etc/neutron/neutron.conf
--config-file /etc/neutron/plugin.ini
```

4. Restart the `neutron-server` service.

Configure nova-scheduler (Controller)

1. On every controller node running the `nova-scheduler` service, add `PciPassthroughFilter` to `[filter_scheduler] enabled_filters` to enable this filter. Ensure `[filter_scheduler] available_filters` is set to the default of `nova.scheduler.filters.all_filters`:

```
[filter_scheduler]
enabled_filters = ComputeFilter, ComputeCapabilitiesFilter,␣
↪ImagePropertiesFilter, ServerGroupAntiAffinityFilter,␣
↪ServerGroupAffinityFilter, PciPassthroughFilter
available_filters = nova.scheduler.filters.all_filters
```

2. Restart the `nova-scheduler` service.

Enable neutron-sriov-nic-agent (Compute)

1. Install the SR-IOV agent, if necessary.

2. Edit the `sriov_agent.ini` file on each compute node. For example:

```
[sriov_nic]
physical_device_mappings = physnet2:eth3
exclude_devices =
```

Note

The `physical_device_mappings` parameter is not limited to be a 1-1 mapping between physical networks and NICs. This enables you to map the same physical network to more than one NIC. For example, if `physnet2` is connected to `eth3` and `eth4`, then `physnet2:eth3, physnet2:eth4` is a valid option.

Note

The SR-IOV agent does not implement any kind of firewall driver.

The `exclude_devices` parameter is empty, therefore, all the VFs associated with eth3 may be configured by the agent. To exclude specific VFs, add them to the `exclude_devices` parameter as follows:

```
exclude_devices = eth1:0000:07:00.2;0000:07:00.3,eth2:0000:05:00.1;
↪0000:05:00.2
```

3. Ensure the SR-IOV agent runs successfully:

```
# neutron-sriov-nic-agent \
  --config-file /etc/neutron/neutron.conf \
  --config-file /etc/neutron/plugins/ml2/sriov_agent.ini
```

4. Enable the neutron SR-IOV agent service.

   If installing from source, you must configure a daemon file for the init system manually.

(Optional) FDB L2 agent extension

Forwarding DataBase (FDB) population is an L2 agent extension to OVS agent. Its objective is to update the FDB table for existing instance using normal port. This enables communication between SR-IOV instances and normal instances. The use cases of the FDB population extension are:

- Direct port and normal port instances reside on the same compute node.
- Direct port instance that uses floating IP address and network node are located on the same host.

For additional information describing the problem, refer to: Virtual switching technologies and Linux bridge.

1. Edit the `ovs_agent.ini` file on each compute node. For example:

```
[agent]
extensions = fdb
```

2. Add the FDB section and the `shared_physical_device_mappings` parameter. This parameter maps each physical port to its physical network name. Each physical network can be mapped to several ports:

```
[FDB]
shared_physical_device_mappings = physnet1:p1p1, physnet1:p1p2
```

Launching instances with SR-IOV ports

Once configuration is complete, you can launch instances with SR-IOV ports.

1. If it does not already exist, create a network and subnet for the chosen physnet. This is the network to which SR-IOV ports will be attached. For example:

```
$ openstack network create --provider-physical-network physnet2 \
    --provider-network-type vlan --provider-segment 1000 \
    sriov-net

$ openstack subnet create --network sriov-net \
    --subnet-pool shared-default-subnetpool-v4 \
    sriov-subnet
```

2. Get the `id` of the network where you want the SR-IOV port to be created:

```
$ net_id=$(openstack network show sriov-net -c id -f value)
```

3. Create the SR-IOV port. `vnic-type=direct` is used here, but other options include `normal`, `direct-physical`, and `macvtap`:

```
$ openstack port create --network $net_id --vnic-type direct \
    sriov-port
```

Alternatively, to request that the SR-IOV port accept trusted capabilities, the binding profile should be enhanced with the `trusted` tag.

```
$ openstack port create --network $net_id --vnic-type direct \
    --binding-profile trusted=true \
    sriov-port
```

4. Get the id of the created port:

```
$ port_id=$(openstack port show sriov-port -c id -f value)
```

5. Create the instance. Specify the SR-IOV port created in step two for the NIC:

```
$ openstack server create --flavor m1.large --image ubuntu_18.04 \
    --nic port-id=$port_id \
    test-sriov
```

> **Note**
>
> There are two ways to attach VFs to an instance. You can create an SR-IOV port or use the `pci_alias` in the Compute service. For more information about using `pci_alias`, refer to nova-api configuration.

SR-IOV with ConnectX-3/ConnectX-3 Pro Dual Port Ethernet

In contrast to Mellanox newer generation NICs, ConnectX-3 family network adapters expose a single PCI device (PF) in the system regardless of the number of physical ports. When the device is **dual port** and SR-IOV is enabled and configured we can observe some inconsistencies in linux networking subsystem.

> **Note**
>
> In the example below `enp4s0` represents PF net device associated with physical port 1 and `enp4s0d1` represents PF net device associated with physical port 2.

Example: A system with ConnectX-3 dual port device and a total of four VFs configured, two VFs assigned to port one and two VFs assigned to port two.

```
$ lspci | grep Mellanox
04:00.0 Network controller: Mellanox Technologies MT27520 Family [ConnectX-3␣
↪Pro]
04:00.1 Network controller: Mellanox Technologies MT27500/MT27520 Family␣
↪[ConnectX-3/ConnectX-3 Pro Virtual Function]
04:00.2 Network controller: Mellanox Technologies MT27500/MT27520 Family␣
↪[ConnectX-3/ConnectX-3 Pro Virtual Function]
04:00.3 Network controller: Mellanox Technologies MT27500/MT27520 Family␣
↪[ConnectX-3/ConnectX-3 Pro Virtual Function]
04:00.4 Network controller: Mellanox Technologies MT27500/MT27520 Family␣
↪[ConnectX-3/ConnectX-3 Pro Virtual Function]
```

Four VFs are available in the system, however,

```
$ ip link show
31: enp4s0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop master ovs-system state␣
↪DOWN mode DEFAULT group default qlen 1000
    link/ether f4:52:14:01:d9:e1 brd ff:ff:ff:ff:ff:ff
    vf 0 MAC 00:00:00:00:00:00, vlan 4095, spoof checking off, link-state auto
    vf 1 MAC 00:00:00:00:00:00, vlan 4095, spoof checking off, link-state auto
    vf 2 MAC 00:00:00:00:00:00, vlan 4095, spoof checking off, link-state auto
    vf 3 MAC 00:00:00:00:00:00, vlan 4095, spoof checking off, link-state auto
32: enp4s0d1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode␣
↪DEFAULT group default qlen 1000
    link/ether f4:52:14:01:d9:e2 brd ff:ff:ff:ff:ff:ff
    vf 0 MAC 00:00:00:00:00:00, vlan 4095, spoof checking off, link-state auto
    vf 1 MAC 00:00:00:00:00:00, vlan 4095, spoof checking off, link-state auto
    vf 2 MAC 00:00:00:00:00:00, vlan 4095, spoof checking off, link-state auto
    vf 3 MAC 00:00:00:00:00:00, vlan 4095, spoof checking off, link-state auto
```

ip command identifies each PF associated net device as having four VFs *each*.

Note

Mellanox `mlx4` driver allows *ip* commands to perform configuration of *all* VFs from either PF associated network devices.

To allow neutron SR-IOV agent to properly identify the VFs that belong to the correct PF network device (thus to the correct network port) Admin is required to provide the `exclude_devices` configuration option in `sriov_agent.ini`

Step 1: derive the VF to Port mapping from mlx4 driver configuration file: `/etc/modprobe.d/mlnx.conf` or `/etc/modprobe.d/mlx4.conf`

```
$ cat /etc/modprobe.d/mlnx.conf | grep "options mlx4_core"
options mlx4_core port_type_array=2,2 num_vfs=2,2,0 probe_vf=2,2,0 log_num_
↪mgm_entry_size=-1
```

Where:

`num_vfs=n1,n2,n3` - The driver will enable `n1` VFs on physical port 1, `n2` VFs on physical port 2 and `n3` dual port VFs (applies only to dual port HCA when all ports are Ethernet ports).

`probe_vfs=m1,m2,m3` - the driver probes `m1` single port VFs on physical port 1, `m2` single port VFs on physical port 2 (applies only if such a port exist) `m3` dual port VFs. Those VFs are attached to the hypervisor. (applies only if all ports are configured as Ethernet).

The VFs will be enumerated in the following order:

1. port 1 VFs
2. port 2 VFs
3. dual port VFs

In our example:

04:00.0 : PF associated to **both** ports.
04:00.1 : VF associated to port **1**
04:00.2 : VF associated to port **1**
04:00.3 : VF associated to port **2**
04:00.4 : VF associated to port **2**

Step 2: Update `exclude_devices` configuration option in `sriov_agent.ini` with the correct mapping

Each PF associated net device shall exclude the **other** ports VFs

```
[sriov_nic]
physical_device_mappings = physnet1:enp4s0,physnet2:enp4s0d1
exclude_devices = enp4s0:0000:04:00.3;0000:04:00.4,enp4s0d1:0000:04:00.1;
↪0000:04:00.2
```

SR-IOV with InfiniBand

The support for SR-IOV with InfiniBand allows a Virtual PCI device (VF) to be directly mapped to the guest, allowing higher performance and advanced features such as RDMA (remote direct memory access). To use this feature, you must:

1. Use InfiniBand enabled network adapters.
2. Run InfiniBand subnet managers to enable InfiniBand fabric.

   All InfiniBand networks must have a subnet manager running for the network to function. This is true even when doing a simple network of two machines with no switch and the cards are plugged in back-to-back. A subnet manager is required for the link on the cards to come up. It is possible to have more than one subnet manager. In this case, one of them will act as the primary, and any other will act as a backup that will take over when the primary subnet manager fails.

3. Install the `ebrctl` utility on the compute nodes.

   Check that `ebrctl` is listed somewhere in `/etc/nova/rootwrap.d/*`:

   ```
   $ grep 'ebrctl' /etc/nova/rootwrap.d/*
   ```

   If `ebrctl` does not appear in any of the rootwrap files, add this to the `/etc/nova/rootwrap.d/compute.filters` file in the `[Filters]` section.

   ```
   [Filters]
   ebrctl: CommandFilter, ebrctl, root
   ```

Known limitations

- When using Quality of Service (QoS), `max_burst_kbps` (burst over `max_kbps`) is not supported. In addition, `max_kbps` is rounded to Mbps.
- Security groups are not supported when using SR-IOV.
- SR-IOV is not integrated into the OpenStack Dashboard (horizon). Users must use the CLI or API to configure SR-IOV interfaces.
- Live migration support has been added to the Libvirt Nova virt-driver in the Train release for instances with neutron SR-IOV ports. Indirect mode SR-IOV interfaces (vnic-type: macvtap or

virtio-forwarder) can now be migrated transparently to the guest. Direct mode SR-IOV interfaces (vnic-type: direct or direct-physical) are detached before the migration and reattached after the migration so this is not transparent to the guest. To avoid loss of network connectivy when live migrating with direct mode sriov the user should create a failover bond in the guest with a transparently live migration port type e.g. vnic-type normal or indirect mode SR-IOV.

> **Note**
>
> SR-IOV features may require a specific NIC driver version, depending on the vendor. Intel NICs, for example, require ixgbe version 4.4.6 or greater, and ixgbevf version 3.2.2 or greater.

- Attaching SR-IOV ports to existing servers is supported starting with the Victoria release.

8.2.34 Service Function Chaining

Service function chain (SFC) essentially refers to the software-defined networking (SDN) version of policy-based routing (PBR). In many cases, SFC involves security, although it can include a variety of other features.

Fundamentally, SFC routes packets through one or more service functions instead of conventional routing that routes packets using destination IP address. Service functions essentially emulate a series of physical network devices with cables linking them together.

A basic example of SFC involves routing packets from one location to another through a firewall that lacks a next hop IP address from a conventional routing perspective. A more complex example involves an ordered series of service functions, each implemented using multiple instances (VMs). Packets must flow through one instance and a hashing algorithm distributes flows across multiple instances at each hop.

Architecture

All OpenStack Networking services and OpenStack Compute instances connect to a virtual network via ports making it possible to create a traffic steering model for service chaining using only ports. Including these ports in a port chain enables steering of traffic through one or more instances providing service functions.

A port chain, or service function path, consists of the following:

- A set of ports that define the sequence of service functions.
- A set of flow classifiers that specify the classified traffic flows entering the chain.

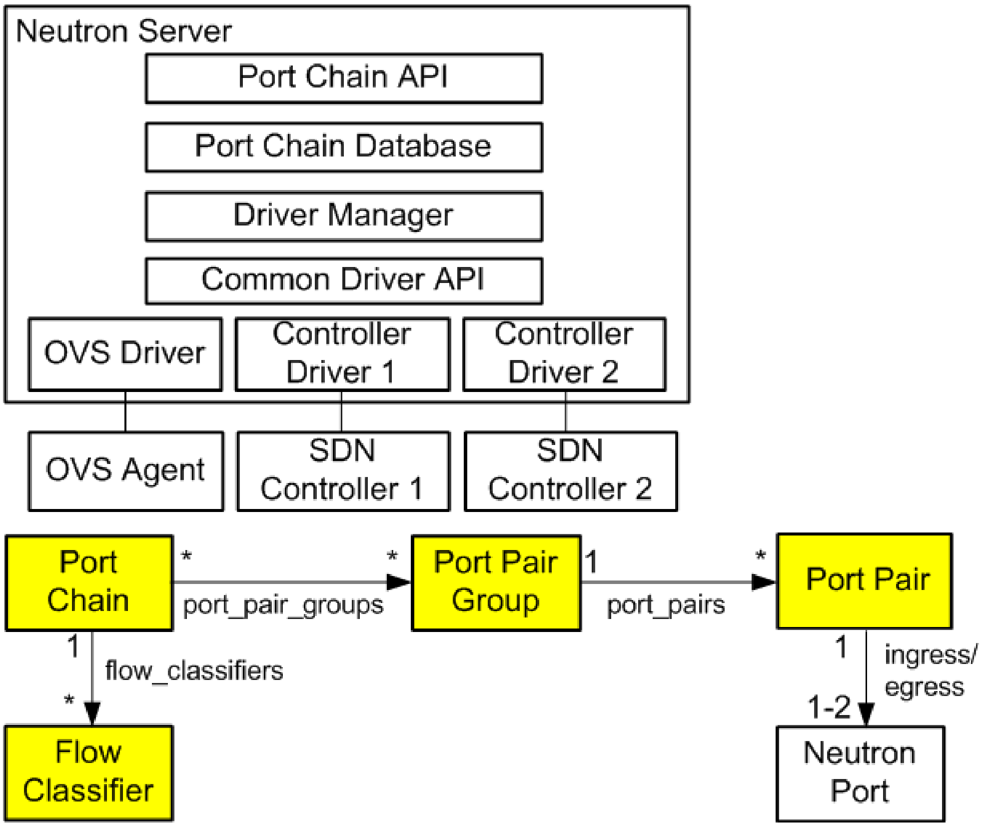
If a service function involves a pair of ports, the first port acts as the ingress port of the service function and the second port acts as the egress port. If both ports use the same value, they function as a single virtual bidirectional port.

A port chain is a unidirectional service chain. The first port acts as the head of the service function chain and the second port acts as the tail of the service function chain. A bidirectional service function chain consists of two unidirectional port chains.

A flow classifier can only belong to one port chain to prevent ambiguity as to which chain should handle packets in the flow. A check prevents such ambiguity. However, you can associate multiple flow classifiers with a port chain because multiple flows can request the same service function path.

Currently, SFC lacks support for multi-project service functions.

The port chain plug-in supports backing service providers including the OVS driver and a variety of SDN controller drivers. The common driver API enables different drivers to provide different implementations for the service chain path rendering.



See the networking-sfc documentation for more information.

Resources

Port chain

- `id` - Port chain ID
- `project_id` - Project ID
- `name` - Readable name
- `description` - Readable description
- `port_pair_groups` - List of port pair group IDs
- `flow_classifiers` - List of flow classifier IDs
- `chain_parameters` - Dictionary of chain parameters

A port chain consists of a sequence of port pair groups. Each port pair group is a hop in the port chain. A group of port pairs represents service functions providing equivalent functionality. For example, a group of firewall service functions.

A flow classifier identifies a flow. A port chain can contain multiple flow classifiers. Omitting the flow classifier effectively prevents steering of traffic through the port chain.

The `chain_parameters` attribute contains one or more parameters for the port chain. Currently, it only supports a correlation parameter that defaults to `mpls` for consistency with Open vSwitch (OVS) capabilities. Future values for the correlation parameter may include the network service header (NSH).

Port pair group

- `id` - Port pair group ID
- `project_id` - Project ID
- `name` - Readable name
- `description` - Readable description
- `port_pairs` - List of service function port pairs

A port pair group may contain one or more port pairs. Multiple port pairs enable load balancing/distribution over a set of functionally equivalent service functions.

Port pair

- `id` - Port pair ID
- `project_id` - Project ID
- `name` - Readable name
- `description` - Readable description
- `ingress` - Ingress port
- `egress` - Egress port
- `service_function_parameters` - Dictionary of service function parameters

A port pair represents a service function instance that includes an ingress and egress port. A service function containing a bidirectional port uses the same ingress and egress port.

The `service_function_parameters` attribute includes one or more parameters for the service function. Currently, it only supports a correlation parameter that determines association of a packet with a chain. This parameter defaults to `none` for legacy service functions that lack support for correlation such as the NSH. If set to `none`, the data plane implementation must provide service function proxy functionality.

Flow classifier

- `id` - Flow classifier ID
- `project_id` - Project ID
- `name` - Readable name
- `description` - Readable description
- `ethertype` - Ethertype (IPv4/IPv6)
- `protocol` - IP protocol
- `source_port_range_min` - Minimum source protocol port
- `source_port_range_max` - Maximum source protocol port
- `destination_port_range_min` - Minimum destination protocol port

- `destination_port_range_max` - Maximum destination protocol port
- `source_ip_prefix` - Source IP address or prefix
- `destination_ip_prefix` - Destination IP address or prefix
- `logical_source_port` - Source port
- `logical_destination_port` - Destination port
- `l7_parameters` - Dictionary of L7 parameters

A combination of the source attributes defines the source of the flow. A combination of the destination attributes defines the destination of the flow. The `l7_parameters` attribute is a place holder that may be used to support flow classification using layer 7 fields, such as a URL. If unspecified, the `logical_source_port` and `logical_destination_port` attributes default to `none`, the `ethertype` attribute defaults to `IPv4`, and all other attributes default to a wildcard value.

Operations

Create a port chain

The following example uses the `openstack` command-line interface (CLI) to create a port chain consisting of three service function instances to handle HTTP (TCP) traffic flows from 192.0.2.11:1000 to 198.51.100.11:80.

- Instance 1
  - Name: vm1
  - Function: Firewall
  - Port pair: [p1, p2]
- Instance 2
  - Name: vm2
  - Function: Firewall
  - Port pair: [p3, p4]
- Instance 3
  - Name: vm3
  - Function: Intrusion detection system (IDS)
  - Port pair: [p5, p6]

Note

The example network `net1` must exist before creating ports on it.

1. Source the credentials of the project that owns the `net1` network.
2. Create ports on network `net1` and record the UUID values.

```
$ openstack port create p1 --network net1
$ openstack port create p2 --network net1
```

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```
$ openstack port create p3 --network net1
$ openstack port create p4 --network net1
$ openstack port create p5 --network net1
$ openstack port create p6 --network net1
```

3. Launch service function instance `vm1` using ports `p1` and `p2`, `vm2` using ports `p3` and `p4`, and `vm3` using ports `p5` and `p6`.

```
$ openstack server create --nic port-id=P1_ID --nic port-id=P2_ID vm1
$ openstack server create --nic port-id=P3_ID --nic port-id=P4_ID vm2
$ openstack server create --nic port-id=P5_ID --nic port-id=P6_ID vm3
```

Replace `P1_ID`, `P2_ID`, `P3_ID`, `P4_ID`, `P5_ID`, and `P6_ID` with the UUIDs of the respective ports.

Note

This command requires additional options to successfully launch an instance. See the CLI reference for more information.

Alternatively, you can launch each instance with one network interface and attach additional ports later.

4. Create flow classifier `FC1` that matches the appropriate packet headers.

```
$ openstack sfc flow classifier create \
  --description "HTTP traffic from 192.0.2.11 to 198.51.100.11" \
  --ethertype IPv4 \
  --source-ip-prefix 192.0.2.11/32 \
  --destination-ip-prefix 198.51.100.11/32 \
  --protocol tcp \
  --source-port 1000:1000 \
  --destination-port 80:80 FC1
```

Note

When using the (default) OVS driver, the `--logical-source-port` parameter is also required

5. Create port pair `PP1` with ports `p1` and `p2`, `PP2` with ports `p3` and `p4`, and `PP3` with ports `p5` and `p6`.

```
$ openstack sfc port pair create \
  --description "Firewall SF instance 1" \
  --ingress p1 \
  --egress p2 PP1

$ openstack sfc port pair create \
  --description "Firewall SF instance 2" \
  --ingress p3 \
  --egress p4 PP2
```

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```
$ openstack sfc port pair create \
  --description "IDS SF instance" \
  --ingress p5 \
  --egress p6 PP3
```

6. Create port pair group PPG1 with port pair PP1 and PP2 and PPG2 with port pair PP3.

```
$ openstack sfc port pair group create \
  --port-pair PP1 --port-pair PP2 PPG1
$ openstack sfc port pair group create \
  --port-pair PP3 PPG2
```

Note

You can repeat the `--port-pair` option for multiple port pairs of functionally equivalent service functions.

7. Create port chain PC1 with port pair groups PPG1 and PPG2 and flow classifier FC1.

```
$ openstack sfc port chain create \
  --port-pair-group PPG1 --port-pair-group PPG2 \
  --flow-classifier FC1 PC1
```

Note

You can repeat the `--port-pair-group` option to specify additional port pair groups in the port chain. A port chain must contain at least one port pair group.

You can repeat the `--flow-classifier` option to specify multiple flow classifiers for a port chain. Each flow classifier identifies a flow.

Update a port chain or port pair group

- Use the **openstack sfc port chain set** command to dynamically add or remove port pair groups or flow classifiers on a port chain.
  - For example, add port pair group PPG3 to port chain PC1:

```
$ openstack sfc port chain set \
  --port-pair-group PPG1 --port-pair-group PPG2 --port-pair-group␣
↪PPG3 \
  --flow-classifier FC1 PC1
```

  - For example, add flow classifier FC2 to port chain PC1:

```
$ openstack sfc port chain set \
  --port-pair-group PPG1 --port-pair-group PPG2 \
  --flow-classifier FC1 --flow-classifier FC2 PC1
```

SFC steers traffic matching the additional flow classifier to the port pair groups in the port chain.

- Use the **openstack sfc port pair group set** command to perform dynamic scale-out or scale-in operations by adding or removing port pairs on a port pair group.

```
$ openstack sfc port pair group set \
  --port-pair PP1 --port-pair PP2 --port-pair PP4 PPG1
```

SFC performs load balancing/distribution over the additional service functions in the port pair group.

8.2.35 Service Subnets

Service subnets enable operators to define valid port types for each subnet on a network without limiting networks to one subnet or manually creating ports with a specific subnet ID. Using this feature, operators can ensure that ports for instances and router interfaces, for example, always use different subnets.

Operation

Define one or more service types for one or more subnets on a particular network. Each service type must correspond to a valid device owner within the port model in order for it to be used.

During IP allocation, the *IPAM* driver returns an address from a subnet with a service type matching the port device owner. If no subnets match, or all matching subnets lack available IP addresses, the IPAM driver attempts to use a subnet without any service types to preserve compatibility. If all subnets on a network have a service type, the IPAM driver cannot preserve compatibility. However, this feature enables strict IP allocation from subnets with a matching device owner. If multiple subnets contain the same service type, or a subnet without a service type exists, the IPAM driver selects the first subnet with a matching service type. For example, a floating IP agent gateway port uses the following selection process:

- `network:floatingip_agent_gateway`
- `None`

Note

Ports with the device owner `network:dhcp` are exempt from the above IPAM logic for subnets with `dhcp_enabled` set to `True`. This preserves the existing automatic DHCP port creation behaviour for DHCP-enabled subnets.

Creating or updating a port with a specific subnet skips this selection process and explicitly uses the given subnet.

Usage

Note

Creating a subnet with a service type requires administrative privileges.

Example 1 - Proof-of-concept

This following example is not typical of an actual deployment. It is shown to allow users to experiment with configuring service subnets.

1. Create a network.

```
$ openstack network create demo-net1
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| description               |                                      |
| headers                   |                                      |
| id                        | b5b729d8-31cc-4d2c-8284-72b3291fec02 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| mtu                       | 1450                                 |
| name                      | demo-net1                            |
| port_security_enabled     | True                                 |
| project_id                | a3db43cd0f224242a847ab84d091217d     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 110                                  |
| revision_number           | 1                                    |
| router:external           | Internal                             |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      | []                                   |
+---------------------------+--------------------------------------+
```

2. Create a subnet on the network with one or more service types. For example, the `compute:nova` service type enables instances to use this subnet.

```
$ openstack subnet create demo-subnet1 --subnet-range 192.0.2.0/24 \
  --service-type 'compute:nova' --network demo-net1
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| id                | 6e38b23f-0b27-4e3c-8e69-fd23a3df1935 |
| ip_version        | 4                                    |
| cidr              | 192.0.2.0/24                         |
| name              | demo-subnet1                         |
| network_id        | b5b729d8-31cc-4d2c-8284-72b3291fec02 |
| revision_number   | 1                                    |
| service_types     | ['compute:nova']                     |
| tags              | []                                   |
| tenant_id         | a8b3054cc1214f18b1186b291525650f     |
+-------------------+--------------------------------------+
```

3. Optionally, create another subnet on the network with a different service type. For example, the `compute:foo` arbitrary service type.

```
$ openstack subnet create demo-subnet2 --subnet-range 198.51.100.0/24 \
  --service-type 'compute:foo' --network demo-net1
+------------------+--------------------------------------+
| Field            | Value                                |
+------------------+--------------------------------------+
| id               | ea139dcd-17a3-4f0a-8cca-dff8b4e03f8a |
| ip_version       | 4                                    |
| cidr             | 198.51.100.0/24                      |
| name             | demo-subnet2                         |
| network_id       | b5b729d8-31cc-4d2c-8284-72b3291fec02 |
| revision_number  | 1                                    |
| service_types    | ['compute:foo']                      |
| tags             | []                                   |
| tenant_id        | a8b3054cc1214f18b1186b291525650f     |
+------------------+--------------------------------------+
```

4. Launch an instance using the network. For example, using the `cirros` image and `m1.tiny` flavor.

```
$ openstack server create demo-instance1 --flavor m1.tiny \
  --image cirros --nic net-id=b5b729d8-31cc-4d2c-8284-72b3291fec02
+--------------------------------------+-------------------------------
↪-------------+
| Field                                | Value                         ␣
↪             |
+--------------------------------------+-------------------------------
↪-------------+
| OS-DCF:diskConfig                    | MANUAL                        ␣
↪             |
| OS-EXT-AZ:availability_zone          |                               ␣
↪             |
| OS-EXT-SRV-ATTR:host                 | None                          ␣
↪             |
| OS-EXT-SRV-ATTR:hypervisor_hostname  | None                          ␣
↪             |
| OS-EXT-SRV-ATTR:instance_name        | instance-00000009             ␣
↪             |
| OS-EXT-STS:power_state               | 0                             ␣
↪             |
| OS-EXT-STS:task_state                | scheduling                    ␣
↪             |
| OS-EXT-STS:vm_state                  | building                      ␣
↪             |
| OS-SRV-USG:launched_at               | None                          ␣
↪             |
| OS-SRV-USG:terminated_at             | None                          ␣
↪             |
| accessIPv4                           |                               ␣
↪             |
```

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```
| accessIPv6                          |                                 ␣
↪                |
| addresses                           |                                 ␣
↪                |
| adminPass                           | Fn85skabdxBL                    ␣
↪                |
| config_drive                        |                                 ␣
↪                |
| created                             | 2016-09-19T15:07:42Z            ␣
↪                |
| flavor                              | m1.tiny (1)                     ␣
↪                |
| hostId                              |                                 ␣
↪                |
| id                                  | 04222b73-1a6e-4c2a-9af4-
↪ef3d17d521ff         |
| image                               | cirros (4aaec87d-c655-4856-8618-
↪b2dada3a2b11) |
| key_name                            | None                            ␣
↪                |
| name                                | demo-instance1                  ␣
↪                |
| os-extended-volumes:volumes_attached | []                             ␣
↪                |
| progress                            | 0                               ␣
↪                |
| project_id                          | d44c19e056674381b86430575184b167␣
↪                |
| properties                          |                                 ␣
↪                |
| security_groups                     | [{u'name': u'default'}]         ␣
↪                |
| status                              | BUILD                           ␣
↪                |
| updated                             | 2016-09-19T15:07:42Z            ␣
↪                |
| user_id                             | 331afbeb322d4c559a181e19051ae362␣
↪                |
+--------------------------------------+--------------------------------
↪-------------+
```

5. Check the instance status. The `Networks` field contains an IP address from the subnet having the `compute:nova` service type.

```
$ openstack server list
+--------------------------------------+-----------------+---------+------
↪---------------+--------+---------+
| ID                                   | Name            | Status  |␣
↪Networks              | Image  | Flavor  |
+--------------------------------------+-----------------+---------+------
```

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```
↪--------------+--------+---------+
| 20181f46-5cd2-4af8-9af0-f4cf5c983008 | demo-instance1  | ACTIVE  | demo-
↪net1=192.0.2.3 | cirros | m1.tiny |
+--------------------------------------+-----------------+---------+------
↪--------------+--------+---------+
```

Example 2 - DVR configuration

The following example outlines how you can configure service subnets in a DVR-enabled deployment, with the goal of minimizing public IP address consumption. This example uses three subnets on the same external network:

- 192.0.2.0/24 for instance floating IP addresses
- 198.51.100.0/24 for floating IP agent gateway IPs configured on compute nodes
- 203.0.113.0/25 for all other IP allocations on the external network

This example uses again the private network, `demo-net1` (b5b729d8-31cc-4d2c-8284-72b3291fec02) which was created in *Example 1 - Proof-of-concept*.

1. Create an external network:

```
$ openstack network create --external demo-ext-net
```

2. Create a subnet on the external network for the instance floating IP addresses. This uses the `network:floatingip` service type.

```
$ openstack subnet create demo-floating-ip-subnet \
  --subnet-range 192.0.2.0/24 --no-dhcp \
  --service-type 'network:floatingip' --network demo-ext-net
```

3. Create a subnet on the external network for the floating IP agent gateway IP addresses, which are configured by DVR on compute nodes. This will use the `network:floatingip_agent_gateway` service type.

```
$ openstack subnet create demo-floating-ip-agent-gateway-subnet \
  --subnet-range 198.51.100.0/24 --no-dhcp \
  --service-type 'network:floatingip_agent_gateway' \
  --network demo-ext-net
```

4. Create a subnet on the external network for all other IP addresses allocated on the external network. This will not use any service type. It acts as a fall back for allocations that do not match either of the above two service subnets.

```
$ openstack subnet create demo-other-subnet \
  --subnet-range 203.0.113.0/25 --no-dhcp \
  --network demo-ext-net
```

5. Create a router:

```
$ openstack router create demo-router
```

6. Add an interface to the router on demo-subnet1:

```
$ openstack router add subnet demo-router demo-subnet1
```

7. Set the external gateway for the router, which will create an interface and allocate an IP address on demo-ext-net:

```
$ openstack router set --external-gateway demo-ext-net demo-router
```

8. Launch an instance on a private network and retrieve the neutron port ID that was allocated. As above, use the `cirros` image and `m1.tiny` flavor:

```
$ openstack server create demo-instance1 --flavor m1.tiny \
  --image cirros --nic net-id=b5b729d8-31cc-4d2c-8284-72b3291fec02
$ openstack port list --server demo-instance1
+--------------------------------------+------+-------------------+-------
↪-----------------------------------------------+--------+
| ID                                   | Name | MAC Address       | Fixed␣
↪IP Addresses                                     | Status |
+--------------------------------------+------+-------------------+-------
↪-----------------------------------------------+--------+
| a752bb24-9bf2-4d37-b9d6-07da69c86f19 |      | fa:16:3e:99:54:32 | ip_
↪address='203.0.113.130',                          | ACTIVE |
|                                      |      |                   |␣
↪subnet_id='6e38b23f-0b27-4e3c-8e69-fd23a3df1935' |        |
+--------------------------------------+------+-------------------+-------
↪-----------------------------------------------+--------+
```

9. Associate a floating IP with the instance port and verify it was allocated an IP address from the correct subnet:

```
$ openstack floating ip create --port \
  a752bb24-9bf2-4d37-b9d6-07da69c86f19 demo-ext-net
+---------------------+--------------------------------------+
| Field               | Value                                |
+---------------------+--------------------------------------+
| fixed_ip_address    | 203.0.113.130                        |
| floating_ip_address | 192.0.2.12                           |
| floating_network_id | 02d236d5-dad9-4082-bb6b-5245f9f84d13 |
| id                  | f15cae7f-5e05-4b19-bd25-4bb71edcf3de |
| port_id             | a752bb24-9bf2-4d37-b9d6-07da69c86f19 |
| project_id          | d44c19e056674381b86430575184b167     |
| revision_number     | 1                                    |
| router_id           | 5a8ca19f-3703-4f81-bc29-db6bc2f528d6 |
| status              | ACTIVE                               |
| tags                | []                                   |
+---------------------+--------------------------------------+
```

10. As the *admin* user, verify the neutron routers are allocated IP addresses from their correct subnets. Use `openstack port list` to find ports associated with the routers.

    First, the router gateway external port:

```
$ openstack port show f148ffeb-3c26-4067-bc5f-5c3dfddae2f5
+-----------------------+-------------------------------------------------------
↪------------------------+
| Field                 | Value                                                 ␣
↪                        |
+-----------------------+-------------------------------------------------------
↪------------------------+
| admin_state_up        | UP                                                    ␣
↪                        |
| device_id             | 5a8ca19f-3703-4f81-bc29-db6bc2f528d6                  ␣
↪                        |
| device_owner          | network:router_gateway                                ␣
↪                        |
| extra_dhcp_opts       |                                                       ␣
↪                        |
| fixed_ips             | ip_address='203.0.113.11',                            ␣
↪                        |
|                       | subnet_id='67c251d9-2b7a-4200-99f6-e13785b0334d
↪'                       |
| id                    | f148ffeb-3c26-4067-bc5f-5c3dfddae2f5                  ␣
↪                        |
| mac_address           | fa:16:3e:2c:0f:69                                     ␣
↪                        |
| network_id            | 02d236d5-dad9-4082-bb6b-5245f9f84d13                  ␣
↪                        |
| revision_number       | 1                                                     ␣
↪                        |
| project_id            |                                                       ␣
↪                        |
| status                | ACTIVE                                                ␣
↪                        |
| tags                  | []                                                    ␣
↪                        |
+-----------------------+-------------------------------------------------------
↪------------------------+
```

Second, the router floating IP agent gateway external port:

```
$ openstack port show a2d1e756-8ae1-4f96-9aa1-e7ea16a6a68a
+-----------------------+-------------------------------------------------------
↪------------------------+
| Field                 | Value                                                 ␣
↪                        |
+-----------------------+-------------------------------------------------------
↪------------------------+
| admin_state_up        | UP                                                    ␣
↪                        |
| device_id             | 3d0c98eb-bca3-45cc-8aa4-90ae3deb0844                  ␣
↪                        |
| device_owner          | network:floatingip_agent_gateway                      ␣
```

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```
↪                                |
| extra_dhcp_opts       |                                                     ␣
↪                                |
| fixed_ips             | ip_address='198.51.100.10',                         ␣
↪                                |
|                       | subnet_id='67c251d9-2b7a-4200-99f6-e13785b0334d
↪'                               |
| id                    | a2d1e756-8ae1-4f96-9aa1-e7ea16a6a68a                ␣
↪                                |
| mac_address           | fa:16:3e:f4:5d:fa                                   ␣
↪                                |
| network_id            | 02d236d5-dad9-4082-bb6b-5245f9f84d13                ␣
↪                                |
| project_id            |                                                     ␣
↪                                |
| revision_number       | 1                                                   ␣
↪                                |
| status                | ACTIVE                                              ␣
↪                                |
| tags                  | []                                                  ␣
↪                                |
+-----------------------+-----------------------------------------------------
↪-------------------------+
```

8.2.36 Subnet Onboarding

The subnet onboard feature allows you to take existing subnets that have been created outside of a subnet pool and move them into an existing subnet pool. This enables you to begin using subnet pools and address scopes if you havent allocated existing subnets from subnet pools. It also allows you to move individual subnets between subnet pools, and by extension, move them between address scopes.

How it works

One of the fundamental constraints of subnet pools is that all subnets of the same address family (IPv4, IPv6) on a network must be allocated from the same subnet pool. Because of this constraint, subnets must be moved, or onboarded, into a subnet pool as a group at the network level rather than being handled individually. As such, the onboarding of subnets requires users to supply the UUID of the network the subnet(s) to onboard are associated with, and the UUID of the target subnet pool to perform the operation.

Does my environment support subnet onboard?

To test that subnet onboard is supported in your environment, execute the following command:

```
$ openstack extension list --network -c Alias -c Description | grep subnet_
↪onboard
| subnet_onboard | Provides support for onboarding subnets into subnet pools
```

Support for subnet onboard exists in the ML2 plugin as of the Stein release. If you require subnet onboard but your current environment does not support it, consider upgrading to a release that supports subnet onboard. When using third-party plugins with neutron, check with the supplier of the plugin regarding support for subnet onboard.

Demo

Suppose an administrator has an existing provider network in their environment that was created without allocating its subnets from a subnet pool.

```
$ openstack network list
+--------------------------------------+---------------+--------------------
↪-----------------+
| ID                                   | Name          | Subnets            ␣
↪                 |
+--------------------------------------+---------------+--------------------
↪-----------------+
| f643a4f5-f8d3-4325-b1fe-6061a9af0f07 | provider-net-1 | 5153cab7-7ab6-4956-
↪8466-39aa85dccc9a |
+--------------------------------------+---------------+--------------------
↪-----------------+

$ openstack subnet show 5153cab7-7ab6-4956-8466-39aa85dccc9a
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 192.168.0.2-192.168.7.254            |
| cidr              | 192.168.0.0/21                       |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 192.168.0.1                          |
| host_routes       |                                      |
| id                | 5153cab7-7ab6-4956-8466-39aa85dccc9a |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| network_id        | f643a4f5-f8d3-4325-b1fe-6061a9af0f07 |
| prefix_length     | None                                 |
| project_id        | 7b80998e5e044cee91c1cdb2e9c63afd     |
| revision_number   | 0                                    |
| segment_id        | None                                 |
| service_types     |                                      |
| subnetpool_id     | None                                 |
| tags              |                                      |
| updated_at        | 2019-03-13T18:24:37Z                 |
+-------------------+--------------------------------------+
```

The administrator has created a subnet pool named `routable-prefixes` and wants to onboard the subnets associated with network `provider-net-1`. The administrator now wants to manage the address space for provider networks using a subnet pool, but doesnt have the prefixes used by these provider networks under the management of a subnet pool or address scope.

```
$ openstack subnet pool list
+--------------------------------------+-------------------+--------------+
| ID                                   | Name              | Prefixes     |
```

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```
+--------------------------------------+-------------------+--------------+
| d05e9f61-248c-43f2-98f4-5142570127e1 | routable-prefixes | 10.10.0.0/16 |
+--------------------------------------+-------------------+--------------+
```

```
$ openstack subnet pool show routable-prefixes
+-------------------+--------------------------------+
| Field             | Value                          |
+-------------------+--------------------------------+
| address_scope_id  | None                           |
| created_at        | 2019-03-102T05:45:01Z          |
| default_prefixlen | 26                             |
| default_quota     | None                           |
| description       | Routable prefixes for projects |
| headers           |                                |
| id                | d3aefb76-2527-43d4-bc21-0ec253 |
|                   | 908545                         |
| ip_version        | 4                              |
| is_default        | False                          |
| max_prefixlen     | 32                             |
| min_prefixlen     | 8                              |
| name              | routable-prefixes              |
| prefixes          | 10.10.0.0/16                   |
| project_id        | cfd1889ac7d64ad891d4f20aef9f8d |
|                   | 7c                             |
| revision_number   | 1                              |
| shared            | True                           |
| tags              | []                             |
| updated_at        | 2019-03-10T05:45:01Z           |
+-------------------+--------------------------------+
```

The administrator can use the following command to bring these subnets under the management of a subnet pool:

```
$ openstack network onboard subnets provider-net-1 routable-prefixes
```

The subnets on `provider-net-1` should now all have their subnetpool_id updated to match the UUID of the `routable-prefixes` subnet pool:

```
$ openstack subnet show 5153cab7-7ab6-4956-8466-39aa85dccc9a
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 192.168.0.2-192.168.7.254            |
| cidr              | 192.168.0.0/21                       |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 192.168.0.1                          |
| host_routes       |                                      |
| id                | 5153cab7-7ab6-4956-8466-39aa85dccc9a |
```

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```
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| network_id        | f643a4f5-f8d3-4325-b1fe-6061a9af0f07 |
| prefix_length     | None                                 |
| project_id        | 7b80998e5e044cee91c1cdb2e9c63afd     |
| revision_number   | 0                                    |
| segment_id        | None                                 |
| service_types     |                                      |
| subnetpool_id     | d3aefb76-2527-43d4-bc21-0ec253908545 |
| updated_at        | 2019-03-13T18:24:37Z                 |
+-------------------+--------------------------------------+
```

The subnet pool will also now show the onboarded prefix(es) in its prefix list:

```
$ openstack subnet pool show routable-prefixes
+-------------------+--------------------------------+
| Field             | Value                          |
+-------------------+--------------------------------+
| address_scope_id  | None                           |
| created_at        | 2019-03-102T05:45:01Z          |
| default_prefixlen | 26                             |
| default_quota     | None                           |
| description       | Routable prefixes for projects |
| headers           |                                |
| id                | d3aefb76-2527-43d4-bc21-0ec253 |
|                   | 908545                         |
| ip_version        | 4                              |
| is_default        | False                          |
| max_prefixlen     | 32                             |
| min_prefixlen     | 8                              |
| name              | routable-prefixes              |
| prefixes          | 10.10.0.0/16, 192.168.0.0/21   |
| project_id        | cfd1889ac7d64ad891d4f20aef9f8d |
|                   | 7c                             |
| revision_number   | 1                              |
| shared            | True                           |
| tags              | []                             |
| updated_at        | 2019-03-12T13:11:037Z          |
+-------------------+--------------------------------+
```

8.2.37 Subnet Pools

Subnet pools have been made available since the Kilo release. It is a simple feature that has the potential to improve your workflow considerably. It also provides a building block from which other new features will be built in to OpenStack Networking.

To see if your cloud has this feature available, you can check that it is listed in the supported aliases. You can do this with the OpenStack client.

```
$ openstack extension list | grep subnet_allocation
| Subnet Allocation | subnet_allocation | Enables allocation of subnets from a subnet pool                                       |
```

Why you need them

Before Kilo, Networking had no automation around the addresses used to create a subnet. To create one, you had to come up with the addresses on your own without any help from the system. There are valid use cases for this but if you are interested in the following capabilities, then subnet pools might be for you.

First, would not it be nice if you could turn your pool of addresses over to Neutron to take care of? When you need to create a subnet, you just ask for addresses to be allocated from the pool. You do not have to worry about what you have already used and what addresses are in your pool. Subnet pools can do this.

Second, subnet pools can manage addresses across projects. The addresses are guaranteed not to overlap. If the addresses come from an externally routable pool then you know that all of the projects have addresses which are *routable* and unique. This can be useful in the following scenarios.

1. IPv6 since OpenStack Networking has no IPv6 floating IPs.
2. Routing directly to a project network from an external network.

How they work

A subnet pool manages a pool of addresses from which subnets can be allocated. It ensures that there is no overlap between any two subnets allocated from the same pool.

As a regular project in an OpenStack cloud, you can create a subnet pool of your own and use it to manage your own pool of addresses. This does not require any admin privileges. Your pool will not be visible to any other project.

If you are an admin, you can create a pool which can be accessed by any regular project. Being a shared resource, there is a quota mechanism to arbitrate access.

Quotas

Subnet pools have a quota system which is a little bit different than other quotas in Neutron. Other quotas in Neutron count discrete instances of an object against a quota. Each time you create something like a router, network, or a port, it uses one from your total quota.

With subnets, the resource is the IP address space. Some subnets take more of it than others. For example, 203.0.113.0/24 uses 256 addresses in one subnet but 198.51.100.224/28 uses only 16. If address space is limited, the quota system can encourage efficient use of the space.

With IPv4, the default_quota can be set to the number of absolute addresses any given project is allowed to consume from the pool. For example, with a quota of 128, I might get 203.0.113.128/26, 203.0.113.224/28, and still have room to allocate 48 more addresses in the future.

With IPv6 it is a little different. It is not practical to count individual addresses. To avoid ridiculously large numbers, the quota is expressed in the number of /64 subnets which can be allocated. For example, with a default_quota of 3, I might get 2001:db8:c18e:c05a::/64, 2001:db8:221c:8ef3::/64, and still have room to allocate one more prefix in the future.

Default subnet pools

Beginning with Mitaka, a subnet pool can be marked as the default. This is handled with a new extension.

```
$ openstack extension list | grep default-subnetpools
| Default Subnetpools | default-subnetpools | Provides ability to mark
and use a subnetpool as the default                                     ␣
↪                                                          |
```

An administrator can mark a pool as default. Only one pool from each address family can be marked default.

```
$ openstack subnet pool set --default 74348864-f8bf-4fc0-ab03-81229d189467
```

If there is a default, it can be requested by passing `--use-default-subnet-pool` instead of `--subnet-pool SUBNETPOOL` when creating a subnet.

Demo

If you have access to an OpenStack Kilo or later based neutron, you can play with this feature now. Give it a try. All of the following commands work equally as well with IPv6 addresses.

First, as admin, create a shared subnet pool:

```
$ openstack subnet pool create --share --pool-prefix 203.0.113.0/24 \
--default-prefix-length 26 demo-subnetpool4
+-------------------+--------------------------------+
| Field             | Value                          |
+-------------------+--------------------------------+
| address_scope_id  | None                           |
| created_at        | 2016-12-14T07:21:26Z           |
| default_prefixlen | 26                             |
| default_quota     | None                           |
| description       |                                |
| headers           |                                |
| id                | d3aefb76-2527-43d4-bc21-0ec253 |
|                   | 908545                         |
| ip_version        | 4                              |
| is_default        | False                          |
| max_prefixlen     | 32                             |
| min_prefixlen     | 8                              |
| name              | demo-subnetpool4               |
| prefixes          | 203.0.113.0/24                 |
| project_id        | cfd1889ac7d64ad891d4f20aef9f8d |
|                   | 7c                             |
| revision_number   | 1                              |
| shared            | True                           |
| tags              | []                             |
| updated_at        | 2016-12-14T07:21:26Z           |
+-------------------+--------------------------------+
```

The `default_prefix_length` defines the subnet size you will get if you do not specify `--prefix-length` when creating a subnet.

Do essentially the same thing for IPv6 and there are now two subnet pools. Regular projects can see them. (the output is trimmed a bit for display)

```
$ openstack subnet pool list
+------------------+------------------+--------------------+
| ID               | Name             | Prefixes           |
+------------------+------------------+--------------------+
| 2b7cc19f-0114-4e | demo-subnetpool  | 2001:db8:a583::/48 |
| f4-ad86-c1bb91fc |                  |                    |
| d1f9             |                  |                    |
| d3aefb76-2527-43 | demo-subnetpool4 | 203.0.113.0/24     |
| d4-bc21-0ec25390 |                  |                    |
| 8545             |                  |                    |
+------------------+------------------+--------------------+
```

Now, use them. It is easy to create a subnet from a pool:

```
$ openstack subnet create --ip-version 4 --subnet-pool \
demo-subnetpool4 --network demo-network1 demo-subnet1
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 203.0.113.194-203.0.113.254          |
| cidr              | 203.0.113.192/26                     |
| created_at        | 2016-12-14T07:33:13Z                 |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 203.0.113.193                        |
| headers           |                                      |
| host_routes       |                                      |
| id                | 8d4fbae3-076c-4c08-b2dd-2d6175115a5e |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | demo-subnet1                         |
| network_id        | 6b377f77-ce00-4ff6-8676-82343817470d |
| project_id        | cfd1889ac7d64ad891d4f20aef9f8d7c     |
| revision_number   | 2                                    |
| service_types     |                                      |
| subnetpool_id     | d3aefb76-2527-43d4-bc21-0ec253908545 |
| tags              | []                                   |
| updated_at        | 2016-12-14T07:33:13Z                 |
+-------------------+--------------------------------------+
```

You can request a specific subnet from the pool. You need to specify a subnet that falls within the pools prefixes. If the subnet is not already allocated, the request succeeds. You can leave off the IP version because it is deduced from the subnet pool.

```
$ openstack subnet create --subnet-pool demo-subnetpool4 \
--network demo-network1 --subnet-range 203.0.113.128/26 subnet2
```

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```
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 203.0.113.130-203.0.113.190          |
| cidr              | 203.0.113.128/26                     |
| created_at        | 2016-12-14T07:27:40Z                 |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 203.0.113.129                        |
| headers           |                                      |
| host_routes       |                                      |
| id                | d32814e3-cf46-4371-80dd-498a80badfba |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | subnet2                              |
| network_id        | 6b377f77-ce00-4ff6-8676-82343817470d |
| project_id        | cfd1889ac7d64ad891d4f20aef9f8d7c     |
| revision_number   | 2                                    |
| service_types     |                                      |
| subnetpool_id     | d3aefb76-2527-43d4-bc21-0ec253908545 |
| tags              | []                                   |
| updated_at        | 2016-12-14T07:27:40Z                 |
+-------------------+--------------------------------------+
```

If the pool becomes exhausted, load some more prefixes:

```
$ openstack subnet pool set --pool-prefix \
198.51.100.0/24 demo-subnetpool4
$ openstack subnet pool show demo-subnetpool4
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| address_scope_id  | None                                 |
| created_at        | 2016-12-14T07:21:26Z                 |
| default_prefixlen | 26                                   |
| default_quota     | None                                 |
| description       |                                      |
| id                | d3aefb76-2527-43d4-bc21-0ec253908545 |
| ip_version        | 4                                    |
| is_default        | False                                |
| max_prefixlen     | 32                                   |
| min_prefixlen     | 8                                    |
| name              | demo-subnetpool4                     |
| prefixes          | 198.51.100.0/24, 203.0.113.0/24      |
| project_id        | cfd1889ac7d64ad891d4f20aef9f8d7c     |
| revision_number   | 2                                    |
| shared            | True                                 |
| tags              | []                                   |
```

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```
| updated_at        | 2016-12-14T07:30:32Z                 |
+-------------------+--------------------------------------+
```

8.2.38 Trunking

The network trunk service allows multiple networks to be connected to an instance using a single virtual NIC (vNIC). Multiple networks can be presented to an instance by connecting it to a single port.

Operation

Network trunking consists of a service plug-in and a set of drivers that manage trunks on different layer-2 mechanism drivers. Users can create a port, associate it with a trunk, and launch an instance on that port. Users can dynamically attach and detach additional networks without disrupting operation of the instance.

Every trunk has a parent port and can have any number of subports. The parent port is the port that the trunk is associated with. Users create instances and specify the parent port of the trunk when launching instances attached to a trunk.

The network presented by the subport is the network of the associated port. When creating a subport, a `segmentation-id` may be required by the driver. `segmentation-id` defines the segmentation ID on which the subport network is presented to the instance. `segmentation-type` may be required by certain drivers like OVS. At this time the following `segmentation-type` values are supported:

- `vlan` uses VLAN for segmentation.
- `inherit` uses the `segmentation-type` from the network the subport is connected to if no `segmentation-type` is specified for the subport. Note that using the `inherit` type requires the `provider` extension to be enabled and only works when the connected networks `segmentation-type` is `vlan`.

> **Note**
>
> The `segmentation-type` and `segmentation-id` parameters are optional in the Networking API. However, all drivers as of the Newton release require both to be provided when adding a subport to a trunk. Future drivers may be implemented without this requirement.

The `segmentation-type` and `segmentation-id` specified by the user on the subports is intentionally decoupled from the `segmentation-type` and ID of the networks. For example, it is possible to configure the Networking service with `project_network_types = vxlan` and still create subports with `segmentation_type = vlan`. The Networking service performs remapping as necessary.

Example configuration

The ML2 plug-in supports trunking with the following mechanism drivers:

- Open vSwitch (OVS)
- Open Virtual Network (OVN)

When using a `segmentation-type` of `vlan`, the OVS driver present the network of the parent port as the untagged VLAN and all subports as tagged VLANs.

Controller node

- In the neutron.conf file, enable the trunk service plug-in:

```
[DEFAULT]
service_plugins = trunk
```

Compute node

- If you are using the Open vSwitch mechanism driver, then you can verify the value of trunk_enabled flag in /etc/neutron/plugins/ml2/openvswitch_agent.ini config file. Note that True is the default value, but you can overwrite this option like this:

```
[OVS]
trunk_enabled = True
```

Verify service operation

1. Source the administrative project credentials and list the enabled extensions.
2. Use the command **openstack extension list --network** to verify that the Trunk Extension and Trunk port details extensions are enabled.

Workflow

At a high level, the basic steps to launching an instance on a trunk are the following:

1. Create networks and subnets for the trunk and subports
2. Create the trunk
3. Add subports to the trunk
4. Launch an instance on the trunk

Create networks and subnets for the trunk and subports

Create the appropriate networks for the trunk and subports that will be added to the trunk. Create subnets on these networks to ensure the desired layer-3 connectivity over the trunk.

Create the trunk

- Create a parent port for the trunk.

```
$ openstack port create --network project-net-A trunk-parent
+-------------------+-------------------------------------------------
↪--------------------+
| Field             | Value                                           ␣
↪                    |
+-------------------+-------------------------------------------------
↪--------------------+
| admin_state_up    | UP                                              ␣
↪                    |
```

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```
| binding_vif_type  | unbound                                                 ␣
↪                   |
| binding_vnic_type | normal                                                  ␣
↪                   |
| fixed_ips         | ip_address='192.0.2.7',subnet_id='8b957198-d3cf-
↪4953-8449-ad4e4dd712cc' |
| id                | 73fb9d54-43a7-4bb1-a8dc-569e0e0a0a38                    ␣
↪                   |
| mac_address       | fa:16:3e:dd:c4:d1                                       ␣
↪                   |
| name              | trunk-parent                                            ␣
↪                   |
| network_id        | 1b47d3e7-cda5-48e4-b0c8-d20bd7e35f55                    ␣
↪                   |
| revision_number   | 1                                                       ␣
↪                   |
| tags              | []                                                      ␣
↪                   |
+-------------------+-------------------------------------------------------------
↪--------------------+
```

- Create the trunk using `--parent-port` to reference the port from the previous step:

```
$ openstack network trunk create --parent-port trunk-parent trunk1
+-----------------+--------------------------------------+
| Field           | Value                                |
+-----------------+--------------------------------------+
| admin_state_up  | UP                                   |
| id              | fdf02fcb-1844-45f1-9d9b-e4c2f522c164 |
| name            | trunk1                               |
| port_id         | 73fb9d54-43a7-4bb1-a8dc-569e0e0a0a38 |
| revision_number | 1                                    |
| sub_ports       |                                      |
+-----------------+--------------------------------------+
```

Add subports to the trunk

Subports can be added to a trunk in two ways: creating the trunk with subports or adding subports to an existing trunk.

- Create trunk with subports:

  This method entails creating the trunk with subports specified at trunk creation.

```
$ openstack port create --network project-net-A trunk-parent
+-------------------+-------------------------------------------------------------
↪--------------------+
| Field             | Value                                                   ␣
↪                   |
+-------------------+-------------------------------------------------------------
```

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```
↪--------------------+
| admin_state_up    | UP                                                          ␣
↪                    |
| binding_vif_type  | unbound                                                     ␣
↪                    |
| binding_vnic_type | normal                                                      ␣
↪                    |
| fixed_ips         | ip_address='192.0.2.7',subnet_id='8b957198-d3cf-
↪4953-8449-ad4e4dd712cc' |
| id                | 73fb9d54-43a7-4bb1-a8dc-569e0e0a0a38                        ␣
↪                    |
| mac_address       | fa:16:3e:dd:c4:d1                                           ␣
↪                    |
| name              | trunk-parent                                                ␣
↪                    |
| network_id        | 1b47d3e7-cda5-48e4-b0c8-d20bd7e35f55                        ␣
↪                    |
| revision_number   | 1                                                           ␣
↪                    |
| tags              | []                                                          ␣
↪                    |
+-------------------+-------------------------------------------------------------
↪--------------------+

$ openstack port create --network trunked-net subport1
+-------------------+-------------------------------------------------------------
↪-----------------------+
| Field             | Value                                                       ␣
↪                       |
+-------------------+-------------------------------------------------------------
↪-----------------------+
| admin_state_up    | UP                                                          ␣
↪                       |
| binding_vif_type  | unbound                                                     ␣
↪                       |
| binding_vnic_type | normal                                                      ␣
↪                       |
| fixed_ips         | ip_address='198.51.100.8',subnet_id='2a860e2c-922b-
↪437b-a149-b269a8c9b120' |
| id                | 91f9dde8-80a4-4506-b5da-c287feb8f5d8                        ␣
↪                       |
| mac_address       | fa:16:3e:ba:f0:4d                                           ␣
↪                       |
| name              | subport1                                                    ␣
↪                       |
| network_id        | aef78ec5-16e3-4445-b82d-b2b98c6a86d9                        ␣
↪                       |
| revision_number   | 1                                                           ␣
↪                       |
```

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```
| tags             | []                                       |
+------------------+------------------------------------------+

$ openstack network trunk create \
  --parent-port trunk-parent \
  --subport port=subport1,segmentation-type=vlan,segmentation-id=100 \
  trunk1
+----------------+-------------------------------------------------------------------------------------------------+
| Field          | Value                                                                                           |
+----------------+-------------------------------------------------------------------------------------------------+
| admin_state_up | UP                                                                                              |
| id             | 61d8e620-fe3a-4d8f-b9e6-e1b0dea6d9e3                                                            |
| name           | trunk1                                                                                          |
| port_id        | 73fb9d54-43a7-4bb1-a8dc-569e0e0a0a38                                                            |
| revision_number| 1                                                                                               |
| sub_ports      | port_id='73fb9d54-43a7-4bb1-a8dc-569e0e0a0a38', segmentation_id='100', segmentation_type='vlan' |
| tags           | []                                                                                              |
+----------------+-------------------------------------------------------------------------------------------------+
```

- Add subports to an existing trunk:

  This method entails creating a trunk, then adding subports to the trunk after it has already been created.

```
$ openstack network trunk set --subport \
  port=subport1,segmentation-type=vlan,segmentation-id=100 \
  trunk1
```

> **Note**
>
> The command provides no output.

```
$ openstack network trunk show trunk1
+----------------+-------------------------------------------------------------------------------------------------+
| Field          | Value                                                                                           |
```

(continues on next page)

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```
↪                                                                    |
+----------------+-------------------------------------------------------------
↪--------------------------------------+
| admin_state_up | UP                                                          ␣
↪                                                                    |
| id             | 61d8e620-fe3a-4d8f-b9e6-e1b0dea6d9e3                        ␣
↪                                                                    |
| name           | trunk1                                                      ␣
↪                                                                    |
| port_id        | 73fb9d54-43a7-4bb1-a8dc-569e0e0a0a38                        ␣
↪                                                                    |
| revision_number| 1                                                           ␣
↪                                                                    |
| sub_ports      | port_id='73fb9d54-43a7-4bb1-a8dc-569e0e0a0a38',␣
↪segmentation_id='100', segmentation_type='vlan' |
| tags           | []                                                          ␣
↪                                                                    |
+----------------+-------------------------------------------------------------
↪--------------------------------------+
```

- When using the OVN driver, additional logical switch port information is available using the following commands:

```
$ ovn-nbctl lsp-get-parent 61d8e620-fe3a-4d8f-b9e6-e1b0dea6d9e3
73fb9d54-43a7-4bb1-a8dc-569e0e0a0a38

$ ovn-nbctl lsp-get-tag 61d8e620-fe3a-4d8f-b9e6-e1b0dea6d9e3
```

Launch an instance on the trunk

- Show trunk details to get the `port_id` of the trunk.

```
$ openstack network trunk show trunk1
+----------------+--------------------------------------+
| Field          | Value                                |
+----------------+--------------------------------------+
| admin_state_up | UP                                   |
| id             | 61d8e620-fe3a-4d8f-b9e6-e1b0dea6d9e3 |
| name           | trunk                                |
| port_id        | 73fb9d54-43a7-4bb1-a8dc-569e0e0a0a38 |
| revision_number| 1                                    |
| sub_ports      |                                      |
| tags           | []                                   |
+----------------+--------------------------------------+
```

- Launch the instance by specifying `port-id` using the value of `port_id` from the trunk details. Launching an instance on a subport is not supported.

Using trunks and subports inside an instance

When configuring instances to use a subport, ensure that the interface on the instance is set to use the MAC address assigned to the port by the Networking service. Instances are not made aware of changes made to the trunk after they are active. For example, when a subport with a `segmentation-type` of `vlan` is added to a trunk, any operations specific to the instance operating system that allow the instance to send and receive traffic on the new VLAN must be handled outside of the Networking service.

When creating subports, the MAC address of the trunk parent port can be set on the subport. This will allow VLAN subinterfaces inside an instance launched on a trunk to be configured without explicitly setting a MAC address. Although unique MAC addresses can be used for subports, this can present issues with ARP spoof protections and the native OVS firewall driver. If the native OVS firewall driver is to be used, we recommend that the MAC address of the parent port be re-used on all subports.

Trunk states

- `ACTIVE`

  The trunk is `ACTIVE` when both the logical and physical resources have been created. This means that all operations within the Networking and Compute services have completed and the trunk is ready for use.

- `DOWN`

  A trunk is `DOWN` when it is first created without an instance launched on it, or when the instance associated with the trunk has been deleted.

- `DEGRADED`

  A trunk can be in a `DEGRADED` state when a temporary failure during the provisioning process is encountered. This includes situations where a subport add or remove operation fails. When in a degraded state, the trunk is still usable and some subports may be usable as well. Operations that cause the trunk to go into a `DEGRADED` state can be retried to fix temporary failures and move the trunk into an `ACTIVE` state.

- `ERROR`

  A trunk is in `ERROR` state if the request leads to a conflict or an error that cannot be fixed by retrying the request. The `ERROR` status can be encountered if the network is not compatible with the trunk configuration or the binding process leads to a persistent failure. When a trunk is in `ERROR` state, it must be brought to a sane state (`ACTIVE`), or else requests to add subports will be rejected.

- `BUILD`

  A trunk is in `BUILD` state while the resources associated with the trunk are in the process of being provisioned. Once the trunk and all of the subports have been provisioned successfully, the trunk transitions to `ACTIVE`. If there was a partial failure, the trunk transitions to `DEGRADED`.

  When `admin_state` is set to `DOWN`, the user is blocked from performing operations on the trunk. `admin_state` is set by the user and should not be used to monitor the health of the trunk.

Limitations and issues

- In `neutron-ovs-agent` the use of `iptables_hybrid` firewall driver and trunk ports are not compatible with each other. The `iptables_hybrid` firewall is not going to filter the traffic of subports. Instead use other firewall drivers like `openvswitch`.
- See bugs for more information.

8.2.39 WSGI Usage with the Neutron API

This document is a guide to deploying Neutron using WSGI. There are two ways to deploy using WSGI: `uwsgi` and Apache `mod_wsgi`.

Please note that if you intend to use mode uwsgi, you should install the `mode_proxy_uwsgi` module. For example on deb-based system:

```
# sudo apt-get install libapache2-mod-proxy-uwsgi
# sudo a2enmod proxy
# sudo a2enmod proxy_uwsgi
```

WSGI Application

The function `neutron.server.get_application` will setup a WSGI application to run behind a WSGI server like uwsgi or mod_wsgi.

Neutron API behind uwsgi

Create a `/etc/neutron/neutron-api-uwsgi.ini` file with the content below:

```
[uwsgi]
chmod-socket = 666
socket = /var/run/uwsgi/neutron-api.socket
start-time = %t
lazy-apps = true
add-header = Connection: close
buffer-size = 65535
hook-master-start = unix_signal:15 gracefully_kill_them_all
thunder-lock = true
plugins = http,python3
enable-threads = true
worker-reload-mercy = 80
exit-on-reload = false
die-on-term = true
master = true
processes = 2
module = neutron.wsgi.api:application
```

Start neutron-api:

```
# uwsgi --procname-prefix neutron-api --ini /etc/neutron/neutron-api-uwsgi.ini
```

Start Neutron RPC server

When Neutron API is served by a web server (like Apache2) it is difficult to start an rpc listener thread. So start the Neutron RPC server process to serve this job:

```
# /usr/bin/neutron-rpc-server --config-file /etc/neutron/neutron.conf --
↪config-file /etc/neutron/plugins/ml2/ml2_conf.ini
```

Neutron Worker Processes

Neutron will attempt to spawn a number of child processes for handling API and RPC requests. The number of API workers is set to the number of CPU cores, further limited by available memory, and the number of RPC workers is set to half that number.

It is strongly recommended that all deployers set these values themselves, via the api_workers and rpc_workers configuration parameters.

For a cloud with a high load to a relatively small number of objects, a smaller value for api_workers will provide better performance than many (somewhere around 4-8.) For a cloud with a high load to lots of different objects, then the more the better. Budget neutron-server using about 2GB of RAM in steady-state.

For rpc_workers, there needs to be enough to keep up with incoming events from the various neutron agents. Signs that there are too few can be agent heartbeats arriving late, nova vif bindings timing out on the hypervisors, or rpc message timeout exceptions in agent logs (for example, broken pipe errors).

There is also the rpc_state_report_workers option, which determines the number fo RPC worker processes dedicated to process state reports from the various agents. This may be increased to resolve frequent delay in processing agents heartbeats.

Note

If OVN ML2 plugin is used without any additional agents, neutron requires no worker for RPC message processing. Set both rpc_workers and rpc_state_report_workers to 0, to disable RPC workers.

Note

ML2/OVN uses the `[uwsgi]start-time = %t` parameter to create the OVN hash ring registers during the initialization process. This value is populated by the uWSGi process with the start time. For more information, check *Configuring uWSGI <https://uwsgi-docs.readthedocs.io/en/latest/Configuration.html>_*.

Note

For general configuration, see the Configuration Reference.

8.3 Deployment examples

The following deployment examples provide building blocks of increasing architectural complexity using the Networking service reference architecture which implements the Modular Layer 2 (ML2) plug-in with the Open vSwitch (OVS) mechanism driver. The mechanism driver supports basic features such as provider networks, self-service networks, and routers. However, more complex features often require a particular mechanism driver. Thus, you should consider the requirements (or goals) of your cloud before choosing a mechanism driver.

After choosing a *mechanism driver*, the deployment examples generally include the following building blocks:

1. Provider (public/external) networks using IPv4 and IPv6

2. Self-service (project/private/internal) networks including routers using IPv4 and IPv6
3. High-availability features
4. Other features such as BGP dynamic routing

8.3.1 Prerequisites

Prerequisites, typically hardware requirements, generally increase with each building block. Each building block depends on proper deployment and operation of prior building blocks. For example, the first building block (provider networks) only requires one controller and two compute nodes, the second building block (self-service networks) adds a network node, and the high-availability building blocks typically add a second network node for a total of five nodes. Each building block could also require additional infrastructure or changes to existing infrastructure such as networks.

For basic configuration of prerequisites, see the latest Install Tutorials and Guides.

Note

Example commands using the `openstack` client assume version 3.2.0 or higher.

Nodes

The deployment examples refer one or more of the following nodes:

- Controller: Contains control plane components of OpenStack services and their dependencies.
  - Two network interfaces: management and provider.
  - Operational SQL server with databases necessary for each OpenStack service.
  - Operational message queue service.
  - Operational OpenStack Identity (keystone) service.
  - Operational OpenStack Image Service (glance).
  - Operational management components of the OpenStack Compute (nova) service with appropriate configuration to use the Networking service.
  - OpenStack Networking (neutron) server service and ML2 plug-in.
- Network: Contains the OpenStack Networking service layer-3 (routing) component. High availability options may include additional components.
  - Three network interfaces: management, overlay, and provider.
  - OpenStack Networking layer-2 (switching) agent, layer-3 agent, and any dependencies.
- Compute: Contains the hypervisor component of the OpenStack Compute service and the OpenStack Networking layer-2, DHCP, and metadata components. High-availability options may include additional components.
  - Two network interfaces: management and provider.
  - Operational hypervisor components of the OpenStack Compute (nova) service with appropriate configuration to use the Networking service.
  - OpenStack Networking layer-2 agent, DHCP agent, metadata agent, and any dependencies.

Each building block defines the quantity and types of nodes including the components on each node.

Note

You can virtualize these nodes for demonstration, training, or proof-of-concept purposes. However, you must use physical hosts for evaluation of performance or scaling.

Networks and network interfaces

The deployment examples refer to one or more of the following networks and network interfaces:

- Management: Handles API requests from clients and control plane traffic for OpenStack services including their dependencies.
- Overlay: Handles self-service networks using an overlay protocol such as VXLAN or GRE.
- Provider: Connects virtual and physical networks at layer-2. Typically uses physical network infrastructure for switching/routing traffic to external networks such as the Internet.

Note

For best performance, 10+ Gbps physical network infrastructure should support jumbo frames.

For illustration purposes, the configuration examples typically reference the following IP address ranges:

- Provider network 1:
  - IPv4: 203.0.113.0/24
  - IPv6: fd00:203:0:113::/64
- Provider network 2:
  - IPv4: 192.0.2.0/24
  - IPv6: fd00:192:0:2::/64
- Self-service networks:
  - IPv4: 198.51.100.0/24 in /24 segments
  - IPv6: fd00:198:51::/48 in /64 segments

You may change them to work with your particular network infrastructure.

8.3.2 Mechanism drivers

Open vSwitch mechanism driver

The Open vSwitch (OVS) mechanism driver uses a combination of OVS and Linux bridges as interconnection devices. However, optionally enabling the OVS native implementation of security groups removes the dependency on Linux bridges.

We recommend using Open vSwitch version 2.4 or higher. Optional features may require a higher minimum version.

Open vSwitch: Provider networks

This architecture example provides layer-2 connectivity between instances and the physical network infrastructure using VLAN (802.1q) tagging. It supports one untagged (flat) network and up to 4095 tagged (VLAN) networks. The actual quantity of VLAN networks depends on the physical network infrastructure. For more information on provider networks, see *Provider networks*.

Warning

Linux distributions often package older releases of Open vSwitch that can introduce issues during operation with the Networking service. We recommend using at least the latest long-term stable (LTS) release of Open vSwitch for the best experience and support from Open vSwitch. See http://www.openvswitch.org for available releases and the installation instructions for more details.

Prerequisites

One controller node with the following components:

- Two network interfaces: management and provider.
- OpenStack Networking server service and ML2 plug-in.

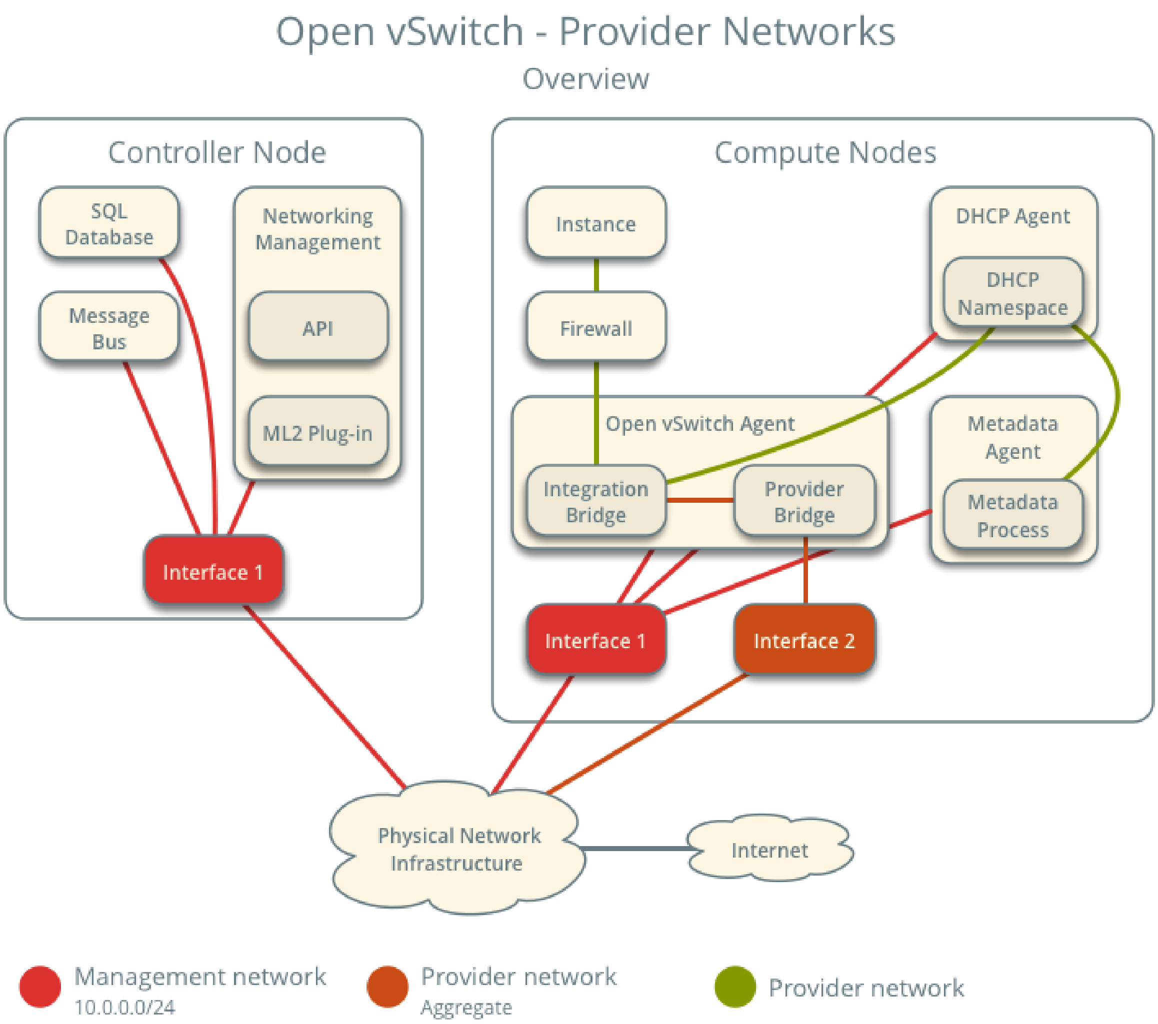
Two compute nodes with the following components:

- Two network interfaces: management and provider.
- OpenStack Networking Open vSwitch (OVS) layer-2 agent, DHCP agent, metadata agent, and any dependencies including OVS.

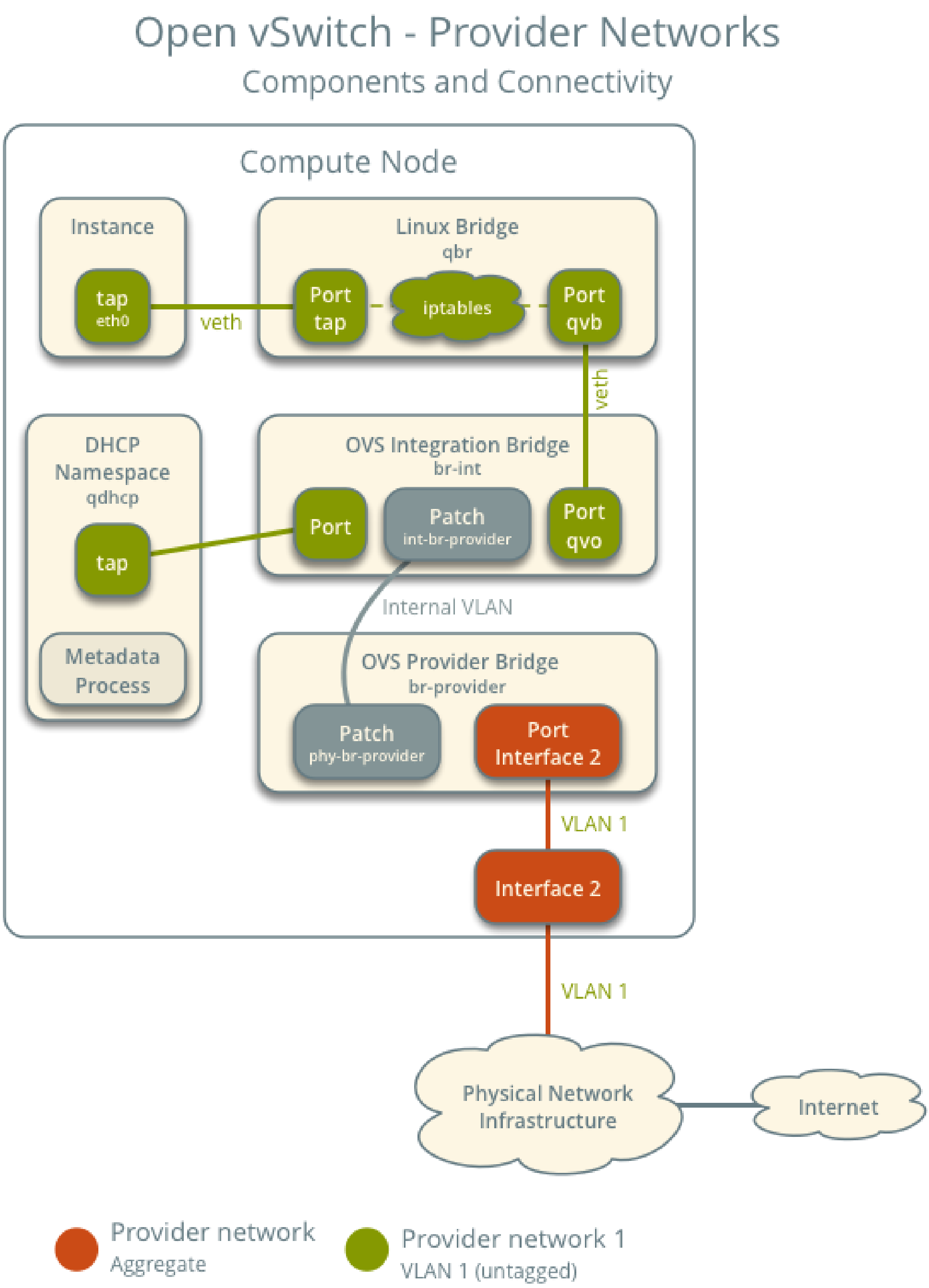
Note

Larger deployments typically deploy the DHCP and metadata agents on a subset of compute nodes to increase performance and redundancy. However, too many agents can overwhelm the message bus. Also, to further simplify any deployment, you can omit the metadata agent and use a configuration drive to provide metadata to instances.

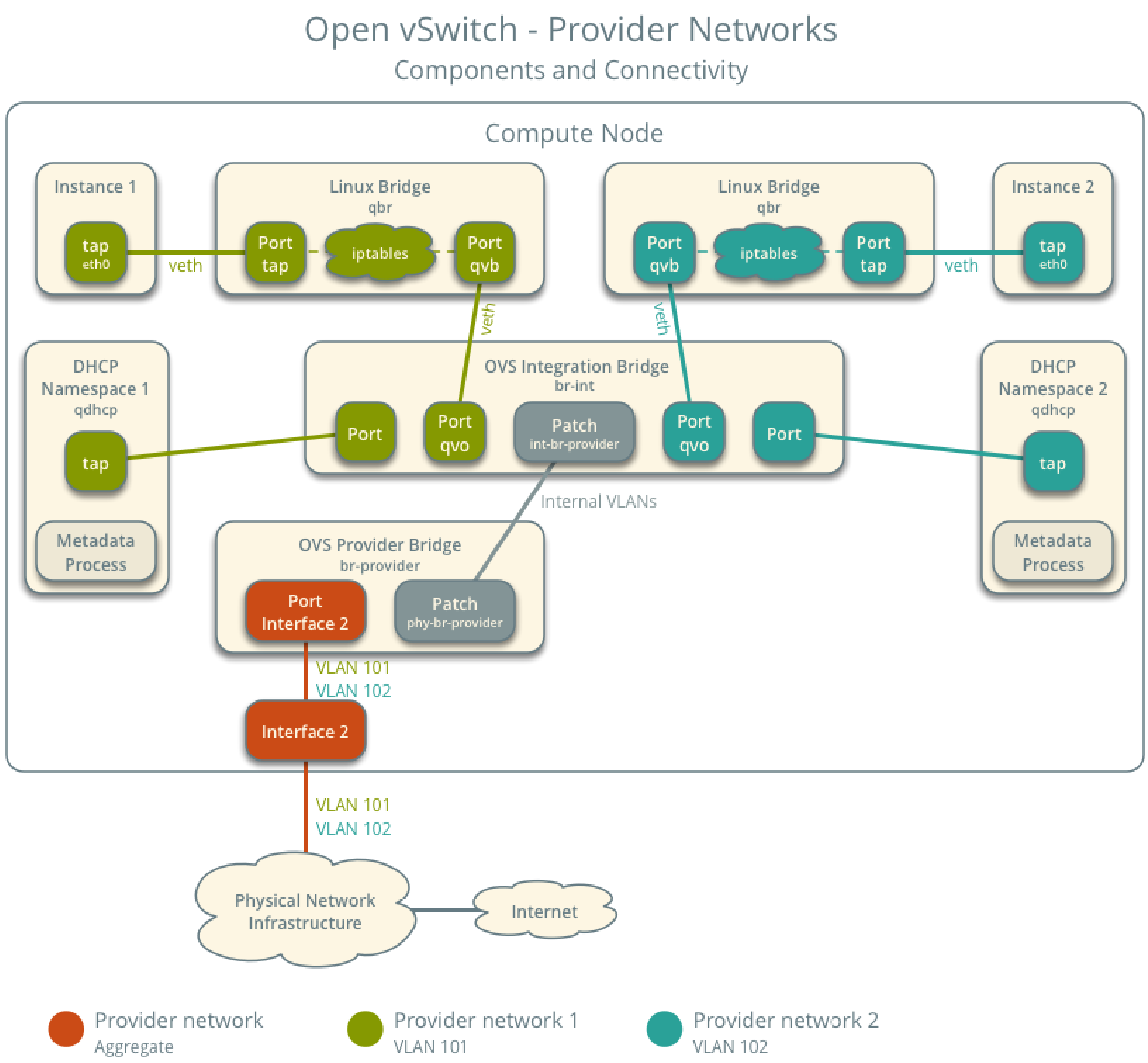
Architecture



The following figure shows components and connectivity for one untagged (flat) network. In this particular case, the instance resides on the same compute node as the DHCP agent for the network. If the DHCP agent resides on another compute node, the latter only contains a DHCP namespace with a port on the OVS integration bridge.



The following figure describes virtual connectivity among components for two tagged (VLAN) networks. Essentially, all networks use a single OVS integration bridge with different internal VLAN tags. The internal VLAN tags almost always differ from the network VLAN assignment in the Networking service. Similar to the untagged network case, the DHCP agent may reside on a different compute node.



Note

These figures omit the controller node because it does not handle instance network traffic.

Example configuration

Use the following example configuration as a template to deploy provider networks in your environment.

Controller node

1. Install the Networking service components that provide the `neutron-server` service and ML2 plug-in.
2. In the `neutron.conf` file:
   - Configure common options:

```
[DEFAULT]
core_plugin = ml2
```

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```
auth_strategy = keystone

[database]
# ...

[keystone_authtoken]
# ...

[nova]
# ...

[agent]
# ...
```

See the Installation Tutorials and Guides and Configuration Reference for your OpenStack release to obtain the appropriate additional configuration for the `[DEFAULT]`, `[database]`, `[keystone_authtoken]`, `[nova]`, and `[agent]` sections.

- Disable service plug-ins because provider networks do not require any. However, this breaks portions of the dashboard that manage the Networking service. See the latest Install Tutorials and Guides for more information.

```
[DEFAULT]
service_plugins =
```

- Enable two DHCP agents per network so both compute nodes can provide DHCP service provider networks.

```
[DEFAULT]
dhcp_agents_per_network = 2
```

- If necessary, *configure MTU*.

3. In the `ml2_conf.ini` file:

- Configure drivers and network types:

```
[ml2]
type_drivers = flat,vlan
project_network_types =
mechanism_drivers = openvswitch
extension_drivers = port_security
```

- Configure network mappings:

```
[ml2_type_flat]
flat_networks = provider

[ml2_type_vlan]
network_vlan_ranges = provider
```

Note

The `project_network_types` option contains no value because the architecture does not support self-service networks.

Note

The `provider` value in the `network_vlan_ranges` option lacks VLAN ID ranges to support use of arbitrary VLAN IDs.

4. Populate the database.

```
# su -s /bin/sh -c "neutron-db-manage --config-file /etc/neutron/neutron.
→conf \
  --config-file /etc/neutron/plugins/ml2/ml2_conf.ini upgrade head"␣
→neutron
```

5. Start the following services:

   - Server

Compute nodes

1. Install the Networking service OVS layer-2 agent, DHCP agent, and metadata agent.

2. Install OVS.

3. In the `neutron.conf` file, configure common options:

```
[DEFAULT]
core_plugin = ml2
auth_strategy = keystone

[database]
# ...

[keystone_authtoken]
# ...

[nova]
# ...

[agent]
# ...
```

   See the Installation Tutorials and Guides and Configuration Reference for your OpenStack release to obtain the appropriate additional configuration for the `[DEFAULT]`, `[database]`, `[keystone_authtoken]`, `[nova]`, and `[agent]` sections.

4. In the `openvswitch_agent.ini` file, configure the OVS agent:

```
[ovs]
bridge_mappings = provider:br-provider

[securitygroup]
firewall_driver = iptables_hybrid
```

5. In the `dhcp_agent.ini` file, configure the DHCP agent:

```
[DEFAULT]
enable_isolated_metadata = True
force_metadata = True
```

> **Note**
>
> The `force_metadata` option forces the DHCP agent to provide a host route to the metadata service on `169.254.169.254` regardless of whether the subnet contains an interface on a router, thus maintaining similar and predictable metadata behavior among subnets.

6. In the `metadata_agent.ini` file, configure the metadata agent:

```
[DEFAULT]
nova_metadata_host = controller
metadata_proxy_shared_secret = METADATA_SECRET
```

The value of `METADATA_SECRET` must match the value of the same option in the `[neutron]` section of the `nova.conf` file.

7. Start the following services:
   - OVS
8. Create the OVS provider bridge `br-provider`:

```
$ ovs-vsctl add-br br-provider
```

9. Add the provider network interface as a port on the OVS provider bridge `br-provider`:

```
$ ovs-vsctl add-port br-provider PROVIDER_INTERFACE
```

Replace `PROVIDER_INTERFACE` with the name of the underlying interface that handles provider networks. For example, `eth1`.

10. Start the following services:
    - OVS agent
    - DHCP agent
    - Metadata agent

Verify service operation

1. Source the administrative project credentials.
2. Verify presence and operation of the agents:

```
$ openstack network agent list
+--------------------------------------+--------------------+----------+---
↪---------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host     |␣
↪Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+----------+---
↪---------------+-------+-------+---------------------------+
| 1236bbcb-e0ba-48a9-80fc-81202ca4fa51 | Metadata agent     | compute2 |␣
↪None              | True  | UP    | neutron-metadata-agent    |
| 457d6898-b373-4bb3-b41f-59345dcfb5c5 | Open vSwitch agent | compute2 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
| 71f15e84-bc47-4c2a-b9fb-317840b2d753 | DHCP agent         | compute2 |␣
↪nova              | True  | UP    | neutron-dhcp-agent        |
| a6c69690-e7f7-4e56-9831-1282753e5007 | Metadata agent     | compute1 |␣
↪None              | True  | UP    | neutron-metadata-agent    |
| af11f22f-a9f4-404f-9fd8-cd7ad55c0f68 | DHCP agent         | compute1 |␣
↪nova              | True  | UP    | neutron-dhcp-agent        |
| bcfc977b-ec0e-4ba9-be62-9489b4b0e6f1 | Open vSwitch agent | compute1 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
+--------------------------------------+--------------------+----------+---
↪---------------+-------+-------+---------------------------+
```

Create initial networks

The configuration supports one flat or multiple VLAN provider networks. For simplicity, the following procedure creates one flat provider network.

1. Source the administrative project credentials.
2. Create a flat network.

```
$ openstack network create --share --provider-physical-network provider \
  --provider-network-type flat provider1
+---------------------------+-----------+-
| Field                     | Value     |
+---------------------------+-----------+
| admin_state_up            | UP        |
| mtu                       | 1500      |
| name                      | provider1 |
| port_security_enabled     | True      |
| provider:network_type     | flat      |
| provider:physical_network | provider  |
| provider:segmentation_id  | None      |
| router:external           | Internal  |
| shared                    | True      |
| status                    | ACTIVE    |
+---------------------------+-----------+
```

Note

The `share` option allows any project to use this network. To limit access to provider networks, see *Role-Based Access Control (RBAC)*.

Note

To create a VLAN network instead of a flat network, change `--provider-network-type flat` to `--provider-network-type vlan` and add `--provider-segment` with a value referencing the VLAN ID.

3. Create a IPv4 subnet on the provider network.

```
$ openstack subnet create --subnet-range 203.0.113.0/24 --gateway 203.0.
↪113.1 \
  --network provider1 --allocation-pool start=203.0.113.11,end=203.0.113.
↪250 \
  --dns-nameserver 8.8.4.4 provider1-v4
+-------------------+---------------------------+
| Field             | Value                     |
+-------------------+---------------------------+
| allocation_pools  | 203.0.113.11-203.0.113.250 |
| cidr              | 203.0.113.0/24            |
| dns_nameservers   | 8.8.4.4                   |
| enable_dhcp       | True                      |
| gateway_ip        | 203.0.113.1               |
| ip_version        | 4                         |
| name              | provider1-v4              |
+-------------------+---------------------------+
```

Important

Enabling DHCP causes the Networking service to provide DHCP which can interfere with existing DHCP services on the physical network infrastructure. Use the `--no-dhcp` option to have the subnet managed by existing DHCP services.

4. Create a IPv6 subnet on the provider network.

```
$ openstack subnet create --subnet-range fd00:203:0:113::/64 --gateway␣
↪fd00:203:0:113::1 \
  --ip-version 6 --ipv6-address-mode slaac --network provider1 \
  --dns-nameserver 2001:4860:4860::8844 provider1-v6
+-------------------+--------------------------------------------------------
↪-+
| Field             | Value                                                  ␣
↪ |
+-------------------+--------------------------------------------------------
↪-+
```

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```
| allocation_pools  | fd00:203:0:113::2-
↪fd00:203:0:113:ffff:ffff:ffff:ffff |
| cidr              | fd00:203:0:113::/64                                                ␣
↪ |
| dns_nameservers   | 2001:4860:4860::8844                                               ␣
↪ |
| enable_dhcp       | True                                                               ␣
↪ |
| gateway_ip        | fd00:203:0:113::1                                                  ␣
↪ |
| ip_version        | 6                                                                  ␣
↪ |
| ipv6_address_mode | slaac                                                              ␣
↪ |
| ipv6_ra_mode      | None                                                               ␣
↪ |
| name              | provider1-v6                                                       ␣
↪ |
+-------------------+------------------------------------------------------------------
↪-+
```

Note

The Networking service uses the layer-3 agent to provide router advertisement. Provider networks rely on physical network infrastructure for layer-3 services rather than the layer-3 agent. Thus, the physical network infrastructure must provide router advertisement on provider networks for proper operation of IPv6.

Verify network operation

1. On each compute node, verify creation of the `qdhcp` namespace.

```
# ip netns
qdhcp-8b868082-e312-4110-8627-298109d4401c
```

2. Source a regular (non-administrative) project credentials.

3. Create the appropriate security group rules to allow `ping` and SSH access instances using the network.

```
$ openstack security group rule create --proto icmp default
+------------------+-----------+
| Field            | Value     |
+------------------+-----------+
| direction        | ingress   |
| ethertype        | IPv4      |
| protocol         | icmp      |
| remote_ip_prefix | 0.0.0.0/0 |
+------------------+-----------+
```

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```
$ openstack security group rule create --ethertype IPv6 \
  --proto ipv6-icmp default
+-----------+-----------+
| Field     | Value     |
+-----------+-----------+
| direction | ingress   |
| ethertype | IPv6      |
| protocol  | ipv6-icmp |
+-----------+-----------+

$ openstack security group rule create --proto tcp --dst-port 22 default
+------------------+-----------+
| Field            | Value     |
+------------------+-----------+
| direction        | ingress   |
| ethertype        | IPv4      |
| port_range_max   | 22        |
| port_range_min   | 22        |
| protocol         | tcp       |
| remote_ip_prefix | 0.0.0.0/0 |
+------------------+-----------+

$ openstack security group rule create --ethertype IPv6 --proto tcp \
  --dst-port 22 default
+----------------+---------+
| Field          | Value   |
+----------------+---------+
| direction      | ingress |
| ethertype      | IPv6    |
| port_range_max | 22      |
| port_range_min | 22      |
| protocol       | tcp     |
+----------------+---------+
```

4. Launch an instance with an interface on the provider network. For example, a CirrOS image using flavor ID 1.

```
$ openstack server create --flavor 1 --image cirros \
  --nic net-id=NETWORK_ID provider-instance1
```

Replace NETWORK_ID with the ID of the provider network.

5. Determine the IPv4 and IPv6 addresses of the instance.

```
$ openstack server list
+--------------------------------------+--------------------+--------+----
↪--------------------------------------------------------+--------+------
↪---+
| ID                                   | Name               | Status |␣
```

(continues on next page)

(continued from previous page)

```
↪Networks                                               | Image    |␣
↪Flavor   |
+--------------------------------------+------------------+--------+----
↪-------------------------------------------------------+--------+------
↪---+
| 018e0ae2-b43c-4271-a78d-62653dd03285 | provider-instance1 | ACTIVE |␣
↪provider1=203.0.113.13, fd00:203:0:113:f816:3eff:fe58:be4e | cirros |␣
↪m1.tiny |
+--------------------------------------+------------------+--------+----
↪-------------------------------------------------------+--------+------
↪---+
```

6. On the controller node or any host with access to the provider network, `ping` the IPv4 and IPv6 addresses of the instance.

```
$ ping -c 4 203.0.113.13
PING 203.0.113.13 (203.0.113.13) 56(84) bytes of data.
64 bytes from 203.0.113.13: icmp_req=1 ttl=63 time=3.18 ms
64 bytes from 203.0.113.13: icmp_req=2 ttl=63 time=0.981 ms
64 bytes from 203.0.113.13: icmp_req=3 ttl=63 time=1.06 ms
64 bytes from 203.0.113.13: icmp_req=4 ttl=63 time=0.929 ms

--- 203.0.113.13 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 0.929/1.539/3.183/0.951 ms

$ ping6 -c 4 fd00:203:0:113:f816:3eff:fe58:be4e
PING␣
↪fd00:203:0:113:f816:3eff:fe58:be4e(fd00:203:0:113:f816:3eff:fe58:be4e)␣
↪56 data bytes
64 bytes from fd00:203:0:113:f816:3eff:fe58:be4e icmp_seq=1 ttl=64 time=1.
↪25 ms
64 bytes from fd00:203:0:113:f816:3eff:fe58:be4e icmp_seq=2 ttl=64 time=0.
↪683 ms
64 bytes from fd00:203:0:113:f816:3eff:fe58:be4e icmp_seq=3 ttl=64 time=0.
↪762 ms
64 bytes from fd00:203:0:113:f816:3eff:fe58:be4e icmp_seq=4 ttl=64 time=0.
↪486 ms

--- fd00:203:0:113:f816:3eff:fe58:be4e ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2999ms
rtt min/avg/max/mdev = 0.486/0.796/1.253/0.282 ms
```

7. Obtain access to the instance.

8. Test IPv4 and IPv6 connectivity to the Internet or other external network.

Network traffic flow

The following sections describe the flow of network traffic in several common scenarios. *North-south* network traffic travels between an instance and external network such as the Internet. *East-west* network traffic travels between instances on the same or different networks. In all scenarios, the physical network infrastructure handles switching and routing among provider networks and external networks such as the Internet. Each case references one or more of the following components:

- Provider network 1 (VLAN)
  - VLAN ID 101 (tagged)
  - IP address ranges 203.0.113.0/24 and fd00:203:0:113::/64
  - Gateway (via physical network infrastructure)
    - IP addresses 203.0.113.1 and fd00:203:0:113:0::1
- Provider network 2 (VLAN)
  - VLAN ID 102 (tagged)
  - IP address range 192.0.2.0/24 and fd00:192:0:2::/64
  - Gateway
    - IP addresses 192.0.2.1 and fd00:192:0:2::1
- Instance 1
  - IP addresses 203.0.113.101 and fd00:203:0:113:0::101
- Instance 2
  - IP addresses 192.0.2.101 and fd00:192:0:2:0::101

North-south

- The instance resides on compute node 1 and uses provider network 1.
- The instance sends a packet to a host on the Internet.

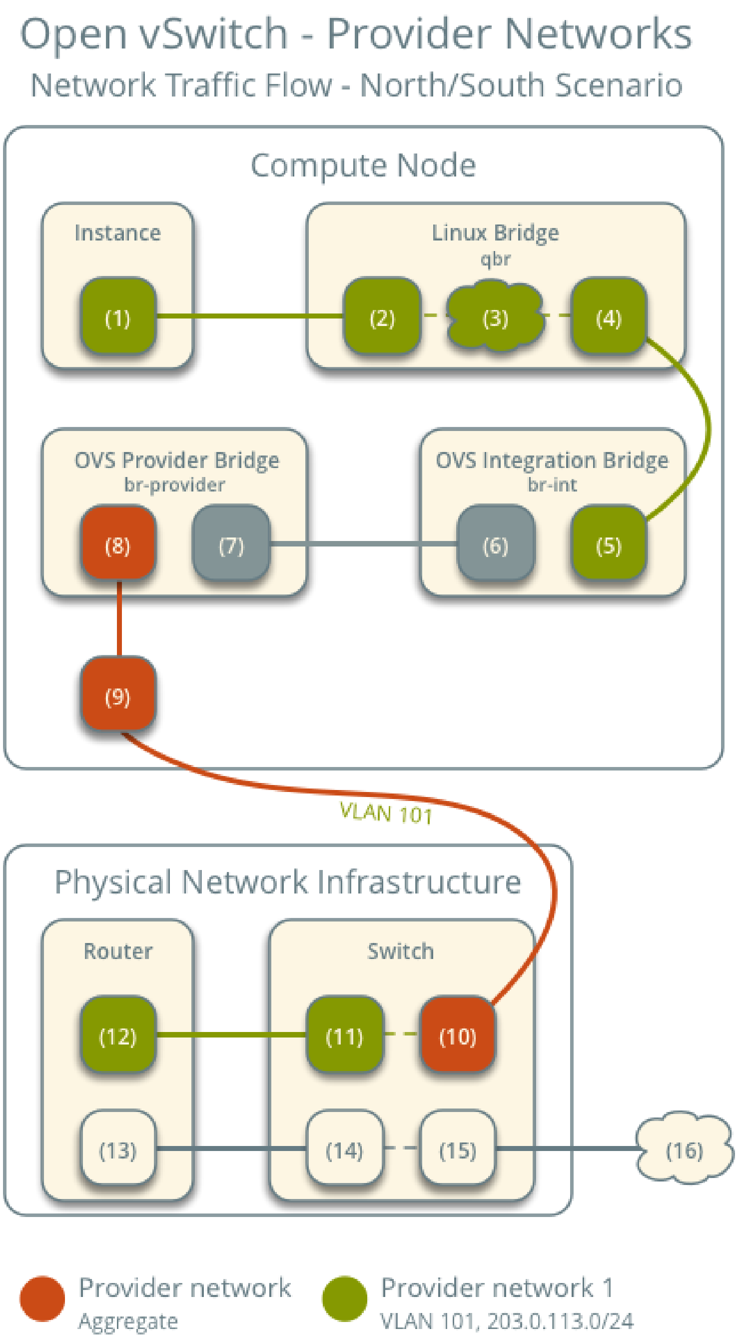
The following steps involve compute node 1.

1. The instance interface (1) forwards the packet to the security group bridge instance port (2) via `veth` pair.
2. Security group rules (3) on the security group bridge handle firewalling and connection tracking for the packet.
3. The security group bridge OVS port (4) forwards the packet to the OVS integration bridge security group port (5) via `veth` pair.
4. The OVS integration bridge adds an internal VLAN tag to the packet.
5. The OVS integration bridge `int-br-provider` patch port (6) forwards the packet to the OVS provider bridge `phy-br-provider` patch port (7).
6. The OVS provider bridge swaps the internal VLAN tag with actual VLAN tag 101.
7. The OVS provider bridge provider network port (8) forwards the packet to the physical network interface (9).

8. The physical network interface forwards the packet to the physical network infrastructure switch (10).

The following steps involve the physical network infrastructure:

1. The switch removes VLAN tag 101 from the packet and forwards it to the router (11).
2. The router routes the packet from the provider network (12) to the external network (13) and forwards the packet to the switch (14).
3. The switch forwards the packet to the external network (15).
4. The external network (16) receives the packet.



Note

Return traffic follows similar steps in reverse.

East-west scenario 1: Instances on the same network

Instances on the same network communicate directly between compute nodes containing those instances.

- Instance 1 resides on compute node 1 and uses provider network 1.
- Instance 2 resides on compute node 2 and uses provider network 1.
- Instance 1 sends a packet to instance 2.

The following steps involve compute node 1:

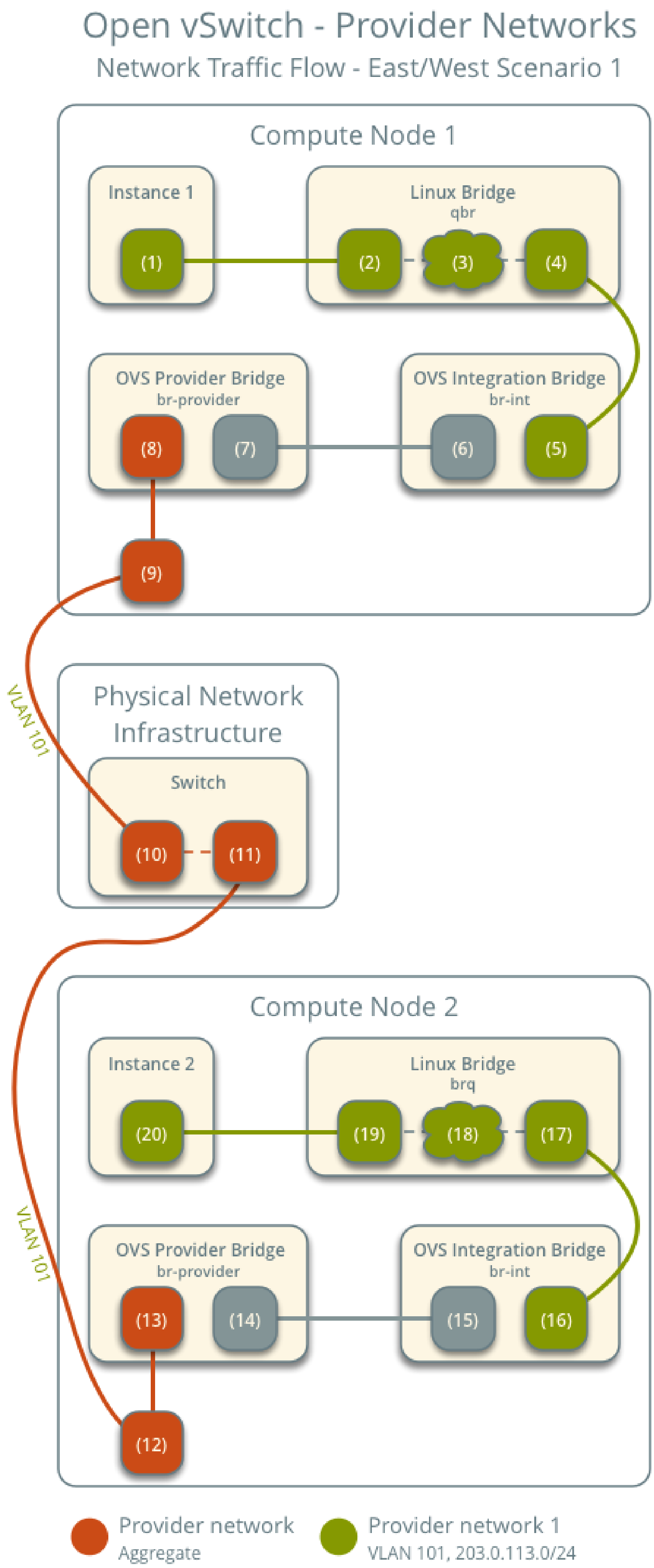
1. The instance 1 interface (1) forwards the packet to the security group bridge instance port (2) via `veth` pair.
2. Security group rules (3) on the security group bridge handle firewalling and connection tracking for the packet.
3. The security group bridge OVS port (4) forwards the packet to the OVS integration bridge security group port (5) via `veth` pair.
4. The OVS integration bridge adds an internal VLAN tag to the packet.
5. The OVS integration bridge `int-br-provider` patch port (6) forwards the packet to the OVS provider bridge `phy-br-provider` patch port (7).
6. The OVS provider bridge swaps the internal VLAN tag with actual VLAN tag 101.
7. The OVS provider bridge provider network port (8) forwards the packet to the physical network interface (9).
8. The physical network interface forwards the packet to the physical network infrastructure switch (10).

The following steps involve the physical network infrastructure:

1. The switch forwards the packet from compute node 1 to compute node 2 (11).

The following steps involve compute node 2:

1. The physical network interface (12) forwards the packet to the OVS provider bridge provider network port (13).
2. The OVS provider bridge `phy-br-provider` patch port (14) forwards the packet to the OVS integration bridge `int-br-provider` patch port (15).
3. The OVS integration bridge swaps the actual VLAN tag 101 with the internal VLAN tag.
4. The OVS integration bridge security group port (16) forwards the packet to the security group bridge OVS port (17).
5. Security group rules (18) on the security group bridge handle firewalling and connection tracking for the packet.
6. The security group bridge instance port (19) forwards the packet to the instance 2 interface (20) via `veth` pair.


Open vSwitch - Provider Networks
Network Traffic Flow - East/West Scenario 1
Compute Node 1
Instance 1
(1)
Linux Bridge
qbr
(2)
(3)
(4)
OVS Provider Bridge
br-provider
(8)
(7)
OVS Integration Bridge
br-int
(6)
(5)
(9)
VLAN 101
Physical Network
Infrastructure
Switch
(10)
(11)
Compute Node 2
Instance 2
(20)
Linux Bridge
brq
(19)
(18)
(17)
OVS Provider Bridge
br-provider
(13)
(14)
OVS Integration Bridge
br-int
(15)
(16)
VLAN 101
(12)
Provider network
Aggregate
Provider network 1
VLAN 101, 203.0.113.0/24

Note

Return traffic follows similar steps in reverse.

East-west scenario 2: Instances on different networks

Instances communicate via router on the physical network infrastructure.

- Instance 1 resides on compute node 1 and uses provider network 1.
- Instance 2 resides on compute node 1 and uses provider network 2.
- Instance 1 sends a packet to instance 2.

Note

Both instances reside on the same compute node to illustrate how VLAN tagging enables multiple logical layer-2 networks to use the same physical layer-2 network.

The following steps involve the compute node:

1. The instance 1 interface (1) forwards the packet to the security group bridge instance port (2) via `veth` pair.
2. Security group rules (3) on the security group bridge handle firewalling and connection tracking for the packet.
3. The security group bridge OVS port (4) forwards the packet to the OVS integration bridge security group port (5) via `veth` pair.
4. The OVS integration bridge adds an internal VLAN tag to the packet.
5. The OVS integration bridge `int-br-provider` patch port (6) forwards the packet to the OVS provider bridge `phy-br-provider` patch port (7).
6. The OVS provider bridge swaps the internal VLAN tag with actual VLAN tag 101.
7. The OVS provider bridge provider network port (8) forwards the packet to the physical network interface (9).
8. The physical network interface forwards the packet to the physical network infrastructure switch (10).

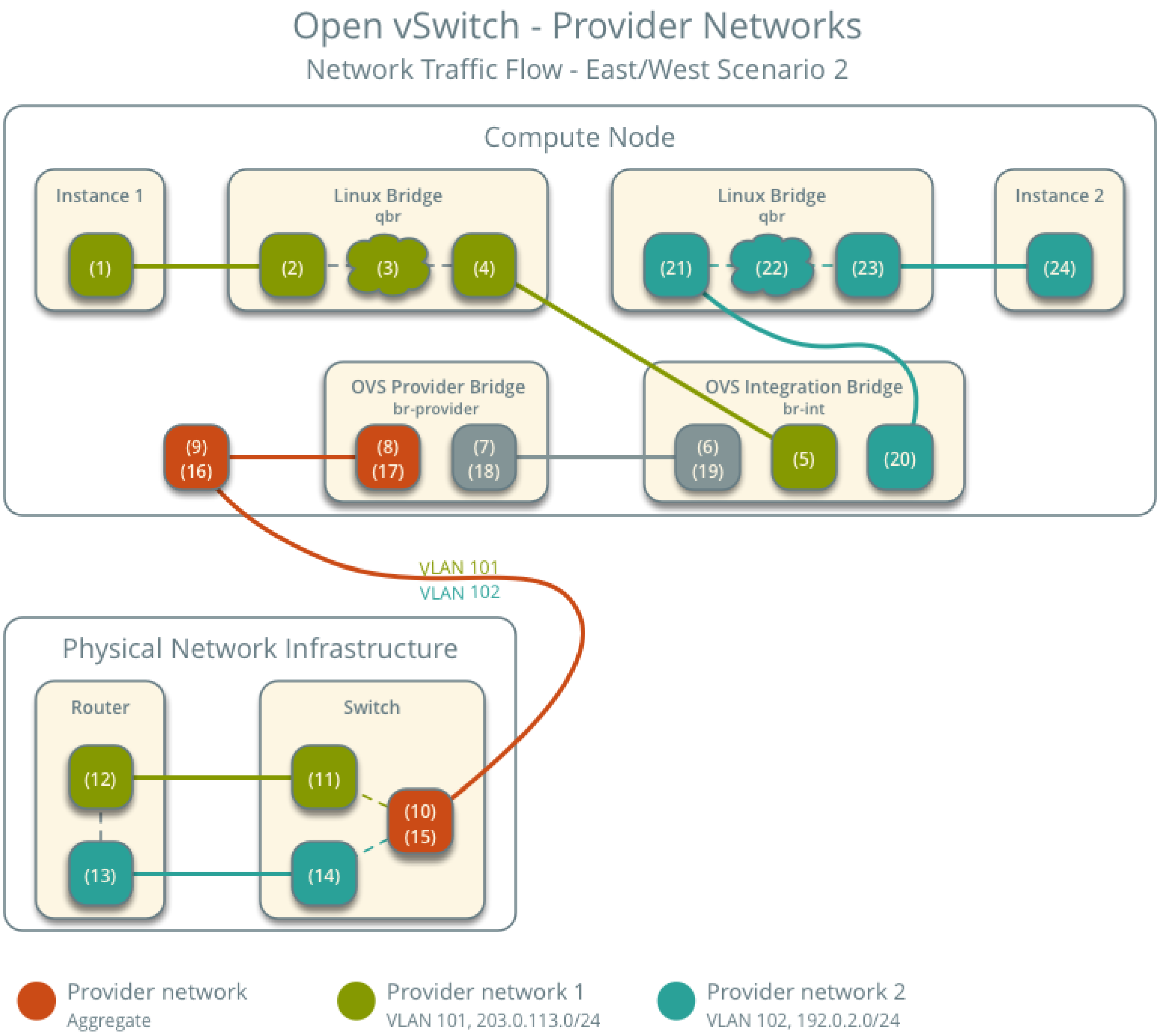
The following steps involve the physical network infrastructure:

1. The switch removes VLAN tag 101 from the packet and forwards it to the router (11).
2. The router routes the packet from provider network 1 (12) to provider network 2 (13).
3. The router forwards the packet to the switch (14).
4. The switch adds VLAN tag 102 to the packet and forwards it to compute node 1 (15).

The following steps involve the compute node:

1. The physical network interface (16) forwards the packet to the OVS provider bridge provider network port (17).

2. The OVS provider bridge `phy-br-provider` patch port (18) forwards the packet to the OVS integration bridge `int-br-provider` patch port (19).
3. The OVS integration bridge swaps the actual VLAN tag 102 with the internal VLAN tag.
4. The OVS integration bridge security group port (20) removes the internal VLAN tag and forwards the packet to the security group bridge OVS port (21).
5. Security group rules (22) on the security group bridge handle firewalling and connection tracking for the packet.
6. The security group bridge instance port (23) forwards the packet to the instance 2 interface (24) via `veth` pair.



Note

Return traffic follows similar steps in reverse.

Open vSwitch: Self-service networks

This architecture example augments *Open vSwitch: Provider networks* to support a nearly limitless quantity of entirely virtual networks. Although the Networking service supports VLAN self-service networks, this example focuses on VXLAN self-service networks. For more information on self-service networks, see *Self-service networks*.

Prerequisites

Add one network node with the following components:

- Three network interfaces: management, provider, and overlay.
- OpenStack Networking Open vSwitch (OVS) layer-2 agent, layer-3 agent, and any including OVS.

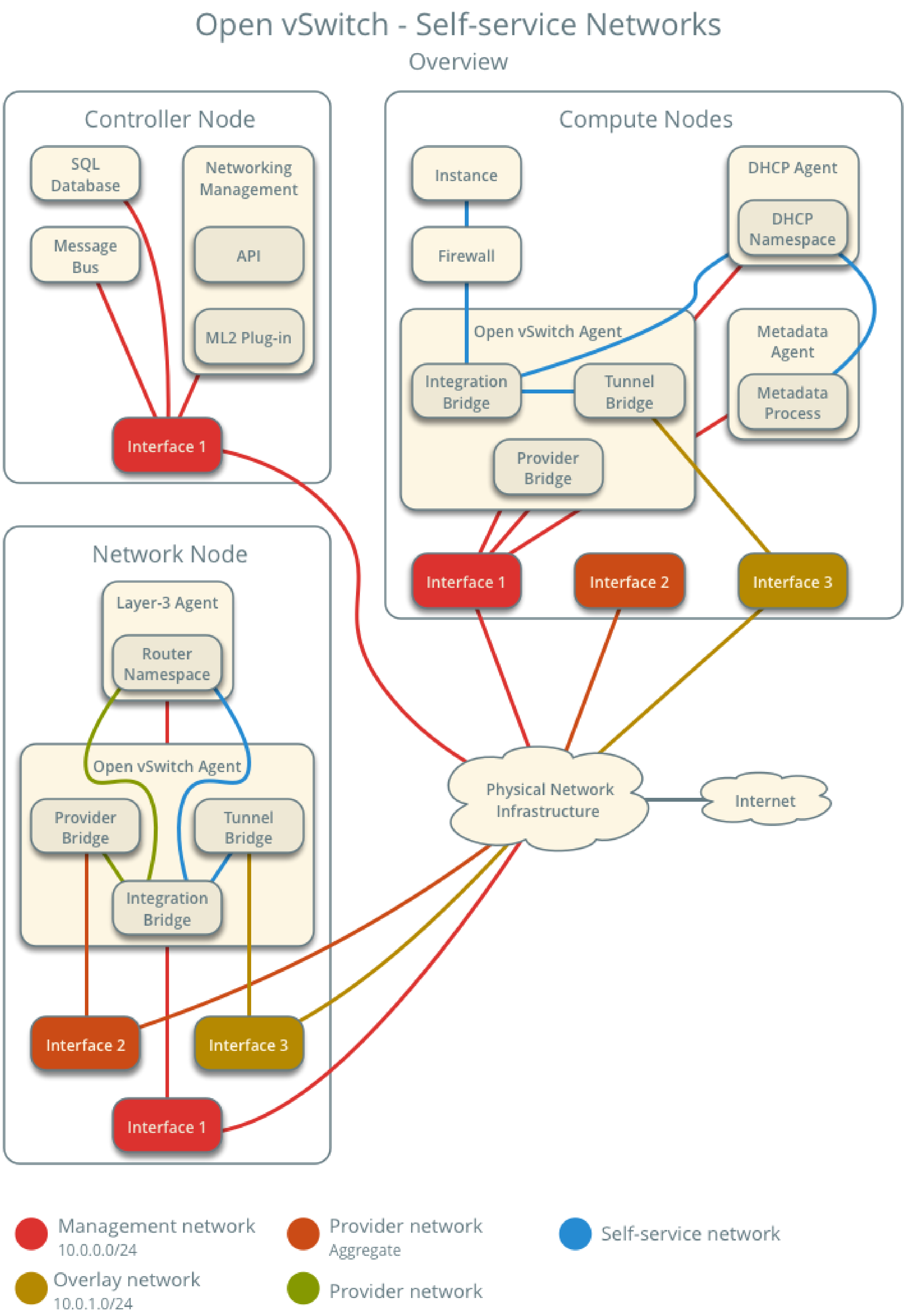
Modify the compute nodes with the following components:

- Add one network interface: overlay.

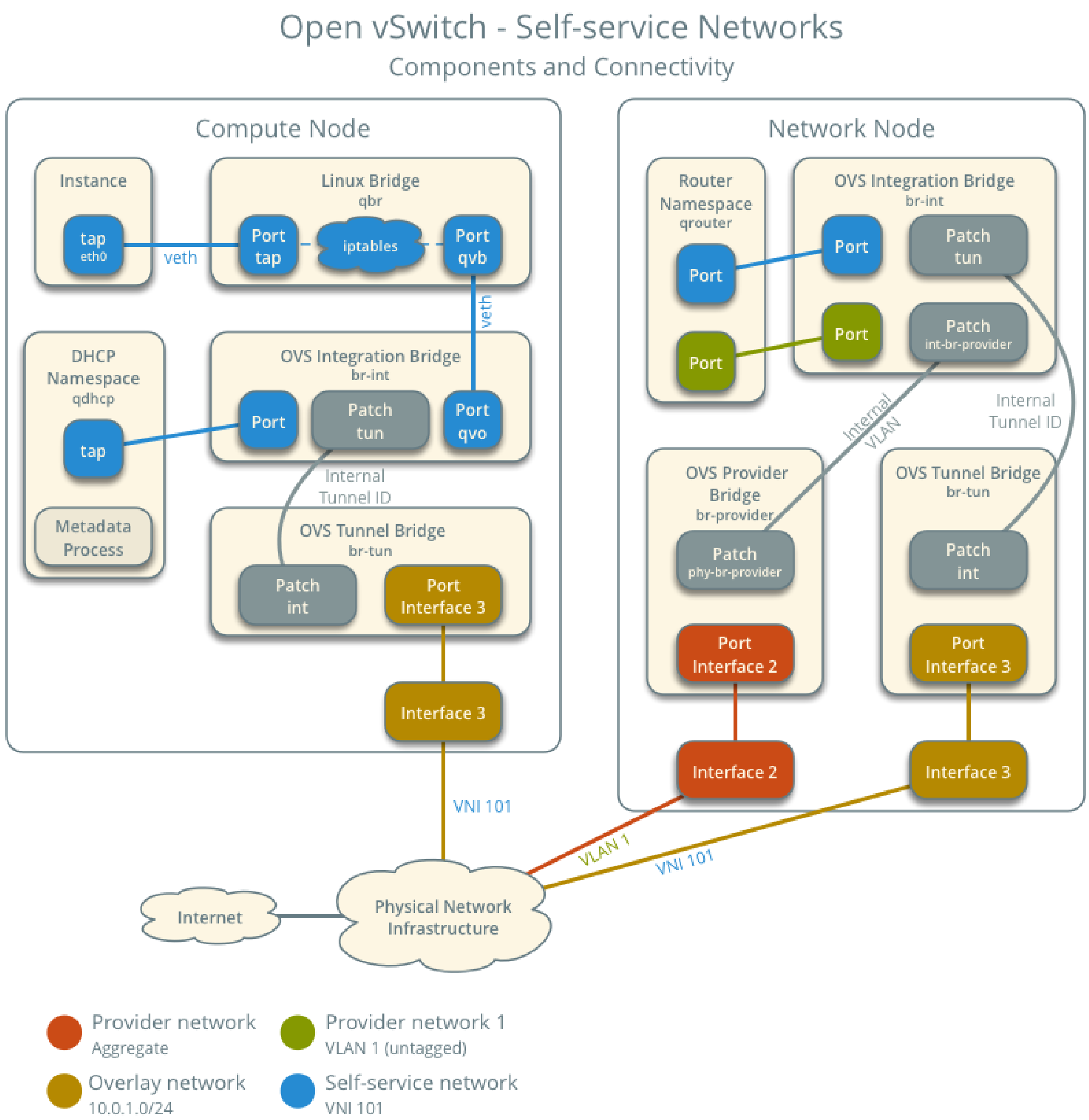
Note

You can keep the DHCP and metadata agents on each compute node or move them to the network node.

Architecture



The following figure shows components and connectivity for one self-service network and one untagged (flat) provider network. In this particular case, the instance resides on the same compute node as the DHCP agent for the network. If the DHCP agent resides on another compute node, the latter only contains a DHCP namespace and with a port on the OVS integration bridge.



Example configuration

Use the following example configuration as a template to add support for self-service networks to an existing operational environment that supports provider networks.

Controller node

1. In the `neutron.conf` file:

   - Enable routing and allow overlapping IP address ranges.

```
[DEFAULT]
service_plugins = router
```

2. In the `ml2_conf.ini` file:

   - Add `vxlan` to type drivers and project network types.

     ```
     [ml2]
     type_drivers = flat,vlan,vxlan
     project_network_types = vxlan
     ```

   - Enable the layer-2 population mechanism driver.

     ```
     [ml2]
     mechanism_drivers = openvswitch,l2population
     ```

   - Configure the VXLAN network ID (VNI) range.

     ```
     [ml2_type_vxlan]
     vni_ranges = VNI_START:VNI_END
     ```

     Replace `VNI_START` and `VNI_END` with appropriate numerical values.

3. Restart the following services:

   - Neutron Server
   - Open vSwitch agent

Network node

1. Install the Networking service OVS layer-2 agent and layer-3 agent.
2. Install OVS.
3. In the `neutron.conf` file, configure common options:

```
[DEFAULT]
core_plugin = ml2
auth_strategy = keystone

[database]
# ...

[keystone_authtoken]
# ...

[nova]
# ...

[agent]
# ...
```

See the Installation Tutorials and Guides and Configuration Reference for your OpenStack release to obtain the appropriate additional configuration for the [DEFAULT], [database], [keystone_authtoken], [nova], and [agent] sections.

4. Start the following services:
   - OVS
5. Create the OVS provider bridge br-provider:

```
$ ovs-vsctl add-br br-provider
```

6. Add the provider network interface as a port on the OVS provider bridge br-provider:

```
$ ovs-vsctl add-port br-provider PROVIDER_INTERFACE
```

   Replace PROVIDER_INTERFACE with the name of the underlying interface that handles provider networks. For example, eth1.

7. In the openvswitch_agent.ini file, configure the layer-2 agent.

```
[ovs]
bridge_mappings = provider:br-provider
local_ip = OVERLAY_INTERFACE_IP_ADDRESS

[agent]
tunnel_types = vxlan
l2_population = True

[securitygroup]
firewall_driver = iptables_hybrid
```

   Replace OVERLAY_INTERFACE_IP_ADDRESS with the IP address of the interface that handles VXLAN overlays for self-service networks.

8. Start the following services:
   - Open vSwitch agent
   - Layer-3 agent

Compute nodes

1. In the openvswitch_agent.ini file, enable VXLAN support including layer-2 population.

```
[ovs]
local_ip = OVERLAY_INTERFACE_IP_ADDRESS

[agent]
tunnel_types = vxlan
l2_population = True
```

   Replace OVERLAY_INTERFACE_IP_ADDRESS with the IP address of the interface that handles VXLAN overlays for self-service networks.

2. Restart the following services:

- Open vSwitch agent

Verify service operation

1. Source the administrative project credentials.
2. Verify presence and operation of the agents.

```
$ openstack network agent list
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host     |␣
↪Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
| 1236bbcb-e0ba-48a9-80fc-81202ca4fa51 | Metadata agent     | compute2 |␣
↪None              | True  | UP    | neutron-metadata-agent    |
| 457d6898-b373-4bb3-b41f-59345dcfb5c5 | Open vSwitch agent | compute2 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
| 71f15e84-bc47-4c2a-b9fb-317840b2d753 | DHCP agent         | compute2 |␣
↪nova              | True  | UP    | neutron-dhcp-agent        |
| 8805b962-de95-4e40-bdc2-7a0add7521e8 | L3 agent           | network1 |␣
↪nova              | True  | UP    | neutron-l3-agent          |
| a33cac5a-0266-48f6-9cac-4cef4f8b0358 | Open vSwitch agent | network1 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
| a6c69690-e7f7-4e56-9831-1282753e5007 | Metadata agent     | compute1 |␣
↪None              | True  | UP    | neutron-metadata-agent    |
| af11f22f-a9f4-404f-9fd8-cd7ad55c0f68 | DHCP agent         | compute1 |␣
↪nova              | True  | UP    | neutron-dhcp-agent        |
| bcfc977b-ec0e-4ba9-be62-9489b4b0e6f1 | Open vSwitch agent | compute1 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
```

Create initial networks

The configuration supports multiple VXLAN self-service networks. For simplicity, the following procedure creates one self-service network and a router with a gateway on the flat provider network. The router uses NAT for IPv4 network traffic and directly routes IPv6 network traffic.

Note

IPv6 connectivity with self-service networks often requires addition of static routes to nodes and physical network infrastructure.

1. Source the administrative project credentials.
2. Update the provider network to support external connectivity for self-service networks.

```
$ openstack network set --external provider1
```

Note

This command provides no output.

3. Source a regular (non-administrative) project credentials.
4. Create a self-service network.

```
$ openstack network create selfservice1
+-------------------------+--------------+
| Field                   | Value        |
+-------------------------+--------------+
| admin_state_up          | UP           |
| mtu                     | 1450         |
| name                    | selfservice1 |
| port_security_enabled   | True         |
| router:external         | Internal     |
| shared                  | False        |
| status                  | ACTIVE       |
+-------------------------+--------------+
```

Note

If you are using an MTU value on your network below 1280, please read the warning listed in the IPv6 configuration guide before creating any subnets.

5. Create a IPv4 subnet on the self-service network.

```
$ openstack subnet create --subnet-range 192.0.2.0/24 \
  --network selfservice1 --dns-nameserver 8.8.4.4 selfservice1-v4
+-------------------+---------------------------+
| Field             | Value                     |
+-------------------+---------------------------+
| allocation_pools  | 192.0.2.2-192.0.2.254     |
| cidr              | 192.0.2.0/24              |
| dns_nameservers   | 8.8.4.4                   |
| enable_dhcp       | True                      |
| gateway_ip        | 192.0.2.1                 |
| ip_version        | 4                         |
| name              | selfservice1-v4           |
+-------------------+---------------------------+
```

6. Create a IPv6 subnet on the self-service network.

```
$ openstack subnet create --subnet-range fd00:192:0:2::/64 \
  --ip-version 6 --ipv6-ra-mode slaac --ipv6-address-mode slaac \
  --network selfservice1 --dns-nameserver 2001:4860:4860::8844 \
  selfservice1-v6
+-------------------+------------------------------------------------------+
| Field             | Value                                                |
```

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```
+-------------------+-----------------------------------------------------+
| allocation_pools  | fd00:192:0:2::2-fd00:192:0:2:ffff:ffff:ffff:ffff |
| cidr              | fd00:192:0:2::/64                                   |
| dns_nameservers   | 2001:4860:4860::8844                                |
| enable_dhcp       | True                                                |
| gateway_ip        | fd00:192:0:2::1                                     |
| ip_version        | 6                                                   |
| ipv6_address_mode | slaac                                               |
| ipv6_ra_mode      | slaac                                               |
| name              | selfservice1-v6                                     |
+-------------------+-----------------------------------------------------+
```

7. Create a router.

```
$ openstack router create router1
+-----------------------+---------+
| Field                 | Value   |
+-----------------------+---------+
| admin_state_up        | UP      |
| name                  | router1 |
| status                | ACTIVE  |
+-----------------------+---------+
```

8. Add the IPv4 and IPv6 subnets as interfaces on the router.

```
$ openstack router add subnet router1 selfservice1-v4
$ openstack router add subnet router1 selfservice1-v6
```

Note

These commands provide no output.

9. Add the provider network as the gateway on the router.

```
$ openstack router set --external-gateway provider1 router1
```

Verify network operation

1. On each compute node, verify creation of a second `qdhcp` namespace.

```
# ip netns
qdhcp-8b868082-e312-4110-8627-298109d4401c
qdhcp-8fbc13ca-cfe0-4b8a-993b-e33f37ba66d1
```

2. On the network node, verify creation of the `qrouter` namespace.

```
# ip netns
qrouter-17db2a15-e024-46d0-9250-4cd4d336a2cc
```

3. Source a regular (non-administrative) project credentials.

4. Create the appropriate security group rules to allow `ping` and SSH access instances using the network.

```
$ openstack security group rule create --proto icmp default
+------------------+-----------+
| Field            | Value     |
+------------------+-----------+
| direction        | ingress   |
| ethertype        | IPv4      |
| protocol         | icmp      |
| remote_ip_prefix | 0.0.0.0/0 |
+------------------+-----------+

$ openstack security group rule create --ethertype IPv6 \
  --proto ipv6-icmp default
+-----------+-----------+
| Field     | Value     |
+-----------+-----------+
| direction | ingress   |
| ethertype | IPv6      |
| protocol  | ipv6-icmp |
+-----------+-----------+

$ openstack security group rule create --proto tcp --dst-port 22 default
+------------------+-----------+
| Field            | Value     |
+------------------+-----------+
| direction        | ingress   |
| ethertype        | IPv4      |
| port_range_max   | 22        |
| port_range_min   | 22        |
| protocol         | tcp       |
| remote_ip_prefix | 0.0.0.0/0 |
+------------------+-----------+

$ openstack security group rule create --ethertype IPv6 --proto tcp \
  --dst-port 22 default
+----------------+---------+
| Field          | Value   |
+----------------+---------+
| direction      | ingress |
| ethertype      | IPv6    |
| port_range_max | 22      |
| port_range_min | 22      |
| protocol       | tcp     |
+----------------+---------+
```

5. Launch an instance with an interface on the self-service network. For example, a CirrOS image using flavor ID 1.

```
$ openstack server create --flavor 1 --image cirros --nic net-id=NETWORK_
```

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```
↪ID selfservice-instance1
```

Replace NETWORK_ID with the ID of the self-service network.

6. Determine the IPv4 and IPv6 addresses of the instance.

```
$ openstack server list
+--------------------------------------+-----------------------+--------+-
↪------------------------------------------------------------+--------+-----
↪----+
| ID                                   | Name                  | Status |␣
↪Networks                                                    | Image  |␣
↪Flavor  |
+--------------------------------------+-----------------------+--------+-
↪------------------------------------------------------------+--------+-----
↪----+
| c055cdb0-ebb4-4d65-957c-35cbdbd59306 | selfservice-instance1 | ACTIVE |␣
↪selfservice1=192.0.2.4, fd00:192:0:2:f816:3eff:fe30:9cb0 | cirros | m1.
↪tiny |
+--------------------------------------+-----------------------+--------+-
↪------------------------------------------------------------+--------+-----
↪----+
```

Warning

The IPv4 address resides in a private IP address range (RFC1918). Thus, the Networking service performs source network address translation (SNAT) for the instance to access external networks such as the Internet. Access from external networks such as the Internet to the instance requires a floating IPv4 address. The Networking service performs destination network address translation (DNAT) from the floating IPv4 address to the instance IPv4 address on the self-service network. On the other hand, the Networking service architecture for IPv6 lacks support for NAT due to the significantly larger address space and complexity of NAT. Thus, floating IP addresses do not exist for IPv6 and the Networking service only performs routing for IPv6 subnets on self-service networks. In other words, you cannot rely on NAT to hide instances with IPv4 and IPv6 addresses or only IPv6 addresses and must properly implement security groups to restrict access.

7. On the controller node or any host with access to the provider network, ping the IPv6 address of the instance.

```
$ ping6 -c 4 fd00:192:0:2:f816:3eff:fe30:9cb0
PING fd00:192:0:2:f816:3eff:fe30:9cb0(fd00:192:0:2:f816:3eff:fe30:9cb0)␣
↪56 data bytes
64 bytes from fd00:192:0:2:f816:3eff:fe30:9cb0: icmp_seq=1 ttl=63 time=2.
↪08 ms
64 bytes from fd00:192:0:2:f816:3eff:fe30:9cb0: icmp_seq=2 ttl=63 time=1.
↪88 ms
64 bytes from fd00:192:0:2:f816:3eff:fe30:9cb0: icmp_seq=3 ttl=63 time=1.
↪55 ms
```

(continues on next page)

(continued from previous page)

```
64 bytes from fd00:192:0:2:f816:3eff:fe30:9cb0: icmp_seq=4 ttl=63 time=1.
↪62 ms

--- fd00:192:0:2:f816:3eff:fe30:9cb0 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.557/1.788/2.085/0.217 ms
```

8. Optionally, enable IPv4 access from external networks such as the Internet to the instance.

   1. Create a floating IPv4 address on the provider network.

      ```
      $ openstack floating ip create provider1
      +-------------+--------------------------------------+
      | Field       | Value                                |
      +-------------+--------------------------------------+
      | fixed_ip    | None                                 |
      | id          | 22a1b088-5c9b-43b4-97f3-970ce5df77f2 |
      | instance_id | None                                 |
      | ip          | 203.0.113.16                         |
      | pool        | provider1                            |
      +-------------+--------------------------------------+
      ```

   2. Associate the floating IPv4 address with the instance.

      ```
      $ openstack server add floating ip selfservice-instance1 203.0.113.16
      ```

      Note

      This command provides no output.

   3. On the controller node or any host with access to the provider network, `ping` the floating IPv4 address of the instance.

      ```
      $ ping -c 4 203.0.113.16
      PING 203.0.113.16 (203.0.113.16) 56(84) bytes of data.
      64 bytes from 203.0.113.16: icmp_seq=1 ttl=63 time=3.41 ms
      64 bytes from 203.0.113.16: icmp_seq=2 ttl=63 time=1.67 ms
      64 bytes from 203.0.113.16: icmp_seq=3 ttl=63 time=1.47 ms
      64 bytes from 203.0.113.16: icmp_seq=4 ttl=63 time=1.59 ms

      --- 203.0.113.16 ping statistics ---
      4 packets transmitted, 4 received, 0% packet loss, time 3005ms
      rtt min/avg/max/mdev = 1.473/2.040/3.414/0.798 ms
      ```

9. Obtain access to the instance.

10. Test IPv4 and IPv6 connectivity to the Internet or other external network.

Network traffic flow

The following sections describe the flow of network traffic in several common scenarios. *North-south* network traffic travels between an instance and external network such as the Internet. *East-west* network traffic travels between instances on the same or different networks. In all scenarios, the physical network infrastructure handles switching and routing among provider networks and external networks such as the Internet. Each case references one or more of the following components:

- Provider network (VLAN)
  - VLAN ID 101 (tagged)
- Self-service network 1 (VXLAN)
  - VXLAN ID (VNI) 101
- Self-service network 2 (VXLAN)
  - VXLAN ID (VNI) 102
- Self-service router
  - Gateway on the provider network
  - Interface on self-service network 1
  - Interface on self-service network 2
- Instance 1
- Instance 2

North-south scenario 1: Instance with a fixed IP address

For instances with a fixed IPv4 address, the network node performs SNAT on north-south traffic passing from self-service to external networks such as the Internet. For instances with a fixed IPv6 address, the network node performs conventional routing of traffic between self-service and external networks.

- The instance resides on compute node 1 and uses self-service network 1.
- The instance sends a packet to a host on the Internet.

The following steps involve compute node 1:

1. The instance interface (1) forwards the packet to the security group bridge instance port (2) via `veth` pair.
2. Security group rules (3) on the security group bridge handle firewalling and connection tracking for the packet.
3. The security group bridge OVS port (4) forwards the packet to the OVS integration bridge security group port (5) via `veth` pair.
4. The OVS integration bridge adds an internal VLAN tag to the packet.
5. The OVS integration bridge exchanges the internal VLAN tag for an internal tunnel ID.
6. The OVS integration bridge patch port (6) forwards the packet to the OVS tunnel bridge patch port (7).
7. The OVS tunnel bridge (8) wraps the packet using VNI 101.

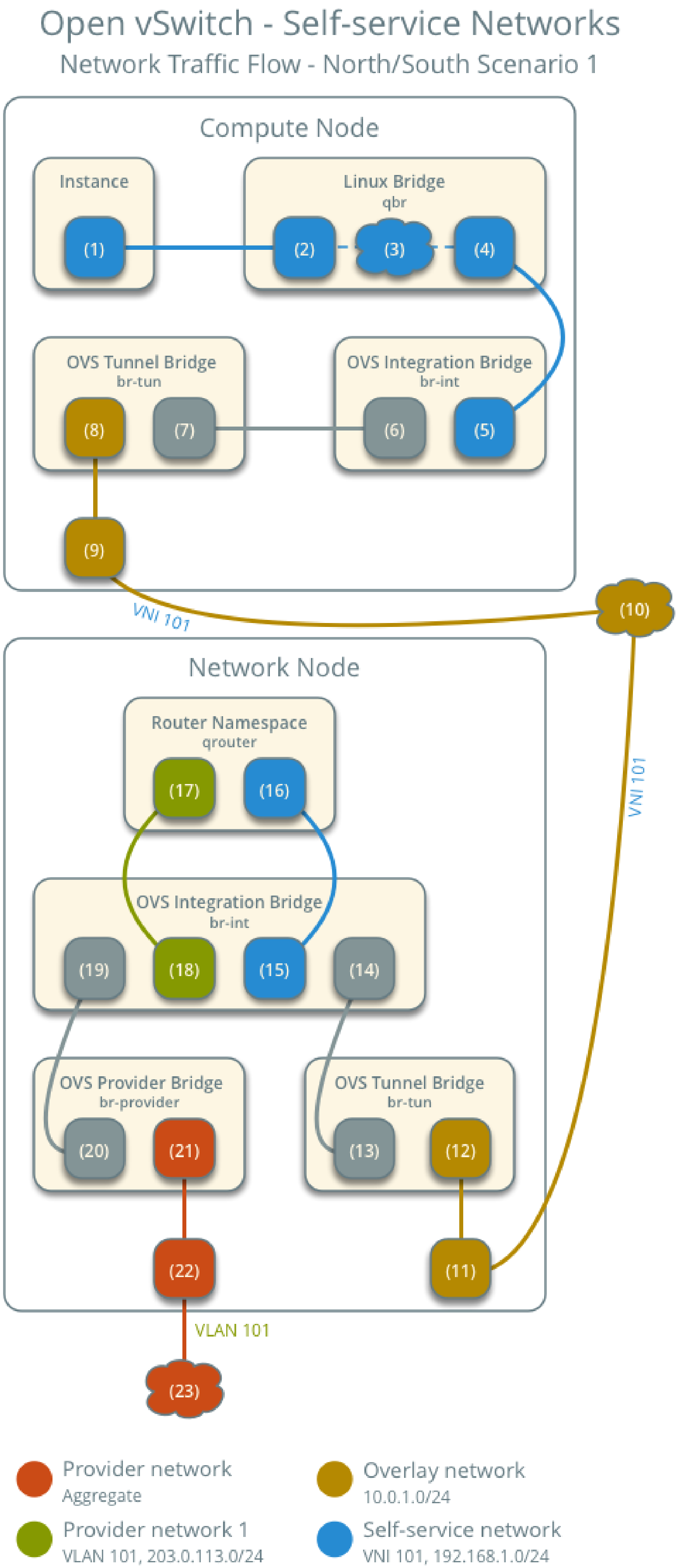
8. The underlying physical interface (9) for overlay networks forwards the packet to the network node via the overlay network (10).

The following steps involve the network node:

1. The underlying physical interface (11) for overlay networks forwards the packet to the OVS tunnel bridge (12).
2. The OVS tunnel bridge unwraps the packet and adds an internal tunnel ID to it.
3. The OVS tunnel bridge exchanges the internal tunnel ID for an internal VLAN tag.
4. The OVS tunnel bridge patch port (13) forwards the packet to the OVS integration bridge patch port (14).
5. The OVS integration bridge port for the self-service network (15) removes the internal VLAN tag and forwards the packet to the self-service network interface (16) in the router namespace.
   - For IPv4, the router performs SNAT on the packet which changes the source IP address to the router IP address on the provider network and sends it to the gateway IP address on the provider network via the gateway interface on the provider network (17).
   - For IPv6, the router sends the packet to the next-hop IP address, typically the gateway IP address on the provider network, via the provider gateway interface (17).
6. The router forwards the packet to the OVS integration bridge port for the provider network (18).
7. The OVS integration bridge adds the internal VLAN tag to the packet.
8. The OVS integration bridge `int-br-provider` patch port (19) forwards the packet to the OVS provider bridge `phy-br-provider` patch port (20).
9. The OVS provider bridge swaps the internal VLAN tag with actual VLAN tag 101.
10. The OVS provider bridge provider network port (21) forwards the packet to the physical network interface (22).
11. The physical network interface forwards the packet to the Internet via physical network infrastructure (23).

Note

Return traffic follows similar steps in reverse. However, without a floating IPv4 address, hosts on the provider or external networks cannot originate connections to instances on the self-service network.


Open vSwitch - Self-service Networks
Network Traffic Flow - North/South Scenario 1
Compute Node
Instance
(1)
Linux Bridge
qbr
(2)
(3)
(4)
OVS Tunnel Bridge
br-tun
(8)
(7)
OVS Integration Bridge
br-int
(6)
(5)
(9)
VNI 101
(10)
Network Node
Router Namespace
qrouter
(17)
(16)
OVS Integration Bridge
br-int
(19)
(18)
(15)
(14)
VNI 101
OVS Provider Bridge
br-provider
(20)
(21)
OVS Tunnel Bridge
br-tun
(13)
(12)
(22)
(11)
VLAN 101
(23)
Provider network
Aggregate
Overlay network
10.0.1.0/24
Provider network 1
VLAN 101, 203.0.113.0/24
Self-service network
VNI 101, 192.168.1.0/24

North-south scenario 2: Instance with a floating IPv4 address

For instances with a floating IPv4 address, the network node performs SNAT on north-south traffic passing from the instance to external networks such as the Internet and DNAT on north-south traffic passing from external networks to the instance. Floating IP addresses and NAT do not apply to IPv6. Thus, the network node routes IPv6 traffic in this scenario.

- The instance resides on compute node 1 and uses self-service network 1.
- A host on the Internet sends a packet to the instance.

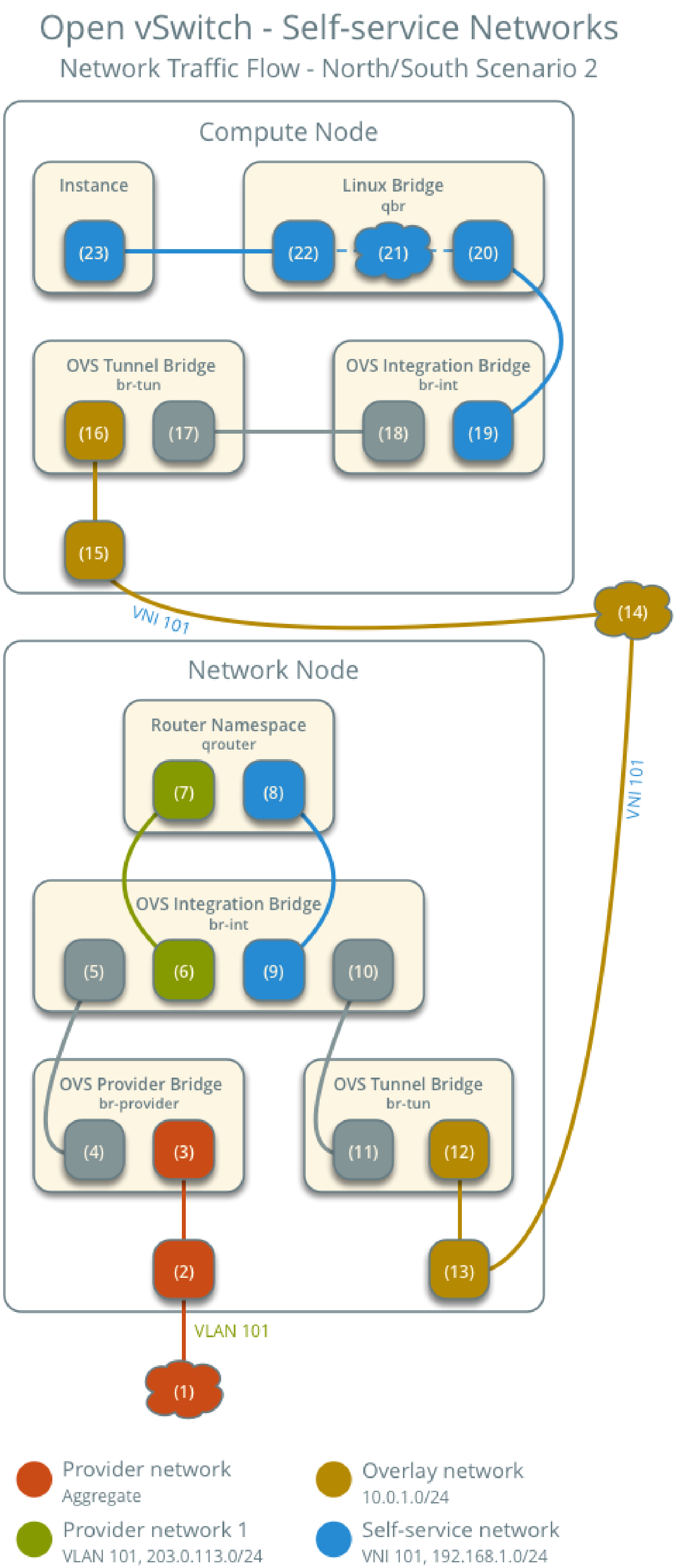
The following steps involve the network node:

1. The physical network infrastructure (1) forwards the packet to the provider physical network interface (2).
2. The provider physical network interface forwards the packet to the OVS provider bridge provider network port (3).
3. The OVS provider bridge swaps actual VLAN tag 101 with the internal VLAN tag.
4. The OVS provider bridge `phy-br-provider` port (4) forwards the packet to the OVS integration bridge `int-br-provider` port (5).
5. The OVS integration bridge port for the provider network (6) removes the internal VLAN tag and forwards the packet to the provider network interface (6) in the router namespace.
   - For IPv4, the router performs DNAT on the packet which changes the destination IP address to the instance IP address on the self-service network and sends it to the gateway IP address on the self-service network via the self-service interface (7).
   - For IPv6, the router sends the packet to the next-hop IP address, typically the gateway IP address on the self-service network, via the self-service interface (8).
6. The router forwards the packet to the OVS integration bridge port for the self-service network (9).
7. The OVS integration bridge adds an internal VLAN tag to the packet.
8. The OVS integration bridge exchanges the internal VLAN tag for an internal tunnel ID.
9. The OVS integration bridge `patch-tun` patch port (10) forwards the packet to the OVS tunnel bridge `patch-int` patch port (11).
10. The OVS tunnel bridge (12) wraps the packet using VNI 101.
11. The underlying physical interface (13) for overlay networks forwards the packet to the network node via the overlay network (14).

The following steps involve the compute node:

1. The underlying physical interface (15) for overlay networks forwards the packet to the OVS tunnel bridge (16).
2. The OVS tunnel bridge unwraps the packet and adds an internal tunnel ID to it.
3. The OVS tunnel bridge exchanges the internal tunnel ID for an internal VLAN tag.
4. The OVS tunnel bridge `patch-int` patch port (17) forwards the packet to the OVS integration bridge `patch-tun` patch port (18).
5. The OVS integration bridge removes the internal VLAN tag from the packet.

6. The OVS integration bridge security group port (19) forwards the packet to the security group bridge OVS port (20) via `veth` pair.
7. Security group rules (21) on the security group bridge handle firewalling and connection tracking for the packet.
8. The security group bridge instance port (22) forwards the packet to the instance interface (23) via `veth` pair.


Open vSwitch - Self-service Networks
Network Traffic Flow - North/South Scenario 2
Compute Node
Instance
(23)
Linux Bridge
qbr
(22)
(21)
(20)
OVS Tunnel Bridge
br-tun
(16)
(17)
OVS Integration Bridge
br-int
(18)
(19)
(15)
VNI 101
(14)
Network Node
Router Namespace
qrouter
(7)
(8)
VNI 101
OVS Integration Bridge
br-int
(5)
(6)
(9)
(10)
OVS Provider Bridge
br-provider
(4)
(3)
OVS Tunnel Bridge
br-tun
(11)
(12)
(2)
(13)
VLAN 101
(1)
Provider network
Aggregate
Overlay network
10.0.1.0/24
Provider network 1
VLAN 101, 203.0.113.0/24
Self-service network
VNI 101, 192.168.1.0/24

Note

Egress instance traffic flows similar to north-south scenario 1, except SNAT changes the source IP address of the packet to the floating IPv4 address rather than the router IP address on the provider network.

East-west scenario 1: Instances on the same network

Instances with a fixed IPv4/IPv6 address or floating IPv4 address on the same network communicate directly between compute nodes containing those instances.

By default, the VXLAN protocol lacks knowledge of target location and uses multicast to discover it. After discovery, it stores the location in the local forwarding database. In large deployments, the discovery process can generate a significant amount of network that all nodes must process. To eliminate the latter and generally increase efficiency, the Networking service includes the layer-2 population mechanism driver that automatically populates the forwarding database for VXLAN interfaces. The example configuration enables this driver. For more information, see *ML2 Plug-in*.

- Instance 1 resides on compute node 1 and uses self-service network 1.
- Instance 2 resides on compute node 2 and uses self-service network 1.
- Instance 1 sends a packet to instance 2.

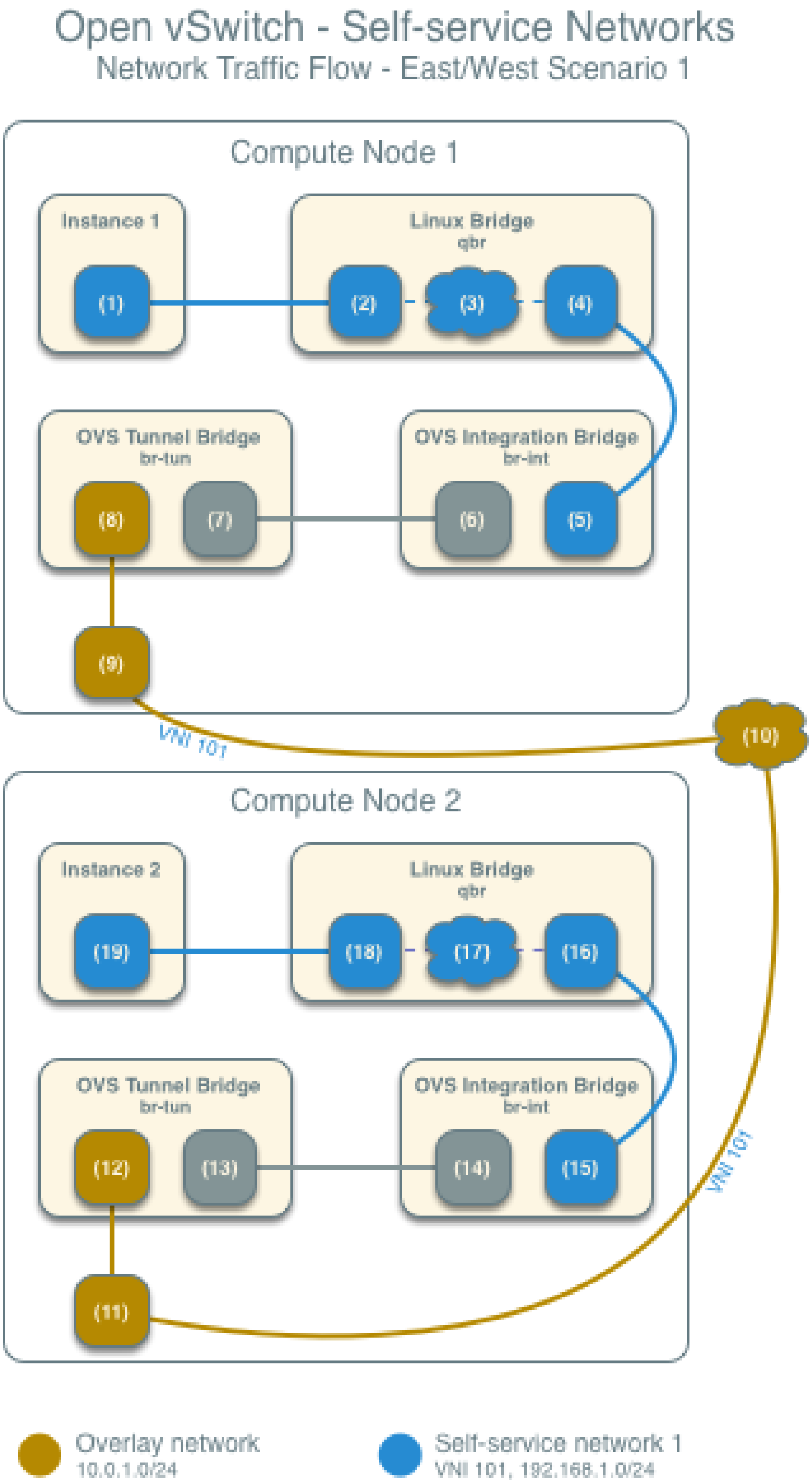
The following steps involve compute node 1:

1. The instance 1 interface (1) forwards the packet to the security group bridge instance port (2) via `veth` pair.
2. Security group rules (3) on the security group bridge handle firewalling and connection tracking for the packet.
3. The security group bridge OVS port (4) forwards the packet to the OVS integration bridge security group port (5) via `veth` pair.
4. The OVS integration bridge adds an internal VLAN tag to the packet.
5. The OVS integration bridge exchanges the internal VLAN tag for an internal tunnel ID.
6. The OVS integration bridge patch port (6) forwards the packet to the OVS tunnel bridge patch port (7).
7. The OVS tunnel bridge (8) wraps the packet using VNI 101.
8. The underlying physical interface (9) for overlay networks forwards the packet to compute node 2 via the overlay network (10).

The following steps involve compute node 2:

1. The underlying physical interface (11) for overlay networks forwards the packet to the OVS tunnel bridge (12).
2. The OVS tunnel bridge unwraps the packet and adds an internal tunnel ID to it.
3. The OVS tunnel bridge exchanges the internal tunnel ID for an internal VLAN tag.
4. The OVS tunnel bridge `patch-int` patch port (13) forwards the packet to the OVS integration bridge `patch-tun` patch port (14).
5. The OVS integration bridge removes the internal VLAN tag from the packet.

6. The OVS integration bridge security group port (15) forwards the packet to the security group bridge OVS port (16) via `veth` pair.
7. Security group rules (17) on the security group bridge handle firewalling and connection tracking for the packet.
8. The security group bridge instance port (18) forwards the packet to the instance 2 interface (19) via `veth` pair.



Note

Return traffic follows similar steps in reverse.

East-west scenario 2: Instances on different networks

Instances using a fixed IPv4/IPv6 address or floating IPv4 address communicate via router on the network node. The self-service networks must reside on the same router.

- Instance 1 resides on compute node 1 and uses self-service network 1.
- Instance 2 resides on compute node 1 and uses self-service network 2.
- Instance 1 sends a packet to instance 2.

Note

Both instances reside on the same compute node to illustrate how VXLAN enables multiple overlays to use the same layer-3 network.

The following steps involve the compute node:

1. The instance interface (1) forwards the packet to the security group bridge instance port (2) via `veth` pair.
2. Security group rules (3) on the security group bridge handle firewalling and connection tracking for the packet.
3. The security group bridge OVS port (4) forwards the packet to the OVS integration bridge security group port (5) via `veth` pair.
4. The OVS integration bridge adds an internal VLAN tag to the packet.
5. The OVS integration bridge exchanges the internal VLAN tag for an internal tunnel ID.
6. The OVS integration bridge `patch-tun` patch port (6) forwards the packet to the OVS tunnel bridge `patch-int` patch port (7).
7. The OVS tunnel bridge (8) wraps the packet using VNI 101.
8. The underlying physical interface (9) for overlay networks forwards the packet to the network node via the overlay network (10).

The following steps involve the network node:

1. The underlying physical interface (11) for overlay networks forwards the packet to the OVS tunnel bridge (12).
2. The OVS tunnel bridge unwraps the packet and adds an internal tunnel ID to it.
3. The OVS tunnel bridge exchanges the internal tunnel ID for an internal VLAN tag.
4. The OVS tunnel bridge `patch-int` patch port (13) forwards the packet to the OVS integration bridge `patch-tun` patch port (14).
5. The OVS integration bridge port for self-service network 1 (15) removes the internal VLAN tag and forwards the packet to the self-service network 1 interface (16) in the router namespace.
6. The router sends the packet to the next-hop IP address, typically the gateway IP address on self-service network 2, via the self-service network 2 interface (17).
7. The router forwards the packet to the OVS integration bridge port for self-service network 2 (18).
8. The OVS integration bridge adds the internal VLAN tag to the packet.

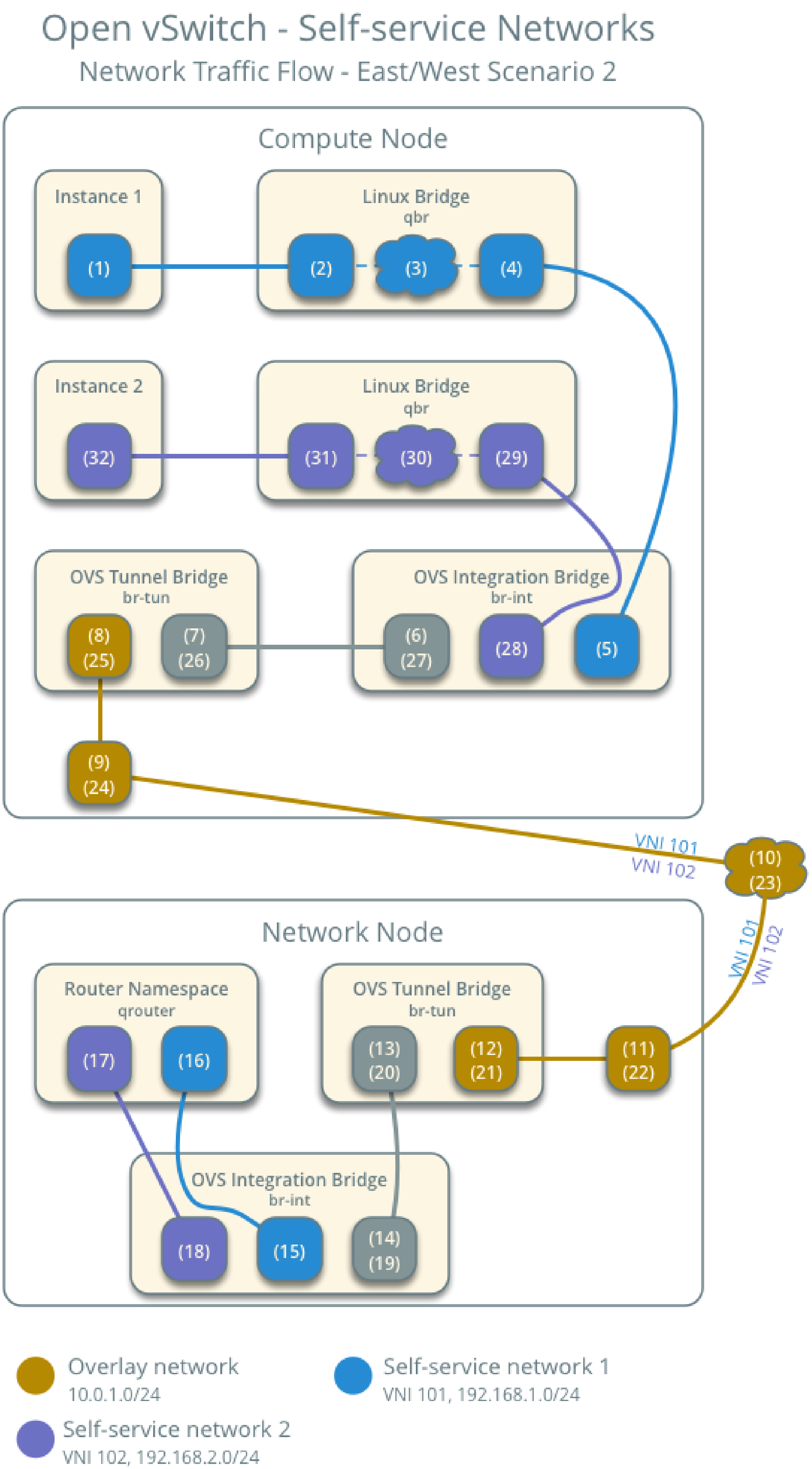
9. The OVS integration bridge exchanges the internal VLAN tag for an internal tunnel ID.
10. The OVS integration bridge `patch-tun` patch port (19) forwards the packet to the OVS tunnel bridge `patch-int` patch port (20).
11. The OVS tunnel bridge (21) wraps the packet using VNI 102.
12. The underlying physical interface (22) for overlay networks forwards the packet to the compute node via the overlay network (23).

The following steps involve the compute node:

1. The underlying physical interface (24) for overlay networks forwards the packet to the OVS tunnel bridge (25).
2. The OVS tunnel bridge unwraps the packet and adds an internal tunnel ID to it.
3. The OVS tunnel bridge exchanges the internal tunnel ID for an internal VLAN tag.
4. The OVS tunnel bridge `patch-int` patch port (26) forwards the packet to the OVS integration bridge `patch-tun` patch port (27).
5. The OVS integration bridge removes the internal VLAN tag from the packet.
6. The OVS integration bridge security group port (28) forwards the packet to the security group bridge OVS port (29) via `veth` pair.
7. Security group rules (30) on the security group bridge handle firewalling and connection tracking for the packet.
8. The security group bridge instance port (31) forwards the packet to the instance interface (32) via `veth` pair.

Note

Return traffic follows similar steps in reverse.


Open vSwitch - Self-service Networks
Network Traffic Flow - East/West Scenario 2
Compute Node
Instance 1
(1)
Linux Bridge
qbr
(2)
(3)
(4)
Instance 2
(32)
Linux Bridge
qbr
(31)
(30)
(29)
OVS Tunnel Bridge
br-tun
(8)
(25)
(7)
(26)
OVS Integration Bridge
br-int
(6)
(27)
(28)
(5)
(9)
(24)
VNI 101
VNI 102
(10)
(23)
Network Node
Router Namespace
qrouter
(17)
(16)
OVS Tunnel Bridge
br-tun
(13)
(20)
(12)
(21)
(11)
(22)
VNI 101
VNI 102
OVS Integration Bridge
br-int
(18)
(15)
(14)
(19)
Overlay network
10.0.1.0/24
Self-service network 1
VNI 101, 192.168.1.0/24
Self-service network 2
VNI 102, 192.168.2.0/24

Open vSwitch: High availability using VRRP

This architecture example augments the self-service deployment example with a high-availability mechanism using the Virtual Router Redundancy Protocol (VRRP) via `keepalived` and provides failover of routing for self-service networks. It requires a minimum of two network nodes because VRRP creates one master (active) instance and at least one backup instance of each router.

During normal operation, `keepalived` on the master router periodically transmits *heartbeat* packets over a hidden network that connects all VRRP routers for a particular project. Each project with VRRP routers uses a separate hidden network. By default this network uses the first value in the `project_network_types` option in the `ml2_conf.ini` file. For additional control, you can specify the self-service network type and physical network name for the hidden network using the `l3_ha_network_type` and `l3_ha_network_name` options in the `neutron.conf` file.

If `keepalived` on the backup router stops receiving *heartbeat* packets, it assumes failure of the master router and promotes the backup router to master router by configuring IP addresses on the interfaces in the `qrouter` namespace. In environments with more than one backup router, `keepalived` on the backup router with the next highest priority promotes that backup router to master router.

Note

This high-availability mechanism configures VRRP using the same priority for all routers. Therefore, VRRP promotes the backup router with the highest IP address to the master router.

Warning

There is a known bug with `keepalived` v1.2.15 and earlier which can cause packet loss when `max_l3_agents_per_router` is set to 3 or more. Therefore, we recommend that you upgrade to `keepalived` v1.2.16 or greater when using this feature.

Interruption of VRRP *heartbeat* traffic between network nodes, typically due to a network interface or physical network infrastructure failure, triggers a failover. Restarting the layer-3 agent, or failure of it, does not trigger a failover providing `keepalived` continues to operate.

Consider the following attributes of this high-availability mechanism to determine practicality in your environment:

- Instance network traffic on self-service networks using a particular router only traverses the master instance of that router. Thus, resource limitations of a particular network node can impact all master instances of routers on that network node without triggering failover to another network node. However, you can configure the scheduler to distribute the master instance of each router uniformly across a pool of network nodes to reduce the chance of resource contention on any particular network node.
- Only supports self-service networks using a router. Provider networks operate at layer-2 and rely on physical network infrastructure for redundancy.
- For instances with a floating IPv4 address, maintains state of network connections during failover as a side effect of 1:1 static NAT. The mechanism does not actually implement connection tracking.

For production deployments, we recommend at least three network nodes with sufficient resources to handle network traffic for the entire environment if one network node fails. Also, the remaining two nodes can continue to provide redundancy.

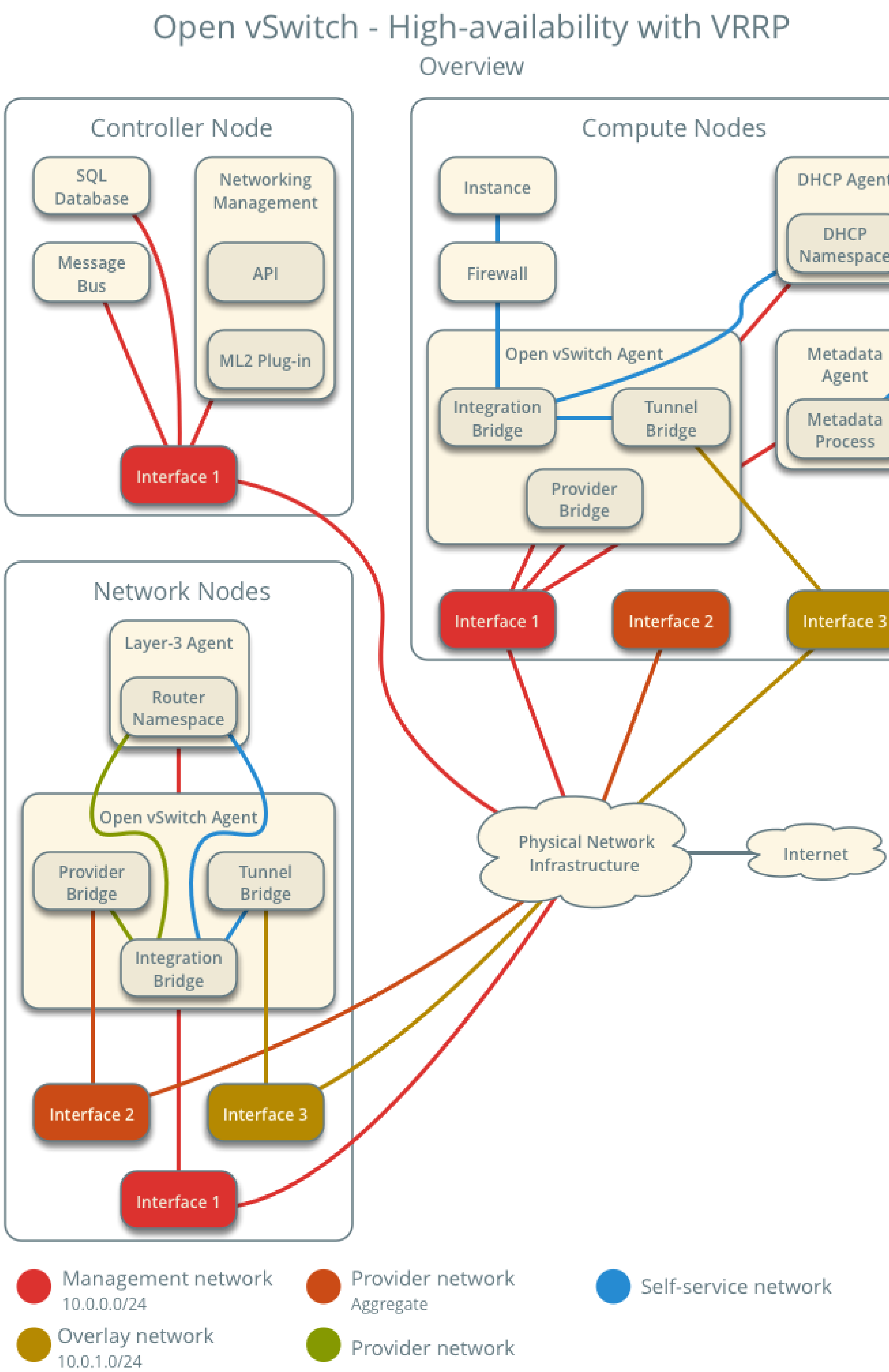
Prerequisites

Add one network node with the following components:

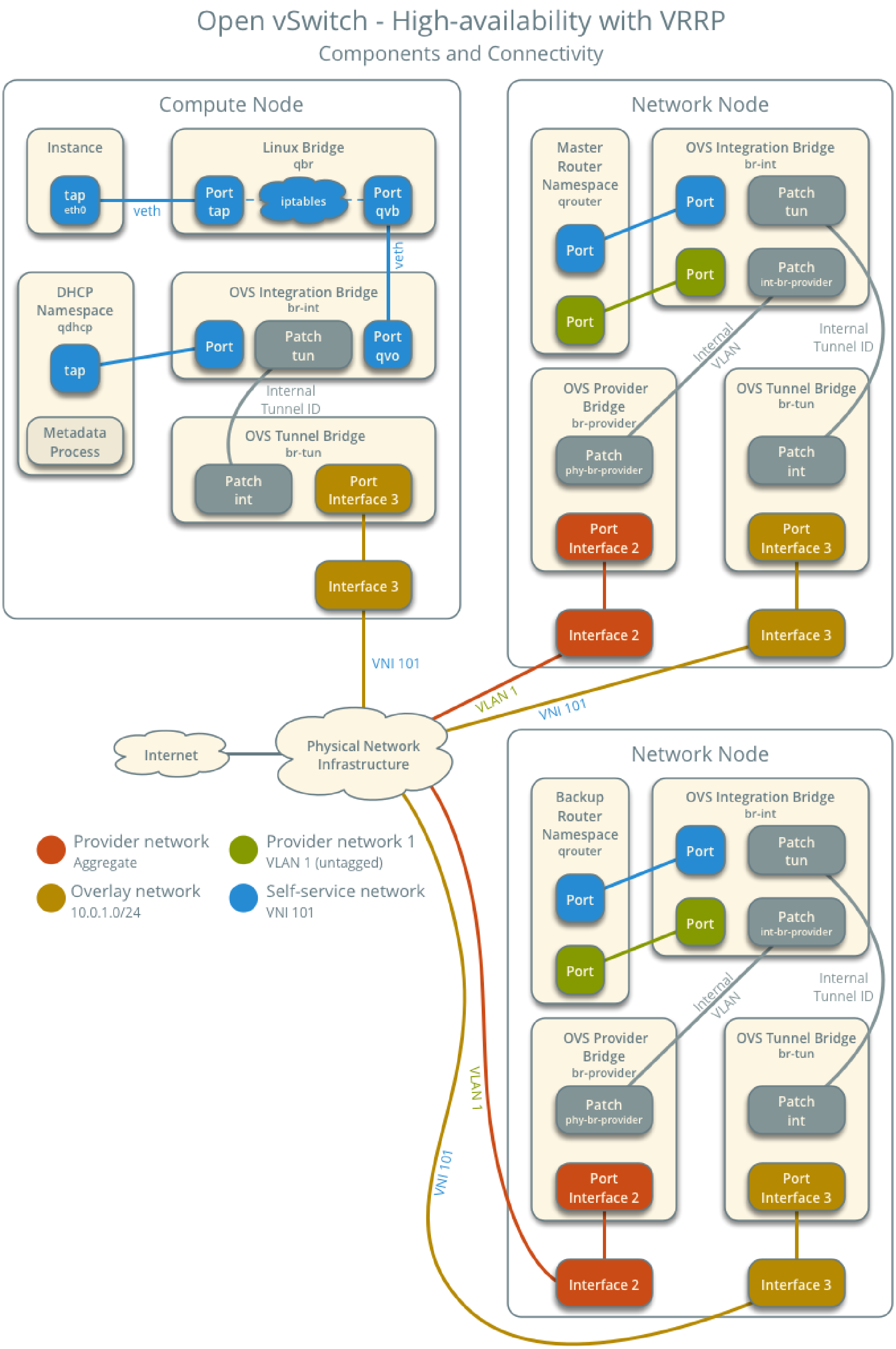
- Three network interfaces: management, provider, and overlay.
- OpenStack Networking layer-2 agent, layer-3 agent, and any dependencies.

> **Note**
>
> You can keep the DHCP and metadata agents on each compute node or move them to the network nodes.

Architecture



The following figure shows components and connectivity for one self-service network and one untagged (flat) network. The primary router resides on network node 1. In this particular case, the instance resides on the same compute node as the DHCP agent for the network. If the DHCP agent resides on another compute node, the latter only contains a DHCP namespace and Linux bridge with a port on the overlay physical network interface.


Open vSwitch - High-availability with VRRP
Components and Connectivity
Compute Node
Instance
tap
eth0
veth
Linux Bridge
qbr
Port
tap
iptables
Port
qvb
veth
DHCP
Namespace
qdhcp
tap
Metadata
Process
OVS Integration Bridge
br-int
Port
Patch
tun
Port
qvo
Internal
Tunnel ID
OVS Tunnel Bridge
br-tun
Patch
int
Port
Interface 3
Interface 3
VNI 101
Network Node
Master
Router
Namespace
qrouter
Port
Port
OVS Integration Bridge
br-int
Port
Patch
tun
Port
Patch
int-br-provider
Internal
VLAN
Internal
Tunnel ID
OVS Provider
Bridge
br-provider
Patch
phy-br-provider
Port
Interface 2
OVS Tunnel Bridge
br-tun
Patch
int
Port
Interface 3
Interface 2
Interface 3
VLAN 1
VNI 101
Internet
Physical Network
Infrastructure
Network Node
Backup
Router
Namespace
qrouter
Port
Port
OVS Integration Bridge
br-int
Port
Patch
tun
Port
Patch
int-br-provider
Internal
VLAN
Internal
Tunnel ID
OVS Provider
Bridge
br-provider
Patch
phy-br-provider
Port
Interface 2
OVS Tunnel Bridge
br-tun
Patch
int
Port
Interface 3
Interface 2
Interface 3
VLAN 1
VNI 101
Provider network
Aggregate
Provider network 1
VLAN 1 (untagged)
Overlay network
10.0.1.0/24
Self-service network
VNI 101

Example configuration

Use the following example configuration as a template to add support for high-availability using VRRP to an existing operational environment that supports self-service networks.

Controller node

1. In the neutron.conf file:

   - Enable VRRP.

   ```
   [DEFAULT]
   l3_ha = True
   ```

2. Restart the following services:

   - Server

Network node 1

No changes.

Network node 2

1. Install the Networking service OVS layer-2 agent and layer-3 agent.
2. Install OVS.
3. In the neutron.conf file, configure common options:

   ```
   [DEFAULT]
   core_plugin = ml2
   auth_strategy = keystone

   [database]
   # ...

   [keystone_authtoken]
   # ...

   [nova]
   # ...

   [agent]
   # ...
   ```

   See the Installation Tutorials and Guides and Configuration Reference for your OpenStack release to obtain the appropriate additional configuration for the [DEFAULT], [database], [keystone_authtoken], [nova], and [agent] sections.

4. Start the following services:

   - OVS

5. Create the OVS provider bridge br-provider:

```
$ ovs-vsctl add-br br-provider
```

6. Add the provider network interface as a port on the OVS provider bridge `br-provider`:

```
$ ovs-vsctl add-port br-provider PROVIDER_INTERFACE
```

Replace `PROVIDER_INTERFACE` with the name of the underlying interface that handles provider networks. For example, `eth1`.

7. In the `openvswitch_agent.ini` file, configure the layer-2 agent.

```
[ovs]
bridge_mappings = provider:br-provider
local_ip = OVERLAY_INTERFACE_IP_ADDRESS

[agent]
tunnel_types = vxlan
l2_population = true

[securitygroup]
firewall_driver = iptables_hybrid
```

Replace `OVERLAY_INTERFACE_IP_ADDRESS` with the IP address of the interface that handles VXLAN overlays for self-service networks.

8. Start the following services:

   - Open vSwitch agent
   - Layer-3 agent

Compute nodes

No changes.

Verify service operation

1. Source the administrative project credentials.
2. Verify presence and operation of the agents.

```
$ openstack network agent list
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host     |␣
↪Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
| 1236bbcb-e0ba-48a9-80fc-81202ca4fa51 | Metadata agent     | compute2 |␣
↪None              | True  | UP    | neutron-metadata-agent    |
| 457d6898-b373-4bb3-b41f-59345dcfb5c5 | Open vSwitch agent | compute2 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
| 71f15e84-bc47-4c2a-b9fb-317840b2d753 | DHCP agent         | compute2 |␣
```

(continues on next page)

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```
↪nova                  | True   | UP     | neutron-dhcp-agent        |
| 8805b962-de95-4e40-bdc2-7a0add7521e8 | L3 agent           | network1 |␣
↪nova                  | True   | UP     | neutron-l3-agent          |
| a33cac5a-0266-48f6-9cac-4cef4f8b0358 | Open vSwitch agent | network1 |␣
↪None                  | True   | UP     | neutron-openvswitch-agent |
| a6c69690-e7f7-4e56-9831-1282753e5007 | Metadata agent     | compute1 |␣
↪None                  | True   | UP     | neutron-metadata-agent    |
| af11f22f-a9f4-404f-9fd8-cd7ad55c0f68 | DHCP agent         | compute1 |␣
↪nova                  | True   | UP     | neutron-dhcp-agent        |
| bcfc977b-ec0e-4ba9-be62-9489b4b0e6f1 | Open vSwitch agent | compute1 |␣
↪None                  | True   | UP     | neutron-openvswitch-agent |
| 7f00d759-f2c9-494a-9fbf-fd9118104d03 | Open vSwitch agent | network2 |␣
↪None                  | True   | UP     | neutron-openvswitch-agent |
| b28d8818-9e32-4888-930b-29addbdd2ef9 | L3 agent           | network2 |␣
↪nova                  | True   | UP     | neutron-l3-agent          |
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
```

Create initial networks

Similar to the self-service deployment example, this configuration supports multiple VXLAN self-service networks. After enabling high-availability, all additional routers use VRRP. The following procedure creates an additional self-service network and router. The Networking service also supports adding high-availability to existing routers. However, the procedure requires administratively disabling and enabling each router which temporarily interrupts network connectivity for self-service networks with interfaces on that router.

1. Source a regular (non-administrative) project credentials.
2. Create a self-service network.

```
$ openstack network create selfservice2
+-------------------------+--------------+
| Field                   | Value        |
+-------------------------+--------------+
| admin_state_up          | UP           |
| mtu                     | 1450         |
| name                    | selfservice2 |
| port_security_enabled   | True         |
| router:external         | Internal     |
| shared                  | False        |
| status                  | ACTIVE       |
+-------------------------+--------------+
```

3. Create a IPv4 subnet on the self-service network.

```
$ openstack subnet create --subnet-range 198.51.100.0/24 \
  --network selfservice2 --dns-nameserver 8.8.4.4 selfservice2-v4
+-------------------+------------------------------+
| Field             | Value                        |
```

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```
+-------------------+-------------------------------+
| allocation_pools  | 198.51.100.2-198.51.100.254   |
| cidr              | 198.51.100.0/24               |
| dns_nameservers   | 8.8.4.4                       |
| enable_dhcp       | True                          |
| gateway_ip        | 198.51.100.1                  |
| ip_version        | 4                             |
| name              | selfservice2-v4               |
+-------------------+-------------------------------+
```

4. Create a IPv6 subnet on the self-service network.

```
$ openstack subnet create --subnet-range fd00:198:51:100::/64 --ip-
↪version 6 \
  --ipv6-ra-mode slaac --ipv6-address-mode slaac --network selfservice2 \
  --dns-nameserver 2001:4860:4860::8844 selfservice2-v6
+-------------------+------------------------------------------------------
↪---+
| Field             | Value                                                ␣
↪   |
+-------------------+------------------------------------------------------
↪---+
| allocation_pools  | fd00:198:51:100::2-
↪fd00:198:51:100:ffff:ffff:ffff:ffff |
| cidr              | fd00:198:51:100::/64                                 ␣
↪   |
| dns_nameservers   | 2001:4860:4860::8844                                 ␣
↪   |
| enable_dhcp       | True                                                 ␣
↪   |
| gateway_ip        | fd00:198:51:100::1                                   ␣
↪   |
| ip_version        | 6                                                    ␣
↪   |
| ipv6_address_mode | slaac                                                ␣
↪   |
| ipv6_ra_mode      | slaac                                                ␣
↪   |
| name              | selfservice2-v6                                      ␣
↪   |
+-------------------+------------------------------------------------------
↪---+
```

5. Create a router.

```
$ openstack router create router2
+-----------------------+---------+
| Field                 | Value   |
+-----------------------+---------+
| admin_state_up        | UP      |
```

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```
| name                    | router2 |
| status                  | ACTIVE  |
+-------------------------+---------+
```

6. Add the IPv4 and IPv6 subnets as interfaces on the router.

```
$ openstack router add subnet router2 selfservice2-v4
$ openstack router add subnet router2 selfservice2-v6
```

Note

These commands provide no output.

7. Add the provider network as a gateway on the router.

```
$ openstack router set --external-gateway provider1 router2
```

Verify network operation

1. Source the administrative project credentials.
2. Verify creation of the internal high-availability network that handles VRRP *heartbeat* traffic.

```
$ openstack network list
+--------------------------------------+----------------------------------
↪-----------------+--------------------------------------+
| ID                                   | Name                             ␣
↪                 | Subnets                              |
+--------------------------------------+----------------------------------
↪-----------------+--------------------------------------+
| 1b8519c1-59c4-415c-9da2-a67d53c68455 | HA network tenant␣
↪f986edf55ae945e2bef3cb4bfd589928 | 6843314a-1e76-4cc9-94f5-c64b7a39364a␣
↪|
+--------------------------------------+----------------------------------
↪-----------------+--------------------------------------+
```

3. On each network node, verify creation of a `qrouter` namespace with the same ID.

   Network node 1:

```
# ip netns
qrouter-b6206312-878e-497c-8ef7-eb384f8add96
```

   Network node 2:

```
# ip netns
qrouter-b6206312-878e-497c-8ef7-eb384f8add96
```

Note

The namespace for router 1 from *Open vSwitch: Self-service networks* should only appear on network node 1 because of creation prior to enabling VRRP.

4. On each network node, show the IP address of interfaces in the `qrouter` namespace. With the exception of the VRRP interface, only one namespace belonging to the master router instance contains IP addresses on the interfaces.

   Network node 1:

```
# ip netns exec qrouter-b6206312-878e-497c-8ef7-eb384f8add96 ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group␣
↪default qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ha-eb820380-40@if21: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc␣
↪noqueue state UP group default qlen 1000
    link/ether fa:16:3e:78:ba:99 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 169.254.192.1/18 brd 169.254.255.255 scope global ha-eb820380-40
       valid_lft forever preferred_lft forever
    inet 169.254.0.1/24 scope global ha-eb820380-40
       valid_lft forever preferred_lft forever
    inet6 fe80::f816:3eff:fe78:ba99/64 scope link
       valid_lft forever preferred_lft forever
3: qr-da3504ad-ba@if24: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc␣
↪noqueue state UP group default qlen 1000
    link/ether fa:16:3e:dc:8e:a8 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 198.51.100.1/24 scope global qr-da3504ad-ba
       valid_lft forever preferred_lft forever
    inet6 fe80::f816:3eff:fedc:8ea8/64 scope link
       valid_lft forever preferred_lft forever
4: qr-442e36eb-fc@if27: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc␣
↪noqueue state UP group default qlen 1000
    link/ether fa:16:3e:ee:c8:41 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fd00:198:51:100::1/64 scope global nodad
       valid_lft forever preferred_lft forever
    inet6 fe80::f816:3eff:feee:c841/64 scope link
       valid_lft forever preferred_lft forever
5: qg-33fedbc5-43@if28: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc␣
↪noqueue state UP group default qlen 1000
    link/ether fa:16:3e:03:1a:f6 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 203.0.113.21/24 scope global qg-33fedbc5-43
       valid_lft forever preferred_lft forever
    inet6 fd00:203:0:113::21/64 scope global nodad
       valid_lft forever preferred_lft forever
    inet6 fe80::f816:3eff:fe03:1af6/64 scope link
       valid_lft forever preferred_lft forever
```

Network node 2:

```
# ip netns exec qrouter-b6206312-878e-497c-8ef7-eb384f8add96 ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group␣
↪default qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ha-7a7ce184-36@if8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc␣
↪noqueue state UP group default qlen 1000
    link/ether fa:16:3e:16:59:84 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 169.254.192.2/18 brd 169.254.255.255 scope global ha-7a7ce184-36
       valid_lft forever preferred_lft forever
    inet6 fe80::f816:3eff:fe16:5984/64 scope link
       valid_lft forever preferred_lft forever
3: qr-da3504ad-ba@if11: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc␣
↪noqueue state UP group default qlen 1000
    link/ether fa:16:3e:dc:8e:a8 brd ff:ff:ff:ff:ff:ff link-netnsid 0
4: qr-442e36eb-fc@if14: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc␣
↪noqueue state UP group default qlen 1000
5: qg-33fedbc5-43@if15: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc␣
↪noqueue state UP group default qlen 1000
    link/ether fa:16:3e:03:1a:f6 brd ff:ff:ff:ff:ff:ff link-netnsid 0
```

Note

The master router may reside on network node 2.

5. Launch an instance with an interface on the additional self-service network. For example, a CirrOS image using flavor ID 1.

```
$ openstack server create --flavor 1 --image cirros --nic net-id=NETWORK_
↪ID selfservice-instance2
```

Replace NETWORK_ID with the ID of the additional self-service network.

6. Determine the IPv4 and IPv6 addresses of the instance.

```
$ openstack server list
+--------------------------------------+-----------------------+--------+-
↪---------------------------------------------------------------+--------
↪+---------+
| ID                                   | Name                  | Status |␣
↪Networks                                                        | Image ␣
↪| Flavor  |
+--------------------------------------+-----------------------+--------+-
↪---------------------------------------------------------------+--------
↪+---------+
| bde64b00-77ae-41b9-b19a-cd8e378d9f8b | selfservice-instance2 | ACTIVE |␣
```

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```
↪selfservice2=fd00:198:51:100:f816:3eff:fe71:e93e, 198.51.100.4 | cirros␣
↪| m1.tiny |
+--------------------------------------+-----------------------+--------+-
↪----------------------------------------------------------------+--------
↪+---------+
```

7. Create a floating IPv4 address on the provider network.

```
$ openstack floating ip create provider1
+-------------+--------------------------------------+
| Field       | Value                                |
+-------------+--------------------------------------+
| fixed_ip    | None                                 |
| id          | 0174056a-fa56-4403-b1ea-b5151a31191f |
| instance_id | None                                 |
| ip          | 203.0.113.17                         |
| pool        | provider1                            |
+-------------+--------------------------------------+
```

8. Associate the floating IPv4 address with the instance.

```
$ openstack server add floating ip selfservice-instance2 203.0.113.17
```

> **Note**
>
> This command provides no output.

Verify failover operation

1. Begin a continuous `ping` of both the floating IPv4 address and IPv6 address of the instance. While performing the next three steps, you should see a minimal, if any, interruption of connectivity to the instance.
2. On the network node with the master router, administratively disable the overlay network interface.
3. On the other network node, verify promotion of the backup router to master router by noting addition of IP addresses to the interfaces in the `qrouter` namespace.
4. On the original network node in step 2, administratively enable the overlay network interface. Note that the master router remains on the network node in step 3.

Keepalived VRRP health check

The health of your `keepalived` instances can be automatically monitored via a bash script that verifies connectivity to all available and configured gateway addresses. In the event that connectivity is lost, the master router is rescheduled to another node.

If all routers lose connectivity simultaneously, the process of selecting a new master router will be repeated in a round-robin fashion until one or more routers have their connectivity restored.

To enable this feature, edit the `l3_agent.ini` file:

```
ha_vrrp_health_check_interval = 30
```

Where `ha_vrrp_health_check_interval` indicates how often in seconds the health check should run. The default value is `0`, which indicates that the check should not run at all.

Network traffic flow

This high-availability mechanism simply augments *Open vSwitch: Self-service networks* with failover of layer-3 services to another router if the primary router fails. Thus, you can reference *Self-service network traffic flow* for normal operation.

Open vSwitch: High availability using DVR

This architecture example augments the self-service deployment example with the Distributed Virtual Router (DVR) high-availability mechanism that provides connectivity between self-service and provider networks on compute nodes rather than network nodes for specific scenarios. For instances with a floating IPv4 address, routing between self-service and provider networks resides completely on the compute nodes to eliminate single point of failure and performance issues with network nodes. Routing also resides completely on the compute nodes for instances with a fixed or floating IPv4 address using self-service networks on the same distributed virtual router. However, instances with a fixed IP address still rely on the network node for routing and SNAT services between self-service and provider networks.

Consider the following attributes of this high-availability mechanism to determine practicality in your environment:

- Only provides connectivity to an instance via the compute node on which the instance resides if the instance resides on a self-service network with a floating IPv4 address. Instances on self-service networks with only an IPv6 address or both IPv4 and IPv6 addresses rely on the network node for IPv6 connectivity.
- The instance of a router on each compute node consumes an IPv4 address on the provider network on which it contains a gateway.

Prerequisites

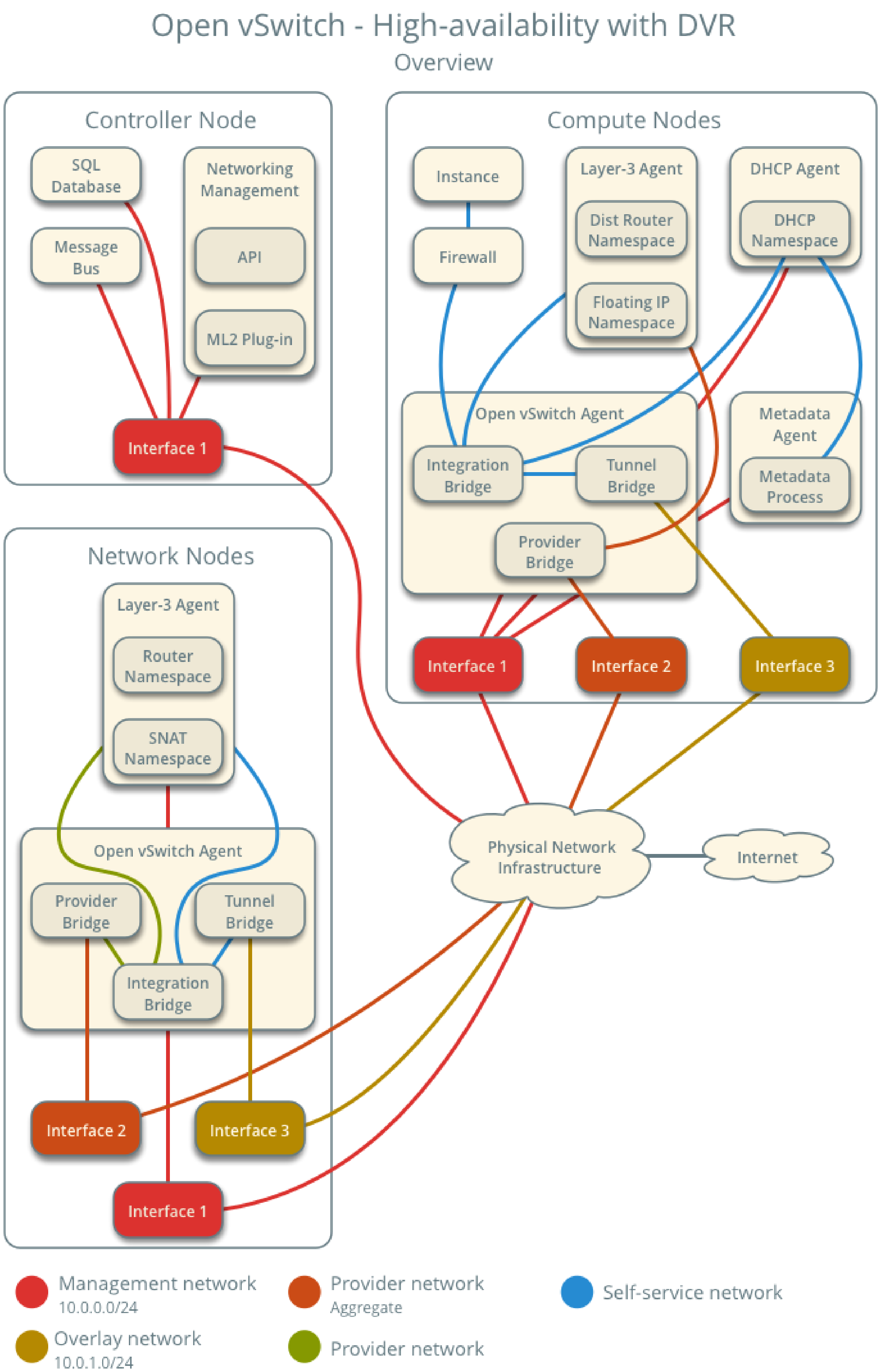
Modify the compute nodes with the following components:

- Install the OpenStack Networking layer-3 agent.

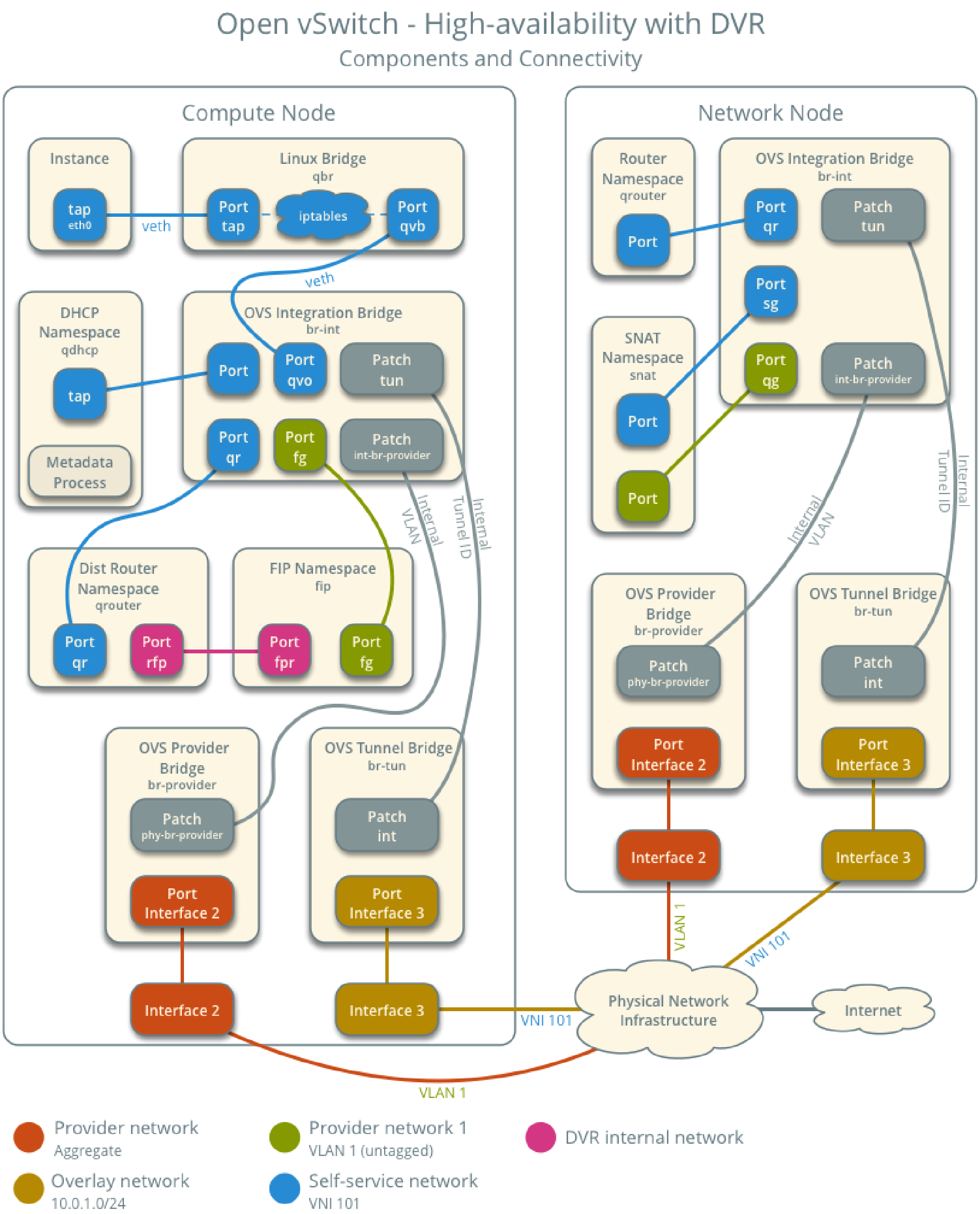
Note

Consider adding at least one additional network node to provide high-availability for instances with a fixed IP address. See See *Distributed Virtual Routing with VRRP* for more information.

Architecture



The following figure shows components and connectivity for one self-service network and one untagged (flat) network. In this particular case, the instance resides on the same compute node as the DHCP agent for the network. If the DHCP agent resides on another compute node, the latter only contains a DHCP namespace with a port on the OVS integration bridge.



Example configuration

Use the following example configuration as a template to add support for high-availability using DVR to an existing operational environment that supports self-service networks.

Controller node

1. In the neutron.conf file:

   - Enable distributed routing by default for all routers.

```
[DEFAULT]
router_distributed = True
```

Note

For a large scale cloud, if your deployment is running DVR with DHCP, we recommend you set host_dvr_for_dhcp=False to achieve higher L3 agent router processing performance. When this is set to False, DNS functionality will not be available via the DHCP namespace (dnsmasq) however, a different nameserver will have to be configured, for example, by specifying a value in dns_nameservers for subnets.

1. Restart the following services:

   - Server

Network node

1. In the openvswitch_agent.ini file, enable distributed routing.

```
[agent]
enable_distributed_routing = True
```

2. In the l3_agent.ini file, configure the layer-3 agent to provide SNAT services.

```
[DEFAULT]
agent_mode = dvr_snat
```

Note

agent_mode = dvr_snat is not supported on compute nodes. For discussion please see: bug #1934666

1. Restart the following services:

   - Open vSwitch agent
   - Layer-3 agent

Compute nodes

1. Install the Networking service layer-3 agent.
2. In the `openswitch_agent.ini` file, enable distributed routing.

```
[agent]
enable_distributed_routing = True
```

3. In the `l3_agent.ini` file, configure the layer-3 agent.

```
[DEFAULT]
agent_mode = dvr
```

4. Restart the following services:
   - Open vSwitch agent
   - Layer-3 agent

Verify service operation

1. Source the administrative project credentials.
2. Verify presence and operation of the agents.

```
$ openstack network agent list
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
| ID                                   | Agent Type         | Host     |␣
↪Availability Zone | Alive | State | Binary                    |
+--------------------------------------+--------------------+----------+--
↪-----------------+-------+-------+---------------------------+
| 05d980f2-a4fc-4815-91e7-a7f7e118c0db | L3 agent           | compute1 |␣
↪nova              | True  | UP    | neutron-l3-agent          |
| 1236bbcb-e0ba-48a9-80fc-81202ca4fa51 | Metadata agent     | compute2 |␣
↪None              | True  | UP    | neutron-metadata-agent    |
| 2a2e9a90-51b8-4163-a7d6-3e199ba2374b | L3 agent           | compute2 |␣
↪nova              | True  | UP    | neutron-l3-agent          |
| 457d6898-b373-4bb3-b41f-59345dcfb5c5 | Open vSwitch agent | compute2 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
| 71f15e84-bc47-4c2a-b9fb-317840b2d753 | DHCP agent         | compute2 |␣
↪nova              | True  | UP    | neutron-dhcp-agent        |
| 8805b962-de95-4e40-bdc2-7a0add7521e8 | L3 agent           | network1 |␣
↪nova              | True  | UP    | neutron-l3-agent          |
| a33cac5a-0266-48f6-9cac-4cef4f8b0358 | Open vSwitch agent | network1 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
| a6c69690-e7f7-4e56-9831-1282753e5007 | Metadata agent     | compute1 |␣
↪None              | True  | UP    | neutron-metadata-agent    |
| af11f22f-a9f4-404f-9fd8-cd7ad55c0f68 | DHCP agent         | compute1 |␣
↪nova              | True  | UP    | neutron-dhcp-agent        |
| bcfc977b-ec0e-4ba9-be62-9489b4b0e6f1 | Open vSwitch agent | compute1 |␣
↪None              | True  | UP    | neutron-openvswitch-agent |
```

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```
+--------------------------------------+---------------------+----------+--
↪----------------+-------+-------+---------------------------+
```

Create initial networks

Similar to the self-service deployment example, this configuration supports multiple VXLAN self-service networks. After enabling high-availability, all additional routers use distributed routing. The following procedure creates an additional self-service network and router. The Networking service also supports adding distributed routing to existing routers.

1. Source a regular (non-administrative) project credentials.
2. Create a self-service network.

```
$ openstack network create selfservice2
+-------------------------+--------------+
| Field                   | Value        |
+-------------------------+--------------+
| admin_state_up          | UP           |
| mtu                     | 1450         |
| name                    | selfservice2 |
| port_security_enabled   | True         |
| revision_number         | 1            |
| router:external         | Internal     |
| shared                  | False        |
| status                  | ACTIVE       |
| tags                    | []           |
+-------------------------+--------------+
```

3. Create a IPv4 subnet on the self-service network.

```
$ openstack subnet create --subnet-range 192.0.2.0/24 \
  --network selfservice2 --dns-nameserver 8.8.4.4 selfservice2-v4
+-------------------+---------------------------+
| Field             | Value                     |
+-------------------+---------------------------+
| allocation_pools  | 192.0.2.2-192.0.2.254     |
| cidr              | 192.0.2.0/24              |
| dns_nameservers   | 8.8.4.4                   |
| enable_dhcp       | True                      |
| gateway_ip        | 192.0.2.1                 |
| ip_version        | 4                         |
| name              | selfservice2-v4           |
| revision_number   | 1                         |
| tags              | []                        |
+-------------------+---------------------------+
```

4. Create a IPv6 subnet on the self-service network.

```
$ openstack subnet create --subnet-range fd00:192:0:2::/64 --ip-version 6␣
↪\
```

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```
  --ipv6-ra-mode slaac --ipv6-address-mode slaac --network selfservice2 \
  --dns-nameserver 2001:4860:4860::8844 selfservice2-v6
+-------------------+-----------------------------------------------------
↪-+
| Field             | Value                                               
↪ |
+-------------------+-----------------------------------------------------
↪-+
| allocation_pools  | fd00:192:0:2::2-fd00:192:0:2:ffff:ffff:ffff:ffff    
↪ |
| cidr              | fd00:192:0:2::/64                                   
↪ |
| dns_nameservers   | 2001:4860:4860::8844                                
↪ |
| enable_dhcp       | True                                                
↪ |
| gateway_ip        | fd00:192:0:2::1                                     
↪ |
| ip_version        | 6                                                   
↪ |
| ipv6_address_mode | slaac                                               
↪ |
| ipv6_ra_mode      | slaac                                               
↪ |
| name              | selfservice2-v6                                     
↪ |
| revision_number   | 1                                                   
↪ |
| tags              | []                                                  
↪ |
+-------------------+-----------------------------------------------------
↪-+
```

5. Create a router.

```
$ openstack router create router2
+-----------------------+---------+
| Field                 | Value   |
+-----------------------+---------+
| admin_state_up        | UP      |
| name                  | router2 |
| revision_number       | 1       |
| status                | ACTIVE  |
| tags                  | []      |
+-----------------------+---------+
```

6. Add the IPv4 and IPv6 subnets as interfaces on the router.

```
$ openstack router add subnet router2 selfservice2-v4
$ openstack router add subnet router2 selfservice2-v6
```

Note

These commands provide no output.

7. Add the provider network as a gateway on the router.

```
$ openstack router set router2 --external-gateway provider1
```

Verify network operation

1. Source the administrative project credentials.
2. Verify distributed routing on the router.

```
$ openstack router show router2
+-------------------------+---------+
| Field                   | Value   |
+-------------------------+---------+
| admin_state_up          | UP      |
| distributed             | True    |
| ha                      | False   |
| name                    | router2 |
| revision_number         | 1       |
| status                  | ACTIVE  |
+-------------------------+---------+
```

3. On each compute node, verify creation of a `qrouter` namespace with the same ID.

   Compute node 1:

```
# ip netns
qrouter-78d2f628-137c-4f26-a257-25fc20f203c1
```

   Compute node 2:

```
# ip netns
qrouter-78d2f628-137c-4f26-a257-25fc20f203c1
```

4. On the network node, verify creation of the `snat` and `qrouter` namespaces with the same ID.

```
# ip netns
snat-78d2f628-137c-4f26-a257-25fc20f203c1
qrouter-78d2f628-137c-4f26-a257-25fc20f203c1
```

Note

The namespace for router 1 from *Open vSwitch: Self-service networks* should also appear on network node 1 because of creation prior to enabling distributed routing.

5. Launch an instance with an interface on the additional self-service network. For example, a CirrOS image using flavor ID 1.

```
$ openstack server create --flavor 1 --image cirros --nic net-id=NETWORK_
↪ID selfservice-instance2
```

Replace `NETWORK_ID` with the ID of the additional self-service network.

6. Determine the IPv4 and IPv6 addresses of the instance.

```
$ openstack server list
+--------------------------------------+-----------------------+--------+-
↪-----------------------------------------------------------+--------+-----
↪----+
| ID                                   | Name                  | Status |␣
↪Networks                                                  | Image  |␣
↪Flavor  |
+--------------------------------------+-----------------------+--------+-
↪-----------------------------------------------------------+--------+-----
↪----+
| bde64b00-77ae-41b9-b19a-cd8e378d9f8b | selfservice-instance2 | ACTIVE |␣
↪selfservice2=fd00:192:0:2:f816:3eff:fe71:e93e, 192.0.2.4 | cirros | m1.
↪tiny |
+--------------------------------------+-----------------------+--------+-
↪-----------------------------------------------------------+--------+-----
↪----+
```

7. Create a floating IPv4 address on the provider network.

```
$ openstack floating ip create provider1
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| fixed_ip          | None                                 |
| id                | 0174056a-fa56-4403-b1ea-b5151a31191f |
| instance_id       | None                                 |
| ip                | 203.0.113.17                         |
| pool              | provider1                            |
| revision_number   | 1                                    |
| tags              | []                                   |
+-------------------+--------------------------------------+
```

8. Associate the floating IPv4 address with the instance.

```
$ openstack server add floating ip selfservice-instance2 203.0.113.17
```

Note

This command provides no output.

9. On the compute node containing the instance, verify creation of the `fip` namespace with the same ID as the provider network.

```
# ip netns
fip-4bfa3075-b4b2-4f7d-b88e-df1113942d43
```

Network traffic flow

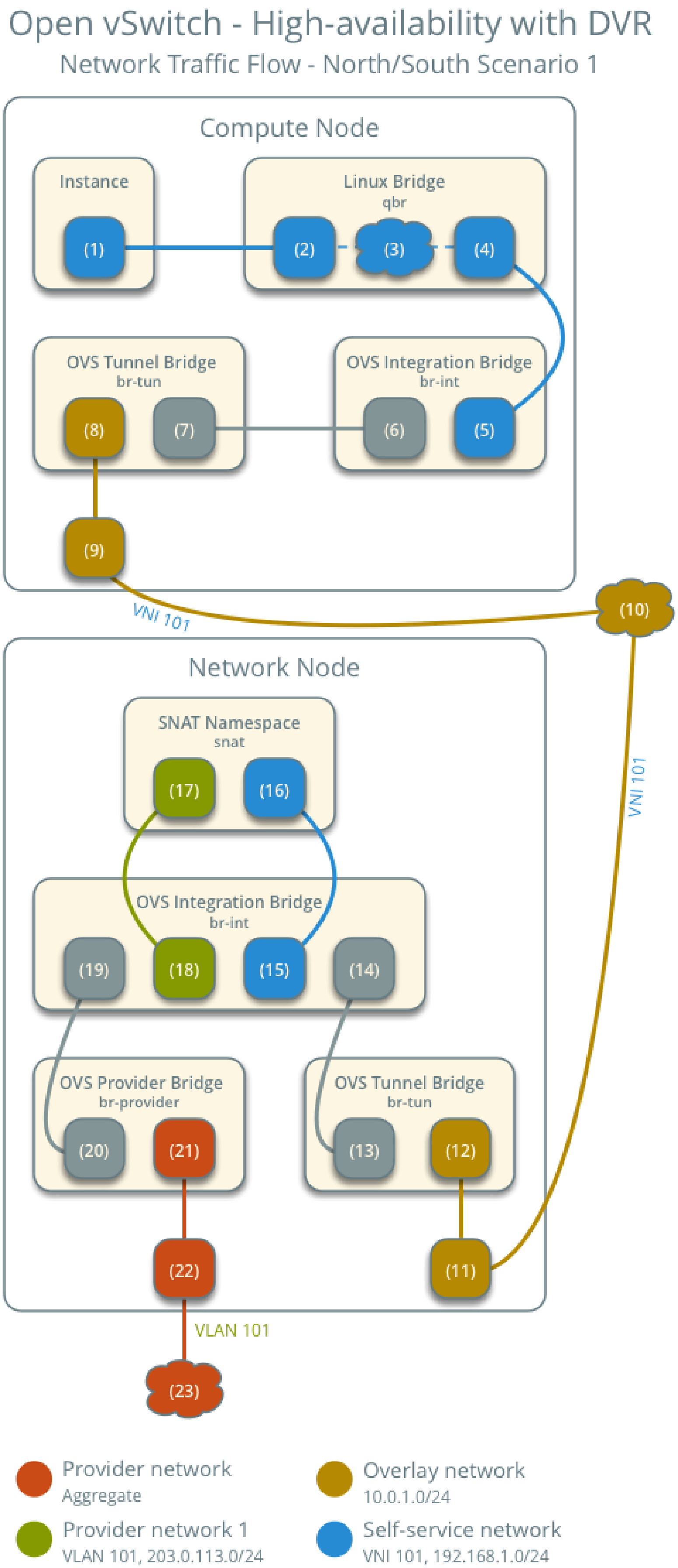
The following sections describe the flow of network traffic in several common scenarios. *North-south* network traffic travels between an instance and external network such as the Internet. *East-west* network traffic travels between instances on the same or different networks. In all scenarios, the physical network infrastructure handles switching and routing among provider networks and external networks such as the Internet. Each case references one or more of the following components:

- Provider network (VLAN)
  - VLAN ID 101 (tagged)
- Self-service network 1 (VXLAN)
  - VXLAN ID (VNI) 101
- Self-service network 2 (VXLAN)
  - VXLAN ID (VNI) 102
- Self-service router
  - Gateway on the provider network
  - Interface on self-service network 1
  - Interface on self-service network 2
- Instance 1
- Instance 2

This section only contains flow scenarios that benefit from distributed virtual routing or that differ from conventional operation. For other flow scenarios, see *Network traffic flow*.

North-south scenario 1: Instance with a fixed IP address

Similar to *North-south scenario 1: Instance with a fixed IP address*, except the router namespace on the network node becomes the SNAT namespace. The network node still contains the router namespace, but it serves no purpose in this case.



Open vSwitch - High-availability with DVR
Network Traffic Flow - North/South Scenario 1
Compute Node
Instance
(1)
Linux Bridge
qbr
(2)
(3)
(4)
OVS Tunnel Bridge
br-tun
(8)
(7)
OVS Integration Bridge
br-int
(6)
(5)
(9)
VNI 101
(10)
Network Node
SNAT Namespace
snat
(17)
(16)
VNI 101
OVS Integration Bridge
br-int
(19)
(18)
(15)
(14)
OVS Provider Bridge
br-provider
(20)
(21)
OVS Tunnel Bridge
br-tun
(13)
(12)
(22)
(11)
VLAN 101
(23)
Provider network
Aggregate
Overlay network
10.0.1.0/24
Provider network 1
VLAN 101, 203.0.113.0/24
Self-service network
VNI 101, 192.168.1.0/24

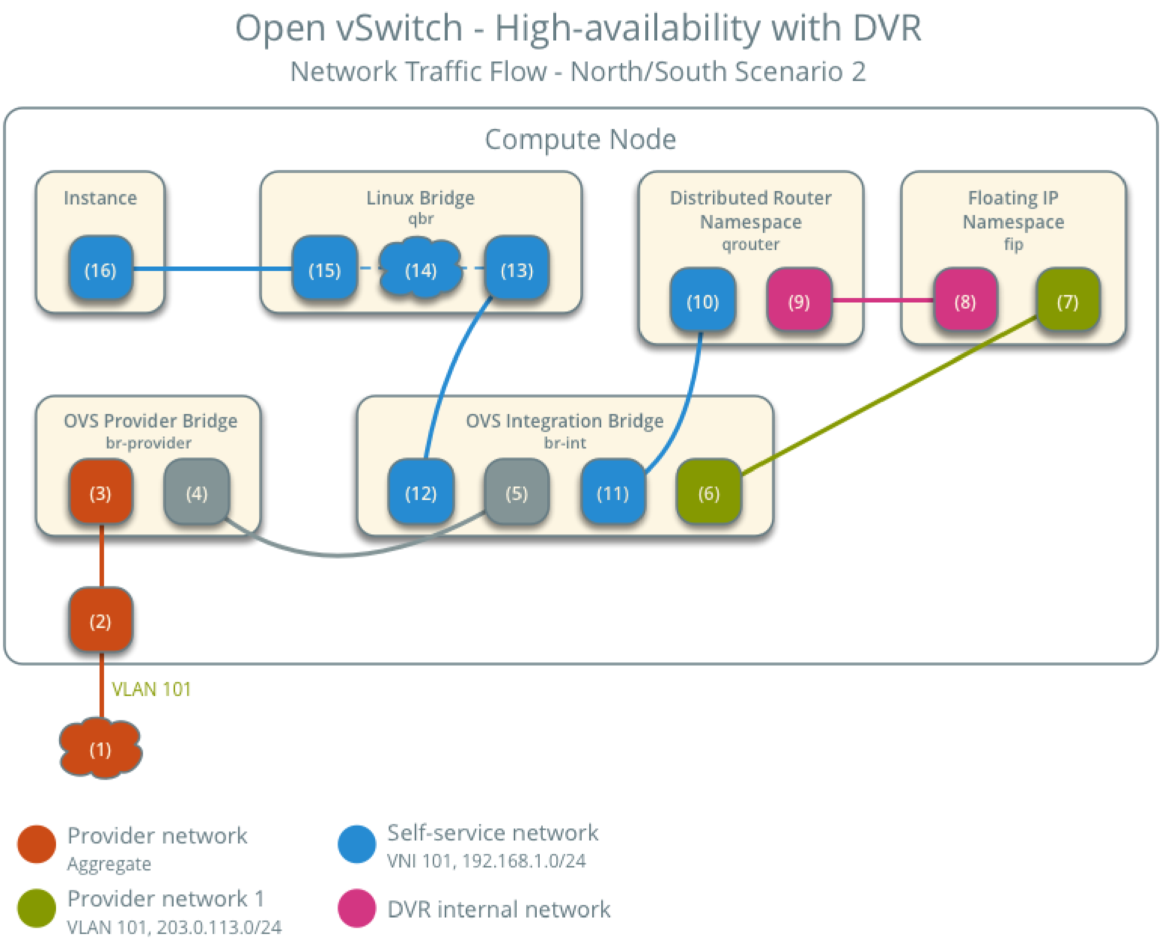
North-south scenario 2: Instance with a floating IPv4 address

For instances with a floating IPv4 address using a self-service network on a distributed router, the compute node containing the instance performs SNAT on north-south traffic passing from the instance to external networks such as the Internet and DNAT on north-south traffic passing from external networks to the instance. Floating IP addresses and NAT do not apply to IPv6. Thus, the network node routes IPv6 traffic in this scenario. north-south traffic passing between the instance and external networks such as the Internet.

- Instance 1 resides on compute node 1 and uses self-service network 1.
- A host on the Internet sends a packet to the instance.

The following steps involve the compute node:

1. The physical network infrastructure (1) forwards the packet to the provider physical network interface (2).
2. The provider physical network interface forwards the packet to the OVS provider bridge provider network port (3).
3. The OVS provider bridge swaps actual VLAN tag 101 with the internal VLAN tag.
4. The OVS provider bridge `phy-br-provider` port (4) forwards the packet to the OVS integration bridge `int-br-provider` port (5).
5. The OVS integration bridge port for the provider network (6) removes the internal VLAN tag and forwards the packet to the provider network interface (7) in the floating IP namespace. This interface responds to any ARP requests for the instance floating IPv4 address.
6. The floating IP namespace routes the packet (8) to the distributed router namespace (9) using a pair of IP addresses on the DVR internal network. This namespace contains the instance floating IPv4 address.
7. The router performs DNAT on the packet which changes the destination IP address to the instance IP address on the self-service network via the self-service network interface (10).
8. The router forwards the packet to the OVS integration bridge port for the self-service network (11).
9. The OVS integration bridge adds an internal VLAN tag to the packet.
10. The OVS integration bridge removes the internal VLAN tag from the packet.
11. The OVS integration bridge security group port (12) forwards the packet to the security group bridge OVS port (13) via `veth` pair.
12. Security group rules (14) on the security group bridge handle firewalling and connection tracking for the packet.
13. The security group bridge instance port (15) forwards the packet to the instance interface (16) via `veth` pair.



Note

Egress traffic follows similar steps in reverse, except SNAT changes the source IPv4 address of the packet to the floating IPv4 address.

East-west scenario 1: Instances on different networks on the same router

Instances with fixed IPv4/IPv6 address or floating IPv4 address on the same compute node communicate via router on the compute node. Instances on different compute nodes communicate via an instance of the router on each compute node.

Note

This scenario places the instances on different compute nodes to show the most complex situation.

The following steps involve compute node 1:

1. The instance interface (1) forwards the packet to the security group bridge instance port (2) via `veth` pair.
2. Security group rules (3) on the security group bridge handle firewalling and connection tracking for the packet.

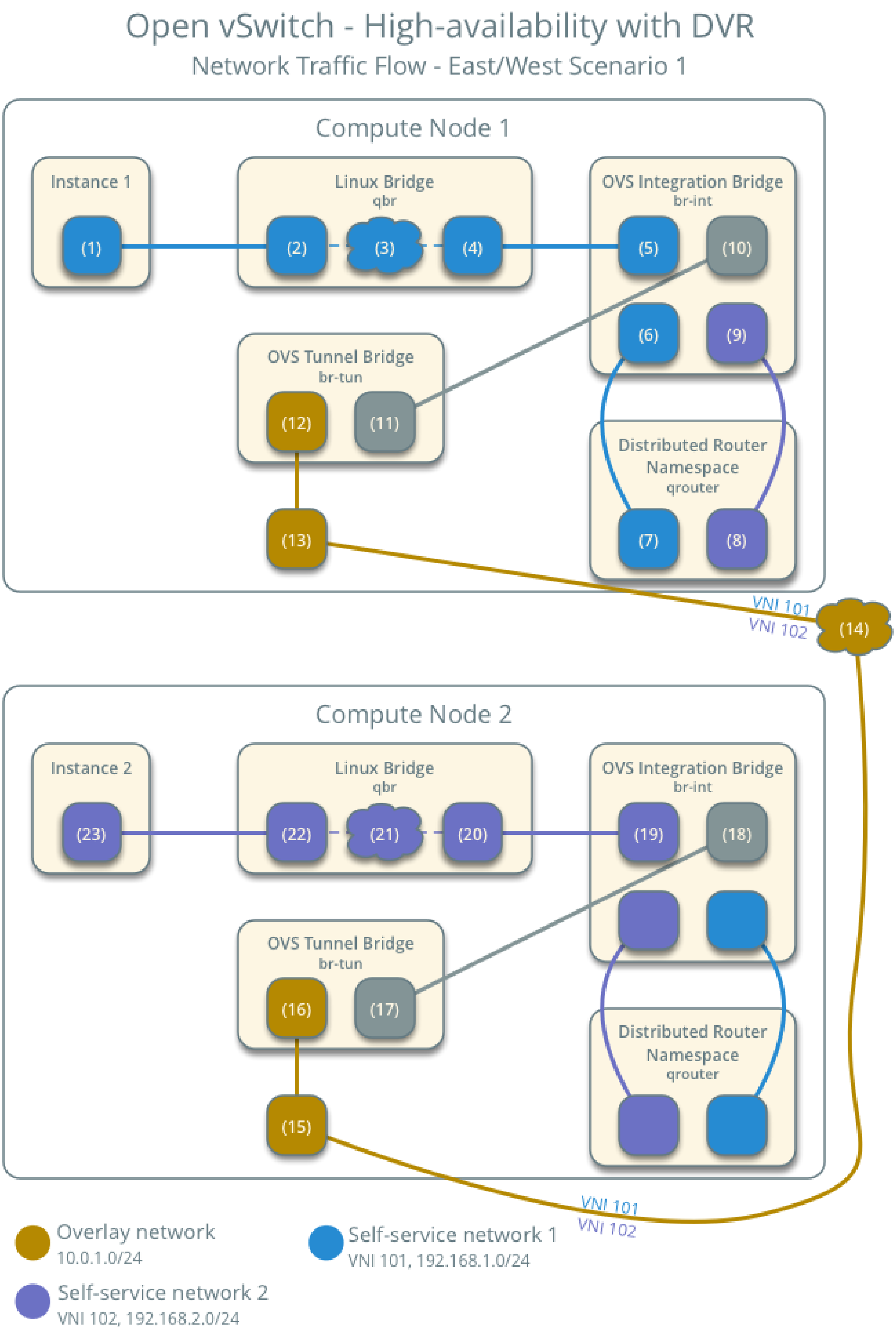
3. The security group bridge OVS port (4) forwards the packet to the OVS integration bridge security group port (5) via `veth` pair.
4. The OVS integration bridge adds an internal VLAN tag to the packet.
5. The OVS integration bridge port for self-service network 1 (6) removes the internal VLAN tag and forwards the packet to the self-service network 1 interface in the distributed router namespace (6).
6. The distributed router namespace routes the packet to self-service network 2.
7. The self-service network 2 interface in the distributed router namespace (8) forwards the packet to the OVS integration bridge port for self-service network 2 (9).
8. The OVS integration bridge adds an internal VLAN tag to the packet.
9. The OVS integration bridge exchanges the internal VLAN tag for an internal tunnel ID.
10. The OVS integration bridge `patch-tun` port (10) forwards the packet to the OVS tunnel bridge `patch-int` port (11).
11. The OVS tunnel bridge (12) wraps the packet using VNI 101.
12. The underlying physical interface (13) for overlay networks forwards the packet to compute node 2 via the overlay network (14).

The following steps involve compute node 2:

1. The underlying physical interface (15) for overlay networks forwards the packet to the OVS tunnel bridge (16).
2. The OVS tunnel bridge unwraps the packet and adds an internal tunnel ID to it.
3. The OVS tunnel bridge exchanges the internal tunnel ID for an internal VLAN tag.
4. The OVS tunnel bridge `patch-int` patch port (17) forwards the packet to the OVS integration bridge `patch-tun` patch port (18).
5. The OVS integration bridge removes the internal VLAN tag from the packet.
6. The OVS integration bridge security group port (19) forwards the packet to the security group bridge OVS port (20) via `veth` pair.
7. Security group rules (21) on the security group bridge handle firewalling and connection tracking for the packet.
8. The security group bridge instance port (22) forwards the packet to the instance 2 interface (23) via `veth` pair.

Note

Routing between self-service networks occurs on the compute node containing the instance sending the packet. In this scenario, routing occurs on compute node 1 for packets from instance 1 to instance 2 and on compute node 2 for packets from instance 2 to instance 1.


Open vSwitch - High-availability with DVR
Network Traffic Flow - East/West Scenario 1
Compute Node 1
Instance 1
(1)
Linux Bridge
qbr
(2)
(3)
(4)
OVS Integration Bridge
br-int
(5)
(10)
(6)
(9)
OVS Tunnel Bridge
br-tun
(12)
(11)
Distributed Router
Namespace
qrouter
(13)
(7)
(8)
VNI 101
VNI 102
(14)
Compute Node 2
Instance 2
(23)
Linux Bridge
qbr
(22)
(21)
(20)
OVS Integration Bridge
br-int
(19)
(18)
OVS Tunnel Bridge
br-tun
(16)
(17)
Distributed Router
Namespace
qrouter
(15)
VNI 101
VNI 102
Overlay network
10.0.1.0/24
Self-service network 1
VNI 101, 192.168.1.0/24
Self-service network 2
VNI 102, 192.168.2.0/24

8.4 Operations

8.4.1 IP availability metrics

Network IP Availability is an information-only API extension that allows a user or process to determine the number of IP addresses that are consumed across networks and the allocation pools of their subnets. This extension was added to neutron in the Mitaka release.

This section illustrates how you can get the Network IP address availability through the command-line interface.

Get Network IP address availability for all IPv4 networks:

```
$ openstack ip availability list

+--------------------------------------+--------------+-----------+----------+
| Network ID                           | Network Name | Total IPs | Used IPs |
+--------------------------------------+--------------+-----------+----------+
| 363a611a-b08b-4281-b64e-198d90cb94fd | private      |       253 |        3 |
| c92d0605-caf2-4349-b1b8-8d5f9ac91df8 | public       |       253 |        1 |
+--------------------------------------+--------------+-----------+----------+
```

Get Network IP address availability for all IPv6 networks:

```
$ openstack ip availability list --ip-version 6

+--------------------------------------+--------------+----------------------
↪+----------+
| Network ID                           | Network Name | Total IPs            ␣
↪| Used IPs |
+--------------------------------------+--------------+----------------------
↪+----------+
| 363a611a-b08b-4281-b64e-198d90cb94fd | private      | 18446744073709551614␣
↪|        3 |
| c92d0605-caf2-4349-b1b8-8d5f9ac91df8 | public       | 18446744073709551614␣
↪|        1 |
+--------------------------------------+--------------+----------------------
↪+----------+
```

Get Network IP address availability statistics for a specific network:

```
$ openstack ip availability show NETWORKUUID

+------------------------+--------------------------------------------------------
↪----------+
| Field                  | Value                                                  ␣
↪          |
+------------------------+--------------------------------------------------------
↪----------+
| network_id             | 0bf90de6-fc0f-4dba-b80d-96670dfb331a                   ␣
↪          |
| network_name           | public                                                 ␣
↪          |
```

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```
| project_id              | 5669caad86a04256994cdf755df4d3c1                    ␣
↪            |
| subnet_ip_availability | cidr='192.0.2.224/28', ip_version='4', subnet_id=
↪'346806ee-   |
|                         | a53e-44fd-968a-ddb2bcd2ba96', subnet_name='public_
↪subnet',    |
|                         | total_ips='13', used_ips='5'                        ␣
↪            |
| total_ips               | 13                                                  ␣
↪            |
| used_ips                | 5                                                   ␣
↪            |
+-------------------------+----------------------------------------------------
↪----------+
```

8.4.2 Resource tags

Various virtual networking resources support tags for use by external systems or any other clients of the Networking service API.

All resources that support standard attributes are applicable for tagging. This includes:

- networks
- subnets
- subnetpools
- ports
- routers
- floatingips
- logs
- security-groups
- security-group-rules
- segments
- policies
- trunks
- network_segment_ranges

Use cases

The following use cases refer to adding tags to networks, but the same can be applicable to any other supported Networking service resource:

1. Ability to map different networks in different OpenStack locations to one logically same network (for multi-site OpenStack).

2. Ability to map IDs from different management/orchestration systems to OpenStack networks in mixed environments. For example, in the Kuryr project, the Docker network ID is mapped to the Neutron network ID.
3. Ability to leverage tags by deployment tools.
4. Ability to tag information about provider networks (for example, high-bandwidth, low-latency, and so on).

Filtering with tags

The API allows searching/filtering of the `GET /v2.0/networks` API. The following query parameters are supported:

- `tags`
- `tags-any`
- `not-tags`
- `not-tags-any`

To request the list of networks that have a single tag, `tags` argument should be set to the desired tag name. Example:

```
GET /v2.0/networks?tags=red
```

To request the list of networks that have two or more tags, the `tags` argument should be set to the list of tags, separated by commas. In this case, the tags given must all be present for a network to be included in the query result. Example that returns networks that have the red and blue tags:

```
GET /v2.0/networks?tags=red,blue
```

To request the list of networks that have one or more of a list of given tags, the `tags-any` argument should be set to the list of tags, separated by commas. In this case, as long as one of the given tags is present, the network will be included in the query result. Example that returns the networks that have the red or the blue tag:

```
GET /v2.0/networks?tags-any=red,blue
```

To request the list of networks that do not have one or more tags, the `not-tags` argument should be set to the list of tags, separated by commas. In this case, only the networks that do not have any of the given tags will be included in the query results. Example that returns the networks that do not have either red or blue tag:

```
GET /v2.0/networks?not-tags=red,blue
```

To request the list of networks that do not have at least one of a list of tags, the `not-tags-any` argument should be set to the list of tags, separated by commas. In this case, only the networks that do not have at least one of the given tags will be included in the query result. Example that returns the networks that do not have the red tag, or do not have the blue tag:

```
GET /v2.0/networks?not-tags-any=red,blue
```

The `tags`, `tags-any`, `not-tags`, and `not-tags-any` arguments can be combined to build more complex queries. Example:

```
GET /v2.0/networks?tags=red,blue&tags-any=green,orange
```

The above example returns any networks that have the red and blue tags, plus at least one of green and orange.

Complex queries may have contradictory parameters. Example:

```
GET /v2.0/networks?tags=blue&not-tags=blue
```

In this case, we should let the Networking service find these networks. Obviously, there are no such networks and the service will return an empty list.

User workflow

Add a tag to a resource:

```
$ openstack network set --tag red ab442634-1cc9-49e5-bd49-0dac9c811f69
$ openstack network show net
+---------------------------+----------------------------------------------
↪--------------------------+
| Field                     | Value                                        ␣
↪                          |
+---------------------------+----------------------------------------------
↪--------------------------+
| admin_state_up            | UP                                           ␣
↪                          |
| availability_zone_hints   |                                              ␣
↪                          |
| availability_zones        | nova                                         ␣
↪                          |
| created_at                | 2018-07-11T09:44:50Z                         ␣
↪                          |
| description               |                                              ␣
↪                          |
| dns_domain                | None                                         ␣
↪                          |
| id                        | ab442634-1cc9-49e5-bd49-0dac9c811f69         ␣
↪                          |
| ipv4_address_scope        | None                                         ␣
↪                          |
| ipv6_address_scope        | None                                         ␣
↪                          |
| is_default                | None                                         ␣
↪                          |
| is_vlan_transparent       | None                                         ␣
↪                          |
| mtu                       | 1450                                         ␣
↪                          |
| name                      | net                                          ␣
↪                          |
| port_security_enabled     | True                                         ␣
```

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```
↪                           |
| project_id                | e6710680bfd14555891f265644e1dd5c                    ␣
↪                           |
| provider:network_type     | vxlan                                               ␣
↪                           |
| provider:physical_network | None                                                ␣
↪                           |
| provider:segmentation_id  | 1047                                                ␣
↪                           |
| qos_policy_id             | None                                                ␣
↪                           |
| revision_number           | 5                                                   ␣
↪                           |
| router:external           | Internal                                            ␣
↪                           |
| segments                  | None                                                ␣
↪                           |
| shared                    | False                                               ␣
↪                           |
| status                    | ACTIVE                                              ␣
↪                           |
| subnets                   |                                                     ␣
↪                           |
| tags                      | red                                                 ␣
↪                           |
| updated_at                | 2018-07-16T06:22:01Z                                ␣
↪                           |
+---------------------------+-------------------------------------------------
↪---------------------------+
```

Remove a tag from a resource:

```
$ openstack network unset --tag red ab442634-1cc9-49e5-bd49-0dac9c811f69
$ openstack network show net
+---------------------------+-------------------------------------------------
↪---------------------------+
| Field                     | Value                                               ␣
↪                           |
+---------------------------+-------------------------------------------------
↪---------------------------+
| admin_state_up            | UP                                                  ␣
↪                           |
| availability_zone_hints   |                                                     ␣
↪                           |
| availability_zones        | nova                                                ␣
↪                           |
| created_at                | 2018-07-11T09:44:50Z                                ␣
↪                           |
| description               |                                                     ␣
↪                           |
```

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```
| dns_domain                | None                                                    ␣
↪                         |
| id                        | ab442634-1cc9-49e5-bd49-0dac9c811f69                    ␣
↪                         |
| ipv4_address_scope        | None                                                    ␣
↪                         |
| ipv6_address_scope        | None                                                    ␣
↪                         |
| is_default                | None                                                    ␣
↪                         |
| is_vlan_transparent       | None                                                    ␣
↪                         |
| mtu                       | 1450                                                    ␣
↪                         |
| name                      | net                                                     ␣
↪                         |
| port_security_enabled     | True                                                    ␣
↪                         |
| project_id                | e6710680bfd14555891f265644e1dd5c                        ␣
↪                         |
| provider:network_type     | vxlan                                                   ␣
↪                         |
| provider:physical_network | None                                                    ␣
↪                         |
| provider:segmentation_id  | 1047                                                    ␣
↪                         |
| qos_policy_id             | None                                                    ␣
↪                         |
| revision_number           | 5                                                       ␣
↪                         |
| router:external           | Internal                                                ␣
↪                         |
| segments                  | None                                                    ␣
↪                         |
| shared                    | False                                                   ␣
↪                         |
| status                    | ACTIVE                                                  ␣
↪                         |
| subnets                   |                                                         ␣
↪                         |
| tags                      |                                                         ␣
↪                         |
| updated_at                | 2018-07-16T06:32:11Z                                    ␣
↪                         |
+---------------------------+--------------------------------------------------
↪-------------------------+
```

Replace all tags on the resource:

```
$ openstack network set --tag red --tag blue ab442634-1cc9-49e5-bd49-
↪0dac9c811f69
$ openstack network show net
+---------------------------+-----------------------------------------------
↪---------------------------+
| Field                     | Value
↪                           |
+---------------------------+-----------------------------------------------
↪---------------------------+
| admin_state_up            | UP
↪                           |
| availability_zone_hints   |
↪                           |
| availability_zones        | nova
↪                           |
| created_at                | 2018-07-11T09:44:50Z
↪                           |
| description               |
↪                           |
| dns_domain                | None
↪                           |
| id                        | ab442634-1cc9-49e5-bd49-0dac9c811f69
↪                           |
| ipv4_address_scope        | None
↪                           |
| ipv6_address_scope        | None
↪                           |
| is_default                | None
↪                           |
| is_vlan_transparent       | None
↪                           |
| mtu                       | 1450
↪                           |
| name                      | net
↪                           |
| port_security_enabled     | True
↪                           |
| project_id                | e6710680bfd14555891f265644e1dd5c
↪                           |
| provider:network_type     | vxlan
↪                           |
| provider:physical_network | None
↪                           |
| provider:segmentation_id  | 1047
↪                           |
| qos_policy_id             | None
↪                           |
| revision_number           | 5
↪                           |
| router:external           | Internal
```

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```
↪                                                    |
| segments                  | None                                             ␣
↪                                                    |
| shared                    | False                                            ␣
↪                                                    |
| status                    | ACTIVE                                           ␣
↪                                                    |
| subnets                   |                                                  ␣
↪                                                    |
| tags                      | blue, red                                        ␣
↪                                                    |
| updated_at                | 2018-07-16T06:50:19Z                             ␣
↪                                                    |
+---------------------------+-----------------------------------------------
↪--------------------------+
```

Clear tags from a resource:

```
$ openstack network unset --all-tag ab442634-1cc9-49e5-bd49-0dac9c811f69
$ openstack network show net
+---------------------------+-----------------------------------------------
↪--------------------------+
| Field                     | Value                                            ␣
↪                                                    |
+---------------------------+-----------------------------------------------
↪--------------------------+
| admin_state_up            | UP                                               ␣
↪                                                    |
| availability_zone_hints   |                                                  ␣
↪                                                    |
| availability_zones        | nova                                             ␣
↪                                                    |
| created_at                | 2018-07-11T09:44:50Z                             ␣
↪                                                    |
| description               |                                                  ␣
↪                                                    |
| dns_domain                | None                                             ␣
↪                                                    |
| id                        | ab442634-1cc9-49e5-bd49-0dac9c811f69             ␣
↪                                                    |
| ipv4_address_scope        | None                                             ␣
↪                                                    |
| ipv6_address_scope        | None                                             ␣
↪                                                    |
| is_default                | None                                             ␣
↪                                                    |
| is_vlan_transparent       | None                                             ␣
↪                                                    |
| mtu                       | 1450                                             ␣
↪                                                    |
```

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```
| name                      | net                                                    ␣
↪                          |
| port_security_enabled     | True                                                   ␣
↪                          |
| project_id                | e6710680bfd14555891f265644e1dd5c                       ␣
↪                          |
| provider:network_type     | vxlan                                                  ␣
↪                          |
| provider:physical_network | None                                                   ␣
↪                          |
| provider:segmentation_id  | 1047                                                   ␣
↪                          |
| qos_policy_id             | None                                                   ␣
↪                          |
| revision_number           | 5                                                      ␣
↪                          |
| router:external           | Internal                                               ␣
↪                          |
| segments                  | None                                                   ␣
↪                          |
| shared                    | False                                                  ␣
↪                          |
| status                    | ACTIVE                                                 ␣
↪                          |
| subnets                   |                                                        ␣
↪                          |
| tags                      |                                                        ␣
↪                          |
| updated_at                | 2018-07-16T07:03:02Z                                   ␣
↪                          |
+---------------------------+-----------------------------------------------------
↪--------------------------+
```

Get list of resources with tag filters from networks. The networks are: test-net1 with red tag, test-net2 with red and blue tags, test-net3 with red, blue, and green tags, and test-net4 with green tag.

Get list of resources with `tags` filter:

```
$ openstack network list --tags red,blue
+--------------------------------------+-----------+---------+
| ID                                   | Name      | Subnets |
+--------------------------------------+-----------+---------+
| 8ca3b9ed-f578-45fa-8c44-c53f13aec05a | test-net3 |         |
| e736e63d-42e4-4f4c-836c-6ad286ffd68a | test-net2 |         |
+--------------------------------------+-----------+---------+
```

Get list of resources with `any-tags` filter:

```
$ openstack network list --any-tags red,blue
+--------------------------------------+-----------+---------+
```

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```
| ID                                   | Name      | Subnets |
+--------------------------------------+-----------+---------+
| 30491224-3855-431f-a688-fb29df004d82 | test-net1 |         |
| 8ca3b9ed-f578-45fa-8c44-c53f13aec05a | test-net3 |         |
| e736e63d-42e4-4f4c-836c-6ad286ffd68a | test-net2 |         |
+--------------------------------------+-----------+---------+
```

Get list of resources with `not-tags` filter:

```
$ openstack network list --not-tags red,blue
+--------------------------------------+-----------+---------+
| ID                                   | Name      | Subnets |
+--------------------------------------+-----------+---------+
| 30491224-3855-431f-a688-fb29df004d82 | test-net1 |         |
| cdb3ed08-ca63-4090-ba12-30b366372993 | test-net4 |         |
+--------------------------------------+-----------+---------+
```

Get list of resources with `not-any-tags` filter:

```
$ openstack network list --not-any-tags red,blue
+--------------------------------------+-----------+---------+
| ID                                   | Name      | Subnets |
+--------------------------------------+-----------+---------+
| cdb3ed08-ca63-4090-ba12-30b366372993 | test-net4 |         |
+--------------------------------------+-----------+---------+
```

Limitations

Filtering resources with a tag whose name contains a comma is not supported. Thus, do not put such a tag name to resources.

Future support

In future releases, the Networking service may support setting tags for additional resources.

8.4.3 Resource purge

The Networking service provides a purge mechanism to delete the following network resources for a project:

- Networks
- Subnets
- Ports
- Router interfaces
- Routers
- Floating IP addresses
- Security groups

Typically, one uses this mechanism to delete networking resources for a defunct project regardless of its existence in the Identity service.

Usage

1. Source the necessary project credentials. The administrative project can delete resources for all other projects. A regular project can delete its own network resources and those belonging to other projects for which it has sufficient access.
2. Delete the network resources for a particular project.

   ```
   $ neutron purge PROJECT_ID
   ```

   Replace `PROJECT_ID` with the project ID.

The command provides output that includes a completion percentage and the quantity of successful or unsuccessful network resource deletions. An unsuccessful deletion usually indicates sharing of a resource with one or more additional projects.

```
Purging resources: 100% complete.
Deleted 1 security_group, 2 ports, 1 router, 1 floatingip, 2 networks.
The following resources could not be deleted: 1 network.
```

The command also indicates if a project lacks network resources.

```
Tenant has no supported resources.
```

8.4.4 Manage Networking service quotas

A quota limits the number of available resources. A default quota might be enforced for all projects. When you try to create more resources than the quota allows, an error occurs:

```
$ openstack network create test_net
 Error while executing command: ConflictException: 409, Quota exceeded for␣
↪resources: ['network'].
```

Per-project quota configuration is also supported by the quota extension API. See *Configure per-project quotas* for details.

Basic quota configuration

In the Networking default quota mechanism, all projects have the same quota values, such as the number of resources that a project can create.

The quota value is defined in the OpenStack Networking `/etc/neutron/neutron.conf` configuration file. This example shows the default quota values:

```
[quotas]
# Default number of resources allowed per project. A negative value means
# unlimited. (integer value)
#default_quota = -1

# Number of networks allowed per project. A negative value means unlimited.
# (integer value)
```

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```
quota_network = 100

# Number of subnets allowed per project, A negative value means unlimited.
# (integer value)
quota_subnet = 100

# Number of ports allowed per project. A negative value means unlimited.
# (integer value)
quota_port = 500

# default driver to use for quota checks
quota_driver = neutron.db.quota.driver_nolock.DbQuotaNoLockDriver

# When set to True, quota usage will be tracked in the Neutron database
# for each resource, by directly mapping to a data model class, for
# example, networks, subnets, ports, etc. When set to False, quota usage
# will be tracked by the quota engine as a count of the object type
# directly. For more information, see the Quota Management and
# Enforcement guide.
# (boolean value)
track_quota_usage = true

#
# From neutron.extensions
#

# Number of routers allowed per project. A negative value means unlimited.
# (integer value)
quota_router = 10

# Number of floating IPs allowed per project. A negative value means
# unlimited.
# (integer value)
quota_floatingip = 50

# Number of security groups allowed per project. A negative value means
# unlimited.
# (integer value)
quota_security_group = 10

# Number of security group rules allowed per project. A negative value means
# unlimited.
# (integer value)
quota_security_group_rule = 100
```

Configure per-project quotas

OpenStack Networking also supports per-project quota limit by quota extension API.

Use these commands to manage per-project quotas:

openstack quota delete
Delete defined quotas for a specified project

openstack quota list
Lists defined quotas for all projects with non-default quota values

openstack quota show
Shows defined quotas for all projects

openstack quota show <project>
Shows quotas for a specified project

openstack quota show default <project>
Show default quotas for a specified project

openstack quota set <resource> <value> <project>
Updates quotas for a specified project

Only users with the `admin` role can change a quota value. By default, the default set of quotas are enforced for all projects, so no **`opentack quota create`** command exists.

1. Configure Networking to show per-project quotas

   Set the `quota_driver` option in the `/etc/neutron/neutron.conf` file.

   ```
   quota_driver = neutron.db.quota.driver.DbQuotaDriver
   ```

   When you set this option, the output for Networking commands shows `quotas`.

2. List Networking extensions.

   To list the Networking extensions, run this command:

   ```
   $ openstack extension list --network
   ```

   The command shows the `quotas` extension, which provides per-project quota management support.

   ```
   +------------------------+------------------------+------------------------
   ↪---+
   | Name                   | Alias                  | Description            ␣
   ↪   |
   +------------------------+------------------------+------------------------
   ↪---+
   | ...                    | ...                    | ...                    ␣
   ↪   |
   | Quota management       | quotas                 | Expose functions for ␣
   ↪   |
   | support                |                        | quotas management per␣
   ↪   |
   |                        |                        | project                ␣
   ↪   |
   ```

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```
| ...                     | ...                     | ...                    ␣
↪    |
+-------------------------+-------------------------+------------------------
↪---+
```

3. Show information for the quotas extension.

   To show information for the `quotas` extension, run this command:

```
$ openstack extension show quotas
+-------------+--------------------------------------------------------------
↪-----------------------------------------------------------------------------
↪---------------------------------------------------------+
| Field       | Value                                                       ␣
↪                                                                            ␣
↪                                                         |
+-------------+--------------------------------------------------------------
↪-----------------------------------------------------------------------------
↪---------------------------------------------------------+
| alias       | quotas                                                      ␣
↪                                                                            ␣
↪                                                         |
| description | Expose functions for quotas management per project          ␣
↪                                                                            ␣
↪                                                         |
| id          | quotas                                                      ␣
↪                                                                            ␣
↪                                                         |
| links       | []                                                          ␣
↪                                                                            ␣
↪                                                         |
| location    | Munch({'cloud': '', 'region_name': 'RegionOne', 'zone':␣
↪None, 'project': Munch({'id': 'afc55714081b4ef29f99ec128cb1fa30', 'name
↪': 'demo', 'domain_id': 'default', 'domain_name': None})}) |
| name        | Quota management support                                    ␣
↪                                                                            ␣
↪                                                         |
| updated     | 2012-07-29T10:00:00-00:00                                   ␣
↪                                                                            ␣
↪                                                         |
+-------------+--------------------------------------------------------------
↪-----------------------------------------------------------------------------
↪---------------------------------------------------------+
```

> **Note**
>
> Only some plug-ins support per-project quotas. Specifically, OVN and Open vSwitch support them, but new versions of other plug-ins might bring additional functionality. See the documentation for each plug-in.

4. List projects who have per-project quota support.

   The **openstack quota list** command lists projects for which the per-project quota is enabled. The command does not list projects with default quota support. You must be an administrative user to run this command:

```
$ openstack quota list --network
+----------------------------------+--------------+----------+-------+----
↪----------+---------+-----------------+----------------------+---------
↪+-------------+
| Project ID                       | Floating IPs | Networks | Ports |␣
↪RBAC Policies | Routers | Security Groups | Security Group Rules |␣
↪Subnets | Subnet Pools |
+----------------------------------+--------------+----------+-------+----
↪----------+---------+-----------------+----------------------+---------
↪+-------------+
| 6f88036c45344d9999a1f971e4882723 |           50 |      100 |   500 |   ␣
↪        10 |      20 |              10 |                  100 |     100␣
↪|          -1 |
| bff5c9455ee24231b5bc713c1b96d422 |          100 |      100 |   500 |   ␣
↪        10 |      10 |              10 |                  100 |     100␣
↪|          -1 |
+----------------------------------+--------------+----------+-------+----
↪----------+---------+-----------------+----------------------+---------
↪+-------------+
```

5. Show per-project quota values.

   The **openstack quota show** command reports the current set of quota limits for the specified project. Non-administrative users can run this command without the `<project>` argument. If per-project quota limits are not enabled for the project, the command shows the default set of quotas.

```
$ openstack quota show 6f88036c45344d9999a1f971e4882723
+----------------+-------+
| Resource       | Limit |
+----------------+-------+
| networks       |   100 |
| ports          |   500 |
| rbac_policies  |    10 |
| routers        |    20 |
| subnets        |   100 |
| subnet_pools   |    -1 |
| floating-ips   |    50 |
| secgroup-rules |   100 |
| secgroups      |    10 |
+----------------+-------+
```

   The following command shows the command output for a non-administrative user.

```
$ openstack quota show
+----------------+-------+
| Resource       | Limit |
```

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```
+----------------+-------+
| networks       |   100 |
| ports          |   500 |
| rbac_policies  |    10 |
| routers        |    20 |
| subnets        |   100 |
| subnet_pools   |    -1 |
| floating-ips   |    50 |
| secgroup-rules |   100 |
| secgroups      |    10 |
+----------------+-------+
```

6. Update quota values for a specified project.

   Use the **openstack quota set** command to update a quota for a specified project.

   ```
   $ openstack quota set --routers 20 6f88036c45344d9999a1f971e4882723
   ```

   You can update quotas for multiple resources through one command.

   ```
   $ openstack quota set --subnets 50 --ports 100␣
   ↪6f88036c45344d9999a1f971e4882723
   ```

   You can update the limits of multiple resources through one command:

   ```
   $ openstack quota set --networks 50 --subnets 50 --ports 100 \
     --floating-ips 20 --routers 5 6f88036c45344d9999a1f971e4882723
   ```

7. Delete per-project quota values.

   To clear per-project quota limits, use the **openstack quota delete** command.

   ```
   $ openstack quota delete 6f88036c45344d9999a1f971e4882723
   ```

   After you run this command, you can see that quota values for the project are reset to the default values.

   ```
   $ openstack quota show --network 6f88036c45344d9999a1f971e4882723
   +----------------+-------+
   | Resource       | Limit |
   +----------------+-------+
   | networks       |   100 |
   | ports          |   500 |
   | rbac_policies  |    10 |
   | routers        |    20 |
   | subnets        |   100 |
   | subnet_pools   |    -1 |
   | floating-ips   |    50 |
   | secgroup-rules |   100 |
   | secgroups      |    10 |
   +----------------+-------+
   ```

Note

Listing default quotas with the OpenStack command line client will provide all quotas for networking and other services.

8.5 Migration

8.5.1 Database

The upgrade of the Networking service database is implemented with Alembic migration chains. The migrations in the `alembic/versions` contain the changes needed to migrate from older Networking service releases to newer ones.

Since Liberty, Networking maintains two parallel Alembic migration branches.

The first branch is called expand and is used to store expansion-only migration rules. These rules are strictly additive and can be applied while the Neutron server is running.

The second branch is called contract and is used to store those migration rules that are not safe to apply while Neutron server is running.

The intent of separate branches is to allow invoking those safe migrations from the expand branch while the Neutron server is running and therefore reducing downtime needed to upgrade the service.

A database management command-line tool uses the Alembic library to manage the migration.

Database management command-line tool

The database management command-line tool is called **neutron-db-manage**. Pass the `--help` option to the tool for usage information.

The tool takes some options followed by some commands:

```
$ neutron-db-manage <options> <commands>
```

The tool needs to access the database connection string, which is provided in the `neutron.conf` configuration file in an installation. The tool automatically reads from `/etc/neutron/neutron.conf` if it is present. If the configuration is in a different location, use the following command:

```
$ neutron-db-manage --config-file /path/to/neutron.conf <commands>
```

Multiple `--config-file` options can be passed if needed.

Instead of reading the DB connection from the configuration file(s), you can use the `--database-connection` option:

```
$ neutron-db-manage --database-connection
  mysql+pymysql://root:secret@127.0.0.1/neutron?charset=utf8 <commands>
```

The *branches*, *current*, and *history* commands all accept a `--verbose` option, which, when passed, will instruct **neutron-db-manage** to display more verbose output for the specified command:

```
$ neutron-db-manage current --verbose
```

Note

The tool usage examples below do not show the options. It is assumed that you use the options that you need for your environment.

In new deployments, you start with an empty database and then upgrade to the latest database version using the following command:

```
$ neutron-db-manage upgrade heads
```

After installing a new version of the Neutron server, upgrade the database using the following command:

```
$ neutron-db-manage upgrade heads
```

In existing deployments, check the current database version using the following command:

```
$ neutron-db-manage current
```

To apply the expansion migration rules, use the following command:

```
$ neutron-db-manage upgrade --expand
```

To apply the non-expansive migration rules, use the following command:

```
$ neutron-db-manage upgrade --contract
```

To check if any contract migrations are pending and therefore if offline migration is required, use the following command:

```
$ neutron-db-manage has_offline_migrations
```

Note

Offline migration requires all Neutron server instances in the cluster to be shutdown before you apply any contract scripts.

To generate a script of the command instead of operating immediately on the database, use the following command:

```
$ neutron-db-manage upgrade heads --sql

.. note::

   The `--sql` option causes the command to generate a script.  The script
   can be run later (online or offline), perhaps after verifying and/or
   modifying it.
```

To migrate between specific migration versions, use the following command:

```
$ neutron-db-manage upgrade <start version>:<end version>
```

To upgrade the database incrementally, use the following command:

```
$ neutron-db-manage upgrade --delta <# of revs>
```

Note

Database downgrade is not supported.

To look for differences between the schema generated by the upgrade command and the schema defined by the models, use the **revision --autogenerate** command:

```
neutron-db-manage revision -m REVISION_DESCRIPTION --autogenerate
```

Note

This generates a prepopulated template with the changes needed to match the database state with the models.

8.5.2 Legacy nova-network to OpenStack Networking (neutron)

Two networking models exist in OpenStack. The first is called legacy networking (nova-network) and it is a sub-process embedded in the Compute project (nova). This model has some limitations, such as creating complex network topologies, extending its back-end implementation to vendor-specific technologies, and providing project-specific networking elements. These limitations are the main reasons the OpenStack Networking (neutron) model was created.

This section describes the process of migrating clouds based on the legacy networking model to the OpenStack Networking model. This process requires additional changes to both compute and networking to support the migration. This document describes the overall process and the features required in both Networking and Compute.

The current process as designed is a minimally viable migration with the goal of deprecating and then removing legacy networking. Both the Compute and Networking teams agree that a one-button migration process from legacy networking to OpenStack Networking (neutron) is not an essential requirement for the deprecation and removal of the legacy networking at a future date. This section includes a process and tools which are designed to solve a simple use case migration.

Users are encouraged to take these tools, test them, provide feedback, and then expand on the feature set to suit their own deployments; deployers that refrain from participating in this process intending to wait for a path that better suits their use case are likely to be disappointed.

Impact and limitations

The migration process from the legacy nova-network networking service to OpenStack Networking (neutron) has some limitations and impacts on the operational state of the cloud. It is critical to understand them in order to decide whether or not this process is acceptable for your cloud and all users.

Management impact

The Networking REST API is publicly read-only until after the migration is complete. During the migration, Networking REST API is read-write only to nova-api, and changes to Networking are only allowed via nova-api.

The Compute REST API is available throughout the entire process, although there is a brief period where it is made read-only during a database migration. The Networking REST API will need to expose (to nova-api) all details necessary for reconstructing the information previously held in the legacy networking database.

Compute needs a per-hypervisor has_transitioned boolean change in the data model to be used during the migration process. This flag is no longer required once the process is complete.

Operations impact

In order to support a wide range of deployment options, the migration process described here requires a rolling restart of hypervisors. The rate and timing of specific hypervisor restarts is under the control of the operator.

The migration may be paused, even for an extended period of time (for example, while testing or investigating issues) with some hypervisors on legacy networking and some on Networking, and Compute API remains fully functional. Individual hypervisors may be rolled back to legacy networking during this stage of the migration, although this requires an additional restart.

In order to support the widest range of deployer needs, the process described here is easy to automate but is not already automated. Deployers should expect to perform multiple manual steps or write some simple scripts in order to perform this migration.

Performance impact

During the migration, nova-network API calls will go through an additional internal conversion to Networking calls. This will have different and likely poorer performance characteristics compared with either the pre-migration or post-migration APIs.

Migration process overview

1. Start neutron-server in intended final config, except with REST API restricted to read-write only by nova-api.
2. Make the Compute REST API read-only.
3. Run a DB dump/restore tool that creates Networking data structures representing current legacy networking config.
4. Enable a nova-api proxy that recreates internal Compute objects from Networking information (via the Networking REST API).
5. Make Compute REST API read-write again. This means legacy networking DB is now unused, new changes are now stored in the Networking DB, and no rollback is possible from here without losing those new changes.

Note

At this moment the Networking DB is the source of truth, but nova-api is the only public read-write API.

Next, youll need to migrate each hypervisor. To do that, follow these steps:

1. Disable the hypervisor. This would be a good time to live migrate or evacuate the compute node, if supported.
2. Disable nova-compute.
3. Enable the Networking agent.
4. Set the has_transitioned flag in the Compute hypervisor database/config.
5. Reboot the hypervisor (or run smart live transition tool if available).
6. Re-enable the hypervisor.

At this point, all compute nodes have been migrated, but they are still using the nova-api API and Compute gateways. Finally, enable OpenStack Networking by following these steps:

1. Bring up the Networking (l3) nodes. The new routers will have identical MAC+IPs as old Compute gateways so some sort of immediate cutover is possible, except for stateful connections issues such as NAT.
2. Make the Networking API read-write and disable legacy networking.

Migration Completed!

8.5.3 Add VRRP to an existing router

This section describes the process of migrating from a classic router to an L3 HA router, which is available starting from the Mitaka release.

Similar to the classic scenario, all network traffic on a project network that requires routing actively traverses only one network node regardless of the quantity of network nodes providing HA for the router. Therefore, this high-availability implementation primarily addresses failure situations instead of bandwidth constraints that limit performance. However, it supports random distribution of routers on different network nodes to reduce the chances of bandwidth constraints and to improve scaling.

This section references parts of *Open vSwitch: High availability using VRRP*. For details regarding needed infrastructure and configuration to allow actual L3 HA deployment, read the relevant guide before continuing with the migration process.

Migration

The migration process is quite simple, it involves turning down the router by setting the routers `admin_state_up` attribute to `False`, upgrading the router to L3 HA and then setting the routers `admin_state_up` attribute back to `True`.

Warning

Once starting the migration, south-north connections (instances to internet) will be severed. New connections will be able to start only when the migration is complete.

Here is the router we have used in our demonstration:

```
$ openstack router show router1
+-------------------------+-------------------------------------------+
| Field                   | Value                                     |
+-------------------------+-------------------------------------------+
| admin_state_up          | UP                                        |
| distributed             | False                                     |
| external_gateway_info   |                                           |
| ha                      | False                                     |
| id                      | 6b793b46-d082-4fd5-980f-a6f80cbb0f2a      |
| name                    | router1                                   |
| project_id              | bb8b84ab75be4e19bd0dfe02f6c3f5c1          |
| routes                  |                                           |
| status                  | ACTIVE                                    |
| tags                    | []                                        |
+-------------------------+-------------------------------------------+
```

1. Source the administrative project credentials.
2. Set the admin_state_up to `False`. This will severe south-north connections until admin_state_up is set to `True` again.

   ```
   $ openstack router set router1 --disable
   ```

3. Set the `ha` attribute of the router to `True`.

   ```
   $ openstack router set router1 --ha
   ```

4. Set the admin_state_up to `True`. After this, south-north connections can start.

   ```
   $ openstack router set router1 --enable
   ```

5. Make sure that the routers `ha` attribute has changed to `True`.

   ```
   $ openstack router show router1
   +-------------------------+-------------------------------------------+
   | Field                   | Value                                     |
   +-------------------------+-------------------------------------------+
   | admin_state_up          | UP                                        |
   | distributed             | False                                     |
   | external_gateway_info   |                                           |
   | ha                      | True                                      |
   | id                      | 6b793b46-d082-4fd5-980f-a6f80cbb0f2a      |
   | name                    | router1                                   |
   | project_id              | bb8b84ab75be4e19bd0dfe02f6c3f5c1          |
   | routes                  |                                           |
   | status                  | ACTIVE                                    |
   | tags                    | []                                        |
   +-------------------------+-------------------------------------------+
   ```

L3 HA to Legacy

To return to classic mode, turn down the router again, turning off L3 HA and starting the router again.

Warning

Once starting the migration, south-north connections (instances to internet) will be severed. New connections will be able to start only when the migration is complete.

Here is the router we have used in our demonstration:

```
$ openstack router show router1
+-------------------------+-------------------------------------------+
| Field                   | Value                                     |
+-------------------------+-------------------------------------------+
| admin_state_up          | DOWN                                      |
| distributed             | False                                     |
| external_gateway_info   |                                           |
| ha                      | True                                      |
| id                      | 6b793b46-d082-4fd5-980f-a6f80cbb0f2a      |
| name                    | router1                                   |
| project_id              | bb8b84ab75be4e19bd0dfe02f6c3f5c1          |
| routes                  |                                           |
| status                  | ACTIVE                                    |
| tags                    | []                                        |
+-------------------------+-------------------------------------------+
```

1. Source the administrative project credentials.
2. Set the admin_state_up to `False`. This will severe south-north connections until admin_state_up is set to `True` again.

   ```
   $ openstack router set router1 --disable
   ```

3. Set the `ha` attribute of the router to `True`.

   ```
   $ openstack router set router1 --no-ha
   ```

4. Set the admin_state_up to `True`. After this, south-north connections can start.

   ```
   $ openstack router set router1 --enable
   ```

5. Make sure that the routers `ha` attribute has changed to `False`.

   ```
   $ openstack router show router1
   +-------------------------+-------------------------------------------+
   | Field                   | Value                                     |
   +-------------------------+-------------------------------------------+
   | admin_state_up          | UP                                        |
   | distributed             | False                                     |
   | external_gateway_info   |                                           |
   | ha                      | False                                     |
   ```

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```
| id                      | 6b793b46-d082-4fd5-980f-a6f80cbb0f2a      |
| name                    | router1                                   |
| project_id              | bb8b84ab75be4e19bd0dfe02f6c3f5c1          |
| routes                  |                                           |
| status                  | ACTIVE                                    |
| tags                    | []                                        |
+-------------------------+-------------------------------------------+
```

8.6 Miscellaneous

8.6.1 Disable libvirt networking

Most OpenStack deployments use the libvirt toolkit for interacting with the hypervisor. Specifically, OpenStack Compute uses libvirt for tasks such as booting and terminating virtual machine instances. When OpenStack Compute boots a new instance, libvirt provides OpenStack with the VIF associated with the instance, and OpenStack Compute plugs the VIF into a virtual device provided by OpenStack Network. The libvirt toolkit itself does not provide any networking functionality in OpenStack deployments.

However, libvirt is capable of providing networking services to the virtual machines that it manages. In particular, libvirt can be configured to provide networking functionality akin to a simplified, single-node version of OpenStack. Users can use libvirt to create layer 2 networks that are similar to OpenStack Networkings networks, confined to a single node.

libvirt network implementation

By default, libvirts networking functionality is enabled, and libvirt creates a network when the system boots. To implement this network, libvirt leverages some of the same technologies that OpenStack Network does. In particular, libvirt uses:

- Linux bridging for implementing a layer 2 network
- dnsmasq for providing IP addresses to virtual machines using DHCP
- iptables to implement SNAT so instances can connect out to the public internet, and to ensure that virtual machines are permitted to communicate with dnsmasq using DHCP

By default, libvirt creates a network named *default*. The details of this network may vary by distribution; on Ubuntu this network involves:

- a Linux bridge named `virbr0` with an IP address of `192.0.2.1/24`
- a dnsmasq process that listens on the `virbr0` interface and hands out IP addresses in the range `192.0.2.2-192.0.2.254`
- a set of iptables rules

When libvirt boots a virtual machine, it places the machines VIF in the bridge `virbr0` unless explicitly told not to.

On Ubuntu, the iptables ruleset that libvirt creates includes the following rules:

```
*nat
-A POSTROUTING -s 192.0.2.0/24 -d 224.0.0.0/24 -j RETURN
```

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```
-A POSTROUTING -s 192.0.2.0/24 -d 255.255.255.255/32 -j RETURN
-A POSTROUTING -s 192.0.2.0/24 ! -d 192.0.2.0/24 -p tcp -j MASQUERADE --to-
↪ports 1024-65535
-A POSTROUTING -s 192.0.2.0/24 ! -d 192.0.2.0/24 -p udp -j MASQUERADE --to-
↪ports 1024-65535
-A POSTROUTING -s 192.0.2.0/24 ! -d 192.0.2.0/24 -j MASQUERADE
*mangle
-A POSTROUTING -o virbr0 -p udp -m udp --dport 68 -j CHECKSUM --checksum-fill
*filter
-A INPUT -i virbr0 -p udp -m udp --dport 53 -j ACCEPT
-A INPUT -i virbr0 -p tcp -m tcp --dport 53 -j ACCEPT
-A INPUT -i virbr0 -p udp -m udp --dport 67 -j ACCEPT
-A INPUT -i virbr0 -p tcp -m tcp --dport 67 -j ACCEPT
-A FORWARD -d 192.0.2.0/24 -o virbr0 -m conntrack --ctstate RELATED,
↪ESTABLISHED -j ACCEPT
-A FORWARD -s 192.0.2.0/24 -i virbr0 -j ACCEPT
-A FORWARD -i virbr0 -o virbr0 -j ACCEPT
-A FORWARD -o virbr0 -j REJECT --reject-with icmp-port-unreachable
-A FORWARD -i virbr0 -j REJECT --reject-with icmp-port-unreachable
-A OUTPUT -o virbr0 -p udp -m udp --dport 68 -j ACCEPT
```

The following shows the dnsmasq process that libvirt manages as it appears in the output of **ps**:

```
2881 ?        S      0:00 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/
↪dnsmasq/default.conf
```

How to disable libvirt networks

Although OpenStack does not make use of libvirts networking, this networking will not interfere with OpenStacks behavior, and can be safely left enabled. However, libvirts networking can be a nuisance when debugging OpenStack networking issues. Because libvirt creates an additional bridge, dnsmasq process, and iptables ruleset, these may distract an operator engaged in network troubleshooting. Unless you need to start up virtual machines using libvirt directly, you can safely disable libvirts network.

To view the defined libvirt networks and their state:

```
# virsh net-list
 Name                 State      Autostart     Persistent
----------------------------------------------------------
 default              active     yes           yes
```

To deactivate the libvirt network named `default`:

```
# virsh net-destroy default
```

Deactivating the network will remove the `virbr0` bridge, terminate the dnsmasq process, and remove the iptables rules.

To prevent the network from automatically starting on boot:

```
# virsh net-autostart --network default --disable
```

To activate the network after it has been deactivated:

```
# virsh net-start default
```

8.6.2 Virtual Private Network-as-a-Service (VPNaaS) scenario

Enabling VPNaaS

This section describes the setting for the reference implementation. Vendor plugins or drivers can have different setup procedure and perhaps they provide their version of manuals.

1. Enable the VPNaaS plug-in in the `/etc/neutron/neutron.conf` file by appending `vpnaas` to `service_plugins` in `[DEFAULT]`:

   ```
   [DEFAULT]
   # ...
   service_plugins = vpnaas
   ```

   Note

   `vpnaas` is just example of reference implementation. It depends on a plugin that you are going to use. Consider to set suitable plugin for your own deployment.

2. Configure the VPNaaS service provider by creating the `/etc/neutron/neutron_vpnaas.conf` file as follows, `strongswan` used in Ubuntu distribution:

   ```
   [service_providers]
   service_provider = VPN:strongswan:neutron_vpnaas.services.vpn.service_
   ↪drivers.ipsec.IPsecVPNDriver:default
   ```

   Note

   There are several kinds of service drivers. Depending upon the Linux distribution, you may need to override this value. Select `libreswan` for RHEL/CentOS, the config will like this: `service_provider = VPN:openswan:neutron_vpnaas.services.vpn.service_drivers.ipsec.IPsecVPNDriver:default`. Consider to use the appropriate one for your deployment.

3. Configure the VPNaaS plugin for the L3 agent by adding to `/etc/neutron/l3_agent.ini` the following section, `StrongSwanDriver` used in Ubuntu distribution:

   ```
   [AGENT]
   extensions = vpnaas

   [vpnagent]
   vpn_device_driver = neutron_vpnaas.services.vpn.device_drivers.strongswan_
   ↪ipsec.StrongSwanDriver
   ```

Note

There are several kinds of device drivers. Depending upon the Linux distribution, you may need to override this value. Select `LibreSwanDriver` for RHEL/CentOS, the config will like this: `vpn_device_driver = neutron_vpnaas.services.vpn.device_drivers.libreswan_ipsec.LibreSwanDriver`. Consider to use the appropriate drivers for your deployment.

4. Create the required tables in the database:

```
# neutron-db-manage --subproject neutron-vpnaas upgrade head
```

Note

In order to run the above command, you need to have neutron-vpnaas package installed on controller node.

5. Restart the `neutron-server` in controller node to apply the settings.

6. Restart the `neutron-l3-agent` in network node to apply the settings.

Using VPNaaS with endpoint group (recommended)

IPsec site-to-site connections will support multiple local subnets, in addition to the current multiple peer CIDRs. The multiple local subnet feature is triggered by not specifying a local subnet, when creating a VPN service. Backwards compatibility is maintained with single local subnets, by providing the subnet in the VPN service creation.

To support multiple local subnets, a new capability called End Point Groups has been added. Each endpoint group will define one or more endpoints of a specific type, and can be used to specify both local and peer endpoints for IPsec connections. The endpoint groups separate the what gets connected from the how to connect for a VPN service, and can be used for different flavors of VPN, in the future.

Refer Multiple Local Subnets for more detail.

Create the IKE policy, IPsec policy, VPN service, local endpoint group and peer endpoint group. Then, create an IPsec site connection that applies the above policies and service.

1. Create an IKE policy:

```
$ openstack vpn ike policy create ikepolicy
  +-------------------------------+----------------------------------------
↪-+
  | Field                         | Value                                  ␣
↪ |
  +-------------------------------+----------------------------------------
↪-+
  | Authentication Algorithm      | sha1                                   ␣
↪ |
  | Description                   |                                        ␣
↪ |
  | Encryption Algorithm          | aes-128                                ␣
```

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```
↪ |
  | ID                            | 735f4691-3670-43b2-b389-f4d81a60ed56 ␣
↪ |
  | IKE Version                   | v1                                   ␣
↪ |
  | Lifetime                      | {u'units': u'seconds', u'value': 3600}
↪ |
  | Name                          | ikepolicy                            ␣
↪ |
  | Perfect Forward Secrecy (PFS) | group5                               ␣
↪ |
  | Phase1 Negotiation Mode       | main                                 ␣
↪ |
  | Project                       | 095247cb2e22455b9850c6efff407584     ␣
↪ |
  | project_id                    | 095247cb2e22455b9850c6efff407584     ␣
↪ |
  +-------------------------------+--------------------------------------
↪-+
```

2. Create an IPsec policy:

```
$ openstack vpn ipsec policy create ipsecpolicy
  +-------------------------------+--------------------------------------
↪-+
  | Field                         | Value                                ␣
↪ |
  +-------------------------------+--------------------------------------
↪-+
  | Authentication Algorithm      | sha1                                 ␣
↪ |
  | Description                   |                                      ␣
↪ |
  | Encapsulation Mode            | tunnel                               ␣
↪ |
  | Encryption Algorithm          | aes-128                              ␣
↪ |
  | ID                            | 4f3f46fc-f2dc-4811-a642-9601ebae310f ␣
↪ |
  | Lifetime                      | {u'units': u'seconds', u'value': 3600}
↪ |
  | Name                          | ipsecpolicy                          ␣
↪ |
  | Perfect Forward Secrecy (PFS) | group5                               ␣
↪ |
  | Project                       | 095247cb2e22455b9850c6efff407584     ␣
↪ |
  | Transform Protocol            | esp                                  ␣
↪ |
  | project_id                    | 095247cb2e22455b9850c6efff407584     ␣
```

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```
↪ |
  +----------------------------+-----------------------------------
↪-+
```

3. Create a VPN service:

```
$ openstack vpn service create vpn \
  --router 9ff3f20c-314f-4dac-9392-defdbbb36a66
  +----------------+--------------------------------------+
  | Field          | Value                                |
  +----------------+--------------------------------------+
  | Description    |                                      |
  | Flavor         | None                                 |
  | ID             | 9f499f9f-f672-4ceb-be3c-d5ff3858c680 |
  | Name           | vpn                                  |
  | Project        | 095247cb2e22455b9850c6efff407584     |
  | Router         | 9ff3f20c-314f-4dac-9392-defdbbb36a66 |
  | State          | True                                 |
  | Status         | PENDING_CREATE                       |
  | Subnet         | None                                 |
  | external_v4_ip | 192.168.20.7                         |
  | external_v6_ip | 2001:db8::7                          |
  | project_id     | 095247cb2e22455b9850c6efff407584     |
  +----------------+--------------------------------------+
```

Note

Please do not specify `--subnet` option in this case.

The Networking openstackclient requires a router (Name or ID) and name.

4. Create local endpoint group:

```
$ openstack vpn endpoint group create ep_subnet \
  --type subnet \
  --value 1f888dd0-2066-42a1-83d7-56518895e47d
  +-------------+-------------------------------------------+
  | Field       | Value                                     |
  +-------------+-------------------------------------------+
  | Description |                                           |
  | Endpoints   | [u'1f888dd0-2066-42a1-83d7-56518895e47d'] |
  | ID          | 667296d0-67ca-4d0f-b676-7650cf96e7b1      |
  | Name        | ep_subnet                                 |
  | Project     | 095247cb2e22455b9850c6efff407584          |
  | Type        | subnet                                    |
  | project_id  | 095247cb2e22455b9850c6efff407584          |
  +-------------+-------------------------------------------+
```

Note

The type of a local endpoint group must be `subnet`.

5. Create peer endpoint group:

```
$ openstack vpn endpoint group create ep_cidr \
  --type cidr \
  --value 192.168.1.0/24
  +-------------+--------------------------------------+
  | Field       | Value                                |
  +-------------+--------------------------------------+
  | Description |                                      |
  | Endpoints   | [u'192.168.1.0/24']                  |
  | ID          | 5c3d7f2a-4a2a-446b-9fcf-9a2557cfc641 |
  | Name        | ep_cidr                              |
  | Project     | 095247cb2e22455b9850c6efff407584     |
  | Type        | cidr                                 |
  | project_id  | 095247cb2e22455b9850c6efff407584     |
  +-------------+--------------------------------------+
```

Note

The type of a peer endpoint group must be `cidr`.

6. Create an ipsec site connection:

```
$ openstack vpn ipsec site connection create conn \
  --vpnservice vpn \
  --ikepolicy ikepolicy \
  --ipsecpolicy ipsecpolicy \
  --peer-address 192.168.20.9 \
  --peer-id 192.168.20.9 \
  --psk secret \
  --local-endpoint-group ep_subnet \
  --peer-endpoint-group ep_cidr
  +--------------------------+--------------------------------------------
↪-----------+
  | Field                    | Value                                      ␣
↪           |
  +--------------------------+--------------------------------------------
↪-----------+
  | Authentication Algorithm | psk                                        ␣
↪           |
  | Description              |                                            ␣
↪           |
  | ID                       | 07e400b7-9de3-4ea3-a9d0-90a185e5b00d       ␣
↪           |
  | IKE Policy               | 735f4691-3670-43b2-b389-f4d81a60ed56       ␣
```

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```
↪             |
 | IPSec Policy           | 4f3f46fc-f2dc-4811-a642-9601ebae310f       ␣
↪             |
 | Initiator              | bi-directional                             ␣
↪             |
 | Local Endpoint Group ID | 667296d0-67ca-4d0f-b676-7650cf96e7b1      ␣
↪             |
 | Local ID               |                                            ␣
↪             |
 | MTU                    | 1500                                       ␣
↪             |
 | Name                   | conn                                       ␣
↪             |
 | Peer Address           | 192.168.20.9                               ␣
↪             |
 | Peer CIDRs             |                                            ␣
↪             |
 | Peer Endpoint Group ID | 5c3d7f2a-4a2a-446b-9fcf-9a2557cfc641       ␣
↪             |
 | Peer ID                | 192.168.20.9                               ␣
↪             |
 | Pre-shared Key         | secret                                     ␣
↪             |
 | Project                | 095247cb2e22455b9850c6efff407584           ␣
↪             |
 | Route Mode             | static                                     ␣
↪             |
 | State                  | True                                       ␣
↪             |
 | Status                 | PENDING_CREATE                             ␣
↪             |
 | VPN Service            | 9f499f9f-f672-4ceb-be3c-d5ff3858c680       ␣
↪             |
 | dpd                    | {u'action': u'hold', u'interval': 30, u
↪'timeout': 120} |
 | project_id             | 095247cb2e22455b9850c6efff407584           ␣
↪             |
 +------------------------+-------------------------------------------
↪------------+
```

Note

Please do not specify `--peer-cidr` option in this case. Peer CIDR(s) are provided by a peer endpoint group.

Configure VPNaaS without endpoint group (the legacy way)

Create the IKE policy, IPsec policy, VPN service. Then, create an ipsec site connection that applies the above policies and service.

1. Create an IKE policy:

```
$ openstack vpn ike policy create ikepolicy1
 +-------------------------------+----------------------------------------
↪-+
 | Field                         | Value                                  ␣
↪ |
 +-------------------------------+----------------------------------------
↪-+
 | Authentication Algorithm      | sha1                                   ␣
↪ |
 | Description                   |                                        ␣
↪ |
 | Encryption Algorithm          | aes-128                                ␣
↪ |
 | ID                            | 99e4345d-8674-4d73-acb4-0e2524425e34 ␣
↪ |
 | IKE Version                   | v1                                     ␣
↪ |
 | Lifetime                      | {u'units': u'seconds', u'value': 3600}
↪ |
 | Name                          | ikepolicy1                             ␣
↪ |
 | Perfect Forward Secrecy (PFS) | group5                                 ␣
↪ |
 | Phase1 Negotiation Mode       | main                                   ␣
↪ |
 | Project                       | 095247cb2e22455b9850c6efff407584     ␣
↪ |
 | project_id                    | 095247cb2e22455b9850c6efff407584     ␣
↪ |
 +-------------------------------+----------------------------------------
↪-+
```

2. Create an IPsec policy:

```
$ openstack vpn ipsec policy create ipsecpolicy1
 +-------------------------------+----------------------------------------
↪-+
 | Field                         | Value                                  ␣
↪ |
 +-------------------------------+----------------------------------------
↪-+
 | Authentication Algorithm      | sha1                                   ␣
↪ |
 | Description                   |                                        ␣
↪ |
```

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```
| Encapsulation Mode            | tunnel                                  ␣
↪ |
| Encryption Algorithm          | aes-128                                 ␣
↪ |
| ID                            | e6f547af-4a1d-4c28-b40b-b97cce746459 ␣
↪ |
| Lifetime                      | {u'units': u'seconds', u'value': 3600}
↪ |
| Name                          | ipsecpolicy1                            ␣
↪ |
| Perfect Forward Secrecy (PFS) | group5                                  ␣
↪ |
| Project                       | 095247cb2e22455b9850c6efff407584        ␣
↪ |
| Transform Protocol            | esp                                     ␣
↪ |
| project_id                    | 095247cb2e22455b9850c6efff407584        ␣
↪ |
+-------------------------------+-----------------------------------------
↪-+
```

3. Create a VPN service:

```
$ openstack vpn service create vpn \
  --router 66ca673a-cbbd-48b7-9fb6-bfa7ee3ef724 \
  --subnet cdfb411e-e818-466a-837c-7f96fc41a6d9
  +----------------+--------------------------------------+
  | Field          | Value                                |
  +----------------+--------------------------------------+
  | Description    |                                      |
  | Flavor         | None                                 |
  | ID             | 79ef6250-ddc3-428f-88c2-0ec8084f4e9a |
  | Name           | vpn                                  |
  | Project        | 095247cb2e22455b9850c6efff407584     |
  | Router         | 66ca673a-cbbd-48b7-9fb6-bfa7ee3ef724 |
  | State          | True                                 |
  | Status         | PENDING_CREATE                       |
  | Subnet         | cdfb411e-e818-466a-837c-7f96fc41a6d9 |
  | external_v4_ip | 192.168.20.2                         |
  | external_v6_ip | 2001:db8::d                          |
  | project_id     | 095247cb2e22455b9850c6efff407584     |
  +----------------+--------------------------------------+
```

Note

The `--subnet` option is required in this scenario.

4. Create an ipsec site connection:

```
$ openstack vpn ipsec site connection create conn \
  --vpnservice vpn \
  --ikepolicy ikepolicy1 \
  --ipsecpolicy ipsecpolicy1 \
  --peer-address 192.168.20.11 \
  --peer-id 192.168.20.11 \
  --peer-cidr 192.168.1.0/24 \
  --psk secret
  +--------------------------+----------------------------------------------
↪------------+
  | Field                    | Value                                        ␣
↪            |
  +--------------------------+----------------------------------------------
↪------------+
  | Authentication Algorithm | psk                                          ␣
↪            |
  | Description              |                                              ␣
↪            |
  | ID                       | 5b2935e6-b2f0-423a-8156-07ed48703d13         ␣
↪            |
  | IKE Policy               | 99e4345d-8674-4d73-acb4-0e2524425e34         ␣
↪            |
  | IPSec Policy             | e6f547af-4a1d-4c28-b40b-b97cce746459         ␣
↪            |
  | Initiator                | bi-directional                               ␣
↪            |
  | Local Endpoint Group ID  | None                                         ␣
↪            |
  | Local ID                 |                                              ␣
↪            |
  | MTU                      | 1500                                         ␣
↪            |
  | Name                     | conn                                         ␣
↪            |
  | Peer Address             | 192.168.20.11                                ␣
↪            |
  | Peer CIDRs               | 192.168.1.0/24                               ␣
↪            |
  | Peer Endpoint Group ID   | None                                         ␣
↪            |
  | Peer ID                  | 192.168.20.11                                ␣
↪            |
  | Pre-shared Key           | secret                                       ␣
↪            |
  | Project                  | 095247cb2e22455b9850c6efff407584             ␣
↪            |
  | Route Mode               | static                                       ␣
↪            |
  | State                    | True                                         ␣
↪            |
```

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```
| Status                    | PENDING_CREATE                                ␣
↪            |
| VPN Service               | 79ef6250-ddc3-428f-88c2-0ec8084f4e9a          ␣
↪            |
| dpd                       | {u'action': u'hold', u'interval': 30, u
↪'timeout': 120} |
| project_id                | 095247cb2e22455b9850c6efff407584              ␣
↪            |
+---------------------------+---------------------------------------------
↪------------+
```

Note

Please do not specify `--local-endpoint-group` and `--peer-endpoint-group` options in this case.

8.7 OVN Driver Administration Guide

8.7.1 OVN information

The original OVN project announcement can be found here:

- https://networkheresy.wordpress.com/2015/01/13/ovn-bringing-native-virtual-networking-to-ovs/

The OVN architecture is described here:

- http://www.ovn.org/support/dist-docs/ovn-architecture.7.html

Here are two tutorials that help with learning different aspects of OVN:

- https://blog.spinhirne.com/posts/an-introduction-to-ovn/a-primer-on-ovn/
- https://docs.ovn.org/en/stable/tutorials/ovn-sandbox.html

There is also an in depth tutorial on using OVN with OpenStack:

- https://docs.ovn.org/en/stable/tutorials/ovn-openstack.html

OVN DB schemas and other man pages:

- http://www.ovn.org/support/dist-docs/ovn-nb.5.html
- http://www.ovn.org/support/dist-docs/ovn-sb.5.html
- http://www.ovn.org/support/dist-docs/ovn-nbctl.8.html
- http://www.ovn.org/support/dist-docs/ovn-sbctl.8.html
- http://www.ovn.org/support/dist-docs/ovn-northd.8.html
- http://www.ovn.org/support/dist-docs/ovn-controller.8.html
- http://www.ovn.org/support/dist-docs/ovn-controller-vtep.8.html

or find a full list of OVS and OVN man pages here:

- http://docs.ovn.org/en/latest/ref/

The openvswitch web page includes a list of presentations, some of which are about OVN:

- http://openvswitch.org/support/

Here are some direct links to past OVN presentations:

- OVN talk at OpenStack Summit in Boston, Spring 2017
- OVN talk at OpenStack Summit in Barcelona, Fall 2016
- OVN talk at OpenStack Summit in Austin, Spring 2016
- OVN Project Update at the OpenStack Summit in Tokyo, Fall 2015 - Slides - Video
- OVN at OpenStack Summit in Vancouver, Sping 2015 - Slides - Video
- OVS Conference 2015

These blog resources may also help with testing and understanding OVN:

- http://networkop.co.uk/blog/2016/11/27/ovn-part1/
- http://networkop.co.uk/blog/2016/12/10/ovn-part2/
- https://blog.russellbryant.net/2016/12/19/comparing-openstack-neutron-ml2ovs-and-ovn-control-plane/
- https://blog.russellbryant.net/2016/11/11/ovn-logical-flows-and-ovn-trace/
- https://blog.russellbryant.net/2016/09/29/ovs-2-6-and-the-first-release-of-ovn/
- http://galsagie.github.io/2015/11/23/ovn-l3-deepdive/
- http://blog.russellbryant.net/2015/10/22/openstack-security-groups-using-ovn-acls/
- http://galsagie.github.io/sdn/openstack/ovs/2015/05/30/ovn-deep-dive/
- http://blog.russellbryant.net/2015/05/14/an-ez-bake-ovn-for-openstack/
- http://galsagie.github.io/sdn/openstack/ovs/2015/04/26/ovn-containers/
- http://blog.russellbryant.net/2015/04/21/ovn-and-openstack-status-2015-04-21/
- http://blog.russellbryant.net/2015/04/08/ovn-and-openstack-integration-development-update/
- http://dani.foroselectronica.es/category/openstack/ovn/

8.7.2 Features

Open Virtual Network (OVN) offers the following virtual network services:

- Layer-2 (switching)

  Native implementation. Replaces the conventional Open vSwitch (OVS) agent.

- Layer-3 (routing)

  Native implementation that supports distributed routing. Replaces the conventional Neutron L3 agent. This includes transparent L3HA :doc::*routing* support, based on BFD monitorization integrated in core OVN.

- DHCP

  Native distributed implementation. Replaces the conventional Neutron DHCP agent. Note that the native implementation does not yet support DNS features.

- DPDK

  OVN and the OVN mechanism driver may be used with OVS using either the Linux kernel datapath or the DPDK datapath.

- Trunk driver

  Uses OVNs functionality of parent port and port tagging to support trunk service plugin. One has to enable the trunk service plugin in neutron configuration files to use this feature.

- VLAN tenant networks

  The OVN driver does support VLAN tenant networks when used with OVN version 2.11 (or higher).

- DNS

  Native implementation. Since the version 2.8 OVN contains a built-in DNS implementation.

- Port Forwarding

  The OVN driver supports port forwarding as an extension of floating IPs. Enable the port_forwarding service plugin in neutron configuration files to use this feature.

- Packet Logging

  Packet logging service is designed as a Neutron plug-in that captures network packets for relevant resources when the registered events occur. OVN supports this feature based on security groups.

- Segments

  Allows for Network segments ranges to be used with OVN. Requires OVN version 20.06 or higher.

- Routed provider networks

  Allows for multiple localnet ports to be attached to a single Logical Switch entry. This work also assumes that only a single localnet port (of the same Logical Switch) is actually mapped to a given hypervisor. Requires OVN version 20.06 or higher.

The following Neutron API extensions are supported with OVN:

Extension Name	Extension Alias
Allowed Address Pairs	allowed-address-pairs
Auto Allocated Topology Services	auto-allocated-topology
Availability Zone	availability_zone
Default Subnetpools	default-subnetpools
DNS Integration	dns-integration
DNS domain for ports	dns-domain-ports
DNS domain names with keywords	dns-integration-domain-keywords
Subnet DNS publish fixed IP	subnet-dns-publish-fixed-ip
Multi Provider Network	multi-provider
Network IP Availability	network-ip-availability
Network Segment	segment
Neutron external network	external-net
Neutron Extra DHCP opts	extra_dhcp_opt
Neutron Extra Route	extraroute
Neutron L3 external gateway	ext-gw-mode

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Table 11 – continued from previous page

Extension Name	Extension Alias
Neutron L3 Router	router
Network MTU	net-mtu
Packet Logging	logging
Port Binding	binding
Port Bindings Extended	binding-extended
Port Forwarding	port_forwarding
Port MAC address Regenerate	port-mac-address-regenerate
Port Security	port-security
Provider Network	provider
Quality of Service	qos
Quota management support	quotas
RBAC Policies	rbac-policies
Resource revision numbers	standard-attr-revisions
security-group	security-group
standard-attr-description	standard-attr-description
Subnet Allocation	subnet_allocation
Subnet service types	subnet-service-types
Tag support	standard-attr-tag
Time Stamp Fields	standard-attr-timestamp

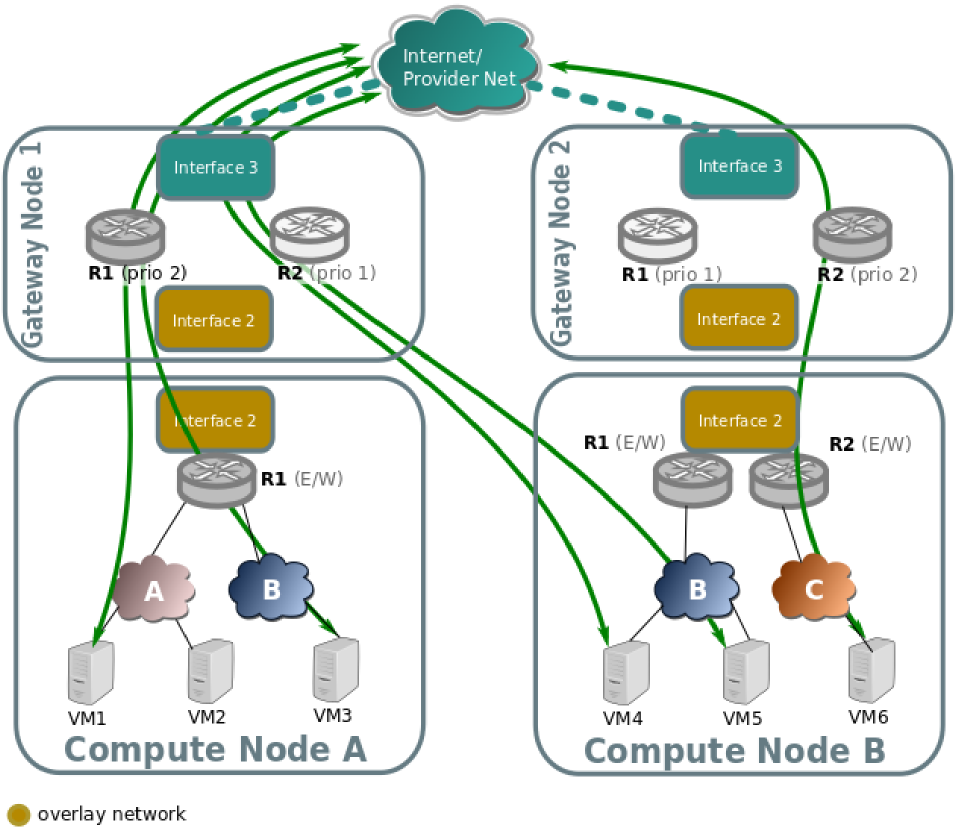
8.7.3 Routing

North/South

The different configurations are detailed in the *Reference architecture*

Non distributed FIP

North/South traffic flows through the active chassis for each router for SNAT traffic, and also for FIPs.



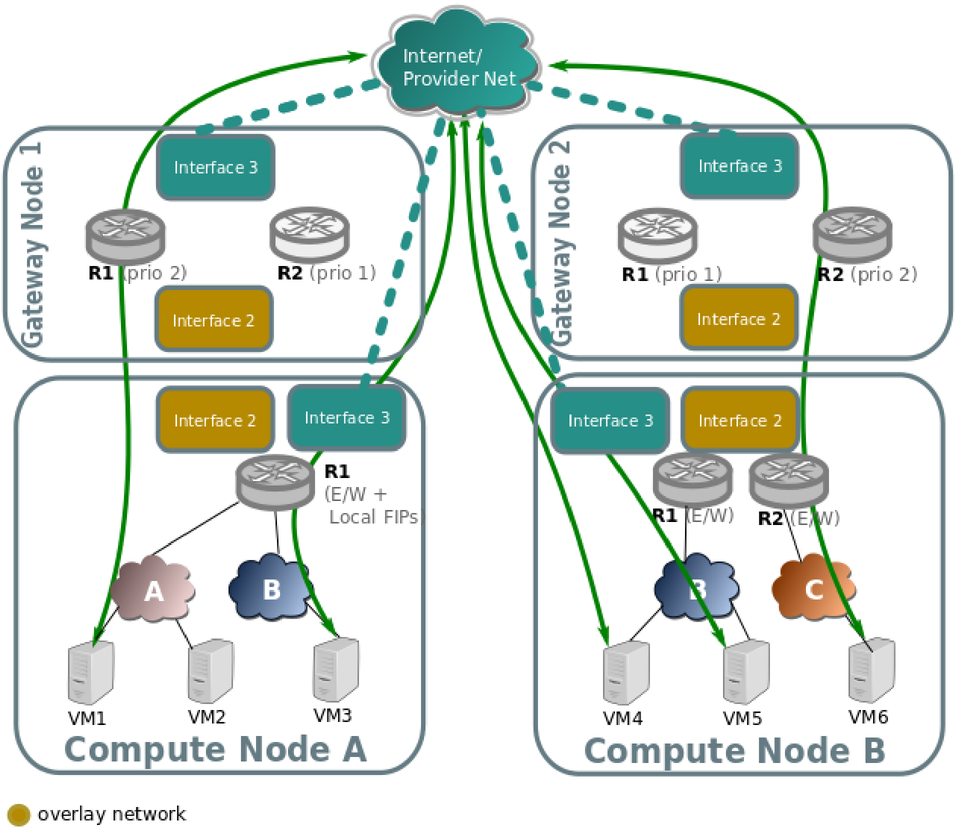
When an external network connected to the router is represented by FLAT or VLAN network type, active chassis is identified by the external Logical Router Port. In practice this means, that LRP will have `hosting-chassis` property set in a `status` row for the external LRP. You can also check Chassis priorities for the LRP with `lrp-get-gateway-chassis` command. Changing the priority will result in traffic failover to another Chassis.

In case of connecting another Geneve network to the router as external network (by creating `access_as_external` RBAC rule for such network), router itself will be pinned to Chassis rather than its LRP. In this scenario Logical Router does have `chassis` property defined inside the `options` row. With that `GATEWAY_PORT` will not be defined for dnat_and_snat rules which are created for FIPs as this will make traffic to pass through the LRP that is not bound to any Chassis.

Distributed Floating IP

In the following diagram we can see how VMs with no Floating IP (VM1, VM6) still communicate throught the gateway nodes using SNAT on the edge routers R1 and R2.

While VM3, VM4, and VM5 have an assigned floating IP, and its traffic flows directly through the local provider bridge/interface to the external network.



L3HA support

Ovn driver implements L3 high availability in a transparent way. You dont need to enable any config flags. As soon as you have more than one chassis capable of acting as an l3 gateway to the specific external network attached to the router it will schedule the router gateway port to multiple chassis, making use of the `gateway_chassis` column on OVNs `Logical_Router_Port` table.

In order to have external connectivity, either:

- Some gateway nodes have `ovn-cms-options` with the value `enable-chassis-as-gw` in Open_vSwitch tables external_ids column, or
- if no gateway node exists with the external ids column set with that value, then all nodes would be eligible to host gateway chassis.

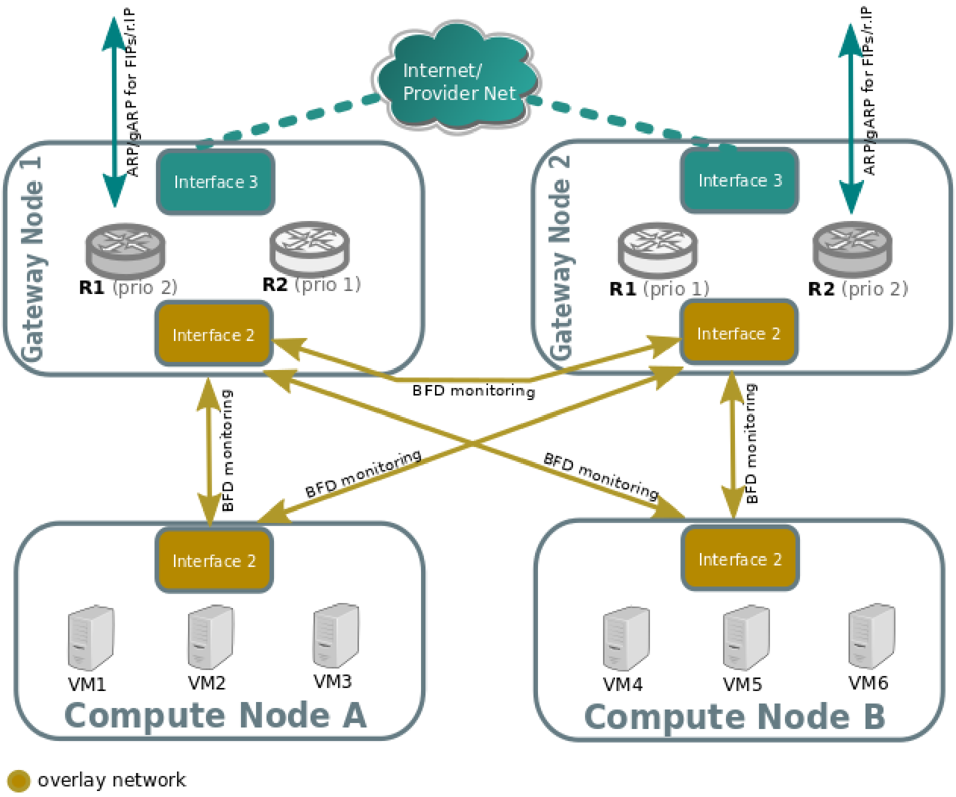
Example to how to enabled chassis to host gateways:

```
$ ovs-vsctl set open . external-ids:ovn-cms-options="enable-chassis-as-gw"
```

At the low level, functionality is all implemented mostly by OpenFlow rules with bundle active_passive outputs. The ARP responder and router enablement/disablement is handled by ovn-controller. Gratuitous ARPs for FIPs and router external addresses are periodically sent by ovn-controller itself.

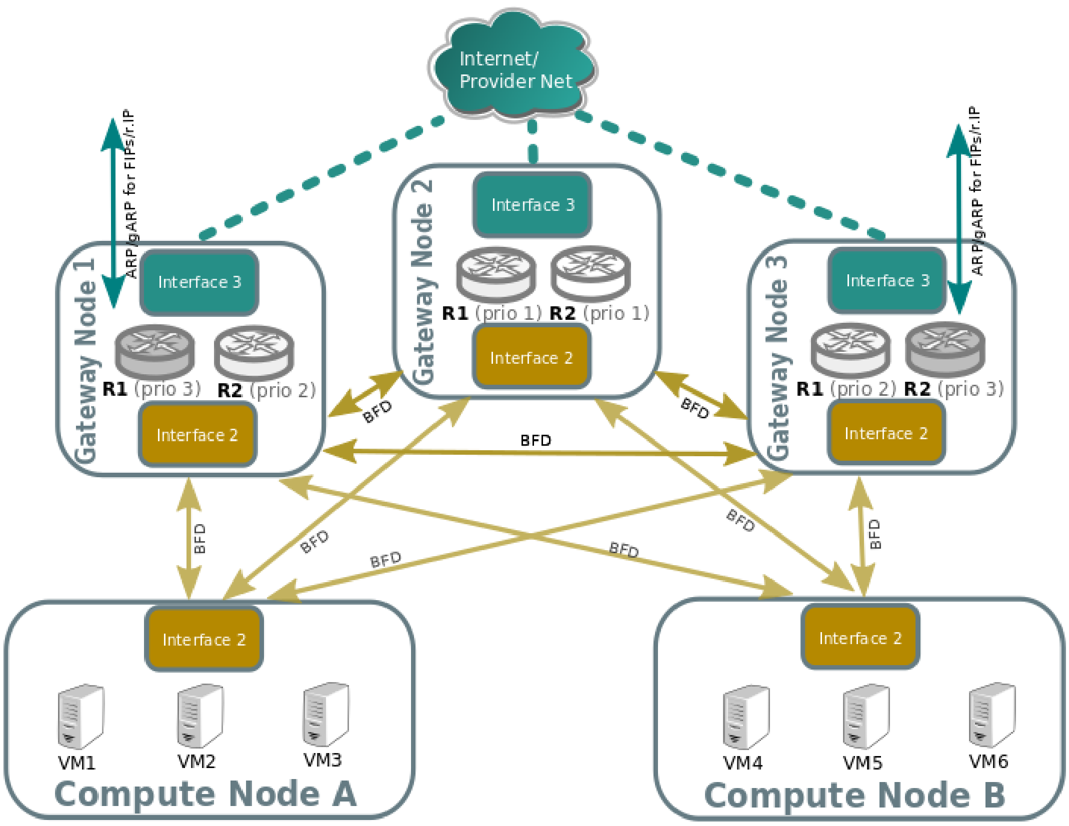
BFD monitoring

OVN monitors the availability of the chassis via the BFD protocol, which is encapsulated on top of the Geneve tunnels established from chassis to chassis.



Each chassis that is marked as a gateway chassis will monitor all the other gateway chassis in the deployment as well as compute node chassis, to let the gateways enable/disable routing of packets and ARP responses / announcements.

Each compute node chassis will monitor each gateway chassis via BFD to automatically steer external traffic (snat/dnat) through the active chassis for a given router.



The gateway nodes monitor each other in star topology. Compute nodes dont monitor each other because thats not necessary.

Neutron exposes, via configuration file, a set of options to configure the OVS BFD options. That allows the administrator to configure the HA failover timeout that is defined as:

$$\text{detection time} = max(\text{bfd-min-rx}, \text{bfd-min-tx}) * \text{bfd-mult}$$

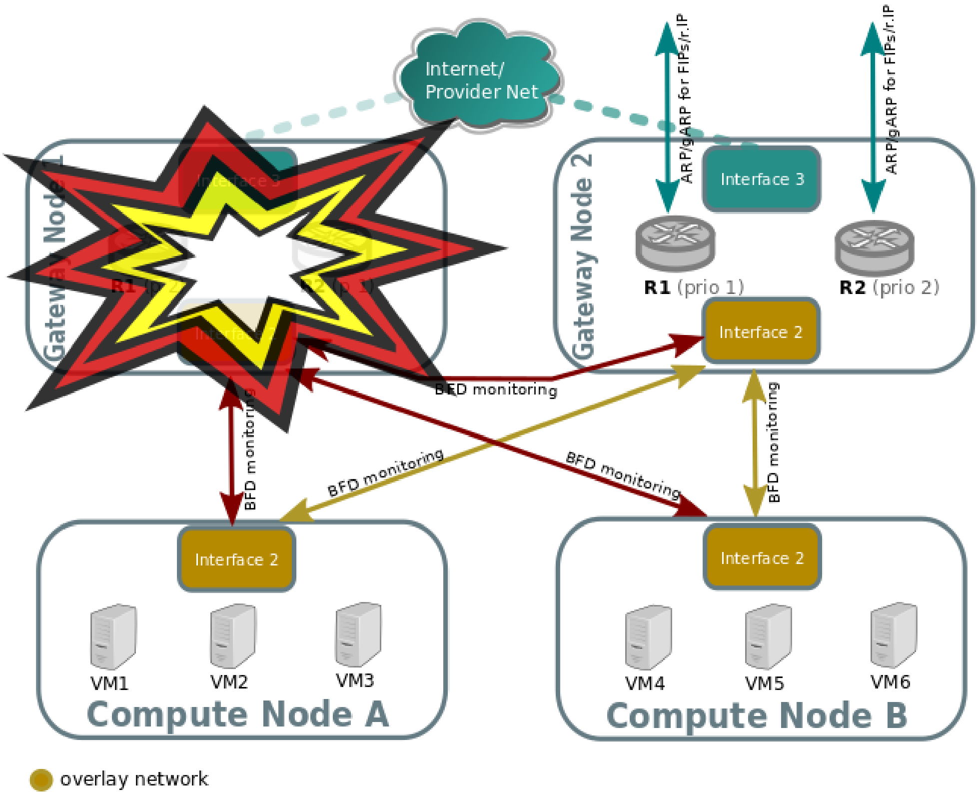
The Neutron configuration provide a list of three preset values and an extra manual one, where the specific parameters can be defined:

- `[ovn]ha_failover_strategy`: this option has 4 possible values:
  - normal: the default OVN NB_Global options for `bfd-min-rx`, `bfd-min-tx` and `bfd-mult` are set[1][2][3].
  - aggressive: the `bfd-min-rx` is halved by 2.
  - conservative: the `bfd-mult` is multiplied by 2.
  - manual: the administrator can manually define the following values.
- `[ovn]bfd_min_rx`: BFD minimum RX interval in milliseconds, default 1000 ms.
- `[ovn]bfd_min_tx`: BFD minimum TX interval in milliseconds, default 100 ms.
- `[ovn]bfd_mult`: BFD detection multiplier, default 3.

[1] https://github.com/openvswitch/ovs/blob/b43ed9036e6fb38fe09ff4746b603edaef5319da/vswitchd/vswitch.xml#L4147
[2] https://github.com/openvswitch/ovs/blob/b43ed9036e6fb38fe09ff4746b603edaef5319da/vswitchd/vswitch.xml#L4155
[3] https://github.com/openvswitch/ovs/blob/b43ed9036e6fb38fe09ff4746b603edaef5319da/vswitchd/vswitch.xml#L4236

Failover (detected by BFD)

Look at the following example:



Compute nodes BFD monitoring of the gateway nodes will detect that tunnel endpoint going to gateway node 1 is down, so. So traffic output that needs to get into the external network through the router will be directed to the lower priority chassis for R1. R2 stays the same because Gateway Node 2 was already the highest priority chassis for R2.

Gateway node 2 will detect that tunnel endpoint to gateway node 1 is down, so it will become responsible for the external leg of R1, and its ovn-controller will populate flows for the external ARP responder, traffic forwarding (N/S) and periodic gratuitous ARPs.

Gateway node 2 will also bind the external port of the router (represented as a chassis-redirect port on the South Bound database).

If Gateway node 1 is still alive, failure over interface 2 will be detected because its not seeing any other nodes.

No mechanisms are still present to detect external network failure, so as good practice to detect network failure we recommend that all interfaces are handled over a single bonded interface with VLANs.

Supported failure modes are:

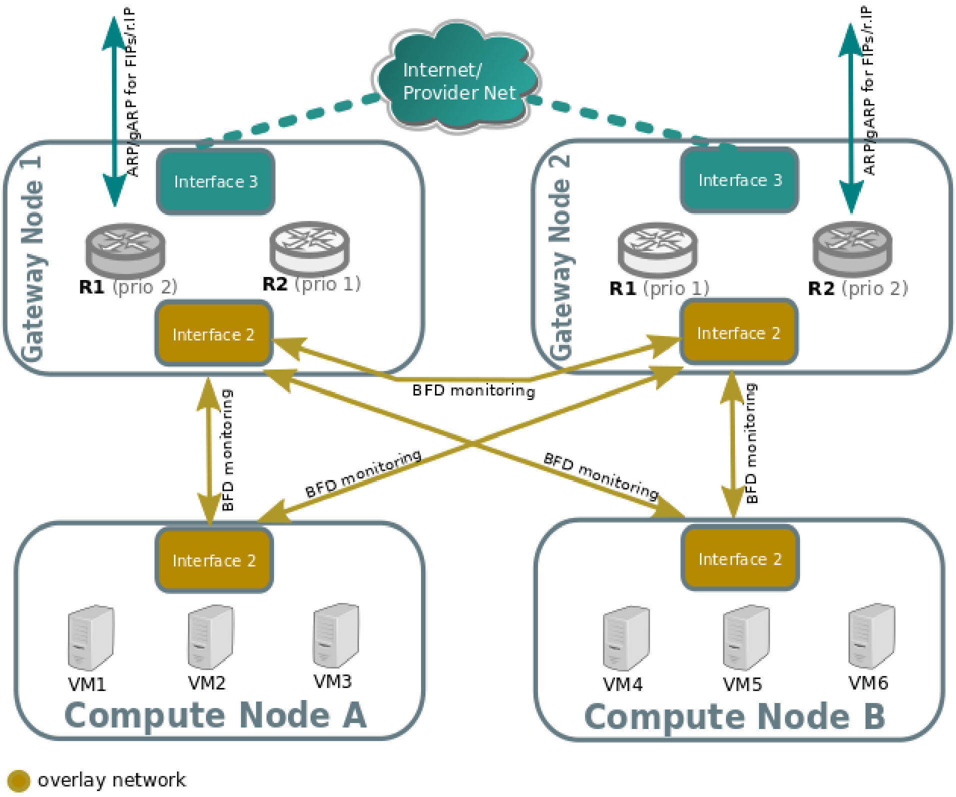
- gateway chassis becomes disconnected from network (tunneling interface)
- ovs-vswitchd is stopped (its responsible for BFD signaling)
- ovn-controller is stopped, as ovn-controller will remove himself as a registered chassis.

Note

As a side note, its also important to understand, that as for VRRP or CARP protocols, this detection mechanism only works for link failures, but not for routing failures.

Failback

L3HA behaviour is preemptive in OVN (at least for the time being) since that would balance back the routers to the original chassis, avoiding any of the gateway nodes becoming a bottleneck.

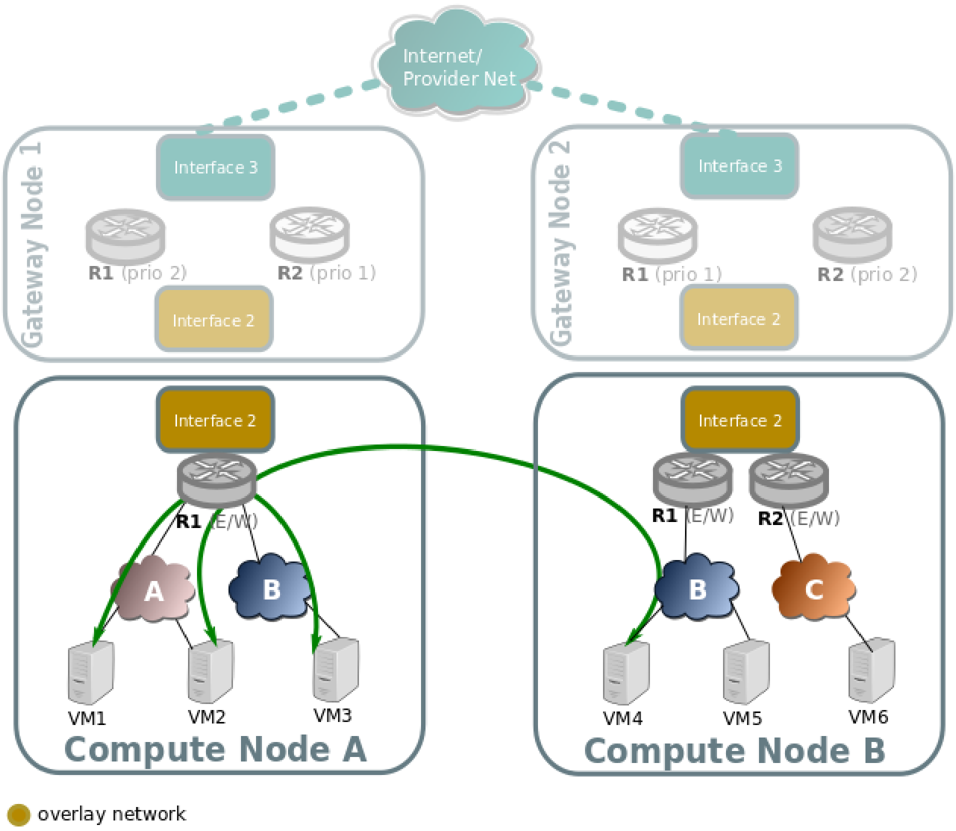


East/West

East/West traffic on ovn driver is completely distributed, that means that routing will happen internally on the compute nodes without the need to go through the gateway nodes.

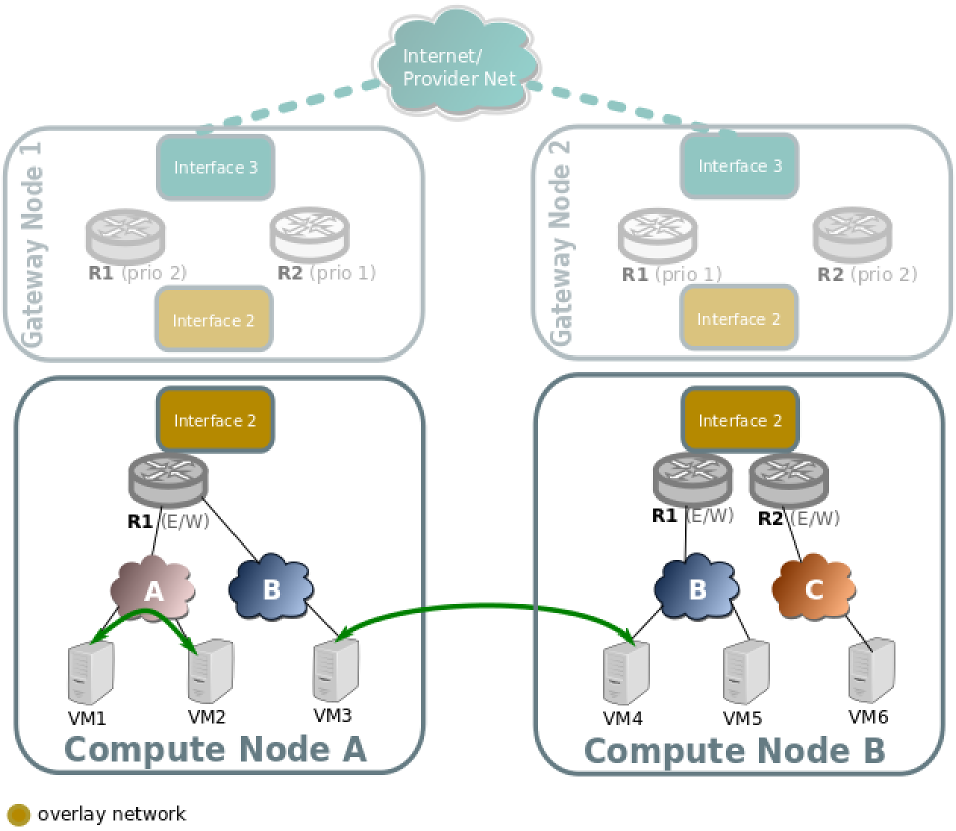
Traffic going through a virtual router, different subnets

Traffic going through a virtual router, and going from a virtual network/subnet to another will flow directly from compute to compute node encapsulated as usual, while all the routing operations like decreasing TTL or switching MAC addresses will be handled in OpenFlow at the source host of the packet.



Traffic across the same subnet

Traffic across a subnet will happen as described in the following diagram, although this kind of communication doesnt make use of routing at all (just encapsulation) its been included for completeness.



Traffic goes directly from instance to instance through br-int in the case of both instances living in the same host (VM1 and VM2), or via encapsulation when living on different hosts (VM3 and VM4).

Packet fragmentation

The Neutron configuration variable `[ovn]ovn_emit_need_to_frag` configures OVN to emit the need to frag packets in case of MTU mismatches. This configuration option allows Neutron to set, in the router gateway `Logical_Router_Port`, the option gateway_mtu. If a packet from any network reaches the gateway `Logical_Router_Port`, OVN will send the need for frag message.

In order to allow any E/W or N/S traffic to cross the router, the value of gateway_mtu will have the lowest MTU value off all networks connected to the router. This could impact the performance of the traffic using the networks connected to the router if the MTU defined is low. But the user can unset the Neutron configuration flag in order to avoid the fragmentation, at the cost of limiting the communication between networks with different MTUs.

References

8.7.4 IP Multicast: IGMP snooping configuration guide for OVN

How to enable it

In order to enable IGMP snooping with the OVN driver the following configuration needs to be set in the `/etc/neutron/neutron.conf` file of the controller nodes:

```
# OVN does reuse the OVS option, therefore the option group is [ovs]
[ovs]
igmp_snooping_enable = True
```

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```
...
```

Upon restarting the Neutron service all existing networks (Logical_Switch, in OVN terms) will be updated in OVN to enable or disable IGMP snooping based on the `igmp_snooping_enable` configuration value.

Note

Currently the OVN driver does not configure IGMP querier in OVN so ovn-controller will not send IGMP group memberships IP querier to retrieve IGMP membership reports from active members.

OVN Database information

The `igmp_snooping_enable` configuration from Neutron is translated into the `mcast_snoop` option set in the `other_config` column from the `Logical_Switch` table in the OVN Northbound Database:

```
$ ovn-nbctl list Logical_Switch
_uuid               : d6a2fbcd-aaa4-4b9e-8274-184238d66a15
other_config        : {mcast_flood_unregistered="false", mcast_snoop="true"}
...
```

To find more information about the learnt IGMP groups by OVN use the command below (populated only when igmp_snooping_enable is True):

```
$ ovn-sbctl list IGMP_group
_uuid               : 2d6cae4c-bd82-4b31-9c63-2d17cbeadc4e
address             : "225.0.0.120"
chassis             : 34e25681-f73f-43ac-a3a4-7da2a710ecd3
datapath            : eaf0f5cc-a2c8-4c30-8def-2bc1ec9dcabc
ports               : [5eaf9dd5-eae5-4749-ac60-4c1451901c56, 8a69efc5-38c5-
↪48fb-bbab-30f2bf9b8d45]
...
```

Note

Since IGMP querier is not yet supported in the OVN driver, restarting the ovn-controller service(s) will result in OVN unlearning the IGMP groups and broadcast all the multicast traffic. This behavior can impact when updating/upgrading the OVN services.

Extra information

When multicast IP traffic is sent to a multicast group address which is in the **224.0.0.X** range, the multicast traffic will be flooded, even when IGMP snooping is enabled. See the RFC 4541 session 2.1.2:

```
2) Packets with a destination IP (DIP) address in the 224.0.0.X range
   which are not IGMP must be forwarded on all ports.
```

The permutations from different configurations are:

- With IGMP snooping disabled: IP Multicast traffic flooded to all ports.

- With IGMP snooping enabled and multicast group address **not in** the 224.0.0.X range: IP Multicast traffic **is not** flooded.
- With IGMP snooping enabled and multicast group address **is in** the 224.0.0.X range: IP Multicast traffic **is** flooded.
- Apart from the `igmp_snooping_enable` configuration option mentioned before, there are 3 other configuration options supported by the OVN driver: `igmp_flood`, `igmp_flood_reports` and `igmp_flood_unregistered`. Check the *ML2 configuration reference page* for more information.

8.7.5 OpenStack and OVN Tutorial

The OVN project documentation includes an in depth tutorial of using OVN with OpenStack.

OpenStack and OVN Tutorial

8.7.6 Reference architecture

The reference architecture defines the minimum environment necessary to deploy OpenStack with Open Virtual Network (OVN) integration for the Networking service in production with sufficient expectations of scale and performance. For evaluation purposes, you can deploy this environment using the *Installation Guide* or Vagrant. Any scaling or performance evaluations should use bare metal instead of virtual machines.

Layout

The reference architecture includes a minimum of four nodes.

The controller node contains the following components that provide enough functionality to launch basic instances:

- One network interface for management
- Identity service
- Image service
- Networking management with ML2 mechanism driver for OVN (control plane)
- Compute management (control plane)

The database node contains the following components:

- One network interface for management
- OVN northbound service (`ovn-northd`)
- Open vSwitch (OVS) database service (`ovsdb-server`) for the OVN northbound database (`ovnnb.db`)
- Open vSwitch (OVS) database service (`ovsdb-server`) for the OVN southbound database (`ovnsb.db`)

Note

For functional evaluation only, you can combine the controller and database nodes.

The two compute nodes contain the following components:

- Two or three network interfaces for management, overlay networks, and optionally provider networks
- Compute management (hypervisor)
- Hypervisor (KVM)
- OVN controller service (`ovn-controller`)
- OVS data plane service (`ovs-vswitchd`)
- OVS database service (`ovsdb-server`) with OVS local configuration (`conf.db`) database
- OVN metadata agent (`ovn-metadata-agent`)

The gateway nodes contain the following components:

- Three network interfaces for management, overlay networks and provider networks.
- OVN controller service (`ovn-controller`)
- OVS data plane service (`ovs-vswitchd`)
- OVS database service (`ovsdb-server`) with OVS local configuration (`conf.db`) database

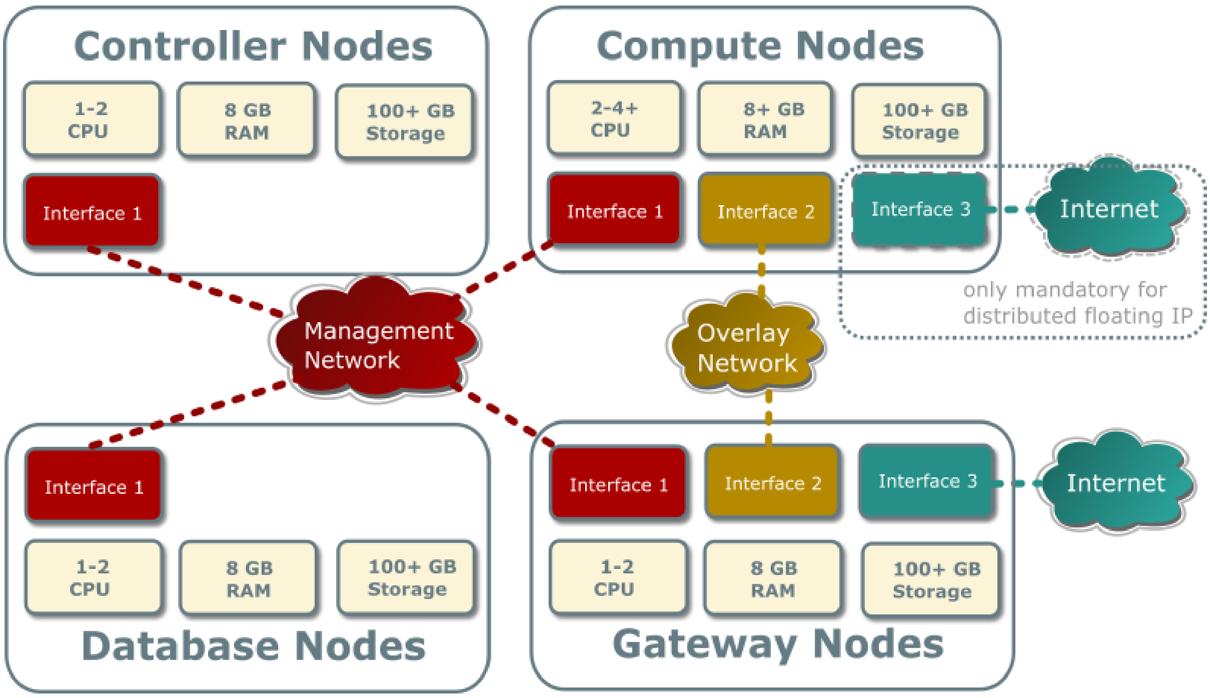
Note

Each OVN metadata agent provides metadata service locally on the compute nodes in a lightweight way. Each network being accessed by the instances of the compute node will have a corresponding metadata ovnmeta-$net_uuid namespace, and inside an haproxy will funnel the requests to the ovn-metadata-agent over a unix socket.

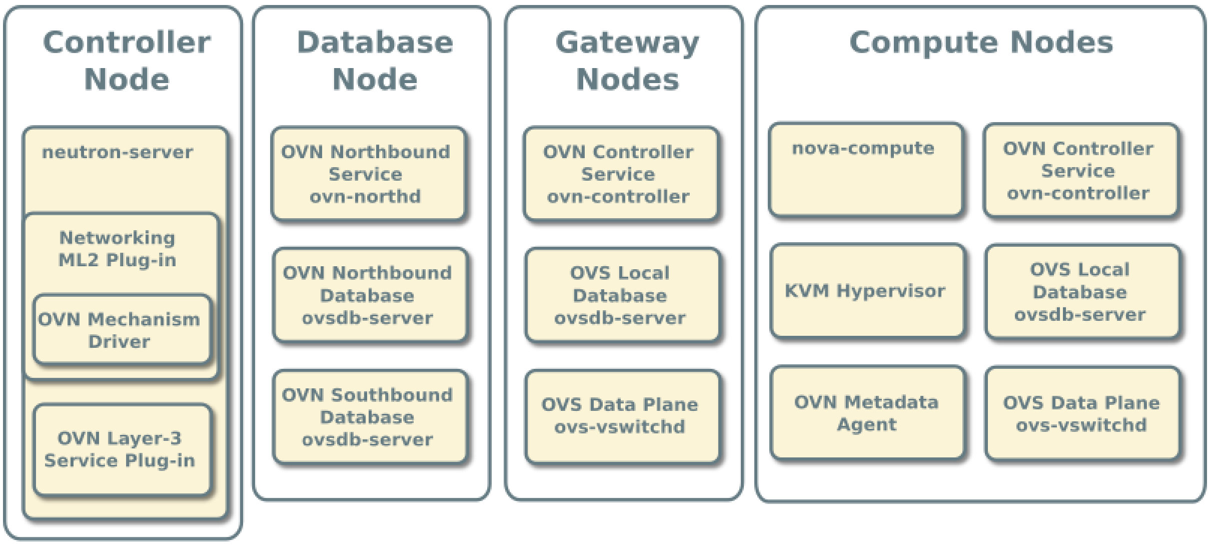
Such namespace can be very helpful for debug purposes to access the local instances on the compute node. If you login as root on such compute node you can execute:

ip netns ovnmeta-$net_uuid exec ssh user@my.instance.ip.address

Hardware layout

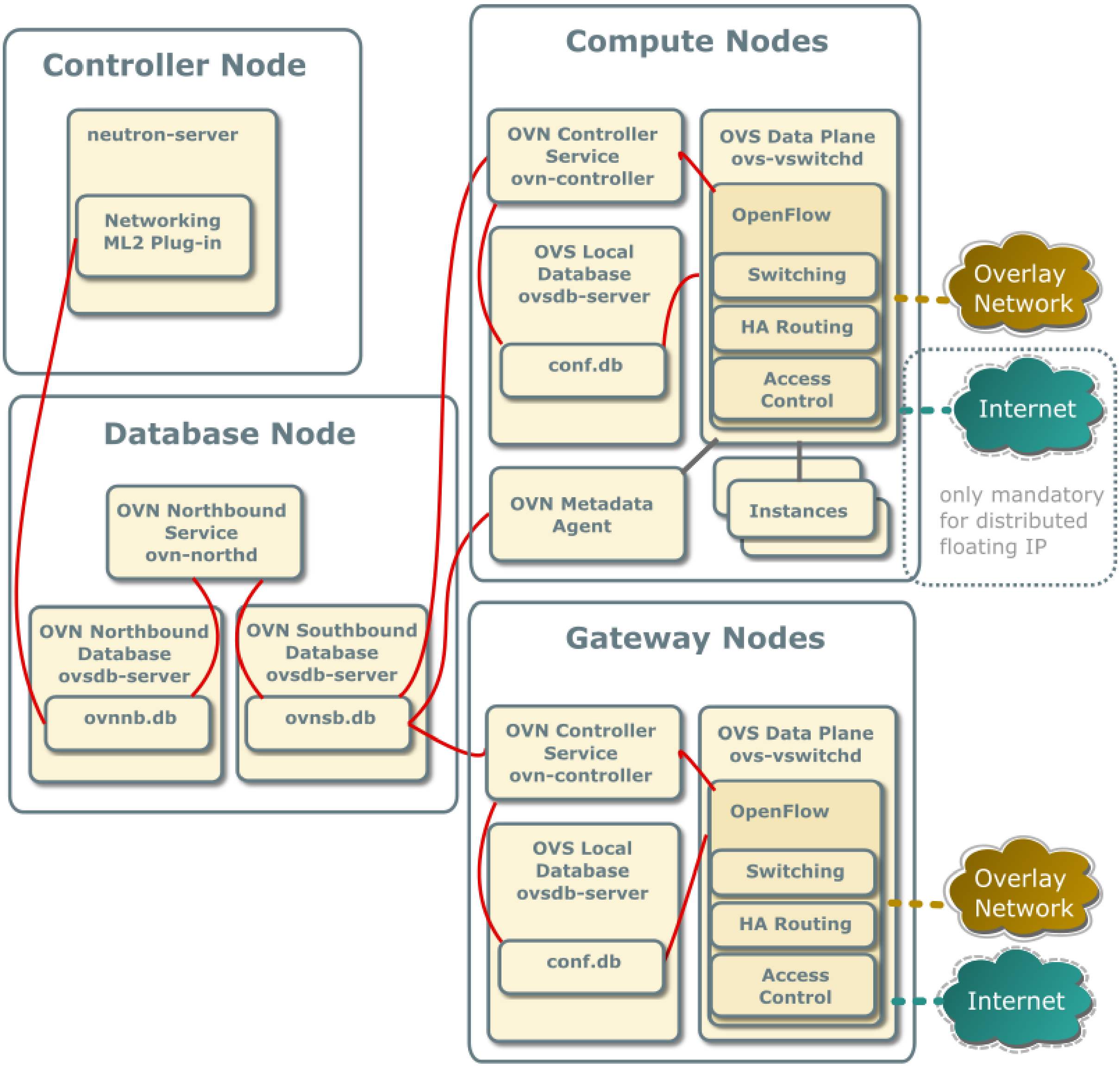


Service layout



Networking service with OVN integration

The reference architecture deploys the Networking service with OVN integration as described in the following scenarios:



With ovn driver, all the E/W traffic which traverses a virtual router is completely distributed, going from compute to compute node without passing through the gateway nodes.

N/S traffic that needs SNAT (without floating IPs) will always pass through the centralized gateway nodes, although, as soon as you have more than one gateway node ovn driver will make use of the HA capabilities of ovn.

Centralized Floating IPs

In this architecture, all the N/S router traffic (snat and floating IPs) goes through the gateway nodes.

The compute nodes dont need connectivity to the external network, although it could be provided if we wanted to have direct connectivity to such network from some instances.

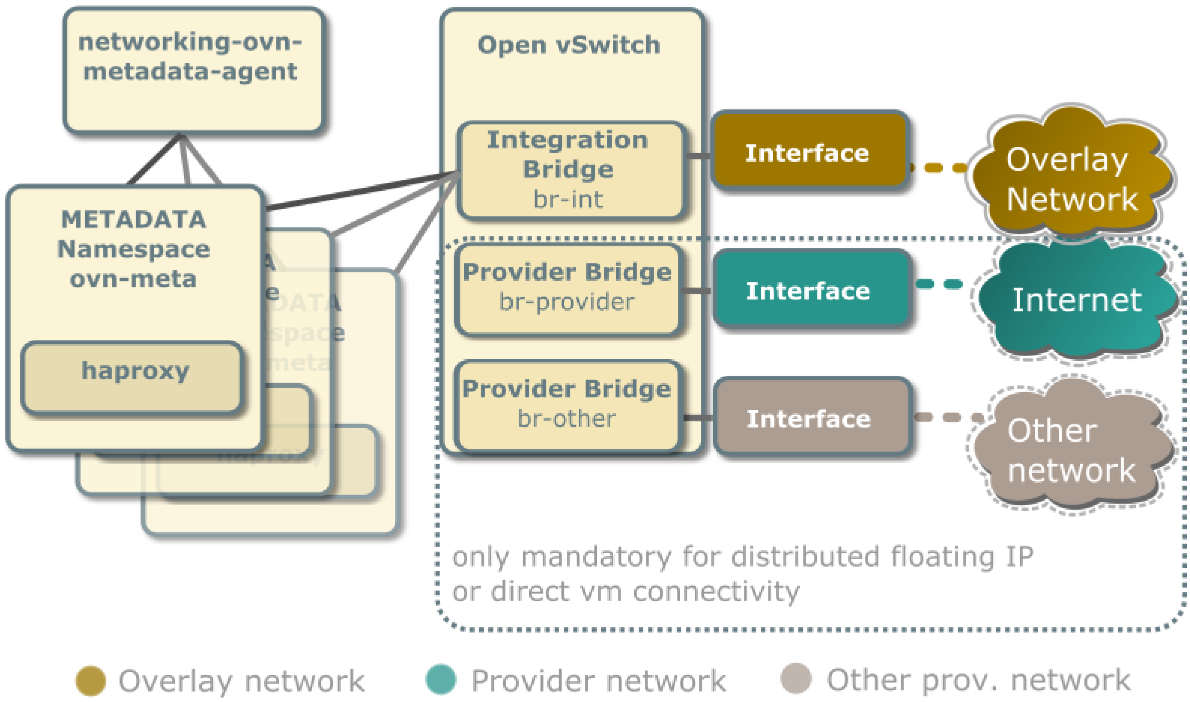
For external connectivity, gateway nodes have to set `ovn-cms-options` with `enable-chassis-as-gw` in Open_vSwitch tables external_ids column, for example:

```
$ ovs-vsctl set open . external-ids:ovn-cms-options="enable-chassis-as-gw"
```

Distributed Floating IPs (DVR)

In this architecture, the floating IP N/S traffic flows directly from/to the compute nodes through the specific provider network bridge. In this case compute nodes need connectivity to the external network.

Each compute node contains the following network components:



> **Note**
>
> The Networking service creates a unique network namespace for each virtual network that enables the metadata service.

Several external connections can be optionally created via provider bridges. Those can be used for direct vm connectivity to the specific networks or the use of distributed floating ips.

Accessing OVN database content

OVN stores configuration data in a collection of OVS database tables. The following commands show the contents of the most common database tables in the northbound and southbound databases. The example database output in this section uses these commands with various output filters.

```
$ ovn-nbctl list Logical_Switch
$ ovn-nbctl list Logical_Switch_Port
$ ovn-nbctl list ACL
$ ovn-nbctl list Address_Set
$ ovn-nbctl list Logical_Router
$ ovn-nbctl list Logical_Router_Port
$ ovn-nbctl list Gateway_Chassis

$ ovn-sbctl list Chassis
```

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```
$ ovn-sbctl list Encap
$ ovn-nbctl list Address_Set
$ ovn-sbctl lflow-list
$ ovn-sbctl list Multicast_Group
$ ovn-sbctl list Datapath_Binding
$ ovn-sbctl list Port_Binding
$ ovn-sbctl list MAC_Binding
$ ovn-sbctl list Gateway_Chassis
```

Note

By default, you must run these commands from the node containing the OVN databases.

Adding a compute node

When you add a compute node to the environment, the OVN controller service on it connects to the OVN southbound database and registers the node as a chassis.

```
_uuid                : 9be8639d-1d0b-4e3d-9070-03a655073871
encaps               : [2fcefdf4-a5e7-43ed-b7b2-62039cc7e32e]
external_ids         : {ovn-bridge-mappings=""}
hostname             : "compute1"
name                 : "410ee302-850b-4277-8610-fa675d620cb7"
vtep_logical_switches: []
```

The encaps field value refers to tunnel endpoint information for the compute node.

```
_uuid                : 2fcefdf4-a5e7-43ed-b7b2-62039cc7e32e
ip                   : "10.0.0.32"
options              : {}
type                 : geneve
```

Security Groups/Rules

When a Neutron Security Group is created, the equivalent Port Group in OVN (pg-<security_group_id> is created). This Port Group references Neutron SG id in its external_ids column.

When a Neutron Port is created, the equivalent Logical Port in OVN is added to those Port Groups associated to the Neutron Security Groups this port belongs to.

When a Neutron Port is deleted, the associated Logical Port in OVN is deleted. Since the schema includes a weak reference to the port, when the LSP gets deleted, it is automatically deleted from any Port Group entry where it was previously present.

Every time a security group rule is created, instead of figuring out the ports affected by its SG and inserting an ACL row which will be referenced by different Logical Switches, we just reference it from the associated Port Group.

OVN operations

1. Creating a security group will cause the OVN mechanism driver to create a port group in the Port_Group table of the northbound DB:

```
_uuid               : e96c5994-695d-4b9c-a17b-c7375ad281e2
acls                : [33c3c2d0-bc7b-421b-ace9-10884851521a, c22170ec-
↪da5d-4a59-b118-f7f0e370ebc4]
external_ids        : {"neutron:security_group_id"="ccbeffee-7b98-4b6f-
↪adf7-d42027ca6447"}
name                : pg_ccbeffee_7b98_4b6f_adf7_d42027ca6447
ports               : []
```

And it also creates the default ACLs for egress traffic in the ACL table of the northbound DB:

```
_uuid               : 33c3c2d0-bc7b-421b-ace9-10884851521a
action              : allow-related
direction           : from-lport
external_ids        : {"neutron:security_group_rule_id"="655b0d7e-144e-
↪4bd8-9243-10a261b91041"}
log                 : false
match               : "inport == @pg_ccbeffee_7b98_4b6f_adf7_d42027ca6447␣
↪&& ip4"
meter               : []
name                : []
priority            : 1002
severity            : []

_uuid               : c22170ec-da5d-4a59-b118-f7f0e370ebc4
action              : allow-related
direction           : from-lport
external_ids        : {"neutron:security_group_rule_id"="a303a34f-5f19-
↪494f-a9e2-e23f246bfcad"}
log                 : false
match               : "inport == @pg_ccbeffee_7b98_4b6f_adf7_d42027ca6447␣
↪&& ip6"
meter               : []
name                : []
priority            : 1002
severity            : []
```

Ports with no security groups

When a port doesnt belong to any Security Group and port security is enabled, we, by default, drop all the traffic to/from that port. In order to implement this through Port Groups, well create a special Port Group with a fixed name (`neutron_pg_drop`) which holds the ACLs to drop all the traffic.

This PG is created automatically once before neutron-server forks into workers.

Networks

Provider networks

A provider (external) network bridges instances to physical network infrastructure that provides layer-3 services. In most cases, provider networks implement layer-2 segmentation using VLAN IDs. A provider network maps to a provider bridge on each compute node that supports launching instances on the provider network. You can create more than one provider bridge, each one requiring a unique name and underlying physical network interface to prevent switching loops. Provider networks and bridges can use arbitrary names, but each mapping must reference valid provider network and bridge names. Each provider bridge can contain one `flat` (untagged) network and up to the maximum number of `vlan` (tagged) networks that the physical network infrastructure supports, typically around 4000.

Creating a provider network involves several commands at the host, OVS, and Networking service levels that yield a series of operations at the OVN level to create the virtual network components. The following example creates a `flat` provider network `provider` using the provider bridge `br-provider` and binds a subnet to it.

Create a provider network

1. On each compute node, create the provider bridge, map the provider network to it, and add the underlying physical or logical (typically a bond) network interface to it.

```
# ovs-vsctl --may-exist add-br br-provider -- set bridge br-provider \
  protocols=OpenFlow13
# ovs-vsctl set Open_vSwitch . external-ids:ovn-bridge-
↪mappings=provider:br-provider
# ovs-vsctl --may-exist add-port br-provider INTERFACE_NAME
```

   Replace `INTERFACE_NAME` with the name of the underlying network interface.

   Note

   These commands provide no output if successful.

2. On the controller node, source the administrative project credentials.

3. On the controller node, to enable this chassis to host gateway routers for external connectivity, set ovn-cms-options to enable-chassis-as-gw.

```
# ovs-vsctl set Open_vSwitch . external-ids:ovn-cms-options="enable-
↪chassis-as-gw"
```

   Note

   This command provide no output if successful.

4. On the controller node, create the provider network in the Networking service. In this case, instances and routers in other projects can use the network.

```
$ openstack network create --external --share \
  --provider-physical-network provider --provider-network-type flat \
  provider
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        | nova                                 |
| created_at                | 2016-06-15 15:50:37+00:00            |
| description               |                                      |
| id                        | 0243277b-4aa8-46d8-9e10-5c9ad5e01521 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| mtu                       | 1500                                 |
| name                      | provider                             |
| project_id                | b1ebf33664df402693f729090cfab861     |
| provider:network_type     | flat                                 |
| provider:physical_network | provider                             |
| provider:segmentation_id  | None                                 |
| qos_policy_id             | None                                 |
| router:external           | External                             |
| shared                    | True                                 |
| status                    | ACTIVE                               |
| subnets                   | 32a61337-c5a3-448a-a1e7-c11d6f062c21 |
| tags                      | []                                   |
| updated_at                | 2016-06-15 15:50:37+00:00            |
+---------------------------+--------------------------------------+
```

Note

The value of `--provider-physical-network` must refer to the provider network name in the mapping.

OVN operations

The OVN mechanism driver and OVN perform the following operations during creation of a provider network.

1. The mechanism driver translates the network into a logical switch in the OVN northbound database.

```
_uuid               : 98edf19f-2dbc-4182-af9b-79cafa4794b6
acls                : []
external_ids        : {"neutron:network_name"=provider}
load_balancer       : []
name                : "neutron-e4abf6df-f8cf-49fd-85d4-3ea399f4d645"
ports               : [92ee7c2f-cd22-4cac-a9d9-68a374dc7b17]
```

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```
.. note::

   The ``neutron:network_name`` field in ``external_ids`` contains
   the network name and ``name`` contains the network UUID.
```

2. In addition, because the provider network is handled by a separate bridge, the following logical port is created in the OVN northbound database.

```
_uuid                : 92ee7c2f-cd22-4cac-a9d9-68a374dc7b17
addresses            : [unknown]
enabled              : []
external_ids         : {}
name                 : "provnet-e4abf6df-f8cf-49fd-85d4-3ea399f4d645"
options              : {network_name=provider}
parent_name          : []
port_security        : []
tag                  : []
type                 : localnet
up                   : false
```

3. The OVN northbound service translates these objects into datapath bindings, port bindings, and the appropriate multicast groups in the OVN southbound database.

   - Datapath bindings

```
_uuid                : f1f0981f-a206-4fac-b3a1-dc2030c9909f
external_ids         : {logical-switch="98edf19f-2dbc-4182-af9b-
↪79cafa4794b6"}
tunnel_key           : 109
```

   - Port bindings

```
_uuid                : 8427506e-46b5-41e5-a71b-a94a6859e773
chassis              : []
datapath             : f1f0981f-a206-4fac-b3a1-dc2030c9909f
logical_port         : "provnet-e4abf6df-f8cf-49fd-85d4-3ea399f4d645"
mac                  : [unknown]
options              : {network_name=provider}
parent_port          : []
tag                  : []
tunnel_key           : 1
type                 : localnet
```

   - Logical flows

```
Datapath: f1f0981f-a206-4fac-b3a1-dc2030c9909f  Pipeline: ingress
  table= 0( ls_in_port_sec_l2), priority=  100, match=(eth.src[40]),
    action=(drop;)
  table= 0( ls_in_port_sec_l2), priority=  100, match=(vlan.
↪present),
    action=(drop;)
```

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```
  table= 0(  ls_in_port_sec_l2), priority=   50,
    match=(inport == "provnet-e4abf6df-f8cf-49fd-85d4-3ea399f4d645"),
    action=(next;)
  table= 1(  ls_in_port_sec_ip), priority=    0, match=(1),
    action=(next;)
  table= 2(  ls_in_port_sec_nd), priority=    0, match=(1),
    action=(next;)
  table= 3(      ls_in_pre_acl), priority=    0, match=(1),
    action=(next;)
  table= 4(       ls_in_pre_lb), priority=    0, match=(1),
    action=(next;)
  table= 5( ls_in_pre_stateful), priority=  100, match=(reg0[0] ==␣
↪1),
    action=(ct_next;)
  table= 5( ls_in_pre_stateful), priority=    0, match=(1),
    action=(next;)
  table= 6(          ls_in_acl), priority=    0, match=(1),
    action=(next;)
  table= 7(           ls_in_lb), priority=    0, match=(1),
    action=(next;)
  table= 8(     ls_in_stateful), priority=  100, match=(reg0[1] ==␣
↪1),
    action=(ct_commit; next;)
  table= 8(     ls_in_stateful), priority=  100, match=(reg0[2] ==␣
↪1),
    action=(ct_lb;)
  table= 8(     ls_in_stateful), priority=    0, match=(1),
    action=(next;)
  table= 9(      ls_in_arp_rsp), priority=  100,
    match=(inport == "provnet-e4abf6df-f8cf-49fd-85d4-3ea399f4d645"),
    action=(next;)
  table= 9(      ls_in_arp_rsp), priority=    0, match=(1),
    action=(next;)
  table=10(      ls_in_l2_lkup), priority=  100, match=(eth.mcast),
    action=(outport = "_MC_flood"; output;)
  table=10(      ls_in_l2_lkup), priority=    0, match=(1),
    action=(outport = "_MC_unknown"; output;)
Datapath: f1f0981f-a206-4fac-b3a1-dc2030c9909f  Pipeline: egress
  table= 0(      ls_out_pre_lb), priority=    0, match=(1),
    action=(next;)
  table= 1(     ls_out_pre_acl), priority=    0, match=(1),
    action=(next;)
  table= 2(ls_out_pre_stateful), priority=  100, match=(reg0[0] ==␣
↪1),
    action=(ct_next;)
  table= 2(ls_out_pre_stateful), priority=    0, match=(1),
    action=(next;)
  table= 3(          ls_out_lb), priority=    0, match=(1),
    action=(next;)
```

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```
  table= 4(         ls_out_acl), priority=    0, match=(1),
    action=(next;)
  table= 5(    ls_out_stateful), priority=  100, match=(reg0[1] ==␣
↪1),
    action=(ct_commit; next;)
  table= 5(    ls_out_stateful), priority=  100, match=(reg0[2] ==␣
↪1),
    action=(ct_lb;)
  table= 5(    ls_out_stateful), priority=    0, match=(1),
    action=(next;)
  table= 6( ls_out_port_sec_ip), priority=    0, match=(1),
    action=(next;)
  table= 7( ls_out_port_sec_l2), priority=  100, match=(eth.mcast),
    action=(output;)
  table= 7( ls_out_port_sec_l2), priority=   50,
    match=(outport == "provnet-e4abf6df-f8cf-49fd-85d4-3ea399f4d645
↪"),
    action=(output;)
```

- Multicast groups

```
_uuid               : 0102f08d-c658-4d0a-a18a-ec8adcaddf4f
datapath            : f1f0981f-a206-4fac-b3a1-dc2030c9909f
name                : _MC_unknown
ports               : [8427506e-46b5-41e5-a71b-a94a6859e773]
tunnel_key          : 65534

_uuid               : fbc38e51-ac71-4c57-a405-e6066e4c101e
datapath            : f1f0981f-a206-4fac-b3a1-dc2030c9909f
name                : _MC_flood
ports               : [8427506e-46b5-41e5-a71b-a94a6859e773]
tunnel_key          : 65535
```

Create a subnet on the provider network

The provider network requires at least one subnet that contains the IP address allocation available for instances, default gateway IP address, and metadata such as name resolution.

1. On the controller node, create a subnet bound to the provider network `provider`.

```
$ openstack subnet create --network provider --subnet-range \
  203.0.113.0/24 --allocation-pool start=203.0.113.101,end=203.0.113.250 \
  --dns-nameserver 8.8.8.8,8.8.4.4 --gateway 203.0.113.1 provider-v4
  +-------------------+--------------------------------------+
  | Field             | Value                                |
  +-------------------+--------------------------------------+
  | allocation_pools  | 203.0.113.101-203.0.113.250          |
  | cidr              | 203.0.113.0/24                       |
  | created_at        | 2016-06-15 15:50:45+00:00            |
  | description       |                                      |
```

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```
| dns_nameservers   | 8.8.8.8, 8.8.4.4                     |
| enable_dhcp       | True                                 |
| gateway_ip        | 203.0.113.1                          |
| host_routes       |                                      |
| id                | 32a61337-c5a3-448a-a1e7-c11d6f062c21 |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | provider-v4                          |
| network_id        | 0243277b-4aa8-46d8-9e10-5c9ad5e01521 |
| project_id        | b1ebf33664df402693f729090cfab861     |
| subnetpool_id     | None                                 |
| updated_at        | 2016-06-15 15:50:45+00:00            |
+-------------------+--------------------------------------+
```

If using DHCP to manage instance IP addresses, adding a subnet causes a series of operations in the Networking service and OVN.

- The Networking service schedules the network on appropriate number of DHCP agents. The example environment contains three DHCP agents.
- Each DHCP agent spawns a network namespace with a `dnsmasq` process using an IP address from the subnet allocation.
- The OVN mechanism driver creates a logical switch port object in the OVN northbound database for each `dnsmasq` process.

OVN operations

The OVN mechanism driver and OVN perform the following operations during creation of a subnet on the provider network.

1. If the subnet uses DHCP for IP address management, create logical ports ports for each DHCP agent serving the subnet and bind them to the logical switch. In this example, the subnet contains two DHCP agents.

```
_uuid                : 5e144ab9-3e08-4910-b936-869bbbf254c8
addresses            : ["fa:16:3e:57:f9:ca 203.0.113.101"]
enabled              : true
external_ids         : {"neutron:port_name"=""}
name                 : "6ab052c2-7b75-4463-b34f-fd3426f61787"
options              : {}
parent_name          : []
port_security        : []
tag                  : []
type                 : ""
up                   : true

_uuid                : 38cf8b52-47c4-4e93-be8d-06bf71f6a7c9
addresses            : ["fa:16:3e:e0:eb:6d 203.0.113.102"]
enabled              : true
external_ids         : {"neutron:port_name"=""}
```

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```
name                : "94aee636-2394-48bc-b407-8224ab6bb1ab"
options             : {}
parent_name         : []
port_security       : []
tag                 : []
type                : ""
up                  : true

_uuid               : 924500c4-8580-4d5f-a7ad-8769f6e58ff5
acls                : []
external_ids        : {"neutron:network_name"=provider}
load_balancer       : []
name                : "neutron-670efade-7cd0-4d87-8a04-27f366eb8941"
ports               : [38cf8b52-47c4-4e93-be8d-06bf71f6a7c9,
                       5e144ab9-3e08-4910-b936-869bbbf254c8,
                       a576b812-9c3e-4cfb-9752-5d8500b3adf9]
```

2. The OVN northbound service creates port bindings for these logical ports and adds them to the appropriate multicast group.

   - Port bindings

```
_uuid               : 030024f4-61c3-4807-859b-07727447c427
chassis             : fc5ab9e7-bc28-40e8-ad52-2949358cc088
datapath            : bd0ab2b3-4cf4-4289-9529-ef430f6a89e6
logical_port        : "6ab052c2-7b75-4463-b34f-fd3426f61787"
mac                 : ["fa:16:3e:57:f9:ca 203.0.113.101"]
options             : {}
parent_port         : []
tag                 : []
tunnel_key          : 2
type                : ""

_uuid               : cc5bcd19-bcae-4e29-8cee-3ec8a8a75d46
chassis             : 6a9d0619-8818-41e6-abef-2f3d9a597c03
datapath            : bd0ab2b3-4cf4-4289-9529-ef430f6a89e6
logical_port        : "94aee636-2394-48bc-b407-8224ab6bb1ab"
mac                 : ["fa:16:3e:e0:eb:6d 203.0.113.102"]
options             : {}
parent_port         : []
tag                 : []
tunnel_key          : 3
type                : ""
```

   - Multicast groups

```
_uuid               : 39b32ccd-fa49-4046-9527-13318842461e
datapath            : bd0ab2b3-4cf4-4289-9529-ef430f6a89e6
name                : _MC_flood
ports               : [030024f4-61c3-4807-859b-07727447c427,
```

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```
                      904c3108-234d-41c0-b93c-116b7e352a75,
                      cc5bcd19-bcae-4e29-8cee-3ec8a8a75d46]
tunnel_key          : 65535
```

3. The OVN northbound service translates the logical ports into additional logical flows in the OVN southbound database.

```
Datapath: bd0ab2b3-4cf4-4289-9529-ef430f6a89e6  Pipeline: ingress
  table= 0(  ls_in_port_sec_l2), priority=   50,
    match=(inport == "94aee636-2394-48bc-b407-8224ab6bb1ab"),
    action=(next;)
  table= 0(  ls_in_port_sec_l2), priority=   50,
    match=(inport == "6ab052c2-7b75-4463-b34f-fd3426f61787"),
    action=(next;)
  table= 9(      ls_in_arp_rsp), priority=   50,
    match=(arp.tpa == 203.0.113.101 && arp.op == 1),
    action=(eth.dst = eth.src; eth.src = fa:16:3e:57:f9:ca;
            arp.op = 2; /* ARP reply */ arp.tha = arp.sha;
            arp.sha = fa:16:3e:57:f9:ca; arp.tpa = arp.spa;
            arp.spa = 203.0.113.101; outport = inport; inport = "";
            /* Allow sending out inport. */ output;)
  table= 9(      ls_in_arp_rsp), priority=   50,
    match=(arp.tpa == 203.0.113.102 && arp.op == 1),
    action=(eth.dst = eth.src; eth.src = fa:16:3e:e0:eb:6d;
            arp.op = 2; /* ARP reply */ arp.tha = arp.sha;
            arp.sha = fa:16:3e:e0:eb:6d; arp.tpa = arp.spa;
            arp.spa = 203.0.113.102; outport = inport;
            inport = ""; /* Allow sending out inport. */ output;)
  table=10(      ls_in_l2_lkup), priority=   50,
    match=(eth.dst == fa:16:3e:57:f9:ca),
    action=(outport = "6ab052c2-7b75-4463-b34f-fd3426f61787"; output;)
  table=10(      ls_in_l2_lkup), priority=   50,
    match=(eth.dst == fa:16:3e:e0:eb:6d),
    action=(outport = "94aee636-2394-48bc-b407-8224ab6bb1ab"; output;)
Datapath: bd0ab2b3-4cf4-4289-9529-ef430f6a89e6  Pipeline: egress
  table= 7( ls_out_port_sec_l2), priority=   50,
    match=(outport == "6ab052c2-7b75-4463-b34f-fd3426f61787"),
    action=(output;)
  table= 7( ls_out_port_sec_l2), priority=   50,
    match=(outport == "94aee636-2394-48bc-b407-8224ab6bb1ab"),
    action=(output;)
```

4. For each compute node without a DHCP agent on the subnet:

   - The OVN controller service translates the logical flows into flows on the integration bridge `br-int`.

```
cookie=0x0, duration=22.303s, table=32, n_packets=0, n_bytes=0,
    idle_age=22, priority=100,reg7=0xffff,metadata=0x4
    actions=load:0x4->NXM_NX_TUN_ID[0..23],
```

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```
        set_field:0xffff/0xffffffff->tun_metadata0,
        move:NXM_NX_REG6[0..14]->NXM_NX_TUN_METADATA0[16..30],
        output:5,output:4,resubmit(,33)
```

5. For each compute node with a DHCP agent on a subnet:

   - Creation of a DHCP network namespace adds two virtual switch ports. The first port connects the DHCP agent with `dnsmasq` process to the integration bridge and the second port patches the integration bridge to the provider bridge `br-provider`.

```
# ovs-ofctl show br-int
OFPT_FEATURES_REPLY (xid=0x2): dpid:000022024a1dc045
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_
↪MATCH_IP
actions: output enqueue set_vlan_vid set_vlan_pcp strip_vlan mod_dl_
↪src mod_dl_dst mod_nw_src mod_nw_dst mod_nw_tos mod_tp_src mod_tp_
↪dst
 7(tap6ab052c2-7b): addr:00:00:00:00:10:7f
     config:     PORT_DOWN
     state:      LINK_DOWN
     speed: 0 Mbps now, 0 Mbps max
 8(patch-br-int-to): addr:6a:8c:30:3f:d7:dd
    config:     0
    state:      0
    speed: 0 Mbps now, 0 Mbps max

# ovs-ofctl -O OpenFlow13 show br-provider
OFPT_FEATURES_REPLY (OF1.3) (xid=0x2): dpid:0000080027137c4a
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS GROUP_STATS QUEUE_
↪STATS
OFPST_PORT_DESC reply (OF1.3) (xid=0x3):
 1(patch-provnet-0): addr:fa:42:c5:3f:d7:6f
     config:     0
     state:      0
     speed: 0 Mbps now, 0 Mbps max
```

   - The OVN controller service translates these logical flows into flows on the integration bridge.

```
cookie=0x0, duration=17.731s, table=0, n_packets=3, n_bytes=258,
    idle_age=16, priority=100,in_port=7
    actions=load:0x2->NXM_NX_REG5[],load:0x4->OXM_OF_METADATA[],
        load:0x2->NXM_NX_REG6[],resubmit(,16)
cookie=0x0, duration=17.730s, table=0, n_packets=15, n_bytes=954,
    idle_age=2, priority=100,in_port=8,vlan_tci=0x0000/0x1000
    actions=load:0x1->NXM_NX_REG5[],load:0x4->OXM_OF_METADATA[],
        load:0x1->NXM_NX_REG6[],resubmit(,16)
cookie=0x0, duration=17.730s, table=0, n_packets=0, n_bytes=0,
    idle_age=17, priority=100,in_port=8,dl_vlan=0
```

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```
    actions=strip_vlan,load:0x1->NXM_NX_REG5[],
        load:0x4->OXM_OF_METADATA[],load:0x1->NXM_NX_REG6[],
        resubmit(,16)
cookie=0x0, duration=17.732s, table=16, n_packets=0, n_bytes=0,
    idle_age=17, priority=100,metadata=0x4,
        dl_src=01:00:00:00:00:00/01:00:00:00:00:00
    actions=drop
cookie=0x0, duration=17.732s, table=16, n_packets=0, n_bytes=0,
    idle_age=17, priority=100,metadata=0x4,vlan_tci=0x1000/0x1000
    actions=drop
cookie=0x0, duration=17.732s, table=16, n_packets=3, n_bytes=258,
    idle_age=16, priority=50,reg6=0x2,metadata=0x4 actions=resubmit(,
↪17)
cookie=0x0, duration=17.732s, table=16, n_packets=0, n_bytes=0,
    idle_age=17, priority=50,reg6=0x3,metadata=0x4 actions=resubmit(,
↪17)
cookie=0x0, duration=17.732s, table=16, n_packets=15, n_bytes=954,
    idle_age=2, priority=50,reg6=0x1,metadata=0x4 actions=resubmit(,
↪17)
cookie=0x0, duration=21.714s, table=17, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,18)
cookie=0x0, duration=21.714s, table=18, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,19)
cookie=0x0, duration=21.714s, table=19, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,20)
cookie=0x0, duration=21.714s, table=20, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,21)
cookie=0x0, duration=21.714s, table=21, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ip,reg0=0x1/0x1,metadata=0x4
    actions=ct(table=22,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=21.714s, table=21, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ipv6,reg0=0x1/0x1,metadata=0x4
    actions=ct(table=22,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=21.714s, table=21, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,22)
cookie=0x0, duration=21.714s, table=22, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,23)
cookie=0x0, duration=21.714s, table=23, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,24)
cookie=0x0, duration=21.714s, table=24, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ipv6,reg0=0x4/0x4,metadata=0x4
    actions=ct(table=25,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=21.714s, table=24, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ip,reg0=0x4/0x4,metadata=0x4
    actions=ct(table=25,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=21.714s, table=24, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ip,reg0=0x2/0x2,metadata=0x4
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,25)
cookie=0x0, duration=21.714s, table=24, n_packets=0, n_bytes=0,
```

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```
    idle_age=21, priority=100,ipv6,reg0=0x2/0x2,metadata=0x4
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,25)
cookie=0x0, duration=21.714s, table=24, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,25)
cookie=0x0, duration=21.714s, table=25, n_packets=15, n_bytes=954,
    idle_age=6, priority=100,reg6=0x1,metadata=0x4 actions=resubmit(,
↪26)
cookie=0x0, duration=21.714s, table=25, n_packets=0, n_bytes=0,
    idle_age=21, priority=50,arp,metadata=0x4,
        arp_tpa=203.0.113.101,arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:f9:5d:f3,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163ef95df3->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a81264->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],
        load:0->NXM_NX_REG6[],load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=21.714s, table=25, n_packets=0, n_bytes=0,
    idle_age=21, priority=50,arp,metadata=0x4,
        arp_tpa=203.0.113.102,arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:f0:a5:9f,
        load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163ef0a59f->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a81265->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],
        load:0->NXM_NX_REG6[],load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=21.714s, table=25, n_packets=3, n_bytes=258,
    idle_age=20, priority=0,metadata=0x4 actions=resubmit(,26)
cookie=0x0, duration=21.714s, table=26, n_packets=18, n_bytes=1212,
    idle_age=6, priority=100,metadata=0x4,
        dl_dst=01:00:00:00:00:00/01:00:00:00:00:00
    actions=load:0xffff->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=21.714s, table=26, n_packets=0, n_bytes=0,
    idle_age=21, priority=50,metadata=0x4,dl_dst=fa:16:3e:f0:a5:9f
    actions=load:0x3->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=21.714s, table=26, n_packets=0, n_bytes=0,
    idle_age=21, priority=50,metadata=0x4,dl_dst=fa:16:3e:f9:5d:f3
    actions=load:0x2->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=21.714s, table=26, n_packets=0, n_bytes=0,
    idle_age=21, priority=0,metadata=0x4
    actions=load:0xfffe->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=17.731s, table=33, n_packets=0, n_bytes=0,
    idle_age=17, priority=100,reg7=0x2,metadata=0x4
    actions=load:0x2->NXM_NX_REG5[],resubmit(,34)
cookie=0x0, duration=118.126s, table=33, n_packets=0, n_bytes=0,
```

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```
    idle_age=118, hard_age=17, priority=100,reg7=0xfffe,metadata=0x4
    actions=load:0x1->NXM_NX_REG5[],load:0x1->NXM_NX_REG7[],
        resubmit(,34),load:0xfffe->NXM_NX_REG7[]
cookie=0x0, duration=118.126s, table=33, n_packets=18, n_bytes=1212,
    idle_age=2, hard_age=17, priority=100,reg7=0xffff,metadata=0x4
    actions=load:0x2->NXM_NX_REG5[],load:0x2->NXM_NX_REG7[],
        resubmit(,34),load:0x1->NXM_NX_REG5[],load:0x1->NXM_NX_
↪REG7[],
        resubmit(,34),load:0xffff->NXM_NX_REG7[]
cookie=0x0, duration=17.730s, table=33, n_packets=0, n_bytes=0,
    idle_age=17, priority=100,reg7=0x1,metadata=0x4
    actions=load:0x1->NXM_NX_REG5[],resubmit(,34)
cookie=0x0, duration=17.697s, table=33, n_packets=0, n_bytes=0,
    idle_age=17, priority=100,reg7=0x3,metadata=0x4
    actions=load:0x1->NXM_NX_REG7[],resubmit(,33)
cookie=0x0, duration=17.731s, table=34, n_packets=3, n_bytes=258,
    idle_age=16, priority=100,reg6=0x2,reg7=0x2,metadata=0x4
    actions=drop
cookie=0x0, duration=17.730s, table=34, n_packets=15, n_bytes=954,
    idle_age=2, priority=100,reg6=0x1,reg7=0x1,metadata=0x4
    actions=drop
cookie=0x0, duration=21.714s, table=48, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,49)
cookie=0x0, duration=21.714s, table=49, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,50)
cookie=0x0, duration=21.714s, table=50, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ip,reg0=0x1/0x1,metadata=0x4
    actions=ct(table=51,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=21.714s, table=50, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ipv6,reg0=0x1/0x1,metadata=0x4
    actions=ct(table=51,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=21.714s, table=50, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,51)
cookie=0x0, duration=21.714s, table=51, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,52)
cookie=0x0, duration=21.714s, table=52, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,53)
cookie=0x0, duration=21.714s, table=53, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ip,reg0=0x4/0x4,metadata=0x4
    actions=ct(table=54,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=21.714s, table=53, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ipv6,reg0=0x4/0x4,metadata=0x4
    actions=ct(table=54,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=21.714s, table=53, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ipv6,reg0=0x2/0x2,metadata=0x4
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,54)
cookie=0x0, duration=21.714s, table=53, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,ip,reg0=0x2/0x2,metadata=0x4
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,54)
```

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```
cookie=0x0, duration=21.714s, table=53, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,54)
cookie=0x0, duration=21.714s, table=54, n_packets=18, n_bytes=1212,
    idle_age=6, priority=0,metadata=0x4 actions=resubmit(,55)
cookie=0x0, duration=21.714s, table=55, n_packets=18, n_bytes=1212,
    idle_age=6, priority=100,metadata=0x4,
        dl_dst=01:00:00:00:00:00/01:00:00:00:00:00
    actions=resubmit(,64)
cookie=0x0, duration=21.714s, table=55, n_packets=0, n_bytes=0,
    idle_age=21, priority=50,reg7=0x3,metadata=0x4
    actions=resubmit(,64)
cookie=0x0, duration=21.714s, table=55, n_packets=0, n_bytes=0,
    idle_age=21, priority=50,reg7=0x2,metadata=0x4
    actions=resubmit(,64)
cookie=0x0, duration=21.714s, table=55, n_packets=0, n_bytes=0,
    idle_age=21, priority=50,reg7=0x1,metadata=0x4
    actions=resubmit(,64)
cookie=0x0, duration=21.712s, table=64, n_packets=15, n_bytes=954,
    idle_age=6, priority=100,reg7=0x3,metadata=0x4 actions=output:7
cookie=0x0, duration=21.711s, table=64, n_packets=3, n_bytes=258,
    idle_age=20, priority=100,reg7=0x1,metadata=0x4 actions=output:8
```

Self-service networks

A self-service (project) network includes only virtual components, thus enabling projects to manage them without additional configuration of the underlying physical network. The OVN mechanism driver supports Geneve and VLAN network types with a preference toward Geneve. Projects can choose to isolate self-service networks, connect two or more together via routers, or connect them to provider networks via routers with appropriate capabilities. Similar to provider networks, self-service networks can use arbitrary names.

Note

Similar to provider networks, self-service VLAN networks map to a unique bridge on each compute node that supports launching instances on those networks. Self-service VLAN networks also require several commands at the host and OVS levels. The following example assumes use of Geneve self-service networks.

Create a self-service network

Creating a self-service network involves several commands at the Networking service level that yield a series of operations at the OVN level to create the virtual network components. The following example creates a Geneve self-service network and binds a subnet to it. The subnet uses DHCP to distribute IP addresses to instances.

1. On the controller node, source the credentials for a regular (non-privileged) project. The following example uses the `demo` project.

2. On the controller node, create a self-service network in the Networking service.

```
$ openstack network create selfservice
+-------------------------+--------------------------------------+
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints |                                      |
| availability_zones      |                                      |
| created_at              | 2016-06-09T15:42:41                  |
| description             |                                      |
| id                      | f49791f7-e653-4b43-99b1-0f5557c313e4 |
| ipv4_address_scope      | None                                 |
| ipv6_address_scope      | None                                 |
| mtu                     | 1442                                 |
| name                    | selfservice                          |
| port_security_enabled   | True                                 |
| project_id              | 1ef26f483b9d44e8ac0c97388d6cb609     |
| router_external         | Internal                             |
| shared                  | False                                |
| status                  | ACTIVE                               |
| subnets                 |                                      |
| tags                    | []                                   |
| updated_at              | 2016-06-09T15:42:41                  |
+-------------------------+--------------------------------------+
```

OVN operations

The OVN mechanism driver and OVN perform the following operations during creation of a self-service network.

1. The mechanism driver translates the network into a logical switch in the OVN northbound database.

```
uuid                   : 0ab40684-7cf8-4d6c-ae8b-9d9143762d37
acls                    : []
external_ids            : {"neutron:network_name"="selfservice"}
name                    : "neutron-d5aadceb-d8d6-41c8-9252-c5e0fe6c26a5"
ports                   : []
```

2. The OVN northbound service translates this object into new datapath bindings and logical flows in the OVN southbound database.

   - Datapath bindings

   ```
   _uuid                : 0b214af6-8910-489c-926a-fd0ed16a8251
   external_ids         : {logical-switch="15e2c80b-1461-4003-9869-
   ↪80416cd97de5"}
   tunnel_key           : 5
   ```

   - Logical flows

   ```
   Datapath: 0b214af6-8910-489c-926a-fd0ed16a8251  Pipeline: ingress
     table= 0(  ls_in_port_sec_l2), priority=  100, match=(eth.src[40]),
   ```

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```
    action=(drop;)
  table= 0(  ls_in_port_sec_l2), priority=  100, match=(vlan.
↪present),
    action=(drop;)
  table= 1(  ls_in_port_sec_ip), priority=    0, match=(1),
    action=(next;)
  table= 2(  ls_in_port_sec_nd), priority=    0, match=(1),
    action=(next;)
  table= 3(      ls_in_pre_acl), priority=    0, match=(1),
    action=(next;)
  table= 4(       ls_in_pre_lb), priority=    0, match=(1),
    action=(next;)
  table= 5( ls_in_pre_stateful), priority=  100, match=(reg0[0] ==␣
↪1),
    action=(ct_next;)
  table= 5( ls_in_pre_stateful), priority=    0, match=(1),
    action=(next;)
  table= 6(          ls_in_acl), priority=    0, match=(1),
    action=(next;)
  table= 7(           ls_in_lb), priority=    0, match=(1),
    action=(next;)
  table= 8(     ls_in_stateful), priority=  100, match=(reg0[2] ==␣
↪1),
    action=(ct_lb;)
  table= 8(     ls_in_stateful), priority=  100, match=(reg0[1] ==␣
↪1),
    action=(ct_commit; next;)
  table= 8(     ls_in_stateful), priority=    0, match=(1),
    action=(next;)
  table= 9(      ls_in_arp_rsp), priority=    0, match=(1),
    action=(next;)
  table=10(      ls_in_l2_lkup), priority=  100, match=(eth.mcast),
    action=(outport = "_MC_flood"; output;)
Datapath: 0b214af6-8910-489c-926a-fd0ed16a8251  Pipeline: egress
  table= 0(      ls_out_pre_lb), priority=    0, match=(1),
    action=(next;)
  table= 1(     ls_out_pre_acl), priority=    0, match=(1),
    action=(next;)
  table= 2(ls_out_pre_stateful), priority=  100, match=(reg0[0] ==␣
↪1),
    action=(ct_next;)
  table= 2(ls_out_pre_stateful), priority=    0, match=(1),
    action=(next;)
  table= 3(          ls_out_lb), priority=    0, match=(1),
    action=(next;)
  table= 4(         ls_out_acl), priority=    0, match=(1),
    action=(next;)
  table= 5(    ls_out_stateful), priority=  100, match=(reg0[1] ==␣
↪1),
```

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```
    action=(ct_commit; next;)
  table= 5(     ls_out_stateful), priority=  100, match=(reg0[2] ==␣
↪1),
    action=(ct_lb;)
  table= 5(     ls_out_stateful), priority=    0, match=(1),
    action=(next;)
  table= 6( ls_out_port_sec_ip), priority=    0, match=(1),
    action=(next;)
  table= 7( ls_out_port_sec_l2), priority=  100, match=(eth.mcast),
    action=(output;)
```

Note

These actions do not create flows on any nodes.

Create a subnet on the self-service network

A self-service network requires at least one subnet. In most cases, the environment provides suitable values for IP address allocation for instances, default gateway IP address, and metadata such as name resolution.

1. On the controller node, create a subnet bound to the self-service network `selfservice`.

```
$ openstack subnet create --network selfservice --subnet-range 192.168.1.
↪0/24 selfservice-v4
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 192.168.1.2-192.168.1.254            |
| cidr              | 192.168.1.0/24                       |
| created_at        | 2016-06-16 00:19:08+00:00            |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 192.168.1.1                          |
| headers           |                                      |
| host_routes       |                                      |
| id                | 8f027f25-0112-45b9-a1b9-2f8097c57219 |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | selfservice-v4                       |
| network_id        | 8ed4e43b-63ef-41ed-808b-b59f1120aec0 |
| project_id        | b1ebf33664df402693f729090cfab861     |
| subnetpool_id     | None                                 |
| updated_at        | 2016-06-16 00:19:08+00:00            |
+-------------------+--------------------------------------+
```

OVN operations

The OVN mechanism driver and OVN perform the following operations during creation of a subnet on a self-service network.

1. If the subnet uses DHCP for IP address management, create logical ports ports for each DHCP agent serving the subnet and bind them to the logical switch. In this example, the subnet contains two DHCP agents.

```
_uuid               : 1ed7c28b-dc69-42b8-bed6-46477bb8b539
addresses           : ["fa:16:3e:94:db:5e 192.168.1.2"]
enabled             : true
external_ids        : {"neutron:port_name"=""}
name                : "0cfbbdca-ff58-4cf8-a7d3-77daaebe3056"
options             : {}
parent_name         : []
port_security       : []
tag                 : []
type                : ""
up                  : true

_uuid               : ae10a5e0-db25-4108-b06a-d2d5c127d9c4
addresses           : ["fa:16:3e:90:bd:f1 192.168.1.3"]
enabled             : true
external_ids        : {"neutron:port_name"=""}
name                : "74930ace-d939-4bca-b577-fccba24c3fca"
options             : {}
parent_name         : []
port_security       : []
tag                 : []
type                : ""
up                  : true

_uuid               : 0ab40684-7cf8-4d6c-ae8b-9d9143762d37
acls                : []
external_ids        : {"neutron:network_name"="selfservice"}
name                : "neutron-d5aadceb-d8d6-41c8-9252-c5e0fe6c26a5"
ports               : [1ed7c28b-dc69-42b8-bed6-46477bb8b539,
                      ae10a5e0-db25-4108-b06a-d2d5c127d9c4]
```

2. The OVN northbound service creates port bindings for these logical ports and adds them to the appropriate multicast group.

   - Port bindings

```
_uuid               : 3e463ca0-951c-46fd-b6cf-05392fa3aa1f
chassis             : 6a9d0619-8818-41e6-abef-2f3d9a597c03
datapath            : 0b214af6-8910-489c-926a-fd0ed16a8251
logical_port        : "a203b410-97c1-4e4a-b0c3-558a10841c16"
mac                 : ["fa:16:3e:a1:dc:58 192.168.1.3"]
options             : {}
parent_port         : []
```

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```
tag                 : []
tunnel_key          : 2
type                : ""

_uuid               : fa7b294d-2a62-45ae-8de3-a41c002de6de
chassis             : d63e8ae8-caf3-4a6b-9840-5c3a57febcac
datapath            : 0b214af6-8910-489c-926a-fd0ed16a8251
logical_port        : "39b23721-46f4-4747-af54-7e12f22b3397"
mac                 : ["fa:16:3e:1a:b4:23 192.168.1.2"]
options             : {}
parent_port         : []
tag                 : []
tunnel_key          : 1
type                : ""
```

- Multicast groups

```
_uuid               : c08d0102-c414-4a47-98d9-dd3fa9f9901c
datapath            : 0b214af6-8910-489c-926a-fd0ed16a8251
name                : _MC_flood
ports               : [3e463ca0-951c-46fd-b6cf-05392fa3aa1f,
                       fa7b294d-2a62-45ae-8de3-a41c002de6de]
tunnel_key          : 65535
```

3. The OVN northbound service translates the logical ports into logical flows in the OVN southbound database.

```
Datapath: 0b214af6-8910-489c-926a-fd0ed16a8251  Pipeline: ingress
  table= 0(  ls_in_port_sec_l2), priority=   50,
    match=(inport == "39b23721-46f4-4747-af54-7e12f22b3397"),
    action=(next;)
  table= 0(  ls_in_port_sec_l2), priority=   50,
    match=(inport == "a203b410-97c1-4e4a-b0c3-558a10841c16"),
    action=(next;)
  table= 9(      ls_in_arp_rsp), priority=   50,
    match=(arp.tpa == 192.168.1.2 && arp.op == 1),
    action=(eth.dst = eth.src; eth.src = fa:16:3e:1a:b4:23;
            arp.op = 2; /* ARP reply */ arp.tha = arp.sha;
            arp.sha = fa:16:3e:1a:b4:23; arp.tpa = arp.spa;
            arp.spa = 192.168.1.2; outport = inport;
            inport = ""; /* Allow sending out inport. */ output;)
  table= 9(      ls_in_arp_rsp), priority=   50,
    match=(arp.tpa == 192.168.1.3 && arp.op == 1),
    action=(eth.dst = eth.src; eth.src = fa:16:3e:a1:dc:58;
            arp.op = 2; /* ARP reply */ arp.tha = arp.sha;
            arp.sha = fa:16:3e:a1:dc:58; arp.tpa = arp.spa;
            arp.spa = 192.168.1.3; outport = inport;
            inport = ""; /* Allow sending out inport. */ output;)
  table=10(      ls_in_l2_lkup), priority=   50,
    match=(eth.dst == fa:16:3e:a1:dc:58),
```

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```
    action=(outport = "a203b410-97c1-4e4a-b0c3-558a10841c16"; output;)
  table=10(      ls_in_l2_lkup), priority=   50,
    match=(eth.dst == fa:16:3e:1a:b4:23),
    action=(outport = "39b23721-46f4-4747-af54-7e12f22b3397"; output;)
Datapath: 0b214af6-8910-489c-926a-fd0ed16a8251  Pipeline: egress
  table= 7( ls_out_port_sec_l2), priority=   50,
    match=(outport == "39b23721-46f4-4747-af54-7e12f22b3397"),
    action=(output;)
  table= 7( ls_out_port_sec_l2), priority=   50,
    match=(outport == "a203b410-97c1-4e4a-b0c3-558a10841c16"),
    action=(output;)
```

4. For each compute node without a DHCP agent on the subnet:

   - The OVN controller service translates these objects into flows on the integration bridge `br-int`.

```
# ovs-ofctl dump-flows br-int
cookie=0x0, duration=9.054s, table=32, n_packets=0, n_bytes=0,
    idle_age=9, priority=100,reg7=0xffff,metadata=0x5
    actions=load:0x5->NXM_NX_TUN_ID[0..23],
        set_field:0xffff/0xffffffff->tun_metadata0,
        move:NXM_NX_REG6[0..14]->NXM_NX_TUN_METADATA0[16..30],
        output:4,output:3
```

5. For each compute node with a DHCP agent on the subnet:

   - Creation of a DHCP network namespace adds a virtual switch ports that connects the DHCP agent with the `dnsmasq` process to the integration bridge.

```
# ovs-ofctl show br-int
OFPT_FEATURES_REPLY (xid=0x2): dpid:000022024a1dc045
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_
↪MATCH_IP
actions: output enqueue set_vlan_vid set_vlan_pcp strip_vlan mod_dl_
↪src mod_dl_dst mod_nw_src mod_nw_dst mod_nw_tos mod_tp_src mod_tp_
↪dst
 9(tap39b23721-46): addr:00:00:00:00:b0:5d
     config:     PORT_DOWN
     state:      LINK_DOWN
     speed: 0 Mbps now, 0 Mbps max
```

   - The OVN controller service translates these objects into flows on the integration bridge.

```
cookie=0x0, duration=21.074s, table=0, n_packets=8, n_bytes=648,
    idle_age=11, priority=100,in_port=9
    actions=load:0x2->NXM_NX_REG5[],load:0x5->OXM_OF_METADATA[],
        load:0x1->NXM_NX_REG6[],resubmit(,16)
cookie=0x0, duration=21.076s, table=16, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,metadata=0x5,
```

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```
        dl_src=01:00:00:00:00:00/01:00:00:00:00:00
    actions=drop
cookie=0x0, duration=21.075s, table=16, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,metadata=0x5,vlan_tci=0x1000/0x1000
    actions=drop
cookie=0x0, duration=21.076s, table=16, n_packets=0, n_bytes=0,
    idle_age=21, priority=50,reg6=0x2,metadata=0x5
    actions=resubmit(,17)
cookie=0x0, duration=21.075s, table=16, n_packets=8, n_bytes=648,
    idle_age=11, priority=50,reg6=0x1,metadata=0x5
    actions=resubmit(,17)
cookie=0x0, duration=21.075s, table=17, n_packets=8, n_bytes=648,
    idle_age=11, priority=0,metadata=0x5
    actions=resubmit(,18)
cookie=0x0, duration=21.076s, table=18, n_packets=8, n_bytes=648,
    idle_age=11, priority=0,metadata=0x5
    actions=resubmit(,19)
cookie=0x0, duration=21.076s, table=19, n_packets=8, n_bytes=648,
    idle_age=11, priority=0,metadata=0x5
    actions=resubmit(,20)
cookie=0x0, duration=21.075s, table=20, n_packets=8, n_bytes=648,
    idle_age=11, priority=0,metadata=0x5
    actions=resubmit(,21)
cookie=0x0, duration=5.398s, table=21, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ipv6,reg0=0x1/0x1,metadata=0x5
    actions=ct(table=22,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=5.398s, table=21, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ip,reg0=0x1/0x1,metadata=0x5
    actions=ct(table=22,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=5.398s, table=22, n_packets=6, n_bytes=508,
    idle_age=2, priority=0,metadata=0x5
    actions=resubmit(,23)
cookie=0x0, duration=5.398s, table=23, n_packets=6, n_bytes=508,
    idle_age=2, priority=0,metadata=0x5
    actions=resubmit(,24)
cookie=0x0, duration=5.398s, table=24, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ipv6,reg0=0x4/0x4,metadata=0x5
    actions=ct(table=25,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=5.398s, table=24, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ip,reg0=0x4/0x4,metadata=0x5
    actions=ct(table=25,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=5.398s, table=24, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ipv6,reg0=0x2/0x2,metadata=0x5
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,25)
cookie=0x0, duration=5.398s, table=24, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ip,reg0=0x2/0x2,metadata=0x5
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,25)
cookie=0x0, duration=5.399s, table=24, n_packets=6, n_bytes=508,
    idle_age=2, priority=0,metadata=0x5 actions=resubmit(,25)
```

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```
cookie=0x0, duration=5.398s, table=25, n_packets=0, n_bytes=0,
    idle_age=5, priority=50,arp,metadata=0x5,
        arp_tpa=192.168.1.2,arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:82:8b:0e,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163e828b0e->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a80102->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],load:0->NXM_NX_REG6[],
        load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=5.378s, table=25, n_packets=0, n_bytes=0,
    idle_age=5, priority=50,arp,metadata=0x5,arp_tpa=192.168.1.3,
        arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:d5:00:02,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163ed50002->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a80103->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],load:0->NXM_NX_REG6[],
        load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=5.399s, table=25, n_packets=6, n_bytes=508,
    idle_age=2, priority=0,metadata=0x5
    actions=resubmit(,26)
cookie=0x0, duration=5.399s, table=26, n_packets=6, n_bytes=508,
    idle_age=2, priority=100,metadata=0x5,
        dl_dst=01:00:00:00:00:00/01:00:00:00:00:00
    actions=load:0xffff->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=5.398s, table=26, n_packets=0, n_bytes=0,
    idle_age=5, priority=50,metadata=0x5,dl_dst=fa:16:3e:d5:00:02
    actions=load:0x2->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=5.398s, table=26, n_packets=0, n_bytes=0,
    idle_age=5, priority=50,metadata=0x5,dl_dst=fa:16:3e:82:8b:0e
    actions=load:0x1->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=21.038s, table=32, n_packets=0, n_bytes=0,
    idle_age=21, priority=100,reg7=0x2,metadata=0x5
    actions=load:0x5->NXM_NX_TUN_ID[0..23],
        set_field:0x2/0xffffffff->tun_metadata0,
        move:NXM_NX_REG6[0..14]->NXM_NX_TUN_METADATA0[16..30],
↪output:4
cookie=0x0, duration=21.038s, table=32, n_packets=8, n_bytes=648,
    idle_age=11, priority=100,reg7=0xffff,metadata=0x5
    actions=load:0x5->NXM_NX_TUN_ID[0..23],
        set_field:0xffff/0xffffffff->tun_metadata0,
        move:NXM_NX_REG6[0..14]->NXM_NX_TUN_METADATA0[16..30],
        output:4,resubmit(,33)
cookie=0x0, duration=5.397s, table=33, n_packets=12, n_bytes=1016,
    idle_age=2, priority=100,reg7=0xffff,metadata=0x5
```

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```
    actions=load:0x1->NXM_NX_REG7[],resubmit(,34),
       load:0xffff->NXM_NX_REG7[]
cookie=0x0, duration=5.397s, table=33, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,reg7=0x1,metadata=0x5
    actions=resubmit(,34)
cookie=0x0, duration=21.074s, table=34, n_packets=8, n_bytes=648,
    idle_age=11, priority=100,reg6=0x1,reg7=0x1,metadata=0x5
    actions=drop
cookie=0x0, duration=21.076s, table=48, n_packets=8, n_bytes=648,
    idle_age=11, priority=0,metadata=0x5 actions=resubmit(,49)
cookie=0x0, duration=21.075s, table=49, n_packets=8, n_bytes=648,
    idle_age=11, priority=0,metadata=0x5 actions=resubmit(,50)
cookie=0x0, duration=5.398s, table=50, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ipv6,reg0=0x1/0x1,metadata=0x5
    actions=ct(table=51,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=5.398s, table=50, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ip,reg0=0x1/0x1,metadata=0x5
    actions=ct(table=51,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=5.398s, table=50, n_packets=6, n_bytes=508,
    idle_age=3, priority=0,metadata=0x5
    actions=resubmit(,51)
cookie=0x0, duration=5.398s, table=51, n_packets=6, n_bytes=508,
    idle_age=3, priority=0,metadata=0x5
    actions=resubmit(,52)
cookie=0x0, duration=5.398s, table=52, n_packets=6, n_bytes=508,
    idle_age=3, priority=0,metadata=0x5
    actions=resubmit(,53)
cookie=0x0, duration=5.399s, table=53, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ipv6,reg0=0x4/0x4,metadata=0x5
    actions=ct(table=54,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=5.398s, table=53, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ip,reg0=0x4/0x4,metadata=0x5
    actions=ct(table=54,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=5.398s, table=53, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ip,reg0=0x2/0x2,metadata=0x5
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,54)
cookie=0x0, duration=5.398s, table=53, n_packets=0, n_bytes=0,
    idle_age=5, priority=100,ipv6,reg0=0x2/0x2,metadata=0x5
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,54)
cookie=0x0, duration=5.398s, table=53, n_packets=6, n_bytes=508,
    idle_age=3, priority=0,metadata=0x5
    actions=resubmit(,54)
cookie=0x0, duration=5.398s, table=54, n_packets=6, n_bytes=508,
    idle_age=3, priority=0,metadata=0x5
    actions=resubmit(,55)
cookie=0x0, duration=5.398s, table=55, n_packets=6, n_bytes=508,
    idle_age=3, priority=100,metadata=0x5,
        dl_dst=01:00:00:00:00:00/01:00:00:00:00:00
    actions=resubmit(,64)
```

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```
cookie=0x0, duration=5.398s, table=55, n_packets=0, n_bytes=0,
    idle_age=5, priority=50,reg7=0x1,metadata=0x5
    actions=resubmit(,64)
cookie=0x0, duration=5.398s, table=55, n_packets=0, n_bytes=0,
    idle_age=5, priority=50,reg7=0x2,metadata=0x5
    actions=resubmit(,64)
cookie=0x0, duration=5.397s, table=64, n_packets=6, n_bytes=508,
    idle_age=3, priority=100,reg7=0x1,metadata=0x5
    actions=output:9
```

Routers

Routers

Routers pass traffic between layer-3 networks.

Create a router

1. On the controller node, source the credentials for a regular (non-privileged) project. The following example uses the demo project.
2. On the controller node, create router in the Networking service.

```
$ openstack router create router
+-----------------------+--------------------------------------+
| Field                 | Value                                |
+-----------------------+--------------------------------------+
| admin_state_up        | UP                                   |
| description           |                                      |
| external_gateway_info | null                                 |
| headers               |                                      |
| id                    | 24addfcd-5506-405d-a59f-003644c3d16a |
| name                  | router                               |
| project_id            | b1ebf33664df402693f729090cfab861     |
| routes                |                                      |
| status                | ACTIVE                               |
+-----------------------+--------------------------------------+
```

OVN operations

The OVN mechanism driver and OVN perform the following operations when creating a router.

1. The OVN mechanism driver translates the router into a logical router object in the OVN northbound database.

```
_uuid               : 1c2e340d-dac9-496b-9e86-1065f9dab752
default_gw          : []
enabled             : []
external_ids        : {"neutron:router_name"="router"}
name                : "neutron-a24fd760-1a99-4eec-9f02-24bb284ff708"
```

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```
ports               : []
static_routes       : []
```

2. The OVN northbound service translates this object into logical flows and datapath bindings in the OVN southbound database.

   - Datapath bindings

   ```
   _uuid               : 4a7485c6-a1ef-46a5-b57c-5ddb6ac15aaa
   external_ids        : {logical-router="1c2e340d-dac9-496b-9e86-
   ↪1065f9dab752"}
   tunnel_key          : 3
   ```

   - Logical flows

   ```
   Datapath: 4a7485c6-a1ef-46a5-b57c-5ddb6ac15aaa  Pipeline: ingress
     table= 0(    lr_in_admission), priority=  100,
       match=(vlan.present || eth.src[40]),
       action=(drop;)
     table= 1(     lr_in_ip_input), priority=  100,
       match=(ip4.mcast || ip4.src == 255.255.255.255 ||
              ip4.src == 127.0.0.0/8 || ip4.dst == 127.0.0.0/8 ||
              ip4.src == 0.0.0.0/8 || ip4.dst == 0.0.0.0/8),
       action=(drop;)
     table= 1(     lr_in_ip_input), priority=   50, match=(ip4.mcast),
       action=(drop;)
     table= 1(     lr_in_ip_input), priority=   50, match=(eth.bcast),
       action=(drop;)
     table= 1(     lr_in_ip_input), priority=   30,
       match=(ip4 && ip.ttl == {0, 1}), action=(drop;)
     table= 1(     lr_in_ip_input), priority=    0, match=(1),
       action=(next;)
     table= 2(       lr_in_unsnat), priority=    0, match=(1),
       action=(next;)
     table= 3(         lr_in_dnat), priority=    0, match=(1),
       action=(next;)
     table= 5(  lr_in_arp_resolve), priority=    0, match=(1),
       action=(get_arp(outport, reg0); next;)
     table= 6(  lr_in_arp_request), priority=  100,
       match=(eth.dst == 00:00:00:00:00:00),
       action=(arp { eth.dst = ff:ff:ff:ff:ff:ff; arp.spa = reg1;
               arp.op = 1; output; };)
     table= 6(  lr_in_arp_request), priority=    0, match=(1),
       action=(output;)
   Datapath: 4a7485c6-a1ef-46a5-b57c-5ddb6ac15aaa  Pipeline: egress
     table= 0(        lr_out_snat), priority=    0, match=(1),
       action=(next;)
   ```

3. The OVN controller service on each compute node translates these objects into flows on the integration bridge `br-int`.

```
# ovs-ofctl dump-flows br-int
cookie=0x0, duration=6.402s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,metadata=0x5,vlan_tci=0x1000/0x1000
    actions=drop
cookie=0x0, duration=6.402s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,metadata=0x5,
    dl_src=01:00:00:00:00:00/01:00:00:00:00:00
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x5,nw_dst=127.0.0.0/8
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x5,nw_dst=0.0.0.0/8
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x5,nw_dst=224.0.0.0/4
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,ip,metadata=0x5,nw_dst=224.0.0.0/4
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x5,nw_src=255.255.255.255
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x5,nw_src=127.0.0.0/8
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x5,nw_src=0.0.0.0/8
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=90,arp,metadata=0x5,arp_op=2
    actions=push:NXM_NX_REG0[],push:NXM_OF_ETH_SRC[],
        push:NXM_NX_ARP_SHA[],push:NXM_OF_ARP_SPA[],
        pop:NXM_NX_REG0[],pop:NXM_OF_ETH_SRC[],
        controller(userdata=00.00.00.01.00.00.00.00),
        pop:NXM_OF_ETH_SRC[],pop:NXM_NX_REG0[]
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,metadata=0x5,dl_dst=ff:ff:ff:ff:ff:ff
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=30,ip,metadata=0x5,nw_ttl=0
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=30,ip,metadata=0x5,nw_ttl=1
    actions=drop
cookie=0x0, duration=6.402s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x5
    actions=resubmit(,18)
cookie=0x0, duration=6.402s, table=18, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x5
```

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```
    actions=resubmit(,19)
cookie=0x0, duration=6.402s, table=19, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x5
    actions=resubmit(,20)
cookie=0x0, duration=6.402s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x5
    actions=resubmit(,32)
cookie=0x0, duration=6.402s, table=48, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x5
    actions=resubmit(,49)
```

Attach a self-service network to the router

Self-service networks, particularly subnets, must interface with a router to enable connectivity with other self-service and provider networks.

1. On the controller node, add the self-service network subnet `selfservice-v4` to the router `router`.

   ```
   $ openstack router add subnet router selfservice-v4
   ```

   Note

   This command provides no output.

OVN operations

The OVN mechanism driver and OVN perform the following operations when adding a subnet as an interface on a router.

1. The OVN mechanism driver translates the operation into logical objects and devices in the OVN northbound database and performs a series of operations on them.

   - Create a logical port.

     ```
     _uuid               : 4c9e70b1-fff0-4d0d-af8e-42d3896eb76f
     addresses           : ["fa:16:3e:0c:55:62 192.168.1.1"]
     enabled             : true
     external_ids        : {"neutron:port_name"=""}
     name                : "5b72d278-5b16-44a6-9aa0-9e513a429506"
     options             : {router-port="lrp-5b72d278-5b16-44a6-9aa0-
     ↪9e513a429506"}
     parent_name         : []
     port_security       : []
     tag                 : []
     type                : router
     up                  : false
     ```

   - Add the logical port to logical switch.

```
_uuid               : 0ab40684-7cf8-4d6c-ae8b-9d9143762d37
acls                : []
external_ids        : {"neutron:network_name"="selfservice"}
name                : "neutron-d5aadceb-d8d6-41c8-9252-c5e0fe6c26a5"
ports               : [1ed7c28b-dc69-42b8-bed6-46477bb8b539,
                       4c9e70b1-fff0-4d0d-af8e-42d3896eb76f,
                       ae10a5e0-db25-4108-b06a-d2d5c127d9c4]
```

- Create a logical router port object.

```
_uuid               : f60ccb93-7b3d-4713-922c-37104b7055dc
enabled             : []
external_ids        : {}
mac                 : "fa:16:3e:0c:55:62"
name                : "lrp-5b72d278-5b16-44a6-9aa0-9e513a429506"
network             : "192.168.1.1/24"
peer                : []
```

- Add the logical router port to the logical router object.

```
_uuid               : 1c2e340d-dac9-496b-9e86-1065f9dab752
default_gw          : []
enabled             : []
external_ids        : {"neutron:router_name"="router"}
name                : "neutron-a24fd760-1a99-4eec-9f02-24bb284ff708"
ports               : [f60ccb93-7b3d-4713-922c-37104b7055dc]
static_routes       : []
```

2. The OVN northbound service translates these objects into logical flows, datapath bindings, and the appropriate multicast groups in the OVN southbound database.

   - Logical flows in the logical router datapath

```
Datapath: 4a7485c6-a1ef-46a5-b57c-5ddb6ac15aaa  Pipeline: ingress
  table= 0(    lr_in_admission), priority=   50,
    match=((eth.mcast || eth.dst == fa:16:3e:0c:55:62) &&
           inport == "lrp-5b72d278-5b16-44a6-9aa0-9e513a429506"),
    action=(next;)
  table= 1(     lr_in_ip_input), priority=  100,
    match=(ip4.src == {192.168.1.1, 192.168.1.255}), action=(drop;)
  table= 1(     lr_in_ip_input), priority=   90,
    match=(ip4.dst == 192.168.1.1 && icmp4.type == 8 &&
           icmp4.code == 0),
    action=(ip4.dst = ip4.src; ip4.src = 192.168.1.1; ip.ttl = 255;
            icmp4.type = 0;
            inport = ""; /* Allow sending out inport. */ next; )
  table= 1(     lr_in_ip_input), priority=   90,
    match=(inport == "lrp-5b72d278-5b16-44a6-9aa0-9e513a429506" &&
           arp.tpa == 192.168.1.1 && arp.op == 1),
    action=(eth.dst = eth.src; eth.src = fa:16:3e:0c:55:62;
            arp.op = 2; /* ARP reply */ arp.tha = arp.sha;
```

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```
            arp.sha = fa:16:3e:0c:55:62; arp.tpa = arp.spa;
            arp.spa = 192.168.1.1;
            outport = "lrp-5b72d278-5b16-44a6-9aa0-9e513a429506";
            inport = ""; /* Allow sending out inport. */ output;)
  table= 1(     lr_in_ip_input), priority=   60,
    match=(ip4.dst == 192.168.1.1), action=(drop;)
  table= 4(   lr_in_ip_routing), priority=   24,
    match=(ip4.dst == 192.168.1.0/255.255.255.0),
    action=(ip.ttl--; reg0 = ip4.dst; reg1 = 192.168.1.1;
            eth.src = fa:16:3e:0c:55:62;
            outport = "lrp-5b72d278-5b16-44a6-9aa0-9e513a429506";
            next;)
Datapath: 4a7485c6-a1ef-46a5-b57c-5ddb6ac15aaa  Pipeline: egress
  table= 1(    lr_out_delivery), priority=  100,
    match=(outport == "lrp-5b72d278-5b16-44a6-9aa0-9e513a429506),
    action=(output;)
```

- Logical flows in the logical switch datapath

```
Datapath: 611d35e8-b1e1-442c-bc07-7c6192ad6216  Pipeline: ingress
  table= 0(  ls_in_port_sec_l2), priority=   50,
    match=(inport == "5b72d278-5b16-44a6-9aa0-9e513a429506"),
    action=(next;)
  table= 3(      ls_in_pre_acl), priority=  110,
    match=(ip && inport == "5b72d278-5b16-44a6-9aa0-9e513a429506"),
    action=(next;)
  table= 9(      ls_in_arp_rsp), priority=   50,
    match=(arp.tpa == 192.168.1.1 && arp.op == 1),
    action=(eth.dst = eth.src; eth.src = fa:16:3e:0c:55:62;
            arp.op = 2; /* ARP reply */ arp.tha = arp.sha;
            arp.sha = fa:16:3e:0c:55:62; arp.tpa = arp.spa;
            arp.spa = 192.168.1.1; outport = inport;
            inport = ""; /* Allow sending out inport. */ output;)
  table=10(      ls_in_l2_lkup), priority=   50,
    match=(eth.dst == fa:16:3e:fa:76:8f),
    action=(outport = "f112b99a-8ccc-4c52-8733-7593fa0966ea"; output;
↪)
Datapath: 611d35e8-b1e1-442c-bc07-7c6192ad6216  Pipeline: egress
  table= 1(     ls_out_pre_acl), priority=  110,
    match=(ip && outport == "f112b99a-8ccc-4c52-8733-7593fa0966ea"),
    action=(next;)
  table= 7( ls_out_port_sec_l2), priority=   50,
    match=(outport == "f112b99a-8ccc-4c52-8733-7593fa0966ea"),
    action=(output;)
```

- Port bindings

```
_uuid                   : 0f86395b-a0d8-40fd-b22c-4c9e238a7880
chassis                 : []
datapath                : 4a7485c6-a1ef-46a5-b57c-5ddb6ac15aaa
```

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```
logical_port         : "lrp-5b72d278-5b16-44a6-9aa0-9e513a429506"
mac                  : []
options              : {peer="5b72d278-5b16-44a6-9aa0-9e513a429506"}
parent_port          : []
tag                  : []
tunnel_key           : 1
type                 : patch

_uuid                : 8d95ab8c-c2ea-4231-9729-7ecbfc2cd676
chassis              : []
datapath             : 4aef86e4-e54a-4c83-bb27-d65c670d4b51
logical_port         : "5b72d278-5b16-44a6-9aa0-9e513a429506"
mac                  : ["fa:16:3e:0c:55:62 192.168.1.1"]
options              : {peer="lrp-5b72d278-5b16-44a6-9aa0-9e513a429506
↪"}
parent_port          : []
tag                  : []
tunnel_key           : 3
type                 : patch
```

- Multicast groups

```
_uuid                : 4a6191aa-d8ac-4e93-8306-b0d8fbbe4e35
datapath             : 4aef86e4-e54a-4c83-bb27-d65c670d4b51
name                 : _MC_flood
ports                : [8d95ab8c-c2ea-4231-9729-7ecbfc2cd676,
                        be71fac3-9f04-41c9-9951-f3f7f1fa1ec5,
                        da5c1269-90b7-4df2-8d76-d4575754b02d]
tunnel_key           : 65535
```

In addition, if the self-service network contains ports with IP addresses (typically instances or DHCP servers), OVN creates a logical flow for each port, similar to the following example.

```
Datapath: 4a7485c6-a1ef-46a5-b57c-5ddb6ac15aaa  Pipeline: ingress
  table= 5(  lr_in_arp_resolve), priority=  100,
    match=(outport == "lrp-f112b99a-8ccc-4c52-8733-7593fa0966ea" &&
           reg0 == 192.168.1.11),
    action=(eth.dst = fa:16:3e:b6:91:70; next;)
```

3. On each compute node, the OVN controller service creates patch ports, similar to the following example.

```
7(patch-f112b99a-): addr:4e:01:91:2a:73:66
    config:     0
    state:      0
    speed: 0 Mbps now, 0 Mbps max
8(patch-lrp-f112b): addr:be:9d:7b:31:bb:87
    config:     0
    state:      0
    speed: 0 Mbps now, 0 Mbps max
```

4. On all compute nodes, the OVN controller service creates the following additional flows:

```
cookie=0x0, duration=6.667s, table=0, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,in_port=8
    actions=load:0x9->OXM_OF_METADATA[],load:0x1->NXM_NX_REG6[],
        resubmit(,16)
cookie=0x0, duration=6.667s, table=0, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,in_port=7
    actions=load:0x7->OXM_OF_METADATA[],load:0x4->NXM_NX_REG6[],
        resubmit(,16)
cookie=0x0, duration=6.674s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg6=0x4,metadata=0x7
    actions=resubmit(,17)
cookie=0x0, duration=6.674s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg6=0x1,metadata=0x9,
        dl_dst=fa:16:3e:fa:76:8f
    actions=resubmit(,17)
cookie=0x0, duration=6.674s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg6=0x1,metadata=0x9,
        dl_dst=01:00:00:00:00:00/01:00:00:00:00:00
    actions=resubmit(,17)
cookie=0x0, duration=6.674s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x9,nw_src=192.168.1.1
    actions=drop
cookie=0x0, duration=6.673s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x9,nw_src=192.168.1.255
    actions=drop
cookie=0x0, duration=6.673s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=90,arp,reg6=0x1,metadata=0x9,
        arp_tpa=192.168.1.1,arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:fa:76:8f,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163efa768f->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a80101->NXM_OF_ARP_SPA[],load:0x1->NXM_NX_REG7[],
        load:0->NXM_NX_REG6[],load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=6.673s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=90,icmp,metadata=0x9,nw_dst=192.168.1.1,
        icmp_type=8,icmp_code=0
    actions=move:NXM_OF_IP_SRC[]->NXM_OF_IP_DST[],mod_nw_src:192.168.1.1,
        load:0xff->NXM_NX_IP_TTL[],load:0->NXM_OF_ICMP_TYPE[],
        load:0->NXM_NX_REG6[],load:0->NXM_OF_IN_PORT[],resubmit(,18)
cookie=0x0, duration=6.674s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=60,ip,metadata=0x9,nw_dst=192.168.1.1
    actions=drop
cookie=0x0, duration=6.674s, table=20, n_packets=0, n_bytes=0,
    idle_age=6, priority=24,ip,metadata=0x9,nw_dst=192.168.1.0/24
    actions=dec_ttl(),move:NXM_OF_IP_DST[]->NXM_NX_REG0[],
        load:0xc0a80101->NXM_NX_REG1[],mod_dl_src:fa:16:3e:fa:76:8f,
        load:0x1->NXM_NX_REG7[],resubmit(,21)
```

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```
cookie=0x0, duration=6.674s, table=21, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg0=0xc0a80103,reg7=0x1,metadata=0x9
    actions=mod_dl_dst:fa:16:3e:d5:00:02,resubmit(,22)
cookie=0x0, duration=6.674s, table=21, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg0=0xc0a80102,reg7=0x1,metadata=0x9
    actions=mod_dl_dst:fa:16:3e:82:8b:0e,resubmit(,22)
cookie=0x0, duration=6.673s, table=21, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg0=0xc0a8010b,reg7=0x1,metadata=0x9
    actions=mod_dl_dst:fa:16:3e:b6:91:70,resubmit(,22)
cookie=0x0, duration=6.673s, table=25, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,arp,metadata=0x7,arp_tpa=192.168.1.1,
        arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:fa:76:8f,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163efa768f->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a80101->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],load:0->NXM_NX_REG6[],
        load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=6.674s, table=26, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,metadata=0x7,dl_dst=fa:16:3e:fa:76:8f
    actions=load:0x4->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=6.667s, table=33, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg7=0x4,metadata=0x7
    actions=resubmit(,34)
cookie=0x0, duration=6.667s, table=33, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg7=0x1,metadata=0x9
    actions=resubmit(,34)
cookie=0x0, duration=6.667s, table=34, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg6=0x4,reg7=0x4,metadata=0x7
    actions=drop
cookie=0x0, duration=6.667s, table=34, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg6=0x1,reg7=0x1,metadata=0x9
    actions=drop
cookie=0x0, duration=6.674s, table=49, n_packets=0, n_bytes=0,
    idle_age=6, priority=110,ipv6,reg7=0x4,metadata=0x7
    actions=resubmit(,50)
cookie=0x0, duration=6.673s, table=49, n_packets=0, n_bytes=0,
    idle_age=6, priority=110,ip,reg7=0x4,metadata=0x7
    actions=resubmit(,50)
cookie=0x0, duration=6.673s, table=49, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg7=0x1,metadata=0x9
    actions=resubmit(,64)
cookie=0x0, duration=6.673s, table=55, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg7=0x4,metadata=0x7
    actions=resubmit(,64)
cookie=0x0, duration=6.667s, table=64, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg7=0x4,metadata=0x7
```

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```
    actions=output:7
cookie=0x0, duration=6.667s, table=64, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg7=0x1,metadata=0x9
    actions=output:8
```

5. On compute nodes not containing a port on the network, the OVN controller also creates additional flows.

```
cookie=0x0, duration=6.673s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,metadata=0x7,
        dl_src=01:00:00:00:00:00/01:00:00:00:00:00
    actions=drop
cookie=0x0, duration=6.674s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,metadata=0x7,vlan_tci=0x1000/0x1000
    actions=drop
cookie=0x0, duration=6.674s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg6=0x3,metadata=0x7,
        dl_src=fa:16:3e:b6:91:70
    actions=resubmit(,17)
cookie=0x0, duration=6.674s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg6=0x2,metadata=0x7
    actions=resubmit(,17)
cookie=0x0, duration=6.674s, table=16, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg6=0x1,metadata=0x7
    actions=resubmit(,17)
cookie=0x0, duration=6.674s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=90,ip,reg6=0x3,metadata=0x7,
        dl_src=fa:16:3e:b6:91:70,nw_src=192.168.1.11
    actions=resubmit(,18)
cookie=0x0, duration=6.674s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=90,udp,reg6=0x3,metadata=0x7,
        dl_src=fa:16:3e:b6:91:70,nw_src=0.0.0.0,
        nw_dst=255.255.255.255,tp_src=68,tp_dst=67
    actions=resubmit(,18)
cookie=0x0, duration=6.674s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=80,ip,reg6=0x3,metadata=0x7,
        dl_src=fa:16:3e:b6:91:70
    actions=drop
cookie=0x0, duration=6.673s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=80,ipv6,reg6=0x3,metadata=0x7,
        dl_src=fa:16:3e:b6:91:70
    actions=drop
cookie=0x0, duration=6.670s, table=17, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,18)
cookie=0x0, duration=6.674s, table=18, n_packets=0, n_bytes=0,
    idle_age=6, priority=90,arp,reg6=0x3,metadata=0x7,
        dl_src=fa:16:3e:b6:91:70,arp_spa=192.168.1.11,
        arp_sha=fa:16:3e:b6:91:70
    actions=resubmit(,19)
```

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```
cookie=0x0, duration=6.673s, table=18, n_packets=0, n_bytes=0,
    idle_age=6, priority=80,icmp6,reg6=0x3,metadata=0x7,icmp_type=135,
        icmp_code=0
    actions=drop
cookie=0x0, duration=6.673s, table=18, n_packets=0, n_bytes=0,
    idle_age=6, priority=80,icmp6,reg6=0x3,metadata=0x7,icmp_type=136,
        icmp_code=0
    actions=drop
cookie=0x0, duration=6.673s, table=18, n_packets=0, n_bytes=0,
    idle_age=6, priority=80,arp,reg6=0x3,metadata=0x7
    actions=drop
cookie=0x0, duration=6.673s, table=18, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,19)
cookie=0x0, duration=6.673s, table=19, n_packets=0, n_bytes=0,
    idle_age=6, priority=110,icmp6,metadata=0x7,icmp_type=136,icmp_code=0
    actions=resubmit(,20)
cookie=0x0, duration=6.673s, table=19, n_packets=0, n_bytes=0,
    idle_age=6, priority=110,icmp6,metadata=0x7,icmp_type=135,icmp_code=0
    actions=resubmit(,20)
cookie=0x0, duration=6.674s, table=19, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=6.670s, table=19, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ipv6,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=6.674s, table=19, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,20)
cookie=0x0, duration=6.673s, table=20, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,21)
cookie=0x0, duration=6.674s, table=21, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ipv6,reg0=0x1/0x1,metadata=0x7
    actions=ct(table=22,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=6.670s, table=21, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,reg0=0x1/0x1,metadata=0x7
    actions=ct(table=22,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=6.674s, table=21, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,22)
cookie=0x0, duration=6.674s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,ct_state=-new+est-rel-inv+trk,metadata=0x7
    actions=resubmit(,23)
cookie=0x0, duration=6.673s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,ct_state=-new-est+rel-inv+trk,metadata=0x7
    actions=resubmit(,23)
cookie=0x0, duration=6.673s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,ct_state=+inv+trk,metadata=0x7
```

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```
    actions=drop
cookie=0x0, duration=6.673s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,icmp6,metadata=0x7,icmp_type=135,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=6.673s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,icmp6,metadata=0x7,icmp_type=136,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=6.674s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=2002,udp,reg6=0x3,metadata=0x7,
        nw_dst=255.255.255.255,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=6.674s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=2002,udp,reg6=0x3,metadata=0x7,
        nw_dst=192.168.1.0/24,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=6.673s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=2002,ct_state=+new+trk,ipv6,reg6=0x3,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=6.673s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=2002,ct_state=+new+trk,ip,reg6=0x3,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=6.674s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=2001,ip,reg6=0x3,metadata=0x7
    actions=drop
cookie=0x0, duration=6.673s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=2001,ipv6,reg6=0x3,metadata=0x7
    actions=drop
cookie=0x0, duration=6.674s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=1,ipv6,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=6.673s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=1,ip,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=6.673s, table=22, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,23)
cookie=0x0, duration=6.673s, table=23, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,24)
cookie=0x0, duration=6.674s, table=24, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ipv6,reg0=0x2/0x2,metadata=0x7
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,25)
cookie=0x0, duration=6.674s, table=24, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,reg0=0x2/0x2,metadata=0x7
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,25)
cookie=0x0, duration=6.673s, table=24, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ipv6,reg0=0x4/0x4,metadata=0x7
```

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```
    actions=ct(table=25,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=6.670s, table=24, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,reg0=0x4/0x4,metadata=0x7
    actions=ct(table=25,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=6.674s, table=24, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,25)
cookie=0x0, duration=6.673s, table=25, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,arp,metadata=0x7,arp_tpa=192.168.1.11,
        arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:b6:91:70,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163eb69170->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a8010b->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],load:0->NXM_NX_REG6[],
        load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=6.670s, table=25, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,arp,metadata=0x7,arp_tpa=192.168.1.3,arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:d5:00:02,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163ed50002->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a80103->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],load:0->NXM_NX_REG6[],
        load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=6.670s, table=25, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,arp,metadata=0x7,arp_tpa=192.168.1.2,
        arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:82:8b:0e,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163e828b0e->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a80102->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],load:0->NXM_NX_REG6[],
        load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=6.674s, table=25, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,26)
cookie=0x0, duration=6.674s, table=26, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,metadata=0x7,
        dl_dst=01:00:00:00:00:00/01:00:00:00:00:00
    actions=load:0xffff->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=6.674s, table=26, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,metadata=0x7,dl_dst=fa:16:3e:d5:00:02
    actions=load:0x2->NXM_NX_REG7[],resubmit(,32)
```

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```
cookie=0x0, duration=6.673s, table=26, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,metadata=0x7,dl_dst=fa:16:3e:b6:91:70
    actions=load:0x3->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=6.670s, table=26, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,metadata=0x7,dl_dst=fa:16:3e:82:8b:0e
    actions=load:0x1->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=6.674s, table=32, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg7=0x3,metadata=0x7
    actions=load:0x7->NXM_NX_TUN_ID[0..23],
        set_field:0x3/0xffffffff->tun_metadata0,
        move:NXM_NX_REG6[0..14]->NXM_NX_TUN_METADATA0[16..30],output:3
cookie=0x0, duration=6.673s, table=32, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg7=0x2,metadata=0x7
    actions=load:0x7->NXM_NX_TUN_ID[0..23],
        set_field:0x2/0xffffffff->tun_metadata0,
        move:NXM_NX_REG6[0..14]->NXM_NX_TUN_METADATA0[16..30],output:3
cookie=0x0, duration=6.670s, table=32, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,reg7=0x1,metadata=0x7
    actions=load:0x7->NXM_NX_TUN_ID[0..23],
        set_field:0x1/0xffffffff->tun_metadata0,
        move:NXM_NX_REG6[0..14]->NXM_NX_TUN_METADATA0[16..30],output:5
cookie=0x0, duration=6.674s, table=48, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,49)
cookie=0x0, duration=6.674s, table=49, n_packets=0, n_bytes=0,
    idle_age=6, priority=110,icmp6,metadata=0x7,icmp_type=135,icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=6.673s, table=49, n_packets=0, n_bytes=0,
    idle_age=6, priority=110,icmp6,metadata=0x7,icmp_type=136,icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=6.674s, table=49, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ipv6,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=6.673s, table=49, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=6.674s, table=49, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,50)
cookie=0x0, duration=6.674s, table=50, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,reg0=0x1/0x1,metadata=0x7
    actions=ct(table=51,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=6.673s, table=50, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ipv6,reg0=0x1/0x1,metadata=0x7
    actions=ct(table=51,zone=NXM_NX_REG5[0..15])
cookie=0x0, duration=6.673s, table=50, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,51)
cookie=0x0, duration=6.670s, table=51, n_packets=0, n_bytes=0,
```

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```
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,52)
cookie=0x0, duration=6.674s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,ct_state=+inv+trk,metadata=0x7
    actions=drop
cookie=0x0, duration=6.674s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,ct_state=-new+est-rel-inv+trk,metadata=0x7
    actions=resubmit(,53)
cookie=0x0, duration=6.673s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,ct_state=-new-est+rel-inv+trk,metadata=0x7
    actions=resubmit(,53)
cookie=0x0, duration=6.673s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,icmp6,metadata=0x7,icmp_type=136,
        icmp_code=0
    actions=resubmit(,53)
cookie=0x0, duration=6.673s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=65535,icmp6,metadata=0x7,icmp_type=135,
        icmp_code=0
    actions=resubmit(,53)
cookie=0x0, duration=6.674s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=2002,ct_state=+new+trk,ip,reg7=0x3,metadata=0x7,
        nw_src=192.168.1.11
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=6.670s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=2002,ct_state=+new+trk,ip,reg7=0x3,metadata=0x7,
        nw_src=192.168.1.11
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=6.670s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=2002,udp,reg7=0x3,metadata=0x7,
        nw_src=192.168.1.0/24,tp_src=67,tp_dst=68
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=6.670s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=2002,ct_state=+new+trk,ipv6,reg7=0x3,
        metadata=0x7
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=6.673s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=2001,ip,reg7=0x3,metadata=0x7
    actions=drop
cookie=0x0, duration=6.673s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=2001,ipv6,reg7=0x3,metadata=0x7
    actions=drop
cookie=0x0, duration=6.674s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=1,ip,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=6.674s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=1,ipv6,metadata=0x7
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=6.674s, table=52, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
```

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```
    actions=resubmit(,53)
cookie=0x0, duration=6.674s, table=53, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ipv6,reg0=0x4/0x4,metadata=0x7
    actions=ct(table=54,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=6.674s, table=53, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,reg0=0x4/0x4,metadata=0x7
    actions=ct(table=54,zone=NXM_NX_REG5[0..15],nat)
cookie=0x0, duration=6.673s, table=53, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ipv6,reg0=0x2/0x2,metadata=0x7
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,54)
cookie=0x0, duration=6.673s, table=53, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,ip,reg0=0x2/0x2,metadata=0x7
    actions=ct(commit,zone=NXM_NX_REG5[0..15]),resubmit(,54)
cookie=0x0, duration=6.674s, table=53, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,54)
cookie=0x0, duration=6.674s, table=54, n_packets=0, n_bytes=0,
    idle_age=6, priority=90,ip,reg7=0x3,metadata=0x7,
        dl_dst=fa:16:3e:b6:91:70,nw_dst=255.255.255.255
    actions=resubmit(,55)
cookie=0x0, duration=6.673s, table=54, n_packets=0, n_bytes=0,
    idle_age=6, priority=90,ip,reg7=0x3,metadata=0x7,
        dl_dst=fa:16:3e:b6:91:70,nw_dst=192.168.1.11
    actions=resubmit(,55)
cookie=0x0, duration=6.673s, table=54, n_packets=0, n_bytes=0,
    idle_age=6, priority=90,ip,reg7=0x3,metadata=0x7,
        dl_dst=fa:16:3e:b6:91:70,nw_dst=224.0.0.0/4
    actions=resubmit(,55)
cookie=0x0, duration=6.670s, table=54, n_packets=0, n_bytes=0,
    idle_age=6, priority=80,ip,reg7=0x3,metadata=0x7,
        dl_dst=fa:16:3e:b6:91:70
    actions=drop
cookie=0x0, duration=6.670s, table=54, n_packets=0, n_bytes=0,
    idle_age=6, priority=80,ipv6,reg7=0x3,metadata=0x7,
        dl_dst=fa:16:3e:b6:91:70
    actions=drop
cookie=0x0, duration=6.674s, table=54, n_packets=0, n_bytes=0,
    idle_age=6, priority=0,metadata=0x7
    actions=resubmit(,55)
cookie=0x0, duration=6.673s, table=55, n_packets=0, n_bytes=0,
    idle_age=6, priority=100,metadata=0x7,
        dl_dst=01:00:00:00:00:00/01:00:00:00:00:00
    actions=resubmit(,64)
cookie=0x0, duration=6.674s, table=55, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg7=0x3,metadata=0x7,
        dl_dst=fa:16:3e:b6:91:70
    actions=resubmit(,64)
cookie=0x0, duration=6.673s, table=55, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg7=0x1,metadata=0x7
```

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```
    actions=resubmit(,64)
cookie=0x0, duration=6.670s, table=55, n_packets=0, n_bytes=0,
    idle_age=6, priority=50,reg7=0x2,metadata=0x7
    actions=resubmit(,64)
```

6. On compute nodes containing a port on the network, the OVN controller also creates an additional flow.

```
cookie=0x0, duration=13.358s, table=52, n_packets=0, n_bytes=0,
    idle_age=13, priority=2002,ct_state=+new+trk,ipv6,reg7=0x3,
        metadata=0x7,ipv6_src=::
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
```

Instances

Launching an instance causes the same series of operations regardless of the network. The following example uses the `provider` provider network, `cirros` image, `m1.tiny` flavor, `default` security group, and `mykey` key.

Launch an instance on a provider network

1. On the controller node, source the credentials for a regular (non-privileged) project. The following example uses the `demo` project.
2. On the controller node, launch an instance using the UUID of the provider network.

```
$ openstack server create --flavor m1.tiny --image cirros \
  --nic net-id=0243277b-4aa8-46d8-9e10-5c9ad5e01521 \
  --security-group default --key-name mykey provider-instance
+--------------------------------------+-------------------------------
↪-------------+
| Property                             | Value                         ␣
↪             |
+--------------------------------------+-------------------------------
↪-------------+
| OS-DCF:diskConfig                    | MANUAL                        ␣
↪             |
| OS-EXT-AZ:availability_zone          | nova                          ␣
↪             |
| OS-EXT-STS:power_state               | 0                             ␣
↪             |
| OS-EXT-STS:task_state                | scheduling                    ␣
↪             |
| OS-EXT-STS:vm_state                  | building                      ␣
↪             |
| OS-SRV-USG:launched_at               | -                             ␣
↪             |
| OS-SRV-USG:terminated_at             | -                             ␣
↪             |
| accessIPv4                           |                               ␣
```

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```
↪             |
| accessIPv6                          |                                ␣
↪             |
| adminPass                           | hdF4LMQqC5PB                   ␣
↪             |
| config_drive                        |                                ␣
↪             |
| created                             | 2015-09-17T21:58:18Z           ␣
↪             |
| flavor                              | m1.tiny (1)                    ␣
↪             |
| hostId                              |                                ␣
↪             |
| id                                  | 181c52ba-aebc-4c32-a97d-
↪2e8e82e4eaaf         |
| image                               | cirros (38047887-61a7-41ea-9b49-
↪27987d5e8bb9) |
| key_name                            | mykey                          ␣
↪             |
| metadata                            | {}                             ␣
↪             |
| name                                | provider-instance              ␣
↪             |
| os-extended-volumes:volumes_attached | []                            ␣
↪             |
| progress                            | 0                              ␣
↪             |
| security_groups                     | default                        ␣
↪             |
| status                              | BUILD                          ␣
↪             |
| tenant_id                           | f5b2ccaa75ac413591f12fcaa096aa5c␣
↪             |
| updated                             | 2015-09-17T21:58:18Z           ␣
↪             |
| user_id                             | 684286a9079845359882afc3aa5011fb␣
↪             |
+--------------------------------------+--------------------------------
↪-------------+
```

OVN operations

The OVN mechanism driver and OVN perform the following operations when launching an instance.

1. The OVN mechanism driver creates a logical port for the instance.

```
_uuid               : cc891503-1259-47a1-9349-1c0293876664
addresses           : ["fa:16:3e:1c:ca:6a 203.0.113.103"]
enabled             : true
external_ids        : {"neutron:port_name"=""}
```

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```
name                    : "cafd4862-c69c-46e4-b3d2-6141ce06b205"
options                 : {}
parent_name             : []
port_security           : ["fa:16:3e:1c:ca:6a 203.0.113.103"]
tag                     : []
type                    : ""
up                      : true
```

2. The OVN mechanism driver updates the appropriate Address Set entry with the address of this instance:

```
_uuid                   : d0becdea-e1ed-48c4-9afc-e278cdef4629
addresses               : ["203.0.113.103"]
external_ids            : {"neutron:security_group_name"=default}
name                    : "as_ip4_90a78a43_b549_4bee_8822_21fcccab58dc"
```

3. The OVN mechanism driver creates ACL entries for this port and any other ports in the project.

```
_uuid                   : f8d27bfc-4d74-4e73-8fac-c84585443efd
action                  : drop
direction               : from-lport
external_ids            : {"neutron:lport"="cafd4862-c69c-46e4-b3d2-
↪6141ce06b205"}
log                     : false
match                   : "inport == \"cafd4862-c69c-46e4-b3d2-6141ce06b205\"␣
↪&& ip"
priority                : 1001

_uuid                   : a61d0068-b1aa-4900-9882-e0671d1fc131
action                  : allow
direction               : to-lport
external_ids            : {"neutron:lport"="cafd4862-c69c-46e4-b3d2-
↪6141ce06b205"}
log                     : false
match                   : "outport == \"cafd4862-c69c-46e4-b3d2-6141ce06b205\
↪" && ip4 && ip4.src == 203.0.113.0/24 && udp && udp.src == 67 && udp.
↪dst == 68"
priority                : 1002

_uuid                   : a5a787b8-7040-4b63-a20a-551bd73eb3d1
action                  : allow-related
direction               : from-lport
external_ids            : {"neutron:lport"="cafd4862-c69c-46e4-b3d2-
↪6141ce06b205"}
log                     : false
match                   : "inport == \"cafd4862-c69c-46e4-b3d2-6141ce06b205\"␣
↪&& ip6"
priority                : 1002

_uuid                   : 7b3f63b8-e69a-476c-ad3d-37de043232b2
```

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```
action              : allow-related
direction           : to-lport
external_ids        : {"neutron:lport"="cafd4862-c69c-46e4-b3d2-
↪6141ce06b205"}
log                 : false
match               : "outport == \"cafd4862-c69c-46e4-b3d2-6141ce06b205\
↪" && ip4 && ip4.src = $as_ip4_90a78a43_b5649_4bee_8822_21fcccab58dc"
priority            : 1002

_uuid               : 36dbb1b1-cd30-4454-a0bf-923646eb7c3f
action              : allow
direction           : from-lport
external_ids        : {"neutron:lport"="cafd4862-c69c-46e4-b3d2-
↪6141ce06b205"}
log                 : false
match               : "inport == \"cafd4862-c69c-46e4-b3d2-6141ce06b205\"␣
↪&& ip4 && (ip4.dst == 255.255.255.255 || ip4.dst == 203.0.113.0/24) &&␣
↪udp && udp.src == 68 && udp.dst == 67"
priority            : 1002

_uuid               : 05a92f66-be48-461e-a7f1-b07bfbd3e667
action              : allow-related
direction           : from-lport
external_ids        : {"neutron:lport"="cafd4862-c69c-46e4-b3d2-
↪6141ce06b205"}
log                 : false
match               : "inport == \"cafd4862-c69c-46e4-b3d2-6141ce06b205\"␣
↪&& ip4"
priority            : 1002

_uuid               : 37f18377-d6c3-4c44-9e4d-2170710e50ff
action              : drop
direction           : to-lport
external_ids        : {"neutron:lport"="cafd4862-c69c-46e4-b3d2-
↪6141ce06b205"}
log                 : false
match               : "outport == \"cafd4862-c69c-46e4-b3d2-6141ce06b205\
↪" && ip"
priority            : 1001

_uuid               : 6d4db3cf-c1f1-4006-ad66-ae582a6acd21
action              : allow-related
direction           : to-lport
external_ids        : {"neutron:lport"="cafd4862-c69c-46e4-b3d2-
↪6141ce06b205"}
log                 : false
match               : "outport == \"cafd4862-c69c-46e4-b3d2-6141ce06b205\
↪" && ip6 && ip6.src = $as_ip6_90a78a43_b5649_4bee_8822_21fcccab58dc"
priority            : 1002
```

4. The OVN mechanism driver updates the logical switch information with the UUIDs of these objects.

```
_uuid               : 924500c4-8580-4d5f-a7ad-8769f6e58ff5
acls                : [05a92f66-be48-461e-a7f1-b07bfbd3e667,
                       36dbb1b1-cd30-4454-a0bf-923646eb7c3f,
                       37f18377-d6c3-4c44-9e4d-2170710e50ff,
                       7b3f63b8-e69a-476c-ad3d-37de043232b2,
                       a5a787b8-7040-4b63-a20a-551bd73eb3d1,
                       a61d0068-b1aa-4900-9882-e0671d1fc131,
                       f8d27bfc-4d74-4e73-8fac-c84585443efd]
external_ids        : {"neutron:network_name"=provider}
name                : "neutron-670efade-7cd0-4d87-8a04-27f366eb8941"
ports               : [38cf8b52-47c4-4e93-be8d-06bf71f6a7c9,
                       5e144ab9-3e08-4910-b936-869bbbf254c8,
                       a576b812-9c3e-4cfb-9752-5d8500b3adf9,
                       cc891503-1259-47a1-9349-1c0293876664]
```

5. The OVN northbound service creates port bindings for the logical ports and adds them to the appropriate multicast group.

   - Port bindings

   ```
   _uuid               : e73e3fcd-316a-4418-bbd5-a8a42032b1c3
   chassis             : fc5ab9e7-bc28-40e8-ad52-2949358cc088
   datapath            : bd0ab2b3-4cf4-4289-9529-ef430f6a89e6
   logical_port        : "cafd4862-c69c-46e4-b3d2-6141ce06b205"
   mac                 : ["fa:16:3e:1c:ca:6a 203.0.113.103"]
   options             : {}
   parent_port         : []
   tag                 : []
   tunnel_key          : 4
   type                : ""
   ```

   - Multicast groups

   ```
   _uuid               : 39b32ccd-fa49-4046-9527-13318842461e
   datapath            : bd0ab2b3-4cf4-4289-9529-ef430f6a89e6
   name                : _MC_flood
   ports               : [030024f4-61c3-4807-859b-07727447c427,
                          904c3108-234d-41c0-b93c-116b7e352a75,
                          cc5bcd19-bcae-4e29-8cee-3ec8a8a75d46,
                          e73e3fcd-316a-4418-bbd5-a8a42032b1c3]
   tunnel_key          : 65535
   ```

6. The OVN northbound service translates the Address Set change into the new Address Set in the OVN southbound database.

```
_uuid               : 2addbee3-7084-4fff-8f7b-15b1efebdaff
addresses           : ["203.0.113.103"]
name                : "as_ip4_90a78a43_b549_4bee_8822_21fcccab58dc"
```

7. The OVN northbound service translates the ACL and logical port objects into logical flows in the

OVN southbound database.

```
Datapath: bd0ab2b3-4cf4-4289-9529-ef430f6a89e6  Pipeline: ingress
  table= 0(  ls_in_port_sec_l2), priority=   50,
    match=(inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
           eth.src == {fa:16:3e:1c:ca:6a}),
    action=(next;)
  table= 1(  ls_in_port_sec_ip), priority=   90,
    match=(inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
           eth.src == fa:16:3e:1c:ca:6a && ip4.src == {203.0.113.103}),
    action=(next;)
  table= 1(  ls_in_port_sec_ip), priority=   90,
    match=(inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
           eth.src == fa:16:3e:1c:ca:6a && ip4.src == 0.0.0.0 &&
           ip4.dst == 255.255.255.255 && udp.src == 68 && udp.dst == 67),
    action=(next;)
  table= 1(  ls_in_port_sec_ip), priority=   80,
    match=(inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
           eth.src == fa:16:3e:1c:ca:6a && ip),
    action=(drop;)
  table= 2(  ls_in_port_sec_nd), priority=   90,
    match=(inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
           eth.src == fa:16:3e:1c:ca:6a &&
           arp.sha == fa:16:3e:1c:ca:6a && (arp.spa == 203.0.113.103 )),
    action=(next;)
  table= 2(  ls_in_port_sec_nd), priority=   80,
    match=(inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
           (arp || nd)),
    action=(drop;)
  table= 3(      ls_in_pre_acl), priority=  110,
    match=(nd),
    action=(next;)
  table= 3(      ls_in_pre_acl), priority=  100,
    match=(ip),
    action=(reg0[0] = 1; next;)
  table= 6(          ls_in_acl), priority=65535,
    match=(ct.inv),
    action=(drop;)
  table= 6(          ls_in_acl), priority=65535,
    match=(nd),
    action=(next;)
  table= 6(          ls_in_acl), priority=65535,
    match=(ct.est && !ct.rel && !ct.new && !ct.inv),
    action=(next;)
  table= 6(          ls_in_acl), priority=65535,
    match=(!ct.est && ct.rel && !ct.new && !ct.inv),
    action=(next;)
  table= 6(          ls_in_acl), priority= 2002,
    match=(ct.new && (inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205"
           && ip6)),
    action=(reg0[1] = 1; next;)
```

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```
  table= 6(          ls_in_acl), priority= 2002,
    match=(inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" && ip4 &&
           (ip4.dst == 255.255.255.255 || ip4.dst == 203.0.113.0/24) &&
           udp && udp.src == 68 && udp.dst == 67),
    action=(reg0[1] = 1; next;)
  table= 6(          ls_in_acl), priority= 2002,
    match=(ct.new && (inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
           ip4)),
    action=(reg0[1] = 1; next;)
  table= 6(          ls_in_acl), priority= 2001,
    match=(inport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" && ip),
    action=(drop;)
  table= 6(          ls_in_acl), priority=    1,
    match=(ip),
    action=(reg0[1] = 1; next;)
  table= 9(      ls_in_arp_rsp), priority=   50,
    match=(arp.tpa == 203.0.113.103 && arp.op == 1),
    action=(eth.dst = eth.src; eth.src = fa:16:3e:1c:ca:6a;
            arp.op = 2; /* ARP reply */ arp.tha = arp.sha;
            arp.sha = fa:16:3e:1c:ca:6a; arp.tpa = arp.spa;
            arp.spa = 203.0.113.103; outport = inport;
            inport = ""; /* Allow sending out inport. */ output;)
  table=10(      ls_in_l2_lkup), priority=   50,
    match=(eth.dst == fa:16:3e:1c:ca:6a),
    action=(outport = "cafd4862-c69c-46e4-b3d2-6141ce06b205"; output;)
Datapath: bd0ab2b3-4cf4-4289-9529-ef430f6a89e6  Pipeline: egress
  table= 1(     ls_out_pre_acl), priority=  110,
    match=(nd),
    action=(next;)
  table= 1(     ls_out_pre_acl), priority=  100,
    match=(ip),
    action=(reg0[0] = 1; next;)
  table= 4(         ls_out_acl), priority=65535,
    match=(!ct.est && ct.rel && !ct.new && !ct.inv),
    action=(next;)
  table= 4(         ls_out_acl), priority=65535,
    match=(ct.est && !ct.rel && !ct.new && !ct.inv),
    action=(next;)
  table= 4(         ls_out_acl), priority=65535,
    match=(ct.inv),
    action=(drop;)
  table= 4(         ls_out_acl), priority=65535,
    match=(nd),
    action=(next;)
  table= 4(         ls_out_acl), priority= 2002,
    match=(ct.new &&
           (outport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" && ip6 &&
            ip6.src == $as_ip6_90a78a43_b549_4bee_8822_21fcccab58dc)),
    action=(reg0[1] = 1; next;)
```

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```
table= 4(         ls_out_acl), priority= 2002,
  match=(ct.new &&
         (outport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" && ip4 &&
          ip4.src == $as_ip4_90a78a43_b549_4bee_8822_21fcccab58dc)),
  action=(reg0[1] = 1; next;)
table= 4(         ls_out_acl), priority= 2002,
  match=(outport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" && ip4 &&
         ip4.src == 203.0.113.0/24 && udp && udp.src == 67 &&
         udp.dst == 68),
  action=(reg0[1] = 1; next;)
table= 4(         ls_out_acl), priority= 2001,
  match=(outport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" && ip),
  action=(drop;)
table= 4(         ls_out_acl), priority=    1,
  match=(ip),
  action=(reg0[1] = 1; next;)
table= 6( ls_out_port_sec_ip), priority=   90,
  match=(outport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
         eth.dst == fa:16:3e:1c:ca:6a &&
         ip4.dst == {255.255.255.255, 224.0.0.0/4, 203.0.113.103}),
  action=(next;)
table= 6( ls_out_port_sec_ip), priority=   80,
  match=(outport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
         eth.dst == fa:16:3e:1c:ca:6a && ip),
  action=(drop;)
table= 7( ls_out_port_sec_l2), priority=   50,
  match=(outport == "cafd4862-c69c-46e4-b3d2-6141ce06b205" &&
         eth.dst == {fa:16:3e:1c:ca:6a}),
  action=(output;)
```

8. The OVN controller service on each compute node translates these objects into flows on the integration bridge `br-int`. Exact flows depend on whether the compute node containing the instance also contains a DHCP agent on the subnet.

   - On the compute node containing the instance, the Compute service creates a port that connects the instance to the integration bridge and OVN creates the following flows:

```
# ovs-ofctl show br-int
OFPT_FEATURES_REPLY (xid=0x2): dpid:000022024a1dc045
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_
↪MATCH_IP
actions: output enqueue set_vlan_vid set_vlan_pcp strip_vlan mod_dl_
↪src mod_dl_dst mod_nw_src mod_nw_dst mod_nw_tos mod_tp_src mod_tp_
↪dst
 9(tapcafd4862-c6): addr:fe:16:3e:1c:ca:6a
     config:     0
     state:      0
     current:    10MB-FD COPPER
     speed: 10 Mbps now, 0 Mbps max
```

```
cookie=0x0, duration=184.992s, table=0, n_packets=175, n_bytes=15270,
    idle_age=15, priority=100,in_port=9
    actions=load:0x3->NXM_NX_REG5[],load:0x4->OXM_OF_METADATA[],
        load:0x4->NXM_NX_REG6[],resubmit(,16)
cookie=0x0, duration=191.687s, table=16, n_packets=175, n_
↪bytes=15270,
    idle_age=15, priority=50,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a
    actions=resubmit(,17)
cookie=0x0, duration=191.687s, table=17, n_packets=2, n_bytes=684,
    idle_age=112, priority=90,udp,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a,nw_src=0.0.0.0,
        nw_dst=255.255.255.255,tp_src=68,tp_dst=67
    actions=resubmit(,18)
cookie=0x0, duration=191.687s, table=17, n_packets=146, n_
↪bytes=12780,
    idle_age=20, priority=90,ip,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a,nw_src=203.0.113.103
    actions=resubmit(,18)
cookie=0x0, duration=191.687s, table=17, n_packets=17, n_bytes=1386,
    idle_age=92, priority=80,ipv6,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a
    actions=drop
cookie=0x0, duration=191.687s, table=17, n_packets=0, n_bytes=0,
    idle_age=191, priority=80,ip,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a
    actions=drop
cookie=0x0, duration=191.687s, table=18, n_packets=10, n_bytes=420,
    idle_age=15, priority=90,arp,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a,arp_spa=203.0.113.103,
        arp_sha=fa:16:3e:1c:ca:6a
    actions=resubmit(,19)
cookie=0x0, duration=191.687s, table=18, n_packets=0, n_bytes=0,
    idle_age=191, priority=80,icmp6,reg6=0x4,metadata=0x4,
        icmp_type=136,icmp_code=0
    actions=drop
cookie=0x0, duration=191.687s, table=18, n_packets=0, n_bytes=0,
    idle_age=191, priority=80,icmp6,reg6=0x4,metadata=0x4,
        icmp_type=135,icmp_code=0
    actions=drop
cookie=0x0, duration=191.687s, table=18, n_packets=0, n_bytes=0,
    idle_age=191, priority=80,arp,reg6=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=75.033s, table=19, n_packets=0, n_bytes=0,
    idle_age=75, priority=110,icmp6,metadata=0x4,icmp_type=135,
        icmp_code=0
    actions=resubmit(,20)
cookie=0x0, duration=75.032s, table=19, n_packets=0, n_bytes=0,
    idle_age=75, priority=110,icmp6,metadata=0x4,icmp_type=136,
        icmp_code=0
```

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```
    actions=resubmit(,20)
cookie=0x0, duration=75.032s, table=19, n_packets=34, n_bytes=5170,
    idle_age=49, priority=100,ip,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=75.032s, table=19, n_packets=0, n_bytes=0,
    idle_age=75, priority=100,ipv6,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=75.032s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=65535,icmp6,metadata=0x4,icmp_type=136,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=75.032s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=65535,icmp6,metadata=0x4,icmp_type=135,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=75.032s, table=22, n_packets=13, n_bytes=1118,
    idle_age=49, priority=65535,ct_state=-new+est-rel-inv+trk,
        metadata=0x4
    actions=resubmit(,23)
cookie=0x0, duration=75.032s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=65535,ct_state=-new-est+rel-inv+trk,
        metadata=0x4
    actions=resubmit(,23)
cookie=0x0, duration=75.032s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=65535,ct_state=+inv+trk,metadata=0x4
    actions=drop
cookie=0x0, duration=75.033s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=2002,ct_state=+new+trk,ipv6,reg6=0x4,
        metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=75.032s, table=22, n_packets=15, n_bytes=1816,
    idle_age=49, priority=2002,ct_state=+new+trk,ip,reg6=0x4,
        metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=75.032s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=2002,udp,reg6=0x4,metadata=0x4,
        nw_dst=203.0.113.0/24,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=75.032s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=2002,udp,reg6=0x4,metadata=0x4,
        nw_dst=255.255.255.255,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=75.033s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=2001,ip,reg6=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=75.032s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=2001,ipv6,reg6=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=75.032s, table=22, n_packets=6, n_bytes=2236,
```

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```
    idle_age=54, priority=1,ip,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=75.032s, table=22, n_packets=0, n_bytes=0,
    idle_age=75, priority=1,ipv6,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=67.064s, table=25, n_packets=0, n_bytes=0,
    idle_age=67, priority=50,arp,metadata=0x4,arp_tpa=203.0.113.103,
        arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:1c:ca:6a,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163ed63dca->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a81268->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],load:0->NXM_NX_REG6[],
        load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=75.033s, table=26, n_packets=19, n_bytes=2776,
    idle_age=44, priority=50,metadata=0x4,dl_dst=fa:16:3e:1c:ca:6a
    actions=load:0x4->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=221031.310s, table=33, n_packets=72, n_
↪bytes=6292,
    idle_age=20, hard_age=65534, priority=100,reg7=0x3,metadata=0x4
    actions=load:0x1->NXM_NX_REG7[],resubmit(,33)
cookie=0x0, duration=184.992s, table=34, n_packets=2, n_bytes=684,
    idle_age=112, priority=100,reg6=0x4,reg7=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=75.034s, table=49, n_packets=0, n_bytes=0,
    idle_age=75, priority=110,icmp6,metadata=0x4,icmp_type=135,
        icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=75.033s, table=49, n_packets=0, n_bytes=0,
    idle_age=75, priority=110,icmp6,metadata=0x4,icmp_type=136,
        icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=75.033s, table=49, n_packets=38, n_bytes=6566,
    idle_age=49, priority=100,ip,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=75.033s, table=49, n_packets=0, n_bytes=0,
    idle_age=75, priority=100,ipv6,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=75.033s, table=52, n_packets=0, n_bytes=0,
    idle_age=75, priority=65535,ct_state=-new-est+rel-inv+trk,
        metadata=0x4
    actions=resubmit(,53)
cookie=0x0, duration=75.033s, table=52, n_packets=13, n_bytes=1118,
    idle_age=49, priority=65535,ct_state=-new+est-rel-inv+trk,
        metadata=0x4
    actions=resubmit(,53)
cookie=0x0, duration=75.033s, table=52, n_packets=0, n_bytes=0,
```

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```
    idle_age=75, priority=65535,icmp6,metadata=0x4,icmp_type=135,
        icmp_code=0
    actions=resubmit(,53)
cookie=0x0, duration=75.033s, table=52, n_packets=0, n_bytes=0,
    idle_age=75, priority=65535,icmp6,metadata=0x4,icmp_type=136,
        icmp_code=0
    actions=resubmit(,53)
cookie=0x0, duration=75.033s, table=52, n_packets=0, n_bytes=0,
    idle_age=75, priority=65535,ct_state=+inv+trk,metadata=0x4
    actions=drop
cookie=0x0, duration=75.034s, table=52, n_packets=4, n_bytes=1538,
    idle_age=54, priority=2002,udp,reg7=0x4,metadata=0x4,
        nw_src=203.0.113.0/24,tp_src=67,tp_dst=68
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=75.033s, table=52, n_packets=0, n_bytes=0,
    idle_age=75, priority=2002,ct_state=+new+trk,ip,reg7=0x4,
        metadata=0x4,nw_src=203.0.113.103
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=2.041s, table=52, n_packets=0, n_bytes=0,
    idle_age=2, priority=2002,ct_state=+new+trk,ipv6,reg7=0x4,
        metadata=0x4,ipv6_src=::2/::2
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=75.033s, table=52, n_packets=2, n_bytes=698,
    idle_age=54, priority=2001,ip,reg7=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=75.033s, table=52, n_packets=0, n_bytes=0,
    idle_age=75, priority=2001,ipv6,reg7=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=75.034s, table=52, n_packets=0, n_bytes=0,
    idle_age=75, priority=1,ipv6,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=75.033s, table=52, n_packets=19, n_bytes=3212,
    idle_age=49, priority=1,ip,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=75.034s, table=54, n_packets=17, n_bytes=2656,
    idle_age=49, priority=90,ip,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a,nw_dst=203.0.113.103
    actions=resubmit(,55)
cookie=0x0, duration=75.033s, table=54, n_packets=0, n_bytes=0,
    idle_age=75, priority=90,ip,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a,nw_dst=255.255.255.255
    actions=resubmit(,55)
cookie=0x0, duration=75.033s, table=54, n_packets=0, n_bytes=0,
    idle_age=75, priority=90,ip,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a,nw_dst=224.0.0.0/4
    actions=resubmit(,55)
cookie=0x0, duration=75.033s, table=54, n_packets=0, n_bytes=0,
    idle_age=75, priority=80,ip,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a
```

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```
    actions=drop
cookie=0x0, duration=75.033s, table=54, n_packets=0, n_bytes=0,
    idle_age=75, priority=80,ipv6,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a
    actions=drop
cookie=0x0, duration=75.033s, table=55, n_packets=21, n_bytes=2860,
    idle_age=44, priority=50,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a
    actions=resubmit(,64)
cookie=0x0, duration=184.992s, table=64, n_packets=166, n_
↪bytes=15088,
    idle_age=15, priority=100,reg7=0x4,metadata=0x4
    actions=output:9
```

- For each compute node that only contains a DHCP agent on the subnet, OVN creates the following flows:

```
cookie=0x0, duration=189.649s, table=16, n_packets=0, n_bytes=0,
    idle_age=189, priority=50,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a
    actions=resubmit(,17)
cookie=0x0, duration=189.650s, table=17, n_packets=0, n_bytes=0,
    idle_age=189, priority=90,udp,reg6=0x4,metadata=0x4,
        dl_src=fa:14:3e:1c:ca:6a,nw_src=0.0.0.0,
        nw_dst=255.255.255.255,tp_src=68,tp_dst=67
    actions=resubmit(,18)
cookie=0x0, duration=189.649s, table=17, n_packets=0, n_bytes=0,
    idle_age=189, priority=90,ip,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a,nw_src=203.0.113.103
    actions=resubmit(,18)
cookie=0x0, duration=189.650s, table=17, n_packets=0, n_bytes=0,
    idle_age=189, priority=80,ipv6,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a
    actions=drop
cookie=0x0, duration=189.650s, table=17, n_packets=0, n_bytes=0,
    idle_age=189, priority=80,ip,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a
    actions=drop
cookie=0x0, duration=189.650s, table=18, n_packets=0, n_bytes=0,
    idle_age=189, priority=90,arp,reg6=0x4,metadata=0x4,
        dl_src=fa:16:3e:1c:ca:6a,arp_spa=203.0.113.103,
        arp_sha=fa:16:3e:1c:ca:6a
    actions=resubmit(,19)
cookie=0x0, duration=189.650s, table=18, n_packets=0, n_bytes=0,
    idle_age=189, priority=80,icmp6,reg6=0x4,metadata=0x4,
        icmp_type=136,icmp_code=0
    actions=drop
cookie=0x0, duration=189.650s, table=18, n_packets=0, n_bytes=0,
    idle_age=189, priority=80,icmp6,reg6=0x4,metadata=0x4,
        icmp_type=135,icmp_code=0
```

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```
    actions=drop
cookie=0x0, duration=189.649s, table=18, n_packets=0, n_bytes=0,
    idle_age=189, priority=80,arp,reg6=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=79.452s, table=19, n_packets=0, n_bytes=0,
    idle_age=79, priority=110,icmp6,metadata=0x4,icmp_type=135,
        icmp_code=0
    actions=resubmit(,20)
cookie=0x0, duration=79.450s, table=19, n_packets=0, n_bytes=0,
    idle_age=79, priority=110,icmp6,metadata=0x4,icmp_type=136,
        icmp_code=0
    actions=resubmit(,20)
cookie=0x0, duration=79.452s, table=19, n_packets=0, n_bytes=0,
    idle_age=79, priority=100,ipv6,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=79.450s, table=19, n_packets=18, n_bytes=3164,
    idle_age=57, priority=100,ip,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=79.450s, table=22, n_packets=6, n_bytes=510,
    idle_age=57, priority=65535,ct_state=-new+est-rel-inv+trk,
        metadata=0x4
    actions=resubmit(,23)
cookie=0x0, duration=79.450s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=65535,ct_state=-new-est+rel-inv+trk,
        metadata=0x4
    actions=resubmit(,23)
cookie=0x0, duration=79.450s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=65535,icmp6,metadata=0x4,icmp_type=136,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=79.450s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=65535,icmp6,metadata=0x4,icmp_type=135,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=79.450s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=65535,ct_state=+inv+trk,metadata=0x4
    actions=drop
cookie=0x0, duration=79.453s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=2002,ct_state=+new+trk,ipv6,reg6=0x4,
        metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=79.450s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=2002,ct_state=+new+trk,ip,reg6=0x4,
        metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=79.450s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=2002,udp,reg6=0x4,metadata=0x4,
        nw_dst=203.0.113.0/24,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
```

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```
cookie=0x0, duration=79.450s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=2002,udp,reg6=0x4,metadata=0x4,
        nw_dst=255.255.255.255,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=79.452s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=2001,ip,reg6=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=79.450s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=2001,ipv6,reg6=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=79.450s, table=22, n_packets=0, n_bytes=0,
    idle_age=79, priority=1,ipv6,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=79.450s, table=22, n_packets=12, n_bytes=2654,
    idle_age=57, priority=1,ip,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=71.483s, table=25, n_packets=0, n_bytes=0,
    idle_age=71, priority=50,arp,metadata=0x4,arp_tpa=203.0.113.103,
        arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:1c:ca:6a,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163ed63dca->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a81268->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],load:0->NXM_NX_REG6[],
        load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=79.450s, table=26, n_packets=8, n_bytes=1258,
    idle_age=57, priority=50,metadata=0x4,dl_dst=fa:16:3e:1c:ca:6a
    actions=load:0x4->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=182.952s, table=33, n_packets=74, n_bytes=7040,
    idle_age=18, priority=100,reg7=0x4,metadata=0x4
    actions=load:0x1->NXM_NX_REG7[],resubmit(,33)
cookie=0x0, duration=79.451s, table=49, n_packets=0, n_bytes=0,
    idle_age=79, priority=110,icmp6,metadata=0x4,icmp_type=135,
        icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=79.450s, table=49, n_packets=0, n_bytes=0,
    idle_age=79, priority=110,icmp6,metadata=0x4,icmp_type=136,
        icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=79.450s, table=49, n_packets=18, n_bytes=3164,
    idle_age=57, priority=100,ip,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=79.450s, table=49, n_packets=0, n_bytes=0,
    idle_age=79, priority=100,ipv6,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=79.450s, table=52, n_packets=0, n_bytes=0,
    idle_age=79, priority=65535,ct_state=-new-est+rel-inv+trk,
```

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```
        metadata=0x4
    actions=resubmit(,53)
cookie=0x0, duration=79.450s, table=52, n_packets=6, n_bytes=510,
    idle_age=57, priority=65535,ct_state=-new+est-rel-inv+trk,
        metadata=0x4
    actions=resubmit(,53)
cookie=0x0, duration=79.450s, table=52, n_packets=0, n_bytes=0,
    idle_age=79, priority=65535,icmp6,metadata=0x4,icmp_type=135,
        icmp_code=0
    actions=resubmit(,53)
cookie=0x0, duration=79.450s, table=52, n_packets=0, n_bytes=0,
    idle_age=79, priority=65535,icmp6,metadata=0x4,icmp_type=136,
        icmp_code=0
    actions=resubmit(,53)
cookie=0x0, duration=79.450s, table=52, n_packets=0, n_bytes=0,
    idle_age=79, priority=65535,ct_state=+inv+trk,metadata=0x4
    actions=drop
cookie=0x0, duration=79.452s, table=52, n_packets=0, n_bytes=0,
    idle_age=79, priority=2002,udp,reg7=0x4,metadata=0x4,
        nw_src=203.0.113.0/24,tp_src=67,tp_dst=68
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=79.450s, table=52, n_packets=0, n_bytes=0,
    idle_age=79, priority=2002,ct_state=+new+trk,ip,reg7=0x4,
        metadata=0x4,nw_src=203.0.113.103
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=71.483s, table=52, n_packets=0, n_bytes=0,
    idle_age=71, priority=2002,ct_state=+new+trk,ipv6,reg7=0x4,
        metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=79.450s, table=52, n_packets=0, n_bytes=0,
    idle_age=79, priority=2001,ipv6,reg7=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=79.450s, table=52, n_packets=0, n_bytes=0,
    idle_age=79, priority=2001,ip,reg7=0x4,metadata=0x4
    actions=drop
cookie=0x0, duration=79.453s, table=52, n_packets=0, n_bytes=0,
    idle_age=79, priority=1,ipv6,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=79.450s, table=52, n_packets=12, n_bytes=2654,
    idle_age=57, priority=1,ip,metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=79.452s, table=54, n_packets=0, n_bytes=0,
    idle_age=79, priority=90,ip,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a,nw_dst=255.255.255.255
    actions=resubmit(,55)
cookie=0x0, duration=79.452s, table=54, n_packets=0, n_bytes=0,
    idle_age=79, priority=90,ip,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a,nw_dst=203.0.113.103
    actions=resubmit(,55)
```

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```
cookie=0x0, duration=79.452s, table=54, n_packets=0, n_bytes=0,
    idle_age=79, priority=90,ip,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a,nw_dst=224.0.0.0/4
    actions=resubmit(,55)
cookie=0x0, duration=79.450s, table=54, n_packets=0, n_bytes=0,
    idle_age=79, priority=80,ip,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a
    actions=drop
cookie=0x0, duration=79.450s, table=54, n_packets=0, n_bytes=0,
    idle_age=79, priority=80,ipv6,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a
    actions=drop
cookie=0x0, duration=79.450s, table=55, n_packets=0, n_bytes=0,
    idle_age=79, priority=50,reg7=0x4,metadata=0x4,
        dl_dst=fa:16:3e:1c:ca:6a
    actions=resubmit(,64)
```

Launch an instance on a self-service network

To launch an instance on a self-service network, follow the same steps as *launching an instance on the provider network*, but using the UUID of the self-service network.

OVN operations

The OVN mechanism driver and OVN perform the following operations when launching an instance.

1. The OVN mechanism driver creates a logical port for the instance.

```
_uuid                : c754d1d2-a7fb-4dd0-b14c-c076962b06b9
addresses            : ["fa:16:3e:15:7d:13 192.168.1.5"]
enabled              : true
external_ids         : {"neutron:port_name"=""}
name                 : "eaf36f62-5629-4ec4-b8b9-5e562c40e7ae"
options              : {}
parent_name          : []
port_security        : ["fa:16:3e:15:7d:13 192.168.1.5"]
tag                  : []
type                 : ""
up                   : true
```

2. The OVN mechanism driver updates the appropriate Address Set object(s) with the address of the new instance:

```
_uuid                : d0becdea-e1ed-48c4-9afc-e278cdef4629
addresses            : ["192.168.1.5", "203.0.113.103"]
external_ids         : {"neutron:security_group_name"=default}
name                 : "as_ip4_90a78a43_b549_4bee_8822_21fcccab58dc"
```

3. The OVN mechanism driver creates ACL entries for this port and any other ports in the project.

```
_uuid               : 00ecbe8f-c82a-4e18-b688-af2a1941cff7
action              : allow
direction           : from-lport
external_ids        : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                 : false
match               : "inport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\"␣
↪&& ip4 && (ip4.dst == 255.255.255.255 || ip4.dst == 192.168.1.0/24) &&␣
↪udp && udp.src == 68 && udp.dst == 67"
priority            : 1002

_uuid               : 2bf5b7ed-008e-4676-bba5-71fe58897886
action              : allow-related
direction           : from-lport
external_ids        : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                 : false
match               : "inport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\"␣
↪&& ip4"
priority            : 1002

_uuid               : 330b4e27-074f-446a-849b-9ab0018b65c5
action              : allow
direction           : to-lport
external_ids        : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                 : false
match               : "outport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\
↪" && ip4 && ip4.src == 192.168.1.0/24 && udp && udp.src == 67 && udp.
↪dst == 68"
priority            : 1002

_uuid               : 683f52f2-4be6-4bd7-a195-6c782daa7840
action              : allow-related
direction           : from-lport
external_ids        : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                 : false
match               : "inport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\"␣
↪&& ip6"
priority            : 1002

_uuid               : 8160f0b4-b344-43d5-bbd4-ca63a71aa4fc
action              : drop
direction           : to-lport
external_ids        : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                 : false
match               : "outport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\
↪" && ip"
```

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```
priority             : 1001

_uuid                : 97c6b8ca-14ea-4812-8571-95d640a88f4f
action               : allow-related
direction            : to-lport
external_ids         : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                  : false
match                : "outport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\
↪" && ip6"
priority             : 1002

_uuid                : 9cfd8eb5-5daa-422e-8fe8-bd22fd7fa826
action               : allow-related
direction            : to-lport
external_ids         : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                  : false
match                : "outport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\
↪" && ip4 && ip4.src == 0.0.0.0/0 && icmp4"
priority             : 1002

_uuid                : f72c2431-7a64-4cea-b84a-118bdc761be2
action               : drop
direction            : from-lport
external_ids         : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                  : false
match                : "inport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\"␣
↪&& ip"
priority             : 1001

_uuid                : f94133fa-ed27-4d5e-a806-0d528e539cb3
action               : allow-related
direction            : to-lport
external_ids         : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                  : false
match                : "outport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\
↪" && ip4 && ip4.src == $as_ip4_90a78a43_b549_4bee_8822_21fcccab58dc"
priority             : 1002

_uuid                : 7f7a92ff-b7e9-49b0-8be0-0dc388035df3
action               : allow-related
direction            : to-lport
external_ids         : {"neutron:lport"="eaf36f62-5629-4ec4-b8b9-
↪5e562c40e7ae"}
log                  : false
match                : "outport == \"eaf36f62-5629-4ec4-b8b9-5e562c40e7ae\
```

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```
↪" && ip6 && ip6.src == $as_ip4_90a78a43_b549_4bee_8822_21fcccab58dc"
priority            : 1002
```

4. The OVN mechanism driver updates the logical switch information with the UUIDs of these objects.

```
_uuid               : 15e2c80b-1461-4003-9869-80416cd97de5
acls                : [00ecbe8f-c82a-4e18-b688-af2a1941cff7,
                       2bf5b7ed-008e-4676-bba5-71fe58897886,
                       330b4e27-074f-446a-849b-9ab0018b65c5,
                       683f52f2-4be6-4bd7-a195-6c782daa7840,
                       7f7a92ff-b7e9-49b0-8be0-0dc388035df3,
                       8160f0b4-b344-43d5-bbd4-ca63a71aa4fc,
                       97c6b8ca-14ea-4812-8571-95d640a88f4f,
                       9cfd8eb5-5daa-422e-8fe8-bd22fd7fa826,
                       f72c2431-7a64-4cea-b84a-118bdc761be2,
                       f94133fa-ed27-4d5e-a806-0d528e539cb3]
external_ids        : {"neutron:network_name"="selfservice"}
name                : "neutron-6cc81cae-8c5f-4c09-aaf2-35d0aa95c084"
ports               : [2df457a5-f71c-4a2f-b9ab-d9e488653872,
                       67c2737c-b380-492b-883b-438048b48e56,
                       c754d1d2-a7fb-4dd0-b14c-c076962b06b9]
```

5. With address sets, it is no longer necessary for the OVN mechanism driver to create separate ACLs for other instances in the project. That is handled automagically via address sets.

6. The OVN northbound service translates the updated Address Set object(s) into updated Address Set objects in the OVN southbound database:

```
_uuid               : 2addbee3-7084-4fff-8f7b-15b1efebdaff
addresses           : ["192.168.1.5", "203.0.113.103"]
name                : "as_ip4_90a78a43_b549_4bee_8822_21fcccab58dc"
```

7. The OVN northbound service adds a Port Binding for the new Logical Switch Port object:

```
_uuid               : 7a558e7b-ed7a-424f-a0cf-ab67d2d832d7
chassis             : b67d6da9-0222-4ab1-a852-ab2607610bf8
datapath            : 3f6e16b5-a03a-48e5-9b60-7b7a0396c425
logical_port        : "e9cb7857-4cb1-4e91-aae5-165a7ab5b387"
mac                 : ["fa:16:3e:b6:91:70 192.168.1.5"]
options             : {}
parent_port         : []
tag                 : []
tunnel_key          : 3
type                : ""
```

8. The OVN northbound service updates the flooding multicast group for the logical datapath with the new port binding:

```
_uuid               : c08d0102-c414-4a47-98d9-dd3fa9f9901c
datapath            : 0b214af6-8910-489c-926a-fd0ed16a8251
```

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```
name                : _MC_flood
ports               : [3e463ca0-951c-46fd-b6cf-05392fa3aa1f,
                        794a6f03-7941-41ed-b1c6-0e00c1e18da0,
                        fa7b294d-2a62-45ae-8de3-a41c002de6de]
tunnel_key          : 65535
```

9. The OVN northbound service adds Logical Flows based on the updated Address Set, ACL and Logical_Switch_Port objects:

```
Datapath: 3f6e16b5-a03a-48e5-9b60-7b7a0396c425  Pipeline: ingress
  table= 0(  ls_in_port_sec_l2), priority=   50,
    match=(inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
      eth.src == {fa:16:3e:b6:a3:54}),
    action=(next;)
  table= 1(  ls_in_port_sec_ip), priority=   90,
    match=(inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
      eth.src == fa:16:3e:b6:a3:54 && ip4.src == 0.0.0.0 &&
      ip4.dst == 255.255.255.255 && udp.src == 68 && udp.dst == 67),
    action=(next;)
  table= 1(  ls_in_port_sec_ip), priority=   90,
    match=(inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
      eth.src == fa:16:3e:b6:a3:54 && ip4.src == {192.168.1.5}),
    action=(next;)
  table= 1(  ls_in_port_sec_ip), priority=   80,
    match=(inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
      eth.src == fa:16:3e:b6:a3:54 && ip),
    action=(drop;)
  table= 2(  ls_in_port_sec_nd), priority=   90,
    match=(inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
      eth.src == fa:16:3e:b6:a3:54 && arp.sha == fa:16:3e:b6:a3:54 &&
      (arp.spa == 192.168.1.5 )),
    action=(next;)
  table= 2(  ls_in_port_sec_nd), priority=   80,
    match=(inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
      (arp || nd)),
    action=(drop;)
  table= 3(      ls_in_pre_acl), priority=  110, match=(nd),
    action=(next;)
  table= 3(      ls_in_pre_acl), priority=  100, match=(ip),
    action=(reg0[0] = 1; next;)
  table= 6(          ls_in_acl), priority=65535,
    match=(!ct.est && ct.rel && !ct.new && !ct.inv),
    action=(next;)
  table= 6(          ls_in_acl), priority=65535,
    match=(ct.est && !ct.rel && !ct.new && !ct.inv),
    action=(next;)
  table= 6(          ls_in_acl), priority=65535, match=(ct.inv),
    action=(drop;)
  table= 6(          ls_in_acl), priority=65535, match=(nd),
    action=(next;)
```

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```
  table= 6(          ls_in_acl), priority= 2002,
    match=(ct.new && (inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
      ip6)),
    action=(reg0[1] = 1; next;)
  table= 6(          ls_in_acl), priority= 2002,
    match=(inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" && ip4 &&
      (ip4.dst == 255.255.255.255 || ip4.dst == 192.168.1.0/24) &&
      udp && udp.src == 68 && udp.dst == 67),
    action=(reg0[1] = 1; next;)
  table= 6(          ls_in_acl), priority= 2002,
    match=(ct.new && (inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
      ip4)),
    action=(reg0[1] = 1; next;)
  table= 6(          ls_in_acl), priority= 2001,
    match=(inport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" && ip),
    action=(drop;)
  table= 6(          ls_in_acl), priority=    1, match=(ip),
    action=(reg0[1] = 1; next;)
  table= 9(   ls_in_arp_nd_rsp), priority=   50,
    match=(arp.tpa == 192.168.1.5 && arp.op == 1),
    action=(eth.dst = eth.src; eth.src = fa:16:3e:b6:a3:54; arp.op = 2; /
↪* ARP reply */ arp.tha = arp.sha; arp.sha = fa:16:3e:b6:a3:54; arp.tpa␣
↪= arp.spa; arp.spa = 192.168.1.5; outport = inport; inport = ""; /*␣
↪Allow sending out inport. */ output;)
  table=10(      ls_in_l2_lkup), priority=   50,
    match=(eth.dst == fa:16:3e:b6:a3:54),
    action=(outport = "e9cb7857-4cb1-4e91-aae5-165a7ab5b387"; output;)
Datapath: 3f6e16b5-a03a-48e5-9b60-7b7a0396c425  Pipeline: egress
  table= 1(     ls_out_pre_acl), priority=  110, match=(nd),
    action=(next;)
  table= 1(     ls_out_pre_acl), priority=  100, match=(ip),
    action=(reg0[0] = 1; next;)
  table= 4(         ls_out_acl), priority=65535, match=(nd),
    action=(next;)
  table= 4(         ls_out_acl), priority=65535,
    match=(!ct.est && ct.rel && !ct.new && !ct.inv),
    action=(next;)
  table= 4(         ls_out_acl), priority=65535,
    match=(ct.est && !ct.rel && !ct.new && !ct.inv),
    action=(next;)
  table= 4(         ls_out_acl), priority=65535, match=(ct.inv),
    action=(drop;)
  table= 4(         ls_out_acl), priority= 2002,
    match=(ct.new &&
      (outport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" && ip6 &&
      ip6.src == $as_ip6_90a78a43_b549_4bee_8822_21fcccab58dc)),
    action=(reg0[1] = 1; next;)
  table= 4(         ls_out_acl), priority= 2002,
    match=(ct.new &&
```

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```
    (outport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" && ip4 &&
    ip4.src == $as_ip4_90a78a43_b549_4bee_8822_21fcccab58dc)),
  action=(reg0[1] = 1; next;)
table= 4(         ls_out_acl), priority= 2002,
  match=(outport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" && ip4 &&
    ip4.src == 192.168.1.0/24 && udp && udp.src == 67 && udp.dst == 68),
  action=(reg0[1] = 1; next;)
table= 4(         ls_out_acl), priority= 2001,
  match=(outport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" && ip),
  action=(drop;)
table= 4(         ls_out_acl), priority=    1, match=(ip),
  action=(reg0[1] = 1; next;)
table= 6( ls_out_port_sec_ip), priority=   90,
  match=(outport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
    eth.dst == fa:16:3e:b6:a3:54 &&
    ip4.dst == {255.255.255.255, 224.0.0.0/4, 192.168.1.5}),
  action=(next;)
table= 6( ls_out_port_sec_ip), priority=   80,
  match=(outport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
    eth.dst == fa:16:3e:b6:a3:54 && ip),
  action=(drop;)
table= 7( ls_out_port_sec_l2), priority=   50,
  match=(outport == "e9cb7857-4cb1-4e91-aae5-165a7ab5b387" &&
    eth.dst == {fa:16:3e:b6:a3:54}),
  action=(output;)
```

10. The OVN controller service on each compute node translates these objects into flows on the integration bridge `br-int`. Exact flows depend on whether the compute node containing the instance also contains a DHCP agent on the subnet.

    - On the compute node containing the instance, the Compute service creates a port that connects the instance to the integration bridge and OVN creates the following flows:

```
# ovs-ofctl show br-int
OFPT_FEATURES_REPLY (xid=0x2): dpid:000022024a1dc045
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_
↪MATCH_IP
actions: output enqueue set_vlan_vid set_vlan_pcp strip_vlan mod_dl_
↪src mod_dl_dst mod_nw_src mod_nw_dst mod_nw_tos mod_tp_src mod_tp_
↪dst
 12(tapeaf36f62-56): addr:fe:16:3e:15:7d:13
     config:     0
     state:      0
     current:    10MB-FD COPPER
```

```
cookie=0x0, duration=179.460s, table=0, n_packets=122, n_bytes=10556,
    idle_age=1, priority=100,in_port=12
    actions=load:0x4->NXM_NX_REG5[],load:0x5->OXM_OF_METADATA[],
        load:0x3->NXM_NX_REG6[],resubmit(,16)
```

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```
cookie=0x0, duration=187.408s, table=16, n_packets=122, n_
↪bytes=10556,
    idle_age=1, priority=50,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13
    actions=resubmit(,17)
cookie=0x0, duration=187.408s, table=17, n_packets=2, n_bytes=684,
    idle_age=84, priority=90,udp,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13,nw_src=0.0.0.0,nw_dst=255.255.255.
↪255,
        tp_src=68,tp_dst=67
    actions=resubmit(,18)
cookie=0x0, duration=187.408s, table=17, n_packets=98, n_bytes=8276,
    idle_age=1, priority=90,ip,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13,nw_src=192.168.1.5
    actions=resubmit(,18)
cookie=0x0, duration=187.408s, table=17, n_packets=17, n_bytes=1386,
    idle_age=55, priority=80,ipv6,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13
    actions=drop
cookie=0x0, duration=187.408s, table=17, n_packets=0, n_bytes=0,
    idle_age=187, priority=80,ip,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13
    actions=drop
cookie=0x0, duration=187.408s, table=18, n_packets=5, n_bytes=210,
    idle_age=10, priority=90,arp,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13,arp_spa=192.168.1.5,
    arp_sha=fa:16:3e:15:7d:13
    actions=resubmit(,19)
cookie=0x0, duration=187.408s, table=18, n_packets=0, n_bytes=0,
    idle_age=187, priority=80,icmp6,reg6=0x3,metadata=0x5,
        icmp_type=135,icmp_code=0
    actions=drop
cookie=0x0, duration=187.408s, table=18, n_packets=0, n_bytes=0,
    idle_age=187, priority=80,icmp6,reg6=0x3,metadata=0x5,
        icmp_type=136,icmp_code=0
    actions=drop
cookie=0x0, duration=187.408s, table=18, n_packets=0, n_bytes=0,
    idle_age=187, priority=80,arp,reg6=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=47.068s, table=19, n_packets=0, n_bytes=0,
    idle_age=47, priority=110,icmp6,metadata=0x5,icmp_type=135,
        icmp_code=0
    actions=resubmit(,20)
cookie=0x0, duration=47.068s, table=19, n_packets=0, n_bytes=0,
    idle_age=47, priority=110,icmp6,metadata=0x5,icmp_type=136,
        icmp_code=0
    actions=resubmit(,20)
cookie=0x0, duration=47.068s, table=19, n_packets=33, n_bytes=4081,
    idle_age=0, priority=100,ip,metadata=0x5
```

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```
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=47.068s, table=19, n_packets=0, n_bytes=0,
    idle_age=47, priority=100,ipv6,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=47.068s, table=22, n_packets=15, n_bytes=1392,
    idle_age=0, priority=65535,ct_state=-new+est-rel-inv+trk,
        metadata=0x5
    actions=resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,ct_state=-new-est+rel-inv+trk,
        metadata=0x5
    actions=resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,icmp6,metadata=0x5,icmp_type=135,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,icmp6,metadata=0x5,icmp_type=136,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,ct_state=+inv+trk,metadata=0x5
    actions=drop
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2002,ct_state=+new+trk,ipv6,reg6=0x3,
        metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=16, n_bytes=1922,
    idle_age=2, priority=2002,ct_state=+new+trk,ip,reg6=0x3,
        metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2002,udp,reg6=0x3,metadata=0x5,
        nw_dst=255.255.255.255,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2002,udp,reg6=0x3,metadata=0x5,
        nw_dst=192.168.1.0/24,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=47.069s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2001,ipv6,reg6=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2001,ip,reg6=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=47.068s, table=22, n_packets=2, n_bytes=767,
    idle_age=27, priority=1,ip,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
```

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```
    idle_age=47, priority=1,ipv6,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=179.457s, table=25, n_packets=2, n_bytes=84,
    idle_age=33, priority=50,arp,metadata=0x5,arp_tpa=192.168.1.5,
        arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:15:7d:13,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163e157d13->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a80105->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],
        load:0->NXM_NX_REG6[],load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=187.408s, table=26, n_packets=50, n_bytes=4806,
    idle_age=1, priority=50,metadata=0x5,dl_dst=fa:16:3e:15:7d:13
    actions=load:0x3->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=469.575s, table=33, n_packets=74, n_bytes=7040,
    idle_age=305, priority=100,reg7=0x4,metadata=0x4
    actions=load:0x1->NXM_NX_REG7[],resubmit(,33)
cookie=0x0, duration=179.460s, table=34, n_packets=2, n_bytes=684,
    idle_age=84, priority=100,reg6=0x3,reg7=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=47.069s, table=49, n_packets=0, n_bytes=0,
    idle_age=47, priority=110,icmp6,metadata=0x5,icmp_type=135,
        icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=47.068s, table=49, n_packets=0, n_bytes=0,
    idle_age=47, priority=110,icmp6,metadata=0x5,icmp_type=136,
        icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=47.068s, table=49, n_packets=34, n_bytes=4455,
    idle_age=0, priority=100,ip,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=47.068s, table=49, n_packets=0, n_bytes=0,
    idle_age=47, priority=100,ipv6,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=47.069s, table=52, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,ct_state=+inv+trk,metadata=0x5
    actions=drop
cookie=0x0, duration=47.069s, table=52, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,icmp6,metadata=0x5,icmp_type=136,
        icmp_code=0
    actions=resubmit(,53)
cookie=0x0, duration=47.068s, table=52, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,icmp6,metadata=0x5,icmp_type=135,
        icmp_code=0
    actions=resubmit(,53)
cookie=0x0, duration=47.068s, table=52, n_packets=22, n_bytes=2000,
    idle_age=0, priority=65535,ct_state=-new+est-rel-inv+trk,
```

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```
        metadata=0x5
    actions=resubmit(,53)
cookie=0x0, duration=47.068s, table=52, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,ct_state=-new-est+rel-inv+trk,
        metadata=0x5
    actions=resubmit(,53)
cookie=0x0, duration=47.068s, table=52, n_packets=0, n_bytes=0,
    idle_age=47, priority=2002,ct_state=+new+trk,ip,reg7=0x3,
        metadata=0x5,nw_src=192.168.1.5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=47.068s, table=52, n_packets=0, n_bytes=0,
    idle_age=47, priority=2002,ct_state=+new+trk,ip,reg7=0x3,
        metadata=0x5,nw_src=203.0.113.103
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=47.068s, table=52, n_packets=3, n_bytes=1141,
    idle_age=27, priority=2002,udp,reg7=0x3,metadata=0x5,
        nw_src=192.168.1.0/24,tp_src=67,tp_dst=68
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=39.497s, table=52, n_packets=0, n_bytes=0,
    idle_age=39, priority=2002,ct_state=+new+trk,ipv6,reg7=0x3,
        metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=47.068s, table=52, n_packets=0, n_bytes=0,
    idle_age=47, priority=2001,ip,reg7=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=47.068s, table=52, n_packets=0, n_bytes=0,
    idle_age=47, priority=2001,ipv6,reg7=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=47.068s, table=52, n_packets=9, n_bytes=1314,
    idle_age=2, priority=1,ip,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=47.068s, table=52, n_packets=0, n_bytes=0,
    idle_age=47, priority=1,ipv6,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=47.068s, table=54, n_packets=23, n_bytes=2945,
    idle_age=0, priority=90,ip,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13,nw_dst=192.168.1.11
    actions=resubmit(,55)
cookie=0x0, duration=47.068s, table=54, n_packets=0, n_bytes=0,
    idle_age=47, priority=90,ip,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13,nw_dst=255.255.255.255
    actions=resubmit(,55)
cookie=0x0, duration=47.068s, table=54, n_packets=0, n_bytes=0,
    idle_age=47, priority=90,ip,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13,nw_dst=224.0.0.0/4
    actions=resubmit(,55)
cookie=0x0, duration=47.068s, table=54, n_packets=0, n_bytes=0,
    idle_age=47, priority=80,ip,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13
```

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```
    actions=drop
cookie=0x0, duration=47.068s, table=54, n_packets=0, n_bytes=0,
    idle_age=47, priority=80,ipv6,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13
    actions=drop
cookie=0x0, duration=47.068s, table=55, n_packets=25, n_bytes=3029,
    idle_age=0, priority=50,reg7=0x3,metadata=0x7,
        dl_dst=fa:16:3e:15:7d:13
    actions=resubmit(,64)
cookie=0x0, duration=179.460s, table=64, n_packets=116, n_
↪bytes=10623,
    idle_age=1, priority=100,reg7=0x3,metadata=0x5
    actions=output:12
```

- For each compute node that only contains a DHCP agent on the subnet, OVN creates the following flows:

```
cookie=0x0, duration=192.587s, table=16, n_packets=0, n_bytes=0,
    idle_age=192, priority=50,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13
    actions=resubmit(,17)
cookie=0x0, duration=192.587s, table=17, n_packets=0, n_bytes=0,
    idle_age=192, priority=90,ip,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13,nw_src=192.168.1.5
    actions=resubmit(,18)
cookie=0x0, duration=192.587s, table=17, n_packets=0, n_bytes=0,
    idle_age=192, priority=90,udp,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13,nw_src=0.0.0.0,
        nw_dst=255.255.255.255,tp_src=68,tp_dst=67
    actions=resubmit(,18)
cookie=0x0, duration=192.587s, table=17, n_packets=0, n_bytes=0,
    idle_age=192, priority=80,ipv6,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13
    actions=drop
cookie=0x0, duration=192.587s, table=17, n_packets=0, n_bytes=0,
    idle_age=192, priority=80,ip,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13
    actions=drop
cookie=0x0, duration=192.587s, table=18, n_packets=0, n_bytes=0,
    idle_age=192, priority=90,arp,reg6=0x3,metadata=0x5,
        dl_src=fa:16:3e:15:7d:13,arp_spa=192.168.1.5,
        arp_sha=fa:16:3e:15:7d:13
    actions=resubmit(,19)
cookie=0x0, duration=192.587s, table=18, n_packets=0, n_bytes=0,
    idle_age=192, priority=80,arp,reg6=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=192.587s, table=18, n_packets=0, n_bytes=0,
    idle_age=192, priority=80,icmp6,reg6=0x3,metadata=0x5,
        icmp_type=135,icmp_code=0
    actions=drop
```

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```
cookie=0x0, duration=192.587s, table=18, n_packets=0, n_bytes=0,
    idle_age=192, priority=80,icmp6,reg6=0x3,metadata=0x5,
        icmp_type=136,icmp_code=0
    actions=drop
cookie=0x0, duration=47.068s, table=19, n_packets=0, n_bytes=0,
    idle_age=47, priority=110,icmp6,metadata=0x5,icmp_type=135,
        icmp_code=0
    actions=resubmit(,20)
cookie=0x0, duration=47.068s, table=19, n_packets=0, n_bytes=0,
    idle_age=47, priority=110,icmp6,metadata=0x5,icmp_type=136,
        icmp_code=0
    actions=resubmit(,20)
cookie=0x0, duration=47.068s, table=19, n_packets=33, n_bytes=4081,
    idle_age=0, priority=100,ip,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=47.068s, table=19, n_packets=0, n_bytes=0,
    idle_age=47, priority=100,ipv6,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,20)
cookie=0x0, duration=47.068s, table=22, n_packets=15, n_bytes=1392,
    idle_age=0, priority=65535,ct_state=-new+est-rel-inv+trk,
        metadata=0x5
    actions=resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,ct_state=-new-est+rel-inv+trk,
        metadata=0x5
    actions=resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,icmp6,metadata=0x5,icmp_type=135,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,icmp6,metadata=0x5,icmp_type=136,
        icmp_code=0
    actions=resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=65535,ct_state=+inv+trk,metadata=0x5
    actions=drop
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2002,ct_state=+new+trk,ipv6,reg6=0x3,
        metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=16, n_bytes=1922,
    idle_age=2, priority=2002,ct_state=+new+trk,ip,reg6=0x3,
        metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2002,udp,reg6=0x3,metadata=0x5,
        nw_dst=255.255.255.255,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
```

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```
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2002,udp,reg6=0x3,metadata=0x5,
        nw_dst=192.168.1.0/24,tp_src=68,tp_dst=67
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=47.069s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2001,ipv6,reg6=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=2001,ip,reg6=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=47.068s, table=22, n_packets=2, n_bytes=767,
    idle_age=27, priority=1,ip,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=47.068s, table=22, n_packets=0, n_bytes=0,
    idle_age=47, priority=1,ipv6,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,23)
cookie=0x0, duration=179.457s, table=25, n_packets=2, n_bytes=84,
    idle_age=33, priority=50,arp,metadata=0x5,arp_tpa=192.168.1.5,
        arp_op=1
    actions=move:NXM_OF_ETH_SRC[]->NXM_OF_ETH_DST[],
        mod_dl_src:fa:16:3e:15:7d:13,load:0x2->NXM_OF_ARP_OP[],
        move:NXM_NX_ARP_SHA[]->NXM_NX_ARP_THA[],
        load:0xfa163e157d13->NXM_NX_ARP_SHA[],
        move:NXM_OF_ARP_SPA[]->NXM_OF_ARP_TPA[],
        load:0xc0a80105->NXM_OF_ARP_SPA[],
        move:NXM_NX_REG6[]->NXM_NX_REG7[],
        load:0->NXM_NX_REG6[],load:0->NXM_OF_IN_PORT[],resubmit(,32)
cookie=0x0, duration=192.587s, table=26, n_packets=61, n_bytes=5607,
    idle_age=6, priority=50,metadata=0x5,dl_dst=fa:16:3e:15:7d:13
    actions=load:0x3->NXM_NX_REG7[],resubmit(,32)
cookie=0x0, duration=184.640s, table=32, n_packets=61, n_bytes=5607,
    idle_age=6, priority=100,reg7=0x3,metadata=0x5
    actions=load:0x5->NXM_NX_TUN_ID[0..23],
        set_field:0x3/0xffffffff->tun_metadata0,
        move:NXM_NX_REG6[0..14]->NXM_NX_TUN_METADATA0[16..30],
↪output:4
cookie=0x0, duration=47.069s, table=49, n_packets=0, n_bytes=0,
    idle_age=47, priority=110,icmp6,metadata=0x5,icmp_type=135,
        icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=47.068s, table=49, n_packets=0, n_bytes=0,
    idle_age=47, priority=110,icmp6,metadata=0x5,icmp_type=136,
        icmp_code=0
    actions=resubmit(,50)
cookie=0x0, duration=47.068s, table=49, n_packets=34, n_bytes=4455,
    idle_age=0, priority=100,ip,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=47.068s, table=49, n_packets=0, n_bytes=0,
    idle_age=47, priority=100,ipv6,metadata=0x5
```

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```
    actions=load:0x1->NXM_NX_REG0[0],resubmit(,50)
cookie=0x0, duration=192.587s, table=52, n_packets=0, n_bytes=0,
    idle_age=192, priority=65535,ct_state=+inv+trk,
        metadata=0x5
    actions=drop
cookie=0x0, duration=192.587s, table=52, n_packets=0, n_bytes=0,
    idle_age=192, priority=65535,ct_state=-new-est+rel-inv+trk,
        metadata=0x5
    actions=resubmit(,50)
cookie=0x0, duration=192.587s, table=52, n_packets=27, n_bytes=2316,
    idle_age=6, priority=65535,ct_state=-new+est-rel-inv+trk,
        metadata=0x5
    actions=resubmit(,50)
cookie=0x0, duration=192.587s, table=52, n_packets=0, n_bytes=0,
    idle_age=192, priority=2002,ct_state=+new+trk,icmp,reg7=0x3,
        metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,50)
cookie=0x0, duration=192.587s, table=52, n_packets=0, n_bytes=0,
    idle_age=192, priority=2002,ct_state=+new+trk,ipv6,reg7=0x3,
        metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,50)
cookie=0x0, duration=192.587s, table=52, n_packets=0, n_bytes=0,
    idle_age=192, priority=2002,udp,reg7=0x3,metadata=0x5,
        nw_src=192.168.1.0/24,tp_src=67,tp_dst=68
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,50)
cookie=0x0, duration=192.587s, table=52, n_packets=0, n_bytes=0,
    idle_age=192, priority=2002,ct_state=+new+trk,ip,reg7=0x3,
        metadata=0x5,nw_src=203.0.113.103
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,50)
cookie=0x0, duration=192.587s, table=52, n_packets=0, n_bytes=0,
    idle_age=192, priority=2001,ip,reg7=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=192.587s, table=52, n_packets=0, n_bytes=0,
    idle_age=192, priority=2001,ipv6,reg7=0x3,metadata=0x5
    actions=drop
cookie=0x0, duration=192.587s, table=52, n_packets=25, n_bytes=2604,
    idle_age=6, priority=1,ip,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=192.587s, table=52, n_packets=0, n_bytes=0,
    idle_age=192, priority=1,ipv6,metadata=0x5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=192.587s, table=54, n_packets=0, n_bytes=0,
    idle_age=192, priority=90,ip,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13,nw_dst=224.0.0.0/4
    actions=resubmit(,55)
cookie=0x0, duration=192.587s, table=54, n_packets=0, n_bytes=0,
    idle_age=192, priority=90,ip,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13,nw_dst=255.255.255.255
    actions=resubmit(,55)
```

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```
cookie=0x0, duration=192.587s, table=54, n_packets=0, n_bytes=0,
    idle_age=192, priority=90,ip,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13,nw_dst=192.168.1.5
    actions=resubmit(,55)
cookie=0x0, duration=192.587s, table=54, n_packets=0, n_bytes=0,
    idle_age=192, priority=80,ipv6,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13
    actions=drop
cookie=0x0, duration=192.587s, table=54, n_packets=0, n_bytes=0,
    idle_age=192, priority=80,ip,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13
    actions=drop
cookie=0x0, duration=192.587s, table=55, n_packets=0, n_bytes=0,
    idle_age=192, priority=50,reg7=0x3,metadata=0x5,
        dl_dst=fa:16:3e:15:7d:13
    actions=resubmit(,64)
```

- For each compute node that contains neither the instance nor a DHCP agent on the subnet, OVN creates the following flows:

```
cookie=0x0, duration=189.763s, table=52, n_packets=0, n_bytes=0,
    idle_age=189, priority=2002,ct_state=+new+trk,ipv6,reg7=0x4,
        metadata=0x4
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
cookie=0x0, duration=189.763s, table=52, n_packets=0, n_bytes=0,
    idle_age=189, priority=2002,ct_state=+new+trk,ip,reg7=0x4,
        metadata=0x4,nw_src=192.168.1.5
    actions=load:0x1->NXM_NX_REG0[1],resubmit(,53)
```

8.7.7 DPDK Support in OVN

Configuration Settings

The following configuration parameter needs to be set in the Neutron ML2 plugin configuration file under the ovn section to enable DPDK support.

vhost_sock_dir
: This is the directory path in which vswitch daemon in all the compute nodes creates the virtio socket. Follow the instructions in INSTALL.DPDK.md in openvswitch source tree to know how to configure DPDK support in vswitch daemons.

Configuration Settings in compute hosts

Compute nodes configured with OVS DPDK should set the datapath_type as netdev for the integration bridge (managed by OVN) and all other bridges if connected to the integration bridge via patch ports. The below command can be used to set the datapath_type.

```
$ sudo ovs-vsctl set Bridge br-int datapath_type=netdev
```

8.7.8 Troubleshooting

The following section describe common problems that you might encounter after/during the installation of the OVN ML2 driver with Devstack and possible solutions to these problems.

Launching VMs failure

Disable AppArmor

Using Ubuntu you might encounter libvirt permission errors when trying to create OVS ports after launching a VM (from the nova compute log). Disabling AppArmor might help with this problem, check out https://help.ubuntu.com/community/AppArmor for instructions on how to disable it.

Multi-Node setup not working

Geneve kernel module not supported

By default OVN creates tunnels between compute nodes using the Geneve protocol. Older kernels (< 3.18) dont support the Geneve module and hence tunneling cant work. You can check it with this command lsmod | grep openvswitch (geneve should show up in the result list)

For more information about which upstream Kernel version is required for support of each tunnel type, see the answer to Why do tunnels not work when using a kernel module other than the one packaged with Open vSwitch? in the OVS FAQ.

MTU configuration

This problem is not unique to OVN but is amplified due to the possible larger size of geneve header compared to other common tunneling protocols (VXLAN). If you are using VMs as compute nodes make sure that you either lower the MTU size on the virtual interface or enable fragmentation on it.

Duplicated or deleted OVN agents

The ovn-controller process is the local controller daemon for OVN. It runs in every host belonging to the OVN network and is in charge of registering the host to the OVN database by creating the corresponding Chassis and Chassis_Private registers in the Southbound database. At the same time, when the process is gracefully stopped, it deletes both registers. These registers are used by Neutron to control the OVN agents.

```
$ openstack network agent list -c ID -c "Agent Type" -c Host -c Alive -c State
+--------------------------------------+------------------------------+-------
↪-+-------+-------+
| ID                                   | Agent Type                   | Host ␣
↪ | Alive | State |
+--------------------------------------+------------------------------+-------
↪-+-------+-------+
| a55c8d85-2071-4452-92cb-95d15c29bde7 | OVN Controller Gateway agent |␣
↪u20ovn | :-)   | UP    |
| 62e29a01-a0ac-55c9-b4ec-e223d5c90853 | OVN Metadata agent           |␣
↪u20ovn | :-)   | UP    |
| ce9a1471-79c1-4472-adfc-9e5ce86eba07 | OVN Controller Gateway agent |␣
↪u20ovn | XXX   | DOWN  |
| 3755938f-9aac-4f08-a1ab-32fcff56d1ce | OVN Metadata agent           |␣
```

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```
↪u20ovn | XXX   | DOWN  |
+-----------------------------------+----------------------------+-------
↪-+-------+-------+
```

If during a system upgrade the OVS system-id changes, the Chassis and Chassis_Private registers will be created again but with a different UUID. If the previous registers are not deleted (that should happen if the ovn-controller process is gracefully stopped), Neutron will show duplicated agents from the same host. In this case, only one agent will be alive and the other one will be down because the Chassis_Private.nb_cfg_timestamp is not updated. In this case, the administrator should manually delete from the OVN Southbound database the stale registers. For example:

- List the Chassis registers, filtering by hostname and name (OVS system-id):

```
$ sudo ovn-sbctl list Chassis | grep name
hostname            : u20ovn
name                : "a55c8d85-2071-4452-92cb-95d15c29bde7"
hostname            : u20ovn
name                : "ce9a1471-79c1-4472-adfc-9e5ce86eba07"
```

- Delete the stale Chassis register:

```
$ sudo ovn-sbctl destroy Chassis ce9a1471-79c1-4472-adfc-9e5ce86eba07
```

- List the Chassis_Private registers, filtering by name:

```
$ sudo ovn-sbctl list Chassis_Private | grep name
name                : "a55c8d85-2071-4452-92cb-95d15c29bde7"
name                : "ce9a1471-79c1-4472-adfc-9e5ce86eba07"
```

- Delete the stale Chassis_Private register:

```
$ sudo ovn-sbctl destroy Chassis_Private ce9a1471-79c1-4472-adfc-
↪9e5ce86eba07
```

If the host name is also updated during the system upgrade, the Neutron agent list could present entries from different host names, but the older ones will be down too. The procedure is the same.

It could also happen that during a node decommission, the Chassis register is deleted but not the Chassis_Private one. In that case, the OVN agent list will present the corresponding agents with the following message: (Chassis register deleted). Again, the procedure is the same: the administrator should manually delete the orphaned OVN Southbound database register. Neutron will receive this event and will delete the associated OVN agents.

Recovering from an OVN Chassis crash

If the ovn-controller process, running on an OVN Chassis host, is killed without being gracefully stopped or the host running the ovn-controller process crashed, the corresponding Chassis and Chassis_Private registers are not deleted. Before restarting the ovn-controller or ovn-controller host after the crash, the administrator should follow the procedure described in the *Duplicated or deleted OVN agents* section, and manually delete stale Chassis and Chassis_Private registers. Neutron will be notified of the deletion and will remove the corresponding Gateway_Chassis registers in the OVN Northbound database.

8.7.9 SR-IOV guide for OVN

The purpose of this page is to describe how SR-IOV works with OVN. Prior to reading this document, it is recommended to first read *the basic guide for SR-IOV*.

External ports

The SR-IOV feature is leverage by OVN `external` ports. For more information about external ports, its scheduling and troubleshoot, please check the *External Ports guide*.

Environment setup for OVN SR-IOV

There are a very few differences between setting up an environment for SR-IOV for the OVS and OVN Neutron drivers. As mentioned at the beginning of this document, the instructions from the *the basic guide for SR-IOV* are required for getting SR-IOV working with the OVN driver.

The only differences required for an OVN deployment are:

- When configuring the `mechanism_drivers` in the *ml2_conf.ini* file we should specify `ovn` driver instead of the `openvswitch` driver
- Disabling the Neutron DHCP agent
- Deploying the OVN Metadata agent on the gateway nodes (controller or networker nodes)

North/South routing

A network with an external port (SR-IOV, baremetal), will create a `HA_Chassis_Group` register to schedule these external ports in gateway chassis. The routers (still) use a set of `Gateway_Chassis` registers to execute the scheduling of the gateway port. Now, since the implementation of[1], when a network is connected as an internal network, the Neutron API will sync the network `HA_Chassis_Group` with the gateway port `Gateway_Chassis` set. That will make both scheduling methods to be in sync and will collocate the gateway router port and the external port in the same OVN gateway chassis; that allows OVN to route the traffic from the external port through the gateway port.

When the network `HA_Chassis_Group` is updated, it could be possible that the currently assigned gateway chassis changes. However, before connecting the network to the router, this port is used only for DHCP and metadata; the external port binding change wont interrupt the traffic.

References

8.7.10 Availability Zones guide for OVN

The purpose of this page is to describe how the availability zones works with OVN. Prior to reading this document, it is recommended to first read *ML2/OVS driver Availability Zones guide*.

There are two types of availability zones available in Neutron: Router and Network. For ML2/OVS, this is related to the scheduling of the L3 agent and DHCP agent respectively. For ML2/OVN, its about the scheduling of logical router ports and external ports respectively.

More details about each type of availability zones can be found later in this document but first lets go over the common parts between them:

[1] https://review.opendev.org/q/topic:%22bug/2125553%22

How to configure it

Different from the ML2/OVS driver for Neutron the availability zones for the OVN driver is not configured via a configuration file. Since ML2/OVN does not rely on an external agent such as the L3 agent, certain nodes (e.g gateway/networker node) wont have any Neutron configuration file present. For this reason, OVN uses the local OVSDB for configuring the availability zones that instance of `ovn-controller` running on that hypervisor belongs to.

The configuration is done via the `ovn-cms-options` entry in *external_ids* column of the local *Open_vSwitch* table:

```
$ ovs-vsctl set Open_vSwitch . external-ids:ovn-cms-options="enable-chassis-
↪as-gw,availability-zones=az-0:az-1:az-2"
```

The above command is adding two configurations to the `ovn-cms-options` option, the `enable-chassis-as-gw` option which tells the OVN driver that this is a gateway/networker node and the `availability-zones` option specifying three availability zones: **az-0**, **az-1** and **az-2**.

Note

Specifying the enable-chassis-as-gw option is not required for the Availability Zones **however** ML2/OVN will only consider nodes that are gateway (the ones with the enable-chassis-as-gw option) when scheduling both `router` and `external` ports. So, even tho the availability-zones option can be set own their own, the ML2/OVN driver does not have a use case for it at the moment.

Note that, the syntax used to specify the availability zones is the `availability-zones` word, followed by an equal sign (=) and a **colon** separated list of the availability zones that this local `ovn-controller` instance belongs to.

To confirm the specific `ovn-controller` availability zones, check the **Availability Zone** column in the output of the command below:

```
$ openstack network agent list
+--------------------------------------+-------------------------------+-------
↪---------+------------------+-------+-------+----------------+
| ID                                   | Agent Type                    | Host ␣
↪          | Availability Zone | Alive | State | Binary         |
+--------------------------------------+-------------------------------+-------
↪---------+------------------+-------+-------+----------------+
| 2d1924b2-99a4-4c6c-a4f2-0be64c0cec8c | OVN Controller Gateway agent |␣
↪gateway-host-0 | az0, az1, az2     | :-)   | UP    | ovn-controller |
+--------------------------------------+-------------------------------+-------
↪---------+------------------+-------+-------+----------------+
```

Note

If you know the UUID of the agent the **openstack network agent show <UUID>** command can also be used.

To confirm the availability zones defined in the system as a whole:

```
$ openstack availability zone list --network
+-----------+-------------+
| Zone Name | Zone Status |
+-----------+-------------+
| az0       | available   |
| az1       | available   |
| az2       | available   |
+-----------+-------------+
```

Router Availability Zones

In order to create a router with availability zones the `--availability-zone-hint` should be passed to the create command, note that this parameter can be specified multiple times in case the router belongs to more than one availability zone. For example:

```
$ openstack router create --availability-zone-hint az-0 --availability-zone-
↪hint az-1 router-0
+-------------------------+--------------------------------------+
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints | az-0, az-1                           |
| availability_zones      |                                      |
| created_at              | 2020-06-04T08:29:33Z                 |
| description             |                                      |
| external_gateway_info   | null                                 |
| flavor_id               | None                                 |
| id                      | 8fd6d01a-57ad-4e91-a788-ebe48742d000 |
| name                    | router-0                             |
| project_id              | 2a364ced6c084888be0919450629de1c     |
| revision_number         | 1                                    |
| routes                  |                                      |
| status                  | ACTIVE                               |
| tags                    |                                      |
| updated_at              | 2020-06-04T08:29:33Z                 |
+-------------------------+--------------------------------------+
```

Its also possible to set the default availability zones via the */etc/neutron/neutron.conf* configuration file:

```
[DEFAULT]
default_availability_zones = az-0,az-2
...
```

When scheduling the gateway ports of a router, the OVN driver will take into consideration the router availability zones and make sure that the ports are scheduled on the nodes belonging to those availability zones.

Note that in the router object we have two attributes related to availability zones: `availability_zones` and `availability_zone_hints`:

```
| availability_zone_hints | az-0, az-1                          |
| availability_zones      |                                     |
```

This distinction makes more sense in the **ML2/OVS** driver which relies on the L3 agent for its router placement (see the *ML2/OVS driver Availability Zones guide* for more information). In **ML2/OVN** the `ovn-controller` service will be running on all nodes of the cluster so the `availability_zone_hints` will always match the `availability_zones` attribute, as below:

```
| availability_zone_hints | az-0, az-1                          |
| availability_zones      | az-0, az-1                          |
```

OVN Database information

In order to check the availability zones of a router via the OVN Northbound database, one can look for the `neutron:availability_zone_hints` key in the `external_ids` column for its entry in the `Logical_Router` table:

```
$ ovn-nbctl list Logical_Router
_uuid               : 4df68f1e-17dd-4b9a-848d-b6152ae19203
external_ids        : {"neutron:availability_zone_hints"="az-0,az-1",
↪"neutron:gw_port_id"="", "neutron:revision_number"="1", "neutron:router_name
↪"=router-0}
name                : neutron-8fd6d01a-57ad-4e91-a788-ebe48742d000
...
```

To check the availability zones of the Chassis, look at the `ovn-cms-options` key in the `other_config` column (or `external_ids` for an older version of OVN) of the `Chassis` table in the OVN Southbound database:

```
$ ovn-sbctl list Chassis
_uuid               : abaa9f07-9988-40c0-bd1a-8d8326af08b0
name                : "2d1924b2-99a4-4c6c-a4f2-0be64c0cec8c"
other_config        : {..., ovn-cms-options="enable-chassis-as-gw,
↪availability-zones=az-0:az-1:az-2"}
...
```

As mentioned in the *Router availability zones* section, the scheduling of the gateway router ports will take into consideration the availability zones that the router belongs to. We can confirm this behavior by looking in the `Gateway_Chassis` table from the OVN Northbound database:

```
$ ovn-nbctl list Gateway_Chassis
_uuid               : ac61b70f-ff51-43d9-830b-f9bc6d74090a
chassis_name        : "2d1924b2-99a4-4c6c-a4f2-0be64c0cec8c"
external_ids        : {}
name                : lrp-5a40eeca-5233-4029-a470-9018aa8b3de9_2d1924b2-99a4-
↪4c6c-a4f2-0be64c0cec8c
options             : {}
priority            : 2

_uuid               : c1b7763b-1784-4e5a-a948-853662faeddc
chassis_name        : "1cde2542-69f9-4598-b20b-d4f68304deb0"
```

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```
external_ids        : {}
name                : lrp-5a40eeca-5233-4029-a470-9018aa8b3de9_1cde2542-69f9-
↪4598-b20b-d4f68304deb0
options             : {}
priority            : 1
```

Each entry on this table represents an instance of the gateway port (L3 HA, for more information see *Routing in OVN*), the `chassis_name` column indicates which Chassis that port instance is scheduled onto. If we co-relate each entry and their `chassis_name` we will see that this port has been only scheduled to Chassis matching with the routers availability zones and with priority to distribute over each zones.

Network Availability Zones

Since OVN has a distributed DHCP server model (see the ovn-architecture document for more information), one may think that theres no need for Ml2/OVN to support Network Availability Zones as theres no need to co-locate a DHCP agent within the same zones to serve the VMs but, in ML2/OVN theres a special case which are the `external` ports and those need to be aware of the Availability Zones for its scheduling.

These `external` ports are ports that are located on a different node than the one that the VM is running. At the moment, ML2/OVN only supports one case that makes use of these ports which is the *SR-IOV support*.

In order to create a network with availability zones the `--availability-zone-hint` should be passed to the create command, note that this parameter can be specified multiple times in case the network belongs to more than one availability zone. For example:

```
$ openstack network create --availability-zone-hint az-0 --availability-zone-
↪hint az-1 network-0
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   | az-0, az-1                           |
| availability_zones        |                                      |
| created_at                | 2021-04-26T14:04:51Z                 |
| description               |                                      |
| dns_domain                |                                      |
| id                        | ba584cdb-b866-4744-85d3-6e38718055cc |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| is_vlan_transparent       | None                                 |
| mtu                       | 1442                                 |
| name                      | network-0                            |
| port_security_enabled     | True                                 |
| project_id                | ffd9e4a60af34b0599f1d50aed20dde0     |
| provider:network_type     | None                                 |
| provider:physical_network | None                                 |
| provider:segmentation_id  | None                                 |
| qos_policy_id             | None                                 |
```

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```
| revision_number           | 1                                    |
| router:external           | Internal                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      |                                      |
| updated_at                | 2021-04-26T14:04:52Z                 |
+---------------------------+--------------------------------------+
```

OVN Database information

Upon creating the first `external` port to a network with Availability Zones set a HA Chassis Group correspondent to that network will also be created in the OVN Northbound Database:

```
$ openstack port create --network network-0 --vnic-type direct port-0
+-------------------------+--------------------------------------+
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| id                      | 2523d7f5-c7ca-40b8-83c5-ac37e5b126ea |
| name                    | port-0                               |
| network_id              | ba584cdb-b866-4744-85d3-6e38718055cc |
...
+-------------------------+--------------------------------------+
```

To find the corresponding HA Chassis Group we need to look for a group named as *neutron-<Neutron Network UUID>*, for example:

```
$ ovn-nbctl list HA_Chassis_Group neutron-ba584cdb-b866-4744-85d3-6e38718055cc
_uuid               : f6a49abb-dc97-4e2a-955a-6f8e8be4865e
external_ids        : {"neutron:availability_zone_hints"="az-0,az-1"}
ha_chassis          : [46850075-7383-4da9-b0b2-5ded2858f681, ce1da6a5-77d3-
↪4945-b218-c0ae35403b80]
name                : neutron-ba584cdb-b866-4744-85d3-6e38718055cc
```

In the output above is possible to see that the HA Chassis Group for the Neutron network `ba584cdb-b866-4744-85d3-6e38718055cc` includes two Chassis (the `ha_chassis` column) that are part of the Availability Zones that this network is also part of.

We can inspect these members to see which one has the **highest** priority, which means that when the `external` port is bound it will first bound to the HA Chassis with the **highest** priority in the Group. In case that Chassis goes down the port will move on to the next Chassis with the **highest** priority and so on. To check these HA Chassis do:

```
$ ovn-nbctl list HA_Chassis 46850075-7383-4da9-b0b2-5ded2858f681
_uuid               : 46850075-7383-4da9-b0b2-5ded2858f681
chassis_name        : "2c5c4479-0e2b-4742-a1d7-df10be020143"
external_ids        : {}
priority            : 32766
```

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```
$ ovn-nbctl list HA_Chassis ce1da6a5-77d3-4945-b218-c0ae35403b8
_uuid               : ce1da6a5-77d3-4945-b218-c0ae35403b80
chassis_name        : "159970f0-71f7-4d3d-9a9e-92e37c5f03c5"
external_ids        : {}
priority            : 32767
```

In this case, the **active** Chassis is the `159970f0-71f7-4d3d-9a9e-92e37c5f03c5`.

And lastly, to find which HA Chassis Group an external port belongs to by looking into the OVN Northbound Database do:

```
$ sudo ovn-nbctl list Logical_Switch_Port 2523d7f5-c7ca-40b8-83c5-ac37e5b126ea
_uuid               : 382d8cd8-575f-4a3f-93ba-a01cb9c2c265
ha_chassis_group    : f6a49abb-dc97-4e2a-955a-6f8e8be4865e
name                : "2523d7f5-c7ca-40b8-83c5-ac37e5b126ea"
type                : external
...
```

The `ha_chassis_group` column will point to the UUID (in the OVN database) of the HA Chassis Group it belongs to.

8.7.11 Routed Provider Networks for OVN

The Routed Provider Networks feature is used to present a multi-segmented layer-3 network as a single entity in Neutron.

After creating a provider network with multiple segments as described in the *Neutron documentation*, each segment connects to a provider `Local_Switch` entry as `Logical_Switch_Port` entries with the `localnet` port type.

For example, in the OVN Northbound database, this is how a VLAN Provider Network with two segments (VLAN: 100, 200) is related to their `Logical_Switch` counterpart:

```
$ ovn-nbctl list logical_switch public
_uuid               : 983719e5-4f32-4fb0-926d-46291457ca41
acls                : []
dns_records         : []
external_ids        : {"neutron:mtu"="1450", "neutron:network_name"=public,
↪"neutron:revision_number"="3"}
forwarding_groups   : []
load_balancer       : []
name                : neutron-6c8be12a-9ed0-4ac4-8130-cb8fad83cd46
other_config        : {mcast_flood_unregistered="false", mcast_snoop="true"}
ports               : [81bce1ab-87f8-4ed5-8163-f16701499dfe, b23d0c2e-773b-
↪4ecb-8306-53d117006a7b]
qos_rules           : []

$ ovn-nbctl list logical_switch_port 81bce1ab-87f8-4ed5-8163-f16701499dfe
_uuid               : 81bce1ab-87f8-4ed5-8163-f16701499dfe
addresses           : [unknown]
dhcpv4_options      : []
dhcpv6_options      : []
```

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```
dynamic_addresses   : []
enabled             : []
external_ids        : {}
ha_chassis_group    : []
name                : provnet-96f663af-19fa-4c7e-a1b8-1dfdc9cd9e82
options             : {network_name=phys-net-1}
parent_name         : []
port_security       : []
tag                 : 100
tag_request         : []
type                : localnet
up                  : false

$ ovn-nbctl list logical_switch_port b23d0c2e-773b-4ecb-8306-53d117006a7b
_uuid               : b23d0c2e-773b-4ecb-8306-53d117006a7b
addresses           : [unknown]
dhcpv4_options      : []
dhcpv6_options      : []
dynamic_addresses   : []
enabled             : []
external_ids        : {}
ha_chassis_group    : []
name                : provnet-469cbc3d-8e06-4a8f-be3a-3fcdadfd398a
options             : {network_name=phys-net-2}
parent_name         : []
port_security       : []
tag                 : 200
tag_request         : []
type                : localnet
up                  : false
```

As you can see, the two `localnet` ports are configured with a VLAN tag and are related to a single `Logical_Switch` entry. When *ovn-controller* sees that a port in that network has been bound to the node its running on it will create a patch port to the provider bridge accordingly to the bridge mappings configuration.

```
compute-1: bridge-mappings = segment-1:br-provider1
compute-2: bridge-mappings = segment-2:br-provider2
```

For example, when a port in the multisegment network gets bound to compute-1, ovn-controller will create a patch-port between br-int and br-provider1.

An important note here is that, on a given hypervisor only ports belonging to **the same segment** should be present. **It is not allowed to mix ports from different segments on the same hypervisor for the same network (Logical_Switch).**

8.7.12 Off-path SmartNIC DPUs with OVN

The purpose of this page is to describe how off-path SmartNIC DPU hardware can be integrated with Neutron when OVN mechanism driver is used. For an in-depth discussion of underlying mechanisms it is recommended to get familiar with the following specifications

- Neutron Off-path SmartNIC DPU Port Binding with OVN specification;
- Nova Integration With Off-path Network Backends specification.

Overview

A class of devices collectively referred to as off-path SmartNIC DPUs introduces an important change to earlier architectures where compute and networking agents used to coexist at the hypervisor host: networking control plane components are now moved to the SmartNIC DPUs CPU side which includes `ovs-vswitchd` and `ovn-controller`. The following diagram provides an overview of the components involved:

```
                         Hypervisor                            LoM Ports
                                    (on-board,
                          Instance           Nova       optional)
                         (e.g. QEMU)       Compute






                            Instance VF               Control Traffic
                                                      PF associated
                                                      with an uplink
                                                      port or a VF.
                                                      (used to replace
                                                       LoM)

      SmartNIC DPU Board

      Control traffic
        App. CPU    via PFs or VFs
       (DC Fabric)
     ovn-controller

     ovs-vswitchd                   NIC Switch
                    ASIC/FPGA
      Neutron OVN
     metadata agent
     Port representors
         br-int

     Optional port for
OOB Port initial NIC switch config        uplink
```

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```
                         DC Fabric



                    OVN SB   Neutron Placement
                      Server
```

Prerequisites

- OpenStack Yoga or newer;
- Open vSwitch >= 2.17;
- Open Virtual Network >= 21.12.0;
- OVN VIF >= 21.12.0;
- A SmartNIC DPU with the following characteristics:
  - A NIC that exposes a card serial number via a PCIe VPD capability on its physical or virtual function PCIe endpoints to both the hypervisor host and the DPU host;
  - Exposes the information about representor ports to applications running on the SmartNIC DPUs CPU in a manner supported by one of the OVN VIF Plug Providers.

Nova configuration

Hypervisor hosts need to be configured to enable:

- Nova PCI passthrough for Nova Compute;

> **Important**
>
> For more information on other version requirements and limitations check the SR-IOV section of the Nova networking guide.

- SR-IOV virtual functions on selected physical functions provided by DPUs to the hypervisor hosts.

In addition to the regular PCI device allow list configuration, the PCI device specification must include the `remote_managed` tag as in the following examples:

- Virtual networks without physical segments;

```
[pci]
passthrough_whitelist = {"vendor_id": "15b3", "product_id": "101e",
↪"physical_network": null, "remote_managed": "true"}
```

- Physical networks (flat, VLAN) with a label:

```
[pci]
passthrough_whitelist = {"vendor_id": "15b3", "product_id": "101e",
↪"physical_network": "dcfabric", "remote_managed": "true"}
```

> **Note**
>
> dcfabric is an arbitrary physnet name. In order for this to work it must be specified consistenly in Nova config, during OVN configuraton when specifying `external_ids:ovn-bridge-mappings` and during Neutron provider network segment creation.

Auto-Discovery

When an instance with a `remote-managed` port is scheduled to a compute host with a free remote-managed device, it claims it and supplies additional information from that device about the NIC to Neutron so that it knows which OVN chassis needs to handle an representor interface plugging and flow programming. For PCI VFs this additional information includes:

- A card serial number from the NICs VPD;
- PF mac address;
- VF logical number.

Neutron uses the card serial number to look up a chassis host name which is needed for port binding to succeed and the rest is used by `ovn-vif` to set up the matching representor port.

As a result, no direct communication or configuration is required between the SmartNIC DPU host and the compute host in order to handle matching of compute hosts to SmartNIC DPUs.

> **Note**
>
> Multiple DPUs per hypervisor host are possible to use, however, at the time of writing, there is no way to indicate to Nova which VFs to choose via Neutron port object attributes.

Having the OVN controller expose the SmartNIC DPU serial number is accomplished by providing the serial number via the `ovn-cms-options` entry in *external_ids* column of the SmartNIC DPU local *Open_vSwitch* table:

```
$ ovs-vsctl set Open_vSwitch . external-ids:ovn-cms-options="card-serial-
↪number=AB0123XX0042"
```

Launch an instance with remote managed port

```
$ openstack port create \
    --network network \
    --vnic-type remote-managed \
    port1
```

```
$ openstack server create \
    --flavor 1 \
    --nic port-id=port1
```

8.7.13 Baremetal provisioning guide for OVN

The purpose of this page is to describe how the baremetal provisioning can be configured with ML2/OVN.

Baremetal provisioning with ML2/OVN can be achieved using the OVNs built-in DHCP server or Neutrons DHCP agent.

How to configure it

Scheduling baremetal ports

The first thing to know is that when a port with VNIC `baremetal` is created, ML2/OVN will create an OVN port of the type `external`. These ports will be bound to nodes that have external connectivity and are responsible to responding to the ARP requests on behalf of the baremetal node.

For more information about external ports, its scheduling and troubleshoot please check the *External Ports guide*.

Metadata access

Different from ML2/OVS, ML2/OVN requires to have the `ovn-metadata-agent` running on the node that the virtual machines are running onto. Since baremetal requires an external port that will be bound to another node, as explained *Scheduling baremetal ports* section, it is required that the `ovn-metadata-agent` is also deployed on the nodes marked with the `enable-chassis-as-gw` option so it can serve metadata to the baremetal nodes booting off those external ports.

Using OVN built-in DHCP for PXE booting

This feature requires OVN running the version **23.06** or above for IPv4 and IPv6 support. If only IPv4 is needed, the OVN version **22.06** is the minimum required.

The version of OVN used for baremetal provisioning should include the following commits [[1]] [[2]].

And last, make sure that configuration option `[ovn]/disable_ovn_dhcp_for_baremetal_ports` is set to **False** (the default).

Using Neutron DHCP Agent for PXE booting

If using the OVN built-in DHCP server is not desirable, the operator will need to deploy Neutrons DHCP agents on the controller nodes and also disable the OVNs DHCP server for the baremetal ports by setting the `[ovn]/disable_ovn_dhcp_for_baremetal_ports` configuration option to **True** (defaults to False).

[1] https://github.com/ovn-org/ovn/commit/0057cde2a64749bd2dbbaff525f7a1edccbd9c8a
[2] https://github.com/ovn-org/ovn/commit/9cbd79c9ebbd0b6d0ea08c2cc70e234e56bb0415

8.7.14 OVN External Ports

The purpose of this page is to describe how ML2/OVN leverages the use of OVNs external ports feature.

What is it

The external ports feature in OVN allows for setting up a port that lives externally to the instance and is reponsible for replying to ARP requests (DHCP, internal DNS, IPv6 router solicitation requests, etc) on its behalf. At the moment this feature is used in two use cases for ML2/OVN:

1. *SR-IOV*
2. *Baremetal provisioning*

ML2/OVN will create a port of the type `external` for ports with the following VNICs:

- direct
- direct-physical
- macvtap
- baremetal

These ports can be listed in OVN with following command:

```
$ ovn-nbctl find Logical_Switch_Port type=external
_uuid               : 105e83ae-252d-401b-a1a7-8d28ec28a359
ha_chassis_group    : [43047e7b-4c78-4984-9788-6263fcc69885]
type                : external
...
```

The next section will talk more about the different configurations for scheduling these ports and how they are represented in the OVN database.

Scheduling and database information

Ports of the type `external` will be scheduled on nodes marked to host these type of ports via the ovn-cms-options configuration. There are two supported configurations for these nodes:

1. `enable-chassis-as-extport-host`
2. `enable-chassis-as-gw`

These options can be set by running the following command locally on each node that will act as a candidate to host these ports:

```
$ ovs-vsctl set Open_vSwitch . external-ids:ovn-cms-options=\"enable-chassis-
↪as-extport-host\"

$ ovs-vsctl set Open_vSwitch . external-ids:ovn-cms-options=\"enable-chassis-
↪as-gw\"
```

The sections below will explain the differences between the two configuration values.

Configuration: `enable-chassis-as-extport-host`

When nodes in the cluster are marked with the `enable-chassis-as-extport-host` configuration, the ML2/OVN driver will schedule the external ports onto these nodes. This configuration takes precedence over `enable-chassis-as-gw`.

With this configuration, the ML2/OVN driver will create one `HA_Chassis_Group` per external port and it will be named as `neutron-extport-<Neutron Port UUID>`. For example:

```
$ ovn-sbctl list Chassis
_uuid                   : fa24d475-9664-4a62-bb1c-52a6fa4966f7
external_ids            : {ovn-cms-options=enable-chassis-as-extport-host, ...}
hostname                : compute-0
name                    : "6fd9cef6-4e9d-4bde-ab82-016c2461957b"
...
_uuid                   : a29ee8f6-5301-45f5-b280-a43e533d4d65
external_ids            : {ovn-cms-options=enable-chassis-as-extport-host, ...}
hostname                : compute-1
name                    : "4fa76c10-c6ea-4ae9-b31c-bc69103fe6f9"
...
```

```
$ ovn-nbctl list HA_Chassis_Group neutron-extport-392a77f9-7c48-4ad0-bd06-
↪8b55bba00bd1
_uuid                   : 1249b761-24e3-414e-ae10-7e880e9d3cf8
external_ids            : {"neutron:availability_zone_hints"=""}
ha_chassis              : [0d6b9718-7718-45d2-a838-1deb40131442, ae6e64e7-f948-
↪49b3-a171-c9cfb58c8b31]
name                    : neutron-extport-392a77f9-7c48-4ad0-bd06-8b55bba00bd1
```

Also, for HA, there will be a limit of five Chassis per `HA_Chassis_Group`, meaning that even if there are more nodes marked with the `enable-chassis-as-extport-host` option, each group will contain up to five members. This limit has been imposed because OVN uses BFD to monitor the connectivity of each member in the group, and having an unlimited number of members can potentially put a lot of stress on OVN.

In general, this option is used when there are specific requirements for `external` ports and they can not be scheduled on controllers or gateway nodes. The next configuration does the opposite and uses the nodes marked as gateway to schedule the `external` ports.

Configuration: `enable-chassis-as-gw`

For the majority of use cases where there are no special requirements for the `external` ports and they can be co-located with gateway ports, this configuration should be used.

Gateway nodes are identified by the `enable-chassis-as-gw` and ovn-bridge-mappings configurations:

```
$ ovn-sbctl list Chassis
_uuid                   : 12b13aff-a821-4cde-a4ac-d9cf8e2c91bc
external_ids            : {ovn-cms-options=enable-chassis-as-gw, ovn-bridge-
↪mappings="public:br-ex", ...}
hostname                : controller-0
name                    : "1a462946-ccfd-46a6-8abf-9dca9eb558fb"
...
```

As mentioned in the *What is it* section, every time a Neutron port with a certain VNIC is created the OVN driver will create a port of the type `external` in the OVN Northbound database.

When the `enable-chassis-as-gw` configuration is used, the ML2/OVN driver will create one `HA_Chassis_Group` per network (instead of one per external port in the previous case) and it will be named as `neutron-<Neutron Network UUID>`.

All `external` ports belonging to this network will share the same `HA_Chassis_Group` and the group is also limited to a maximum of five members for HA.

```
$ ovn-nbctl list HA_Chassis_Group
_uuid               : 43047e7b-4c78-4984-9788-6263fcc69885
external_ids        : {"neutron:availability_zone_hints"=""}
ha_chassis          : [3005bf84-fc95-4361-866d-bfa1c980adc8, 72c7671e-dd48-
↪4100-9741-c47221672961]
name                : neutron-4b2944ca-c7a3-4cf6-a9c8-6aa541a20535
```

High availability

As hinted above, the ML2/OVN driver does provide high availability to the `external` ports. This is done via the `HA_Chassis_Group` mechanism from OVN.

On every `external` port there will be a column called `ha_chassis_group` which points to the `HA_Chassis_Group` that the port belongs to:

```
$ ovn-nbctl find logical_switch_port type=external
ha_chassis_group    : 924fd0fe-3e84-4eaa-aa1d-41103ec511e5
name                : "287040d6-0936-4363-ae0a-2d5a239e55fa"
type                : external
...
```

In the `HA_Chassis_Group`, the members of each group are listed in the `ha_chassis` column:

```
$ ovn-nbctl list HA_Chassis_Group 924fd0fe-3e84-4eaa-aa1d-41103ec511e5
_uuid               : 924fd0fe-3e84-4eaa-aa1d-41103ec511e5
external_ids        : {"neutron:availability_zone_hints"=""}
ha_chassis          : [3005bf84-fc95-4361-866d-bfa1c980adc8, 72c7671e-dd48-
↪4100-9741-c47221672961]
name                : neutron-extport-287040d6-0936-4363-ae0a-2d5a239e55fa
```

Note

There will be a maximum of five members for each group, this limit has been imposed because OVN uses BFD to monitor the connectivity of each member in the group, and having an unlimited number of members can potentially put a lot of stress on OVN.

When listing the members of a group there will be a column called `priority` that contains a numerical value, the member with the highest `priority` is the chassis where the ports will be scheduled on. OVN will monitor each member via BFD protocol, and if the chassis that is hosting the ports goes down, the ports will be automatically scheduled on the next chassis with the highest priority that is alive.

```
$ ovn-nbctl list HA_Chassis 3005bf84-fc95-4361-866d-bfa1c980adc8 72c7671e-
↪dd48-4100-9741-c47221672961
_uuid               : 3005bf84-fc95-4361-866d-bfa1c980adc8
chassis_name        : "1a462946-ccfd-46a6-8abf-9dca9eb558fb"
external_ids        : {}
priority            : 32767

_uuid               : 72c7671e-dd48-4100-9741-c47221672961
chassis_name        : "a0cb9d55-a6da-4f84-857f-d4b674088c8c"
external_ids        : {}
priority            : 32766
```

In the example above, the Chassis with the UUID 1a462946-ccfd-46a6-8abf-9dca9eb558fb is the one that is hosting the external port 287040d6-0936-4363-ae0a-2d5a239e55fa.

8.7.15 RPC messages in OVN

ML2/OVN driver uses the OVN NB tables Port_Group and ACL to implement security groups. Security groups and security group rules are directly sent to OVN NB via the OVSDB protocol. Neutron doesnt send any RPC messages related to these topics when using the ML2/OVN mechanism driver.

However, other RPC topics are kept in case other drivers are being used, for example ML2/SRIOV, DHCP agents (for baremetal ports), etc. If the configuration variable [DEFAULT]rpc_workers is set to 0, that means no Neutron agent needs an RPC server; in that case, the ML2plugin will not initialize any RPC client and no RPC notifications will be sent.

The OVNL3RouterPlugin class instantiates the RPC notifier handler but doesnt assign an RPC instance. The ovn-router plugin doesnt have any associated agent that requires RPC information.

8.7.16 OVN L3 scheduler

Introduction

The OVN L3 scheduler assigns the router gateway ports to a list of chassis. Having more than one chassis assigned allows the service to have high availability: if the Logical_Router_Port acting as gateway is assigned to a failed chassis, OVN will bind this port to the next chassis in the list. This list of chassis is prioritized; the Logical_Router_Port will be bound to the chassis in the defined order.

This is done by associating multiple Gateway_Chassis rows with a Logical_Router_Port in the OVN Northbound database. A Gateway_Chassis register is just a link to a Chassis register and a priority. For the same Logical_Router_Port, all Gateway_Chassis assigned will have a different priority, starting from 1 (the lowest priority) up to the number of Gateway_Chassis assigned.

The maximum number of Gateway_Chassis that can be assigned to a Logical_Router_Port is 5. This number is hardcoded. That means in Neutron the highest priority a Gateway_Chassis will have is 5.

If no gateway chassis are available during the Logical_Router_Port scheduling, no Gateway_Chassis will be assigned and no value will be set in the options column of the Logical_Router register; that will be used to detect an unhosted router gateway port.

Types of schedulers

The OVN L3 scheduler is configurable and allows us to implement several types of algorithms. There are currently two implemented in the in-tree repository:

- `OVNGatewayChanceScheduler`
- `OVNGatewayLeastLoadedScheduler`

`OVNGatewayChanceScheduler`

The scheduler algorithm in this class is very simple: from the list of gateway chassis provided (candidate chassis), it shuffles and returns the list.

`OVNGatewayLeastLoadedScheduler`

The goal of this scheduler is to balance the available chassis to host the same number of `Logical_Router_Port`. Since[1], the scheduler will retrieve the list of available candidates and will assign, per priority, the least loaded chassis. That means this scheduler will not only consider the chassis with bound `Logical_Router_Port` (highest priority gateway chassis), but it will balance also the lower priority assignations. This is done by (1) iterating over the list of priorities (from 1 to the number of chassis to schedule), (2) creating a list of `Logical_Router_Port` assigned to each `Chassis` **on the select priority** and (3) selecting the least loaded `Chassis`

Re-schedule `Logical_Router_Port` if a `Chassis` is removed

When a gateway `Chassis` is removed from the environment, it creates a hole in the `Gateway_Chassis` assignation for a `Logical_Router_Port`. The `Gateway_Chassis` register associated to the removed `Chassis` is deleted and removed from the list of HA assigned `Chassis`. This event is captured by Neutron, which re-schedules `Gateway_Chassis` to create a balanced list of assignations, same as done in `OVNGatewayLeastLoadedScheduler`. This was implemented in[2].

This process only applies to the lower priority `Gateway_Chassis` registers, never the upper one; this is because the `Logical_Router_Port` is bound to this `Chassis` and could be transmitting. If the highest `Gateway_Chassis` is changed, the `Logical_Router_Port` is bound to the new `Chassis` and could break any active sessions.

> **Note**
>
> Neutron does not support adding or modifying the `Gateway_Chassis` registers with the ovn-nbctl lrp-set-gateway-chassis or the ovn-nbctl set commands. Operators should not use these commands to modify the `Gateway_Chassis` registers because Neutron will not be able to re-schedule the corresponding `Logical_Router_Port` properly.

Availability Zones (AZ) distribution

Both the `OVNGatewayChanceScheduler` and the `OVNGatewayLeastLoadedScheduler` schedulers have the Availability Zones (AZ) in consideration. If a router has any AZ defined, the schedulers will select only those chassis located in the AZs. If no chassis meets this condition, the `Logical_Router_Port` wont be assigned to any chassis and wont be bound.

[1] https://review.opendev.org/c/openstack/neutron/+/893653
[2] https://review.opendev.org/c/openstack/neutron/+/893654

Once the list of candidate `Chassis` (depending on the scheduler selected) is created, this list is reordered to prioritize these `Chassis` from different AZs. That will spread the allocation choices to all AZs; if the current (and highest) `Chassis` binding fails, the next `Chassis` in the list will belong to another AZ.

This improvement was implemented in[3].

Soft Anti-Affinity for `Logical_Router` with multiple `Logical_Router_Port`

Support for multiple gateway ports[4] was implemented to support configurations that provide resiliency and load sharing across multiple router ports at the layer 3 level.

In addition to external dependencies such as BFD for liveness detection and ECMP for load sharing accross default routes, the feature required changes to the scheduler, the goal being that each `Logical_Router_Port` record for a `Logical_Router` would have a different set of `Chassis` for each priority.

The Anti-Affinity is accomplished by having the OVN driver provide the router object subject to scheduling to the scheduler. The scheduler then checks whether there already exists `Logical_Router_Port` records for the target router, and makes any `Chassis` involed in the already existing ports appear as having higher load, making it less likely that the already used `Chassis` gets picked for a new `Logical_Router_Port`.

Since the algorithm is based on load and priority, Anti-Affinity is only supported for the `OVNGatewayLeastLoadedScheduler`.

This improvement was implemented in[5].

References

8.7.17 OVN Maintenance Worker

The maintenance worker is needed by the ML2/OVN mechanism driver to sync the Neutron and OVN databases. Periodic tasks for these inconsistency checks are implemented in `DBInconsistenciesPeriodics`.

On initialization of the maintenance thread the ML2/OVN mechanism driver will add periodics objects, at least `DBInconsistenciesPeriodics`. Plugins may need to add their own periodic tasks to the OVN maintenance worker. If a plugin implements the `ovn_maintenance_periodics` method it should return a list of periodics objects. The mechanism driver will add the returned periodics to the maintenance thread.

[3] https://review.opendev.org/c/openstack/neutron/+/892604
[4] https://review.opendev.org/c/openstack/neutron-specs/+/870030
[5] https://review.opendev.org/c/openstack/neutron/+/873699

CHAPTER

NINE

CONFIGURATION GUIDE

9.1 Configuration Reference

This section provides a list of all configuration options for various neutron services. These are auto-generated from neutron code when this documentation is built.

Configuration filenames used below are filenames usually used, but there is no restriction on configuration filename in neutron and you can use arbitrary file names.

9.1.1 neutron.conf

DEFAULT

`state_path`

Type
string

Default
`/var/lib/neutron`

Where to store Neutron state files. This directory must be writable by the agent.

`bind_host`

Type
host address

Default
`0.0.0.0`

The host IP to bind to.

`bind_port`

Type
port number

Default
`9696`

Minimum Value
0

Maximum Value
65535

The port to bind to

`api_extensions_path`

Type
string

Default
`''`

The path for API extensions. Note that this can be a colon-separated list of paths. For example: api_extensions_path = extensions:/path/to/more/exts:/even/more/exts. The __path__ of neutron.extensions is appended to this, so if your extensions are in there you do not need to specify them here.

`auth_strategy`

Type
string

Default
`keystone`

The type of authentication to use

`core_plugin`

Type
string

Default
`<None>`

The core plugin Neutron will use

`service_plugins`

Type
list

Default
`[]`

The service plugins Neutron will use

`base_mac`

Type
string

Default
`fa:16:3e:00:00:00`

The base MAC address Neutron will use for VIFs. The first 3 octets will remain unchanged. If the 4th octet is not 00, it will also be used. The others will be randomly generated.

`allow_bulk`

Type
boolean

Default
`True`

Allow the usage of the bulk API

`pagination_max_limit`

Type
string

Default
`-1`

The maximum number of items returned in a single response, value of infinite or negative integer means no limit

`default_availability_zones`

Type
list

Default
`[]`

Default value of availability zone hints. The availability zone aware schedulers use this when the resources availability_zone_hints is empty. Multiple availability zones can be specified by a comma separated string. This value can be empty. In this case, even if availability_zone_hints for a resource is empty, availability zone is considered for high availability while scheduling the resource.

`max_dns_nameservers`

Type
integer

Default
`5`

Maximum number of DNS nameservers per subnet

`max_subnet_host_routes`

Type
integer

Default
`20`

Maximum number of host routes per subnet

`ipv6_pd_enabled`

Type
boolean

Default
`False`

Warning: This feature is experimental with low test coverage. Enables IPv6 Prefix Delegation for automatic subnet CIDR allocation. Set to True to enable IPv6 Prefix Delegation for subnet allocation in a PD-capable environment. Users making subnet creation requests for IPv6 subnets without providing a CIDR or subnetpool ID will be given a CIDR via the Prefix Delegation mechanism. Note that enabling PD will override the behavior of the default IPv6 subnetpool.

Warning

This option is deprecated for removal since 2023.2. Its value may be silently ignored in the future.

Reason
There is no reference implementation for the feature for any of in-tree drivers.

`dhcp_lease_duration`

Type
integer

Default
`86400`

DHCP lease duration (in seconds). Use -1 to tell dnsmasq to use infinite lease times.

`dns_domain`

Type
string

Default
`openstacklocal`

Domain to use for building the hostnames

`external_dns_driver`

Type
string

Default
`<None>`

Driver for external DNS integration.

`dhcp_agent_notification`

Type
boolean

Default
`True`

Allow sending resource operation notification to DHCP agent

`host`

Type
host address

Default
`example.domain`

This option has a sample default set, which means that its actual default value may vary from the one documented above.

Hostname to be used by the Neutron server, agents and services running on this machine. All the agents and services running on this machine must use the same host value.

`network_link_prefix`

Type
string

Default
`<None>`

This string is prepended to the normal URL that is returned in links to the OpenStack Network API. If it is empty (the default), the URLs are returned unchanged.

`notify_nova_on_port_status_changes`

Type
boolean

Default
`True`

Send notification to Nova when port status changes

`notify_nova_on_port_data_changes`

Type
boolean

Default
`True`

Send notification to Nova when port data (fixed_ips/floatingip) changes so Nova can update its cache.

`send_events_interval`

Type
integer

Default
`2`

Number of seconds between sending events to Nova if there are any events to send.

`setproctitle`

Type
string

Default
`on`

Set process name to match child worker role. Available options are: off - retains the previous behavior; on - renames processes to neutron-server: role (original string); brief - renames the same as on, but without the original string, such as neutron-server: role.

`ipam_driver`

Type
string

Default
`internal`

Neutron IPAM (IP address management) driver to use. By default, the reference implementation of the Neutron IPAM driver is used.

`vlan_transparent`

Type
boolean

Default
`<None>`

If True, then allow plugins that support it to create VLAN transparent networks.

> **Warning**
>
> This option is deprecated for removal since 2025.2. Its value may be silently ignored in the future.
>
> **Reason**
> This option is going to be removed as availability of the *vlan_transparency* in the deployment is now calculated automatically based on the loaded mechanism drivers.

`vlan_qinq`

Type
boolean

Default
`<None>`

If True, then allow plugins that support it to create VLAN transparent networks using 0x8a88 ethertype.

> **Warning**
>
> This option is deprecated for removal since 2025.2. Its value may be silently ignored in the future.
>
> **Reason**
> This option is going to be removed as availability of the *vlan_qinq* in the deployment is now calculated automatically based on the loaded mechanism drivers.

`filter_validation`

Type
boolean

Default
`True`

If True, then allow plugins to decide whether to perform validations on filter parameters. Filter validation is enabled if this config is turned on and it is supported by all plugins

`global_physnet_mtu`

Type
integer

Default
`1500`

MTU of the underlying physical network. Neutron uses this value to calculate MTU for all virtual network components. For flat and VLAN networks, neutron uses this value without modification. For overlay networks such as VXLAN, neutron automatically subtracts the overlay protocol overhead from this value. Defaults to 1500, the standard value for Ethernet.

`http_retries`

Type
integer

Default
`3`

Minimum Value
0

Number of times client connections (Nova, Ironic) should be retried on a failed HTTP call. 0 (zero) means connection is attempted only once (not retried). Setting to any positive integer means that on failure the connection is retried that many times. For example, setting to 3 means total attempts to connect will be 4.

`enable_traditional_dhcp`

Type
boolean

Default
`True`

If False, neutron-server will disable the following DHCP-agent related functions: 1. DHCP provisioning block 2. DHCP scheduler API extension 3. Network scheduling mechanism 4. DHCP RPC/notification

`my_ip`

Type
string

Default
`10.0.19.190`

IPv4 address of this host. If no address is provided and one cannot be determined, 127.0.0.1 will be used.

`my_ipv6`

Type
string

Default
`::1`

IPv6 address of this host. If no address is provided and one cannot be determined, ::1 will be used.

`backlog`

Type
integer

Default
`4096`

Number of backlog requests to configure the socket with

Warning

This option is deprecated for removal since 2026.1. Its value may be silently ignored in the future.

Reason
This option has no effect.

`retry_until_window`

Type
integer

Default
`30`

Number of seconds to keep retrying to listen

Warning

This option is deprecated for removal since 2026.1. Its value may be silently ignored in the future.

Reason
This option has no effect.

`use_ssl`

Type
boolean

Default
`False`

Enable SSL on the API server

Warning

This option is deprecated for removal since 2026.1. Its value may be silently ignored in the future.

Reason
This option has no effect.

`periodic_interval`

Type
integer

Default
`40`

Seconds between running periodic tasks.

`api_workers`

Type
integer

Default
`<None>`

Minimum Value
1

Number of separate API worker processes for service. If not specified, the default is equal to the number of CPUs available for best performance, capped by potential RAM usage.

`rpc_workers`

Type
integer

Default
`<None>`

Minimum Value
0

Number of RPC worker processes for service. If not specified, the default is equal to half the number of API workers. If set to 0, no RPC worker is launched.

`rpc_state_report_workers`

Type
integer

Default
`1`

Minimum Value
0

Number of RPC worker processes dedicated to the state reports queue. If set to 0, no dedicated RPC worker for state reports queue is launched.

`periodic_fuzzy_delay`

Type
integer

Default
`5`

Range of seconds to randomly delay when starting the periodic task scheduler to reduce stampeding. (Disable by setting to 0)

`rpc_response_max_timeout`

Type
integer

Default
`600`

Maximum seconds to wait for a response from an RPC call.

`interface_driver`

Type
string

Default
`openvswitch`

The driver used to manage virtual interfaces.

`metadata_proxy_socket`

Type
string

Default
`$state_path/metadata_proxy`

Location for Metadata Proxy UNIX domain socket.

`metadata_proxy_user`

Type
string

Default
`''`

User (uid or name) running metadata proxy after its initialization (if empty: agent effective user).

`metadata_proxy_group`

Type
string

Default
`''`

Group (gid or name) running metadata proxy after its initialization (if empty: agent effective group).

`agent_down_time`

Type
integer

Default
`75`

Maximum Value
2147483.0

Seconds to regard the agent as down; should be at least twice report_interval, to be sure the agent is down for good.

`dhcp_load_type`

Type
string

Default
`networks`

Valid Values
networks, subnets, ports

Representing the resource type whose load is being reported by the agent. This can be networks, subnets or ports. When specified (Default is networks), the server will extract particular load sent as part of its agent configuration object from the agent report state, which is the number of resources being consumed, at every report_interval. dhcp_load_type can be used in combination with network_scheduler_driver = neutron.scheduler.dhcp_agent_scheduler.WeightScheduler When the network_scheduler_driver is WeightScheduler, dhcp_load_type can be configured to represent the choice for the resource being balanced. Example: dhcp_load_type=networks

`enable_new_agents`

Type
boolean

Default
`True`

Agents start with admin_state_up=False when enable_new_agents=False. In this case, a users resources will not be scheduled automatically to an agent until an admin sets admin_state_up to True.

`rpc_resources_processing_step`

Type
integer

Default
`20`

Minimum Value
1

Number of resources for neutron to divide a large RPC call into data sets. It can be reduced if RPC timeouts occur. The best value should be determined empirically in your environment.

`max_routes`

Type
integer

Default
`30`

Maximum number of routes per router

`enable_snat_by_default`

Type
boolean

Default
`True`

Define the default value of enable_snat if not provided in external_gateway_info.

`network_scheduler_driver`

Type
string

Default
`neutron.scheduler.dhcp_agent_scheduler.WeightScheduler`

Driver to use for scheduling networks to a DHCP agent

`network_auto_schedule`

Type
boolean

Default
`True`

Allow auto scheduling networks to a DHCP agent.

`allow_automatic_dhcp_failover`

Type
boolean

Default
`True`

Automatically remove networks from offline DHCP agents.

`dhcp_agents_per_network`

Type
integer

Default
`1`

Minimum Value
1

Number of DHCP agents scheduled to host a project network. If this number is greater than 1, the scheduler automatically assigns multiple DHCP agents for a given project network, providing high availability for the DHCP service. However this does not provide high availability for the IPv6 metadata service in isolated networks.

`enable_services_on_agents_with_admin_state_down`

Type
boolean

Default
`False`

Enable services on an agent with admin_state_up False. If this option is False, when admin_state_up of an agent is turned False, services on it will be disabled. Agents with admin_state_up False are not selected for automatic scheduling regardless of this option. But manual scheduling to such agents is available if this option is True.

`dvr_base_mac`

Type
string

Default
`fa:16:3f:00:00:00`

The base MAC address used for unique DVR instances by Neutron. The first 3 octets will remain unchanged. If the 4th octet is not 00, it will also be used. The others will be randomly generated. The dvr_base_mac *must* be different from base_mac to avoid mixing it up with MACs allocated for project ports. A 4-octet example would be dvr_base_mac = fa:16:3f:4f:00:00. The default is 3 octets

`router_distributed`

Type
boolean

Default
`False`

System-wide flag to determine the type of router that projects can create. Only admin can override.

`enable_dvr`

Type
boolean

Default
`True`

Determine if setup is configured for DVR. If False, the DVR API extension will be disabled.

`host_dvr_for_dhcp`

Type
boolean

Default
`True`

Flag to determine if hosting a DVR local router to the DHCP agent is desired. If False, any L3 function supported by the DHCP agent instance will not be possible, for instance: DNS.

`router_scheduler_driver`

Type
string

Default
`neutron.scheduler.l3_agent_scheduler.LeastRoutersScheduler`

Driver to use for scheduling a router to a default L3 agent

`router_auto_schedule`

Type
boolean

Default
`True`

Allow auto scheduling of routers to L3 agents.

`allow_automatic_l3agent_failover`

Type
boolean

Default
`False`

Automatically reschedule routers from offline L3 agents to online L3 agents.

`l3_ha`

Type
boolean

Default
`False`

Enable HA mode for virtual routers.

`max_l3_agents_per_router`

Type
integer

Default
`3`

Maximum number of L3 agents which a HA router will be scheduled on. If it is set to 0 then the router will be scheduled on every agent.

`l3_ha_net_cidr`

Type
string

Default
`169.254.192.0/18`

Subnet used for the L3 HA admin network.

`l3_ha_network_type`

Type
string

Default
`''`

The network type to use when creating the L3 HA network for an HA router. By default, or if empty, the first project_network_types value is used. This is helpful when the VRRP traffic should use a specific network which is not the default one.

`l3_ha_network_physical_name`

Type
string

Default
`''`

The physical network name with which the L3 HA network should be created.

`enable_default_route_ecmp`

Type
boolean

Default
`False`

Define the default value for enable_default_route_ecmp if not specified on the router.

`enable_default_route_bfd`

Type
boolean

Default
`False`

Define the default value for enable_default_route_bfd if not specified on the router.

`max_allowed_address_pair`

Type
integer

Default
`10`

Maximum number of allowed address pairs

`allowed_conntrack_helpers`

Type
list

Default
`[{'tftp': 'udp'}, {'ftp': 'tcp'}, {'sip': 'tcp'}, {'sip': 'udp'}]`

This option has a sample default set, which means that its actual default value may vary from the one documented above.

Defines the allowed conntrack helpers, and conntrack helper module protocol constraints.

executor_thread_pool_size

Type
integer

Default
`64`

Size of executor thread pool when executor is threading or eventlet.

Table 1: Deprecated Variations

Group	Name
DEFAULT	rpc_thread_pool_size

rpc_response_timeout

Type
integer

Default
`60`

Seconds to wait for a response from a call.

transport_url

Type
string

Default
`rabbit://`

The network address and optional user credentials for connecting to the messaging backend, in URL format. The expected format is:

driver://[user:pass@]host:port[,[userN:passN@]hostN:portN]/virtual_host?query

Example: rabbit://rabbitmq:password@127.0.0.1:5672//

For full details on the fields in the URL see the documentation of oslo_messaging.TransportURL at https://docs.openstack.org/oslo.messaging/latest/reference/transport.html

control_exchange

Type
string

Default
`openstack`

The default exchange under which topics are scoped. May be overridden by an exchange name specified in the transport_url option.

`rpc_ping_enabled`

Type
boolean

Default
`False`

Add an endpoint to answer to ping calls. Endpoint is named oslo_rpc_server_ping

`debug`

Type
boolean

Default
`False`

Mutable
This option can be changed without restarting.

If set to true, the logging level will be set to DEBUG instead of the default INFO level.

`log_config_append`

Type
string

Default
`<None>`

Mutable
This option can be changed without restarting.

The name of a logging configuration file. This file is appended to any existing logging configuration files. For details about logging configuration files, see the Python logging module documentation. Note that when logging configuration files are used then all logging configuration is set in the configuration file and other logging configuration options are ignored (for example, log-date-format).

Table 2: Deprecated Variations

Group	Name
DEFAULT	log-config
DEFAULT	log_config

`log_date_format`

Type
string

Default
`%Y-%m-%d %H:%M:%S`

Defines the format string for %(asctime)s in log records. Default: the value above . This option is ignored if log_config_append is set.

`log_file`

Type
string

Default
`<None>`

(Optional) Name of log file to send logging output to. If no default is set, logging will go to stderr as defined by use_stderr. This option is ignored if log_config_append is set.

Table 3: Deprecated Variations

Group	Name
DEFAULT	logfile

`log_dir`

Type
string

Default
`<None>`

(Optional) The base directory used for relative log_file paths. This option is ignored if log_config_append is set.

Table 4: Deprecated Variations

Group	Name
DEFAULT	logdir

`use_syslog`

Type
boolean

Default
`False`

Use syslog for logging. Existing syslog format is DEPRECATED and will be changed later to honor RFC5424. This option is ignored if log_config_append is set.

`use_journal`

Type
boolean

Default
`False`

Enable journald for logging. If running in a systemd environment you may wish to enable journal support. Doing so will use the journal native protocol which includes structured metadata in addition to log messages.This option is ignored if log_config_append is set.

`syslog_log_facility`

Type
string

Default
`LOG_USER`

Syslog facility to receive log lines. This option is ignored if log_config_append is set.

`use_json`

Type
boolean

Default
`False`

Use JSON formatting for logging. This option is ignored if log_config_append is set.

`use_stderr`

Type
boolean

Default
`False`

Log output to standard error. This option is ignored if log_config_append is set.

`log_color`

Type
boolean

Default
`False`

(Optional) Set the color key according to log levels. This option takes effect only when logging to stderr or stdout is used. This option is ignored if log_config_append is set.

`log_rotate_interval`

Type
integer

Default
`1`

The amount of time before the log files are rotated. This option is ignored unless log_rotation_type is set to interval.

`log_rotate_interval_type`

Type
string

Default
`days`

Valid Values
Seconds, Minutes, Hours, Days, Weekday, Midnight

Rotation interval type. The time of the last file change (or the time when the service was started) is used when scheduling the next rotation.

`max_logfile_count`

Type
integer

Default
`30`

Maximum number of rotated log files.

`max_logfile_size_mb`

Type
integer

Default
`200`

Log file maximum size in MB. This option is ignored if log_rotation_type is not set to size.

`log_rotation_type`

Type
string

Default
`none`

Valid Values
interval, size, none

Log rotation type.

Possible values

interval
Rotate logs at predefined time intervals.

size
Rotate logs once they reach a predefined size.

none
Do not rotate log files.

`logging_context_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [%(global_request_id)s %(request_id)s %(user_identity)s] %(instance)s%(message)s`

Format string to use for log messages with context. Used by oslo_log.formatters.ContextFormatter `logging_default_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [-] %(instance)s%(message)s`

Format string to use for log messages when context is undefined. Used by oslo_log.formatters.ContextFormatter

`logging_debug_format_suffix`

Type
string

Default
`%(funcName)s %(pathname)s:%(lineno)d`

Additional data to append to log message when logging level for the message is DEBUG. Used by oslo_log.formatters.ContextFormatter

`logging_exception_prefix`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d ERROR %(name)s %(instance)s`

Prefix each line of exception output with this format. Used by oslo_log.formatters.ContextFormatter

`logging_user_identity_format`

Type
string

Default
`%(user)s %(project)s %(domain)s %(system_scope)s %(user_domain)s %(project_domain)s`

Defines the format string for %(user_identity)s that is used in logging_context_format_string. Used by oslo_log.formatters.ContextFormatter

`default_log_levels`

Type
list

Default
`['amqp=WARN', 'boto=WARN', 'sqlalchemy=WARN', 'suds=INFO', 'oslo.messaging=INFO', 'oslo_messaging=INFO', 'iso8601=WARN', 'requests.packages.urllib3.connectionpool=WARN', 'urllib3.connectionpool=WARN', 'websocket=WARN', 'requests.packages.urllib3.util.retry=WARN', 'urllib3.util.retry=WARN', 'keystonemiddleware=WARN', 'routes.middleware=WARN',`

'stevedore=WARN', 'taskflow=WARN', 'keystoneauth=WARN', 'oslo.cache=INFO', 'oslo_policy=INFO', 'dogpile.core.dogpile=INFO']

List of package logging levels in logger=LEVEL pairs. This option is ignored if log_config_append is set.

`publish_errors`

Type
boolean

Default
`False`

Enables or disables publication of error events.

`instance_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance that is passed with the log message.

`instance_uuid_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance UUID that is passed with the log message.

`rate_limit_interval`

Type
integer

Default
`0`

Interval, number of seconds, of log rate limiting.

`rate_limit_burst`

Type
integer

Default
`0`

Maximum number of logged messages per rate_limit_interval.

`rate_limit_except_level`

Type
string

Default
`CRITICAL`

Valid Values
CRITICAL, ERROR, INFO, WARNING, DEBUG,

Log level name used by rate limiting. Logs with level greater or equal to rate_limit_except_level are not filtered. An empty string means that all levels are filtered.

`fatal_deprecations`

Type
boolean

Default
`False`

Enables or disables fatal status of deprecations.

`run_external_periodic_tasks`

Type
boolean

Default
`True`

Some periodic tasks can be run in a separate process. Should we run them here?

`backdoor_port`

Type
string

Default
`<None>`

Enable eventlet backdoor. Acceptable values are 0, <port>, and <start>:<end>, where 0 results in listening on a random tcp port number; <port> results in listening on the specified port number (and not enabling backdoor if that port is in use); and <start>:<end> results in listening on the smallest unused port number within the specified range of port numbers. The chosen port is displayed in the services log file.

Warning

This option is deprecated for removal. Its value may be silently ignored in the future.

Reason
The backdoor_port option is deprecated and will be removed in a future release.

`backdoor_socket`

Type
string

Default
`<None>`

Enable eventlet backdoor, using the provided path as a unix socket that can receive connections. This option is mutually exclusive with backdoor_port in that only one should be provided. If both are provided then the existence of this option overrides the usage of that option. Inside the path {pid} will be replaced with the PID of the current process.

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> The backdoor_socket option is deprecated and will be removed in a future release.

`log_options`

Type
boolean

Default
`True`

Enables or disables logging values of all registered options when starting a service (at DEBUG level).

`graceful_shutdown_timeout`

Type
integer

Default
`60`

Specify a timeout after which a gracefully shutdown server will exit. Zero value means endless wait.

`api_paste_config`

Type
string

Default
`api-paste.ini`

File name for the paste.deploy config for api service

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> The api_paste_config option is deprecated and will be removed in a future release.

`wsgi_log_format`

Type
string

Default
`%(client_ip)s "%(request_line)s" status: %(status_code)s len: %(body_length)s time: %(wall_seconds).7f`

A python format string that is used as the template to generate log lines. The following values can beformatted into it: client_ip, date_time, request_line, status_code, body_length, wall_seconds.

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> The wsgi_log_format option is deprecated and will be removed in a future release.

`tcp_keepidle`

Type
integer

Default
`600`

Sets the value of TCP_KEEPIDLE in seconds for each server socket. Not supported on OS X.

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> The tcp_keepidle option is deprecated and will be removed in a future release.

`wsgi_default_pool_size`

Type
integer

Default
`100`

Size of the pool of greenthreads used by wsgi

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> The wsgi_default_pool_size option is deprecated and will be removed in a future release.

`max_header_line`

Type
integer

Default
`16384`

Maximum line size of message headers to be accepted. max_header_line may need to be increased when using large tokens (typically those generated when keystone is configured to use PKI tokens with big service catalogs).

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> The max_header_line option is deprecated and will be removed in a future release.

`wsgi_keep_alive`

Type
boolean

Default
`True`

If False, closes the client socket connection explicitly.

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> The wsgi_keep_alive option is deprecated and will be removed in a future release.

`client_socket_timeout`

Type
integer

Default
`900`

Timeout for client connections socket operations. If an incoming connection is idle for this number of seconds it will be closed. A value of 0 means wait forever.

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.

Reason
The client_socket_timeout option is deprecated and will be removed in a future release.

wsgi_server_debug

Type
boolean

Default
`False`

True if the server should send exception tracebacks to the clients on 500 errors. If False, the server will respond with empty bodies.

Warning

This option is deprecated for removal. Its value may be silently ignored in the future.

Reason
The wsgi_server_debug option is deprecated and will be removed in a future release.

agent

root_helper

Type
string

Default
`sudo`

Root helper application. Use sudo neutron-rootwrap /etc/neutron/rootwrap.conf to use the real root filter facility. Change to sudo to skip the filtering and just run the command directly.

use_helper_for_ns_read

Type
boolean

Default
`True`

Use the root helper when listing the namespaces on a system. This may not be required depending on the security configuration. If the root helper is not required, set this to False for a performance improvement.

root_helper_daemon

Type
string

Default
`<None>`

Root helper daemon application to use when possible.

Use sudo neutron-rootwrap-daemon /etc/neutron/rootwrap.conf to run rootwrap in daemon mode which has been reported to improve performance at scale. For more information on running rootwrap in daemon mode, see:

https://docs.openstack.org/oslo.rootwrap/latest/user/usage.html#daemon-mode

`report_interval`

Type
floating point

Default
`30`

Seconds between nodes reporting state to server; should be less than agent_down_time, best if it is half or less than agent_down_time.

`log_agent_heartbeats`

Type
boolean

Default
`False`

Log agent heartbeats

`comment_iptables_rules`

Type
boolean

Default
`True`

Add comments to iptables rules. Set to false to disallow the addition of comments to generated iptables rules that describe each rules purpose. System must support the iptables comments module for addition of comments.

`debug_iptables_rules`

Type
boolean

Default
`False`

Duplicate every iptables difference calculation to ensure the format being generated matches the format of iptables-save. This option should not be turned on for production systems because it imposes a performance penalty.

`use_random_fully`

Type
boolean

Default
`True`

Use random-fully in SNAT masquerade rules.

`check_child_processes_action`

Type
string

Default
`respawn`

Valid Values
respawn, exit

Action to be executed when a child process dies

`check_child_processes_interval`

Type
integer

Default
`60`

Interval between checks of child process liveness, in seconds, use 0 to disable

`kill_scripts_path`

Type
string

Default
`/etc/neutron/kill_scripts/`

Location of scripts used to kill external processes. Names of scripts here must follow the pattern: <process-name>-kill where <process-name> is name of the process which should be killed using this script. For example, kill script for dnsmasq process should be named dnsmasq-kill. If path is set to None, then default kill command will be used to stop processes.

`availability_zone`

Type
string

Default
`nova`

Availability zone of this node

cache

`config_prefix`

Type
string

Default
`cache.oslo`

Prefix for building the configuration dictionary for the cache region. This should not need to be changed unless there is another dogpile.cache region with the same configuration name.

`expiration_time`

Type
integer

Default
`600`

Minimum Value
1

Default TTL, in seconds, for any cached item in the dogpile.cache region. This applies to any cached method that doesnt have an explicit cache expiration time defined for it.

`backend_expiration_time`

Type
integer

Default
`<None>`

Minimum Value
1

Expiration time in cache backend to purge expired records automatically. This should be greater than expiration_time and all cache_time options

`backend`

Type
string

Default
`dogpile.cache.null`

Valid Values
oslo_cache.memcache_pool, oslo_cache.dict, oslo_cache.etcd3gw, dogpile.cache.pymemcache, dogpile.cache.memcached, dogpile.cache.pylibmc, dogpile.cache.bmemcached, dogpile.cache.dbm, dogpile.cache.redis, dogpile.cache.redis_sentinel, dogpile.cache.memory, dogpile.cache.memory_pickle, dogpile.cache.null

Cache backend module. For eventlet-based or environments with hundreds of threaded servers, Memcache with pooling (oslo_cache.memcache_pool) is recommended. For environments with less than 100 threaded servers, Memcached (dogpile.cache.memcached) or Redis (dogpile.cache.redis) is recommended. Test environments with a single instance of the server can use the dogpile.cache.memory backend.

`backend_argument`

Type
multi-valued

Default
`''`

Arguments supplied to the backend module. Specify this option once per argument to be passed to the dogpile.cache backend. Example format: <argname>:<value>.

`proxies`

Type
list

Default
`[]`

Proxy classes to import that will affect the way the dogpile.cache backend functions. See the dogpile.cache documentation on changing-backend-behavior.

`enabled`

Type
boolean

Default
`False`

Global toggle for caching.

`debug_cache_backend`

Type
boolean

Default
`False`

Extra debugging from the cache backend (cache keys, get/set/delete/etc calls). This is only really useful if you need to see the specific cache-backend get/set/delete calls with the keys/values. Typically this should be left set to false.

`memcache_servers`

Type
list

Default
`['localhost:11211']`

Memcache servers in the format of host:port. This is used by backends dependent on Memcached.If `dogpile.cache.memcached` or `oslo_cache.memcache_pool` is used and a given host refer to an IPv6 or a given domain refer to IPv6 then you should prefix the given address with the address family (`inet6`) (e.g `inet6:[::1]:11211`, `inet6:[fd12:3456:789a:1::1]:11211`, `inet6:[controller-0.internalapi]:11211`). If the address family is not given then these backends will use the default `inet` address family which corresponds to IPv4

`memcache_dead_retry`

Type
integer

Default
`300`

Number of seconds memcached server is considered dead before it is tried again. (dogpile.cache.memcache and oslo_cache.memcache_pool backends only).

`memcache_pool_maxsize`

Type
integer

Default
`10`

Max total number of open connections to every memcached server. (oslo_cache.memcache_pool backend only).

`memcache_pool_unused_timeout`

Type
integer

Default
`60`

Number of seconds a connection to memcached is held unused in the pool before it is closed. (oslo_cache.memcache_pool backend only).

`memcache_pool_connection_get_timeout`

Type
integer

Default
`10`

Number of seconds that an operation will wait to get a memcache client connection.

`memcache_pool_flush_on_reconnect`

Type
boolean

Default
`False`

Global toggle if memcache will be flushed on reconnect. (oslo_cache.memcache_pool backend only).

`memcache_sasl_enabled`

Type
boolean

Default
`False`

Enable the SASL(Simple Authentication and SecurityLayer) if the SASL_enable is true, else disable.

`redis_server`

Type
string

Default
`localhost:6379`

Redis server in the format of host:port

`redis_db`

Type
integer

Default
`0`

Minimum Value
0

Database id in Redis server

`redis_sentinels`

Type
list

Default
`['localhost:26379']`

Redis sentinel servers in the format of host:port

`redis_sentinel_service_name`

Type
string

Default
`mymaster`

Service name of the redis sentinel cluster.

`username`

Type
string

Default
<None>

the user name for authentication to backend.

Table 5: Deprecated Variations

Group	Name
cache	memcache_username
cache	redis_username

`password`

Type
string

Default
<None>

the password for authentication to backend.

Table 6: Deprecated Variations

Group	Name
cache	memcache_password
cache	redis_password

`tls_enabled`

Type
boolean

Default
`False`

Global toggle for TLS usage when communicating with the caching servers. Currently supported by `dogpile.cache.bmemcache`, `dogpile.cache.pymemcache`, `oslo_cache.memcache_pool`, `dogpile.cache.redis` and `dogpile.cache.redis_sentinel`.

`tls_cafile`

Type
string

Default
`<None>`

Path to a file of concatenated CA certificates in PEM format necessary to establish the caching servers authenticity. If tls_enabled is False, this option is ignored.

`tls_certfile`

Type
string

Default
`<None>`

Path to a single file in PEM format containing the clients certificate as well as any number of CA certificates needed to establish the certificates authenticity. This file is only required when client side authentication is necessary. If tls_enabled is False, this option is ignored.

`tls_keyfile`

Type
string

Default
`<None>`

Path to a single file containing the clients private key in. Otherwise the private key will be taken from the file specified in tls_certfile. If tls_enabled is False, this option is ignored.

`tls_allowed_ciphers`

Type
string

Default
`<None>`

Set the available ciphers for sockets created with the TLS context. It should be a string in the OpenSSL cipher list format. If not specified, all OpenSSL enabled ciphers will be available. Currently supported by `dogpile.cache.bmemcache`, `dogpile.cache.pymemcache` and `oslo_cache.memcache_pool`.

`socket_timeout`

Type
floating point

Default
`1.0`

Timeout in seconds for every call to a server. Currently supported by `dogpile.cache.memcache`, `oslo_cache.memcache_pool`, `dogpile.cache.redis` and `dogpile.cache.redis_sentinel`.

Table 7: Deprecated Variations

Group	Name
cache	memcache_socket_timeout
cache	redis_socket_timeout

`enable_socket_keepalive`

Type
boolean

Default
`False`

Global toggle for the socket keepalive of dogpiles pymemcache backend

`socket_keepalive_idle`

Type
integer

Default
1

Minimum Value
0

The time (in seconds) the connection needs to remain idle before TCP starts sending keepalive probes. Should be a positive integer most greater than zero.

`socket_keepalive_interval`

Type
integer

Default
1

Minimum Value
0

The time (in seconds) between individual keepalive probes. Should be a positive integer greater than zero.

`socket_keepalive_count`

Type
integer

Default
`1`

Minimum Value
0

The maximum number of keepalive probes TCP should send before dropping the connection. Should be a positive integer greater than zero.

`enable_retry_client`

Type
boolean

Default
`False`

Enable retry client mechanisms to handle failure. Those mechanisms can be used to wrap all kind of pymemcache clients. The wrapper allows you to define how many attempts to make and how long to wait between attempts.

`retry_attempts`

Type
integer

Default
`2`

Minimum Value
1

Number of times to attempt an action before failing.

`retry_delay`

Type
floating point

Default
`0`

Number of seconds to sleep between each attempt.

`hashclient_retry_attempts`

Type
integer

Default
`2`

Minimum Value
1

Amount of times a client should be tried before it is marked dead and removed from the pool in the HashClients internal mechanisms.

`hashclient_retry_timeout`

Type
floating point

Default
`1`

Time in seconds that should pass between retry attempts in the HashClients internal mechanisms.

Table 8: Deprecated Variations

Group	Name
cache	hashclient_retry_delay

`hashclient_dead_timeout`

Type
floating point

Default
`60`

Time in seconds before attempting to add a node back in the pool in the HashClients internal mechanisms.

Table 9: Deprecated Variations

Group	Name
cache	dead_timeout

`enforce_fips_mode`

Type
boolean

Default
`False`

Global toggle for enforcing the OpenSSL FIPS mode. This feature requires Python support. This is available in Python 3.9 in all environments and may have been backported to older Python versions on select environments. If the Python executable used does not support OpenSSL FIPS mode, an exception will be raised. Currently supported by `dogpile.cache.bmemcache`, `dogpile.cache.pymemcache` and `oslo_cache.memcache_pool`.

Warning

This option is deprecated for removal. Its value may be silently ignored in the future.

Reason
FIPS_mode_set API was removed in OpenSSL 3.0.0. This option has no effect now.

cors

allowed_origin

Type
list

Default
`<None>`

Indicate whether this resource may be shared with the domain received in the requests origin header. Format: <protocol>://<host>[:<port>], no trailing slash. Example: https://horizon.example.com

allow_credentials

Type
boolean

Default
`True`

Indicate that the actual request can include user credentials

expose_headers

Type
list

Default
`['X-Auth-Token', 'X-Subject-Token', 'X-Service-Token', 'X-OpenStack-Request-ID', 'OpenStack-Volume-microversion']`

Indicate which headers are safe to expose to the API. Defaults to HTTP Simple Headers.

max_age

Type
integer

Default
`3600`

Maximum cache age of CORS preflight requests.

allow_methods

Type
list

Default
`['GET', 'PUT', 'POST', 'DELETE', 'PATCH']`

Indicate which methods can be used during the actual request.

`allow_headers`

Type
list

Default
`['X-Auth-Token', 'X-Identity-Status', 'X-Roles', 'X-Service-Catalog', 'X-User-Id', 'X-Tenant-Id', 'X-OpenStack-Request-ID']`

Indicate which header field names may be used during the actual request.

database

`engine`

Type
string

Default
`''`

Database engine for which script will be generated when using offline migration.

`sqlite_synchronous`

Type
boolean

Default
`True`

If True, SQLite uses synchronous mode.

`backend`

Type
string

Default
`sqlalchemy`

The back end to use for the database.

`connection`

Type
string

Default
`<None>`

The SQLAlchemy connection string to use to connect to the database.

`slave_connection`

Type
string

Default
`<None>`

The SQLAlchemy connection string to use to connect to the slave database.

`asyncio_connection`

Type
string

Default
`<None>`

The SQLAlchemy asyncio connection string to use to connect to the database.

`asyncio_slave_connection`

Type
string

Default
`<None>`

The SQLAlchemy asyncio connection string to use to connect to the slave database.

`synchronous_reader`

Type
boolean

Default
`True`

Whether or not to assume a reader context needs to guarantee it can read data committed by a writer assuming replication lag is present; defaults to True. When False, a reader context works the same as async_reader and will select the slave database if present. When using a galera cluster, this can be set to False only if you set mysql_wsrep_sync_wait to 1 (this will guarantee that the reader will wait until writesets are committed).Note that this may incur a performance degradation within the galera cluster. Note also that this parameter has no effect if you do not set any slave_connection.

`mysql_sql_mode`

Type
string

Default
`TRADITIONAL`

The SQL mode to be used for MySQL sessions. This option, including the default, overrides any server-set SQL mode. To use whatever SQL mode is set by the server configuration, set this to no value. Example: mysql_sql_mode=

`mysql_wsrep_sync_wait`

Type
integer

Default
`<None>`

For Galera only, configure wsrep_sync_wait causality checks on new connections. Default is None, meaning dont configure any setting.

`connection_recycle_time`

Type
integer

Default
`3600`

Connections which have been present in the connection pool longer than this number of seconds will be replaced with a new one the next time they are checked out from the pool.

`max_pool_size`

Type
integer

Default
`5`

Maximum number of SQL connections to keep open in a pool. Setting a value of 0 indicates no limit.

`max_retries`

Type
integer

Default
`10`

Maximum number of database connection retries during startup. Set to -1 to specify an infinite retry count.

`retry_interval`

Type
integer

Default
`10`

Interval between retries of opening a SQL connection.

`max_overflow`

Type
integer

Default
`50`

If set, use this value for max_overflow with SQLAlchemy.

connection_debug

Type
integer

Default
`0`

Minimum Value
0

Maximum Value
100

Verbosity of SQL debugging information: 0=None, 100=Everything.

connection_trace

Type
boolean

Default
`False`

Add Python stack traces to SQL as comment strings.

pool_timeout

Type
integer

Default
`<None>`

If set, use this value for pool_timeout with SQLAlchemy.

use_db_reconnect

Type
boolean

Default
`False`

Enable the experimental use of database reconnect on connection lost.

db_retry_interval

Type
integer

Default
`1`

Seconds between retries of a database transaction.

db_inc_retry_interval

Type
boolean

Default
`True`

If True, increases the interval between retries of a database operation up to db_max_retry_interval.

`db_max_retry_interval`

Type
integer

Default
`10`

If db_inc_retry_interval is set, the maximum seconds between retries of a database operation.

`db_max_retries`

Type
integer

Default
`20`

Maximum retries in case of connection error or deadlock error before error is raised. Set to -1 to specify an infinite retry count.

`connection_parameters`

Type
string

Default
`''`

Optional URL parameters to append onto the connection URL at connect time; specify as param1=value1&param2=value2&

designate

`auth_url`

Type
unknown type

Default
`<None>`

Authentication URL

`auth_type`

Type
unknown type

Default
`<None>`

Authentication type to load

Table 10: Deprecated Variations

Group	Name
designate	auth_plugin

`cafile`

Type
string

Default
`<None>`

PEM encoded Certificate Authority to use when verifying HTTPs connections.

`certfile`

Type
string

Default
`<None>`

PEM encoded client certificate cert file

`collect_timing`

Type
boolean

Default
`False`

Collect per-API call timing information.

`default_domain_id`

Type
unknown type

Default
`<None>`

Optional domain ID to use with v3 and v2 parameters. It will be used for both the user and project domain in v3 and ignored in v2 authentication.

`default_domain_name`

Type
unknown type

Default
`<None>`

Optional domain name to use with v3 API and v2 parameters. It will be used for both the user and project domain in v3 and ignored in v2 authentication.

domain_id

Type
unknown type

Default
<None>

Domain ID to scope to

domain_name

Type
unknown type

Default
<None>

Domain name to scope to

insecure

Type
boolean

Default
False

Verify HTTPS connections.

keyfile

Type
string

Default
<None>

PEM encoded client certificate key file

password

Type
unknown type

Default
<None>

Users password

project_domain_id

Type
unknown type

Default
<None>

Domain ID containing project

project_domain_name

Type
unknown type

Default
<None>

Domain name containing project

project_id

Type
unknown type

Default
<None>

Project ID to scope to

Table 11: Deprecated Variations

Group	Name
designate	tenant-id
designate	tenant_id

project_name

Type
unknown type

Default
<None>

Project name to scope to

Table 12: Deprecated Variations

Group	Name
designate	tenant-name
designate	tenant_name

split_loggers

Type
boolean

Default
False

Log requests to multiple loggers.

system_scope

Type
unknown type

Default
<None>

Scope for system operations

`tenant_id`

Type
unknown type

Default
<None>

Tenant ID

`tenant_name`

Type
unknown type

Default
<None>

Tenant Name

`timeout`

Type
integer

Default
<None>

Timeout value for http requests

`trust_id`

Type
unknown type

Default
<None>

ID of the trust to use as a trustee use

`user_domain_id`

Type
unknown type

Default
<None>

Users domain id

`user_domain_name`

Type
unknown type

Default
<None>

Users domain name

user_id

Type
unknown type

Default
<None>

User id

username

Type
unknown type

Default
<None>

Username

Table 13: Deprecated Variations

Group	Name
designate	user-name
designate	user_name

url

Type
URI

Default
<None>

URL for connecting to designate

allow_reverse_dns_lookup

Type
boolean

Default
True

Allow the creation of PTR records

ipv4_ptr_zone_prefix_size

Type
unknown type

Default
24

Minimum Value
8

Maximum Value
24

Number of bits in an IPv4 PTR zone that will be considered network prefix. It has to align to byte boundary. Minimum value is 8. Maximum value is 24. As a consequence, range of values is 8, 16 and 24

`ipv6_ptr_zone_prefix_size`

Type
unknown type

Default
`120`

Minimum Value
4

Maximum Value
124

Number of bits in an IPv6 PTR zone that will be considered network prefix. It has to align to nyble boundary. Minimum value is 4. Maximum value is 124. As a consequence, range of values is 4, 8, 12, 16,, 124

`ptr_zone_email`

Type
string

Default
`''`

The email address to be used when creating PTR zones. If not specified, the email address will be admin@<dns_domain>

experimental

`ipv6_pd_enabled`

Type
boolean

Default
`False`

Enable execution of the experimental IPv6 Prefix Delegation functionality in the plugin.

healthcheck

`detailed`

Type
boolean

Default
`False`

Show more detailed information as part of the response. Security note: Enabling this option may expose sensitive details about the service being monitored. Be sure to verify that it will not violate your security policies.

`backends`

Type
list

Default
`[]`

Additional backends that can perform health checks and report that information back as part of a request.

`allowed_source_ranges`

Type
list

Default
`[]`

A list of network addresses to limit source ip allowed to access healthcheck information. Any request from ip outside of these network addresses are ignored.

`ignore_proxied_requests`

Type
boolean

Default
`False`

Ignore requests with proxy headers.

`disable_by_file_path`

Type
string

Default
`<None>`

Check the presence of a file to determine if an application is running on a port. Used by DisableByFileHealthcheck plugin.

`disable_by_file_paths`

Type
list

Default
`[]`

Check the presence of a file based on a port to determine if an application is running on a port. Expects a port:path list of strings. Used by DisableByFilesPortsHealthcheck plugin.

enable_by_file_paths

Type
list

Default
`[]`

Check the presence of files. Used by EnableByFilesHealthcheck plugin.

ironic

auth_url

Type
unknown type

Default
`<None>`

Authentication URL

auth_type

Type
unknown type

Default
`<None>`

Authentication type to load

Table 14: Deprecated Variations

Group	Name
ironic	auth_plugin

cafile

Type
string

Default
`<None>`

PEM encoded Certificate Authority to use when verifying HTTPs connections.

certfile

Type
string

Default
`<None>`

PEM encoded client certificate cert file `collect_timing`

Type
boolean

Default
`False`

Collect per-API call timing information.

`default_domain_id`

Type
unknown type

Default
`<None>`

Optional domain ID to use with v3 and v2 parameters. It will be used for both the user and project domain in v3 and ignored in v2 authentication.

`default_domain_name`

Type
unknown type

Default
`<None>`

Optional domain name to use with v3 API and v2 parameters. It will be used for both the user and project domain in v3 and ignored in v2 authentication.

`domain_id`

Type
unknown type

Default
`<None>`

Domain ID to scope to

`domain_name`

Type
unknown type

Default
`<None>`

Domain name to scope to

`insecure`

Type
boolean

Default
`False`

Verify HTTPS connections.

keyfile

Type
string

Default
<None>

PEM encoded client certificate key file

password

Type
unknown type

Default
<None>

Users password

project_domain_id

Type
unknown type

Default
<None>

Domain ID containing project

project_domain_name

Type
unknown type

Default
<None>

Domain name containing project

project_id

Type
unknown type

Default
<None>

Project ID to scope to

Table 15: Deprecated Variations

Group	Name
ironic	tenant-id
ironic	tenant_id

project_name

Type
unknown type

Default
`<None>`

Project name to scope to

Table 16: Deprecated Variations

Group	Name
ironic	tenant-name
ironic	tenant_name

`split_loggers`

Type
boolean

Default
`False`

Log requests to multiple loggers.

`system_scope`

Type
unknown type

Default
`<None>`

Scope for system operations

`tenant_id`

Type
unknown type

Default
`<None>`

Tenant ID

`tenant_name`

Type
unknown type

Default
`<None>`

Tenant Name

`timeout`

Type
integer

Default
`<None>`

Timeout value for http requests

`trust_id`

Type
unknown type

Default
<None>

ID of the trust to use as a trustee use

`user_domain_id`

Type
unknown type

Default
<None>

Users domain id

`user_domain_name`

Type
unknown type

Default
<None>

Users domain name

`user_id`

Type
unknown type

Default
<None>

User id

`username`

Type
unknown type

Default
<None>

Username

Table 17: Deprecated Variations

Group	Name
ironic	user-name
ironic	user_name

`enable_notifications`

Type
boolean

Default
`False`

Send notification events to Ironic. (For example on relevant port status changes.)

keystone_authtoken

`www_authenticate_uri`

Type
string

Default
`<None>`

Complete public Identity API endpoint. This endpoint should not be an admin endpoint, as it should be accessible by all end users. Unauthenticated clients are redirected to this endpoint to authenticate. Although this endpoint should ideally be unversioned, client support in the wild varies. If youre using a versioned v2 endpoint here, then this should *not* be the same endpoint the service user utilizes for validating tokens, because normal end users may not be able to reach that endpoint.

Table 18: Deprecated Variations

Group	Name
keystone_authtoken	auth_uri

`auth_uri`

Type
string

Default
`<None>`

Complete public Identity API endpoint. This endpoint should not be an admin endpoint, as it should be accessible by all end users. Unauthenticated clients are redirected to this endpoint to authenticate. Although this endpoint should ideally be unversioned, client support in the wild varies. If youre using a versioned v2 endpoint here, then this should *not* be the same endpoint the service user utilizes for validating tokens, because normal end users may not be able to reach that endpoint. This option is deprecated in favor of www_authenticate_uri and will be removed in the S release.

> **Warning**
>
> This option is deprecated for removal since Queens. Its value may be silently ignored in the future.
>
> **Reason**
> The auth_uri option is deprecated in favor of www_authenticate_uri and will be removed in the S release.

`auth_version`

Type
string

Default
`<None>`

API version of the Identity API endpoint.

`interface`

Type
string

Default
`internal`

Interface to use for the Identity API endpoint. Valid values are public, internal (default) or admin.

`delay_auth_decision`

Type
boolean

Default
`False`

Do not handle authorization requests within the middleware, but delegate the authorization decision to downstream WSGI components.

`http_connect_timeout`

Type
integer

Default
`<None>`

Request timeout value for communicating with Identity API server.

`http_request_max_retries`

Type
integer

Default
`3`

How many times are we trying to reconnect when communicating with Identity API Server.

`cache`

Type
string

Default
`<None>`

Request environment key where the Swift cache object is stored. When auth_token middleware is deployed with a Swift cache, use this option to have the middleware share a caching backend with swift. Otherwise, use the `memcached_servers` option instead.

certfile

Type
string

Default
`<None>`

Required if identity server requires client certificate

keyfile

Type
string

Default
`<None>`

Required if identity server requires client certificate

cafile

Type
string

Default
`<None>`

A PEM encoded Certificate Authority to use when verifying HTTPs connections. Defaults to system CAs.

insecure

Type
boolean

Default
`False`

Verify HTTPS connections.

region_name

Type
string

Default
`<None>`

The region in which the identity server can be found.

memcached_servers

Type
list

Default
`<None>`

Optionally specify a list of memcached server(s) to use for caching. If left undefined, tokens will instead be cached in-process.

Table 19: Deprecated Variations

Group	Name
keystone_authtoken	memcache_servers

`token_cache_time`

Type
integer

Default
`300`

In order to prevent excessive effort spent validating tokens, the middleware caches previously-seen tokens for a configurable duration (in seconds). Set to -1 to disable caching completely.

`memcache_security_strategy`

Type
string

Default
`None`

Valid Values
None, MAC, ENCRYPT

(Optional) If defined, indicate whether token data should be authenticated or authenticated and encrypted. If MAC, token data is authenticated (with HMAC) in the cache. If ENCRYPT, token data is encrypted and authenticated in the cache. If the value is not one of these options or empty, auth_token will raise an exception on initialization.

`memcache_secret_key`

Type
string

Default
`<None>`

(Optional, mandatory if memcache_security_strategy is defined) This string is used for key derivation.

`memcache_tls_enabled`

Type
boolean

Default
`False`

(Optional) Global toggle for TLS usage when comunicating with the caching servers.

`memcache_tls_cafile`

Type
string

Default
<None>

(Optional) Path to a file of concatenated CA certificates in PEM format necessary to establish the caching servers authenticity. If tls_enabled is False, this option is ignored.

`memcache_tls_certfile`

Type
string

Default
<None>

(Optional) Path to a single file in PEM format containing the clients certificate as well as any number of CA certificates needed to establish the certificates authenticity. This file is only required when client side authentication is necessary. If tls_enabled is False, this option is ignored.

`memcache_tls_keyfile`

Type
string

Default
<None>

(Optional) Path to a single file containing the clients private key in. Otherwhise the private key will be taken from the file specified in tls_certfile. If tls_enabled is False, this option is ignored.

`memcache_tls_allowed_ciphers`

Type
string

Default
<None>

(Optional) Set the available ciphers for sockets created with the TLS context. It should be a string in the OpenSSL cipher list format. If not specified, all OpenSSL enabled ciphers will be available.

`memcache_pool_dead_retry`

Type
integer

Default
300

(Optional) Number of seconds memcached server is considered dead before it is tried again.

`memcache_pool_maxsize`

Type
integer

Default
10

(Optional) Maximum total number of open connections to every memcached server.

`memcache_pool_socket_timeout`

Type
integer

Default
`3`

(Optional) Socket timeout in seconds for communicating with a memcached server.

`memcache_pool_unused_timeout`

Type
integer

Default
`60`

(Optional) Number of seconds a connection to memcached is held unused in the pool before it is closed.

`memcache_pool_conn_get_timeout`

Type
integer

Default
`10`

(Optional) Number of seconds that an operation will wait to get a memcached client connection from the pool.

`memcache_use_advanced_pool`

Type
boolean

Default
`True`

(Optional) Use the advanced (eventlet safe) memcached client pool.

`include_service_catalog`

Type
boolean

Default
`True`

(Optional) Indicate whether to set the X-Service-Catalog header. If False, middleware will not ask for service catalog on token validation and will not set the X-Service-Catalog header.

`enforce_token_bind`

Type
string

Default
`permissive`

Used to control the use and type of token binding. Can be set to: disabled to not check token binding. permissive (default) to validate binding information if the bind type is of a form known to the server and ignore it if not. strict like permissive but if the bind type is unknown the token will be rejected. required any form of token binding is needed to be allowed. Finally the name of a binding method that must be present in tokens.

`service_token_roles`

Type
list

Default
`['service']`

A choice of roles that must be present in a service token. Service tokens are allowed to request that an expired token can be used and so this check should tightly control that only actual services should be sending this token. Roles here are applied as an ANY check so any role in this list must be present. For backwards compatibility reasons this currently only affects the allow_expired check.

`service_token_roles_required`

Type
boolean

Default
`False`

For backwards compatibility reasons we must let valid service tokens pass that dont pass the service_token_roles check as valid. Setting this true will become the default in a future release and should be enabled if possible.

`service_type`

Type
string

Default
`<None>`

The name or type of the service as it appears in the service catalog. This is used to validate tokens that have restricted access rules.

`memcache_sasl_enabled`

Type
boolean

Default
`False`

Enable the SASL(Simple Authentication and Security Layer) if the SASL_enable is true, else disable.

`memcache_username`

Type
string

Default
''

the user name for the SASL

`memcache_password`

Type
string

Default
''

the username password for SASL

`auth_type`

Type
unknown type

Default
<None>

Authentication type to load

Table 20: Deprecated Variations

Group	Name
keystone_authtoken	auth_plugin

`auth_section`

Type
unknown type

Default
<None>

Config Section from which to load plugin specific options

nova

`region_name`

Type
string

Default
<None>

Name of Nova region to use. Useful if Keystone manages more than one region.

`endpoint_type`

Type
string

Default
`public`

Valid Values
public, admin, internal

Type of the Nova endpoint to use. This endpoint will be looked up in the Keystone catalog and should be one of public, internal or admin.

auth_url

Type
unknown type

Default
`<None>`

Authentication URL

auth_type

Type
unknown type

Default
`<None>`

Authentication type to load

Table 21: Deprecated Variations

Group	Name
nova	auth_plugin

cafile

Type
string

Default
`<None>`

PEM encoded Certificate Authority to use when verifying HTTPs connections.

certfile

Type
string

Default
`<None>`

PEM encoded client certificate cert file

collect_timing

Type
boolean

Default
`False`

Collect per-API call timing information.

`default_domain_id`

Type
unknown type

Default
`<None>`

Optional domain ID to use with v3 and v2 parameters. It will be used for both the user and project domain in v3 and ignored in v2 authentication.

`default_domain_name`

Type
unknown type

Default
`<None>`

Optional domain name to use with v3 API and v2 parameters. It will be used for both the user and project domain in v3 and ignored in v2 authentication.

`domain_id`

Type
unknown type

Default
`<None>`

Domain ID to scope to

`domain_name`

Type
unknown type

Default
`<None>`

Domain name to scope to

`insecure`

Type
boolean

Default
`False`

Verify HTTPS connections.

`keyfile`

Type
string

Default
`<None>`

PEM encoded client certificate key file

`password`

Type
unknown type

Default
`<None>`

Users password

`project_domain_id`

Type
unknown type

Default
`<None>`

Domain ID containing project

`project_domain_name`

Type
unknown type

Default
`<None>`

Domain name containing project

`project_id`

Type
unknown type

Default
`<None>`

Project ID to scope to

Table 22: Deprecated Variations

Group	Name
nova	tenant-id
nova	tenant_id

`project_name`

Type
unknown type

Default
`<None>`

Project name to scope to

Table 23: Deprecated Variations

Group	Name
nova	tenant-name
nova	tenant_name

split_loggers

Type
boolean

Default
`False`

Log requests to multiple loggers.

system_scope

Type
unknown type

Default
`<None>`

Scope for system operations

tenant_id

Type
unknown type

Default
`<None>`

Tenant ID

tenant_name

Type
unknown type

Default
`<None>`

Tenant Name

timeout

Type
integer

Default
`<None>`

Timeout value for http requests

trust_id

Type
unknown type

Default
<None>

ID of the trust to use as a trustee use

user_domain_id

Type
unknown type

Default
<None>

Users domain id

user_domain_name

Type
unknown type

Default
<None>

Users domain name

user_id

Type
unknown type

Default
<None>

User id

username

Type
unknown type

Default
<None>

Username

Table 24: Deprecated Variations

Group	Name
nova	user-name
nova	user_name

oslo_concurrency

disable_process_locking

Type
boolean

Default
False

Enables or disables inter-process locks.

lock_path

Type
string

Default
<None>

Directory to use for lock files. For security, the specified directory should only be writable by the user running the processes that need locking. Defaults to environment variable OSLO_LOCK_PATH. If external locks are used, a lock path must be set.

oslo_messaging_kafka

kafka_max_fetch_bytes

Type
integer

Default
1048576

Max fetch bytes of Kafka consumer

kafka_consumer_timeout

Type
floating point

Default
1.0

Default timeout(s) for Kafka consumers

consumer_group

Type
string

Default
oslo_messaging_consumer

Group id for Kafka consumer. Consumers in one group will coordinate message consumption

producer_batch_timeout

Type
floating point

Default
0.0

Upper bound on the delay for KafkaProducer batching in seconds

producer_batch_size

Type
integer

Default
`16384`

Size of batch for the producer async send

`compression_codec`

Type
string

Default
`none`

Valid Values
none, gzip, snappy, lz4, zstd

The compression codec for all data generated by the producer. If not set, compression will not be used. Note that the allowed values of this depend on the kafka version

`enable_auto_commit`

Type
boolean

Default
`False`

Enable asynchronous consumer commits

`max_poll_records`

Type
integer

Default
`500`

The maximum number of records returned in a poll call

`security_protocol`

Type
string

Default
`PLAINTEXT`

Valid Values
PLAINTEXT, SASL_PLAINTEXT, SSL, SASL_SSL

Protocol used to communicate with brokers

`sasl_mechanism`

Type
string

Default
`PLAIN`

Mechanism when security protocol is SASL `ssl_cafile`

Type
string

Default
`''`

CA certificate PEM file used to verify the server certificate

`ssl_client_cert_file`

Type
string

Default
`''`

Client certificate PEM file used for authentication.

`ssl_client_key_file`

Type
string

Default
`''`

Client key PEM file used for authentication.

`ssl_client_key_password`

Type
string

Default
`''`

Client key password file used for authentication.

oslo_messaging_notifications

`driver`

Type
multi-valued

Default
`''`

The Drivers(s) to handle sending notifications. Possible values are messaging, messagingv2, routing, log, test, noop

`transport_url`

Type
string

Default
`<None>`

A URL representing the messaging driver to use for notifications. If not set, we fall back to the same configuration used for RPC.

`topics`

Type
list

Default
`['notifications']`

AMQP topic used for OpenStack notifications.

`retry`

Type
integer

Default
`-1`

The maximum number of attempts to re-send a notification message which failed to be delivered due to a recoverable error. 0 - No retry, -1 - indefinite

oslo_messaging_rabbit

`amqp_durable_queues`

Type
boolean

Default
`False`

Use durable queues in AMQP. If rabbit_quorum_queue is enabled, queues will be durable and this value will be ignored.

`amqp_auto_delete`

Type
boolean

Default
`False`

Auto-delete queues in AMQP.

`rpc_conn_pool_size`

Type
integer

Default
`30`

Minimum Value
1

Size of RPC connection pool.

`conn_pool_min_size`

Type
integer

Default
`2`

The pool size limit for connections expiration policy

`conn_pool_ttl`

Type
integer

Default
`1200`

The time-to-live in sec of idle connections in the pool

`ssl`

Type
boolean

Default
`False`

Connect over SSL.

`ssl_version`

Type
string

Default
`''`

SSL version to use (valid only if SSL enabled). Valid values are TLSv1 and SSLv23. SSLv2, SSLv3, TLSv1_1, and TLSv1_2 may be available on some distributions.

`ssl_key_file`

Type
string

Default
`''`

SSL key file (valid only if SSL enabled).

`ssl_cert_file`

Type
string

Default
`''`

SSL cert file (valid only if SSL enabled).

ssl_ca_file

Type
string

Default
`''`

SSL certification authority file (valid only if SSL enabled).

ssl_enforce_fips_mode

Type
boolean

Default
`False`

Global toggle for enforcing the OpenSSL FIPS mode. This feature requires Python support. This is available in Python 3.9 in all environments and may have been backported to older Python versions on select environments. If the Python executable used does not support OpenSSL FIPS mode, an exception will be raised.

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> FIPS_mode_set API was removed in OpenSSL 3.0.0. This option has no effect now.

heartbeat_in_pthread

Type
boolean

Default
`False`

(DEPRECATED) It is recommend not to use this option anymore. Run the health check heartbeat thread through a native python thread by default. If this option is equal to False then the health check heartbeat will inherit the execution model from the parent process. For example if the parent process has monkey patched the stdlib by using eventlet/greenlet then the heartbeat will be run through a green thread. This option should be set to True only for the wsgi services.

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> The option is related to Eventlet which will be removed. In addition this has never worked as expected with services using eventlet for core service framework.

`kombu_reconnect_delay`

Type
floating point

Default
`1.0`

Minimum Value
0.0

Maximum Value
4.5

How long to wait (in seconds) before reconnecting in response to an AMQP consumer cancel notification.

`kombu_reconnect_splay`

Type
floating point

Default
`0.0`

Minimum Value
0.0

Random time to wait for when reconnecting in response to an AMQP consumer cancel notification.

`kombu_compression`

Type
string

Default
`<None>`

EXPERIMENTAL: Possible values are: gzip, bz2. If not set compression will not be used. This option may not be available in future versions.

`kombu_missing_consumer_retry_timeout`

Type
integer

Default
`60`

How long to wait a missing client before abandoning to send it its replies. This value should not be longer than rpc_response_timeout.

Table 25: Deprecated Variations

Group	Name
oslo_messaging_rabbit	kombu_reconnect_timeout

`kombu_failover_strategy`

Type
string

Default
`round-robin`

Valid Values
round-robin, shuffle

Determines how the next RabbitMQ node is chosen in case the one we are currently connected to becomes unavailable. Takes effect only if more than one RabbitMQ node is provided in config.

`rabbit_login_method`

Type
string

Default
`AMQPLAIN`

Valid Values
PLAIN, AMQPLAIN, EXTERNAL, RABBIT-CR-DEMO

The RabbitMQ login method.

`rabbit_retry_interval`

Type
integer

Default
`1`

Minimum Value
1

How frequently to retry connecting with RabbitMQ.

`rabbit_retry_backoff`

Type
integer

Default
`2`

Minimum Value
0

How long to backoff for between retries when connecting to RabbitMQ.

`rabbit_interval_max`

Type
integer

Default
`30`

Minimum Value
1

Maximum interval of RabbitMQ connection retries.

`rabbit_ha_queues`

Type
boolean

Default
`False`

Try to use HA queues in RabbitMQ (x-ha-policy: all). If you change this option, you must wipe the RabbitMQ database. In RabbitMQ 3.0, queue mirroring is no longer controlled by the x-ha-policy argument when declaring a queue. If you just want to make sure that all queues (except those with auto-generated names) are mirrored across all nodes, run: rabbitmqctl set_policy HA ^(?!amq.).* {ha-mode: all}

`rabbit_quorum_queue`

Type
boolean

Default
`False`

Use quorum queues in RabbitMQ (x-queue-type: quorum). The quorum queue is a modern queue type for RabbitMQ implementing a durable, replicated FIFO queue based on the Raft consensus algorithm. It is available as of RabbitMQ 3.8.0. If set this option will conflict with the HA queues (`rabbit_ha_queues`) aka mirrored queues, in other words the HA queues should be disabled. Quorum queues are also durable by default so the amqp_durable_queues option is ignored when this option is enabled.

`rabbit_transient_quorum_queue`

Type
boolean

Default
`False`

Use quorum queues for transients queues in RabbitMQ. Enabling this option will then make sure those queues are also using quorum kind of rabbit queues, which are HA by default.

`rabbit_quorum_delivery_limit`

Type
integer

Default
`0`

Each time a message is redelivered to a consumer, a counter is incremented. Once the redelivery count exceeds the delivery limit the message gets dropped or dead-lettered (if a DLX exchange has been configured) Used only when rabbit_quorum_queue is enabled, Default 0 which means dont set a limit.

`rabbit_quorum_max_memory_length`

Type
integer

Default
`0`

By default all messages are maintained in memory if a quorum queue grows in length it can put memory pressure on a cluster. This option can limit the number of messages in the quorum queue. Used only when rabbit_quorum_queue is enabled, Default 0 which means dont set a limit.

`rabbit_quorum_max_memory_bytes`

Type
integer

Default
`0`

By default all messages are maintained in memory if a quorum queue grows in length it can put memory pressure on a cluster. This option can limit the number of memory bytes used by the quorum queue. Used only when rabbit_quorum_queue is enabled, Default 0 which means dont set a limit.

`rabbit_transient_queues_ttl`

Type
integer

Default
`1800`

Minimum Value
0

Positive integer representing duration in seconds for queue TTL (x-expires). Queues which are unused for the duration of the TTL are automatically deleted. The parameter affects only reply and fanout queues. Setting 0 as value will disable the x-expires. If doing so, make sure you have a rabbitmq policy to delete the queues or you deployment will create an infinite number of queue over time.In case rabbit_stream_fanout is set to True, this option will control data retention policy (x-max-age) for messages in the fanout queue rather then the queue duration itself. So the oldest data in the stream queue will be discarded from it once reaching TTL Setting to 0 will disable x-max-age for stream which make stream grow indefinitely filling up the diskspace

`rabbit_qos_prefetch_count`

Type
integer

Default
`0`

Specifies the number of messages to prefetch. Setting to zero allows unlimited messages.

`heartbeat_timeout_threshold`

Type
integer

Default
`60`

Number of seconds after which the Rabbit broker is considered down if heartbeats keep-alive fails (0 disables heartbeat).

`heartbeat_rate`

Type
integer

Default
`3`

How often times during the heartbeat_timeout_threshold we check the heartbeat.

`direct_mandatory_flag`

Type
boolean

Default
`True`

(DEPRECATED) Enable/Disable the RabbitMQ mandatory flag for direct send. The direct send is used as reply, so the MessageUndeliverable exception is raised in case the client queue does not exist.MessageUndeliverable exception will be used to loop for a timeout to lets a chance to sender to recover.This flag is deprecated and it will not be possible to deactivate this functionality anymore

Warning

This option is deprecated for removal. Its value may be silently ignored in the future.

Reason
Mandatory flag no longer deactivable.

`enable_cancel_on_failover`

Type
boolean

Default
`False`

Enable x-cancel-on-ha-failover flag so that rabbitmq server will cancel and notify consumerswhen queue is down

`use_queue_manager`

Type
boolean

Default
`False`

Should we use consistant queue names or random ones

hostname

Type
string

Default
`node1.example.com`

This option has a sample default set, which means that its actual default value may vary from the one documented above.

Hostname used by queue manager. Defaults to the value returned by socket.gethostname().

processname

Type
string

Default
`nova-api`

This option has a sample default set, which means that its actual default value may vary from the one documented above.

Process name used by queue manager

rabbit_stream_fanout

Type
boolean

Default
`False`

Use stream queues in RabbitMQ (x-queue-type: stream). Streams are a new persistent and replicated data structure (queue type) in RabbitMQ which models an append-only log with non-destructive consumer semantics. It is available as of RabbitMQ 3.9.0. If set this option will replace all fanout queues with only one stream queue.

oslo_middleware

enable_proxy_headers_parsing

Type
boolean

Default
`False`

Whether the application is behind a proxy or not. This determines if the middleware should parse the headers or not.

oslo_policy

enforce_scope

Type
boolean

Default
`True`

This option controls whether or not to enforce scope when evaluating policies. If `True`, the scope of the token used in the request is compared to the `scope_types` of the policy being enforced. If the scopes do not match, an `InvalidScope` exception will be raised. If `False`, a message will be logged informing operators that policies are being invoked with mismatching scope.

> **Warning**
>
> This option is deprecated for removal. Its value may be silently ignored in the future.
>
> **Reason**
> This configuration was added temporarily to facilitate a smooth transition to the new RBAC. OpenStack will always enforce scope checks. This configuration option is deprecated and will be removed in the 2025.2 cycle.

enforce_new_defaults

Type
boolean

Default
`True`

This option controls whether or not to use old deprecated defaults when evaluating policies. If `True`, the old deprecated defaults are not going to be evaluated. This means if any existing token is allowed for old defaults but is disallowed for new defaults, it will be disallowed. It is encouraged to enable this flag along with the `enforce_scope` flag so that you can get the benefits of new defaults and `scope_type` together. If `False`, the deprecated policy check string is logically ORd with the new policy check string, allowing for a graceful upgrade experience between releases with new policies, which is the default behavior.

policy_file

Type
string

Default
`policy.yaml`

The relative or absolute path of a file that maps roles to permissions for a given service. Relative paths must be specified in relation to the configuration file setting this option.

policy_default_rule

Type
string

Default
`default`

Default rule. Enforced when a requested rule is not found.

policy_dirs

Type
multi-valued

Default
`policy.d`

Directories where policy configuration files are stored. They can be relative to any directory in the search path defined by the config_dir option, or absolute paths. The file defined by policy_file must exist for these directories to be searched. Missing or empty directories are ignored.

`remote_content_type`

Type
string

Default
`application/x-www-form-urlencoded`

Valid Values
application/x-www-form-urlencoded, application/json

Content Type to send and receive data for REST based policy check

`remote_ssl_verify_server_crt`

Type
boolean

Default
`False`

server identity verification for REST based policy check

`remote_ssl_ca_crt_file`

Type
string

Default
`<None>`

Absolute path to ca cert file for REST based policy check

`remote_ssl_client_crt_file`

Type
string

Default
`<None>`

Absolute path to client cert for REST based policy check

`remote_ssl_client_key_file`

Type
string

Default
`<None>`

Absolute path client key file REST based policy check

remote_timeout

Type
floating point

Default
`60`

Minimum Value
0

Timeout in seconds for REST based policy check

oslo_reports

log_dir

Type
string

Default
`<None>`

Path to a log directory where to create a file

file_event_handler

Type
string

Default
`<None>`

The path to a file to watch for changes to trigger the reports, instead of signals. Setting this option disables the signal trigger for the reports. If application is running as a WSGI application it is recommended to use this instead of signals.

file_event_handler_interval

Type
integer

Default
`1`

How many seconds to wait between polls when file_event_handler is set

oslo_versionedobjects

fatal_exception_format_errors

Type
boolean

Default
`False`

Make exception message format errors fatal

placement

region_name

Type
string

Default
<None>

Name of placement region to use. Useful if Keystone manages more than one region.

endpoint_type

Type
string

Default
public

Valid Values
public, admin, internal

Type of the placement endpoint to use. This endpoint will be looked up in the Keystone catalog and should be one of public, internal or admin.

auth_url

Type
unknown type

Default
<None>

Authentication URL

auth_type

Type
unknown type

Default
<None>

Authentication type to load

Table 26: Deprecated Variations

Group	Name
placement	auth_plugin

cafile

Type
string

Default
<None>

PEM encoded Certificate Authority to use when verifying HTTPs connections.

`certfile`

Type
string

Default
`<None>`

PEM encoded client certificate cert file

`collect_timing`

Type
boolean

Default
`False`

Collect per-API call timing information.

`default_domain_id`

Type
unknown type

Default
`<None>`

Optional domain ID to use with v3 and v2 parameters. It will be used for both the user and project domain in v3 and ignored in v2 authentication.

`default_domain_name`

Type
unknown type

Default
`<None>`

Optional domain name to use with v3 API and v2 parameters. It will be used for both the user and project domain in v3 and ignored in v2 authentication.

`domain_id`

Type
unknown type

Default
`<None>`

Domain ID to scope to

`domain_name`

Type
unknown type

Default
`<None>`

Domain name to scope to

`insecure`

Type
boolean

Default
`False`

Verify HTTPS connections.

`keyfile`

Type
string

Default
`<None>`

PEM encoded client certificate key file

`password`

Type
unknown type

Default
`<None>`

Users password

`project_domain_id`

Type
unknown type

Default
`<None>`

Domain ID containing project

`project_domain_name`

Type
unknown type

Default
`<None>`

Domain name containing project

`project_id`

Type
unknown type

Default
`<None>`

Project ID to scope to

Table 27: Deprecated Variations

Group	Name
placement	tenant-id
placement	tenant_id

project_name

Type
unknown type

Default
`<None>`

Project name to scope to

Table 28: Deprecated Variations

Group	Name
placement	tenant-name
placement	tenant_name

split_loggers

Type
boolean

Default
`False`

Log requests to multiple loggers.

system_scope

Type
unknown type

Default
`<None>`

Scope for system operations

tenant_id

Type
unknown type

Default
`<None>`

Tenant ID

tenant_name

Type
unknown type

Default
`<None>`

Tenant Name

`timeout`

Type
integer

Default
`<None>`

Timeout value for http requests

`trust_id`

Type
unknown type

Default
`<None>`

ID of the trust to use as a trustee use

`user_domain_id`

Type
unknown type

Default
`<None>`

Users domain id

`user_domain_name`

Type
unknown type

Default
`<None>`

Users domain name

`user_id`

Type
unknown type

Default
`<None>`

User id

`username`

Type
unknown type

Default
`<None>`

Username

Table 29: Deprecated Variations

Group	Name
placement	user-name
placement	user_name

privsep

Configuration options for the oslo.privsep daemon. Note that this group name can be changed by the consuming service. Check the services docs to see if this is the case.

`user`

Type
string

Default
`<None>`

User that the privsep daemon should run as.

`group`

Type
string

Default
`<None>`

Group that the privsep daemon should run as.

`capabilities`

Type
unknown type

Default
`[]`

List of Linux capabilities retained by the privsep daemon.

`thread_pool_size`

Type
integer

Default
`multiprocessing.cpu_count()`

Minimum Value
1

This option has a sample default set, which means that its actual default value may vary from the one documented above.

The number of threads available for privsep to concurrently run processes. Defaults to the number of CPU cores in the system.

`helper_command`

Type
string

Default
`<None>`

Command to invoke to start the privsep daemon if not using the fork method. If not specified, a default is generated using sudo privsep-helper and arguments designed to recreate the current configuration. This command must accept suitable privsep_context and privsep_sock_path arguments.

`logger_name`

Type
string

Default
`oslo_privsep.daemon`

Logger name to use for this privsep context. By default all contexts log with oslo_privsep.daemon.

`log_daemon_traceback`

Type
boolean

Default
`False`

Print the exception traceback happened in the daemon in the client logger

profiler

`enabled`

Type
boolean

Default
`False`

Enable the profiling for all services on this node.

Default value is False (fully disable the profiling feature).

Possible values:

- True: Enables the feature
- False: Disables the feature. The profiling cannot be started via this project operations. If the profiling is triggered by another project, this project part will be empty.

Table 30: Deprecated Variations

Group	Name
profiler	profiler_enabled

`trace_sqlalchemy`

Type
boolean

Default
`False`

Enable SQL requests profiling in services.

Default value is False (SQL requests wont be traced).

Possible values:

- True: Enables SQL requests profiling. Each SQL query will be part of the trace and can the be analyzed by how much time was spent for that.
- False: Disables SQL requests profiling. The spent time is only shown on a higher level of operations. Single SQL queries cannot be analyzed this way.

`trace_requests`

Type
boolean

Default
`False`

Enable python requests package profiling.

Supported drivers: jaeger+otlp

Default value is False.

Possible values:

- True: Enables requests profiling.
- False: Disables requests profiling.

`hmac_keys`

Type
string

Default
`SECRET_KEY`

Secret key(s) to use for encrypting context data for performance profiling.

This string value should have the following format: <key1>[,<key2>,<keyn>], where each key is some random string. A user who triggers the profiling via the REST API has to set one of these keys in the headers of the REST API call to include profiling results of this node for this particular project.

Both enabled flag and hmac_keys config options should be set to enable profiling. Also, to generate correct profiling information across all services at least one key needs to be consistent between OpenStack projects. This ensures it can be used from client side to generate the trace, containing information from all possible resources.

connection_string

Type
string

Default
`messaging://`

Connection string for a notifier backend.

Default value is `messaging://` which sets the notifier to oslo_messaging.

Examples of possible values:

- `messaging://` - use oslo_messaging driver for sending spans.
- `redis://127.0.0.1:6379` - use redis driver for sending spans.
- `mongodb://127.0.0.1:27017` - use mongodb driver for sending spans.
- `elasticsearch://127.0.0.1:9200` - use elasticsearch driver for sending spans.
- `jaeger://127.0.0.1:6831` - use jaeger tracing as driver for sending spans.

es_doc_type

Type
string

Default
`notification`

Document type for notification indexing in elasticsearch.

es_scroll_time

Type
string

Default
`2m`

This parameter is a time value parameter (for example: es_scroll_time=2m), indicating for how long the nodes that participate in the search will maintain relevant resources in order to continue and support it.

es_scroll_size

Type
integer

Default
`10000`

Elasticsearch splits large requests in batches. This parameter defines maximum size of each batch (for example: es_scroll_size=10000).

socket_timeout

Type
floating point

Default
`0.1`

Redissentinel provides a timeout option on the connections. This parameter defines that timeout (for example: socket_timeout=0.1).

`sentinel_service_name`

Type
string

Default
`mymaster`

Redissentinel uses a service name to identify a master redis service. This parameter defines the name (for example: `sentinal_service_name=mymaster`).

`filter_error_trace`

Type
boolean

Default
`False`

Enable filter traces that contain error/exception to a separated place.

Default value is set to False.

Possible values:

- True: Enable filter traces that contain error/exception.
- False: Disable the filter.

profiler_jaeger

`service_name_prefix`

Type
string

Default
`<None>`

Set service name prefix to Jaeger service name.

`process_tags`

Type
dict

Default
`{}`

Set process tracer tags.

profiler_otlp

service_name_prefix

Type
string

Default
`<None>`

Set service name prefix to OTLP exporters.

quotas

default_quota

Type
integer

Default
`-1`

Default number of resources allowed per project. A negative value means unlimited.

quota_network

Type
integer

Default
`100`

Number of networks allowed per project. A negative value means unlimited.

quota_subnet

Type
integer

Default
`100`

Number of subnets allowed per project, A negative value means unlimited.

quota_port

Type
integer

Default
`500`

Number of ports allowed per project. A negative value means unlimited.

quota_driver

Type
string

Default
`neutron.db.quota.driver_nolock.DbQuotaNoLockDriver`

Default driver to use for quota checks.

track_quota_usage

Type
boolean

Default
`True`

When set to True, quota usage will be tracked in the Neutron database for each resource, by directly mapping to a data model class, for example, networks, subnets, ports, etc. When set to False, quota usage will be tracked by the quota engine as a count of the object type directly. For more information, see the Quota Management and Enforcement guide.

quota_router

Type
integer

Default
`10`

Number of routers allowed per project. A negative value means unlimited.

quota_floatingip

Type
integer

Default
`50`

Number of floating IPs allowed per project. A negative value means unlimited.

quota_security_group

Type
integer

Default
`10`

Number of security groups allowed per project. A negative value means unlimited.

quota_security_group_rule

Type
integer

Default
`100`

Number of security group rules allowed per project. A negative value means unlimited.

service_providers

service_provider

Type
multi-valued

Default
''

Defines providers for advanced services using the format: <service_type>:<name>:<driver>[:default]

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DEFAULT

debug

Type
boolean

Default
`False`

Mutable
This option can be changed without restarting.

If set to true, the logging level will be set to DEBUG instead of the default INFO level.

log_config_append

Type
string

Default
`<None>`

Mutable
This option can be changed without restarting.

The name of a logging configuration file. This file is appended to any existing logging configuration files. For details about logging configuration files, see the Python logging module documentation. Note that when logging configuration files are used then all logging configuration is set in the configuration file and other logging configuration options are ignored (for example, log-date-format).

Table 31: Deprecated Variations

Group	Name
DEFAULT	log-config
DEFAULT	log_config

log_date_format

Type
string

Default
`%Y-%m-%d %H:%M:%S`

Defines the format string for %(asctime)s in log records. Default: the value above . This option is ignored if log_config_append is set.

`log_file`

Type
string

Default
`<None>`

(Optional) Name of log file to send logging output to. If no default is set, logging will go to stderr as defined by use_stderr. This option is ignored if log_config_append is set.

Table 32: Deprecated Variations

Group	Name
DEFAULT	logfile

`log_dir`

Type
string

Default
`<None>`

(Optional) The base directory used for relative log_file paths. This option is ignored if log_config_append is set.

Table 33: Deprecated Variations

Group	Name
DEFAULT	logdir

`use_syslog`

Type
boolean

Default
`False`

Use syslog for logging. Existing syslog format is DEPRECATED and will be changed later to honor RFC5424. This option is ignored if log_config_append is set.

`use_journal`

Type
boolean

Default
`False`

Enable journald for logging. If running in a systemd environment you may wish to enable journal support. Doing so will use the journal native protocol which includes structured metadata in addition to log messages.This option is ignored if log_config_append is set.

`syslog_log_facility`

Type
string

Default
`LOG_USER`

Syslog facility to receive log lines. This option is ignored if log_config_append is set.

`use_json`

Type
boolean

Default
`False`

Use JSON formatting for logging. This option is ignored if log_config_append is set.

`use_stderr`

Type
boolean

Default
`False`

Log output to standard error. This option is ignored if log_config_append is set.

`log_color`

Type
boolean

Default
`False`

(Optional) Set the color key according to log levels. This option takes effect only when logging to stderr or stdout is used. This option is ignored if log_config_append is set.

`log_rotate_interval`

Type
integer

Default
`1`

The amount of time before the log files are rotated. This option is ignored unless log_rotation_type is set to interval.

`log_rotate_interval_type`

Type
string

Default
`days`

Valid Values
Seconds, Minutes, Hours, Days, Weekday, Midnight

Rotation interval type. The time of the last file change (or the time when the service was started) is used when scheduling the next rotation.

`max_logfile_count`

Type
integer

Default
`30`

Maximum number of rotated log files.

`max_logfile_size_mb`

Type
integer

Default
`200`

Log file maximum size in MB. This option is ignored if log_rotation_type is not set to size.

`log_rotation_type`

Type
string

Default
`none`

Valid Values
interval, size, none

Log rotation type.

Possible values

interval
Rotate logs at predefined time intervals.

size
Rotate logs once they reach a predefined size.

none
Do not rotate log files.

`logging_context_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [%(global_request_id)s %(request_id)s %(user_identity)s] %(instance)s%(message)s`

Format string to use for log messages with context. Used by oslo_log.formatters.ContextFormatter `logging_default_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [-] %(instance)s%(message)s`

Format string to use for log messages when context is undefined. Used by oslo_log.formatters.ContextFormatter

`logging_debug_format_suffix`

Type
string

Default
`%(funcName)s %(pathname)s:%(lineno)d`

Additional data to append to log message when logging level for the message is DEBUG. Used by oslo_log.formatters.ContextFormatter

`logging_exception_prefix`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d ERROR %(name)s %(instance)s`

Prefix each line of exception output with this format. Used by oslo_log.formatters.ContextFormatter

`logging_user_identity_format`

Type
string

Default
`%(user)s %(project)s %(domain)s %(system_scope)s %(user_domain)s %(project_domain)s`

Defines the format string for %(user_identity)s that is used in logging_context_format_string. Used by oslo_log.formatters.ContextFormatter

`default_log_levels`

Type
list

Default
`['amqp=WARN', 'boto=WARN', 'sqlalchemy=WARN', 'suds=INFO', 'oslo.messaging=INFO', 'oslo_messaging=INFO', 'iso8601=WARN', 'requests.packages.urllib3.connectionpool=WARN', 'urllib3.connectionpool=WARN', 'websocket=WARN', 'requests.packages.urllib3.util.retry=WARN', 'urllib3.util.retry=WARN', 'keystonemiddleware=WARN', 'routes.middleware=WARN',`

'stevedore=WARN', 'taskflow=WARN', 'keystoneauth=WARN', 'oslo.cache=INFO', 'oslo_policy=INFO', 'dogpile.core.dogpile=INFO']

List of package logging levels in logger=LEVEL pairs. This option is ignored if log_config_append is set.

`publish_errors`

Type
boolean

Default
`False`

Enables or disables publication of error events.

`instance_format`

Type
string

Default
`"[instance: %(uuid)s] "`

The format for an instance that is passed with the log message.

`instance_uuid_format`

Type
string

Default
`"[instance: %(uuid)s] "`

The format for an instance UUID that is passed with the log message.

`rate_limit_interval`

Type
integer

Default
`0`

Interval, number of seconds, of log rate limiting.

`rate_limit_burst`

Type
integer

Default
`0`

Maximum number of logged messages per rate_limit_interval.

`rate_limit_except_level`

Type
string

Default
`CRITICAL`

Valid Values
CRITICAL, ERROR, INFO, WARNING, DEBUG,

Log level name used by rate limiting. Logs with level greater or equal to rate_limit_except_level are not filtered. An empty string means that all levels are filtered.

`fatal_deprecations`

Type
boolean

Default
`False`

Enables or disables fatal status of deprecations.

ml2

`type_drivers`

Type
list

Default
`['local', 'flat', 'vlan', 'gre', 'vxlan', 'geneve']`

List of network type driver entrypoints to be loaded from the neutron.ml2.type_drivers namespace.

`project_network_types`

Type
list

Default
`['local']`

Ordered list of network_types to allocate as project networks. The default value local is useful for single-box testing but provides no connectivity between hosts.

Table 34: Deprecated Variations

Group	Name
ml2	tenant_network_types

`mechanism_drivers`

Type
list

Default
`[]`

An ordered list of networking mechanism driver entrypoints to be loaded from the neutron.ml2.mechanism_drivers namespace.

`extension_drivers`

Type
list

Default
`[]`

An ordered list of extension driver entrypoints to be loaded from the neutron.ml2.extension_drivers namespace. For example: extension_drivers = port_security,qos

`path_mtu`

Type
integer

Default
`0`

Maximum size of an IP packet (MTU) that can traverse the underlying physical network infrastructure without fragmentation when using an overlay/tunnel protocol. This option allows specifying a physical network MTU value that differs from the default global_physnet_mtu value.

`physical_network_mtus`

Type
dict

Default
`{}`

Mappings of physical networks to MTU values. The format of the mapping is <physnet>:<mtu val>. This mapping allows specifying a physical network MTU value that differs from the default global_physnet_mtu value.

`external_network_type`

Type
string

Default
`<None>`

Default network type for external networks when no provider attributes are specified. By default it is None, which means that if provider attributes are not specified while creating external networks then they will have the same type as project networks. Allowed values for external_network_type config option depend on the network type values configured in type_drivers config option.

`overlay_ip_version`

Type
integer

Default
4

Valid Values
4, 6

IP version of all overlay (tunnel) network endpoints.

Possible values

4
IPv4

6
IPv6

tunnelled_network_rp_name

Type
string

Default
`rp_tunnelled`

Resource provider name for the host with tunnelled networks. This resource provider represents the available bandwidth for all tunnelled networks in a compute node. NOTE: this parameter is used both by the Neutron server and the mechanism driver agents; it is recommended not to change it once any resource provider register has been created.

ml2_type_flat

flat_networks

Type
list

Default
`*`

List of physical_network names with which flat networks can be created. Use default * to allow flat networks with arbitrary physical_network names. Use an empty list to disable flat networks.

ml2_type_geneve

vni_ranges

Type
list

Default
`[]`

Comma-separated list of <vni_min>:<vni_max> tuples enumerating ranges of Geneve VNI IDs that are available for project network allocation. Note OVN does not use the actual values.

max_header_size

Type
integer

Default
`30`

The maximum allowed Geneve encapsulation header size (in bytes). Geneve header is extensible, this value is used to calculate the maximum MTU for Geneve-based networks. The default is 30, which is the size of the Geneve header without any additional option headers. Note the default is not enough for OVN which requires at least 38.

ml2_type_gre

tunnel_id_ranges

Type
list

Default
`[]`

Comma-separated list of <tun_min>:<tun_max> tuples enumerating ranges of GRE tunnel IDs that are available for project network allocation

ml2_type_vlan

network_vlan_ranges

Type
list

Default
`[]`

List of <physical_network>:<vlan_min>:<vlan_max> or <physical_network> specifying physical_network names usable for VLAN provider and project networks, as well as ranges of VLAN tags on each available for allocation to project networks. If no range is defined, the whole valid VLAN ID set [1, 4094] will be assigned.

ml2_type_vxlan

vni_ranges

Type
list

Default
`[]`

Comma-separated list of <vni_min>:<vni_max> tuples enumerating ranges of VXLAN VNI IDs that are available for project network allocation

vxlan_group

Type
string

Default
`<None>`

Multicast group for VXLAN. When configured, will enable sending all broadcast traffic to this multicast group. When left unconfigured, will disable multicast VXLAN mode.

ovn

ovn_nb_connection

Type
list

Default
`['tcp:127.0.0.1:6641']`

The connection string for the OVN_Northbound OVSDB. Use tcp:IP:PORT for TCP connection. Use ssl:IP:PORT for SSL connection. The ovn_nb_private_key, ovn_nb_certificate and ovn_nb_ca_cert are mandatory. Use unix:FILE for unix domain socket connection. Multiple connections can be specified by a comma separated string. See also: https://github.com/openvswitch/ovs/blob/ab4d3bfbef37c31331db5a9dbe7c22eb8d5e5e5f/python/ovs/db/idl.py#L2
L216

`ovn_nb_private_key`

Type
string

Default
`''`

The PEM file with private key for SSL connection to OVN-NB-DB

`ovn_nb_certificate`

Type
string

Default
`''`

The PEM file with certificate that certifies the private key specified in ovn_nb_private_key

`ovn_nb_ca_cert`

Type
string

Default
`''`

The PEM file with CA certificate that OVN should use to verify certificates presented to it by SSL peers

`ovn_sb_connection`

Type
list

Default
`['tcp:127.0.0.1:6642']`

The connection string for the OVN_Southbound OVSDB. Use tcp:IP:PORT for TCP connection. Use ssl:IP:PORT for SSL connection. The ovn_sb_private_key, ovn_sb_certificate and ovn_sb_ca_cert are mandatory. Use unix:FILE for unix domain socket connection. Multiple connections can be specified by a comma separated string. See also: https://github.com/openvswitch/ovs/blob/ab4d3bfbef37c31331db5a9dbe7c22eb8d5e5e5f/python/ovs/db/idl.py#L2
L216

`ovn_sb_private_key`

Type
string

Default
`''`

The PEM file with private key for SSL connection to OVN-SB-DB

`ovn_sb_certificate`

Type
string

Default
`''`

The PEM file with certificate that certifies the private key specified in ovn_sb_private_key

`ovn_sb_ca_cert`

Type
string

Default
`''`

The PEM file with CA certificate that OVN should use to verify certificates presented to it by SSL peers

`ovsdb_connection_timeout`

Type
integer

Default
`180`

Timeout, in seconds, for the OVSDB connection transaction

`ovsdb_retry_max_interval`

Type
integer

Default
`180`

Max interval, in seconds ,between each retry to get the OVN NB and SB IDLs

`ovsdb_probe_interval`

Type
integer

Default
`60000`

Minimum Value
0

The probe interval for the OVSDB session, in milliseconds. If this is zero, it disables the connection keepalive feature. If non-zero the value will be forced to at least 1000 milliseconds.

`neutron_sync_mode`

Type
string

Default
`log`

Valid Values
off, log, repair, migrate

The synchronization mode of OVN_Northbound OVSDB with Neutron DB.

Possible values

off
Synchronization is off.

log
During neutron-server startup, check to see if OVN is in sync with the Neutron database. Log warnings for any inconsistencies found so that an admin can investigate.

repair
During neutron-server startup, automatically create resources found in Neutron but not in OVN. Also remove resources from OVN that are no longer found in Neutron.

migrate
This mode is to OVS to OVN migration. It will sync the DB just like repair mode but it will additionally fix the Neutron DB resource from OVS to OVN.

`ovn_l3_scheduler`

Type
string

Default
`leastloaded`

Valid Values
leastloaded, chance

The OVN L3 Scheduler type used to schedule router gateway ports on hypervisors/chassis.

Possible values

leastloaded
Select chassis with fewest gateway ports.

chance
Select chassis randomly.

`enable_distributed_floating_ip`

Type
boolean

Default
`False`

Enable distributed floating IP support. If True, the NAT action for floating IPs will be done locally and not in the centralized gateway. This saves the path to the external network. This requires the user to configure the physical network map (i.e. ovn-bridge-mappings) on each compute node.

`vhost_sock_dir`

Type
string

Default
`/var/run/openvswitch`

The directory in which vhost virtio sockets are created by all the vswitch daemons

`dhcp_default_lease_time`

Type
integer

Default
`43200`

Default lease time (in seconds) to use with OVNs native DHCP service.

`ovsdb_log_level`

Type
string

Default
`INFO`

Valid Values
CRITICAL, ERROR, WARNING, INFO, DEBUG

The log level used for OVSDB

`ovn_metadata_enabled`

Type
boolean

Default
`False`

Whether to use metadata service.

`dns_servers`

Type
list

Default
`[]`

Comma-separated list of the DNS servers which will be used as forwarders if a subnets dns_nameservers field is empty. If both subnets dns_nameservers and this option are empty, then the DNS resolvers on the host running the neutron server will be used.

`dns_records_ovn_owned`

Type
boolean

Default
`False`

Whether to consider DNS records local to OVN or not. For OVN version 24.03 and above if this option is set to True, DNS records will be treated local to the OVN controller and it will respond to the queries for the records and record types known to it, else it will forward them to the configured DNS server(s).

`ovn_dhcp4_global_options`

Type
dict

Default
`{}`

Dictionary of global DHCPv4 options which will be automatically set on each subnet upon creation and on all existing subnets when Neutron starts. An empty value for a DHCP option will cause that option to be unset globally. Multiple values should be separated by semi-colon. EXAMPLES: - ntp_server:1.2.3.4,wpad:1.2.3.5;1.2.3.6 - Set ntp_server and wpad - ntp_server:,wpad:1.2.3.5 - Unset ntp_server and set wpad See the ovn-nb(5) man page for available options.

`ovn_dhcp6_global_options`

Type
dict

Default
`{}`

Dictionary of global DHCPv6 options which will be automatically set on each subnet upon creation and on all existing subnets when Neutron starts. An empty value for a DHCPv6 option will cause that option to be unset globally. Multiple values should be separated by semi-colon. See the ovn-nb(5) man page for available options.

`disable_ovn_dhcp_for_baremetal_ports`

Type
boolean

Default
`False`

Disable OVNs built-in DHCP for baremetal ports (VNIC type baremetal). This allows operators to plug their own DHCP server of choice for PXE booting baremetal nodes. OVN 23.06.0 and newer also supports baremetal `PXE` based provisioning over IPv6. If an older version of OVN is used for baremetal provisioning over IPv6 this option should be set to True and neutron-dhcp-agent should be used instead.

`localnet_learn_fdb`

Type
boolean

Default
`False`

If enabled it will allow localnet ports to learn MAC addresses and store them in FDB SB table. This avoids flooding for traffic towards unknown IPs when port security is disabled. It requires OVN 22.09 or newer.

`fdb_age_threshold`

Type
integer

Default
`300`

Minimum Value
0

The number of seconds to keep FDB entries in the OVN DB. A value of 0 means disabled. This is supported by OVN >= 23.09.

`mac_binding_age_threshold`

Type
integer

Default
`0`

Minimum Value
0

The number of seconds to keep MAC_Binding entries in the OVN DB. 0 to disable aging.

`broadcast_arps_to_all_routers`

Type
boolean

Default
`True`

If enabled (default) OVN will flood ARP requests to all attached ports on a network. If set to False, ARP requests are only sent to routers on that network if the target MAC address matches. ARP requests that do not match a router will only be forwarded to non-router ports. Supported by OVN >= 23.06.

`ovn_router_indirect_snat`

Type
boolean

Default
`False`

Whether to configure SNAT for all nested subnets connected to the router through any other routers, similar to the default ML2/OVS behavior.

`live_migration_activation_strategy`

Type
string

Default
`rarp`

Valid Values
rarp,

Activation strategy to use for live migration.

Possible values

rarp
Expect the hypervisor to send a Reverse ARP request through the migrated port after migration is complete.

A migrated port is immediately activated on the destination host.

`stateless_nat_enabled`

Type
boolean

Default
`False`

If enabled, the floating IP NAT rules will be stateless, instead of using the conntrack OVN actions. This strategy is faster in some environments, like for example DPDK deployments.

`ovs_create_tap`

Type
boolean

Default
`False`

If enabled, os-vif will create the TAP devices for ports. This controls the `ovs_create_tap` parameter in the vif-details dictionary.

> **Warning**
>
> This option is deprecated for removal since 2026.1. Its value may be silently ignored in the future.
>
> **Reason**
> This flag was added for backwards compatibility during the development of the functionality in Nova (LP#2069718

`ha_failover_strategy`

Type
string

Default
`normal`

Valid Values
normal, aggressive, conservative, manual

HA chassis group failover strategy. Determines how aggressively OVN fails over to a new chassis when the current one becomes unavailable. Supported values: normal,aggressive,conservative,manual

`bfd_min_rx`

Type
integer

Default
`1000`

Minimum Value
1

BFD minimum RX interval in milliseconds. This sets the minimum interval between BFD control packets received by the local system. This option will apply only if ha_failover_strategy=manual

`bfd_min_tx`

Type
integer

Default
`100`

Minimum Value
1

BFD minimum TX interval in milliseconds. This sets the minimum interval between BFD control packets sent by the local system. This option will apply only if ha_failover_strategy=manual

`bfd_mult`

Type
integer

Default
`3`

Minimum Value
1

Maximum Value
255

BFD detection multiplier. The number of BFD control packets that must be missed in a row before the session is declared down. This option will apply only if ha_failover_strategy=manual

ovn_nb_global

ignore_lsp_down

Type
boolean

Default
`False`

If set to False, ARP/ND reply flows for logical switch ports will be installed only if the port is UP, i.e. claimed by a Chassis. If set to True, these flows are installed regardless of the status of the port, which can result in a situation that an ARP request to an IP is resolved even before the relevant VM/container is running. For environments where this is not an issue, setting it to True can reduce the load and latency of the control plane. The default value is False.

fdb_removal_limit

Type
integer

Default
`0`

Minimum Value
0

FDB aging bulk removal limit. This limits how many rows can expire in a single transaction. Zero (0) means number of removals is unlimited. When the limit is reached, the next batch removal is delayed by 5 seconds. This is supported by OVN >= 23.09.

mac_binding_removal_limit

Type
integer

Default
`0`

Minimum Value
0

MAC binding aging bulk removal limit. This limits how many entries can expire in a single transaction. The default is 0 which is unlimited. When the limit is reached, the next batch removal is delayed by 5 seconds.

ovs

ovsdb_timeout

Type
integer

Default
`10`

Timeout in seconds for OVSDB commands. If the timeout expires, OVSDB commands will fail with ALARMCLOCK error.

`bridge_mac_table_size`

Type
integer

Default
`50000`

The maximum number of MAC addresses to learn on a bridge managed by the Neutron OVS agent. Values outside a reasonable range (10 to 1,000,000) might be overridden by Open vSwitch according to the documentation.

`igmp_snooping_enable`

Type
boolean

Default
`False`

Enable IGMP snooping for integration bridge. If this option is set to True, support for Internet Group Management Protocol (IGMP) is enabled in integration bridge.

`igmp_flood`

Type
boolean

Default
`False`

Multicast packets (except reports) are unconditionally forwarded to the ports bridging a logical network to a physical network.

`igmp_flood_reports`

Type
boolean

Default
`True`

Multicast reports are unconditionally forwarded to the ports bridging a logical network to a physical network.

`igmp_flood_unregistered`

Type
boolean

Default
`False`

This option enables or disables flooding of unregistered multicast packets to all ports. If False, The switch will send unregistered multicast packets only to ports connected to multicast routers.

ovs_driver

vnic_type_prohibit_list

Type
list

Default
`[]`

Comma-separated list of VNIC types for which support is administratively prohibited by the mechanism driver. Please note that the supported vnic_types depend on your network interface card, on the kernel version of your operating system, and on other factors, like OVS version. In case of ovs mechanism driver the valid vnic types are normal and direct. Note that direct is supported only from kernel 4.8, and from ovs 2.8.0. Bind DIRECT (SR-IOV) port allows to offload the OVS flows using tc to the SR-IOV NIC. This allows to support hardware offload via tc and that allows us to manage the VF by OpenFlow control plane using representor net-device.

securitygroup

firewall_driver

Type
string

Default
`<None>`

Driver for security groups firewall in the L2 agent

enable_security_group

Type
boolean

Default
`True`

Controls whether the neutron security group API is enabled in the server. It should be false when using no security groups or using the Nova security group API.

enable_ipset

Type
boolean

Default
`True`

Use IPsets to speed-up the iptables based security groups. Enabling IPset support requires that ipset is installed on the L2 agent node.

permitted_ethertypes

Type
list

Default
`[]`

Comma-separated list of ethertypes to be permitted, in hexadecimal (starting with 0x). For example, 0x4008 to permit InfiniBand.

sriov_driver

vnic_type_prohibit_list

Type
list

Default
`[]`

Comma-separated list of VNIC types for which support is administratively prohibited by the mechanism driver. Please note that the supported vnic_types depend on your network interface card, on the kernel version of your operating system, and on other factors. In the case of SRIOV mechanism drivers the valid VNIC types are direct, macvtap and direct-physical.

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DEFAULT

debug

Type
boolean

Default
`False`

Mutable
This option can be changed without restarting.

If set to true, the logging level will be set to DEBUG instead of the default INFO level.

log_config_append

Type
string

Default
`<None>`

Mutable
This option can be changed without restarting.

The name of a logging configuration file. This file is appended to any existing logging configuration files. For details about logging configuration files, see the Python logging module documentation. Note that when logging configuration files are used then all logging configuration is set in the configuration file and other logging configuration options are ignored (for example, log-date-format).

Table 35: Deprecated Variations

Group	Name
DEFAULT	log-config
DEFAULT	log_config

`log_date_format`

Type
string

Default
`%Y-%m-%d %H:%M:%S`

Defines the format string for %(asctime)s in log records. Default: the value above . This option is ignored if log_config_append is set.

`log_file`

Type
string

Default
`<None>`

(Optional) Name of log file to send logging output to. If no default is set, logging will go to stderr as defined by use_stderr. This option is ignored if log_config_append is set.

Table 36: Deprecated Variations

Group	Name
DEFAULT	logfile

`log_dir`

Type
string

Default
`<None>`

(Optional) The base directory used for relative log_file paths. This option is ignored if log_config_append is set.

Table 37: Deprecated Variations

Group	Name
DEFAULT	logdir

`use_syslog`

Type
boolean

Default
`False`

Use syslog for logging. Existing syslog format is DEPRECATED and will be changed later to honor RFC5424. This option is ignored if log_config_append is set.

`use_journal`

Type
boolean

Default
`False`

Enable journald for logging. If running in a systemd environment you may wish to enable journal support. Doing so will use the journal native protocol which includes structured metadata in addition to log messages.This option is ignored if log_config_append is set.

`syslog_log_facility`

Type
string

Default
`LOG_USER`

Syslog facility to receive log lines. This option is ignored if log_config_append is set.

`use_json`

Type
boolean

Default
`False`

Use JSON formatting for logging. This option is ignored if log_config_append is set.

`use_stderr`

Type
boolean

Default
`False`

Log output to standard error. This option is ignored if log_config_append is set.

`log_color`

Type
boolean

Default
`False`

(Optional) Set the color key according to log levels. This option takes effect only when logging to stderr or stdout is used. This option is ignored if log_config_append is set.

`log_rotate_interval`

Type
integer

Default
1

The amount of time before the log files are rotated. This option is ignored unless log_rotation_type is set to interval.

`log_rotate_interval_type`

Type
string

Default
`days`

Valid Values
Seconds, Minutes, Hours, Days, Weekday, Midnight

Rotation interval type. The time of the last file change (or the time when the service was started) is used when scheduling the next rotation.

`max_logfile_count`

Type
integer

Default
`30`

Maximum number of rotated log files.

`max_logfile_size_mb`

Type
integer

Default
`200`

Log file maximum size in MB. This option is ignored if log_rotation_type is not set to size.

`log_rotation_type`

Type
string

Default
`none`

Valid Values
interval, size, none

Log rotation type.

Possible values

interval
Rotate logs at predefined time intervals.

size
Rotate logs once they reach a predefined size.

none
Do not rotate log files.

`logging_context_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [%(global_request_id)s %(request_id)s %(user_identity)s] %(instance)s%(message)s`

Format string to use for log messages with context. Used by oslo_log.formatters.ContextFormatter

`logging_default_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [-] %(instance)s%(message)s`

Format string to use for log messages when context is undefined. Used by oslo_log.formatters.ContextFormatter

`logging_debug_format_suffix`

Type
string

Default
`%(funcName)s %(pathname)s:%(lineno)d`

Additional data to append to log message when logging level for the message is DEBUG. Used by oslo_log.formatters.ContextFormatter

`logging_exception_prefix`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d ERROR %(name)s %(instance)s`

Prefix each line of exception output with this format. Used by oslo_log.formatters.ContextFormatter

`logging_user_identity_format`

Type
string

Default
`%(user)s %(project)s %(domain)s %(system_scope)s %(user_domain)s %(project_domain)s`

Defines the format string for %(user_identity)s that is used in logging_context_format_string. Used by oslo_log.formatters.ContextFormatter `default_log_levels`

Type
list

Default
`['amqp=WARN', 'boto=WARN', 'sqlalchemy=WARN', 'suds=INFO', 'oslo.messaging=INFO', 'oslo_messaging=INFO', 'iso8601=WARN', 'requests.packages.urllib3.connectionpool=WARN', 'urllib3.connectionpool=WARN', 'websocket=WARN', 'requests.packages.urllib3.util.retry=WARN', 'urllib3.util.retry=WARN', 'keystonemiddleware=WARN', 'routes.middleware=WARN', 'stevedore=WARN', 'taskflow=WARN', 'keystoneauth=WARN', 'oslo.cache=INFO', 'oslo_policy=INFO', 'dogpile.core.dogpile=INFO']`

List of package logging levels in logger=LEVEL pairs. This option is ignored if log_config_append is set.

`publish_errors`

Type
boolean

Default
`False`

Enables or disables publication of error events.

`instance_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance that is passed with the log message.

`instance_uuid_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance UUID that is passed with the log message.

`rate_limit_interval`

Type
integer

Default
`0`

Interval, number of seconds, of log rate limiting.

`rate_limit_burst`

Type
integer

Default
`0`

Maximum number of logged messages per rate_limit_interval.

`rate_limit_except_level`

Type
string

Default
`CRITICAL`

Valid Values
CRITICAL, ERROR, INFO, WARNING, DEBUG,

Log level name used by rate limiting. Logs with level greater or equal to rate_limit_except_level are not filtered. An empty string means that all levels are filtered.

`fatal_deprecations`

Type
boolean

Default
`False`

Enables or disables fatal status of deprecations.

agent

`polling_interval`

Type
integer

Default
`2`

The number of seconds the agent will wait between polling for local device changes.

`quitting_rpc_timeout`

Type
integer

Default
`10`

Set new timeout in seconds for new RPC calls after agent receives SIGTERM. If value is set to 0, RPC timeout will not be changed

`dscp`

Type
integer

Default
<None>

Minimum Value
0

Maximum Value
63

The DSCP value to use for outer headers during tunnel encapsulation.

`dscp_inherit`

Type
boolean

Default
`False`

If set to True, the DSCP value of tunnel interfaces is overwritten and set to inherit. The DSCP value of the inner header is then copied to the outer header.

macvtap

`physical_interface_mappings`

Type
list

Default
`[]`

Comma-separated list of <physical_network>:<physical_interface> tuples mapping physical network names to the agents node-specific physical network interfaces to be used for flat and VLAN networks. All physical networks listed in network_vlan_ranges on the server should have mappings to appropriate interfaces on each agent.

securitygroup

`firewall_driver`

Type
string

Default
<None>

Driver for security groups firewall in the L2 agent

`enable_security_group`

Type
boolean

Default
`True`

Controls whether the neutron security group API is enabled in the server. It should be false when using no security groups or using the Nova security group API.

`enable_ipset`

Type
boolean

Default
`True`

Use IPsets to speed-up the iptables based security groups. Enabling IPset support requires that ipset is installed on the L2 agent node.

`permitted_ethertypes`

Type
list

Default
`[]`

Comma-separated list of ethertypes to be permitted, in hexadecimal (starting with 0x). For example, 0x4008 to permit InfiniBand.

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DEFAULT

`rpc_response_max_timeout`

Type
integer

Default
`600`

Maximum seconds to wait for a response from an RPC call.

`debug`

Type
boolean

Default
`False`

Mutable
This option can be changed without restarting.

If set to true, the logging level will be set to DEBUG instead of the default INFO level.

`log_config_append`

Type
string

Default
`<None>`

Mutable
This option can be changed without restarting.

The name of a logging configuration file. This file is appended to any existing logging configuration files. For details about logging configuration files, see the Python logging module documentation. Note that when logging configuration files are used then all logging configuration is set in the configuration file and other logging configuration options are ignored (for example, log-date-format).

Table 38: Deprecated Variations

Group	Name
DEFAULT	log-config
DEFAULT	log_config

`log_date_format`

Type
string

Default
`%Y-%m-%d %H:%M:%S`

Defines the format string for %(asctime)s in log records. Default: the value above . This option is ignored if log_config_append is set.

`log_file`

Type
string

Default
`<None>`

(Optional) Name of log file to send logging output to. If no default is set, logging will go to stderr as defined by use_stderr. This option is ignored if log_config_append is set.

Table 39: Deprecated Variations

Group	Name
DEFAULT	logfile

`log_dir`

Type
string

Default
`<None>`

(Optional) The base directory used for relative log_file paths. This option is ignored if log_config_append is set.

Table 40: Deprecated Variations

Group	Name
DEFAULT	logdir

`use_syslog`

Type
boolean

Default
`False`

Use syslog for logging. Existing syslog format is DEPRECATED and will be changed later to honor RFC5424. This option is ignored if log_config_append is set.

`use_journal`

Type
boolean

Default
`False`

Enable journald for logging. If running in a systemd environment you may wish to enable journal support. Doing so will use the journal native protocol which includes structured metadata in addition to log messages.This option is ignored if log_config_append is set.

`syslog_log_facility`

Type
string

Default
`LOG_USER`

Syslog facility to receive log lines. This option is ignored if log_config_append is set.

`use_json`

Type
boolean

Default
`False`

Use JSON formatting for logging. This option is ignored if log_config_append is set.

`use_stderr`

Type
boolean

Default
`False`

Log output to standard error. This option is ignored if log_config_append is set.

`log_color`

Type
boolean

Default
`False`

(Optional) Set the color key according to log levels. This option takes effect only when logging to stderr or stdout is used. This option is ignored if log_config_append is set.

`log_rotate_interval`

Type
integer

Default
`1`

The amount of time before the log files are rotated. This option is ignored unless log_rotation_type is set to interval.

`log_rotate_interval_type`

Type
string

Default
`days`

Valid Values
Seconds, Minutes, Hours, Days, Weekday, Midnight

Rotation interval type. The time of the last file change (or the time when the service was started) is used when scheduling the next rotation.

`max_logfile_count`

Type
integer

Default
`30`

Maximum number of rotated log files.

`max_logfile_size_mb`

Type
integer

Default
`200`

Log file maximum size in MB. This option is ignored if log_rotation_type is not set to size.

`log_rotation_type`

Type
string

Default
`none`

Valid Values
interval, size, none

Log rotation type.

Possible values

interval
Rotate logs at predefined time intervals.

size
Rotate logs once they reach a predefined size.

none
Do not rotate log files.

`logging_context_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [%(global_request_id)s %(request_id)s %(user_identity)s] %(instance)s%(message)s`

Format string to use for log messages with context. Used by oslo_log.formatters.ContextFormatter

`logging_default_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [-] %(instance)s%(message)s`

Format string to use for log messages when context is undefined. Used by oslo_log.formatters.ContextFormatter

`logging_debug_format_suffix`

Type
string

Default
`%(funcName)s %(pathname)s:%(lineno)d`

Additional data to append to log message when logging level for the message is DEBUG. Used by oslo_log.formatters.ContextFormatter

`logging_exception_prefix`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d ERROR %(name)s %(instance)s`

Prefix each line of exception output with this format. Used by oslo_log.formatters.ContextFormatter

logging_user_identity_format

Type
string

Default
`%(user)s %(project)s %(domain)s %(system_scope)s %(user_domain)s %(project_domain)s`

Defines the format string for %(user_identity)s that is used in logging_context_format_string. Used by oslo_log.formatters.ContextFormatter

default_log_levels

Type
list

Default
`['amqp=WARN', 'boto=WARN', 'sqlalchemy=WARN', 'suds=INFO', 'oslo.messaging=INFO', 'oslo_messaging=INFO', 'iso8601=WARN', 'requests.packages.urllib3.connectionpool=WARN', 'urllib3.connectionpool=WARN', 'websocket=WARN', 'requests.packages.urllib3.util.retry=WARN', 'urllib3.util.retry=WARN', 'keystonemiddleware=WARN', 'routes.middleware=WARN', 'stevedore=WARN', 'taskflow=WARN', 'keystoneauth=WARN', 'oslo.cache=INFO', 'oslo_policy=INFO', 'dogpile.core.dogpile=INFO']`

List of package logging levels in logger=LEVEL pairs. This option is ignored if log_config_append is set.

publish_errors

Type
boolean

Default
`False`

Enables or disables publication of error events.

instance_format

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance that is passed with the log message.

instance_uuid_format

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance UUID that is passed with the log message.

`rate_limit_interval`

Type
integer

Default
`0`

Interval, number of seconds, of log rate limiting.

`rate_limit_burst`

Type
integer

Default
`0`

Maximum number of logged messages per rate_limit_interval.

`rate_limit_except_level`

Type
string

Default
`CRITICAL`

Valid Values
CRITICAL, ERROR, INFO, WARNING, DEBUG,

Log level name used by rate limiting. Logs with level greater or equal to rate_limit_except_level are not filtered. An empty string means that all levels are filtered.

`fatal_deprecations`

Type
boolean

Default
`False`

Enables or disables fatal status of deprecations.

agent

`minimize_polling`

Type
boolean

Default
`True`

Minimize polling by monitoring OVSDB for interface changes.

`ovsdb_monitor_respawn_interval`

Type
integer

Default
`30`

The number of seconds to wait before respawning the OVSDB monitor after losing communication with it.

`tunnel_types`

Type
list

Default
`[]`

Network types supported by the agent (gre, vxlan and/or geneve).

`vxlan_udp_port`

Type
port number

Default
`4789`

Minimum Value
0

Maximum Value
65535

The UDP port to use for VXLAN tunnels.

`l2_population`

Type
boolean

Default
`False`

Use ML2 l2population mechanism driver to learn remote MAC and IPs and improve tunnel scalability.

`arp_responder`

Type
boolean

Default
`False`

Enable local ARP responder if it is supported. Requires OVS 2.1 and ML2 l2population driver. Allows the switch (when supporting an overlay) to respond to an ARP request locally without performing a costly ARP broadcast into the overlay. NOTE: If enable_distributed_routing is set to True then arp_responder will automatically be set to True in the agent, regardless of the setting in the config file.

`dont_fragment`

Type
boolean

Default
`True`

Set or un-set the do not fragment (DF) bit on outgoing IP packet carrying GRE/VXLAN tunnel.

`enable_distributed_routing`

Type
boolean

Default
`False`

Make the l2 agent run in DVR mode.

`drop_flows_on_start`

Type
boolean

Default
`False`

Reset flow table on start. Setting this to True will cause brief traffic interruption.

`tunnel_csum`

Type
boolean

Default
`False`

Set or un-set the tunnel header checksum on outgoing IP packet carrying GRE/VXLAN tunnel.

`baremetal_smartnic`

Type
boolean

Default
`False`

Enable the agent to process Smart NIC ports.

`explicitly_egress_direct`

Type
boolean

Default
`False`

When set to True, the accepted egress unicast traffic will not use action NORMAL. The accepted egress packets will be taken care of in the final egress tables direct output flows for unicast traffic. This will aslo change the pipleline for ingress traffic to ports without security, the final output action will be hit in table 94.

`extensions`

Type
list

Default
`[]`

Extensions list to use

dhcp

enable_ipv6

Type
boolean

Default
`True`

When set to True, the OVS agent DHCP extension will add related flows for DHCPv6 packets.

dhcp_renewal_time

Type
integer

Default
`0`

DHCP renewal time T1 (in seconds). If set to 0, it will default to half of the lease time.

dhcp_rebinding_time

Type
integer

Default
`0`

DHCP rebinding time T2 (in seconds). If set to 0, it will default to 7/8 of the lease time.

metadata

provider_cidr

Type
string

Default
`240.0.0.0/16`

Local metadata CIDR for VMs metadata traffic, will be used as the IP range to generate the VMs metadata IP.

provider_vlan_id

Type
integer

Default
1

The metadata tap device local vlan ID. This is only available on the metadata bridge device.

`provider_base_mac`

Type
string

Default
`fa:16:ee:00:00:00`

The base MAC address Neutron Openvswitch agent will use for metadata traffic.

`host_proxy_listen_port`

Type
integer

Default
`80`

Host haproxy listen port for metadata path. This is transparent for metadata traffic, VMs still try to access 169.254.169.254:80 for metadata. But in the metadata datapath flow pipeline, the destination TCP port 80 will be changed to the value of *host_proxy_listen_port* which the host haproxy will listen on. For return traffic, the TCP source port will be changed back to 80.

`auth_ca_cert`

Type
string

Default
`<None>`

Certificate Authority public key (CA cert) file for ssl

`nova_metadata_host`

Type
host address

Default
`127.0.0.1`

IP address or DNS name of Nova metadata server.

`nova_metadata_port`

Type
port number

Default
`8775`

Minimum Value
0

Maximum Value
65535

TCP Port used by Nova metadata server.

`metadata_proxy_shared_secret`

Type
string

Default
`''`

When proxying metadata requests, Neutron signs the Instance-ID header with a shared secret to prevent spoofing. You may select any string for a secret, but it must match here and in the configuration used by the Nova metadata server. NOTE: Nova uses the same config key, but in [neutron] section.

`nova_metadata_protocol`

Type
string

Default
`http`

Valid Values
http, https

Protocol to access Nova metadata, http or https

`nova_metadata_insecure`

Type
boolean

Default
`False`

Allow to perform insecure SSL (https) requests to Nova metadata

`nova_client_cert`

Type
string

Default
`''`

Client certificate for Nova metadata api server.

`nova_client_priv_key`

Type
string

Default
`''`

Private key of client certificate.

network_log

rate_limit

Type
integer

Default
`100`

Minimum Value
100

Maximum packets logging per second.

burst_limit

Type
integer

Default
`25`

Minimum Value
25

Maximum number of packets per rate_limit.

local_output_log_base

Type
string

Default
`<None>`

Output logfile path on agent side, default syslog file.

ovs

integration_bridge

Type
string

Default
`br-int`

Integration bridge to use. Do not change this parameter unless you have a good reason to. This is the name of the OVS integration bridge. There is one per hypervisor. The integration bridge acts as a virtual patch bay. All VM VIFs are attached to this bridge and then patched according to their network connectivity.

tunnel_bridge

Type
string

Default
`br-tun`

Tunnel bridge to use.

`int_peer_patch_port`

Type
string

Default
`patch-tun`

Peer patch port in integration bridge for tunnel bridge.

`tun_peer_patch_port`

Type
string

Default
`patch-int`

Peer patch port in tunnel bridge for integration bridge.

`local_ip`

Type
ip address

Default
`<None>`

IP address of local overlay (tunnel) network endpoint. Use either an IPv4 or IPv6 address that resides on one of the host network interfaces. The IP version of this value must match the value of the overlay_ip_version option in the ML2 plug-in configuration file on the neutron server node(s).

`bridge_mappings`

Type
list

Default
`[]`

Comma-separated list of <physical_network>:<bridge> tuples mapping physical network names to the agents node-specific Open vSwitch bridge names to be used for flat and VLAN networks. The length of bridge names should be no more than 11. Each bridge must exist, and should have a physical network interface configured as a port. All physical networks configured on the server should have mappings to appropriate bridges on each agent. Note: If you remove a bridge from this mapping, make sure to disconnect it from the integration bridge as it wont be managed by the agent anymore.

`resource_provider_bandwidths`

Type
list

Default
`[]`

Comma-separated list of <bridge>:<egress_bw>:<ingress_bw> tuples, showing the available bandwidth for the given bridge in the given direction. The direction is meant from VM perspective.

Bandwidth is measured in kilobits per second (kbps). The bridge must appear in bridge_mappings as the value. But not all bridges in bridge_mappings must be listed here. For a bridge not listed here we neither create a resource provider in placement nor report inventories against. An omitted direction means we do not report an inventory for the corresponding class.

`resource_provider_hypervisors`

Type
dict

Default
`{}`

Mapping of bridges to hypervisors: <bridge>:<hypervisor>, hypervisor name is used to locate the parent of the resource provider tree. Only needs to be set in the rare case when the hypervisor name is different from the resource_provider_default_hypervisor config option value as known by the nova-compute managing that hypervisor.

`resource_provider_packet_processing_without_direction`

Type
list

Default
`[]`

Comma-separated list of <hypervisor>:<packet_rate> tuples, defining the minimum packet rate the OVS backend can guarantee in kilo (1000) packet per second. The hypervisor name is used to locate the parent of the resource provider tree. Only needs to be set in the rare case when the hypervisor name is different from the DEFAULT.host config option value as known by the nova-compute managing that hypervisor or if multiple hypervisors are served by the same OVS backend. The default is :0 which means no packet processing capacity is guaranteed on the hypervisor named according to DEFAULT.host.

`resource_provider_packet_processing_with_direction`

Type
list

Default
`[]`

Similar to the resource_provider_packet_processing_without_direction but used in case the OVS backend has hardware offload capabilities. In this case the format is <hypervisor>:<egress_pkt_rate>:<ingress_pkt_rate> which allows defining packet processing capacity per traffic direction. The direction is meant from the VM perspective. Note that the resource_provider_packet_processing_without_direction and the resource_provider_packet_processing_with_direction are mutually exclusive options.

`resource_provider_default_hypervisor`

Type
string

Default
`<None>`

The default hypervisor name used to locate the parent of the resource provider. If this option is not set, canonical name is used `resource_provider_inventory_defaults`

Type
dict

Default
`{'allocation_ratio': 1.0, 'min_unit': 1, 'step_size': 1, 'reserved': 0}`

Key:value pairs to specify defaults used while reporting resource provider inventories. Possible keys with their types: allocation_ratio:float, max_unit:int, min_unit:int, reserved:int, step_size:int, See also: https://docs.openstack.org/api-ref/placement/#update-resource-provider-inventories

`resource_provider_packet_processing_inventory_defaults`

Type
dict

Default
`{'allocation_ratio': 1.0, 'min_unit': 1, 'step_size': 1, 'reserved': 0}`

Key:value pairs to specify defaults used while reporting packet rate inventories. Possible keys with their types: allocation_ratio:float, max_unit:int, min_unit:int, reserved:int, step_size:int, See also: https://docs.openstack.org/api-ref/placement/#update-resource-provider-inventories

`datapath_type`

Type
string

Default
`system`

Valid Values
system, netdev

OVS datapath to use.

Possible values

system
Kernel datapath

netdev
Userspace datapath

`vhostuser_socket_dir`

Type
string

Default
`/var/run/openvswitch`

OVS vhost-user socket directory.

`of_listen_address`

Type
ip address

Default
`127.0.0.1`

Address to listen on for OpenFlow connections.

`of_listen_port`

Type
port number

Default
`6633`

Minimum Value
0

Maximum Value
65535

Port to listen on for OpenFlow connections.

`of_connect_timeout`

Type
integer

Default
`300`

Timeout in seconds to wait for the local switch connecting the controller.

`of_request_timeout`

Type
integer

Default
`300`

Timeout in seconds to wait for a single OpenFlow request.

`of_inactivity_probe`

Type
integer

Default
`10`

The inactivity_probe interval in seconds for the local switch connection to the controller. A value of 0 disables inactivity probes.

`openflow_processed_per_port`

Type
boolean

Default
`False`

If enabled, all OpenFlow rules associated to a port are processed at once, in one single transaction. That avoids possible inconsistencies during OVS agent restart and port updates. If disabled, the flows will be processed in batches of `_constants.AGENT_RES_PROCESSING_STEP` number of OpenFlow rules.

`qos_meter_bandwidth`

Type
boolean

Default
`False`

Whether to enable the Openvswitch meter bandwidth limit features which will add meter kbps rules and apply them to the OpenFlow flow table BANDWIDTH_RATE_LIMIT for VM ports.

`trunk_enabled`

Type
boolean

Default
`True`

Enable agent side trunk extension. If you do not need the trunk extension, you can safely set config option to False, it will avoid the agent to declare some queues.

`ovsdb_connection`

Type
string

Default
`tcp:127.0.0.1:6640`

The connection string for the OVSDB backend. Will be used for all OVSDB commands and by ovsdb-client when monitoring

`ssl_key_file`

Type
string

Default
`<None>`

The SSL private key file to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ssl_cert_file`

Type
string

Default
`<None>`

The SSL certificate file to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ssl_ca_cert_file`

Type
string

Default
`<None>`

The Certificate Authority (CA) certificate to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ovsdb_debug`

Type
boolean

Default
`False`

Enable OVSDB debug logs

securitygroup

`firewall_driver`

Type
string

Default
`<None>`

Driver for security groups firewall in the L2 agent

`enable_security_group`

Type
boolean

Default
`True`

Controls whether the neutron security group API is enabled in the server. It should be false when using no security groups or using the Nova security group API.

`enable_ipset`

Type
boolean

Default
`True`

Use IPsets to speed-up the iptables based security groups. Enabling IPset support requires that ipset is installed on the L2 agent node.

permitted_ethertypes

Type
list

Default
`[]`

Comma-separated list of ethertypes to be permitted, in hexadecimal (starting with 0x). For example, 0x4008 to permit InfiniBand.

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DEFAULT

rpc_response_max_timeout

Type
integer

Default
`600`

Maximum seconds to wait for a response from an RPC call.

debug

Type
boolean

Default
`False`

Mutable
This option can be changed without restarting.

If set to true, the logging level will be set to DEBUG instead of the default INFO level.

log_config_append

Type
string

Default
`<None>`

Mutable
This option can be changed without restarting.

The name of a logging configuration file. This file is appended to any existing logging configuration files. For details about logging configuration files, see the Python logging module documentation. Note that when logging configuration files are used then all logging configuration is set in the configuration file and other logging configuration options are ignored (for example, log-date-format).

Table 41: Deprecated Variations

Group	Name
DEFAULT	log-config
DEFAULT	log_config

`log_date_format`

Type
string

Default
`%Y-%m-%d %H:%M:%S`

Defines the format string for %(asctime)s in log records. Default: the value above . This option is ignored if log_config_append is set.

`log_file`

Type
string

Default
`<None>`

(Optional) Name of log file to send logging output to. If no default is set, logging will go to stderr as defined by use_stderr. This option is ignored if log_config_append is set.

Table 42: Deprecated Variations

Group	Name
DEFAULT	logfile

`log_dir`

Type
string

Default
`<None>`

(Optional) The base directory used for relative log_file paths. This option is ignored if log_config_append is set.

Table 43: Deprecated Variations

Group	Name
DEFAULT	logdir

`use_syslog`

Type
boolean

Default
`False`

Use syslog for logging. Existing syslog format is DEPRECATED and will be changed later to honor RFC5424. This option is ignored if log_config_append is set.

`use_journal`

Type
boolean

Default
`False`

Enable journald for logging. If running in a systemd environment you may wish to enable journal support. Doing so will use the journal native protocol which includes structured metadata in addition to log messages.This option is ignored if log_config_append is set.

`syslog_log_facility`

Type
string

Default
`LOG_USER`

Syslog facility to receive log lines. This option is ignored if log_config_append is set.

`use_json`

Type
boolean

Default
`False`

Use JSON formatting for logging. This option is ignored if log_config_append is set.

`use_stderr`

Type
boolean

Default
`False`

Log output to standard error. This option is ignored if log_config_append is set.

`log_color`

Type
boolean

Default
`False`

(Optional) Set the color key according to log levels. This option takes effect only when logging to stderr or stdout is used. This option is ignored if log_config_append is set.

`log_rotate_interval`

Type
integer

Default
`1`

The amount of time before the log files are rotated. This option is ignored unless log_rotation_type is set to interval.

`log_rotate_interval_type`

Type
string

Default
`days`

Valid Values
Seconds, Minutes, Hours, Days, Weekday, Midnight

Rotation interval type. The time of the last file change (or the time when the service was started) is used when scheduling the next rotation.

`max_logfile_count`

Type
integer

Default
`30`

Maximum number of rotated log files.

`max_logfile_size_mb`

Type
integer

Default
`200`

Log file maximum size in MB. This option is ignored if log_rotation_type is not set to size.

`log_rotation_type`

Type
string

Default
`none`

Valid Values
interval, size, none

Log rotation type.

Possible values

interval
Rotate logs at predefined time intervals.

size
Rotate logs once they reach a predefined size.

none
Do not rotate log files.

`logging_context_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [%(global_request_id)s %(request_id)s %(user_identity)s] %(instance)s%(message)s`

Format string to use for log messages with context. Used by oslo_log.formatters.ContextFormatter

`logging_default_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [-] %(instance)s%(message)s`

Format string to use for log messages when context is undefined. Used by oslo_log.formatters.ContextFormatter

`logging_debug_format_suffix`

Type
string

Default
`%(funcName)s %(pathname)s:%(lineno)d`

Additional data to append to log message when logging level for the message is DEBUG. Used by oslo_log.formatters.ContextFormatter

`logging_exception_prefix`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d ERROR %(name)s %(instance)s`

Prefix each line of exception output with this format. Used by oslo_log.formatters.ContextFormatter

`logging_user_identity_format`

Type
string

Default
`%(user)s %(project)s %(domain)s %(system_scope)s %(user_domain)s %(project_domain)s`

Defines the format string for %(user_identity)s that is used in logging_context_format_string. Used by oslo_log.formatters.ContextFormatter `default_log_levels`

Type
list

Default
`['amqp=WARN', 'boto=WARN', 'sqlalchemy=WARN', 'suds=INFO', 'oslo.messaging=INFO', 'oslo_messaging=INFO', 'iso8601=WARN', 'requests.packages.urllib3.connectionpool=WARN', 'urllib3.connectionpool=WARN', 'websocket=WARN', 'requests.packages.urllib3.util.retry=WARN', 'urllib3.util.retry=WARN', 'keystonemiddleware=WARN', 'routes.middleware=WARN', 'stevedore=WARN', 'taskflow=WARN', 'keystoneauth=WARN', 'oslo.cache=INFO', 'oslo_policy=INFO', 'dogpile.core.dogpile=INFO']`

List of package logging levels in logger=LEVEL pairs. This option is ignored if log_config_append is set.

`publish_errors`

Type
boolean

Default
`False`

Enables or disables publication of error events.

`instance_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance that is passed with the log message.

`instance_uuid_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance UUID that is passed with the log message.

`rate_limit_interval`

Type
integer

Default
`0`

Interval, number of seconds, of log rate limiting.

`rate_limit_burst`

Type
integer

Default
`0`

Maximum number of logged messages per rate_limit_interval.

`rate_limit_except_level`

Type
string

Default
`CRITICAL`

Valid Values
CRITICAL, ERROR, INFO, WARNING, DEBUG,

Log level name used by rate limiting. Logs with level greater or equal to rate_limit_except_level are not filtered. An empty string means that all levels are filtered.

`fatal_deprecations`

Type
boolean

Default
`False`

Enables or disables fatal status of deprecations.

agent

`extensions`

Type
list

Default
`[]`

Extensions list to use

sriov_nic

`physical_device_mappings`

Type
list

Default
`[]`

Comma-separated list of <physical_network>:<network_device> tuples mapping physical network names to the agents node-specific physical network device interfaces of SR-IOV physical function to be used for VLAN networks. All physical networks listed in network_vlan_ranges on the server should have mappings to appropriate interfaces on each agent.

`exclude_devices`

Type
list

Default
`[]`

Comma-separated list of <network_device>:<vfs_to_exclude> tuples, mapping network_device to the agents node-specific list of virtual functions that should not be used for virtual networking. vfs_to_exclude is a semicolon-separated list of virtual functions to exclude from network_device. The network_device in the mapping should appear in the physical_device_mappings list.

`resource_provider_bandwidths`

Type
list

Default
`[]`

Comma-separated list of <network_device>:<egress_bw>:<ingress_bw> tuples, showing the available bandwidth for the given device in the given direction. The direction is meant from VM perspective. Bandwidth is measured in kilobits per second (kbps). The device must appear in physical_device_mappings as the value. But not all devices in physical_device_mappings must be listed here. For a device not listed here we neither create a resource provider in placement nor report inventories against. An omitted direction means we do not report an inventory for the corresponding class.

`resource_provider_hypervisors`

Type
dict

Default
`{}`

Mapping of network devices to hypervisors: <network_device>:<hypervisor>, hypervisor name is used to locate the parent of the resource provider tree. Only needs to be set in the rare case when the hypervisor name is different from the resource_provider_default_hypervisor config option value as known by the nova-compute managing that hypervisor.

`resource_provider_default_hypervisor`

Type
string

Default
`<None>`

The default hypervisor name used to locate the parent of the resource provider. If this option is not set, canonical name is used

`resource_provider_inventory_defaults`

Type
dict

Default
`{'allocation_ratio': 1.0, 'min_unit': 1, 'step_size': 1, 'reserved': 0}`

Key:value pairs to specify defaults used while reporting resource provider inventories. Possible keys with their types: allocation_ratio:float, max_unit:int, min_unit:int, reserved:int, step_size:int, See also: https://docs.openstack.org/api-ref/placement/#update-resource-provider-inventories

9.1.6 dhcp_agent.ini

DEFAULT

`ovs_use_veth`

Type
boolean

Default
`False`

Uses veth for an OVS interface or not. Support kernels with limited namespace support (e.g. RHEL 6.5) and rate limiting on routers gateway port so long as ovs_use_veth is set to True.

`interface_driver`

Type
string

Default
`openvswitch`

The driver used to manage virtual interfaces.

`rpc_response_max_timeout`

Type
integer

Default
`600`

Maximum seconds to wait for a response from an RPC call.

`resync_interval`

Type
integer

Default
`5`

The DHCP agent will resync its state with Neutron to recover from any transient notification or RPC errors. The interval is the maximum number of seconds between attempts. The resync can be done more often based on the events triggered.

`resync_throttle`

Type
integer

Default
`1`

Throttle the number of resync state events between the local DHCP state and Neutron to only once per resync_throttle seconds. The value of throttle introduces a minimum interval between resync state events. Otherwise the resync may end up in a busy-loop. The value must be less than resync_interval.

`dhcp_driver`

Type
string

Default
`neutron.agent.linux.dhcp.Dnsmasq`

The driver used to manage the DHCP server.

`enable_isolated_metadata`

Type
boolean

Default
`False`

The DHCP server can assist with providing metadata support on isolated networks. Setting this value to True will cause the DHCP server to append specific host routes to the DHCP request. The metadata service will only be activated when the subnet does not contain any router port. The guest instance must be configured to request host routes via DHCP (Option 121). This option does not have any effect when force_metadata is set to True.

`force_metadata`

Type
boolean

Default
`False`

In some cases the Neutron router is not present to provide the metadata IP but the DHCP server can be used to provide this info. Setting this value will force the DHCP server to append specific host routes to the DHCP request. If this option is set, then the metadata service will be activated for all of the networks.

`enable_metadata_network`

Type
boolean

Default
`False`

Allows for serving metadata requests coming from a dedicated metadata access network whose CIDR is 169.254.169.254/16 (or larger prefix), and is connected to a Neutron router from which the VMs send metadata:1 request. In this case DHCP Option 121 will not be injected in VMs, as they will be able to reach 169.254.169.254 through a router. This option requires enable_isolated_metadata = True.

`num_sync_threads`

Type
integer

Default
`4`

Number of threads to use during sync process. Should not exceed connection pool size configured on server.

`bulk_reload_interval`

Type
integer

Default
`0`

Minimum Value
0

Time to sleep between reloading the DHCP allocations. This will only be invoked if the value is not 0. If a network has N updates in X seconds then it will reload once and not N times.

`dhcp_confs`

Type
string

Default
`$state_path/dhcp`

Location to store DHCP server config files.

`dnsmasq_config_file`

Type
string

Default
`''`

Override the default dnsmasq settings with this file.

`dnsmasq_dns_servers`

Type
list

Default
`[]`

Comma-separated list of the DNS servers which will be used as forwarders.

`dnsmasq_base_log_dir`

Type
string

Default
`<None>`

Base log dir for dnsmasq logging. The log contains DHCP and DNS log information and is useful for debugging issues with either DHCP or DNS. If this section is null, disable dnsmasq log.

`dnsmasq_local_resolv`

Type
boolean

Default
`False`

Enables the dnsmasq service to provide name resolution for instances via DNS resolvers on the host running the DHCP agent. Effectively removes the no-resolv option from the dnsmasq process arguments. Adding custom DNS resolvers to the dnsmasq_dns_servers option disables this feature.

`dnsmasq_lease_max`

Type
integer

Default
`16777216`

Limit number of leases to prevent a denial-of-service.

`dhcp_broadcast_reply`

Type
boolean

Default
`False`

Use broadcast in DHCP replies.

`dnsmasq_enable_addr6_list`

Type
boolean

Default
`False`

Enable dhcp-host entry with list of addresses when port has multiple IPv6 addresses in the same subnet.

`dnsmasq_txt_record`

Type
string

Default
`''`

Return a TXT DNS record (option format: <name>[[,<text>],<text>]). The value of the TXT record is a set of strings, so any number may be included, delimited by commas; use quotes to put commas into a string. Note that the maximum length of a single string is 255 characters, longer strings are split into 255 character chunks.

`debug`

Type
boolean

Default
`False`

Mutable
This option can be changed without restarting.

If set to true, the logging level will be set to DEBUG instead of the default INFO level.

`log_config_append`

Type
string

Default
`<None>`

Mutable
This option can be changed without restarting.

The name of a logging configuration file. This file is appended to any existing logging configuration files. For details about logging configuration files, see the Python logging module documentation. Note that when logging configuration files are used then all logging configuration is set in the configuration file and other logging configuration options are ignored (for example, log-date-format).

Table 44: Deprecated Variations

Group	Name
DEFAULT	log-config
DEFAULT	log_config

`log_date_format`

Type
string

Default
`%Y-%m-%d %H:%M:%S`

Defines the format string for %(asctime)s in log records. Default: the value above . This option is ignored if log_config_append is set.

`log_file`

Type
string

Default
`<None>`

(Optional) Name of log file to send logging output to. If no default is set, logging will go to stderr as defined by use_stderr. This option is ignored if log_config_append is set.

Table 45: Deprecated Variations

Group	Name
DEFAULT	logfile

`log_dir`

Type
string

Default
`<None>`

(Optional) The base directory used for relative log_file paths. This option is ignored if log_config_append is set.

Table 46: Deprecated Variations

Group	Name
DEFAULT	logdir

`use_syslog`

Type
boolean

Default
`False`

Use syslog for logging. Existing syslog format is DEPRECATED and will be changed later to honor RFC5424. This option is ignored if log_config_append is set.

`use_journal`

Type
boolean

Default
`False`

Enable journald for logging. If running in a systemd environment you may wish to enable journal support. Doing so will use the journal native protocol which includes structured metadata in addition to log messages.This option is ignored if log_config_append is set.

`syslog_log_facility`

Type
string

Default
`LOG_USER`

Syslog facility to receive log lines. This option is ignored if log_config_append is set.

`use_json`

Type
boolean

Default
`False`

Use JSON formatting for logging. This option is ignored if log_config_append is set.

`use_stderr`

Type
boolean

Default
`False`

Log output to standard error. This option is ignored if log_config_append is set.

`log_color`

Type
boolean

Default
`False`

(Optional) Set the color key according to log levels. This option takes effect only when logging to stderr or stdout is used. This option is ignored if log_config_append is set.

`log_rotate_interval`

Type
integer

Default
`1`

The amount of time before the log files are rotated. This option is ignored unless log_rotation_type is set to interval.

`log_rotate_interval_type`

Type
string

Default
`days`

Valid Values
Seconds, Minutes, Hours, Days, Weekday, Midnight

Rotation interval type. The time of the last file change (or the time when the service was started) is used when scheduling the next rotation.

`max_logfile_count`

Type
integer

Default
`30`

Maximum number of rotated log files.

`max_logfile_size_mb`

Type
integer

Default
`200`

Log file maximum size in MB. This option is ignored if log_rotation_type is not set to size.

`log_rotation_type`

Type
string

Default
`none`

Valid Values
interval, size, none

Log rotation type.

Possible values

interval
Rotate logs at predefined time intervals.

size
Rotate logs once they reach a predefined size.

none
Do not rotate log files.

`logging_context_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [%(global_request_id)s %(request_id)s %(user_identity)s] %(instance)s%(message)s`

Format string to use for log messages with context. Used by oslo_log.formatters.ContextFormatter

`logging_default_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [-] %(instance)s%(message)s`

Format string to use for log messages when context is undefined. Used by oslo_log.formatters.ContextFormatter

`logging_debug_format_suffix`

Type
string

Default
`%(funcName)s %(pathname)s:%(lineno)d`

Additional data to append to log message when logging level for the message is DEBUG. Used by oslo_log.formatters.ContextFormatter

`logging_exception_prefix`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d ERROR %(name)s %(instance)s`

Prefix each line of exception output with this format. Used by oslo_log.formatters.ContextFormatter

`logging_user_identity_format`

Type
string

Default
`%(user)s %(project)s %(domain)s %(system_scope)s %(user_domain)s %(project_domain)s`

Defines the format string for %(user_identity)s that is used in logging_context_format_string. Used by oslo_log.formatters.ContextFormatter

`default_log_levels`

Type
list

Default
`['amqp=WARN', 'boto=WARN', 'sqlalchemy=WARN', 'suds=INFO', 'oslo.messaging=INFO', 'oslo_messaging=INFO', 'iso8601=WARN', 'requests.packages.urllib3.connectionpool=WARN', 'urllib3.connectionpool=WARN', 'websocket=WARN', 'requests.packages.urllib3.util.retry=WARN', 'urllib3.util.retry=WARN', 'keystonemiddleware=WARN', 'routes.middleware=WARN', 'stevedore=WARN', 'taskflow=WARN', 'keystoneauth=WARN', 'oslo.cache=INFO', 'oslo_policy=INFO', 'dogpile.core.dogpile=INFO']`

List of package logging levels in logger=LEVEL pairs. This option is ignored if log_config_append is set.

`publish_errors`

Type
boolean

Default
`False`

Enables or disables publication of error events.

`instance_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance that is passed with the log message.

`instance_uuid_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance UUID that is passed with the log message.

`rate_limit_interval`

Type
integer

Default
`0`

Interval, number of seconds, of log rate limiting.

`rate_limit_burst`

Type
integer

Default
`0`

Maximum number of logged messages per rate_limit_interval.

`rate_limit_except_level`

Type
string

Default
`CRITICAL`

Valid Values
CRITICAL, ERROR, INFO, WARNING, DEBUG,

Log level name used by rate limiting. Logs with level greater or equal to rate_limit_except_level are not filtered. An empty string means that all levels are filtered.

`fatal_deprecations`

Type
boolean

Default
`False`

Enables or disables fatal status of deprecations.

agent

`availability_zone`

Type
string

Default
`nova`

Availability zone of this node

`report_interval`

Type
floating point

Default
`30`

Seconds between nodes reporting state to server; should be less than agent_down_time, best if it is half or less than agent_down_time.

`log_agent_heartbeats`

Type
boolean

Default
`False`

Log agent heartbeats

metadata_rate_limiting

`rate_limit_enabled`

Type
boolean

Default
`False`

Enable rate limiting on the metadata API.

`ip_versions`

Type
list

Default
`[4]`

Comma separated list of the metadata address IP versions (4, 6) for which rate limiting will be enabled. The default is to rate limit only for the metadata IPv4 address. NOTE: at the moment, the open source version of HAProxy only allows us to rate limit for IPv4 or IPv6, but not both at the same time.

`base_window_duration`

Type
integer

Default
`10`

Duration (seconds) of the base window on the metadata API.

`base_query_rate_limit`

Type
integer

Default
`10`

Max number of queries to accept during the base window.

`burst_window_duration`

Type
integer

Default
`10`

Duration (seconds) of the burst window on the metadata API.

`burst_query_rate_limit`

Type
integer

Default
`10`

Max number of queries to accept during the burst window.

ovs

`ovsdb_connection`

Type
string

Default
`tcp:127.0.0.1:6640`

The connection string for the OVSDB backend. Will be used for all OVSDB commands and by ovsdb-client when monitoring `ssl_key_file`

Type
string

Default
`<None>`

The SSL private key file to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ssl_cert_file`

Type
string

Default
`<None>`

The SSL certificate file to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ssl_ca_cert_file`

Type
string

Default
`<None>`

The Certificate Authority (CA) certificate to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ovsdb_debug`

Type
boolean

Default
`False`

Enable OVSDB debug logs

`ovsdb_timeout`

Type
integer

Default
`10`

Timeout in seconds for OVSDB commands. If the timeout expires, OVSDB commands will fail with ALARMCLOCK error.

`bridge_mac_table_size`

Type
integer

Default
`50000`

The maximum number of MAC addresses to learn on a bridge managed by the Neutron OVS agent. Values outside a reasonable range (10 to 1,000,000) might be overridden by Open vSwitch according to the documentation.

`igmp_snooping_enable`

Type
boolean

Default
`False`

Enable IGMP snooping for integration bridge. If this option is set to True, support for Internet Group Management Protocol (IGMP) is enabled in integration bridge.

`igmp_flood`

Type
boolean

Default
`False`

Multicast packets (except reports) are unconditionally forwarded to the ports bridging a logical network to a physical network.

`igmp_flood_reports`

Type
boolean

Default
`True`

Multicast reports are unconditionally forwarded to the ports bridging a logical network to a physical network.

`igmp_flood_unregistered`

Type
boolean

Default
`False`

This option enables or disables flooding of unregistered multicast packets to all ports. If False, The switch will send unregistered multicast packets only to ports connected to multicast routers.

9.1.7 l3_agent.ini

DEFAULT

`ovs_use_veth`

Type
boolean

Default
`False`

Uses veth for an OVS interface or not. Support kernels with limited namespace support (e.g. RHEL 6.5) and rate limiting on routers gateway port so long as ovs_use_veth is set to True.

`interface_driver`

Type
string

Default
`openvswitch`

The driver used to manage virtual interfaces.

`rpc_response_max_timeout`

Type
integer

Default
`600`

Maximum seconds to wait for a response from an RPC call.

`agent_mode`

Type
string

Default
`legacy`

Valid Values
dvr, dvr_snat, legacy, dvr_no_external

The working mode for the agent.

Possible values

dvr
Enable DVR functionality and must be used for an L3 agent that runs on a compute host.

dvr_snat
Enable centralized SNAT support in conjunction with DVR. This mode must be used for an L3 agent running on a centralized node (or in single-host deployments, e.g. devstack).

legacy
Preserve the existing behavior where the L3 agent is deployed on a centralized networking node to provide L3 services like DNAT and SNAT. Use this mode if you do not want to adopt DVR.

dvr_no_external
Enable only East/West DVR routing functionality for an L3 agent that runs on a compute host, while the North/South functionality such as DNAT and SNAT will be provided by the centralized network node that is running in dvr_snat mode. This mode should be used when there is no external network connectivity on the compute host.

`metadata_port`

Type
port number

Default
`9697`

Minimum Value
0

Maximum Value
65535

TCP Port used by Neutron metadata namespace proxy.

`handle_internal_only_routers`

Type
boolean

Default
`True`

Indicates that this L3 agent should also handle routers that do not have an external network gateway configured. This option should be True only for a single agent in a Neutron deployment, and may be False for all agents if all routers must have an external network gateway.

`ipv6_gateway`

Type
string

Default
`''`

With IPv6, the network used for the external gateway does not need to have an associated subnet, since the automatically assigned link-local address (LLA) can be used. However, an IPv6 gateway address is needed for use as the next-hop for the default route. If no IPv6 gateway address is configured here, (and only then) the Neutron router will be configured to get its default route from Router Advertisements (RAs) from the upstream router; in which case the upstream router must also be configured to send these RAs. The ipv6_gateway, when configured, should be the LLA of the interface on the upstream router. If a next-hop using a global unique address (GUA) is desired, it needs to be done via a subnet allocated to the network and not through this parameter.

`enable_metadata_proxy`

Type
boolean

Default
`True`

Allow running metadata proxy.

`metadata_access_mark`

Type
string

Default
`0x1`

Iptables mangle mark used to mark metadata valid requests. This mark will be masked with 0xffff so that only the lower 16 bits will be used.

`external_ingress_mark`

Type
string

Default
`0x2`

Iptables mangle mark used to mark ingress from an external network. This mark will be masked with 0xffff so that only the lower 16 bits will be used.

`radvd_user`

Type
string

Default
`''`

The username passed to radvd, used to drop root privileges and change user ID to username and group ID of the primary group of username. If no user specified (default), the user executing the L3 agent will be passed. If root is specified, because radvd is spawned as root, no username parameter will be passed.

`cleanup_on_shutdown`

Type
boolean

Default
`False`

Delete all routers on L3 agent shutdown. For L3 HA routers it includes a shutdown of keepalived and the state change monitor. NOTE: Setting to True could affect the data plane when stopping or restarting the L3 agent.

`periodic_interval`

Type
integer

Default
`40`

Seconds between running periodic tasks.

`api_workers`

Type
integer

Default
`<None>`

Minimum Value
1

Number of separate API worker processes for service. If not specified, the default is equal to the number of CPUs available for best performance, capped by potential RAM usage.

`rpc_workers`

Type
integer

Default
`<None>`

Minimum Value
0

Number of RPC worker processes for service. If not specified, the default is equal to half the number of API workers. If set to 0, no RPC worker is launched.

`rpc_state_report_workers`

Type
integer

Default
`1`

Minimum Value
0

Number of RPC worker processes dedicated to the state reports queue. If set to 0, no dedicated RPC worker for state reports queue is launched.

`periodic_fuzzy_delay`

Type
integer

Default
`5`

Range of seconds to randomly delay when starting the periodic task scheduler to reduce stampeding. (Disable by setting to 0)

`ha_confs_path`

Type
string

Default
`$state_path/ha_confs`

Location to store keepalived config files

`ha_vrrp_auth_type`

Type
string

Default
`PASS`

Valid Values
AH, PASS

VRRP authentication type

`ha_vrrp_auth_password`

Type
string

Default
`<None>`

VRRP authentication password

`ha_vrrp_advert_int`

Type
integer

Default
`2`

The advertisement interval in seconds

`ha_keepalived_state_change_server_threads`

Type
integer

Default
`(1 + <num_of_cpus>) / 2`

Minimum Value
1

This option has a sample default set, which means that its actual default value may vary from the one documented above.

Number of concurrent threads for keepalived server connection requests. More threads create a higher CPU load on the agent node.

`ha_vrrp_health_check_interval`

Type
integer

Default
`0`

The VRRP health check interval in seconds. Values > 0 enable VRRP health checks. Setting it to 0 disables VRRP health checks. Recommended value is 5. This will cause pings to be sent to the gateway IP address(es) - requires ICMP_ECHO_REQUEST to be enabled on the gateway(s). If a gateway fails, all routers will be reported as primary, and a primary election will be repeated in a round-robin fashion, until one of the routers restores the gateway connection.

`ha_conntrackd_enabled`

Type
boolean

Default
`False`

Enable conntrackd to synchronize connection tracking states between HA routers.

`ha_conntrackd_hashsize`

Type
integer

Default
`32768`

Number of buckets in the cache hashtable

`ha_conntrackd_hashlimit`

Type
integer

Default
`131072`

Maximum number of conntracks

`ha_conntrackd_unix_backlog`

Type
integer

Default
`20`

Unix socket backlog

`ha_conntrackd_socketbuffersize`

Type
integer

Default
`262142`

Socket buffer size for events

`ha_conntrackd_socketbuffersize_max_grown`

Type
integer

Default
`655355`

Maximum size of socket buffer

`ha_conntrackd_ipv4_mcast_addr`

Type
string

Default
`225.0.0.50`

Multicast address: The address that you use as destination in the synchronization messages

`ha_conntrackd_group`

Type
integer

Default
`3780`

The multicast base port number. The generated virtual router ID added to this value.

`ha_conntrackd_sndsocketbuffer`

Type
integer

Default
`24985600`

Buffer used to enqueue the packets that are going to be transmitted

`ha_conntrackd_rcvsocketbuffer`

Type
integer

Default
`24985600`

Buffer used to enqueue the packets that the socket is pending to handle

`ra_confs`

Type
string

Default
`$state_path/ra`

Location to store IPv6 Router Advertisement config files

`min_rtr_adv_interval`

Type
integer

Default
`30`

MinRtrAdvInterval setting for radvd.conf

`max_rtr_adv_interval`

Type
integer

Default
`100`

MaxRtrAdvInterval setting for radvd.conf

agent

availability_zone

Type
string

Default
`nova`

Availability zone of this node

report_interval

Type
floating point

Default
`30`

Seconds between nodes reporting state to server; should be less than agent_down_time, best if it is half or less than agent_down_time.

log_agent_heartbeats

Type
boolean

Default
`False`

Log agent heartbeats

extensions

Type
list

Default
`[]`

Extensions list to use

metadata_rate_limiting

rate_limit_enabled

Type
boolean

Default
`False`

Enable rate limiting on the metadata API.

ip_versions

Type
list

Default
`[4]`

Comma separated list of the metadata address IP versions (4, 6) for which rate limiting will be enabled. The default is to rate limit only for the metadata IPv4 address. NOTE: at the moment, the open source version of HAProxy only allows us to rate limit for IPv4 or IPv6, but not both at the same time.

`base_window_duration`

Type
integer

Default
`10`

Duration (seconds) of the base window on the metadata API.

`base_query_rate_limit`

Type
integer

Default
`10`

Max number of queries to accept during the base window.

`burst_window_duration`

Type
integer

Default
`10`

Duration (seconds) of the burst window on the metadata API.

`burst_query_rate_limit`

Type
integer

Default
`10`

Max number of queries to accept during the burst window.

network_log

`rate_limit`

Type
integer

Default
`100`

Minimum Value
100

Maximum packets logging per second.

`burst_limit`

Type
integer

Default
`25`

Minimum Value
25

Maximum number of packets per rate_limit.

`local_output_log_base`

Type
string

Default
`<None>`

Output logfile path on agent side, default syslog file.

ovs

`ovsdb_connection`

Type
string

Default
`tcp:127.0.0.1:6640`

The connection string for the OVSDB backend. Will be used for all OVSDB commands and by ovsdb-client when monitoring

`ssl_key_file`

Type
string

Default
`<None>`

The SSL private key file to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ssl_cert_file`

Type
string

Default
`<None>`

The SSL certificate file to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ssl_ca_cert_file`

Type
string

Default
`<None>`

The Certificate Authority (CA) certificate to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ovsdb_debug`

Type
boolean

Default
`False`

Enable OVSDB debug logs

`ovsdb_timeout`

Type
integer

Default
`10`

Timeout in seconds for OVSDB commands. If the timeout expires, OVSDB commands will fail with ALARMCLOCK error.

`bridge_mac_table_size`

Type
integer

Default
`50000`

The maximum number of MAC addresses to learn on a bridge managed by the Neutron OVS agent. Values outside a reasonable range (10 to 1,000,000) might be overridden by Open vSwitch according to the documentation.

`igmp_snooping_enable`

Type
boolean

Default
`False`

Enable IGMP snooping for integration bridge. If this option is set to True, support for Internet Group Management Protocol (IGMP) is enabled in integration bridge.

`igmp_flood`

Type
boolean

Default
`False`

Multicast packets (except reports) are unconditionally forwarded to the ports bridging a logical network to a physical network.

igmp_flood_reports

Type
boolean

Default
`True`

Multicast reports are unconditionally forwarded to the ports bridging a logical network to a physical network.

igmp_flood_unregistered

Type
boolean

Default
`False`

This option enables or disables flooding of unregistered multicast packets to all ports. If False, The switch will send unregistered multicast packets only to ports connected to multicast routers.

9.1.8 metadata_agent.ini

DEFAULT

metadata_proxy_socket

Type
string

Default
`$state_path/metadata_proxy`

Location for Metadata Proxy UNIX domain socket.

metadata_proxy_user

Type
string

Default
`''`

User (uid or name) running metadata proxy after its initialization (if empty: agent effective user).

metadata_proxy_group

Type
string

Default
`''`

Group (gid or name) running metadata proxy after its initialization (if empty: agent effective group).

`auth_ca_cert`

Type
string

Default
`<None>`

Certificate Authority public key (CA cert) file for ssl

`nova_metadata_host`

Type
host address

Default
`127.0.0.1`

IP address or DNS name of Nova metadata server.

`nova_metadata_port`

Type
port number

Default
`8775`

Minimum Value
0

Maximum Value
65535

TCP Port used by Nova metadata server.

`metadata_proxy_shared_secret`

Type
string

Default
`''`

When proxying metadata requests, Neutron signs the Instance-ID header with a shared secret to prevent spoofing. You may select any string for a secret, but it must match here and in the configuration used by the Nova metadata server. NOTE: Nova uses the same config key, but in [neutron] section.

`nova_metadata_protocol`

Type
string

Default
`http`

Valid Values
http, https

Protocol to access Nova metadata, http or https

`nova_metadata_insecure`

Type
boolean

Default
`False`

Allow to perform insecure SSL (https) requests to Nova metadata

`nova_client_cert`

Type
string

Default
`''`

Client certificate for Nova metadata api server.

`nova_client_priv_key`

Type
string

Default
`''`

Private key of client certificate.

`metadata_proxy_socket_mode`

Type
string

Default
`deduce`

Valid Values
deduce, user, group, all

Metadata Proxy UNIX domain socket mode, 4 values allowed: deduce: deduce mode from metadata_proxy_user/group values, user: set metadata proxy socket mode to 0o644, to use when metadata_proxy_user is agent effective user or root, group: set metadata proxy socket mode to 0o664, to use when metadata_proxy_group is agent effective group or root, all: set metadata proxy socket mode to 0o666, to use otherwise.

`metadata_workers`

Type
integer

Default
`<num_of_cpus> / 2`

This option has a sample default set, which means that its actual default value may vary from the one documented above.

Number of separate worker threads for metadata server (defaults to 0 when used with ML2/OVN and half of the number of CPUs with other backend drivers)

`metadata_backlog`

Type
integer

Default
`4096`

Minimum Value
1

Number of backlog requests to configure the metadata server socket with

`rpc_response_max_timeout`

Type
integer

Default
`600`

Maximum seconds to wait for a response from an RPC call.

`debug`

Type
boolean

Default
`False`

Mutable
This option can be changed without restarting.

If set to true, the logging level will be set to DEBUG instead of the default INFO level.

`log_config_append`

Type
string

Default
`<None>`

Mutable
This option can be changed without restarting.

The name of a logging configuration file. This file is appended to any existing logging configuration files. For details about logging configuration files, see the Python logging module documentation. Note that when logging configuration files are used then all logging configuration is set in the configuration file and other logging configuration options are ignored (for example, log-date-format).

Table 47: Deprecated Variations

Group	Name
DEFAULT	log-config
DEFAULT	log_config

`log_date_format`

Type
string

Default
`%Y-%m-%d %H:%M:%S`

Defines the format string for %(asctime)s in log records. Default: the value above . This option is ignored if log_config_append is set.

`log_file`

Type
string

Default
`<None>`

(Optional) Name of log file to send logging output to. If no default is set, logging will go to stderr as defined by use_stderr. This option is ignored if log_config_append is set.

Table 48: Deprecated Variations

Group	Name
DEFAULT	logfile

`log_dir`

Type
string

Default
`<None>`

(Optional) The base directory used for relative log_file paths. This option is ignored if log_config_append is set.

Table 49: Deprecated Variations

Group	Name
DEFAULT	logdir

`use_syslog`

Type
boolean

Default
`False`

Use syslog for logging. Existing syslog format is DEPRECATED and will be changed later to honor RFC5424. This option is ignored if log_config_append is set.

`use_journal`

Type
boolean

Default
`False`

Enable journald for logging. If running in a systemd environment you may wish to enable journal support. Doing so will use the journal native protocol which includes structured metadata in addition to log messages.This option is ignored if log_config_append is set.

`syslog_log_facility`

Type
string

Default
`LOG_USER`

Syslog facility to receive log lines. This option is ignored if log_config_append is set.

`use_json`

Type
boolean

Default
`False`

Use JSON formatting for logging. This option is ignored if log_config_append is set.

`use_stderr`

Type
boolean

Default
`False`

Log output to standard error. This option is ignored if log_config_append is set.

`log_color`

Type
boolean

Default
`False`

(Optional) Set the color key according to log levels. This option takes effect only when logging to stderr or stdout is used. This option is ignored if log_config_append is set.

`log_rotate_interval`

Type
integer

Default
`1`

The amount of time before the log files are rotated. This option is ignored unless log_rotation_type is set to interval.

`log_rotate_interval_type`

Type
string

Default
`days`

Valid Values
Seconds, Minutes, Hours, Days, Weekday, Midnight

Rotation interval type. The time of the last file change (or the time when the service was started) is used when scheduling the next rotation.

`max_logfile_count`

Type
integer

Default
`30`

Maximum number of rotated log files.

`max_logfile_size_mb`

Type
integer

Default
`200`

Log file maximum size in MB. This option is ignored if log_rotation_type is not set to size.

`log_rotation_type`

Type
string

Default
`none`

Valid Values
interval, size, none

Log rotation type.

Possible values

interval
Rotate logs at predefined time intervals.

size
Rotate logs once they reach a predefined size.

none
Do not rotate log files.

`logging_context_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [%(global_request_id)s %(request_id)s %(user_identity)s] %(instance)s%(message)s`

Format string to use for log messages with context. Used by oslo_log.formatters.ContextFormatter

`logging_default_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [-] %(instance)s%(message)s`

Format string to use for log messages when context is undefined. Used by oslo_log.formatters.ContextFormatter

`logging_debug_format_suffix`

Type
string

Default
`%(funcName)s %(pathname)s:%(lineno)d`

Additional data to append to log message when logging level for the message is DEBUG. Used by oslo_log.formatters.ContextFormatter

`logging_exception_prefix`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d ERROR %(name)s %(instance)s`

Prefix each line of exception output with this format. Used by oslo_log.formatters.ContextFormatter `logging_user_identity_format`

Type
string

Default
`%(user)s %(project)s %(domain)s %(system_scope)s %(user_domain)s %(project_domain)s`

Defines the format string for %(user_identity)s that is used in logging_context_format_string. Used by oslo_log.formatters.ContextFormatter

`default_log_levels`

Type
list

Default
`['amqp=WARN', 'boto=WARN', 'sqlalchemy=WARN', 'suds=INFO', 'oslo.messaging=INFO', 'oslo_messaging=INFO', 'iso8601=WARN', 'requests.packages.urllib3.connectionpool=WARN', 'urllib3.connectionpool=WARN', 'websocket=WARN', 'requests.packages.urllib3.util.retry=WARN', 'urllib3.util.retry=WARN', 'keystonemiddleware=WARN', 'routes.middleware=WARN', 'stevedore=WARN', 'taskflow=WARN', 'keystoneauth=WARN', 'oslo.cache=INFO', 'oslo_policy=INFO', 'dogpile.core.dogpile=INFO']`

List of package logging levels in logger=LEVEL pairs. This option is ignored if log_config_append is set.

`publish_errors`

Type
boolean

Default
`False`

Enables or disables publication of error events.

`instance_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance that is passed with the log message.

`instance_uuid_format`

Type
string

Default
`"[instance:  %(uuid)s] "`

The format for an instance UUID that is passed with the log message.

`rate_limit_interval`

Type
integer

Default
`0`

Interval, number of seconds, of log rate limiting.

`rate_limit_burst`

Type
integer

Default
`0`

Maximum number of logged messages per rate_limit_interval.

`rate_limit_except_level`

Type
string

Default
`CRITICAL`

Valid Values
CRITICAL, ERROR, INFO, WARNING, DEBUG,

Log level name used by rate limiting. Logs with level greater or equal to rate_limit_except_level are not filtered. An empty string means that all levels are filtered.

`fatal_deprecations`

Type
boolean

Default
`False`

Enables or disables fatal status of deprecations.

agent

`report_interval`

Type
floating point

Default
`30`

Seconds between nodes reporting state to server; should be less than agent_down_time, best if it is half or less than agent_down_time.

`log_agent_heartbeats`

Type
boolean

Default
`False`

Log agent heartbeats

cache

config_prefix

Type
string

Default
`cache.oslo`

Prefix for building the configuration dictionary for the cache region. This should not need to be changed unless there is another dogpile.cache region with the same configuration name.

expiration_time

Type
integer

Default
`600`

Minimum Value
1

Default TTL, in seconds, for any cached item in the dogpile.cache region. This applies to any cached method that doesnt have an explicit cache expiration time defined for it.

backend_expiration_time

Type
integer

Default
`<None>`

Minimum Value
1

Expiration time in cache backend to purge expired records automatically. This should be greater than expiration_time and all cache_time options

backend

Type
string

Default
`dogpile.cache.null`

Valid Values
oslo_cache.memcache_pool, oslo_cache.dict, oslo_cache.etcd3gw, dogpile.cache.pymemcache, dogpile.cache.memcached, dogpile.cache.pylibmc, dogpile.cache.bmemcached, dogpile.cache.dbm, dogpile.cache.redis, dogpile.cache.redis_sentinel, dogpile.cache.memory, dogpile.cache.memory_pickle, dogpile.cache.null

Cache backend module. For eventlet-based or environments with hundreds of threaded servers, Memcache with pooling (oslo_cache.memcache_pool) is recommended. For environments with less than 100 threaded servers, Memcached (dogpile.cache.memcached) or Redis (dogpile.cache.redis) is recommended. Test environments with a single instance of the server can use the dogpile.cache.memory backend.

`backend_argument`

Type
multi-valued

Default
`''`

Arguments supplied to the backend module. Specify this option once per argument to be passed to the dogpile.cache backend. Example format: <argname>:<value>.

`proxies`

Type
list

Default
`[]`

Proxy classes to import that will affect the way the dogpile.cache backend functions. See the dogpile.cache documentation on changing-backend-behavior.

`enabled`

Type
boolean

Default
`False`

Global toggle for caching.

`debug_cache_backend`

Type
boolean

Default
`False`

Extra debugging from the cache backend (cache keys, get/set/delete/etc calls). This is only really useful if you need to see the specific cache-backend get/set/delete calls with the keys/values. Typically this should be left set to false.

`memcache_servers`

Type
list

Default
`['localhost:11211']`

Memcache servers in the format of host:port. This is used by backends dependent on Memcached.If `dogpile.cache.memcached` or `oslo_cache.memcache_pool` is used and a given host refer to an IPv6 or a given domain refer to IPv6 then you should prefix the given address with the address family (`inet6`) (e.g `inet6:[::1]:11211`, `inet6:[fd12:3456:789a:1::1]:11211`, `inet6:[controller-0.internalapi]:11211`). If the address family is not given then these backends will use the default `inet` address family which corresponds to IPv4

`memcache_dead_retry`

Type
integer

Default
`300`

Number of seconds memcached server is considered dead before it is tried again. (dogpile.cache.memcache and oslo_cache.memcache_pool backends only).

`memcache_pool_maxsize`

Type
integer

Default
`10`

Max total number of open connections to every memcached server. (oslo_cache.memcache_pool backend only).

`memcache_pool_unused_timeout`

Type
integer

Default
`60`

Number of seconds a connection to memcached is held unused in the pool before it is closed. (oslo_cache.memcache_pool backend only).

`memcache_pool_connection_get_timeout`

Type
integer

Default
`10`

Number of seconds that an operation will wait to get a memcache client connection.

`memcache_pool_flush_on_reconnect`

Type
boolean

Default
`False`

Global toggle if memcache will be flushed on reconnect. (oslo_cache.memcache_pool backend only).

`memcache_sasl_enabled`

Type
boolean

Default
`False`

Enable the SASL(Simple Authentication and SecurityLayer) if the SASL_enable is true, else disable.

`redis_server`

Type
string

Default
`localhost:6379`

Redis server in the format of host:port

`redis_db`

Type
integer

Default
`0`

Minimum Value
0

Database id in Redis server

`redis_sentinels`

Type
list

Default
`['localhost:26379']`

Redis sentinel servers in the format of host:port

`redis_sentinel_service_name`

Type
string

Default
`mymaster`

Service name of the redis sentinel cluster.

`username`

Type
string

Default
<None>

the user name for authentication to backend.

Table 50: Deprecated Variations

Group	Name
cache	memcache_username
cache	redis_username

password

Type
string

Default
`<None>`

the password for authentication to backend.

Table 51: Deprecated Variations

Group	Name
cache	memcache_password
cache	redis_password

tls_enabled

Type
boolean

Default
`False`

Global toggle for TLS usage when communicating with the caching servers. Currently supported by `dogpile.cache.bmemcache`, `dogpile.cache.pymemcache`, `oslo_cache.memcache_pool`, `dogpile.cache.redis` and `dogpile.cache.redis_sentinel`.

tls_cafile

Type
string

Default
`<None>`

Path to a file of concatenated CA certificates in PEM format necessary to establish the caching servers authenticity. If tls_enabled is False, this option is ignored.

tls_certfile

Type
string

Default
`<None>`

Path to a single file in PEM format containing the clients certificate as well as any number of CA certificates needed to establish the certificates authenticity. This file is only required when client side authentication is necessary. If tls_enabled is False, this option is ignored.

`tls_keyfile`

Type
string

Default
`<None>`

Path to a single file containing the clients private key in. Otherwise the private key will be taken from the file specified in tls_certfile. If tls_enabled is False, this option is ignored.

`tls_allowed_ciphers`

Type
string

Default
`<None>`

Set the available ciphers for sockets created with the TLS context. It should be a string in the OpenSSL cipher list format. If not specified, all OpenSSL enabled ciphers will be available. Currently supported by `dogpile.cache.bmemcache`, `dogpile.cache.pymemcache` and `oslo_cache.memcache_pool`.

`socket_timeout`

Type
floating point

Default
`1.0`

Timeout in seconds for every call to a server. Currently supported by `dogpile.cache.memcache`, `oslo_cache.memcache_pool`, `dogpile.cache.redis` and `dogpile.cache.redis_sentinel`.

Table 52: Deprecated Variations

Group	Name
cache	memcache_socket_timeout
cache	redis_socket_timeout

`enable_socket_keepalive`

Type
boolean

Default
`False`

Global toggle for the socket keepalive of dogpiles pymemcache backend `socket_keepalive_idle`

Type
integer

Default
`1`

Minimum Value
0

The time (in seconds) the connection needs to remain idle before TCP starts sending keepalive probes. Should be a positive integer most greater than zero.

`socket_keepalive_interval`

Type
integer

Default
`1`

Minimum Value
0

The time (in seconds) between individual keepalive probes. Should be a positive integer greater than zero.

`socket_keepalive_count`

Type
integer

Default
`1`

Minimum Value
0

The maximum number of keepalive probes TCP should send before dropping the connection. Should be a positive integer greater than zero.

`enable_retry_client`

Type
boolean

Default
`False`

Enable retry client mechanisms to handle failure. Those mechanisms can be used to wrap all kind of pymemcache clients. The wrapper allows you to define how many attempts to make and how long to wait between attempts.

`retry_attempts`

Type
integer

Default
`2`

Minimum Value
1

Number of times to attempt an action before failing.

`retry_delay`

Type
floating point

Default
`0`

Number of seconds to sleep between each attempt.

`hashclient_retry_attempts`

Type
integer

Default
`2`

Minimum Value
1

Amount of times a client should be tried before it is marked dead and removed from the pool in the HashClients internal mechanisms.

`hashclient_retry_timeout`

Type
floating point

Default
`1`

Time in seconds that should pass between retry attempts in the HashClients internal mechanisms.

Table 53: Deprecated Variations

Group	Name
cache	hashclient_retry_delay

`hashclient_dead_timeout`

Type
floating point

Default
`60`

Time in seconds before attempting to add a node back in the pool in the HashClients internal mechanisms.

Table 54: Deprecated Variations

Group	Name
cache	dead_timeout

enforce_fips_mode

Type
boolean

Default
`False`

Global toggle for enforcing the OpenSSL FIPS mode. This feature requires Python support. This is available in Python 3.9 in all environments and may have been backported to older Python versions on select environments. If the Python executable used does not support OpenSSL FIPS mode, an exception will be raised. Currently supported by `dogpile.cache.bmemcache`, `dogpile.cache.pymemcache` and `oslo_cache.memcache_pool`.

Warning

This option is deprecated for removal. Its value may be silently ignored in the future.

Reason
FIPS_mode_set API was removed in OpenSSL 3.0.0. This option has no effect now.

9.1.9 Neutron Metering system

The Neutron metering service enables operators to account the traffic in/out of the OpenStack environment. The concept is quite simple, operators can create metering labels, and decide if the labels are applied to all projects (tenants) or if they are applied to a specific one. Then, the operator needs to create traffic rules in the metering labels. The traffic rules are used to match traffic in/out of the OpenStack environment, and the accounting of packets and bytes is sent to the notification queue for further processing by Ceilometer (or some other system that is consuming that queue). The message sent in the queue is of type `event`. Therefore, it requires an event processing configuration to be added/enabled in Ceilometer.

The metering agent has the following configurations:

- `driver`: the driver used to implement the metering rules. The default is `neutron.services.metering.drivers.noop`, which means, we do not execute anything in the networking host. The only driver implemented so far is `neutron.services.metering.drivers.iptables.iptables_driver.IptablesMeteringDriver`. Therefore, only `iptables` is supported so far;
- `measure_interval`: the interval in seconds used to gather the bytes and packets information from the network plane. The default value is `30` seconds;
- `report_interval`: the interval in seconds used to generated the report (message) of the data that is gathered. The default value is `300` seconds.
- `granular_traffic_data`: Defines if the metering agent driver should present traffic data in a granular fashion, instead of grouping all of the traffic data for all projects and routers where the labels were assigned to. The default value is `False` for backward compatibility.

Non-granular traffic messages

The non-granular (`granular_traffic_data = False`) traffic messages (here also called as legacy) have the following format; bear in mind that if labels are shared, then the counters are for all routers of all projects where the labels were applied.

```
{
"pkts": "<the number of packets that matched the rules of the labels>",
"bytes": "<the number of bytes that matched the rules of the labels>",
"time": "<seconds between the first data collection and the last one>",
"first_update": "timeutils.utcnow_ts() of the first collection",
"last_update": "timeutils.utcnow_ts() of the last collection",
"host": "<neutron metering agent host name>",
"label_id": "<the label id>",
"tenant_id": "<the tenant id>"
}
```

The `first_update` and `last_update` timestamps represent the moment when the first and last data collection happened within the report interval. On the other hand, the `time` represents the difference between those two timestamp.

The `tenant_id` is only consistent when labels are not shared. Otherwise, they will contain the project id of the last router of the last project processed when the agent is started up. In other words, it is better not use it when dealing with shared labels.

All of the messages generated in this configuration mode are sent to the message bus as `l3.meter` events.

Granular traffic messages

The granular (`granular_traffic_data = True`) traffic messages allow operators to obtain granular information for shared metering labels. Therefore, a single label, when configured as `shared=True` and applied in all projects/routers of the environment, it will generate data in a granular fashion.

It (the metering agent) will account the traffic counter data in the following granularities.

- `label` all of the traffic counter for a given label. One must bear in mind that a label can be assigned to multiple routers. Therefore, this granularity represents all aggregation for all data for all routers of all projects where the label has been applied.
- `router` all of the traffic counter for all labels that are assigned to the router.
- `project` all of the traffic counters for all labels of all routers that a project has.
- `router-label` all of the traffic counters for a router and the given label.
- `project-label` all of the traffic counters for all routers of a project that have a given label.

Each granularity presented here is sent to the message bus with different events types that vary according to the granularity. The mapping between granularity and event type is presented as follows.

- `label` event type `l3.meter.label`.
- `router` event type `l3.meter.router`.
- `project` event type `l3.meter.project`..
- `router-label` event type `l3.meter.label_router`.
- `project-label` event type `l3.meter.label_project`.

Furthermore, we have metadata that is appended to the messages depending on the granularity. As follows we present the mapping between the granularities and the metadata that will be available.

- `label`, `router-label`, and `project-label` granularities have the metadata `label_id`, `label_name`, `label_shared`, `project_id` (if shared, this value will come with `all` for the `label` granularity), and `router_id` (only for `router-label` granularity).
- The `router` granularity has the `router_id` and `project_id` metadata.
- The `project` granularity only has the `project_id` metadata.

The message will also contain some attributes that can be found in the legacy mode such as `bytes`, `pkts`, `time`, `first_update`, `last_update`, and `host`. As follows we present an example of JSON message with all of the possible attributes.


```
{
    "resource_id": "router-f0f745d9a59c47fdbbdd187d718f9e41-label-00c714f1-
↪49c8-462c-8f5d-f05f21e035c7",
    "project_id": "f0f745d9a59c47fdbbdd187d718f9e41",
    "first_update": 1591058790,
    "bytes": 0,
    "label_id": "00c714f1-49c8-462c-8f5d-f05f21e035c7",
    "label_name": "test1",
    "last_update": 1591059037,
    "host": "<hostname>",
    "time": 247,
    "pkts": 0,
    "label_shared": true
}
```


The `resource_id` is a unique identified for the resource being monitored. Here we consider a resource to be any of the granularities that we handle.

Sample of metering_agent.ini

As follows we present all of the possible configuration one can use in the metering agent init file.

DEFAULT

ovs_use_veth

Type
boolean

Default
`False`

Uses veth for an OVS interface or not. Support kernels with limited namespace support (e.g. RHEL 6.5) and rate limiting on routers gateway port so long as ovs_use_veth is set to True.

interface_driver

Type
string

Default
`openvswitch`

The driver used to manage virtual interfaces.

`rpc_response_max_timeout`

Type
integer

Default
`600`

Maximum seconds to wait for a response from an RPC call.

`driver`

Type
string

Default
`neutron.services.metering.drivers.noop.noop_driver.NoopMeteringDriver`

Metering driver

`measure_interval`

Type
integer

Default
`30`

Interval between two metering measures

`report_interval`

Type
integer

Default
`300`

Interval between two metering reports

`granular_traffic_data`

Type
boolean

Default
`False`

Defines if the metering agent driver should present traffic data in a granular fashion, instead of grouping all of the traffic data for all projects and routers where the labels were assigned to. The default value is *False* for backward compatibility.

`debug`

Type
boolean

Default
`False`

Mutable
This option can be changed without restarting.

If set to true, the logging level will be set to DEBUG instead of the default INFO level.

`log_config_append`

Type
string

Default
`<None>`

Mutable
This option can be changed without restarting.

The name of a logging configuration file. This file is appended to any existing logging configuration files. For details about logging configuration files, see the Python logging module documentation. Note that when logging configuration files are used then all logging configuration is set in the configuration file and other logging configuration options are ignored (for example, log-date-format).

Table 55: Deprecated Variations

Group	Name
DEFAULT	log-config
DEFAULT	log_config

`log_date_format`

Type
string

Default
`%Y-%m-%d %H:%M:%S`

Defines the format string for %(asctime)s in log records. Default: the value above . This option is ignored if log_config_append is set.

`log_file`

Type
string

Default
`<None>`

(Optional) Name of log file to send logging output to. If no default is set, logging will go to stderr as defined by use_stderr. This option is ignored if log_config_append is set.

Table 56: Deprecated Variations

Group	Name
DEFAULT	logfile

log_dir

Type
string

Default
<None>

(Optional) The base directory used for relative log_file paths. This option is ignored if log_config_append is set.

Table 57: Deprecated Variations

Group	Name
DEFAULT	logdir

use_syslog

Type
boolean

Default
False

Use syslog for logging. Existing syslog format is DEPRECATED and will be changed later to honor RFC5424. This option is ignored if log_config_append is set.

use_journal

Type
boolean

Default
False

Enable journald for logging. If running in a systemd environment you may wish to enable journal support. Doing so will use the journal native protocol which includes structured metadata in addition to log messages.This option is ignored if log_config_append is set.

syslog_log_facility

Type
string

Default
LOG_USER

Syslog facility to receive log lines. This option is ignored if log_config_append is set.

use_json

Type
boolean

Default
False

Use JSON formatting for logging. This option is ignored if log_config_append is set.

`use_stderr`

Type
boolean

Default
`False`

Log output to standard error. This option is ignored if log_config_append is set.

`log_color`

Type
boolean

Default
`False`

(Optional) Set the color key according to log levels. This option takes effect only when logging to stderr or stdout is used. This option is ignored if log_config_append is set.

`log_rotate_interval`

Type
integer

Default
`1`

The amount of time before the log files are rotated. This option is ignored unless log_rotation_type is set to interval.

`log_rotate_interval_type`

Type
string

Default
`days`

Valid Values
Seconds, Minutes, Hours, Days, Weekday, Midnight

Rotation interval type. The time of the last file change (or the time when the service was started) is used when scheduling the next rotation.

`max_logfile_count`

Type
integer

Default
`30`

Maximum number of rotated log files.

`max_logfile_size_mb`

Type
integer

Default
`200`

Log file maximum size in MB. This option is ignored if log_rotation_type is not set to size.

`log_rotation_type`

Type
string

Default
`none`

Valid Values
interval, size, none

Log rotation type.

Possible values

interval
Rotate logs at predefined time intervals.

size
Rotate logs once they reach a predefined size.

none
Do not rotate log files.

`logging_context_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [%(global_request_id)s %(request_id)s %(user_identity)s] %(instance)s%(message)s`

Format string to use for log messages with context. Used by oslo_log.formatters.ContextFormatter

`logging_default_format_string`

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d %(levelname)s %(name)s [-] %(instance)s%(message)s`

Format string to use for log messages when context is undefined. Used by oslo_log.formatters.ContextFormatter

`logging_debug_format_suffix`

Type
string

Default
`%(funcName)s %(pathname)s:%(lineno)d`

Additional data to append to log message when logging level for the message is DEBUG. Used by oslo_log.formatters.ContextFormatter

logging_exception_prefix

Type
string

Default
`%(asctime)s.%(msecs)03d %(process)d ERROR %(name)s %(instance)s`

Prefix each line of exception output with this format. Used by oslo_log.formatters.ContextFormatter

logging_user_identity_format

Type
string

Default
`%(user)s %(project)s %(domain)s %(system_scope)s %(user_domain)s %(project_domain)s`

Defines the format string for %(user_identity)s that is used in logging_context_format_string. Used by oslo_log.formatters.ContextFormatter

default_log_levels

Type
list

Default
`['amqp=WARN', 'boto=WARN', 'sqlalchemy=WARN', 'suds=INFO', 'oslo.messaging=INFO', 'oslo_messaging=INFO', 'iso8601=WARN', 'requests.packages.urllib3.connectionpool=WARN', 'urllib3.connectionpool=WARN', 'websocket=WARN', 'requests.packages.urllib3.util.retry=WARN', 'urllib3.util.retry=WARN', 'keystonemiddleware=WARN', 'routes.middleware=WARN', 'stevedore=WARN', 'taskflow=WARN', 'keystoneauth=WARN', 'oslo.cache=INFO', 'oslo_policy=INFO', 'dogpile.core.dogpile=INFO']`

List of package logging levels in logger=LEVEL pairs. This option is ignored if log_config_append is set.

publish_errors

Type
boolean

Default
`False`

Enables or disables publication of error events.

instance_format

Type
string

Default
`"[instance: %(uuid)s] "`

The format for an instance that is passed with the log message.

`instance_uuid_format`

Type
string

Default
`"[instance: %(uuid)s] "`

The format for an instance UUID that is passed with the log message.

`rate_limit_interval`

Type
integer

Default
`0`

Interval, number of seconds, of log rate limiting.

`rate_limit_burst`

Type
integer

Default
`0`

Maximum number of logged messages per rate_limit_interval.

`rate_limit_except_level`

Type
string

Default
`CRITICAL`

Valid Values
CRITICAL, ERROR, INFO, WARNING, DEBUG,

Log level name used by rate limiting. Logs with level greater or equal to rate_limit_except_level are not filtered. An empty string means that all levels are filtered.

`fatal_deprecations`

Type
boolean

Default
`False`

Enables or disables fatal status of deprecations.

agent

report_interval

Type
floating point

Default
30

Seconds between nodes reporting state to server; should be less than agent_down_time, best if it is half or less than agent_down_time.

log_agent_heartbeats

Type
boolean

Default
False

Log agent heartbeats

ovs

ovsdb_connection

Type
string

Default
tcp:127.0.0.1:6640

The connection string for the OVSDB backend. Will be used for all OVSDB commands and by ovsdb-client when monitoring

ssl_key_file

Type
string

Default
<None>

The SSL private key file to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

ssl_cert_file

Type
string

Default
<None>

The SSL certificate file to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ssl_ca_cert_file`

Type
string

Default
`<None>`

The Certificate Authority (CA) certificate to use when interacting with OVSDB. Required when using an ssl: prefixed ovsdb_connection

`ovsdb_debug`

Type
boolean

Default
`False`

Enable OVSDB debug logs

`ovsdb_timeout`

Type
integer

Default
`10`

Timeout in seconds for OVSDB commands. If the timeout expires, OVSDB commands will fail with ALARMCLOCK error.

`bridge_mac_table_size`

Type
integer

Default
`50000`

The maximum number of MAC addresses to learn on a bridge managed by the Neutron OVS agent. Values outside a reasonable range (10 to 1,000,000) might be overridden by Open vSwitch according to the documentation.

`igmp_snooping_enable`

Type
boolean

Default
`False`

Enable IGMP snooping for integration bridge. If this option is set to True, support for Internet Group Management Protocol (IGMP) is enabled in integration bridge.

`igmp_flood`

Type
boolean

Default
`False`

Multicast packets (except reports) are unconditionally forwarded to the ports bridging a logical network to a physical network.

`igmp_flood_reports`

Type
boolean

Default
`True`

Multicast reports are unconditionally forwarded to the ports bridging a logical network to a physical network.

`igmp_flood_unregistered`

Type
boolean

Default
`False`

This option enables or disables flooding of unregistered multicast packets to all ports. If False, The switch will send unregistered multicast packets only to ports connected to multicast routers.

9.2 Policy Reference

Warning

JSON formatted policy file is deprecated since Neutron 18.0.0 (Wallaby). This oslopolicy-convert-json-to-yaml tool will migrate your existing JSON-formatted policy file to YAML in a backward-compatible way.

Neutron, like most OpenStack projects, uses a policy language to restrict permissions on REST API actions.

The following is an overview of all available policies in neutron.

9.2.1 neutron

`context_is_admin`

Default
`role:admin`

Rule for cloud admin access

`context_with_global_access`

Default
`!`

Rule for context with global access to the resources `service_api`

Default
`role:service`

Default rule for the service-to-service APIs.

`owner`

Default
`project_id:%(project_id)s`

Rule for resource owner access

`admin_or_owner`

Default
`rule:context_is_admin or rule:owner`

Rule for admin or owner access

`context_is_advsvc`

Default
`role:advsvc`

Rule for advsvc role access

`admin_or_network_owner`

Default
`rule:context_is_admin or project_id:%(network:project_id)s`

Rule for admin or network owner access

`admin_owner_or_network_owner`

Default
`rule:owner or rule:admin_or_network_owner`

Rule for resource owner, admin or network owner access

`network_owner`

Default
`project_id:%(network:project_id)s`

Rule for network owner access

`admin_only`

Default
`rule:context_is_admin`

Rule for admin-only access

`regular_user`

Default
<empty string>

Rule for regular user access

`shared`

Default
`field:networks:shared=True`

Rule of shared network

`default`

Default
`rule:admin_or_owner`

Default access rule

`admin_or_ext_parent_owner`

Default
`rule:context_is_admin or project_id:%(ext_parent:project_id)s`

Rule for common parent owner check

`ext_parent_owner`

Default
`project_id:%(ext_parent:project_id)s`

Rule for common parent owner check

`sg_owner`

Default
`project_id:%(security_group:project_id)s`

Rule for security group owner access

`shared_address_groups`

Default
`field:address_groups:shared=True`

Definition of a shared address group

`get_address_group`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s) or rule:shared_address_groups`

Operations

- **GET** `/address-groups`
- **GET** `/address-groups/{id}`

Scope Types

- **project**

Get an address group

`shared_address_scopes`

Default
`field:address_scopes:shared=True`

Definition of a shared address scope

`create_address_scope`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/address-scopes`

Scope Types

- **project**

Create an address scope

`create_address_scope:shared`

Default
`rule:admin_only`

Operations

- **POST** `/address-scopes`

Scope Types

- **project**

Create a shared address scope

`get_address_scope`

Default
`rule:admin_only or role:reader and project_id:%(project_id)s or rule:shared_address_scopes`

Operations

- **GET** `/address-scopes`
- **GET** `/address-scopes/{id}`

Scope Types

- **project**

Get an address scope

`update_address_scope`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/address-scopes/{id}`

Scope Types

- **project**

Update an address scope

`update_address_scope:shared`

Default
`rule:admin_only`

Operations

- **PUT** `/address-scopes/{id}`

Scope Types

- **project**

Update `shared` attribute of an address scope

`delete_address_scope`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/address-scopes/{id}`

Scope Types

- **project**

Delete an address scope

`create_agent`

Default
`rule:admin_only`

Operations

- **POST** `/agents/{id}`

Scope Types

- **project**

Create an agent

`get_agent`

Default
`rule:admin_only`

Operations

- **GET** `/agents`
- **GET** `/agents/{id}`

Scope Types

- **project**

Get an agent

`update_agent`

Default
`rule:admin_only`

Operations

- **PUT** `/agents/{id}`

Scope Types

- **project**

Update an agent

`delete_agent`

Default
`rule:admin_only`

Operations

- **DELETE** `/agents/{id}`

Scope Types

- **project**

Delete an agent

`create_dhcp-network`

Default
`rule:admin_only`

Operations

- **POST** `/agents/{agent_id}/dhcp-networks`

Scope Types

- **project**

Add a network to a DHCP agent

`get_dhcp-networks`

Default
`rule:admin_only`

Operations

- **GET** `/agents/{agent_id}/dhcp-networks`

Scope Types

- **project**

List networks on a DHCP agent

`delete_dhcp-network`

Default
`rule:admin_only`

Operations

- **DELETE** `/agents/{agent_id}/dhcp-networks/{network_id}`

Scope Types

- **project**

Remove a network from a DHCP agent

`create_l3-router`

Default
`rule:admin_only`

Operations

- **POST** `/agents/{agent_id}/l3-routers`

Scope Types

- **project**

Add a router to an L3 agent

`get_l3-routers`

Default
`rule:admin_only`

Operations

- **GET** `/agents/{agent_id}/l3-routers`

Scope Types

- **project**

List routers on an L3 agent

`delete_l3-router`

Default
`rule:admin_only`

Operations

- **DELETE** `/agents/{agent_id}/l3-routers/{router_id}`

Scope Types

- **project**

Remove a router from an L3 agent

`get_dhcp-agents`

Default
`rule:admin_only`

Operations

- **GET** `/networks/{network_id}/dhcp-agents`

Scope Types

- **project**

List DHCP agents hosting a network

`get_l3-agents`

Default
`rule:admin_only`

Operations

- **GET** `/routers/{router_id}/l3-agents`

Scope Types

- **project**

List L3 agents hosting a router

`get_auto_allocated_topology`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/auto-allocated-topology/{project_id}`

Scope Types

- **project**

Get a projects auto-allocated topology

`delete_auto_allocated_topology`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/auto-allocated-topology/{project_id}`

Scope Types

- **project**

Delete a projects auto-allocated topology

`get_availability_zone`

Default
`role:reader`

Operations

- **GET** `/availability_zones`

Scope Types

- **project**

List availability zones

`create_default_security_group_rule`

Default
`rule:admin_only`

Operations

- **POST** `/default-security-group-rules`

Scope Types

- **project**

Create a templated of the security group rule `get_default_security_group_rule`

Default
`role:reader`

Operations

- **GET** `/default-security-group-rules`
- **GET** `/default-security-group-rules/{id}`

Scope Types

- **project**

Get a templated of the security group rule

`delete_default_security_group_rule`

Default
`rule:admin_only`

Operations

- **DELETE** `/default-security-group-rules/{id}`

Scope Types

- **project**

Delete a templated of the security group rule

`create_flavor`

Default
`rule:admin_only`

Operations

- **POST** `/flavors`

Scope Types

- **project**

Create a flavor

`get_flavor`

Default
`role:reader`

Operations

- **GET** `/flavors`
- **GET** `/flavors/{id}`

Scope Types

- **project**

Get a flavor

`update_flavor`

Default
`rule:admin_only`

Operations

- **PUT** `/flavors/{id}`

Scope Types

- **project**

Update a flavor

`delete_flavor`

Default
`rule:admin_only`

Operations

- **DELETE** `/flavors/{id}`

Scope Types

- **project**

Delete a flavor

`create_service_profile`

Default
`rule:admin_only`

Operations

- **POST** `/service_profiles`

Scope Types

- **project**

Create a service profile

`get_service_profile`

Default
`rule:admin_only`

Operations

- **GET** `/service_profiles`
- **GET** `/service_profiles/{id}`

Scope Types

- **project**

Get a service profile

`update_service_profile`

Default
`rule:admin_only`

Operations

- **PUT** `/service_profiles/{id}`

Scope Types

- **project**

Update a service profile

`delete_service_profile`

Default
`rule:admin_only`

Operations

- **DELETE** `/service_profiles/{id}`

Scope Types

- **project**

Delete a service profile

`get_flavor_service_profile`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Scope Types

- **project**

Get a flavor associated with a given service profiles. There is no corresponding GET operations in API currently. This rule is currently referred only in the DELETE of flavor_service_profile.

`create_flavor_service_profile`

Default
`rule:admin_only`

Operations

- **POST** `/flavors/{flavor_id}/service_profiles`

Scope Types

- **project**

Associate a flavor with a service profile

`delete_flavor_service_profile`

Default
`rule:admin_only`

Operations

- **DELETE** `/flavors/{flavor_id}/service_profiles/{profile_id}`

Scope Types

- **project**

Disassociate a flavor with a service profile

`create_floatingip`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/floatingips`

Scope Types

- **project**

Create a floating IP

`create_floatingip:floating_ip_address`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **POST** `/floatingips`

Scope Types

- **project**

Create a floating IP with a specific IP address

`create_floatingip:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/floatingips/{id}/tags`

Scope Types

- **project**

Create the floating IP tags

`get_floatingip`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/floatingips`
- **GET** `/floatingips/{id}`

Scope Types

- **project**

Get a floating IP

`get_floatingip:tags`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/floatingips/{id}/tags`
- **GET** `/floatingips/{id}/tags/{tag_id}`

Scope Types

- **project**

Get the floating IP tags

`update_floatingip`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/floatingips/{id}`

Scope Types

- **project**

Update a floating IP

`update_floatingip:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/floatingips/{id}/tags`
- **PUT** `/floatingips/{id}/tags/{tag_id}`

Scope Types

- **project**

Update the floating IP tags

`delete_floatingip`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/floatingips/{id}`

Scope Types

- **project**

Delete a floating IP

`delete_floatingip:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/floatingips/{id}/tags`

- **DELETE** `/floatingips/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the floating IP tags

`get_floatingip_pool`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/floatingip_pools`

Scope Types

- **project**

Get floating IP pools

`create_floatingip_port_forwarding`

Default
`(rule:admin_only) or (role:member and rule:ext_parent_owner)`

Operations

- **POST** `/floatingips/{floatingip_id}/port_forwardings`

Scope Types

- **project**

Create a floating IP port forwarding

`get_floatingip_port_forwarding`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/floatingips/{floatingip_id}/port_forwardings`
- **GET** `/floatingips/{floatingip_id}/port_forwardings/{port_forwarding_id}`

Scope Types

- **project**

Get a floating IP port forwarding

`update_floatingip_port_forwarding`

Default
`(rule:admin_only) or (role:member and rule:ext_parent_owner)`

Operations

- **PUT** `/floatingips/{floatingip_id}/port_forwardings/{port_forwarding_id}`

Scope Types

- **project**

Update a floating IP port forwarding

`delete_floatingip_port_forwarding`

Default
`(rule:admin_only) or (role:member and rule:ext_parent_owner)`

Operations

- **DELETE** `/floatingips/{floatingip_id}/port_forwardings/{port_forwarding_id}`

Scope Types

- **project**

Delete a floating IP port forwarding

`create_router_conntrack_helper`

Default
`(rule:admin_only) or (role:member and rule:ext_parent_owner)`

Operations

- **POST** `/routers/{router_id}/conntrack_helpers`

Scope Types

- **project**

Create a router conntrack helper

`get_router_conntrack_helper`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/routers/{router_id}/conntrack_helpers`
- **GET** `/routers/{router_id}/conntrack_helpers/{conntrack_helper_id}`

Scope Types

- **project**

Get a router conntrack helper

`update_router_conntrack_helper`

Default
`(rule:admin_only) or (role:member and rule:ext_parent_owner)`

Operations

- **PUT** `/routers/{router_id}/conntrack_helpers/{conntrack_helper_id}`

Scope Types

- **project**

Update a router conntrack helper

`delete_router_conntrack_helper`

Default
`(rule:admin_only) or (role:member and rule:ext_parent_owner)`

Operations

- **DELETE** `/routers/{router_id}/conntrack_helpers/{conntrack_helper_id}`

Scope Types

- **project**

Delete a router conntrack helper

`create_local_ip`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/local-ips`

Scope Types

- **project**

Create a Local IP

`get_local_ip`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/local-ips`
- **GET** `/local-ips/{id}`

Scope Types

- **project**

Get a Local IP

`update_local_ip`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/local-ips/{id}`

Scope Types

- **project**

Update a Local IP

`delete_local_ip`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/local-ips/{id}`

Scope Types

- **project**

Delete a Local IP

`create_local_ip_port_association`

Default
`(rule:admin_only) or (role:member and rule:ext_parent_owner)`

Operations

- **POST** `/local_ips/{local_ip_id}/port_associations`

Scope Types

- **project**

Create a Local IP port association

`get_local_ip_port_association`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/local_ips/{local_ip_id}/port_associations`
- **GET** `/local_ips/{local_ip_id}/port_associations/{fixed_port_id}`

Scope Types

- **project**

Get a Local IP port association

`delete_local_ip_port_association`

Default
`(rule:admin_only) or (role:member and rule:ext_parent_owner)`

Operations

- **DELETE** `/local_ips/{local_ip_id}/port_associations/{fixed_port_id}`

Scope Types

- **project**

Delete a Local IP port association

`get_loggable_resource`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **GET** `/log/loggable-resources`

Scope Types

- **project**

Get loggable resources

`create_log`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **POST** `/log/logs`

Scope Types

- **project**

Create a network log

`get_log`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **GET** `/log/logs`
- **GET** `/log/logs/{id}`

Scope Types

- **project**

Get a network log

`update_log`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **PUT** `/log/logs/{id}`

Scope Types

- **project**

Update a network log

`delete_log`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **DELETE** `/log/logs/{id}`

Scope Types

- **project**

Delete a network log

`create_metering_label`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **POST** `/metering/metering-labels`

Scope Types

- **project**

Create a metering label

`get_metering_label`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/metering/metering-labels`
- **GET** `/metering/metering-labels/{id}`

Scope Types

- **project**

Get a metering label

`delete_metering_label`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **DELETE** `/metering/metering-labels/{id}`

Scope Types

- **project**

Delete a metering label

`create_metering_label_rule`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **POST** `/metering/metering-label-rules`

Scope Types

- **project**

Create a metering label rule

`get_metering_label_rule`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/metering/metering-label-rules`
- **GET** `/metering/metering-label-rules/{id}`

Scope Types

- **project**

Get a metering label rule

`delete_metering_label_rule`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **DELETE** `/metering/metering-label-rules/{id}`

Scope Types

- **project**

Delete a metering label rule

`create_ndp_proxy`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/ndp_proxies`

Scope Types

- **project**

Create a ndp proxy

`get_ndp_proxy`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/ndp_proxies`
- **GET** `/ndp_proxies/{id}`

Scope Types

- **project**

Get a ndp proxy

`update_ndp_proxy`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/ndp_proxies/{id}`

Scope Types

- **project**

Update a ndp proxy

`delete_ndp_proxy`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/ndp_proxies/{id}`

Scope Types

- **project**

Delete a ndp proxy

`external`

Default
`field:networks:router:external=True`

Definition of an external network

`create_network`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/networks`

Scope Types

- **project**

Create a network

`create_network:shared`

Default
`rule:admin_only`

Operations

- **POST** `/networks`

Scope Types

- **project**

Create a shared network

`create_network:router:external`

Default
`rule:admin_only`

Operations

- **POST** `/networks`

Scope Types

- **project**

Create an external network

`create_network:is_default`

Default
`rule:admin_only`

Operations

- **POST** `/networks`

Scope Types

- **project**

Specify `is_default` attribute when creating a network

`create_network:port_security_enabled`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/networks`

Scope Types

- **project**

Specify `port_security_enabled` attribute when creating a network

`create_network:segments`

Default
`rule:admin_only`

Operations

- **POST** `/networks`

Scope Types

- **project**

Specify `segments` attribute when creating a network

`create_network:provider:network_type`

Default
`rule:admin_only`

Operations

- **POST** `/networks`

Scope Types

- **project**

Specify `provider:network_type` when creating a network

`create_network:provider:physical_network`

Default
`rule:admin_only`

Operations

- **POST** `/networks`

Scope Types

- **project**

Specify `provider:physical_network` when creating a network

`create_network:provider:segmentation_id`

Default
`rule:admin_only`

Operations

- **POST** `/networks`

Scope Types

- **project**

Specify `provider:segmentation_id` when creating a network

`create_network:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/networks/{id}/tags`

Scope Types

- **project**

Create the network tags

`get_network`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s) or rule:service_api or rule:shared or rule:external or rule:context_is_advsvc`

Operations

- **GET** `/networks`
- **GET** `/networks/{id}`

Scope Types

- **project**

Get a network

get_network:segments

Default
`rule:admin_only`

Operations

- **GET** `/networks`
- **GET** `/networks/{id}`

Scope Types

- **project**

Get `segments` attribute of a network

get_network:provider:network_type

Default
`rule:admin_only`

Operations

- **GET** `/networks`
- **GET** `/networks/{id}`

Scope Types

- **project**

Get `provider:network_type` attribute of a network

get_network:provider:physical_network

Default
`rule:admin_only`

Operations

- **GET** `/networks`
- **GET** `/networks/{id}`

Scope Types

- **project**

Get `provider:physical_network` attribute of a network

`get_network:provider:segmentation_id`

Default
`rule:admin_only`

Operations

- **GET** `/networks`
- **GET** `/networks/{id}`

Scope Types

- **project**

Get `provider:segmentation_id` attribute of a network

`get_network:tags`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s) or rule:shared or rule:external or rule:context_is_advsvc`

Operations

- **GET** `/networks/{id}/tags`
- **GET** `/networks/{id}/tags/{tag_id}`

Scope Types

- **project**

Get the network tags

`update_network`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/networks/{id}`

Scope Types

- **project**

Update a network

`update_network:segments`

Default
`rule:admin_only`

Operations

- **PUT** `/networks/{id}`

Scope Types

- **project**

Update `segments` attribute of a network

update_network:shared

Default
`rule:admin_only`

Operations

- **PUT** `/networks/{id}`

Scope Types

- **project**

Update `shared` attribute of a network

update_network:provider:network_type

Default
`rule:admin_only`

Operations

- **PUT** `/networks/{id}`

Scope Types

- **project**

Update `provider:network_type` attribute of a network

update_network:provider:physical_network

Default
`rule:admin_only`

Operations

- **PUT** `/networks/{id}`

Scope Types

- **project**

Update `provider:physical_network` attribute of a network

update_network:provider:segmentation_id

Default
`rule:admin_only`

Operations

- **PUT** `/networks/{id}`

Scope Types

- **project**

Update `provider:segmentation_id` attribute of a network

update_network:router:external

Default
`rule:admin_only`

Operations

- **PUT** /networks/{id}

Scope Types

- **project**

Update router:external attribute of a network

update_network:is_default

Default
rule:admin_only

Operations

- **PUT** /networks/{id}

Scope Types

- **project**

Update is_default attribute of a network

update_network:port_security_enabled

Default
(rule:admin_only) or (role:member and project_id:%(project_id)s)

Operations

- **PUT** /networks/{id}

Scope Types

- **project**

Update port_security_enabled attribute of a network

update_network:tags

Default
(rule:admin_only) or (role:member and project_id:%(project_id)s)

Operations

- **PUT** /networks/{id}/tags
- **PUT** /networks/{id}/tags/{tag_id}

Scope Types

- **project**

Update the network tags

delete_network

Default
(rule:admin_only) or (role:member and project_id:%(project_id)s)

Operations

- **DELETE** /networks/{id}

Scope Types

- **project**

Delete a network

`delete_network:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/networks/{id}/tags`
- **DELETE** `/networks/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the network tags

`get_network_ip_availability`

Default
`(rule:admin_only) or (rule:service_api)`

Operations

- **GET** `/network-ip-availabilities`
- **GET** `/network-ip-availabilities/{network_id}`

Scope Types

- **project**

Get network IP availability

`create_network_segment_range`

Default
`rule:admin_only`

Operations

- **POST** `/network_segment_ranges`

Scope Types

- **project**

Create a network segment range

`create_network_segment_range:tags`

Default
`rule:admin_only`

Operations

- **POST** `/network_segment_ranges/{id}/tags`

Scope Types

- **project**

Create the network segment range tags

`get_network_segment_range`

Default
`rule:admin_only`

Operations

- **GET** `/network_segment_ranges`
- **GET** `/network_segment_ranges/{id}`

Scope Types

- **project**

Get a network segment range

`get_network_segment_range:tags`

Default
`rule:admin_only`

Operations

- **GET** `/network_segment_ranges/{id}/tags`
- **GET** `/network_segment_ranges/{id}/tags/{tag_id}`

Scope Types

- **project**

Get the network segment range tags

`update_network_segment_range`

Default
`rule:admin_only`

Operations

- **PUT** `/network_segment_ranges/{id}`

Scope Types

- **project**

Update a network segment range

`update_network_segment_range:tags`

Default
`rule:admin_only`

Operations

- **PUT** `/network_segment_ranges/{id}/tags`
- **PUT** `/network_segment_ranges/{id}/tags/{tag_id}`

Scope Types

- **project**

Update the network segment range tags

`delete_network_segment_range`

Default
`rule:admin_only`

Operations

- **DELETE** `/network_segment_ranges/{id}`

Scope Types

- **project**

Delete a network segment range

`delete_network_segment_range:tags`

Default
`rule:admin_only`

Operations

- **DELETE** `/network_segment_ranges/{id}/tags`
- **DELETE** `/network_segment_ranges/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the network segment range tags

`get_port_binding`

Default
`(rule:admin_only) or (rule:service_api)`

Operations

- **GET** `/ports/{port_id}/bindings/`

Scope Types

- **project**

Get port binding information

`create_port_binding`

Default
`rule:service_api`

Operations

- **POST** `/ports/{port_id}/bindings/`

Scope Types

- **project**

Create port binding on the host

`delete_port_binding`

Default
`rule:service_api`

Operations

- **DELETE** `/ports/{port_id}/bindings/`

Scope Types

- **project**

Delete port binding on the host

`activate`

Default
`rule:service_api`

Operations

- **PUT** `/ports/{port_id}/bindings/{host}`

Scope Types

- **project**

Activate port binding on the host

`network_device`

Default
`field:port:device_owner=~^network:`

Definition of port with network device_owner

`admin_or_data_plane_int`

Default
`rule:context_is_admin or role:data_plane_integrator`

Rule for data plane integration

`create_port`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s) or rule:service_api`

Operations

- **POST** `/ports`

Scope Types

- **project**

Create a port

`create_port:device_id`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s) or rule:service_api`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `device_id` attribute when creating a port

`create_port:device_owner`

Default
`not rule:network_device or (rule:admin_only) or (rule:service_api) or role:member and rule:network_owner`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `device_owner` attribute when creating a port

`create_port:mac_address`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `mac_address` attribute when creating a port

`create_port:fixed_ips`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner or rule:shared`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `fixed_ips` information when creating a port

`create_port:fixed_ips:ip_address`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify IP address in `fixed_ips` when creating a port `create_port:fixed_ips:subnet_id`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner or rule:shared`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify subnet ID in `fixed_ips` when creating a port

`create_port:port_security_enabled`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `port_security_enabled` attribute when creating a port

`create_port:binding:host_id`

Default
`(rule:admin_only) or (rule:service_api)`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `binding:host_id` attribute when creating a port

`create_port:binding:profile`

Default
`rule:service_api`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `binding:profile` attribute when creating a port

`create_port:binding:vnic_type`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s) or rule:service_api`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `binding:vnic_type` attribute when creating a port

`create_port:allowed_address_pairs`

Default
`(rule:admin_only) or (role:member and rule:network_owner) or rule:service_api`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `allowed_address_pairs` attribute when creating a port

`create_port:allowed_address_pairs:mac_address`

Default
`(rule:admin_only) or (role:member and rule:network_owner) or rule:service_api`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify ``mac_address` of `allowed_address_pairs`` attribute when creating a port

`create_port:allowed_address_pairs:ip_address`

Default
`(rule:admin_only) or (role:member and rule:network_owner) or rule:service_api`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `ip_address` of `allowed_address_pairs` attribute when creating a port

`create_port:hints`

Default
`rule:admin_only`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `hints` attribute when creating a port

`create_port:trusted`

Default
`rule:admin_only`

Operations

- **POST** `/ports`

Scope Types

- **project**

Specify `trusted` attribute when creating a port

`create_port:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s) or rule:context_is_advsvc`

Operations

- **POST** `/ports/{id}/tags`

Scope Types

- **project**

Create the port tags

`get_port`

Default
`(rule:admin_only) or (rule:service_api) or role:reader and rule:network_owner or role:reader and project_id:%(project_id)s`

Operations

- **GET** `/ports`
- **GET** `/ports/{id}`

Scope Types

- **project**

Get a port

`get_port:binding:vif_type`

Default
`(rule:admin_only) or (rule:service_api)`

Operations

- **GET** `/ports`
- **GET** `/ports/{id}`

Scope Types

- **project**

Get `binding:vif_type` attribute of a port

`get_port:binding:vif_details`

Default
`(rule:admin_only) or (rule:service_api)`

Operations

- **GET** `/ports`
- **GET** `/ports/{id}`

Scope Types

- **project**

Get `binding:vif_details` attribute of a port

`get_port:binding:host_id`

Default
`(rule:admin_only) or (rule:service_api)`

Operations

- **GET** `/ports`
- **GET** `/ports/{id}`

Scope Types

- **project**

Get `binding:host_id` attribute of a port

`get_port:binding:profile`

Default
`(rule:admin_only) or (rule:service_api)`

Operations

- **GET** `/ports`
- **GET** `/ports/{id}`

Scope Types

- **project**

Get `binding:profile` attribute of a port

`get_port:resource_request`

Default
`rule:admin_only`

Operations

- **GET** `/ports`
- **GET** `/ports/{id}`

Scope Types

- **project**

Get `resource_request` attribute of a port

`get_port:hints`

Default
`rule:admin_only`

Operations

- **GET** `/ports`
- **GET** `/ports/{id}`

Scope Types

- **project**

Get `hints` attribute of a port

`get_port:trusted`

Default
`rule:admin_only`

Operations

- **GET** `/ports`
- **GET** `/ports/{id}`

Scope Types

- **project**

Get `trusted` attribute of a port

`get_port:tags`

Default
`rule:context_is_advsvc or (rule:admin_only) or (role:reader and rule:network_owner) or role:reader and project_id:%(project_id)s`

Operations

- **GET** `/ports/{id}/tags`
- **GET** `/ports/{id}/tags/{tag_id}`

Scope Types

- **project**

Get the port tags

`update_port`

Default
`(rule:admin_only) or (rule:service_api) or role:member and project_id:%(project_id)s`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Update a port

`update_port:device_id`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s) or rule:service_api`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Update `device_id` attribute of a port

`update_port:device_owner`

Default
`not rule:network_device or (rule:admin_only) or (rule:service_api) or role:member and rule:network_owner`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Update `device_owner` attribute of a port

`update_port:mac_address`

Default
`(rule:admin_only) or (rule:service_api) or role:manager and rule:network_owner`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Update `mac_address` attribute of a port `update_port:fixed_ips`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Specify `fixed_ips` information when updating a port

`update_port:fixed_ips:ip_address`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Specify IP address in `fixed_ips` information when updating a port

`update_port:fixed_ips:subnet_id`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner or rule:shared`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Specify subnet ID in `fixed_ips` information when updating a port

`update_port:port_security_enabled`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Update `port_security_enabled` attribute of a port

`update_port:binding:host_id`

Default
`(rule:admin_only) or (rule:service_api)`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Update `binding:host_id` attribute of a port

`update_port:binding:profile`

Default
`rule:service_api`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Update `binding:profile` attribute of a port

`update_port:binding:vnic_type`

Default
`(rule:admin_only) or (rule:service_api) or role:member and project_id:%(project_id)s`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Update `binding:vnic_type` attribute of a port

`update_port:allowed_address_pairs`

Default
`(rule:admin_only) or (role:member and rule:network_owner) or rule:service_api`

Operations

- **PUT** `/ports/{id}`

Scope Types

- **project**

Update `allowed_address_pairs` attribute of a port

`update_port:allowed_address_pairs:mac_address`

Default
`(rule:admin_only) or (role:member and rule:network_owner) or rule:service_api`

Operations

- **PUT** /ports/{id}

Scope Types

- **project**

Update mac_address of allowed_address_pairs attribute of a port

`update_port:allowed_address_pairs:ip_address`

Default
`(rule:admin_only) or (role:member and rule:network_owner) or rule:service_api`

Operations

- **PUT** /ports/{id}

Scope Types

- **project**

Update ip_address of allowed_address_pairs attribute of a port

`update_port:data_plane_status`

Default
`rule:admin_only or role:data_plane_integrator`

Operations

- **PUT** /ports/{id}

Scope Types

- **project**

Update data_plane_status attribute of a port

`update_port:hints`

Default
`rule:admin_only`

Operations

- **PUT** /ports/{id}

Scope Types

- **project**

Update hints attribute of a port

`update_port:trusted`

Default
`rule:admin_only`

Operations

- **PUT** /ports/{id}

Scope Types

- **project**

Update `trusted` attribute of a port

`update_port:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s) or rule:context_is_advsvc`

Operations

- **PUT** `/ports/{id}/tags`
- **PUT** `/ports/{id}/tags/{tag_id}`

Scope Types

- **project**

Update the port tags

`delete_port`

Default
`(rule:admin_only) or (rule:service_api) or role:member and rule:network_owner or role:member and project_id:%(project_id)s`

Operations

- **DELETE** `/ports/{id}`

Scope Types

- **project**

Delete a port

`delete_port:tags`

Default
`rule:context_is_advsvc or role:member and project_id:%(project_id)s or (rule:admin_only) or (role:member and rule:network_owner)`

Operations

- **DELETE** `/ports/{id}/tags`
- **DELETE** `/ports/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the port tags

`shared_qos_policy`

Default
`field:policies:shared=True`

Rule of shared qos policy

`get_policy`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s) or rule:shared_qos_policy`

Operations

- **GET** `/qos/policies`
- **GET** `/qos/policies/{id}`

Scope Types

- **project**

Get QoS policies

`get_policy:tags`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s) or rule:shared_qos_policy`

Operations

- **GET** `/qos/policies/{id}/tags`
- **GET** `/qos/policies/{id}/tags/{tag_id}`

Scope Types

- **project**

Get QoS policy tags

`create_policy`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **POST** `/qos/policies`

Scope Types

- **project**

Create a QoS policy

`create_policy:tags`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **POST** `/qos/policies/{id}/tags`

Scope Types

- **project**

Create the QoS policy tags

`update_policy`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **PUT** `/qos/policies/{id}`

Scope Types

- **project**

Update a QoS policy

`update_policy:tags`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **PUT** `/qos/policies/{id}/tags`
- **PUT** `/qos/policies/{id}/tags/{tag_id}`

Scope Types

- **project**

Update the QoS policy tags

`delete_policy`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **DELETE** `/qos/policies/{id}`

Scope Types

- **project**

Delete a QoS policy

`delete_policy:tags`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **DELETE** `/qos/policies/{id}/tags`
- **DELETE** `/qos/policies/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the QoS policy tags

`get_rule_type`

Default
`role:reader`

Operations

- **GET** `/qos/rule-types`
- **GET** `/qos/rule-types/{rule_type}`

Scope Types

- **project**

Get available QoS rule types

`get_policy_bandwidth_limit_rule`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/qos/policies/{policy_id}/bandwidth_limit_rules`
- **GET** `/qos/policies/{policy_id}/bandwidth_limit_rules/{rule_id}`

Scope Types

- **project**

Get a QoS bandwidth limit rule

`create_policy_bandwidth_limit_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **POST** `/qos/policies/{policy_id}/bandwidth_limit_rules`

Scope Types

- **project**

Create a QoS bandwidth limit rule

`update_policy_bandwidth_limit_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **PUT** `/qos/policies/{policy_id}/bandwidth_limit_rules/{rule_id}`

Scope Types

- **project**

Update a QoS bandwidth limit rule

`delete_policy_bandwidth_limit_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **DELETE** `/qos/policies/{policy_id}/bandwidth_limit_rules/{rule_id}`

Scope Types

- **project**

Delete a QoS bandwidth limit rule

`get_policy_packet_rate_limit_rule`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/qos/policies/{policy_id}/packet_rate_limit_rules`
- **GET** `/qos/policies/{policy_id}/packet_rate_limit_rules/{rule_id}`

Scope Types

- **project**

Get a QoS packet rate limit rule

`create_policy_packet_rate_limit_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **POST** `/qos/policies/{policy_id}/packet_rate_limit_rules`

Scope Types

- **project**

Create a QoS packet rate limit rule

`update_policy_packet_rate_limit_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **PUT** `/qos/policies/{policy_id}/packet_rate_limit_rules/{rule_id}`

Scope Types

- **project**

Update a QoS packet rate limit rule

`delete_policy_packet_rate_limit_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **DELETE** `/qos/policies/{policy_id}/packet_rate_limit_rules/{rule_id}`

Scope Types

- **project**

Delete a QoS packet rate limit rule

`get_policy_dscp_marking_rule`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/qos/policies/{policy_id}/dscp_marking_rules`
- **GET** `/qos/policies/{policy_id}/dscp_marking_rules/{rule_id}`

Scope Types

- **project**

Get a QoS DSCP marking rule

`create_policy_dscp_marking_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **POST** `/qos/policies/{policy_id}/dscp_marking_rules`

Scope Types

- **project**

Create a QoS DSCP marking rule

`update_policy_dscp_marking_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **PUT** `/qos/policies/{policy_id}/dscp_marking_rules/{rule_id}`

Scope Types

- **project**

Update a QoS DSCP marking rule

`delete_policy_dscp_marking_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **DELETE** `/qos/policies/{policy_id}/dscp_marking_rules/{rule_id}`

Scope Types

- **project**

Delete a QoS DSCP marking rule

`get_policy_minimum_bandwidth_rule`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/qos/policies/{policy_id}/minimum_bandwidth_rules`
- **GET** `/qos/policies/{policy_id}/minimum_bandwidth_rules/{rule_id}`

Scope Types

- **project**

Get a QoS minimum bandwidth rule

`create_policy_minimum_bandwidth_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **POST** `/qos/policies/{policy_id}/minimum_bandwidth_rules`

Scope Types

- **project**

Create a QoS minimum bandwidth rule

`update_policy_minimum_bandwidth_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **PUT** `/qos/policies/{policy_id}/minimum_bandwidth_rules/{rule_id}`

Scope Types

- **project**

Update a QoS minimum bandwidth rule

`delete_policy_minimum_bandwidth_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **DELETE** `/qos/policies/{policy_id}/minimum_bandwidth_rules/{rule_id}`

Scope Types

- **project**

Delete a QoS minimum bandwidth rule

`get_policy_minimum_packet_rate_rule`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/qos/policies/{policy_id}/minimum_packet_rate_rules`
- **GET** `/qos/policies/{policy_id}/minimum_packet_rate_rules/{rule_id}`

Scope Types

- **project**

Get a QoS minimum packet rate rule

`create_policy_minimum_packet_rate_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **POST** `/qos/policies/{policy_id}/minimum_packet_rate_rules`

Scope Types

- **project**

Create a QoS minimum packet rate rule

`update_policy_minimum_packet_rate_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **PUT** `/qos/policies/{policy_id}/minimum_packet_rate_rules/{rule_id}`

Scope Types

- **project**

Update a QoS minimum packet rate rule

`delete_policy_minimum_packet_rate_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **DELETE** `/qos/policies/{policy_id}/minimum_packet_rate_rules/{rule_id}`

Scope Types

- **project**

Delete a QoS minimum packet rate rule

`get_alias_bandwidth_limit_rule`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/qos/alias_bandwidth_limit_rules/{rule_id}/`

Scope Types

- **project**

Get a QoS bandwidth limit rule through alias

`update_alias_bandwidth_limit_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **PUT** `/qos/alias_bandwidth_limit_rules/{rule_id}/`

Scope Types

- **project**

Update a QoS bandwidth limit rule through alias

`delete_alias_bandwidth_limit_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **DELETE** `/qos/alias_bandwidth_limit_rules/{rule_id}/`

Scope Types

- **project**

Delete a QoS bandwidth limit rule through alias

`get_alias_dscp_marking_rule`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/qos/alias_dscp_marking_rules/{rule_id}/`

Scope Types

- **project**

Get a QoS DSCP marking rule through alias

`update_alias_dscp_marking_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **PUT** `/qos/alias_dscp_marking_rules/{rule_id}/`

Scope Types

- **project**

Update a QoS DSCP marking rule through alias

`delete_alias_dscp_marking_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **DELETE** `/qos/alias_dscp_marking_rules/{rule_id}/`

Scope Types

- **project**

Delete a QoS DSCP marking rule through alias

`get_alias_minimum_bandwidth_rule`

Default
`(rule:admin_only) or (role:reader and rule:ext_parent_owner)`

Operations

- **GET** `/qos/alias_minimum_bandwidth_rules/{rule_id}/`

Scope Types

- **project**

Get a QoS minimum bandwidth rule through alias

`update_alias_minimum_bandwidth_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **PUT** `/qos/alias_minimum_bandwidth_rules/{rule_id}/`

Scope Types

- **project**

Update a QoS minimum bandwidth rule through alias `delete_alias_minimum_bandwidth_rule`

Default
`(rule:admin_only) or (role:manager and rule:ext_parent_owner)`

Operations

- **DELETE** `/qos/alias_minimum_bandwidth_rules/{rule_id}/`

Scope Types

- **project**

Delete a QoS minimum bandwidth rule through alias

`get_alias_minimum_packet_rate_rule`

Default
`rule:get_policy_minimum_packet_rate_rule`

Operations

- **GET** `/qos/alias_minimum_packet_rate_rules/{rule_id}/`

Scope Types

- **project**

Get a QoS minimum packet rate rule through alias

`update_alias_minimum_packet_rate_rule`

Default
`rule:update_policy_minimum_packet_rate_rule`

Operations

- **PUT** `/qos/alias_minimum_packet_rate_rules/{rule_id}/`

Scope Types

- **project**

Update a QoS minimum packet rate rule through alias

`delete_alias_minimum_packet_rate_rule`

Default
`rule:delete_policy_minimum_packet_rate_rule`

Operations

- **DELETE** `/qos/alias_minimum_packet_rate_rules/{rule_id}/`

Scope Types

- **project**

Delete a QoS minimum packet rate rule through alias

`get_quota`

Default
`(rule:admin_only) or (role:manager and project_id:%(project_id)s)`

Operations

- **GET** `/quota`
- **GET** `/quota/{id}`

Scope Types

- **project**

Get a resource quota

`update_quota`

Default
`rule:admin_only`

Operations

- **PUT** `/quota/{id}`

Scope Types

- **project**

Update a resource quota

`delete_quota`

Default
`rule:admin_only`

Operations

- **DELETE** `/quota/{id}`

Scope Types

- **project**

Delete a resource quota

`restrict_wildcard`

Default
`(not field:rbac_policy:target_tenant=* and not field:rbac_policy:target_project=*) or rule:admin_only`

Definition of a wildcard target_project

`create_rbac_policy`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/rbac-policies`

Scope Types

- **project**

Create an RBAC policy

`create_rbac_policy:target_tenant`

Default
`rule:admin_only or (not field:rbac_policy:target_tenant=* and not field:rbac_policy:target_project=*)`

Operations

- **POST** `/rbac-policies`

Scope Types

- **project**

Specify `target_tenant` when creating an RBAC policy

`create_rbac_policy:target_project`

Default
`rule:admin_only or not field:rbac_policy:target_project=*`

Operations

- **POST** `/rbac-policies`

Scope Types

- **project**

Specify `target_project` when creating an RBAC policy

`update_rbac_policy`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/rbac-policies/{id}`

Scope Types

- **project**

Update an RBAC policy

`update_rbac_policy:target_tenant`

Default
`rule:admin_only or (not field:rbac_policy:target_tenant=* and not field:rbac_policy:target_project=*)`

Operations

- **PUT** `/rbac-policies/{id}`

Scope Types

- **project**

Update `target_tenant` attribute of an RBAC policy

`update_rbac_policy:target_project`

Default
`rule:admin_only or not field:rbac_policy:target_project=*`

Operations

- **PUT** /rbac-policies/{id}

Scope Types

- **project**

Update `target_project` attribute of an RBAC policy

`get_rbac_policy`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** /rbac-policies
- **GET** /rbac-policies/{id}

Scope Types

- **project**

Get an RBAC policy

`delete_rbac_policy`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** /rbac-policies/{id}

Scope Types

- **project**

Delete an RBAC policy

`create_router`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** /routers

Scope Types

- **project**

Create a router

`create_router:distributed`

Default
`rule:admin_only`

Operations

- **POST** /routers

Scope Types

- **project**

Specify `distributed` attribute when creating a router

`create_router:ha`

Default
`rule:admin_only`

Operations

- **POST** `/routers`

Scope Types

- **project**

Specify `ha` attribute when creating a router

`create_router:external_gateway_info`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/routers`

Scope Types

- **project**

Specify `external_gateway_info` information when creating a router

`create_router:external_gateway_info:network_id`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/routers`

Scope Types

- **project**

Specify `network_id` in `external_gateway_info` information when creating a router

`create_router:external_gateway_info:enable_snat`

Default
`rule:admin_only`

Operations

- **POST** `/routers`

Scope Types

- **project**

Specify `enable_snat` in `external_gateway_info` information when creating a router

`create_router:external_gateway_info:external_fixed_ips`

Default
`rule:admin_only`

Operations

- **POST** /routers

Scope Types

- **project**

Specify external_fixed_ips in external_gateway_info information when creating a router

`create_router:enable_default_route_bfd`

Default
`rule:admin_only`

Operations

- **POST** /routers

Scope Types

- **project**

Specify enable_default_route_bfd attribute when creating a router

`create_router:enable_default_route_ecmp`

Default
`rule:admin_only`

Operations

- **POST** /routers

Scope Types

- **project**

Specify enable_default_route_ecmp attribute when creating a router

`create_router:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** /routers/{id}/tags

Scope Types

- **project**

Create the router tags

`get_router`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** /routers
- **GET** /routers/{id}

Scope Types

- **project**

Get a router

`get_router:distributed`

Default
`rule:admin_only`

Operations

- **GET** `/routers`
- **GET** `/routers/{id}`

Scope Types

- **project**

Get `distributed` attribute of a router

`get_router:ha`

Default
`rule:admin_only`

Operations

- **GET** `/routers`
- **GET** `/routers/{id}`

Scope Types

- **project**

Get `ha` attribute of a router

`get_router:tags`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/routers/{id}/tags`
- **GET** `/routers/{id}/tags/{tag_id}`

Scope Types

- **project**

Get the router tags

`update_router`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update a router

update_router:distributed

Default
`rule:admin_only`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update `distributed` attribute of a router

update_router:ha

Default
`rule:admin_only`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update `ha` attribute of a router

update_router:external_gateway_info

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update `external_gateway_info` information of a router

update_router:external_gateway_info:network_id

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update `network_id` attribute of `external_gateway_info` information of a router

update_router:external_gateway_info:enable_snat

Default
`rule:admin_only`

Operations

- **PUT** /routers/{id}

Scope Types

- **project**

Update enable_snat attribute of external_gateway_info information of a router

update_router:external_gateway_info:external_fixed_ips

Default
rule:admin_only

Operations

- **PUT** /routers/{id}

Scope Types

- **project**

Update external_fixed_ips attribute of external_gateway_info information of a router

update_router:enable_default_route_bfd

Default
rule:admin_only

Operations

- **PUT** /routers/{id}

Scope Types

- **project**

Specify enable_default_route_bfd attribute when updating a router

update_router:enable_default_route_ecmp

Default
rule:admin_only

Operations

- **PUT** /routers/{id}

Scope Types

- **project**

Specify enable_default_route_ecmp attribute when updating a router

update_router:tags

Default
(rule:admin_only) or (role:member and project_id:%(project_id)s)

Operations

- **PUT** /routers/{id}/tags
- **PUT** /routers/{id}/tags/{tag_id}

Scope Types

- **project**

Update the router tags

`delete_router`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/routers/{id}`

Scope Types

- **project**

Delete a router

`delete_router:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/routers/{id}/tags`
- **DELETE** `/routers/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the router tags

`add_router_interface`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}/add_router_interface`

Scope Types

- **project**

Add an interface to a router

`remove_router_interface`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}/remove_router_interface`

Scope Types

- **project**

Remove an interface from a router

`add_extraroutes`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}/add_extraroutes`

Scope Types

- **project**

Add extra route to a router

`remove_extraroutes`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}/remove_extraroutes`

Scope Types

- **project**

Remove extra route from a router

`add_external_gateways`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Add router external gateways

`add_external_gateways:external_gateways`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Add router external gateways

`add_external_gateways:external_gateways:network_id`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Add router external gateways with defined network ID

`add_external_gateways:external_gateways:enable_snat`

Default
`rule:admin_only`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Add router external gateways specifying SNAT flag

`add_external_gateways:external_gateways:external_fixed_ips`

Default
`rule:admin_only`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Add router external gateways specifying the fixed IPs

`update_external_gateways`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update router external gateways

`update_external_gateways:external_gateways`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update router external gateways

`update_external_gateways:external_gateways:network_id`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update router external gateways network ID

`update_external_gateways:external_gateways:enable_snat`

Default
`rule:admin_only`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update router external gateways SNAT flag

`update_external_gateways:external_gateways:external_fixed_ips`

Default
`rule:admin_only`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Update router external gateways fixed IPs

`remove_external_gateways`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Remove router external gateways

`remove_external_gateways:external_gateways`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/routers/{id}`

Scope Types

- **project**

Remove router external gateways

`admin_or_sg_owner`

Default
`rule:context_is_admin or project_id:%(security_group:project_id)s`

Rule for admin or security group owner access

`admin_owner_or_sg_owner`

Default
`rule:owner or rule:admin_or_sg_owner`

Rule for resource owner, admin or security group owner access

`shared_security_group`

Default
`field:security_groups:shared=True`

Definition of a shared security group

`rule_default_sg`

Default
`field:security_group_rules:belongs_to_default_sg=True`

Definition of a security group rule that belongs to the project default security group

`create_security_group`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/security-groups`

Scope Types

- **project**

Create a security group

`create_security_group:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/security-groups/{id}/tags`

Scope Types

- **project**

Create the security group tags

`get_security_group`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s) or rule:shared_security_group`

Operations

- **GET** `/security-groups`
- **GET** `/security-groups/{id}`

Scope Types

- **project**

Get a security group

`get_security_group:tags`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s) or rule:shared_security_group`

Operations

- **GET** `/security-groups/{id}/tags`
- **GET** `/security-groups/{id}/tags/{tag_id}`

Scope Types

- **project**

Get the security group tags

`update_security_group`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/security-groups/{id}`

Scope Types

- **project**

Update a security group

`update_security_group:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/security-groups/{id}/tags`
- **PUT** `/security-groups/{id}/tags/{tag_id}`

Scope Types

- **project**

Update the security group tags

`delete_security_group`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/security-groups/{id}`

Scope Types

- **project**

Delete a security group

`delete_security_group:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/security-groups/{id}/tags`
- **DELETE** `/security-groups/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the security group tags

`create_security_group_rule`

Default
`(rule:admin_only) or (role:member and rule:sg_owner)`

Operations

- **POST** `/security-group-rules`

Scope Types

- **project**

Create a security group rule

`get_security_group_rule`

Default
`(rule:admin_only) or (role:reader and rule:sg_owner)`

Operations

- **GET** `/security-group-rules`
- **GET** `/security-group-rules/{id}`

Scope Types

- **project**

Get a security group rule

`delete_security_group_rule`

Default
`(rule:admin_only) or (role:member and rule:sg_owner)`

Operations

- **DELETE** `/security-group-rules/{id}`

Scope Types

- **project**

Delete a security group rule

`create_segment`

Default
`rule:admin_only`

Operations

- **POST** `/segments`

Scope Types

- **project**

Create a segment

`create_segments_tags`

Default
`rule:admin_only`

Operations

- **POST** `/segments/{id}/tags`

Scope Types

- **project**

Create the segment tags

`get_segment`

Default
`rule:admin_only`

Operations

- **GET** `/segments`
- **GET** `/segments/{id}`

Scope Types

- **project**

Get a segment

`get_segments_tags`

Default
`rule:admin_only`

Operations

- **GET** `/segments/{id}/tags`
- **GET** `/segments/{id}/tags/{tag_id}`

Scope Types

- **project**

Get the segment tags

`update_segment`

Default
`rule:admin_only`

Operations

- **PUT** `/segments/{id}`

Scope Types

- **project**

Update a segment

`update_segments_tags`

Default
`rule:admin_only`

Operations

- **PUT** `/segments/{id}/tags`
- **PUT** `/segments/{id}/tags/{tag_id}`

Scope Types

- **project**

Update the segment tags

`delete_segment`

Default
`rule:admin_only`

Operations

- **DELETE** `/segments/{id}`

Scope Types

- **project**

Delete a segment

`delete_segments_tags`

Default
`rule:admin_only`

Operations

- **DELETE** `/segments/{id}/tags`
- **DELETE** `/segments/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the segment tags

`get_service_provider`

Default
`role:reader`

Operations

- **GET** `/service-providers`

Scope Types

- **project**

Get service providers

`external_network`

Default
`field:subnets:router:external=True`

Definition of a subnet that belongs to an external network

`create_subnet`

Default
`(rule:admin_only) or (role:member and rule:network_owner)`

Operations

- **POST** `/subnets`

Scope Types

- **project**

Create a subnet

`create_subnet:segment_id`

Default
`rule:admin_only`

Operations

- **POST** `/subnets`

Scope Types

- **project**

Specify `segment_id` attribute when creating a subnet

`create_subnet:service_types`

Default
`rule:admin_only`

Operations

- **POST** `/subnets`

Scope Types

- **project**

Specify `service_types` attribute when creating a subnet

`create_subnet:tags`

Default
`role:member and project_id:%(project_id)s or (rule:admin_only) or (role:member and rule:network_owner)`

Operations

- **POST** `/subnets/{id}/tags`

Scope Types

- **project**

Create the subnet tags

`get_subnet`

Default
`role:reader and project_id:%(project_id)s or rule:shared or rule:external_network or (rule:admin_only) or (role:reader and rule:network_owner) or rule:service_api`

Operations

- **GET** `/subnets`
- **GET** `/subnets/{id}`

Scope Types

- **project**

Get a subnet

`get_subnet:segment_id`

Default
`rule:admin_only`

Operations

- **GET** `/subnets`
- **GET** `/subnets/{id}`

Scope Types

- **project**

Get `segment_id` attribute of a subnet

`get_subnet:tags`

Default
`role:reader and project_id:%(project_id)s or rule:shared or rule:external_network or (rule:admin_only) or (role:reader and rule:network_owner)`

Operations

- **GET** `/subnets/{id}/tags`
- **GET** `/subnets/{id}/tags/{tag_id}`

Scope Types

- **project**

Get the subnet tags

`update_subnet`

Default
`role:member and project_id:%(project_id)s or (rule:admin_only) or (role:member and rule:network_owner)`

Operations

- **PUT** `/subnets/{id}`

Scope Types

- **project**

Update a subnet

`update_subnet:segment_id`

Default
`rule:admin_only`

Operations

- **PUT** `/subnets/{id}`

Scope Types

- **project**

Update `segment_id` attribute of a subnet

`update_subnet:service_types`

Default
`rule:admin_only`

Operations

- **PUT** `/subnets/{id}`

Scope Types

- **project**

Update `service_types` attribute of a subnet

`update_subnet:tags`

Default
`role:member and project_id:%(project_id)s or (rule:admin_only) or (role:member and rule:network_owner)`

Operations

- **PUT** `/subnets/{id}/tags`
- **PUT** `/subnets/{id}/tags/{tag_id}`

Scope Types

- **project**

Update the subnet tags

`delete_subnet`

Default
`role:member and project_id:%(project_id)s or (rule:admin_only) or (role:member and rule:network_owner)`

Operations

- **DELETE** `/subnets/{id}`

Scope Types

- **project**

Delete a subnet

`delete_subnet:tags`

Default
`role:member and project_id:%(project_id)s or (rule:admin_only) or (role:member and rule:network_owner)`

Operations

- **DELETE** `/subnets/{id}/tags`
- **DELETE** `/subnets/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the subnet tags

`shared_subnetpools`

Default
`field:subnetpools:shared=True`

Definition of a shared subnetpool

`create_subnetpool`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/subnetpools`

Scope Types

- **project**

Create a subnetpool

create_subnetpool:shared

Default
`rule:admin_only`

Operations

- **POST** `/subnetpools`

Scope Types

- **project**

Create a shared subnetpool

create_subnetpool:is_default

Default
`rule:admin_only`

Operations

- **POST** `/subnetpools`

Scope Types

- **project**

Specify `is_default` attribute when creating a subnetpool

create_subnetpool:tags

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** `/subnetpools/{id}/tags`

Scope Types

- **project**

Create the subnetpool tags

get_subnetpool

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s) or rule:shared_subnetpools`

Operations

- **GET** `/subnetpools`
- **GET** `/subnetpools/{id}`

Scope Types

- **project**

Get a subnetpool

get_subnetpool:tags

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s) or rule:shared_subnetpools`

Operations

- **GET** `/subnetpools/{id}/tags`
- **GET** `/subnetpools/{id}/tags/{tag_id}`

Scope Types

- **project**

Get the subnetpool tags

`update_subnetpool`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/subnetpools/{id}`

Scope Types

- **project**

Update a subnetpool

`update_subnetpool:is_default`

Default
`rule:admin_only`

Operations

- **PUT** `/subnetpools/{id}`

Scope Types

- **project**

Update `is_default` attribute of a subnetpool

`update_subnetpool:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/subnetpools/{id}/tags`
- **PUT** `/subnetpools/{id}/tags/{tag_id}`

Scope Types

- **project**

Update the subnetpool tags

`delete_subnetpool`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/subnetpools/{id}`

Scope Types

- **project**

Delete a subnetpool

`delete_subnetpool:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/subnetpools/{id}/tags`
- **DELETE** `/subnetpools/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete the subnetpool tags

`onboard_network_subnets`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/subnetpools/{id}/onboard_network_subnets`

Scope Types

- **project**

Onboard existing subnet into a subnetpool

`add_prefixes`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/subnetpools/{id}/add_prefixes`

Scope Types

- **project**

Add prefixes to a subnetpool

`remove_prefixes`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** /subnetpools/{id}/remove_prefixes

Scope Types

- **project**

Remove unallocated prefixes from a subnetpool

`create_trunk`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** /trunks

Scope Types

- **project**

Create a trunk

`create_trunk:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **POST** /trunks/{id}/tags

Scope Types

- **project**

Create the trunk tags

`get_trunk`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** /trunks
- **GET** /trunks/{id}

Scope Types

- **project**

Get a trunk

`get_trunk:tags`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** /trunks/{id}/tags
- **GET** /trunks/{id}/tags/{tag_id}

Scope Types

- **project**

Get the trunk tags

`update_trunk`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/trunks/{id}`

Scope Types

- **project**

Update a trunk

`update_trunk:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/trunks/{id}/tags`
- **PUT** `/trunks/{id}/tags/{tag_id}`

Scope Types

- **project**

Update the trunk tags

`delete_trunk`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/trunks/{id}`

Scope Types

- **project**

Delete a trunk

`delete_trunk:tags`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **DELETE** `/trunks/{id}/tags`
- **DELETE** `/trunks/{id}/tags/{tag_id}`

Scope Types

- **project**

Delete a trunk

`get_subports`

Default
`(rule:admin_only) or (role:reader and project_id:%(project_id)s)`

Operations

- **GET** `/trunks/{id}/get_subports`

Scope Types

- **project**

List subports attached to a trunk

`add_subports`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/trunks/{id}/add_subports`

Scope Types

- **project**

Add subports to a trunk

`remove_subports`

Default
`(rule:admin_only) or (role:member and project_id:%(project_id)s)`

Operations

- **PUT** `/trunks/{id}/remove_subports`

Scope Types

- **project**

Delete subports from a trunk

9.3 Custom Policy Roles

Besides the *default policy roles*, Neutron also supports using custom roles. Using custom roles with for example read only access to all of the resources requires to configure the policy rule which allows `global access` to the resources.

To grant the `auditor` role access to fetch all of the resources from the database, following rule should be added to the `policy.yaml` file:

```
"context_with_global_access": "role:auditor"
```

This will make all SQL queries made by neutron with the `auditor` role in the context to not be scoped by the project ID. This however dont grant the `auditor` role to receive all of the resources from the Neutron API yet. To grant such permissions for example for the `get_network` action, following rule should be added to the `policy.yaml` file:

```
"get_network": "role:admin_only or (role:reader and project_id:%(project_
↪id)s) or rule:shared or rule:external or rule:context_is_advsvc or␣
↪role:auditor"
```

With those 2 rules in place, the `auditor` role will be able to fetch all of the networks from the Neutron API.

CHAPTER

TEN

COMMAND-LINE INTERFACE REFERENCE

10.1 neutron-sanity-check

The **neutron-sanity-check** client is a tool that checks various sanity about the Networking service.

This chapter documents **neutron-sanity-check** version `10.0.0`.

10.1.1 neutron-sanity-check usage

```
usage: neutron-sanity-check [-h] [--arp_header_match] [--arp_responder]
                            [--bridge_firewalling] [--config-dir DIR]
                            [--config-file PATH] [--debug] [--dhcp_release6]
                            [--dnsmasq_version]
                            [--ebtables_installed] [--icmpv6_header_match]
                            [--ip6tables_installed] [--ip_nonlocal_bind]
                            [--iproute2_vxlan] [--ipset_installed]
                            [--keepalived_ipv6_support]
                            [--log-config-append PATH]
                            [--log-date-format DATE_FORMAT]
                            [--log-dir LOG_DIR] [--log-file PATH]
                            [--noarp_header_match] [--noarp_responder]
                            [--nobridge_firewalling] [--nodebug]
                            [--nodhcp_release6]
                            [--nodnsmasq_version] [--noebtables_installed]
                            [--noicmpv6_header_match]
                            [--noip6tables_installed] [--noip_nonlocal_bind]
                            [--noiproute2_vxlan] [--noipset_installed]
                            [--nokeepalived_ipv6_support] [--nonova_notify]
                            [--noovs_conntrack] [--noovs_geneve]
                            [--noovs_patch] [--noovs_vxlan] [--noovsdb_native]
                            [--noread_netns] [--nouse-syslog] [--nova_notify]
                            [--noverbose] [--nowatch-log-file]
                            [--ovs_conntrack] [--ovs_geneve] [--ovs_patch]
                            [--ovs_vxlan] [--ovsdb_native] [--read_netns]
                            [--state_path STATE_PATH]
                            [--syslog-log-facility SYSLOG_LOG_FACILITY]
                            [--use-syslog] [--verbose] [--version]
                            [--watch-log-file]
```

10.1.2 neutron-sanity-check optional arguments

-h, --help
show this help message and exit

--arp_header_match
Check for ARP header match support

--arp_responder
Check for ARP responder support

--bridge_firewalling
Check bridge firewalling

--ip_nonlocal_bind
Check ip_nonlocal_bind kernel option works with network namespaces.

--config-dir DIR
Path to a config directory to pull `*.conf` files from. This file set is sorted, so as to provide a predictable parse order if individual options are over-ridden. The set is parsed after the file(s) specified via previous config-file, arguments hence over-ridden options in the directory take precedence.

--config-file PATH
Path to a config file to use. Multiple config files can be specified, with values in later files taking precedence. Dafaults to `None`.

--debug, -d
Print debugging output (set logging level to `DEBUG` instead of default `INFO` level).

--dhcp_release6
Check dhcp_release6 installation

--dnsmasq_version
Check minimal dnsmasq version

--ebtables_installed
Check ebtables installation

--icmpv6_header_match
Check for ICMPv6 header match support

--ip6tables_installed
Check ip6tables installation

--iproute2_vxlan
Check for iproute2 vxlan support

--ipset_installed
Check ipset installation

--keepalived_ipv6_support
Check keepalived IPv6 support

--log-config-append PATH, --log_config PATH
The name of a logging configuration file. This file is appended to any existing logging configuration files. For details about logging configuration files, see the Python logging module documentation. Note that when logging configuration files are used then all logging configuration is set in the configuration file and other logging configuration options are ignored (for example, `logging_context_format_string`).

--log-date-format DATE_FORMAT
Format string for %(asctime)s in log records. Default: None. This option is ignored if `log_config_append` is set.

--log-dir LOG_DIR, --logdir LOG_DIR
(Optional) The base directory used for relative `log-file` paths. This option is ignored if `log_config_append` is set.

--log-file PATH, --logfile PATH
(Optional) Name of log file to output to. If no default is set, logging will go to stderr as defined by `use_stderr`. This option is ignored if `log_config_append` is set.

--noarp_header_match
The inverse of arp_header_match

--noarp_responder
The inverse of arp_responder

--nobridge_firewalling
The inverse of bridge_firewalling

--nodebug
The inverse of debug

--nodhcp_release6
The inverse of dhcp_release6

--nodnsmasq_version
The inverse of dnsmasq_version

--noebtables_installed
The inverse of ebtables_installed

--noicmpv6_header_match
The inverse of icmpv6_header_match

--noip6tables_installed
The inverse of ip6tables_installed

--noip_nonlocal_bind
The inverse of ip_nonlocal_bind

--noiproute2_vxlan
The inverse of iproute2_vxlan

--noipset_installed
The inverse of ipset_installed

--nokeepalived_ipv6_support
The inverse of keepalived_ipv6_support

--nonova_notify
The inverse of nova_notify

--noovs_conntrack
The inverse of ovs_conntrack

--noovs_geneve
The inverse of ovs_geneve

--noovs_patch
The inverse of ovs_patch

--noovs_vxlan
The inverse of ovs_vxlan

--noovsdb_native
The inverse of ovsdb_native

--noread_netns
The inverse of read_netns

--nouse-syslog
The inverse of use-syslog

--nova_notify
Check for nova notification support

--noverbose
The inverse of verbose

--nowatch-log-file
The inverse of watch-log-file

--ovs_geneve
Check for OVS Geneve support

--ovs_patch
Check for patch port support

--ovs_vxlan
Check for OVS vxlan support

--ovsdb_native
Check ovsdb native interface support

--read_netns
Check netns permission settings

--state_path STATE_PATH
Where to store Neutron state files. This directory must be writable by the agent.

--syslog-log-facility SYSLOG_LOG_FACILITY
Syslog facility to receive log lines. This option is ignored if `log_config_append` is set.

--use-syslog
Use syslog for logging. Existing syslog format is **DEPRECATED** and will be changed later to honor RFC5424. This option is ignored if `log_config_append` is set.

--verbose, -v
If set to `false`, the logging level will be set to `WARNING` instead of the default `INFO` level.

--version
show programs version number and exit

--watch-log-file
Uses logging handler designed to watch file system. When log file is moved or removed this handler will open a new log file with specified path instantaneously. It makes sense only if `log_file` option is specified and Linux platform is used. This option is ignored if `log_config_append` is set.

10.2 neutron-status

The **neutron-status** provides routines for checking the status of Neutron deployment.

10.2.1 neutron-status usage

```
usage: neutron-status [-h] [--config-dir DIR] [--config-file PATH]
                      <category> <command>
```

Categories are:

- upgrade

Detailed descriptions are below.

You can also run with a category argument such as upgrade to see a list of all commands in that category:

```
neutron-status upgrade
```

These sections describe the available categories and arguments for **neutron-status**.

Command details

neutron-status upgrade check

Performs a release-specific readiness check before restarting services with new code. This command expects to have complete configuration and access to databases and services.

Return Codes

Return code	Description
0	All upgrade readiness checks passed successfully and there is nothing to do.
1	At least one check encountered an issue and requires further investigation. This is considered a warning but the upgrade may be OK.
2	There was an upgrade status check failure that needs to be investigated. This should be considered something that stops an upgrade.
255	An unexpected error occurred.

History of Checks

21.0.0 (Ussuri)

- A Check was added for NIC Switch agents to ensure nodes are running with kernel 3.13 or newer. This check serves as a notification for operators to ensure this requirement is fullfiled on relevant nodes.

CHAPTER

ELEVEN

OVN DRIVER

11.1 Gaps from ML2/OVS

This is a list of some of the currently known gaps between ML2/OVS and OVN. It is not a complete list, but is enough to be used as a starting point for implementors working on closing these gaps. A TODO list for OVN is located at[1].

- DHCP service for instances

  ML2/OVS adds packet filtering rules to every instance that allow DHCP queries from instances to reach the DHCP agent. For OVN this traffic has to be explicitly allowed by security group rules attached to the instance. Note that the default security group does allow all outgoing traffic, so this only becomes relevant when using custom security groups[2]. Proposed patch is[3] but it needs to be revived and updated.

- DNS resolution for instances

  OVN cannot use the hosts networking for DNS resolution, so Case 2b in[4] can only be used when additional DHCP agents are deployed. For Case 2a a different configuration option has to be used in `ml2_conf.ini`:

  ```
  [ovn]
  dns_servers = 203.0.113.8, 198.51.100.53
  ```

  OVN answers queries for hosts and IP addresses in tenant networks by spoofing responses from the configured DNS servers. This may lead to confusion in debugging.

  OVN can only answer queries that are sent via UDP, queries that use TCP will be ignored by OVN and forwarded to the configured resolvers.

  OVN can only answer queries with no additional options being set (EDNS). Such queries depending on the OVN version will either get broken responses or will also be forwarded to the configured resolvers.

- IPv6 NDP proxy

  The NDP proxy functionality for IPv6 addresses is not supported by OVN.

- East/West Fragmentation

[1] https://github.com/ovn-org/ovn/blob/master/TODO.rst
[2] https://bugs.launchpad.net/neutron/+bug/1926515
[3] https://review.opendev.org/c/openstack/neutron/+/788594
[4] https://docs.openstack.org/neutron/latest/admin/config-dns-res.html

The core OVN implementation does not support fragmentation of East/West traffic using an OVN router between two private networks. This is being tracked in[5] and[6].

- Traffic metering

  Currently `neutron-metering-agent` can only work with the Neutron L3 agent. It is not supported by the `ovn-router` service plugin nor by the `neutron-ovn-agent`. This is being reported and tracked in[7].

- Floating IP Port Forwarding in provider networks and with distributed routing

  Currently, when provider network types like `vlan` or `flat` are plugged to a router as internal networks while the `enable_distributed_floating_ip` configuration option is enabled, Floating IP port forwardings which are using such router will not work properly. Due to an incompatible setting of the router to make traffic in the vlan/flat networks to be distributed but port forwardings are always centralized in ML2/OVN backend. This is being reported in[8].

- QoS packet rate limit and minimum packet rate rules are not supported *Quality of Service (QoS)*.

11.1.1 References

11.2 OVN supported DHCP options

This is a list of the current supported DHCP options in ML2/OVN:

11.2.1 IP version 4

Option name / code	OVN value
arp-timeout	arp_cache_timeout
bootfile-name	bootfile_name
classless-static-route	classless_static_route
default-ttl	default_ttl
dns-server	dns_server
domain-name	domain_name
domain-search	domain_search_list
ethernet-encap	ethernet_encap
ip-forward-enable	ip_forward_enable
lease-time	lease_time
log-server	log_server
lpr-server	lpr_server
ms-classless-static-route	ms_classless_static_route
mtu	mtu
netmask	netmask
nis-server	nis_server
ntp-server	ntp_server
path-prefix	path_prefix
policy-filter	policy_filter
router-discovery	router_discovery

continues on next page

[5] https://bugs.launchpad.net/neutron/+bug/2032817
[6] https://bugzilla.redhat.com/show_bug.cgi?id=2238494
[7] https://bugs.launchpad.net/neutron/+bug/2048773
[8] https://bugs.launchpad.net/neutron/+bug/2028846

Table 1 – continued from previous page

Option name / code	OVN value
router	router
router-solicitation	router_solicitation
server-id	server_id
server-ip-address	tftp_server_address
swap-server	swap_server
T1	T1
T2	T2
tcp-ttl	tcp_ttl
tcp-keepalive	tcp_keepalive_interval
tftp-server-address	tftp_server_address
tftp-server	tftp_server
wpad	wpad
1	netmask
3	router
6	dns_server
7	log_server
9	lpr_server
15	domain_name
16	swap_server
19	ip_forward_enable
21	policy_filter
23	default_ttl
26	mtu
31	router_discovery
32	router_solicitation
35	arp_cache_timeout
36	ethernet_encap
37	tcp_ttl
38	tcp_keepalive_interval
41	nis_server
42	ntp_server
51	lease_time
54	server_id
58	T1
59	T2
66	tftp_server
67	bootfile_name
119	domain_search_list
121	classless_static_route
150	tftp_server_address
210	path_prefix
249	ms_classless_static_route
252	wpad

11.2.2 IP version 6

Option name / code	OVN value
dns-server	dns_server
domain-search	domain_search
ia-addr	ia_addr
server-id	server_id
2	server_id
5	ia_addr
23	dns_server
24	domain_search

11.2.3 OVN Database information

In OVN the DHCP options are stored on a table called `DHCP_Options` in the OVN Northbound database.

Lets add a DHCP option to a Neutron port:

```
$ openstack port set --extra-dhcp-option name='server-ip-address',value='10.0.
↪0.1' b4c3f265-369e-4bf5-8789-7caa9a1efb9c
```

To find that port in OVN we can use command below:

```
$ ovn-nbctl find Logical_Switch_Port name=b4c3f265-369e-4bf5-8789-7caa9a1efb9c
...
dhcpv4_options      : 5f00d1a2-c57d-4d1f-83ea-09bf8be13288
dhcpv6_options      : []
...
```

For DHCP, the columns that we care about are the `dhcpv4_options` and `dhcpv6_options`. These columns has the uuids of entries in the `DHCP_Options` table with the DHCP information for this port.

```
$ ovn-nbctl list DHCP_Options 5f00d1a2-c57d-4d1f-83ea-09bf8be13288
_uuid               : 5f00d1a2-c57d-4d1f-83ea-09bf8be13288
cidr                : "10.0.0.0/26"
external_ids        : {"neutron:revision_number"="0", port_id="b4c3f265-369e-
↪4bf5-8789-7caa9a1efb9c", subnet_id="5157ed8b-e7f1-4c56-b789-fa420098a687"}
options             : {classless_static_route="{169.254.169.254/32,10.0.0.2,␣
↪0.0.0.0/0,10.0.0.1}", dns_server="{8.8.8.8}", domain_name="\"openstackgate.
↪local\"", lease_time="43200", log_server="127.0.0.3", mtu="1442", router=
↪"10.0.0.1", server_id="10.0.0.1", server_mac="fa:16:3e:dc:57:22", tftp_
↪server_address="10.0.0.1"}
```

Here you can see that the option `tftp_server_address` has been set in the **options** column. Note that, the `tftp_server_address` option is the OVN translated name for `server-ip-address` (option 150). Take a look at the table in this document to find out more about the supported options and their counterpart names in OVN.

11.3 ml2ovn-trace

This is a simple wrapper around ovn-trace that will fill in datapath, inport, eth.src, ip.src, eth.dst, and ip.dst based on values pulled from openstack objects.

11.3.1 Usage

```
Usage: ml2ovn-trace [OPTIONS] [OVNTRACE_ARGS]...

Options:
  -c, --cloud TEXT                  Cloud from clouds.yaml to connect to
  -n, --net TEXT                    Network to limit interfaces lookups to
  --from-net TEXT                   Network to limit src interface lookups to
  --to-net TEXT                     Network to limit dst interface lookups to
  -f, --from [server|router]=value
                                    Fill eth-src/ip-src from the same object,
                                    e.g. server=vm1

  --eth-src [mac|server|router]=value
                                    Object from which to fill eth.src
                                    [required]

  --ip-src [ip|server|router]=value
                                    Object from which to fill ip.src  [required]
  -t, --to [server|router]=value    Fill eth-dst/ip-dst from the same object,
                                    e.g. server=vm2

  -v, --eth-dst, --via [mac|server|router]=value
                                    Object from which to fill eth.dst
                                    [required]

  --ip-dst [ip|server|router]=value
                                    Object from which to fill ip.dst  [required]
  -m, --microflow TEXT              Additional microflow text to append to the
                                    one generated

  -v, --verbose                     Enables verbose mode
  --dry-run                         Print ovn-trace output, but don't run it
  --help                            Show this message and exit.
```

11.3.2 Examples

If vm1 and vm2 only have one network interface and you want to trace between them:

```
$ sudo ml2ovn-trace --from server=vm1 --to server=vm2
```

Or if you want to limit to a specific network:

```
$ sudo ml2ovn-trace --net net1 --from server=vm1 --to server=vm2
```

Or if you want to go from vm1 to the floating IP of vm2 via vm1s router:

```
$ sudo ml2ovn-trace --net net1 --from server=vm1 --to ip=172.18.1.7 --via␣
↪router=net1-router
```

To add to the generated microflow, use -m. For example, for SSH:

```
$ sudo ml2ovn-trace --net net1 --from server=vm1 --to server=vm2 -m "tcp.
↪dst==22"
```

To pass arbitrary (non microflow) arguments to ovn-trace, place them after :

```
$ sudo ml2ovn-trace --net net1 --from server=vm1 --to server=vm2 -- --summary
```

11.4 Frequently Asked Questions

Q: What are the key differences between ML2/ovs and ML2/ovn?

Detail	ml2/ovs	ml2/ovn
agent/server communication	rabbit mq messaging + RPC.	ovsdb protocol on the NorthBound and SouthBound databases.
l3ha API	routers expose an ha field that can be disabled or enabled by admin with a deployment default.	routers dont expose an ha field, and will make use of HA as soon as there is more than one network node available.
l3ha dataplane	qrouter namespace with keepalive process and an internal ha network for VRRP traffic.	ovn-controller configures specific OpenFlow rules, and enables BFD protocol over tunnel endpoints to detect connectivity issues to nodes.
DVR API	exposes the distributed flag on routers only modifiable by admin.	exposes the distributed flag based on the configuration option enable_distributed_floating_ip
DVR dataplane	uses namespaces, veths, ip routing, ip rules and iptables on the compute nodes.	Uses OpenFlow rules on the compute nodes.
E/W traffic	goes through network nodes when the router is not distributed (DVR).	completely distributed in all cases.
Metadata Service	Metadata service is provided by the qrouters or dhcp namespaces in the network nodes.	Metadata is completely distributed across compute nodes, and served from the ovnmeta-xxxxx-xxxx namespace.
DHCP Service	DHCP is provided via qdhcp-xxxxx-xxx namespaces which run dnsmasq inside.	DHCP is provided by OpenFlow and ovn-controller, being distributed across computes.
Trunk Ports	Trunk ports are built by creating br-trunk-xxx bridges and patch ports.	Trunk ports live in br-int as OpenFlow rules, while subports are directly attached to br-int.

Q: Why cant I use the distributed or ha flags of routers?

Networking OVN implements HA and distributed in a transparent way for the administrator and users.

HA will be automatically used on routers as soon as more than two gateway nodes are detected. And distributed floating IPs will be used as soon as its configured (see next question).

Q: Does OVN support DVR or distributed L3 routing?

Yes, its controlled by a single flag in configuration.

DVR will be used for floating IPs if the ovn / enable_distributed_floating_ip flag is configured to True in the neutron server configuration, being a deployment wide setting. In contrast to ML2/ovs which was able to specify this setting per router (only admin).

Although ovn driver does not expose the distributed flag of routers throught the API.

Q: Does OVN support integration with physical switches?

OVN currently integrates with physical switches by optionally using them as VTEP gateways from logical to physical networks and via integrations provided by the Neutron ML2 framework, hierarchical port binding.

Q: Whats the status of HA for ovn driver and OVN?

Typically, multiple copies of neutron-server are run across multiple servers and uses a load balancer. The neutron ML2 mechanism driver provided by ovn driver supports this deployment model. DHCP and metadata services are distributed across compute nodes, and dont depend on the network nodes.

The network controller portion of OVN is distributed - an instance of the ovn-controller service runs on every hypervisor. OVN also includes some central components for control purposes.

ovn-northd is a centralized service that does some translation between the northbound and southbound databases in OVN. Currently, you only run this service once. You can manage it in an active/passive HA mode using something like Pacemaker. The OVN project plans to allow this service to be horizontally scaled both for scaling and HA reasons. This will allow it to be run in an active/active HA mode.

OVN also makes use of ovsdb-server for the OVN northbound and southbound databases. ovsdb-server supports active/passive HA using replication. For more information, see: http://docs.openvswitch.org/en/latest/topics/ovsdb-replication/

A typical deployment would use something like Pacemaker to manage the active/passive HA process. Clients would be pointed at a virtual IP address. When the HA manager detects a failure of the master, the virtual IP would be moved and the passive replica would become the new master.

Q: Which core OVN version should I use for my OpenStack installation?

OpenStack doesnt set explicit version requirements for OVN installation, but its recommended to follow at least the version that is used in upstream CI, e.g.: https://github.com/openstack/neutron/blob/4d31284373e89cb2b29539d6718f90a4c4d8284b/zuul.d/tempest-singlenode.yaml#L310

Some new features may require the latest core OVN version to work. For example, to be able to use VXLAN network type, one must run OVN 20.09+.

See *OVN information* for links to more details on OVNs architecture.

11.5 OVN agent

The OVN agent is a service that could be executed in any node running the ovn-controller service. This agent provides additional functionalities not provided by OVN; for example, a metadata proxy between the virtual machines and the Nova metadata service. This agent will replace the need of the OVN metadata agent.

11.5.1 OVN and OVS database connectivity

The OVN agent can access the local OVS database where the service is running. It also has access to the Northbound and Southbound OVN databases. The connection strings to these databases are defined in the agent configuration file:

```
[ovn]
ovn_nb_connection = tcp:192.168.10.100:6641
ovn_sb_connection = tcp:192.168.10.100:6642
[ovs]
ovsdb_connection = tcp:127.0.0.1:6640
```

11.5.2 Plugable extensions

The OVN agent provides functionalities via extensions. When the agent is started, the `OVNAgentExtensionManager` instance loads the configured extensions. The extensions are defined in the stevedore entry points, under the section neutron.agent.ovn.extensions. The extensions are defined in the agent configuration file in the extensions parameter:

```
[agent]
extensions = metadata
```

In `devstack`, the `[agent]extensions` configuration parameter is set by `OVN_AGENT_EXTENSIONS`.

Each extension will inherit from `OVNAgentExtension`, which provides the API for an OVN agent extension. The extensions are loaded in two steps:

- Initialization: this phase involves the call of `OVNAgentExtension.consume_api` and `OVNAgentExtension.initialize` (in this order). The first one assigns the extension API to the instance. In this case, the OVN agent has a specific instance `OVNAgentExtensionAPI` that gives to the extensions the needed access to OVS and OVN databases, using the same IDL instance. The second one is not currently used in the base class; it could be used, for example as in the metadata extension, to spawn the process monitor.
- Start: in this phase, the OVN and OVS database connections are established and can be accessed. The extension manager will call each extension `OVNAgentExtension.start` method.

Each extension should define a set of OVS, OVN Northbound and OVN Southbound tables to monitor, and a set of events related to these databases. The OVN agent will create the corresponding IDL connections using the conjunction of these tables and events.

11.5.3 Event-driven service

The OVN agent is a `oslo_service.service.Service` type class, that is launched when the script is executed. Once initialized, the service is waiting for new events that will trigger actions. As mentioned in the previous section, each extension will subscribe to a set of events from the OVN and OVS databases; these events will trigger a set of actions executed on the OVN agent.

11.5.4 Zuul CI testing

In order to enable this new agent, it is needed:

- To disable the default OVN Metadata agent (devstack service `q-ovn-metadata-agent`).
- To enable the OVN agent (devstack service `q-ovn-agent`).

Check the Neutron CI job `neutron-tempest-plugin-ovn` definition and [OVN] Use the OVN agent in neutron-tempest-plugin-ovn for more information.

11.6 Virtual IPs

It is common practice to create an unbound port in a Neutron network to allocate (reserve) an IP address that will be used as a Virtual IP (VIP) by other ports in the same network. Such IP addresses are then added as `allowed_address_pairs` to the ports used by Virtual Machines.

Applications, such as keepalived, running inside these Virtual Machines can then configure the VIP on one of the VMs and move it between VMs dynamically.

11.6.1 Implementation in OVN

For Virtual IP addresses to work properly in the OVN backend, Neutron needs to mark the `Logical Switch Port` corresponding to the port with the Virtual IP as `virtual`. Neutron does this for ports that are unbound and have a fixed IP address that is also configured in the `allowed_address_pairs` of any other port in the same network.

11.6.2 Limitations

- In the case when a Virtual IP address is going to be used in Virtual Machines and configured as `allowed_address_pairs`, it is necessary to also create such an unbound port in Neutron in order to:
  - reserve that IP address for that use case so that it will not be later allocated for another port in the same network as the fixed IP,
  - let OVN know that this IP and `Logical Switch Port` is `virtual` so that OVN can configure it accordingly.
- A port created in Neutron in order to allocate virtual IP address has to be `unbound`, it can not be attached directly to any Virtual Machine.
- Because of how Virtual IP addresses are implemented in the ML2/OVN backend, the Virtual IP address must be set in the `allowed_address_pairs` of the VM port as a single IP address (/32 for IPv4 or /128 for IPv6). Setting a larger CIDR as `allowed_address_pairs`, even if it contains the Virtual IP address, will not mark the `Logical Switch Port` corresponding to the port with that IP address as `virtual`.
- Another limitation is that setting an IP address that belongs to the distributed metadata port in the same network as `allowed_address_pairs` is not allowed.

11.6.3 Usage example

To use a Virtual IP address in Neutron, you need to create an unbound port in a Neutron network and add the Virtual IP address to the `allowed_address_pairs` of the port(s) that belong to the Virtual Machine(s).

- Create an unbound port in the Neutron network to allocate the Virtual IP address:

```
$ openstack port create --network private virtual-ip-port
+-------------------------+---------------------------------------------
↪-----------------------------------------------------------+
| Field                   | Value                                       ␣
```

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```
↪                                                         |
+-------------------------+-----------------------------------------------
↪----------------------------------------------------------+
| admin_state_up          | UP                                            ␣
↪                                                         |
| allowed_address_pairs   |                                               ␣
↪                                                         |
| binding_host_id         |                                               ␣
↪                                                         |
| binding_profile         |                                               ␣
↪                                                         |
| binding_vif_details     |                                               ␣
↪                                                         |
| binding_vif_type        | unbound                                       ␣
↪                                                         |
| binding_vnic_type       | normal                                        ␣
↪                                                         |
| created_at              | 2025-11-28T14:39:06Z                          ␣
↪                                                         |
| data_plane_status       | None                                          ␣
↪                                                         |
| description             |                                               ␣
↪                                                         |
| device_id               |                                               ␣
↪                                                         |
| device_owner            |                                               ␣
↪                                                         |
| device_profile          | None                                          ␣
↪                                                         |
| dns_assignment          |                                               ␣
↪                                                         |
| dns_domain              | None                                          ␣
↪                                                         |
| dns_name                | None                                          ␣
↪                                                         |
| extra_dhcp_opts         |                                               ␣
↪                                                         |
| fixed_ips               | ip_address='10.0.0.20', subnet_id='866305cc-
↪26db-48d7-8471-cbd267321b8b'                                 |
|                         | ip_address=
↪'fde7:7c8e:8883:0:f816:3eff:feb6:559f', subnet_id='b8b0a413-6229-4c64-
↪9d6e-65906a33b056' |
| hardware_offload_type   | None                                          ␣
↪                                                         |
| hints                   |                                               ␣
↪                                                         |
| id                      | 3f078d1b-2f6e-41d8-99d7-70bc801f3979          ␣
↪                                                         |
| ip_allocation           | None                                          ␣
```

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```
↪                                                          |
| mac_address             | fa:16:3e:b6:55:9f                                  ␣
↪                                                          |
| name                    | virtual-ip-port                                    ␣
↪                                                          |
| network_id              | c8e5e81c-d318-43f6-a45e-056f22a518e6               ␣
↪                                                          |
| numa_affinity_policy    | None                                               ␣
↪                                                          |
| port_security_enabled   | True                                               ␣
↪                                                          |
| project_id              | b7907ac4c9794e5787a8d6bac0e5b80b                   ␣
↪                                                          |
| propagate_uplink_status | None                                               ␣
↪                                                          |
| resource_request        | None                                               ␣
↪                                                          |
| revision_number         | 1                                                  ␣
↪                                                          |
| qos_network_policy_id   | None                                               ␣
↪                                                          |
| qos_policy_id           | None                                               ␣
↪                                                          |
| security_group_ids      | 876d4c44-e2fd-48fc-bbd4-4bd295676a0e               ␣
↪                                                          |
| status                  | DOWN                                               ␣
↪                                                          |
| tags                    |                                                    ␣
↪                                                          |
| trunk_details           | None                                               ␣
↪                                                          |
| trusted                 | None                                               ␣
↪                                                          |
| updated_at              | 2025-11-28T14:39:06Z                               ␣
↪                                                          |
+-------------------------+--------------------------------------------------
↪-------------------------------------------------------+
```

- Create a Virtual Machine

```
openstack server create --flavor m1.micro --image cirros-0.5.1-x86_64-
↪disk --network private virtual-machine
+-------------------------------------+------------------------------------
↪------------------------------------------------------------------------
↪---------------------------------------+
| Field                               | Value                              ␣
↪                                                                        ␣
↪                                       |
+-------------------------------------+------------------------------------
↪------------------------------------------------------------------------
```

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```
↪-------------------------------------------+
| OS-DCF:diskConfig                    | MANUAL                                     ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-AZ:availability_zone          | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:host                 | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:hostname             | virtual-machine                            ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:hypervisor_hostname  | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:instance_name        | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:kernel_id            | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:launch_index         | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:ramdisk_id           | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:reservation_id       | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:root_device_name     | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-SRV-ATTR:user_data            | None                                       ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-STS:power_state               | N/A                                        ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-STS:task_state                | scheduling                                 ␣
↪                                                                                   ␣
↪                                          |
| OS-EXT-STS:vm_state                  | building                                   ␣
↪                                                                                   ␣
↪                                          |
| OS-SRV-USG:launched_at               | None                                       ␣
↪                                                                                   ␣
↪                                          |
```

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```
| OS-SRV-USG:terminated_at          | None                                                    ␣
↪                                                                                             ␣
↪                                    |
| accessIPv4                        | None                                                    ␣
↪                                                                                             ␣
↪                                    |
| accessIPv6                        | None                                                    ␣
↪                                                                                             ␣
↪                                    |
| addresses                         | N/A                                                     ␣
↪                                                                                             ␣
↪                                    |
| adminPass                         | QNkLbpeZ72LF                                            ␣
↪                                                                                             ␣
↪                                    |
| config_drive                      | None                                                    ␣
↪                                                                                             ␣
↪                                    |
| created                           | 2025-11-28T14:41:22Z                                    ␣
↪                                                                                             ␣
↪                                    |
| description                       | None                                                    ␣
↪                                                                                             ␣
↪                                    |
| flavor                            | description=, disk='1', ephemeral=
↪'0', extra_specs.hw_rng:allowed='True', id='m1.micro', is_disabled=, is_
↪public='True', location=, name='m1.micro', |
|                                   | original_name='m1.micro', ram='256
↪', rxtx_factor=, swap='0', vcpus='1'                                                        ␣
↪                                    |
| hostId                            | None                                                    ␣
↪                                                                                             ␣
↪                                    |
| host_status                       | None                                                    ␣
↪                                                                                             ␣
↪                                    |
| id                                | d2573702-b79c-46a3-bd7a-
↪d8aa50341082                                                                                 ␣
↪                                             |
| image                             | cirros-0.5.1-x86_64-disk␣
↪(7b920c82-0879-4526-9ee8-7e3b77e7fe28)                                                       ␣
↪                                            |
| key_name                          | None                                                    ␣
↪                                                                                             ␣
↪                                    |
| locked                            | None                                                    ␣
↪                                                                                             ␣
↪                                    |
| locked_reason                     | None                                                    ␣
```

(continues on next page)

(continued from previous page)

```
↪                                                                                                   ␣
↪                                                                       |
| name                                | virtual-machine                                             ␣
↪                                                                                                   ␣
↪                                                                       |
| pinned_availability_zone            | None                                                        ␣
↪                                                                                                   ␣
↪                                                                       |
| progress                            | None                                                        ␣
↪                                                                                                   ␣
↪                                                                       |
| project_id                          | b7907ac4c9794e5787a8d6bac0e5b80b                            ␣
↪                                                                                                   ␣
↪                                                                       |
| properties                          | None                                                        ␣
↪                                                                                                   ␣
↪                                                                       |
| scheduler_hints                     |                                                             ␣
↪                                                                                                   ␣
↪                                                                       |
| security_groups                     | name='default'                                              ␣
↪                                                                                                   ␣
↪                                                                       |
| server_groups                       | None                                                        ␣
↪                                                                                                   ␣
↪                                                                       |
| status                              | BUILD                                                       ␣
↪                                                                                                   ␣
↪                                                                       |
| tags                                |                                                             ␣
↪                                                                                                   ␣
↪                                                                       |
| trusted_image_certificates          | None                                                        ␣
↪                                                                                                   ␣
↪                                                                       |
| updated                             | 2025-11-28T14:41:22Z                                        ␣
↪                                                                                                   ␣
↪                                                                       |
| user_id                             | d46c7955bea644c9a45e5d95bb462e29                            ␣
↪                                                                                                   ␣
↪                                                                       |
| volumes_attached                    |                                                             ␣
↪                                                                                                   ␣
↪                                                                       |
+-------------------------------------+-------------------------------------------------------------
↪--------------------------------------------------------------------------------------------------
↪---------------------------------------+
```

- List ports of the Virtual Machine

```
$ openstack port list --device-id d2573702-b79c-46a3-bd7a-d8aa50341082
+--------------------------------------+------+-------------------+-------
↪-----------------------------------------------------------------------
↪----------------------+--------+
| ID                                   | Name | MAC Address       | Fixed␣
↪IP Addresses                                                          ␣
↪                      | Status |
+--------------------------------------+------+-------------------+-------
↪-----------------------------------------------------------------------
↪----------------------+--------+
| 692c7f41-0497-4d4c-9766-3d71ffd229df |      | fa:16:3e:b6:44:9a | ip_
↪address='10.0.0.30', subnet_id='866305cc-26db-48d7-8471-cbd267321b8b'  ␣
↪                         | ACTIVE |
|                                      |      |                   | ip_
↪address='fde7:7c8e:8883:0:f816:3eff:feb6:449a', subnet_id='b8b0a413-
↪6229-4c64-9d6e-65906a33b056' |         |
+--------------------------------------+------+-------------------+-------
↪-----------------------------------------------------------------------
↪----------------------+--------+
```

- Set the Virtual IP address as an allowed address pair to the port of the Virtual Machine

```
$ openstack port set --allowed-address ip-address=10.0.0.20 692c7f41-0497-
↪4d4c-9766-3d71ffd229df
```

After these steps, the Virtual IP address will be available on the port of the Virtual Machine.

If a CIDR such as `10.0.0.0/24` is set in the `allowed_address_pairs` instead of the IP address `10.0.0.20`, then the `Logical Switch Port` related to the port with IP address `10.0.0.20` would not be marked as a Virtual IP address due to the limitations mentioned above.

CHAPTER

TWELVE

API REFERENCE

The reference of the OpenStack networking API is found at https://docs.openstack.org/api-ref/network/.

CHAPTER

THIRTEEN

NEUTRON FEATURE CLASSIFICATION

13.1 Introduction

This document describes how features are listed in *General Feature Support* and *Provider Network Support*.

13.1.1 Goals

The object of this document is to inform users whether or not features are complete, well documented, stable, and tested. This approach ensures good user experience for those well maintained features.

Note

Tests are specific to particular combinations of technologies. The plugins chosen for deployment make a big difference to whether or not features will work.

13.1.2 Concepts

These definitions clarify the terminology used throughout this document.

13.1.3 Feature status

- Immature
- Mature
- Required
- Deprecated (scheduled to be removed in a future release)

Immature

Immature features do not have enough functionality to satisfy real world use cases.

An immature feature is a feature being actively developed, which is only partially functional and upstream tested, most likely introduced in a recent release, and that will take time to mature thanks to feedback from downstream QA.

Users of these features will likely identify gaps and/or defects that were not identified during specification and code review.

Mature

A feature is considered mature if it satisfies the following criteria:

- Complete API documentation including concept and REST call definition.
- Complete Administrator documentation.
- Tempest tests that define the correct functionality of the feature.
- Enough functionality and reliability to be useful in real world scenarios.
- Low probability of support for the feature being dropped.

Required

Required features are core networking principles that have been thoroughly tested and have been implemented in real world use cases.

In addition they satisfy the same criteria for any mature features.

Note

Any new drivers must prove that they support all required features before they are merged into neutron.

Deprecated

Deprecated features are no longer supported and only security related fixes or development will happen towards them.

Deployment rating of features

The deployment rating shows only the state of the tests for each feature on a particular deployment.

Important

Despite the obvious parallels that could be drawn, this list is unrelated to the Interop effort. See InteropWG

13.2 General Feature Support

Warning

Please note, while this document is still being maintained, this is slowly being updated to re-group and classify features using the definitions described in here: *Introduction*.

This document covers the maturity and support of the Neutron API and its API extensions. Details about the API can be found at Networking API v2.0.

When considering which capabilities should be marked as mature the following general guiding principles were applied:

- **Inclusivity** - people have shown ability to make effective use of a wide range of network plugins and drivers with broadly varying feature sets. Aiming to keep the requirements as inclusive as possible, avoids second-guessing how a user wants to use their networks.
- **Bootstrapping** - a practical use case test is to consider that starting point for the network deploy is an empty data center with new machines and network connectivity. Then look at what are the minimum features required of the network service, in order to get user instances running and connected over the network.
- **Reality** - there are many networking drivers and plugins compatible with neutron. Each with their own supported feature set.

Summary

Feature	*Status*	**OVN**	**Open vSwitch**
Networks	mandatory	✓	✓
Subnets	mandatory	✓	✓
Ports	mandatory	✓	✓
Routers	mandatory	✓	✓
Security Groups	mature	✓	✓
External Networks	mature	✓	✓
Distributed Virtual Routers	immature	✓	✓
L3 High Availability	immature	✓	✓
Quality of Service	mature	✓	✓
Border Gateway Protocol	immature	?	✓
DNS	mature	✓	✓
Trunk Ports	mature	✓	✓
Metering	mature	?	✓

Details

- **Networks Status: mandatory.**

  API Alias: core

  CLI commands:

  - `openstack network *`

  Notes: The ability to create, modify and delete networks. https://docs.openstack.org/api-ref/network/v2/#networks

  Driver Support:

  - **OVN:** `complete`
  - **Open vSwitch:** `complete`

- **Subnets Status: mandatory.**

  API Alias: core

  CLI commands:

  - `openstack subnet *`

  Notes: The ability to create and manipulate subnets and subnet pools. https://docs.openstack.org/api-ref/network/v2/#subnets

Driver Support:

- **OVN:** `complete`
- **Open vSwitch:** `complete`

- **Ports Status: mandatory.**

  API Alias: core

  CLI commands:

  - `openstack port *`

  Notes: The ability to create and manipulate ports. https://docs.openstack.org/api-ref/network/v2/#ports

  Driver Support:

  - **OVN:** `complete`
  - **Open vSwitch:** `complete`

- **Routers Status: mandatory.**

  API Alias: router

  CLI commands:

  - `openstack router *`

  Notes: The ability to create and manipulate routers. https://docs.openstack.org/api-ref/network/v2/#routers-routers

  Driver Support:

  - **OVN:** `complete`
  - **Open vSwitch:** `complete`

- **Security Groups Status: mature.**

  API Alias: security-group

  CLI commands:

  - `openstack security group *`

  Notes: Security groups are set by default, and can be modified to control ingress & egress traffic. https://docs.openstack.org/api-ref/network/v2/#security-groups-security-groups

  Driver Support:

  - **OVN:** `complete`
  - **Open vSwitch:** `complete`

- **External Networks Status: mature.**

  API Alias: external-net

  Notes: The ability to create an external network to provide internet access to and from instances using floating IP addresses and security group rules.

  Driver Support:

  - **OVN:** `complete`

- **Open vSwitch:** `complete`

- **Distributed Virtual Routers Status: immature.**

  API Alias: dvr

  Notes: The ability to support the distributed virtual routers. https://wiki.openstack.org/wiki/Neutron/DVR

  Driver Support:

  - **OVN:** `partial`
  - **Open vSwitch:** `complete`

- **L3 High Availability Status: immature.**

  API Alias: l3-ha

  Notes: The ability to support the High Availability features and extensions. https://wiki.openstack.org/wiki/Neutron/L3_High_Availability_VRRP.

  Driver Support:

  - **OVN:** `partial`
  - **Open vSwitch:** `complete`

- **Quality of Service Status: mature.**

  API Alias: qos

  Notes: Support for Neutron Quality of Service policies and API. https://docs.openstack.org/api-ref/network/v2/#qos-policies-qos

  Driver Support:

  - **OVN:** `complete`
  - **Open vSwitch:** `complete`

- **Border Gateway Protocol Status: immature.**

  Notes: https://docs.openstack.org/api-ref/network/v2/#bgp-mpls-vpn-interconnection

  Driver Support:

  - **OVN:** `unknown`
  - **Open vSwitch:** `complete`

- **DNS Status: mature.**

  API Alias: dns-integration

  Notes: The ability to integrate with an external DNS as a Service. https://docs.openstack.org/neutron/latest/admin/config-dns-int.html

  Driver Support:

  - **OVN:** `complete`
  - **Open vSwitch:** `complete`

- **Trunk Ports Status: mature.**

  API Alias: trunk

  Notes: Neutron extension to access lots of neutron networks over a single vNIC as tagged/encapsulated traffic. https://docs.openstack.org/api-ref/network/v2/#trunk-networking

  Driver Support:

  - **OVN:** `complete`
  - **Open vSwitch:** `complete`

- **Metering Status: mature.**

  API Alias: metering

  Notes: Meter traffic at the L3 router levels. https://docs.openstack.org/api-ref/network/v2/#metering-labels-and-rules-metering-labels-metering-label-rules

  Driver Support:

  - **OVN:** `unknown`
  - **Open vSwitch:** `complete`

Notes:

- **This document is a continuous work in progress**

13.3 Provider Network Support

Warning

Please note, while this document is still being maintained, this is slowly being updated to re-group and classify features using the definitions described in here: *Introduction*.

This document covers the maturity and support for various network isolation technologies.

When considering which capabilities should be marked as mature the following general guiding principles were applied:

- **Inclusivity** - people have shown ability to make effective use of a wide range of network plugins and drivers with broadly varying feature sets. Aiming to keep the requirements as inclusive as possible, avoids second-guessing how a user wants to use their networks.
- **Bootstrapping** - a practical use case test is to consider that starting point for the network deploy is an empty data center with new machines and network connectivity. Then look at what are the minimum features required of the network service, in order to get user instances running and connected over the network.
- **Reality** - there are many networking drivers and plugins compatible with neutron. Each with their own supported feature set.

Summary

Feature	*Status*	OVN	Open vSwitch
VLAN provider network support	mature	✓	✓
VXLAN provider network support	mature	×	✓
GRE provider network support	immature	×	✓
Geneve provider network support	immature	✓	✓

Details

- **VLAN provider network support Status: mature.**

  Driver Support:

  - **OVN:** `complete`
  - **Open vSwitch:** `complete`

- **VXLAN provider network support Status: mature.**

  Driver Support:

  - **OVN:** `missing`
  - **Open vSwitch:** `complete`

- **GRE provider network support Status: immature.**

  Driver Support:

  - **OVN:** `missing`
  - **Open vSwitch:** `complete`

- **Geneve provider network support Status: immature.**

  Driver Support:

  - **OVN:** `complete`
  - **Open vSwitch:** `complete`

Notes:

- **This document is a continuous work in progress**

CHAPTER

FOURTEEN

CONTRIBUTOR GUIDE

This document describes Neutron for contributors of the project, and assumes that you are already familiar with Neutron from an *end-user perspective*.

14.1 Basic Information

14.1.1 So You Want to Contribute

For general information on contributing to OpenStack, please check out the contributor guide to get started. It covers all the basics that are common to all OpenStack projects: the accounts you need, the basics of interacting with our Gerrit review system, how we communicate as a community, etc.

Below will cover the more project specific information you need to get started with Neutron.

Communication

- IRC channel: #openstack-neutron
- Mailing lists prefix: [neutron]
- Team Meeting:

  This is general Neutron team meeting. The discussion in this meeting is about all things related to the Neutron project, like community goals, progress with blueprints, bugs, etc. There is also `On Demand Agenda` at the end of this meeting, where anyone can add a topic to discuss with the Neutron team.

  - time: http://eavesdrop.openstack.org/#Neutron_Team_Meeting
  - agenda: https://wiki.openstack.org/wiki/Network/Meetings
- Drivers team meeting:

  This is the meeting where Neutron drivers discuss about new RFEs.

  - time: http://eavesdrop.openstack.org/#Neutron_drivers_Meeting
  - agenda: https://wiki.openstack.org/wiki/Meetings/NeutronDrivers
- Neutron CI team meeting:

  This is the meeting where upstream CI issues are discussed every week. If You are interested in helping our CI to be green, thats good place to join and help.

  - time: http://eavesdrop.openstack.org/#Neutron_CI_team
  - agenda: https://etherpad.openstack.org/p/neutron-ci-meetings

Contacting the Core Team

The list of current Neutron core reviewers is available on gerrit. Overall structure of Neutron team is available in *Neutron teams*.

New Feature Planning

Neutron team uses `RFE (Request for Enhancements)` to propose new features. RFE should be submitted as a Launchpad bug first (see section *Reporting a Bug*). The title of RFE bug should starts with `[RFE]` tag. Such RFEs need to be discussed and approved by the *Neutron drivers team*. In some cases an additional spec proposed to the Neutron specs repo may be necessary. The complete process is described in detail in *Blueprints guide*.

Task Tracking

We track our tasks in Launchpad. If youre looking for some smaller, easier work item to pick up and get started on, search for the Low hanging fruit tag. List of all official tags which Neutron team is using is available on *bugs*. Every week, one of our team members is the *bug deputy* and at the end of the week such person usually sends report about new bugs to the mailing list openstack-discuss@lists.openstack.org or talks about it on our team meeting. This is also good place to look for some work to do.

Reporting a Bug

You found an issue and want to make sure we are aware of it? You can do so on Launchpad. More info about Launchpad usage can be found on OpenStack docs page.

Getting Your Patch Merged

All changes proposed to the Neutron or one of the Neutron stadium projects require two +2 votes from Neutron core reviewers before one of the core reviewers can approve patch by giving `Workflow +1` vote. More detailed guidelines for reviewers of Neutron patches are available at *Code reviews guide*.

Project Team Lead Duties

Neutrons PTL duties are described very well in the All common PTL duties guide. Additionally to what is described in this guide, Neutrons PTL duties are:

- triage new RFEs and prepare Neutron drivers team meeting,
- maintain list of the *stadium projects* health - if each project has gotten active team members and if it is following community and Neutrons guidelines and goals,
- maintain list of the *stadium projects lieutenants* - check if those people are still active in the projects, if their contact data are correct, maybe there is someone new who is active in the stadium project and could be added to this list.

Over the past few years, the Neutron team has followed a mentoring approach for:

- new contributors,
- potential new core reviewers,
- future PTLs.

The Neutron PTLs responsibility is to identify potential new core reviewers and help with their mentoring process. Mentoring of new contributors and potential core reviewers can be of course delegated to the other members of the Neutron team. Mentoring of future PTLs is responibility of the Neutron PTL.

14.2 Neutron Policies

14.2.1 Neutron Policies

In the Policies Guide, you will find documented policies for developing with Neutron. This includes the processes we use for blueprints and specs, bugs, contributor onboarding, core reviewer memberships, and other procedural items.

Blueprints and Specs

The Neutron team uses the neutron-specs repository for its specification reviews. Detailed information can be found on the wiki. Please also find additional information in the reviews.rst file.

The Neutron team does not enforce deadlines for specs. These can be submitted throughout the release cycle. The drivers team will review this on a regular basis throughout the release, and based on the load for the milestones, will assign these into milestones or move them to the backlog for selection into a future release.

Please note that we use a template for spec submissions. It is not required to fill out all sections in the template. Review of the spec may require filling in information left out by the submitter.

Sub-Projects and Specs

The neutron-specs repository is only meant for specs from Neutron itself, and the advanced services repositories as well. This includes FWaaS and VPNaaS. Other sub-projects are encouraged to fold their specs into their own devref code in their sub-project gerrit repositories. Please see additional comments in the Neutron teams *section* for reviewer requirements of the neutron-specs repository.

Neutron Request for Feature Enhancements

In Liberty the team introduced the concept of feature requests. Feature requests are tracked as Launchpad bugs, by tagging them with a set of tags starting with *rfe*, enabling the submission and review of feature requests before code is submitted. This allows the team to verify the validity of a feature request before the process of submitting a neutron-spec is undertaken, or code is written. It also allows the community to express interest in a feature by subscribing to the bug and posting a comment in Launchpad. The rfe tag should not be used for work that is already well-defined and has an assignee. If you are intending to submit code immediately, a simple bug report will suffice. Note the temptation to game the system exists, but given the history in Neutron for this type of activity, it will not be tolerated and will be called out as such in public on the mailing list.

RFEs can be submitted by anyone and by having the community vote on them in Launchpad, we can gauge interest in features. The drivers team will evaluate these on a weekly basis along with the specs. RFEs will be evaluated in the current cycle against existing project priorities and available resources.

The workflow for the life an RFE in Launchpad is as follows:

- The bug is submitted and will by default land in the New state. Anyone can make a bug an RFE by adding the *rfe* tag.
- As soon as a member of the neutron-drivers team acknowledges the bug, the *rfe* tag will be replaced with the *rfe-confirmed* tag. No assignee, or milestone is set at this time. The importance will be set to Wishlist to signal the fact that the report is indeed a feature or enhancement and there is no severity associated to it.

- A member of the neutron-drivers team replaces the *rfe-confirmed* tag with the *rfe-triaged* tag when he/she thinks its ready to be discussed in the drivers meeting. The bug will be in this state while the discussion is ongoing.
- The neutron-drivers team will evaluate the RFE and may advise the submitter to file a spec in neutron-specs to elaborate on the feature request, in case the RFE requires extra scrutiny, more design discussion, etc.
- The PTL will work with the Lieutenant for the area being identified by the RFE to evaluate resources against the current workload.
- A member of the Neutron release team (or the PTL) will register a matching Launchpad blueprint to be used for milestone tracking purposes, and for identifying the responsible assignee and approver. If the RFE has a spec the blueprint will have a pointer to the spec document, which will become available on specs.o.o. once it is approved and merged. The blueprint will then be linked to the original RFE bug report as a pointer to the discussion that led to the approval of the RFE. The blueprint submitter will also need to identify the following:
  - Priority: there will be only two priorities to choose from, High and Low. It is worth noting that priority is not to be confused with importance, which is a property of Launchpad Bugs. Priority gives an indication of how promptly a work item should be tackled to allow it to complete. High priority is to be chosen for work items that must make substantial progress in the span of the targeted release, and deal with the following aspects:
    * OpenStack cross-project interaction and interoperability issues;
    * Issues that affect the existing systems usability;
    * Stability and testability of the platform;
    * Risky implementations that may require complex and/or pervasive changes to API and the logical model;

    Low priority is to be chosen for everything else. RFEs without an associated blueprint are effectively equivalent to low priority items. Bear in mind that, even though staffing should take priorities into account (i.e. by giving more resources to high priority items over low priority ones), the open source reality is that they can both proceed at their own pace and low priority items can indeed complete faster than high priority ones, even though they are given fewer resources.
  - Drafter: who is going to submit and iterate on the spec proposal; he/she may be the RFE submitter.
  - Assignee: who is going to develop the bulk of the code, or the go-to contributor, if more people are involved. Typically this is the RFE submitter, but not necessarily.
  - Approver: a member of the Neutron team who can commit enough time during the ongoing release cycle to ensure that code posted for review does not languish, and that all aspects of the feature development are taken care of (client, server changes and/or support from other projects if needed - tempest, nova, openstack-infra, devstack, etc.), as well as comprehensive testing. This is typically a core member who has enough experience with what it takes to get code merged, but other resources amongst the wider team can also be identified. Approvers are volunteers who show a specific interest in the blueprint specification, and have enough insight in the area of work so that they can make effective code reviews and provide design feedback. An approver will not work in isolation, as he/she can and will reach out for help to get the job done; however he/she is the main point of contact with the following responsibilities:

- Pair up with the drafter/assignee in order to help skip development blockers.
- Review patches associated with the blueprint: approver and assignee should touch base regularly and ping each other when new code is available for review, or if review feedback goes unaddressed.
- Reach out to other reviewers for feedback in areas that may step out of the zone of her/his confidence.
- Escalate issues, and raise warnings to the release team/PTL if the effort shows slow progress. Approver and assignee are key parts to land a blueprint: should the approver and/or assignee be unable to continue the commitment during the release cycle, it is the Approvers responsibility to reach out the release team/PTL so that replacements can be identified.
- Provide a status update during the Neutron IRC meeting, if required.

Approver assignments must be carefully identified to ensure that no-one overcommits. A Neutron contributor develops code himself/herself, and if he/she is an approver of more than a couple of blueprints in a single cycle/milestone (depending on the complexity of the spec), it may mean that he/she is clearly oversubscribed.

The Neutron team will review the status of blueprints targeted for the milestone during their weekly meeting to ensure a smooth progression of the work planned. Blueprints for which resources cannot be identified will have to be deferred.

- In either case (a spec being required or not), once the discussion has happened and there is positive consensus on the RFE, the report is approved, and its tag will move from *rfe-triaged* to *rfe-approved*.
- An RFE can be occasionally marked as rfe-postponed if the team identifies a dependency between the proposed RFE and other pending tasks that prevent the RFE from being worked on immediately.
- Once an RFE is approved, it needs volunteers. Approved RFEs that do not have an assignee but sound relatively simple or limited in scope (e.g. the addition of a new API with no ramification in the plugin backends), should be promoted during team meetings or the ML so that volunteers can pick them up and get started with neutron development. The team will regularly scan *rfe-approved* or *rfe-postponed* RFEs to see what their latest status is and mark them incomplete if no assignees can be found, or they are no longer relevant.
- As for setting the milestone (both for RFE bugs or blueprints), the current milestone is always chosen, assuming that work will start as soon as the feature is approved. Work that fails to complete by the defined milestone will roll over automatically until it gets completed or abandoned.
- If the code fails to merge, the bug report may be marked as incomplete, unassigned and untargeted, and it will be garbage collected by the Launchpad Janitor if no-one takes over in time. Renewed interest in the feature will have to go through RFE submission process once again.

In summary:

State	Meaning
New	This is where all RFEs start, as filed by the community
Incomplete	**Drivers/LTs - Move to this state to mean,** more information needed before proceeding
Confirmed	**Drivers/LTs - Move to this state to mean,** yes, I see that you filed it
Triaged	**Drivers/LTs - Move to this state to mean,** discussion is ongoing
Wont Fix	Drivers/LTs - Move to this state to reject an RFE

Once the triaging (discussion is complete) and the RFE is approved, the tag goes from rfe to rfe-approved, and at this point the bug report goes through the usual state transition. Note, that the importance will be set to wishlist, to reflect the fact that the bug report is indeed not a bug, but a new feature or enhancement. This will also help have RFEs that are not followed up by a blueprint standout in the Launchpad milestone dashboards.

The drivers team will be discussing the following bug reports during their IRC meeting:

- New RFEs
- Incomplete RFEs
- Confirmed RFEs
- Triaged RFEs

RFE Submission Guidelines

Before we dive into the guidelines for writing a good RFE, it is worth mentioning that depending on your level of engagement with the Neutron project and your role (user, developer, deployer, operator, etc.), you are more than welcome to have a preliminary discussion of a potential RFE by reaching out to other people involved in the project. This usually happens by posting mails on the relevant mailing lists (e.g. openstack-discuss - include [neutron] in the subject) or on #openstack-neutron IRC channel on OFTC. If current ongoing code reviews are related to your feature, posting comments/questions on gerrit may also be a way to engage. Some amount of interaction with Neutron developers will give you an idea of the plausibility and form of your RFE before you submit it. That said, this is not mandatory.

When you submit a bug report on https://bugs.launchpad.net/neutron/+filebug, there are two fields that must be filled: summary and further information. The summary must be brief enough to fit in one line: if you cant describe it in a few words it may mean that you are either trying to capture more than one RFE at once, or that you are having a hard time defining what you are trying to solve at all.

The further information section must be a description of what you would like to see implemented in Neutron. The description should provide enough details for a knowledgeable developer to understand what is the existing problem in the current platform that needs to be addressed, or what is the enhancement that would make the platform more capable, both for a functional and a non-functional standpoint. To this aim it is important to describe why you believe the RFE should be accepted, and motivate the reason why without it Neutron is a poorer platform. The description should be self contained, and no external references should be necessary to further explain the RFE.

In other words, when you write an RFE you should ask yourself the following questions:

- What is that I (specify what user - a user can be a human or another system) cannot do today when interacting with Neutron? On the other hand, is there a Neutron component X that is unable to accomplish something?
- Is there something that you would like Neutron handle better, ie. in a more scalable, or in a more reliable way?
- What is that I would like to see happen after the RFE is accepted and implemented?
- Why do you think it is important?

Once you are happy with what you wrote, add rfe as tag, and submit. Do not worry, we are here to help you get it right! Happy hacking.

Missing your target

There are occasions when a spec will be approved and the code will not land in the cycle it was targeted at. For these cases, the work flow to get the spec into the next release is as follows:

- During the RC window, the PTL will create a directory named <release> under the backlog directory in the neutron specs repo, and he/she will move all specs that did not make the release to this directory.
- Anyone can propose a patch to neutron-specs which moves a spec from the previous release into the new release directory.

The specs which are moved in this way can be fast-tracked into the next release. Please note that it is required to re-propose the spec for the new release.

Documentation

The above process involves two places where any given feature can start to be documented - namely in the RFE bug, and in the spec - and in addition to those Neutron has a substantial *developer reference guide* (aka devref), and user-facing docs such as the *networking guide*. So it might be asked:

- What is the relationship between all of those?
- What is the point of devref documentation, if everything has already been described in the spec?

The answers have been beautifully expressed in an openstack-dev post:

1. RFE: I want X
2. Spec: I plan to implement X like this
3. devref: How X is implemented and how to extend it
4. OS docs: API and guide for using X

Once a feature X has been implemented, we shouldnt have to go to back to its RFE bug or spec to find information on it. The devref may reuse a lot of content from the spec, but the spec is not maintained and the implementation may differ in some ways from what was intended when the spec was agreed. The devref should be kept current with refactorings, etc., of the implementation.

Devref content should be added as part of the implementation of a new feature. Since the spec is not maintained after the feature is implemented, the devref should include a maintained version of the information from the spec.

If a feature requires OS docs (4), the feature patch shall include the new, or updated, documentation changes. If the feature is purely a developer facing thing, (4) is not needed.

Bugs

Neutron (client, core, FwaaS, VPNaaS) maintains all of its bugs in the following Launchpad projects:

- Launchpad Neutron
- Launchpad python-neutronclient

Neutron Bugs Team In Launchpad

The Neutron Bugs team in Launchpad is used to allow access to the projects above. Members of the above group have the ability to set bug priorities, target bugs to releases, and other administrative tasks around bugs. The administrators of this group are the members of the neutron-drivers-core gerrit group. Non administrators of this group include anyone who is involved with the Neutron project and has a desire to assist with bug triage.

If you would like to join this Launchpad group, its best to reach out to a member of the above mentioned neutron-drivers-core team in #openstack-neutron on OFTC and let them know why you would like to be a member. The team is more than happy to add additional bug triage capability, but it helps to know who is requesting access, and IRC is a quick way to make the connection.

As outlined below the bug deputy is a volunteer who wants to help with defect management. Permissions will have to be granted assuming that people sign up on the deputy role. The permission wont be given freely, a person must show some degree of prior involvement.

Neutron Bug Deputy

Neutron maintains the notion of a bug deputy. The bug deputy plays an important role in the Neutron community. As a large project, Neutron is routinely fielding many bug reports. The bug deputy is responsible for acting as a first contact for these bug reports and performing initial screening/triaging. The bug deputy is expected to communicate with the various Neutron teams when a bug has been triaged. In addition, the bug deputy should be reporting High and Critical priority bugs.

To avoid burnout, and to give a chance to everyone to gain experience in defect management, the Neutron bug deputy is a rotating role. The rotation will be set on a period (typically one or two weeks) determined by the team during the weekly Neutron IRC meeting and/or according to holidays. During the Neutron IRC meeting we will expect a volunteer to step up for the period. Members of the Neutron core team are invited to fill in the role, however non-core Neutron contributors who are interested are also encouraged to take up the role.

This contributor is going to be the bug deputy for the period, and he/she will be asked to report to the team during the subsequent IRC meeting. The PTL will also work with the team to assess that everyone gets his/her fair share at fulfilling this duty. It is reasonable to expect some imbalance from time to time, and the team will work together to resolve it to ensure that everyone is 100% effective and well rounded in their role as _custodian_ of Neutron quality. Should the duty load be too much in busy times of the release, the PTL and the team will work together to assess whether more than one deputy is necessary in a given period.

The presence of a bug deputy does not mean the rest of the team is simply off the hook for the period, in fact the bug deputy will have to actively work with the Lieutenants/Drivers, and these should help in getting the bug report moving down the resolution pipeline.

During the period a member acts as bug deputy, he/she is expected to watch bugs filed against the Neutron projects (as listed above) and do a first screening to determine potential severity, tagging, opensearch queries, other affected projects, affected releases, etc.

From time to time bugs will be filed and auto-assigned by members of the core team to get them to a swift resolution. Obviously, the deputy is exempt from screening these.

Finally, the PTL will work with the deputy to produce a brief summary of the issues of the week to be shared with the larger team during the weekly IRC meeting and tracked in the meeting notes. If for some reason the deputy is not going to attend the team meeting to report, the deputy should consider sending a brief report to the openstack-discuss@ mailing list in advance of the meeting.

Getting Ready to Serve as the Neutron Bug Deputy

If you are interested in serving as the Neutron bug deputy, there are several steps you will need to follow in order to be prepared.

- Request to be added to the neutron-bugs team in Launchpad. This request will be approved when you are assigned a bug deputy slot.
- Read this page in full. Keep this document in mind at all times as it describes the duties of the bug deputy and how to triage bugs particularly around setting the importance and tags of bugs.
- Sign up for neutron bug emails from LaunchPad.
  - Navigate to the LaunchPad Neutron bug list.
  - On the right hand side, click on Subscribe to bug mail.
  - In the pop-up that is displayed, keep the recipient as Yourself, and your subscription something useful like Neutron Bugs. You can choose either option for how much mail you get, but keep in mind that getting mail for all changes - while informative - will result in several dozen emails per day at least.
  - Do the same for the LaunchPad python-neutronclient bug list.
- Configure the information you get from LaunchPad to make visible additional information, especially the age of the bugs. You accomplish that by clicking the little gear on the left hand side of the screen at the top of the bugs list. This provides an overview of information for each bug on a single page.
- Optional: Set up your mail client to highlight bug email that indicates a new bug has been filed, since those are the ones you will be wanting to triage. Filter based on email from @bugs.launchpad.net with [NEW] in the subject line.
- Volunteer during the course of the Neutron team meeting, when volunteers to be bug deputy are requested (usually towards the beginning of the meeting).
- View your scheduled week on the Neutron Meetings page.
- During your shift, if it is feasible for your timezone, plan on attending the Neutron Drivers meeting. That way if you have tagged any bugs as RFE, you can be present to discuss them.

Bug Deputy routines in your week

- Scan New bugs to triage. If it doesnt have enough info to triage, ask more info and mark it Incomplete. If you could confirm it by yourself, mark it Confirmed. Otherwise, find someone familiar with the topic and ask his/her help.
- Scan Incomplete bugs to see if it got more info. If it was, make it back to New.
- Repeat the above routines for bugs filed in your week at least. If you can, do the same for older bugs.
- Take a note of bugs you processed. At the end of your week, post a report on openstack-discuss mailing list.

Plugin and Driver Repositories

Many plugins and drivers have backend code that exists in another repository. These repositories may have their own Launchpad projects for bugs. The teams working on the code in these repos assume full responsibility for bug handling in those projects. For this reason, bugs whose solution would exist solely in the plugin/driver repo should not have Neutron in the affected projects section. However, you should add Neutron (Or any other project) to that list only if you expect that a patch is needed to that repo in order to solve the bug.

Its also worth adding that some of these projects are part of the so called Neutron stadium. Because of that, their release is managed centrally by the Neutron release team; requests for releases need to be funnelled and screened properly before they can happen. Release request process is described *here*.

Bug Screening Best Practices

When screening bug reports, the first step for the bug deputy is to assess how well written the bug report is, and whether there is enough information for anyone else besides the bug submitter to reproduce the bug and come up with a fix. There is plenty of information on the OpenStack Bugs on how to write a good bug report and to learn how to tell a good bug report from a bad one. Should the bug report not adhere to these best practices, the bug deputys first step would be to redirect the submitter to this section, invite him/her to supply the missing information, and mark the bug report as Incomplete. For future submissions, the reporter can then use the template provided below to ensure speedy triaging. Done often enough, this practice should (ideally) ensure that in the long run, only good bug reports are going to be filed.

Bug Report Template

The more information you provide, the higher the chance of speedy triaging and resolution: identifying the problem is half the solution. To this aim, when writing a bug report, please consider supplying the following details and following these suggestions:

- Summary (Bug title): keep it small, possibly one line. If you cannot describe the issue in less than 100 characters, you are probably submitting more than one bug at once.
- Further information (Bug description): conversely from other bug trackers, Launchpad does not provide a structured way of submitting bug-related information, but everything goes in this section. Therefore, you are invited to break down the description in the following fields:
  - High level description: provide a brief sentence (a couple of lines) of what are you trying to accomplish, or would like to accomplish differently; the why is important, but can be omitted if obvious (not to you of course).

  - Pre-conditions: what is the initial state of your system? Please consider enumerating resources available in the system, if useful in diagnosing the problem. Who are you? A regular user or a super-user? Are you describing service-to-service interaction?
  - Step-by-step reproduction steps: these can be actual neutron client commands or raw API requests; Grab the output if you think it is useful. Please, consider using paste.o.o for long outputs as Launchpad poorly format the description field, making the reading experience somewhat painful.
  - Expected output: what did you hope to see? How would you have expected the system to behave? A specific error/success code? The output in a specific format? Or more than a user was supposed to see, or less?
  - Actual output: did the system silently fail (in this case log traces are useful)? Did you get a different response from what you expected?
  - Version:
    * OpenStack version (Specific stable branch, or git hash if from trunk);
    * Linux distro, kernel. For a distro, its also worth knowing specific versions of client and server, not just major release;
    * Relevant underlying processes such as openvswitch, iproute etc;
    * DevStack or other _deployment_ mechanism?
  - Environment: what services are you running (core services like DB and AMQP broker, as well as Nova/hypervisor if it matters), and which type of deployment (clustered servers); if you are running DevStack, is it a single node? Is it multi-node? Are you reporting an issue in your own environment or something you encountered in the OpenStack CI Infrastructure, aka the Gate?
  - Perceived severity: what would you consider the importance to be?
- Tags (Affected component): try to use the existing tags by relying on auto-completion. Please, refrain from creating new ones, if you need new official *tags*, please reach out to the PTL. If you would like a fix to be backported, please add a backport-potential tag. This does not mean you are gonna get the backport, as the stable team needs to follow the stable branch policy for merging fixes to stable branches.
- Attachments: consider attaching logs, truncated log snippets are rarely useful. Be proactive, and consider attaching redacted configuration files if you can, as that will speed up the resolution process greatly.

Bug Triage Process

The process of bug triaging consists of the following steps:

- Check if a bug was filed for a correct component (project). If not, either change the project or mark it as Invalid.
- For bugs that affect documentation proceed like this. If documentation affects:
  - the ReST API, add the api-ref tag to the bug.
  - the OpenStack manuals, like the Networking Guide or the Configuration Reference, create a patch for the affected files in the documentation directory in this repository. For a layout of the how the documentation directory is structured see the effective neutron guide

  - developer documentation (devref), set the bug to Confirmed for the project Neutron, otherwise set it to Invalid.

- Check if a similar bug was filed before. Rely on your memory if Launchpad is not clever enough to spot a duplicate upon submission. You may also check already verified bugs for Neutron and python-neutronclient to see if the bug has been reported. If so, mark it as a duplicate of the previous bug.
- Check if the bug meets the requirements of a good bug report, by checking that the *guidelines* are being followed. Omitted information is still acceptable if the issue is clear nonetheless; use your good judgement and your experience. Consult another core member/PTL if in doubt. If the bug report needs some love, mark the bug as Incomplete, point the submitter to this document and hope he/she turns around quickly with the missing information.

If the bug report is sound, move next:

- Revise tags as recommended by the submitter. Ensure they are official tags. If the bug report talks about deprecating features or config variables, add a deprecation tag to the list.
- As deputy one is usually excused not to process RFE bugs which are the responsibility of the drivers team members.
- Depending on ease of reproduction (or if the issue can be spotted in the code), mark it as Confirmed. If you are unable to assess/triage the issue because you do not have access to a repro environment, consider reaching out the *Lieutenant*, go-to person for the affected component; he/she may be able to help: assign the bug to him/her for further screening. If the bug already has an assignee, check that a patch is in progress. Sometimes more than one patch is required to address an issue, make sure that there is at least one patch that Closes the bug or document/question what it takes to mark the bug as fixed.
- If the bug indicates test or gate failure, look at the failures for that test over time using OpenStack OpenSearch. This can help to validate whether the bug identifies an issue that is occurring all of the time, some of the time, or only for the bug submitter. To use OpenSearch please check documentation.
- If the bug is the result of a misuse of the system, mark the bug either as Wont fix, or Opinion if you are still on the fence and need other peoples input.
- Assign the importance after reviewing the proposed severity. Bugs that obviously break core and widely used functionality should get assigned as High or Critical importance. The same applies to bugs that were filed for gate failures.
- Choose a milestone, if you can. Targeted bugs are especially important close to the end of the release.
- (Optional). Add comments explaining the issue and possible strategy of fixing/working around the bug. Also, as good as some are at adding all thoughts to bugs, it is still helpful to share the in-progress items that might not be captured in a bug description or during our weekly meeting. In order to provide some guidance and reduce ramp up time as we rotate, tagging bugs with needs-attention can be useful to quickly identify what reports need further screening/eyes on.

Check for Bugs with the timeout-abandon tag:

- Search for any bugs with the timeout abandon tag: Timeout abandon. This tag indicates that the bug had a patch associated with it that was automatically abandoned after a timing out with negative feedback.

- For each bug with this tag, determine if the bug is still valid and update the status accordingly. For example, if another patch fixed the bug, ensure its marked as Fix Released. Or, if that was the only patch for the bug and its still valid, mark it as Confirmed.
- After ensuring the bug report is in the correct state, remove the timeout-abandon tag.

You are done! Iterate.

Bug Expiration Policy and Bug Squashing

More can be found at this Launchpad page. In a nutshell, in order to make a bug report expire automatically, it needs to be unassigned, untargeted, and marked as Incomplete.

The OpenStack community has had Bug Days but they have not been wildly successful. In order to keep the list of open bugs set to a manageable number (more like <100+, rather than closer to 1000+), at the end of each release (in feature freeze and/or during less busy times), the PTL with the help of team will go through the list of open (namely new, opinion, in progress, confirmed, triaged) bugs, and do a major sweep to have the Launchpad Janitor pick them up. This gives 60 days grace period to reporters/assignees to come back and revive the bug. Assuming that at regime, bugs are properly reported, acknowledged and fix-proposed, losing unaddressed issues is not going to be a major issue, but brief stats will be collected to assess how the team is doing over time.

Tagging Bugs

Launchpads Bug Tracker allows you to create ad-hoc groups of bugs with tagging.

In the Neutron team, we have a list of agreed tags that we may apply to bugs reported against various aspects of Neutron itself. The list of approved tags used to be available on the wiki, however the section has been moved here, to improve collaborative editing, and keep the information more current. By using a standard set of tags, each explained on this page, we can avoid confusion. A bug report can have more than one tag at any given time.

Proposing New Tags

New tags, or changes in the meaning of existing tags (or deletion), are to be proposed via patch to this section. After discussion, and approval, a member of the bug team will create/delete the tag in Launchpad. Each tag covers an area with an identified go-to contact or *Lieutenant*, who can provide further insight. Bug queries are provided below for convenience, more will be added over time if needed.

Tag	Description	Contact
access-control	A bug affecting RBAC and policy.yaml	Slawek Kaplonski
api	A bug affecting the API layer	Akihiro Motoki
api-ref	A bug affecting the API reference	Akihiro Motoki
auto-allocated-topology	A bug affecting get-me-a-network	N/A
baremetal	A bug affecting Ironic support	N/A
db	A bug affecting the DB layer	Rodolfo Alonso Hernandez
deprecation	To track config/feature deprecations	Neutron PTL/drivers
dns	A bug affecting DNS integration	Miguel Lavalle
doc	A bug affecting in-tree doc	Akihiro Motoki
fullstack	A bug in the fullstack subtree	Rodolfo Alonso Hernandez
functional-tests	A bug in the functional tests subtree	Rodolfo Alonso Hernandez
gate-failure	A bug affecting gate stability	Slawek Kaplonski

continues on next pag

Table 1 – continued from previous page

Tag	Description	Contact
ipv6	A bug affecting IPv6 support	Brian Haley
l2-pop	A bug in L2 Population mech driver	Miguel Lavalle
l3-bgp	A bug affecting neutron-dynamic-routing	Tobias Urdin/ Jens Harbott
l3-dvr-backlog	A bug affecting distributed routing	Yulong Liu/ Brian Haley
l3-ha	A bug affecting L3 HA (vrrp)	Brian Haley
l3-ipam-dhcp	A bug affecting L3/DHCP/metadata	Miguel Lavalle
lib	An issue affecting neutron-lib	Neutron PTL
loadimpact	Performance penalty/improvements	Miguel Lavalle/ Oleg Bondarev
logging	An issue with logging guidelines	N/A
low-hanging-fruit	Starter bugs for new contributors	Miguel Lavalle
metering	A bug affecting the metering layer	N/A
needs-attention	A bug that needs further screening	PTL/Bug Deputy
opnfv	Reported by/affecting OPNFV initiative	Drivers team
ops	Reported by or affecting operators	Drivers Team
oslo	An interop/cross-project issue	Bernard Cafarelli/ Rodolfo Alonso Hernand
ovn	A bug affecting ML2/OVN	Jakub Libosvar/ Lucas Alvares Gomes
ovn-bgp-agent	A bug affecting OVN BGP agent	Luis Tomas Bolivar/ Lucas Alvares Gomes
ovn-octavia-provider	A bug affecting OVN Octavia provider driver	Fernando Royo
ovs	A bug affecting ML2/OVS	Miguel Lavalle
ovs-fw	A bug affecting OVS firewall	Miguel Lavalle
ovsdb-lib	A bug affecting OVSDB library	Terry Wilson
pyroute2	A bug affecting pyroute2 library	Rodolfo Alonso Hernandez
qos	A bug affecting ML2/QoS	Rodolfo Alonso Hernandez
rfe	Feature enhancements being screened	Drivers Team
rfe-confirmed	Confirmed feature enhancements	Drivers Team
rfe-triaged	Triaged feature enhancements	Drivers Team
rfe-approved	Approved feature enhancements	Drivers Team
rfe-postponed	Postponed feature enhancements	Drivers Team
sg-fw	A bug affecting security groups	Brian Haley
sriov-pci-pt	A bug affecting Sriov/PCI PassThrough	Moshe Levi
stable	A bug affecting only stable branches	Bernard Cafarelli
tempest	A bug in tempest subtree tests	Rodolfo Alonso Hernandez
troubleshooting	An issue affecting ease of debugging	PTL/Drivers Team
unittest	A bug affecting the unit test subtree	Rodolfo Alonso Hernandez
usability	UX, interoperability, feature parity	PTL/Drivers Team
vpnaas	A bug affecting neutron-vpnaas	Dongcan Ye
xxx-backport-potential	Cherry-pick request for stable team	Bernard Cafarelli/ Brian Haley

Access Control

- Access Control - All bugs
- Access Control - In progress

API

- API - All bugs
- API - In progress

API Reference

- API Reference - All bugs
- API Reference - In progress

Auto Allocated Topology

- Auto Allocated Topology - All bugs
- Auto Allocated Topology - In progress

Baremetal

- Baremetal - All bugs
- Baremetal - In progress

DB

- DB - All bugs
- DB - In progress

Deprecation

- Deprecation - All bugs
- DeprecationB - In progress

DNS

- DNS - All bugs
- DNS - In progress

DOC

- DOC - All bugs
- DOC - In progress

Fullstack

- Fullstack - All bugs
- Fullstack - In progress

Functional Tests

- Functional tests - All bugs
- Functional tests - In progress

FWAAS

- FWaaS - All bugs
- FWaaS - In progress

Gate Failure

- Gate failure - All bugs
- Gate failure - In progress

IPV6

- IPv6 - All bugs
- IPv6 - In progress

L2 Population

- L2 Pop - All bugs
- L2 Pop - In progress

L3 BGP

- L3 BGP - All bugs
- L3 BGP - In progress

L3 DVR Backlog

- L3 DVR - All bugs
- L3 DVR - In progress

L3 HA

- L3 HA - All bugs
- L3 HA - In progress

L3 IPAM DHCP

- L3 IPAM DHCP - All bugs
- L3 IPAM DHCP - In progress

Lib

- Lib - All bugs

Load Impact

- Load Impact - All bugs
- Load Impact - In progress

Logging

- Logging - All bugs
- Logging - In progress

Low hanging fruit

- Low hanging fruit - All bugs
- Low hanging fruit - In progress

Metering

- Metering - All bugs
- Metering - In progress

Needs Attention

- Needs Attention - All bugs

OPNFV

- OPNFV - All bugs

Operators/Operations (ops)

- Ops - All bugs

OSLO

- Oslo - All bugs
- Oslo - In progress

OVN

- OVN - All bugs
- OVN - In progress

OVN BGP Agent

- OVN BGP Agent - All bugs
- OVN BGP Agent - In progress

OVN Octavia Provider driver

- OVN Octavia Provider driver - All bugs
- OVN Octavia Provider driver - In progress

OVS

- OVS - All bugs
- OVS - In progress

OVS Firewall

- OVS Firewall - All bugs
- OVS Firewall - In progress

OVSDB Lib

- OVSDB Lib - All bugs
- OVSDB Lib - In progress

pyroute2

- Pyroute2 Lib - All bugs
- Pyroute2 Lib - In progress

QoS

- QoS - All bugs
- QoS - In progress

RFE

- RFE - All bugs
- RFE - In progress

RFE-Confirmed

- RFE-Confirmed - All bugs

RFE-Triaged

- RFE-Triaged - All bugs

RFE-Approved

- RFE-Approved - All bugs
- RFE-Approved - In progress

RFE-Postponed

- RFE-Postponed - All bugs
- RFE-Postponed - In progress

SRIOV-PCI PASSTHROUGH

- SRIOV/PCI-PT - All bugs
- SRIOV/PCI-PT - In progress

SG-FW

- Security groups - All bugs
- Security groups - In progress

Stable

- Stable - All bugs
- Stable - In progress bugs

Tempest

- Tempest - All bugs
- Tempest - In progress

Troubleshooting

- Troubleshooting - All bugs
- Troubleshooting - In progress

Unit test

- Unit test - All bugs
- Unit test - In progress

Usability

- UX - All bugs
- UX - In progress

VPNAAS

- VPNaaS - All bugs
- VPNaaS - In progress

Backport/RC potential

List of all `Backport/RC potential` bugs for stable releases can be found on launchpad. Pointer to Launchpads page with list of such bugs for any stable release can be built by using link:

https://bugs.launchpad.net/neutron/+bugs?field.tag={STABLE_BRANCH}-backport-potential

where `STABLE_BRANCH` is always name of one of the 3 latest releases.

Code Reviews

Code reviews are a critical component of all OpenStack projects. Neutron accepts patches from many diverse people with diverse backgrounds, employers, and experience levels. Code reviews provide a way to enforce a level of consistency across the project, and also allow for the careful on boarding of contributions from new contributors.

Neutron Code Review Practices

Neutron follows the code review guidelines as set forth for all OpenStack projects. It is expected that all reviewers are following the guidelines set forth on that page.

In addition to that, the following rules are to follow:

- Any change that requires a new feature from Neutron runtime dependencies requires special review scrutiny to make sure such a change does not break a supported platform (examples of those platforms are latest Ubuntu LTS or CentOS). Runtime dependencies include but are not limited to: kernel, daemons and tools as defined in `oslo.rootwrap` filter files, runlevel management systems, as well as other elements of Neutron execution environment.

  > **Note**
  >
  > For some components, the list of supported platforms can be wider than usual. For example, Open vSwitch agent is expected to run successfully in Win32 runtime environment.

  1. All such changes must be tagged with `UpgradeImpact` in their commit messages.
  2. Reviewers are then advised to make an effort to check if the newly proposed runtime dependency is fulfilled on supported platforms.
  3. Specifically, reviewers and authors are advised to use existing gate and experimental platform specific jobs to validate those patches. To trigger experimental jobs, use the usual protocol (posting `check experimental` comment in Gerrit). CI will then execute and report back a baseline of Neutron tests for platforms of interest and will provide feedback on the effect of the runtime change required.

4. If review identifies that the proposed change would break a supported platform, advise to rework the patch so that its no longer breaking the platform. One of the common ways of achieving that is gracefully falling back to alternative means on older platforms, another is hiding the new code behind a conditional, potentially controlled with a `oslo.config` option.

> **Note**
>
> Neutron team retains the right to remove any platform conditionals in future releases. Platform owners are expected to accommodate in due course, or otherwise see their platforms broken. The team also retains the right to discontinue support for unresponsive platforms.

5. The change should also include a new sanity check that would help interested parties to identify their platform limitation in timely manner.

- Special attention should also be paid to changes in Neutron that can impact the Stadium and the wider family of networking-related projects (referred to as sub-projects below). These changes include:

  1. Renaming or removal of methods.
  2. Addition or removal of positional arguments.
  3. Renaming or removal of constants.

  To mitigate the risk of impacting the sub-projects with these changes, the following measures are suggested:

  1. Use of the online tool codesearch to ascertain how the proposed changes will affect the code of the sub-projects.
  2. Review the results of the non-voting check and 3rd party CI jobs executed by the sub-projects against the proposed change, which are returned by Zuul in the changes Gerrit page.

  When impacts are identified as a result of the above steps, every effort must be made to work with the affected sub-projects to resolve the issues.

- Any change that modifies or introduces a new API should have test coverage in neutron-tempest-plugin or tempest test suites. There should be at least one API test added for a new feature, but it is preferred that both API and scenario tests be added where it is appropriate.

  Scenario tests should cover not only the base level of new functionality, but also standard ways in which the functionality can be used. For example, if the feature adds a new kind of networking (like e.g. trunk ports) then tests should make sure that instances can use IPs provided by that networking, can be migrated, etc.

  It is also preferred that some negative test cases, like API tests to ensure that correct HTTP error is returned when wrong data is provided, will be added where it is appropriate.

- It is usually enough for any mechanical changes, like e.g. translation imports or imports of updated CI templates, to have only one +2 Code-Review vote to be approved. If there is any uncertainty about a specific patch, it is better to wait for review from another core reviewer before approving the patch.

Neutron Spec Review Practices

In addition to code reviews, Neutron also maintains a BP specification git repository. Detailed instructions for the use of this repository are provided here. It is expected that Neutron core team members are actively reviewing specifications which are pushed out for review to the specification repository. In addition, there is a neutron-drivers team, composed of a handful of Neutron core reviewers, who can approve and merge Neutron specs.

Some guidelines around this process are provided below:

- Once a specification has been pushed, it is expected that it will not be approved for at least 3 days after a first Neutron core reviewer has reviewed it. This allows for additional cores to review the specification.
- For blueprints which the core team deems of High or Critical importance, core reviewers may be assigned based on their subject matter expertise.
- Specification priority will be set by the PTL with review by the core team once the specification is approved.

Tracking Review Statistics

Stackalytics provides some nice interfaces to track review statistics. The links are provided below. These statistics are used to track not only Neutron core reviewer statistics, but also to track review statistics for potential future core members.

- 30 day review stats
- 60 day review stats
- 90 day review stats
- 180 day review stats

Contributor Onboarding

For new contributors, the following are useful onboarding information.

Contributing to Neutron

Work within Neutron is discussed on the openstack-discuss mailing list, as well as in the #openstack-neutron IRC channel. While these are great channels for engaging Neutron, the bulk of discussion of patches and code happens in gerrit itself.

With regards to gerrit, code reviews are a great way to learn about the project. There is also a list of low or wishlist priority bugs which are ideal for a new contributor to take on. If you havent done so you should setup a Neutron development environment so you can actually run the code. Devstack is the usual convenient environment to setup such an environment. See devstack.org or NeutronDevstack for more information on using Neutron with devstack.

Helping with documentation can also be a useful first step for a newcomer. Here is a list of tagged documentation and API reference bugs:

- Documentation bugs
- Api-ref bugs

IRC Information and Etiquette

The main IRC channel for Neutron is #openstack-neutron.

Gate Failure Triage

This page provides guidelines for spotting and assessing neutron gate failures. Some hints for triaging failures are also provided.

Spotting Gate Failures

This can be achieved using several tools:

- Grafana dashboard
- OpenSearch

For checking gate failures with opensearch please see documentation. The following query will return failures for a specific job:

> build_status:FAILURE AND message:Finished AND
build_name:check-tempest-dsvm-neutron AND build_queue:gate

And divided by the total number of jobs executed:

> message:Finished AND build_name:check-tempest-dsvm-neutron AND
build_queue:gate

It will return the failure rate in the selected period for a given job. It is important to remark that failures in the check queue might be misleading as the problem causing the failure is most of the time in the patch being checked. Therefore it is always advisable to work on failures occurred in the gate queue. However, these failures are a precious resource for assessing frequency and determining root cause of failures which manifest in the gate queue.

The step above will provide a quick outlook of where things stand. When the failure rate raises above 10% for a job in 24 hours, its time to be on alert. 25% is amber alert. 33% is red alert. Anything above 50% means that probably somebody from the infra team has already a contract out on you. Whether you are relaxed, in alert mode, or freaking out because you see a red dot on your chest, it is always a good idea to check on daily bases the elastic-recheck pages.

Under the gate pipeline tab, you can see gate failure rates for already known bugs. The bugs in this page are ordered by decreasing failure rates (for the past 24 hours). If one of the bugs affecting Neutron is among those on top of that list, you should check that the corresponding bug is already assigned and somebody is working on it. If not, and there is not a good reason for that, it should be ensured somebody gets a crack at it as soon as possible. The other part of the story is to check for uncategorized failures. This is where failures for new (unknown) gate breaking bugs end up; on the other hand also infra error causing job failures end up here. It should be duty of the diligent Neutron developer to ensure the classification rate for neutron jobs is as close as possible to 100%. To this aim, the diligent Neutron developer should adopt the procedure outlined in the following sections.

Troubleshooting Tempest jobs

1. Open logs for failed jobs and look for logs/testr_results.html.gz.
2. **If that file is missing, check console.html and see where the job failed.**
   1. If there is a failure in devstack-gate-cleanup-host.txt its likely to be an infra issue.

    2. If the failure is in devstacklog.txt it could a devstack, neutron, or infra issue.

3. However, most of the time the failure is in one of the tempest tests. Take note of the error message and go to opensearch.
4. On opensearch, search for occurrences of this error message, and try to identify the root cause for the failure (see below).
5. File a bug for this failure, and push an *Elastic Recheck Query* for it.
6. If you are confident with the area of this bug, and you have time, assign it to yourself; otherwise look for an assignee or talk to the Neutrons bug deputy to find an assignee.

Troubleshooting functional/fullstack job

1. Go to the job link provided by Zuul CI.
2. Look at logs/testr_results.html.gz for which particular test failed.
3. More logs from a particular test are stored at logs/dsvm-functional-logs/<path_of_the_test> (or dsvm-fullstack-logs for fullstack job).
4. Find the error in the logs and search for similar errors in existing launchpad bugs. If no bugs were reported, create a new bug report. Dont forget to put a snippet of the trace into the new launchpad bug. If the log file for a particular job doesnt contain any trace, pick the one from testr_results.html.gz.
5. Create an *Elastic Recheck Query*

Troubleshooting Grenade jobs

Grenade is used in the Neutron gate to test every patch proposed to Neutron to ensure it will not break the upgrade process. Upgrading from the N-1 to the N branch is constantly being tested. So if you send patch to the Neutron `master` branch Grenade jobs will first deploy Neutron from the last stable release and then upgrade it to the master branch with your patch. Details about how Grenade works are available in the documentation.

In Neutron CI jobs that use Grenade are run in the multinode jobs configuration which means that we have deployed OpenStack on 2 VMs:

- one called `controller` which is in an all in one node so it runs neutron-server, as well as the neutron-ovs-agent and nova-compute services,
- one called `compute1` which runs only services like nova-compute and neutron-ovs-agent.

Neutron supports that neutron-server in N version will always work with the agents which runs in N-1 version. To test such scenario all our Grenade jobs upgrade OpenStack services only on the `controller` node. Services which run on the `compute1` node are always run with the old release during that job.

Debugging of failures in the Grenade job is very similar to debugging any other Tempest based job. The difference is that in the logs of the Grenade job, there is always logs/old and logs/new directories which contain Devstack logs from each run of the Devstacks stack.sh script. In the logs/grenade.sh_log.txt file there is a full log of the grenade.sh run and you should always start checking failures from that file. Logs of the Neutron services for old and new versions are in the same files, like, for example, logs/screen-q-svc.txt for neutron-server logs. You will find in that log when the service was restarted - that is the moment when it was upgraded by Grenade and it is now running the new version.

Advanced Troubleshooting of Gate Jobs

As a first step of troubleshooting a failing gate job, you should always check the logs of the job as described above. Unfortunately, sometimes when a tempest/functional/fullstack job is failing, it might be hard to reproduce it in a local environment, and might also be hard to understand the reason of such a failure from only reading the logs of the failed job. In such cases there are some additional ways to debug the job directly on the test node in a `live` setting.

This can be done in two ways:

1. Using the remote_pdb python module and `telnet` to directly access the python debugger while in the failed test.

   To achieve this, you need to send a `Do not merge` patch to gerrit with changes as described below:

   - Add an iptables rule to accept incoming telnet connections to remote_pdb. This can be done in one of the ansible roles used in the test job. Like for example in `neutron/roles/configure_functional_tests` file for functional tests:

     ```
     sudo iptables -I openstack-INPUT -p tcp -m state --state NEW -m tcp -
     ↪-dport 44444 -j ACCEPT
     ```

   - Increase the `OS_TEST_TIMEOUT` value to make the test wait longer when remote_pdb is active to make debugging easier. This change can also be done in the ansible role mentioned above:

     ```
     export OS_TEST_TIMEOUT=999999
     ```

     Please note that the overall job will be limited by the job timeout, and that cannot be changed from within the job.

   - To make it easier to find the IP address of the test node, you should add to the ansible role so it prints the IPs configured on the test node. For example:

     ```
     hostname -I
     ```

   - Add the package `remote_pdb` to the `test-requirements.txt` file. That way it will be automatically installed in the venv of the test before it is run:

     ```
     $ tail -1 test-requirements.txt
     remote_pdb
     ```

   - Finally, you need to import and call the remote_pdb module in the part of your test code where you want to start the debugger:

     ```
     $ diff --git a/neutron/tests/fullstack/test_connectivity.py b/
     ↪neutron/tests/fullstack/test_connectivity.py
     index c8650b0..260207b 100644
     --- a/neutron/tests/fullstack/test_connectivity.py
     +++ b/neutron/tests/fullstack/test_connectivity.py
     @@ -189,6 +189,8 @@ class
     TestLinuxBridgeConnectivitySameNetwork(BaseConnectivitySameNetworkTest):
              ]

          def test_connectivity(self):
     ```

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```
+        import remote_pdb; remote_pdb.set_trace('0.0.0.0',␣
↪port=44444)
+
self._test_connectivity()
```

Please note that discovery of public IP addresses is necessary because by default remote_pdb will only bind to the `127.0.0.1` IP address. Above is just an example of one of possible method, there could be other ways to do this as well.

When all the above changes are done, you must commit them and go to the Zuul status page to find the status of the tests for your `Do not merge` patch. Open the console log for your job and wait there until `remote_pdb` is started. You then need to find the IP address of the test node in the console log. This is necessary to connect via `telnet` and start debugging. It will be something like:

```
RemotePdb session open at 172.99.68.50:44444, waiting for connection ...
```

An example of such a `Do not merge` patch described above can be found at https://review.opendev.org/#/c/558259/.

Please note that after adding new packages to the `requirements.txt` file, the `requirements-check` job for your test patch will fail, but it is not important for debugging.

2. If root access to the test node is necessary, for example, to check if VMs have really been spawned, or if router/dhcp namespaces have been configured properly, etc., you can ask a member of the infra-team to hold the job for troubleshooting. You can ask someone to help with that on the `openstack-infra` IRC channel. In that case, the infra-team will need to add your SSH key to the test node, and configure things so that if the job fails, the node will not be destroyed. You will then be able to SSH to it and debug things further. Please remember to tell the infra-team when you finish debugging so they can unlock and destroy the node being held.

The above two solutions can be used together. For example, you should be able to connect to the test node with both methods:

- using `remote_pdb` to connect via `telnet`;
- using SSH to connect as a root to the test node.

You can then ask the infra-team to add your key to the specific node on which you have already started your `remote_pdb` session.

Root Causing a Gate Failure

Time-based identification, i.e. find the naughty patch by log scavenging.

Filing An Elastic Recheck Query

The elastic recheck page has all the current open ER queries. To file one, please see the ER Wiki.

Pre-release check list

This page lists things to cover before a Neutron release and will serve as a guide for next release managers.

Server

Major release

A Major release is cut off once per development cycle and has an assigned name (Victoria, Wallaby,)

Prior to major release,

1. consider blocking all patches that are not targeted for the new release;
2. consider blocking trivial patches to keep the gate clean;
3. revise the current list of blueprints and bugs targeted for the release; roll over anything that does not fit there, or wont make it (note that no new features land in master after so called feature freeze is claimed by release team; there is a feature freeze exception (FFE) process described in release engineering documentation in more details: http://docs.openstack.org/project-team-guide/release-management.html);
4. start collecting state for targeted features from the team. For example, propose a post-mortem patch for neutron-specs as in: https://review.opendev.org/c/openstack/neutron-specs/+/286413/
5. revise deprecation warnings collected in latest Zuul runs: some of them may indicate a problem that should be fixed prior to release (see deprecations.txt file in those log directories); also, check whether any Launchpad bugs with the deprecation tag need a clean-up or a follow-up in the context of the release being planned;
6. update the ovs-ovn minimum requirement table in `doc/source/install/ovs-ovn-requirements.rst`;
7. check that release notes and sample configuration files render correctly, arrange clean-up if needed;
8. ensure all doc links are valid by running `tox -e linkcheck` and addressing any broken links.

New major release process contains several phases:

1. master branch is blocked for patches that are not targeted for the release;
2. the whole team is expected to work on closing remaining pieces targeted for the release;
3. once the team is ready to release the first release candidate (RC1), either PTL or one of release liaisons proposes a patch for openstack/releases repo. For example, see: https://review.opendev.org/c/openstack/releases/+/753039/
4. once the openstack/releases patch lands, release team creates a new stable branch using hash values specified in the patch;
5. at this point, master branch is open for patches targeted to the next release; PTL unblocks all patches that were blocked in step 1;
6. if additional patches are identified that are critical for the release and must be shipped in the final major build, corresponding bugs are tagged with <release>-rc-potential in Launchpad, fixes are prepared and land in master branch, and are then backported to the newly created stable branch;
7. if patches landed in the release stable branch as per the previous step, a new release candidate that would include those patches should be requested by PTL in openstack/releases repo;

8. eventually, the latest release candidate requested by PTL becomes the final major release of the project.

Release candidate (RC) process allows for stabilization of the final release.

The following technical steps should be taken before the final release is cut off:

1. the latest alembic script of the version that is being released is tagged with a milestone label; for example, in https://review.opendev.org/c/openstack/neutron/+/944804 the latest script is being tagged with `RELEASE_2025_1`
2. the new release tag must be created; using the previous example, the tag `RELEASE_2025_2` is created.
3. add the released version tag to the `NEUTRON_MILESTONES` list; in the previous example, the tag `RELEASE_2025_1` is added to this list.
4. update the `CURRENT_RELEASE` variable with the new tag created and add it to the `RELEASES` tuple; in the previous example, `RELEASE_2025_2` is added to the `RELEASES` tuple and used as the current release milestone.

In the new stable branch, you should make sure that:

1. .gitreview file points to the new branch; https://review.opendev.org/c/openstack/neutron/+/754738/
2. if the branch uses constraints to manage gated dependency versions, the default constraints file name points to corresponding stable branch in openstack/requirements repo; https://review.opendev.org/c/openstack/neutron/+/754739/
3. job templates are updated to use versions for that branch; https://review.opendev.org/c/openstack/neutron-tempest-plugin/+/756585/ and https://review.opendev.org/c/openstack/neutron/+/759856/
4. all CI jobs running against master branch of another project are dropped; https://review.opendev.org/c/openstack/neutron/+/756695/
5. neutron itself is capped in requirements in the new branch; https://review.opendev.org/c/openstack/requirements/+/764022/
6. all new Neutron features without an API extension which have new tempest tests (in `tempest` or in `neutron-tempest-plugin`) must have a new item in `available_features` list under `network-feature-enabled` section in `tempest.conf`. To make stable jobs execute only the necessary tests the list in devstack (devstack/lib/tempest) must be checked and filled; https://review.opendev.org/c/openstack/devstack/+/769885
7. Grafana dashboards for stable branches should be updated to point to the latest releases; https://review.opendev.org/c/openstack/project-config/+/757102
8. Check API extensions list in devstack: https://review.opendev.org/c/openstack/devstack/+/811485 (Full list of QA related release checks can be found here: https://wiki.openstack.org/wiki/QA/releases#Projects_with_only_Branches

Note that some of those steps are covered by the OpenStack release team and its release bot.

While preparing the next release and even in the middle of development, its worth keeping the infrastructure clean. Consider using these tools to declutter the project infrastructure:

1. declutter Gerrit:

```
<neutron>/tools/abandon_old_reviews.sh
```

2. declutter Launchpad:

```
<release-tools>/pre_expire_bugs.py neutron --day <back-to-the-beginning-
↪of-the-release>
```

Minor release

A Minor release is created from an existing stable branch after the initial major release, and usually contains bug fixes and small improvements only. The minor release frequency should follow the release schedule for the current series. For example, assuming the current release is Rocky, stable branch releases should coincide with milestones R1, R2, R3 and the final release. Stable branches can be also released more frequently if needed, for example, if there is a major bug fix that has merged recently.

The following steps should be taken before claiming a successful minor release:

1. a patch for openstack/releases repo is proposed and merged.

Minor version number should be bumped always in cases when new release contains a patch which introduces for example:

1. new OVO version for an object,
2. new configuration option added,
3. requirement change,
4. API visible change,

The above list doesnt cover all possible cases. Those are only examples of fixes which require bump of minor version number but there can be also other types of changes requiring the same.

Changes that require the minor version number to be bumped should always have a release note added.

In other cases only patch number can be bumped.

Client

Most tips from the Server section apply to client releases too. Several things to note though:

1. when preparing for a major release, pay special attention to client bits that are targeted for the release. Global openstack/requirements freeze happens long before first RC release of server components. So if you plan to land server patches that depend on a new client, make sure you dont miss the requirements freeze. After the freeze is in action, there is no easy way to land more client patches for the planned target. All this may push an affected feature to the next development cycle.

Team Structure

Neutron Core Reviewers

The Neutron Core Reviewer Team is responsible for many things related to Neutron. A lot of these things include mundane tasks such as the following:

- Ensuring the bug count is low
- Curating the gate and triaging failures

- Working on integrating shared code from projects such as Oslo
- Ensuring documentation is up to date and remains relevant
- Ensuring the level of testing for Neutron is adequate and remains relevant as features are added
- Helping new contributors with questions as they peel back the covers of Neutron
- Answering questions and participating in mailing list discussions
- Interfacing with other OpenStack teams and ensuring they are going in the same parallel direction
- Reviewing and merging code into the neutron tree

In essence, core reviewers share the following common ideals:

1. They share responsibility in the projects success.
2. They have made a long-term, recurring time investment to improve the project.
3. They spend their time doing what needs to be done to ensure the projects success, not necessarily what is the most interesting or fun.

A core reviewers responsibility doesnt end up with merging code. The above lists are adding context around these responsibilities.

Core Review Hierarchy

As Neutron has grown in complexity, it has become impossible for any one person to know enough to merge changes across the entire codebase. Areas of expertise have developed organically, and it is not uncommon for existing cores to defer to these experts when changes are proposed. Existing cores should be aware of the implications when they do merge changes outside the scope of their knowledge. It is with this in mind we propose a new system built around Lieutenants through a model of trust.

In order to scale development and responsibility in Neutron, we have adopted a Lieutenant system. The PTL is the leader of the Neutron project, and ultimately responsible for decisions made in the project. The PTL has designated Lieutenants in place to help run portions of the Neutron project. The Lieutenants are in charge of their own areas, and they can propose core reviewers for their areas as well. The core reviewer addition and removal polices are in place below. The Lieutenants for each system, while responsible for their area, ultimately report to the PTL. The PTL may opt to have regular one on one meetings with the lieutenants. The PTL will resolve disputes in the project that arise between areas of focus, core reviewers, and other projects. Please note Lieutenants should be leading their own area of focus, not doing all the work themselves.

As was mentioned in the previous section, a cores responsibilities do not end with merging code. They are responsible for bug triage and gate issues among other things. Lieutenants have an increased responsibility to ensure gate and bug triage for their area of focus is under control.

Neutron Lieutenants

The following are the current Neutron Lieutenants.

Area	Lieutenant	IRC nick
API	Akihiro Motoki	amotoki
	Slawomir Kaplonski	slaweq
DB	Rodolfo Alonso Hernandez	ralonsoh
Built-In Control Plane	Miguel Lavalle	mlavalle
Client	Akihiro Motoki	amotoki
	Slawomir Kaplonski	slaweq
	Lajos Katona	lajoskatona
Docs	Akihiro Motoki	amotoki
	Lajos Katona	lajoskatona
Infra	Rodolfo Alonso Hernandez	ralonsoh
	Jens Harbott	frickler
L3	Miguel Lavalle	mlavalle
	Yulong Liu	liuyulong
Testing	Lajos Katona	lajoskatona
	Slawomir Kaplonski	slaweq

Some notes on the above:

- Built-In Control Plane means the L2 agents, DHCP agents, SGs, metadata agents and ML2.
- The client includes commands installed server side.
- L3 includes the L3 agent, DVR, Dynamic routing and IPAM.
- Note these areas may change as the project evolves due to code refactoring, new feature areas, and libification of certain pieces of code.
- Infra means interactions with infra from a neutron perspective

Sub-project Lieutenants

Neutron also consists of several plugins, drivers, and agents that are developed effectively as sub-projects within Neutron in their own git repositories. Lieutenants are also named for these sub-projects to identify a clear point of contact and leader for that area. The Lieutenant is also responsible for updating the core review team for the sub-projects repositories.

Area	Lieutenant	IRC nick
networking-bgpvpn / networking-bagpipe	Lajos Katona	lajoskatona
	Thomas Morin	tmorin
neutron-dynamic-routing	Tobias Urdin	tobias-urdin
	Jens Harbott	frickler
neutron-fwaas	ZhouHeng	zhouhenglc
neutron-vpnaas	YAMAMOTO Takashi	yamamoto
	Dongcan Ye	yedongcan
networking-sfc	Dharmendra Kushwaha	dkushwaha
ovn-octavia-provider	Luis Tomas Bolivar	ltomasbo
	Fernando Royo	froyo
ovsdbapp	Terry Wilson	otherwiseguy
os-ken	Rodolfo Alonso	ralonsoh

Existing Core Reviewers

Existing core reviewers have been reviewing code for a varying degree of cycles. With the new plan of Lieutenants and ownership, its fair to try to understand how they fit into the new model. Existing core reviewers seem to mostly focus in particular areas and are cognizant of their own strengths and weaknesses. These members may not be experts in all areas, but know their limits, and will not exceed those limits when reviewing changes outside their area of expertise. The model is built on trust, and when that trust is broken, responsibilities will be taken away.

Lieutenant Responsibilities

In the hierarchy of Neutron responsibilities, Lieutenants are expected to partake in the following additional activities compared to other core reviewers:

- Ensuring feature requests for their areas have adequate testing and documentation coverage.
- Gate triage and resolution. Lieutenants are expected to work to keep the Neutron gate running smoothly by triaging issues, filing elastic recheck queries, and closing gate bugs.
- Triaging bugs for the specific areas.

Neutron Teams

Given all of the above, Neutron has a number of core reviewer teams with responsibility over the areas of code listed below:

Neutron Core Reviewer Team

Neutron core reviewers have merge rights to the following git repositories:

- openstack/neutron
- openstack/python-neutronclient

Please note that as we adopt to the system above with core specialty in particular areas, we expect this broad core team to shrink as people naturally evolve into an area of specialization.

Core Reviewer Teams for Plugins and Drivers

The plugin decomposition effort has led to having many drivers with code in separate repositories with their own core reviewer teams. For each one of these repositories in the following repository list, there is a core team associated with it:

- Neutron project team

These teams are also responsible for handling their own specs/RFEs/features if they choose to use them. However, by choosing to be a part of the Neutron project, they submit to oversight and veto by the Neutron PTL if any issues arise.

Neutron Specs Core Reviewer Team

Neutron specs core reviewers have +2 rights to the following git repositories:

- openstack/neutron-specs

The Neutron specs core reviewer team is responsible for reviewing specs targeted to all Neutron git repositories (Neutron + Advanced Services). It is worth noting that specs reviewers have the following attributes which are potentially different than code reviewers:

- Broad understanding of cloud and networking technologies
- Broad understanding of core OpenStack projects and technologies
- An understanding of the effect approved specs have on the teams development capacity for each cycle

Specs core reviewers may match core members of the above mentioned groups, but the group can be extended to other individuals, if required.

Drivers Team

The drivers team is the group of people who have full rights to the specs repo. This team, which matches Launchpad Neutron Drivers team, is instituted to ensure a consistent architectural vision for the Neutron project, and to continue to disaggregate and share the responsibilities of the Neutron PTL. The team is in charge of reviewing and commenting on *RFEs*, and working with specification contributors to provide guidance on the process that govern contributions to the Neutron project as a whole. The team meets regularly to go over RFEs and discuss the project roadmap. Anyone is welcome to join and/or read the meeting notes.

Release Team

The release team is a group of people with some additional gerrit permissions primarily aimed at allowing release management of Neutron sub-projects. These permissions include:

- Ability to push signed tags to sub-projects whose releases are managed by the Neutron release team as opposed to the OpenStack release team.
- Ability to push merge commits for Neutron or other sub-projects.
- Ability to approve changes in all Neutron git repositories. This is required as the team needs to be able to quickly unblock things if needed, especially at release time.

Code Merge Responsibilities

While everyone is encouraged to review changes for these repositories, members of the Neutron core reviewer group have the ability to +2/-2 and +A changes to these repositories. This is an extra level of responsibility not to be taken lightly. Correctly merging code requires not only understanding the code itself, but also how the code affects things like documentation, testing, and interactions with other projects. It also means you pay attention to release milestones and understand if a patch youre merging is marked for the release, especially critical during the feature freeze.

The bottom line here is merging code is a responsibility Neutron core reviewers have.

Adding or Removing Core Reviewers

A new Neutron core reviewer may be proposed at anytime on the openstack-discuss mailing list. Typically, the Lieutenant for a given area will propose a new core reviewer for their specific area of coverage, though the Neutron PTL may propose new core reviewers as well. The proposal is typically made after discussions with existing core reviewers. Once a proposal has been made, three existing Neutron core reviewers from the Lieutenants area of focus must respond to the email with a +1. If the member is being added by a Lieutenant from an area of focus with less than three members, a simple majority will be used to determine if the vote is successful. Another Neutron core reviewer from the same area of focus can vote -1 to veto the proposed new core reviewer. The PTL will mediate all disputes for core reviewer additions.

The PTL may remove a Neutron core reviewer at any time. Typically when a member has decreased their involvement with the project through a drop in reviews and participation in general project development, the PTL will propose their removal and remove them. Please note there is no voting or vetoing of core reviewer removal. Members who have previously been a core reviewer may be fast-tracked back into a core reviewer role if their involvement picks back up and the existing core reviewers support their re-instatement.

Neutron Core Reviewer Membership Expectations

Neutron core reviewers have the following expectations:

- Reasonable attendance at the weekly Neutron IRC meetings.
- **Participation in Neutron discussions on the mailing list, as well as**
  in-channel in #openstack-neutron.
- Participation in Neutron related design summit sessions at the OpenStack Summits.

Please note in-person attendance at design summits, mid-cycles, and other code sprints is not a requirement to be a Neutron core reviewer. The Neutron team will do its best to facilitate virtual attendance at all events. Travel is not to be taken lightly, and we realize the costs involved for those who partake in attending these events.

In addition to the above, code reviews are the most important requirement of Neutron core reviewers. Neutron follows the documented OpenStack code review guidelines. We encourage all people to review Neutron patches, but core reviewers are required to maintain a level of review numbers relatively close to other core reviewers. There are no hard statistics around code review numbers, but in general we use 30, 60, 90 and 180 day stats when examining review stats.

- 30 day review stats
- 60 day review stats
- 90 day review stats
- 180 day review stats

There are soft-touch items around being a Neutron core reviewer as well. Gaining trust with the existing Neutron core reviewers is important. Being able to work together with the existing Neutron core reviewer team is critical as well. Being a Neutron core reviewer means spending a significant amount of time with the existing Neutron core reviewers team on IRC, the mailing list, at Summits, and in reviews. Ensuring you participate and engage here is critical to becoming and remaining a core reviewer.

Third-party CI

What Is Expected of Third Party CI System for Neutron

As of the Liberty summit, Neutron no longer *requires* a third-party CI, but it is strongly encouraged, as internal neutron refactoring can break external plugins and drivers at any time.

Neutron expects any Third Party CI system that interacts with gerrit to follow the requirements set by the Infrastructure team[1] as well as the Neutron Third Party CI guidelines below. Please ping the PTL

[1] http://ci.openstack.org/third_party.html

in #openstack-neutron or send an email to the openstack-discuss ML (with subject [neutron]) with any questions. Be aware that the Infrastructure documentation as well as this document are living documents and undergo changes. Track changes to the infrastructure documentation using this url[2] (and please review the patches) and check this doc on a regular basis for updates.

What Changes to Run Against

If your code is a neutron plugin or driver, you should run against every neutron change submitted, except for docs, tests, tools, and top-level setup files. You can skip your CI runs for such exceptions by using `skip-if` and `all-files-match-any` directives in Zuul. You can see a programmatic example of the exceptions here[3].

If your code is in a neutron-*aas repo, you should run against the tests for that repo. You may also run against every neutron change, if your service driver is using neutron interfaces that are not provided by your service plugin (e.g. firewall/fwaas_plugin_v2.py). If you are using only plugin interfaces, it should be safe to test against only the service repo tests.

What Tests To Run

Network API tests (git link). Network scenario tests (The test_network_* tests here). Any tests written specifically for your setup. http://opendev.org/openstack/tempest/src/tempest/api/network

Run with the test filter: network. This will include all neutron specific tests as well as any other tests that are tagged as requiring networking. An example tempest setup for devstack-gate:

```
export DEVSTACK_GATE_NEUTRON=1
export DEVSTACK_GATE_TEMPEST_REGEX='(?!.*\[.*\bslow\b.*\
↪])((network)|(neutron))'
```

Third Party CI Voting

The Neutron team encourages you to NOT vote -1 with a third-party CI. False negatives are noisy to the community, and have given -1 from third-party CIs a bad reputation. Really bad, to the point of people ignoring them all. Failure messages are useful to those doing refactors, and provide you feedback on the state of your plugin.

If you insist on voting, by default, the infra team will not allow voting by new 3rd party CI systems. The way to get your 3rd party CI system to vote is to talk with the Neutron PTL, who will let infra know the system is ready to vote. The requirements for a new system to be given voting rights are as follows:

- A new system must be up and running for a month, with a track record of voting on the sandbox system.
- A new system must correctly run and pass tests on patches for the third party driver/plugin for a month.
- A new system must have a logfile setup and retention setup similar to the below.

Once the system has been running for a month, the owner of the third party CI system can contact the Neutron PTL to have a conversation about getting voting rights upstream.

The general process to get these voting rights is outlined here. Please follow that, taking note of the guidelines Neutron also places on voting for its CI systems.

[2] https://review.opendev.org/#/q/status:open+project:openstack-infra/system-config+branch:master+topic:third-party,n,z
[3] https://github.com/openstack-infra/project-config/blob/master/dev/zuul/layout.yaml

A third party system can have its voting rights removed as well. If the system becomes unstable (stops running, voting, or start providing inaccurate results), the Neutron PTL or any core reviewer will make an attempt to contact the owner and copy the openstack-discuss mailing list. If no response is received within 2 days, the Neutron PTL will remove voting rights for the third party CI system. If a response is received, the owner will work to correct the issue. If the issue cannot be addressed in a reasonable amount of time, the voting rights will be temporarily removed.

Log & Test Results Filesystem Layout

Third-Party CI systems MUST provide logs and configuration data to help developers troubleshoot test failures. A third-party CI that DOES NOT post logs should be a candidate for removal, and new CI systems MUST post logs before they can be awarded voting privileges.

Third party CI systems should follow the filesystem layout convention of the OpenStack CI system. Please store your logs as viewable in a web browser, in a directory structure. Requiring the user to download a giant tarball is not acceptable, and will be reason to not allow your system to vote from the start, or cancel its voting rights if this changes while the system is running.

At the root of the results - there should be the following:

- console.html.gz - contains the output of stdout of the test run
- local.conf / localrc - contains the setup used for this run
- logs - contains the output of detail test log of the test run

The above logs must be a directory, which contains the following:

- Log files for each screen session that DevStack creates and launches an OpenStack component in
- Test result files
- testr_results.html.gz
- tempest.txt.gz

List of existing plugins and drivers

https://wiki.openstack.org/wiki/Neutron_Plugins_and_Drivers#Existing_Plugin_and_Drivers

References

14.3 Gerrit Rechecks

14.3.1 Recheck Failed CI jobs in Neutron

This document provides guidelines on what to do in case your patch fails one of the Zuul CI jobs. In order to discover potential bugs hidden in the code or tests themselves, its very helpful to check failed scenarios to investigate the cause of the failure. Sometimes the failure will be caused by the patch being tested, while other times the failure can be caused by a previously untracked bug. Such failures are usually related to tests that interact with a live system, like functional, fullstack and tempest jobs.

Unnecessary rechecks lead to wasted resources as well as longer result times for patches in other projects. As a consequence, before issuing a recheck, make sure that the gate failure is not caused by your patch. A failed job can also be caused by some infra issue, for example the inability to fetch things from external resources like git or pip due to an outage. Such failures outside of the OpenStack world are not worth tracking in launchpad and you can recheck by leaving a short comment indicating what went wrong. Data about gate stability is collected and visualized via Grafana.

Please, do not recheck without providing the bug number for the failed job. For example, do not just put an empty recheck comment but find the related bug number and put a recheck bug ###### comment instead. If a bug does not exist yet, create one so other team members can have a look. It helps us maintain better visibility of gate failures. You can find how to troubleshoot gate failures in the *Gate Failure Triage* documentation.

Here are some real examples of proper rechecks:

- Spurious issue in other component: **recheck tempest-integrated-storage : intermittent failure nova bug #1836754**
- Deployment issue on the job: **recheck cinder-plugin-ceph-tempest timed out, errors all over the place**
- External service failure: **recheck Third party grenade : Failed to retrieve .deb packages**

14.4 Neutron Stadium

14.4.1 Neutron Stadium

This section contains information on policies and procedures for the so called Neutron Stadium. The Neutron Stadium is the list of projects that show up in the OpenStack Governance Document.

The list includes projects that the Neutron PTL and core team are directly involved in, and manage on a day to day basis. To do so, the PTL and team ensure that common practices and guidelines are followed throughout the Stadium, for all aspects that pertain software development, from inception, to coding, testing, documentation and more.

The Stadium is not to be intended as a VIP club for OpenStack networking projects, or an upper tier within OpenStack. It is simply the list of projects the Neutron team and PTL claim responsibility for when producing Neutron deliverables throughout the release cycles.

For more details on the Stadium, and what it takes for a project to be considered an integral part of the Stadium, please read on.

Stadium Governance

Background

Neutron grew to become a big monolithic codebase, and its core team had a tough time making progress on a number of fronts, like adding new features, ensuring stability, etc. During the Kilo timeframe, a decomposition effort started, where the codebase got disaggregated into separate repos, like the high level services, and the various third-party solutions for L2 and L3 services, and the Stadium was officially born.

These initiatives enabled the various individual teams in charge of the smaller projects the opportunity to iterate faster and reduce the time to feature. This has been due to the increased autonomy and implicit trust model that made the lack of oversight of the PTL and the Neutron drivers/core team acceptable for a small number of initiatives. When the proposed arrangement allowed projects to be automatically enlisted as a Neutron project based simply on description, and desire for affiliation, the number of projects included in the Stadium started to grow rapidly, which created a number of challenges for the PTL and the drivers team.

In fact, it became harder and harder to ensure consistency in the APIs, architecture, design, implementation and testing of the overarching project; all aspects of software development, like documentation, integration, release management, maintenance, and upgrades started to being neglected for some projects and that led to some unhappy experiences.

The point about uniform APIs is particularly important, because the Neutron platform is so flexible that a project can take a totally different turn in the way it exposes functionality, that it is virtually impossible for the PTL and the drivers team to ensure that good API design principles are being followed over time. In a situation where each project is on its own, that might be acceptable, but allowing independent API evolution while still under the Neutron umbrella is counterproductive.

These challenges led the Neutron team to find a better balance between autonomy and consistency and lay down criteria that more clearly identify when a project can be eligible for inclusion in the Neutron governance.

This document describes these criteria, and document the steps involved to maintain the integrity of the Stadium, and how to ensure this integrity be maintained over time when modifications to the governance are required.

When is a project considered part of the Stadium?

In order to be considered part of the Stadium, a project must show a track record of alignment with the Neutron core project. This means showing proof of adoption of practices as led by the Neutron core team. Some of these practices are typically already followed by the most mature OpenStack projects:

- Exhaustive documentation: it is expected that each project will have a *developer*, *user/operator* and API documentations available.
- Exhaustive OpenStack CI coverage: unit, functional, and tempest coverage using OpenStack CI (upstream) resources so that Grafana support is available. Access to CI resources and historical data by the team is key to ensuring stability and robustness of a project. In particular, it is of paramount importance to ensure that DB models/migrations are tested functionally to prevent data inconsistency issues or unexpected DB logic errors due to schema/models mismatch. For more details, please look at the following resources:
  - https://review.opendev.org/#/c/346091/
  - https://review.opendev.org/#/c/346272/
  - https://review.opendev.org/#/c/346083/

  More Database related information can be found on:
  - *Alembic Migrations*
  - *Database Layer*

  Bear in mind that many projects have been transitioning their codebase and tests to fully support Python 3+, and it is important that each Stadium project supports Python 3+ the same way Neutron core does. For more information on how to do testing, please refer to the *Neutron testing documentation*.
- Good release footprint, according to the chosen release model.
- Adherence to deprecation and stable backports policies.
- Demonstrated ability to do upgrades and/or rolling upgrades, where applicable. This means having grenade support on top of the CI coverage as described above.

- Client bindings and CLI developed according to the OpenStack Client plugin model.

On top of the above mentioned criteria, the following also are taken into consideration:

- A project must use, adopt and implement open software and technologies.
- A project must integrate with Neutron via one of the supported, advertised and maintained public Python APIs. REST API does not qualify (the project python-neutronclient is an exception).
- It adopts neutron-lib (with related hacking rules applied), and has proof of good decoupling from Neutron core internals.
- It provides an API that adopts API guidelines as set by the Neutron core team, and that relies on an open implementation.
- It adopts modular interfaces to provide networking services: this means that L2/7 services are provided in the form of ML2 mech drivers and service plugins respectively. A service plugin can expose a driver interface to support multiple backend technologies, and/or adopt the flavor framework as necessary.

Adding or removing projects to the Stadium

When a project is to be considered part of the Stadium, proof of compliance to the aforementioned practices will have to be demonstrated typically for at least two OpenStack releases. Application for inclusion is to be considered only within the first milestone of each OpenStack cycle, which is the time when the PTL and Neutron team do release planning, and have the most time available to discuss governance issues.

Projects part of the Neutron Stadium have typically the first milestone to get their house in order, during which time reassessment happens; if removed, because of substantial lack of meeting the criteria, a project cannot reapply within the same release cycle it has been evicted.

The process for proposing a repo into openstack/ and under the Neutron governance is to propose a patch to the openstack/governance repository. For example, to propose networking-foo, one would add the following entry under Neutron in reference/projects.yaml:

```
- repo: openstack/networking-foo
  tags:
    - name: release:independent
```

Typically this is a patch that the PTL, in collaboration with the projects point of contact, will shepherd through the review process. This step is undertaken once it is clear that all criteria are met. The next section provides an informal checklist that shows what steps a project needs to go through in order to enable the PTL and the TC to vote positively on the proposed inclusion.

Once a project is included, it abides by the Neutron *RFE submission process*, where specifications to neutron-specs are required for major API as well as major architectural changes that may require core Neutron platform enhancements.

Checklist

- How to integrate documentation into docs.o.o: The documentation website has a section for project developer documentation. Each project in the Neutron Stadium must have an entry under the Networking Sub Projects section that points to the developer documentation for the project, available at `https://docs.openstack.org/<your-project>/latest/`. This is a two step process that involves the following:

  - Build the artefacts: this can be done by following example https://review.opendev.org/#/c/293399/.
  - Publish the artefacts: this can be done by following example https://review.opendev.org/#/c/216448/.

  More information can also be found on the project creator guide.

- How to integrate into Grafana: Grafana is a great tool that provides the ability to display historical series, like failure rates of OpenStack CI jobs. A few examples that added dashboards over time are:
  - Neutron.
  - Networking-OVN.
  - Networking-Midonet.

  Any subproject must have a Grafana dashboard that shows failure rates for at least Gate and Check queues.

- How to integrate into neutron-libs CI: there are a number of steps required to integrate with neutron-lib CI and adopt neutron-lib in general. One step is to validate that neutron-lib master is working with the master of a given project that uses neutron-lib. For example patch introduced such support for the Neutron project. Any subproject that wants to do the same would need to adopt the following few lines:
  1. https://review.opendev.org/#/c/338603/4/jenkins/jobs/projects.yaml@4685
  2. https://review.opendev.org/#/c/338603/3/zuul/layout.yaml@8501
  3. https://review.opendev.org/#/c/338603/4/grafana/neutron.yaml@39

  Line 1 and 2 respectively add a job to the periodic queue for the project, whereas line 3 introduced the failure rate trend for the periodic job to spot failure spikes etc. Make sure your project has the following:
  1. https://review.opendev.org/#/c/357086/
  2. https://review.opendev.org/#/c/359143/

- How to port api-ref over to neutron-lib: to publish the subproject API reference into the Networking API guide you must contribute the API documentation into neutron-libs api-ref directory as done in the WADL/REST transition patch. Once this is done successfully, a link to the subproject API will show under the published table of content. An RFE bug tracking this effort effectively initiates the request for Stadium inclusion, where all the aspects as outlined in this documented are reviewed by the PTL.

- How to port API definitions over the neutron-lib: the most basic steps to port API definitions over to neutron-lib are demonstrated in the following patches:
  - https://review.opendev.org/#/c/353131/
  - https://review.opendev.org/#/c/353132/

  The neutron-lib patch introduces the elements that define the API, and testing coverage validates that the resource and actions maps use valid keywords. API reference documentation is provided alongside the definition to keep everything in one place. The neutron patch uses the Neutron extension framework to plug the API definition on top of the Neutron API backbone. The change can only merge when there is a released version of neutron-lib.

- How to integrate into the openstack release: every project in the Stadium must have release notes. In order to set up release notes, please see the patches below for an example on how to set up reno:
  - https://review.opendev.org/#/c/320904/
  - https://review.opendev.org/#/c/243085/

  For release documentation related to Neutron, please check the *Neutron Policies*. Once, everything is set up and your project is released, make sure you see an entry on the release page (e.g. Pike. Make sure you release according to the project declared release model.
- How to port OpenStack Client over to python-neutronclient: client API bindings and client command line interface support must be developed in python-neutronclient under osc module. If your project requires one or both, consider looking at the following example on how to contribute these two python-neutronclient according to the OSC framework and guidelines:
  - https://review.opendev.org/#/c/340624/
  - https://review.opendev.org/#/c/340763/
  - https://review.opendev.org/#/c/352653/

  More information on how to develop python-openstackclient plugins can be found on the following links:
  - https://docs.openstack.org/python-openstackclient/latest/contributor/plugins.html
  - https://docs.openstack.org/python-openstackclient/latest/contributor/humaninterfaceguide.html

  It is worth prefixing the commands being added with the keyword network to avoid potential clash with other commands with similar names. This is only required if the command object name is highly likely to have an ambiguous meaning.

Sub-Project Guidelines

This document provides guidance for those who maintain projects that consume main neutron or neutron advanced services repositories as a dependency. It is not meant to describe projects that are not tightly coupled with Neutron code.

Code Reuse

At all times, avoid using any Neutron symbols that are explicitly marked as private (those have an underscore at the start of their names).

Try to avoid copy pasting the code from Neutron to extend it. Instead, rely on enormous number of different plugin entry points provided by Neutron (L2 agent extensions, API extensions, service plugins, core plugins, ML2 mechanism drivers, etc.)

Requirements

Neutron dependency

Subprojects usually depend on neutron repositories, by using -e https:// schema to define such a dependency. The dependency *must not* be present in requirements lists though, and instead belongs to tox.ini deps section. This is because next pbr library releases do not guarantee -e https:// dependencies will work.

You may still put some versioned neutron dependency in your requirements list to indicate the dependency for anyone who packages your subproject.

Explicit dependencies

Each neutron project maintains its own lists of requirements. Subprojects that depend on neutron while directly using some of those libraries that neutron maintains as its dependencies must not rely on the fact that neutron will pull the needed dependencies for them. Direct library usage requires that this library is mentioned in requirements lists of the subproject.

The reason to duplicate those dependencies is that neutron team does not stick to any backwards compatibility strategy in regards to requirements lists, and is free to drop any of those dependencies at any time, breaking anyone who could rely on those libraries to be pulled by neutron itself.

Automated requirements updates

At all times, subprojects that use neutron as a dependency should make sure their dependencies do not conflict with neutrons ones.

Core neutron projects maintain their requirements lists by utilizing a so-called proposal bot. To keep your subproject in sync with neutron, it is highly recommended that you register your project in openstack/requirements:projects.txt file to enable the bot to update requirements for you.

Once a subproject opts in global requirements synchronization, it should enable check-requirements jobs in project-config. For example, see this patch.

Stable branches

Stable branches for subprojects should be created at the same time when corresponding neutron stable branches are created. This is to avoid situations when a postponed cut-off results in a stable branch that contains some patches that belong to the next release. This would require reverting patches, and this is something you should avoid.

Make sure your neutron dependency uses corresponding stable branch for neutron, not master.

Note that to keep requirements in sync with core neutron repositories in stable branches, you should make sure that your project is registered in openstack/requirements:projects.txt *for the branch in question*.

Subproject stable branches are supervised by horizontal neutron-stable-maint team.

More info on stable branch process can be found on the following page.

Stable merge requirements

Merges into stable branches are handled by members of the neutron-stable-maint gerrit group. The reason for this is to ensure consistency among stable branches, and compliance with policies for stable backports.

For sub-projects who participate in the Neutron Stadium effort and who also create and utilize stable branches, there is an expectation around what is allowed to be merged in these stable branches. The Stadium projects should be following the stable branch policies as defined by on the Stable Branch wiki. This means that, among other things, no features are allowed to be backported into stable branches.

Releases

It is suggested that sub-projects cut off new releases from time to time, especially for stable branches. It will make the life of packagers and other consumers of your code easier.

Sub-Project Release Process

All subproject releases are managed by global OpenStack Release Managers team. The neutron-release team handles only the following operations:

- Make stable branches end of life

To release a sub-project, follow the following steps:

- For projects which have not moved to post-versioning, we need to push an alpha tag to avoid pbr complaining. A member of the neutron-release group will handle this.
- A sub-project owner should modify setup.cfg to remove the version (if you have one), which moves your project to post-versioning, similar to all the other Neutron projects. You can skip this step if you dont have a version in setup.cfg.
- A sub-project owner proposes a patch to openstack/releases repository with the intended git hash. The Neutron release liaison should be added in Gerrit to the list of reviewers for the patch.

  Note

  New major tag versions should conform to SemVer requirements, meaning no year numbers should be used as a major version. The switch to SemVer is advised at earliest convenience for all new major releases.

  Note

  Before Ocata, when releasing the very first release in a stable series, a sub-project owner would need to request a new stable branch creation during Gerrit review, but not anymore. See the following email for more details.

- The Neutron release liaison votes with +1 for the openstack/releases patch.
- The releases will now be on PyPI. A sub-project owner should verify this by going to an URL similar to this.
- A sub-project owner should next go to Launchpad and release this version using the Release Now button for the release itself.
- If a sub-project uses the delay-release option, a sub-project owner should update any bugs that were fixed with this release to Fix Released in Launchpad. This step is not necessary if the sub-project uses the direct-release option, which is the default.[1]
- The new release will be available on OpenStack Releases.
- A sub-project owner should add the next milestone to the Launchpad series, or if a new series is required, create the new series and a new milestone.

[1] http://lists.openstack.org/pipermail/openstack-dev/2015-December/081724.html

Note

You need to be careful when picking a git commit to base new releases on. In most cases, youll want to tag the *merge* commit that merges your last commit in to the branch. This bug shows an instance where this mistake was caught. Notice the difference between the incorrect commit and the correct one which is the merge commit. `git log 6191994..22dd683 --oneline` shows that the first one misses a handful of important commits that the second one catches. This is the nature of merging to master.

To make a branch end of life, follow the following steps:

- A member of neutron-release will abandon all open change reviews on the branch.
- A member of neutron-release will push an EOL tag on the branch. (eg. icehouse-eol)
- A sub-project owner should request the infrastructure team to delete the branch by sending an email to the infrastructure mailing list, not by bothering the infrastructure team on IRC.
- A sub-project owner should tweak zuul jobs in project-config if any.

References

14.5 Developer Guide

In the Developer Guide, you will find information on Neutrons lower level programming APIs. There are sections that cover the core pieces of Neutron, including its database, message queue, and scheduler components. There are also subsections that describe specific plugins inside Neutron. Finally, the developer guide includes information about Neutron testing infrastructure.

14.5.1 Effective Neutron: 100 specific ways to improve your Neutron contributions

There are a number of skills that make a great Neutron developer: writing good code, reviewing effectively, listening to peer feedback, etc. The objective of this document is to describe, by means of examples, the pitfalls, the good and bad practices that we as project encounter on a daily basis and that make us either go slower or accelerate while contributing to Neutron.

By reading and collaboratively contributing to such a knowledge base, your development and review cycle becomes shorter, because you will learn (and teach to others after you) what to watch out for, and how to be proactive in order to prevent negative feedback, minimize programming errors, writing better tests, and so on and so forthin a nutshell, how to become an effective Neutron developer.

The notes below are meant to be free-form and brief by design. They are not meant to replace or duplicate OpenStack documentation, or any project-wide documentation initiative like peer-review notes or the team guide. For this reason, references are acceptable and should be favored, if the shortcut is deemed useful to expand on the distilled information. We will try to keep these notes tidy by breaking them down into sections if it makes sense. Feel free to add, adjust, remove as you see fit. Please do so, taking into consideration yourself and other Neutron developers as readers. Capture your experience during development and review and add any comment that you believe will make your life and others easier.

Happy hacking!

Developing better software

Plugin development

Document common pitfalls as well as good practices done during plugin development.

- Use mixin classes as last resort. They can be a powerful tool to add behavior but their strength is also a weakness, as they can introduce unpredictable behavior to the MRO, amongst other issues.
- In lieu of mixins, if you need to add behavior that is relevant for ML2, consider using the extension manager.
- If you make changes to the DB class methods, like calling methods that can be inherited, think about what effect that may have to plugins that have controller backends.
- If you make changes to the ML2 plugin or components used by the ML2 plugin, think about the effect that may have to other plugins.
- When adding behavior to the L2 and L3 db base classes, do not assume that there is an agent on the other side of the message broker that interacts with the server. Plugins may not rely on agents at all.
- Be mindful of required capabilities when you develop plugin extensions. The Extension description provides the ability to specify the list of required capabilities for the extension you are developing. By declaring this list, the server will not start up if the requirements are not met, thus avoiding leading the system to experience undetermined behavior at runtime.

Database interaction

Document common pitfalls as well as good practices done during database development.

- first() does not raise an exception.
- Do not use delete() to remove objects. A delete query does not load the object so no sqlalchemy events can be triggered that would do things like recalculate quotas or update revision numbers of parent objects. For more details on all of the things that can go wrong using bulk delete operations, see the Warning sections in the link above.
- Beware of the InvalidRequestError exception. There is even a Neutron bug registered for it. Bear in mind that this error may also occur when nesting transaction blocks, and the innermost block raises an error without proper rollback. Consider if savepoints can fit your use case.
- When designing data models that are related to each other, be careful to how you model the relationships loading strategy. For instance a joined relationship can be very efficient over others (some examples include router gateways or network availability zones).
- If you add a relationship to a Neutron object that will be referenced in the majority of cases where the object is retrieved, be sure to use the lazy=joined parameter to the relationship so the related objects are loaded as part of the same query. Otherwise, the default method is select, which emits a new DB query to retrieve each related object adversely impacting performance. For example, see patch 88665 which resulted in a significant improvement since router retrieval functions always include the gateway interface.
- Conversely, do not use lazy=joined if the relationship is only used in corner cases because the JOIN statement comes at a cost that may be significant if the relationship contains many objects. For example, see patch 168214 which reduced a subnet retrieval by ~90% by avoiding a join to the IP allocation table.

- When writing extensions to existing objects (e.g. Networks), ensure that they are written in a way that the data on the object can be calculated without additional DB lookup. If thats not possible, ensure the DB lookup is performed once in bulk during a list operation. Otherwise a list call for a 1000 objects will change from a constant small number of DB queries to 1000 DB queries. For example, see patch 257086 which changed the availability zone code from the incorrect style to a database friendly one.
- Beware of ResultProxy.inserted_primary_key which returns a list of last inserted primary keys not the last inserted primary key:

```
result = session.execute(mymodel.insert().values(**values))
# result.inserted_primary_key is a list even if we inserted a unique row!
```

System development

Document common pitfalls as well as good practices done when invoking system commands and interacting with linux utils.

- When a patch requires a new platform tool or a new feature in an existing tool, check if common platforms ship packages with the aforementioned feature. Also, tag such a patch with `UpgradeImpact` to raise its visibility (as these patches are brought up to the attention of the core team during team meetings). More details in *review guidelines*.
- When a patch or the code depends on a new feature in the kernel or in any platform tools (dnsmasq, ip, Open vSwitch etc.), consider introducing a new sanity check to validate deployments for the expected features. Note that sanity checks *must not* check for version numbers of underlying platform tools because distributions may decide to backport needed features into older versions. Instead, sanity checks should validate actual features by attempting to use them.

Mocking and testing

Document common pitfalls as well as good practices done when writing tests, any test. For anything more elaborate, please visit the testing section.

- Preferring low level testing versus full path testing (e.g. not testing database via client calls). The former is to be favored in unit testing, whereas the latter is to be favored in functional testing.
- Prefer specific assertions (assert(Not)In, assert(Not)IsInstance, assert(Not)IsNone, etc) over generic ones (assertTrue/False, assertEqual) because they raise more meaningful errors:

```
def test_specific(self):
    self.assertIn(3, [1, 2])
    # raise meaningful error: "MismatchError: 3 not in [1, 2]"

def test_generic(self):
    self.assertTrue(3 in [1, 2])
    # raise meaningless error: "AssertionError: False is not true"
```

- Use the pattern self.assertEqual(expected, observed) not the opposite, it helps reviewers to understand which one is the expected/observed value in non-trivial assertions. The expected and observed values are also labeled in the output when the assertion fails.
- Prefer specific assertions (assertTrue, assertFalse) over assertEqual(True/False, observed).

- Dont write tests that dont test the intended code. This might seem silly but it is easy to do with a lot of mocks in place. Ensure that your tests break as expected before your code change.
- Avoid heavy use of the mock library to test your code. If your code requires more than one mock to ensure that it does the correct thing, it needs to be refactored into smaller, testable units. Otherwise we depend on fullstack/tempest/api tests to test all of the real behavior and we end up with code containing way too many hidden dependencies and side effects.
- All behavior changes to fix bugs should include a test that prevents a regression. If you made a change and it didnt break a test, it means the code was not adequately tested in the first place, its not an excuse to leave it untested.
- Test the failure cases. Use a mock side effect to throw the necessary exceptions to test your except clauses.
- Dont mimic existing tests that violate these guidelines. We are attempting to replace all of these so more tests like them create more work. If you need help writing a test, reach out to the testing lieutenants and the team on IRC.
- Mocking open() is a dangerous practice because it can lead to unexpected bugs like bug 1503847. In fact, when the built-in open method is mocked during tests, some utilities (like debtcollector) may still rely on the real thing, and may end up using the mock rather what they are really looking for. If you must, consider using OpenFixture, but it is better not to mock open() at all.

Documentation

The documenation for Neutron that exists in this repository is broken down into the following directories based on content:

- doc/source/admin/ - feature-specific configuration documentation aimed at operators.
- doc/source/configuration - stubs for auto-generated configuration files. Only needs updating if new config files are added.
- doc/source/contributor/internals - developer documentation for lower-level technical details.
- doc/source/contributor/policies - neutron team policies and best practices.
- doc/source/install - install-specific documentation for standing-up network-enabled nodes.

Additional documentation resides in the neutron-lib repository:

- api-ref - API reference documentation for Neutron resource and API extensions.

Backward compatibility

Document common pitfalls as well as good practices done when extending the RPC Interfaces.

- Make yourself familiar with *Upgrade review guidelines*.

Deprecation

Sometimes we want to refactor things in a non-backward compatible way. In most cases you can use debtcollector to mark things for deprecation. Config items have deprecation options supported by oslo.config.

The deprecation process must follow the standard deprecation requirements. In terms of neutron development, this means:

- A launchpad bug to track the deprecation.

- A patch to mark the deprecated items. If the deprecation affects users (config items, API changes) then a release note must be included.
- Wait at least one cycle and at least three months linear time.
- A patch that removes the deprecated items. Make sure to refer to the original launchpad bug in the commit message of this patch.

Scalability issues

Document common pitfalls as well as good practices done when writing code that needs to process a lot of data.

Translation and logging

Document common pitfalls as well as good practices done when instrumenting your code.

- Make yourself familiar with OpenStack logging guidelines to avoid littering the logs with traces logged at inappropriate levels.
- The logger should only be passed unicode values. For example, do not pass it exceptions or other objects directly (LOG.error(exc), LOG.error(port), etc.). See https://docs.openstack.org/oslo.log/latest/user/migration.html#no-more-implicit-conversion-to-unicode-str for more details.
- Dont pass exceptions into LOG.exception: it is already implicitly included in the log message by Python logging module.
- Dont use LOG.exception when there is no exception registered in current thread context: Python 3.x versions before 3.5 are known to fail on it.

Project interfaces

Document common pitfalls as well as good practices done when writing code that is used to interface with other projects, like Keystone or Nova.

Documenting your code

Document common pitfalls as well as good practices done when writing docstrings.

Landing patches more rapidly

Scoping your patch appropriately

- Do not make multiple changes in one patch unless absolutely necessary. Cleaning up nearby functions or fixing a small bug you noticed while working on something else makes the patch very difficult to review. It also makes cherry-picking and reverting very difficult. Even apparently minor changes such as reformatting whitespace around your change can burden reviewers and cause merge conflicts.
- If a fix or feature requires code refactoring, submit the refactoring as a separate patch than the one that changes the logic. Otherwise its difficult for a reviewer to tell the difference between mistakes in the refactor and changes required for the fix/feature. If its a bug fix, try to implement the fix before the refactor to avoid making cherry-picks to stable branches difficult.
- Consider your reviewers time before submitting your patch. A patch that requires many hours or days to review will sit in the todo list until someone has many hours or days free (which may never

happen.) If you can deliver your patch in small but incrementally understandable and testable pieces you will be more likely to attract reviewers.

Nits and pedantic comments

Document common nits and pedantic comments to watch out for.

- Make sure you spell correctly, the best you can, no-one wants rebase generators at the end of the release cycle!
- The odd pep8 error may cause an entire CI run to be wasted. Consider running validation (pep8 and/or tests) before submitting your patch. If you keep forgetting consider installing a git hook so that Git will do it for you.
- Sometimes, new contributors want to dip their toes with trivial patches, but we at OpenStack *love* bike shedding and their patches may sometime stall. In some extreme cases, the more trivial the patch, the higher the chances it fails to merge. To ensure we as a team provide/have a frustration-free experience new contributors should be redirected to fixing low-hanging-fruit bugs that have a tangible positive impact to the codebase. Spelling mistakes, and docstring are fine, but there is a lot more that is relatively easy to fix and has a direct impact to Neutron users.

Reviewer comments

- Acknowledge them one by one by either clicking Done or by replying extensively. If you do not, the reviewer wont know whether you thought it was not important, or you simply forgot. If the reply satisfies the reviewer, consider capturing the input in the code/document itself so that its for reviewers of newer patchsets to see (and other developers when the patch merges).
- Watch for the feedback on your patches. Acknowledge it promptly and act on it quickly, so that the reviewer remains engaged. If you disappear for a week after you posted a patchset, it is very likely that the patch will end up being neglected.
- Do not take negative feedback personally. Neutron is a large project with lots of contributors with different opinions on how things should be done. Many come from widely varying cultures and languages so the English, text-only feedback can unintentionally come across as harsh. Getting a -1 means reviewers are trying to help get the patch into a state that can be merged, it doesnt just mean they are trying to block it. Its very rare to get a patch merged on the first iteration that makes everyone happy.

Code Review

- You should visit OpenStack How To Review wiki
- Stay focussed and review what matters for the release. Please check out the Neutron section for the Gerrit dashboard. The output is generated by this tool.

IRC

- IRC is a place where you can speak with many of the Neutron developers and core reviewers. For more information you should visit OpenStack IRC wiki Neutron IRC channel is #openstack-neutron
- There are weekly IRC meetings related to many different projects/teams in Neutron. A full list of these meetings and their date/time can be found in OpenStack IRC Meetings. It is important to attend these meetings in the area of your contribution and possibly mention your work and patches.

- When you have questions regarding an idea or a specific patch of yours, it can be helpful to find a relevant person in IRC and speak with them about it. You can find a users IRC nickname in their launchpad account.
- Being available on IRC is useful, since reviewers can contact you directly to quickly clarify a review issue. This speeds up the feedback loop.
- Each area of Neutron or sub-project of Neutron has a specific lieutenant in charge of it. You can most likely find these lieutenants on IRC, it is advised however to try and send public questions to the channel rather then to a specific person if possible. (This increase the chances of getting faster answers to your questions). A list of the areas and lieutenants nicknames can be found at *Core Reviewers*.

Commit messages

Document common pitfalls as well as good practices done when writing commit messages. For more details see Git commit message best practices. This is the TL;DR version with the important points for committing to Neutron.

- One liners are bad, unless the change is trivial.
- Use `UpgradeImpact` when the change could cause issues during the upgrade from one version to the next.
- `APIImpact` should be used when the api-ref in neutron-lib must be updated to reflect the change, and only as a last resort. Rather, the ideal workflow includes submitting a corresponding neutron-lib api-ref change along with the implementation, thereby removing the need to use `APIImpact`.
- Make sure the commit message doesnt have any spelling/grammar errors. This is the first thing reviewers read and they can be distracting enough to invite -1s.
- Describe what the change accomplishes. If its a bug fix, explain how this code will fix the problem. If its part of a feature implementation, explain what component of the feature the patch implements. Do not just describe the bug, thats what launchpad is for.
- Use the Closes-Bug: #BUG-NUMBER tag if the patch addresses a bug. Submitting a bugfix without a launchpad bug reference is unacceptable, even if its trivial. Launchpad is how bugs are tracked so fixes without a launchpad bug are a nightmare when users report the bug from an older version and the Neutron team cant tell if/why/how its been fixed. Launchpad is also how backports are identified and tracked so patches without a bug report cannot be picked to stable branches.
- Use the Implements: blueprint NAME-OF-BLUEPRINT or Partially-Implements: blueprint NAME-OF-BLUEPRINT for features so reviewers can determine if the code matches the spec that was agreed upon. This also updates the blueprint on launchpad so its easy to see all patches that are related to a feature.
- If its not immediately obvious, explain what the previous code was doing that was incorrect. (e.g. code assumed it would never get None from a function call)
- Be specific in your commit message about what the patch does and why it does this. For example, Fixes incorrect logic in security groups is not helpful because the code diff already shows that you are modifying security groups. The message should be specific enough that a reviewer looking at the code can tell if the patch does what the commit says in the most appropriate manner. If the reviewer has to guess why you did something, lots of your time will be wasted explaining why certain changes were made.

Dealing with Zuul

Document common pitfalls as well as good practices done when dealing with OpenStack CI.

- When you submit a patch, consider checking its status in the queue. If you see a job failures, you might as well save time and try to figure out in advance why it is failing.
- Excessive use of recheck to get test to pass is discouraged. Please examine the logs for the failing test(s) and make sure your change has not tickled anything that might be causing a new failure or race condition. Getting your change in could make it even harder to debug what is actually broken later on.

14.5.2 Setting Up a Development Environment

This page describes how to setup a working Python development environment that can be used in developing Neutron on Ubuntu, Fedora or Mac OS X. These instructions assume youre already familiar with Git and Gerrit, which is a code repository mirror and code review toolset , however if you arent please see this Git tutorial for an introduction to using Git and this guide for a tutorial on using Gerrit and Git for code contribution to OpenStack projects.

Following these instructions will allow you to run the Neutron unit tests. If you want to be able to run Neutron in a full OpenStack environment, you can use the excellent DevStack project to do so. There is a wiki page that describes setting up Neutron using DevStack.

Getting the code

Grab the code:

```
git clone https://opendev.org/openstack/neutron.git
cd neutron
```

About ignore files

In the .gitignore files, add patterns to exclude files created by tools integrated, such as test frameworks from the projects recommended workflow, rendered documentation and package builds.

Dont add patterns to exclude files created by preferred personal like for example editors, IDEs or operating system. These should instead be maintained outside the repository, for example in a ~/.gitignore file added with:

```
git config --global core.excludesfile '~/.gitignore'
```

Ignores files for all repositories that you work with.

Testing Neutron

See *Testing Neutron*.

14.5.3 Deploying an OVN Development Environment with vagrant

The vagrant directory contains a set of vagrant configurations which will help you deploy Neutron with the OVN driver for testing or development purposes.

We provide a sparse multinode architecture with clear separation between services. In the future we will include all-in-one and multi-gateway architectures.

Vagrant prerequisites

Those are the prerequisites for using the vagrant file definitions

1. Install VirtualBox and Vagrant. Alternatively you can use parallels or libvirt vagrant plugin.
2. Install plug-ins for Vagrant:

```
$ vagrant plugin install vagrant-cachier
$ vagrant plugin install vagrant-vbguest
```

3. On Linux hosts, you can enable instances to access external networks such as the Internet by enabling IP forwarding and configuring SNAT from the IP address range of the provider network interface (typically vboxnet1) on the host to the external network interface on the host. For example, if the `eth0` network interface on the host provides external network connectivity:

```
# sysctl -w net.ipv4.ip_forward=1
# sysctl -p
# iptables -t nat -A POSTROUTING -s 10.10.0.0/16 -o eth0 -j MASQUERADE
```

Note: These commands do not persist after rebooting the host.

Sparse architecture

The Vagrant scripts deploy OpenStack with Open Virtual Network (OVN) using four nodes (five if you use the optional ovn-vtep node) to implement a minimal variant of the reference architecture:

1. ovn-db: Database node containing the OVN northbound (NB) and southbound (SB) databases via the Open vSwitch (OVS) database and `ovn-northd` services.
2. ovn-controller: Controller node containing the Identity service, Image service, control plane portion of the Compute service, control plane portion of the Networking service including the `ovn` ML2 driver, and the dashboard. In addition, the controller node is configured as an NFS server to support instance live migration between the two compute nodes.
3. ovn-compute1 and ovn-compute2: Two compute nodes containing the Compute hypervisor, `ovn-controller` service for OVN, metadata agents for the Networking service, and OVS services. In addition, the compute nodes are configured as NFS clients to support instance live migration between them.
4. ovn-vtep: Optional. A node to run the HW VTEP simulator. This node is not started by default but can be started by running vagrant up ovn-vtep after doing a normal vagrant up.

During deployment, Vagrant creates three VirtualBox networks:

1. Vagrant management network for deployment and VM access to external networks such as the Internet. Becomes the VM `eth0` network interface.
2. OpenStack management network for the OpenStack control plane, OVN control plane, and OVN overlay networks. Becomes the VM `eth1` network interface.
3. OVN provider network that connects OpenStack instances to external networks such as the Internet. Becomes the VM `eth2` network interface.

Requirements

The default configuration requires approximately 12 GB of RAM and supports launching approximately four OpenStack instances using the `m1.tiny` flavor. You can change the amount of resources for each VM in the `instances.yml` file.

Deployment

1. Follow the pre-requisites described in *Vagrant prerequisites*
2. Clone the `neutron` repository locally and change to the `neutron/vagrant/ovn/sparse` directory:

```
$ git clone https://opendev.org/openstack/neutron.git
$ cd neutron/vagrant/ovn/sparse
```

3. If necessary, adjust any configuration in the `instances.yml` file.
   - If you change any IP addresses or networks, avoid conflicts with the host.
   - For evaluating large MTUs, adjust the `mtu` option. You must also change the MTU on the equivalent `vboxnet` interfaces on the host to the same value after Vagrant creates them. For example:

   ```
   # ip link set dev vboxnet0 mtu 9000
   # ip link set dev vboxnet1 mtu 9000
   ```

4. Launch the VMs and grab some coffee:

```
$ vagrant up
```

5. After the process completes, you can use the `vagrant status` command to determine the VM status:

```
$ vagrant status
Current machine states:

ovn-db                    running (virtualbox)
ovn-controller            running (virtualbox)
ovn-vtep                  running (virtualbox)
ovn-compute1              running (virtualbox)
ovn-compute2              running (virtualbox)
```

6. You can access the VMs using the following commands:

```
$ vagrant ssh ovn-db
$ vagrant ssh ovn-controller
$ vagrant ssh ovn-vtep
$ vagrant ssh ovn-compute1
$ vagrant ssh ovn-compute2
```

Note: If you prefer to use the VM console, the password for the `root`
account is `vagrant`. Since ovn-controller is set as the primary in the Vagrantfile, the command `vagrant ssh` (without specifying the name) will connect ssh to that virtual machine.

7. Access OpenStack services via command-line tools on the `ovn-controller` node or via the dashboard from the host by pointing a web browser at the IP address of the `ovn-controller` node.

   Note: By default, OpenStack includes two accounts: `admin` and `demo`,
   both using password `password`.

8. After completing your tasks, you can destroy the VMs:

```
$ vagrant destroy
```

14.5.4 Contributing new extensions to Neutron

Introduction

Neutron has a pluggable architecture, with a number of extension points. This documentation covers aspects relevant to contributing new Neutron v2 core (aka monolithic) plugins, ML2 mechanism drivers, and L3 service plugins. This document will initially cover a number of process-oriented aspects of the contribution process, and proceed to provide a how-to guide that shows how to go from 0 LOCs to successfully contributing new extensions to Neutron. In the remainder of this guide, we will try to use practical examples as much as we can so that people have working solutions they can start from.

This guide is for a developer who wants to have a degree of visibility within the OpenStack Networking project. If you are a developer who wants to provide a Neutron-based solution without interacting with the Neutron community, you are free to do so, but you can stop reading now, as this guide is not for you.

Plugins and drivers for non-reference implementations are known as third-party code. This includes code for supporting vendor products, as well as code for supporting open-source networking implementations.

Before the Kilo release these plugins and drivers were included in the Neutron tree. During the Kilo cycle the third-party plugins and drivers underwent the first phase of a process called decomposition. During this phase, each plugin and driver moved the bulk of its logic to a separate git repository, while leaving a thin shim in the neutron tree together with the DB models and migrations (and perhaps some config examples).

During the Liberty cycle the decomposition concept was taken to its conclusion by allowing third-party code to exist entirely out of tree. Further extension mechanisms have been provided to better support external plugins and drivers that alter the API and/or the data model.

In the Mitaka cycle we will **require** all third-party code to be moved out of the neutron tree completely.

Outside the tree can be anything that is publicly available: it may be a repo on opendev.org for instance, a tarball, a pypi package, etc. A plugin/drivers maintainer team self-governs in order to promote sharing, reuse, innovation, and release of the out-of-tree deliverable. It should not be required for any member of the core team to be involved with this process, although core members of the Neutron team can participate in whichever capacity is deemed necessary to facilitate out-of-tree development.

This guide is aimed at you as the maintainer of code that integrates with Neutron but resides in a separate repository.

Contribution Process

If you want to extend OpenStack Networking with your technology, and you want to do it within the visibility of the OpenStack project, follow the guidelines and examples below. Well describe best practices for:

- Design and Development;
- Testing and Continuous Integration;

- Defect Management;
- Backport Management for plugin specific code;
- DevStack Integration;
- Documentation;

Once you have everything in place you may want to add your project to the list of Neutron sub-projects. See *Adding or removing projects to the Stadium* for details.

Design and Development

Assuming you have a working repository, any development to your own repo does not need any blueprint, specification or bugs against Neutron. However, if your project is a part of the Neutron Stadium effort, you are expected to participate in the principles of the Four Opens, meaning your design should be done in the open. Thus, it is encouraged to file documentation for changes in your own repository.

If your code is hosted on opendev.org then the gerrit review system is automatically provided. Contributors should follow the review guidelines similar to those of Neutron. However, you as the maintainer have the flexibility to choose who can approve/merge changes in your own repo.

It is recommended (but not required, see *policies*) that you set up a third-party CI system. This will provide a vehicle for checking the third-party code against Neutron changes. See *Testing and Continuous Integration* below for more detailed recommendations.

Design documents can still be supplied in form of Restructured Text (RST) documents, within the same third-party library repo. If changes to the common Neutron code are required, an *RFE* may need to be filed. However, every case is different and you are invited to seek guidance from Neutron core reviewers about what steps to follow.

Testing and Continuous Integration

The following strategies are recommendations only, since third-party CI testing is not an enforced requirement. However, these strategies are employed by the majority of the plugin/driver contributors that actively participate in the Neutron development community, since they have learned from experience how quickly their code can fall out of sync with the rapidly changing Neutron core code base.

- You should run unit tests in your own external library (e.g. on opendev.org where Zuul setup is for free).
- Your third-party CI should validate third-party integration with Neutron via functional testing. The third-party CI is a communication mechanism. The objective of this mechanism is as follows:
  - it communicates to you when someone has contributed a change that potentially breaks your code. It is then up to you maintaining the affected plugin/driver to determine whether the failure is transient or real, and resolve the problem if it is.
  - it communicates to a patch author that they may be breaking a plugin/driver. If they have the time/energy/relationship with the maintainer of the plugin/driver in question, then they can (at their discretion) work to resolve the breakage.
  - it communicates to the community at large whether a given plugin/driver is being actively maintained.
  - A maintainer that is perceived to be responsive to failures in their third-party CI jobs is likely to generate community goodwill.

It is worth noting that if the plugin/driver repository is hosted on opendev.org, due to current openstack-infra limitations, it is not possible to have third-party CI systems participating in the gate pipeline for the repo. This means that the only validation provided during the merge process to the repo is through unit tests. Post-merge hooks can still be exploited to provide third-party CI feedback, and alert you of potential issues. As mentioned above, third-party CI systems will continue to validate Neutron core commits. This will allow them to detect when incompatible changes occur, whether they are in Neutron or in the third-party repo.

Defect Management

Bugs affecting third-party code should *not* be filed in the Neutron project on launchpad. Bug tracking can be done in any system you choose, but by creating a third-party project in launchpad, bugs that affect both Neutron and your code can be more easily tracked using launchpads also affects project feature.

Security Issues

Here are some answers to how to handle security issues in your repo, taken from this mailing list message:

- How should security your issues be managed?

The OpenStack Vulnerability Management Team (VMT) follows a documented process which can basically be reused by any project-team when needed.

- Should the OpenStack security team be involved?

The OpenStack VMT directly oversees vulnerability reporting and disclosure for a subset of OpenStack source code repositories. However, they are still quite happy to answer any questions you might have about vulnerability management for your own projects even if theyre not part of that set. Feel free to reach out to the VMT in public or in private.

Also, the VMT is an autonomous subgroup of the much larger OpenStack Security project-team. Theyre a knowledgeable bunch and quite responsive if you want to get their opinions or help with security-related issues (vulnerabilities or otherwise).

- Does a CVE need to be filed?

It can vary widely. If a commercial distribution such as Red Hat is redistributing a vulnerable version of your software, then they may assign one anyway even if you dont request one yourself. Or the reporter may request one; the reporter may even be affiliated with an organization who has already assigned/obtained a CVE before they initiate contact with you.

- Do the maintainers need to publish OSSN or equivalent documents?

OpenStack Security Advisories (OSSA) are official publications of the OpenStack VMT and only cover VMT-supported software. OpenStack Security Notes (OSSN) are published by editors within the OpenStack Security project-team on more general security topics and may even cover issues in non-OpenStack software commonly used in conjunction with OpenStack, so its at their discretion as to whether they would be able to accommodate a particular issue with an OSSN.

However, these are all fairly arbitrary labels, and what really matters in the grand scheme of things is that vulnerabilities are handled seriously, fixed with due urgency and care, and announced widely not just on relevant OpenStack mailing lists but also preferably somewhere with broader distribution like the Open Source Security mailing list. The goal is to get information on your vulnerabilities, mitigating measures and fixes into the hands of the people using your software in a timely manner.

- Anything else to consider here?

The OpenStack VMT is in the process of trying to reinvent itself so that it can better scale within the context of the Big Tent. This includes making sure the policy/process documentation is more consumable and reusable even by project-teams working on software outside the scope of our charter. Its a work in progress, and any input is welcome on how we can make this function well for everyone.

Backport Management Strategies

This section applies only to third-party maintainers who had code in the Neutron tree during the Kilo and earlier releases. It will be obsolete once the Kilo release is no longer supported.

If a change made to out-of-tree third-party code needs to be back-ported to in-tree code in a stable branch, you may submit a review without a corresponding master branch change. The change will be evaluated by core reviewers for stable branches to ensure that the backport is justified and that it does not affect Neutron core code stability.

DevStack Integration Strategies

When developing and testing a new or existing plugin or driver, the aid provided by DevStack is incredibly valuable: DevStack can help get all the software bits installed, and configured correctly, and more importantly in a predictable way.

For DevStack integration there are a few options available, and they may or may not make sense depending on whether you are contributing a new or existing plugin or driver.

If you are contributing a new plugin, the approach to choose should be based on Extras.d Hooks externally hosted plugins. With the extra.d hooks, the DevStack integration is co-located with the third-party integration library, and it leads to the greatest level of flexibility when dealing with DevStack based dev/test deployments.

One final consideration is worth making for third-party CI setups: if Devstack Gate is used, it does provide hook functions that can be executed at specific times of the devstack-gate-wrap script run. For more details see devstack-vm-gate-wrap.sh.

Documentation

For a layout of the how the documentation directory is structured see the effective neutron guide

Project Initial Setup

The how-to below assumes that the third-party library will be hosted on opendev.org. This lets you tap in the entire OpenStack CI infrastructure and can be a great place to start from to contribute your new or existing driver/plugin. The list of steps below are summarized version of what you can find on http://docs.openstack.org/infra/manual/creators.html. They are meant to be the bare minimum you have to complete in order to get you off the ground.

- Create a public repository: this can be a personal opendev.org repo or any publicly available git repo, e.g. `https://github.com/john-doe/foo.git`. This would be a temporary buffer to be used to feed the one on opendev.org.
- Initialize the repository: if you are starting afresh, you may *optionally* want to use cookiecutter to get a skeleton project. You can learn how to use cookiecutter on https://opendev.org/openstack-dev/cookiecutter. If you want to build the repository from an existing Neutron module, you may want to skip this step now, build the history first (next step), and come back here to initialize the remainder of the repository with other files being generated by the cookiecutter (like tox.ini, setup.cfg, setup.py, etc.).

- Create a repository on opendev.org. For this you need the help of the OpenStack infra team. It is worth noting that you only get one shot at creating the repository on opendev.org. This is the time you get to choose whether you want to start from a clean slate, or you want to import the repo created during the previous step. In the latter case, you can do so by specifying the upstream section for your project in project-config/gerrit/project.yaml. Steps are documented on the Repository Creators Guide.
- Ask for a Launchpad user to be assigned to the core team created. Steps are documented in this section.
- Fix, fix, fix: at this point you have an external base to work on. You can develop against the new opendev.org project, the same way you work with any other OpenStack project: you have pep8, docs, and python CI jobs that validate your patches when posted to Gerrit. For instance, one thing you would need to do is to define an entry point for your plugin or driver in your own setup.cfg similarly as to how it is done in the setup.cfg for ODL.
- Define an entry point for your plugin or driver in setup.cfg
- Create third-party CI account: if you do not already have one, follow instructions for third-party CI to get one.

Internationalization support

OpenStack is committed to broad international support. Internationalization (I18n) is one of important areas to make OpenStack ubiquitous. Each project is recommended to support i18n.

This section describes how to set up translation support. The description in this section uses the following variables:

- repository : `openstack/${REPOSITORY}` (e.g., `openstack/networking-foo`)
- top level python path : `${MODULE_NAME}` (e.g., `networking_foo`)

oslo.i18n

- Each subproject repository should have its own oslo.i18n integration wrapper module `${MODULE_NAME}/_i18n.py`. The detail is found at https://docs.openstack.org/oslo.i18n/latest/user/usage.html.

  Note

  DOMAIN name should match your **module** name `${MODULE_NAME}`.

- Import `_()` from your `${MODULE_NAME}/_i18n.py`.

  Warning

  Do not use `_()` in the builtins namespace which is registered by **gettext.install()** in `neutron/__init__.py`. It is now deprecated as described in oslo.18n documentation.

Enable Translation

To update and import translations, you need to make a change in project-config. A good example is found at https://review.opendev.org/#/c/224222/. After doing this, the necessary jobs will be run and push/pull a message catalog to/from the translation infrastructure.

Integrating with the Neutron system

Configuration Files

The `data_files` in the `[files]` section of `setup.cfg` of Neutron shall not contain any third-party references. These shall be located in the same section of the third-party repos own `setup.cfg` file.

- Note: Care should be taken when naming sections in configuration files. When the Neutron service or an agent starts, oslo.config loads sections from all specified config files. This means that if a section [foo] exists in multiple config files, duplicate settings will collide. It is therefore recommended to prefix section names with a third-party string, e.g. [vendor_foo].

Since Mitaka, configuration files are not maintained in the git repository but should be generated as follows:

```
``tox -e genconfig``
```

If a tox environment is unavailable, then you can run the following script instead to generate the configuration files:

```
./tools/generate_config_file_samples.sh
```

It is advised that subprojects do not keep their configuration files in their respective trees and instead generate them using a similar approach as Neutron does.

ToDo: Inclusion in OpenStack documentation?
Is there a recommended way to have third-party config options listed in the configuration guide in docs.openstack.org?

Database Models and Migrations

A third-party repo may contain database models for its own tables. Although these tables are in the Neutron database, they are independently managed entirely within the third-party code. Third-party code shall **never** modify neutron core tables in any way.

Each repo has its own *expand* and *contract* alembic migration branches. A third-party repos alembic migration branches may operate only on tables that are owned by the repo.

- Note: Care should be taken when adding new tables. To prevent collision of table names it is **required** to prefix them with a vendor/plugin string.
- Note: A third-party maintainer may opt to use a separate database for their tables. This may complicate cases where there are foreign key constraints across schemas for DBMS that do not support this well. Third-party maintainer discretion advised.

The database tables owned by a third-party repo can have references to fields in neutron core tables. However, the alembic branch for a plugin/driver repo shall never update any part of a table that it does not own.

Note: What happens when a referenced item changes?

- **Q:** If a drivers table has a reference (for example a foreign key) to a neutron core table, and the referenced item is changed in neutron, what should you do?
- **A:** Fortunately, this should be an extremely rare occurrence. Neutron core reviewers will not allow such a change unless there is a very carefully thought-out design decision behind it. That design will include how to address any third-party code affected. (This is another good reason why you should stay actively involved with the Neutron developer community.)

The `neutron-db-manage` alembic wrapper script for neutron detects alembic branches for installed third-party repos, and the upgrade command automatically applies to all of them. A third-party repo must register its alembic migrations at installation time. This is done by providing an entrypoint in setup.cfg as follows:

For a third-party repo named `networking-foo`, add the alembic_migrations directory as an entrypoint in the `neutron.db.alembic_migrations` group:

```
[entry_points]
neutron.db.alembic_migrations =
    networking-foo = networking_foo.db.migration:alembic_migrations
```

ToDo: neutron-db-manage autogenerate

The alembic autogenerate command needs to support branches in external repos. Bug #1471333 has been filed for this.

DB Model/Migration Testing

Here is a *template functional test* third-party maintainers can use to develop tests for model-vs-migration sync in their repos. It is recommended that each third-party CI sets up such a test, and runs it regularly against Neutron master.

Entry Points

The Python setuptools installs all entry points for packages in one global namespace for an environment. Thus each third-party repo can define its packages own `[entry_points]` in its own `setup.cfg` file.

For example, for the `networking-foo` repo:

```
[entry_points]
console_scripts =
    neutron-foo-agent = networking_foo.cmd.eventlet.agents.foo:main
neutron.core_plugins =
    foo_monolithic = networking_foo.plugins.monolithic.plugin:FooPluginV2
neutron.service_plugins =
    foo_l3 = networking_foo.services.l3_router.l3_foo:FooL3ServicePlugin
neutron.ml2.type_drivers =
    foo_type = networking_foo.plugins.ml2.drivers.foo:FooType
neutron.ml2.mechanism_drivers =
    foo_ml2 = networking_foo.plugins.ml2.drivers.foo:FooDriver
neutron.ml2.extension_drivers =
    foo_ext = networking_foo.plugins.ml2.drivers.foo:FooExtensionDriver
```

- Note: It is advisable to include `foo` in the names of these entry points to avoid conflicts with other third-party packages that may get installed in the same environment.

API Extensions

Extensions can be loaded in two ways:

1. Use the `append_api_extensions_path()` library API. This method is defined in `neutron/api/extensions.py` in the neutron tree.
2. Leverage the `api_extensions_path` config variable when deploying. See the example config file `etc/neutron.conf` in the neutron tree where this variable is commented.

Service Providers

If your project uses service provider(s) the same way VPNAAS does, you specify your service provider in your `project_name.conf` file like so:

```
[service_providers]
# Must be in form:
# service_provider=<service_type>:<name>:<driver>[:default][,...]
```

In order for Neutron to load this correctly, make sure you do the following in your code:

```
from neutron.db import servicetype_db
service_type_manager = servicetype_db.ServiceTypeManager.get_instance()
service_type_manager.add_provider_configuration(
    YOUR_SERVICE_TYPE,
    pconf.ProviderConfiguration(YOUR_SERVICE_MODULE, YOUR_SERVICE_TYPE))
```

This is typically required when you instantiate your service plugin class.

Interface Drivers

Interface (VIF) drivers for the reference implementations are defined in `neutron/agent/linux/interface.py`. Third-party interface drivers shall be defined in a similar location within their own repo.

The entry point for the interface driver is a Neutron config option. It is up to the installer to configure this item in the `[DEFAULT]` section. For example:

```
[DEFAULT]
interface_driver = networking_foo.agent.linux.interface.FooInterfaceDriver
```

The `openvswitch` interface driver is used by default in case the option is not explicitly set.

ToDo: Interface Driver port bindings.

`VIF_TYPE_*` constants in `neutron_lib/api/definitions/portbindings.py` should be moved from neutron core to the repositories where their drivers are implemented. We need to provide some config or hook mechanism for VIF types to be registered by external interface drivers. For Nova, selecting the VIF driver can be done outside of Neutron (using the new os-vif python library?). Armando and Akihiro to discuss.

Rootwrap Filters

If a third-party repo needs a rootwrap filter for a command that is not used by Neutron core, then the filter shall be defined in the third-party repo.

For example, to add a rootwrap filters for commands in repo `networking-foo`:

- In the repo, create the file: `etc/neutron/rootwrap.d/foo.filters`
- In the repos `setup.cfg` add the filters to data_files:

```
[files]
data_files =
    etc/neutron/rootwrap.d =
        etc/neutron/rootwrap.d/foo.filters
```

Extending python-neutronclient

The maintainer of a third-party component may wish to add extensions to the Neutron CLI client. Thanks to https://review.opendev.org/148318 this can now be accomplished. See Client Command Extensions.

Other repo-split items

(These are still TBD.)

- Splitting policy.yaml? **ToDo** Armando will investigate.
- Generic instructions (or a template) for installing an out-of-tree plugin or driver for Neutron. Possibly something for the networking guide, and/or a template that plugin/driver maintainers can modify and include with their package.

14.5.5 Neutron public API

Neutron main tree serves as a library for multiple subprojects that rely on different modules from neutron.* namespace to accommodate their needs. Specifically, advanced service repositories and open source or vendor plugin/driver repositories do it.

Neutron modules differ in their API stability a lot, and there is no part of it that is explicitly marked to be consumed by other projects.

That said, there are modules that other projects should definitely avoid relying on.

Breakages

Neutron API is not very stable, and there are cases when a desired change in neutron tree is expected to trigger breakage for one or more external repositories under the neutron tent. Below you can find a list of known incompatible changes that could or are known to trigger those breakages. The changes are listed in reverse chronological order (newer at the top).

- change: QoS plugin refactor
  - commit: I863f063a0cfbb464cedd00bddc15dd853cbb6389
  - **solution: implement the new abstract methods in**
    neutron.extensions.qos.QoSPluginBase.
  - severity: Low (some out-of-tree plugins might be affected).

- change: Consume ConfigurableMiddleware from oslo_middleware.
  - commit: If7360608f94625b7d0972267b763f3e7d7624fee
  - **solution: switch to oslo_middleware.base.ConfigurableMiddleware;**
    stop using neutron.wsgi.Middleware and neutron.wsgi.Debug.
  - severity: Low (some out-of-tree plugins might be affected).
- change: Consume sslutils and wsgi modules from oslo.service.
  - commit: Ibfdf07e665fcfcd093a0e31274e1a6116706aec2
  - **solution: switch using oslo_service.wsgi.Router; stop using**
    neutron.wsgi.Router.
  - severity: Low (some out-of-tree plugins might be affected).
- change: oslo.service adopted.
  - commit: 6e693fc91dd79cfbf181e3b015a1816d985ad02c
  - **solution: switch using oslo_service.* namespace; stop using ANY**
    neutron.openstack.* contents.
  - severity: low (plugins must not rely on that subtree).
- change: oslo.utils.fileutils adopted.
  - commit: I933d02aa48260069149d16caed02b020296b943a
  - **solution: switch using oslo_utils.fileutils module; stop using**
    neutron.openstack.fileutils module.
  - severity: low (plugins must not rely on that subtree).
- change: Reuse callers session in DB methods.
  - commit: 47dd65cf986d712e9c6ca5dcf4420dfc44900b66
  - solution: Add context to args and reuse.
  - severity: High (mostly undetected, as 3rd party CI run Tempest tests only).
- change: switches to oslo.log, removes neutron.openstack.common.log.
  - commit: 22328baf1f60719fcaa5b0fbd91c0a3158d09c31
  - **solution: a) switch to oslo.log; b) copy log module into your tree and**
    use it (may not work due to conflicts between the module and oslo.log configuration options).
  - severity: High (most CI systems are affected).
- change: Implements reorganize-unit-test-tree spec.
  - commit: 1105782e3914f601b8f4be64939816b1afe8fb54
  - **solution: Code affected needs to update existing unit tests to reflect**
    new locations.
  - severity: High (mostly undetected, as 3rd party CI run Tempest tests only).
- change: drop linux/ovs_lib compat layer.
  - commit: 3bbf473b49457c4afbfc23fd9f59be8aa08a257d

- solution: switch to using neutron/agent/common/ovs_lib.py.
- severity: High (most CI systems are affected).

14.5.6 Client command extension support

The client command extension adds support for extending the neutron client while considering ease of creation.

The full document can be found in the python-neutronclient repository: https://docs.openstack.org/python-neutronclient/latest/contributor/client_command_extensions.html

14.5.7 Alembic Migrations

Introduction

The migrations in the alembic/versions contain the changes needed to migrate from older Neutron releases to newer versions. A migration occurs by executing a script that details the changes needed to upgrade the database. The migration scripts are ordered so that multiple scripts can run sequentially to update the database.

The Migration Wrapper

The scripts are executed by Neutrons migration wrapper `neutron-db-manage` which uses the Alembic library to manage the migration. Pass the `--help` option to the wrapper for usage information.

The wrapper takes some options followed by some commands:

```
neutron-db-manage <options> <commands>
```

The wrapper needs to be provided with the database connection string, which is usually provided in the `neutron.conf` configuration file in an installation. The wrapper automatically reads from `/etc/neutron/neutron.conf` if it is present. If the configuration is in a different location:

```
neutron-db-manage --config-file /path/to/neutron.conf <commands>
```

Multiple `--config-file` options can be passed if needed.

Instead of reading the DB connection from the configuration file(s) the `--database-connection` option can be used:

```
neutron-db-manage --database-connection mysql+pymysql://root:secret@127.0.0.1/
→neutron?charset=utf8 <commands>
```

The `branches`, `current`, and `history` commands all accept a `--verbose` option, which, when passed, will instruct `neutron-db-manage` to display more verbose output for the specified command:

```
neutron-db-manage current --verbose
```

For some commands the wrapper needs to know the entrypoint of the core plugin for the installation. This can be read from the configuration file(s) or specified using the `--core_plugin` option:

```
neutron-db-manage --core_plugin neutron.plugins.ml2.plugin.Ml2Plugin
→<commands>
```

When giving examples below of using the wrapper the options will not be shown. It is assumed you will use the options that you need for your environment.

For new deployments you will start with an empty database. You then upgrade to the latest database version via:

```
neutron-db-manage upgrade heads
```

For existing deployments the database will already be at some version. To check the current database version:

```
neutron-db-manage current
```

After installing a new version of Neutron server, upgrading the database is the same command:

```
neutron-db-manage upgrade heads
```

To create a script to run the migration offline:

```
neutron-db-manage upgrade heads --sql
```

To run the offline migration between specific migration versions:

```
neutron-db-manage upgrade <start version>:<end version> --sql
```

Upgrade the database incrementally:

```
neutron-db-manage upgrade --delta <# of revs>
```

NOTE: Database downgrade is not supported.

Migration Branches

Neutron makes use of alembic branches for two purposes.

1. Independent Sub-Project Tables

Various Sub-Projects can be installed with Neutron. Each sub-project registers its own alembic branch which is responsible for migrating the schemas of the tables owned by the sub-project.

The neutron-db-manage script detects which sub-projects have been installed by enumerating the `neutron.db.alembic_migrations` entrypoints. For more details see the Entry Points section of Contributing extensions to Neutron.

The neutron-db-manage script runs the given alembic command against all installed sub-projects. (An exception is the `revision` command, which is discussed in the *Developers* section below.)

2. Offline/Online Migrations

Since Liberty, Neutron maintains two parallel alembic migration branches.

The first one, called expand, is used to store expansion-only migration rules. Those rules are strictly additive and can be applied while neutron-server is running. Examples of additive database schema changes are: creating a new table, adding a new table column, adding a new index, etc.

The second branch, called contract, is used to store those migration rules that are not safe to apply while neutron-server is running. Those include: column or table removal, moving data from one part of the database into another (renaming a column, transforming single table into multiple, etc.), introducing or modifying constraints, etc.

The intent of the split is to allow invoking those safe migrations from expand branch while neutron-server is running, reducing downtime needed to upgrade the service.

For more details, see the *Expand and Contract Scripts* section below.

Developers

A database migration script is required when you submit a change to Neutron or a sub-project that alters the database model definition. The migration script is a special python file that includes code to upgrade the database to match the changes in the model definition. Alembic will execute these scripts in order to provide a linear migration path between revisions. The neutron-db-manage command can be used to generate migration scripts for you to complete. The operations in the template are those supported by the Alembic migration library.

The neutron-db-manage command is designed for sequential database upgrades. Typically, its executed once to reach a specific schema version. Subsequent executions should then resume from that achieved point. This design theoretically prevents scenarios where the same upgrade script runs multiple times, such as attempting to create an already existing table. However, due to potential real-world issues such as loss of the recorded database version or backporting of scripts, Alembic migration scripts must be idempotent. Therefore, scripts that modify the schema, such as adding a new table, should first check for the objects existence, for example:

```
import sqlalchemy as sa

from neutron.db import migration

def upgrade():
    migration.add_column_if_not_exists(
        'networks',
        sa.Column('qinq', sa.Boolean(), server_default=None)
    )
```

Module *neutron.db.migration* provides helper functions to create table or add column to the existing table in the idempotent way.

Running neutron-db-manage without devstack

When, as a developer, you want to work with the Neutron DB schema and alembic migrations only, it can be rather tedious to rely on devstack just to get an up-to-date neutron-db-manage installed. This section describes how to work on the schema and migration scripts with just the unit test virtualenv and mysql. You can also operate on a separate test database so you dont mess up the installed Neutron database.

Setting up the environment

Install mysql service

This only needs to be done once since it is a system install. If you have run devstack on your system before, then the mysql service is already installed and you can skip this step.

Mysql must be configured as installed by devstack, and the following script accomplishes this without actually running devstack:

```
INSTALL_MYSQL_ONLY=True ./tools/configure_for_func_testing.sh ../devstack
```

Run this from the root of the neutron repo. It assumes an up-to-date clone of the devstack repo is in `../devstack`.

Note that you must know the mysql root password. It is derived from (in order of precedence):

- `$MYSQL_PASSWORD` in your environment
- `$MYSQL_PASSWORD` in `../devstack/local.conf`
- `$MYSQL_PASSWORD` in `../devstack/localrc`
- default of secretmysql from `tools/configure_for_func_testing.sh`

Work on a test database

Rather than using the neutron database when working on schema and alembic migration script changes, we can work on a test database. In the examples below, we use a database named `testdb`.

To create the database:

```
mysql -e "create database testdb;"
```

You will often need to clear it to re-run operations from a blank database:

```
mysql -e "drop database testdb; create database testdb;"
```

To work on the test database instead of the neutron database, point to it with the `--database-connection` option:

```
neutron-db-manage --database-connection mysql+pymysql://root:secretmysql@127.
↪0.0.1/testdb?charset=utf8 <commands>
```

You may find it convenient to set up an alias (in your .bashrc) for this:

```
alias test-db-manage='neutron-db-manage --database-connection mysql+pymysql://
↪root:secretmysql@127.0.0.1/testdb?charset=utf8'
```

Create and activate the virtualenv

From the root of the neutron (or sub-project) repo directory, run:

```
tox --notest -r -e py38
source .tox/py38/bin/activate
```

Now you can use the `test-db-manage` alias in place of `neutron-db-manage` in the script auto-generation instructions below.

When you are done, exit the virtualenv:

```
deactivate
```

Script Auto-generation

This section describes how to auto-generate an alembic migration script for a model change. You may either use the system installed devstack environment, or a virtualenv + testdb environment as described in *Running neutron-db-manage without devstack*.

Stop the neutron service. Work from the base directory of the neutron (or sub-project) repo. Check out the master branch and do `git pull` to ensure it is fully up to date. Check out your development branch and rebase to master.

NOTE: Make sure you have not updated the `CONTRACT_HEAD` or `EXPAND_HEAD` yet at this point.

Start with an empty database and upgrade to heads:

```
mysql -e "drop database neutron; create database neutron;"
neutron-db-manage upgrade heads
```

The database schema is now created without your model changes. The alembic `revision --autogenerate` command will look for differences between the schema generated by the upgrade command and the schema defined by the models, including your model updates:

```
neutron-db-manage revision -m "description of revision" --autogenerate
```

This generates a prepopulated template with the changes needed to match the database state with the models. You should inspect the autogenerated template to ensure that the proper models have been altered. When running the above command you will probably get the following error message:

```
Multiple heads are present; please specify the head revision on which the
new revision should be based, or perform a merge.
```

This is alembic telling you that it does not know which branch (contract or expand) to generate the revision for. You must decide, based on whether you are doing contracting or expanding changes to the schema, and provide either the `--contract` or `--expand` option. If you have both types of changes, you must run the command twice, once with each option, and then manually edit the generated revision scripts to separate the migration operations.

In rare circumstances, you may want to start with an empty migration template and manually author the changes necessary for an upgrade. You can create a blank file for a branch via:

```
neutron-db-manage revision -m "description of revision" --expand
neutron-db-manage revision -m "description of revision" --contract
```

NOTE: If you use above command you should check that migration is created in a directory that is named as current release. If not, please raise the issue with the development team (IRC, mailing list, launchpad bug).

NOTE: The description of revision text should be a simple English sentence. The first 30 characters of the description will be used in the file name for the script, with underscores substituted for spaces. If the truncation occurs at an awkward point in the description, you can modify the script file name manually before committing.

The timeline on each alembic branch should remain linear and not interleave with other branches, so that there is a clear path when upgrading. To verify that alembic branches maintain linear timelines, you can run this command:

```
neutron-db-manage check_migration
```

If this command reports an error, you can troubleshoot by showing the migration timelines using the `history` command:

```
neutron-db-manage history
```

Expand and Contract Scripts

The obsolete branchless design of a migration script included that it indicates a specific version of the schema, and includes directives that apply all necessary changes to the database at once. If we look for example at the script `2d2a8a565438_hierarchical_binding.py`, we will see:

```
# .../alembic_migrations/versions/2d2a8a565438_hierarchical_binding.py

def upgrade():

    # .. inspection code ...

    op.create_table(
        'ml2_port_binding_levels',
        sa.Column('port_id', sa.String(length=36), nullable=False),
        sa.Column('host', sa.String(length=255), nullable=False),
        # ... more columns ...
    )

    for table in port_binding_tables:
        op.execute((
            "INSERT INTO ml2_port_binding_levels "
            "SELECT port_id, host, 0 AS level, driver, segment AS segment_id "
            "FROM %s "
            "WHERE host <> '' "
            "AND driver <> '';"
        ) % table)

    op.drop_constraint(fk_name_dvr[0], 'ml2_dvr_port_bindings', 'foreignkey')
    op.drop_column('ml2_dvr_port_bindings', 'cap_port_filter')
    op.drop_column('ml2_dvr_port_bindings', 'segment')
    op.drop_column('ml2_dvr_port_bindings', 'driver')

    # ... more DROP instructions ...
```

The above script contains directives that are both under the expand and contract categories, as well as some data migrations. The `op.create_table` directive is an expand; it may be run safely while the old version of the application still runs, as the old code simply doesnt look for this table. The `op.drop_constraint` and `op.drop_column` directives are contract directives (the drop column more so than the drop constraint); running at least the `op.drop_column` directives means that the old version of the application will fail, as it will attempt to access these columns which no longer exist.

The data migrations in this script are adding new rows to the newly added `ml2_port_binding_levels` table.

Under the new migration script directory structure, the above script would be stated as two scripts; an expand and a contract script:

```
# expansion operations
# .../alembic_migrations/versions/liberty/expand/2bde560fc638_hierarchical_
↪binding.py

def upgrade():

    op.create_table(
        'ml2_port_binding_levels',
        sa.Column('port_id', sa.String(length=36), nullable=False),
        sa.Column('host', sa.String(length=255), nullable=False),
        # ... more columns ...
    )


# contraction operations
# .../alembic_migrations/versions/liberty/contract/4405aedc050e_hierarchical_
↪binding.py

def upgrade():

    for table in port_binding_tables:
        op.execute((
            "INSERT INTO ml2_port_binding_levels "
            "SELECT port_id, host, 0 AS level, driver, segment AS segment_id "
            "FROM %s "
            "WHERE host <> '' "
            "AND driver <> '';"
        ) % table)

    op.drop_constraint(fk_name_dvr[0], 'ml2_dvr_port_bindings', 'foreignkey')
    op.drop_column('ml2_dvr_port_bindings', 'cap_port_filter')
    op.drop_column('ml2_dvr_port_bindings', 'segment')
    op.drop_column('ml2_dvr_port_bindings', 'driver')

    # ... more DROP instructions ...
```

The two scripts would be present in different subdirectories and also part of entirely separate versioning streams. The expand operations are in the expand script, and the contract operations are in the contract script.

For the time being, data migration rules also belong to contract branch. There is expectation that eventually live data migrations move into middleware that will be aware about different database schema elements to converge on, but Neutron is still not there.

Scripts that contain only expansion or contraction rules do not require a split into two parts.

If a contraction script depends on a script from expansion stream, the following directive should be added in the contraction script:

```
depends_on = ('<expansion-revision>',)
```

Expand and Contract Branch Exceptions

In some cases, we have to have expand operations in contract migrations. For example, table networksegments was renamed in contract migration, so all operations with this table are required to be in contract branch as well. For such cases, we use the `contract_creation_exceptions` that should be implemented as part of such migrations. This is needed to get functional tests pass.

Usage:

```
def contract_creation_exceptions():
    """Docstring should explain why we allow such exception for contract
    branch.
    """
    return {
        sqlalchemy_obj_type: ['name']
        # For example: sa.Column: ['subnets.segment_id']
    }
```

HEAD files for conflict management

In directory `neutron/db/migration/alembic_migrations/versions` there are two files, `CONTRACT_HEAD` and `EXPAND_HEAD`. These files contain the ID of the head revision in each branch. The purpose of these files is to validate the revision timelines and prevent non-linear changes from entering the merge queue.

When you create a new migration script by neutron-db-manage these files will be updated automatically. But if another migration script is merged while your change is under review, you will need to resolve the conflict manually by changing the `down_revision` in your migration script.

Applying database migration rules

To apply just expansion rules, execute:

```
neutron-db-manage upgrade --expand
```

After the first step is done, you can stop neutron-server, apply remaining non-expansive migration rules, if any:

```
neutron-db-manage upgrade --contract
```

and finally, start your neutron-server again.

If you have multiple neutron-server instances in your cloud, and there are pending contract scripts not applied to the database, full shutdown of all those services is required before upgrade contract is executed. You can determine whether there are any pending contract scripts by checking the message returned from the following command:

```
neutron-db-manage has_offline_migrations
```

If you are not interested in applying safe migration rules while the service is running, you can still upgrade database the old way, by stopping the service, and then applying all available rules:

```
neutron-db-manage upgrade head[s]
```

It will apply all the rules from both the expand and the contract branches, in proper order.

Tagging milestone revisions

When named release (liberty, mitaka, etc.) is done for neutron or a sub-project, the alembic revision scripts at the head of each branch for that release must be tagged. This is referred to as a milestone revision tag.

For example, here is a patch that tags the liberty milestone revisions for the neutron-fwaas sub-project. Note that each branch (expand and contract) is tagged.

Tagging milestones allows neutron-db-manage to upgrade the schema to a milestone release, e.g.:

```
neutron-db-manage upgrade liberty
```

Generation of comparable metadata with current database schema

Directory `neutron/db/migration/models` contains module `head.py`, which provides all database models at current HEAD. Its purpose is to create comparable metadata with the current database schema. The database schema is generated by alembic migration scripts. The models must match, and this is verified by a model-migration sync test in Neutrons functional test suite. That test requires all modules containing DB models to be imported by head.py in order to make a complete comparison.

When adding new database models, developers must update this module, otherwise the change will fail to merge.

14.5.8 Upgrade checks

Introduction

CLI tool `neutron-status upgrade check` contains checks which perform a release-specific readiness check before restarting services with new code. For more details see neutron-status command-line client page.

3rd party plugins checks

Neutron upgrade checks script allows to add checks by stadium and 3rd party projects. The `neutron-status` script detects which sub-projects have been installed by enumerating the `neutron.status.upgrade.checks` entrypoints. For more details see the Entry Points section of Contributing extensions to Neutron. Checks can be run in random order and should be independent from each other.

The recommended entry point name is a repository name: For example, neutron-fwaas for FWaaS and networking-sfc for SFC:

```
neutron.status.upgrade.checks =
    neutron-fwaas = neutron_fwaas.upgrade.checks:Checks
```

Entrypoint should be class which inherits from `neutron.cmd.upgrade_checks.base.BaseChecks`.

An example of a checks class can be found in `neutron.cmd.upgrade_checks.checks.CoreChecks`.

14.5.9 Testing

Testing Neutron

Why Should You Care

Theres two ways to approach testing:

1) Write unit tests because theyre required to get your patch merged. This typically involves mock heavy tests that assert that your code is as written.

2) Putting as much thought into your testing strategy as you do to the rest of your code. Use different layers of testing as appropriate to provide high *quality* coverage. Are you touching an agent? Test it against an actual system! Are you adding a new API? Test it for race conditions against a real database! Are you adding a new cross-cutting feature? Test that it does what its supposed to do when run on a real cloud!

Do you feel the need to verify your change manually? If so, the next few sections attempt to guide you through Neutrons different test infrastructures to help you make intelligent decisions and best exploit Neutrons test offerings.

Definitions

We will talk about three classes of tests: unit, functional and integration. Each respective category typically targets a larger scope of code. Other than that broad categorization, here are a few more characteristic:

- Unit tests - Should be able to run on your laptop, directly following a git clone of the project. The underlying system must not be mutated, mocks can be used to achieve this. A unit test typically targets a function or class.
- Functional tests - Run against a pre-configured environment (tools/configure_for_func_testing.sh). Typically test a component such as an agent using no mocks.
- Integration tests - Run against a running cloud, often target the API level, but also scenarios, user stories or grenade. You may find such tests under tests/fullstack, and in the Tempest, Rally, Grenade and neutron-tempest-plugin(neutron_tempest_plugin/api|scenario) projects.

Tests in the Neutron tree are typically organized by the testing infrastructure used, and not by the scope of the test. For example, many tests under the unit directory invoke an API call and assert that the expected output was received. The scope of such a test is the entire Neutron server stack, and clearly not a specific function such as in a typical unit test.

Testing Frameworks

The different frameworks are listed below. The intent is to list the capabilities of each testing framework as to help the reader understand when should each tool be used. Remember that when adding code that touches many areas of Neutron, each area should be tested with the appropriate framework. Overlap between different test layers is often desirable and encouraged.

Unit Tests

Unit tests (neutron/tests/unit/) are meant to cover as much code as possible. They are designed to test the various pieces of the Neutron tree to make sure any new changes dont break existing functionality. Unit tests have no requirements nor make changes to the system they are running on. They use an in-memory sqlite database to test DB interaction.

At the start of each test run:

- RPC listeners are mocked away.
- The fake Oslo messaging driver is used.

At the end of each test run:

- Mocks are automatically reverted.
- The in-memory database is cleared of content, but its schema is maintained.
- The global Oslo configuration object is reset.

The unit testing framework can be used to effectively test database interaction, for example, distributed routers allocate a MAC address for every host running an OVS agent. One of DVRs DB mixins implements a method that lists all host MAC addresses. Its test looks like this:

```
def test_get_dvr_mac_address_list(self):
    self._create_dvr_mac_entry('host_1', 'mac_1')
    self._create_dvr_mac_entry('host_2', 'mac_2')
    mac_list = self.mixin.get_dvr_mac_address_list(self.ctx)
    self.assertEqual(2, len(mac_list))
```

It inserts two new host MAC address, invokes the method under test and asserts its output. The test has many things going for it:

- It targets the method under test correctly, not taking on a larger scope than is necessary.
- It does not use mocks to assert that methods were called, it simply invokes the method and asserts its output (In this case, that the list method returns two records).

This is allowed by the fact that the method was built to be testable - The method has clear input and output with no side effects.

You can get oslo.db to generate a file-based sqlite database by setting OS_TEST_DBAPI_ADMIN_CONNECTION to a file based URL as described in this mailing list post. This file will be created but (confusingly) wont be the actual file used for the database. To find the actual file, set a break point in your test method and inspect self.engine.url.

```
$ OS_TEST_DBAPI_ADMIN_CONNECTION=sqlite:///sqlite.db .tox/py38/bin/python -m \
    testtools.run neutron.tests.unit...
...
(Pdb) self.engine.url
sqlite:////tmp/iwbgvhbshp.db
```

Now, you can inspect this file using sqlite3.

```
$ sqlite3 /tmp/iwbgvhbshp.db
```

Functional Tests

Functional tests (neutron/tests/functional/) are intended to validate actual system interaction. Mocks should be used sparingly, if at all. Care should be taken to ensure that existing system resources are not modified and that resources created in tests are properly cleaned up both on test success and failure.

Lets examine the benefits of the functional testing framework. Neutron offers a library called ip_lib that wraps around the ip binary. One of its methods is called device_exists which accepts a device name and a namespace and returns True if the device exists in the given namespace. Its easy building a test that targets the method directly, and such a test would be considered a unit test. However, what framework should such a test use? A test using the unit tests framework could not mutate state on the system, and so could not actually create a device and assert that it now exists. Such a test would look roughly like this:

- It would mock execute, a method that executes shell commands against the system to return an IP device named foo.
- It would then assert that when device_exists is called with foo, it returns True, but when called with a different device name it returns False.
- It would most likely assert that execute was called using something like: ip link show foo.

The value of such a test is arguable. Remember that new tests are not free, they need to be maintained. Code is often refactored, reimplemented and optimized.

- There are other ways to find out if a device exists (Such as by looking at /sys/class/net), and in such a case the test would have to be updated.
- Methods are mocked using their name. When methods are renamed, moved or removed, their mocks must be updated. This slows down development for avoidable reasons.
- Most importantly, the test does not assert the behavior of the method. It merely asserts that the code is as written.

When adding a functional test for device_exists, several framework level methods were added. These methods may now be used by other tests as well. One such method creates a virtual device in a namespace, and ensures that both the namespace and the device are cleaned up at the end of the test run regardless of success or failure using the addCleanup method. The test generates details for a temporary device, asserts that a device by that name does not exist, creates that device, asserts that it now exists, deletes it, and asserts that it no longer exists. Such a test avoids all three issues mentioned above if it were written using the unit testing framework.

Functional tests are also used to target larger scope, such as agents. Many good examples exist: See the OVS, L3 and DHCP agents functional tests. Such tests target a top level agent method and assert that the system interaction that was supposed to be performed was indeed performed. For example, to test the DHCP agents top level method that accepts network attributes and configures dnsmasq for that network, the test:

- Instantiates an instance of the DHCP agent class (But does not start its process).
- Calls its top level function with prepared data.
- Creates a temporary namespace and device, and calls dhclient from that namespace.
- Assert that the device successfully obtained the expected IP address.

Test exceptions

Test neutron.tests.functional.agent.test_ovs_flows.OVSFlowTestCase.test_install_flood_to_tun is currently skipped if openvswitch version is less than 2.5.1. This version contains bug where appctl command prints wrong output for Final flow. Its been fixed in openvswitch 2.5.1 in this commit. If openvswitch version meets the test requirement then the test is triggered normally.

Fullstack Tests

Why?

The idea behind fullstack testing is to fill a gap between unit + functional tests and Tempest. Tempest tests are expensive to run, and target black box API tests exclusively. Tempest requires an OpenStack deployment to be run against, which can be difficult to configure and setup. Full stack testing addresses these issues by taking care of the deployment itself, according to the topology that the test requires. Developers further benefit from full stack testing as it can sufficiently simulate a real environment and provide a rapidly reproducible way to verify code while youre still writing it.

More details can be found in *FullStack Testing* guide.

Integration - tempest tests

Tempest is the integration test suit of Openstack, more details can be found in *Tempest testing*

API Tests

API tests (neutron-tempest-plugin/neutron_tempest_plugin/api/) are intended to ensure the function and stability of the Neutron API. As much as possible, changes to this path should not be made at the same time as changes to the code to limit the potential for introducing backwards-incompatible changes, although the same patch that introduces a new API should include an API test.

Since API tests target a deployed Neutron daemon that is not test-managed, they should not depend on controlling the runtime configuration of the target daemon. API tests should be black-box - no assumptions should be made about implementation. Only the contract defined by Neutrons REST API should be validated, and all interaction with the daemon should be via a REST client.

The neutron-tempest-plugin/neutron_tempest_plugin directory was copied from the Tempest project around the Kilo timeframe. At the time, there was an overlap of tests between the Tempest and Neutron repositories. This overlap was then eliminated by carving out a subset of resources that belong to Tempest, with the rest in Neutron.

API tests that belong to Tempest deal with a subset of Neutrons resources:

- Port
- Network
- Subnet
- Security Group
- Router
- Floating IP

These resources were chosen for their ubiquity. They are found in most Neutron deployments regardless of plugin, and are directly involved in the networking and security of an instance. Together, they form the bare minimum needed by Neutron.

This is excluding extensions to these resources (For example: Extra DHCP options to subnets, or snat_gateway mode to routers) that are not mandatory in the majority of cases.

Tests for other resources should be contributed to the Neutron repository. Scenario tests should be similarly split up between Tempest and Neutron according to the API theyre targeting.

To create an API test, the testing class must at least inherit from neutron_tempest_plugin.api.base.BaseNetworkTest base class. As some of tests may require certain extensions to be enabled, the base class provides `required_extensions` class attribute which can be used by subclasses to define a list of required extensions for particular test class.

Scenario Tests

Scenario tests (neutron-tempest-plugin/neutron_tempest_plugin/scenario), like API tests, use the Tempest test infrastructure and have the same requirements. Guidelines for writing a good scenario test may be found at the Tempest developer guide: https://docs.openstack.org/tempest/latest/field_guide/scenario.html

Scenario tests, like API tests, are split between the Tempest and Neutron repositories according to the Neutron API the test is targeting.

Some scenario tests require advanced `Glance` images (for example, `Ubuntu` or `CentOS`) in order to pass. Those tests are skipped by default. To enable them, include the following in `tempest.conf`:

```
[compute]
image_ref = <uuid of advanced image>
[neutron_plugin_options]
image_is_advanced = True
```

Specific test requirements for advanced images are:

1. `test_trunk` requires `802.11q` kernel module loaded.

Rally Tests

Rally tests (rally-jobs/plugins) use the rally infrastructure to exercise a neutron deployment. Guidelines for writing a good rally test can be found in the rally plugin documentation. There are also some examples in tree; the process for adding rally plugins to neutron requires three steps: 1) write a plugin and place it under rally-jobs/plugins/. This is your rally scenario; 2) (optional) add a setup file under rally-jobs/extra/. This is any devstack configuration required to make sure your environment can successfully process your scenario requests; 3) edit neutron-neutron.yaml. This is your scenario contract or SLA.

Grenade Tests

Grenade is a tool to test upgrade process between OpenStack releases. It actually not introduces any new tests but it is a tool which uses Tempest tests to verify upgrade process between releases. Neutron runs couple of Grenade jobs in check and gate queue - see *CI Testing* summary.

You can run Grenade tests locally on the virtual machine(s). It is pretty similar to deploying OpenStack using Devstack. All is described in the Projects wiki and documentation.

More info about how to troubleshoot Grenade failures in the CI jobs can be found in the *Troubleshooting Grenade jobs* document.

Development Process

It is expected that any new changes that are proposed for merge come with tests for that feature or code area. Any bugs fixes that are submitted must also have tests to prove that they stay fixed! In addition, before proposing for merge, all of the current tests should be passing.

Structure of the Unit Test Tree

The structure of the unit test tree should match the structure of the code tree, e.g.

```
- target module: neutron.agent.utils

- test module: neutron.tests.unit.agent.test_utils
```

Unit test modules should have the same path under neutron/tests/unit/ as the module they target has under neutron/, and their name should be the name of the target module prefixed by *test_*. This requirement is intended to make it easier for developers to find the unit tests for a given module.

Similarly, when a test module targets a package, that modules name should be the name of the package prefixed by *test_* with the same path as when a test targets a module, e.g.

```
- target package: neutron.ipam

- test module: neutron.tests.unit.test_ipam
```

The following command can be used to validate whether the unit test tree is structured according to the above requirements:

```
./tools/check_unit_test_structure.sh
```

Where appropriate, exceptions can be added to the above script. If code is not part of the Neutron namespace, for example, its probably reasonable to exclude their unit tests from the check.

Note

At no time should the production code import anything from testing subtree (neutron.tests). There are distributions that split out neutron.tests modules in a separate package that is not installed by default, making any code that relies on presence of the modules to fail. For example, RDO is one of those distributions.

Running Tests

Before submitting a patch for review you should always ensure all tests pass; a tox run is triggered by the zuul jobs executed on gerrit for each patch pushed for review.

Neutron, like other OpenStack projects, uses tox for managing the virtual environments for running test cases. It uses Testr for managing the running of the test cases.

Tox handles the creation of a series of virtualenvs that target specific versions of Python.

Testr handles the parallel execution of series of test cases as well as the tracking of long-running tests and other things.

For more information on the standard Tox-based test infrastructure used by OpenStack and how to do some common test/debugging procedures with Testr, see this wiki page: https://wiki.openstack.org/wiki/Testr

PEP8 and Unit Tests

Running pep8 and unit tests is as easy as executing this in the root directory of the Neutron source code:

```
tox
```

To run only pep8:

```
tox -e pep8
```

Since pep8 includes running pylint on all files, it can take quite some time to run.

To restrict the pylint check to only the files altered by the latest patch changes:

```
tox -e pep8 -- HEAD~1
```

To run only the unit tests:

```
tox -e py38
```

Many changes span across both the neutron and neutron-lib repos, and tox will always build the test environment using the published module versions specified in requirements.txt. To run tox tests against a different version of neutron-lib, use the TOX_ENV_SRC_MODULES environment variable to point at a local package repo.

For example, to run against the master branch of neutron-lib:

```
cd $SRC
git clone https://opendev.org/openstack/neutron-lib
cd $NEUTRON_DIR
env TOX_ENV_SRC_MODULES=$SRC/neutron-lib tox -r -e py38
```

To run against a change of your own, repeat the same steps, but use the directory with your changes, not a fresh clone.

To run against a particular gerrit change of the lib (substituting the desired gerrit refs for this example):

```
cd $SRC
git clone https://opendev.org/openstack/neutron-lib
cd neutron-lib
git fetch https://opendev.org/openstack/neutron-lib refs/changes/13/635313/6 &
↪& git checkout FETCH_HEAD
cd $NEUTRON_DIR
env TOX_ENV_SRC_MODULES=$SRC/neutron-lib tox -r -e py38
```

Note that the -r is needed to re-create the tox virtual envs, and will also be needed to restore them to standard when not using this method.

Any pip installable package can be overriden with this environment variable, not just neutron-lib. To specify multiple packages to override, specify them as a space separated list to TOX_ENV_SRC_MODULES. Example:

```
env TOX_ENV_SRC_MODULES="$SRC/neutron-lib $SRC/oslo.db" tox -r -e py38
```

Functional Tests

To run functional tests that do not require sudo privileges or specific-system dependencies:

```
tox -e functional
```

To run all the functional tests, including those requiring sudo privileges and system-specific dependencies, the procedure defined by tools/configure_for_func_testing.sh should be followed.

IMPORTANT: configure_for_func_testing.sh relies on DevStack to perform extensive modification to the underlying host. Execution of the script requires sudo privileges and it is recommended that the following commands be invoked only on a clean and disposable VM. A VM that has had DevStack previously installed on it is also fine.

```
git clone https://opendev.org/openstack/devstack ../devstack
./tools/configure_for_func_testing.sh ../devstack -i
tox -e dsvm-functional
```

The -i option is optional and instructs the script to use DevStack to install and configure all of Neutrons package dependencies. It is not necessary to provide this option if DevStack has already been used to deploy Neutron to the target host.

Fullstack Tests

See *FullStack Testing* guide.

API & Scenario Tests

To run the api or scenario tests, deploy Tempest, neutron-tempest-plugin and Neutron with DevStack and then run the following command, from the tempest directory:

```
$ export DEVSTACK_GATE_TEMPEST_REGEX="neutron"
$ tox -e venv-tempest -- pip install (path to the neutron-tempest-plugin␣
↪directory)
$ tox -e all -- $DEVSTACK_GATE_TEMPEST_REGEX
```

If you want to limit the amount of tests, or run an individual test, you can do, for instance:

```
$ tox -e all -- neutron_tempest_plugin.api.admin.test_routers_ha
$ tox -e all -- neutron_tempest_plugin.api.test_qos.QosTestJSON.test_create_
↪policy
```

If you want to use special config for Neutron, like use advanced images (Ubuntu or CentOS) testing advanced features, you may need to add config in tempest/etc/tempest.conf:

```
[neutron_plugin_options]
image_is_advanced = True
```

The Neutron tempest plugin configs are under `neutron_plugin_options` scope of `tempest.conf`.

Running Individual Tests

For running individual test modules, cases or tests, you just need to pass the dot-separated path you want as an argument to it.

For example, the following would run only a single test or test case:

```
$ tox -e py38 neutron.tests.unit.test_manager
$ tox -e py38 neutron.tests.unit.test_manager.NeutronManagerTestCase
$ tox -e py38 neutron.tests.unit.test_manager.NeutronManagerTestCase.test_
↪service_plugin_is_loaded
```

If you want to pass other arguments to stestr, you can do the following:

```
$ tox -e py38 -- neutron.tests.unit.test_manager --serial
```

Coverage

Neutron has a fast growing code base and there are plenty of areas that need better coverage.

To get a grasp of the areas where tests are needed, you can check current unit tests coverage by running:

```
$ tox -ecover
```

Since the coverage command can only show unit test coverage, a coverage document is maintained that shows test coverage per area of code in: doc/source/devref/testing_coverage.rst. You could also rely on Zuul logs, that are generated post-merge (not every project builds coverage results). To access them, do the following:

- Check out the latest merge commit
- Go to: http://logs.openstack.org/<first-2-digits-of-sha1>/<sha1>/post/neutron-coverage/.
- Spec is a work in progress to provide a better landing page.

Debugging

By default, calls to pdb.set_trace() will be ignored when tests are run. For pdb statements to work, invoke tox as follows:

```
$ tox -e venv -- python -m testtools.run [test module path]
```

Tox-created virtual environments (venvs) can also be activated after a tox run and reused for debugging:

```
$ tox -e venv
$ . .tox/venv/bin/activate
$ python -m testtools.run [test module path]
```

Tox packages and installs the Neutron source tree in a given venv on every invocation, but if modifications need to be made between invocation (e.g. adding more pdb statements), it is recommended that the source tree be installed in the venv in editable mode:

```
# run this only after activating the venv
$ pip install --editable .
```

Editable mode ensures that changes made to the source tree are automatically reflected in the venv, and that such changes are not overwritten during the next tox run.

Post-mortem Debugging

TBD: how to do this with tox.

References

Full Stack Testing

> **Warning**
>
> Full Stack testing was disabled as a side effect of the Eventlet removal process, see the related patch: https://review.opendev.org/c/openstack/neutron/+/953295

How?

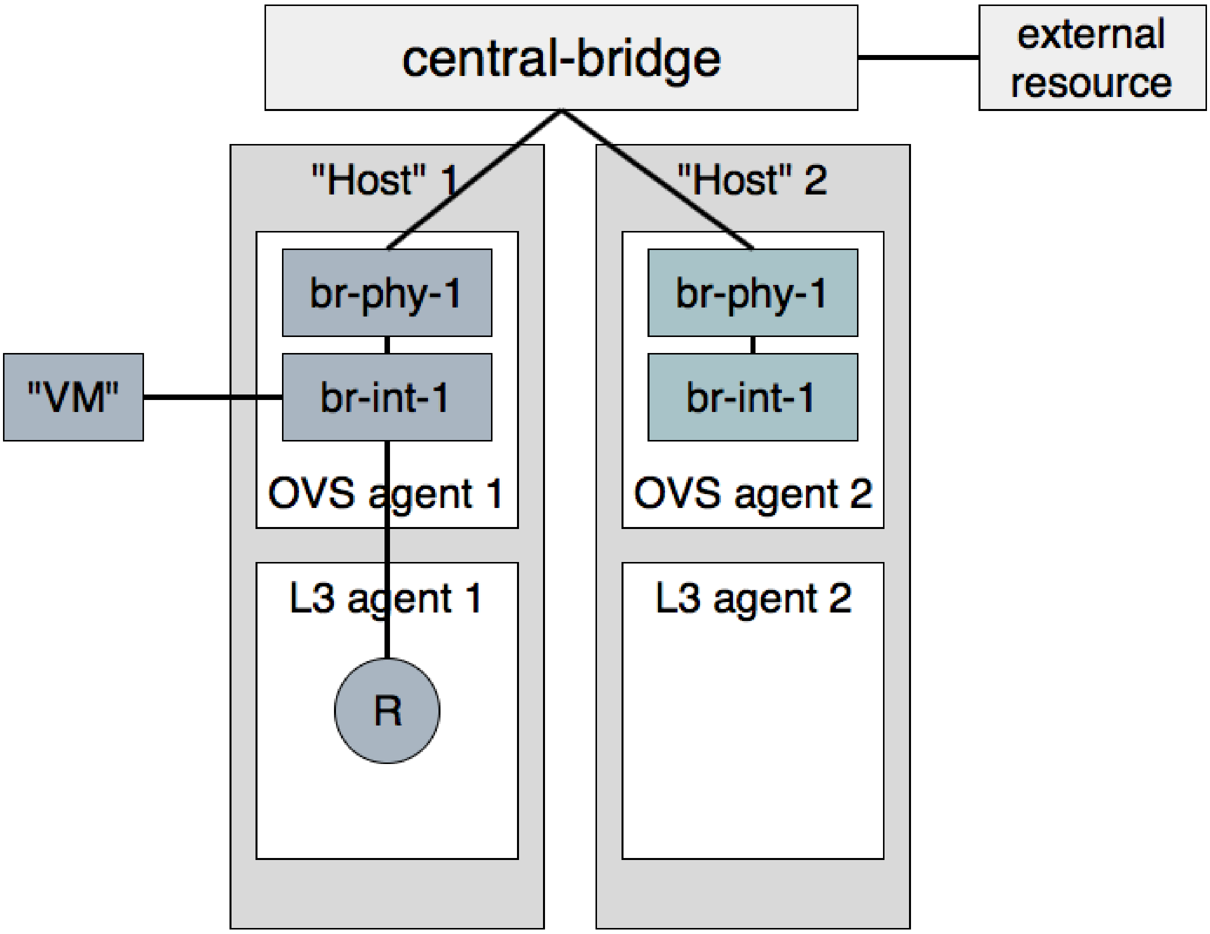
Full stack tests set up their own Neutron processes (Server & agents). They assume a working Rabbit and MySQL server before the run starts. Instructions on how to run fullstack tests on a VM are available below.

Each test defines its own topology (What and how many servers and agents should be running).

Since the test runs on the machine itself, full stack testing enables white box testing. This means that you can, for example, create a router through the API and then assert that a namespace was created for it.

Full stack tests run in the Neutron tree with Neutron resources alone. You may use the Neutron API (The Neutron server is set to NOAUTH so that Keystone is out of the picture). VMs may be simulated with a container-like class: neutron.tests.fullstack.resources.machine.FakeFullstackMachine. An example of its usage may be found at: neutron/tests/fullstack/test_connectivity.py.

Full stack testing can simulate multi node testing by starting an agent multiple times. Specifically, each node would have its own copy of the OVS/DHCP/L3 agents, all configured with the same host value. Each OVS agent is connected to its own pair of br-int/br-ex, and those bridges are then interconnected.



Segmentation at the database layer is guaranteed by creating a database per test. The messaging layer achieves segmentation by utilizing a RabbitMQ feature called vhosts. In short, just like a MySQL server serve multiple databases, so can a RabbitMQ server serve multiple messaging domains. Exchanges and queues in one vhost are segmented from those in another vhost.

Please note that if the change you would like to test using fullstack tests involves a change to python-neutronclient as well as neutron, then you should make sure your fullstack tests are in a separate third change that depends on the python-neutronclient change using the Depends-On tag in the commit message. You will need to wait for the next release of python-neutronclient, and a minimum version bump for python-neutronclient in the global requirements, before your fullstack tests will work in the gate. This is because tox uses the version of python-neutronclient listed in the upper-constraints.txt file in the openstack/requirements repository.

When?

1) Youd like to test the interaction between Neutron components (Server and agents) and have already tested each component in isolation via unit or functional tests. You should have many unit tests, fewer tests to test a component and even fewer to test their interaction. Edge cases should not be tested with full stack testing.

2) Youd like to increase coverage by testing features that require multi node testing such as l2pop, L3 HA and DVR.

3) Youd like to test agent restarts. Weve found bugs in the OVS, DHCP and L3 agents and havent found an effective way to test these scenarios. Full stack testing can help here as the full stack infrastructure can restart an agent during the test.

Example

Neutron offers a Quality of Service API, initially offering bandwidth capping at the port level. In the reference implementation, it does this by utilizing an OVS feature. neutron.tests.fullstack.test_qos.TestBwLimitQoSOvs.test_bw_limit_qos_policy_rule_lifecycle is a positive example of how the fullstack testing infrastructure should be used. It creates a network, subnet, QoS policy & rule and a port utilizing that policy. It then asserts that the expected bandwidth limitation is present on the OVS bridge connected to that port. The test is a true integration test, in the sense that it invokes the API and then asserts that Neutron interacted with the hypervisor appropriately.

How to run fullstack tests locally?

Fullstack tests can be run locally. That makes it much easier to understand exactly how it works, debug issues in the existing tests or write new ones.

Before proceeding, please make sure that the machine runs the latest kernel from your distibution repositories (reboot the machine, if needed). Otherwise, you may experience issues with the *openvswitch* built from source during the environment preparation.

To run fullstack tests locally, you should clone the following repositories under */opt/stack/* directory (you may have to create it first with *mkdir -p /opt/stack*):

- Devstack
- Neutron
- Requirements

When repositories are available locally, the first thing which needs to be done is preparation of the environment. There is a simple script in Neutron to do that:

```
$ export VENV=dsvm-fullstack
$ tools/configure_for_func_testing.sh /opt/stack/devstack -i
```

This will prepare needed files, install required packages, etc. When it is done you should see a message like:

```
Phew, we're done!
```

That means that all went well and you should be ready to run fullstack tests locally.

Fullstack tests execute a custom dhclient-script. From kernel version 4.14 onward, apparmor on certain distros could deny the execution of this script. To be sure, check journalctl

```
sudo journalctl | grep DENIED | grep fullstack-dhclient-script
```

To execute these tests, the easiest workaround is to disable apparmor

```
sudo systemctl stop apparmor
sudo systemctl disable apparmor
```

A more granular solution could be to disable apparmor only for dhclient

```
sudo ln -s /etc/apparmor.d/sbin.dhclient /etc/apparmor.d/disable/
```

Now that your environment is ready for tests, you can try to run just one:

```
$ tox -e dsvm-fullstack neutron.tests.fullstack.test_qos.TestBwLimitQoSOvs.
↪test_bw_limit_qos_policy_rule_lifecycle
dsvm-fullstack create: /opt/stack/neutron/.tox/dsvm-fullstack
dsvm-fullstack installdeps: -chttps://releases.openstack.org/constraints/
↪upper/master, -r/opt/stack/neutron/requirements.txt, -r/opt/stack/neutron/
↪test-requirements.txt, -r/opt/stack/neutron/neutron/tests/functional/
↪requirements.txt
dsvm-fullstack develop-inst: /opt/stack/neutron
{0} neutron.tests.fullstack.test_qos.TestBwLimitQoSOvs.test_bw_limit_qos_
↪policy_rule_lifecycle(ingress) [40.395436s] ... ok
{1} neutron.tests.fullstack.test_qos.TestBwLimitQoSOvs.test_bw_limit_qos_
↪policy_rule_lifecycle(egress) [43.277898s] ... ok
Stopping rootwrap daemon process with pid=12657
Running upgrade for neutron ...
OK
/usr/lib/python3.8/subprocess.py:942: ResourceWarning: subprocess 13475 is␣
↪still running
  _warn("subprocess %s is still running" % self.pid,
ResourceWarning: Enable tracemalloc to get the object allocation traceback
Stopping rootwrap daemon process with pid=12669
Running upgrade for neutron ...
OK
/usr/lib/python3.8/subprocess.py:942: ResourceWarning: subprocess 13477 is␣
↪still running
  _warn("subprocess %s is still running" % self.pid,
ResourceWarning: Enable tracemalloc to get the object allocation traceback

======
Totals
======
Ran: 2 tests in 43.3367 sec.
 - Passed: 2
 - Skipped: 0
 - Expected Fail: 0
 - Unexpected Success: 0
 - Failed: 0
Sum of execute time for each test: 83.6733 sec.

==============
Worker Balance
==============
 - Worker 0 (1 tests) => 0:00:40.395436
 - Worker 1 (1 tests) => 0:00:43.277898
______________________________________________________________________________
↪______________________________________________________________________________
↪_ summary ____________________________________________________________________
↪______________________________________________________________________________
↪_____________
  dsvm-fullstack: commands succeeded
  congratulations :)
```

That means that our test was run successfully. Now you can start hacking, write new fullstack tests or debug failing ones as needed.

Debugging tests locally

If you need to debug a fullstack test locally you can use the `remote_pdb` module for that. First need to install remote_pdb module in the virtual environment created for fullstack testing by tox.

```
$ .tox/dsvm-fullstack/bin/pip install remote_pdb
```

Then you need to install a breakpoint in your code. For example, lets do that in the neutron.tests.fullstack.test_qos.TestBwLimitQoSOvs.test_bw_limit_qos_policy_rule_lifecycle module:

```
def test_bw_limit_qos_policy_rule_lifecycle(self):
    import remote_pdb; remote_pdb.set_trace(port=1234)
    new_limit = BANDWIDTH_LIMIT + 100
```

Now you can run the test again:

```
$ tox -e dsvm-fullstack neutron.tests.fullstack.test_qos.TestBwLimitQoSOvs.
↪test_bw_limit_qos_policy_rule_lifecycle
```

It will pause with message like:

```
RemotePdb session open at 127.0.0.1:1234, waiting for connection ...
```

And now you can start debugging using `telnet` tool:

```
$ telnet 127.0.0.1 1234
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
>
/opt/stack/neutron/neutron/tests/fullstack/test_qos.py(208)test_bw_limit_qos_
↪policy_rule_lifecycle()
-> new_limit = BANDWIDTH_LIMIT + 100
(Pdb)
```

From that point you can start debugging your code in the same way you usually do with `pdb` module.

Checking test logs

Each fullstack test is spawning its own, isolated environment with needed services. So, for example, it can be `neutron-server`, `neutron-ovs-agent` or `neutron-dhcp-agent`. And often there is a need to check logs of some of those processes. That is of course possible when running fullstack tests locally. By default, logs are stored in `/opt/stack/logs/dsvm-fullstack-logs`. The logs directory can be defined by the environment variable `OS_LOG_PATH`. In that directory there are directories with names matching names of the tests, for example:

```
$ ls -l
total 224
drwxr-xr-x 2 vagrant vagrant   4096 Nov 26 16:49 TestBwLimitQoSOvs.test_bw_
```

(continues on next page)

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```
↪limit_qos_policy_rule_lifecycle_egress_
-rw-rw-r-- 1 vagrant vagrant  94928 Nov 26 16:50 TestBwLimitQoSOvs.test_bw_
↪limit_qos_policy_rule_lifecycle_egress_.txt
drwxr-xr-x 2 vagrant vagrant   4096 Nov 26 16:49 TestBwLimitQoSOvs.test_bw_
↪limit_qos_policy_rule_lifecycle_ingress_
-rw-rw-r-- 1 vagrant vagrant 121027 Nov 26 16:54 TestBwLimitQoSOvs.test_bw_
↪limit_qos_policy_rule_lifecycle_ingress_.txt
```

For each test there is a directory and txt file with the same name. This txt file contains the log from the test runner. So you can check exactly what was done by the test when it was run. This file contains logs from all runs of the same test. So if you run the test 10 times, you will have the logs from all 10 runs of the test. In the directory with same name there are logs from the neutron services run during the test, for example:

```
$ ls -l TestBwLimitQoSOvs.test_bw_limit_qos_policy_rule_lifecycle_ingress_/
total 1836
-rw-rw-r-- 1 vagrant vagrant 333371 Nov 26 16:40 neutron-openvswitch-agent--
↪2020-11-26--16-40-38-818499.log
-rw-rw-r-- 1 vagrant vagrant 552097 Nov 26 16:53 neutron-openvswitch-agent--
↪2020-11-26--16-49-29-716615.log
-rw-rw-r-- 1 vagrant vagrant 461483 Nov 26 16:41 neutron-server--2020-11-26--
↪16-40-35-875937.log
-rw-rw-r-- 1 vagrant vagrant 526070 Nov 26 16:54 neutron-server--2020-11-26--
↪16-49-26-758447.log
```

Here each file is only from one run and one service. In the name of the file there is timestamp of when the service was started.

Debugging fullstack failures in the gate

Sometimes there is a need to investigate reason that a test failed in the gate. After every `neutron-fullstack` job run, on the Zuul job page there are logs available. In the directory `controller/logs/dsvm-fullstack-logs` you can find exactly the same files with logs from each test case as mentioned above.

You can also check, for example, the journal log from the node where the tests were run. All those logs are available in the file `controller/logs/devstack.journal.xz` in the jobs logs. In `controller/logs/devstack.journal.README.txt` there are also instructions on how to download and check those journal logs locally.

ML2 OVS with DevStack

This document describes how to test OpenStack Neutron with ML2 OpenvSwitch using DevStack. We will start by describing how to test on a single host.

Single Node Test Environment

1. Create a test system.

Its best to use a throwaway dev system for running DevStack. Your best bet is to use either CentOS 8 or the latest Ubuntu LTS.

2. Create the stack user.

```
$ git clone https://opendev.org/openstack/devstack.git
$ sudo ./devstack/tools/create-stack-user.sh
```

3. Switch to the stack user, copy Devstack to stack folder and clone Neutron.

```
$ sudo cp -r devstack /opt/stack
$ sudo chown -R stack:stack /opt/stack/devstack
$ sudo su - stack
$ cd /opt/stack
$ git clone https://opendev.org/openstack/neutron.git
```

4. Configure DevStack to use the ML2 OVS driver.

Disable the OVN driver since it is the default ML2 driver for devstack to Neutron. You may want to set some values for the various PASSWORD variables in that file so DevStack doesnt have to prompt you for them. Feel free to edit it if youd like, but it should work as-is.

```
$ cd devstack
$ cp ../neutron/devstack/ml2-ovs-local.conf.sample local.conf
```

5. (Optional) Change the host IP to your local one

```
$ cd devstack
$ sed -i 's/#HOST_IP=.*/HOST_IP=172.16.189.6/g' local.conf
```

5. Run DevStack.

This is going to take a while. It installs a bunch of packages, clones a bunch of git repos, and installs everything from these git repos.

```
$ ./stack.sh
```

Once DevStack completes successfully, you should see output that looks something like this:

```
This is your host IP address: 172.16.189.6
This is your host IPv6 address: ::1
Horizon is now available at http://172.16.189.6/dashboard
Keystone is serving at http://172.16.189.6/identity/
The default users are: admin and demo
The password: password
2017-03-09 15:10:54.117 | stack.sh completed in 2110 seconds.
```

Next Steps

- For Environment Variables please read [Environment Variables]
- For Default Network Configuration please read [Default Network Configuration]
- For Booting VMs please read [Booting VMs]
- For VM Connectivity please read [VM Connectivity]

Adding Another Node

After completing the earlier instructions for setting up devstack, you can use a second VM to emulate an additional compute or network node. Create the `stack` user:

```
$ git clone https://opendev.org/openstack/devstack.git
$ sudo ./devstack/tools/create-stack-user.sh
```

Switch to the `stack` user and clone DevStack and neutron:

```
$ sudo su - stack
$ git clone https://opendev.org/openstack/devstack.git
$ git clone https://opendev.org/openstack/neutron.git
```

Use the compute node sample configuration file to add new node, you can enable some features or extensions like DVR, L2pop in this conf:

```
$ cd devstack
$ cp ../neutron/devstack/ml2-ovs-compute-local.conf.sample local.conf
```

Note

The config differences between compute node and network node are whether run the compute services and the L3 agent mode. So this sample local.conf can be used to add new network node.

You must set SERVICE_HOST in local.conf. The value should be the IP address of the main DevStack host. You must also set HOST_IP to the IP address of this new host. See the text in the sample configuration file for more information. Once that is complete, run DevStack:

```
$ ./stack.sh
```

This should complete in less time than before, as its only running a single OpenStack service (nova-compute) along with neutron-openvswitch-agent, neutron-l3-agent, neutron-dhcp-agent and neutron-metadata-agent. The final output will look something like this:

```
This is your host IP address: 172.16.189.30
This is your host IPv6 address: ::1
2017-03-09 18:39:27.058 | stack.sh completed in 1149 seconds.
```

Now go back to your main DevStack host to verify the install:

```
$ . openrc
$ openstack network agent list
$ openstack compute service list
```

Testing

Then we can following the steps at the testing page to do the following works, for reference please read Testing Neutron's related sections

Neutron Jobs Running in Zuul CI

Different kinds of CI jobs are running against patches proposed for Neutron in Gerrit. They have different purposes and complexity. Some jobs are more lightweight (for example, unit tests or linters), while others are more heavyweight (for example, tempest or grenade jobs).

Mainline Tempest and Grenade jobs

Neutron CI runs a number of tempest and grenade (upgrade) jobs in CI. These jobs are required to pass to merge a patch.

Periodic and experimental jobs

Due to a significant number of jobs, not all of them are run on every patch. Some of them are instead executed periodically or on-demand. The periodic jobs are run on a schedule, while the experimental jobs are run on-demand by making a `check experimental` comment in Gerrit under a patch.

Where to find job definitions

You may inspect the list of jobs defined for the project by either looking under `zuul.d/` in the Neutron repository or by visiting the Zuul web interface at https://zuul.opendev.org/

Some jobs are not defined in this repository, but in the neutron-tempest-plugin repository.

Finally, some jobs are defined through templates. Please consult `zuul.d/project.yaml` for the list of templates used in the Neutron project.

Alternatively, you may also inspect the list of jobs in a recent patch in Gerrit comments. (Note that the list executed for a particular patch may be affected by `irrelevant-files` filters. You may consult these in the `zuul.d/` configuration files.)

OVN with DevStack

This document describes how to test OpenStack with OVN using DevStack. We will start by describing how to test on a single host.

Single Node Test Environment

1. Create a test system.

Its best to use a throwaway dev system for running DevStack. Your best bet is to use either CentOS 8 or the latest Ubuntu LTS (18.04, Bionic).

2. Create the `stack` user.

```
$ git clone https://opendev.org/openstack/devstack.git
$ sudo ./devstack/tools/create-stack-user.sh
```

3. Switch to the `stack` user and clone DevStack and Neutron.

```
$ sudo su - stack
$ git clone https://opendev.org/openstack/devstack.git
$ git clone https://opendev.org/openstack/neutron.git
```

4. Configure DevStack to use the OVN driver.

OVN driver comes with a sample DevStack configuration file you can start with. For example, you may want to set some values for the various PASSWORD variables in that file so DevStack doesnt have to prompt you for them. Feel free to edit it if youd like, but it should work as-is.

```
$ cd devstack
$ cp ../neutron/devstack/ovn-local.conf.sample local.conf
```

5. Run DevStack.

This is going to take a while. It installs a bunch of packages, clones a bunch of git repos, and installs everything from these git repos.

```
$ ./stack.sh
```

Once DevStack completes successfully, you should see output that looks something like this:

```
This is your host IP address: 172.16.189.6
This is your host IPv6 address: ::1
Horizon is now available at http://172.16.189.6/dashboard
Keystone is serving at http://172.16.189.6/identity/
The default users are: admin and demo
The password: password
2017-03-09 15:10:54.117 | stack.sh completed in 2110 seconds.
```

Environment Variables

Once DevStack finishes successfully, were ready to start interacting with OpenStack APIs. OpenStack provides a set of command line tools for interacting with these APIs. DevStack provides a file you can source to set up the right environment variables to make the OpenStack command line tools work.

```
$ . openrc
```

If youre curious what environment variables are set, they generally start with an OS prefix:

```
$ env | grep OS
OS_REGION_NAME=RegionOne
OS_IDENTITY_API_VERSION=2.0
OS_PASSWORD=password
OS_AUTH_URL=http://192.168.122.8:5000/v2.0
OS_USERNAME=demo
OS_TENANT_NAME=demo
OS_VOLUME_API_VERSION=2
OS_CACERT=/opt/stack/data/CA/int-ca/ca-chain.pem
OS_NO_CACHE=1
```

Default Network Configuration

By default, DevStack creates networks called `private` and `public`. Run the following command to see the existing networks:

```
$ openstack network list
+--------------------------------------+---------+----------------------------
```

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```
↪-----------------------------------------------+
| ID                                   | Name    | Subnets                    ␣
↪                                                |
+--------------------------------------+---------+----------------------------
↪-----------------------------------------------+
| 40080dad-0064-480a-b1b0-592ae51c1471 | private | 5ff81545-7939-4ae0-8365-
↪1658d45fa85c, da34f952-3bfc-45bb-b062-d2d973c1a751 |
| 7ec986dd-aae4-40b5-86cf-8668feeeab67 | public  | 60d0c146-a29b-4cd3-bd90-
↪3745603b1a4b, f010c309-09be-4af2-80d6-e6af9c78bae7 |
+--------------------------------------+---------+----------------------------
↪-----------------------------------------------+
```

A Neutron network is implemented as an OVN logical switch. OVN driver creates logical switches with a name in the format neutron-<network UUID>. We can use `ovn-nbctl` to list the configured logical switches and see that their names correlate with the output from `openstack network list`:

```
$ ovn-nbctl ls-list
71206f5c-b0e6-49ce-b572-eb2e964b2c4e (neutron-40080dad-0064-480a-b1b0-
↪592ae51c1471)
8d8270e7-fd51-416f-ae85-16565200b8a4 (neutron-7ec986dd-aae4-40b5-86cf-
↪8668feeeab67)

$ ovn-nbctl get Logical_Switch neutron-40080dad-0064-480a-b1b0-592ae51c1471␣
↪external_ids
{"neutron:network_name"=private}
```

Booting VMs

In this section well go through the steps to create two VMs that have a virtual NIC attached to the `private` Neutron network.

DevStack uses libvirt as the Nova backend by default. If KVM is available, it will be used. Otherwise, it will just run qemu emulated guests. This is perfectly fine for our testing, as we only need these VMs to be able to send and receive a small amount of traffic so performance is not very important.

1. Get the Network UUID.

Start by getting the UUID for the `private` network from the output of `openstack network list` from earlier and save it off:

```
$ PRIVATE_NET_ID=$(openstack network show private -c id -f value)
```

2. Create an SSH keypair.

Next create an SSH keypair in Nova. Later, when we boot a VM, well ask that the public key be put in the VM so we can SSH into it.

```
$ openstack keypair create demo > id_rsa_demo
$ chmod 600 id_rsa_demo
```

3. Choose a flavor.

We need minimal resources for these test VMs, so the `m1.nano` flavor is sufficient.

```
$ openstack flavor list
+----+-----------+-------+------+-----------+-------+-----------+
| ID | Name      |   RAM | Disk | Ephemeral | VCPUs | Is Public |
+----+-----------+-------+------+-----------+-------+-----------+
| 1  | m1.tiny   |   512 |    1 |         0 |     1 | True      |
| 2  | m1.small  |  2048 |   20 |         0 |     1 | True      |
| 3  | m1.medium |  4096 |   40 |         0 |     2 | True      |
| 4  | m1.large  |  8192 |   80 |         0 |     4 | True      |
| 42 | m1.nano   |    64 |    0 |         0 |     1 | True      |
| 5  | m1.xlarge | 16384 |  160 |         0 |     8 | True      |
| 84 | m1.micro  |   128 |    0 |         0 |     1 | True      |
| c1 | cirros256 |   256 |    0 |         0 |     1 | True      |
| d1 | ds512M    |   512 |    5 |         0 |     1 | True      |
| d2 | ds1G      |  1024 |   10 |         0 |     1 | True      |
| d3 | ds2G      |  2048 |   10 |         0 |     2 | True      |
| d4 | ds4G      |  4096 |   20 |         0 |     4 | True      |
+----+-----------+-------+------+-----------+-------+-----------+

$ FLAVOR_ID=$(openstack flavor show m1.nano -c id -f value)
```

4. Choose an image.

DevStack imports the CirrOS image by default, which is perfect for our testing. Its a very small test image.

```
$ openstack image list
+--------------------------------------+--------------------------+--------+
| ID                                   | Name                     | Status |
+--------------------------------------+--------------------------+--------+
| 849a8db2-3754-4cf6-9271-491fa4ff7195 | cirros-0.3.5-x86_64-disk | active |
+--------------------------------------+--------------------------+--------+

$ IMAGE_ID=$(openstack image list -c ID -f value)
```

5. Setup a security rule so that we can access the VMs we will boot up next.

By default, DevStack does not allow users to access VMs, to enable that, we will need to add a rule. We will allow both ICMP and SSH.

```
$ openstack security group rule create --ingress --ethertype IPv4 --dst-port␣
↪22 --protocol tcp default
$ openstack security group rule create --ingress --ethertype IPv4 --protocol␣
↪ICMP default
$ openstack security group rule list
+--------------------------------------+-------------+-----------+------------
↪+--------------------------------------+-------------------------------------
↪--+
| ID                                   | IP Protocol | IP Range  | Port Range␣
↪| Remote Security Group                | Security Group                      ␣
↪  |
+--------------------------------------+-------------+-----------+------------
```

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```
↪+--------------------------------------+-------------------------------------
↪--+
...
| ade97198-db44-429e-9b30-24693d86d9b1 | tcp          | 0.0.0.0/0 | 22:22     ␣
↪| None                               | a47b14da-5607-404a-8de4-
↪3a0f1ad3649c |
| d0861a98-f90e-4d1a-abfb-827b416bc2f6 | icmp         | 0.0.0.0/0 |           ␣
↪| None                               | a47b14da-5607-404a-8de4-
↪3a0f1ad3649c |
...
+--------------------------------------+------------+----------+------------
↪+--------------------------------------+-------------------------------------
↪--+
```

6. Boot some VMs.

Now we will boot two VMs. Well name them `test1` and `test2`.

```
$ openstack server create --nic net-id=$PRIVATE_NET_ID --flavor $FLAVOR_ID --
↪image $IMAGE_ID --key-name demo test1
+-----------------------------+-----------------------------------------------
↪------------------+
| Field                       | Value                                         ␣
↪                  |
+-----------------------------+-----------------------------------------------
↪------------------+
| OS-DCF:diskConfig           | MANUAL                                        ␣
↪                  |
| OS-EXT-AZ:availability_zone |                                               ␣
↪                  |
| OS-EXT-STS:power_state      | NOSTATE                                       ␣
↪                  |
| OS-EXT-STS:task_state       | scheduling                                    ␣
↪                  |
| OS-EXT-STS:vm_state         | building                                      ␣
↪                  |
| OS-SRV-USG:launched_at      | None                                          ␣
↪                  |
| OS-SRV-USG:terminated_at    | None                                          ␣
↪                  |
| accessIPv4                  |                                               ␣
↪                  |
| accessIPv6                  |                                               ␣
↪                  |
| addresses                   |                                               ␣
↪                  |
| adminPass                   | BzAWWA6byGP6                                  ␣
↪                  |
| config_drive                |                                               ␣
↪                  |
```

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```
| created                       | 2017-03-09T16:56:08Z                              ␣
↪                  |
| flavor                        | m1.nano (42)                                      ␣
↪                  |
| hostId                        |                                                   ␣
↪                  |
| id                            | d8b8084e-58ff-44f4-b029-a57e7ef6ba61              ␣
↪                  |
| image                         | cirros-0.3.5-x86_64-disk (849a8db2-3754-4cf6-
↪9271-491fa4ff7195) |
| key_name                      | demo                                              ␣
↪                  |
| name                          | test1                                             ␣
↪                  |
| progress                      | 0                                                 ␣
↪                  |
| project_id                    | b6522570f7344c06b1f24303abf3c479                  ␣
↪                  |
| properties                    |                                                   ␣
↪                  |
| security_groups               | name='default'                                    ␣
↪                  |
| status                        | BUILD                                             ␣
↪                  |
| updated                       | 2017-03-09T16:56:08Z                              ␣
↪                  |
| user_id                       | c68f77f1d85e43eb9e5176380a68ac1f                  ␣
↪                  |
| volumes_attached              |                                                   ␣
↪                  |
+-------------------------------+-----------------------------------------------
↪-------------------+

$ openstack server create --nic net-id=$PRIVATE_NET_ID --flavor $FLAVOR_ID --
↪image $IMAGE_ID --key-name demo test2
+-------------------------------+-----------------------------------------------
↪-------------------+
| Field                         | Value                                             ␣
↪                  |
+-------------------------------+-----------------------------------------------
↪-------------------+
| OS-DCF:diskConfig             | MANUAL                                            ␣
↪                  |
| OS-EXT-AZ:availability_zone |                                                   ␣
↪                  |
| OS-EXT-STS:power_state        | NOSTATE                                           ␣
↪                  |
| OS-EXT-STS:task_state         | scheduling                                        ␣
↪                  |
```

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```
| OS-EXT-STS:vm_state          | building                                           ␣
↪                 |
| OS-SRV-USG:launched_at       | None                                               ␣
↪                 |
| OS-SRV-USG:terminated_at     | None                                               ␣
↪                 |
| accessIPv4                   |                                                    ␣
↪                 |
| accessIPv6                   |                                                    ␣
↪                 |
| addresses                    |                                                    ␣
↪                 |
| adminPass                    | YB8dmt5v88JV                                       ␣
↪                 |
| config_drive                 |                                                    ␣
↪                 |
| created                      | 2017-03-09T16:56:50Z                               ␣
↪                 |
| flavor                       | m1.nano (42)                                       ␣
↪                 |
| hostId                       |                                                    ␣
↪                 |
| id                           | 170d4f37-9299-4a08-b48b-2b90fce8e09b               ␣
↪                 |
| image                        | cirros-0.3.5-x86_64-disk (849a8db2-3754-4cf6-
↪9271-491fa4ff7195) |
| key_name                     | demo                                               ␣
↪                 |
| name                         | test2                                              ␣
↪                 |
| progress                     | 0                                                  ␣
↪                 |
| project_id                   | b6522570f7344c06b1f24303abf3c479                   ␣
↪                 |
| properties                   |                                                    ␣
↪                 |
| security_groups              | name='default'                                     ␣
↪                 |
| status                       | BUILD                                              ␣
↪                 |
| updated                      | 2017-03-09T16:56:51Z                               ␣
↪                 |
| user_id                      | c68f77f1d85e43eb9e5176380a68ac1f                   ␣
↪                 |
| volumes_attached             |                                                    ␣
↪                 |
+------------------------------+----------------------------------------------------
↪-----------------+
```

Once both VMs have been started, they will have a status of `ACTIVE`:

```
$ openstack server list
+--------------------------------------+-------+--------+---------------------
↪-------------------------------------+-------------------------+
| ID                                   | Name  | Status | Networks            ␣
↪                                     | Image Name              |
+--------------------------------------+-------+--------+---------------------
↪-------------------------------------+-------------------------+
| 170d4f37-9299-4a08-b48b-2b90fce8e09b | test2 | ACTIVE |␣
↪private=fd5d:9d1b:457c:0:f816:3eff:fe24:49df, 10.0.0.3  | cirros-0.3.5-x86_
↪64-disk |
| d8b8084e-58ff-44f4-b029-a57e7ef6ba61 | test1 | ACTIVE |␣
↪private=fd5d:9d1b:457c:0:f816:3eff:fe3f:953d, 10.0.0.10 | cirros-0.3.5-x86_
↪64-disk |
+--------------------------------------+-------+--------+---------------------
↪-------------------------------------+-------------------------+
```

Our two VMs have addresses of `10.0.0.3` and `10.0.0.10`. If we list Neutron ports, there are two new ports with these addresses associated with them:

```
$ openstack port list
+--------------------------------------+------+-------------------+-----------
↪-----------------------------------------------------------------------------
↪-------------+--------+
| ID                                   | Name | MAC Address       | Fixed IP␣
↪Addresses                                                                   ␣
↪                | Status |
+--------------------------------------+------+-------------------+-----------
↪-----------------------------------------------------------------------------
↪-------------+--------+
...
| 97c970b0-485d-47ec-868d-783c2f7acde3 |      | fa:16:3e:3f:95:3d | ip_
↪address='10.0.0.10', subnet_id='da34f952-3bfc-45bb-b062-d2d973c1a751'      ␣
↪                       | ACTIVE |
|                                      |      |                   | ip_
↪address='fd5d:9d1b:457c:0:f816:3eff:fe3f:953d', subnet_id='5ff81545-7939-
↪4ae0-8365-1658d45fa85c' |        |
| e003044d-334a-4de3-96d9-35b2d2280454 |      | fa:16:3e:24:49:df | ip_
↪address='10.0.0.3', subnet_id='da34f952-3bfc-45bb-b062-d2d973c1a751'       ␣
↪                       | ACTIVE |
|                                      |      |                   | ip_
↪address='fd5d:9d1b:457c:0:f816:3eff:fe24:49df', subnet_id='5ff81545-7939-
↪4ae0-8365-1658d45fa85c' |        |
...
+--------------------------------------+------+-------------------+-----------
↪-----------------------------------------------------------------------------
↪-------------+--------+
```

Now we can look at OVN using `ovn-nbctl` to see the logical switch ports that were created for these two Neutron ports. The first part of the output is the OVN logical switch port UUID. The second part in parentheses is the logical switch port name. Neutron sets the logical switch port name equal to the Neutron port ID.

```
$ ovn-nbctl lsp-list neutron-$PRIVATE_NET_ID
...
fde1744b-e03b-46b7-b181-abddcbe60bf2 (97c970b0-485d-47ec-868d-783c2f7acde3)
7ce284a8-a48a-42f5-bf84-b2bca62cd0fe (e003044d-334a-4de3-96d9-35b2d2280454)
...
```

These two ports correspond to the two VMs we created.

VM Connectivity

We can connect to our VMs by associating a floating IP address from the public network.

```
$ TEST1_PORT_ID=$(openstack port list --server test1 -c id -f value)
$ openstack floating ip create --port $TEST1_PORT_ID public
+---------------------+--------------------------------------+
| Field               | Value                                |
+---------------------+--------------------------------------+
| created_at          | 2017-03-09T18:58:12Z                 |
| description         |                                      |
| fixed_ip_address    | 10.0.0.10                            |
| floating_ip_address | 172.24.4.8                           |
| floating_network_id | 7ec986dd-aae4-40b5-86cf-8668feeeab67 |
| id                  | 24ff0799-5a72-4a5b-abc0-58b301c9aee5 |
| name                | None                                 |
| port_id             | 97c970b0-485d-47ec-868d-783c2f7acde3 |
| project_id          | b6522570f7344c06b1f24303abf3c479     |
| revision_number     | 1                                    |
| router_id           | ee51adeb-0dd8-4da0-ab6f-7ce60e00e7b0 |
| status              | DOWN                                 |
| updated_at          | 2017-03-09T18:58:12Z                 |
+---------------------+--------------------------------------+
```

Devstack does not wire up the public network by default so we must do that before connecting to this floating IP address.

```
$ sudo ip link set br-ex up
$ sudo ip route add 172.24.4.0/24 dev br-ex
$ sudo ip addr add 172.24.4.1/24 dev br-ex
```

Now you should be able to connect to the VM via its floating IP address. First, ping the address.

```
$ ping -c 1 172.24.4.8
PING 172.24.4.8 (172.24.4.8) 56(84) bytes of data.
64 bytes from 172.24.4.8: icmp_seq=1 ttl=63 time=0.823 ms

--- 172.24.4.8 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.823/0.823/0.823/0.000 ms
```

Now SSH to the VM:

```
$ ssh -i id_rsa_demo cirros@172.24.4.8 hostname
test1
```

Adding Another Compute Node

After completing the earlier instructions for setting up devstack, you can use a second VM to emulate an additional compute node. This is important for OVN testing as it exercises the tunnels created by OVN between the hypervisors.

Just as before, create a throwaway VM but make sure that this VM has a different host name. Having same host name for both VMs will confuse Nova and will not produce two hypervisors when you query nova hypervisor list later. Once the VM is setup, create the `stack` user:

```
$ git clone https://opendev.org/openstack/devstack.git
$ sudo ./devstack/tools/create-stack-user.sh
```

Switch to the `stack` user and clone DevStack and neutron:

```
$ sudo su - stack
$ git clone https://opendev.org/openstack/devstack.git
$ git clone https://opendev.org/openstack/neutron.git
```

OVN comes with another sample configuration file that can be used for this:

```
$ cd devstack
$ cp ../neutron/devstack/ovn-compute-local.conf.sample local.conf
```

You must set SERVICE_HOST in local.conf. The value should be the IP address of the main DevStack host. You must also set HOST_IP to the IP address of this new host. See the text in the sample configuration file for more information. Once that is complete, run DevStack:

```
$ cd devstack
$ ./stack.sh
```

This should complete in less time than before, as its only running a single OpenStack service (nova-compute) along with OVN (ovn-controller, ovs-vswitchd, ovsdb-server). The final output will look something like this:

```
This is your host IP address: 172.16.189.30
This is your host IPv6 address: ::1
2017-03-09 18:39:27.058 | stack.sh completed in 1149 seconds.
```

Now go back to your main DevStack host. You can use admin credentials to verify that the additional hypervisor has been added to the deployment:

```
$ cd devstack
$ . openrc admin
$ ./tools/discover_hosts.sh
$ openstack hypervisor list
+----+------------------------+-----------------+---------------+-------+
| ID | Hypervisor Hostname    | Hypervisor Type | Host IP       | State |
```

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```
+----+-----------------------+----------------+--------------+-------+
|  1 | centos7-ovn-devstack   | QEMU           | 172.16.189.6  | up    |
|  2 | centos7-ovn-devstack-2 | QEMU           | 172.16.189.30 | up    |
+----+-----------------------+----------------+--------------+-------+
```

You can also look at OVN and OVS to see that the second host has shown up. For example, there will be a second entry in the Chassis table of the OVN_Southbound database. You can use the `ovn-sbctl` utility to list chassis, their configuration, and the ports bound to each of them:

```
$ ovn-sbctl show

Chassis "ddc8991a-d838-4758-8d15-71032da9d062"
    hostname: "centos7-ovn-devstack"
    Encap vxlan
        ip: "172.16.189.6"
        options: {csum="true"}
    Encap geneve
        ip: "172.16.189.6"
        options: {csum="true"}
    Port_Binding "97c970b0-485d-47ec-868d-783c2f7acde3"
    Port_Binding "e003044d-334a-4de3-96d9-35b2d2280454"
    Port_Binding "cr-lrp-08d1f28d-cc39-4397-b12b-7124080899a1"
Chassis "b194d07e-0733-4405-b795-63b172b722fd"
    hostname: "centos7-ovn-devstack-2.os1.phx2.redhat.com"
    Encap geneve
        ip: "172.16.189.30"
        options: {csum="true"}
    Encap vxlan
        ip: "172.16.189.30"
        options: {csum="true"}
```

You can also see a tunnel created to the other compute node:

```
$ ovs-vsctl show
...
Bridge br-int
    fail_mode: secure
    ...
    Port "ovn-b194d0-0"
        Interface "ovn-b194d0-0"
            type: geneve
            options: {csum="true", key=flow, remote_ip="172.16.189.30"}
    ...
...
```

Provider Networks

Neutron has a provider networks API extension that lets you specify some additional attributes on a network. These attributes let you map a Neutron network to a physical network in your environment. The OVN ML2 driver is adding support for this API extension. It currently supports flat and vlan networks.

Here is how you can test it:

First you must create an OVS bridge that provides connectivity to the provider network on every host running ovn-controller. For trivial testing this could just be a dummy bridge. In a real environment, you would want to add a local network interface to the bridge, as well.

```
$ ovs-vsctl add-br br-provider
```

ovn-controller on each host must be configured with a mapping between a network name and the bridge that provides connectivity to that network. In this case well create a mapping from the network name providernet to the bridge br-provider.

```
$ ovs-vsctl set open . \
external-ids:ovn-bridge-mappings=providernet:br-provider
```

If you want to enable this chassis to host a gateway router for external connectivity, then set ovn-cms-options to enable-chassis-as-gw.

```
$ ovs-vsctl set open . \
external-ids:ovn-cms-options="enable-chassis-as-gw"
```

Now create a Neutron provider network.

```
$ openstack network create provider --share \
--provider-physical-network providernet \
--provider-network-type flat
```

Alternatively, you can define connectivity to a VLAN instead of a flat network:

```
$ openstack network create provider-101 --share \
--provider-physical-network providernet \
--provider-network-type vlan
--provider-segment 101
```

Observe that the OVN ML2 driver created a special logical switch port of type localnet on the logical switch to model the connection to the physical network.

```
$ ovn-nbctl show
...
 switch 5bbccbbd-f5ca-411b-bad9-01095d6f1316 (neutron-729dbbee-db84-4a3d-afc3-
↪82c0b3701074)
     port provnet-729dbbee-db84-4a3d-afc3-82c0b3701074
         addresses: ["unknown"]
...

$ ovn-nbctl lsp-get-type provnet-729dbbee-db84-4a3d-afc3-82c0b3701074
localnet

$ ovn-nbctl lsp-get-options provnet-729dbbee-db84-4a3d-afc3-82c0b3701074
network_name=providernet
```

If VLAN is used, there will be a VLAN tag shown on the localnet port as well.

Finally, create a Neutron port on the provider network.

```
$ openstack port create --network provider myport
```

or if you followed the VLAN example, it would be:

```
$ openstack port create --network provider-101 myport
```

Skydive

Skydive is an open source real-time network topology and protocols analyzer. It aims to provide a comprehensive way of understanding what is happening in the network infrastructure. Skydive works by utilizing agents to collect host-local information, and sending this information to a central agent for further analysis. It utilizes elasticsearch to store the data.

To enable Skydive support with OVN and devstack, enable it on the control and compute nodes.

On the control node, enable it as follows:

```
enable_plugin skydive https://github.com/skydive-project/skydive.git
enable_service skydive-analyzer
```

On the compute nodes, enable it as follows:

```
enable_plugin skydive https://github.com/skydive-project/skydive.git
enable_service skydive-agent
```

Troubleshooting

If you run into any problems, take a look at our *Troubleshooting* page.

Additional Resources

See the documentation and other references linked from the *OVN information* page.

Tempest Testing

Tempest basics in Networking projects

Tempest is the integration test suite of Openstack, for details see Tempest Testing Project.

Tempest makes it possible to add project-specific plugins, and for networking this is neutron-tempest-plugin.

neutron-tempest-plugin covers API and scenario tests not just for core Neutron functionality, but for stadium projects as well. For reference please read Testing Neutron's related sections

API Tests

API tests (neutron-tempest-plugin/neutron_tempest_plugin/api/) are intended to ensure the function and stability of the Neutron API. As much as possible, changes to this path should not be made at the same time as changes to the code to limit the potential for introducing backwards-incompatible changes, although the same patch that introduces a new API should include an API test.

Since API tests target a deployed Neutron daemon that is not test-managed, they should not depend on controlling the runtime configuration of the target daemon. API tests should be black-box - no assumptions should be made about implementation. Only the contract defined by Neutrons REST API should be validated, and all interaction with the daemon should be via a REST client.

The neutron-tempest-plugin/neutron_tempest_plugin directory was copied from the Tempest project around the Kilo timeframe. At the time, there was an overlap of tests between the Tempest and Neutron repositories. This overlap was then eliminated by carving out a subset of resources that belong to Tempest, with the rest in Neutron.

API tests that belong to Tempest deal with a subset of Neutrons resources:

- Port
- Network
- Subnet
- Security Group
- Router
- Floating IP

These resources were chosen for their ubiquity. They are found in most Neutron deployments regardless of plugin, and are directly involved in the networking and security of an instance. Together, they form the bare minimum needed by Neutron.

This is excluding extensions to these resources (For example: Extra DHCP options to subnets, or snat_gateway mode to routers) that are not mandatory in the majority of cases.

Tests for other resources should be contributed to the neutron-tempest-plugin repository. Scenario tests should be similarly split up between Tempest and Neutron according to the API theyre targeting.

To create an API test, the testing class must at least inherit from neutron_tempest_plugin.api.base.BaseNetworkTest base class. As some of tests may require certain extensions to be enabled, the base class provides `required_extensions` class attribute which can be used by subclasses to define a list of required extensions for a particular test class.

Scenario Tests

Scenario tests (neutron-tempest-plugin/neutron_tempest_plugin/scenario), like API tests, use the Tempest test infrastructure and have the same requirements. Guidelines for writing a good scenario test may be found in the Tempest developer guide: https://docs.openstack.org/tempest/latest/field_guide/scenario.html

Scenario tests, like API tests, are split between the Tempest and Neutron repositories according to the Neutron API the test is targeting.

Some scenario tests require advanced `Glance` images (for example, `Ubuntu` or `CentOS`) in order to pass. Those tests are skipped by default. To enable them, include the following in `tempest.conf`:

```
[compute]
image_ref = <uuid of advanced image>
[neutron_plugin_options]
default_image_is_advanced = True
```

To use the `advanced image` only for the tests that really need it and `cirros` for the rest to keep test execution as fast as possible:

```
[compute]
image_ref = <uuid of cirros image>
[neutron_plugin_options]
advanced_image_ref = <uuid of advanced image>
advanced_image_flavor_ref = <suitable flavor for the advance image>
advanced_image_ssh_user = <username for the advanced image>
```

Specific test requirements for advanced images are:

1. `test_trunk` requires `802.11q` kernel module loaded.
2. `test_metadata` requires capabilty to run `curl` for IPv6 addresses.
3. `test_multicast` needs to execute python script on the VM to open multicast sockets.
4. `test_mtu` requires ping to be able to send packets with specific `mtu`.

Zuul basics & job structure

Zuul is the gating system behind Openstack, for details see: Zuul - A Project Gating System.

Zuul job definitions are in yaml, ansible in the depths. The job definitions can be inherited. Networking projects job definitions parents are coming from devstack zuul job config and from tempest and defined in neutron-tempest-plugin zuul.d folder and in neutron zuul.d folder .

Where to look

Debugging zuul results

Tempest executed with different configurations, for details check this page *Tempest jobs running in Neutron CI*

When zuul reports back job results to a review it gives links to the results as well.

The logs can be checked online if you select `Logs` tab on the logs page.

- `job-output.txt` is the full log which contains not just test execution logs, but devstack console output.
- `test_results.html` is the clickable html test report.
- `controller` and `compute` (in case of multinode job) are a dictionary tree containing the relevant files (configuration files, logs etc) created in the job. For example under controller/logs/etc/neutron/ you can check how Neutron services were configured, or in the file controller/logs/tempest_conf.txt you can check tempest configuration file.
- services log files are the in files `controller/logs/screen-`*`.txt`, so for example neutron l2 agent logs are in the file controller/logs/screen-q-agt.txt.

Downloading logs

There is a possibility to download all logs related to a job.

If you choose this on the zuul logs page select `Artifacts` tab on the logs page and click on `Download all logs`. This will download a script `download-logs.sh`, which when executed downloads all the logs for the job under `/tmp/`:

```
$ chmod +x download-logs.sh
$ ./download-logs.sh
2020-12-07T18:12:09+01:00 | Querying https://zuul.opendev.org/api/tenant/
↪openstack/build/8caed05f5ba441b4be2b061d1d421e4e for manifest
2020-12-07T18:12:11+01:00 | Saving logs to /tmp/zuul-logs.c8ZhLM
2020-12-07T18:12:11+01:00 | Getting logs from https://3612101d6c142bf9c77a-
↪c96c299047b55dcdeaefef8e344ceab6.ssl.cf1.rackcdn.com/694539/11/check/
↪tempest-slow-py3/8caed05/
2020-12-07T18:12:11+01:00 |   compute1/logs/apache/access_log.txt                ␣
↪                                [ 0001/0337 ]
...

$ ls /tmp/zuul-logs.c8ZhLM/
compute1
controller
```

Executing tempest locally

For executing tempest locally you need a working devstack, to make it worse if you have to debug a test executed in a multinode job you need a multinode setup as well.

For devstack documentation please refer to this page: DevStack

To have tempest installed and have a proper configuration file for it in your local.conf file enable tempest as service:

```
ENABLED_SERVICES+=tempest
```

or

```
enable_service tempest
```

To use specific config options for tempest you can add those as well to local.conf:

```
[[test-config|/opt/stack/tempest/etc/tempest.conf]]
[network-feature-enabled]
qos_placement_physnet=physnet1
```

To make devstack setup neutron and neutron-tempest-plugin as well enable their devstack plugin:

```
enable_plugin neutron https://opendev.org/openstack/neutron
enable_plugin neutron-tempest-plugin https://opendev.org/openstack/neutron-
↪tempest-plugin
```

If you need a special image for the tests you can set that too in local.conf:

```
IMAGE_URLS="http://download.cirros-cloud.net/0.3.4/cirros-0.3.4-i386-disk.img,
↪https://cloud-images.ubuntu.com/releases/bionic/release/ubuntu-18.04-server-
↪cloudimg-amd64.img"
ADVANCED_IMAGE_NAME=ubuntu-18.04-server-cloudimg-amd64
ADVANCED_INSTANCE_TYPE=ds512M
ADVANCED_INSTANCE_USER=ubuntu
```

If devstack succeeds you can find tempest and neutron-tempest-plugin under `/opt/stack/` directory (with all other project folders which are set to be installed from git).

Tempests configuration file is under `/opt/stack/tempest/etc/` folder, you can check there if everything is as expected.

You can check if neutron-tempest-plugin is known as a tempest plugin by tempest:

```
$ tempest list-plugins
+--------------------------------+-------------------------------------------
↪-----------------+
|             Name               |                    EntryPoint           ␣
↪                 |
+--------------------------------+-------------------------------------------
↪-----------------+
|         neutron_tests          |     neutron_tempest_plugin.
↪plugin:NeutronTempestPlugin     |
+--------------------------------+-------------------------------------------
↪-----------------+
```

To execute a given test or group of tests you can use a regex, or you can use the idempotent id of a test or the tag associated with the test:

```
tempest run --config etc/tempest.conf --regex tempest.scenario
tempest run --config etc/tempest.conf --regex neutron_tempest_plugin.scenario
tempest run --config etc/tempest.conf smoke
tempest run --config etc/tempest.conf ab40fc48-ca8d-41a0-b2a3-f6679c847bfe
```

Template for ModelMigrationSync for external repos

This section contains a template for a test which checks that the Python models for database tables are synchronized with the alembic migrations that create the database schema. This test should be implemented in all driver/plugin repositories that were split out from Neutron.

What does the test do?

This test compares models with the result of existing migrations. It is based on ModelsMigrationsSync which is provided by oslo.db and was adapted for Neutron. It compares core Neutron models and vendor specific models with migrations from Neutron core and migrations from the driver/plugin repo. This test is functional - it runs against the MySQL dialect. The detailed description of this test can be found in Neutron Database Layer section - *Tests to verify that database migrations and models are in sync*.

Steps for implementing the test

1. Import all models in one place

Create a module `networking_foo/db/models/head.py` with the following content:

```
from neutron_lib.db import model_base

from networking_foo import models  # noqa
# Alternatively, import separate modules here if the models are not in one
```

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```
# models.py file


def get_metadata():
    return model_base.BASEV2.metadata
```

2. Implement the test module

The test uses external.py from Neutron. This file contains lists of table names, which were moved out of Neutron:

```
VPNAAS_TABLES = [...]

FWAAS_TABLES = [...]

# Arista ML2 driver Models moved to openstack/networking-arista
REPO_ARISTA_TABLES = [...]

# Models moved to openstack/networking-cisco
REPO_CISCO_TABLES = [...]

...

TABLES = (FWAAS_TABLES + VPNAAS_TABLES + ...
          + REPO_ARISTA_TABLES + REPO_CISCO_TABLES)
```

Also the test uses **VERSION_TABLE**, it is the name of table in database which contains revision id of head migration. It is preferred to keep this variable in `networking_foo/db/migration/alembic_migrations/__init__.py` so it will be easy to use in test.

Create a module `networking_foo/tests/functional/db/test_migrations.py` with the following content:

```
from oslo_config import cfg

from neutron.db.migration.alembic_migrations import external
from neutron.db.migration import cli as migration
from neutron.tests.functional.db import test_migrations
from neutron.tests.unit import testlib_api

from networking_foo.db.migration import alembic_migrations
from networking_foo.db.models import head

# EXTERNAL_TABLES should contain all names of tables that are not related to
# current repo.
EXTERNAL_TABLES = set(external.TABLES) - set(external.REPO_FOO_TABLES)


class TestModelsMigrations(testlib_api.MySQLTestCaseMixin,
                           testlib_api.SqlTestCaseLight,
```

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```
                                test_migrations.TestModelsMigrations):

    def db_sync(self, engine):
        cfg.CONF.set_override('connection', engine.url, group='database')
        for conf in migration.get_alembic_configs():
            self.alembic_config = conf
            self.alembic_config.neutron_config = cfg.CONF
            migration.do_alembic_command(conf, 'upgrade', 'heads')

    def get_metadata(self):
        return head.get_metadata()

    def include_object(self, object_, name, type_, reflected, compare_to):
        if type_ == 'table' and (name == 'alembic' or
                                 name == alembic_migrations.VERSION_TABLE or
                                 name in EXTERNAL_TABLES):
            return False
        else:
            return True
```

3. Add functional requirements

A separate file `networking_foo/tests/functional/requirements.txt` should be created containing the following requirements that are needed for successful test execution.

```
psutil>=3.2.2 # BSD
PyMySQL>=0.6.2  # MIT License
```

Example implementation in VPNaaS

Test Coverage

The intention is to track merged features or areas of code that lack certain types of tests. This document may be used both by developers that want to contribute tests, and operators that are considering adopting a feature.

Coverage

Note that while both API and scenario tests target a deployed OpenStack cloud, API tests are under the Neutron tree and scenario tests are under the Tempest tree.

It is the expectation that API changes involve API tests, agent features or modifications involve functional tests, and Neutron-wide features involve fullstack or scenario tests as appropriate.

The table references tests that explicitly target a feature, and not a job that is configured to run against a specific backend (Thereby testing it implicitly). So, for example, while the Linux bridge agent has a job that runs the API and scenario tests with the Linux bridge agent configured, it does not have functional tests that target the agent explicitly. The gate column is about running API/scenario tests with Neutron configured in a certain way, such as what L2 agent to use or what type of routers to create.

- V - Merged

- Blank - Not applicable
- X - Absent or lacking
- Patch number - Currently in review
- A name - That person has committed to work on an item
- Implicit - The code is executed, yet no assertions are made

Area	Unit	Functional	API	Fullstack	Scenario	Gate
DVR	V	L3-V OVS-X	V	X	X	V
L3 HA	V	V	X	286087*	X	X
L2pop	V	X		Implicit		
DHCP HA	V					
OVS ARP responder	V	X		Implicit		
OVS agent	V	V		V		V
OVN	V	V				V
Linux Bridge agent	V	X		V		V
Metering	V	X	V	X		
DHCP agent	V	V				V
rpc_workers						X
Ref IPAM driver	V					X
MTU advertisement	V			X		
VLAN transparency	V		X	X		
Prefix delegation	V	X*		X		

- Patch https://review.opendev.org/c/openstack/neutron/+/286087 was abandoned.
- Prefix delegation doesnt have a reference implementation in tree and hence is not covered with functional tests of any sort.

Missing Infrastructure

The following section details missing test *types*. If you want to pick up an action item, please contact amuller for more context and guidance.

- The Neutron team would like Rally to persist results over a window of time, graph and visualize this data, so that reviewers could compare average runs against a proposed patch.
- Its possible to test RPC methods via the unit tests infrastructure. This was proposed in patch 162811. The goal is provide developers a light weight way to rapidly run tests that target the RPC layer, so that a patch that modifies an RPC methods signature could be verified quickly and locally.
- Neutron currently runs a partial-grenade job that verifies that an OVS version from the latest stable release works with neutron-server from master. We would like to expand this to DHCP and L3 agents as well.

Transient DB Failure Injection

Neutron has a service plugin to inject random delays and Deadlock exceptions into normal Neutron operations. The service plugin is called Loki and is located under neutron.services.loki.loki_plugin.

To enable the plugin, just add loki to the list of service_plugins in your neutron-server neutron.conf file.

The plugin will inject a Deadlock exception on database flushes with a 1/50 probability and a delay of 1 second with a 1/200 probability when SQLAlchemy objects are loaded into the persistent state from the DB. The goal is to ensure the code is tolerant of these transient delays/failures that will be experienced in busy production (and Galera) systems.

Neutron WSGI API server

Since OpenStack Epoxy (2025.1), the Neutron API server only runs using a uWSGI server that loads the Neutron WSGI application. The configuration and modules needed and how to execute the Neutron API using WSGI is described in *WSGI Usage with the Neutron API*. The `eventlet` API server is no longer supported and the code will be removed.

With the older implementation (`eventlet` API server) it was easy to create an entry script to start the Neutron API, using the same code as in the generated script defined in the `[entry_points]console_script` section. That script started a python executable; it was possible to add break points (using `pdb`) or create a profile for PyCharm, for example.

In the WSGI case this is a bit more complicated. It is not possible to attach a Python debugger to the running process because this is not a Python executable. The uWSGI server, when a request is received, uses the application entry point to route the request, but the root process is not a Python executable.

rpdb

An alternative to `pdb` is the use of rpdb. This library works the same as `pdb` but opening a TCP port that can be accessed using telnet, netcat, etc. For example:

```
import rpdb
debugger = rpdb.Rpdb(addr='0.0.0.0', port=12345)
debugger.set_trace()
```

To access to the remote PDB console, it is needed to execute the following command:

```
$ telnet localhost 12345
```

14.6 Neutron Internals

14.6.1 Neutron Internals

Address Scopes and Subnet Pools

This page discusses subnet pools and address scopes

Subnet Pools

Learn about subnet pools by watching the summit talk given in Vancouver[1].

Subnet pools were added in Kilo. They are relatively simple. A SubnetPool has any number of SubnetPoolPrefix objects associated to it. These prefixes are in CIDR format. Each CIDR is a piece of the address space that is available for allocation.

Subnet Pools support IPv6 just as well as IPv4.

[1] http://www.youtube.com/watch?v=QqP8yBUUXBM&t=6m12s

The Subnet model object now has a subnetpool_id attribute whose default is null for backward compatibility. The subnetpool_id attribute stores the UUID of the subnet pool that acted as the source for the address range of a particular subnet.

When creating a subnet, the subnetpool_id can be optionally specified. If it is, the cidr field is not required. If cidr is specified, it will be allocated from the pool assuming the pool includes it and hasnt already allocated any part of it. If cidr is left out, then the prefixlen attribute can be specified. If it is not, the default prefix length will be taken from the subnet pool. Think of it this way, the allocation logic always needs to know the size of the subnet desired. It can pull it from a specific CIDR, prefixlen, or default. A specific CIDR is optional and the allocation will try to honor it if provided. The request will fail if it cant honor it.

Subnet pools do not allow overlap of subnets.

Subnet Pool Quotas

A quota mechanism was provided for subnet pools. It is different than other quota mechanisms in Neutron because it doesnt count instances of first class objects. Instead it counts how much of the address space is used.

For IPv4, it made reasonable sense to count quota in terms of individual addresses. So, if youre allowed exactly one /24, your quota should be set to 256. Three /26s would be 192. This mechanism encourages more efficient use of the IPv4 space which will be increasingly important when working with globally routable addresses.

For IPv6, the smallest viable subnet in Neutron is a /64. There is no reason to allocate a subnet of any other size for use on a Neutron network. It would look pretty funny to set a quota of 4611686018427387904 to allow one /64 subnet. To avoid this, we count IPv6 quota in terms of /64s. So, a quota of 3 allows three /64 subnets. When we need to allocate something smaller in the future, we will need to ensure that the code can handle non-integer quota consumption.

Allocation

Allocation is done in a way that aims to minimize fragmentation of the pool. The relevant code is here[2]. First, the available prefixes are computed using a set difference: pool - allocations. The result is compacted[3] and then sorted by size. The subnet is then allocated from the smallest available prefix that is large enough to accommodate the request.

Address Scopes

Before subnet pools or address scopes, it was impossible to tell if a network address was routable in a certain context because the address was given explicitly on subnet create and wasnt validated against any other addresses. Address scopes are meant to solve this by putting control over the address space in the hands of an authority: the address scope owner. It makes use of the already existing SubnetPool concept for allocation.

Address scopes are the thing within which address overlap is not allowed and thus provide more flexible control as well as decoupling of address overlap from tenancy.

Prior to the Mitaka release, there was implicitly only a single shared address scope. Arbitrary address overlap was allowed making it pretty much a free for all. To make things seem somewhat sane, normal

[2] neutron/ipam/subnet_alloc.py (_allocate_any_subnet)

[3] http://pythonhosted.org/netaddr/api.html#netaddr.IPSet.compact

users are not able to use routers to cross-plug networks from different projects and NAT was used between internal networks and external networks. It was almost as if each project had a private address scope.

The problem is that this model cannot support use cases where NAT is not desired or supported (e.g. IPv6) or we want to allow different projects to cross-plug their networks.

An AddressScope covers only one address family. But, they work equally well for IPv4 and IPv6.

Routing

The reference implementation honors address scopes. Within an address scope, addresses route freely (barring any FW rules or other external restrictions). Between scopes, routing is prevented unless address translation is used.

For now, floating IPs are the only place where traffic crosses scope boundaries. When a floating IP is associated to a fixed IP, the fixed IP is allowed to access the address scope of the floating IP by way of a 1:1 NAT rule. That means the fixed IP can access not only the external network, but also any internal networks that are in the same address scope as the external network. This is diagrammed as follows:

```
+----------------------+      +---------------------------+
|    address scope 1   |      |      address scope 2      |
|                      |      |                           |
| +------------------+ |      |   +------------------+    |
| | internal network | |      |   | external network |    |
| +-------------+----+ |      |   +--------+---------+    |
|               |      |      |            |              |
|       +-------+--+   |      |     +------+------+       |
|       | fixed ip +----------------+ floating IP |       |
|       +----------+   |      |     +--+--------+-+       |
+----------------------+      |        |        |         |
                              | +------+---+ +--+-------+ |
                              | | internal | | internal | |
                              | +----------+ +----------+ |
                              +---------------------------+
```

Due to the asymmetric route in DVR, and the fact that DVR local routers do not know the information of the floating IPs that reside in other hosts, there is a limitation in the DVR multiple hosts scenario. With DVR in multiple hosts, when the destination of traffic is an internal fixed IP in a different host, the fixed IP with a floating IP associated cant cross the scope boundary to access the internal networks that are in the same address scope of the external network. See https://bugs.launchpad.net/neutron/+bug/1682228

RPC

The L3 agent in the reference implementation needs to know the address scope for each port on each router in order to map ingress traffic correctly.

Each subnet from the same address family on a network is required to be from the same subnet pool. Therefore, the address scope will also be the same. If this were not the case, it would be more difficult to match ingress traffic on a port with the appropriate scope. It may be counter-intuitive but L3 address scopes need to be anchored to some sort of non-L3 thing (e.g. an L2 interface) in the topology in order to determine the scope of ingress traffic. For now, we use ports/networks. In the future, we may be able to distinguish by something else like the remote MAC address or something.

The address scope id is set on each port in a dict under the address_scopes attribute. The scope is distinct per address family. If the attribute does not appear, it is assumed to be null for both families. A value of null means that the addresses are in the implicit address scope which holds all addresses that dont have an explicit one. All subnets that existed in Neutron before address scopes existed fall here.

Here is an example of how the json will look in the context of a router port:

```
"address_scopes": {
    "4": "d010a0ea-660e-4df4-86ca-ae2ed96da5c1",
    "6": null
},
```

To implement floating IPs crossing scope boundaries, the L3 agent needs to know the target scope of the floating ip. The fixed address is not enough to disambiguate because, theoretically, there could be overlapping addresses from different scopes. The scope is computed[4] from the floating ip fixed port and attached to the floating ip dict under the fixed_ip_address_scope attribute. Heres what the json looks like (trimmed):

```
{
     ...
     "floating_ip_address": "172.24.4.4",
     "fixed_ip_address": "172.16.0.3",
     "fixed_ip_address_scope": "d010a0ea-660e-4df4-86ca-ae2ed96da5c1",
     ...
}
```

Model

The model for subnet pools and address scopes can be found in neutron/db/models_v2.py and neutron/db/address_scope_db.py. This document wont go over all of the details. It is worth noting how they relate to existing Neutron objects. The existing Neutron subnet now optionally references a single subnet pool:

```
+----------------+        +------------------+        +--------------+
| Subnet         |        | SubnetPool       |        | AddressScope |
+----------------+        +------------------+        +--------------+
| subnet_pool_id +------> | address_scope_id +------> |              |
|                |        |                  |        |              |
|                |        |                  |        |              |
|                |        |                  |        |              |
+----------------+        +------------------+        +--------------+
```

L3 Agent

The L3 agent is limited in its support for multiple address scopes. Within a router in the reference implementation, traffic is marked on ingress with the address scope corresponding to the network it is coming from. If that traffic would route to an interface in a different address scope, the traffic is blocked unless an exception is made.

One exception is made for floating IP traffic. When traffic is headed to a floating IP, DNAT is applied and the traffic is allowed to route to the private IP address potentially crossing the address scope boundary.

[4] neutron/db/l3_db.py (_get_sync_floating_ips)

When traffic flows from an internal port to the external network and a floating IP is assigned, that traffic is also allowed.

Another exception is made for traffic from an internal network to the external network when SNAT is enabled. In this case, SNAT to the routers fixed IP address is applied to the traffic. However, SNAT is not used if the external network has an explicit address scope assigned and it matches the internal networks. In that case, traffic routes straight through without NAT. The internal networks addresses are viable on the external network in this case.

The reference implementation has limitations. Even with multiple address scopes, a router implementation is unable to connect to two networks with overlapping IP addresses. There are two reasons for this.

First, a single routing table is used inside the namespace. An implementation using multiple routing tables has been in the works but there are some unresolved issues with it.

Second, the default SNAT feature cannot be supported with the current Linux conntrack implementation unless a double NAT is used (one NAT to get from the address scope to an intermediate address specific to the scope and a second NAT to get from that intermediate address to an external address). Single NAT wont work if there are duplicate addresses across the scopes.

Due to these complications the router will still refuse to connect to overlapping subnets. We can look in to an implementation that overcomes these limitations in the future.

Agent Extensions

All reference agents utilize a common extension mechanism that allows for the introduction and enabling of a core resource extension without needing to change agent code. This mechanism allows multiple agent extensions to be run by a single agent simultaneously. The mechanism may be especially interesting to third parties whose extensions lie outside the neutron tree.

Under this framework, an agent may expose its API to each of its extensions thereby allowing an extension to access resources internal to the agent. At layer 2, for instance, upon each port event the agent is then able to trigger a handle_port method in its extensions.

Interactions with the agent API object are in the following order:

1. The agent initializes the agent API object.
2. The agent passes the agent API object into the extension manager.
3. The manager passes the agent API object into each extension.
4. An extension calls the new agent API object method to receive, for instance, bridge wrappers with cookies allocated.

```
+------------+
| Agent API  +--------------------------------------------------+
+-----+------+                                                  |
      |                                   +------------+        |
      |1                               +--+ Extension  +--+     |
      |                                |  +------------+  |     |
+----+-+-+----+  2  +--------------+  3  |                  |  4  |
|   Agent     +-----+ Ext. manager +-----+--+   ....    +--+-----+
+------------+     +--------------+     |                  |
                                        |  +------------+  |
```
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```
+---+ Extension +---+
+-----------+
```

Each extension is referenced through a stevedore entry point defined within a specific namespace. For example, L2 extensions are referenced through the neutron.agent.l2.extensions namespace.

The relevant modules are:

- neutron_lib.agent.extension: This module defines an abstract extension interface for all agent extensions across L2 and L3.
- neutron_lib.agent.l2_extension:
- neutron_lib.agent.l3_extension: These modules subclass neutron_lib.agent.extension.AgentExtension and define a layer-specific abstract extension interface.
- neutron.agent.agent_extensions_manager: This module contains a manager that allows extensions to load themselves at runtime.
- neutron.agent.l2.l2_agent_extensions_manager:
- neutron.agent.l3.l3_agent_extensions_manager: Each of these modules passes core resource events to loaded extensions.

Agent API object

Every agent can pass an agent API object into its extensions in order to expose its internals to them in a controlled way. To accommodate different agents, each extension may define a consume_api() method that will receive this object.

This agent API object is part of neutrons public interface for third parties. All changes to the interface will be managed in a backwards-compatible way.

At this time, on the L2 side, only the L2 Open vSwitch agent provides an agent API object to extensions. See *L2 agent extensions*. For L3, see *L3 agent extensions*.

The relevant modules are:

- neutron_lib.agent.extension
- neutron_lib.agent.l2_extension
- neutron_lib.agent.l3_extension
- neutron.agent.agent_extensions_manager
- neutron.agent.l2.l2_agent_extensions_manager
- neutron.agent.l3.l3_agent_extensions_manager

API Extensions

API extensions is the standard way of introducing new functionality to the Neutron project, it allows plugins to determine if they wish to support the functionality or not.

Examples

The easiest way to demonstrate how an API extension is written, is by studying an existing API extension and explaining the different layers.

Security Group API

https://wiki.openstack.org/wiki/Neutron/SecurityGroups

API Extension

The API extension is the front end portion of the code, which handles defining a REST-ful API, which is used by projects.

Database API

The Security Group API extension adds a number of methods to the database layer of Neutron

Agent RPC

This portion of the code handles processing requests from projects, after they have been stored in the database. It involves messaging all the L2 agents running on the compute nodes, and modifying the IPTables rules on each hypervisor.

- Plugin RPC classes
  - SecurityGroupServerRpcMixin - defines the RPC API that the plugin uses to communicate with the agents running on the compute nodes
  - SecurityGroupServerRpcMixin - Defines the API methods used to fetch data from the database, in order to return responses to agents via the RPC API
- Agent RPC classes
  - The SecurityGroupServerRpcApi defines the API methods that can be called by agents, back to the plugin that runs on the Neutron controller
  - The SecurityGroupAgentRpcCallbackMixin defines methods that a plugin uses to call back to an agent after performing an action called by an agent.

IPTables Driver

- `prepare_port_filter` takes a `port` argument, which is a `dictionary` object that contains information about the port - including the `security_group_rules`
- `prepare_port_filter` appends the port to an internal dictionary, `filtered_ports` which is used to track the internal state.
- Each security group has a chain in Iptables.
- The `IptablesFirewallDriver` has a method to convert security group rules into iptables statements.

Extensions for Resources with standard attributes

Resources that inherit from the HasStandardAttributes DB class can automatically have the extensions written for standard attributes (e.g. timestamps, revision number, etc) extend their resources by defining the api_collections on their model. These are used by extensions for standard attr resources to generate the extended resources map.

Any new addition of a resource to the standard attributes collection must be accompanied with a new extension to ensure that it is discoverable via the API. If its a completely new resource, the extension describing that resource will suffice. If its an existing resource that was released in a previous cycle having the standard attributes added for the first time, then a dummy extension needs to be added indicating that the resource now has standard attributes. This ensures that an API caller can always discover if an attribute will be available.

For example, if Flavors were migrated to include standard attributes, we need a new flavor-standardattr extension. Then as an API caller, I will know that flavors will have timestamps by checking for flavor-standardattr and timestamps.

Current API resources extended by standard attr extensions:

- subnets: neutron.db.models_v2.Subnet
- trunks: neutron.services.trunk.models.Trunk
- routers: neutron.db.l3_db.Router
- segments: neutron.db.segments_db.NetworkSegment
- security_group_rules: neutron.db.models.securitygroup.SecurityGroupRule
- networks: neutron.db.models_v2.Network
- policies: neutron.db.qos.models.QosPolicy
- subnetpools: neutron.db.models_v2.SubnetPool
- ports: neutron.db.models_v2.Port
- security_groups: neutron.db.models.securitygroup.SecurityGroup
- floatingips: neutron.db.l3_db.FloatingIP
- network_segment_ranges: neutron.db.models.network_segment_range.NetworkSegmentRange

API Layer for Neutron WSGI/HTTP

This section will cover the internals of Neutrons HTTP API, and the classes in Neutron that can be used to create Extensions to the Neutron API.

Python web applications interface with webservers through the Python Web Server Gateway Interface (WSGI) - defined in PEP 333

Startup

Neutrons WSGI server is started from then module pointed in the `neutron-api-uwsgi.ini` file. By default is `neutron.wsgi.api:application` and starts in the API server.

WSGI Application

During the building of the NeutronApiService, the *api_server* function creates a WSGI application using the *load_paste_app* function inside config.py - which parses api-paste.ini - in order to create a WSGI app using Pastes deploy.

The api-paste.ini file defines the WSGI applications and routes - using the Paste INI file format.

The INI file directs paste to instantiate the APIRouter class of Neutron, which contains several methods that map Neutron resources (such as Ports, Networks, Subnets) to URLs, and the controller for each resource.

Further reading

Yong Sheng Gong: Deep Dive into Neutron

Calling the ML2 Plugin

When writing code for an extension, service plugin, or any other part of Neutron you must not call core plugin methods that mutate state while you have a transaction open on the session that you pass into the core plugin method.

The create and update methods for ports, networks, and subnets in ML2 all have a precommit phase and postcommit phase. During the postcommit phase, the data is expected to be fully persisted to the database and ML2 drivers will use this time to relay information to a backend outside of Neutron. Calling the ML2 plugin within a transaction would violate this semantic because the data would not be persisted to the DB; and, were a failure to occur that caused the whole transaction to be rolled back, the backend would become inconsistent with the state in Neutrons DB.

To prevent this, these methods are protected with a decorator that will raise a RuntimeError if they are called with context that has a session in an active transaction. The decorator can be found at neutron.common.utils.transaction_guard and may be used in other places in Neutron to protect functions that are expected to be called outside of a transaction.

Code Profiling

As more functionality is added to Neutron over time, efforts to improve performance become more difficult, given the rising complexity of the code. Identifying performance bottlenecks is frequently not straightforward, because they arise as a result of complex interactions of different code components.

To help community developers to improve Neutron performance, a Python decorator has been implemented. Decorating a method or a function with it will result in profiling data being added to the corresponding Neutron component log file. These data are generated using cProfile which is part of the Python standard library.

Once a method or function has been decorated, every one of its executions will add to the corresponding log file data grouped in 3 sections:

1. The top calls (sorted by CPU cumulative time) made by the decorated method or function. The number of calls included in this section can be controlled by a configuration option, as explained in *Setting up Neutron for code profiling*. Following is a summary example of this section:

```
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: DEBUG neutron.profiling.profiled_decorator [None req-
↪dc2d428f-4531-4f07-a12d-56843b5f9374 c_rally_8af8f2b4_YbhFJ6Ge c_rally_
```

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```
↪8af8f2b4_fqvy1XJp] os-profiler parent trace-id c5b30c7f-100b-4e1c-8f07-
↪b2c38f41ad65 trace-id 6324fa85-ea5f-4ae2-9d89-2aabff0dddfc   16928␣
↪millisecs elapsed for neutron.plugins.ml2.plugin.create_port((<neutron.
↪plugins.ml2.plugin.Ml2Plugin object at 0x7f0b4e6ca978>, <neutron_lib.
↪context.Context object at 0x7f0b4bcee240>, {'port': {'tenant_id':
↪'421ab52e126e45af81a3eb1962613e18', 'network_id': 'dc59577a-9589-4617-
↪82b5-6ee31dbdb15d', 'fixed_ips': [{'ip_address': '1.1.5.177', 'subnet_id
↪': 'e15ec947-9edd-4793-bf0f-c463c7ff2f62'}], 'admin_state_up': True,
↪'device_id': 'f33db890-7958-440e-b07b-432e40bb4049', 'device_owner':
↪'network:router_interface', 'name': '', 'project_id':
↪'421ab52e126e45af81a3eb1962613e18', 'mac_address': <neutron_lib.
↪constants.Sentinel object at 0x7f0b4fc69860>, 'allowed_address_pairs':
↪<neutron_lib.constants.Sentinel object at 0x7f0b4fc69860>, 'extra_dhcp_
↪opts': None, 'binding:vnic_type': 'normal', 'binding:host_id': <neutron_
↪lib.constants.Sentinel object at 0x7f0b4fc69860>, 'binding:profile':
↪<neutron_lib.constants.Sentinel object at 0x7f0b4fc69860>, 'port_
↪security_enabled': <neutron_lib.constants.Sentinel object at␣
↪0x7f0b4fc69860>, 'description': '', 'security_groups': <neutron_lib.
↪constants.Sentinel object at 0x7f0b4fc69860>}}), {}):
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:          247612 function calls (238220 primitive calls)␣
↪in 16.943 seconds
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:    Ordered by: cumulative time
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:    List reduced from 1861 to 100 due to restriction <100>
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:    ncalls  tottime  percall  cumtime  percall␣
↪filename:lineno(function)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:       4/2    0.000    0.000   16.932    8.466 /usr/local/
↪lib/python3.6/dist-packages/neutron_lib/db/api.py:132(wrapped)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:         1    0.000    0.000   16.928   16.928 /opt/stack/
↪neutron/neutron/common/utils.py:678(inner)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:      20/9    0.000    0.000   16.884    1.876 /usr/local/
↪lib/python3.6/dist-packages/sqlalchemy/orm/strategies.py:1317(<genexpr>)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:     37/17    0.000    0.000   16.867    0.992 /opt/stack/
↪osprofiler/osprofiler/sqlalchemy.py:84(handler)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:     37/17    0.000    0.000   16.860    0.992 /opt/stack/
↪osprofiler/osprofiler/profiler.py:86(stop)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:       8/3    0.005    0.001   16.844    5.615 /usr/local/
↪lib/python3.6/dist-packages/neutron_lib/db/api.py:224(wrapped)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:         1    0.000    0.000   16.836   16.836 /opt/stack/
```

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```
↪neutron/neutron/plugins/ml2/plugin.py:1395(_create_port_db)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:          1    0.000    0.000   16.836   16.836 /opt/stack/
↪neutron/neutron/db/db_base_plugin_v2.py:1413(create_port_db)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:          1    0.000    0.000   16.836   16.836 /opt/stack/
↪neutron/neutron/db/db_base_plugin_v2.py:1586(_enforce_device_owner_not_
↪router_intf_or_device_id)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:          1    0.000    0.000   16.836   16.836 /opt/stack/
↪neutron/neutron/db/l3_db.py:522(get_router)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:          1    0.000    0.000   16.836   16.836 /opt/stack/
↪neutron/neutron/db/l3_db.py:186(_get_router)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:      34/22    0.000    0.000   16.745    0.761 /usr/local/
↪lib/python3.6/dist-packages/sqlalchemy/orm/loading.py:35(instances)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:       39/8    0.000    0.000   16.727    2.091 /usr/local/
↪lib/python3.6/dist-packages/sqlalchemy/sql/elements.py:285(_execute_on_
↪connection)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:       39/8    0.001    0.000   16.727    2.091 /usr/local/
↪lib/python3.6/dist-packages/sqlalchemy/engine/base.py:1056(_execute_
↪clauseelement)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:      17/13    0.000    0.000   16.704    1.285 /usr/local/
↪lib/python3.6/dist-packages/sqlalchemy/orm/strategies.py:1310(get)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:      20/14    0.001    0.000   16.704    1.193 /usr/local/
↪lib/python3.6/dist-packages/sqlalchemy/orm/strategies.py:1315(_load)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:      19/14    0.000    0.000   16.703    1.193 /usr/local/
↪lib/python3.6/dist-packages/sqlalchemy/orm/loading.py:88(<listcomp>)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:      76/23    0.001    0.000   16.699    0.726 /opt/stack/
↪osprofiler/osprofiler/profiler.py:426(_notify)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:       39/8    0.001    0.000   16.696    2.087 /usr/local/
↪lib/python3.6/dist-packages/sqlalchemy/engine/base.py:1163(_execute_
↪context)
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:      75/23    0.000    0.000   16.686    0.725 /opt/stack/
↪osprofiler/osprofiler/notifier.py:28(notify)
```

2. Callers section: all functions or methods that called each function or method in the resulting profiling data. This is restricted by the configured number of top calls to log, as explained in *Setting up Neutron for code profiling*. Following is a summary example of this section:

```
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:    Ordered by: cumulative time
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:    List reduced from 1861 to 100 due to restriction
↪<100>
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: Function                                                ␣
↪                                                   was called by...
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                       ncalls ␣
↪tottime  cumtime
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: /usr/local/lib/python3.6/dist-packages/neutron_lib/db/
↪api.py:132(wrapped)                                     <-      2/0    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/neutron_lib/db/api.
↪py:224(wrapped)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: /opt/stack/neutron/neutron/common/utils.py:678(inner)   ␣
↪                                                   <-
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪strategies.py:1317(<genexpr>)                           <-        3    0.
↪000    0.000  /opt/stack/osprofiler/osprofiler/profiler.py:426(_notify)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                            1    0.
↪000   16.883  /usr/local/lib/python3.6/dist-packages/neutron_lib/db/api.
↪py:132(wrapped)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                            2    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/engine/
↪base.py:69(__init__)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                            1    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/engine/
↪base.py:1056(_execute_clauseelement)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                            1    0.
↪000   16.704  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪query.py:3281(one)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                            0    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪query.py:3337(__iter__)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
```

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```
↪server[19578]:                                                               ␣
↪                                                                   1     0.
↪000     0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪query.py:3362(_execute_and_instances)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                               ␣
↪                                                                   1     0.
↪000     0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪session.py:1127(_connection_for_bind)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                               ␣
↪                                                                   1     0.
↪000     0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪strategies.py:1310(get)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                               ␣
↪                                                                   1     0.
↪000     0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪strategies.py:1315(_load)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                               ␣
↪                                                                   1     0.
↪000     0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪strategies.py:2033(load_scalar_from_joined_new_row)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                               ␣
↪                                                                 1/0     0.
↪000     0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/pool/
↪base.py:840(_checkin)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                               ␣
↪                                                                 1/0     0.
↪000     0.000  /usr/local/lib/python3.6/dist-packages/webob/request.
↪py:1294(send)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: /opt/stack/osprofiler/osprofiler/sqlalchemy.
↪py:84(handler)                                                       <-     ␣
↪16/0     0.000     0.000  /usr/local/lib/python3.6/dist-packages/
↪sqlalchemy/event/attr.py:316(__call__)
Oct 20 01:52:40.767003 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: /opt/stack/osprofiler/osprofiler/profiler.py:86(stop)    ␣
↪                                                     <-     16/0     0.
↪000     0.000  /opt/stack/osprofiler/osprofiler/sqlalchemy.py:84(handler)
```

3. Callees section: a list of all functions or methods that were called by the indicated function or method. Again, this is restricted by the configured number of top calls to log. Following is a summary example of this section:

```
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:    Ordered by: cumulative time
```

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```
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:    List reduced from 1861 to 100 due to restriction
↪<100>
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: Function                                                ␣
↪                                                    called...
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                        ncalls ␣
↪tottime  cumtime
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: /usr/local/lib/python3.6/dist-packages/neutron_lib/db/
↪api.py:132(wrapped)                                 ->      1/0    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/oslo_db/api.
↪py:135(wrapper)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                           1    0.
↪000   16.883  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪strategies.py:1317(<genexpr>)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: /opt/stack/neutron/neutron/common/utils.py:678(inner)   ␣
↪                                                   ->      1    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/neutron_lib/
↪context.py:145(session)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                           1    0.
↪000   16.928  /usr/local/lib/python3.6/dist-packages/neutron_lib/db/api.
↪py:224(wrapped)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                           1    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪session.py:2986(is_active)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪strategies.py:1317(<genexpr>)                       ->      1    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/engine/
↪default.py:579(do_execute)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                           2    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/engine/
↪default.py:1078(post_exec)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                         ␣
↪                                                           2    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/engine/
```

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```
↪default.py:1122(get_result_proxy)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                          ␣
↪                                                              0    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/event/
↪attr.py:316(__call__)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                          ␣
↪                                                              1    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/event/
↪base.py:266(__getattr__)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                          ␣
↪                                                           15/3    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪loading.py:35(instances)
Oct 20 01:52:40.788842 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                          ␣
↪                                                              1    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪strategies.py:1317(<listcomp>)
Oct 20 01:52:40.791161 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                          ␣
↪                                                              1    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/orm/
↪strategies.py:1318(<lambda>)
Oct 20 01:52:40.791161 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]:                                                          ␣
↪                                                              3    0.
↪000    0.000  /usr/local/lib/python3.6/dist-packages/sqlalchemy/util/
↪langhelpers.py:852(__get__)
```

Setting up Neutron for code profiling

To start profiling Neutron code, the following steps have to be taken:

1. Add he following line to the `[default]` section of `/etc/neutron/neutron.conf` (code profiling is disabled by default):

   ```
   enable_code_profiling = True
   ```

2. Add the following import line to each module to be profiled:

   ```
   from neutron.profiling import profiled_decorator
   ```

3. Decorate each mehtod or function to be profiled as follows:

   ```
   @profiled_decorator.profile
   def create_subnet(self, context, subnet):
   ```

4. For each decorated method or function execution, only the top 50 calls by cumulative CPU time

are logged. This can be changed adding the following line to the [default] section of /etc/neutron/neutron.conf:

```
code_profiling_calls_to_log = 100
```

Profiling code with the Neutron Rally job

Code profiling is enabled for the neutron-rally-task job in Neutrons check queue in Zuul. Taking advantage of the fact that os-profiler is enabled for this job, the data logged by the profiled_decorator.profile decorator includes the os-profiler parent trace-id and trace-id as can be seen here:

```
Oct 20 01:52:40.759379 ubuntu-bionic-vexxhost-sjc1-0012393267 neutron-
↪server[19578]: DEBUG neutron.profiling.profiled_decorator [None req-
↪dc2d428f-4531-4f07-a12d-56843b5f9374 c_rally_8af8f2b4_YbhFJ6Ge c_rally_
↪8af8f2b4_fqvy1XJp] os-profiler parent trace-id c5b30c7f-100b-4e1c-8f07-
↪b2c38f41ad65 trace-id 6324fa85-ea5f-4ae2-9d89-2aabff0dddfc   16928␣
↪millisecs elapsed for neutron.plugins.ml2.plugin.create_port((<neutron.
↪plugins.ml2.plugin.Ml2Plugin object at 0x7f0b4e6ca978>, <neutron_lib.
↪context.Context object at 0x7f0b4bcee240>, {'port': {'tenant_id':
↪'421ab52e126e45af81a3eb1962613e18', 'network_id': 'dc59577a-9589-4617-82b5-
↪6ee31dbdb15d', 'fixed_ips': [{'ip_address': '1.1.5.177', 'subnet_id':
↪'e15ec947-9edd-4793-bf0f-c463c7ff2f62'}], 'admin_state_up': True, 'device_id
↪': 'f33db890-7958-440e-b07b-432e40bb4049', 'device_owner': 'network:router_
↪interface', 'name': '', 'project_id': '421ab52e126e45af81a3eb1962613e18',
↪'mac_address': <neutron_lib.constants.Sentinel object at 0x7f0b4fc69860>,
↪'allowed_address_pairs': <neutron_lib.constants.Sentinel object at␣
↪0x7f0b4fc69860>, 'extra_dhcp_opts': None, 'binding:vnic_type': 'normal',
↪'binding:host_id': <neutron_lib.constants.Sentinel object at 0x7f0b4fc69860>
↪, 'binding:profile': <neutron_lib.constants.Sentinel object at␣
↪0x7f0b4fc69860>, 'port_security_enabled': <neutron_lib.constants.Sentinel␣
↪object at 0x7f0b4fc69860>, 'description': '', 'security_groups': <neutron_
↪lib.constants.Sentinel object at 0x7f0b4fc69860>}}), {}):
```

Community developers wanting to use this to correlate data from os-profiler and the profiled_decorator.profile decorator can submit a DNM (Do Not Merge) patch, decorating the functions and methods they want to profile and optionally:

1. Configure the number of calls to be logged in the neutron-rally-task job definition, as described in *Setting up Neutron for code profiling*.
2. Increase the timeout parameter value of the neutron-rally-task job in the .zuul yaml file. The value used for the Neutron gate might be too short when logging large quantities of profiling data.

The profiled_decorator.profile and os-profiler data will be found in the neutron-rally-task log files and HTML report respectively.

Database Layer

This section contains some common information that will be useful for developers that need to do some database changes as well as to execute queries using the oslo.db API.

Difference between default and server_default parameters for columns

For columns it is possible to set default or server_default. What is the difference between them and why should they be used?

The explanation is quite simple:

- default - the default value that SQLAlchemy will specify in queries for creating instances of a given model;
- server_default - the default value for a column that SQLAlchemy will specify in DDL.

Summarizing, default is useless in migrations and only server_default should be used. For synchronizing migrations with models server_default parameter should also be added in model. If default value in database is not needed, server_default should not be used. The declarative approach can be bypassed (i.e. default may be omitted in the model) if default is enforced through business logic.

Database migrations

For details on the neutron-db-manage wrapper and alembic migrations, see Alembic Migrations.

Tests to verify that database migrations and models are in sync

class neutron.tests.functional.db.test_migrations.**TestModelsMigrations**(**args*, ***kwds*)

Test for checking of equality models state and migrations.

For the opportunistic testing you need to set up a db named openstack_citest with user openstack_citest and password openstack_citest on localhost. The test will then use that db and user/password combo to run the tests.

For MySQL on Ubuntu this can be done with the following commands:

```
mysql -u root
>create database openstack_citest;
>grant all privileges on openstack_citest.* to
 openstack_citest@localhost identified by 'openstack_citest';
```

Output is a list that contains information about differences between db and models. Output example:

```
[('add_table',
  Table('bat', MetaData(bind=None),
        Column('info', String(), table=<bat>), schema=None)),
 ('remove_table',
  Table(u'bar', MetaData(bind=None),
        Column(u'data', VARCHAR(), table=<bar>), schema=None)),
 ('add_column',
  None,
```

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```
 'foo',
 Column('data', Integer(), table=<foo>)),
('remove_column',
 None,
 'foo',
 Column(u'old_data', VARCHAR(), table=None)),
[('modify_nullable',
  None,
  'foo',
  u'x',
  {'existing_server_default': None,
  'existing_type': INTEGER()},
  True,
  False)]]
```

- `remove_*` means that there is extra table/column/constraint in db;
- `add_*` means that it is missing in db;
- `modify_*` means that on column in db is set wrong type/nullable/server_default. Element contains information:
  - what should be modified,
  - schema,
  - table,
  - column,
  - existing correct column parameters,
  - right value,
  - wrong value.

This class also contains tests for branches, like that correct operations are used in contract and expand branches.

db_sync(*engine*)

Run migration scripts with the given engine instance.

This method must be implemented in subclasses and run migration scripts for a DB the given engine is connected to.

filter_metadata_diff(*diff*)

Filter changes before assert in test_models_sync().

Allow subclasses to whitelist/blacklist changes. By default, no filtering is performed, changes are returned as is.

Parameters
`diff` a list of differences (see *compare_metadata()* docs for details on format)

Returns
a list of differences

get_engine()

Return the engine instance to be used when running tests.

This method must be implemented in subclasses and return an engine instance to be used when running tests.

get_metadata()

Return the metadata instance to be used for schema comparison.

This method must be implemented in subclasses and return the metadata instance attached to the BASE model.

include_object(*object_*, *name*, *type_*, *reflected*, *compare_to*)

Return True for objects that should be compared.

Parameters

- **object** a SchemaItem object such as a Table or Column object
- **name** the name of the object
- **type** a string describing the type of object (e.g. table)
- **reflected** True if the given object was produced based on table reflection, False if its from a local MetaData object
- **compare_to** the object being compared against, if available, else None

setUp()

Hook method for setting up the test fixture before exercising it.

The Standard Attribute Table

There are many attributes that we would like to store in the database which are common across many Neutron objects (e.g. tags, timestamps, rbac entries). We have previously been handling this by duplicating the schema to every table via model mixins. This means that a DB migration is required for each object that wants to adopt one of these common attributes. This becomes even more cumbersome when the relationship between the attribute and the object is many-to-one because each object then needs its own table for the attributes (assuming referential integrity is a concern).

To address this issue, the standardattribute table is available. Any model can add support for this table by inheriting the HasStandardAttributes mixin in neutron.db.standard_attr. This mixin will add a standard_attr_id BigInteger column to the model with a foreign key relationship to the standardattribute table. The model will then be able to access any columns of the standardattribute table and any tables related to it.

A model that inherits HasStandardAttributes must implement the property api_collections, which is a list of API resources that the new object may appear under. In most cases, this will only be one (e.g. ports for the Port model). This is used by all of the service plugins that add standard attribute fields to determine which API responses need to be populated.

A model that supports tag mechanism must implement the property collection_resource_map which is a dict of collection_name and resource_name for API resources. And also the model must implement tag_support with a value True.

The introduction of a new standard attribute only requires one column addition to the standardattribute table for one-to-one relationships or a new table for one-to-many or one-to-zero relationships. Then all of the models using the HasStandardAttribute mixin will automatically gain access to the new attribute.

Any attributes that will apply to every neutron resource (e.g. timestamps) can be added directly to the standardattribute table. For things that will frequently be NULL for most entries (e.g. a column to store an error reason), a new table should be added and joined to in a query to prevent a bunch of NULL entries in the database.

Session handling

The main information reference is in Usage, that provides an initial picture of how to use oslo.db in Neutron. Any request call to the Neutron server API must have a neutron_context parameter, that is an instance of Context. This context holds a *sqlalchemy.orm.session.Session* instance that manages persistence operations for ORM-mapped objects (from SQLAlchemy documentation). A *Session* establishes all conversations with the database and represents a holding zone for all loaded or associated objects during its lifespan.

A *Session* instance establishes a transaction to the database using the defined *Engine*. This transaction represents an SQL transaction that is a logical unit of work that contains one or more SQL statements. Regardless of the number of statements this transaction may have, the execution is atomic; if the transaction fails, any previous SQL statement already executed that implies a change in the database is undone (rollback).

Database transactions

Any Neutron database operation, regardless of the type and the amount, should be executed inside a transaction. There are two type of transactions:

- Reader: for reading operations.
- Writer: for any operation that implies a change in the database, like a register creation, modification or deletion.

The neutron-lib library provides an API wrapper for the oslo.db operations. The *CONTEXT_READER* and *CONTEXT_WRITER* context managers can be used both as decorators or context managers. For example:

```
from neutron_lib.db import api as db_api
from neutron.db import models_v2

def get_ports(context, network_id):
    with db_api.CONTEXT_READER.using(context):
        query = context.session.query(models_v2.Port)
        query.filter(models_v2.Port.network_id == network_id)
        return query.all()

@db_api.CONTEXT_WRITER
def delete_port(context, port_id):
    query = context.session.query(models_v2.Port)
    query.filter(models_v2.Port.id == port_id)
    query.delete()
```

The transaction contexts can be nested. For example, if inside a context a decorated method is called, the current transaction is preserved. There is only one exception on this rule: a reader context cannot be upgraded to writer. That means inside a reader context it is not possible to start a writer context. The following exception will be raised:

```
TypeError: Can't upgrade a READER transaction to a WRITER mid-transaction
```

Another consideration that must be taken when implementing/reviewing new code is that, as commented before, a transaction is an atomic operation on the database. If the database layer (SQLAlchemy, oslo.db) returns a database exception, the current active transaction should end. In other words, we can catch, if needed, the exception raised and retry any needed operation, but any further database command should be executed in a new context. This is needed to allow the context wrapper (writer, reader) to properly finish the operation, for example rolling back the already executed commands. Check the patch https://review.opendev.org/c/openstack/neutron/+/843263 as an example of how to handle database exceptions.

Retry decorators

This is an appendix for *Retrying Operations*

This is also related to the previous section. The neutron-lib library provides a decorator called *retry_if_session_inactive* that can be used to retry any method if the context session is not active; in other words, there is no active transaction when the method is called. The session is retrieved from the context parameter passed into the method (it is a must to have this parameter in the method signature).

This retry decorator can be used along with a transaction decorator but the retry decorator must be declared before the context one. If we first declare the database context (writer or reader) and then the retry decorator, the retry context would be always called from inside an active transaction making it useless. An example of a good implementation (first the retry decorator and then the reader one):

```
@db_api.retry_if_session_inactive()
@db_api.CONTEXT_READER
def get_ports_count(self, context, filters=None):
    return self._get_ports_query(context, filters).count()
```

Database Models Relocation

This document is intended to track and notify developers that db models in neutron will be centralized and moved to a new tree under neutron/db/models. This was discussed in [1]. The reason for relocating db models is to solve the cyclic import issue while implementing oslo versioned objects for resources in neutron.

The reason behind this relocation is Mixin class and db models for some resources in neutron are in same module. In Mixin classes, there are methods which provide functionality of fetching, adding, updating and deleting data via queries. These queries will be replaced with use of versioned objects and definition of versioned object will be using db models. So object files will be importing models and Mixin need to import those objects which will end up in cyclic import.

Structure of Model Definitions

We have decided to move all models definitions to neutron/db/models/ with no further nesting after that point. The deprecation method to move models has already been added to avoid breakage of third party plugins using those models. All relocated models need to use deprecate method that will generate a warning and return new class for use of old class. Some examples of relocated models [2] and [3]. In future if you define new models please make sure they are separated from mixins and are under tree neutron/db/models/ .

References

[1]. https://www.mail-archive.com/openstack-dev@lists.openstack.org/msg88910.html [2]. https://review.opendev.org/#/c/348562/ [3]. https://review.opendev.org/#/c/348757/

DNS Nameserver Order Consistency

In Neutron subnets, DNS nameservers are given priority when created or updated. This means if you create a subnet with multiple DNS servers, the order will be retained and guests will receive the DNS servers in the order you created them in when the subnet was created. The same thing applies for update operations on subnets to add, remove, or update DNS servers.

Get Subnet Details Info

```
$ openstack subnet list
+--------------------------------------+---------+---------------------------
↪----------+-------------+
| ID                                   | Name    | Network                   ␣
↪          | Subnet      |
+--------------------------------------+---------+---------------------------
↪----------+-------------+
| 1a2d261b-b233-3ab9-902e-88576a82afa6 | private | a404518c-800d-2353-9193-
↪57dbb42ac5ee | 10.0.0.0/24 |
+--------------------------------------+---------+---------------------------
↪----------+-------------+

$ openstack subnet show 1a2d261b-b233-3ab9-902e-88576a82afa6
+----------------------+--------------------------------------+
| Field                | Value                                |
+----------------------+--------------------------------------+
| allocation_pools     | 10.0.0.2-10.0.0.254                  |
| cidr                 | 10.0.0.0/24                          |
| created_at           | 2024-02-13T21:42:34Z                 |
| description          |                                      |
| dns_nameservers      | 8.8.4.4, 8.8.8.8                     |
| dns_publish_fixed_ip | None                                 |
| enable_dhcp          | True                                 |
| gateway_ip           | 10.0.0.1                             |
| host_routes          |                                      |
| id                   | 1a2d26fb-b733-4ab3-992e-88554a87afa6 |
| ip_version           | 4                                    |
| ipv6_address_mode    | None                                 |
| ipv6_ra_mode         | None                                 |
| name                 | private                              |
| network_id           | a404518c-800d-2353-9193-57dbb42ac5ee |
| project_id           | 3868290ab10f417390acbb754160dbb2     |
| revision_number      | 0                                    |
| segment_id           | None                                 |
| service_types        |                                      |
| subnetpool_id        |                                      |
| tags                 |                                      |
```

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```
| updated_at            | 2024-02-13T21:42:34Z                 |
+-----------------------+--------------------------------------+
```

Update Subnet DNS Nameservers

Note: `--no-dns-nameserver` must be passed to clear the current list, otherwise a conflict will be raised if there are duplicates.

```
$ openstack subnet set --no-dns-nameserver --dns-nameserver 8.8.8.8 \
  --dns-nameserver 8.8.4.4 1a2d261b-b233-3ab9-902e-88576a82afa6

$ openstack subnet show 1a2d261b-b233-3ab9-902e-88576a82afa6
+-----------------------+--------------------------------------+
| Field                 | Value                                |
+-----------------------+--------------------------------------+
| allocation_pools      | 10.0.0.2-10.0.0.254                  |
| cidr                  | 10.0.0.0/24                          |
| created_at            | 2024-02-13T21:42:34Z                 |
| description           |                                      |
| dns_nameservers       | 8.8.8.8, 8.8.4.4                     |
| dns_publish_fixed_ip  | None                                 |
| enable_dhcp           | True                                 |
| gateway_ip            | 10.0.0.1                             |
| host_routes           |                                      |
| id                    | 1a2d26fb-b733-4ab3-992e-88554a87afa6 |
| ip_version            | 4                                    |
| ipv6_address_mode     | None                                 |
| ipv6_ra_mode          | None                                 |
| name                  | private                              |
| network_id            | a404518c-800d-2353-9193-57dbb42ac5ee |
| project_id            | 3868290ab10f417390acbb754160dbb2     |
| revision_number       | 1                                    |
| segment_id            | None                                 |
| service_types         |                                      |
| subnetpool_id         |                                      |
| tags                  |                                      |
| updated_at            | 2024-02-13T21:42:34Z                 |
+-----------------------+--------------------------------------+
```

As shown in above output, the order of the DNS nameservers has been updated. New virtual machines deployed to this subnet will receive the DNS nameservers in this new priority order. Existing virtual machines that have already been deployed will not be immediately affected by changing the DNS nameserver order on the neutron subnet. Virtual machines that are configured to get their IP address via DHCP will detect the DNS nameserver order change when their DHCP lease expires or when the virtual machine is restarted. Existing virtual machines configured with a static IP address will never detect the updated DNS nameserver order.

External DNS Service Integration

Since the Mitaka release, neutron has an interface defined to interact with an external DNS service. This interface is based on an abstract driver that can be used as the base class to implement concrete drivers to interact with various DNS services. The reference implementation of such a driver integrates neutron with OpenStack Designate.

This integration allows users to publish *dns_name* and *dns_domain* attributes associated with floating IP addresses, ports, and networks in an external DNS service.

Changes to the neutron API

To support integration with an external DNS service, the *dns_name* and *dns_domain* attributes were added to floating ips, ports and networks. The *dns_name* specifies the name to be associated with a corresponding IP address, both of which will be published to an existing domain with the name *dns_domain* in the external DNS service.

Specifically, floating ips, ports and networks are extended as follows:

- Floating ips have a *dns_name* and a *dns_domain* attribute.
- Ports have a *dns_name* attribute.
- Networks have a *dns_domain* attributes.

Pre-configured domains for projects and users

ML2 plugin extension `dns_domain_keywords` provides same dns integration as `dns_domain_ports` and `subnet_dns_publish_fixed_ip` and it also allows to configure networks dns_domain with some specific keywords: <project_id>, <project_name>, <user_id>, <user_name>. Please see example below for more details.

- Create DNS zone. `0511951bd56e4a0aac27ac65e00bddd0` is ID of the project used in the example

```
$ openstack zone create 0511951bd56e4a0aac27ac65e00bddd0.example.com. --
↪email admin@0511951bd56e4a0aac27ac65e00bddd0.example.com
+----------------+--------------------------------------------------+
| Field          | Value                                            |
+----------------+--------------------------------------------------+
| action         | CREATE                                           |
| attributes     |                                                  |
| created_at     | 2021-02-19T14:48:06.000000                       |
| description    | None                                             |
| email          | admin@0511951bd56e4a0aac27ac65e00bddd0.example.com |
| id             | c14a8edc-d0b9-4cdd-93f1-1ab5a5f5ff9d             |
| masters        |                                                  |
| name           | 0511951bd56e4a0aac27ac65e00bddd0.example.com.    |
| pool_id        | 794ccc2c-d751-44fe-b57f-8894c9f5c842             |
| project_id     | 0511951bd56e4a0aac27ac65e00bddd0                 |
| serial         | 1613746085                                       |
| status         | PENDING                                          |
| transferred_at | None                                             |
| ttl            | 3600                                             |
```

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```
| type              | PRIMARY                                                  |
| updated_at        | None                                                     |
| version           | 1                                                        |
+-------------------+----------------------------------------------------------+
```

- Create network with dns_domain

```
$ openstack network create dns-test-network --dns-domain "<project_id>.
↪demo.net."
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2021-02-19T15:16:36Z                 |
| description               |                                      |
| dns_domain                | <project_id>.demo.net.               |
| id                        | fb247287-43aa-4a83-b768-a3b34dc6735a |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| is_vlan_transparent       | None                                 |
| mtu                       | 1450                                 |
| name                      | dns-test-network                     |
| port_security_enabled     | True                                 |
| project_id                | 0511951bd56e4a0aac27ac65e00bddd0     |
| provider:network_type     | vxlan                                |
| provider:physical_network | None                                 |
| provider:segmentation_id  | 1003                                 |
| qos_policy_id             | None                                 |
| revision_number           | 1                                    |
| router:external           | Internal                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      |                                      |
| updated_at                | 2021-02-19T15:16:37Z                 |
+---------------------------+--------------------------------------+

$ openstack subnet create --network dns-test-network --subnet-range 192.
↪168.100.0/24 --dns-publish-fixed-ip dns-test-subnet
+----------------------+--------------------------------------+
| Field                | Value                                |
+----------------------+--------------------------------------+
| allocation_pools     | 192.168.100.2-192.168.100.254        |
| cidr                 | 192.168.100.0/24                     |
| created_at           | 2021-02-19T15:21:50Z                 |
| description          |                                      |
```

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```
| dns_nameservers       |                                      |
| dns_publish_fixed_ip  | True                                 |
| enable_dhcp           | True                                 |
| gateway_ip            | 192.168.100.1                        |
| host_routes           |                                      |
| id                    | 2547a3f2-374f-4262-aed5-3a69af73e732 |
| ip_version            | 4                                    |
| ipv6_address_mode     | None                                 |
| ipv6_ra_mode          | None                                 |
| name                  | dns-test-subnet                      |
| network_id            | fb247287-43aa-4a83-b768-a3b34dc6735a |
| prefix_length         | None                                 |
| project_id            | 0511951bd56e4a0aac27ac65e00bddd0     |
| revision_number       | 0                                    |
| segment_id            | None                                 |
| service_types         |                                      |
| subnetpool_id         | None                                 |
| tags                  |                                      |
| updated_at            | 2021-02-19T15:21:50Z                 |
+-----------------------+--------------------------------------+
```

- Create port in that network

```
$ openstack port create --network dns-test-network --dns-name dns-test-
↪port test-port
+-------------------------+-----------------------------------------------
↪---------------------------------------------------------------------
↪----+
| Field                   | Value                                          ␣
↪                                                                      ␣
↪    |
+-------------------------+-----------------------------------------------
↪---------------------------------------------------------------------
↪----+
| admin_state_up          | UP                                             ␣
↪                                                                      ␣
↪    |
| allowed_address_pairs   |                                                ␣
↪                                                                      ␣
↪    |
| binding_host_id         |                                                ␣
↪                                                                      ␣
↪    |
| binding_profile         |                                                ␣
↪                                                                      ␣
↪    |
| binding_vif_details     |                                                ␣
↪                                                                      ␣
↪    |
| binding_vif_type        | unbound                                        ␣
```

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```
↪                                                                                    ␣
↪    |
| binding_vnic_type       | normal                                                   ␣
↪                                                                                    ␣
↪    |
| created_at              | 2021-02-19T15:22:51Z                                     ␣
↪                                                                                    ␣
↪    |
| data_plane_status       | None                                                     ␣
↪                                                                                    ␣
↪    |
| description             |                                                          ␣
↪                                                                                    ␣
↪    |
| device_id               |                                                          ␣
↪                                                                                    ␣
↪    |
| device_owner            |                                                          ␣
↪                                                                                    ␣
↪    |
| device_profile          | None                                                     ␣
↪                                                                                    ␣
↪    |
| dns_assignment          | fqdn='dns-test-port.
↪0511951bd56e4a0aac27ac65e00bddd0.example.com.', hostname='dns-test-port
↪', ip_address='192.168.100.17' |
| dns_domain              |                                                          ␣
↪                                                                                    ␣
↪    |
| dns_name                | dns-test-port                                            ␣
↪                                                                                    ␣
↪    |
| extra_dhcp_opts         |                                                          ␣
↪                                                                                    ␣
↪    |
| fixed_ips               | ip_address='192.168.100.17', subnet_id=
↪'2547a3f2-374f-4262-aed5-3a69af73e732'                                              ␣
↪           |
| id                      | f30908a1-6ef5-4137-bff4-c1205c6660ee                     ␣
↪                                                                                    ␣
↪    |
| ip_allocation           | None                                                     ␣
↪                                                                                    ␣
↪    |
| mac_address             | fa:16:3e:e8:33:b8                                        ␣
↪                                                                                    ␣
↪    |
| name                    | test-port                                                ␣
↪                                                                                    ␣
```

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```
↪     |
| network_id              | fb247287-43aa-4a83-b768-a3b34dc6735a                 ␣
↪                                                                                ␣
↪     |
| numa_affinity_policy    | None                                                 ␣
↪                                                                                ␣
↪     |
| port_security_enabled   | True                                                 ␣
↪                                                                                ␣
↪     |
| project_id              | 0511951bd56e4a0aac27ac65e00bddd0                     ␣
↪                                                                                ␣
↪     |
| propagate_uplink_status | None                                                 ␣
↪                                                                                ␣
↪     |
| qos_network_policy_id   | None                                                 ␣
↪                                                                                ␣
↪     |
| qos_policy_id           | None                                                 ␣
↪                                                                                ␣
↪     |
| resource_request        | None                                                 ␣
↪                                                                                ␣
↪     |
| revision_number         | 1                                                    ␣
↪                                                                                ␣
↪     |
| security_group_ids      | 4425c3fd-6705-4134-9878-07b333d81314                 ␣
↪                                                                                ␣
↪     |
| status                  | DOWN                                                 ␣
↪                                                                                ␣
↪     |
| tags                    |                                                      ␣
↪                                                                                ␣
↪     |
| trunk_details           | None                                                 ␣
↪                                                                                ␣
↪     |
| updated_at              | 2021-02-19T15:22:51Z                                 ␣
↪                                                                                ␣
↪     |
+-------------------------+-----------------------------------------------------
↪-------------------------------------------------------------------------------
↪----+
```

- Test if recordset was created properly in the DNS zone

```
$ openstack recordset list c14a8edc-d0b9-4cdd-93f1-1ab5a5f5ff9d
+--------------------------------------+------------------------------------
↪---------------------------+------+----------------------------------------
↪-------------------------------------------------------------------+-------
↪--+--------+
| id                                   | name                              ␣
↪                           | type | records                            ␣
↪                                                                   |␣
↪status | action |
+--------------------------------------+------------------------------------
↪---------------------------+------+----------------------------------------
↪-------------------------------------------------------------------+-------
↪--+--------+
| 1c302468-4e30-466e-9330-e4cd9191ff99 | 0511951bd56e4a0aac27ac65e00bddd0.
↪example.com.               | SOA  | ns1.devstack.org. admin.
↪0511951bd56e4a0aac27ac65e00bddd0.example.com. 1613748171 3549 600 86400␣
↪3600 | ACTIVE | NONE   |
| 99ce92d1-8c7a-4193-aeb2-44835048a6fa | 0511951bd56e4a0aac27ac65e00bddd0.
↪example.com.               | NS   | ns1.devstack.org.                  ␣
↪                                                                   |␣
↪ACTIVE | NONE   |
| 01f0569d-ce81-4424-915f-c6fe6229256e | dns-test-port.
↪0511951bd56e4a0aac27ac65e00bddd0.example.com. | A    | 192.168.100.17  ␣
↪                                                                       ␣
↪              | ACTIVE | NONE   |
+--------------------------------------+------------------------------------
↪---------------------------+------+----------------------------------------
↪-------------------------------------------------------------------+-------
↪--+--------+
```

i18n for the Neutron Stadium

- Refer to oslo_i18n documentation for the general mechanisms that should be used: https://docs.openstack.org/oslo.i18n/latest/user/usage.html
- Each stadium project should NOT consume _i18n module from neutron-lib or neutron.
- It is recommended that you create a {package_name}/_i18n.py file in your repo, and use that. Your localization strings will also live in your repo.

L2 Agent Extensions

L2 agent extensions are part of a generalized L2/L3 extension framework. See *agent extensions*.

Open vSwitch agent API

- neutron.plugins.ml2.drivers.openvswitch.agent.ovs_agent_extension_api

Open vSwitch agent API object includes two methods that return wrapped and hardened bridge objects with cookie values allocated for calling extensions:

```
#. request_int_br
#. request_tun_br
```

Bridge objects returned by those methods already have new default cookie values allocated for extension flows. All flow management methods (add_flow, mod_flow,) enforce those allocated cookies.

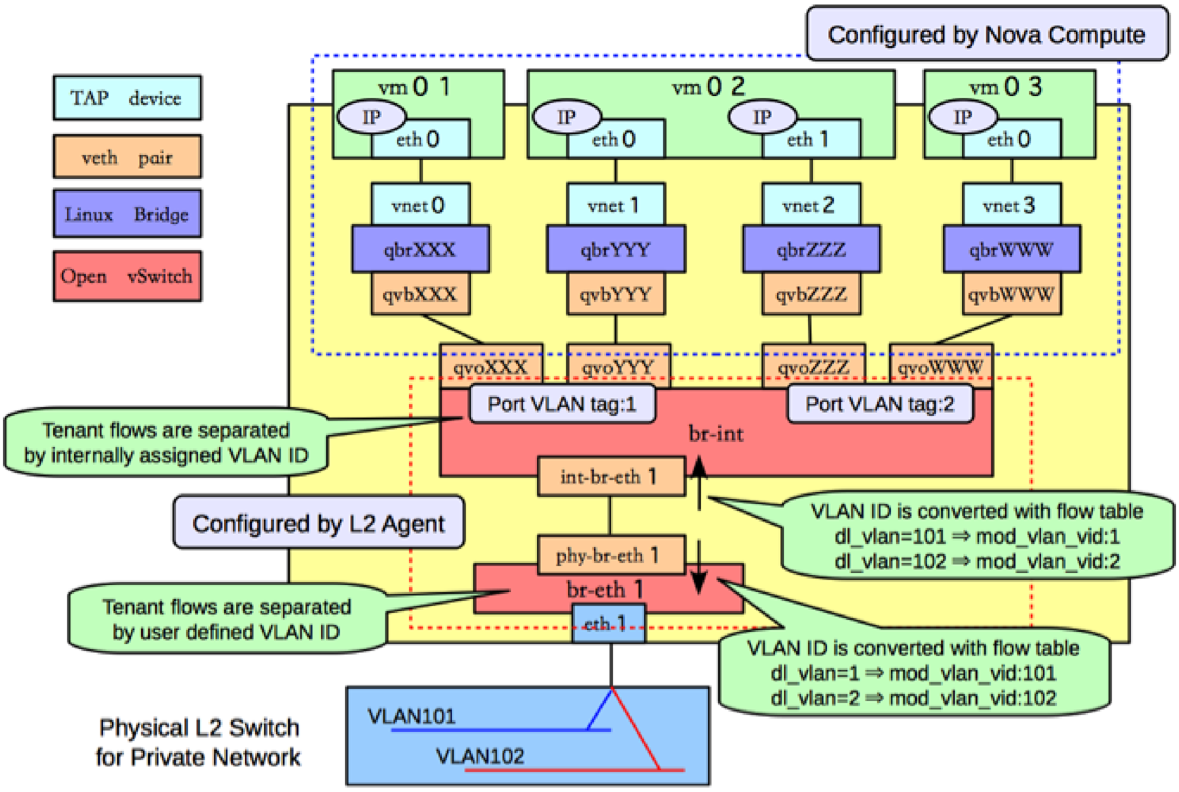
L2 Agent Networking

Open vSwitch L2 Agent

This Agent uses the Open vSwitch virtual switch to create L2 connectivity for instances, along with bridges created in conjunction with OpenStack Nova for filtering.

ovs-neutron-agent can be configured to use different networking technologies to create project isolation. These technologies are implemented as ML2 type drivers which are used in conjunction with the Open vSwitch mechanism driver.

VLAN Tags



GRE Tunnels

GRE Tunneling is documented in depth in the Networking in too much detail by RedHat.

VXLAN Tunnels

VXLAN is an overlay technology which encapsulates MAC frames at layer 2 into a UDP header. More information can be found in The VXLAN wiki page.

Geneve Tunnels

Geneve uses UDP as its transport protocol and is dynamic in size using extensible option headers. It is important to note that currently it is only supported in newer kernels. (kernel >= 3.18, OVS version >=2.4) More information can be found in the Geneve RFC document.

Bridge Management

In order to make the agent capable of handling more than one tunneling technology, to decouple the requirements of segmentation technology from project isolation, and to preserve backward compatibility for OVS agents working without tunneling, the agent relies on a tunneling bridge, or br-tun, and the well known integration bridge, or br-int.

All VM VIFs are plugged into the integration bridge. VM VIFs on a given virtual network share a common local VLAN (i.e. not propagated externally). The VLAN id of this local VLAN is mapped to the physical networking details realizing that virtual network.

For virtual networks realized as VXLAN/GRE tunnels, a Logical Switch (LS) identifier is used to differentiate project traffic on inter-HV tunnels. A mesh of tunnels is created to other Hypervisors in the cloud. These tunnels originate and terminate on the tunneling bridge of each hypervisor, leaving br-int unaffected. Port patching is done to connect local VLANs on the integration bridge to inter-hypervisor tunnels on the tunnel bridge.

For each virtual network realized as a VLAN or flat network, a veth or a pair of patch ports is used to connect the local VLAN on the integration bridge with the physical network bridge, with flow rules adding, modifying, or stripping VLAN tags as necessary, thus preserving backward compatibility with the way the OVS agent used to work prior to the tunneling capability (for more details, please look at https://review.opendev.org/#/c/4367).

Bear in mind, that this design decision may be overhauled in the future to support existing VLAN-tagged traffic (coming from NFV VMs for instance) and/or to deal with potential QinQ support natively available in the Open vSwitch.

OVS Filtering Tables

ovs-neutron-agent and other L2 agent extensions use OVS filtering tables.

For the list of tables and the short name for them used in Neutron see ovs-neutron-agent constants

For a detailed discussion of Open vSwitch firewall driver and how the filtering tables are used for security-groups see *Open vSwitch Firewall Driver*.

Tackling the Network Trunking use case

Rationale

At the time the first design for the OVS agent came up, trunking in OpenStack was merely a pipe dream. Since then, lots has happened in the OpenStack platform, and many deployments have gone into production since early 2012.

In order to address the vlan-aware-vms use case on top of Open vSwitch, the following aspects must be taken into account:

- Design complexity: starting afresh is always an option, but a complete rearchitecture is only desirable under some circumstances. After all, customers want solutionsyesterday. It is noteworthy that the OVS agent design is already relatively complex, as it accommodates a number of deployment options, especially in relation to security rules and/or acceleration.
- Upgrade complexity: being able to retrofit the existing design means that an existing deployment does not need to go through a forklift upgrade in order to expose new functionality; alternatively, the desire of avoiding a migration requires a more complex solution that is able to support multiple modes of operations;
- Design reusability: ideally, a proposed design can easily apply to the various technology backends that the Neutron L2 agent supports: Open vSwitch and Linux Bridge.
- Performance penalty: no solution is appealing enough if it is unable to satisfy the stringent requirement of high packet throughput, at least in the long term.
- Feature compatibility: VLAN transparency is for better or for worse intertwined with vlan awareness. The former is about making the platform not interfere with the tag associated to the packets sent by the VM, and let the underlay figure out where the packet needs to be sent out; the latter is about making the platform use the vlan tag associated to packet to determine where the packet needs to go. Ideally, a design choice to satisfy the awareness use case will not have a negative impact for solving the transparency use case. Having said that, the two features are still meant to be mutually exclusive in their application, and plugging subports into networks whose vlan-transparency flag is set to True might have unexpected results. In fact, it would be impossible from the platforms point of view discerning which tagged packets are meant to be treated transparently and which ones are meant to be used for demultiplexing (in order to reach the right destination). The outcome might only be predictable if two layers of vlan tags are stacked up together, making guest support even more crucial for the combined use case.

It is clear by now that an acceptable solution must be assessed with these issues in mind. The potential solutions worth enumerating are:

- VLAN interfaces: in laymans terms, these interfaces allow to demux the traffic before it hits the integration bridge where the traffic will get isolated and sent off to the right destination. This solution is proven to work for both iptables-based and native ovs security rules (credit to Rawlin Peters). This solution has the following design implications:
  - Design complexity: this requires relative small changes to the existing OVS design, and it can work with both iptables and native ovs security rules.
  - Upgrade complexity: in order to employ this solution no major upgrade is necessary and thus no potential dataplane disruption is involved.
  - Design reusability: VLAN interfaces can easily be employed for both Open vSwitch and Linux Bridge.

  - Performance penalty: using VLAN interfaces means that the kernel must be involved. For Open vSwitch, being able to use a fast path like DPDK would be an unresolved issue (Kernel NIC interfaces are not on the roadmap for distros and OVS, and most likely will never be). Even in the absence of an extra bridge, i.e. when using native ovs firewall, and with the advent of userspace connection tracking that would allow the stateful firewall driver to work with DPDK, the performance gap between a pure userspace DPDK capable solution and a kernel based solution will be substantial, at least under certain traffic conditions.
  - Feature compatibility: in order to keep the design simple once VLAN interfaces are adopted, and yet enable VLAN transparency, Open vSwitch needs to support QinQ, which is currently lacking as of 2.5 and with no ongoing plan for integration.
- Going full openflow: in laymans terms, this means programming the dataplane using OpenFlow in order to provide tenant isolation, and packet processing. This solution has the following design implications:
  - Design complexity: this requires a big rearchitecture of the current Neutron L2 agent solution.
  - Upgrade complexity: existing deployments will be unable to work correctly unless one of the actions take place: a) the agent can handle both the old and new way of wiring the data path; b) a dataplane migration is forced during a release upgrade and thus it may cause (potentially unrecoverable) dataplane disruption.
  - Design reusability: a solution for Linux Bridge will still be required to avoid widening the gap between Open vSwitch (e.g. OVS has DVR but LB does not).
  - Performance penalty: using Open Flow will allow to leverage the user space and fast processing given by DPDK, but at a considerable engineering cost nonetheless. Security rules will have to be provided by a learn based firewall to fully exploit the capabilities of DPDK, at least until user space connection tracking becomes available in OVS.
  - Feature compatibility: with the adoption of Open Flow, tenant isolation will no longer be provided by means of local vlan provisioning, thus making the requirement of QinQ support no longer strictly necessary for Open vSwitch.
- Per trunk port OVS bridge: in laymans terms, this is similar to the first option, in that an extra layer of mux/demux is introduced between the VM and the integration bridge (br-int) but instead of using vlan interfaces, a combination of a new per port OVS bridge and patch ports to wire this new bridge with br-int will be used. This solution has the following design implications:
  - Design complexity: the complexity of this solution can be considered in between the above mentioned options in that some work is already available since Mitaka and the data path wiring logic can be partially reused.
  - Upgrade complexity: if two separate code paths are assumed to be maintained in the OVS agent to handle regular ports and ports participating a trunk with no ability to convert from one to the other (and vice versa), no migration is required. This is done at a cost of some loss of flexibility and maintenance complexity.
  - Design reusability: a solution to support vlan trunking for the Linux Bridge mech driver will still be required to avoid widening the gap with Open vSwitch (e.g. OVS has DVR but LB does not).
  - Performance penalty: from a performance standpoint, the adoption of a trunk bridge relieves the agent from employing kernel interfaces, thus unlocking the full potential of fast packet processing. That said, this is only doable in combination with a native ovs firewall. At the

time of writing the only DPDK enabled firewall driver is the learn based one available in the networking-ovs-dpdk repo;

- Feature compatibility: the existing local provisioning logic will not be affected by the introduction of a trunk bridge, therefore use cases where VMs are connected to a vlan transparent network via a regular port will still require QinQ support from OVS.

To summarize:

- VLAN interfaces (A) are compelling because will lead to a relatively contained engineering cost at the expense of performance. The Open vSwitch community will need to be involved in order to deliver vlan transparency. Irrespective of whether this strategy is chosen for Open vSwitch or not, this is still the only viable approach for Linux Bridge and thus pursued to address Linux Bridge support for VLAN trunking. To some extent, this option can also be considered a fallback strategy for OVS deployments that are unable to adopt DPDK.
- Open Flow (B) is compelling because it will allow Neutron to unlock the full potential of Open vSwitch, at the expense of development and operations effort. The development is confined within the boundaries of the Neutron community in order to address vlan awareness and transparency (as two distinct use cases, ie. to be adopted separately). Stateful firewall (based on ovs conntrack) limits the adoption for DPDK at the time of writing, but a learn-based firewall can be a suitable alternative. Obviously this solution is not compliant with iptables firewall.
- Trunk Bridges (C) tries to bring the best of option A and B together as far as OVS development and performance are concerned, but it comes at the expense of maintenance complexity and loss of flexibility. A Linux Bridge solution would still be required and, QinQ support will still be needed to address vlan transparency.

All things considered, as far as OVS is concerned, option (C) is the most promising in the medium term. Management of trunks and ports within trunks will have to be managed differently and, to start with, it is sensible to restrict the ability to update ports (i.e. convert) once they are bound to a particular bridge (integration vs trunk). Security rules via iptables rules is obviously not supported, and never will be.

Option (A) for OVS could be pursued in conjunction with Linux Bridge support, if the effort is seen particularly low hanging fruit. However, a working solution based on this option positions the OVS agent as a sub-optminal platform for performance sensitive applications in comparison to other accelerated or SDN-controller based solutions. Since further data plane performance improvement is hindered by the extra use of kernel resources, this option is not at all appealing in the long term.

Embracing option (B) in the long run may be complicated by the adoption of option (C). The development and maintenance complexity involved in Option (C) and (B) respectively poses the existential question as to whether investing in the agent-based architecture is an effective strategy, especially if the end result would look a lot like other maturing alternatives.

Implementation VLAN Interfaces (Option A)

This implementation doesnt require any modification of the vif-drivers since Nova will plug the vif of the VM the same way as it does for traditional ports.

Trunk port creation

A VM is spawned passing to Nova the port-id of a parent port associated with a trunk. Nova/libvirt will create the tap interface and will plug it into br-int or into the firewall bridge if using iptables firewall. In the external-ids of the port Nova will store the port ID of the parent port. The OVS agent detects that a new vif has been plugged. It gets the details of the new port and wires it. The agent configures it in the same way as a traditional port: packets coming out from the VM will be tagged using the internal VLAN ID associated to the network, packets going to the VM will be stripped of the VLAN ID. After wiring it successfully the OVS agent will send a message notifying Neutron server that the parent port is up. Neutron will send back to Nova an event to signal that the wiring was successful. If the parent port is associated with one or more subports the agent will process them as described in the next paragraph.

Subport creation

If a subport is added to a parent port but no VM was booted using that parent port yet, no L2 agent will process it (because at that point the parent port is not bound to any host). When a subport is created for a parent port and a VM that uses that parent port is already running, the OVS agent will create a VLAN interface on the VM tap using the VLAN ID specified in the subport segmentation id. Theres a small possibility that a race might occur: the firewall bridge might be created and plugged while the vif is not there yet. The OVS agent needs to check if the vif exists before trying to create a subinterface. Lets see how the models differ when using the iptables firewall or the OVS native firewall.

Iptables Firewall

```
+----------------------------+
|             VM             |
|   eth0            eth0.100 |
+-----+-----------------+----+
      |
      |
  +---+---+       +-----+-----+
  | tap1  |-------|  tap1.100 |
  +---+---+       +-----+-----+
      |                 |
      |                 |
  +---+---+         +---+---+
  | qbr1  |         | qbr2  |
  +---+---+         +---+---+
      |                 |
      |                 |
+-----+-----------------+----+
|    port 1          port 2  |
|   (tag 3)         (tag 5)  |
|           br-int           |
+----------------------------+
```

Lets assume the subport is on network2 and uses segmentation ID 100. In the case of hybrid plugging the OVS agent will have to create the firewall bridge (qbr2), create tap1.100 and plug it into qbr2. It will connect qbr2 to br-int and set the subport ID in the external-ids of port 2.

Inbound traffic from the VM point of view

The untagged traffic will flow from port 1 to eth0 through qbr1. For the traffic coming out of port 2, the internal VLAN ID of network2 will be stripped. The packet will then go untagged through qbr2 where iptables rules will filter the traffic. The tag 100 will be pushed by tap1.100 and the packet will finally get to eth0.100.

Outbound traffic from the VM point of view

The untagged traffic will flow from eth0 to port1 going through qbr1 where firewall rules will be applied. Traffic tagged with VLAN 100 will leave eth0.100, go through tap1.100 where the VLAN 100 is stripped. It will reach qbr2 where iptables rules will be applied and go to port 2. The internal VLAN of network2 will be pushed by br-int when the packet enters port2 because its a tagged port.

OVS Firewall case

```
+----------------------------+
|             VM             |
|   eth0            eth0.100 |
+-----+-----------------+----+
      |
      |
  +---+---+       +-----+-----+
  | tap1  |-------|  tap1.100 |
  +---+---+       +-----+-----+
      |                 |
      |                 |
      |                 |
+-----+-----------------+----+
|    port 1          port 2  |
|   (tag 3)         (tag 5)  |
|           br-int           |
+----------------------------+
```

When a subport is created the OVS agent will create the VLAN interface tap1.100 and plug it into br-int. Lets assume the subport is on network2.

Inbound traffic from the VM point of view

The traffic will flow untagged from port 1 to eth0. The traffic going out from port 2 will be stripped of the VLAN ID assigned to network2. It will be filtered by the rules installed by the firewall and reach tap1.100. tap1.100 will tag the traffic using VLAN 100. It will then reach the VMs eth0.100.

Outbound traffic from the VM point of view

The untagged traffic will flow and reach port 1 where it will be tagged using the VLAN ID associated to the network. Traffic tagged with VLAN 100 will leave eth0.100 and reach tap1.100 where VLAN 100 will be stripped. It will then reach port2. It will be filtered by the rules installed by the firewall on port 2. Then the packets will be tagged using the internal VLAN associated to network2 by br-int since port 2 is a tagged port.

Parent port deletion

Deleting a port that is an active parent in a trunk is forbidden. If the parent port has no trunk associated (its a normal port), it can be deleted. The OVS agent doesnt need to perform any action, the deletion will result in a removal of the port data from the DB.

Trunk deletion

When Nova deletes a VM, it deletes the VMs corresponding Neutron ports only if they were created by Nova when booting the VM. In the vlan-aware-vm case the parent port is passed to Nova, so the port data will remain in the DB after the VM deletion. Nova will delete the VIF of the VM (in the example tap1) as part of the VM termination. The OVS agent will detect that deletion and notify the Neutron server that the parent port is down. The OVS agent will clean up the corresponding subports as explained in the next paragraph.

The deletion of a trunk that is used by a VM is not allowed. The trunk can be deleted (leaving the parent port intact) when the parent port is not used by any VM. After the trunk is deleted, the parent port can also be deleted.

Subport deletion

Removing a subport that is associated with a parent port that was not used to boot any VM is a no op from the OVS agent perspective. When a subport associated with a parent port that was used to boot a VM is deleted, the OVS agent will take care of removing the firewall bridge if using the iptables firewall, and the port on br-int.

Implementation Trunk Bridge (Option C)

This implementation is based on this etherpad. Credits to Bence Romsics. The IDs used for bridge and port names are truncated.

```
+--------------------------------+
|             VM                 |
|   eth0                eth0.100  |
+-----+--------------------+-----+
      |
      |
+-----+--------------------------+
|    tap1                        |
|          tbr-trunk-id          |
|                                |
| tpt-parent-id   spt-subport-id |
|   (tag 0)         (tag 100)    |
+-----+-----------------+--------+
      |                 |
      |                 |
      |                 |
+-----+-----------------+---------+
| tpi-parent-id    spi-subport-id |
|   (tag 3)          (tag 5)      |
|                                 |
|           br-int                |
+---------------------------------+
```

tpt-parent-id: trunk bridge side of the patch port that implements a trunk. tpi-parent-id: int bridge side of the patch port that implements a trunk. spt-subport-id: trunk bridge side of the patch port that implements a subport. spi-subport-id: int bridge side of the patch port that implements a subport.

Trunk creation

A VM is spawned passing to Nova the port-id of a parent port associated with a trunk. Neutron will pass to Nova the bridge where to plug the vif as part of the vif details. The os-vif driver creates the trunk bridge tbr-trunk-id if it does not exist in plug(). It will create the tap interface tap1 and plug it into tbr-trunk-id setting the parent port ID in the external-ids. The trunk driver will wire the parent port via a patch port to connect the trunk bridge to the integration bridge:

```
ovs-vsctl add-port tbr-trunk-id tpt-parent-id -- set Interface tpt-parent-id␣
↪type=patch options:peer=tpi-parent-id -- set Port tpt-parent-id vlan_
↪mode=access tag=0
ovs-vsctl add-port br-int tpi-parent-id -- set Interface tpi-parent-id␣
↪type=patch options:peer=tpt-parent-id
```

tpt-parent-id, the trunk bridge side of the patch will carry untagged traffic (vlan_mode=access tag=0). The OVS agent will be monitoring the creation of ports on the integration bridge. tpi-parent-id, the br-int side the patch port is tagged with VLAN 3 by ovs-agent. We assume that the trunk is on network1 that on this host is associated with VLAN 3. If the parent port is associated with one or more subports the agent will process them as described in the next paragraph.

Subport creation

If a subport is added to a parent port but no VM was booted using that parent port yet, the agent wont process the subport (because at this point theres no node associated with the parent port). When a subport is added to a parent port that is used by a VM the OVS agent will create a new patch port:

```
ovs-vsctl add-port tbr-trunk-id spt-subport-id tag=100 -- set Interface spt-
↪subport-id type=patch options:peer=spi-subport-id
ovs-vsctl add-port br-int spi-subport-id tag=5 -- set Interface spi-subport-
↪id type=patch options:peer=spt-subport-id
```

This patch port connects the trunk bridge to the integration bridge. spt-subport-id, the trunk bridge side of the patch is tagged using VLAN 100. We assume that the segmentation ID of the subport is 100. spi-subport-id, the br-int side of the patch port is tagged with VLAN 5. We assume that the subport is on network2 that on this host uses VLAN 5. The OVS agent will set the subport ID in the external-ids of spt-subport-id and spi-subport-id.

Inbound traffic from the VM point of view

The traffic coming out of tpi-parent-id will be stripped by br-int of VLAN 3. It will reach tpt-parent-id untagged and from there tap1. The traffic coming out of spi-subport-id will be stripped by br-int of VLAN 5. It will reach spt-subport-id where it will be tagged with VLAN 100 and it will then get to tap1 tagged.

Outbound traffic from the VM point of view

The untagged traffic coming from tap1 will reach tpt-parent-id and from there tpi-parent-id where it will be tagged using VLAN 3. The traffic tagged with VLAN 100 from tap1 will reach spt-subport-id. VLAN 100 will be stripped since spt-subport-id is a tagged port and the packet will reach spi-subport-id, where its tagged using VLAN 5.

Parent port deletion

Deleting a port that is an active parent in a trunk is forbidden. If the parent port has no trunk associated, it can be deleted. The OVS agent doesnt need to perform any action.

Trunk deletion

When Nova deletes a VM, it deletes the VMs corresponding Neutron ports only if they were created by Nova when booting the VM. In the vlan-aware-vm case the parent port is passed to Nova, so the port data will remain in the DB after the VM deletion. Nova will delete the port on the trunk bridge where the VM is plugged. The L2 agent will detect that and delete the trunk bridge. It will notify the Neutron server that the parent port is down.

The deletion of a trunk that is used by a VM is not allowed. The trunk can be deleted (leaving the parent port intact) when the parent port is not used by any VM. After the trunk is deleted, the parent port can also be deleted.

Subport deletion

The OVS agent will delete the patch port pair corresponding to the subport deleted.

Agent resync

During resync the agent should check that all the trunk and subports are still valid. It will delete the stale trunk and subports using the procedure specified in the previous paragraphs according to the implementation.

Local IP

Local IP is a new feature added in Yoga release. For details on openvswitch agent impact please see: *Local IPs*.

Further Reading

- Darragh OReilly - The Open vSwitch plugin with VLANs

SR-IOV Networking L2 Agent

SR-IOV (Single Root I/O Virtualization) is a specification that allows a PCIe device to appear to be multiple separate physical PCIe devices. SR-IOV works by introducing the idea of physical functions (PFs) and virtual functions (VFs). Physical functions (PFs) are full-featured PCIe functions. Virtual functions (VFs) are lightweight functions that lack configuration resources.

SR-IOV supports VLANs for L2 network isolation, other networking technologies such as VXLAN/GRE may be supported in the future.

SR-IOV NIC agent manages configuration of SR-IOV Virtual Functions that connect VM instances running on the compute node to the public network.

In most common deployments, there are compute and a network nodes. Compute node can support VM connectivity via SR-IOV enabled NIC. SR-IOV NIC Agent manages Virtual Functions admin state. Quality of service is partially implemented with the bandwidth limit and minimum bandwidth rules. In the future it will manage additional settings, such as additional quality of service rules, rate limit settings, spoofcheck and more.

Network node will be usually deployed with either ML2 Open vSwitch or ML2 OVN to support network node functionality.

The SR-IOV network agent does not implement any port firewalling.

Trusted virtual functions

In order to enable VF (SR-IOV virtual function) to request trusted mode, a new trusted VF concept was introduced in Linux kernel 4.4. It allows VF to become trusted by the Physical Function and perform some privileged operations, such as enabling VF promiscuous mode and changing VF MAC address within the guest.

This last operation (VF MAC change) implies, in many NIC drivers, that the host VF interface changes the MAC address too. The SR-IOV agent will detect this change and declare the port as DOWN; the MAC address must be the same as the one configured by Neutron. If the MAC address is restored, matching the Neutron DB port MAC address, the SR-IOV agent will declare the port as UP again.

It could happen that the MAC change happens during the SR-IOV agent periodic hardware inspection. This event will raise an error in the (MAC, PCI) tuple for this specific port. The SR-IOV agent will declare itself as out of sync and will force a full resync. During this resync process, all ports bound to this agent will set their status first to BUILD and then to ACTIVE again, causing a port status flapping. This event does not affect the user traffic.

Further Reading

Nir Yechiel - SR-IOV Networking Part I: Understanding the Basics

SR-IOV Passthrough For Networking

Trusted Virtual Functions

L3 Agent Extensions

L3 agent extensions are part of a generalized L2/L3 extension framework. See *agent extensions*.

L3 agent extension API

The L3 agent extension API object includes several methods that expose router information to L3 agent extensions:

```
#. get_routers_in_project
#. get_router_hosting_port
#. is_router_in_namespace
#. get_router_info
```

Layer 3 Networking via Layer 3 & OpenVSwitch Agents

This page discusses the usage of Neutron with Layer 3 functionality enabled.

Neutron logical network setup

```
vagrant@bionic64:~/devstack$ openstack network list
+--------------------------------------+---------+----------------------------
↪---------------------------------------------+
| ID                                   | Name    | Subnets                    ␣
↪                                             |
+--------------------------------------+---------+----------------------------
↪---------------------------------------------+
```

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```
| 6ece2847-971b-487a-9c7b-184651ebbc82 | public  | 0d9c4261-4046-462f-9d92-
↪64fb89bc3ae6, 9e90b059-da97-45b8-8cb8-f9370217e181 |
| 713bae25-8276-4e0a-a453-e59a1d65425a | private | 6fa3bab9-103e-45d5-872c-
↪91f21b52ceda, c5c9f5c2-145d-46d2-a513-cf675530eaed |
+--------------------------------------+---------+--------------------------
↪------------------------------------------------+

vagrant@bionic64:~/devstack$ openstack subnet list
+--------------------------------------+---------------------+----------------
↪----------------------+--------------------+
| ID                                   | Name                | Network        ␣
↪                      | Subnet             |
+--------------------------------------+---------------------+----------------
↪----------------------+--------------------+
| 0d9c4261-4046-462f-9d92-64fb89bc3ae6 | public-subnet       | 6ece2847-971b-
↪487a-9c7b-184651ebbc82 | 172.24.4.0/24      |
| 6fa3bab9-103e-45d5-872c-91f21b52ceda | ipv6-private-subnet | 713bae25-8276-
↪4e0a-a453-e59a1d65425a | 2001:db8:8000::/64 |
| 9e90b059-da97-45b8-8cb8-f9370217e181 | ipv6-public-subnet  | 6ece2847-971b-
↪487a-9c7b-184651ebbc82 | 2001:db8::/64      |
| c5c9f5c2-145d-46d2-a513-cf675530eaed | private-subnet      | 713bae25-8276-
↪4e0a-a453-e59a1d65425a | 10.0.0.0/24        |
+--------------------------------------+---------------------+----------------
↪----------------------+--------------------+

vagrant@bionic64:~/devstack$ openstack port list
+--------------------------------------+------+-------------------+-----------
↪-----------------------------------------------------------------------------
↪------------+--------+
| ID                                   | Name | MAC Address       | Fixed IP␣
↪Addresses                                                                    ␣
↪            | Status |
+--------------------------------------+------+-------------------+-----------
↪-----------------------------------------------------------------------------
↪------------+--------+
| 420abb60-2a5a-4e80-90a3-3ff47742dc53 |      | fa:16:3e:2d:5c:4e | ip_
↪address='172.24.4.7', subnet_id='0d9c4261-4046-462f-9d92-64fb89bc3ae6'      ␣
↪                      | ACTIVE |
|                                      |      |                   | ip_
↪address='2001:db8::1', subnet_id='9e90b059-da97-45b8-8cb8-f9370217e181'     ␣
↪                      |        |
| b42d789d-c9ed-48a1-8822-839c4599301e |      | fa:16:3e:0a:ff:24 | ip_
↪address='10.0.0.1', subnet_id='c5c9f5c2-145d-46d2-a513-cf675530eaed'        ␣
↪                      | ACTIVE |
| cfff6574-091c-4d16-a54b-5b7f3eab89ce |      | fa:16:3e:a0:a3:9e | ip_
↪address='10.0.0.2', subnet_id='c5c9f5c2-145d-46d2-a513-cf675530eaed'        ␣
↪                      | ACTIVE |
|                                      |      |                   | ip_
↪address='2001:db8:8000:0:f816:3eff:fea0:a39e', subnet_id='6fa3bab9-103e-
```

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```
↪45d5-872c-91f21b52ceda' |        |
| e3b7fede-277e-4c72-b66c-418a582b61ca |      | fa:16:3e:13:dd:42 | ip_
↪address='2001:db8:8000::1', subnet_id='6fa3bab9-103e-45d5-872c-91f21b52ceda
↪'                    | ACTIVE |
+--------------------------------------+------+-------------------+----------
↪--------------------------------------------------------------------------
↪-------------+--------+


vagrant@bionic64:~/devstack$ openstack subnet show c5c9f5c2-145d-46d2-a513-
↪cf675530eaed
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 10.0.0.2-10.0.0.254                  |
| cidr              | 10.0.0.0/24                          |
| created_at        | 2016-11-08T21:55:22Z                 |
| description       |                                      |
| dns_nameservers   |                                      |
| enable_dhcp       | True                                 |
| gateway_ip        | 10.0.0.1                             |
| host_routes       |                                      |
| id                | c5c9f5c2-145d-46d2-a513-cf675530eaed |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | private-subnet                       |
| network_id        | 713bae25-8276-4e0a-a453-e59a1d65425a |
| project_id        | 35e3820f7490493ca9e3a5e685393298     |
| revision_number   | 2                                    |
| service_types     |                                      |
| subnetpool_id     | b1f81d96-d51d-41f3-96b5-a0da16ad7f0d |
| updated_at        | 2016-11-08T21:55:22Z                 |
+-------------------+--------------------------------------+
```

Neutron logical router setup

```
vagrant@bionic64:~/devstack$ openstack router list
+--------------------------------------+---------+--------+-------+----------
↪--+-------+----------------------------------+
| ID                                   | Name    | Status | State |␣
↪Distributed | HA    | Project                          |
+--------------------------------------+---------+--------+-------+----------
↪--+-------+----------------------------------+
| 82fa9a47-246e-4da8-a864-53ea8daaed42 | router1 | ACTIVE | UP    | False    ␣
↪  | False | 35e3820f7490493ca9e3a5e685393298 |
+--------------------------------------+---------+--------+-------+----------
↪--+-------+----------------------------------+
vagrant@bionic64:~/devstack$ openstack router show router1
```

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```
+-------------------------+-----------------------------------------------------
↪--------------------------------------------------------------------------
↪-----------------------+
| Field                 | Value                                                ␣
↪                                                                            ␣
↪                       |
+-------------------------+-----------------------------------------------------
↪--------------------------------------------------------------------------
↪-----------------------+
| admin_state_up        | UP                                                   ␣
↪                                                                            ␣
↪                       |
| availability_zone_hints |                                                    ␣
↪                                                                            ␣
↪                       |
| availability_zones    | nova                                                 ␣
↪                                                                            ␣
↪                       |
| created_at            | 2016-11-08T21:55:30Z                                 ␣
↪                                                                            ␣
↪                       |
| description           |                                                      ␣
↪                                                                            ␣
↪                       |
| distributed           | False                                                ␣
↪                                                                            ␣
↪                       |
| external_gateway_info   | {"network_id": "6ece2847-971b-487a-9c7b-
↪184651ebbc82", "enable_snat": true, "external_fixed_ips": [{"subnet_id":
↪"0d9c4261-4046-462f-                    |
|                       | 9d92-64fb89bc3ae6", "ip_address": "172.24.4.7"}, {
↪"subnet_id": "9e90b059-da97-45b8-8cb8-f9370217e181", "ip_address":
↪"2001:db8::1"}]}                   |
| flavor_id             | None                                                 ␣
↪                                                                            ␣
↪                       |
| ha                    | False                                                ␣
↪                                                                            ␣
↪                       |
| id                    | 82fa9a47-246e-4da8-a864-53ea8daaed42                 ␣
↪                                                                            ␣
↪                       |
| name                  | router1                                              ␣
↪                                                                            ␣
↪                       |
| project_id            | 35e3820f7490493ca9e3a5e685393298                     ␣
↪                                                                            ␣
↪                       |
| revision_number       | 8                                                    ␣
```

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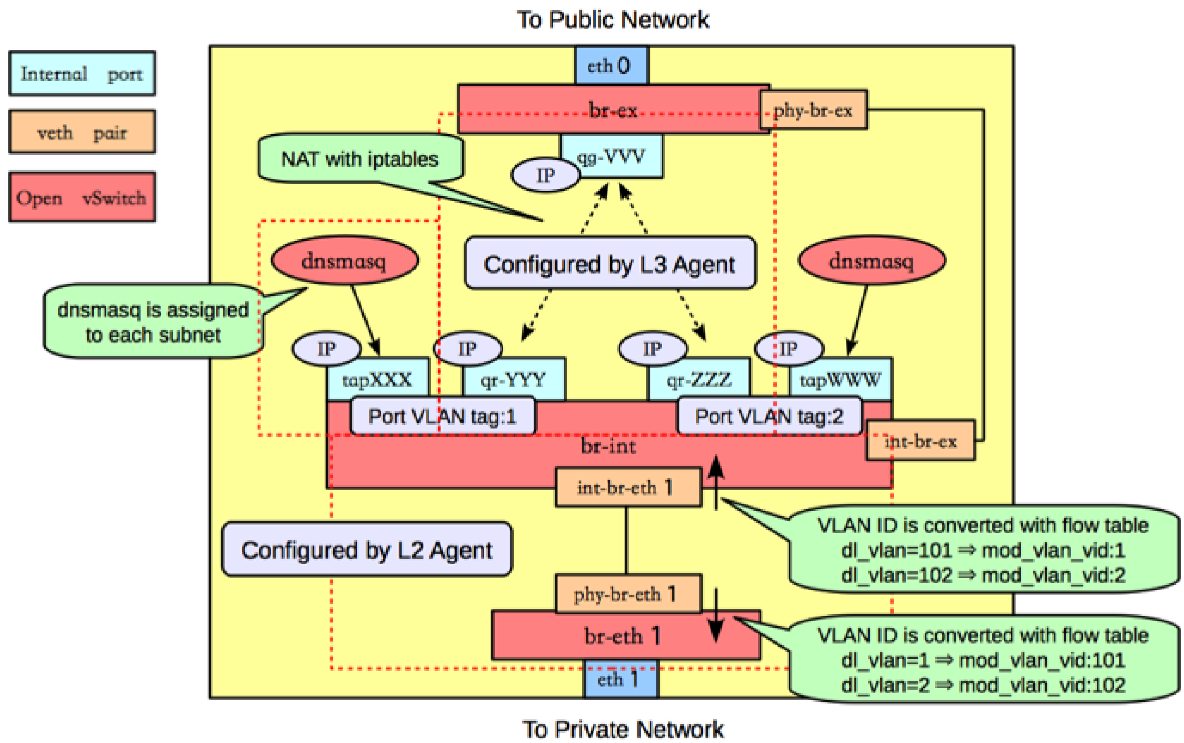
```
↪                                                                              ␣
↪                         |
| routes                  |                                                    ␣
↪                                                                              ␣
↪                         |
| status                  | ACTIVE                                             ␣
↪                                                                              ␣
↪                         |
| updated_at              | 2016-11-08T21:55:51Z                               ␣
↪                                                                              ␣
↪                         |
+-------------------------+----------------------------------------------------
↪---------------------------------------------------------------------------
↪-----------------------+
vagrant@bionic64:~/devstack$ openstack port list --router router1
+--------------------------------------+------+-------------------+-----------
↪---------------------------------------------------------------------+-----
↪---+
| ID                                   | Name | MAC Address       | Fixed IP␣
↪Addresses                                                           |␣
↪Status |
+--------------------------------------+------+-------------------+-----------
↪---------------------------------------------------------------------+-----
↪---+
| 420abb60-2a5a-4e80-90a3-3ff47742dc53 |      | fa:16:3e:2d:5c:4e | ip_
↪address='172.24.4.7', subnet_id='0d9c4261-4046-462f-9d92-64fb89bc3ae6'     ␣
↪ | ACTIVE |
|                                      |      |                   | ip_
↪address='2001:db8::1', subnet_id='9e90b059-da97-45b8-8cb8-f9370217e181'    ␣
↪ |        |
| b42d789d-c9ed-48a1-8822-839c4599301e |      | fa:16:3e:0a:ff:24 | ip_
↪address='10.0.0.1', subnet_id='c5c9f5c2-145d-46d2-a513-cf675530eaed'       ␣
↪ | ACTIVE |
| e3b7fede-277e-4c72-b66c-418a582b61ca |      | fa:16:3e:13:dd:42 | ip_
↪address='2001:db8:8000::1', subnet_id='6fa3bab9-103e-45d5-872c-91f21b52ceda
↪' | ACTIVE |
+--------------------------------------+------+-------------------+-----------
↪---------------------------------------------------------------------+-----
↪---+
```

See the Networking Guide for more detail on the creation of networks, subnets, and routers.

Neutron Routers are realized in OpenVSwitch



router1 in the Neutron logical network is realized through a port (qr-0ba8700e-da) in OpenVSwitch - attached to br-int:

```
vagrant@bionic64:~/devstack$ sudo ovs-vsctl show
b9b27fc3-5057-47e7-ba64-0b6afe70a398
    Bridge br-int
        Port "qr-0ba8700e-da"
            tag: 1
            Interface "qr-0ba8700e-da"
                type: internal
        Port br-int
            Interface br-int
                type: internal
        Port int-br-ex
            Interface int-br-ex
        Port "tapbb60d1bb-0c"
            tag: 1
            Interface "tapbb60d1bb-0c"
                type: internal
        Port "qvob2044570-ad"
            tag: 1
            Interface "qvob2044570-ad"
        Port "int-br-eth1"
            Interface "int-br-eth1"
    Bridge "br-eth1"
        Port "phy-br-eth1"
```

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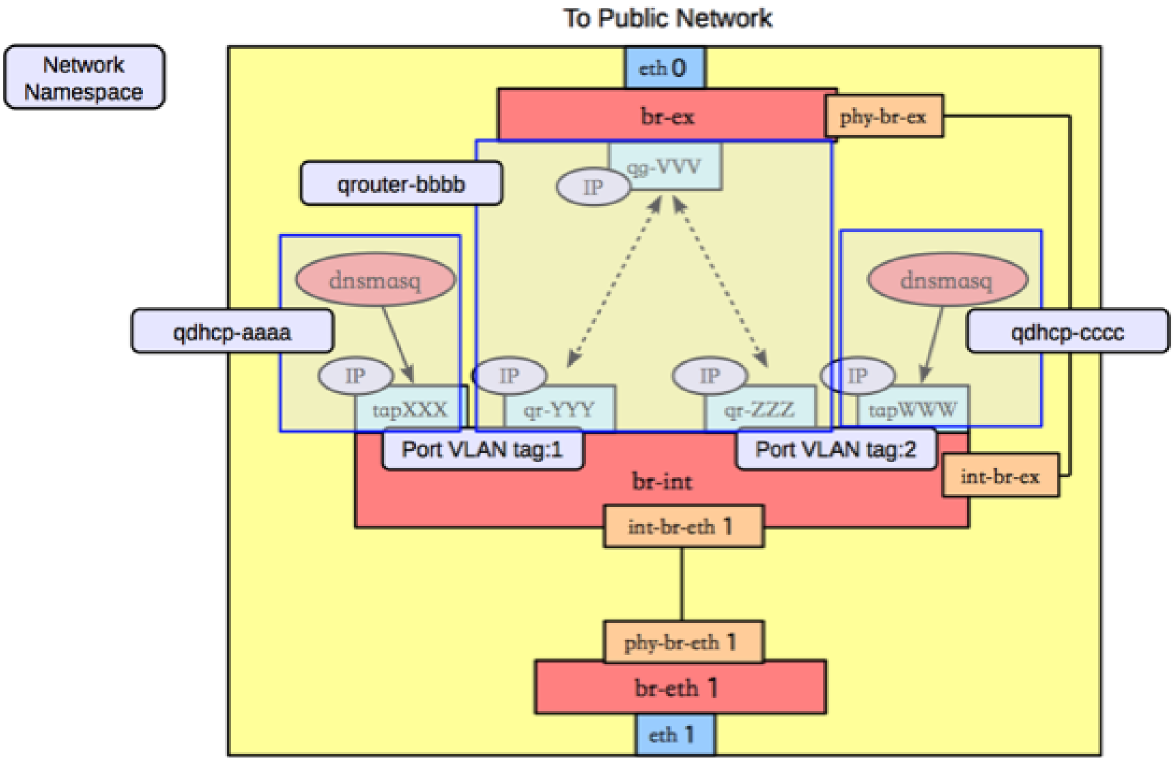
```
            Interface "phy-br-eth1"
        Port "br-eth1"
            Interface "br-eth1"
                type: internal
    Bridge br-ex
        Port phy-br-ex
            Interface phy-br-ex
        Port "qg-0143bce1-08"
            Interface "qg-0143bce1-08"
                type: internal
        Port br-ex
            Interface br-ex
                type: internal
    ovs_version: "1.4.0+build0"


vagrant@bionic64:~/devstack$ brctl show
bridge name         bridge id           STP enabled    interfaces
br-eth1             0000.e2e7fc5ccb4d   no
br-ex               0000.82ee46beaf4d   no             phy-br-ex
                                                       qg-41efb3f9-f0
                                                       qg-77e0666b-cd
br-int              0000.5e46cb509849   no             int-br-ex
                                                       qr-54c9cd83-43
                                                       qvo199abeb2-63
                                                       qvo1abbbb60-b8
                                                       tap74b45335-cc
qbr199abeb2-63      8000.ba06e5f8675c   no             qvb199abeb2-63
                                                       tap199abeb2-63
qbr1abbbb60-b8      8000.46a87ed4fb66   no             qvb1abbbb60-b8
                                                       tap1abbbb60-b8
virbr0              8000.000000000000   yes
```

Finding the router in ip/ipconfig

The neutron-l3-agent uses the Linux IP stack and iptables to perform L3 forwarding and NAT. In order to support multiple routers with potentially overlapping IP addresses, neutron-l3-agent defaults to using Linux network namespaces to provide isolated forwarding contexts. As a result, the IP addresses of routers will not be visible simply by running ip addr list or ifconfig on the node. Similarly, you will not be able to directly ping fixed IPs.

To do either of these things, you must run the command within a particular routers network namespace. The namespace will have the name qrouter-<UUID of the router>.



For example:

```
vagrant@bionic64:~$ openstack router list
+--------------------------------------+---------+--------+---------------------------
↪----------------------------------------------+
| ID                                   | Name    | Status | State |␣
↪Distributed | HA    | Project                          |
+--------------------------------------+---------+--------+---------------------------
↪----------------------------------------------+
| ad948c6e-afb6-422a-9a7b-0fc44cbb3910 | router1 | Active | UP    | True      ␣
↪   | False | 35e3820f7490493ca9e3a5e685393298 |
+--------------------------------------+---------+--------+---------------------------
↪----------------------------------------------+
vagrant@bionic64:~/devstack$ sudo ip netns exec qrouter-ad948c6e-afb6-422a-
↪9a7b-0fc44cbb3910 ip addr list
18: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
19: qr-54c9cd83-43: <BROADCAST,MULTICAST,PROMISC,UP,LOWER_UP> mtu 1500 qdisc␣
↪noqueue state UNKNOWN
    link/ether fa:16:3e:dd:c1:8f brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.1/24 brd 10.0.0.255 scope global qr-54c9cd83-43
    inet6 fe80::f816:3eff:fedd:c18f/64 scope link
       valid_lft forever preferred_lft forever
20: qg-77e0666b-cd: <BROADCAST,MULTICAST,PROMISC,UP,LOWER_UP> mtu 1500 qdisc␣
↪noqueue state UNKNOWN
```

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```
    link/ether fa:16:3e:1f:d3:ec brd ff:ff:ff:ff:ff:ff
    inet 192.168.27.130/28 brd 192.168.27.143 scope global qg-77e0666b-cd
    inet6 fe80::f816:3eff:fe1f:d3ec/64 scope link
       valid_lft forever preferred_lft forever
```

Provider Networking

Neutron can also be configured to create provider networks.

L3 agent extensions

See *L3 Agent Extensions*.

Further Reading

- Packet Pushers - Neutron Network Implementation on Linux
- OpenStack Networking Guide
- Neutron - Layer 3 API extension
- Darragh OReilly - The Quantum L3 router and floating IPs

Live-migration

Lets consider a VM with one port migrating from host1 with nova-compute1, neutron-l2-agent1 and neutron-l3-agent1 to host2 with nova-compute2 and neutron-l2-agent2 and neutron-l3agent2.

Since the VM that is about to migrate is hosted by nova-compute1, nova sends the live-migration order to nova-compute1 through RPC.

Nova Live Migration consists of the following stages:

- Pre-live-migration
- Live-migration-operation
- Post-live-migration

Pre-live-migration actions

Nova-compute1 will first ask nova-compute2 to perform pre-live-migration actions with a synchronous RPC call. Nova-compute2 will use neutron REST API to retrieve the list of VMs ports. Then, it calls its vif driver to create the VMs port (VIF) using plug_vifs().

In the case Open vSwitch Hybrid plug is used, Neutron-l2-agent2 will detect this new VIF, request the device details from the neutron server and configure it accordingly. However, ports status wont change, since this port is not bound to nova-compute2.

Nova-compute1 calls setup_networks_on_hosts. This updates the Neutron ports binding:profile with the information of the target host. The port update RPC message sent out by Neutron server will be received by neutron-l3-agent2, which proactively sets up the DVR router.

If pre-live-migration fails, nova rollbacks and port is removed from host2. If pre-live-migration succeeds, nova proceeds with live-migration-operation.

Potential error cases related to networking

- Plugging the VIFs on host2 fails

  As Live migration operation was not yet started, the instance resides active on host1.

Live-migration-operation

Once nova-compute2 has performed pre-live-migration actions, nova-compute1 can start the live-migration. This results in the creation of the VM and its corresponding tap interface on node 2.

In the case Open vSwitch normal plug, linux bridge or MacVTap is being used, Neutron-l2-agent2 will detect this new tap device and configure it accordingly. However, ports status wont change, since this port is not bound to nova-compute2.

As soon as the instance is active on host2, the original instance on host1 gets removed and with it the corresponding tap device. Assuming OVS-hybrid plug is NOT used, Neutron-l2-agent1 detects the removal and tells the neutron server to set the ports status to DOWN state with RPC messages.

There is no rollback if failure happens in live-migration-operation stage. TBD: Error are handled by the post-live-migration stage.

Potential error cases related to networking

- Some host devices that are specified in the instance definition are not present on the target host. Migration fails before it really started. This can happen with MacVTap agent. See bug https://bugs.launchpad.net/bugs/1550400

Post-live-migration actions

Once live-migration succeeded, both nova-compute1 and nova-compute2 perform post-live-migration actions. Nova-compute1 which is aware of the success will send a RPC cast to nova-compute2 to tell it to perform post-live-migration actions.

On host2, nova-compute2 sends a REST call update_port(binding=host2, profile={}) to the neutron server to tell it to update the ports binding. This will clear the port binding information and move the ports status to DOWN. The ML2 plugin will then try to rebind the port according to its new host. This update_port REST call always triggers a port-update RPC fanout message to every neutron-l2-agent. Since neutron-l2-agent2 is now hosting the port, it will take this message into account and re-synchronize the port by asking the neutron server details about it through RPC messages. This will move the port from DOWN status to BUILD, and then back to ACTIVE. This update also removes the migrating_to value from the portbindng dictionary. Its not clearing it totally, like indicated by {}, but just removing the migrating_to key and value.

On host1, nova-compute1 calls its vif driver to unplug the VMs port.

Assuming, Open vSwitch Hybrid plug is used, Neutron-l2-agent1 detects the removal and tells the neutron server to set the ports status to DOWN state with RPC messages. For all other cases this happens as soon as the instance and its tap device got destroyed on host1, like described in *Live-migration-operation*.

If neutron didnt already processed the REST call update_port(binding=host2), the port status will effectively move to BUILD and then to DOWN. Otherwise, the port is bound to host2, and neutron wont change the port status since its not bound the host that is sending RPC messages.

There is no rollback if failure happens in post-live-migration stage. In the case of an error, the instance is set into ERROR state.

Potential error cases related to networking

- Portbinding for host2 fails

  If this happens, the vif_type of the port is set to binding_failed. When Nova tries to recreated the domain.xml on the migration target it will stumble over this invalid vif_type and fail. The instance is put into ERROR state.

Post-Copy Migration

Usually, Live Migration is executed as pre-copy migration. The instance is active on host1 until nearly all memory has been copied to host2. If a certain threshold of copied memory is met, the instance on the source gets paused, the rest of the memory copied over and the instance started on the target. The challenge with this approach is, that migration might take a infinite amount of time, when the instance is heavily writing to memory.

This issue gets solved with post-copy migration. At some point in time, the instance on host2 will be set to active, although still a huge amount of memory pages reside only on host1. The phase that starts now is called the post_copy phase. If the instance tries to access a memory page that has not yet been transferred, libvirt/qemu takes care of moving this page to the target immediately. New pages will only be written to the source. With this approach the migration operation takes a finite amount of time.

Today, the rebinding of the port from host1 to host2 happens in the post_live_migration phase, after migration finished. This is fine for the pre-copy case, as the time windows between the activation of the instance on the target and the binding of the port to the target is pretty small. This becomes more problematic for the post-copy migration case. The instance becomes active on the target pretty early but the portbinding still happens after migration finished. During this time window, the instance might not be reachable via the network. This should be solved with bug https://bugs.launchpad.net/nova/+bug/1605016

Error recovery

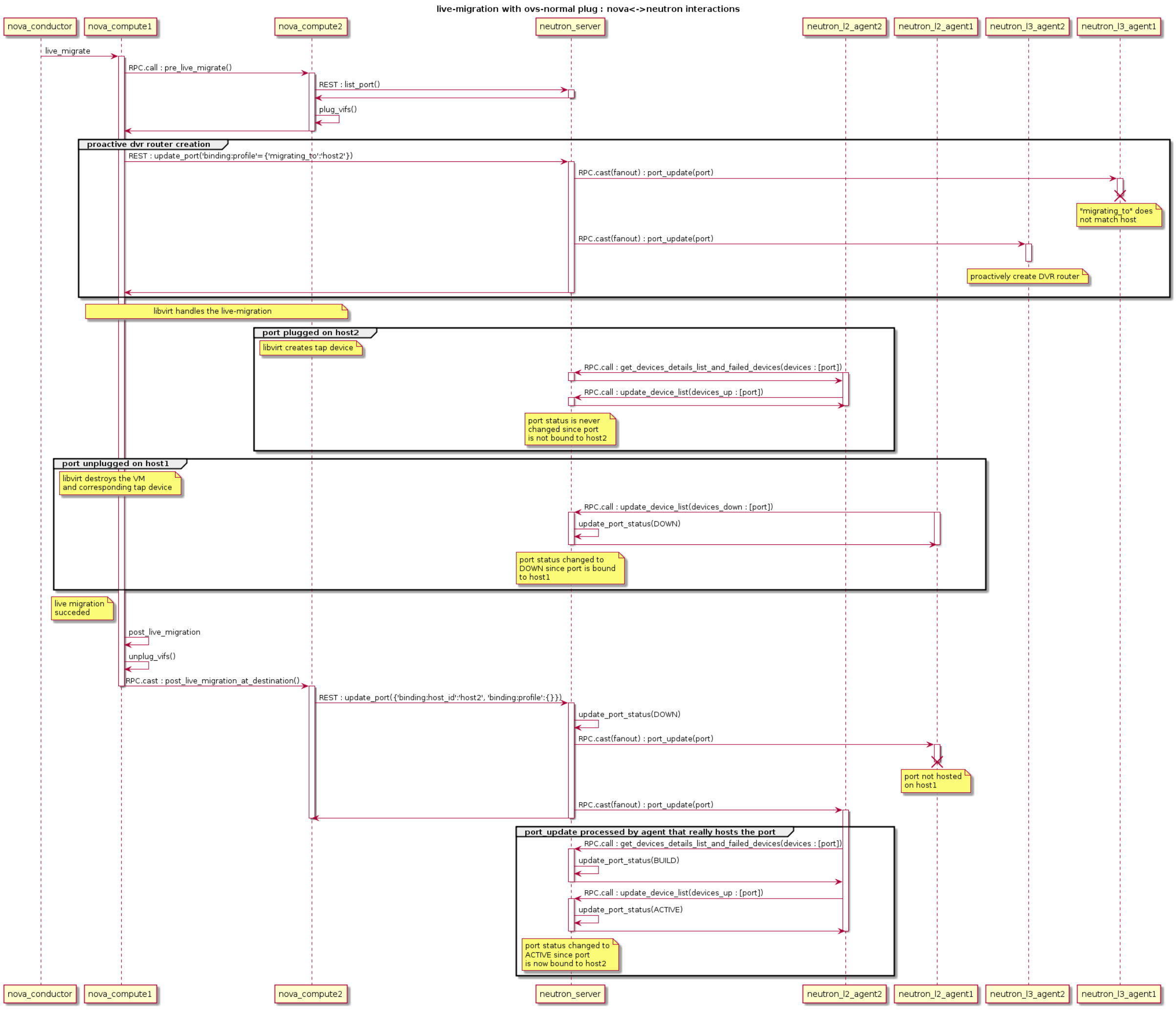
If the Live Migration fails, Nova will revert the operation. That implies deleting any object created in the database or in the destination compute node. However, in some cases have been reported the presence of duplicated port bindings per port. In this state, the port cannot be migrated until the inactive port binding (the failed destination host port binding) has been deleted.

To this end, the script `neutron-remove-duplicated-port-bindings` has been created. This script finds all duplicated port binding (that means, all port bindings that point to the same port) and deletes the inactive one.

> **Note**
>
> This script cannot be executed while a Live Migration or a cross cell Cold Migration. The script will delete the inactive port binding and will break the process.

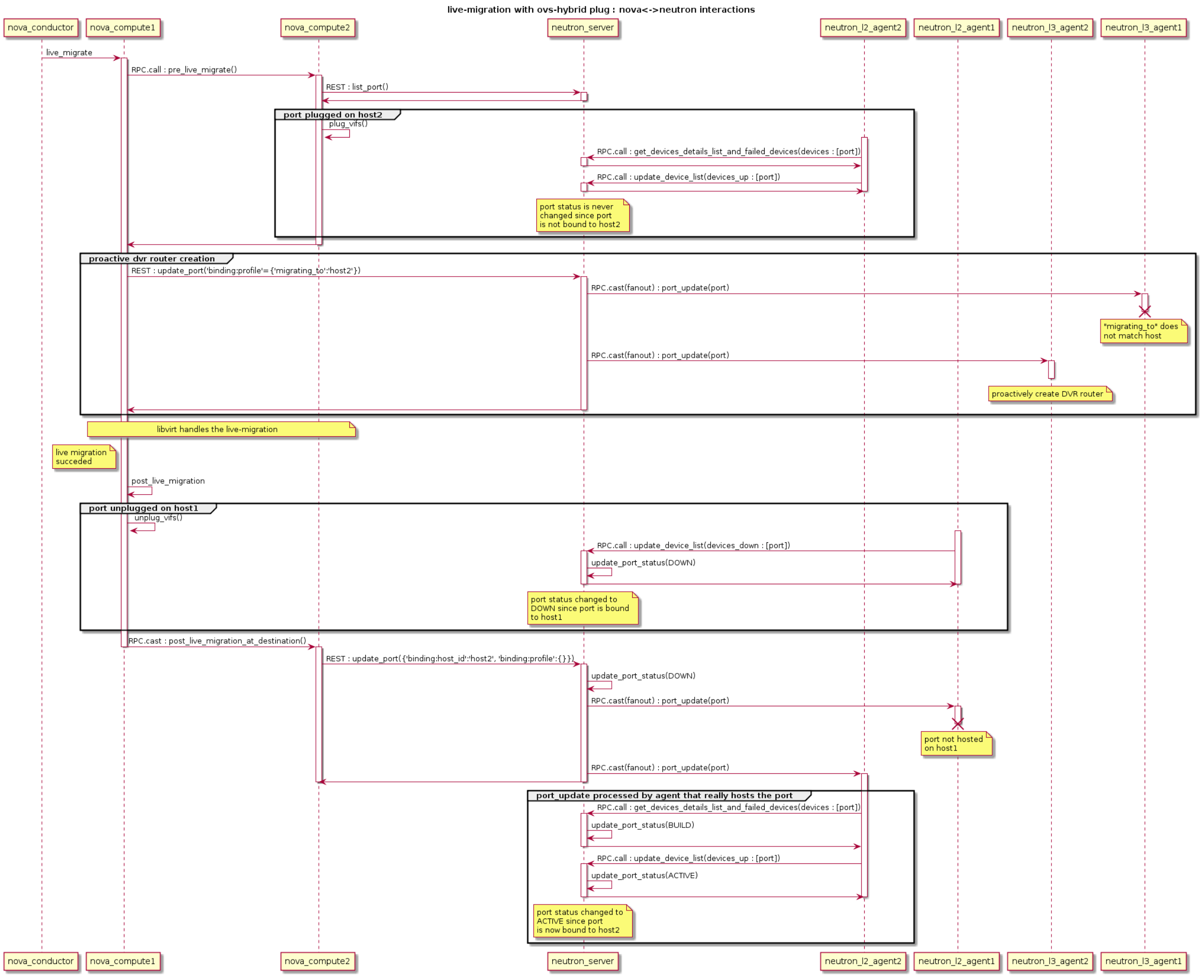
Flow Diagram

OVS Normal plug, Linux bridge, MacVTap, SR-IOV



OVS-Hybrid plug

The sequence with RPC messages from neutron-l2-agent processed first is described in the following UML sequence diagram



Local IP

Local IP is a virtual IP that can be shared across multiple ports/VMs (similar to anycast IP) and is guaranteed to only be reachable within the same physical server/node boundaries. The feature is primarily focused on high efficiency and performance of the networking data plane for very large scale clouds and/or clouds with high network throughput demands. Technically it is Neutron API/DB extension + openvswitch agent extension.

Usage

Usage is similar to Floating IP usage. If you want to assign a virtual Local IP to one of your VMs:

- first create Local IP object using network (name or ID) or local-port (name or ID) input parameter: it will be used to allocate/take IP address

```
$ openstack local ip create --network <network>
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| created_at        | 2021-12-01T13:50:24Z                 |
| description       |                                      |
| id                | b4425e9d-f1d0-4493-a2a8-1d3c7fbe049b |
| ip_mode           | translate                            |
| local_ip_address  | 172.24.4.10                          |
```

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```
| local_port_id     | 13181907-f258-4381-9516-ca07648ea239 |
| name              |                                      |
| network_id        | be0ec407-e341-4efa-a33a-3e0160afeedc |
| project_id        | b8462a1eba47462ea8c3e4e6adc22e63     |
| revision_number   | 0                                    |
| updated_at        | 2021-12-01T13:50:24Z                 |
+-------------------+--------------------------------------+
```

- then create Local IP association object using local-ip (name or ID) and fixed-port-id input parameters, thus assigning Local IP to the needed VM

```
$ openstack local ip association create <local-ip> <fixed-port-id>
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| fixed_ip          | 10.0.0.194                           |
| fixed_port_id     | 8cd37bb6-f7c3-4013-8d97-5c97676678a0 |
| host              |                                      |
| id                | None                                 |
| local_ip_address  | 172.24.4.10                          |
| name              | None                                 |
+-------------------+--------------------------------------+
```

- Unlike Floating IP you can have many Local IP associations: to VMs on different nodes.

Warning

Assigning two or more fixed ports to the same Local IP on the same node is currently not supported. NAT could go either way or not work at all.

All nodes VMs' egress traffic targeting IP address of Local IP object will be DNATed to local VM.

Note: if no Local IP is assigned on a node packets will be redirected to an underlying Neutron port IP address.

Note: in Yoga release only `translate` ip_mode is supported (default) - it means DNAT will be used for packet redirection. Support for `passthrough` mode (no modifications to IP packets) will be added in next releases.

OpenVSwitch Agent Impact

Unconditional changes

- 2 new OF tables are added for br-int:
  - LOCAL_EGRESS_TABLE - to save VLANs of local ports
  - LOCAL_IP_TABLE - for Local IP handling rules
- both tables has default rule to resubmit packets to TRANSIENT_TABLE;
- the only modification to packets flow is that egress packets will first go through empty LOCAL_EGRESS_TABLE before entering TRANSIENT_TABLE. This should be optimized by

OVS to have no impact on performance.

If local_ip agent extension is enabled

- LOCAL_EGRESS_TABLE will have a rule to save ports local VLAN to req6. This is needed in order to distinguish Local IPs from different nets. Then packets will be resubmitted to LOCAL_IP_TABLE which just has one default rule unless some local Port is associated with any Local IP.

If user creates Local IP Association with one of the ports owned by agent

Following rules will be added to LOCAL_SWITCHING table:

- local gARP blocker rule to prevent undesired Local IP ARP updates from other nodes (including real IP address owner)

Following rules will be added to LOCAL_IP_TABLE:

- local arp responder rule to answer local ARP requests for Local IP address
- Local IP translation flows to do actual DNAT (Local IP -> fixed IP)
  - via conntrack using `ct` with `nat` action if `static_nat` config option is *False* (default)
  - via static NAT rules with source/destination (ETH + IP + TCP/UDP ports) tuples used for learning back flows - if `static_nat` config is *True*

Yoga release limitations

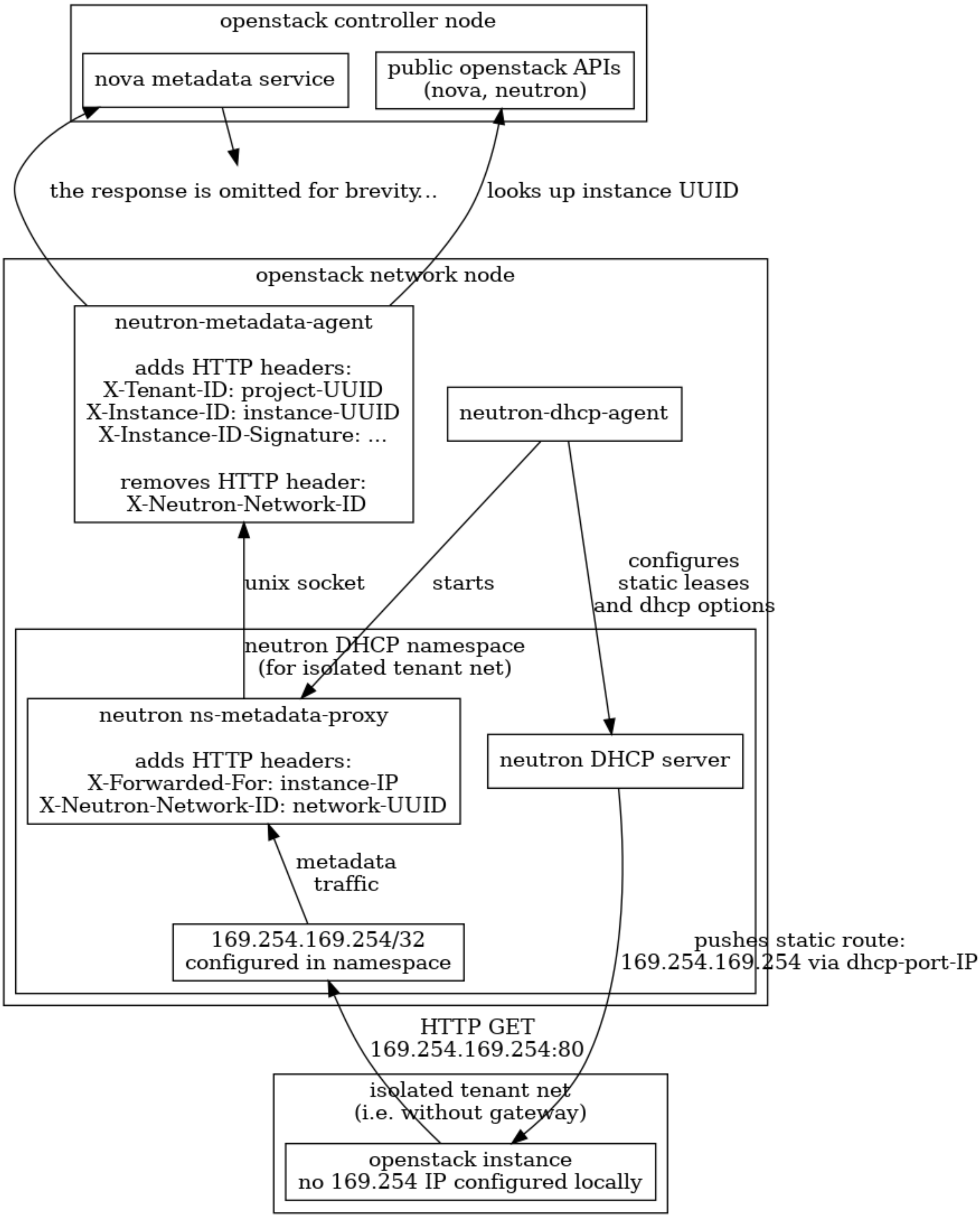
- Only IPv4 is supported. IPv6 support will be considered in future releases
- Only openvswitch ML2 mechanism driver/agent supports the feature
- No deterministic handling of packets if a node contains multiple local ports from same L2 segment associated with the same Local IP

Metadata Service Architectural Overview

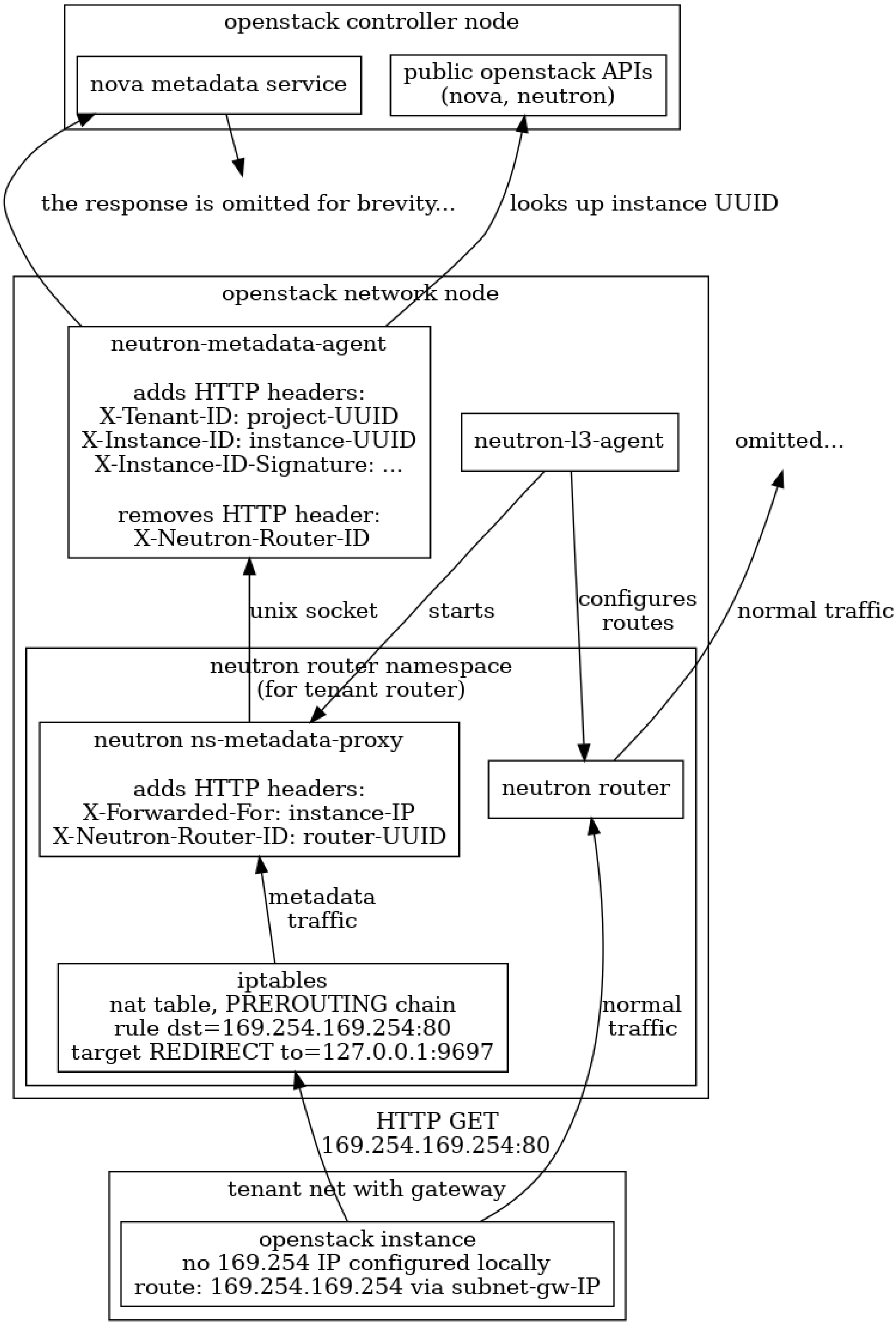
The following figures give an overview of the traditional implementation of the metadata service primarily focusing on the component view and the flow of information. There are two distinct figures depicting the metadata service as implemented in isolated networks or in networks with a router.

Please be aware that these figures are not complete. They do not apply to *ml2/ovn implementation* or to distributed metadata. They also omit details like IPv6 metadata or redundancy in the deployment.

Metadata on isolated networks - DHCP Agent



Metadata on networks with a router - L3 Agent



ML2 Extension Manager

The extension manager for ML2 was introduced in Juno (more details can be found in the approved spec). The features allows for extending ML2 resources without actually having to introduce cross cutting concerns to ML2. The mechanism has been applied for a number of use cases, and extensions that currently use this frameworks are available under ml2/extensions.

Network IP Availability Extension

This extension is an information-only API that allows a user or process to determine the amount of IPs that are consumed across networks and their subnets allocation pools. Each network and embedded subnet returns with values for **used_ips** and **total_ips** making it easy to determine how much of your networks IP space is consumed.

This API provides the ability for network administrators to periodically list usage (manual or automated) in order to preemptively add new network capacity when thresholds are exceeded.

Important Note:

This API tracks a networks consumable IPs. Whats the distinction? After a network and its subnets are created, consumable IPs are:

- Consumed in the subnets allocations (derives used IPs)
- Consumed from the subnets allocation pools (derives total IPs)

This API tracks consumable IPs so network administrators know when their subnets IP pools (and ultimately a networks) IPs are about to run out. This API does not account reserved IPs such as a subnets gateway IP or other reserved or unused IPs of a subnets cidr that are consumed as a result of the subnet creation itself.

API Specification

Availability for all networks

GET /v2.0/network-ip-availabilities

```
Request to url: v2.0/network-ip-availabilities
  headers: {'content-type': 'application/json', 'X-Auth-Token': 'SOME_AUTH_
↪TOKEN'}
```

Example response

```
Response:
  HTTP/1.1 200 OK
  Content-Type: application/json; charset=UTF-8
```

```
{
    "network_ip_availabilities": [
        {
            "network_id": "f944c153-3f46-417b-a3c2-487cd9a456b9",
            "network_name": "net1",
            "subnet_ip_availability": [
                {
                    "cidr": "10.0.0.0/24",
```

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```
                    "ip_version": 4,
                    "subnet_id": "46b1406a-8373-454c-8eb8-500a09eb77fb",
                    "subnet_name": "",
                    "total_ips": 253,
                    "used_ips": 3
                }
            ],
            "tenant_id": "test-project",
            "total_ips": 253,
            "used_ips": 3
        },
        {
            "network_id": "47035bae-4f29-4fef-be2e-2941b72528a8",
            "network_name": "net2",
            "subnet_ip_availability": [],
            "tenant_id": "test-project",
            "total_ips": 0,
            "used_ips": 0
        },
        {
            "network_id": "2e3ea0cd-c757-44bf-bb30-42d038687e3f",
            "network_name": "net3",
            "subnet_ip_availability": [
                {
                    "cidr": "40.0.0.0/24",
                    "ip_version": 4,
                    "subnet_id": "aab6b35c-16b5-489c-a5c7-fec778273495",
                    "subnet_name": "",
                    "total_ips": 253,
                    "used_ips": 2
                }
            ],
            "tenant_id": "test-project",
            "total_ips": 253,
            "used_ips": 2
        }
    ]
}
```


Availability by network ID

GET /v2.0/network-ip-availabilities/{network_uuid}

```
Request to url: /v2.0/network-ip-availabilities/aba3b29b-c119-4b45-afbd-
↪88e500acd970
  headers: {'content-type': 'application/json', 'X-Auth-Token': 'SOME_AUTH_
↪TOKEN'}
```

Example response

```
Response:
  HTTP/1.1 200 OK
  Content-Type: application/json; charset=UTF-8
```


```
{
    "network_ip_availability": {
        "network_id": "f944c153-3f46-417b-a3c2-487cd9a456b9",
        "network_name": "net1",
        "subnet_ip_availability": [
            {
                "cidr": "10.0.0.0/24",
                "ip_version": 4,
                "subnet_name": "",
                "subnet_id": "46b1406a-8373-454c-8eb8-500a09eb77fb",
                "total_ips": 253,
                "used_ips": 3
            }
        ],
        "tenant_id": "test-project",
        "total_ips": 253,
        "used_ips": 3
    }
}
```


Supported Query Filters

This API currently supports the following query parameters:

- **network_id**: Returns availability for the network matching the network ID. Note: This query (?network_id={network_id_guid})is roughly equivalent to *Availability by network ID* section except it returns the plural response form as a list rather than as an item.
- **network_name**: Returns availability for network matching the provided name
- **tenant_id**: Returns availability for all networks owned by the provided project ID.
- **ip_version**: Filters network subnets by those supporting the supplied ip version. Values can be either 4 or 6.

Query filters can be combined to further narrow results and what is returned will match all criteria. When a parameter is specified more than once, it will return results that match both. Examples:

```
# Fetch IPv4 availability for a specific project uuid
GET /v2.0/network-ip-availabilities?ip_version=4&tenant_id=example-project-
↪uuid

# Fetch multiple networks by their ids
GET /v2.0/network-ip-availabilities?network_id=uuid_sample_1&network_id=uuid_
↪sample_2
```

Objects

Object versioning is a key concept in achieving rolling upgrades. Since its initial implementation by the nova community, a versioned object model has been pushed to an oslo library so that its benefits can be shared across projects.

Oslo VersionedObjects (aka OVO) is a database facade, where you define the middle layer between software and the database schema. In this layer, a versioned object per database resource is created with a strict data definition and version number. With OVO, when you change the database schema, the version of the object also changes and a backward compatible translation is provided. This allows different versions of software to communicate with one another (via RPC).

OVO is also commonly used for RPC payload versioning. OVO creates versioned dictionary messages by defining a strict structure and keeping strong typing. Because of it, you can be sure of what is sent and how to use the data on the receiving end.

Usage of objects

CRUD operations

Objects support CRUD operations: `create()`, `get_object()` and `get_objects()` (equivalent of `read`), `update()`, `delete()`, `update_objects()`, and `delete_objects()`. The nature of OVO is, when any change is applied, OVO tracks it. After calling `create()` or `update()`, OVO detects this and changed fields are saved in the database. Please take a look at simple object usage scenarios using example of DNSNameServer:

```
# to create an object, you can pass the attributes in constructor:
dns = DNSNameServer(context, address='asd', subnet_id='xxx', order=1)
dns.create()

# or you can create a dict and pass it as kwargs:
dns_data = {'address': 'asd', 'subnet_id': 'xxx', 'order': 1}
dns = DNSNameServer(context, **dns_data)
dns.create()

# for fetching multiple objects:
dnses = DNSNameServer.get_objects(context)
# will return list of all dns name servers from DB

# for fetching objects with substrings in a string field:
from neutron_lib.objects import utils as obj_utils
dnses = DNSNameServer.get_objects(context, address=obj_utils.StringContains(
↪'10.0.0'))
# will return list of all dns name servers from DB that has '10.0.0' in their␣
↪addresses

# to update fields:
dns = DNSNameServer.get_object(context, address='asd', subnet_id='xxx')
dns.order = 2
dns.update()

# if you don't care about keeping the object, you can execute the update
# without fetch of the object state from the underlying persistent layer
```

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```
count = DNSNameServer.update_objects(
    context, {'order': 3}, address='asd', subnet_id='xxx')

# to remove object with filter arguments:
filters = {'address': 'asd', 'subnet_id': 'xxx'}
DNSNameServer.delete_objects(context, **filters)
```

Filter, sort and paginate

The NeutronDbObject class has strict validation on which field sorting and filtering can happen. When calling get_objects(), count(), update_objects(), delete_objects() and objects_exist(), validate_filters() is invoked, to see if its a supported filter criterion (which is by default non-synthetic fields only). Additional filters can be defined using register_filter_hook_on_model(). This will add the requested string to valid filter names in object implementation. It is optional.

In order to disable filter validation, validate_filters=False needs to be passed as an argument in aforementioned methods. It was added because the default behaviour of the neutron API is to accept everything at API level and filter it out at DB layer. This can be used by out of tree extensions.

register_filter_hook_on_model() is a complementary implementation in the NeutronDbObject layer to DB layers neutron_lib.db.model_query.register_hook(), which adds support for extra filtering during construction of SQL query. When extension defines extra query hook, it needs to be registered using the objects register_filter_hook_on_model(), if it is not already included in the objects fields.

To limit or paginate results, Pager object can be used. It accepts sorts (list of (key, direction) tuples), limit, page_reverse and marker keywords.

```
# filtering

# to get an object based on primary key filter
dns = DNSNameServer.get_object(context, address='asd', subnet_id='xxx')

# to get multiple objects
dnses = DNSNameServer.get_objects(context, subnet_id='xxx')

filters = {'subnet_id': ['xxx', 'yyy']}
dnses = DNSNameServer.get_objects(context, **filters)

# do not validate filters
dnses = DNSNameServer.get_objects(context, validate_filters=False,
                                  fake_filter='xxx')

# count the dns servers for given subnet
dns_count = DNSNameServer.count(context, subnet_id='xxx')

# sorting
# direction True == ASC, False == DESC
direction = False
pager = Pager(sorts=[('order', direction)])
```

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```
dnses = DNSNameServer.get_objects(context, _pager=pager, subnet_id='xxx')
```

Defining your own object

In order to add a new object in neutron, you have to:

1. Create an object derived from `NeutronDbObject` (aka base object)
2. Add/reuse data model
3. Define fields

It is mandatory to define data model using `db_model` attribute from `NeutronDbObject`.

Fields should be defined using `oslo_versionobjects.fields` exposed types. If there is a special need to create a new type of field, you can use `common_types.py` in the `neutron.objects` directory. Example:

```
fields = {
    'id': common_types.UUIDField(),
    'name': obj_fields.StringField(),
    'subnetpool_id': common_types.UUIDField(nullable=True),
    'ip_version': common_types.IPVersionEnumField()
}
```

`VERSION` is mandatory and defines the version of the object. Initially, set the `VERSION` field to 1.0. Change `VERSION` if fields or their types are modified. When you change the version of objects being exposed via RPC, add method `obj_make_compatible(self, primitive, target_version)`. For example, if a new version introduces a new parameter, it needs to be removed for previous versions:

```
from oslo_utils import versionutils

def obj_make_compatible(self, primitive, target_version):
    _target_version = versionutils.convert_version_to_tuple(target_version)
    if _target_version < (1, 1):  # version 1.1 introduces "new_parameter"
        primitive.pop('new_parameter', None)
```

In the following example the object has changed an attribute definition. For example, in version 1.1 `description` is allowed to be `None` but not in version 1.0:

```
from oslo_utils import versionutils
from oslo_versionedobjects import exception

def obj_make_compatible(self, primitive, target_version):
    _target_version = versionutils.convert_version_to_tuple(target_version)
    if _target_version < (1, 1):  # version 1.1 changes "description"
        if primitive['description'] is None:
            # "description" was not nullable before
            raise exception.IncompatibleObjectVersion(
                objver=target_version, objname='OVOName')
```

Using the first example as reference, this is how the unit test can be implemented:

```
def test_object_version_degradation_1_1_to_1_0(self):
    OVO_obj_1_1 = self._method_to_create_this_OVO()
    OVO_obj_1_0 = OVO_obj_1_1.obj_to_primitive(target_version='1.0')

    self.assertNotIn('new_parameter', OVO_obj_1_0['versioned_object.data'])
```

Note

Standard Attributes are automatically added to OVO fields in base class. Attributes[1] like `description`, `created_at`, `updated_at` and `revision_number` are added in[2].

`primary_keys` is used to define the list of fields that uniquely identify the object. In case of database backed objects, its usually mapped onto SQL primary keys. For immutable object fields that cannot be changed, there is a `fields_no_update` list, that contains `primary_keys` by default.

If there is a situation where a field needs to be named differently in an object than in the database schema, you can use `fields_need_translation`. This dictionary contains the name of the field in the object definition (the key) and the name of the field in the database (the value). This allows to have a different object layer representation for database persisted data. For example in IP allocation pools:

```
fields_need_translation = {
    'start': 'first_ip',  # field_ovo: field_db
    'end': 'last_ip'
}
```

The above dictionary is used in `modify_fields_from_db()` and in `modify_fields_to_db()` methods which are implemented in base class and will translate the software layer to database schema naming, and vice versa. It can also be used to rename `orm.relationship` backed object-type fields.

Most object fields are usually directly mapped to database model attributes. Sometimes its useful to expose attributes that are not defined in the model table itself, like relationships and such. In this case, `synthetic_fields` may become handy. This object property can define a list of object fields that dont belong to the object database model and that are hence instead to be implemented in some custom way. Some of those fields map to `orm.relationships` defined on models, while others are completely untangled from the database layer.

When exposing existing `orm.relationships` as an ObjectField-typed field, you can use the `foreign_keys` object property that defines a link between two object types. When used, it allows objects framework to automatically instantiate child objects, and fill the relevant parent fields, based on `orm.relationships` defined on parent models. In order to automatically populate the `synthetic_fields`, the `foreign_keys` property is introduced. `load_synthetic_db_fields()`[3] method from NeutronDbObject uses `foreign_keys` to match the foreign key in related object and local field that the foreign key is referring to. See simplified examples:

```
class DNSNameServerSqlModel(model_base.BASEV2):
    address = sa.Column(sa.String(128), nullable=False, primary_key=True)
    subnet_id = sa.Column(sa.String(36),
```

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[1] https://opendev.org/openstack/neutron/src/tag/ocata-eol/neutron/objects/base.py#L258
[2] https://opendev.org/openstack/neutron/src/tag/ocata-eol/neutron/db/standard_attr.py
[3] https://opendev.org/openstack/neutron/src/tag/ocata-eol/neutron/objects/base.py#L516

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```
                          sa.ForeignKey('subnets.id', ondelete="CASCADE"),
                          primary_key=True)

class SubnetSqlModel(model_base.BASEV2, HasId, HasProject):
    name = sa.Column(sa.String(attr.NAME_MAX_LEN))
    allocation_pools = orm.relationship(IPAllocationPoolSqlModel)
    dns_nameservers = orm.relationship(DNSNameServerSqlModel,
                                       backref='subnet',
                                       cascade='all, delete, delete-orphan',
                                       lazy='subquery')

class IPAllocationPoolSqlModel(model_base.BASEV2, HasId):
    subnet_id = sa.Column(sa.String(36), sa.ForeignKey('subnets.id'))

@obj_base.VersionedObjectRegistry.register
class DNSNameServerOVO(base.NeutronDbObject):
    VERSION = '1.0'
    db_model = DNSNameServerSqlModel

    # Created based on primary_key=True in model definition.
    # The object is uniquely identified by the pair of address and
    # subnet_id fields. Override the default 'id' 1-tuple.
    primary_keys = ['address', 'subnet_id']

    # Allow to link DNSNameServerOVO child objects into SubnetOVO parent
    # object fields via subnet_id child database model attribute.
    # Used during loading synthetic fields in SubnetOVO get_objects.
    foreign_keys = {'SubnetOVO': {'subnet_id': 'id'}}

    fields = {
        'address': obj_fields.StringField(),
        'subnet_id': common_types.UUIDField(),
    }

@obj_base.VersionedObjectRegistry.register
class SubnetOVO(base.NeutronDbObject):
    VERSION = '1.0'
    db_model =  SubnetSqlModel

    fields = {
        'id': common_types.UUIDField(),  # HasId from model class
        'project_id': obj_fields.StringField(nullable=True),  # HasProject␣
↪from model class
        'subnet_name': obj_fields.StringField(nullable=True),
        'dns_nameservers': obj_fields.ListOfObjectsField('DNSNameServer',
                                                         nullable=True),
        'allocation_pools': obj_fields.ListOfObjectsField('IPAllocationPoolOVO
↪',
                                                          nullable=True)
```

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```
    }

    # Claim dns_nameservers field as not directly mapped into the object
    # database model table.
    synthetic_fields = ['allocation_pools', 'dns_nameservers']

    # Rename in-database subnet_name attribute into name object field
    fields_need_translation = {
        'name': 'subnet_name'
    }


@obj_base.VersionedObjectRegistry.register
class IPAllocationPoolOVO(base.NeutronDbObject):
    VERSION = '1.0'
    db_model = IPAllocationPoolSqlModel

    fields = {
        'subnet_id': common_types.UUIDField()
    }

    foreign_keys = {'SubnetOVO': {'subnet_id': 'id'}}
```

The `foreign_keys` is used in `SubnetOVO` to populate the `allocation_pools`[4] synthetic field using the `IPAllocationPoolOVO` class. Single object type may be linked to multiple parent object types, hence `foreign_keys` property may have multiple keys in the dictionary.

Note

`foreign_keys` is declared in related object `IPAllocationPoolOVO`, the same way as its done in the SQL model `IPAllocationPoolSqlModel`: `sa.ForeignKey('subnets.id')`

Note

Only single foreign key is allowed (usually parent ID), you cannot link through multiple model attributes.

It is important to remember about the nullable parameter. In the SQLAlchemy model, the nullable parameter is by default `True`, while for OVO fields, the nullable is set to `False`. Make sure you correctly map database column nullability properties to relevant object fields.

Synthetic fields

`synthetic_fields` is a list of fields, that are not directly backed by corresponding object SQL table attributes. Synthetic fields are not limited in types that can be used to implement them.

[4] https://opendev.org/openstack/neutron/src/tag/ocata-eol/neutron/objects/base.py#L542

```
fields = {
    'dhcp_agents': obj_fields.ObjectField('NetworkDhcpAgentBinding',
                                          nullable=True), # field that␣
↪contains another single NeutronDbObject of NetworkDhcpAgentBinding type
    'shared': obj_fields.BooleanField(default=False),
    'subnets': obj_fields.ListOfObjectsField('Subnet', nullable=True)
}

# All three fields do not belong to corresponding SQL table, and will be
# implemented in some object-specific way.
synthetic_fields = ['dhcp_agents', 'shared', 'subnets']
```

`ObjectField` and `ListOfObjectsField` take the name of object class as an argument.

Implementing custom synthetic fields

Sometimes you may want to expose a field on an object that is not mapped into a corresponding database model attribute, or its `orm.relationship`; or may want to expose a `orm.relationship` data in a format that is not directly mapped onto a child object type. In this case, here is what you need to do to implement custom getters and setters for the custom field. The custom method to load the synthetic fields can be helpful if the field is not directly defined in the database, OVO class is not suitable to load the data or the related object contains only the ID and property of the parent object, for example `subnet_id` and property of it: `is_external`.

In order to implement the custom method to load the synthetic field, you need to provide loading method in the OVO class and override the base class method `from_db_object()` and `obj_load_attr()`. The first one is responsible for loading the fields to object attributes when calling `get_object()` and `get_objects()`, `create()` and `update()`. The second is responsible for loading attribute when it is not set in object. Also, when you need to create related object with attributes passed in constructor, `create()` and `update()` methods need to be overwritten. Additionally `is_external` attribute can be exposed as a boolean, instead of as an object-typed field. When field is changed, but it doesnt need to be saved into database, `obj_reset_changes()` can be called, to tell OVO library to ignore that. Lets see an example:

```
@obj_base.VersionedObjectRegistry.register
class ExternalSubnet(base.NeutronDbObject):
    VERSION = '1.0'
    fields = {'subnet_id': common_types.UUIDField(),
              'is_external': obj_fields.BooleanField()}
    primary_keys = ['subnet_id']
    foreign_keys = {'Subnet': {'subnet_id': 'id'}}


@obj_base.VersionedObjectRegistry.register
class Subnet(base.NeutronDbObject):
    VERSION = '1.0'
    fields = {'external': obj_fields.BooleanField(nullable=True),}
    synthetic_fields = ['external']

    # support new custom 'external=' filter for get_objects family of
    # objects API
```

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```
    def __init__(self, context=None, **kwargs):
        super(Subnet, self).__init__(context, **kwargs)
        self.add_extra_filter_name('external')

    def create(self):
        fields = self.get_changes()
        with db_api.context_manager.writer.using(context):
            if 'external' in fields:
                ExternalSubnet(context, subnet_id=self.id,
                    is_external=fields['external']).create()
            # Call to super() to create the SQL record for the object, and
            # reload its fields from the database, if needed.
            super(Subnet, self).create()

    def update(self):
        fields = self.get_changes()
        with db_api.context_manager.writer.using(context):
            if 'external' in fields:
                # delete the old ExternalSubnet record, if present
                obj_db_api.delete_objects(
                    self.obj_context, ExternalSubnet.db_model,
                    subnet_id=self.id)
                # create the new intended ExternalSubnet object
                ExternalSubnet(context, subnet_id=self.id,
                    is_external=fields['external']).create()
            # calling super().update() will reload the synthetic fields
            # and also will update any changed non-synthetic fields, if any
            super(Subnet, self).update()

    # this method is called when user of an object accesses the attribute
    # and requested attribute is not set.
    def obj_load_attr(self, attrname):
        if attrname == 'external':
            return self._load_external()
        # it is important to call super if attrname does not match
        # because the base implementation is handling the nullable case
        super(Subnet, self).obj_load_attr(attrname)

    def _load_external(self, db_obj=None):
        # do the loading here
        if db_obj:
            # use DB model to fetch the data that may be side-loaded
            external = db_obj.external.is_external if db_obj.external else␣
↪None
        else:
            # perform extra operation to fetch the data from DB
            external_obj = ExternalSubnet.get_object(context,
                subnet_id=self.id)
            external = external_obj.is_external if external_obj else None
```

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```
        # it is important to set the attribute and call obj_reset_changes
        setattr(self, 'external', external)
        self.obj_reset_changes(['external'])

    # this is defined in NeutronDbObject and is invoked during get_object(s)
    # and create/update.
    def from_db_object(self, obj):
        super(Subnet, self).from_db_object(obj)
        self._load_external(obj)
```

In the above example, the `get_object(s)` methods do not have to be overwritten, because `from_db_object()` takes care of loading the synthetic fields in custom way.

Standard attributes

The standard attributes are added automatically in metaclass `DeclarativeObject`. If adding standard attribute, it has to be added in `neutron/objects/extensions/standardattributes.py`. It will be added to all relevant objects that use the `standardattributes` model. Be careful when adding something to the above, because it could trigger a change in the objects `VERSION`. For more on how standard attributes work, check[5].

RBAC handling in objects

The RBAC is implemented currently for resources like: Subnet(*), Network and QosPolicy. Subnet is a special case, because access control of Subnet depends on Network RBAC entries.

The RBAC support for objects is defined in `neutron/objects/rbac_db.py`. It defines new base class `NeutronRbacObject`. The new class wraps standard `NeutronDbObject` methods like `create()`, `update()` and `to_dict()`. It checks if the `shared` attribute is defined in the `fields` dictionary and adds it to `synthetic_fields`. Also, `rbac_db_model` is required to be defined in Network and QosPolicy classes.

`NeutronRbacObject` is a common place to handle all operations on the RBAC entries, like getting the info if resource is shared or not, creation and updates of them. By wrapping the `NeutronDbObject` methods, it is manipulating the shared attribute while `create()` and `update()` methods are called.

The example of defining the Network OVO:

```
class Network(standard_attr.HasStandardAttributes, model_base.BASEV2,
          model_base.HasId, model_base.HasProject):
    """Represents a v2 neutron network."""
    name = sa.Column(sa.String(attr.NAME_MAX_LEN))
    rbac_entries = orm.relationship(rbac_db_models.NetworkRBAC,
                                    backref='network', lazy='joined',
                                    cascade='all, delete, delete-orphan')


# Note the base class for Network OVO:
@obj_base.VersionedObjectRegistry.register
```

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[5] https://docs.openstack.org/neutron/latest/contributor/internals/db_layer.html#the-standard-attribute-table

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```
class Network(rbac_db.NeutronRbacObject):
    # Version 1.0: Initial version
    VERSION = '1.0'

    # rbac_db_model is required to be added here
    rbac_db_model = rbac_db_models.NetworkRBAC
    db_model = models_v2.Network

    fields = {
        'id': common_types.UUIDField(),
        'project_id': obj_fields.StringField(nullable=True),
        'name': obj_fields.StringField(nullable=True),
        # share is required to be added to fields
        'shared': obj_fields.BooleanField(default=False),
    }
```

Note

The `shared` field is not added to the `synthetic_fields`, because `NeutronRbacObject` requires to add it by itself, otherwise `ObjectActionError` is raised.[6]

Extensions to neutron resources

One of the methods to extend neutron resources is to add an arbitrary value to dictionary representing the data by providing `extend_(subnet|port|network)_dict()` function and defining loading method.

From DB perspective, all the data will be loaded, including all declared fields from DB relationships. Current implementation for core resources (Port, Subnet, Network etc.) is that DB result is parsed by `make_<resource>_dict()` and `extend_<resource>_dict()`. When extension is enabled, `extend_<resource>_dict()` takes the DB results and declares new fields in resulting dict. When extension is not enabled, data will be fetched, but will not be populated into resulting dict, because `extend_<resource>_dict()` will not be called.

Plugins can still use objects for some work, but then convert them to dicts and work as they please, extending the dict as they wish.

For example:

```
class TestSubnetExtension(model_base.BASEV2):
    subnet_id = sa.Column(sa.String(36),
                          sa.ForeignKey('subnets.id', ondelete="CASCADE"),
                          primary_key=True)
    value = sa.Column(sa.String(64))
    subnet = orm.relationship(
        models_v2.Subnet,
        # here is the definition of loading the extension with Subnet model:
        backref=orm.backref('extension', cascade='delete', uselist=False))
```

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[6] https://opendev.org/openstack/neutron/src/tag/ocata-eol/neutron/objects/rbac_db.py#L291

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```
@oslo_obj_base.VersionedObjectRegistry.register_if(False)
class TestSubnetExtensionObject(obj_base.NeutronDbObject):
    # Version 1.0: Initial version
    VERSION = '1.0'

    db_model = TestSubnetExtension

    fields = {
        'subnet_id': common_types.UUIDField(),
        'value': obj_fields.StringField(nullable=True)
    }

    primary_keys = ['subnet_id']
    foreign_keys = {'Subnet': {'subnet_id': 'id'}}


@obj_base.VersionedObjectRegistry.register
class Subnet(base.NeutronDbObject):
    # Version 1.0: Initial version
    VERSION = '1.0'

    fields = {
        'id': common_types.UUIDField(),
        'extension': obj_fields.ObjectField(TestSubnetExtensionObject.__name__
↪,
                                            nullable=True),
    }

    synthetic_fields = ['extension']


# when defining the extend_subnet_dict function:
def extend_subnet_dict(self, session, subnet_ovo, result):
    value = subnet_ovo.extension.value if subnet_ovo.extension else ''
    result['subnet_extension'] = value
```

The above example is the ideal situation, where all extensions have objects adopted and enabled in core neutron resources.

By introducing the OVO work in tree, interface between base plugin code and registered extension functions hasnt been changed. Those still receive a SQLAlchemy model, not an object. This is achieved by capturing the corresponding database model on `get_***/create/update`, and exposing it via `<object>.db_obj`

Removal of downgrade checks over time

While the code to check object versions is meant to remain for a long period of time, in the interest of not accruing too much cruft over time, they are not intended to be permanent. OVO downgrade code should account for code that is within the upgrade window of any major OpenStack distribution. The longest currently known is for Ubuntu Cloud Archive which is to upgrade four versions, meaning during the upgrade the control nodes would be running a release that is four releases newer than what is running on the computes.

Known fast forward upgrade windows are:

- Red Hat OpenStack Platform (RHOSP): X -> X+3[7]
- Ubuntu Cloud Archive: X -> X+4[8]

Therefore removal of OVO version downgrade code should be removed in the fifth cycle after the code was introduced. For example, if an object version was introduced in Ocata then it can be removed in Train.

Backward compatibility for tenant_id

All objects can support `tenant_id` and `project_id` filters and fields at the same time; it is automatically enabled for all objects that have a `project_id` field. The base `NeutronDbObject` class has support for exposing `tenant_id` in dictionary access to the object fields (`subnet['tenant_id']`) and in `to_dict()` method. There is a `tenant_id` read-only property for every object that has `project_id` in `fields`. It is not exposed in `obj_to_primitive()` method, so it means that `tenant_id` will not be sent over RPC callback wire. When talking about filtering/sorting by `tenant_id`, the filters should be converted to expose `project_id` field. This means that for the long run, the API layer should translate it, but as temporary workaround it can be done at DB layer before passing filters to objects `get_objects()` method, for example:

```
def convert_filters(result):
    if 'tenant_id' in result:
        result['project_id'] = result.pop('tenant_id')
    return result

def get_subnets(context, filters):
    filters = convert_filters(**filters)
    return subnet_obj.Subnet.get_objects(context, **filters)
```

The `convert_filters` method is available in `neutron_lib.objects.utils`[9].

References

Open vSwitch Firewall Driver

The OVS driver has the same API as the current iptables firewall driver, keeping the state of security groups and ports inside of the firewall. Class `SGPortMap` was created to keep state consistent, and maps from ports to security groups and vice-versa. Every port and security group is represented by its own object encapsulating the necessary information.

> **Note**
>
> Open vSwitch firewall driver uses `register 5` for identifying the port related to the flow and `register 6` which identifies the network, used in particular for conntrack zones.

[7] https://access.redhat.com/support/policy/updates/openstack/platform/
[8] https://www.ubuntu.com/about/release-cycle
[9] https://opendev.org/openstack/neutron-lib/src/neutron_lib/objects/utils.py

Ingress/Egress Terminology

In this document, the terms `ingress` and `egress` are relative to a VM instance connected to OVS (or a netns connected to OVS):

- `ingress` applies to traffic that will ultimately go into a VM (or into a netns), assuming it is not dropped
- `egress` applies to traffic coming from a VM (or from a netns)

```
                  .                                     .
           _______|\                             _______|\
          \ ingress \                           \ ingress \
          /_______  /                           /_______  /
                  |/         .-----------------.        |/
                  '          |                 |        '
                             |                 |-----------( netns interface )
( non-VM, non-netns     )---|       OVS       |
( interface: phy, patch )   |                 |------------( VM interface )
            .                |                 |   .
           /|________        '-----------------'  /|________
          /   egress /                           /   egress /
          \  ________\                           \  ________\
           \|                                     \|
            '                                      '
```

Note that these terms are used differently in OVS code and documentation, where they are relative to the OVS bridge, with `ingress` applying to traffic as it comes into the OVS bridge, and `egress` applying to traffic as it leaves the OVS bridge.

Firewall API calls

There are two main calls performed by the firewall driver in order to either create or update a port with security groups - `prepare_port_filter` and `update_port_filter`. Both methods rely on the security group objects that are already defined in the driver and work similarly to their iptables counterparts. The definition of the objects will be described later in this document. `prepare_port_filter` must be called only once during port creation, and it defines the initial rules for the port. When the port is updated, all filtering rules are removed, and new rules are generated based on the available information about security groups in the driver.

Security group rules can be defined in the firewall driver by calling `update_security_group_rules`, which rewrites all the rules for a given security group. If a remote security group is changed, then `update_security_group_members` is called to determine the set of IP addresses that should be allowed for this remote security group. Calling this method will not have any effect on existing instance ports. In other words, if the port is using security groups and its rules are changed by calling one of the above methods, then no new rules are generated for this port. `update_port_filter` must be called for the changes to take effect.

All the machinery above is controlled by security group RPC methods, which mean the firewall driver doesnt have any logic of which port should be updated based on the provided changes, it only accomplishes actions when called from the controller.

OpenFlow rules

At first, every connection is split into ingress and egress processes based on the input or output port respectively. Each port contains the initial hardcoded flows for ARP, DHCP and established connections, which are accepted by default. To detect established connections, a flow must by marked by conntrack first with an `action=ct()` rule. An accepted flow means that ingress packets for the connection are directly sent to the port, and egress packets are left to be normally switched by the integration bridge.

> **Note**
>
> There is a new config option `explicitly_egress_direct`, if it is set to True, it will direct egress unicast traffic to the local port directly or to the patch bridge port if the destination is in a remote host. So there is no NORMAL for egress in such scenario. This option is used to overcome the egress packet flooding when the openflow firewall is enabled.

Connections that are not matched by the above rules are sent to either the ingress or egress filtering table, depending on its direction. The reason the rules are based on security group rules in separate tables is to make it easy to detect these rules during removal.

Security group rules are treated differently for those without a remote group ID and those with a remote group ID. A security group rule without a remote group ID is expanded into several OpenFlow rules by the method `create_flows_from_rule_and_port`. A security group rule with a remote group ID is expressed by three sets of flows. The first two are conjunctive flows which will be described in the next section. The third set matches on the conjunction IDs and does accept actions.

Flow priorities for security group rules

The OpenFlow spec says a packet should not match against multiple flows at the same priority[1]. The firewall driver uses 8 levels of priorities to achieve this. The method `flow_priority_offset` calculates a priority for a given security group rule. The use of priorities is essential with conjunction flows, which will be described later in the conjunction flows examples.

Uses of conjunctive flows

With a security group rule with a remote group ID, flows that match on nw_src for remote_group_id addresses and match on dl_dst for port MAC addresses are needed (for ingress rules; likewise for egress rules). Without conjunction, this results in O(n*m) flows where n and m are number of ports in the remote group ID and the port security group, respectively.

A conj_id is allocated for each (remote_group_id, security_group_id, direction, ethertype, flow_priority_offset) tuple. The class `ConjIdMap` handles the mapping. The same conj_id is shared between security group rules if multiple rules belong to the same tuple above.

Conjunctive flows consist of 2 dimensions. Flows that belong to the dimension 1 of 2 are generated by the method `create_flows_for_ip_address` and are in charge of IP address based filtering specified by their remote group IDs. Flows that belong to the dimension 2 of 2 are generated by the method `create_flows_from_rule_and_port` and modified by the method `substitute_conjunction_actions`, which represents the portion of the rule other than its remote group ID.

Those dimension 2 of 2 flows are per port and contain no remote group information. When there are

[1] Although OVS seems to magically handle overlapping flows under some cases, we shouldnt rely on that.

multiple security group rules for a port, those flows can overlap. To avoid such a situation, flows are sorted and fed to `merge_port_ranges` or `merge_common_rules` methods to rearrange them.

Rules example with explanation:

The following example presents two ports on the same host. They have different security groups and there is ICMP traffic allowed from the first security group to the second security group. Ports have the following attributes:

```
Port 1
  - plugged to the port 1 in OVS bridge
  - IP address: 192.168.0.1
  - MAC address: fa:16:3e:a4:22:10
  - security group 1: can send ICMP packets out
  - allowed address pair: 10.0.0.1/32, fa:16:3e:8c:84:13

Port 2
  - plugged to the port 2 in OVS bridge
  - IP address: 192.168.0.2
  - MAC address: fa:16:3e:24:57:c7
  - security group 2:
    - can receive ICMP packets from security group 1
    - can receive TCP packets from security group 1
    - can receive TCP packets to port 80 from security group 2
    - can receive IP packets from security group 3
  - allowed address pair: 10.1.0.0/24, fa:16:3e:8c:84:14

Port 3
  - patch bridge port (e.g. patch-tun) in OVS bridge
```

`table 0` (LOCAL_SWITCHING) - `table 59` (PACKET_RATE_LIMIT) contain some low priority rules to continue packet processing in `table 60` (TRANSIENT) aka TRANSIENT table. `table 0` (LOCAL_SWITCHING) - `table 59` (PACKET_RATE_LIMIT) is left for use to other features that take precedence over firewall, e.g. DVR, ARP poison/spoofing prevention, MAC spoof filtering and packet rate limitation etc. The only requirement is that after such a feature is done with its processing, it needs to pass packets for processing to the TRANSIENT table. This TRANSIENT table distinguishes the ingress traffic from the egress traffic and loads into `register 5` a value identifying the port (for egress traffic based on the switch port number, and for ingress traffic based on the network id and destination MAC address); `register 6` contains a value identifying the network (which is also the OVSDB port tag) to isolate connections into separate conntrack zones. For VLAN networks, the physical VLAN tag will be used to act as an extra match rule to do such identifying work as well.

```
table=60,  priority=100,in_port=1 actions=load:0x1->NXM_NX_REG5[],load:0x284->
↪NXM_NX_REG6[],resubmit(,71)
table=60,  priority=100,in_port=2 actions=load:0x2->NXM_NX_REG5[],load:0x284->
↪NXM_NX_REG6[],resubmit(,71)
table=60,  priority=90,dl_vlan=0x284,dl_dst=fa:16:3e:a4:22:10␣
↪actions=load:0x1->NXM_NX_REG5[],load:0x284->NXM_NX_REG6[],resubmit(,81)
table=60,  priority=90,dl_vlan=0x284,dl_dst=fa:16:3e:8c:84:13␣
↪actions=load:0x1->NXM_NX_REG5[],load:0x284->NXM_NX_REG6[],resubmit(,81)
table=60,  priority=90,dl_vlan=0x284,dl_dst=fa:16:3e:24:57:c7␣
```

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```
↪actions=load:0x2->NXM_NX_REG5[],load:0x284->NXM_NX_REG6[],resubmit(,81)
table=60,  priority=90,dl_vlan=0x284,dl_dst=fa:16:3e:8c:84:14␣
↪actions=load:0x2->NXM_NX_REG5[],load:0x284->NXM_NX_REG6[],resubmit(,81)
table=60,  priority=0 actions=NORMAL
```

The following table, `table 71` (BASE_EGRESS) implements ARP spoofing protection, IP spoofing protection, allows traffic related to IP address allocations (DHCP, DHCPv6, SLAAC, NDP) for egress traffic, and allows ARP replies. Also identifies not tracked connections which are processed later with information obtained from conntrack. Notice the `zone=NXM_NX_REG6[0..15]` in `actions` when obtaining information from conntrack. It says every port has its own conntrack zone defined by the value in `register 6` (OVSDB port tag identifying the network). Its there to avoid accepting established traffic that belongs to a different port with the same conntrack parameters.

The very first rule in `table 71` (BASE_EGRESS) is a rule removing conntrack information for a use-case where a Neutron logical port is placed directly to the hypervisor. In such cases the kernel does conntrack lookup before the packet reaches the Open vSwitch bridge. Tracked packets are sent back for processing by the same table after conntrack information is cleared.

```
table=71, priority=110,ct_state=+trk actions=ct_clear,resubmit(,71)
```

Rules below allow ICMPv6 traffic for multicast listeners, neighbour solicitation and neighbour advertisement.

```
table=71, priority=95,icmp6,reg5=0x1,in_port=1,dl_src=fa:16:3e:a4:22:11,ipv6_
↪src=fe80::11,icmp_type=130 actions=resubmit(,94)
table=71, priority=95,icmp6,reg5=0x1,in_port=1,dl_src=fa:16:3e:a4:22:11,ipv6_
↪src=fe80::11,icmp_type=131 actions=resubmit(,94)
table=71, priority=95,icmp6,reg5=0x1,in_port=1,dl_src=fa:16:3e:a4:22:11,ipv6_
↪src=fe80::11,icmp_type=132 actions=resubmit(,94)
table=71, priority=95,icmp6,reg5=0x1,in_port=1,dl_src=fa:16:3e:a4:22:11,ipv6_
↪src=fe80::11,icmp_type=135 actions=resubmit(,94)
table=71, priority=95,icmp6,reg5=0x1,in_port=1,dl_src=fa:16:3e:a4:22:11,ipv6_
↪src=fe80::11,icmp_type=136 actions=resubmit(,94)
table=71, priority=95,icmp6,reg5=0x2,in_port=2,dl_src=fa:16:3e:a4:22:22,ipv6_
↪src=fe80::22,icmp_type=130 actions=resubmit(,94)
table=71, priority=95,icmp6,reg5=0x2,in_port=2,dl_src=fa:16:3e:a4:22:22,ipv6_
↪src=fe80::22,icmp_type=131 actions=resubmit(,94)
table=71, priority=95,icmp6,reg5=0x2,in_port=2,dl_src=fa:16:3e:a4:22:22,ipv6_
↪src=fe80::22,icmp_type=132 actions=resubmit(,94)
table=71, priority=95,icmp6,reg5=0x2,in_port=2,dl_src=fa:16:3e:a4:22:22,ipv6_
↪src=fe80::22,icmp_type=135 actions=resubmit(,94)
table=71, priority=95,icmp6,reg5=0x2,in_port=2,dl_src=fa:16:3e:a4:22:22,ipv6_
↪src=fe80::22,icmp_type=136 actions=resubmit(,94)
```

Following rules implement ARP spoofing protection

```
table=71, priority=95,arp,reg5=0x1,in_port=1,dl_src=fa:16:3e:a4:22:10,arp_
↪spa=192.168.0.1 actions=resubmit(,94)
table=71, priority=95,arp,reg5=0x1,in_port=1,dl_src=fa:16:3e:8c:84:13,arp_
↪spa=10.0.0.1 actions=resubmit(,94)
table=71, priority=95,arp,reg5=0x2,in_port=2,dl_src=fa:16:3e:24:57:c7,arp_
```

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```
↪spa=192.168.0.2 actions=resubmit(,94)
table=71, priority=95,arp,reg5=0x2,in_port=2,dl_src=fa:16:3e:8c:84:14,arp_
↪spa=10.1.0.0/24 actions=resubmit(,94)
```

DHCP and DHCPv6 traffic is allowed to instance but DHCP servers are blocked on instances.

```
table=71, priority=80,udp,reg5=0x1,in_port=1,tp_src=68,tp_dst=67␣
↪actions=resubmit(,73)
table=71, priority=80,udp6,reg5=0x1,in_port=1,tp_src=546,tp_dst=547␣
↪actions=resubmit(,73)
table=71, priority=70,udp,reg5=0x1,in_port=1,tp_src=67,tp_dst=68␣
↪actions=resubmit(,93)
table=71, priority=70,udp6,reg5=0x1,in_port=1,tp_src=547,tp_dst=546␣
↪actions=resubmit(,93)
table=71, priority=80,udp,reg5=0x2,in_port=2,tp_src=68,tp_dst=67␣
↪actions=resubmit(,73)
table=71, priority=80,udp6,reg5=0x2,in_port=2,tp_src=546,tp_dst=547␣
↪actions=resubmit(,73)
table=71, priority=70,udp,reg5=0x2,in_port=2,tp_src=67,tp_dst=68␣
↪actions=resubmit(,93)
table=71, priority=70,udp6,reg5=0x2,in_port=2,tp_src=547,tp_dst=546␣
↪actions=resubmit(,93)
```

Following rules obtain conntrack information for valid IP and MAC address combinations. All other packets are dropped.

```
table=71, priority=65,ip,reg5=0x1,in_port=1,dl_src=fa:16:3e:a4:22:10,nw_
↪src=192.168.0.1 actions=ct(table=72,zone=NXM_NX_REG6[0..15])
table=71, priority=65,ip,reg5=0x1,in_port=1,dl_src=fa:16:3e:8c:84:13,nw_
↪src=10.0.0.1 actions=ct(table=72,zone=NXM_NX_REG6[0..15])
table=71, priority=65,ip,reg5=0x2,in_port=2,dl_src=fa:16:3e:24:57:c7,nw_
↪src=192.168.0.2 actions=ct(table=72,zone=NXM_NX_REG6[0..15])
table=71, priority=65,ip,reg5=0x2,in_port=2,dl_src=fa:16:3e:8c:84:14,nw_
↪src=10.1.0.0/24 actions=ct(table=72,zone=NXM_NX_REG6[0..15])
table=71, priority=65,ipv6,reg5=0x1,in_port=1,dl_src=fa:16:3e:a4:22:10,ipv6_
↪src=fe80::f816:3eff:fea4:2210 actions=ct(table=72,zone=NXM_NX_REG6[0..15])
table=71, priority=65,ipv6,reg5=0x2,in_port=2,dl_src=fa:16:3e:24:57:c7,ipv6_
↪src=fe80::f816:3eff:fe24:57c7 actions=ct(table=72,zone=NXM_NX_REG6[0..15])
table=71, priority=10,reg5=0x1,in_port=1 actions=resubmit(,93)
table=71, priority=10,reg5=0x2,in_port=2 actions=resubmit(,93)
table=71, priority=0 actions=drop
```

`table 72` (RULES_EGRESS) accepts only established or related connections, and implements rules defined by security groups. As this egress connection might also be an ingress connection for some other port, its not switched yet but eventually processed by the ingress pipeline.

All established or new connections defined by security group rules are `accepted`, which will be explained later. All invalid packets are dropped. In the case below we allow all ICMP egress traffic.

```
table=72, priority=75,ct_state=+est-rel-rpl,icmp,reg5=0x1 actions=resubmit(,
↪73)
```

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```
table=72, priority=75,ct_state=+new-est,icmp,reg5=0x1 actions=resubmit(,73)
table=72, priority=50,ct_state=+inv+trk actions=resubmit(,93)
```

Important on the flows below is the `ct_mark=0x1`. Flows that were marked as not existing anymore by rule introduced later will value this value. Those are typically connections that were allowed by some security group rule and the rule was removed.

```
table=72, priority=50,ct_mark=0x1,reg5=0x1 actions=resubmit(,93)
table=72, priority=50,ct_mark=0x1,reg5=0x2 actions=resubmit(,93)
```

All other connections that are not marked and are established or related are allowed.

```
table=72, priority=50,ct_state=+est-rel+rpl,ct_zone=644,ct_mark=0,reg5=0x1␣
↪actions=resubmit(,94)
table=72, priority=50,ct_state=+est-rel+rpl,ct_zone=644,ct_mark=0,reg5=0x2␣
↪actions=resubmit(,94)
table=72, priority=50,ct_state=-new-est+rel-inv,ct_zone=644,ct_mark=0,
↪reg5=0x1 actions=resubmit(,94)
table=72, priority=50,ct_state=-new-est+rel-inv,ct_zone=644,ct_mark=0,
↪reg5=0x2 actions=resubmit(,94)
```

In the following, flows are marked established for connections that werent matched in the previous flows, which means they dont have an accepting security group rule anymore.

```
table=72, priority=40,ct_state=-est,reg5=0x1 actions=resubmit(,93)
table=72, priority=40,ct_state=+est,reg5=0x1 actions=ct(commit,zone=NXM_NX_
↪REG6[0..15],exec(load:0x1->NXM_NX_CT_MARK[]))
table=72, priority=40,ct_state=-est,reg5=0x2 actions=resubmit(,93)
table=72, priority=40,ct_state=+est,reg5=0x2 actions=ct(commit,zone=NXM_NX_
↪REG6[0..15],exec(load:0x1->NXM_NX_CT_MARK[]))
table=72, priority=0 actions=drop
```

In the following `table 73` (ACCEPT_OR_INGRESS) are all detected ingress connections sent to the ingress pipeline. Since the connection was already accepted by the egress pipeline, all remaining egress connections are sent to the normal floodlearn switching in `table 94` (ACCEPTED_EGRESS_TRAFFIC_NORMAL).

```
table=73, priority=100,reg6=0x284,dl_dst=fa:16:3e:a4:22:10 actions=load:0x1->
↪NXM_NX_REG5[],resubmit(,81)
table=73, priority=100,reg6=0x284,dl_dst=fa:16:3e:8c:84:13 actions=load:0x1->
↪NXM_NX_REG5[],resubmit(,81)
table=73, priority=100,reg6=0x284,dl_dst=fa:16:3e:24:57:c7 actions=load:0x2->
↪NXM_NX_REG5[],resubmit(,81)
table=73, priority=100,reg6=0x284,dl_dst=fa:16:3e:8c:84:14 actions=load:0x2->
↪NXM_NX_REG5[],resubmit(,81)
table=73, priority=90,ct_state=+new-est,reg5=0x1 actions=ct(commit,zone=NXM_
↪NX_REG6[0..15]),resubmit(,91)
table=73, priority=90,ct_state=+new-est,reg5=0x2 actions=ct(commit,zone=NXM_
↪NX_REG6[0..15]),resubmit(,91)
table=73, priority=80,reg5=0x1 actions=resubmit(,94)
```

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```
table=73, priority=80,reg5=0x2 actions=resubmit(,94)
table=73, priority=0 actions=drop
```

`table 81` (BASE_INGRESS) is similar to `table 71` (BASE_EGRESS), allows basic ingress traffic for obtaining IP address and ARP queries. Note that the VLAN tag must be removed by adding `strip_vlan` to actions list, prior to injecting packet directly to port. Not tracked packets are sent to obtain conntrack information.

```
table=81, priority=100,arp,reg5=0x1 actions=strip_vlan,output:1
table=81, priority=100,arp,reg5=0x2 actions=strip_vlan,output:2
table=81, priority=100,icmp6,reg5=0x1,icmp_type=130 actions=strip_vlan,
↪output:1
table=81, priority=100,icmp6,reg5=0x1,icmp_type=131 actions=strip_vlan,
↪output:1
table=81, priority=100,icmp6,reg5=0x1,icmp_type=132 actions=strip_vlan,
↪output:1
table=81, priority=100,icmp6,reg5=0x1,icmp_type=135 actions=strip_vlan,
↪output:1
table=81, priority=100,icmp6,reg5=0x1,icmp_type=136 actions=strip_vlan,
↪output:1
table=81, priority=100,icmp6,reg5=0x2,icmp_type=130 actions=strip_vlan,
↪output:2
table=81, priority=100,icmp6,reg5=0x2,icmp_type=131 actions=strip_vlan,
↪output:2
table=81, priority=100,icmp6,reg5=0x2,icmp_type=132 actions=strip_vlan,
↪output:2
table=81, priority=100,icmp6,reg5=0x2,icmp_type=135 actions=strip_vlan,
↪output:2
table=81, priority=100,icmp6,reg5=0x2,icmp_type=136 actions=strip_vlan,
↪output:2
table=81, priority=95,udp,reg5=0x1,tp_src=67,tp_dst=68 actions=strip_vlan,
↪output:1
table=81, priority=95,udp6,reg5=0x1,tp_src=547,tp_dst=546 actions=strip_vlan,
↪output:1
table=81, priority=95,udp,reg5=0x2,tp_src=67,tp_dst=68 actions=strip_vlan,
↪output:2
table=81, priority=95,udp6,reg5=0x2,tp_src=547,tp_dst=546 actions=strip_vlan,
↪output:2
table=81, priority=90,ct_state=-trk,ip,reg5=0x1 actions=ct(table=82,zone=NXM_
↪NX_REG6[0..15])
table=81, priority=90,ct_state=-trk,ipv6,reg5=0x1 actions=ct(table=82,
↪zone=NXM_NX_REG6[0..15])
table=81, priority=90,ct_state=-trk,ip,reg5=0x2 actions=ct(table=82,zone=NXM_
↪NX_REG6[0..15])
table=81, priority=90,ct_state=-trk,ipv6,reg5=0x2 actions=ct(table=82,
↪zone=NXM_NX_REG6[0..15])
table=81, priority=80,ct_state=+trk,reg5=0x1 actions=resubmit(,82)
table=81, priority=80,ct_state=+trk,reg5=0x2 actions=resubmit(,82)
table=81, priority=0 actions=drop
```

Similarly to `table 72` (RULES_EGRESS), `table 82` (RULES_INGRESS) accepts established and related connections. In this case we allow all ICMP traffic coming from `security group 1` which is in this case only `port 1`. The first four flows match on the IP addresses, and the next two flows match on the ICMP protocol. These six flows define conjunction flows, and the next two define actions for them.

```
table=82, priority=71,ct_state=+est-rel-rpl,ip,reg6=0x284,nw_src=192.168.0.1␣
↪actions=conjunction(18,1/2)
table=82, priority=71,ct_state=+est-rel-rpl,ip,reg6=0x284,nw_src=10.0.0.1␣
↪actions=conjunction(18,1/2)
table=82, priority=71,ct_state=+new-est,ip,reg6=0x284,nw_src=192.168.0.1␣
↪actions=conjunction(19,1/2)
table=82, priority=71,ct_state=+new-est,ip,reg6=0x284,nw_src=10.0.0.1␣
↪actions=conjunction(19,1/2)
table=82, priority=71,ct_state=+est-rel-rpl,icmp,reg5=0x2␣
↪actions=conjunction(18,2/2)
table=82, priority=71,ct_state=+new-est,icmp,reg5=0x2 actions=conjunction(19,
↪2/2)
table=82, priority=71,conj_id=18,ct_state=+est-rel-rpl,ip,reg5=0x2␣
↪actions=strip_vlan,output:2
table=82, priority=71,conj_id=19,ct_state=+new-est,ip,reg5=0x2␣
↪actions=ct(commit,zone=NXM_NX_REG6[0..15]),strip_vlan,output:2,resubmit(,92)
table=82, priority=50,ct_state=+inv+trk actions=resubmit(,93)
```

There are some more security group rules with remote group IDs. Next we look at TCP related ones. Excerpt of flows that correspond to those rules are:

```
table=82, priority=73,ct_state=+est-rel-rpl,tcp,reg5=0x2,tp_dst=0x60/0xffe0␣
↪actions=conjunction(22,2/2)
table=82, priority=73,ct_state=+new-est,tcp,reg5=0x2,tp_dst=0x60/0xffe0␣
↪actions=conjunction(23,2/2)
table=82, priority=73,ct_state=+est-rel-rpl,tcp,reg5=0x2,tp_dst=0x40/0xfff0␣
↪actions=conjunction(22,2/2)
table=82, priority=73,ct_state=+new-est,tcp,reg5=0x2,tp_dst=0x40/0xfff0␣
↪actions=conjunction(23,2/2)
table=82, priority=73,ct_state=+est-rel-rpl,tcp,reg5=0x2,tp_dst=0x58/0xfff8␣
↪actions=conjunction(22,2/2)
table=82, priority=73,ct_state=+new-est,tcp,reg5=0x2,tp_dst=0x58/0xfff8␣
↪actions=conjunction(23,2/2)
table=82, priority=73,ct_state=+est-rel-rpl,tcp,reg5=0x2,tp_dst=0x54/0xfffc␣
↪actions=conjunction(22,2/2)
table=82, priority=73,ct_state=+new-est,tcp,reg5=0x2,tp_dst=0x54/0xfffc␣
↪actions=conjunction(23,2/2)
table=82, priority=73,ct_state=+est-rel-rpl,tcp,reg5=0x2,tp_dst=0x52/0xfffe␣
↪actions=conjunction(22,2/2)
table=82, priority=73,ct_state=+new-est,tcp,reg5=0x2,tp_dst=0x52/0xfffe␣
↪actions=conjunction(23,2/2)
table=82, priority=73,ct_state=+est-rel-rpl,tcp,reg5=0x2,tp_dst=80␣
↪actions=conjunction(22,2/2),conjunction(14,2/2)
table=82, priority=73,ct_state=+est-rel-rpl,tcp,reg5=0x2,tp_dst=81␣
↪actions=conjunction(22,2/2)
table=82, priority=73,ct_state=+new-est,tcp,reg5=0x2,tp_dst=80␣
```

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```
↪actions=conjunction(23,2/2),conjunction(15,2/2)
table=82, priority=73,ct_state=+new-est,tcp,reg5=0x2,tp_dst=81␣
↪actions=conjunction(23,2/2)
```

Only dimension 2/2 flows are shown here, as the other are similar to the previous ICMP example. There are many more flows but only the port ranges that cover from 64 to 127 are shown for brevity.

The conjunction IDs 14 and 15 correspond to packets from the security group 1, and the conjunction IDs 22 and 23 correspond to those from the security group 2. These flows are from the following security group rules,

```
- can receive TCP packets from security group 1
- can receive TCP packets to port 80 from security group 2
```

and these rules have been processed by `merge_port_ranges` into:

```
- can receive TCP packets to port != 80 from security group 1
- can receive TCP packets to port 80 from security group 1 or 2
```

before translating to flows so that there is only one matching flow even when the TCP destination port is 80.

The remaining is a L4 protocol agnostic rule.

```
table=82, priority=70,ct_state=+est-rel-rpl,ip,reg5=0x2␣
↪actions=conjunction(24,2/2)
table=82, priority=70,ct_state=+new-est,ip,reg5=0x2 actions=conjunction(25,2/
↪2)
```

Any IP packet that matches the previous TCP flows matches one of these flows, but the corresponding security group rules have different remote group IDs. Unlike the above TCP example, theres no convenient way of expressing `protocol != TCP` or `icmp_code != 1`. So the OVS firewall uses a different priority than the previous TCP flows so as not to mix them up.

The mechanism for dropping connections that are not allowed anymore is the same as in `table 72` (RULES_EGRESS).

```
table=82, priority=50,ct_mark=0x1,reg5=0x1 actions=resubmit(,93)
table=82, priority=50,ct_mark=0x1,reg5=0x2 actions=resubmit(,93)
table=82, priority=50,ct_state=+est-rel+rpl,ct_zone=644,ct_mark=0,reg5=0x1␣
↪actions=strip_vlan,output:1
table=82, priority=50,ct_state=+est-rel+rpl,ct_zone=644,ct_mark=0,reg5=0x2␣
↪actions=strip_vlan,output:2
table=82, priority=50,ct_state=-new-est+rel-inv,ct_zone=644,ct_mark=0,
↪reg5=0x1 actions=strip_vlan,output:1
table=82, priority=50,ct_state=-new-est+rel-inv,ct_zone=644,ct_mark=0,
↪reg5=0x2 actions=strip_vlan,output:2
table=82, priority=40,ct_state=-est,reg5=0x1 actions=resubmit(,93)
table=82, priority=40,ct_state=+est,reg5=0x1 actions=ct(commit,zone=NXM_NX_
↪REG6[0..15],exec(load:0x1->NXM_NX_CT_MARK[]))
table=82, priority=40,ct_state=-est,reg5=0x2 actions=resubmit(,93)
table=82, priority=40,ct_state=+est,reg5=0x2 actions=ct(commit,zone=NXM_NX_
```

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```
↪REG6[0..15],exec(load:0x1->NXM_NX_CT_MARK[]))
table=82, priority=0 actions=drop
```

Note

Conntrack zones on a single node are now based on the network to which a port is plugged in. That makes a difference between traffic on hypervisor only and east-west traffic. For example, if a port has a VIP that was migrated to a port on a different node, then the new port wont contain conntrack information about previous traffic that happened with that VIP.

By default `table 94` (ACCEPTED_EGRESS_TRAFFIC_NORMAL) will have one single flow like this:

```
table=94, priority=1 actions=NORMAL
```

If `explicitly_egress_direct` is set to True, flows of `table 94` (ACCEPTED_EGRESS_TRAFFIC_NORMAL) will be:

```
table=94, priority=12,reg6=0x284,dl_dst=fa:16:3e:a4:22:10 actions=output:1
table=94, priority=12,reg6=0x284,dl_dst=fa:16:3e:24:57:c7 actions=output:2
table=94, priority=10,reg6=0x284,dl_src=fa:16:3e:a4:22:10,dl_
↪dst=00:00:00:00:00:00/01:00:00:00:00:00 actions=push_vlan:0x8100,set_
↪field:0x1->vlan_vid,output:3
table=94, priority=10,reg6=0x284,dl_src=fa:16:3e:24:57:c7,dl_
↪dst=00:00:00:00:00:00/01:00:00:00:00:00 actions=push_vlan:0x8100,set_
↪field:0x1->vlan_vid,output:3
table=94, priority=1 actions=NORMAL
```

The OVS firewall will initialize a default goto table 94 flow on TRANSIENT_TABLE `table 60` (TRANSIENT), if `explicitly_egress_direct` is set to True, which is mainly for ports without security groups and disabled port_security. For instance:

::

table=60, priority=2 actions=resubmit(,94)

Then for packets from the outside to VM without security functionalities (disable-port-security no-security-group) will go to table 94 and do the same direct actions.

OVS firewall integration points

There are three tables where packets are sent once after going through the OVS firewall pipeline. The tables can be used by other mechanisms that are supposed to work with the OVS firewall, typically L2 agent extensions.

Egress pipeline

Packets are sent to `table 91` (ACCEPTED_EGRESS_TRAFFIC) and `table 94` (ACCEPTED_EGRESS_TRAFFIC_NORMAL) when they are considered accepted by the egress pipeline, and they will be processed so that they are forwarded to their destination by being submitted to a NORMAL action, that results in Ethernet flood/learn processing.

Two tables are used to differentiate between the first packets of a connection and the following packets. This was introduced for performance reasons to allow the logging extension to only log the first packets of a connection. Only the first accepted packet of each connection session will go to `table 91` (ACCEPTED_EGRESS_TRAFFIC) and the following ones will go to `table 94` (ACCEPTED_EGRESS_TRAFFIC_NORMAL).

Note that `table 91` (ACCEPTED_EGRESS_TRAFFIC) merely resubmits to `table 94` (ACCEPTED_EGRESS_TRAFFIC_NORMAL) that contains the actual NORMAL action; this allows to have a single place where the NORMAL action can be overridden by other components (currently used by `networking-bagpipe` driver for `networking-bgpvpn`).

Ingress pipeline

The first packet of each connection accepted by the ingress pipeline is sent to `table 92` (ACCEPTED_INGRESS_TRAFFIC). The default action in this table is DROP because at this point the packets have already been delivered to their destination port. This integration point is essentially provided for the logging extension.

Packets are sent to `table 93` (DROPPED_TRAFFIC) if processing by the ingress filtering concluded that they should be dropped.

Upgrade path from iptables hybrid driver

During an upgrade, the agent will need to re-plug each instances tap device into the integration bridge while trying to not break existing connections. One of the following approaches can be taken:

1) Pause the running instance in order to prevent a short period of time where its network interface does not have firewall rules. This can happen due to the firewall driver calling OVS to obtain information about OVS the port. Once the instance is paused and no traffic is flowing, we can delete the qvo interface from integration bridge, detach the tap device from the qbr bridge and plug the tap device back into the integration bridge. Once this is done, the firewall rules are applied for the OVS tap interface and the instance is started from its paused state.

2) Set drop rules for the instances tap interface, delete the qbr bridge and related veths, plug the tap device into the integration bridge, apply the OVS firewall rules and finally remove the drop rules for the instance.

3) Compute nodes can be upgraded one at a time. A free node can be switched to use the OVS firewall, and instances from other nodes can be live-migrated to it. Once the first node is evacuated, its firewall driver can be then be switched to the OVS driver.

4) Once migration is complete, stale iptables rules should be cleaned-up on all nodes where the firewall driver was changed. They can be found by searching for the string neutron, for example:

```
sudo iptables -S | grep neutron
```

Note

During upgrading to openvswitch firewall, the security rules are still working for previous iptables controlled hybrid ports. But it will not work if one tries to replace openvswitch firewall with iptables.

OVN Design Notes

Mapping between Neutron and OVN data models

The primary job of the Neutron OVN ML2 driver is to translate requests for resources into OVNs data model. Resources are created in OVN by updating the appropriate tables in the OVN northbound database (an ovsdb database). This document looks at the mappings between the data that exists in Neutron and what the resulting entries in the OVN northbound DB would look like.

Network

```
Neutron Network:
    id
    name
    subnets
    admin_state_up
    status
    tenant_id
```

Once a network is created, we should create an entry in the Logical Switch table.

```
OVN northbound DB Logical Switch:
    external_ids: {
        'neutron:network_name': network.name
    }
```

Subnet

```
Neutron Subnet:
    id
    name
    ip_version
    network_id
    cidr
    gateway_ip
    allocation_pools
    dns_nameservers
    host_routers
    tenant_id
    enable_dhcp
    ipv6_ra_mode
    ipv6_address_mode
```

Once a subnet is created, we should create an entry in the DHCP Options table with the DHCPv4 or DHCPv6 options.

```
OVN northbound DB DHCP_Options:
    cidr
    options
    external_ids: {
```

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```
        'subnet_id': subnet.id
    }
```

Port

```
Neutron Port:
    id
    name
    network_id
    admin_state_up
    mac_address
    fixed_ips
    device_id
    device_owner
    tenant_id
    status
```

When a port is created, we should create an entry in the Logical Switch Ports table in the OVN northbound DB.

```
OVN Northbound DB Logical Switch Port:
    switch: reference to OVN Logical Switch
    router_port: (empty)
    name: port.id
    up: (read-only)
    macs: [port.mac_address]
    port_security:
    external_ids: {'neutron:port_name': port.name}
```

If the port has extra DHCP options defined, we should create an entry in the DHCP Options table in the OVN northbound DB.

```
OVN northbound DB DHCP_Options:
    cidr
    options
    external_ids: {
        'subnet_id': subnet.id,
        'port_id':  port.id
    }
```

Router

```
Neutron Router:
    id
    name
    admin_state_up
    status
    tenant_id
    external_gw_info:
```

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```
        network_id
        external_fixed_ips: list of dicts
            ip_address
            subnet_id
```

```
OVN Northbound DB Logical Router:
    ip:
    default_gw:
    external_ids:
```

Router Port

```
OVN Northbound DB Logical Router Port:
    router: (reference to Logical Router)
    network: (reference to network this port is connected to)
    mac:
    external_ids:
```

Security Groups

```
Neutron Port:
    id
    security_group: id
    network_id

Neutron Security Group
    id
    name
    tenant_id
    security_group_rules

Neutron Security Group Rule
    id
    tenant_id
    security_group_id
    direction
    remote_group_id
    ethertype
    protocol
    port_range_min
    port_range_max
    remote_ip_prefix
```

```
OVN Northbound DB ACL Rule:
    lswitch:  (reference to Logical Switch - port.network_id)
    priority: (0..65535)
    match: boolean expressions according to security rule
           Translation map (sg_rule  ==> match expression)
```

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```
        ----------------------------------------------
        sg_rule.direction="Ingress" => "inport=port.id"
        sg_rule.direction="Egress" => "outport=port.id"
        sg_rule.ethertype => "eth.type"
        sg_rule.protocol => "ip.proto"
        sg_rule.port_range_min/port_range_max  =>
                "port_range_min &lt;= tcp.src &lt;= port_range_max"
                "port_range_min &lt;= udp.src &lt;= port_range_max"

        sg_rule.remote_ip_prefix => "ip4.src/mask, ip4.dst/mask, ipv6.src/mask, ipv6.dst/mask"

        (all match options for ACL can be found here:
         http://openvswitch.org/support/dist-docs/ovn-nb.5.html)
    action: "allow-related"
    log: true/false
    external_ids: {'neutron:port_id': port.id}
                  {'neutron:security_rule_id': security_rule.id}
```

Security groups maps between three neutron objects to one OVN-NB object, this enable us to do the mapping in various ways, depending on OVN capabilities

The current implementation will use the first option in this list for simplicity, but all options are kept here for future reference

1) For every <neutron port, security rule> pair, define an ACL entry:

```
Leads to many ACL entries.
acl.match = sg_rule converted
example: ((inport==port.id) && (ip.proto == "tcp") &&
         (1024 &lt;= tcp.src &lt;= 4095) && (ip.src==192.168.0.1/16))

external_ids: {'neutron:port_id': port.id}
              {'neutron:security_rule_id': security_rule.id}
```

2) For every <neutron port, security group> pair, define an ACL entry:

```
Reduce the number of ACL entries.
Means we have to manage the match field in case specific rule changes
example: (((inport==port.id) && (ip.proto == "tcp") &&
         (1024 &lt;= tcp.src &lt;= 4095) && (ip.src==192.168.0.1/16)) ||
         ((outport==port.id) && (ip.proto == "udp") && (1024 &lt;= tcp.src &lt;= 4095)) ||
         ((inport==port.id) && (ip.proto == 6) ) ||
         ((inport==port.id) && (eth.type == 0x86dd)))

         (This example is a security group with four security rules)

external_ids: {'neutron:port_id': port.id}
              {'neutron:security_group_id': security_group.id}
```

3) For every <lswitch, security group> pair, define an ACL entry:

```
Reduce even more the number of ACL entries.
Manage complexity increase
example: (((inport==port.id) && (ip.proto == "tcp") && (1024 &lt;= tcp.
↪src &lt;= 4095)
          && (ip.src==192.168.0.1/16)) ||
         ((outport==port.id) && (ip.proto == "udp") && (1024 &lt;= tcp.
↪src &lt;= 4095)) ||
         ((inport==port.id) && (ip.proto == 6) ) ||
         ((inport==port.id) && (eth.type == 0x86dd))) ||

         (((inport==port2.id) && (ip.proto == "tcp") && (1024 &lt;= tcp.
↪src &lt;= 4095)
         && (ip.src==192.168.0.1/16)) ||
         ((outport==port2.id) && (ip.proto == "udp") && (1024 &lt;= tcp.
↪src &lt;= 4095)) ||
         ((inport==port2.id) && (ip.proto == 6) ) ||
         ((inport==port2.id) && (eth.type == 0x86dd)))

external_ids: {'neutron:security_group': security_group.id}
```

Which option to pick depends on OVN match field length capabilities, and the trade off between better performance due to less ACL entries compared to the complexity to manage them.

If the default behaviour is not drop for unmatched entries, a rule with lowest priority must be added to drop all traffic (match==1)

Spoofing protection rules are being added by OVN internally and we need to ignore the automatically added rules in Neutron

Using the native DHCP feature provided by OVN

DHCPv4

OVN implements a native DHCPv4 support which caters to the common use case of providing an IP address to a booting instance by providing stateless replies to DHCPv4 requests based on statically configured address mappings. To do this it allows a short list of DHCPv4 options to be configured and applied at each compute host running ovn-controller.

OVN northbound db provides a table DHCP_Options to store the DHCP options. Logical switch port has a reference to this table.

When a subnet is created and enable_dhcp is True, a new entry is created in this table. The options column stores the DHCPv4 options. These DHCPv4 options are included in the DHCPv4 reply by the ovn-controller when the VIF attached to the logical switch port sends a DHCPv4 request.

In order to map the DHCP_Options row with the subnet, the OVN ML2 driver stores the subnet id in the external_ids column.

When a new port is created, the dhcpv4_options column of the logical switch port refers to the DHCP_Options row created for the subnet of the port. If the port has multiple IPv4 subnets, then the first subnet in the fixed_ips is used.

If the port has extra DHCPv4 options defined, then a new entry is created in the DHCP_Options table for the port. The default DHCP options are obtained from the subnet DHCP_Options table and the extra

DHCPv4 options of the port are overridden. In order to map the port DHCP_Options row with the port, the OVN ML2 driver stores both the subnet id and port id in the external_ids column.

If an admin wants to disable native OVN DHCPv4 for any particular port, then the admin needs to define the dhcp_disabled with the value true in the extra DHCP options.

Ex. openstack port set extra-dhcp-option name=dhcp_disabled,value=true,ip-version=4 <PORT_ID>

DHCPv6

OVN implements a native DHCPv6 support similar to DHCPv4. When a v6 subnet is created, the OVN ML2 driver will insert a new entry into DHCP_Options table only when the subnet ipv6_address_mode is not slaac, and enable_dhcp is True.

OVN Neutron Worker and Port status handling

When the logical switch ports VIF is attached or removed to/from the ovn integration bridge, ovn-northd updates the Logical_Switch_Port.up to True or False accordingly.

In order for the OVN Neutron ML2 driver to update the corresponding neutron ports status to ACTIVE or DOWN in the db, it needs to monitor the OVN Northbound db. A neutron worker is created for this purpose.

The implementation of the ovn worker can be found here - networking_ovn.ovsdb.worker.OvnWorker.

Neutron service will create n api workers and m rpc workers and 1 ovn worker (all these workers are separate processes).

Api workers and rpc workers will create ovsdb idl client object (ovs.db.idl.Idl) to connect to the OVN_Northbound db. See networking_ovn.ovsdb.impl_idl_ovn.OvsdbNbOvnIdl and ovsdbapp.backend.ovs_idl.connection.Connection classes for more details.

Ovn worker will create networking_ovn.ovsdb.ovsdb_monitor.OvnIdl class object (which inherits from ovs.db.idl.Idl) to connect to the OVN_Northbound db. On receiving the OVN_Northbound db updates from the ovsdb-server, notify function of OVnIdl is called by the parent class object.

OvnIdl.notify() function passes the received events to the ovsdb_monitor.OvnDbNotifyHandler class. ovsdb_monitor.OvnDbNotifyHandler checks for any changes in the Logical_Switch_Port.up and updates the neutron ports status accordingly.

If notify_nova_on_port_status_changes configuration is set, then neutron would notify nova on port status changes.

ovsdb locks

If there are multiple neutron servers running, then each neutron server will have one ovn worker which listens for the notify events. When the Logical_Switch_Port.up is updated by ovn-northd, we do not want all the neutron servers to handle the event and update the neutron port status. In order for only one neutron server to handle the events, ovsdb locks are used.

At start, each neutron servers ovn worker will try to acquire a lock with id - neutron_ovn_event_lock. The ovn worker which has acquired the lock will handle the notify events.

In case the neutron server with the lock dies, ovsdb-server will assign the lock to another neutron server in the queue.

More details about the ovsdb locks can be found here [1] and [2]

[1] - https://tools.ietf.org/html/draft-pfaff-ovsdb-proto-04#section-4.1.8 [2] - https://github.com/openvswitch/ovs/blob/branch-2.4/python/ovs/db/idl.py#L67

One thing to note is the ovn worker (with OvnIdl) do not carry out any transactions to the OVN Northbound db.

Since the api and rpc workers are not configured with any locks, using the ovsdb lock on the OVN_Northbound and OVN_Southbound DBs by the ovn workers will not have any side effects to the transactions done by these api and rpc workers.

Handling port status changes when neutron server(s) are down

When neutron server starts, ovn worker would receive a dump of all logical switch ports as events. ovsdb_monitor.OvnDbNotifyHandler would sync up if there are any inconsistencies in the port status.

OVN Southbound DB Access

The OVN Neutron ML2 driver has a need to acquire chassis information (hostname and physnets combinations). This is required initially to support routed networks. Thus, the plugin will initiate and maintain a connection to the OVN SB DB during startup.

OpenStack Metadata API and OVN

Introduction

OpenStack Nova presents a metadata API to VMs similar to what is available on Amazon EC2. Neutron is involved in this process because the source IP address is not enough to uniquely identify the source of a metadata request since networks can have overlapping IP addresses. Neutron is responsible for intercepting metadata API requests and adding HTTP headers which uniquely identify the source of the request before forwarding it to the metadata API server.

The purpose of this document is to propose a design for how to enable this functionality when OVN is used as the backend for OpenStack Neutron.

Neutron and Metadata Today

The following blog post describes how VMs access the metadata API through Neutron today.

https://www.suse.com/c/vms-get-access-metadata-neutron/

In summary, we run a metadata proxy in either the router namespace or DHCP namespace. The DHCP namespace can be used when theres no router connected to the network. The one downside to the DHCP namespace approach is that it requires pushing a static route to the VM through DHCP so that it knows to route metadata requests to the DHCP server IP address.

- Instance sends a HTTP request for metadata to 169.254.169.254
- This request either hits the router or DHCP namespace depending on the route in the instance
- The metadata proxy service in the namespace adds the following info to the request:
  - Instance IP (X-Forwarded-For header)
  - Router or Network-ID (X-Neutron-Network-Id or X-Neutron-Router-Id header)
- The metadata proxy service sends this request to the metadata agent (outside the namespace) via a UNIX domain socket.

- The neutron-metadata-agent service forwards the request to the Nova metadata API service by adding some new headers (instance ID and Tenant ID) to the request [0].

For proper operation, Neutron and Nova must be configured to communicate together with a shared secret. Neutron uses this secret to sign the Instance-ID header of the metadata request to prevent spoofing. This secret is configured through metadata_proxy_shared_secret on both nova and neutron configuration files (optional).

[0] https://opendev.org/openstack/neutron/src/commit/f73f39f2cfcd4eace2bda14c99ead9a8cc8560f4/neutron/agent/metadata/agent.py#L175

Neutron and Metadata with OVN

The current metadata API approach does not translate directly to OVN. There are no Neutron agents in use with OVN. Further, OVN makes no use of its own network namespaces that we could take advantage of like the original implementation makes use of the router and dhcp namespaces.

We must use a modified approach that fits the OVN model. This section details a proposed approach.

Overview of Proposed Approach

The proposed approach would be similar to the *isolated network* case in the current ML2+OVS implementation. Therefore, we would be running a metadata proxy (haproxy) instance on every hypervisor for each network a VM on that host is connected to.

The downside of this approach is that well be running more metadata proxies than were doing now in case of routed networks (one per virtual router) but since haproxy is very lightweight and they will be idling most of the time, it shouldnt be a big issue overall. However, the major benefit of this approach is that we dont have to implement any scheduling logic to distribute metadata proxies across the nodes, nor any HA logic. This, however, can be evolved in the future as explained below in this document.

Also, this approach relies on a new feature in OVN that we must implement first so that an OVN port can be present on *every* chassis (similar to *localnet* ports). This new type of logical port would be *localport* and we will never forward packets over a tunnel for these ports. We would only send packets to the local instance of a *localport*.

Step 1 - Create a port for the metadata proxy

When using the DHCP agent today, Neutron automatically creates a port for the DHCP agent to use. We could do the same thing for use with the metadata proxy (haproxy). Well create an OVN *localport* which will be present on every chassis and this port will have the same MAC/IP address on every host. Eventually, we can share the same neutron port for both DHCP and metadata.

Step 2 - Routing metadata API requests to the correct Neutron port

This works similarly to the current approach.

We would program OVN to include a static route in DHCP responses that routes metadata API requests to the *localport* that is hosting the metadata API proxy.

Also, in case DHCP isnt enabled or the client ignores the route info, we will program a static route in the OVN logical router which will still get metadata requests directed to the right place.

If the DHCP route does not work and the network is isolated, VMs wont get metadata, but this already happens with the current implementation so this approach doesnt introduce a regression.

Step 3 - Management of the namespaces and haproxy instances

We propose a new agent called `neutron-ovn-metadata-agent`. We will run this agent on every hypervisor and it will be responsible for spawning the haproxy instances for managing the OVS interfaces, network namespaces and haproxy processes used to proxy metadata API requests.

Step 4 - Metadata API request processing

Similar to the existing neutron metadata agent, `neutron-ovn-metadata-agent` must act as an intermediary between haproxy and the Nova metadata API service. `neutron-ovn-metadata-agent` is the process that will have access to the host networks where the Nova metadata API exists. Each haproxy will be in a network namespace not able to reach the appropriate host network. Haproxy will add the necessary headers to the metadata API request and then forward it to `neutron-ovn-metadata-agent` over a UNIX domain socket, which matches the behavior of the current metadata agent.

Metadata Proxy Management Logic

In neutron-ovn-metadata-agent.

- On startup:
  - Do a full sync. Ensure we have all the required metadata proxies running. For that, the agent would watch the `Port_Binding` table of the OVN Southbound database and look for all rows with the `chassis` column set to the host the agent is running on. For all those entries, make sure a metadata proxy instance is spawned for every `datapath` (Neutron network) those ports are attached to. Ensure any running metadata proxies no longer needed are torn down.
- Open and maintain a connection to the OVN Northbound database (using the ovsdbapp library). On first connection, and anytime a reconnect happens:
  - Do a full sync.
- Register a callback for creates/updates/deletes to Logical_Switch_Port rows to detect when metadata proxies should be started or torn down. `neutron-ovn-metadata-agent` will watch OVN Southbound database (`Port_Binding` table) to detect when a port gets bound to its chassis. At that point, the agent will make sure that theres a metadata proxy attached to the OVN *localport* for the network which this port is connected to.
- When a new network is created, we must create an OVN *localport* for use as a metadata proxy. This port will be owned by `network:dhcp` so that it gets auto deleted upon the removal of the network and it will remain `DOWN` and not bound to any chassis. The metadata port will be created regardless of the DHCP setting of the subnets within the network as long as the metadata service is enabled.
- When a network is deleted, we must tear down the metadata proxy instance (if present) on the host and delete the corresponding OVN *localport* (which will happen automatically as its owned by `network:dhcp`).

Launching a metadata proxy includes:

- Creating a network namespace:

```
$ sudo ip netns add <ns-name>
```

- Creating a VETH pair (OVS upgrades that upgrade the kernel module will make internal ports go away and then brought back by OVS scripts. This may cause some disruption. Therefore, veth pairs are preferred over internal ports):

```
$ sudo ip link add <iface-name>0 type veth peer name <iface-name>1
```

- Creating an OVS interface and placing one end in that namespace:

```
$ sudo ovs-vsctl add-port br-int <iface-name>0
$ sudo ip link set <iface-name>1 netns <ns-name>
```

- Setting the IP and MAC addresses on that interface:

```
$ sudo ip netns exec <ns-name> \
> ip link set <iface-name>1 address <neutron-port-mac>
$ sudo ip netns exec <ns-name> \
> ip addr add <neutron-port-ip>/<netmask> dev <iface-name>1
```

- Bringing the VETH pair up:

```
$ sudo ip netns exec <ns-name> ip link set <iface-name>1 up
$ sudo ip link set <iface-name>0 up
```

- Set `external-ids:iface-id=NEUTRON_PORT_UUID` on the OVS interface so that OVN is able to correlate this new OVS interface with the correct OVN logical port:

```
$ sudo ovs-vsctl set Interface <iface-name>0 external_ids:iface-id=
↪<neutron-port-uuid>
```

- Starting haproxy in this network namespace.

Tearing down a metadata proxy includes:

- Removing the network UUID from our chassis.
- Stopping haproxy.
- Deleting the OVS interface.
- Deleting the network namespace.

Other considerations

This feature will be enabled by default when using `ovn` driver, but there should be a way to disable it in case operators who dont need metadata dont have to deal with the complexity of it (haproxy instances, network namespaces, etcetera). In this case, the agent would not create the neutron ports needed for metadata.

Right now, the `vif-plugged` event to Nova is sent out when the up column in the OVN Northbound databases Logical_Switch_Port table changes to True, indicating that the VIF is now up. There could be a race condition when the first VM for a certain network boots on a hypervisor if it does so before the metadata proxy instance has been spawned. Fortunately, retries on cloud-init should eventually fetch metadata even when this might happen.

Alternatives Considered

Alternative 1: Build metadata support into ovn-controller

Weve been building some features useful to OpenStack directly into OVN. DHCP and DNS are key examples of things weve replaced by building them into ovn-controller. The metadata API case has some key differences that make this a less attractive solution:

The metadata API is an OpenStack specific feature. DHCP and DNS by contrast are more clearly useful outside of OpenStack. Building metadata API proxy support into ovn-controller means embedding an HTTP and TCP stack into ovn-controller. This is a significant degree of undesired complexity.

This option has been ruled out for these reasons.

Alternative 2: Distributed metadata and High Availability

In this approach, we would spawn a metadata proxy per virtual router or per network (if isolated), thus, improving the number of metadata proxy instances running in the cloud. However, scheduling and HA have to be considered. Also, we wouldnt need the OVN *localport* implementation.

`neutron-ovn-metadata-agent` would run on any host that we wish to be able to host metadata API proxies. These hosts must also be running ovn-controller.

Each of these hosts will have a Chassis record in the OVN southbound database created by ovn-controller. The Chassis table has a column called `external_ids` which can be used for general metadata however we see fit. `neutron-ovn-metadata-agent` will update its corresponding Chassis record with an external-id of `neutron-metadata-proxy-host=true` to indicate that this OVN chassis is one capable of hosting metadata proxy instances.

Once we have a way to determine hosts capable of hosting metadata API proxies, we can add logic to the ovn ML2 driver that schedules metadata API proxies. This would be triggered by Neutron API requests.

The output of the scheduling process would be setting an `external_ids` key on a Logical_Switch_Port in the OVN northbound database that corresponds with a metadata proxy. The key could be something like `neutron-metadata-proxy-chassis=CHASSIS_HOSTNAME`.

`neutron-ovn-metadata-agent` on each host would also be watching for updates to these Logical_Switch_Port rows. When it detects that a metadata proxy has been scheduled locally, it will kick off the process to spawn the local haproxy instance and get it plugged into OVN.

HA must also be considered. We must know when a host goes down so that all metadata proxies scheduled to that host can be rescheduled. This is almost the exact same problem we have with L3 HA. When a host goes down, we need to trigger rescheduling gateways to other hosts. We should ensure that the approach used for rescheduling L3 gateways can be utilized for rescheduling metadata proxies, as well.

In neutron-server (ovn mechanism driver) .

Introduce a new ovn driver configuration option:

- `[ovn] isolated_metadata=[True|False]`

Events that trigger scheduling a new metadata proxy:

- If isolated_metadata is True
  - When a new network is created, we must create an OVN logical port for use as a metadata proxy and then schedule this to one of the `neutron-ovn-metadata-agent` instances.
- If isolated_metadata is False
  - When a network is attached to or removed from a logical router, ensure that at least one of the networks has a metadata proxy port already created. If not, pick a network and create a metadata proxy port and then schedule it to an agent. At this point, we need to update the static route for metadata API.

Events that trigger unscheduling an existing metadata proxy:

- When a network is deleted, delete the metadata proxy port if it exists and unschedule it from a `neutron-ovn-metadata-agent`.

To schedule a new metadata proxy:

- Determine the list of available OVN Chassis that can host metadata proxies by reading the `Chassis` table of the OVN Southbound database. Look for chassis that have an external-id of `neutron-metadata-proxy-host=true`.
- Of the available OVN chassis, choose the one least loaded, or currently hosting the fewest number of metadata proxies.
- Set `neutron-metadata-proxy-chassis=CHASSIS_HOSTNAME` as an external-id on the Logical_Switch_Port in the OVN Northbound database that corresponds to the neutron port used for this metadata proxy. `CHASSIS_HOSTNAME` maps to the hostname row of a Chassis record in the OVN Southbound database.

This approach has been ruled out for its complexity although we have analyzed the details deeply because, eventually, and depending on the implementation of L3 HA, we will want to evolve to it.

Other References

- HAProxy config https://review.openstack.org/#/c/431691/34/neutron/agent/metadata/driver.py
- https://engineeringblog.yelp.com/2015/04/true-zero-downtime-haproxy-reloads.html

Neutron/OVN Database consistency

This document presents the problem and proposes a solution for the data consistency issue between the Neutron and OVN databases. Although the focus of this document is OVN this problem is common enough to be present in other ML2 drivers (e.g OpenDayLight, BigSwitch, etc). Some of them already contain a mechanism in place for dealing with it.

Problem description

In a common Neutron deployment model there could have multiple Neutron API workers processing requests. For each request, the worker will update the Neutron database and then invoke the ML2 driver to translate the information to that specific SDN data model.

There are at least two situations that could lead to some inconsistency between the Neutron and the SDN databases, for example:

Problem 1: Neutron API workers race condition

```
In Neutron:
  with neutron_db_transaction:
       update_neutron_db()
       ml2_driver.update_port_precommit()
  ml2_driver.update_port_postcommit()

In the ML2 driver:
  def update_port_postcommit:
      port = neutron_db.get_port()
      update_port_in_ovn(port)
```

Imagine the case where a port is being updated twice and each request is being handled by a different API worker. The method responsible for updating the resource in the OVN (`update_port_postcommit`) is not atomic and invoked outside of the Neutron database transaction. This could lead to a problem where the order in which the updates are committed to the Neutron database are different than the order that they are committed to the OVN database, resulting in an inconsistency.

This problem has been reported at bug #1605089.

Problem 2: Backend failures

Another situation is when the changes are already committed in Neutron but an exception is raised upon trying to update the OVN database (e.g lost connectivity to the `ovsdb-server`). We currently dont have a good way of handling this problem, obviously it would be possible to try to immediately rollback the changes in the Neutron database and raise an exception but, that rollback itself is an operation that could also fail.

Plus, rollbacks is not very straight forward when it comes to updates or deletes. In a case where a VM is being teared down and OVN fail to delete a port, re-creating that port in Neutron doesnt necessary fix the problem. The decommission of a VM involves many other things, in fact, we could make things even worse by leaving some dirty data around. I believe this is a problem that would be better dealt with by other methods.

Proposed change

In order to fix the problems presented at the *Problem description* section this document proposes a solution based on the Neutrons `revision_number` attribute. In summary, for every resource in Neutron theres an attribute called `revision_number` which gets incremented on each update made on that resource. For example:

```
$ openstack port create --network nettest porttest
...
| revision_number | 2 |
...

$ openstack port set porttest --mac-address 11:22:33:44:55:66

$ mysql -e "use neutron; select standard_attr_id from ports where id=\
↪"91c08021-ded3-4c5a-8d57-5b5c389f8e39\";"
+------------------+
| standard_attr_id |
+------------------+
|             1427 |
+------------------+

$ mysql -e "use neutron; SELECT revision_number FROM standardattributes WHERE␣
↪id=1427;"
+-----------------+
| revision_number |
+-----------------+
|               3 |
+-----------------+
```

This document proposes a solution that will use the *revision_number* attribute for three things:

1. Perform a compare-and-swap operation based on the resource version
2. Guarantee the order of the updates (*Problem 1*)
3. Detecting when resources in Neutron and OVN are out-of-sync

But, before any of points above can be done we need to change the ovn driver code to:

#1 - Store the revision_number referent to a change in OVNDB

To be able to compare the version of the resource in Neutron against the version in OVN we first need to know which version the OVN resource is present at.

Fortunately, each table in the OVNDB contains a special column called `external_ids` which external systems (like Neutron) can use to store information about its own resources that corresponds to the entries in OVNDB.

So, every time a resource is created or updated in OVNDB by ovn driver, the Neutron `revision_number` referent to that change will be stored in the `external_ids` column of that resource. That will allow ovn driver to look at both databases and detect whether the version in OVN is up-to-date with Neutron or not.

#2 - Ensure correctness when updating OVN

As stated in *Problem 1*, simultaneous updates to a single resource will race and, with the current code, the order in which these updates are applied is not guaranteed to be the correct order. That means that, if two or more updates arrives we cant prevent an older version of that update to be applied after a newer one.

This document proposes creating a special `OVSDB command` that runs as part of the same transaction that is updating a resource in OVNDB to prevent changes with a lower `revision_number` to be applied in case the resource in OVN is at a higher `revision_number` already.

This new OVSDB command needs to basically do two things:

1. Add a verify operation to the `external_ids` column in OVNDB so that if another client modifies that column mid-operation the transaction will be restarted.

A better explanation of what verify does is described at the doc string of the Transaction class in the OVS code itself, I quote:

> Because OVSDB handles multiple clients, it can happen that between the time that OVSDB client A reads a column and writes a new value, OVSDB client B has written that column. Client As write should not ordinarily overwrite client Bs, especially if the column in question is a map column that contains several more or less independent data items. If client A adds a verify operation before it writes the column, then the transaction fails in case client B modifies it first. Client A will then see the new value of the column and compose a new transaction based on the new contents written by client B.

2. Compare the `revision_number` from the update against what is presently stored in OVNDB. If the version in OVNDB is already higher than the version in the update, abort the transaction.

So basically this new command is responsible for guarding the OVN resource by not allowing old changes to be applied on top of new ones. Heres a scenario where two concurrent updates comes in the wrong order and how the solution above will deal with it:

Neutron worker 1 (NW-1): Updates a port with address A (revision_number: 2)

Neutron worker 2 (NW-2): Updates a port with address B (revision_number: 3)

TXN 1: NW-2 transaction is committed first and the OVN resource now has RN 3

TXN 2: NW-1 transaction detects the change in the external_ids column and is restarted

TXN 2: NW-1 the new command now sees that the OVN resource is at RN 3, which is higher than the update version (RN 2) and aborts the transaction.

Theres a bit more for the above to work with the current ovn driver code, basically we need to tidy up the code to do two more things.

1. Consolidate changes to a resource in a single transaction.

This is important regardless of this spec, having all changes to a resource done in a single transaction minimizes the risk of having half-changes written to the database in case of an eventual problem. This should be done already but its important to have it here in case we find more examples like that as we code.

2. When doing partial updates, use the OVNDB as the source of comparison to create the deltas.

Being able to do a partial update in a resource is important for performance reasons; its a way to minimize the number of changes that will be performed in the database.

Right now, some of the update() methods in ovn driver creates the deltas using the *current* and *original* parameters that are passed to it. The *current* parameter is, as the name says, the current version of the object present in the Neutron DB. The *original* parameter is the previous version (current - 1) of that object.

The problem of creating the deltas by comparing these two objects is because only the data in the Neutron DB is used for it. We need to stop using the *original* object for it and instead we should create the delta based on the *current* version of the Neutron DB against the data stored in the OVNDB to be able to detect the real differences between the two databases.

So in summary, to guarantee the correctness of the updates this document proposes to:

1. Create a new OVSDB command is responsible for comparing revision numbers and aborting the transaction, when needed.
2. Consolidate changes to a resource in a single transaction (should be done already)
3. When doing partial updates, create the deltas based in the current version in the Neutron DB and the OVNDB.

#3 - Detect and fix out-of-sync resources

When things are working as expected the above changes should ensure that Neutron DB and OVNDB are in sync but, what happens when things go bad ? As per *Problem 2*, things like temporarily losing connectivity with the OVNDB could cause changes to fail to be committed and the databases getting out-of-sync. We need to be able to detect the resources that were affected by these failures and fix them.

We do already have the means to do it, similar to what the ovn_db_sync.py script does we could fetch all the data from both databases and compare each resource. But, depending on the size of the deployment this can be really slow and costy.

This document proposes an optimization for this problem to make it efficient enough so that we can run it periodically (as a periodic task) and not manually as a script anymore.

First, we need to create an additional table in the Neutron database that would serve as a cache for the revision numbers in **OVNDB**.

The new table schema could look this:

Column name	Type	Description
standard_attr_id	Integer	Primary key. The reference ID from the standardattributes table in Neutron for that resource. ONDELETE SET NULL.
resource_uuid	String	The UUID of the resource
resource_type	String	The type of the resource (e.g, Port, Router,)
revision_number	Integer	The version of the object present in OVN
acquired_at	DateTime	The time that the entry was create. For troubleshooting purposes
updated_at	DateTime	The time that the entry was updated. For troubleshooting purposes

For the different actions: Create, update and delete; this table will be used as:

1. Create:

In the create_*_precommit() method, we will create an entry in the new table within the same Neutron transaction. The revision_number column for the new entry will have a placeholder value until the resource is successfully created in OVNDB.

In case we fail to create the resource in OVN (but succeed in Neutron) we still have the entry logged in the new table and this problem can be detected by fetching all resources where the revision_number column value is equal to the placeholder value.

The pseudo-code will look something like this:

```
def create_port_precommit(ctx, port):
    create_initial_revision(port['id'], revision_number=-1,
                            session=ctx.session)

def create_port_postcommit(ctx, port):
    create_port_in_ovn(port)
    bump_revision(port['id'], revision_number=port['revision_number'])
```

2. Update:

For update its simpler, we need to bump the revision number for that resource **after** the OVN transaction is committed in the update_*_postcommit() method. That way, if an update fails to be applied to OVN the inconsistencies can be detected by a JOIN between the new table and the `standardattributes` table where the revision_number columns does not match.

The pseudo-code will look something like this:

```
def update_port_postcommit(ctx, port):
    update_port_in_ovn(port)
    bump_revision(port['id'], revision_number=port['revision_number'])
```

3. Delete:

The `standard_attr_id` column in the new table is a foreign key constraint with a `ONDELETE=SET`

NULL set. That means that, upon Neutron deleting a resource the `standard_attr_id` column in the new table will be set to *NULL*.

If deleting a resource succeeds in Neutron but fails in OVN, the inconsistency can be detect by looking at all resources that has a `standard_attr_id` equals to NULL.

The pseudo-code will look something like this:

```
def delete_port_postcommit(ctx, port):
    delete_port_in_ovn(port)
    delete_revision(port['id'])
```

With the above optimization its possible to create a periodic task that can run quite frequently to detect and fix the inconsistencies caused by random backend failures.

Note

Theres no lock linking both database updates in the postcommit() methods. So, its true that the method bumping the revision_number column in the new table in Neutron DB could still race but, that should be fine because this table acts like a cache and the real revision_number has been written in OVNDB.

The mechanism that will detect and fix the out-of-sync resources should detect this inconsistency as well and, based on the revision_number in OVNDB, decide whether to sync the resource or only bump the revision_number in the cache table (in case the resource is already at the right version).

Refereces

- Theres a chain of patches with a proof of concept for this approach, they start at: https://review.openstack.org/#/c/517049/

Alternatives

Journaling

An alternative solution to this problem is *journaling*. The basic idea is to create another table in the Neutron database and log every operation (create, update and delete) instead of passing it directly to the SDN controller.

A separated thread (or multiple instances of it) is then responsible for reading this table and applying the operations to the SDN backend.

This approach has been used and validated by drivers such as networking-odl.

An attempt to implement this approach in *ovn driver* can be found here.

Some things to keep in mind about this approach:

- The code can get quite complex as this approach is not only about applying the changes to the SDN backend asynchronously. The dependencies between each resource as well as their operations also needs to be computed. For example, before attempting to create a router port the router that this port belongs to needs to be created. Or, before attempting to delete a network all the dependent resources on it (subnets, ports, etc) needs to be processed first.
- The number of journal threads running can cause problems. In my tests I had three controllers, each one with 24 CPU cores (Intel Xeon E5-2620 with hyperthreading enabled) and 64GB RAM. Running 1 journal thread per Neutron API worker has caused `ovsdb-server` to misbehave when

under heavy pressure[1]. Running multiple journal threads seem to be causing other types of problems in other drivers as well.

- When under heavy pressure[1], I noticed that the journal threads could come to a halt (or really slowed down) while the API workers were handling a lot of requests. This resulted in some operations taking more than a minute to be processed. This behaviour can be seem in this screenshot.
- Given that the 1 journal thread per Neutron API worker approach is problematic, determining the right number of journal threads is also difficult. In my tests, Ive noticed that 3 journal threads per controller worked better but that number was pure based on `trial & error`. In production this number should probably be calculated based in the environment.
- At least temporarily, the data in the Neutron database is duplicated between the normal tables and the journal one.
- Some operations like creating a new resource via Neutrons API will return HTTP 201, which indicates that the resource has been created and is ready to be used, but as these resources are created asynchronously one could argue that the HTTP codes are now misleading. As a note, the resource will be created at the Neutron database by the time the HTTP request returns but it may not be present in the SDN backend yet.

Given all considerations, this approach is still valid and the fact that its already been used by other ML2 drivers makes it more open for collaboration and code sharing.

Synchronization of the Neutron and OVN Databases (neutron-ovn-db-sync-util)

Consistency of the data between Neutron and OVN databases, how important it is and what are the problems and potential solutions to ensure it are described in the *Neutron/OVN Database consistency* document. Additionally to all what is described there, Neutron provides CLI tool *neutron-ovn-db-sync-util* which can be used by the cloud operator to sync OVN database with the data stored in Neutron DB.

Usage

The `neutron-ovn-db-sync-util` CLI tool is used to synchronize the OVN Northbound and Southbound databases with the Neutron database. This tool should be used ad-hoc, for example when inconsistencies between databases are detected.

Basic usage:

```
neutron-ovn-db-sync-util --config-file /etc/neutron/neutron.conf \
                         --config-file /etc/neutron/plugins/ml2/ml2_conf.ini
```

The tool supports several command-line options:

- `--config-file`: Specify the configuration file(s) to use. Multiple config files can be specified. If no config file is provided, the tool will attempt to load configuration from `/etc/neutron/neutron.conf`.
- `--ovn-ovn_nb_connection`: OVN Northbound database connection string (usually configured in ml2_conf.ini).
- `--ovn-ovn_sb_connection`: OVN Southbound database connection string (usually configured in ml2_conf.ini).

[1] I ran the tests using Browbeat which is basically orchestrate Openstack Rally and monitor the machines usage of resources.

- `--ovn-sync_mode`: Synchronization mode to use. Available modes:
  - `log`: Only log inconsistencies without making any changes
  - `repair`: Log inconsistencies and repair them by updating OVN database
  - `migrate`: Run in migration mode for OVS to OVN migration (runs as repair mode in the synchronizer)
- `--sync_plugin`: Specify a particular sync plugin to run. If not specified, all registered sync plugins will be loaded and executed.
- `--migration_plugin`: Specify a particular migration plugin to run during OVS to OVN migration. If not specified, all registered migration plugins will be loaded and executed.

Example - Running in log mode to check for inconsistencies:

```
neutron-ovn-db-sync-util --config-file /etc/neutron/neutron.conf \
                         --config-file /etc/neutron/plugins/ml2/ml2_conf.ini \
                         --ovn-sync_mode log
```

Example - Running in repair mode to fix inconsistencies:

```
neutron-ovn-db-sync-util --config-file /etc/neutron/neutron.conf \
                         --config-file /etc/neutron/plugins/ml2/ml2_conf.ini \
                         --ovn-sync_mode repair
```

Important

When running the sync utility, it is strongly recommended to stop the neutron-server service to avoid race conditions where the server might be modifying the database while the sync tool is running.

Sync plugins

Sync plugins are responsible for synchronizing specific aspects of the Neutron database with the OVN databases. Neutron provides two built-in sync plugins:

- **OvnNbSynchronizer**: Synchronizes the OVN Northbound database
- **OvnSbSynchronizer**: Synchronizes the OVN Southbound database

Writing a custom sync plugin

Stadium projects or third-party plugins can provide their own sync plugins by implementing a synchronizer class and registering it via setuptools entry points.

To create a custom sync plugin:

1. Create a class that inherits from `BaseOvnDbSynchronizer` (defined in `neutron_lib.ovn.db_sync`):

```
from neutron_lib.ovn import db_sync

class CustomSynchronizer(db_sync.BaseOvnDbSynchronizer):
    """Custom synchronizer for my plugin."""
```

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```
    # Specify required mechanism drivers
    _required_mechanism_drivers = ['mechanism-driver-required-by-project']

    # Specify required service plugins
    _required_service_plugins = ['service-plugin-required-by-project']

    # Specify required ML2 extension drivers
    _required_ml2_ext_drivers = ['ml2-extension-required-by-project']

    def __init__(self, core_plugin, ovn_driver, mode, is_maintenance=False):
        super().__init__(core_plugin, ovn_driver, mode, is_maintenance)
        # Initialize any plugin-specific resources

    def do_sync(self):
        """Implement the synchronization logic."""
        if self.mode == ovn_const.OVN_DB_SYNC_MODE_OFF:
            return

        # Your synchronization logic here
        # Example:
        ctx = context.get_admin_context()
        self.sync_my_resources(ctx)

    def sync_my_resources(self, ctx):
        """Sync specific resources."""
        # Compare Neutron DB with OVN DB
        # Log inconsistencies
        # If mode is REPAIR, fix the inconsistencies
```

2. Register the plugin via setuptools entry point in your `setup.cfg`:

```
[entry_points]
neutron.ovn.db_sync =
    my-sync-plugin = my_neutron_plugin.ovn.db_sync:MyCustomSynchronizer
```

The entry point group must be `neutron.ovn.db_sync` for the plugin to be discovered by the `neutron-ovn-db-sync-util` tool.

Key implementation guidelines

- **Use class attributes for requirements**: Define `_required_mechanism_drivers`, `_required_service_plugins`, and `_required_ml2_ext_drivers` to ensure all necessary plugins and drivers are loaded before synchronization begins.
- **Implement do_sync() method**: This is the main entry point for your synchronization logic. The method should check the mode and act accordingly:
  - In `log` mode: Only log inconsistencies
  - In `repair` mode: Log and fix inconsistencies

- **Use transactions**: When making changes to OVN database, use transactions to ensure atomicity:

```
with self.ovn_nb_api.transaction(check_error=True) as txn:
    txn.add(self.ovn_nb_api.some_operation(...))
```

- **Access OVN APIs**: The base class provides access to:
  - `self.ovn_nb_api`: OVN Northbound API
  - `self.ovn_sb_api`: OVN Southbound API
  - `self.core_plugin`: Neutron core plugin
  - `self.ovn_driver`: OVN mechanism driver
  - `self.mode`: Current sync mode

OVS to OVN migration

The `neutron-ovn-db-sync-util` tool supports migration from ML2/OVS to ML2/OVN deployments. When run in `migrate` mode, the tool performs the following:

1. Runs all synchronization plugins in `repair` mode to ensure the OVN databases are synchronized with Neutron.
2. Executes all registered migration plugins to perform ML2/OVS specific data transformations.

To run a migration:

```
neutron-ovn-db-sync-util --config-file /etc/neutron/neutron.conf \
                         --config-file /etc/neutron/plugins/ml2/ml2_conf.ini \
                         --ovn-sync_mode migrate
```

The migration process:

1. First, the tool validates that the ML2 core plugin and OVN mechanism driver are properly configured.
2. It then loads all sync plugins and runs them in `repair` mode to synchronize the current state of Neutron with OVN.
3. After synchronization is complete, it loads and executes all migration plugins registered under the `neutron.ovn.db_migration` entry point.

Warning

Migration is a one-time operation and should be carefully planned:

- Stop all neutron-server instances before running migration
- Backup your databases before starting
- Test the migration in a development environment first
- Monitor logs carefully during the migration process

Migration plugins

Migration plugins handle the transformation of ML2/OVS specific data to ML2/OVN format during the migration process. Unlike sync plugins which ensure consistency, migration plugins perform one-time data transformations.

Writing a custom migration plugin

Stadium projects or third-party plugins that stored OVS-specific data in the Neutron database may need to provide migration plugins to transform that data for OVN compatibility.

To create a custom migration plugin:

1. Create a migration function that performs the necessary database transformations:

```
def migrate_my_plugin_data():
    """Migrate ML2/OVS specific data to ML2/OVN format.

    This function is called during the migration process and should
    handle the transformation of any plugin-specific data from OVS
    to OVN format.
    """
    LOG.info("Starting migration of my plugin data")

    # Access the database
    ctx = context.get_admin_context()

    # Perform data transformations
    # Example: Update table records, migrate configuration, etc.
    with db_api.CONTEXT_WRITER.using(ctx):
        # Your migration logic here
        pass

    LOG.info("Completed migration of my plugin data")
```

2. Register the migration function via setuptools entry point in your `setup.cfg`:

```
[entry_points]
neutron.ovn.db_migration =
    my-migration-plugin = my_neutron_plugin.ovn.migration:migrate_my_
↪plugin_data
```

The entry point group must be `neutron.ovn.db_migration` for the plugin to be discovered during migration.

Key implementation guidelines

- **Idempotency**: Migration functions should be idempotent - safe to run multiple times without causing issues if the migration is interrupted and restarted.
- **Error handling**: Use appropriate error handling and logging to track migration progress and handle failures gracefully.
- **Database transactions**: Use database context managers to ensure data integrity:

```
with db_api.CONTEXT_WRITER.using(ctx):
    # Database modifications here
```

- **Logging**: Provide detailed logging at the start and completion of migration, as well as for any significant operations performed during the migration.
- **No return value needed**: Migration functions are called via `ExtensionManager.map()` and dont need to return any value.

OpenStack LoadBalancer API and OVN

Introduction

Load balancing is essential for enabling simple or automatic delivery scaling and availability since application delivery, scaling and availability are considered vital features of any cloud. Octavia is an open source, operator-scale load balancing solution designed to work with OpenStack.

The purpose of this document is to propose a design for how we can use OVN as the backend for OpenStacks LoadBalancer API provided by Octavia.

Octavia LoadBalancers Today

A Detailed design analysis of Octavia is available here:

https://docs.openstack.org/octavia/queens/contributor/design/version0.5/component-design.html

Currently, Octavia uses the in-built Amphorae driver to fulfill the Loadbalancing requests in Openstack. Amphorae can be a Virtual machine, container, dedicated hardware, appliance or device that actually performs the task of load balancing in the Octavia system. More specifically, an amphora takes requests from clients on the front-end and distributes these to back-end systems. Amphorae communicates with its controllers over the LoadBalancers network through a driver interface on the controller.

Amphorae needs a placeholder, such as a separate VM/Container for deployment, so that it can handle the LoadBalancers requests. Along with this, it also needs a separate network (termed as lb-mgmt-network) which handles all Amphorae requests.

Amphorae has the capability to handle L4 (TCP/UDP) as well as L7 (HTTP) LoadBalancer requests and provides monitoring features using HealthMonitors.

Octavia with OVN

OVN native LoadBalancer currently supports L4 protocols, with support for L7 protocols aimed for in future releases. Currently it also does not have any monitoring facility. However, it does not need any extra hardware/VM/Container for deployment, which is a major positive point when compared with Amphorae. Also, it does not need any special network to handle the LoadBalancers requests as they are taken care by OpenFlow rules directly. And, though OVN does not have support for TLS, it is in the works and once implemented can be integrated with Octavia.

This following section details about how OVN can be used as an Octavia driver.

Overview of Proposed Approach

The OVN Driver for Octavia runs under the scope of Octavia. Octavia API receives and forwards calls to the OVN Driver.

Step 1 - Creating a LoadBalancer

Octavia API receives and issues a LoadBalancer creation request on a network to the OVN Provider driver. OVN driver creates a LoadBalancer in the OVN NorthBound DB and asynchronously updates the Octavia DB with the status response. A VIP port is created in Neutron when the LoadBalancer creation is complete. The VIP information however is not updated in the NorthBound DB until the Members are associated with the LoadBalancers Pool.

Step 2 - Creating LoadBalancer entities (Pools, Listeners, Members)

Once a LoadBalancer is created by OVN in its NorthBound DB, users can now create Pools, Listeners and Members associated with the LoadBalancer using the Octavia API. With the creation of each entity, the LoadBalancers *external_ids* column in the NorthBound DB would be updated and corresponding Logical and Openflow rules would be added for handling them.

Step 3 - LoadBalancer request processing

When a user sends a request to the VIP IP address, OVN pipeline takes care of load balancing the VIP request to one of the backend members. More information about this can be found in the ovn-northd man pages.

OVN LoadBalancer Driver Logic

- On startup: Open and maintain a connection to the OVN Northbound DB (using the ovsdbapp library). On first connection, and anytime a reconnect happens:
  - Do a full sync.
- Register a callback when a new interface is added to a router or deleted from a router.
- When a new LoadBalancer L1 is created, create a Row in OVNs `Load_Balancer` table and update its entries for name and network references. If the network on which the LoadBalancer is created, is associated with a router, say R1, then add the router reference to the LoadBalancers *external_ids* and associate the LoadBalancer to the router. Also associate the LoadBalancer L1 with all those networks which have an interface on the router R1. This is required so that Logical Flows for inter-network communication while using the LoadBalancer L1 is possible. Also, during this time, a new port is created via Neutron which acts as a VIP Port. The information of this new port is not visible on the OVNs NorthBound DB till a member is added to the LoadBalancer.
- If a new network interface is added to the router R1 described above, all the LoadBalancers on that network are associated with the router R1 and all the LoadBalancers on the router are associated with the new network.
- If a network interface is removed from the router R1, then all the LoadBalancers which have been solely created on that network (identified using the *ls_ref* attribute in the LoadBalancers *external_ids*) are removed from the router. Similarly those LoadBalancers which are associated with the network but not actually created on that network are removed from the network.
- LoadBalancer can either be deleted with all its children entities using the *cascade* option, or its members/pools/listeners can be individually deleted. When the LoadBalancer is deleted, its references and associations from all networks and routers are removed. This might change in the future once the association of LoadBalancers with networks/routers are changed to *weak* from *strong* [3]. Also the VIP port is deleted when the LoadBalancer is deleted.

OVN LoadBalancer at work

OVN Northbound schema [5] has a table to store LoadBalancers. The table looks like:

```
"Load_Balancer": {
    "columns": {
        "name": {"type": "string"},
        "vips": {
            "type": {"key": "string", "value": "string",
                     "min": 0, "max": "unlimited"}},
        "protocol": {
            "type": {"key": {"type": "string",
                             "enum": ["set", ["tcp", "udp"]]},
                             "min": 0, "max": 1}},
        "external_ids": {
            "type": {"key": "string", "value": "string",
                     "min": 0, "max": "unlimited"}}},
        "isRoot": true},
```

There is a `load_balancer` column in the Logical_Switch table (which corresponds to a Neutron network) as well as the Logical_Router table (which corresponds to a Neutron router) referring back to the Load_Balancer table.

The OVN driver updates the OVN Northbound DB. When a LoadBalancer is created, a row in this table is created. And when the listeners and members are added, vips column is updated accordingly. And the Logical_Switchs `load_balancer` column is also updated accordingly.

ovn-northd service which monitors for changes to the OVN Northbound DB, generates OVN logical flows to enable load balancing and ovn-controller running on each compute node, translates the logical flows into actual OpenFlow rules.

The status of each entity in the Octavia DB is managed according to [4]

Below are few examples on what happens when LoadBalancer commands are executed and what changes in the Load_Balancer Northbound DB table.

1. Create a LoadBalancer:

```
$ openstack loadbalancer create --provider ovn --vip-subnet-id=private lb1

$ ovn-nbctl list load_balancer
_uuid           : 9dd65bae-2501-43f2-b34e-38a9cb7e4251
external_ids    : {
    lr_ref="neutron-52b6299c-6e38-4226-a275-77370296f257",
    ls_refs="{\"neutron-2526c68a-5a9e-484c-8e00-0716388f6563\": 1}",
    neutron:vip="10.0.0.10",
    neutron:vip_port_id="2526c68a-5a9e-484c-8e00-0716388f6563"}
name            : "973a201a-8787-4f6e-9b8f-ab9f93c31f44"
protocol        : []
vips            : {}
```

2. Create a pool:

```
$ openstack loadbalancer pool create --name p1 --loadbalancer lb1
--protocol TCP --lb-algorithm SOURCE_IP_PORT

$ ovn-nbctl list load_balancer
_uuid          : 9dd65bae-2501-43f2-b34e-38a9cb7e4251
external_ids   : {
    lr_ref="neutron-52b6299c-6e38-4226-a275-77370296f257",
    ls_refs="{\"neutron-2526c68a-5a9e-484c-8e00-0716388f6563\": 1}",
    "pool_f2ddf7a6-4047-4cc9-97be-1d1a6c47ece9"="", neutron:vip="10.0.0.10
↪",
    neutron:vip_port_id="2526c68a-5a9e-484c-8e00-0716388f6563"}
name           : "973a201a-8787-4f6e-9b8f-ab9f93c31f44"
protocol       : []
vips           : {}
```

3. Create a member:

```
$ openstack loadbalancer member create --address 10.0.0.107
 --subnet-id 2d54ec67-c589-473b-bc67-41f3d1331fef --protocol-port 80 p1

$ ovn-nbctl list load_balancer
_uuid          : 9dd65bae-2501-43f2-b34e-38a9cb7e4251
external_ids   : {
    lr_ref="neutron-52b6299c-6e38-4226-a275-77370296f257",
    ls_refs="{\"neutron-2526c68a-5a9e-484c-8e00-0716388f6563\": 2}",
    "pool_f2ddf7a6-4047-4cc9-97be-1d1a6c47ece9"=
    "member_579c0c9f-d37d-4ba5-beed-cabf6331032d_10.0.0.107:80",
    neutron:vip="10.0.0.10",
    neutron:vip_port_id="2526c68a-5a9e-484c-8e00-0716388f6563"}
name           : "973a201a-8787-4f6e-9b8f-ab9f93c31f44"
protocol       : []
vips           : {}
```

4. Create another member:

```
$ openstack loadbalancer member create --address 20.0.0.107
 --subnet-id c2e2da10-1217-4fe2-837a-1c45da587df7 --protocol-port 80 p1

$ ovn-nbctl list load_balancer
_uuid          : 9dd65bae-2501-43f2-b34e-38a9cb7e4251
external_ids   : {
    lr_ref="neutron-52b6299c-6e38-4226-a275-77370296f257",
    ls_refs="{\"neutron-2526c68a-5a9e-484c-8e00-0716388f6563\": 2,
          \"neutron-12c42705-3e15-4e2d-8fc0-070d1b80b9ef\": 1}",
    "pool_f2ddf7a6-4047-4cc9-97be-1d1a6c47ece9"=
    "member_579c0c9f-d37d-4ba5-beed-cabf6331032d_10.0.0.107:80,
     member_d100f2ed-9b55-4083-be78-7f203d095561_20.0.0.107:80",
    neutron:vip="10.0.0.10",
    neutron:vip_port_id="2526c68a-5a9e-484c-8e00-0716388f6563"}
name           : "973a201a-8787-4f6e-9b8f-ab9f93c31f44"
```

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```
protocol       : []
vips           : {}
```

5. Create a listener:

```
$ openstack loadbalancer listener create --name l1 --protocol TCP
 --protocol-port 82 --default-pool p1 lb1

$ ovn-nbctl list load_balancer
_uuid          : 9dd65bae-2501-43f2-b34e-38a9cb7e4251
external_ids   : {
    lr_ref="neutron-52b6299c-6e38-4226-a275-77370296f257",
    ls_refs="{\"neutron-2526c68a-5a9e-484c-8e00-0716388f6563\": 2,
              \"neutron-12c42705-3e15-4e2d-8fc0-070d1b80b9ef\": 1}",
    "pool_f2ddf7a6-4047-4cc9-97be-1d1a6c47ece9"="10.0.0.107:80,20.0.0.
↪107:80",
    "listener_12345678-2501-43f2-b34e-38a9cb7e4132"=
        "82:pool_f2ddf7a6-4047-4cc9-97be-1d1a6c47ece9",
    neutron:vip="10.0.0.10",
    neutron:vip_port_id="2526c68a-5a9e-484c-8e00-0716388f6563"}
name           : "973a201a-8787-4f6e-9b8f-ab9f93c31f44"
protocol       : []
vips           : {"10.0.0.10:82"="10.0.0.107:80,20.0.0.107:80"}
```

As explained earlier in the design section:

- If a network N1 has a LoadBalancer LB1 associated to it and one of its interfaces is added to a router R1, LB1 is associated with R1 as well.
- If a network N2 has a LoadBalancer LB2 and one of its interfaces is added to the router R1, then R1 will have both LoadBalancers LB1 and LB2. N1 and N2 will also have both the LoadBalancers associated to them. However, kindly note that though network N1 would have both LB1 and LB2 LoadBalancers associated with it, only LB1 would be the LoadBalancer which has a direct reference to the network N1, since LB1 was created on N1. This is visible in the `ls_ref` key of the `external_ids` column in LB1s entry in the `load_balancer` table.
- If a network N3 is added to the router R1, N3 will also have both LoadBalancers (LB1, LB2) associated to it.
- If the interface to network N2 is removed from R1, network N2 will now only have LB2 associated with it. Networks N1 and N3 and router R1 will have LoadBalancer LB1 associated with them.

Limitations

Following actions are not supported by OVN Driver:

- Creating a LoadBalancer/Listener/Pool with L7 Protocol
- Creating HealthMonitors
- Currently only one algorithm is supported for pool management (Source IP Port)
- Creating Listeners and Pools with different protocols. They should be of the same protocol type.

Following issue exists with OVNs integration with Octavia:

- If creation/deletion of a LoadBalancer, Listener, Pool or Member fails, then the corresponding object will remain in the DB in a PENDING_* state.

Support Matrix

A detailed matrix of the operations supported by OVN Provider driver in Octavia can be found in https://docs.openstack.org/octavia/latest/user/feature-classification/index.html

Other References

[1] Octavia API: https://docs.openstack.org/api-ref/load-balancer/v2/

[2] Octavia Glossary: https://docs.openstack.org/octavia/queens/reference/glossary.html

[3] https://github.com/openvswitch/ovs/commit/612f80fa8ebf88dad2e204364c6c02b451dca36c

[4] https://docs.openstack.org/api-ref/load-balancer/v2/index.html#status-codes

[5] https://github.com/openvswitch/ovs/blob/d1b235d7a6246e00d4afc359071d3b6b3ed244c3/ovn/ovn-nb.ovsschema#L117

Distributed OVSDB events handler

This document presents the problem and proposes a solution for handling OVSDB events in a distributed fashion in ovn driver.

Problem description

In ovn driver, the OVSDB Monitor class is responsible for listening to the OVSDB events and performing certain actions on them. We use it extensively for various tasks including critical ones such as monitoring for port binding events (in order to notify Neutron/Nova that a port has been bound to a certain chassis). Currently, this class uses a distributed OVSDB lock to ensure that only one instance handles those events at a time.

The problem with this approach is that it creates a bottleneck because even if we have multiple Neutron Workers running at the moment, only one is actively handling those events. And, this problem is highlighted even more when working with technologies such as containers which rely on creating multiple ports at a time and waiting for them to be bound.

Proposed change

In order to fix this problem, this document proposes using a Consistent Hash Ring to split the load of handling events across multiple Neutron Workers.

A new table called `ovn_hash_ring` will be created in the Neutron Database where the Neutron Workers capable of handling OVSDB events will be registered. The table will use the following schema:

Column name	Type	Description
node_uuid	String	Primary key. The unique identification of a Neutron Worker.
hostname	String	The hostname of the machine this Node is running on.
created_at	Date-Time	The time that the entry was created. For troubleshooting purposes.
updated_at	Date-Time	The time that the entry was updated. Used as a heartbeat to indicate that the Node is still alive.

This table will be used to form the Consistent Hash Ring. Fortunately, we have an implementation already in the tooz library of OpenStack. It was contributed by the Ironic team which also uses this data structure in order to spread the API request load across multiple Ironic Conductors.

Heres how a Consistent Hash Ring from tooz works:

```
from tooz import hashring

hring = hashring.HashRing({'worker1', 'worker2', 'worker3'})

# Returns set(['worker3'])
hring[b'event-id-1']

# Returns set(['worker1'])
hring[b'event-id-2']
```

How OVSDB Monitor will use the Ring

Every instance of the OVSDB Monitor class will be listening to a series of events from the OVSDB database and each of them will have a unique ID registered in the database which will be part of the *Consistent Hash Ring*.

When an event arrives, each OVSDB Monitor instance will hash that event UUID and the ring will return one instance ID, which will then be compared with its own ID and if it matches that instance will then process the event.

Verifying status of OVSDB Monitor instance

A new maintenance task will be created in ovn driver which will update the `updated_at` column from the `ovn_hash_ring` table for the entries matching its hostname indicating that all Neutron Workers running on that hostname are alive.

Note that only a single maintenance instance runs on each machine so the writes to the Neutron database are optimized.

When forming the ring, the code should check for entries where the value of `updated_at` column is newer than a given timeout. Entries that havent been updated in a certain time wont be part of the ring. If the ring already exists it will be re-balanced.

Clean up and minimizing downtime window

Apart from heartbeating, we need to make sure that we remove the Nodes from the ring when the service is stopped or killed.

By stopping the `neutron-server` service, all Nodes sharing the same hostname as the machine where the service is running will be removed from the `ovn_hash_ring` table. This is done by handling the SIGTERM event. Upon this event arriving, ovn driver should invoke the clean up method and then let the process halt.

Unfortunately nothing can be done in case of a SIGKILL, this will leave the nodes in the database and they will be part of the ring until the timeout is reached or the service is restarted. This can introduce a window of time which can result in some events being lost. The current implementation shares the same problem, if the instance holding the current OVSDB lock is killed abruptly, events will be lost until the lock is moved on to the next instance which is alive. One could argue that the current implementation aggravates the problem because all events will be lost where with the distributed mechanism **some** events will be lost. As far as distributed systems goes, thats a normal scenario and things are soon corrected.

Ideas for future improvements

This section contains some ideas that can be added on top of this work to further improve it:

- Listen to changes to the Chassis table in the OVSDB and force a ring re-balance when a Chassis is added or removed from it.
- Cache the ring for a short while to minimize the database reads when the service is under heavy load.
- To greater minimize/avoid event losses it would be possible to cache the last X events to be reprocessed in case a node times out and the ring re-balances.

L3 HA Scheduling of Gateway Chassis

Problem Description

Currently if a single network node is active in the system, gateway chassis for the routers would be scheduled on that node. However, when a new node is added to the system, neither rescheduling nor rebalancing occur automatically. This makes the router created on the first node to be not in HA mode.

Side-effects of this behavior include:

- Skewed up load on different network nodes due to lack of router rescheduling.
- If the active node, where the gateway chassis for a router is scheduled goes down, then because of lack of HA the North-South traffic from that router will be hampered.

Overview of Proposed Approach

Gateway scheduling has been proposed in [2]. However, rebalancing or rescheduling was not a part of that solution. This specification clarifies what is rescheduling and rebalancing. Rescheduling would automatically happen on every event triggered by addition or deletion of chassis. Rebalancing would be only triggered by manual operator action.

Rescheduling of Gateway Chassis

In order to provide proper rescheduling of the gateway ports during addition or deletion of the chassis, following approach can be considered:

- Identify the number of chassis in which each router has been scheduled
  - Consider router for scheduling if no. of chassis < *MAX_GW_CHASSIS*

MAX_GW_CHASSIS is defined in [0]

- Find a list of chassis where router is scheduled and reschedule it up to *MAX_GW_CHASSIS* gateways using list of available candidates. Do not modify the primary chassis association to not interrupt network flows.

Rescheduling is an event triggered operation which will occur whenever a chassis is added or removed. When it happend, `schedule_unhosted_gateways()` [1] will be called to host the unhosted gateways. Routers without gateway ports are excluded in this operation because those are not connected to provider networks and havent the gateway ports. More information about it can be found in the `gateway_chassis` table definition in OVN NorthBound DB [5].

Chassis which has the flag `enable-chassis-as-gw` enabled in their OVN southbound database table, would be the ones eligible for hosting the routers. Rescheduling of router depends on current prorities set. Each chassis is given a specific priority for the routers gateway and priority increases with increasing value (i.e. 1 < 2 < 3). The highest prioritized chassis hosts gateway port. Other chassis are selected as backups.

There are two approaches for rescheduling supported by ovn driver right now: * Least loaded - select least-loaded chassis first, * Random - select chassis randomly.

Few points to consider for the design:

- If there are 2 Chassis C1 and C2, where the routers are already balanced, and a new chassis C3 is added, then routers should be rescheduled only from C1 to C3 and C2 to C3. Rescheduling from C1 to C2 and vice-versa should not be allowed.
- In order to reschedule the routers chassis, the `primary` chassis for a gateway router will be left untouched. However, for the scenario where all routers are scheduled in only one chassis which is available as gateway, the addition of the second gateway chassis would schedule the router gateway ports at a lower priority on the new chassis.

Following scenarios are possible which have been considered in the design:

- **Case #1:**
  - System has only one chassis C1 and all router gateway ports are scheduled on it. We add a new chassis C2.
  - Behavior: All the routers scheduled on C1 will also be scheduled on C2 with priority 1.
- **Case #2:**
  - System has 2 chassis C1 and C2 during installation. C1 goes down.
  - Behavior: In this case, all routers would be rescheduled to C2. Once C1 is back up, routers would be rescheduled on it. However, since C2 is now the new primary, routers on C1 would have lower priority.
- **Case #3:**
  - System has 2 chassis C1 and C2 during installation. C3 is added to it.

   - Behavior: In this case, routers would not move their primary chassis associations. So routers which have their primary on C1, would remain there, and same for routers on C2. However, lower proritized candidates of existing gateways would be scheduled on the chassis C3, depending on the type of used scheduler (Random or LeastLoaded).

Rebalancing of Gateway Chassis

Rebalancing is the second part of the design and it assigns a new primary to already scheduled router gateway ports. Downtime is expected in this operation. Rebalancing of routers can be achieved using external cli script. Similar approach has been implemeneted for DHCP rescheduling [4]. The primary chassis gateway could be moved only to other, previously scheduled gateway. Rebalancing of chassis occurs only if number of scheduled primary chassis ports per each provider network hosted by given chassis is higher than average number of hosted primary gateway ports per chassis per provider network.

This dependency is determined by formula:

avg_gw_per_chassis = num_gw_by_provider_net / num_chassis_with_provider_net

Where:

- avg_gw_per_chassis - average number of scheduler primary gateway chassis withing same provider network.
- num_gw_by_provider_net - number of primary chassis gateways scheduled in given provider networks.
- num_chassis_with_provider_net - number of chassis that has connectivity to given provider network.

The rebalancing occurs only if:

num_gw_by_provider_net_by_chassis > avg_gw_per_chassis

Where:

- num_gw_by_provider_net_by_chassis - number of hosted primary gateways by given provider network by given chassis
- avg_gw_per_chassis - average number of scheduler primary gateway chassis withing same provider network.

Following scenarios are possible which have been considered in the design:

- **Case #1:**
   - System has only two chassis C1 and C2. Chassis host the same number of gateways.
   - Behavior: Rebalancing doesnt occur.
- **Case #2:**
   - System has only two chassis C1 and C2. C1 hosts 3 gateways. C2 hosts 2 gateways.
   - Behavior: Rebalancing doesnt occur to not continuously move gateways between chassis in loop.
- **Case #3:**
   - System has two chassis C1 and C2. In meantime third chassis C3 has been added to the system.

- Behavior: Rebalancing should occur. Gateways from C1 and C2 should be moved to C3 up to avg_gw_per_chassis.

- **Case #4:**
  - System has two chassis C1 and C2. C1 is connected to provnet1, but C2 is connected to provnet2.
  - Behavior: Rebalancing shouldnt occur because of lack of chassis within same provider network.

References

ML2/OVN Port forwarding

ML2/OVN supports Port Forwarding (PF) across the North/South data plane. Specific L4 Ports of the Floating IP (FIP) can be directed to a specific FixedIP:PortNumber of a VM, so that different services running in a VM can be isolated, and can communicate with external networks easily.

OVNs native load balancing (LB) feature is used for providing this functionality. An OVN load balancer is expressed in the OVN northbound load_balancer table for all mappings for a given FIP+protocol. All PFs for the same FIP+protocol are kept as Virtual IP (VIP) mappings inside a LB entry. See the diagram below for an example of how that looks like:

```
VIP:PORT = MEMBER1:MPORT1, MEMBER2:MPORT2

The same is extended for port forwarding as:

FIP:PORT = PRIVATE_IP:PRIV_PORT

      Neutron DB                                OVN Northbound DB

+---------------------+               +---------------------------------+
| Floating IP AA      |               | Load Balancer AA UDP            |
|                     |               |                                 |
| +-----------------+ |               |                                 |
| | Port Forwarding | |   +----------->AA:portA => internal IP1:portX   |
| |                 | |   |           |                                 |
| | External PortA  +-----+   +-------->AA:portB => internal IP2:portX   |
| | Fixed IP1 PortX | |       |       |                                 |
| | Protocol: UDP   | |       |       +---------------------------------+
| +-----------------+ |       |
|                     |       |       +---------------------------------+
| +-----------------+ |       |       | Load Balancer AA TCP            |
| | Port Forwarding | |       |       |                                 |
| |                 | |       |       |                                 |
| | External PortB  +---------+   +---->AA:portC => internal IP3:portX   |
| | Fixed IP2 PortX | |           |   |                                 |
| | Protocol: UDP   | |           |   +---------------------------------+
| +-----------------+ |           |
|                     |           |
| +-----------------+ |           |
| | Port Forwarding | |           |
```

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```
| |                 | |           |  +---------------------------------+
| | External PortC  | |           |  | Load Balancer BB TCP             |
| | Fixed IP3 PortX +-------------+  |                                 |
| | Protocol: TCP   | |              |                                 |
| +-----------------+ |    +---------->BB:portD => internal IP4:portX  |
|                     |    |         |                                 |
+---------------------+    |         +---------------------------------+
                           |
                           |         +-------------------+
                           |         | Logical Router X1 |
+---------------------+    |         |                   |
| Floating IP BB      |    |         | Load Balancers:   |
|                     |    |         | AA UDP, AA TCP    |
| +-----------------+ |    |         +-------------------+
| | Port Forwarding | |    |
| |                 | |    |         +-------------------+
| | External PortD  | |    |         | Logical Router Z1 |
| | Fixed IP4 PortX +------+         |                   |
| | Protocol: TCP   | |              | Load Balancers:   |
| +-----------------+ |              | BB TCP            |
+---------------------+              +-------------------+
```

The OVN LB entries have names that include the id of the FIP and a protocol suffix. That protocol portion is needed because a single FIP can have multiple UDP and TCP port forwarding entries while a given LB entry can either be one or the other protocol (not both). Based on that, the format used to specify an LB entry is:

```
pf-floatingip-<NEUTRON_FIP_ID>-<PROTOCOL>
```

A revision value is present in external_ids of each OVN load balancer entry. That number is synchronized with floating IP entries (NOT the port forwarding!) of the Neutron database.

In order to differentiate a load balancer entry that was created by port forwarding vs load balancer entries maintained by ovn-octavia-provider, the external_ids field also has an owner value:

```
external_ids = {
   ovn_const.OVN_DEVICE_OWNER_EXT_ID_KEY: PORT_FORWARDING_PLUGIN,
   ovn_const.OVN_FIP_EXT_ID_KEY: pf_obj.floatingip_id,
   ovn_const.OVN_ROUTER_NAME_EXT_ID_KEY: rtr_name,
   neutron:revision_number: fip_obj.revision_number,
}
```

The following registry (API) neutron events trigger the OVN backend to map port forwarding into LB:

```
@registry.receives(PORT_FORWARDING_PLUGIN, [events.AFTER_INIT])
def register(self, resource, event, trigger, payload=None):
   registry.subscribe(self._handle_notification, PORT_FORWARDING, events.
↪AFTER_CREATE)
   registry.subscribe(self._handle_notification, PORT_FORWARDING, events.
↪AFTER_UPDATE)
```

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```
    registry.subscribe(self._handle_notification, PORT_FORWARDING, events.
↪AFTER_DELETE)
```

ML2/OVN Network Logging

ML2/OVN supports network logging, based on security groups. Unlike ML2/OVS, the driver for this functionality leverages the Northbound database to manage affected security group rules. Thus, there is no need for an agent.

It is good to keep in mind that Openstack Security Groups (SG) and their rules (SGR) map 1:1 into OVNs Port Groups (PG) and Access Control Lists (ACL):

```
Openstack Security Group <=> OVN Port Group
Openstack Security Group Rule <=> OVN ACL
```

Just like SGs have a list of SGRs, PGs have a list of ACLs. PGs also have a list of logical ports, but that is not really relevant in this context. With regards to Neutron ports, network logging entries (NLE) can filter on Neutron ports, also known as targets. When that is the case, the underlying implementation finds the corresponding SGs out of the Neutron port. So it is all back to SGs and affected SGRs. Or PGs and ACLs as far as OVN is concerned.

For more info on port groups, see: https://docs.openstack.org/networking-ovn/latest/contributor/design/acl_optimizations.html

In order to enable network logging, the Neutron OVN driver relies on 2 tables of the Northbound database: Meter and ACL.

Meter Table

Meters are how network logging events get throttled, so they do not negatively affect the control plane. Logged events are sent to the ovn-controller that runs locally on each compute node. Thus, the throttle keeps ovn-controller from getting overwhelmed. Note that the meters used for network logging do not rate-limit the datapath; they only affect the logs themselves. With the addition of fair meters, multiple ACLs can refer to the same meter without competing with each other for what logs get rate limited. This attribute is a pre-requisite for this feature, as the design aspires to keep the complexity associated with the management of meters outside Openstack. The benefit of ACLs sharing a fair meter is that a noisy neighbor (ACL) will not consume all the available capacity set for the meter.

For more info on fair meters, see: https://github.com/ovn-org/ovn/commit/880dca99eaf73db7e783999c29386d03c82093bf

Below is an example of a meter configuration in OVN. You can locate the fair, unit, burst_size, and rate attributes:

```
$ ovn-nbctl list meter
_uuid               : 70c76ba9-f303-471b-9d49-25dee299827f
bands               : [f114c205-a170-4425-8ca6-4e71099d1955]
external_ids        : {"neutron:device_owner"=logging-plugin}
fair                : true
name                : acl_log_meter
unit                : pktps
```

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```
$ ovn-nbctl list meter-band
_uuid               : f114c205-a170-4425-8ca6-4e71099d1955
action              : drop
burst_size          : 25
external_ids        : {}
rate                : 100
```

The burst_size and rate attributes are configurable through neutron.conf.services.logging.log_driver_opts. That is not new.

ACL Table

As mentioned before, ACLs are the OVNs counterpart to Openstacks SGRs. Moreover, there are a few attributes in each ACL that makes it able to provide the networking logging feature. Lets use the example below to point out the relevant fields:

```
 $ openstack network log create --resource-type security_group \
   --resource ${SG} --event ACCEPT logme -f value -c ID
 2e456c7f-154e-40a8-bb10-f88ba51b90b5

 $ openstack security group show ${SG} -f json -c rules | jq '.rules | .[2]'␣
↪| grep -v 'null'
 {
  "id": "de4ea1e4-c946-40ed-b5b6-53c59418dc0b",
  "tenant_id": "2600067ea3a446dba332d20a30ed44fa",
  "security_group_id": "c604e984-0789-4c9a-a297-3e7f62fa73fd",
  "ethertype": "IPv4",
  "direction": "egress",
  "standard_attr_id": 48,
  "tags": [],
  "created_at": "2021-02-06T22:17:44Z",
  "updated_at": "2021-02-06T22:17:44Z",
  "revision_number": 0,
  "project_id": "2600067ea3a446dba332d20a30ed44fa"
}

$ ovn-nbctl find acl \
  "external_ids:\"neutron:security_group_rule_id\""="de4ea1e4-c946-40ed-b5b6-
↪53c59418dc0b"
_uuid               : 791679e9-237d-4732-a31e-aa634496e02b
action              : allow-related
direction           : from-lport
external_ids        : {"neutron:security_group_rule_id"="de4ea1e4-c946-40ed-
↪b5b6-53c59418dc0b"}
log                 : true
match               : "inport == @pg_c604e984_0789_4c9a_a297_3e7f62fa73fd &&␣
↪ip4"
meter               : acl_log_meter
name                : neutron-2e456c7f-154e-40a8-bb10-f88ba51b90b5
priority            : 1002
```

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```
severity            : info
```

The first command creates a networking-log for a given SG. The second shows an SGR from that SG. The third shell command is where we can see how the ACL with the meter information gets populated. These are the attributes pertinent to network logging:

- log: a boolean that dictates whether a log will be generated. Even if the NLE applies to the SGR via its associated SG, this may be false if the action is not a match. That would be the case if the NLE specified event DROP, in this example.
- meter: this is the name of the fair meter. It is the same for all ACLs.
- name: This is a string composed of the prefix neutron- and the id of the NLE. It will be part of the generated logs.
- severity: this is the log severity that will be used by the ovn-controller. It is currently hard coded in Neutron, but can be made configurable in future releases.

If we poked the SGR with packets that match its criteria, the ovn-controller local to where the ACLs is enforced will log something that looks like this:

```
2021-02-16T11:59:00.640Z|00045|acl_log(ovn_pinctrl0)|INFO|
name="neutron-2e456c7f-154e-40a8-bb10-f88ba51b90b5",
verdict=allow, severity=info: icmp,vlan_tci=0x0000,dl_src=fa:16:3e:24:dc:88,
dl_dst=fa:16:3e:15:6d:e0,
nw_src=10.0.0.12,nw_dst=10.0.0.11,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,
icmp_code=0
```

It is beyond the scope of this document to talk about what happens after the logs are generated by ovn-controllers. The harvesting of files across compute nodes is something a project like Monasca may be a good fit.

Neutron Open vSwitch vhost-user Support

Neutron supports using Open vSwitch + DPDK vhost-user interfaces directly in the OVS ML2 driver and agent. The current implementation relies on a multiple configuration values and includes runtime verification of Open vSwitchs capability to provide these interfaces.

The OVS agent detects the capability of the underlying Open vSwitch installation and passes that information over RPC via the agent configurations dictionary. The ML2 driver uses this information to select the proper VIF type and binding details.

Platform requirements

- OVS 2.4.0+
- DPDK 2.0+

Configuration

```
[OVS]
datapath_type=netdev
vhostuser_socket_dir=/var/run/openvswitch
```

When OVS is running with DPDK support enabled, and the `datapath_type` is set to `netdev`, then the OVS ML2 driver will use the `vhost-user` VIF type and pass the necessary binding details to use OVS+DPDK and vhost-user sockets. This includes the `vhostuser_socket_dir` setting, which must match the directory passed to `ovs-vswitchd` on startup.

What about the networking-ovs-dpdk repo?

The networking-ovs-dpdk repo will continue to exist and undergo active development. This feature just removes the necessity for a separate ML2 driver and OVS agent in the networking-ovs-dpdk repo. The networking-ovs-dpdk project also provides a devstack plugin which also allows automated CI, a Puppet module, and an OpenFlow-based security group implementation.

Neutron Plugin Architecture

Salvatore Orlando: How to write a Neutron Plugin (if you really need to)

Plugin API

v2 Neutron Plug-in API specification.

NeutronPluginBaseV2 provides the definition of minimum set of methods that needs to be implemented by a v2 Neutron Plug-in.

class `neutron.neutron_plugin_base_v2.`**`NeutronPluginBaseV2`**

abstract create_network(*context*, *network*)

Create a network.

Create a network, which represents an L2 network segment which can have a set of subnets and ports associated with it.

Parameters

- **context** neutron api request context
- **network** dictionary describing the network, with keys as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. All keys will be populated.

abstract create_port(*context*, *port*)

Create a port.

Create a port, which is a connection point of a device (e.g., a VM NIC) to attach to a L2 neutron network.

Parameters

- **context** neutron api request context
- **port** dictionary describing the port, with keys as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. All keys will be populated.

abstract create_subnet(*context*, *subnet*)

Create a subnet.

Create a subnet, which represents a range of IP addresses that can be allocated to devices

Parameters

- **context** neutron api request context
- **subnet** dictionary describing the subnet, with keys as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. All keys will be populated.

`create_subnetpool`(*context*, *subnetpool*)

Create a subnet pool.

Parameters

- **context** neutron api request context
- **subnetpool** Dictionary representing the subnetpool to create.

`abstract delete_network`(*context*, *id*)

Delete a network.

Parameters

- **context** neutron api request context
- **id** UUID representing the network to delete.

`abstract delete_port`(*context*, *id*)

Delete a port.

Parameters

- **context** neutron api request context
- **id** UUID representing the port to delete.

`abstract delete_subnet`(*context*, *id*)

Delete a subnet.

Parameters

- **context** neutron api request context
- **id** UUID representing the subnet to delete.

`delete_subnetpool`(*context*, *id*)

Delete a subnet pool.

Parameters

- **context** neutron api request context
- **id** The UUID of the subnet pool to delete.

`abstract get_network`(*context*, *id*, *fields=None*)

Retrieve a network.

Parameters

- **context** neutron api request context
- **id** UUID representing the network to fetch.
- **fields** a list of strings that are valid keys in a network dictionary as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Only these fields will be returned.

`abstract get_networks`(*context*, *filters=None*, *fields=None*, *sorts=None*, *limit=None*, *marker=None*, *page_reverse=False*)

Retrieve a list of networks.

The contents of the list depends on the identity of the user making the request (as indicated by the context) as well as any filters.

Parameters

- **context** neutron api request context
- **filters** a dictionary with keys that are valid keys for a network as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Values in this dictionary are an iterable containing values that will be used for an exact match comparison for that value. Each result returned by this function will have matched one of the values for each key in filters.
- **fields** a list of strings that are valid keys in a network dictionary as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Only these fields will be returned.

`get_networks_count`(*context*, *filters=None*)

Return the number of networks.

The result depends on the identity of the user making the request (as indicated by the context) as well as any filters.

Parameters

- **context** neutron api request context
- **filters** a dictionary with keys that are valid keys for a network as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Values in this dictionary are an iterable containing values that will be used for an exact match comparison for that value. Each result returned by this function will have matched one of the values for each key in filters.

NOTE: this method is optional, as it was not part of the originally
defined plugin API.

`abstract get_port`(*context*, *id*, *fields=None*)

Retrieve a port.

Parameters

- **context** neutron api request context
- **id** UUID representing the port to fetch.
- **fields** a list of strings that are valid keys in a port dictionary as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Only these fields will be returned.

`abstract get_ports`(*context*, *filters=None*, *fields=None*, *sorts=None*, *limit=None*, *marker=None*, *page_reverse=False*)

Retrieve a list of ports.

The contents of the list depends on the identity of the user making the request (as indicated by the context) as well as any filters.

Parameters

- **context** neutron api request context
- **filters** a dictionary with keys that are valid keys for a port as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Values in this dictionary are an iterable containing values that will be used for an exact match comparison for that value. Each result returned by this function will have matched one of the values for each key in filters.
- **fields** a list of strings that are valid keys in a port dictionary as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Only these fields will be returned.

`get_ports_count`(*context*, *filters=None*)

Return the number of ports.

The result depends on the identity of the user making the request (as indicated by the context) as well as any filters.

Parameters

- **context** neutron api request context
- **filters** a dictionary with keys that are valid keys for a network as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Values in this dictionary are an iterable containing values that will be used for an exact match comparison for that value. Each result returned by this function will have matched one of the values for each key in filters.

Note

this method is optional, as it was not part of the originally defined plugin API.

`abstract get_subnet`(*context*, *id*, *fields=None*)

Retrieve a subnet.

Parameters

- **context** neutron api request context
- **id** UUID representing the subnet to fetch.
- **fields** a list of strings that are valid keys in a subnet dictionary as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Only these fields will be returned.

`get_subnetpool`(*context*, *id*, *fields=None*)

Show a subnet pool.

Parameters

- **context** neutron api request context
- **id** The UUID of the subnetpool to show.

`get_subnetpools`(*context*, *filters=None*, *fields=None*, *sorts=None*, *limit=None*, *marker=None*, *page_reverse=False*)

Retrieve list of subnet pools.

`abstract get_subnets`(*context*, *filters=None*, *fields=None*, *sorts=None*, *limit=None*, *marker=None*, *page_reverse=False*)

Retrieve a list of subnets.

The contents of the list depends on the identity of the user making the request (as indicated by the context) as well as any filters.

Parameters

- **context** neutron api request context
- **filters** a dictionary with keys that are valid keys for a subnet as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Values in this dictionary are an iterable containing values that will be used for an exact match comparison for that value. Each result returned by this function will have matched one of the values for each key in filters.
- **fields** a list of strings that are valid keys in a subnet dictionary as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Only these fields will be returned.

`get_subnets_count`(*context*, *filters=None*)

Return the number of subnets.

The result depends on the identity of the user making the request (as indicated by the context) as well as any filters.

Parameters

- **context** neutron api request context
- **filters** a dictionary with keys that are valid keys for a network as listed in the `RESOURCE_ATTRIBUTE_MAP` object in `neutron/api/v2/attributes.py`. Values in this dictionary are an iterable containing values that will be used for an exact match comparison for that value. Each result returned by this function will have matched one of the values for each key in filters.

Note

this method is optional, as it was not part of the originally defined plugin API.

`has_native_datastore`()

Return True if the plugin uses Neutrons native datastore.

> **Note**
>
> plugins like ML2 should override this method and return True.

rpc_state_report_workers_supported()

Return whether the plugin supports state report RPC workers.

> **Note**
>
> this method is optional, as it was not part of the originally defined plugin API.

rpc_workers_supported()

Return whether the plugin supports multiple RPC workers.

A plugin that supports multiple RPC workers should override the start_rpc_listeners method to ensure that this method returns True and that start_rpc_listeners is called at the appropriate time. Alternately, a plugin can override this method to customize detection of support for multiple rpc workers

> **Note**
>
> this method is optional, as it was not part of the originally defined plugin API.

start_rpc_listeners()

Start the RPC listeners.

Most plugins start RPC listeners implicitly on initialization. In order to support multiple process RPC, the plugin needs to expose control over when this is started.

> **Note**
>
> this method is optional, as it was not part of the originally defined plugin API.

start_rpc_state_reports_listener()

Start the RPC listeners consuming state reports queue.

This optional method creates rpc consumer for REPORTS queue only.

> **Note**
>
> this method is optional, as it was not part of the originally defined plugin API.

abstract update_network(*context*, *id*, *network*)

Update values of a network.

Parameters

- **context** neutron api request context
- **id** UUID representing the network to update.

- **network** dictionary with keys indicating fields to update. valid keys are those that have a value of True for allow_put as listed in the RESOURCE_ATTRIBUTE_MAP object in neutron/api/v2/attributes.py.

abstract update_port(*context*, *id*, *port*)

Update values of a port.

Parameters

- **context** neutron api request context
- **id** UUID representing the port to update.
- **port** dictionary with keys indicating fields to update. valid keys are those that have a value of True for allow_put as listed in the RESOURCE_ATTRIBUTE_MAP object in neutron/api/v2/attributes.py.

abstract update_subnet(*context*, *id*, *subnet*)

Update values of a subnet.

Parameters

- **context** neutron api request context
- **id** UUID representing the subnet to update.
- **subnet** dictionary with keys indicating fields to update. valid keys are those that have a value of True for allow_put as listed in the RESOURCE_ATTRIBUTE_MAP object in neutron/api/v2/attributes.py.

update_subnetpool(*context*, *id*, *subnetpool*)

Update a subnet pool.

Parameters

- **context** neutron api request context
- **subnetpool** Dictionary representing the subnetpool attributes to update.

Policy Enforcement and Authorization

As most OpenStack projects, Neutron leverages oslo_policy[1]. However, since Neutron loves to be special and complicate every developers life, it also augments oslo_policy capabilities by:

- A wrapper module with its own API: neutron.policy;
- The ability of adding fine-grained checks on attributes for resources in request bodies;
- The ability of using the policy engine to filter out attributes in responses;
- Adding some custom rule checks beyond those defined in oslo_policy;

This document discusses Neutron-specific aspects of policy enforcement, and in particular how the enforcement logic is wired into API processing. For any other information please refer to the developer documentation for oslo_policy[2].

[1] Oslo policy module
[2] Oslo policy developer

Authorization workflow

The Neutron API controllers perform policy checks in two phases during the processing of an API request:

- Request authorization, immediately before dispatching the request to the plugin layer for POST, PUT, and DELETE, and immediately after returning from the plugin layer for GET requests;
- Response filtering, when building the response to be returned to the API consumer.

Request authorization

The aim of this step is to authorize processing for a request or reject it with an error status code. This step uses the `neutron.policy.enforce` routine. This routine raises `oslo_policy.PolicyNotAuthorized` when policy enforcement fails. The Neutron REST API controllers catch this exception and return:

- A 403 response code on a POST request or an PUT request for an object owned by the project submitting the request;
- A 403 response for failures while authorizing API actions such as `add_router_interface`;
- A 404 response for DELETE, GET and all other PUT requests.

For DELETE operations the resource must first be fetched. This is done invoking the same `_item`[3] method used for processing GET requests. This is also true for PUT operations, since the Neutron API implements PATCH semantics for PUTs. The criteria to evaluate are built in the `_build_match_rule`[4] routine. This routine takes in input the following parameters:

- The action to be performed, in the `<operation>_<resource>` form, `e.g.: create_network`
- The data to use for performing checks. For POST operations this could be a partial specification of the object, whereas it is always a full specification for GET, PUT, and DELETE requests, as resource data are retrieved before dispatching the call to the plugin layer.
- The collection name for the resource specified in the previous parameter; for instance, for a network it would be the networks.

The `_build_match_rule` routine returns a `oslo_policy.RuleCheck` instance built in the following way:

- Always add a check for the action being performed. This will match a policy like create_network in `policy.yaml`;
- Return for GET operations; more detailed checks will be performed anyway when building the response;
- For each attribute which has been explicitly specified in the request create a rule matching policy names in the form `<operation>_<resource>:<attribute>` rule, and link it with the previous rule with an And relationship (using `oslo_policy.AndCheck`); this step will be performed only if the `enforce_policy` flag is set to `True` in the resource attribute descriptor (usually found in a data structure called `RESOURCE_ATTRIBUTE_MAP`);
- If the attribute is a composite one then further rules will be created; These will match policy names in the form `<operation>_<resource>:<attribute>:<sub_attribute>`. An And relationship will be used in this case too.

[3] API controller item method

[4] Policy engines build_match_rule method

As all the rules to verify are linked by And relationships, all the policy checks should succeed in order for a request to be authorized. Rule verification is performed by `oslo_policy` with no customization from the Neutron side.

Response Filtering

Some Neutron extensions, like the provider networks one, add some attribute to resources which are however not meant to be consumed by all clients. This might be because these attributes contain implementation details, or are meant only to be used when exchanging information between services, such as Nova and Neutron;

For this reason the policy engine is invoked again when building API responses. This is achieved by the `_exclude_attributes_by_policy`[5] method in `neutron.api.v2.base.Controller`;

This method, for each attribute in the response returned by the plugin layer, first checks if the `is_visible` flag is True. In that case it proceeds to checking policies for the attribute; if the policy check fails the attribute is added to a list of attributes that should be removed from the response before returning it to the API client.

The neutron.policy API

The `neutron.policy` module exposes a simple API whose main goal if to allow the REST API controllers to implement the authorization workflow discussed inu this document. It is a bad practice to call the policy engine from within the plugin layer, as this would make request authorization dependent on configured plugins, and therefore make API behaviour dependent on the plugin itself, which defies Neutron tenet of being backend agnostic.

The neutron.policy API exposes the following routines:

- `init` Initializes the policy engine loading rules from the json policy (files). This method can safely be called several times.
- `reset` Clears all the rules currently configured in the policy engine. It is called in unit tests and at the end of the initialization of core API router[6] in order to ensure rules are loaded after all the extensions are loaded.
- `refresh` Combines init and reset. Called when a SIGHUP signal is sent to an API worker.
- `set_rules` Explicitly set policy engines rules. Used only in unit tests.
- `check` Perform a check using the policy engine. Builds match rules as described in this document, and then evaluates the resulting rule using oslo_policys policy engine. Returns True if the checks succeeds, false otherwise.
- `enforce` Operates like the check routine but raises if the check in oslo_policy fails.

Neutron specific policy rules

Neutron provides two additional policy rule classes in order to support the augmented authorization capabilities it provides. They both extend `oslo_policy.RuleCheck` and are registered using the `oslo_policy.register` decorator.

[5] exclude_attributes_by_policy method
[6] Policy reset in neutron.api.v2.router

OwnerCheck: Extended Checks for Resource Ownership

This class is registered for rules matching the `tenant_id` keyword and overrides the generic check performed by oslo_policy in this case. It uses for those cases where neutron needs to check whether the project submitting a request for a new resource owns the parent resource of the one being created. Current usages of `OwnerCheck` include, for instance, creating and updating a subnet. This class supports the extension parent resources owner check which the parent resource introduced by service plugins. Such as router and floatingip owner check for `router` service plugin. Developers can register the extension resource name and service plugin name which were registered in neutron-lib into `EXT_PARENT_RESOURCE_MAPPING` which is located in `neutron_lib.services.constants`.

The check, performed in the `__call__` method, works as follows:

- verify if the target field is already in the target data. If yes, then simply verify whether the value for the target field in target data is equal to value for the same field in credentials, just like `oslo_policy.GenericCheck` would do. This is also the most frequent case as the target field is usually `tenant_id`;
- if the previous check failed, extract a parent resource type and a parent field name from the target field. For instance `networks:tenant_id` identifies the `tenant_id` attribute of the `network` resource. For extension parent resource case, `ext_parent:tenant_id` identifies the `tenant_id` attribute of the registered extension resource in `EXT_PARENT_RESOURCE_MAPPING`;
- if no parent resource or target field could be identified raise a `PolicyCheckError` exception;
- Retrieve a parent foreign key from the `_RESOURCE_FOREIGN_KEYS` data structure in `neutron.policy`. This foreign key is simply the attribute acting as a primary key in the parent resource. A `PolicyCheckError` exception will be raised if such parent foreign key cannot be retrieved;
- Using the core plugin, retrieve an instance of the resource having parent foreign key as an identifier;
- Finally, verify whether the target field in this resource matches the one in the initial request data. For instance, for a port create request, verify whether the `tenant_id` of the port data structure matches the `tenant_id` of the network where this port is being created.

FieldCheck: Verify Resource Attributes

This class is registered with the policy engine for rules matching the field keyword, and provides a way to perform fine grained checks on resource attributes. For instance, using this class of rules it is possible to specify a rule for granting every project read access to shared resources.

In policy.yaml, a FieldCheck rules is specified in the following way:

```
> field: <resource>:<field>=<value>
```

This will result in the initialization of a FieldCheck that will check for `<field>` in the target resource data, and return `True` if it is equal to `<value>` or return `False` is the `<field>` either is not equal to `<value>` or does not exist at all.

Guidance for Neutron API developers

When developing REST APIs for Neutron it is important to be aware of how the policy engine will authorize these requests. This is true both for APIs served by Neutron core and for the APIs served by the various Neutron stadium services.

- If an attribute of a resource might be subject to authorization checks then the `enforce_policy` attribute should be set to `True`. While setting this flag to `True` for each attribute is a viable strategy, it is worth noting that this will require a call to the policy engine for each attribute, thus consistently increasing the time required to complete policy checks for a resource. This could result in a scalability issue, especially in the case of list operations retrieving a large number of resources;
- Some resource attributes, even if not directly used in policy checks might still be required by the policy engine. This is for instance the case of the `tenant_id` attribute. For these attributes the `required_by_policy` attribute should always set to `True`. This will ensure that the attribute is included in the resource data sent to the policy engine for evaluation;
- The `tenant_id` attribute is a fundamental one in Neutron API request authorization. The default policy, `admin_or_owner`, uses it to validate if a project owns the resource it is trying to operate on. To this aim, if a resource without a tenant_id is created, it is important to ensure that ad-hoc authZ policies are specified for this resource.
- There is still only one check which is hardcoded in Neutrons API layer: the check to verify that a project owns the network on which it is creating a port. This check is hardcoded and is always executed when creating a port, unless the network is shared. Unfortunately a solution for performing this check in an efficient way through the policy engine has not yet been found. Due to its nature, there is no way to override this check using the policy engine.
- It is strongly advised to not perform policy checks in the plugin or in the database management classes. This might lead to divergent API behaviours across plugins. Also, it might leave the Neutron DB in an inconsistent state if a request is not authorized after it has already been dispatched to the backend.

Notes

- No authorization checks are performed for requests coming from the RPC over AMQP channel. For all these requests a neutron admin context is built, and the plugins will process them as such.
- For `PUT` and `DELETE` requests a 404 error is returned on request authorization failures rather than a 403, unless the project submitting the request own the resource to update or delete. This is to avoid conditions in which an API client might try and find out other projects resource identifiers by sending out `PUT` and `DELETE` requests for random resource identifiers.
- There is no way at the moment to specify an `OR` relationship between two attributes of a given resource (eg.: `port.name == 'meh' or port.status == 'DOWN'`), unless the rule with the or condition is explicitly added to the policy.yaml file.
- `OwnerCheck` performs a plugin access; this will likely require a database access, but since the behaviour is implementation specific it might also imply a round-trip to the backend. This class of checks, when involving retrieving attributes for parent resources should be used very sparingly.
- In order for `OwnerCheck` rules to work, parent resources should have an entry in `neutron.policy._RESOURCE_FOREIGN_KEYS`; moreover the resource must be managed by the core plugin (ie: the one defined in the core_plugin configuration variable)

Policy-in-Code support

Guideline on defining in-code policies

The following is the guideline of policy definitions.

Ideally we should define all available policies, but in the neutron policy enforcement it is not practical to define all policies because we check all attributes of a target resource in the *Response Filtering*. Considering this, we have the special guidelines for get operation.

- All policies of `<action>_<resource>` must be defined for all types of operations. Valid actions are `create`, `update`, `delete` and `get`.
- `get_<resourceS>` (get plural) is unnecessary. The neutron API layer use a single form policy `get_<resource>` when listing resources[7][8].
- Member actions for individual resources must be defined. For example, `add_router_interface` of `router` resource.
- All policies with attributes on create, update and delete actions must be defined. `<action>_<resource>:<attribute>(:<sub_attribute>)` policy is required for attributes with `enforce_policy` in the API definitions. Note that it is recommended to define even if a rule is same as for `<action>_<resource>` from the documentation perspective.
- For a policy with attributes of get actions like `get_<resource>:<attribute>(:<sub_attribute>)`, the following guideline is applied:
  - A policy with an attribute must be defined if the policy is different from the policy for `get_<resource>` (without attributes).
  - If a policy with an attribute is same as for `get_<resource>`, there is no need to define it explicitly. This is for simplicity. We check all attributes of a target resource in the process of *Response Filtering* so it leads to a long long policy definitions for get actions in our documentation. It is not happy for operators either.
  - If an attribute is marked as `enforce_policy`, it is recommended to define the corresponding policy with the attribute. This is for clarification. If an attribute is marked as `enforce_policy` in the API definitions, for example, the neutron API limits to set such attribute only to admin users but allows to retrieve a value for regular users. If policies for the attribute are different across the types of operations, it is better to define all of them explicitly.

Registering policies in neutron related projects

Policy-in-code support in neutron is a bit different from other projects because the neutron server needs to load policies in code from multiple projects. Each neutron related project should register the following two entry points `oslo.policy.policies` and `neutron.policies` in `setup.cfg` like below:

```
oslo.policy.policies =
    neutron = neutron.conf.policies:list_rules
neutron.policies =
    neutron = neutron.conf.policies:list_rules
```

The above two entries are same, but they have different purposes.

- The first entry point is a normal entry point defined by oslo.policy and it is used to generate a sample policy file[9][10].

[7] https://github.com/openstack/neutron/blob/051b6b40f3921b9db4f152a54f402c402cbf138c/neutron/pecan_wsgi/hooks/policy_enforcement.py#L173

[8] https://github.com/openstack/neutron/blob/051b6b40f3921b9db4f152a54f402c402cbf138c/neutron/pecan_wsgi/hooks/policy_enforcement.py#L143

[9] https://docs.openstack.org/oslo.policy/latest/user/usage.html#sample-file-generation

[10] https://docs.openstack.org/oslo.policy/latest/cli/index.html#oslopolicy-sample-generator

- The second one is specific to neutron. It is used by `neutron.policy` module to load policies of neutron related projects.

`oslo.policy.policies` entry point is used by all projects which adopt oslo.policy, so we cannot determine which projects are neutron related projects, so the second entry point is required.

The recommended entry point name is a repository name: For example, neutron-fwaas for FWaaS and networking-sfc for SFC:

```
oslo.policy.policies =
    neutron-fwaas = neutron_fwaas.policies:list_rules
neutron.policies =
    neutron-fwaas = neutron_fwaas.policies:list_rules
```

Except registering the `neutron.policies` entry point, other steps to be done in each neutron related project for policy-in-code support are same for all OpenStack projects.

References

Provisioning Blocks in relation to Composite Object Status

We use the STATUS field on objects to indicate when a resource is ready by setting it to ACTIVE so external systems know when its safe to use that resource. Knowing when to set the status to ACTIVE is simple when there is only one entity responsible for provisioning a given object. When that entity has finishing provisioning, we just update the STATUS directly to active. However, there are resources in Neutron that require provisioning by multiple asynchronous entities before they are ready to be used so managing the transition to the ACTIVE status becomes more complex. To handle these cases, Neutron has the provisioning_blocks module to track the entities that are still provisioning a resource.

The main example of this is with ML2, the L2 agents and the DHCP agents. When a port is created and bound to a host, its placed in the DOWN status. The L2 agent now has to setup flows, security group rules, etc for the port and the DHCP agent has to setup a DHCP reservation for the ports IP and MAC. Before the transition to ACTIVE, both agents must complete their work or the port user (e.g. Nova) may attempt to use the port and not have connectivity. To solve this, the provisioning_blocks module is used to track the provisioning state of each agent and the status is only updated when both complete.

High Level View

To make use of the provisioning_blocks module, provisioning components should be added whenever there is work to be done by another entity before an objects status can transition to ACTIVE. This is accomplished by calling the add_provisioning_component method for each entity. Then as each entity finishes provisioning the object, the provisioning_complete must be called to lift the provisioning block.

When the last provisioning block is removed, the provisioning_blocks module will trigger a callback notification containing the object ID for the objects resource type with the event PROVISIONING_COMPLETE. A subscriber to this event can now update the status of this object to ACTIVE or perform any other necessary actions.

A normal state transition will look something like the following:

1. Request comes in to create an object
2. Logic on the Neutron server determines which entities are required to provision the object and adds a provisioning component for each entity for that object.
3. A notification is emitted to the entities so they start their work.

4. Object is returned to the API caller in the DOWN (or BUILD) state.
5. Each entity tells the server when it has finished provisioning the object. The server calls provisioning_complete for each entity that finishes.
6. When provisioning_complete is called on the last remaining entity, the provisioning_blocks module will emit an event indicating that provisioning has completed for that object.
7. A subscriber to this event on the server will then update the status of the object to ACTIVE to indicate that it is fully provisioned.

For a more concrete example, see the section below.

ML2, L2 agents, and DHCP agents

ML2 makes use of the provisioning_blocks module to prevent the status of ports from being transitioned to ACTIVE until both the L2 agent and the DHCP agent have finished wiring a port.

When a port is created or updated, the following happens to register the DHCP agents provisioning blocks:

1. The subnet_ids are extracted from the fixed_ips field of the port and then ML2 checks to see if DHCP is enabled on any of the subnets.
2. The configuration for the DHCP agents hosting the network are looked up to ensure that at least one of them is new enough to report back that it has finished setting up the port reservation.
3. If either of the preconditions above fail, a provisioning block for the DHCP agent is not added and any existing DHCP agent blocks for that port are cleared to ensure the port isnt blocked waiting for an event that will never happen.
4. If the preconditions pass, a provisioning block is added for the port under the DHCP entity.

When a port is created or updated, the following happens to register the L2 agents provisioning blocks:

1. If the port is not bound, nothing happens because we dont know yet if an L2 agent is involved so we have to wait until a port update that binds it.
2. Once the port is bound, the agent based mechanism drivers will check if they have an agent on the bound host and if the VNIC type belongs to the mechanism driver, a provisioning block is added for the port under the L2 Agent entity.

Once the DHCP agent has finished setting up the reservation, it calls dhcp_ready_on_ports via the RPC API with the port ID. The DHCP RPC handler receives this and calls provisioning_complete in the provisioning module with the port ID and the DHCP entity to remove the provisioning block.

Once the L2 agent has finished setting up the reservation, it calls the normal update_device_list (or update_device_up) via the RPC API. The RPC callbacks handler calls provisioning_complete with the port ID and the L2 Agent entity to remove the provisioning block.

On the provisioning_complete call that removes the last record, the provisioning_blocks module emits a callback PROVISIONING_COMPLETE event with the port ID. A function subscribed to this in ML2 then calls update_port_status to set the port to ACTIVE.

At this point the normal notification is emitted to Nova allowing the VM to be unpaused.

In the event that the DHCP or L2 agent is down, the port will not transition to the ACTIVE status (as is the case now if the L2 agent is down). Agents must account for this by telling the server that wiring has been completed after configuring everything during startup. This ensures that ports created on offline agents (or agents that crash and restart) eventually become active.

To account for server instability, the notifications about port wiring be complete must use RPC calls so the agent gets a positive acknowledgement from the server and it must keep retrying until either the port is deleted or it is successful.

If an ML2 driver immediately places a bound port in the ACTIVE state (e.g. after calling a backend in update_port_postcommit), this patch will not have any impact on that process.

Quality of Service

Quality of Service advanced service is designed as a service plugin. The service is decoupled from the rest of Neutron code on multiple levels (see below).

QoS extends core resources (ports, networks) without using mixins inherited from plugins but through an ml2 extension driver.

Details about the DB models, API extension, and use cases can be found here: qos spec .

Service side design

- neutron.extensions.qos: base extension + API controller definition. Note that rules are subattributes of policies and hence embedded into their URIs.
- neutron.extensions.qos_fip: base extension + API controller definition. Adds qos_policy_id to floating IP, enabling users to set/update the binding QoS policy of a floating IP.
- neutron.services.qos.qos_plugin: QoSPlugin, service plugin that implements qos extension, receiving and handling API calls to create/modify policies and rules.
- neutron.services.qos.drivers.manager: the manager that passes object actions down to every enabled QoS driver and issues RPC calls when any of the drivers require RPC push notifications.
- neutron.services.qos.drivers.base: the interface class for pluggable QoS drivers that are used to update backends about new {create, update, delete} events on any rule or policy change, including precommit events that some backends could need for synchronization reason. The drivers also declare which QoS rules, VIF drivers and VNIC types are supported.
- neutron.core_extensions.base: Contains an interface class to implement core resource (port/network) extensions. Core resource extensions are then easily integrated into interested plugins. We may need to have a core resource extension manager that would utilize those extensions, to avoid plugin modifications for every new core resource extension.
- neutron.core_extensions.qos: Contains QoS core resource extension that conforms to the interface described above.
- neutron.plugins.ml2.extensions.qos: Contains ml2 extension driver that handles core resource updates by reusing the core_extensions.qos module mentioned above. In the future, we would like to see a plugin-agnostic core resource extension manager that could be integrated into other plugins with ease.

QoS plugin implementation guide

The neutron.extensions.qos.QoSPluginBase class uses method proxies for methods relating to QoS policy rules. Each of these such methods is generic in the sense that it is intended to handle any rule type. For example, QoSPluginBase has a create_policy_rule method instead of both create_policy_dscp_marking_rule and create_policy_bandwidth_limit_rule methods. The logic behind the proxies allows a call to a plugins create_policy_dscp_marking_rule to be handled by the create_policy_rule method, which will receive a QosDscpMarkingRule object as an argument in order to execute behavior specific to the DSCP marking rule type. This approach allows new rule types to be introduced without requiring a plugin to modify code as a result. As would be expected, any subclass of QoSPluginBase must override the base classs abc.abstractmethod methods, even if to raise NotImplemented.

Supported QoS rule types

Each QoS driver has a member called `supported_rules`, where the driver exposes the rules its able to handle.

For a list of all rule types, see: neutron.services.qos.qos_consts.VALID_RULE_TYPES.

The list of supported QoS rule types exposed by neutron is calculated as the common subset of rules supported by all active QoS drivers.

Note: the list of supported rule types reported by core plugin is not enforced when accessing QoS rule resources. This is mostly because then we would not be able to create rules while at least one of the QoS driver in gate lacks support for the rules were trying to test.

Database models

QoS design defines the following two conceptual resources to apply QoS rules for a port, a network or a floating IP:

- QoS policy
- QoS rule (type specific)

Each QoS policy contains zero or more QoS rules. A policy is then applied to a network or a port, making all rules of the policy applied to the corresponding Neutron resource.

When applied through a network association, policy rules could apply or not to neutron internal ports (like router, dhcp, etc..). The QosRule base object provides a default should_apply_to_port method which could be overridden. In the future we may want to have a flag in QoSNetworkPolicyBinding or QosRule to enforce such type of application (for example when limiting all the ingress of routers devices on an external network automatically).

Each project can have at most one default QoS policy, although is not mandatory. If a default QoS policy is defined, all new networks created within this project will have assigned this policy, as long as no other QoS policy is explicitly attached during the creation process. If the default QoS policy is unset, no change to existing networks will be made.

From database point of view, following objects are defined in schema:

- QosPolicy: directly maps to the conceptual policy resource.
- QosNetworkPolicyBinding, QosPortPolicyBinding, QosFIPPolicyBinding: define attachment between a Neutron resource and a QoS policy.
- QosPolicyDefault: defines a default QoS policy per project.
- QosBandwidthLimitRule: defines the rule to limit the maximum egress bandwidth.
- QosDscpMarkingRule: defines the rule that marks the Differentiated Service bits for egress traffic.
- QosMinimumBandwidthRule: defines the rule that creates a minimum bandwidth constraint.
- QosMinimumPacketRateRule: defines the rule that creates a minimum packet rate constraint.

All database models are defined under:

- neutron.db.qos.models

QoS versioned objects

For QoS, the following neutron objects are implemented:

- QosPolicy: directly maps to the conceptual policy resource, as defined above.
- QosPolicyDefault: defines a default QoS policy per project.
- QosBandwidthLimitRule: defines the instance bandwidth limit rule type, characterized by a max kbps and a max burst kbits. This rule has also a direction parameter to set the traffic direction, from the instances point of view.
- QosDscpMarkingRule: defines the DSCP rule type, characterized by an even integer between 0 and 56. These integers are the result of the bits in the DiffServ section of the IP header, and only certain configurations are valid. As a result, the list of valid DSCP rule types is: 0, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 46, 48, and 56.
- QosMinimumBandwidthRule: defines the minimum assured bandwidth rule type, characterized by a min_kbps parameter. This rule has also a direction parameter to set the traffic direction, from the instance point of view. The only direction now implemented is egress.
- QosMinimumPacketRateRule: defines the minimum assured packet rate rule type, characterized by a min_kpps parameter. This rule has also a direction parameter to set the traffic direction, from the instance point of view.

Those are defined in:

- neutron.objects.qos.policy
- neutron.objects.qos.rule

For QosPolicy neutron object, the following public methods were implemented:

- get_network_policy/get_port_policy/get_fip_policy: returns a policy object that is attached to the corresponding Neutron resource.
- attach_network/attach_port/attach_floatingip: attach a policy to the corresponding Neutron resource.
- detach_network/detach_port/detach_floatingip: detach a policy from the corresponding Neutron resource.

In addition to the fields that belong to QoS policy database object itself, synthetic fields were added to the object that represent lists of rules that belong to the policy. To get a list of all rules for a specific policy, a consumer of the object can just access the corresponding attribute via:

- policy.rules

Implementation is done in a way that will allow adding a new rule list field with little or no modifications in the policy object itself. This is achieved by smart introspection of existing available rule object definitions and automatic definition of those fields on the policy class.

Note that rules are loaded in a non lazy way, meaning they are all fetched from the database on policy fetch.

For Qos<type>Rule objects, an extendable approach was taken to allow easy addition of objects for new rule types. To accommodate this, fields common to all types are put into a base class called QosRule that is then inherited into type-specific rule implementations that, ideally, only define additional fields and some other minor things.

Note that the QosRule base class is not registered with oslo.versionedobjects registry, because its not expected that generic rules should be instantiated (and to suggest just that, the base rule class is marked as ABC).

QoS objects rely on some primitive database API functions that are added in:

- neutron_lib.db.api: those can be reused to fetch other models that do not have corresponding versioned objects yet, if needed.
- neutron.db.qos.api: contains database functions that are specific to QoS models.

RPC communication

Details on RPC communication implemented in reference backend driver are discussed in a separate page.

The flow of updates is as follows:

- if a port that is bound to the agent is attached to a QoS policy, then ML2 plugin detects the change by relying on ML2 QoS extension driver, and notifies the agent about a port change. The agent proceeds with the notification by calling to get_device_details() and getting the new port dict that contains a new qos_policy_id. Each device details dict is passed into l2 agent extension manager that passes it down into every enabled extension, including QoS. QoS extension sees that there is a new unknown QoS policy for a port, so it uses ResourcesPullRpcApi to fetch the current state of the policy (with all the rules included) from the server. After that, the QoS extension applies the rules by calling into QoS driver that corresponds to the agent.
- For floating IPs, a `fip_qos` L3 agent extension was implemented. This extension receives and processes router updates. For each update, it goes over each floating IP associated to the router. If a floating IP has a QoS policy associated to it, the extension uses ResourcesPullRpcApi to fetch the policy details from the Neutron server. If the policy includes `bandwidth_limit` rules, the extension applies them to the appropriate router device by directly calling the l3_tc_lib.
- on existing QoS policy update (it includes any policy or its rules change), server pushes the new policy object state through ResourcesPushRpcApi interface. The interface fans out the serialized (dehydrated) object to any agent that is listening for QoS policy updates. If an agent have seen the policy before (it is attached to one of the ports/floating IPs it maintains), then it goes with applying the updates to the port/floating IP. Otherwise, the agent silently ignores the update.

Agent side design

Reference agents implement QoS functionality using an L2 agent extension.

- neutron.agent.l2.extensions.qos defines QoS L2 agent extension. It receives handle_port and delete_port events and passes them down into QoS agent backend driver (see below). The file also defines the QosAgentDriver interface. Note: each backend implements its own driver. The driver handles low level interaction with the underlying networking technology, while the QoS extension handles operations that are common to all agents.

For L3 agent:

- neutron.agent.l3.extensions.fip_qos defines QoS L3 agent extension. It implements the L3 agent side of floating IP rate limit. For all routers, if floating IP has QoS `bandwidth_limit` rules, the corresponding TC filters will be added to the appropriate router device, depending on the router type.

Agent backends

At the moment, QoS is supported by Open vSwitch, SR-IOV and Linux bridge ml2 drivers.

Each agent backend defines a QoS driver that implements the QosAgentDriver interface:

- Open vSwitch (QosOVSAgentDriver);
- SR-IOV (QosSRIOVAgentDriver);

For the Networking back ends, QoS supported rules, and traffic directions (from the VM point of view), please see the table: Networking back ends, supported rules, and traffic direction.

Open vSwitch

Open vSwitch implementation relies on the new ovs_lib OVSBridge functions:

- get_egress_bw_limit_for_port
- create_egress_bw_limit_for_port
- delete_egress_bw_limit_for_port
- get_ingress_bw_limit_for_port
- update_ingress_bw_limit_for_port
- delete_ingress_bw_limit_for_port

An egress bandwidth limit is effectively configured on the port by setting the port Interface parameters ingress_policing_rate and ingress_policing_burst.

That approach is less flexible than linux-htb, Queues and OvS QoS profiles, which we may explore in the future, but which will need to be used in combination with openflow rules.

An ingress bandwidth limit is effectively configured on the port by setting Queue and OvS QoS profile with linux-htb type for port.

The Open vSwitch DSCP marking implementation relies on the recent addition of the ovs_agent_extension_api OVSAgentExtensionAPI to request access to the integration bridge functions:

- add_flow
- mod_flow
- delete_flows
- dump_flows_for

The DSCP markings are in fact configured on the port by means of openflow rules.

Note

As of Ussuri release, the QoS rules can be applied for direct ports with hardware offload capability (switchdev), this requires Open vSwitch version 2.11.0 or newer and Linux kernel based on kernel 5.4.0 or newer.

SR-IOV

SR-IOV bandwidth limit and minimum bandwidth implementation relies on the new pci_lib function:

- set_vf_rate

As the name of the function suggests, the limit is applied on a Virtual Function (VF). This function has a parameter called rate_type and its value can be set to rate or min_tx_rate, which is for enforcing bandwidth limit or minimum bandwidth respectively.

ip link interface has the following limitation for bandwidth limit: it uses Mbps as units of bandwidth measurement, not kbps, and does not support float numbers. So in case the limit is set to something less than 1000 kbps, its set to 1 Mbps only. If the limit is set to something that does not divide to 1000 kbps chunks, then the effective limit is rounded to the nearest integer Mbps value.

Linux bridge

The Linux bridge implementation relies on the new tc_lib functions.

For egress bandwidth limit rule:

- set_filters_bw_limit
- update_filters_bw_limit
- delete_filters_bw_limit

The egress bandwidth limit is configured on the tap port by setting traffic policing on tc ingress queueing discipline (qdisc). Details about ingress qdisc can be found on lartc how-to. The reason why ingress qdisc is used to configure egress bandwidth limit is that tc is working on traffic which is visible from inside bridge perspective. So traffic incoming to bridge via tap interface is in fact outgoing from Neutrons port. This implementation is the same as what Open vSwitch is doing when ingress_policing_rate and ingress_policing_burst are set for port.

For ingress bandwidth limit rule:

- set_tbf_bw_limit
- update_tbf_bw_limit
- delete_tbf_bw_limit

The ingress bandwidth limit is configured on the tap port by setting a simple tc-tbf queueing discipline (qdisc) on the port. It requires a value of HZ parameter configured in kernel on the host. This value is necessary to calculate the minimal burst value which is set in tc. Details about how it is calculated can be found in here. This solution is similar to Open vSwitch implementation.

QoS driver design

QoS framework is flexible enough to support any third-party vendor. To integrate a third party driver (that just wants to be aware of the QoS create/update/delete API calls), one needs to implement neutron.services.qos.drivers.base, and register the driver during the core plugin or mechanism driver load, see

neutron.services.qos.drivers.openvswitch.driver register method for an example.

Note

All the functionality MUST be implemented by the vendor, neutrons QoS framework will just act as an interface to bypass the received QoS API request and help with database persistence for the API operations.

Note

L3 agent `fip_qos` extension does not have a driver implementation, it directly uses the `l3_tc_lib` for all types of routers.

Configuration

To enable the service, the following steps should be followed:

On server side:

- enable qos service in service_plugins;
- for ml2, add qos to extension_drivers in [ml2] section;
- for L3 floating IP QoS, add qos and router to service_plugins.

On agent side (OVS):

- add qos to extensions in [agent] section.

On L3 agent side:

- For for floating IPs QoS support, add fip_qos to extensions in [agent] section.

Testing strategy

All the code added or extended as part of the effort got reasonable unit test coverage.

Neutron objects

Base unit test classes to validate neutron objects were implemented in a way that allows code reuse when introducing a new object type.

There are two test classes that are utilized for that:

- BaseObjectIfaceTestCase: class to validate basic object operations (mostly CRUD) with database layer isolated.
- BaseDbObjectTestCase: class to validate the same operations with models in place and database layer unmocked.

Every new object implemented on top of one of those classes is expected to either inherit existing test cases as is, or reimplement it, if it makes sense in terms of how those objects are implemented. Specific test classes can obviously extend the set of test cases as they see needed (f.e. you need to define new test cases for those additional methods that you may add to your object implementations on top of base semantics common to all neutron objects).

Functional tests

Additions to ovs_lib to set bandwidth limits on ports are covered in:

- neutron.tests.functional.agent.test_ovs_lib

New functional tests for tc_lib to set bandwidth limits on ports are in:

- neutron.tests.functional.agent.linux.test_tc_lib

New functional tests for test_l3_tc_lib to set TC filters on router floating IP related device are covered in:

- neutron.tests.functional.agent.linux.test_l3_tc_lib

New functional tests for L3 agent floating IP rate limit:

- neutron.tests.functional.agent.l3.extensions.test_fip_qos_extension

API tests

API tests for basic CRUD operations for ports, networks, policies, and rules were added in:

- neutron-tempest-plugin.api.test_qos

Quota Management and Enforcement

Most resources exposed by the Neutron API are subject to quota limits. The Neutron API exposes an extension for managing such quotas. Quota limits are enforced at the API layer, before the request is dispatched to the plugin.

Default values for quota limits are specified in neutron.conf. Admin users can override those defaults values on a per-project basis. Limits are stored in the Neutron database; if no limit is found for a given resource and project, then the default value for such resource is used. Configuration-based quota management, where every project gets the same quota limit specified in the configuration file, has been deprecated as of the Liberty release.

Please note that Neutron does not support both specification of quota limits per user and quota management for hierarchical multitenancy (as a matter of fact Neutron does not support hierarchical multitenancy at all). Also, quota limits are currently not enforced on RPC interfaces listening on the AMQP bus.

Plugin and ML2 drivers are not supposed to enforce quotas for resources they manage. However, the `subnet_allocation`[1] extension is an exception and will be discussed below.

The quota management and enforcement mechanisms discussed here apply to every resource which has been registered with the Quota engine, regardless of whether such resource belongs to the core Neutron API or one of its extensions.

High Level View

There are two main components in the Neutron quota system:

- The Quota API extensions.
- The Quota Engine.

Both components rely on a quota driver. The neutron codebase currently defines three quota drivers:

- `neutron.db.quota.driver.DbQuotaDriver`

[1] Subnet allocation extension: http://opendev.org/openstack/neutron/src/neutron/extensions/subnetallocation.py

- `neutron.db.quota.driver_nolock.DbQuotaNoLockDriver` (default)

The `DbQuotaNoLockDriver` is the default quota driver, defined in the configuration option `quota_driver`.

The Quota API extension handles quota management, whereas the Quota Engine component handles quota enforcement. This API extension is loaded like any other extension. For this reason plugins must explicitly support it by including quotas in the supported_extension_aliases attribute.

In the Quota API simple CRUD operations are used for managing project quotas. Please note that the current behaviour when deleting a project quota is to reset quota limits for that project to configuration defaults. The API extension does not validate the project identifier with the identity service.

In addition, the Quota Detail API extension complements the Quota API extension by allowing users (typically admins) the ability to retrieve details about quotas per project. Quota details include the used/limit/reserved count for the projects resources (networks, ports, etc.).

Performing quota enforcement is the responsibility of the Quota Engine. RESTful API controllers, before sending a request to the plugin, try to obtain a reservation from the quota engine for the resources specified in the client request. If the reservation is successful, then it proceeds to dispatch the operation to the plugin.

For a reservation to be successful, the total amount of resources requested, plus the total amount of resources reserved, plus the total amount of resources already stored in the database should not exceed the projects quota limit.

Finally, both quota management and enforcement rely on a quota driver[2], whose task is basically to perform database operations.

Quota Management

The quota management component is fairly straightforward.

However, unlike the vast majority of Neutron extensions, it uses it own controller class[3]. This class does not implement the POST operation. List, get, update, and delete operations are implemented by the usual index, show, update and delete methods. These method simply call into the quota driver for either fetching project quotas or updating them.

The `_update_attributes` method is called only once in the controller lifetime. This method dynamically updates Neutrons resource attribute map[4] so that an attribute is added for every resource managed by the quota engine. Request authorisation is performed in this controller, and only admin users are allowed to modify quotas for projects. As the neutron policy engine is not used, it is not possible to configure which users should be allowed to manage quotas using `policy.yaml`.

The driver operations dealing with quota management are:

- `delete_tenant_quota`, which simply removes all entries from the quotas table for a given project identifier;
- `update_quota_limit`, which adds or updates an entry in the quotas project for a given project identifier and a given resource name;
- `_get_quotas`, which fetches limits for a set of resource and a given project identifier;

[2] DB Quota driver class: http://opendev.org/openstack/neutron/src/neutron/db/quota/driver.py#L30

[3] Quota API extension controller: https://opendev.org/openstack/neutron/src/tag/19.0.0/neutron/extensions/quotasv2.py#L56

[4] Neutron resource attribute map: https://opendev.org/openstack/neutron-lib/src/tag/2.17.0/neutron_lib/api/attributes.py#L299

- `_get_all_quotas`, which behaves like `_get_quotas`, but for all projects

Resource Usage Info

Neutron has two ways of tracking resource usage info:

- `CountableResource`, where resource usage is calculated every time quotas limits are enforced by counting rows in the resource table or resources tables and reservations for that resource.
- `TrackedResource`, depends on the selected driver:
  - `DbQuotaDriver`: the resource usage relies on a specific table tracking usage data, and performs explicitly counting only when the data in this table are not in sync with actual used and reserved resources.
  - `DbQuotaNoLockDriver`: the resource usage is counted directly from the database table associated to the resource. In this new driver, `CountableResource` and `TrackedResource` could look similar but `TrackedResource` depends on one single database model (table) and the resource count is done directly on this table only.

Another difference between `CountableResource` and `TrackedResource` is that the former invokes a plugin method to count resources. `CountableResource` should be therefore employed for plugins which do not leverage the Neutron database. The actual class that the Neutron quota engine will use is determined by the `track_quota_usage` variable in the quota configuration section. If `True`, `TrackedResource` instances will be created, otherwise the quota engine will use `CountableResource` instances. Resource creation is performed by the `create_resource_instance` factory method in the `neutron.quota.resource` module.

DbQuotaDriver description

From a performance perspective, having a table tracking resource usage has some advantages, albeit not fundamental. Indeed the time required for executing queries to explicitly count objects will increase with the number of records in the table. On the other hand, using `TrackedResource` will fetch a single record, but has the drawback of having to execute an UPDATE statement once the operation is completed. Nevertheless, `CountableResource` instances do not simply perform a SELECT query on the relevant table for a resource, but invoke a plugin method, which might execute several statements and sometimes even interacts with the backend before returning. Resource usage tracking also becomes important for operational correctness when coupled with the concept of resource reservation, discussed in another section of this chapter.

Tracking quota usage is not as simple as updating a counter every time resources are created or deleted. Indeed a quota-limited resource in Neutron can be created in several ways. While a RESTful API request is the most common one, resources can be created by RPC handlers listing on the AMQP bus, such as those which create DHCP ports, or by plugin operations, such as those which create router ports.

To this aim, `TrackedResource` instances are initialised with a reference to the model class for the resource for which they track usage data. During object initialisation, SqlAlchemy event handlers are installed for this class. The event handler is executed after a record is inserted or deleted. As result usage data for that resource and will be marked as dirty once the operation completes, so that the next time usage data is requested, it will be synchronised counting resource usage from the database. Even if this solution has some drawbacks, listed in the exceptions and caveats section, it is more reliable than solutions such as:

- Updating the usage counters with the new correct value every time an operation completes.
- Having a periodic task synchronising quota usage data with actual data in the Neutron DB.

DbQuotaNoLockDriver description

The strategy of this quota driver is the opposite to `DbQuotaDriver`. Instead of tracking the usage quota of each resource in a specific table, this driver retrieves the used resources directly form the database. Each `TrackedResource` is linked to a database table that stores the tracked resources. This driver claims that a trivial query on the resource table, filtering by project ID, is faster than attending to the DB events and tracking the quota usage in an independent table.

This driver relays on the database engine transactionality isolation. Each time a new resource is requested, the quota driver opens a database transaction to:

- Clean up the expired reservations. The amount of expired reservations is always limited because of the short timeout set (2 minutes).
- Retrieve the used resources for a specific project. This query retrieves only the project_id column of the resource to avoid backref requests; that limits the scope of the query and speeds up it.
- Retrieve the reserved resources, created by other concurrent operations.
- If there is enough quota, create a new reservation register.

Those operations, executed in the same transaction, are fast enough to avoid another concurrent resource reservation, exceeding the available quota. At the same time, this driver does not create a lock per resource and project ID, allowing concurrent requests that wont be blocked by the resource lock. Because the quota reservation process, described before, is a fast operation, the chances of overcommiting resources over the quota limits are low. Neutron does not enforce quota in such way that a quota limit violation could never occur[5].

Regardless of whether `CountableResource` or `TrackedResource` is used, the quota engine always invokes its `count()` method to retrieve resource usage. Therefore, from the perspective of the Quota engine there is absolutely no difference between `CountableResource` and `TrackedResource`.

Quota Enforcement in DbQuotaDriver

Before dispatching a request to the plugin, the Neutron base controller[6] attempts to make a reservation for requested resource(s). Reservations are made by calling the `make_reservation` method in `neutron.quota.QuotaEngine`. The process of making a reservation is fairly straightforward:

- Get current resource usages. This is achieved by invoking the count method on every requested resource, and then retrieving the amount of reserved resources.
- Fetch current quota limits for requested resources, by invoking the `_get_project_quotas` method.
- Fetch expired reservations for selected resources. This amount will be subtracted from resource usage. As in most cases there wont be any expired reservation, this approach actually requires less DB operations than doing a sum of non-expired, reserved resources for each request.
- For each resource calculate its headroom, and verify the requested amount of resource is less than the headroom.
- If the above is true for all resource, the reservation is saved in the DB, otherwise an `OverQuotaLimit` exception is raised.

The quota engine is able to make a reservation for multiple resources. However, it is worth noting that because of the current structure of the Neutron API layer, there will not be any practical case in which a

[5] Quota limit exceeded: https://bugs.launchpad.net/neutron/+bug/1862050/

[6] Base controller class: https://opendev.org/openstack/neutron/src/tag/19.0.0/neutron/api/v2/base.py#L44

reservation for multiple resources is made. For this reason performance optimisation avoiding repeating queries for every resource are not part of the current implementation.

In order to ensure correct operations, a row-level lock is acquired in the transaction which creates the reservation. The lock is acquired when reading usage data. In case of write-set certification failures, which can occur in active/active clusters such as MySQL galera, the decorator `neutron_lib.db.api.retry_db_errors` will retry the transaction if a DBDeadLock exception is raised. While non-locking approaches are possible, it has been found out that, since a non-locking algorithms increases the chances of collision, the cost of handling a `DBDeadlock` is still lower than the cost of retrying the operation when a collision is detected. A study in this direction was conducted for IP allocation operations, but the same principles apply here as well[7]. Nevertheless, moving away for DB-level locks is something that must happen for quota enforcement in the future.

Committing and cancelling a reservation is as simple as deleting the reservation itself. When a reservation is committed, the resources which were committed are now stored in the database, so the reservation itself should be deleted. The Neutron quota engine simply removes the record when cancelling a reservation (i.e. the request failed to complete), and also marks quota usage info as dirty when the reservation is committed (i.e. the request completed correctly). Reservations are committed or cancelled by respectively calling the `commit_reservation` and `cancel_reservation` methods in `neutron.quota.QuotaEngine`.

Reservations are not perennial. Eternal reservation would eventually exhaust projects quotas because they would never be removed when an API worker crashes whilst in the middle of an operation. Reservation expiration is currently set to 120 seconds, and is not configurable, not yet at least. Expired reservations are not counted when calculating resource usage. While creating a reservation, if any expired reservation is found, all expired reservation for that project and resource will be removed from the database, thus avoiding build-up of expired reservations.

Setting up Resource Tracking for a Plugin

By default plugins do not leverage resource tracking. Having the plugin explicitly declare which resources should be tracked is a precise design choice aimed at limiting as much as possible the chance of introducing errors in existing plugins.

For this reason a plugin must declare which resource it intends to track. This can be achieved using the tracked_resources decorator available in the neutron.quota.resource_registry module. The decorator should ideally be applied to the plugins __init__ method.

The decorator accepts in input a list of keyword arguments. The name of the argument must be a resource name, and the value of the argument must be a DB model class. For example:

```
@resource_registry.tracked_resources(network=models_v2.Network,
                                     port=models_v2.Port,
                                     subnet=models_v2.Subnet,
                                     subnetpool=models_v2.SubnetPool)
```

Will ensure network, port, subnet and subnetpool resources are tracked. In theory, it is possible to use this decorator multiple times, and not exclusively to `__init__` methods. However, this would eventually lead to code readability and maintainability problems, so developers are strongly encourage to apply this decorator exclusively to the plugins `__init__` method (or any other method which is called by the plugin only once during its initialization).

[7] http://lists.openstack.org/pipermail/openstack-dev/2015-February/057534.html

Notes for Implementors of RPC Interfaces and RESTful Controllers

Neutron unfortunately does not have a layer which is called before dispatching the operation from the plugin which can be leveraged both from RESTful and RPC over AMQP APIs. In particular the RPC handlers call straight into the plugin, without doing any request authorisation or quota enforcement.

Therefore RPC handlers must explicitly indicate if they are going to call the plugin to create or delete any sort of resources. This is achieved in a simple way, by ensuring modified resources are marked as dirty after the RPC handler execution terminates. To this aim developers can use the mark_resources_dirty decorator available in the module neutron.quota.resource_registry.

The decorator would scan the whole list of registered resources, and store the dirty status for their usage trackers in the database for those resources for which items have been created or destroyed during the plugin operation.

Exceptions and Caveats

Please be aware of the following limitations of the quota enforcement engine:

- Subnet allocation from subnet pools, in particularly shared pools, is also subject to quota limit checks. However this checks are not enforced by the quota engine, but trough a mechanism implemented in the `neutron.ipam.subnetalloc` module. This is because the quota engine is not able to satisfy the requirements for quotas on subnet allocation.
- The quota engine also provides a `limit_check` routine which enforces quota checks without creating reservations. This way of doing quota enforcement is extremely unreliable and superseded by the reservation mechanism. It has not been removed to ensure off-tree plugins and extensions which leverage are not broken.
- SqlAlchemy events might not be the most reliable way for detecting changes in resource usage. Since the event mechanism monitors the data model class, it is paramount for a correct quota enforcement, that resources are always created and deleted using object relational mappings. For instance, deleting a resource with a `query.delete` call will not trigger the event. SQLAlchemy events should be considered as a temporary measure adopted as Neutron lacks persistent API objects.
- As `CountableResource` instance do not track usage data, when making a reservation no write-intent lock is acquired. Therefore the quota engine with `CountableResource` is not concurrency-safe.
- The mechanism for specifying for which resources enable usage tracking relies on the fact that the plugin is loaded before quota-limited resources are registered. For this reason it is not possible to validate whether a resource actually exists or not when enabling tracking for it. Developers should pay particular attention into ensuring resource names are correctly specified.
- The code assumes usage trackers are a trusted source of truth: if they report a usage counter and the dirty bit is not set, that counter is correct. If its dirty than surely that counter is out of sync. This is not very robust, as there might be issues upon restart when toggling the use_tracked_resources configuration variable, as stale counters might be trusted upon for making reservations. Also, the same situation might occur if a server crashes after the API operation is completed but before the reservation is committed, as the actual resource usage is changed but the corresponding usage tracker is not marked as dirty.

References

Retrying Operations

Inside of the neutron_lib.db.api module there is a decorator called retry_if_session_inactive. This should be used to protect any functions that perform DB operations. This decorator will capture any deadlock errors, RetryRequests, connection errors, and unique constraint violations that are thrown by the function it is protecting.

This decorator will not retry an operation if the function it is applied to is called within an active session. This is because the majority of the exceptions it captures put the session into a partially rolled back state so it is no longer usable. It is important to ensure there is a decorator outside of the start of the transaction. The decorators are safe to nest if a function is sometimes called inside of another transaction.

If a function is being protected that does not take context as an argument the retry_db_errors decorator function may be used instead. It retries the same exceptions and has the same anti-nesting behavior as retry_if_session_active, but it does not check if a session is attached to any context keywords. (retry_if_session_active just uses retry_db_errors internally after checking the session)

Idempotency on Failures

The function that is being decorated should always fully cleanup whenever it encounters an exception so its safe to retry the operation. So if a function creates a DB object, commits, then creates another, the function must have a cleanup handler to remove the first DB object in the case that the second one fails. Assume any DB operation can throw a retriable error.

You may see some retry decorators at the API layers in Neutron; however, we are trying to eliminate them because each API operation has many independent steps that makes ensuring idempotency on partial failures very difficult.

Argument Mutation

A decorated function should not mutate any complex arguments which are passed into it. If it does, it should have an exception handler that reverts the change so its safe to retry.

The decorator will automatically create deep copies of sets, lists, and dicts which are passed through it, but it will leave the other arguments alone.

Retrying to Handle Race Conditions

One of the difficulties with detecting race conditions to create a DB record with a unique constraint is determining where to put the exception handler because a constraint violation can happen immediately on flush or it may not happen all of the way until the transaction is being committed on the exit of the session context manager. So we would end up with code that looks something like this:

```
from neutron.db import api as db_api

def create_port(context, ip_address, mac_address):
    _ensure_mac_not_in_use(context, mac_address)
    _ensure_ip_not_in_use(context, ip_address)
    try:
        with db_api.CONTEXT_READER.using(context):
            port_obj = Port(ip=ip_address, mac=mac_address)
            do_expensive_thing(...)
```

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```
            do_extra_other_thing(...)
            return port_obj
    except DBDuplicateEntry as e:
        # code to parse columns
        if 'mac' in e.columns:
            raise MacInUse(mac_address)
        if 'ip' in e.columns:
            raise IPAddressInUse(ip)

def _ensure_mac_not_in_use(context, mac):
    if context.session.query(Port).filter_by(mac=mac).count():
        raise MacInUse(mac)

def _ensure_ip_not_in_use(context, ip):
    if context.session.query(Port).filter_by(ip=ip).count():
        raise IPAddressInUse(ip)
```

So we end up with an exception handler that has to understand where things went wrong and convert them into appropriate exceptions for the end-users. This distracts significantly from the main purpose of create_port.

Since the retry decorator will automatically catch and retry DB duplicate errors for us, we can allow it to retry on this race condition which will give the original validation logic to be re-executed and raise the appropriate error. This keeps validation logic in one place and makes the code cleaner.

```
from neutron.db import api as db_api

@db_api.retry_if_session_inactive()
def create_port(context, ip_address, mac_address):
    _ensure_mac_not_in_use(context, mac_address)
    _ensure_ip_not_in_use(context, ip_address)
    with db_api.CONTEXT_READER.using(context):
       port_obj = Port(ip=ip_address, mac=mac_address)
       do_expensive_thing(...)
       do_extra_other_thing(...)
       return port_obj

def _ensure_mac_not_in_use(context, mac):
    if context.session.query(Port).filter_by(mac=mac).count():
        raise MacInUse(mac)

def _ensure_ip_not_in_use(context, ip):
    if context.session.query(Port).filter_by(ip=ip).count():
        raise IPAddressInUse(ip)
```

Nesting

Once the decorator retries an operation the maximum number of times, it will attach a flag to the exception it raises further up that will prevent decorators around the calling functions from retrying the error again. This prevents an exponential increase in the number of retries if they are layered.

Usage

Here are some usage examples:

```
from neutron.db import api as db_api

@db_api.retry_if_session_inactive()
def create_elephant(context, elephant_details):
    ....

@db_api.retry_if_session_inactive()
def atomic_bulk_create_elephants(context, elephants):
    with db_api.CONTEXT_WRITER.using(context):
        for elephant in elephants:
            # note that if create_elephant throws a retriable
            # exception, the decorator around it will not retry
            # because the session is active. The decorator around
            # atomic_bulk_create_elephants will be responsible for
            # retrying the entire operation.
            create_elephant(context, elephant)

# sample usage when session is attached to a var other than 'context'
@db_api.retry_if_session_inactive(context_var_name='ctx')
def some_function(ctx):
    ...
```

RPC API Layer

Neutron uses the oslo.messaging library to provide an internal communication channel between Neutron services. This communication is typically done via AMQP, but those details are mostly hidden by the use of oslo.messaging and it could be some other protocol in the future.

RPC APIs are defined in Neutron in two parts: client side and server side.

Client Side

Here is an example of an rpc client definition:

```
import oslo_messaging

from neutron.common import rpc as n_rpc


class ClientAPI(object):
    """Client side RPC interface definition.

    API version history:
        1.0 - Initial version
        1.1 - Added my_remote_method_2
    """

    def __init__(self, topic):
```

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```
        target = oslo_messaging.Target(topic=topic, version='1.0')
        self.client = n_rpc.get_client(target)

    def my_remote_method(self, context, arg1, arg2):
        cctxt = self.client.prepare()
        return cctxt.call(context, 'my_remote_method', arg1=arg1, arg2=arg2)

    def my_remote_method_2(self, context, arg1):
        cctxt = self.client.prepare(version='1.1')
        return cctxt.call(context, 'my_remote_method_2', arg1=arg1)
```

This class defines the client side interface for an rpc API. The interface has 2 methods. The first method existed in version 1.0 of the interface. The second method was added in version 1.1. When the newer method is called, it specifies that the remote side must implement at least version 1.1 to handle this request.

Server Side

The server side of an rpc interface looks like this:

```
import oslo_messaging


class ServerAPI(object):

    target = oslo_messaging.Target(version='1.1')

    def my_remote_method(self, context, arg1, arg2):
        return 'foo'

    def my_remote_method_2(self, context, arg1):
        return 'bar'
```

This class implements the server side of the interface. The oslo_messaging.Target() defined says that this class currently implements version 1.1 of the interface.

Versioning

Note that changes to rpc interfaces must always be done in a backwards compatible way. The server side should always be able to handle older clients (within the same major version series, such as 1.X).

It is possible to bump the major version number and drop some code only needed for backwards compatibility. For more information about how to do that, see https://wiki.openstack.org/wiki/RpcMajorVersionUpdates.

Example Change

As an example minor API change, lets assume we want to add a new parameter to my_remote_method_2. First, we add the argument on the server side. To be backwards compatible, the new argument must have a default value set so that the interface will still work even if the argument is not supplied. Also, the interfaces minor version number must be incremented. So, the new server side code would look like this:

```
import oslo_messaging


class ServerAPI(object):

    target = oslo_messaging.Target(version='1.2')

    def my_remote_method(self, context, arg1, arg2):
        return 'foo'

    def my_remote_method_2(self, context, arg1, arg2=None):
        if not arg2:
            # Deal with the fact that arg2 was not specified if needed.
        return 'bar'
```

We can now update the client side to pass the new argument. The client must also specify that version 1.2 is required for this method call to be successful. The updated client side would look like this:

```
import oslo_messaging

from neutron.common import rpc as n_rpc


class ClientAPI(object):
    """Client side RPC interface definition.

    API version history:
        1.0 - Initial version
        1.1 - Added my_remote_method_2
        1.2 - Added arg2 to my_remote_method_2
    """

    def __init__(self, topic):
        target = oslo_messaging.Target(topic=topic, version='1.0')
        self.client = n_rpc.get_client(target)

    def my_remote_method(self, context, arg1, arg2):
        cctxt = self.client.prepare()
        return cctxt.call(context, 'my_remote_method', arg1=arg1, arg2=arg2)

    def my_remote_method_2(self, context, arg1, arg2):
        cctxt = self.client.prepare(version='1.2')
        return cctxt.call(context, 'my_remote_method_2',
                          arg1=arg1, arg2=arg2)
```

Neutron RPC APIs

As discussed before, RPC APIs are defined in two parts: a client side and a server side. Several of these pairs exist in the Neutron code base. The code base is being updated with documentation on every rpc interface implementation that indicates where the corresponding server or client code is located.

Example: DHCP

The DHCP agent includes a client API, neutron.agent.dhcp.agent.DhcpPluginAPI. The DHCP agent uses this class to call remote methods back in the Neutron server. The server side is defined in neutron.api.rpc.handlers.dhcp_rpc.DhcpRpcCallback. It is up to the Neutron plugin in use to decide whether the DhcpRpcCallback interface should be exposed.

Similarly, there is an RPC interface defined that allows the Neutron plugin to remotely invoke methods in the DHCP agent. The client side is defined in neutron.api.rpc.agentnotifiers.dhcp_rpc_agent_api.DhcpAgentNotifyAPI. The server side of this interface that runs in the DHCP agent is neutron.agent.dhcp.agent.DhcpAgent.

More Info

For more information, see the oslo.messaging documentation: https://docs.openstack.org/oslo.messaging/latest/.

RPC Messaging Callback System

Neutron already has a callback system for in-process resource callbacks where publishers and subscribers are able to publish and subscribe for resource events.

This system is different, and is intended to be used for inter-process callbacks, via the messaging fanout mechanisms.

In Neutron, agents may need to subscribe to specific resource details which may change over time. And the purpose of this messaging callback system is to allow agent subscription to those resources without the need to extend modify existing RPC calls, or creating new RPC messages.

A few resource which can benefit of this system:

- QoS policies;
- Security Groups.

Using a remote publisher/subscriber pattern, the information about such resources could be published using fanout messages to all interested nodes, minimizing messaging requests from agents to server since the agents get subscribed for their whole lifecycle (unless they unsubscribe).

Within an agent, there could be multiple subscriber callbacks to the same resource events, the resources updates would be dispatched to the subscriber callbacks from a single message. Any update would come in a single message, doing only a single oslo versioned objects deserialization on each receiving agent.

This publishing/subscription mechanism is highly dependent on the format of the resources passed around. This is why the library only allows versioned objects to be published and subscribed. Oslo versioned objects allow object version down/up conversion.[2][3]

For the VOs versioning schema look here:[4]

[2] https://github.com/openstack/oslo.versionedobjects/blob/ce00f18f7e9143b5175e889970564813189e3e6d/oslo_versionedobjects/base.py#L474

[3] https://github.com/openstack/oslo.versionedobjects/blob/ce00f18f7e9143b5175e889970564813189e3e6d/oslo_versionedobjects/tests/test_objects.py#L114

[4] https://github.com/openstack/oslo.versionedobjects/blob/ce00f18f7e9143b5175e889970564813189e3e6d/oslo_versionedobjects/base.py#L248

versioned_objects serialization/deserialization with the obj_to_primitive(target_version=..) and primitive_to_obj()[1] methods is used internally to convert/retrieve objects before/after messaging.

Serialized versioned objects look like:

```
{'versioned_object.version': '1.0',
 'versioned_object.name': 'QoSPolicy',
 'versioned_object.data': {'rules': [
                                     {'versioned_object.version': '1.0',
                                      'versioned_object.name':
↪'QoSBandwidthLimitRule',
                                      'versioned_object.data': {'name': u'a'},
                                      'versioned_object.namespace':
↪'versionedobjects'}
                                     ],
                           'uuid': u'abcde',
                           'name': u'aaa'},
 'versioned_object.namespace': 'versionedobjects'}
```

Rolling upgrades strategy

In this section we assume the standard Neutron upgrade process, which means upgrade the server first and then upgrade the agents:

More information about the upgrade strategy.

We provide an automatic method which avoids manual pinning and unpinning of versions by the administrator which could be prone to error.

Resource pull requests

Resource pull requests will always be ok because the underlying resource RPC does provide the version of the requested resource id / ids. The server will be upgraded first, so it will always be able to satisfy any version the agents request.

Resource push notifications

Agents will subscribe to the neutron-vo-<resource_type>-<version> fanout queue which carries updated objects for the version they know about. The versions they know about depend on the runtime Neutron versioned objects they started with.

When the server upgrades, it should be able to instantly calculate a census of agent versions per object (we will define a mechanism for this in a later section). It will use the census to send fanout messages on all the version span a resource type has.

For example, if neutron-server knew it has rpc-callback aware agents with versions 1.0, and versions 1.2 of resource type A, any update would be sent to neutron-vo-A_1.0 and neutron-vo-A_1.2.

TODO(mangelajo): Verify that after upgrade is finished any unused messaging resources (queues, exchanges, and so on) are released as older agents go away and neutron-server stops producing new message casts. Otherwise document the need for a neutron-server restart after rolling upgrade has finished if we want the queues cleaned up.

[1] https://github.com/openstack/oslo.versionedobjects/blob/ce00f18f7e9143b5175e889970564813189e3e6d/oslo_versionedobjects/tests/test_objects.py#L410

Leveraging agent state reports for object version discovery

We add a row to the agent db for tracking agent known objects and version numbers. This resembles the implementation of the configuration column.

Agents report at start not only their configuration now, but also their subscribed object type / version pairs, that are stored in the database and made available to any neutron-server requesting it:

```
'resource_versions': {'QosPolicy': '1.1',
                      'SecurityGroup': '1.0',
                      'Port': '1.0'}
```

There was a subset of Liberty agents depending on QosPolicy that required QosPolicy: 1.0 if the qos plugin is installed. We were able to identify those by the binary name (included in the report):

- neutron-openvswitch-agent
- neutron-sriov-nic-agent

This transition was handled in the Mitaka version, but its not handled anymore in Newton, since only one major version step upgrades are supported.

Version discovery

With the above mechanism in place and considering the exception of neutron-openvswitch-agent and neutron-sriov-agent requiring QoSpolicy 1.0, we discover the subset of versions to be sent on every push notification.

Agents that are in down state are excluded from this calculation. We use an extended timeout for agents in this calculation to make sure were on the safe side, specially if deployer marked agents with low timeouts.

Starting at Mitaka, any agent interested in versioned objects via this API should report their resource/version tuples of interest (the resource type/ version pairs theyre subscribed to).

The plugins interested in this RPC mechanism must inherit AgentDbMixin, since this mechanism is only intended to be used from agents at the moment, while it could be extended to be consumed from other components if necessary.

The AgentDbMixin provides:

```
def get_agents_resource_versions(self, tracker):
    ...
```

Caching mechanism

The version subset per object is cached to avoid DB requests on every push given that we assume that all old agents are already registered at the time of upgrade.

Cached subset is re-evaluated (to cut down the version sets as agents upgrade) after neutron.api.rpc.callbacks.version_manager.VERSIONS_TTL.

As a fast path to update this cache on all neutron-servers when upgraded agents come up (or old agents revive after a long timeout or even a downgrade) the server registering the new status update notifies the other servers about the new consumer resource versions via cast.

All notifications for all calculated version sets must be sent, as non-upgraded agents would otherwise not receive them.

It is safe to send notifications to any fanout queue as they will be discarded if no agent is listening.

Topic names for every resource type RPC endpoint

neutron-vo-<resource_class_name>-<version>

In the future, we may want to get oslo messaging to support subscribing topics dynamically, then we may want to use:

neutron-vo-<resource_class_name>-<resource_id>-<version> instead,

or something equivalent which would allow fine granularity for the receivers to only get interesting information to them.

Subscribing to resources

Imagine that you have agent A, which just got to handle a new port, which has an associated security group, and QoS policy.

The agent code processing port updates may look like:

```
from neutron.api.rpc.callbacks.consumer import registry
from neutron.api.rpc.callbacks import events
from neutron.api.rpc.callbacks import resources


def process_resource_updates(context, resource_type, resource_list, event_
↪type):

    # send to the right handler which will update any control plane
    # details related to the updated resources...


def subscribe_resources():
    registry.register(process_resource_updates, resources.SEC_GROUP)

    registry.register(process_resource_updates, resources.QOS_POLICY)

def port_update(port):

    # here we extract sg_id and qos_policy_id from port..

    sec_group = registry.pull(resources.SEC_GROUP, sg_id)
    qos_policy = registry.pull(resources.QOS_POLICY, qos_policy_id)
```

The relevant function is:

- register(callback, resource_type): subscribes callback to a resource type.

The callback function will receive the following arguments:

- context: the neutron context that triggered the notification.
- resource_type: the type of resource which is receiving the update.
- resource_list: list of resources which have been pushed by server.

- event_type: will be one of CREATED, UPDATED, or DELETED, see neutron.api.rpc.callbacks.events for details.

With the underlying oslo_messaging support for dynamic topics on the receiver we cannot implement a per resource type + resource id topic, rabbitmq seems to handle 10000s of topics without suffering, but creating 100s of oslo_messaging receivers on different topics seems to crash.

We may want to look into that later, to avoid agents receiving resource updates which are uninteresting to them.

Unsubscribing from resources

To unsubscribe registered callbacks:

- unsubscribe(callback, resource_type): unsubscribe from specific resource type.
- unsubscribe_all(): unsubscribe from all resources.

Sending resource events

On the server side, resource updates could come from anywhere, a service plugin, an extension, anything that updates, creates, or destroys the resources and that is of any interest to subscribed agents.

A callback is expected to receive a list of resources. When resources in the list belong to the same resource type, a single push RPC message is sent; if the list contains objects of different resource types, resources of each type are grouped and sent separately, one push RPC message per type. On the receiver side, resources in a list always belong to the same type. In other words, a server-side push of a list of heterogeneous objects will result into N messages on bus and N client-side callback invocations, where N is the number of unique resource types in the given list, e.g. L(A, A, B, C, C, C) would be fragmented into L1(A, A), L2(B), L3(C, C, C), and each list pushed separately.

Note: there is no guarantee in terms of order in which separate resource lists will be delivered to consumers.

The server/publisher side may look like:

```
from neutron.api.rpc.callbacks.producer import registry
from neutron.api.rpc.callbacks import events

def create_qos_policy(...):
    policy = fetch_policy(...)
    update_the_db(...)
    registry.push([policy], events.CREATED)

def update_qos_policy(...):
    policy = fetch_policy(...)
    update_the_db(...)
    registry.push([policy], events.UPDATED)

def delete_qos_policy(...):
    policy = fetch_policy(...)
    update_the_db(...)
    registry.push([policy], events.DELETED)
```

References

Segments Extension

Neutron has an extension that allows CRUD operations on the `/segments` resource in the API, that corresponds to the `NetworkSegment` entity in the DB layer. The extension is implemented as a service plug-in.

Details about the DB models, API extension, and use cases can be found here: routed networks spec

Note

The `segments` service plug-in is not configured by default. To configure it, add `segments` to the `service_plugins` parameter in `neutron.conf`

Core plug-ins can coordinate with the `segments` service plug-in by subscribing callbacks to events associated to the `SEGMENT` resource. Currently, the segments plug-in notifies subscribers of the following events:

- `PRECOMMIT_CREATE`
- `AFTER_CREATE`
- `BEFORE_DELETE`
- `PRECOMMIT_DELETE`
- `AFTER_DELETE`

As of this writing, `ML2` and `OVN` register callbacks to receive events from the `segments` service plug-in. The `ML2` plug-in defines the callback `_handle_segment_change` to process all the relevant segments events.

Segments extension relevant modules

- `neutron/extensions/segment.py` defines the extension
- `neutron/db/models/segment.py` defines the DB models for segments and for the segment host mapping, that is used in the implementation of routed networks.
- `neutron/db/segments_db.py` has functions to add, retrieve and delete segments from the DB.
- `neutron/services/segments/db.py` defines a mixin class with the methods that perform API CRUD operations for the `segments` plug-in. It also has a set of functions to create and maintain the mapping of segments to hosts, which is necessary in the implementation of routed networks.
- `neutron/services/segments/plugin.py` defines the `segments` service plug-in.

Service Extensions

Historically, Neutron supported the following advanced services:

1. **FWaaS** (*Firewall-as-a-Service*): runs as part of the L3 agent.
2. **VPNaaS** (*VPN-as-a-Service*): derives from L3 agent to add VPNaaS functionality.

Starting with the Kilo release, these services are split into separate repositories, and more extensions are being developed as well. Service plugins are a clean way of adding functionality in a cohesive manner and yet, keeping them decoupled from the guts of the framework. The aforementioned features are developed as extensions (also known as service plugins), and more capabilities are being added to Neutron following the same pattern. For those that are deemed orthogonal to any network service (e.g. tags, timestamps, auto_allocate, etc), there is an informal mechanism to have these loaded automatically at server startup. If you consider adding an entry to the dictionary, please be kind and reach out to your PTL or a member of the drivers team for approval.

1. http://opendev.org/openstack/neutron-fwaas/
2. http://opendev.org/openstack/neutron-vpnaas/

Calling the Core Plugin from Services

There are many cases where a service may want to create a resource managed by the core plugin (e.g. ports, networks, subnets). This can be achieved by importing the plugins directory and getting a direct reference to the core plugin:

```
from neutron_lib.plugins import directory

plugin = directory.get_plugin()
plugin.create_port(context, port_dict)
```

However, there is an important caveat. Calls to the core plugin in almost every case should not be made inside of an ongoing transaction. This is because many plugins (including ML2), can be configured to make calls to a backend after creating or modifying an object. If the call is made inside of a transaction and the transaction is rolled back after the core plugin call, the backend will not be notified that the change was undone. This will lead to consistency errors between the core plugin and its configured backend(s).

ML2 has a guard against certain methods being called with an active DB transaction to help prevent developers from accidentally making this mistake. It will raise an error that says explicitly that the method should not be called within a transaction.

Services and Agents

A usual Neutron setup consists of multiple services and agents running on one or multiple nodes (though some exotic setups potentially may not need any agents). Each of those services provides some of the networking or API services. Among those of special interest:

1. neutron-server that provides API endpoints and serves as a single point of access to the database. It usually runs on nodes called Controllers.
2. Layer2 agent that can utilize Open vSwitch or other vendor specific technology to provide network segmentation and isolation for project networks. The L2 agent should run on every node where it is deemed responsible for wiring and securing virtual interfaces (usually both Compute and Network nodes).
3. Layer3 agent that runs on Network node and provides East-West and North-South routing plus some advanced services such as FWaaS or VPNaaS.

For the purpose of this document, we call all services, servers and agents that run on any node as just services.

Entry points

Entry points for services are defined in setup.cfg under console_scripts section. Those entry points should generally point to main() functions located under neutron/cmd/ path.

Note: some existing vendor/plugin agents still maintain their entry points in other locations. Developers responsible for those agents are welcome to apply the guideline above.

Connecting to the Database

Only the neutron-server connects to the neutron database. Agents may never connect directly to the database, as this would break the ability to do rolling upgrades.

Configuration Options

In addition to database access, configuration options are segregated between neutron-server and agents. Both services and agents may load the main ``neutron.conf`` since this file should contain the oslo.messaging configuration for internal Neutron RPCs and may contain host specific configuration such as file paths. In addition ``neutron.conf`` contains the database, Keystone, and Nova credentials and endpoints strictly for neutron-server to use.

In addition neutron-server may load a plugin specific configuration file, yet the agents should not. As the plugin configuration is primarily site wide options and the plugin provides the persistence layer for Neutron, agents should be instructed to act upon these values via RPC.

Each individual agent may have its own configuration file. This file should be loaded after the main ``neutron.conf`` file, so the agent configuration takes precedence. The agent specific configuration may contain configurations which vary between hosts in a Neutron deployment such as the `local_ip` for an L2 agent. If any agent requires access to additional external services beyond the neutron RPC, those endpoints should be defined in the agent-specific configuration file (e.g. nova metadata for metadata agent).

Tags in Neutron Resources

Tag service plugin allows users to set tags on their resources. Tagging resources can be used by external systems or any other clients of the Neutron REST API (and NOT backend drivers).

The following use cases refer to adding tags to networks, but the same can be applicable to any other Neutron resource:

1) Ability to map different networks in different OpenStack locations to one logically same network (for Multi site OpenStack)
2) Ability to map Ids from different management/orchestration systems to OpenStack networks in mixed environments, for example for project Kuryr, map docker network id to neutron network id
3) Leverage tags by deployment tools
4) allow operators to tag information about provider networks (e.g. high-bandwidth, low-latency, etc)
5) new features like get-me-a-network or a similar port scheduler could choose a network for a port based on tags

Which Resources

Tag system uses standardattr mechanism so its targeting to resources that have the mechanism. Some resources with standard attribute dont suit fit tag support usecases (e.g. security_group_rule). If new tag support resource is added, the resource model should inherit HasStandardAttributes and then it must implement the property api_parent and tag_support. And also the change must include a release note for API user.

Current API resources extended by tag extensions:

- floatingips
- networks
- network_segment_ranges
- policies
- ports
- routers
- security_groups
- subnetpools
- subnets
- trunks

Model

Tag is not standalone resource. Tag is always related to existing resources. The following shows tag model:

```
+------------------+        +------------------+
|     Network      |        |       Tag        |
+------------------+        +------------------+
| standard_attr_id +------> | standard_attr_id |
|                  |        | tag              |
|                  |        |                  |
+------------------+        +------------------+
```

Tag has two columns only and tag column is just string. These tags are defined per resource. Tag is unique in a resource but it can be overlapped throughout.

API

The following shows basic API for tag. Tag is regarded as a subresource of resource so API always includes id of resource related to tag.

Add a single tag on a network

```
PUT /v2.0/networks/{network_id}/tags/{tag}
```

Returns *201 Created*. If the tag already exists, no error is raised, it just returns the *201 Created* because the OpenStack Development Mailing List discussion told us that PUT should be no issue updating an existing tag.

Replace set of tags on a network

```
PUT /v2.0/networks/{network_id}/tags
```

with request payload

```
{
    'tags': ['foo', 'bar', 'baz']
}
```

Response

```
{
    'tags': ['foo', 'bar', 'baz']
}
```

Check if a tag exists or not on a network

```
GET /v2.0/networks/{network_id}/tags/{tag}
```

Remove a single tag on a network

```
DELETE /v2.0/networks/{network_id}/tags/{tag}
```

Remove all tags on a network

```
DELETE /v2.0/networks/{network_id}/tags
```

PUT and DELETE for collections are the motivation of extending the API framework.

Note

Much of this document discusses upgrade considerations for the Neutron reference implementation using Neutrons agents. Its expected that each Neutron plugin provides its own documentation that discusses upgrade considerations specific to that choice of backend. For example, OVN does not use Neutron agents, but does have a local controller that runs on each compute node. OVN supports rolling upgrades, but information about how that works should be covered in the documentation for the OVN Neutron plugin.

Upgrade strategy

There are two general upgrade scenarios supported by Neutron:

1. All services are shut down, code upgraded, then all services are started again.
2. Services are upgraded gradually, based on operator service windows.

The latter is the preferred way to upgrade an OpenStack cloud, since it allows for more granularity and less service downtime. This scenario is usually called rolling upgrade.

Rolling upgrade

Rolling upgrades imply that during some interval of time there will be services of different code versions running and interacting in the same cloud. It puts multiple constraints onto the software.

1. older services should be able to talk with newer services.
2. older services should not require the database to have older schema (otherwise newer services that require the newer schema would not work).

More info on rolling upgrades in OpenStack.

Those requirements are achieved in Neutron by:

1. If the Neutron backend makes use of Neutron agents, the Neutron server have backwards compatibility code to deal with older messaging payloads.
2. isolating a single service that accesses database (neutron-server).

To simplify the matter, its always assumed that the order of service upgrades is as following:

1. first, all neutron-servers are upgraded.
2. then, if applicable, neutron agents are upgraded.

This approach allows us to avoid backwards compatibility code on agent side and is in line with other OpenStack projects that support rolling upgrades (specifically, nova).

Server upgrade

Neutron-server is the very first component that should be upgraded to the new code. Its also the only component that relies on new database schema to be present, other components communicate with the cloud through AMQP and hence do not depend on particular database state.

Database upgrades are implemented with alembic migration chains.

Database upgrade is split into two parts:

1. `neutron-db-manage upgrade --expand`
2. `neutron-db-manage upgrade --contract`

Each part represents a separate alembic branch.

The former step can be executed while old neutron-server code is running. The latter step requires *all* neutron-server instances to be shut down. Once its complete, neutron-servers can be started again.

Note

Full shutdown of neutron-server instances can be skipped depending on whether there are pending contract scripts not applied to the database:

```
$ neutron-db-manage has_offline_migrations
Command will return a message if there are pending contract scripts.
```

More info on alembic scripts.

Agents upgrade

Note

This section does not apply when the cloud does not use AMQP agents to provide networking services to instances. In that case, other backend specific upgrade instructions may also apply.

Once neutron-server services are restarted with the new database schema and the new code, its time to upgrade Neutron agents.

Note that in the meantime, neutron-server should be able to serve AMQP messages sent by older versions of agents which are part of the cloud.

The recommended order of agent upgrade (per node) is:

1. first, L2 agents (openvswitch, sr-iov).
2. then, all other agents (L3, DHCP, Metadata,).

The rationale of the agent upgrade order is that L2 agent is usually responsible for wiring ports for other agents to use, so its better to allow it to do its job first and then proceed with other agents that will use the already configured ports for their needs.

Each network/compute node can have its own upgrade schedule that is independent of other nodes.

AMQP considerations

Since its always assumed that neutron-server component is upgraded before agents, only the former should handle both old and new RPC versions.

The implication of that is that no code that handles UnsupportedVersion oslo.messaging exceptions belongs to agent code.

Notifications

For notifications that are issued by neutron-server to listening agents, special consideration is needed to support rolling upgrades. In this case, a newer controller sends newer payload to older agents.

Until we have proper RPC version pinning feature to enforce older payload format during upgrade (as its implemented in other projects like nova), we leave our agents resistant against unknown arguments sent as part of server notifications. This is achieved by consistently capturing those unknown arguments with keyword arguments and ignoring them on agent side; and by not enforcing newer RPC entry point versions on server side.

This approach is not ideal, because it makes RPC API less strict. Thats why other approaches should be considered for notifications in the future.

More information about RPC versioning.

Interface signature

An RPC interface is defined by its name, version, and (named) arguments that it accepts. There are no strict guarantees that arguments will have expected types or meaning, as long as they are serializable.

Message content versioning

To provide better compatibility guarantees for rolling upgrades, RPC interfaces could also define specific format for arguments they accept. In OpenStack world, its usually implemented using oslo.versionedobjects library, and relying on the library to define serialized form for arguments that are passed through AMQP wire.

Note that Neutron has *not* adopted oslo.versionedobjects library for its RPC interfaces yet (except for QoS feature).

More information about RPC callbacks used for QoS.

Networking backends

Backend software upgrade should not result in any data plane disruptions. Meaning, e.g. Open vSwitch L2 agent should not reset flows or rewire ports; Neutron L3 agent should not delete namespaces left by older version of the agent; Neutron DHCP agent should not require immediate DHCP lease renewal; etc.

The same considerations apply to setups that do not rely on agents. Meaning, f.e. OpenDaylight or OVN controller should not break data plane connectivity during its upgrade process.

Upgrade testing

Grenade is the OpenStack project that is designed to validate upgrade scenarios.

Currently, only offline (non-rolling) upgrade scenario is validated in Neutron gate. The upgrade scenario follows the following steps:

1. the old cloud is set up using latest stable release code
2. all services are stopped
3. code is updated to the patch under review
4. new database migration scripts are applied, if needed
5. all services are started
6. the new cloud is validated with a subset of tempest tests

The scenario validates that no configuration option names are changed in one cycle. More generally, it validates that the new cloud is capable of running using the old configuration files. It also validates that database migration scripts can be executed.

The scenario does *not* validate AMQP versioning compatibility.

Other projects (for example Nova) have so called partial grenade jobs where some services are left running using the old version of code. Such a job would be needed in Neutron gate to validate rolling upgrades for the project. Till that time, its all up to reviewers to catch compatibility issues in patches on review.

Another hole in testing belongs to split migration script branches. Its assumed that an old cloud can successfully run after expand migration scripts from the new cloud are applied to its database; but its not validated in gate.

Review guidelines

There are several upgrade related gotchas that should be tracked by reviewers.

First things first, a general advice to reviewers: make sure new code does not violate requirements set by global OpenStack deprecation policy.

Now to specifics:

1. Configuration options:
   - options should not be dropped from the tree without waiting for deprecation period (currently its one development cycle long) and a deprecation message issued if the deprecated option is used.
   - option values should not change their meaning between releases.
2. Data plane:
   - agent restart should not result in data plane disruption (no Open vSwitch ports reset; no network namespaces deleted; no device names changed).
3. RPC versioning:
   - no RPC version major number should be bumped before all agents had a chance to upgrade (meaning, at least one release cycle is needed before compatibility code to handle old clients is stripped from the tree).
   - no compatibility code should be added to agent side of AMQP interfaces.
   - server code should be able to handle all previous versions of agents, unless the major version of an interface is bumped.
   - no RPC interface arguments should change their meaning, or names.
   - new arguments added to RPC interfaces should not be mandatory. It means that server should be able to handle old requests, without the new argument specified. Also, if the argument is not passed, the old behaviour before the addition of the argument should be retained.
   - minimal client version must not be bumped for server initiated notification changes for at least one cycle.
4. Database migrations:
   - migration code should be split into two branches (contract, expand) as needed. No code that is unsafe to execute while neutron-server is running should be added to expand branch.
   - if possible, contract migrations should be minimized or avoided to reduce the time when API endpoints must be down during database upgrade.

14.6.2 Module Reference

Todo

Add in all the big modules as automodule indexes.

14.7 OVN Driver

14.7.1 OVN backend

OVN Tools

This document offers details on Neutron tools available for assisting with using the Open Virtual Network (OVN) backend.

Patches and Cherry-picks

Overview

As described in the ovn-migration blueprint, Neutrons OVN ML2 plugin has merged to the Neutron repository as of the Ussuri release. With that, special care must be taken to apply Neutron changes to the proper stable branches of the networking-ovn repo.

Note

These scripts are generic enough to work on any patch file, but particularly handy with the networking-ovn migration.

tools/files_in_patch.py

Use this to show files that are changed in a patch file.

```
$ # Make a patch to use as example
$ git show > /tmp/commit.patch

$ ./tools/files_in_patch.py /tmp/commit.patch | grep .py
tools/download_gerrit_change.py
tools/files_in_patch.py
tools/migrate_names.py
```

tools/download_gerrit_change.py

This tool is needed by `migrate_names.py` (see below), but it can be used independently. Given a Gerrit change id, it will fetch the latest patchset of the change from review.opendev.org as a patch file. The output can be stdout or an optional filename.

```
$ ./tools/download_gerrit_change.py --help
Usage: download_gerrit_change.py [OPTIONS] GERRIT_CHANGE

Options:
  -o, --output_patch TEXT  Output patch file. Default: stdout
  -g, --gerrit_url TEXT    The url to Gerrit server  [default:
                           https://review.opendev.org/]
  -t, --timeout INTEGER    Timeout, in seconds  [default: 10]
  --help                   Show this message and exit.

$ ./tools/download_gerrit_change.py 698863 -o /tmp/change.patch
```

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```
$ ./tools/files_in_patch.py /tmp/change.patch
networking_ovn/ml2/mech_driver.py
networking_ovn/ml2/trunk_driver.py
networking_ovn/tests/unit/ml2/test_mech_driver.py
networking_ovn/tests/unit/ml2/test_trunk_driver.py
```

tools/migrate_names.py

Use this tool to modify the name of the files in a patchfile so it can be converted to/from the legacy networking-ovn and Neutron repositories.

The mapping of how the files are renamed is based on `migrate_names.txt`, which is located in the same directory where `migrate_names.py` is installed. That behavior can be modified via the `--mapfile` option. More information on how the map is parsed is provided in the header section of that file.

```
$ ./tools/migrate_names.py --help
Usage: migrate_names.py [OPTIONS]

Options:
  -i, --input_patch TEXT    input_patch patch file or gerrit change
  -o, --output_patch TEXT   Output patch file. Default: stdout
  -m, --mapfile PATH        Data file that specifies mapping to be applied to
                            input  [default: /home/user/openstack/neutron.git
                            /tools/migrate_names.txt]
  --reverse / --no-reverse  Map filenames from networking-ovn to Neutron repo
  --help                    Show this message and exit.
$ ./tools/migrate_names.py -i 701646 > /tmp/ovn_change.patch
$ ./tools/migrate_names.py -o /tmp/reverse.patch -i /tmp/ovn_change.patch --
↪reverse
$ diff /tmp/reverse.patch /tmp/ovn_change.patch | grep .py
< --- a/neutron/plugins/ml2/drivers/ovn/mech_driver/mech_driver.py
< +++ b/neutron/plugins/ml2/drivers/ovn/mech_driver/mech_driver.py
> --- a/networking_ovn/ml2/mech_driver.py
> +++ b/networking_ovn/ml2/mech_driver.py
<... snip ...>

$ ./tools/files_in_patch.py /tmp/ovn_change.patch
networking_ovn/ml2/mech_driver.py
networking_ovn/ml2/trunk_driver.py
networking_ovn/tests/unit/ml2/test_mech_driver.py
networking_ovn/tests/unit/ml2/test_trunk_driver.py
```

14.8 Dashboards

There is a collection of dashboards to help developers and reviewers located here.

14.8.1 CI Status Dashboards

Gerrit Dashboards

- Neutron priority reviews
- Neutron master branch reviews
- Neutron subproject reviews (master branch)
- Neutron stable branch reviews
- Neutron Infra reviews

These dashboard links can be generated by Gerrit Dashboard Creator. Useful dashboard definitions are found in `dashboards` directory.

Grafana Dashboards

Look for neutron and networking-* dashboard by names by going to the following link:

Grafana

For instance:

- Neutron
- Neutron-lib